48L

process design review · 578 steps · generated by CarbonEDA v0 on 2026-11-05 14:34 EST

541.7

kgCO2e per wafer start

380.6
EPW energy · 70%

42.4
MPW materials · 8%

118.8
GPW gases · 22%

wafer 300 mm · yield 1 · carbon intensity 0.43 kgCO2e/kWh · facility overhead 0.4 x tool energy
EPW 380.6 = tool 271.8 + facility overhead 108.7 kgCO2e (632 + 253 kWh)

flow technologies/flash/48L/48L_final.csv
layout technologies/flash/48L/48L_layout.oas
structure images from the spx run of 2026-10-31 10:07:32

CarbonEDA parameters

config/carbon_config.yaml and materials.yaml as used for every number in this report

EPW energy

(tool + 0.4 x tool facility overhead) x 0.43 kgCO2e/kWh

process area	kWh / step	steps
Dry_etch	1.33	45
RIE	1.33	146
HAR_etch	1.33	28
WLP3_etch	5.33	2
WLP6_etch	1.33	2
Ion_implantation	1.33	30
Diffusion	1.33	16
Metallization	2	15
Wet_etch	0.28	9
Wafer_clean	0.28	77
Resist	0.28	1
CMP	0.28	21
Deposition	1.09	14
Deposition_stack	0.545	55
CVD	1.09	73
ArF-i	0.36	6
ArF	0.33	9
KrF	0.33	18
Hg	0.33	11

GPW process gases

kg x (1 - utilization) x (1 - abatement) x GWP, plus byproducts kg/kg x (1 - abatement) x GWP; defaults: utilization 1.0, abatement 0.9

area	gas	g / step	upstream*	utilization	abatement	byproducts (kg per kg gas)
CVD	NF3	2.54	12.5	0.8	0.95	CF4 0.046, C2F6 0.045, CHF3 0.025, CH2F2 0.00086,...
Deposition_stack	NF3	1.27	12.5	0.8	0.95	CF4 0.046, C2F6 0.045, CHF3 0.025, CH2F2 0.00086,...
HAR_etch	C4F8	0.16	12.1	0.82	0.98	CF4 0.045, C2F6 0.031, C4F6 0.018, CHF3 0.012, CH...
HAR_etch	SF6	0.439	125.6	0.71	0.95	CF4 0.04, C2F6 0.044, CHF3 0.0037, CH2F2 2.9e-05
WLP3_etch	C4F8	0.641	12.1	0.82	0.98	CF4 0.045, C2F6 0.031, C4F6 0.018, CHF3 0.012, CH...
WLP3_etch	SF6	1.76	125.6	0.71	0.95	CF4 0.04, C2F6 0.044, CHF3 0.0037, CH2F2 2.9e-05
WLP6_etch	C4F8	0.16	12.1	0.82	0.98	CF4 0.045, C2F6 0.031, C4F6 0.018, CHF3 0.012, CH...
WLP6_etch	SF6	0.439	125.6	0.71	0.95	CF4 0.04, C2F6 0.044, CHF3 0.0037, CH2F2 2.9e-05
RIE	C4F8	0.16	12.1	0.82	0.98	CF4 0.045, C2F6 0.031, C4F6 0.018, CHF3 0.012, CH...
RIE	SF6	0.439	125.6	0.71	0.95	CF4 0.04, C2F6 0.044, CHF3 0.0037, CH2F2 2.9e-05

* kgCO2e per kg of gas made, counted in MPW

GWP (kgCO2e per kg, 100-yr) and abatement of every gas emitted

gas	GWP	abatement
CF4	7390	0.89
C2F6	12200	0.95
C4F6	0	0.98
C4F8	10300	0.98
CHF3	14800	0.98

gas	GWP	abatement
CH2F2	675	0.98
CH3F	116	0.98
NF3	17200	0.95
SF6	22800	0.95

MPW materials

blanket film over a 300 mm wafer: thickness x area x density x GWP; the wafer itself 26000 gCO2e (300 mm); yield 1

material	g/cm³	gCO2e/g
Ti	4.5	47.63

material	g/cm³	gCO2e/g
W	19.3	75.81

material	g/cm³	gCO2e/g
TaN	13.7	25.82

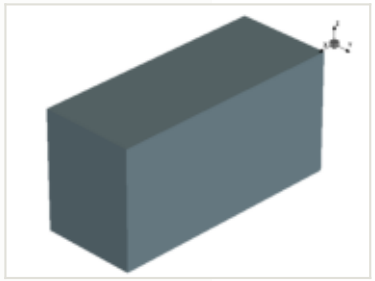
Step #20: Wafer

substrate · Wafer

26	26	4.8%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



this step

process area	Wafer
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	26 kgCO2e
GPW	0 kgCO2e
embodied carbon	**26 kgCO2e**

Wafer step: not counted as a process step (uncounted_process_areas)

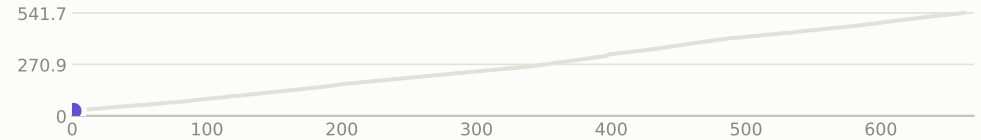
carbon so far, of the whole flow (kgCO2e)

EPW energy — 0 of 380.6

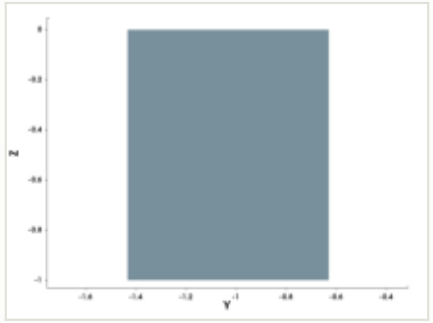
MPW materials — 26 of 42.4 (this step +26)

GPW gases — 0 of 118.8

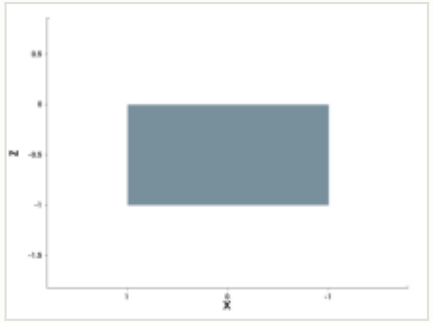
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



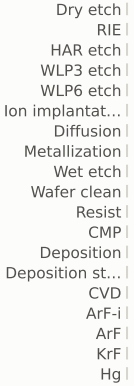
y cut at -1.032 µm



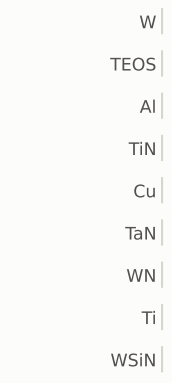
mask for this step

no mask for this step

steps by process area



materials deposited (g)



gases emitted (g)

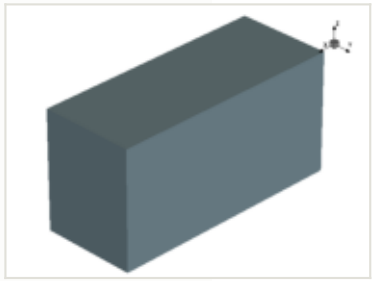
Step #30: Dry_etch

marking · Dry_etch

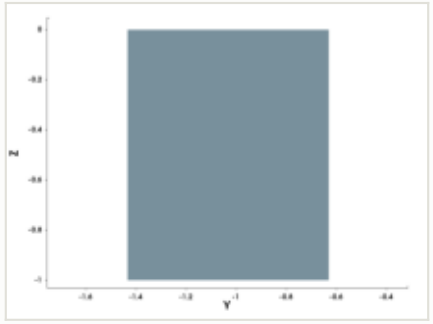
0.801	26.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

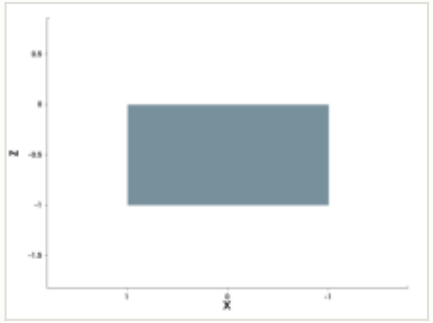
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

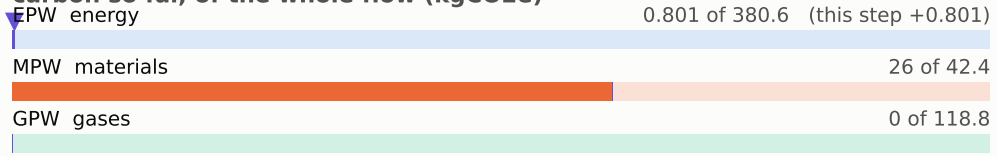
no mask for this step

this step

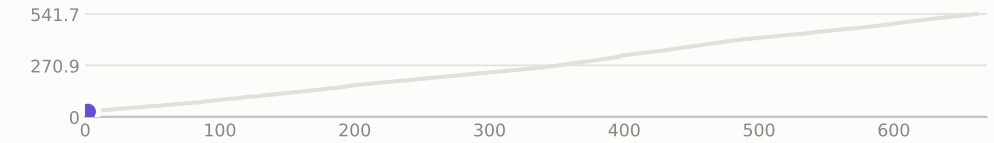
process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

carbon so far, of the whole flow (kgCO2e)

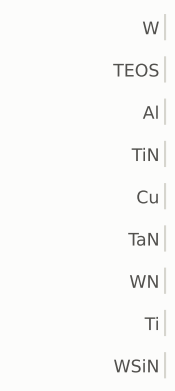


embodied carbon up to each step (kgCO2e per wafer)



steps by process area

materials deposited (g)



gases emitted (g)

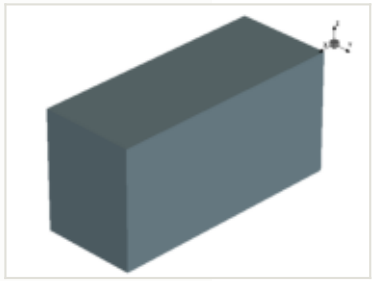
Step #40: Wafer_clean

clean · Wafer_clean

0.169	27	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



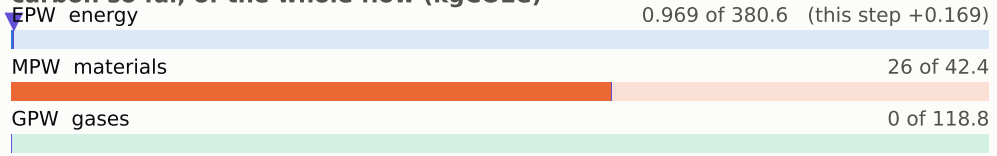
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

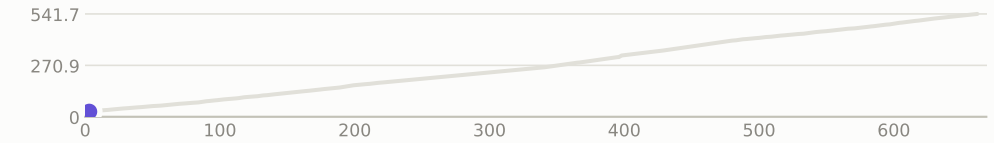
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

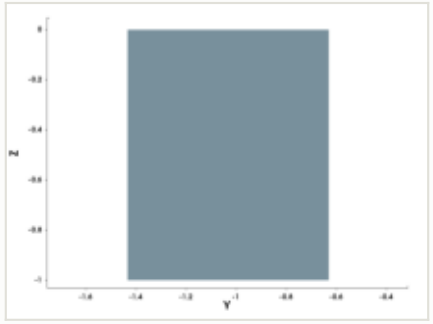
carbon so far, of the whole flow (kgCO2e)



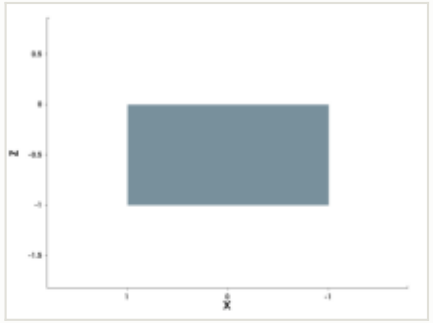
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



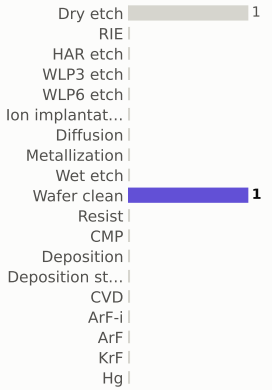
y cut at -1.032 µm



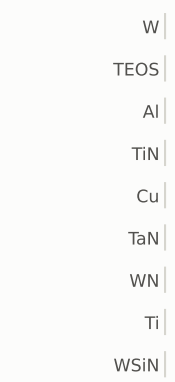
mask for this step

no mask for this step

steps by process area



materials deposited (g)



gases emitted (g)

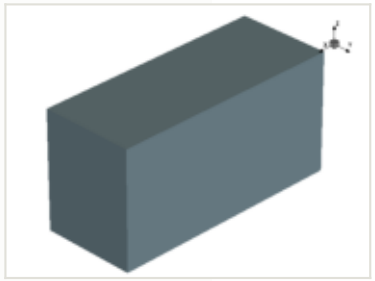
Step #60: Wafer_clean

clean · Wafer_clean

0.169	27.1	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



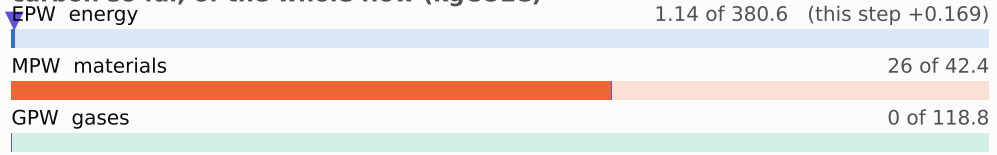
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

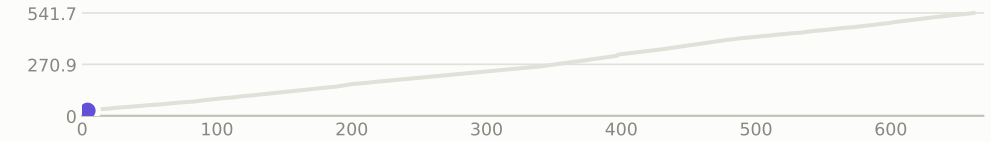
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

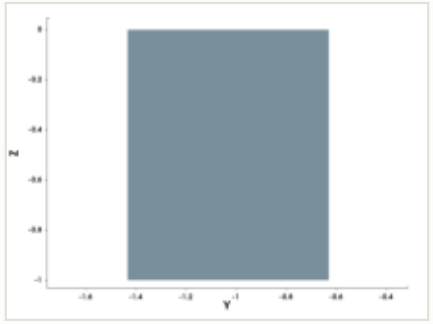
carbon so far, of the whole flow (kgCO2e)



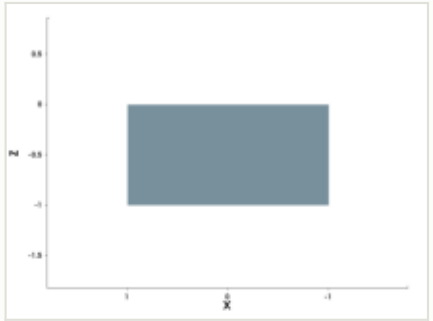
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



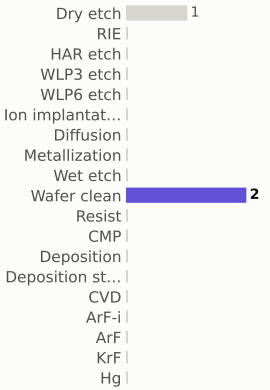
y cut at -1.032 µm



mask for this step

no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)

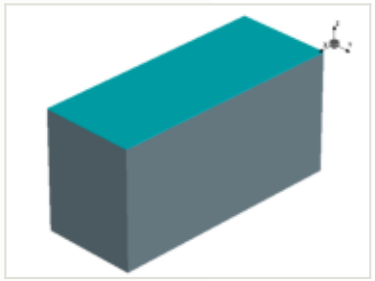
Step #70: Diffusion

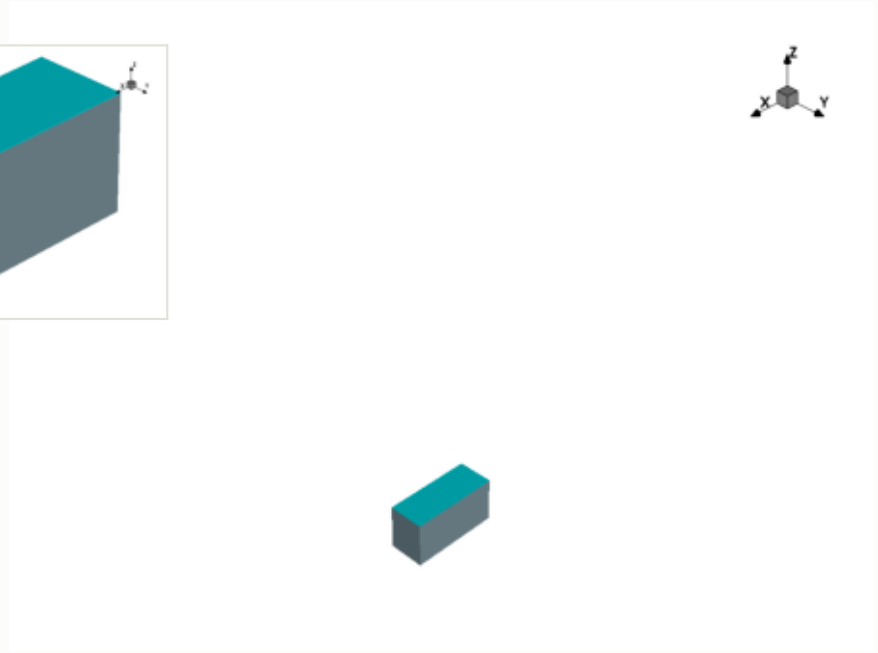
depo_isotropic · Diffusion

0.801	27.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



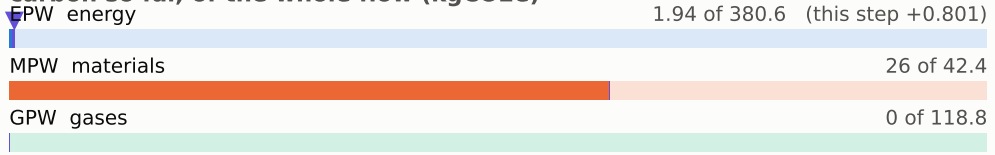


this step

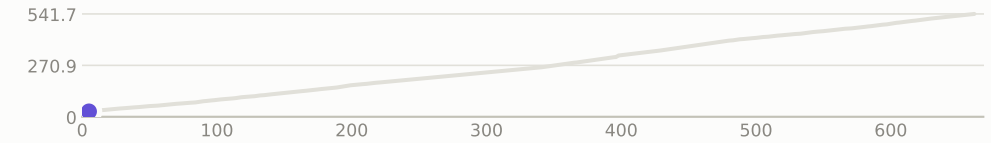
process area	Diffusion
material	—
thickness	0.0064 µm
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

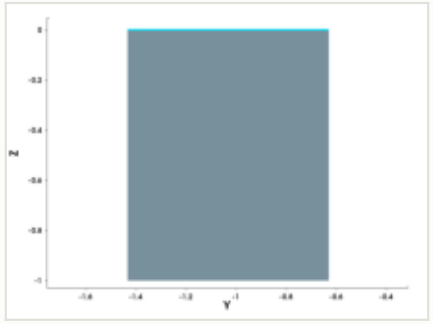
carbon so far, of the whole flow (kgCO2e)



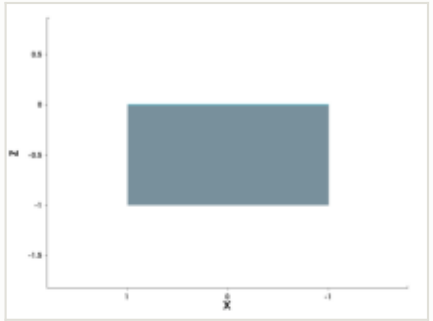
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



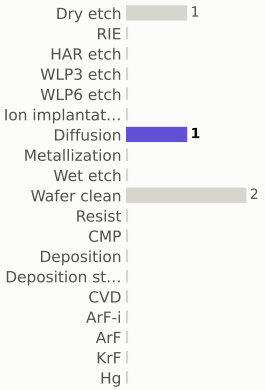
y cut at -1.032 µm



mask for this step

no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)

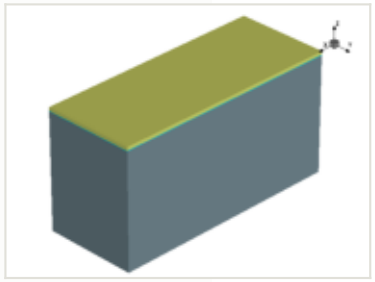
Step #80: CVD

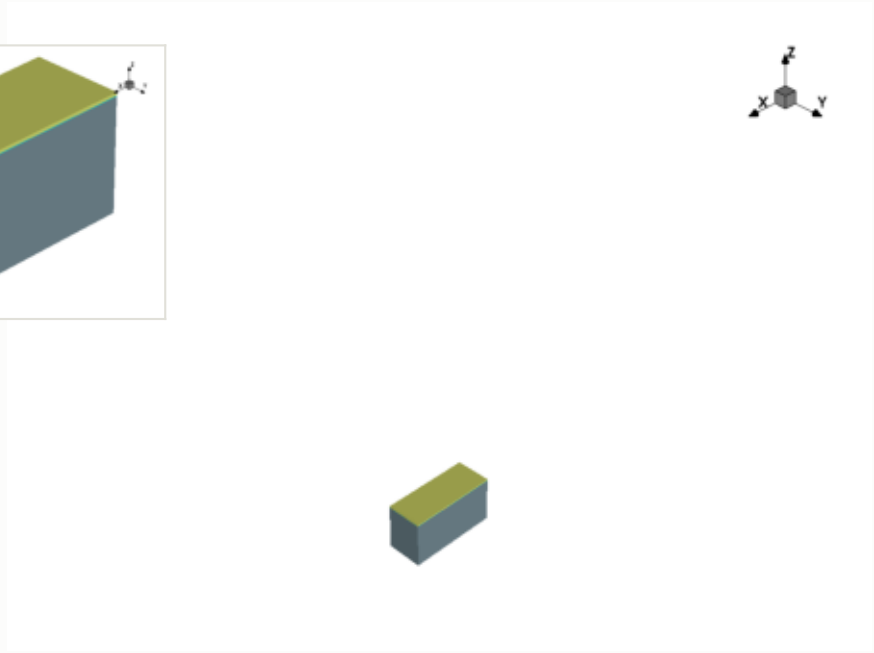
depo_isotropic · CVD

1.31	29.2	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



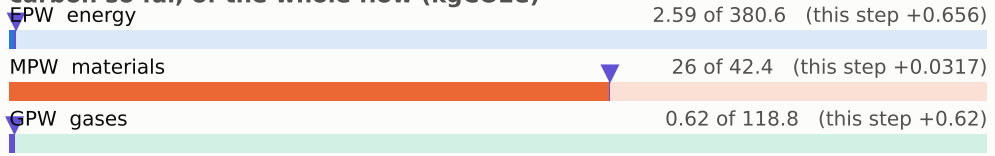


this step

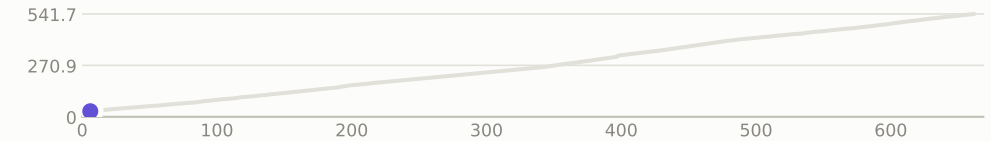
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

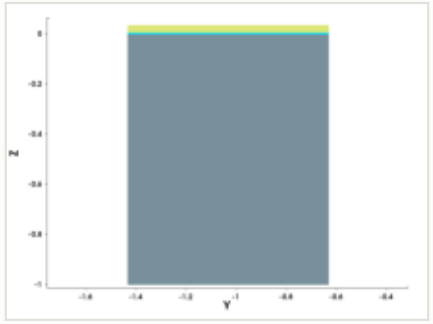
carbon so far, of the whole flow (kgCO2e)



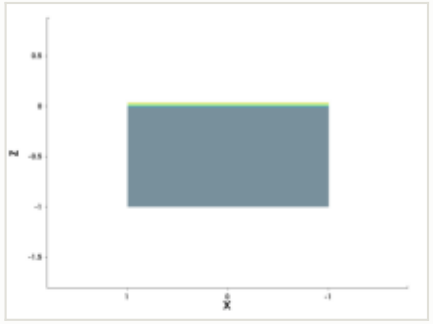
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



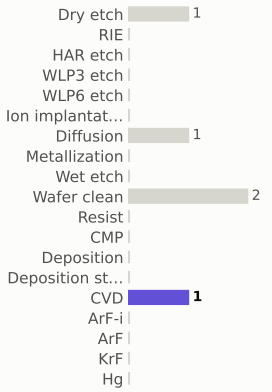
y cut at -1.032 µm



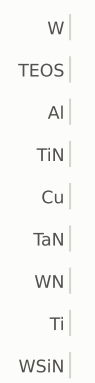
mask for this step

no mask for this step

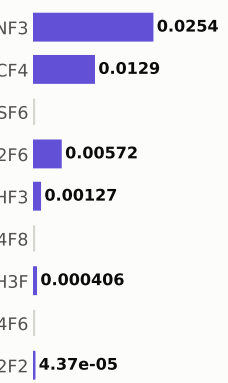
steps by process area



materials deposited (g)



gases emitted (g)



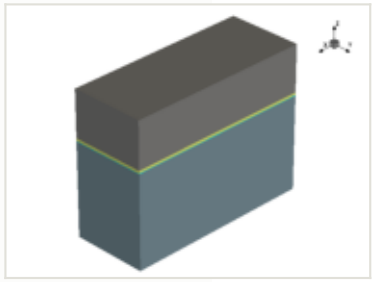
Step #90: KrF

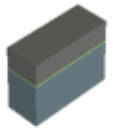
litho_patterning · KrF

0.199	29.4	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



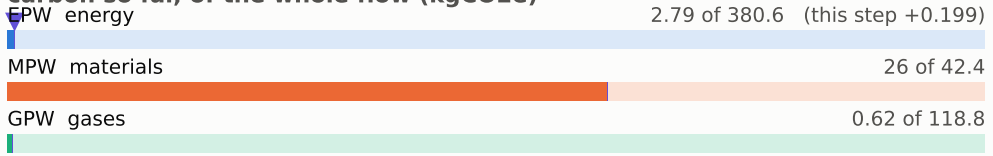


this step

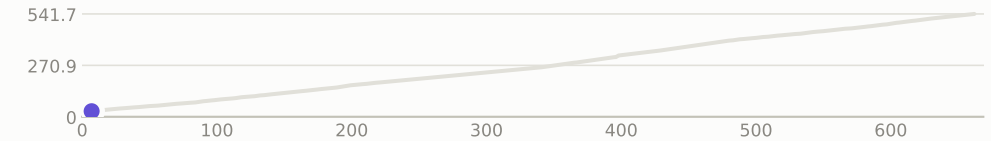
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

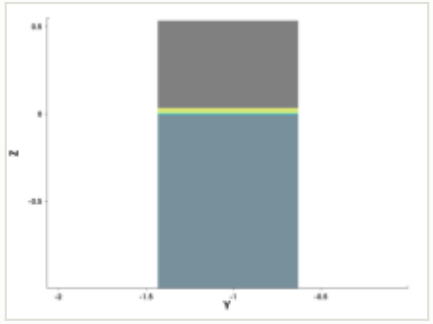
carbon so far, of the whole flow (kgCO2e)



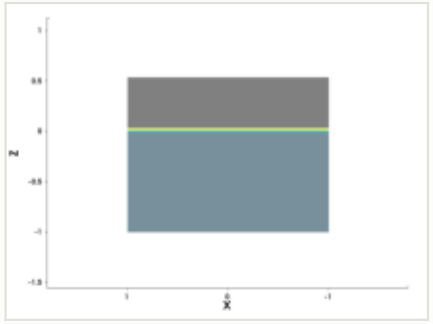
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

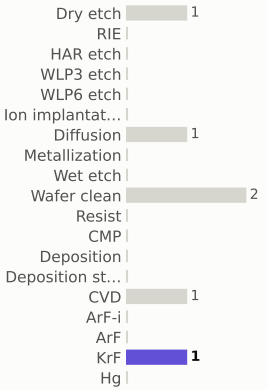


y cut at -1.032 µm

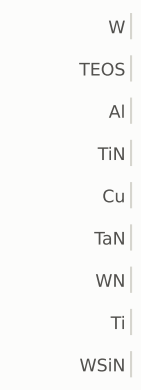


mask: layer 1:0, positive (OASIS)

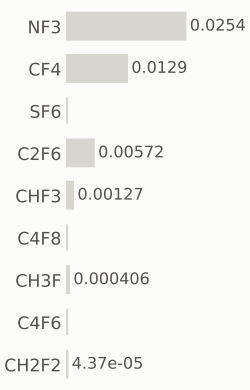
steps by process area



materials deposited (g)



gases emitted (g)



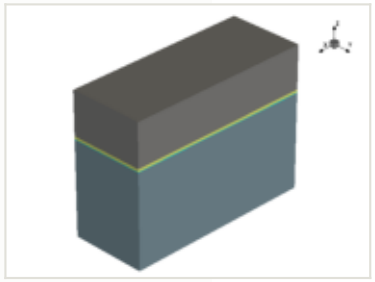
Step #100: RIE

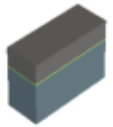
etch_anisotropic · RIE

1.04	30.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



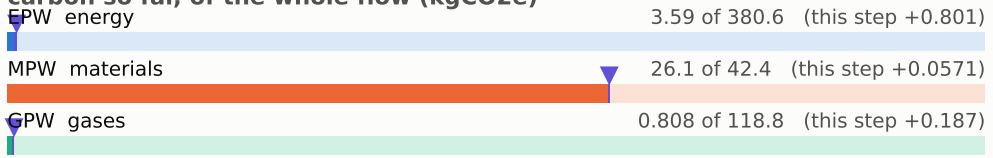


this step

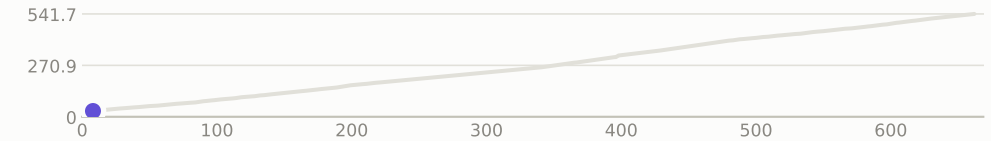
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

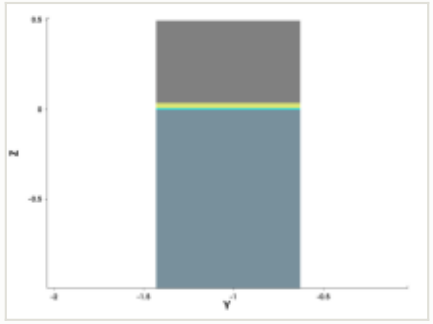
carbon so far, of the whole flow (kgCO2e)



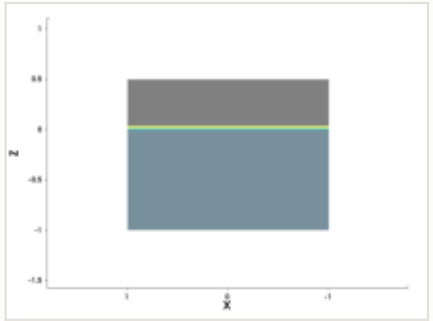
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



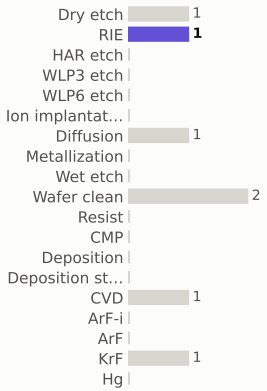
y cut at -1.032 µm



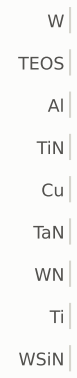
mask for this step

no mask for this step

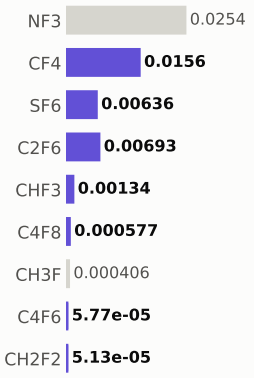
steps by process area



materials deposited (g)



gases emitted (g)



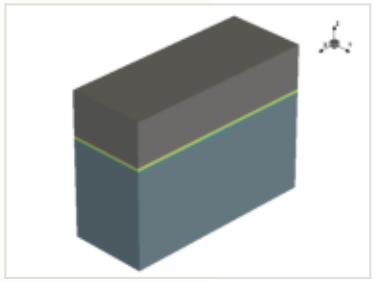
Step #110: RIE

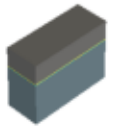
etch_anisotropic · RIE

1.04	31.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



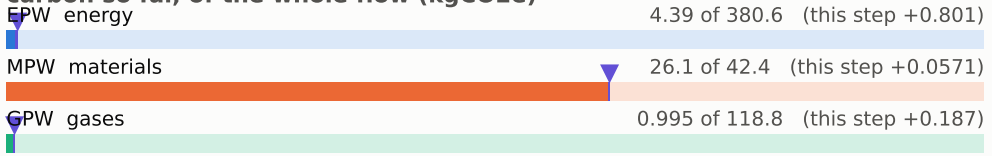


this step

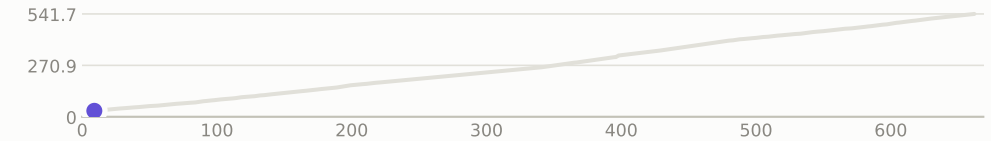
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

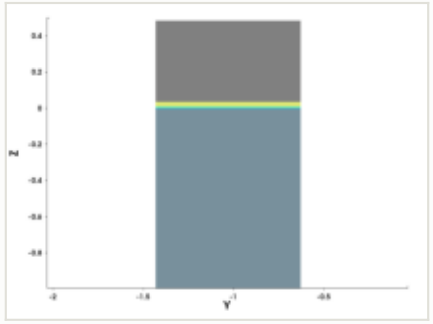
carbon so far, of the whole flow (kgCO2e)



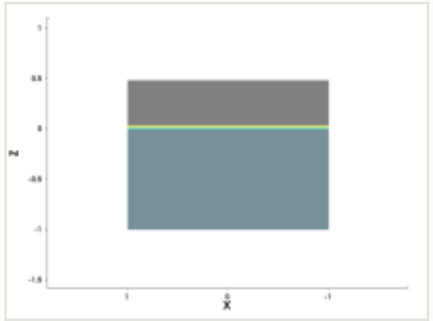
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



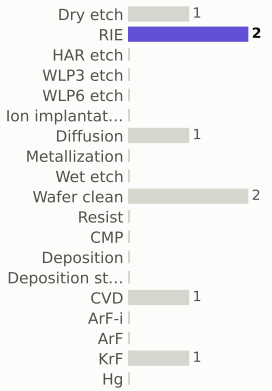
y cut at -1.032 µm



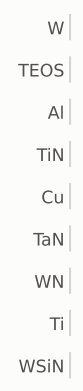
mask for this step

no mask for this step

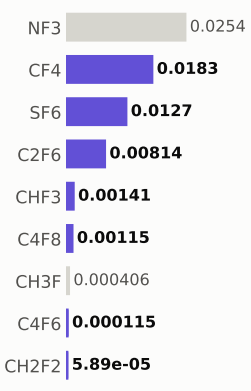
steps by process area



materials deposited (g)



gases emitted (g)



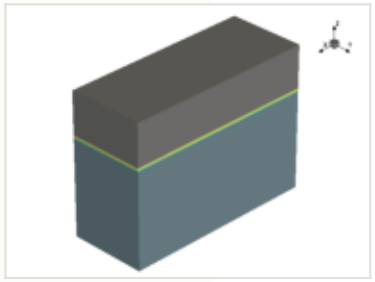
Step #120: RIE

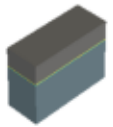
etch_anisotropic · RIE

1.04	32.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



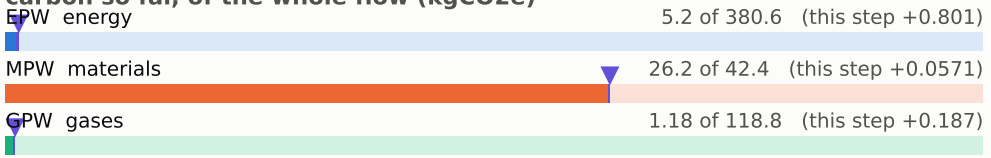


this step

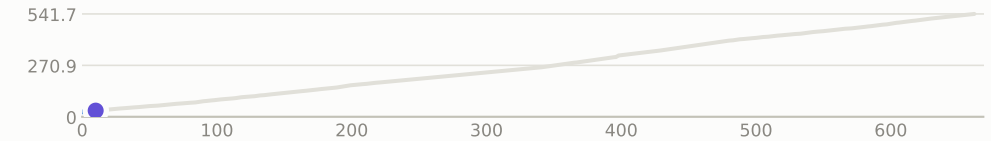
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

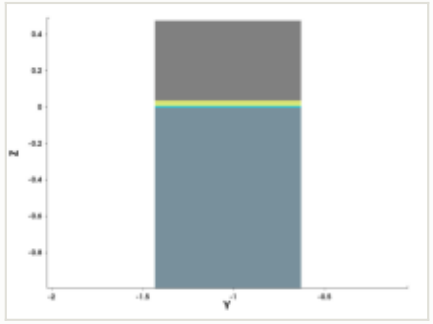
carbon so far, of the whole flow (kgCO2e)



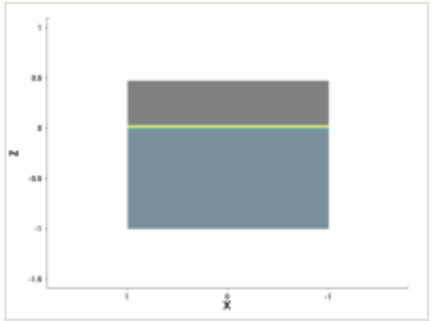
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



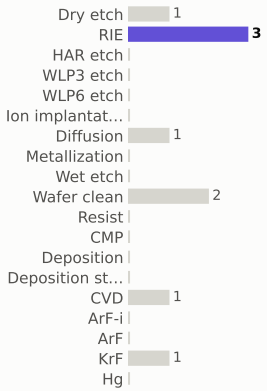
y cut at -1.032 µm



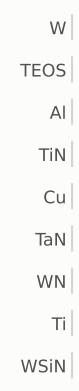
mask for this step

no mask for this step

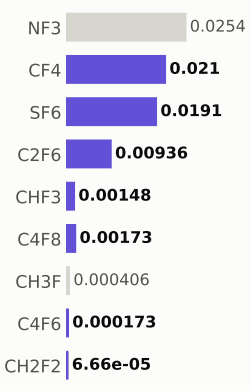
steps by process area



materials deposited (g)



gases emitted (g)



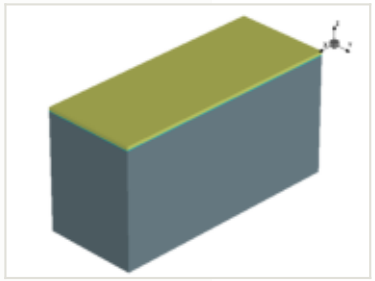
Step #130: Dry_etch

etch_strip · Dry_etch

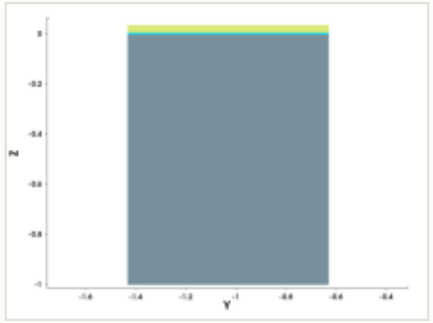
0.801	33.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

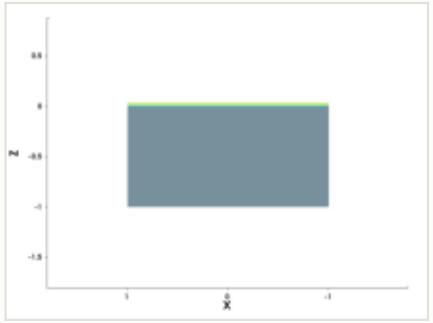
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



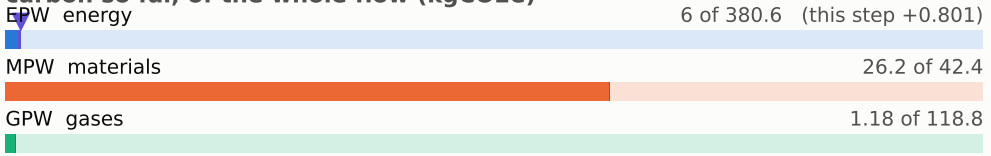
mask for this step

no mask for this step

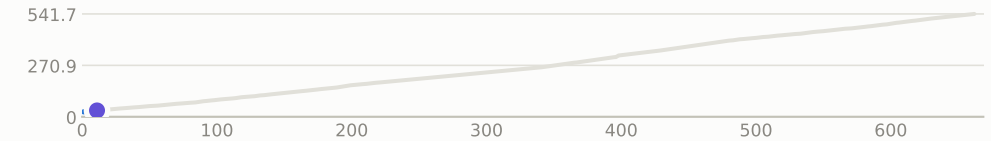
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

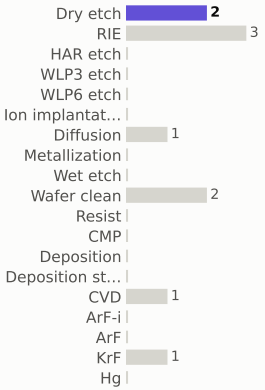
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

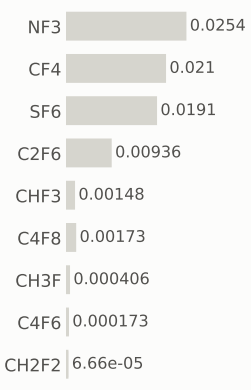


steps by process area



materials deposited (g)

gases emitted (g)



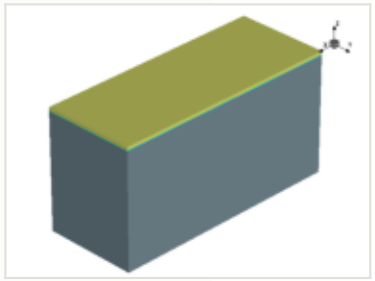
Step #140: Wafer_clean

clean · Wafer_clean

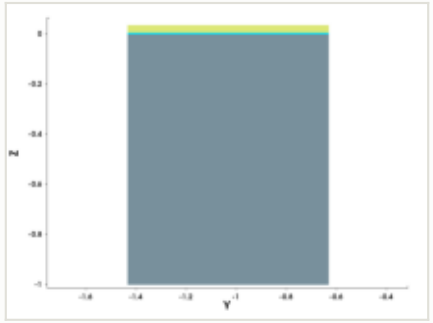
0.169	33.5	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

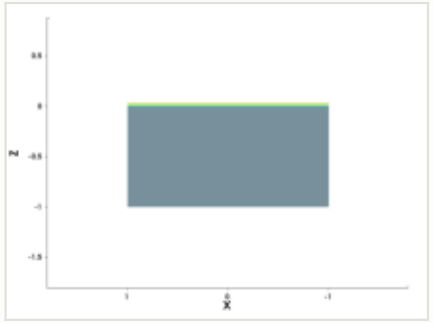
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

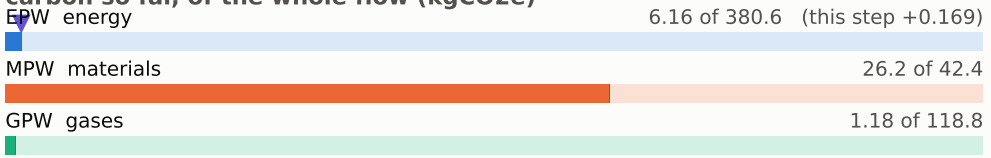
no mask for this step

this step

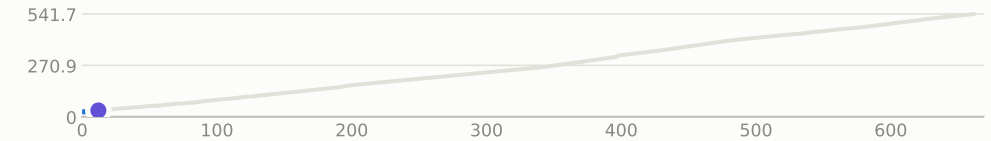
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

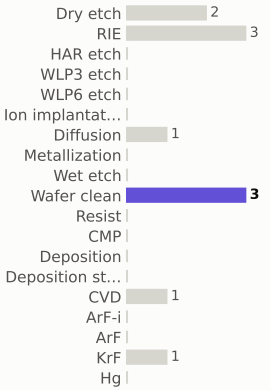
carbon so far, of the whole flow (kgCO2e)



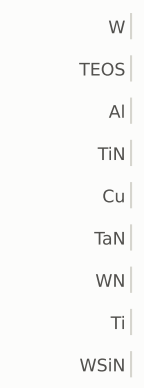
embodied carbon up to each step (kgCO2e per wafer)



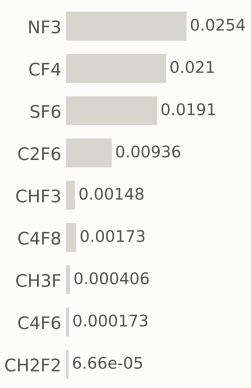
steps by process area



materials deposited (g)



gases emitted (g)



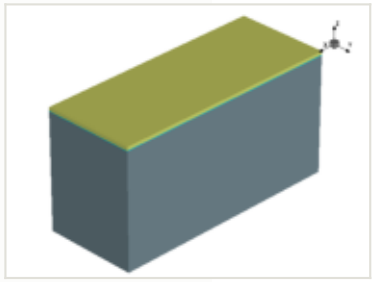
Step #150: Diffusion

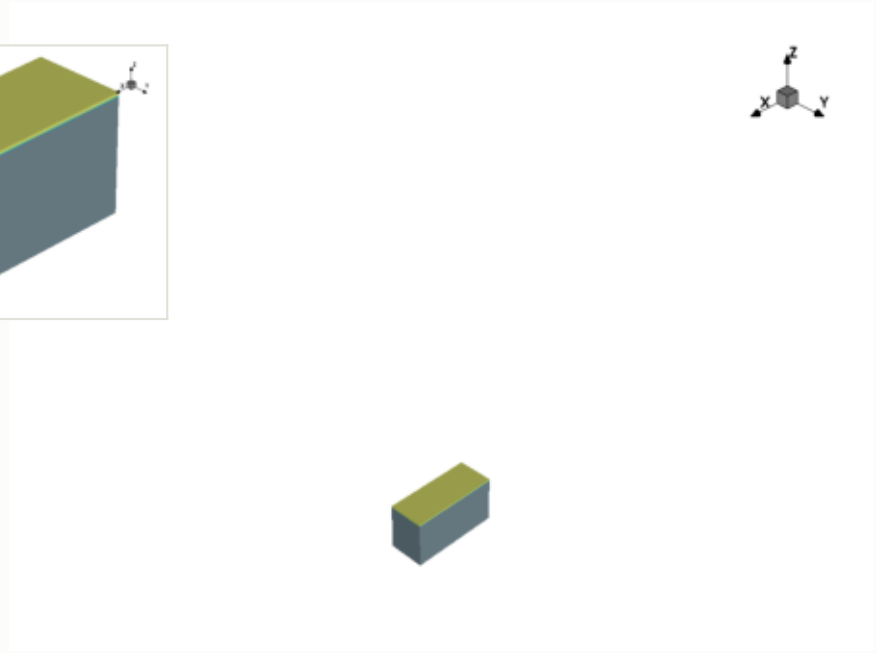
depo_isotropic · Diffusion

0.801	34.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



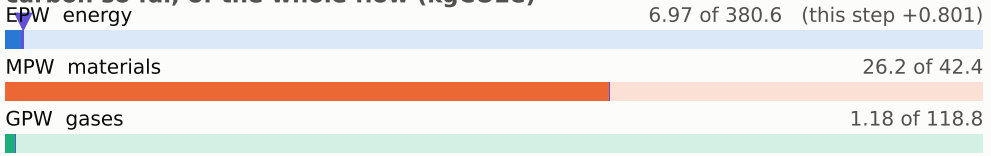


this step

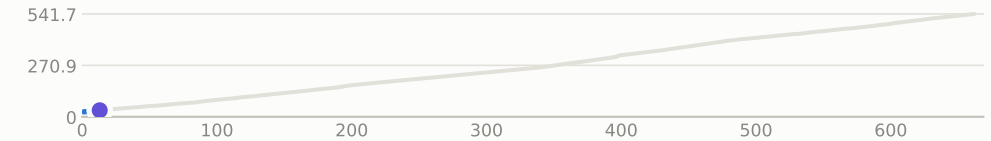
process area	Diffusion
material	—
thickness	0.042 µm
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

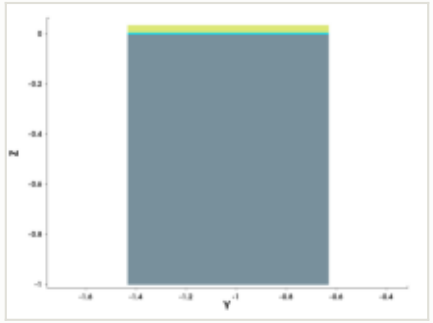
carbon so far, of the whole flow (kgCO2e)



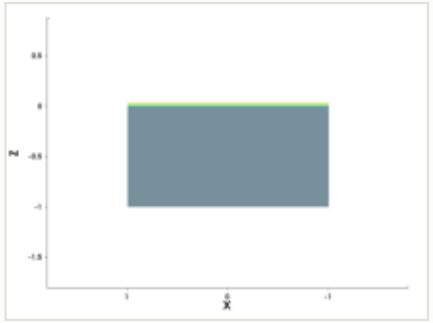
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



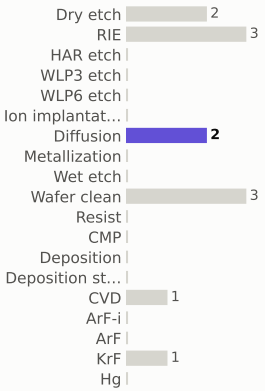
y cut at -1.032 µm



mask for this step

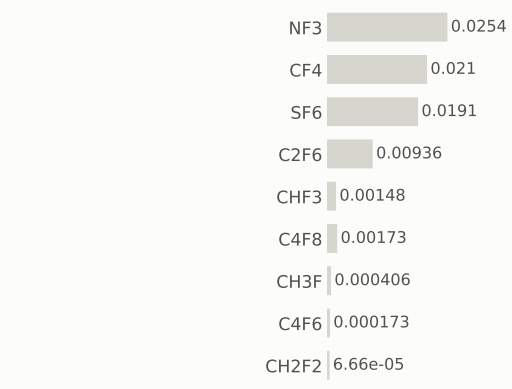
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



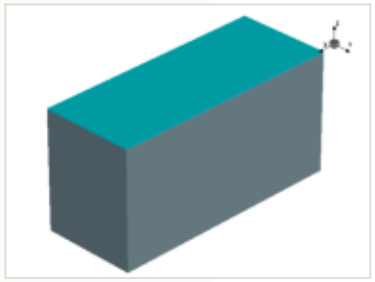
Step #160: Wet_etch

etch_isotropic · Wet_etch

0.169	34.5	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

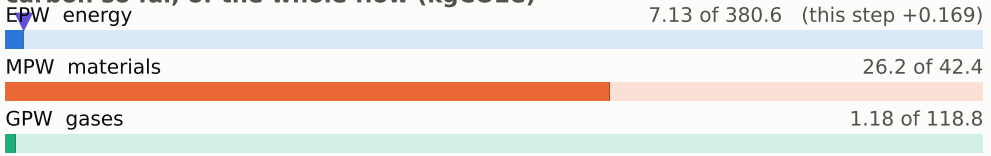


this step

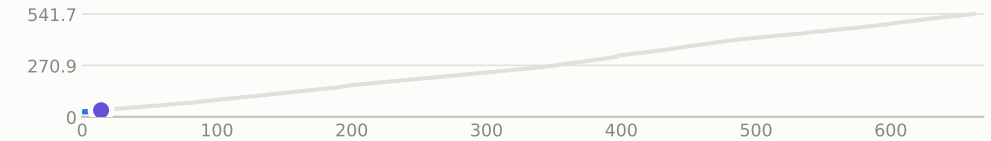
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

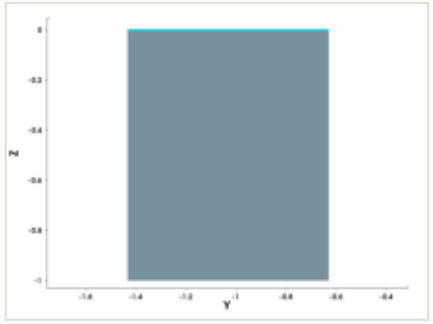
carbon so far, of the whole flow (kgCO2e)



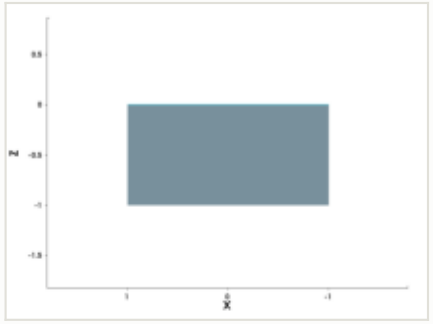
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



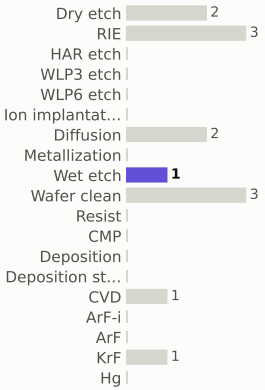
y cut at -1.032 µm



mask for this step

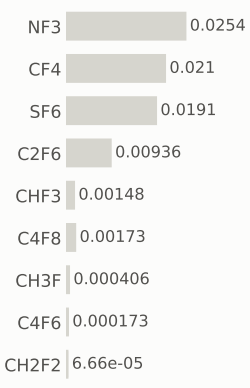
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



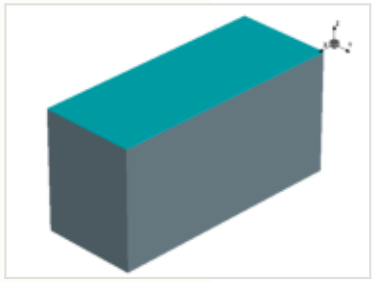
Step #170: Wafer_clean

clean · Wafer_clean

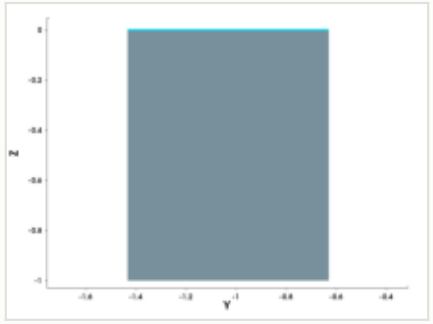
0.169	34.7	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

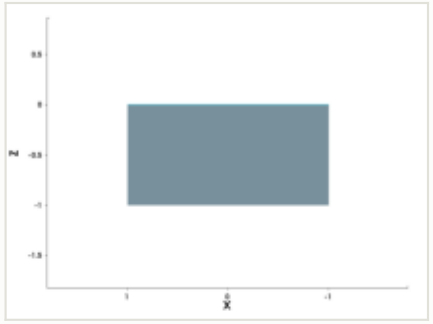
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

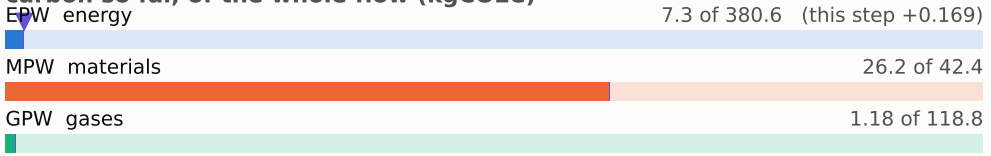
no mask for this step

this step

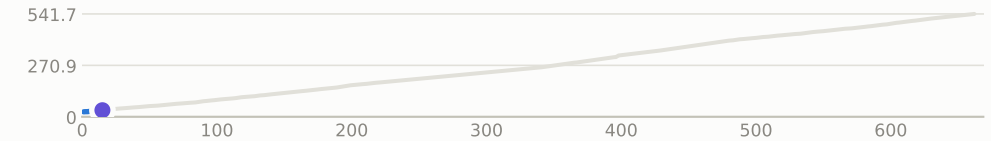
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

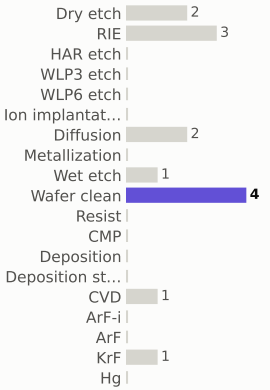
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

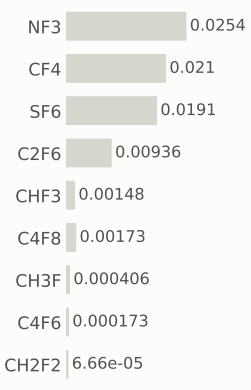


steps by process area



materials deposited (g)

gases emitted (g)



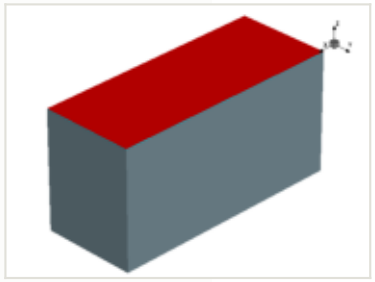
Step #180: Diffusion

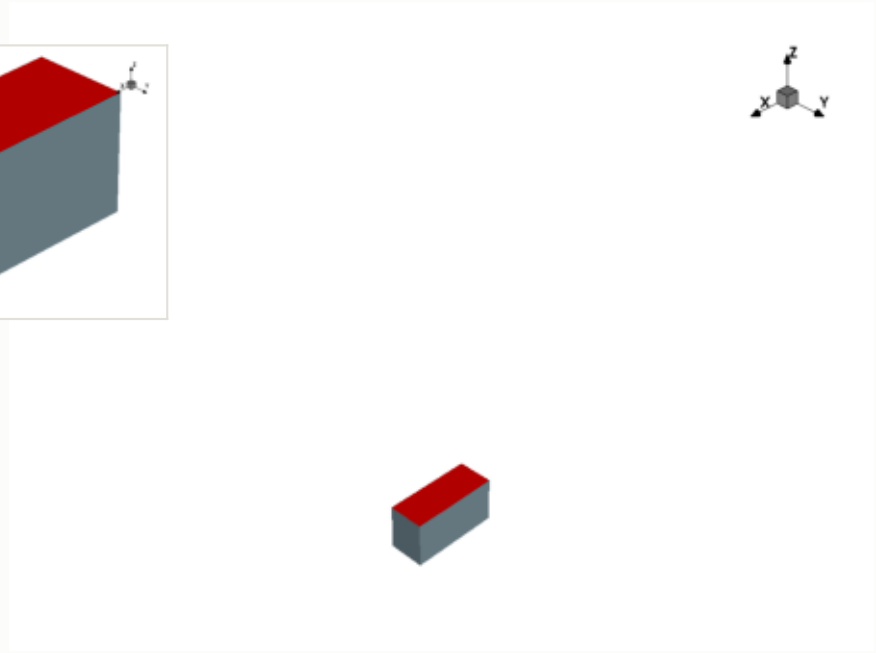
implant_ellipsoid · Diffusion

0.801	35.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



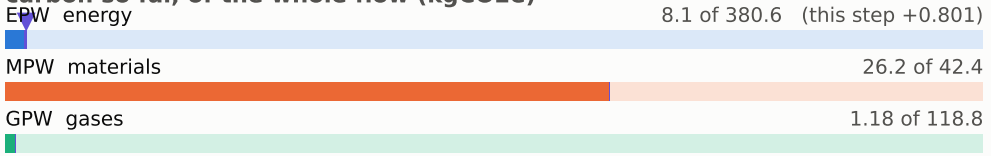


this step

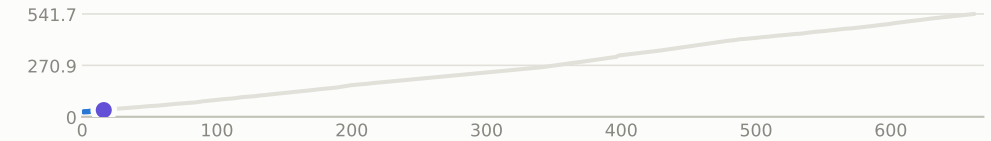
process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

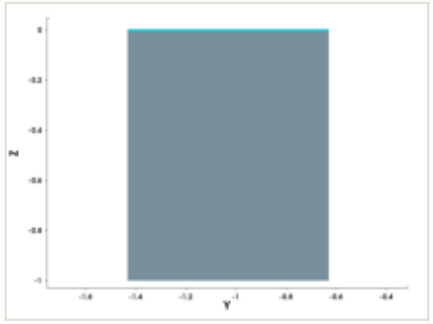
carbon so far, of the whole flow (kgCO2e)



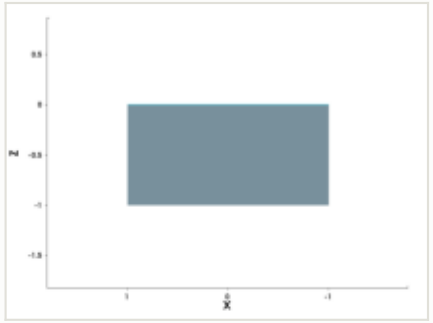
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



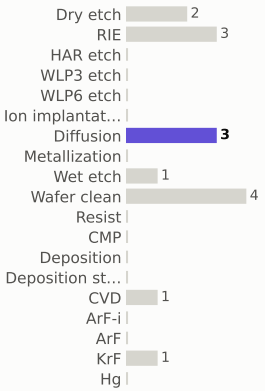
y cut at -1.032 µm



mask for this step

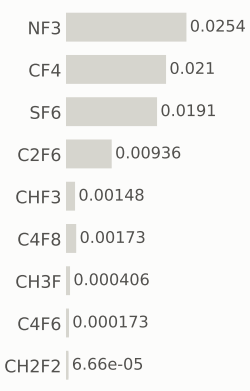
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



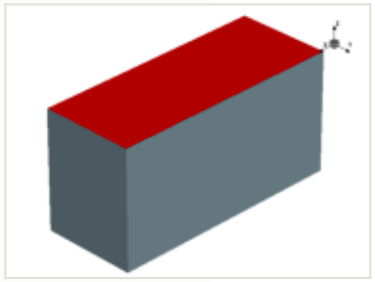
Step #190: Diffusion

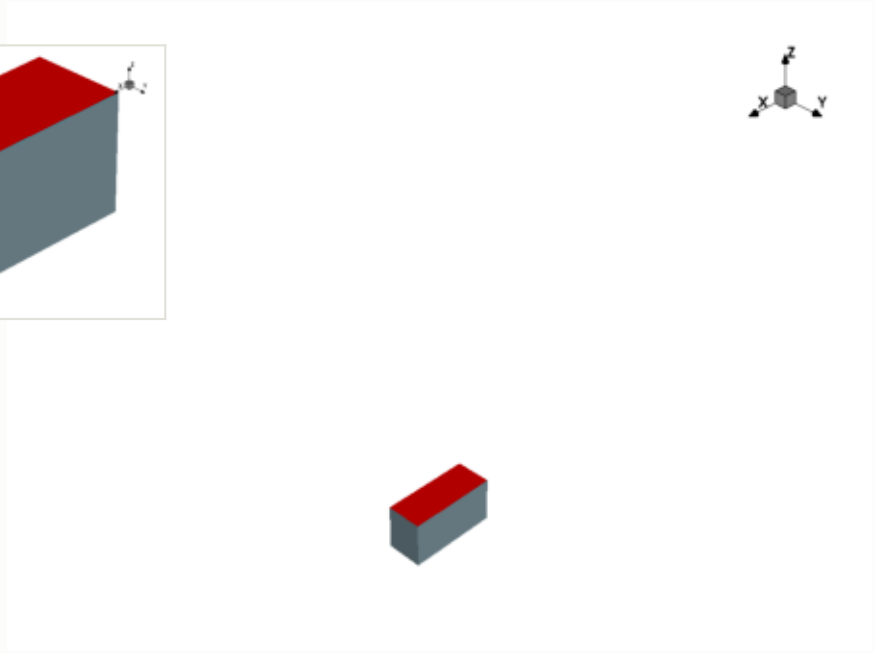
anneal · Diffusion

0.801	36.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

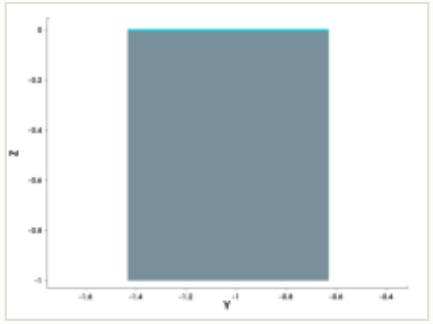
3D structure after this step

zoomed to fit

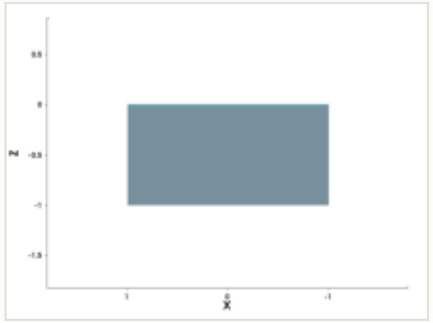




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

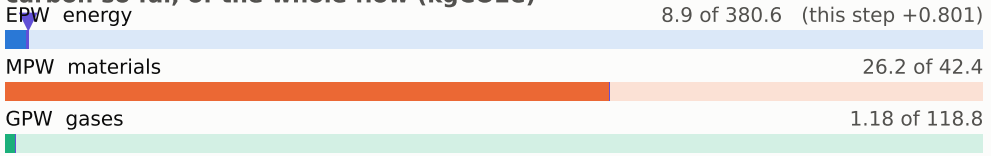
no mask for this step

this step

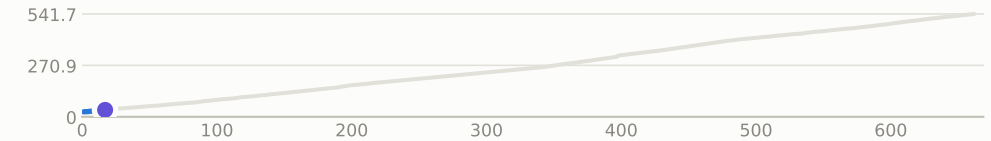
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

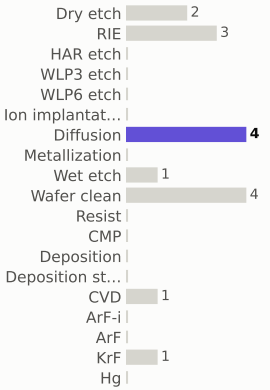
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

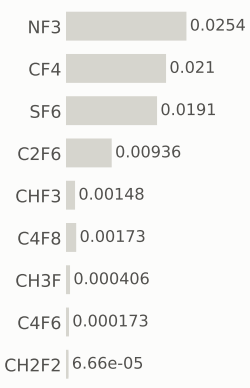


steps by process area



materials deposited (g)

gases emitted (g)



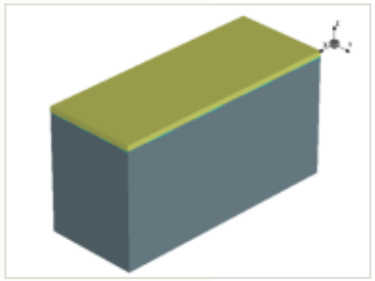
Step #200: CVD

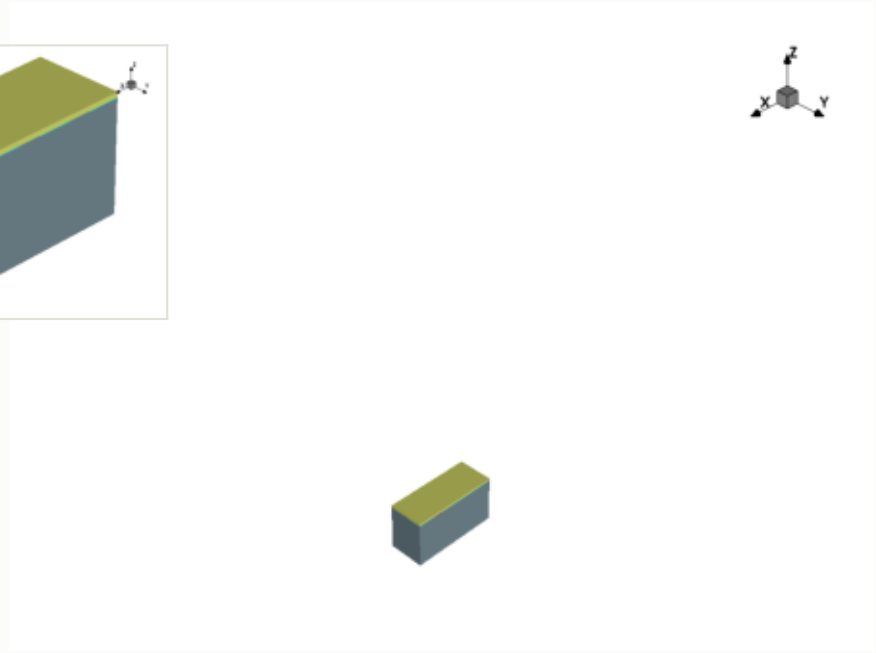
depo_isotropic · CVD

1.31	37.6	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



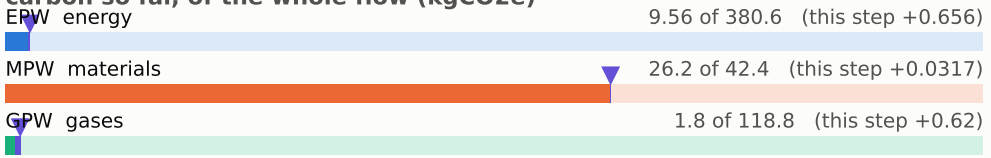


this step

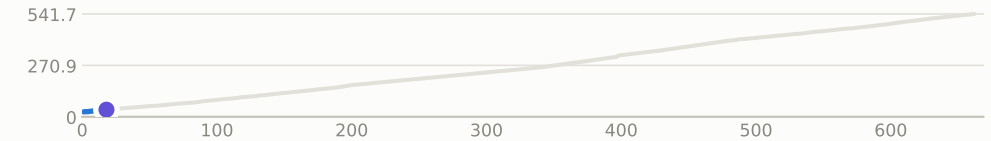
process area	CVD
material	—
thickness	0.05 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

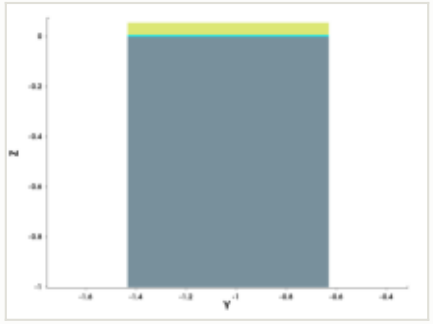
carbon so far, of the whole flow (kgCO2e)



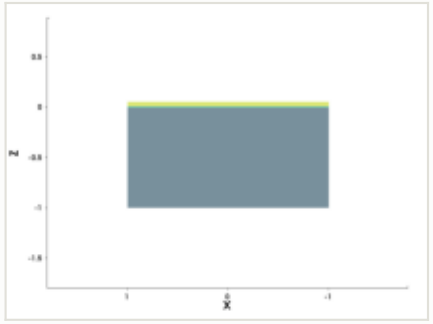
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



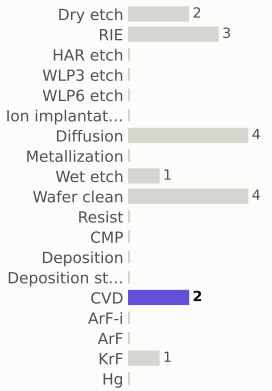
y cut at -1.032 µm



mask for this step

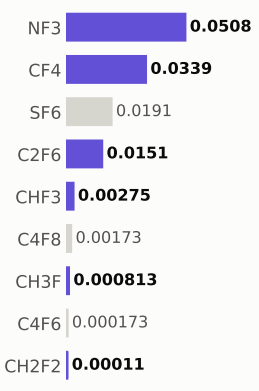
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



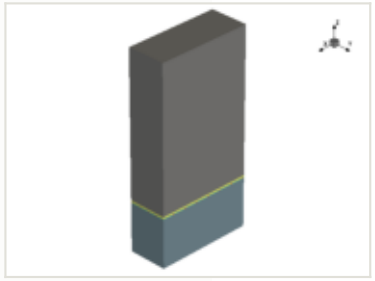
Step #220: Hg

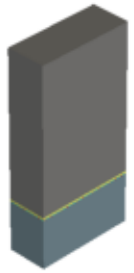
litho_patterning · Hg

0.199	37.8	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



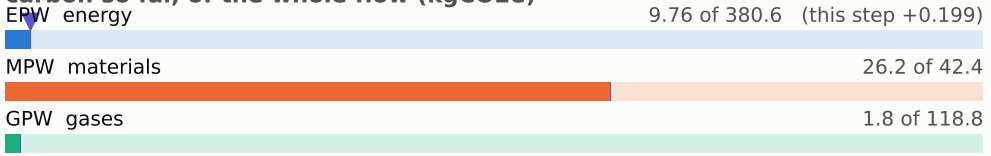


this step

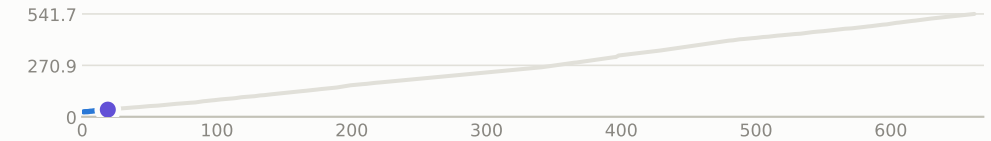
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

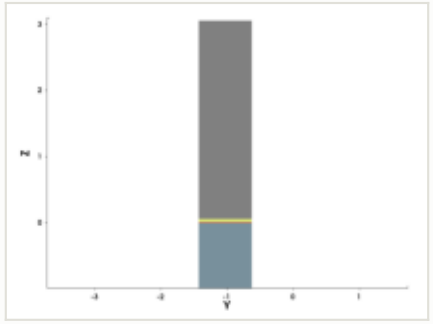
carbon so far, of the whole flow (kgCO2e)



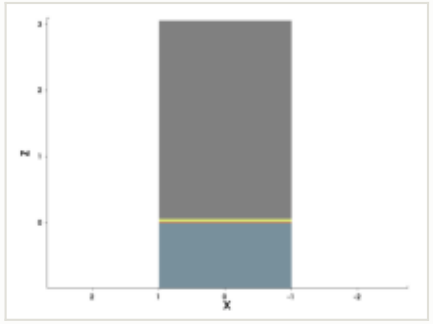
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

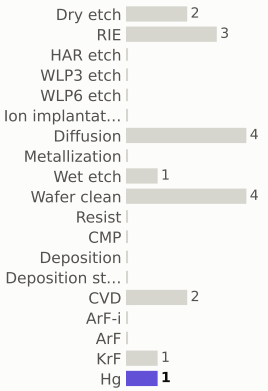


y cut at -1.032 µm



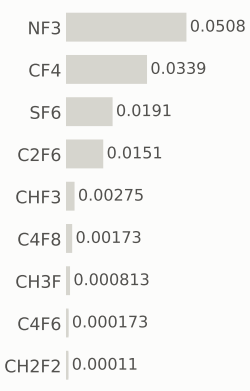
mask: layer 2:0, positive (OASIS)

steps by process area



materials deposited (g)

gases emitted (g)



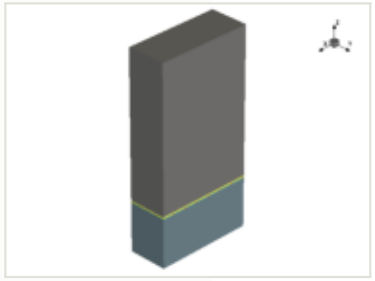
Step #230: Ion_implantation

implant_ellipsoid · Ion_implantation

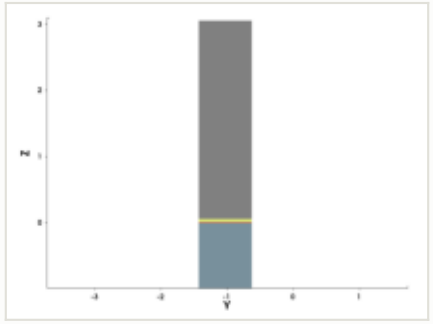
0.801	38.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

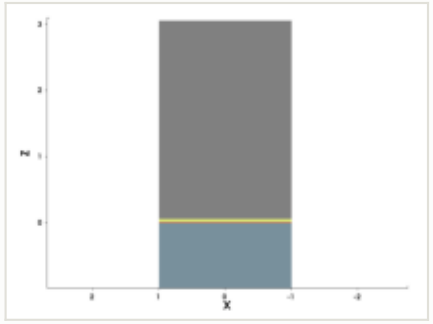
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

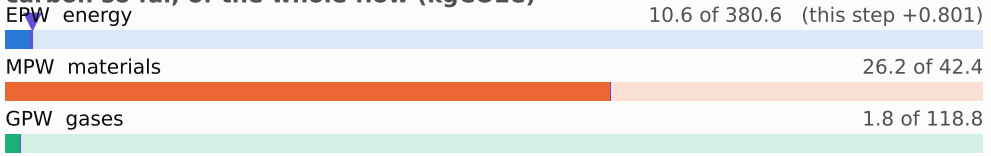
no mask for this step

this step

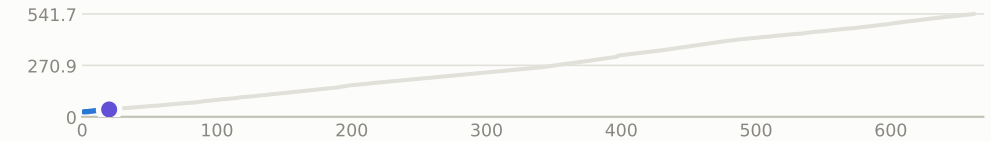
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

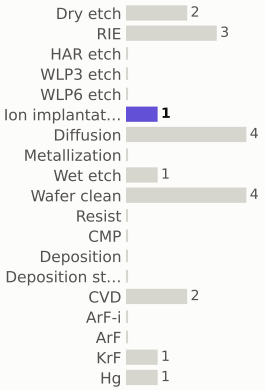
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

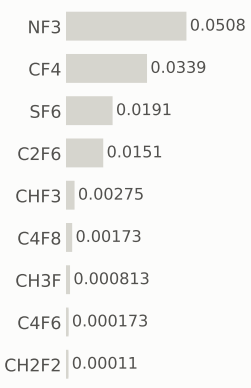


steps by process area



materials deposited (g)

gases emitted (g)



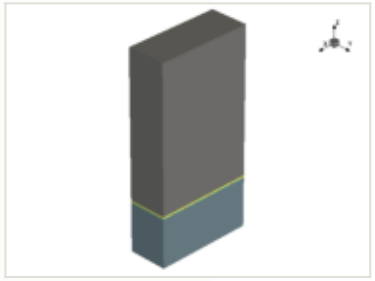
Step #240: Ion_implantation

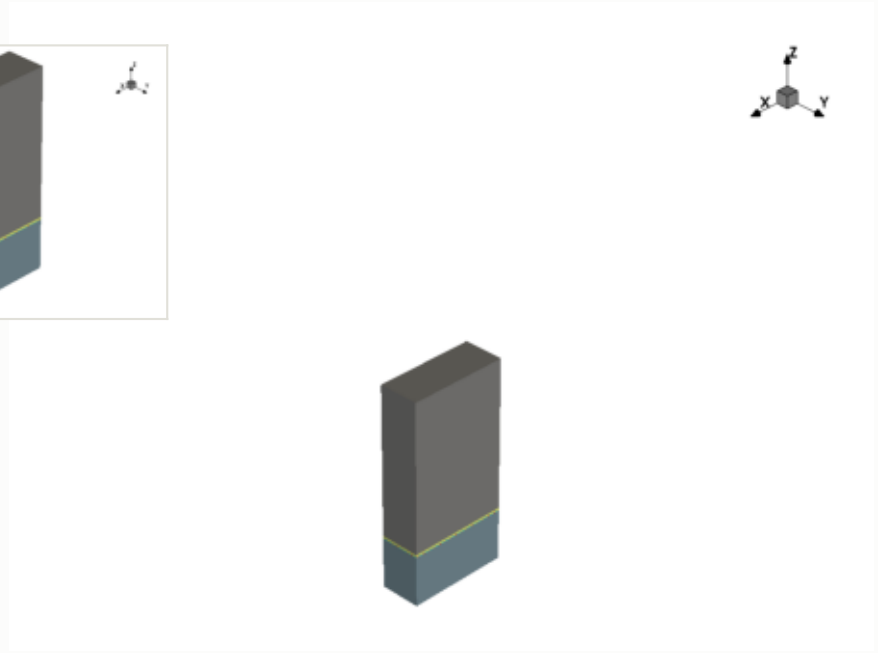
implant_ellipsoid · Ion_implantation

0.801	39.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

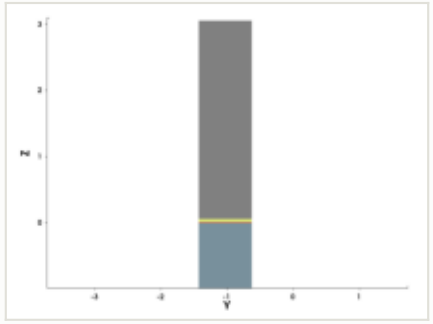
3D structure after this step

zoomed to fit

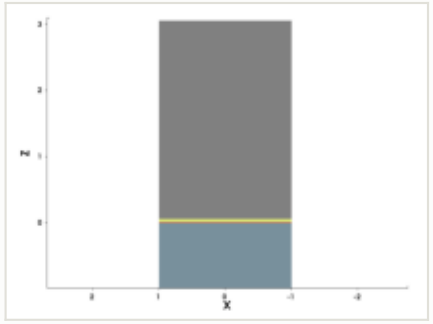




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

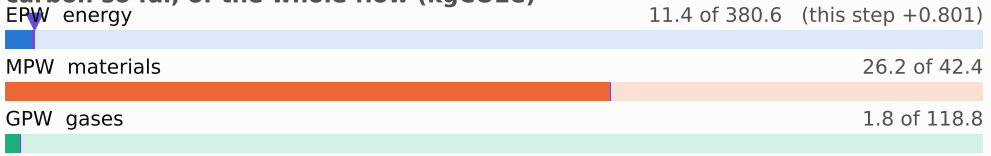
no mask for this step

this step

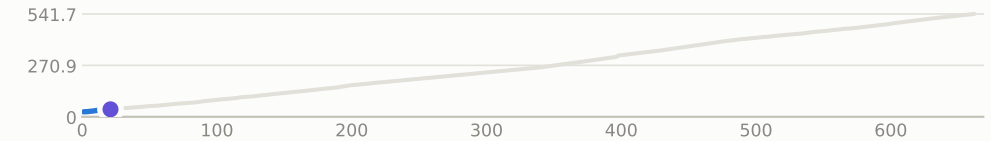
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

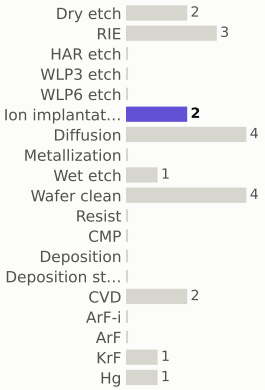
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

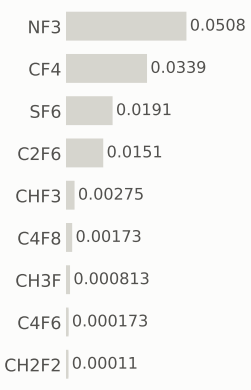


steps by process area



materials deposited (g)

gases emitted (g)



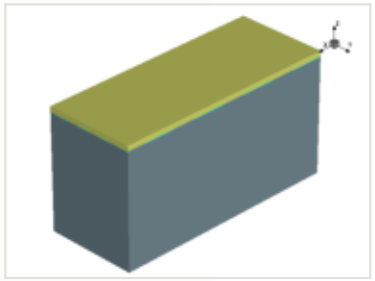
Step #250: Dry_etch

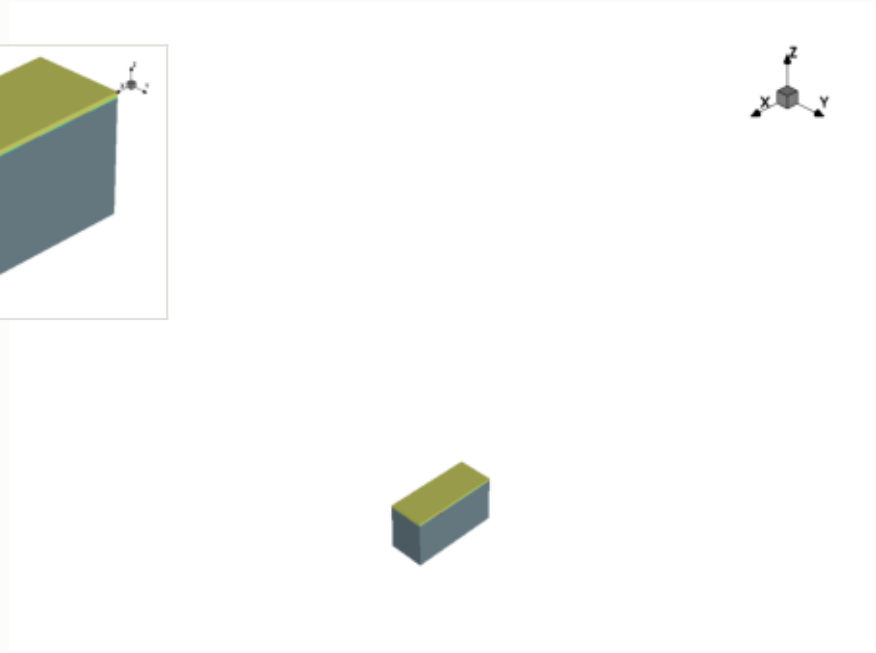
etch_strip · Dry_etch

0.801	40.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

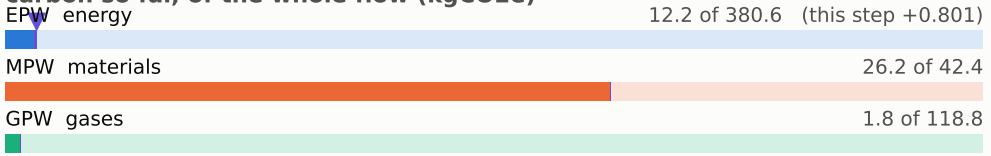




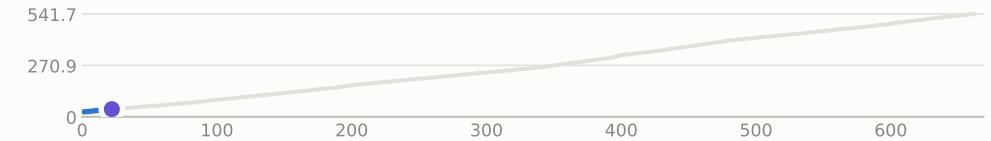
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

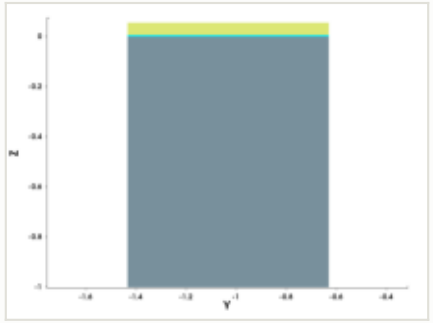
carbon so far, of the whole flow (kgCO2e)



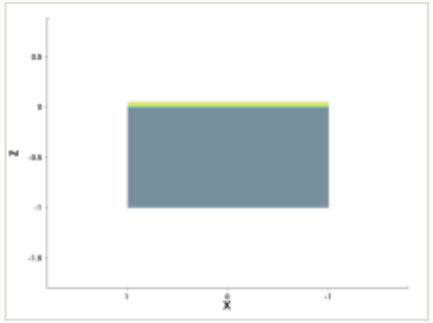
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



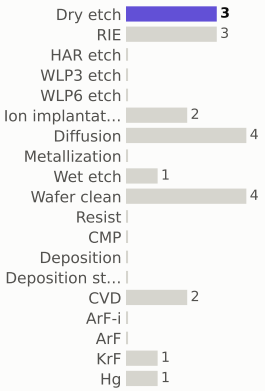
y cut at -1.032 µm



mask for this step

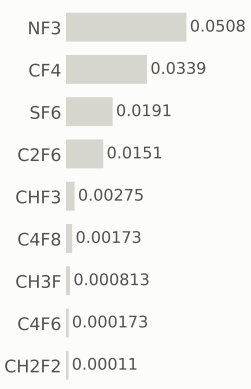
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



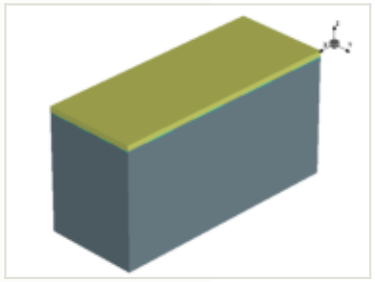
Step #260: Wafer_clean

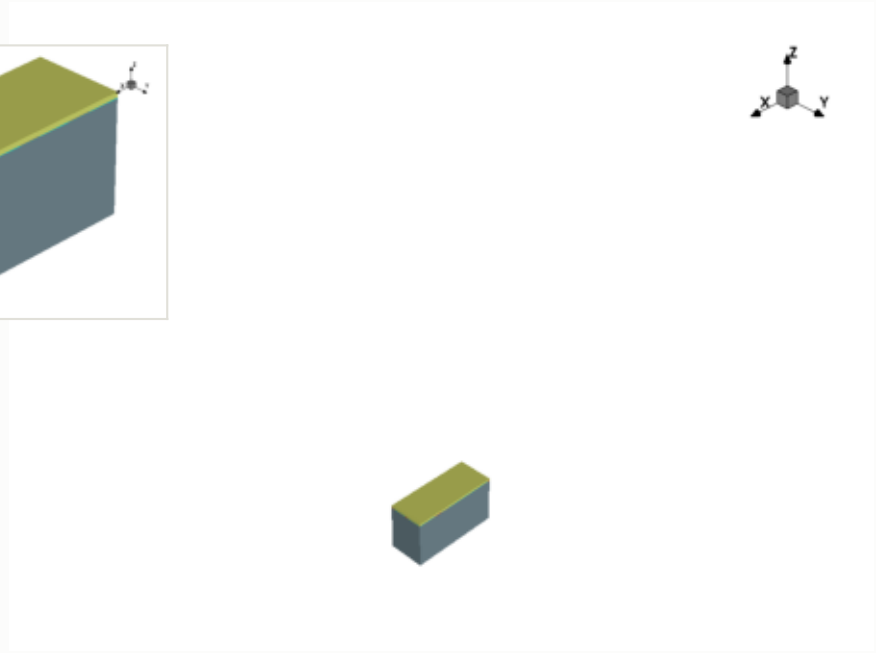
clean · Wafer_clean

0.169	40.4	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





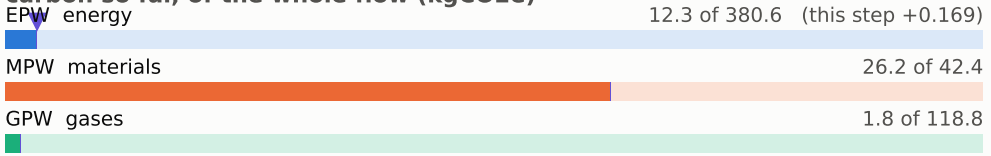
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

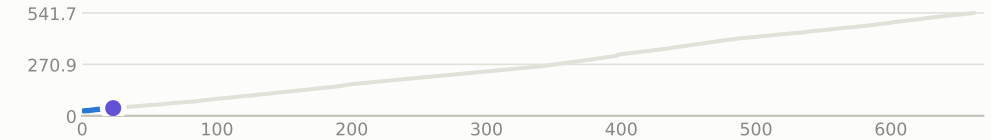
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

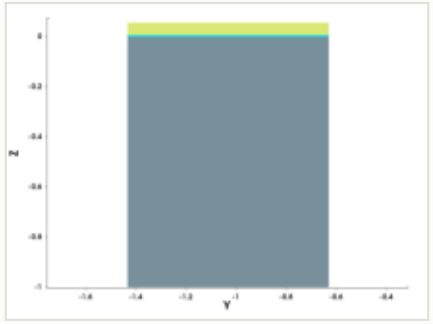
carbon so far, of the whole flow (kgCO2e)



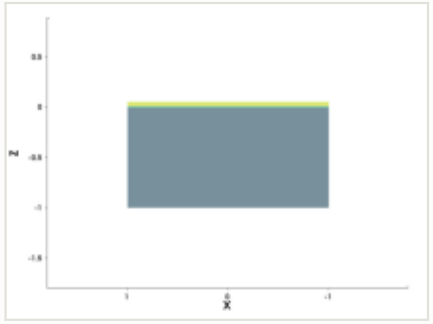
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



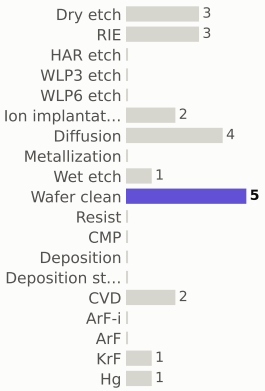
y cut at -1.032 µm



mask for this step

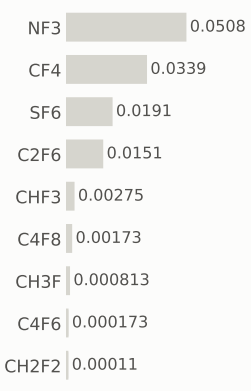
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



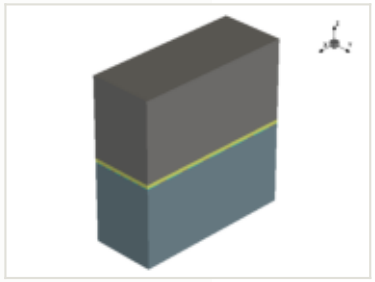
Step #270: Hg

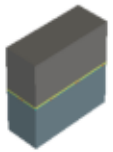
litho_patterning · Hg

0.199	40.6	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



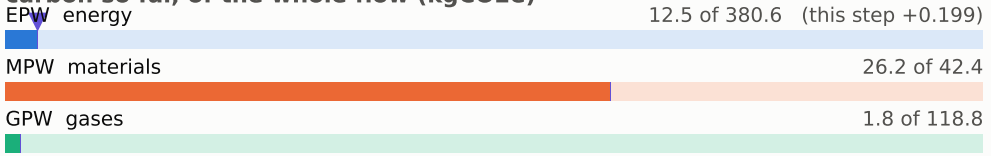


this step

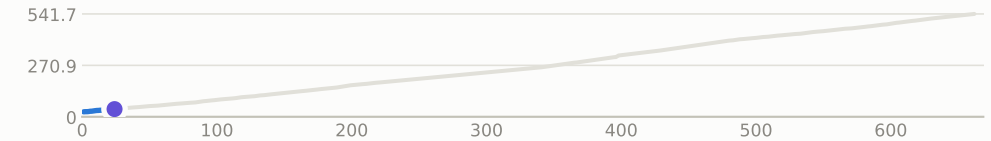
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

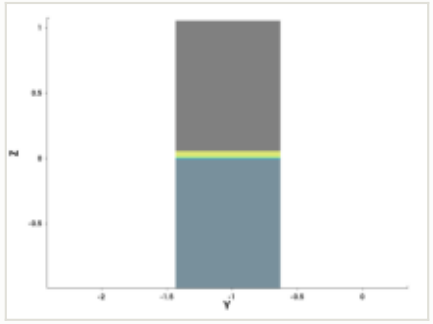
carbon so far, of the whole flow (kgCO2e)



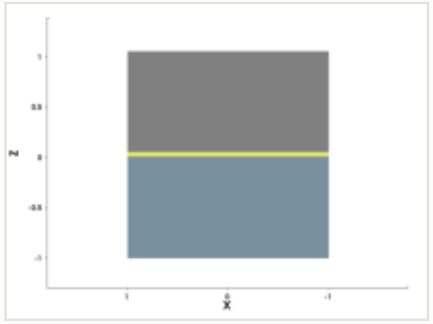
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

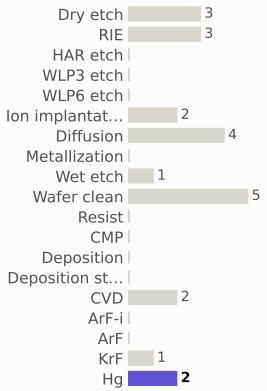


y cut at -1.032 µm



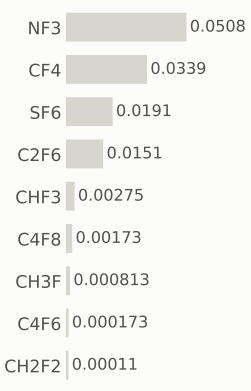
mask: layer 3:0, positive (OASIS)

steps by process area



materials deposited (g)

gases emitted (g)



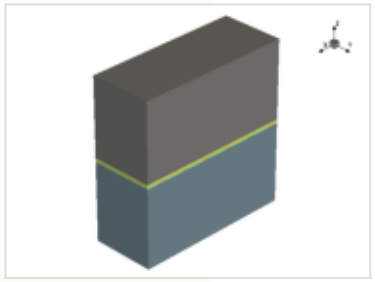
Step #280: Ion_implantation

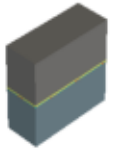
implant_ellipsoid · Ion_implantation

0.801	41.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

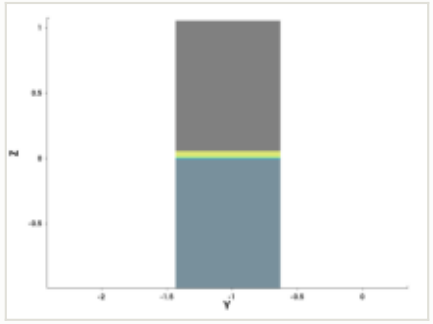
3D structure after this step

zoomed to fit

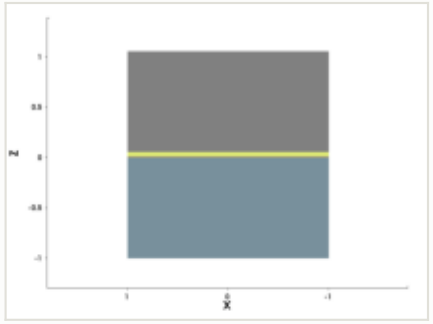




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

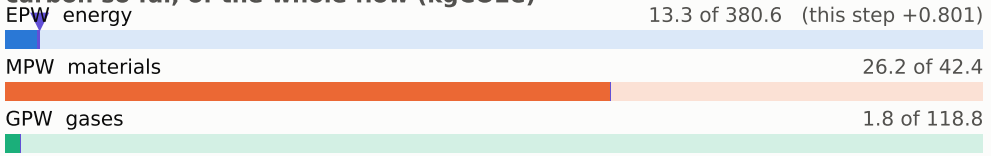
no mask for this step

this step

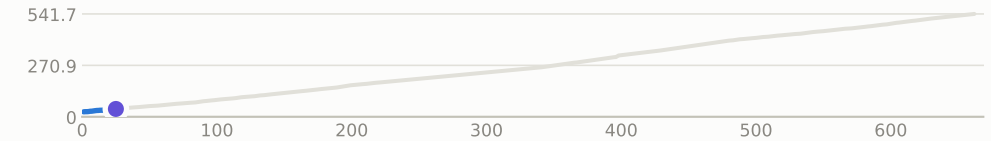
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

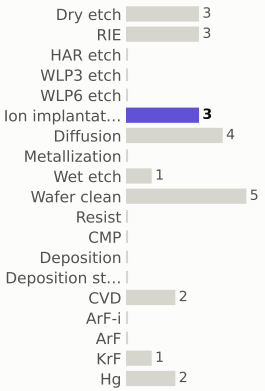
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

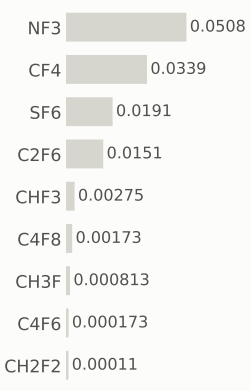


steps by process area



materials deposited (g)

gases emitted (g)



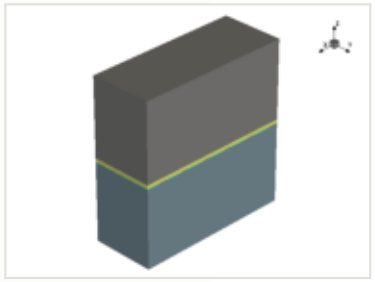
Step #290: Ion_implantation

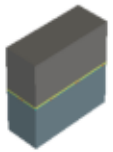
implant_ellipsoid · Ion_implantation

0.801	42.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

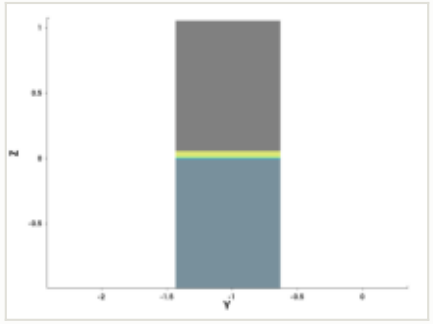
3D structure after this step

zoomed to fit

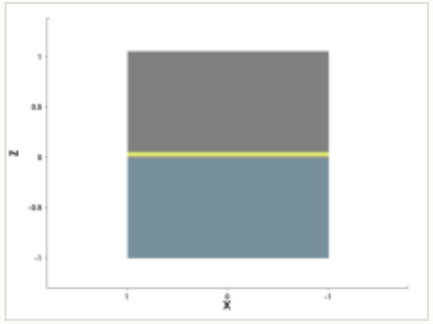




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

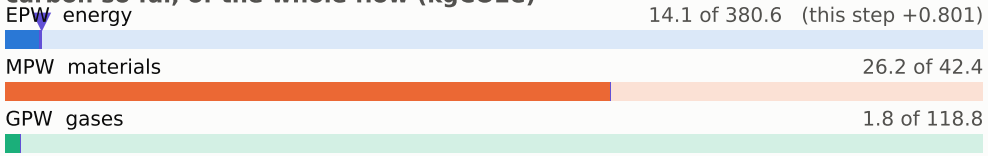
no mask for this step

this step

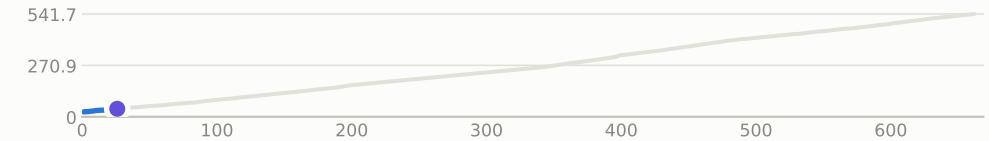
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

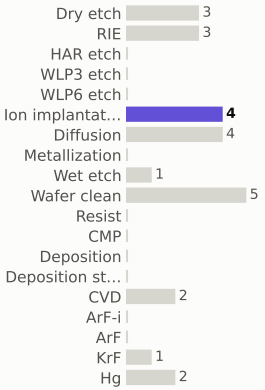
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

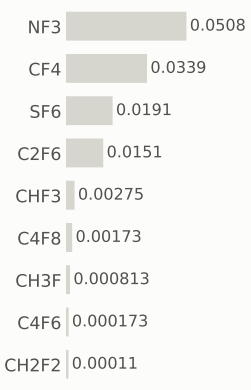


steps by process area



materials deposited (g)

gases emitted (g)



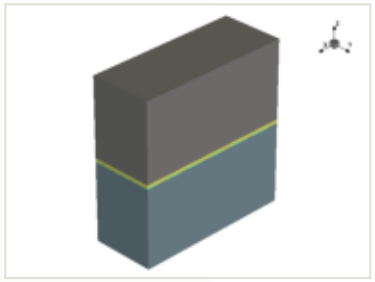
Step #300: Ion_implantation

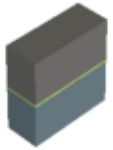
implant_ellipsoid · Ion_implantation

0.801	43	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

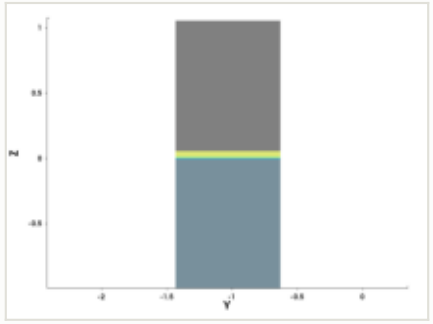
3D structure after this step

zoomed to fit

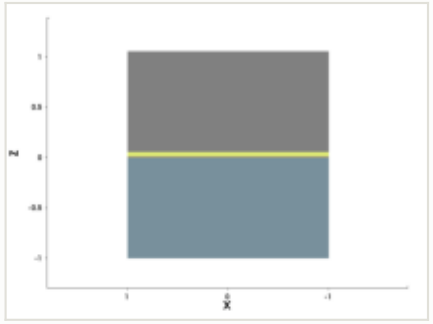




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

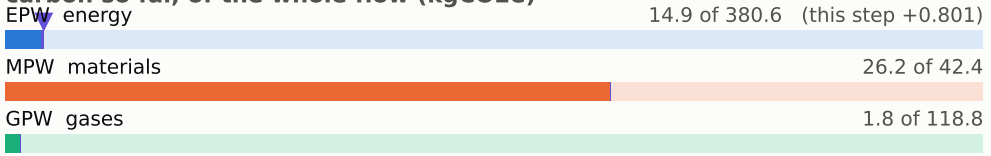
no mask for this step

this step

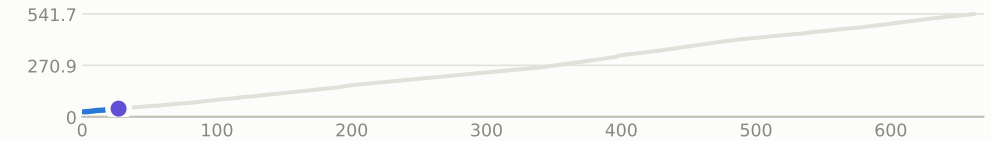
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

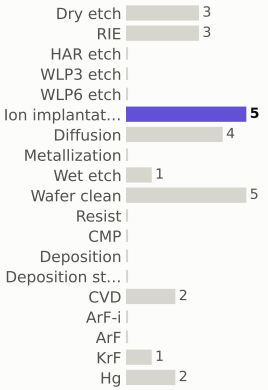
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

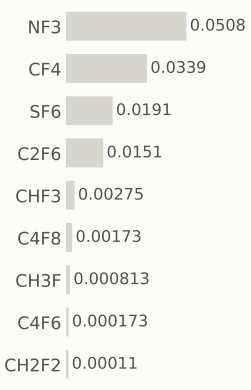


steps by process area



materials deposited (g)

gases emitted (g)



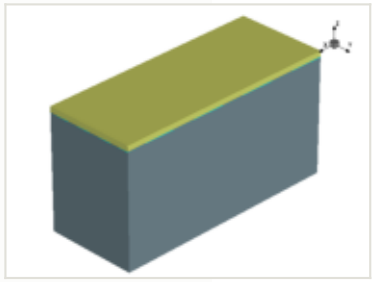
Step #310: Dry_etch

etch_strip · Dry_etch

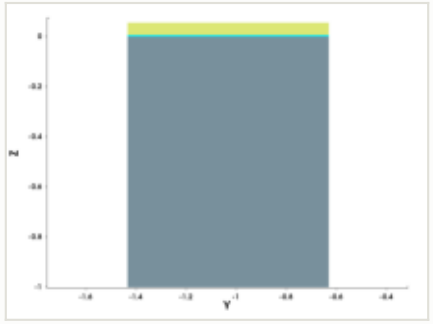
0.801	43.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

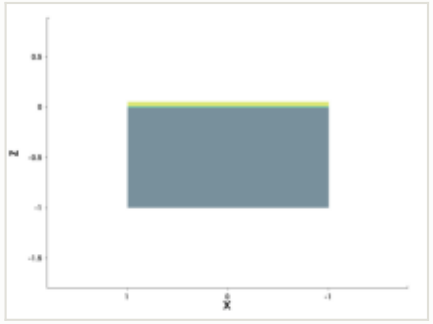
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



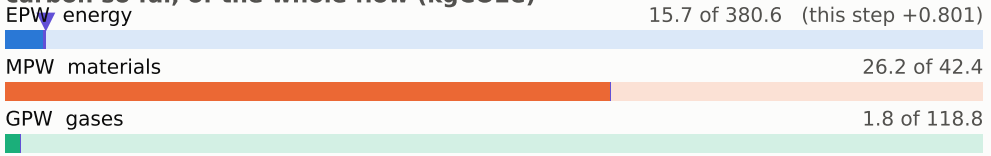
mask for this step

no mask for this step

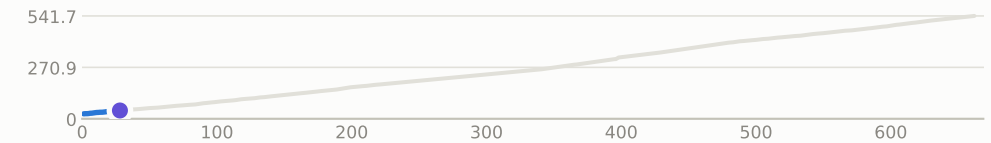
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

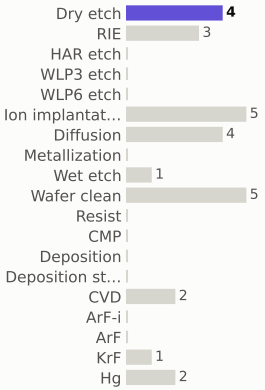
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

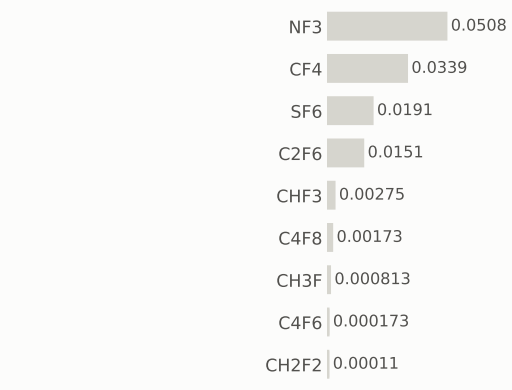


steps by process area



materials deposited (g)

gases emitted (g)



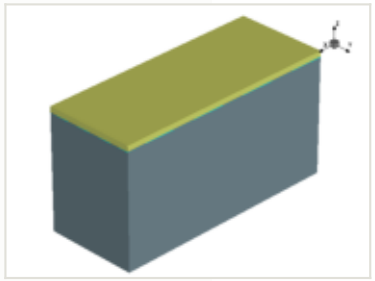
Step #320: Wafer_clean

clean · Wafer_clean

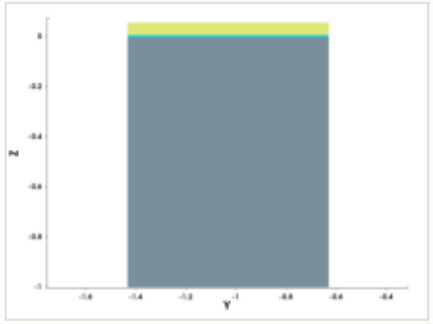
0.169	43.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

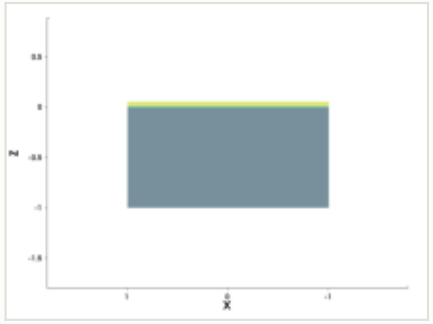
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

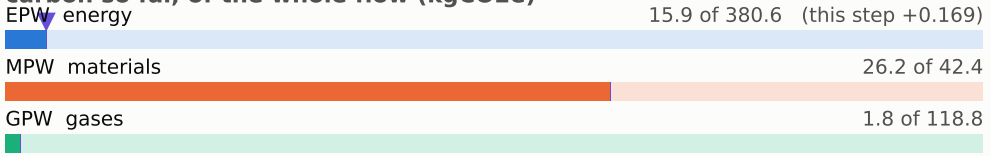
no mask for this step

this step

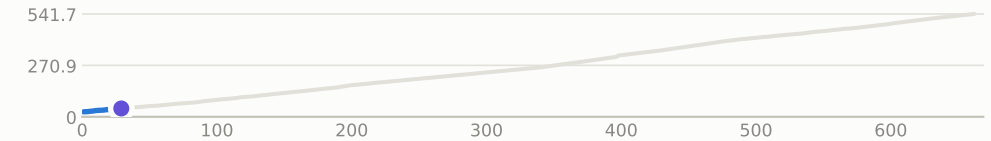
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

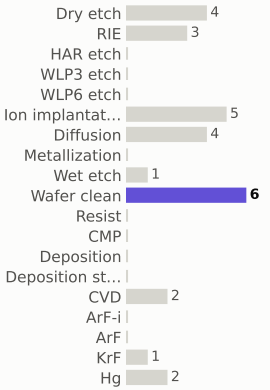
carbon so far, of the whole flow (kgCO2e)



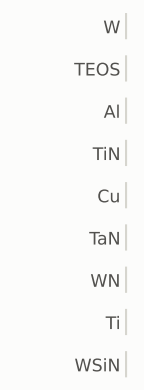
embodied carbon up to each step (kgCO2e per wafer)



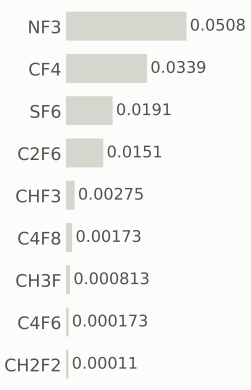
steps by process area



materials deposited (g)



gases emitted (g)



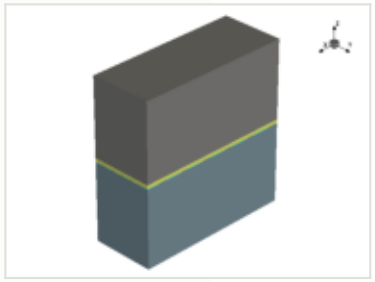
Step #330: Hg

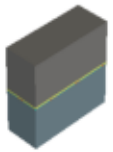
litho_patterning · Hg

0.199	44.1	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



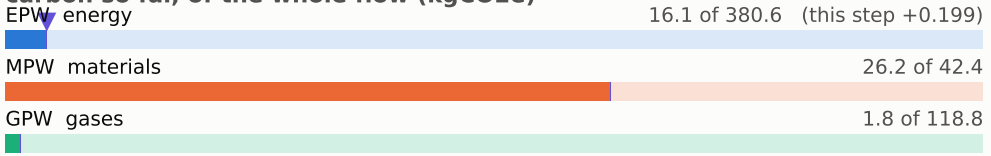


this step

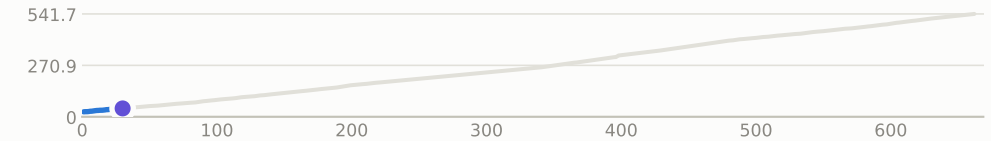
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

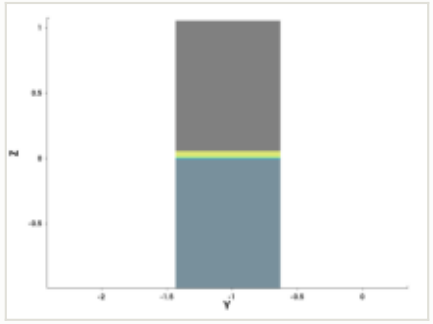
carbon so far, of the whole flow (kgCO2e)



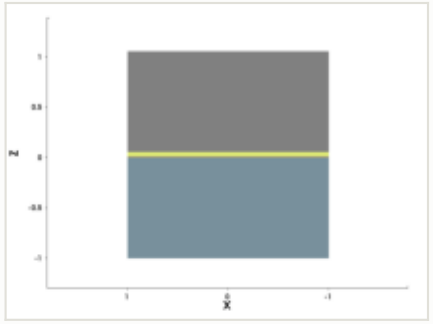
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

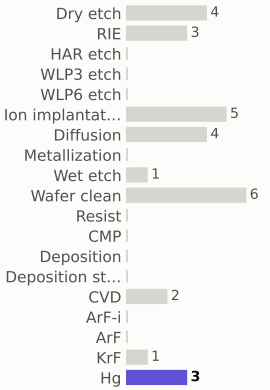


y cut at -1.032 µm

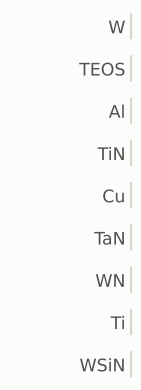


mask: layer 4:0, positive (OASIS)

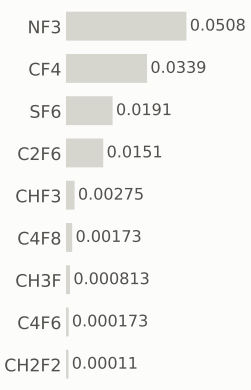
steps by process area



materials deposited (g)



gases emitted (g)



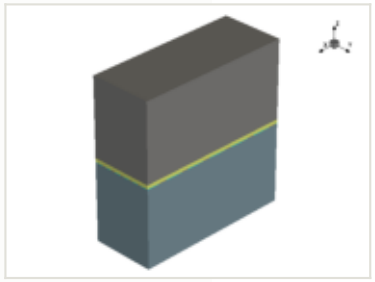
Step #340: Ion_implantation

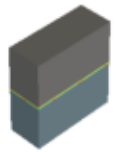
implant_ellipsoid · Ion_implantation

0.801	44.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

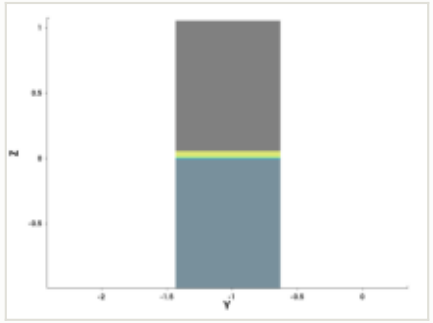
3D structure after this step

zoomed to fit

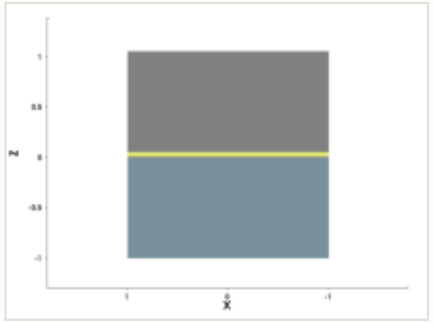




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

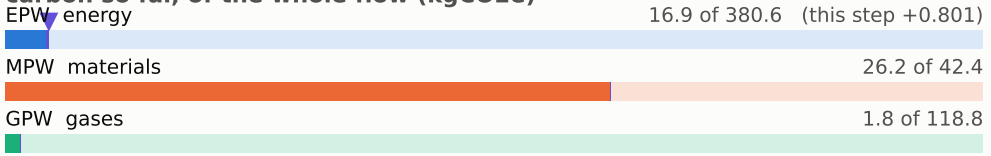
no mask for this step

this step

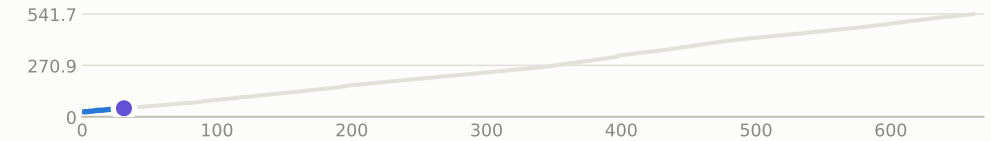
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

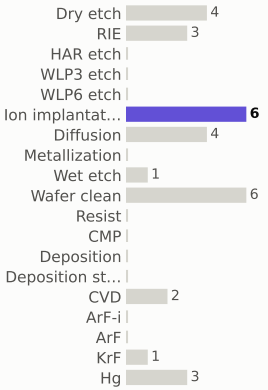
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

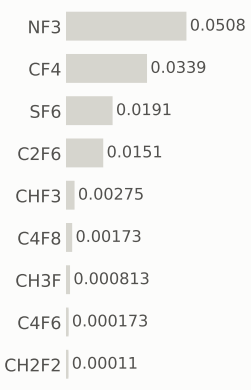


steps by process area



materials deposited (g)

gases emitted (g)



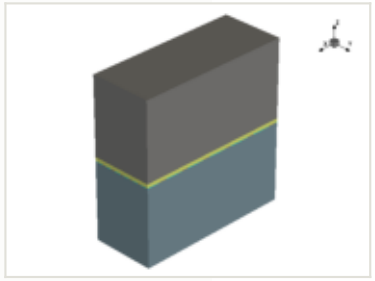
Step #350: Ion_implantation

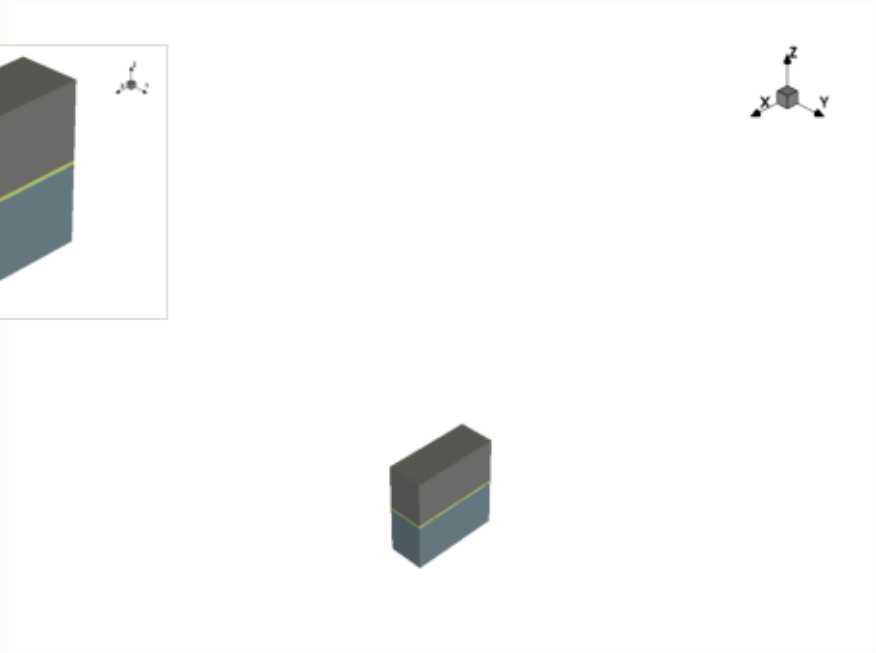
implant_ellipsoid · Ion_implantation

0.801	45.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

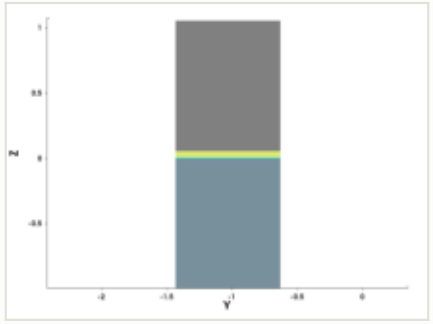
3D structure after this step

zoomed to fit

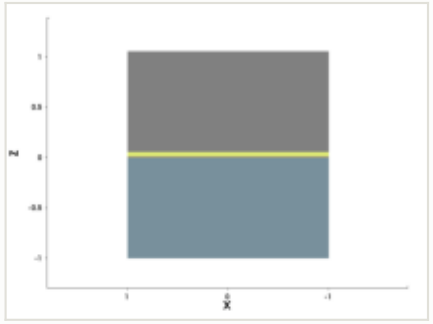




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

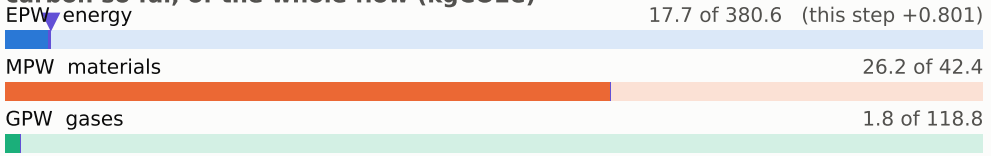
no mask for this step

this step

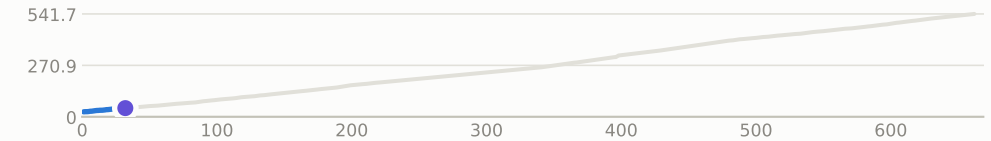
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

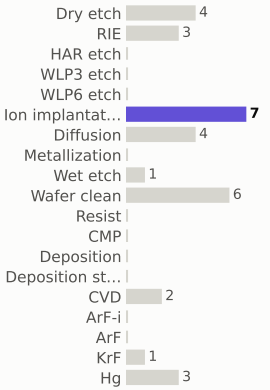
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

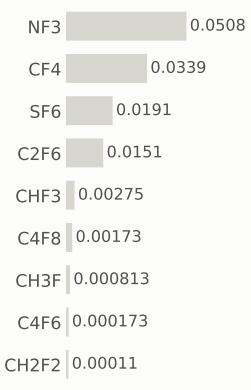


steps by process area



materials deposited (g)

gases emitted (g)



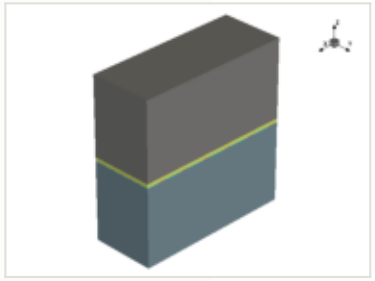
Step #360: Ion_implantation

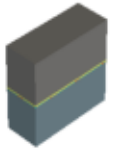
implant_ellipsoid · Ion_implantation

0.801	46.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

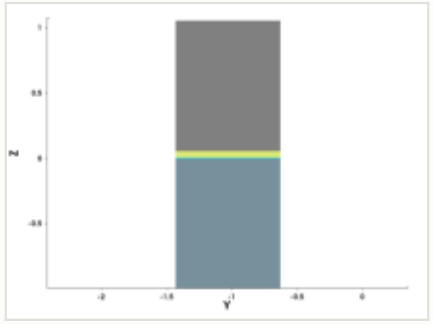
3D structure after this step

zoomed to fit

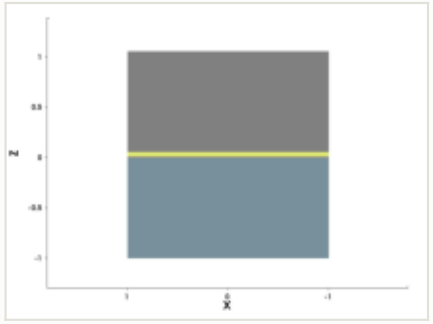




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

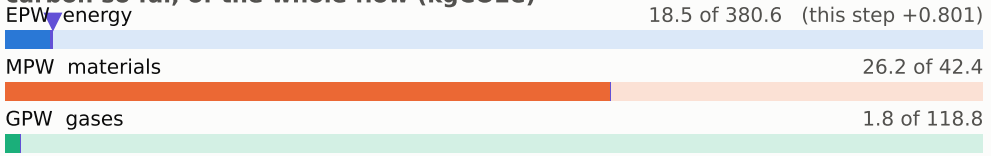
no mask for this step

this step

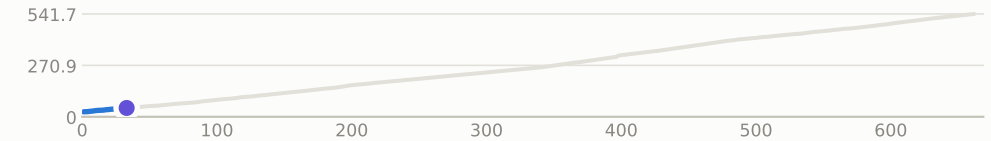
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

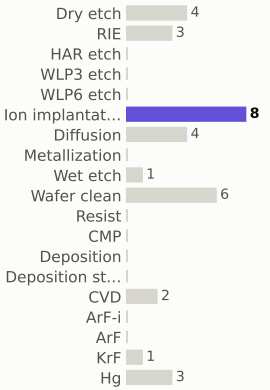
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

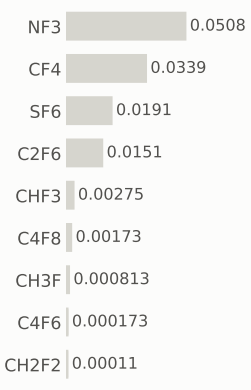


steps by process area



materials deposited (g)

gases emitted (g)



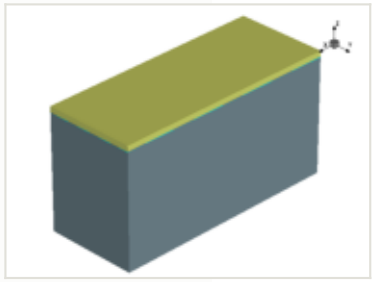
Step #370: Dry_etch

etch_strip · Dry_etch

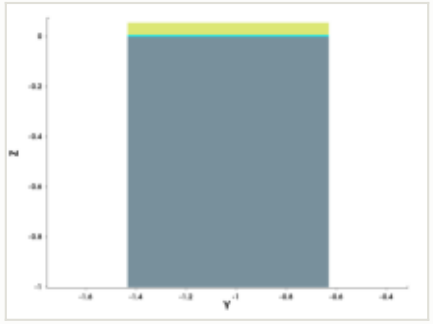
0.801	47.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

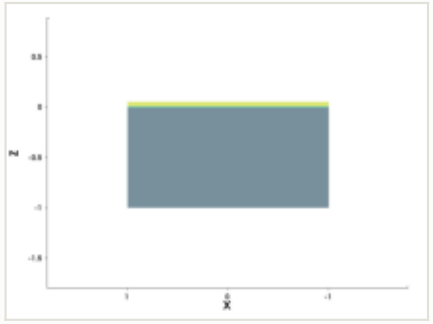
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



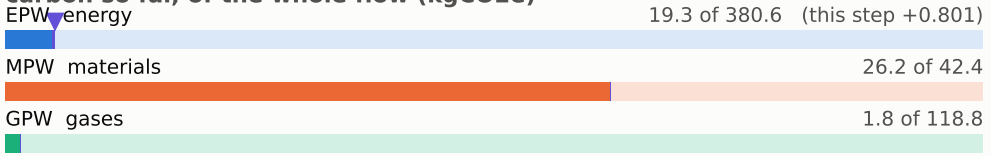
mask for this step

no mask for this step

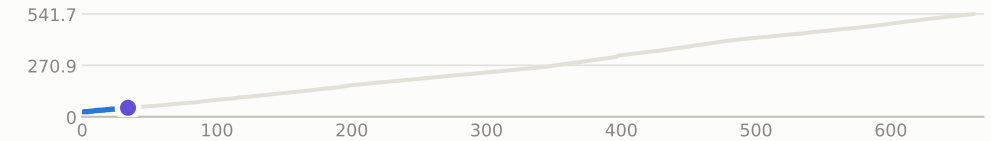
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

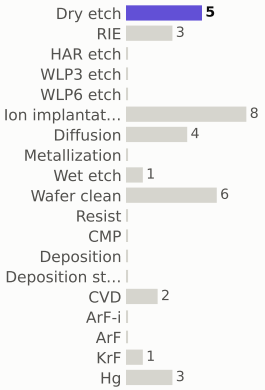
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

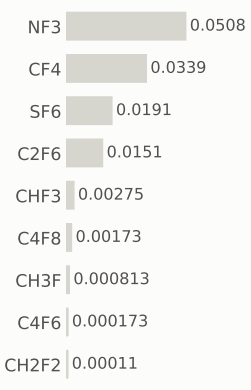


steps by process area



materials deposited (g)

gases emitted (g)



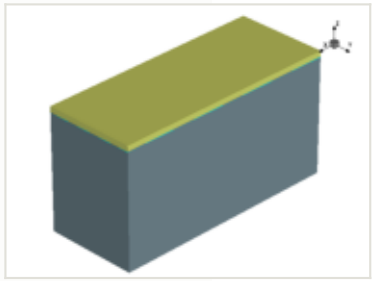
Step #380: Wafer_clean

clean · Wafer_clean

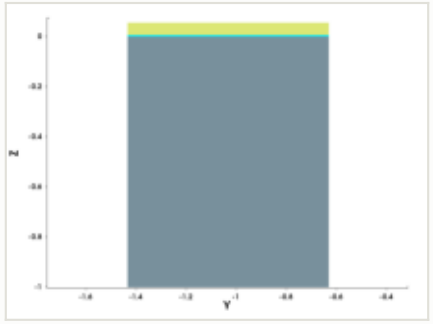
0.169	47.5	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

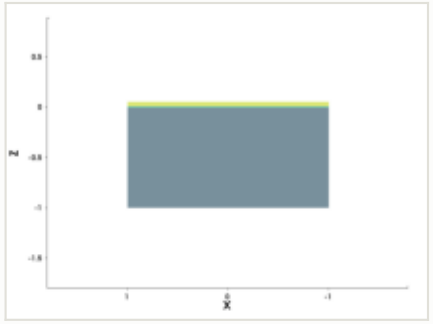
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

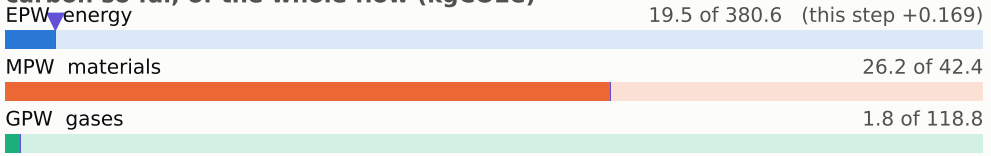
no mask for this step

this step

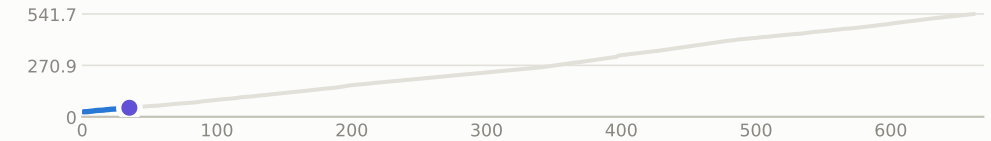
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

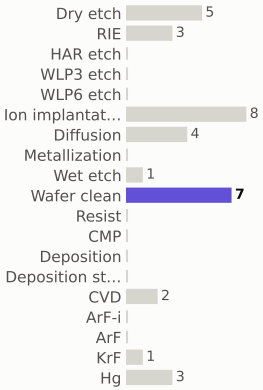
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

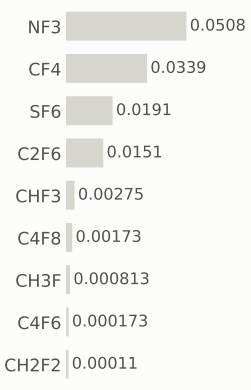


steps by process area



materials deposited (g)

gases emitted (g)



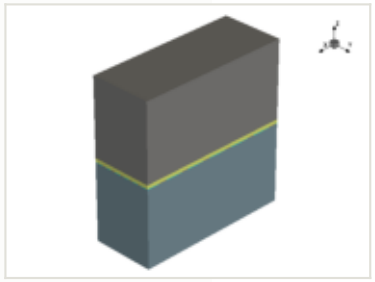
Step #390: Hg

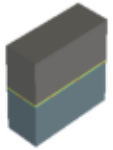
litho_patterning · Hg

0.199	47.7	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



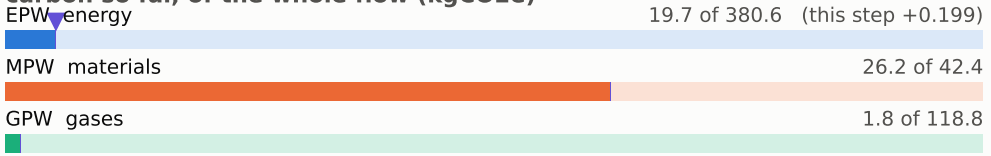


this step

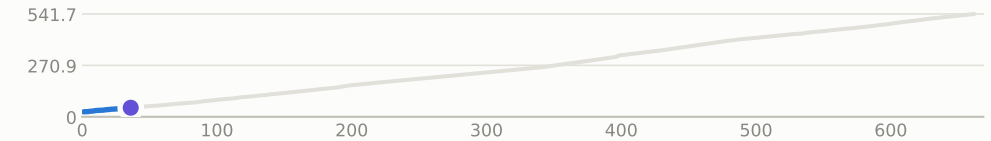
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

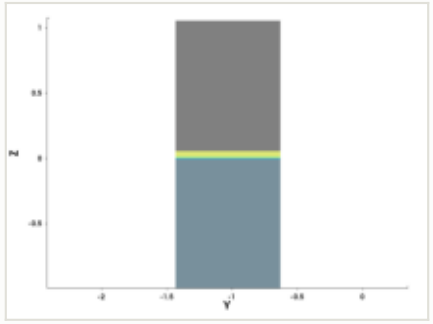
carbon so far, of the whole flow (kgCO2e)



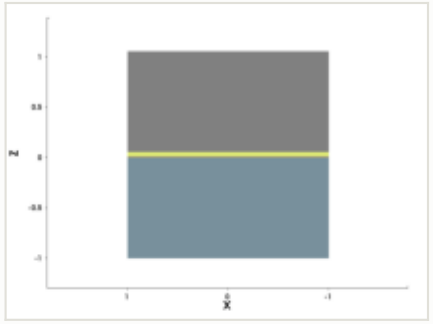
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

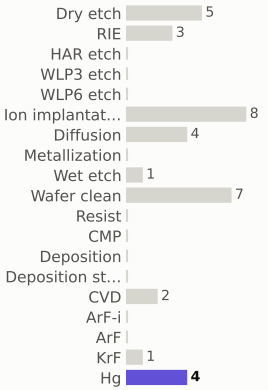


y cut at -1.032 µm

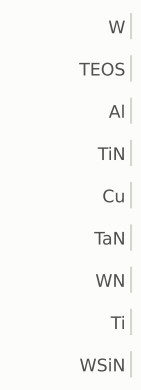


mask: layer 5:0, positive (OASIS)

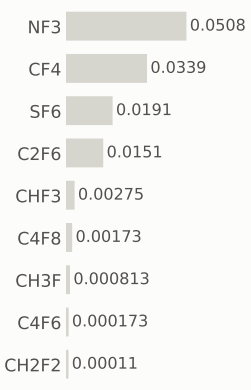
steps by process area



materials deposited (g)



gases emitted (g)



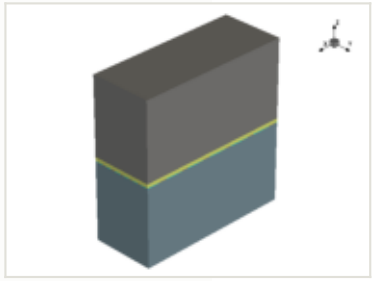
Step #400: Ion_implantation

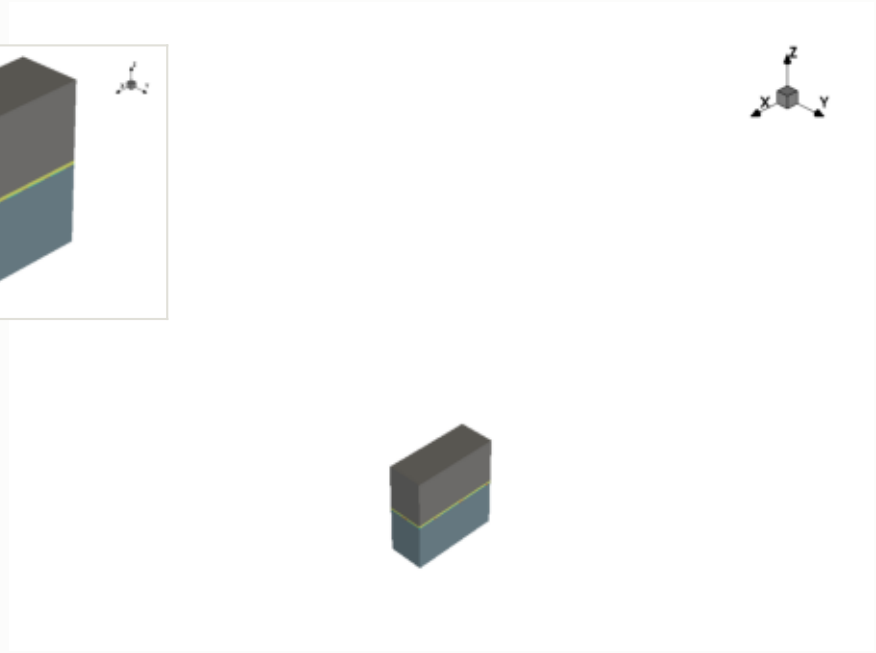
implant_ellipsoid · Ion_implantation

0.801	48.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

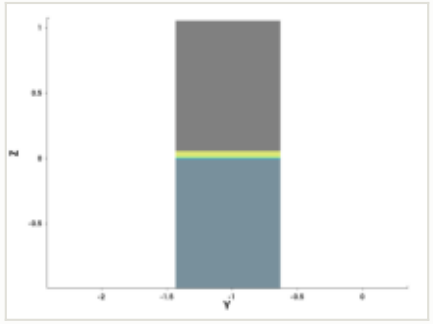
3D structure after this step

zoomed to fit

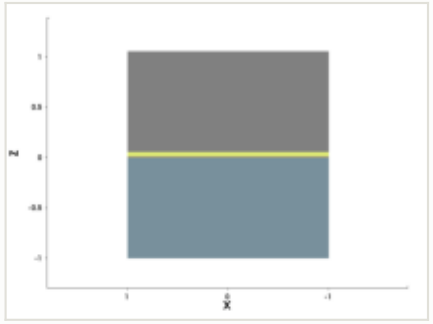




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

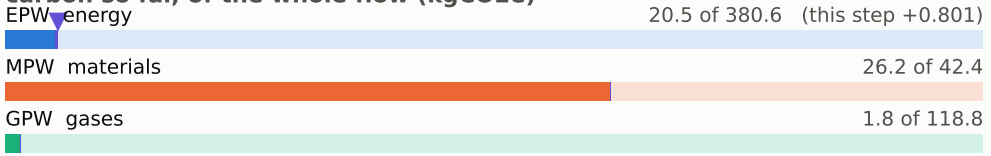
no mask for this step

this step

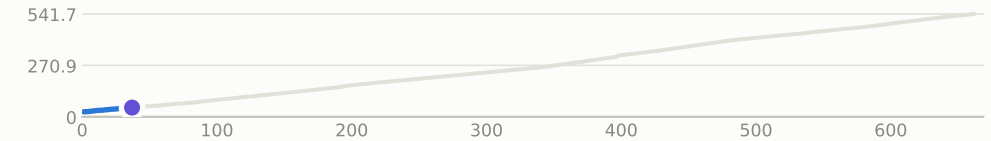
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

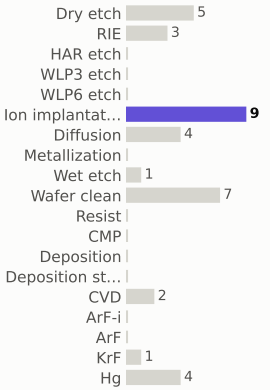
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

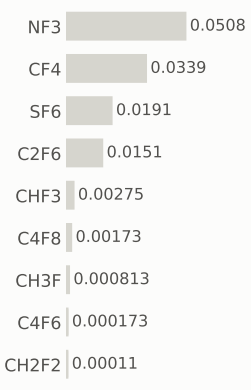


steps by process area



materials deposited (g)

gases emitted (g)



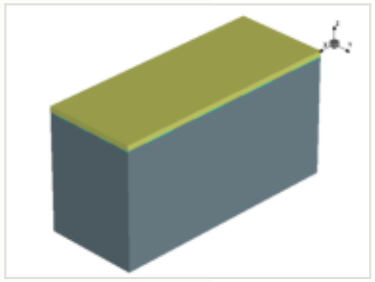
Step #410: Dry_etch

etch_strip · Dry_etch

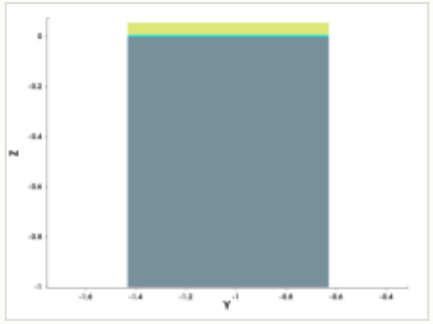
0.801	49.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

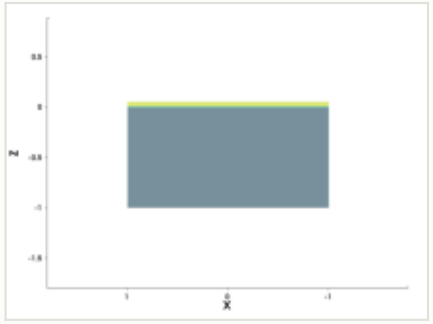
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



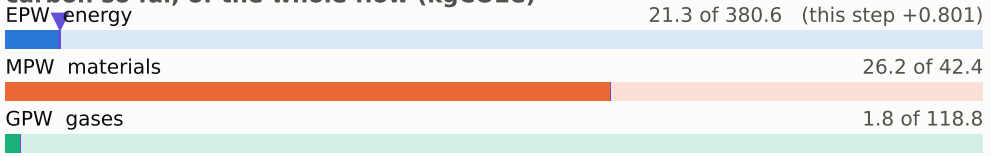
mask for this step

no mask for this step

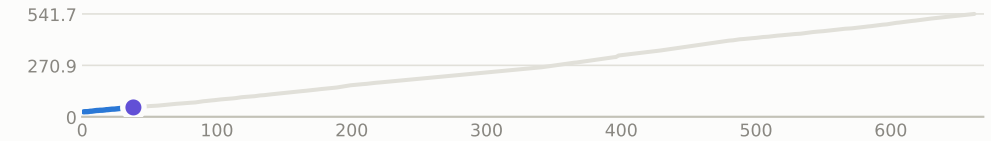
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

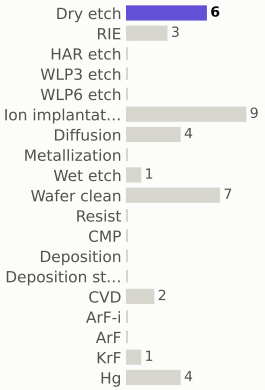
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

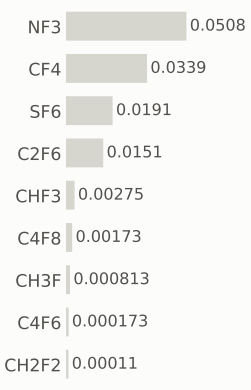


steps by process area



materials deposited (g)

gases emitted (g)



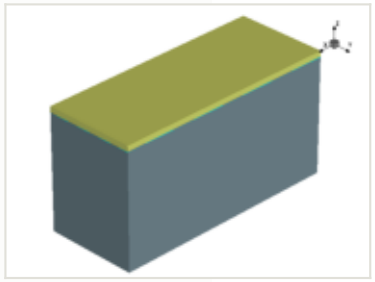
Step #420: Wafer_clean

clean · Wafer_clean

0.169	49.5	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



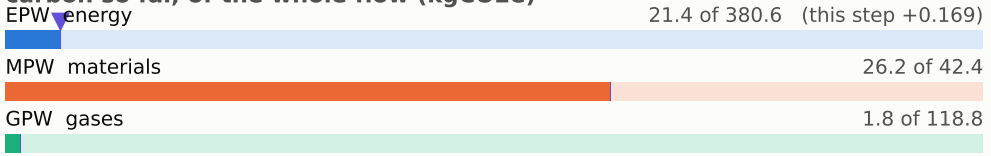
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

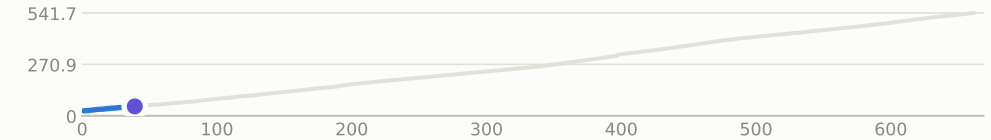
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

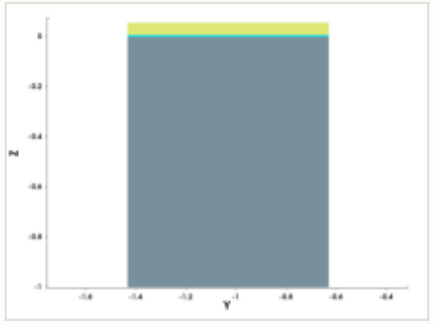
carbon so far, of the whole flow (kgCO2e)



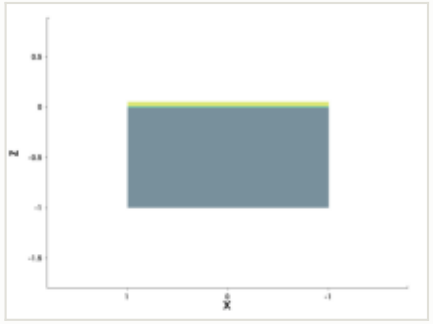
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



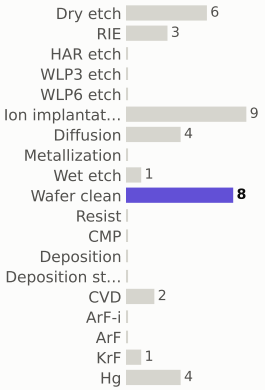
y cut at -1.032 µm



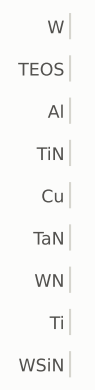
mask for this step

no mask for this step

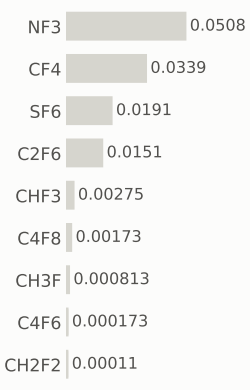
steps by process area



materials deposited (g)



gases emitted (g)



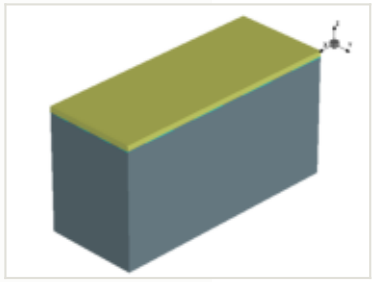
Step #430: Hg

litho_patterning · Hg

0.199	49.7	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

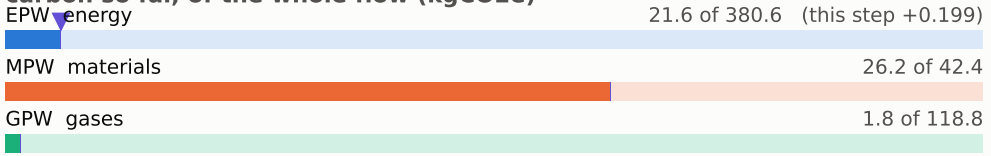


this step

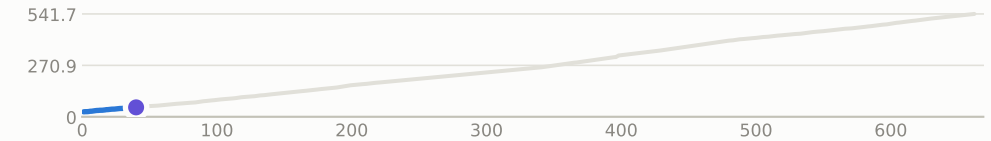
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

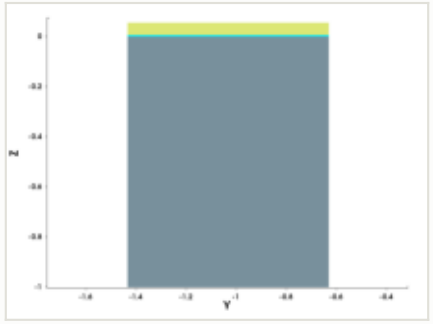
carbon so far, of the whole flow (kgCO2e)



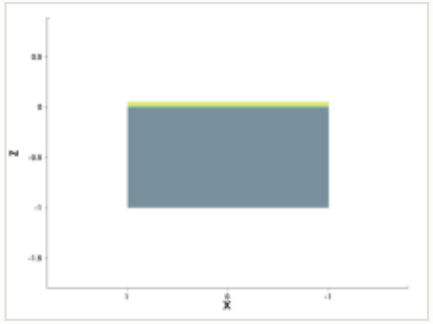
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

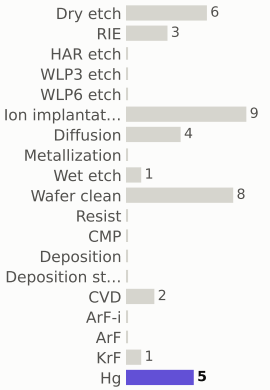


y cut at -1.032 µm

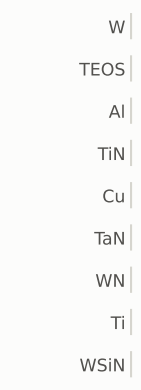


mask: layer 6:0 (OASIS)

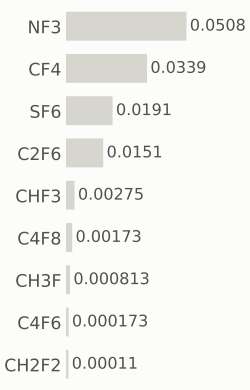
steps by process area



materials deposited (g)



gases emitted (g)



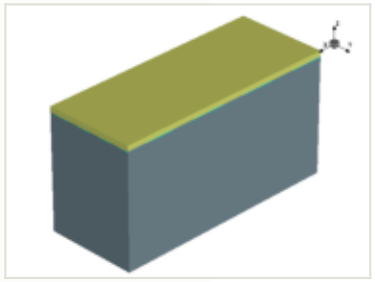
Step #440: Ion_implantation

implant_ellipsoid · Ion_implantation

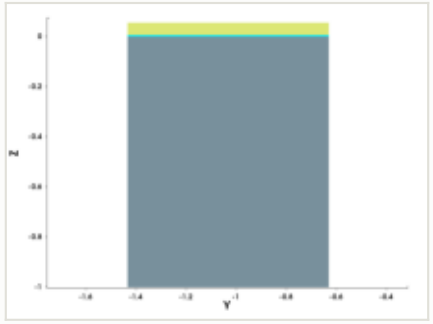
0.801	50.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

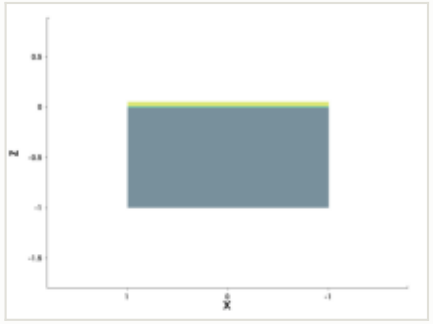
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

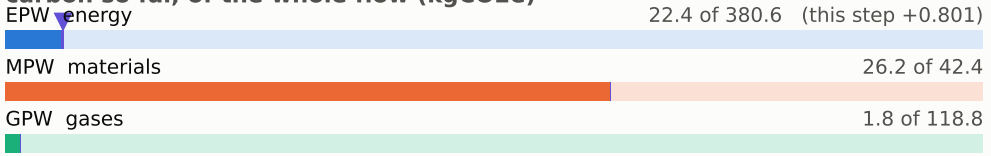
no mask for this step

this step

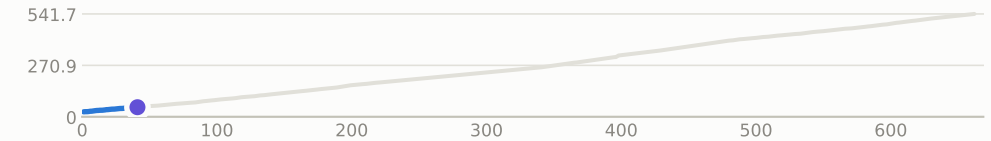
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

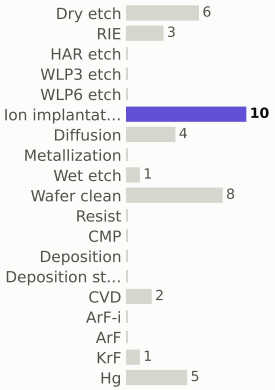
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

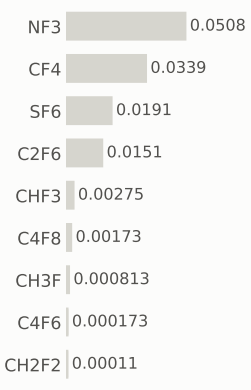


steps by process area



materials deposited (g)

gases emitted (g)



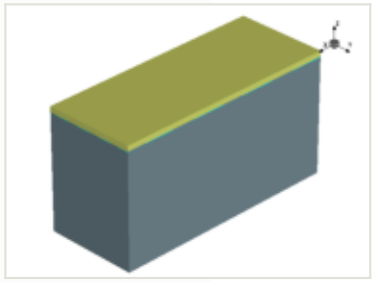
Step #450: Ion_implantation

implant_ellipsoid · Ion_implantation

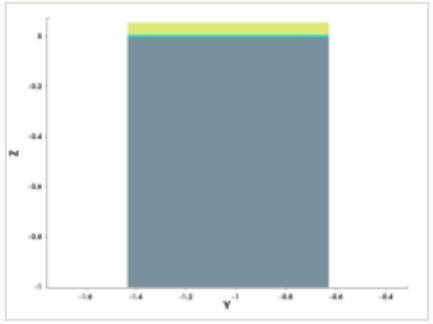
0.801	51.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

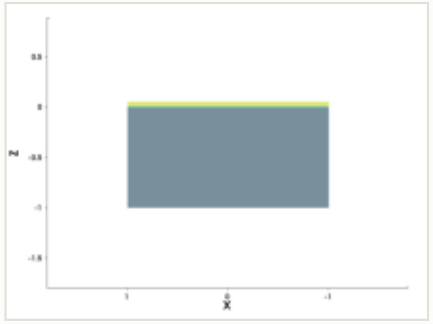
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

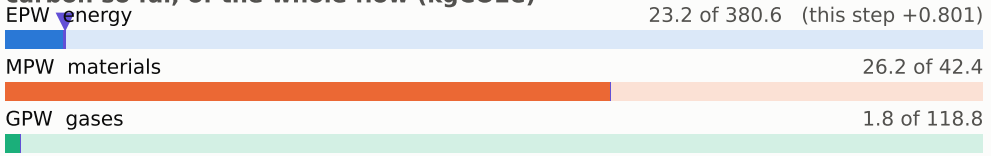
no mask for this step

this step

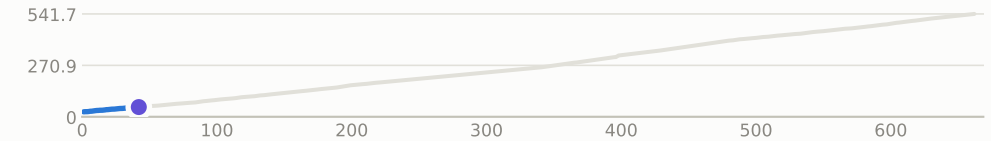
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

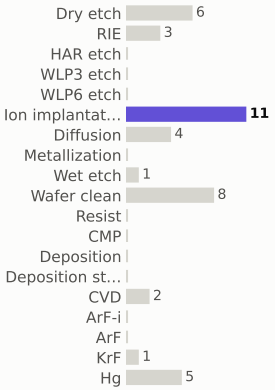
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

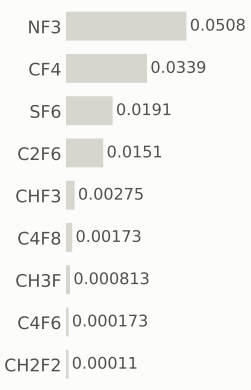


steps by process area



materials deposited (g)

gases emitted (g)



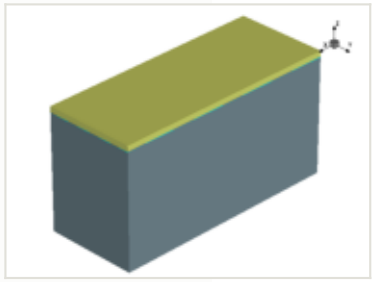
Step #460: Ion_implantation

implant_ellipsoid · Ion_implantation

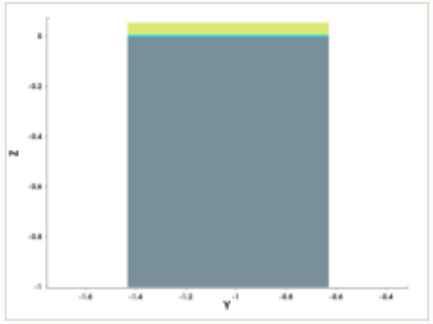
0.801	52.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

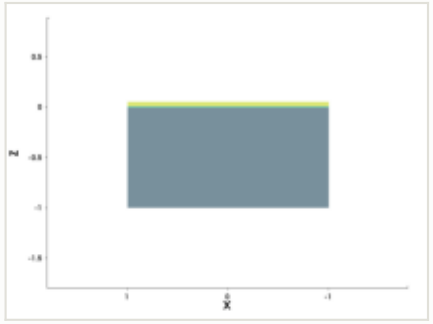
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

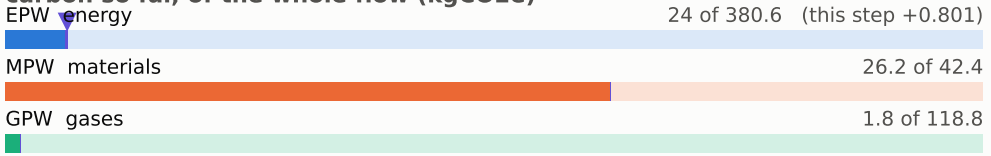
no mask for this step

this step

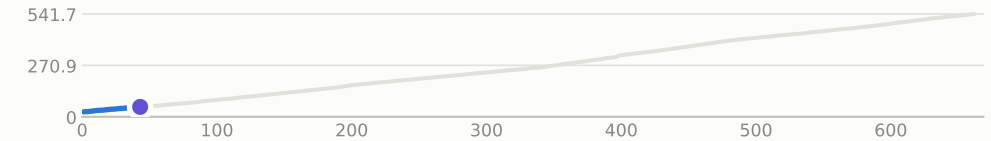
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

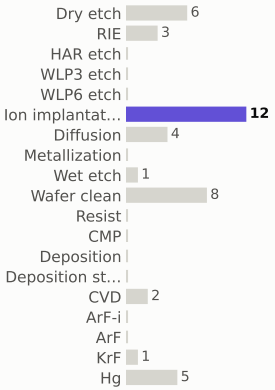
carbon so far, of the whole flow (kgCO2e)



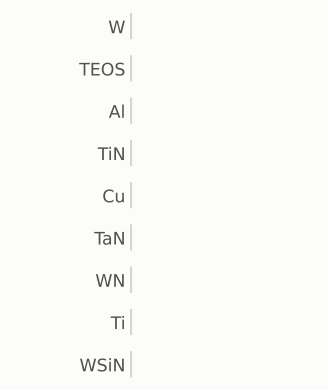
embodied carbon up to each step (kgCO2e per wafer)



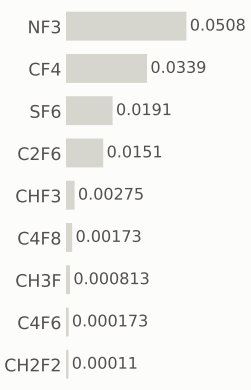
steps by process area



materials deposited (g)



gases emitted (g)



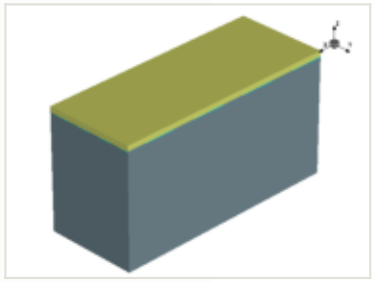
Step #470: Dry_etch

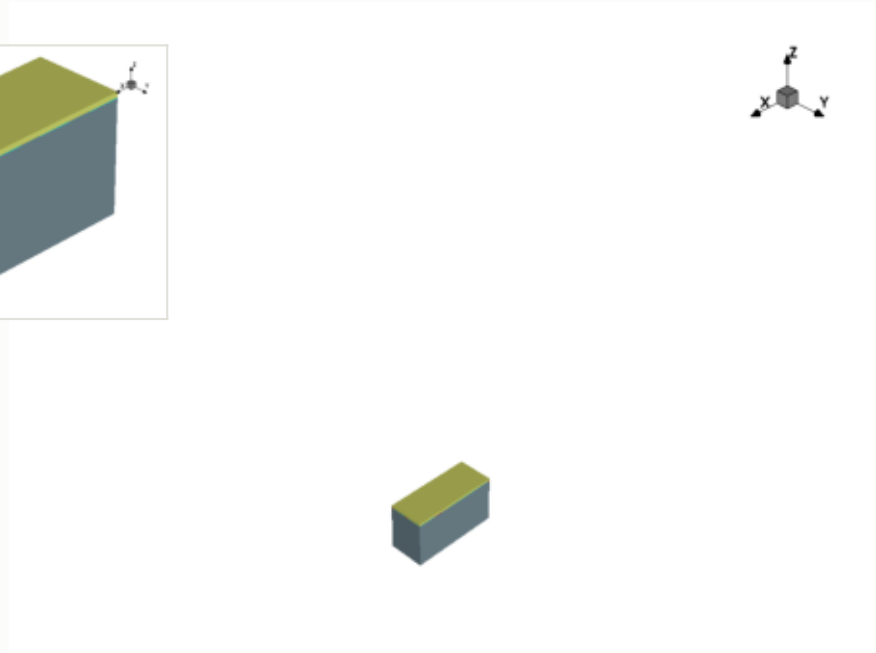
etch_strip · Dry_etch

0.801	52.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



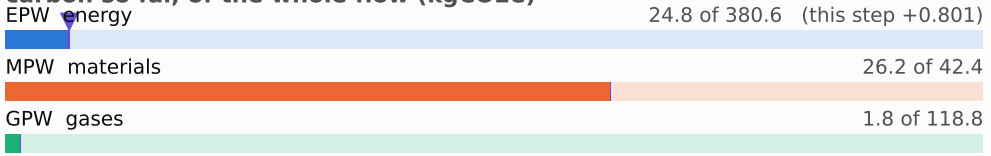


this step

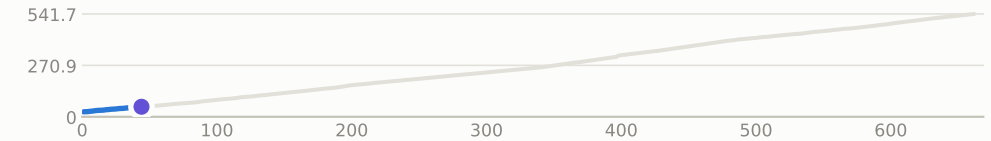
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

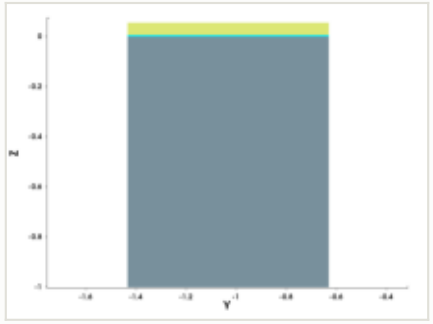
carbon so far, of the whole flow (kgCO2e)



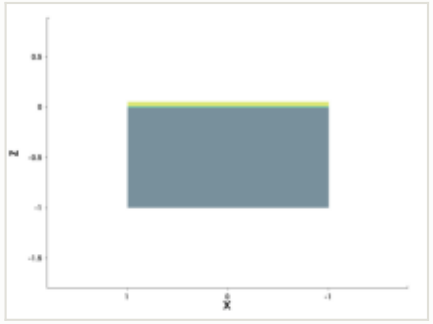
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



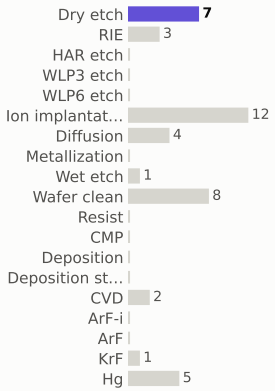
y cut at -1.032 µm



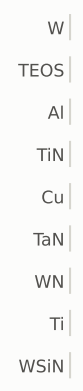
mask for this step

no mask for this step

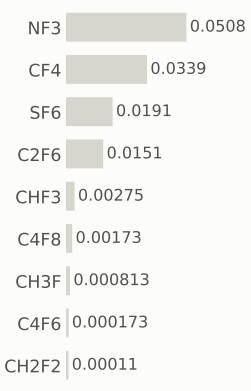
steps by process area



materials deposited (g)



gases emitted (g)



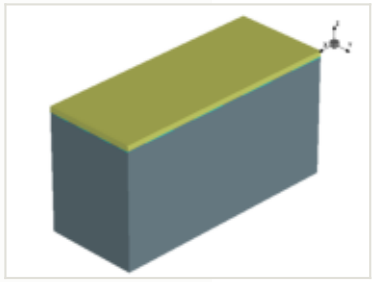
Step #480: Wafer_clean

clean · Wafer_clean

0.169	53	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



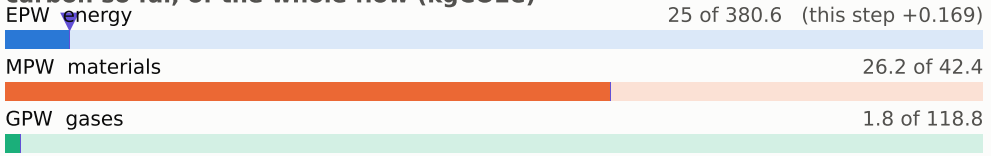
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

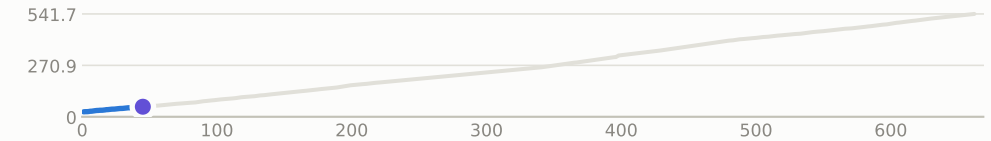
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

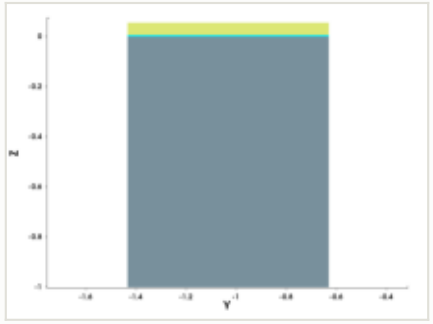
carbon so far, of the whole flow (kgCO2e)



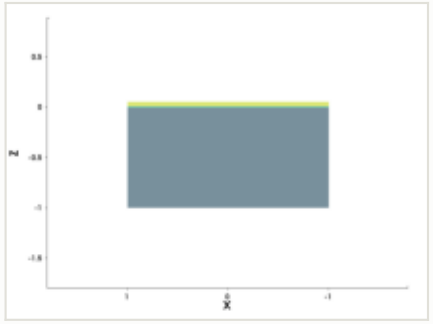
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



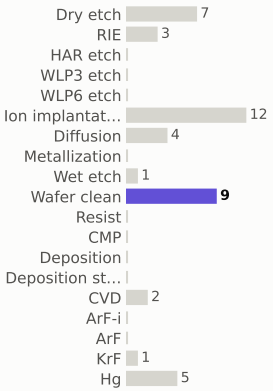
y cut at -1.032 µm



mask for this step

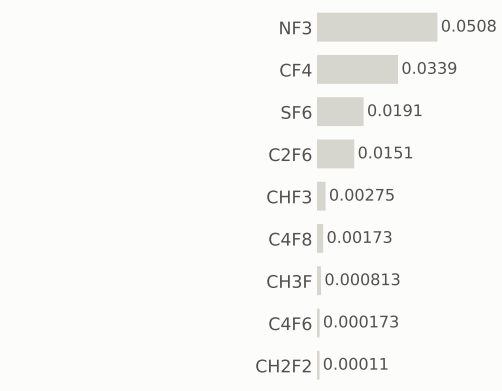
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



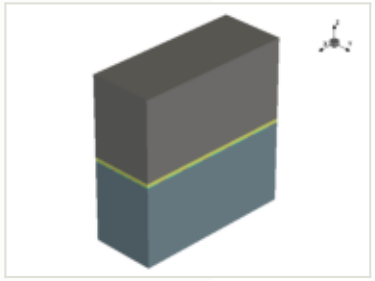
Step #490: Hg

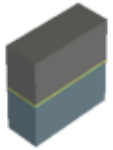
litho_patterning · Hg

0.199	53.2	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



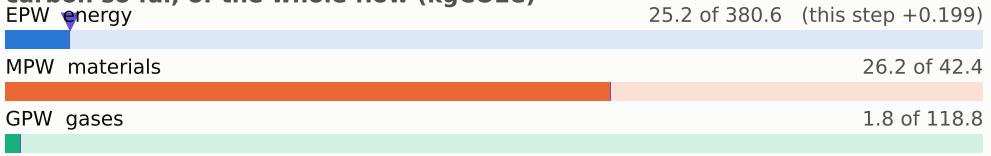


this step

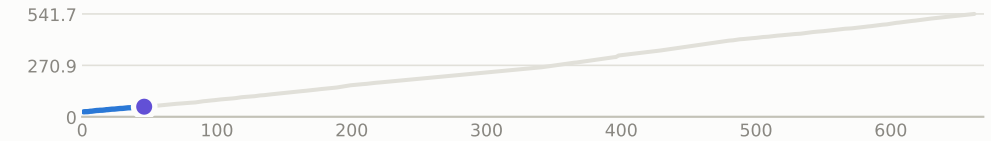
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

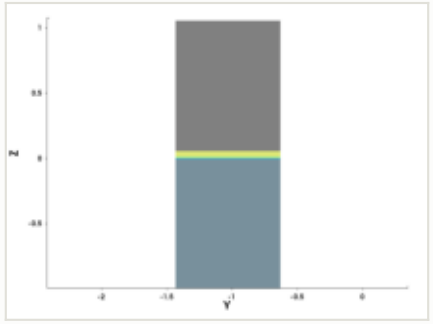
carbon so far, of the whole flow (kgCO2e)



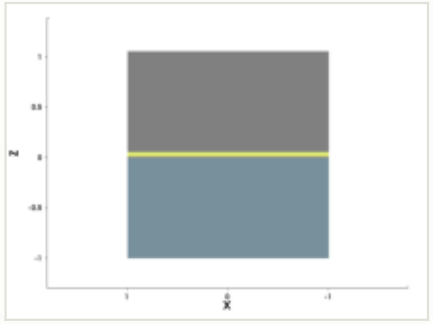
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

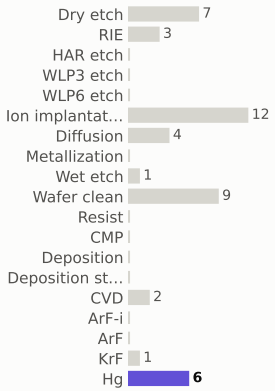


y cut at -1.032 µm



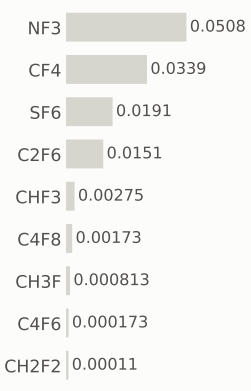
mask: layer 7:0, positive (OASIS)

steps by process area



materials deposited (g)

gases emitted (g)



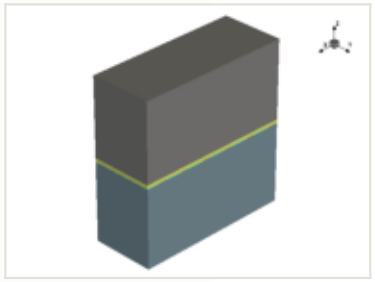
Step #500: Ion_implantation

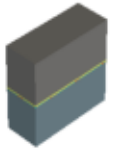
implant_ellipsoid · Ion_implantation

0.801	54	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

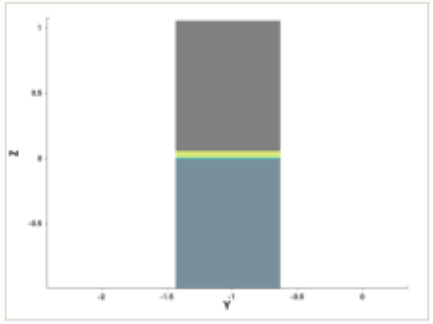
3D structure after this step

zoomed to fit

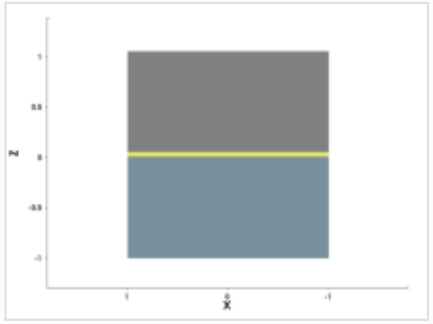




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

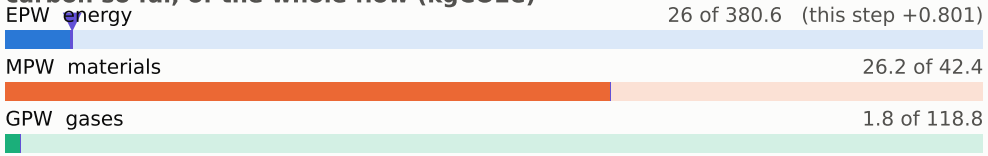
no mask for this step

this step

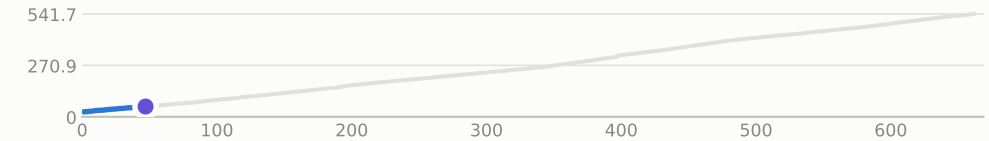
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

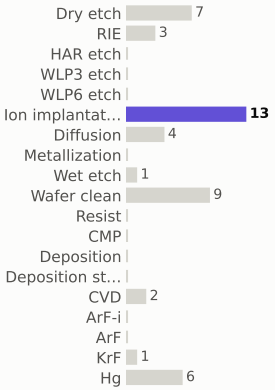
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

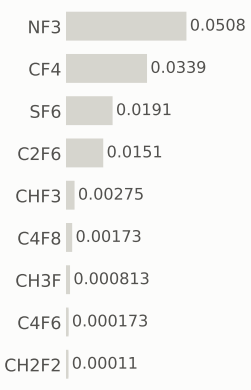


steps by process area



materials deposited (g)

gases emitted (g)



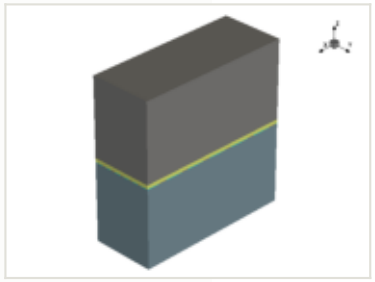
Step #510: Ion_implantation

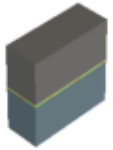
implant_ellipsoid · Ion_implantation

0.801	54.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

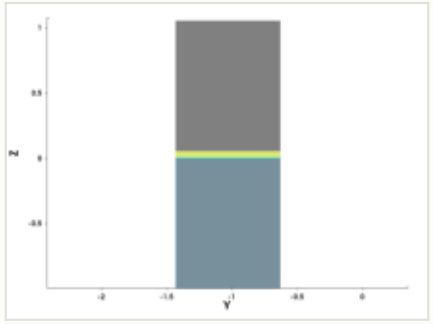
3D structure after this step

zoomed to fit

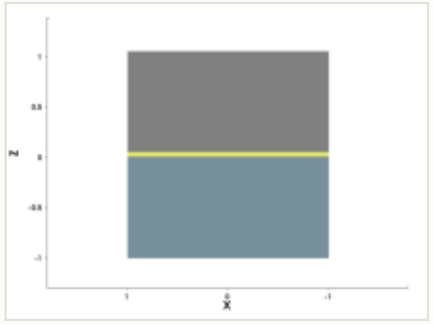




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

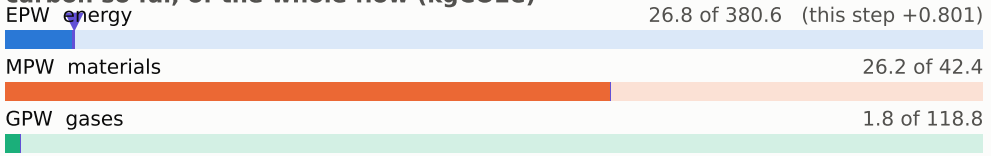
no mask for this step

this step

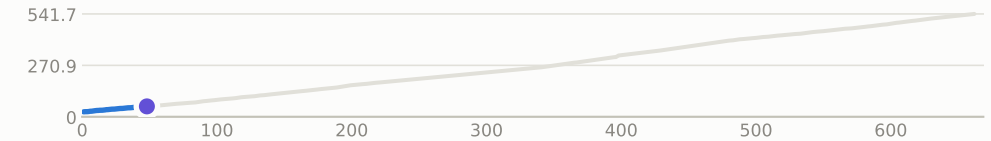
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

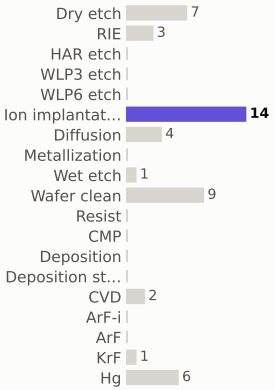
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

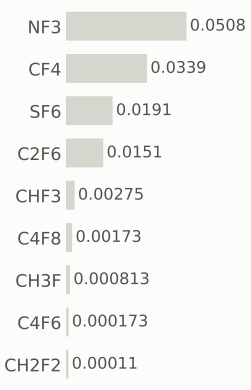


steps by process area



materials deposited (g)

gases emitted (g)



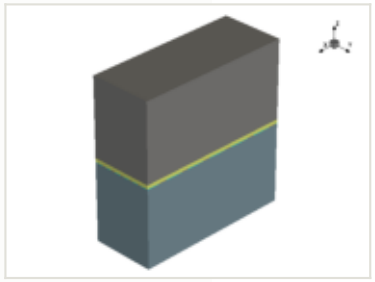
Step #520: Ion_implantation

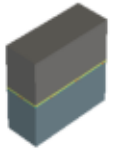
implant_ellipsoid · Ion_implantation

0.801	55.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

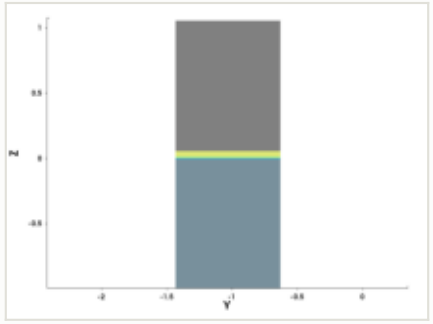
3D structure after this step

zoomed to fit

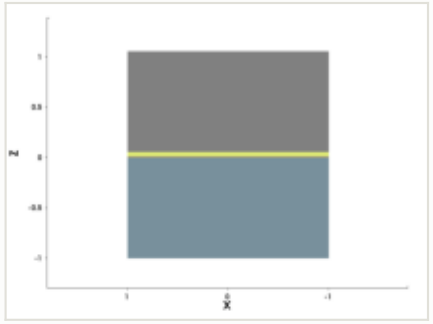




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

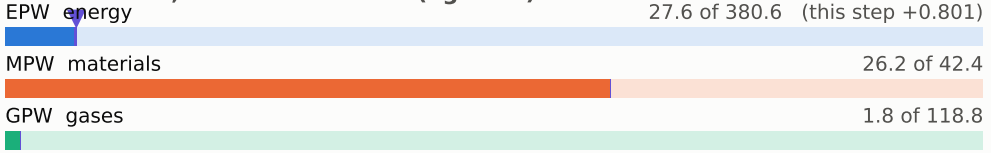
no mask for this step

this step

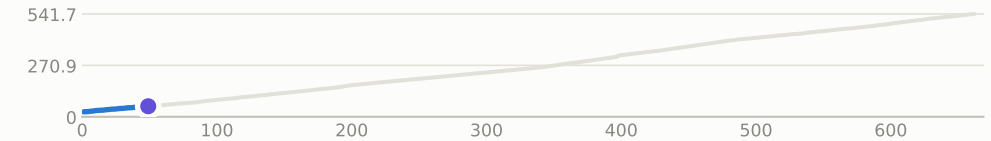
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

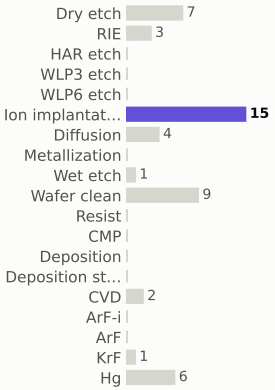
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

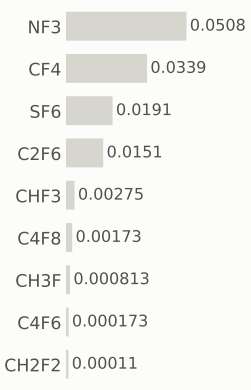


steps by process area



materials deposited (g)

gases emitted (g)



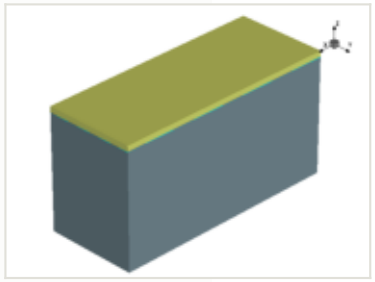
Step #530: Dry_etch

etch_strip · Dry_etch

0.801	56.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

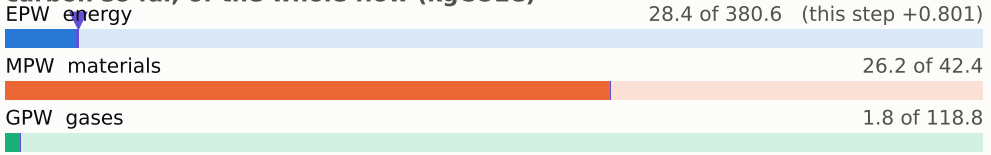


this step

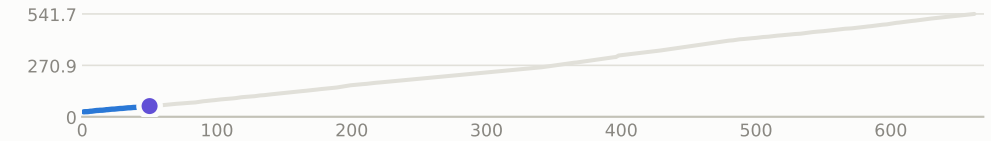
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

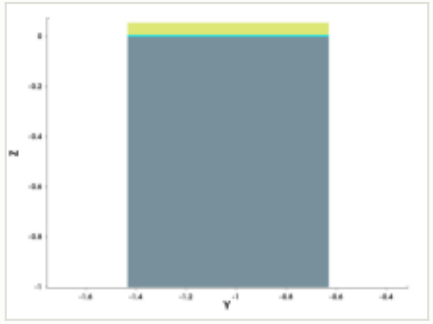
carbon so far, of the whole flow (kgCO2e)



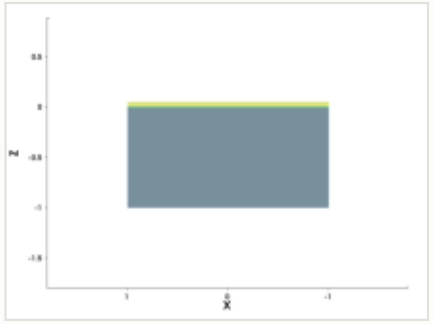
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



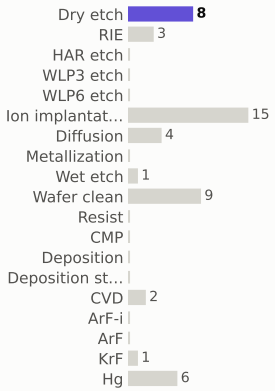
y cut at -1.032 µm



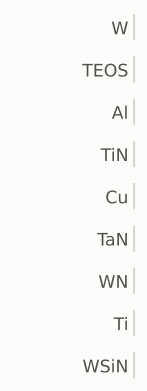
mask for this step

no mask for this step

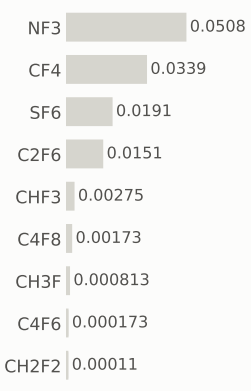
steps by process area



materials deposited (g)



gases emitted (g)



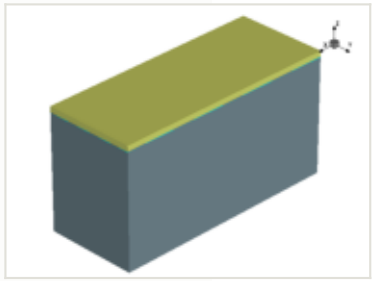
Step #540: Wafer_clean

clean · Wafer_clean

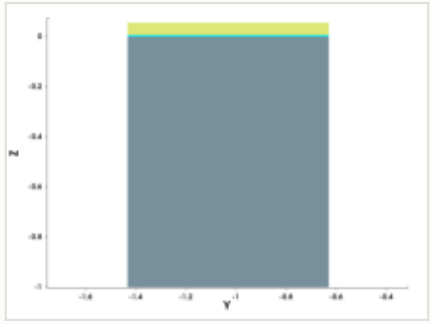
0.169	56.6	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

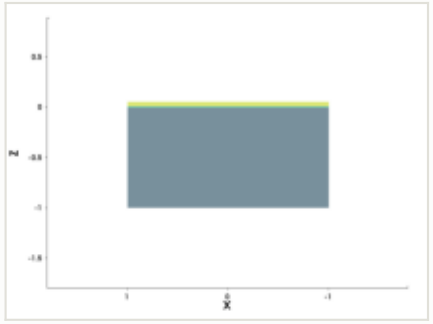
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

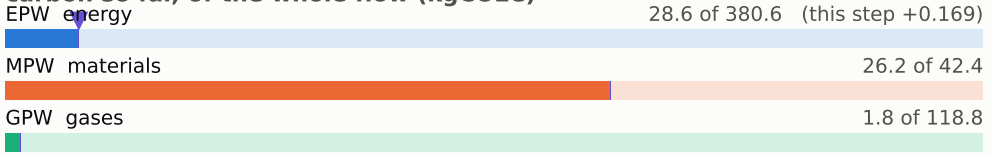
no mask for this step

this step

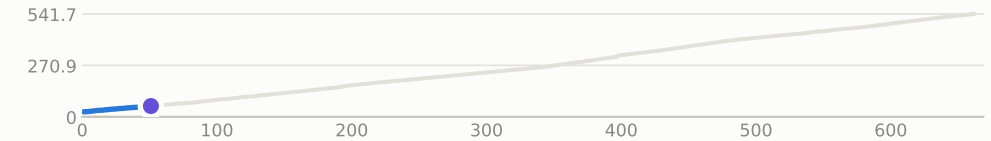
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

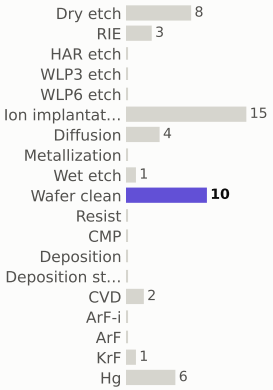
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

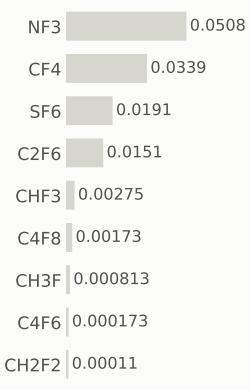


steps by process area



materials deposited (g)

gases emitted (g)



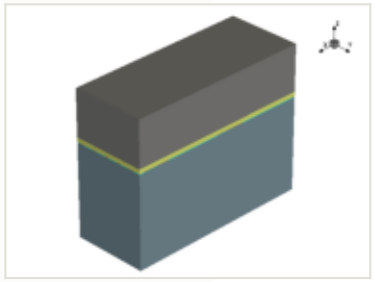
Step #550: Hg

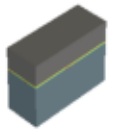
litho_patterning · Hg

0.199	56.8	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



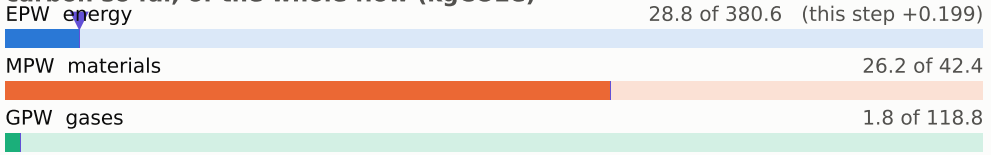


this step

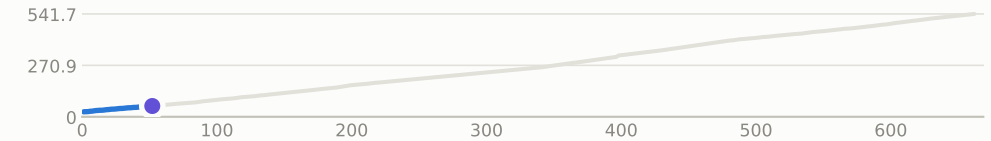
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

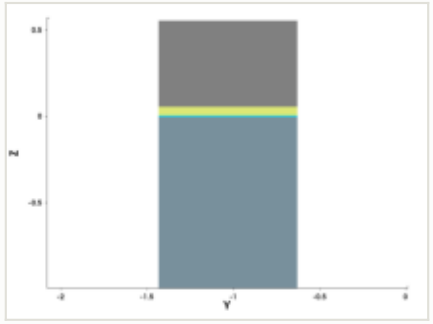
carbon so far, of the whole flow (kgCO2e)



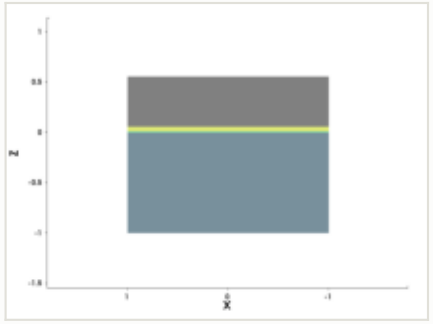
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

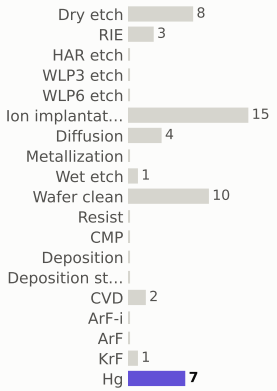


y cut at -1.032 µm

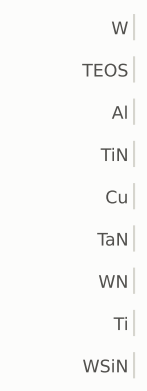


mask: layer 8:0, positive (OASIS)

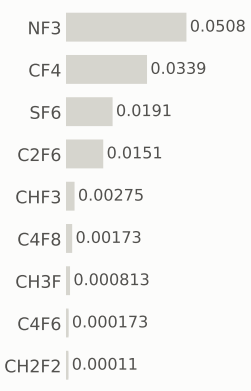
steps by process area



materials deposited (g)



gases emitted (g)



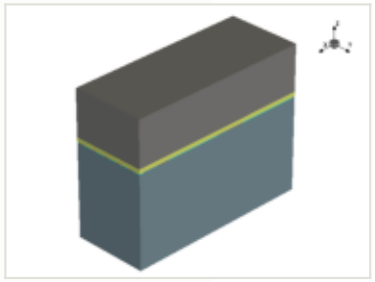
Step #560: Ion_implantation

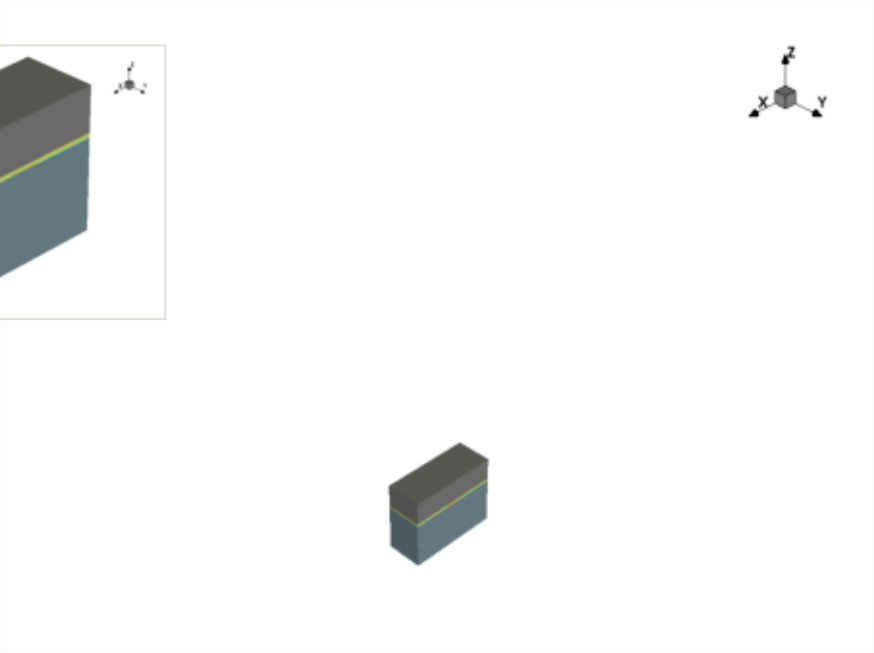
implant_ellipsoid · Ion_implantation

0.801	57.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

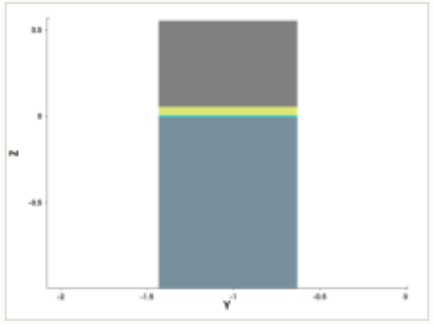
3D structure after this step

zoomed to fit

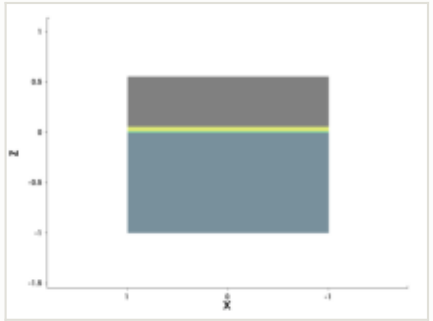




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

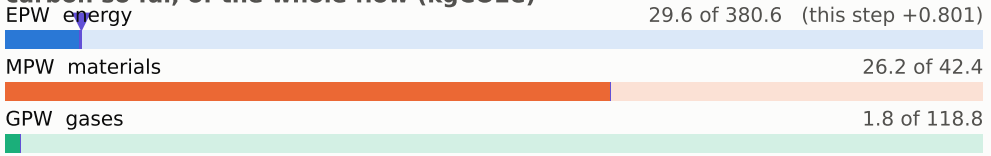
no mask for this step

this step

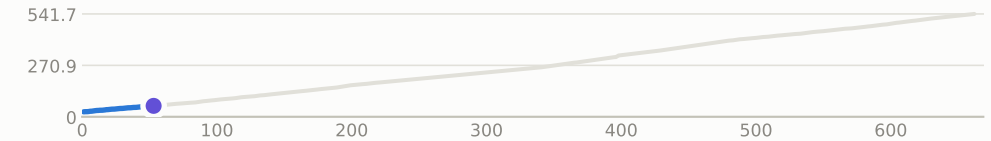
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

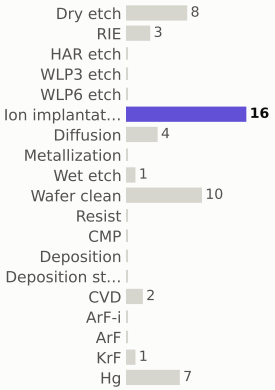
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

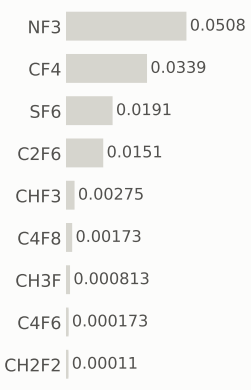


steps by process area



materials deposited (g)

gases emitted (g)



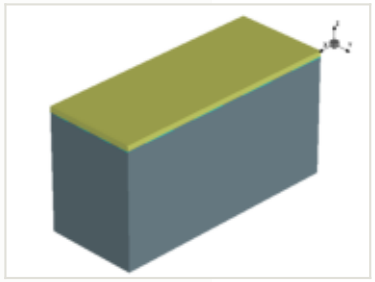
Step #570: Dry_etch

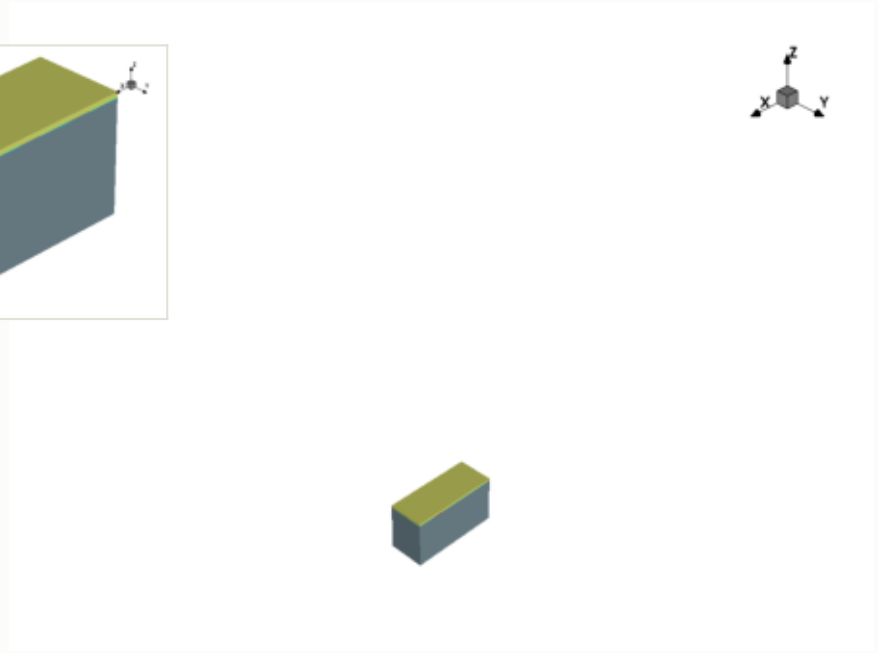
etch_strip · Dry_etch

0.801	58.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



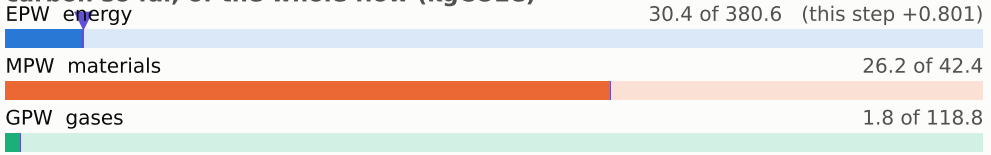


this step

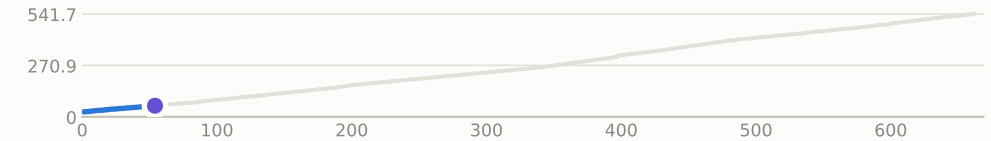
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

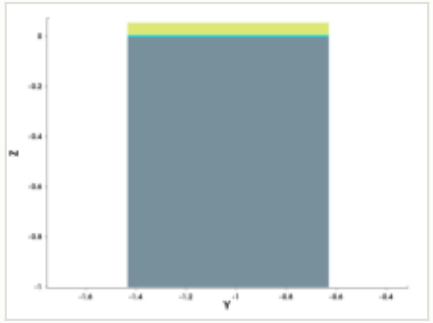
carbon so far, of the whole flow (kgCO2e)



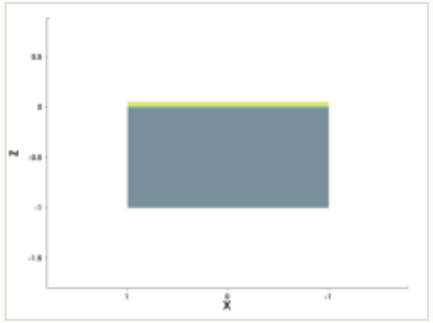
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



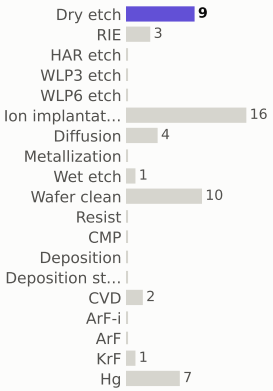
y cut at -1.032 µm



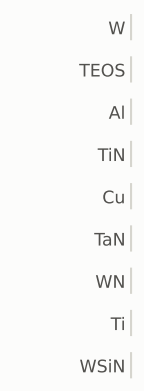
mask for this step

no mask for this step

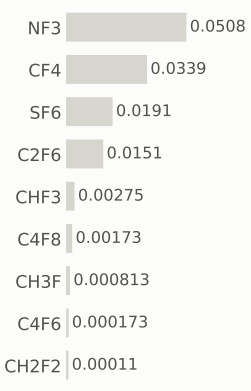
steps by process area



materials deposited (g)



gases emitted (g)



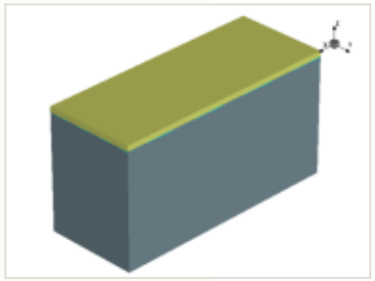
Step #580: Wafer_clean

clean · Wafer_clean

0.169	58.6	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



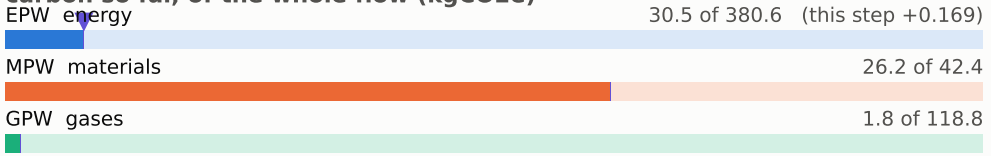
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

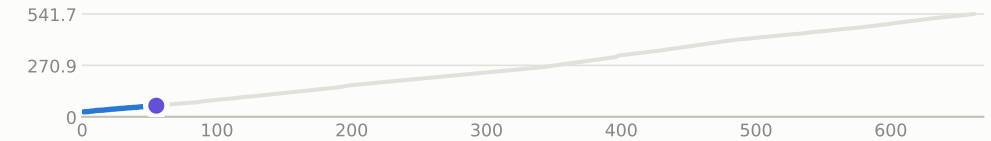
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

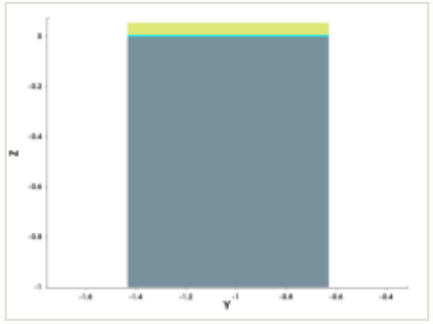
carbon so far, of the whole flow (kgCO2e)



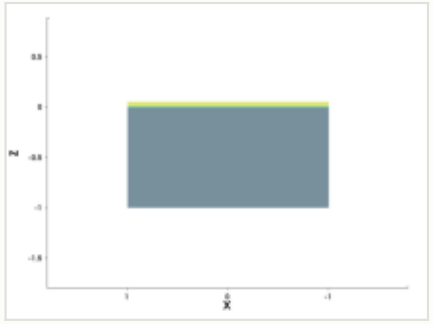
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



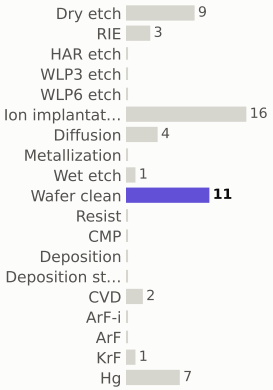
y cut at -1.032 µm



mask for this step

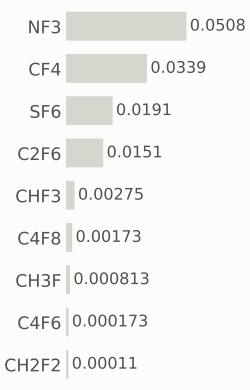
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



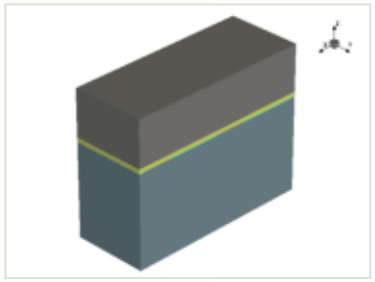
Step #590: Hg

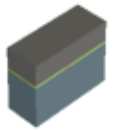
litho_patterning · Hg

0.199	58.8	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



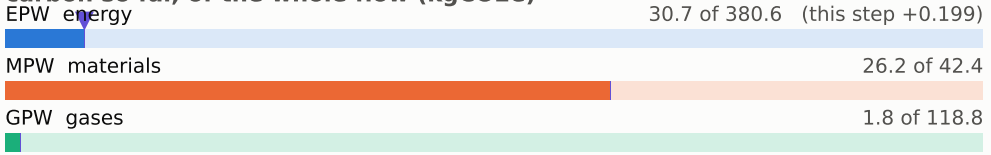


this step

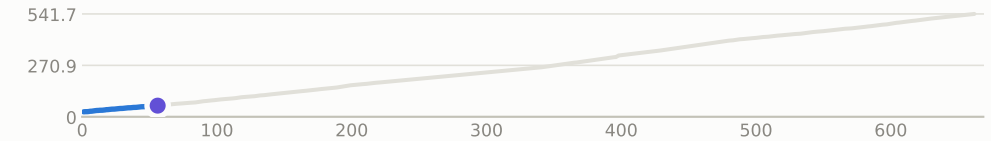
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

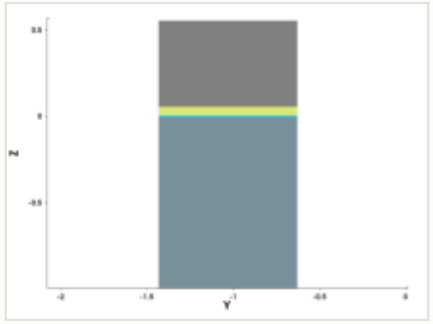
carbon so far, of the whole flow (kgCO2e)



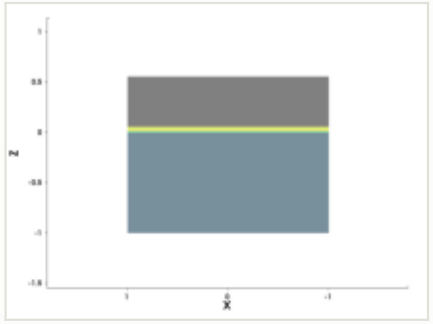
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

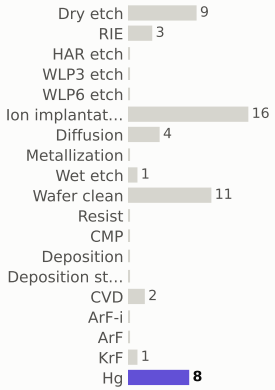


y cut at -1.032 µm

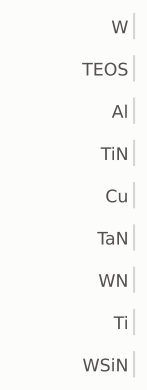


mask: layer 9:0, positive (OASIS)

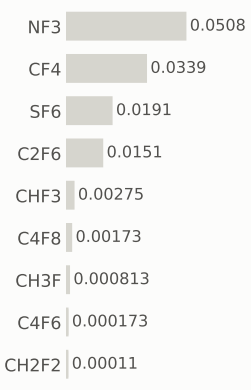
steps by process area



materials deposited (g)



gases emitted (g)



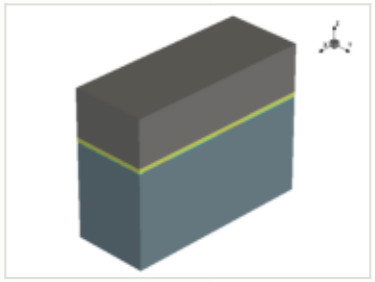
Step #600: Ion_implantation

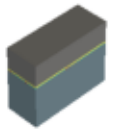
implant_ellipsoid · Ion_implantation

0.801	59.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

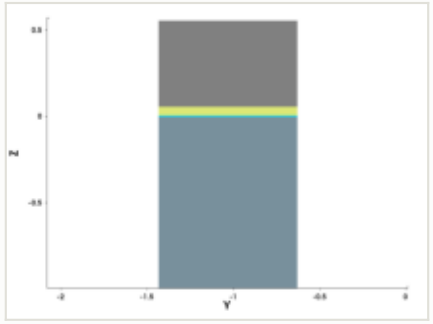
3D structure after this step

zoomed to fit

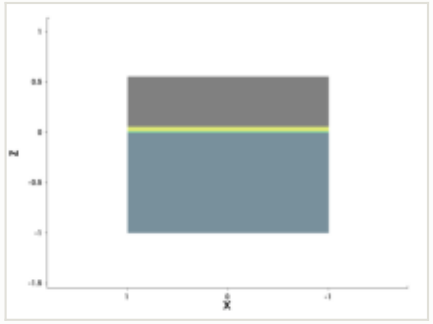




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

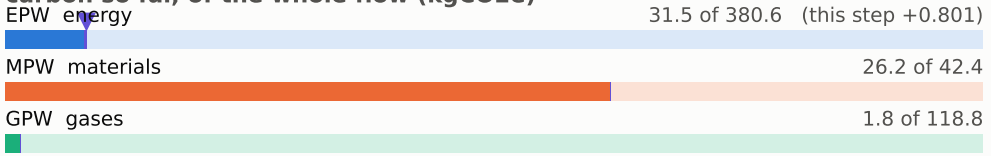
no mask for this step

this step

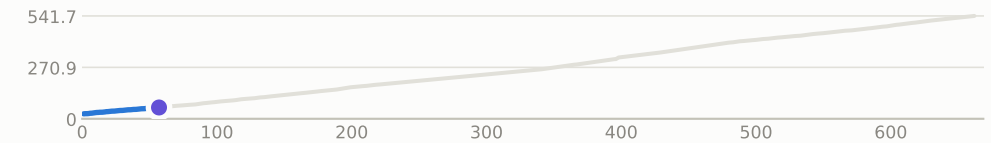
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

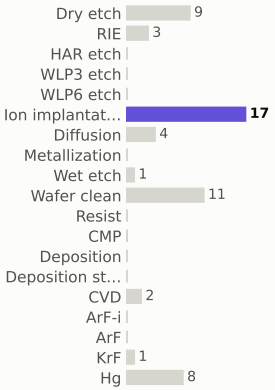
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

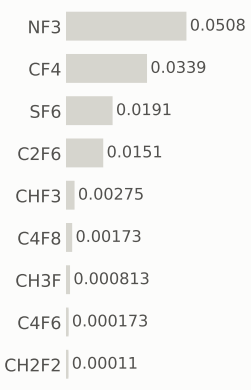


steps by process area



materials deposited (g)

gases emitted (g)



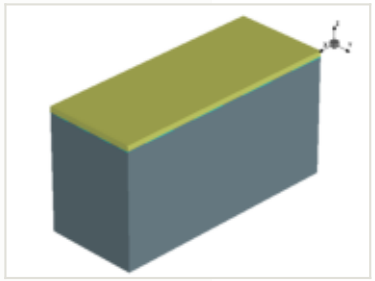
Step #610: Dry_etch

etch_strip · Dry_etch

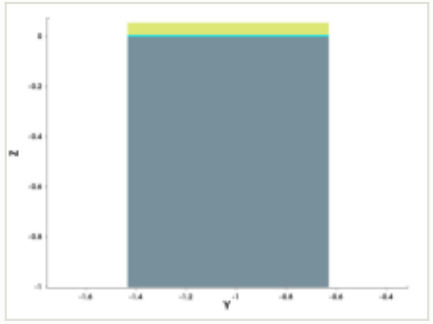
0.801	60.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

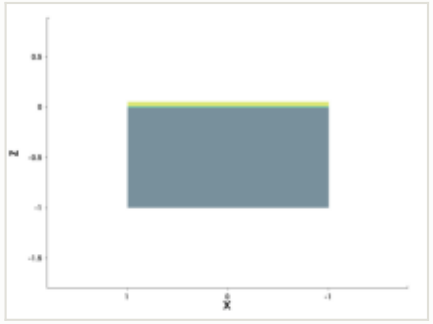
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



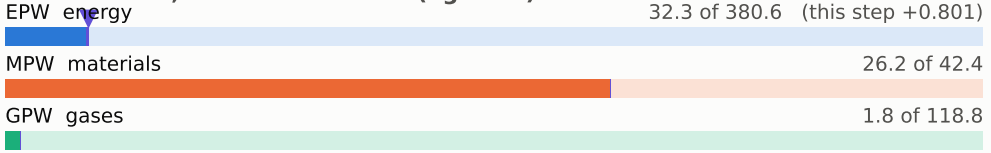
mask for this step

no mask for this step

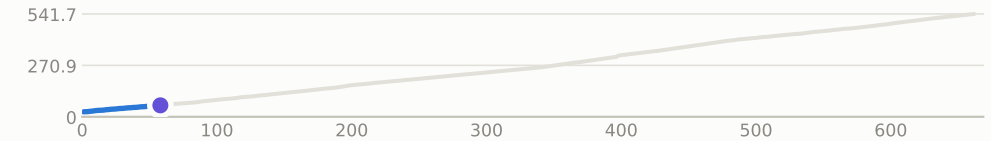
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

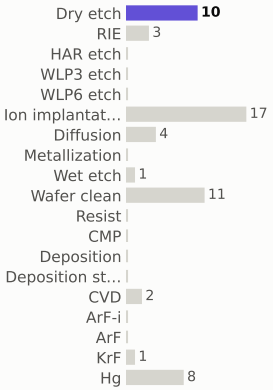
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

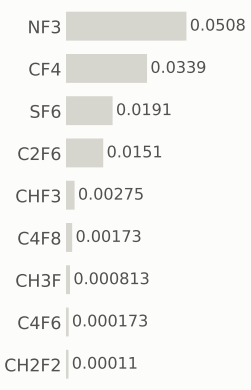


steps by process area



materials deposited (g)

gases emitted (g)



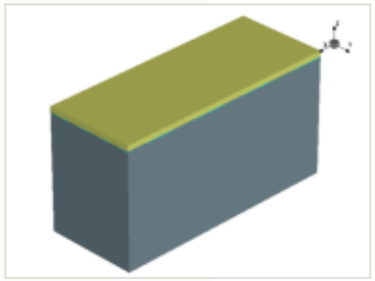
Step #620: Wafer_clean

clean · Wafer_clean

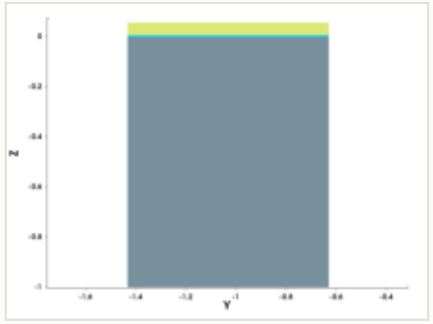
0.169	60.6	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

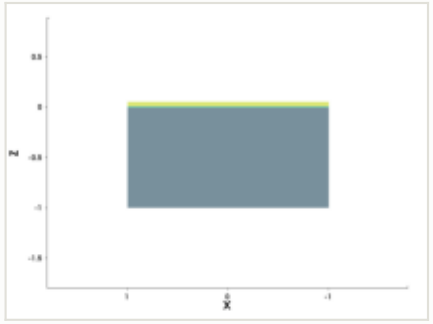
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

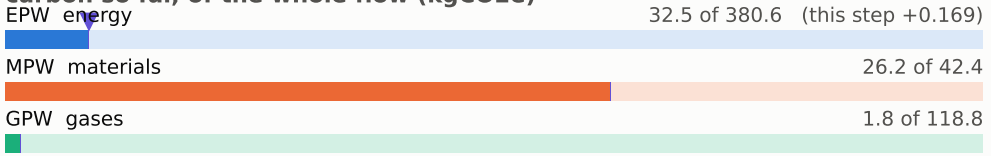
no mask for this step

this step

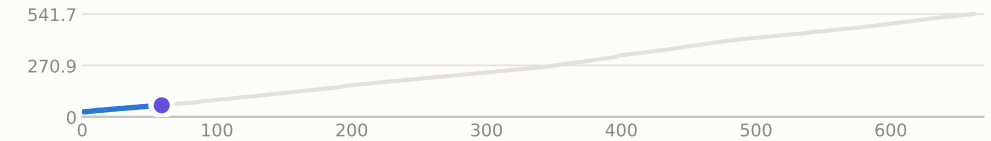
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

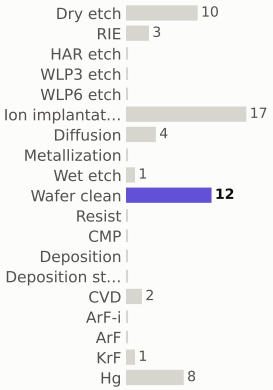
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

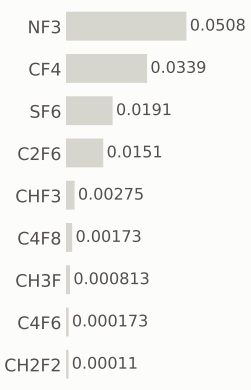


steps by process area



materials deposited (g)

gases emitted (g)



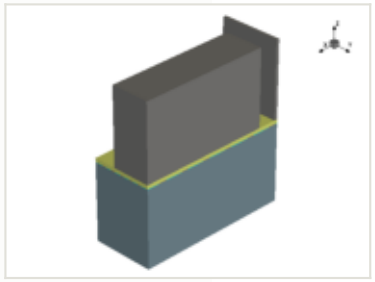
Step #640: ArF

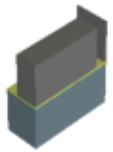
litho_patterning · ArF

0.199	60.7	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

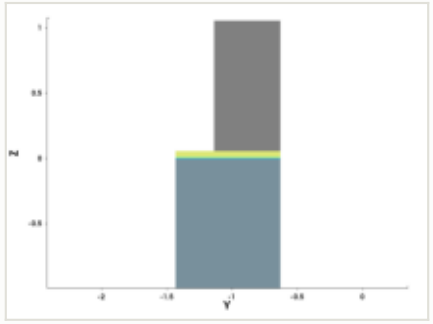
3D structure after this step

zoomed to fit

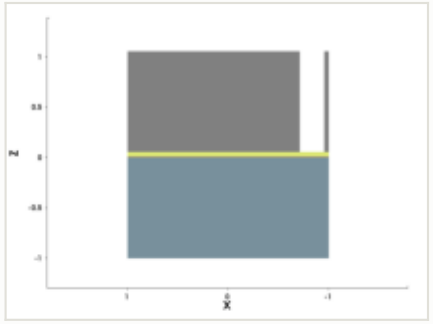




x cut at 0.000 µm



y cut at -1.032 µm

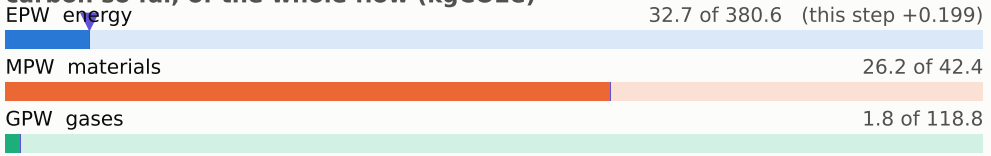


mask: layer 10:0, positive (OASIS)

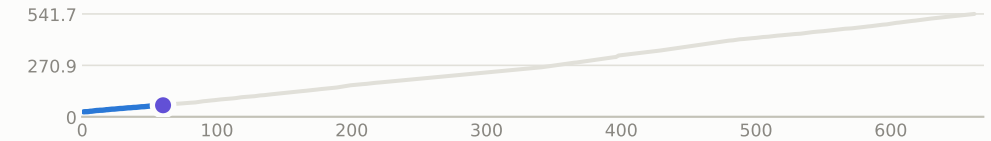
this step

process area	ArF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

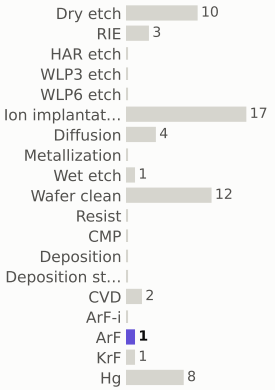
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

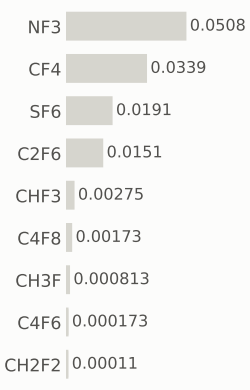


steps by process area



materials deposited (g)

gases emitted (g)



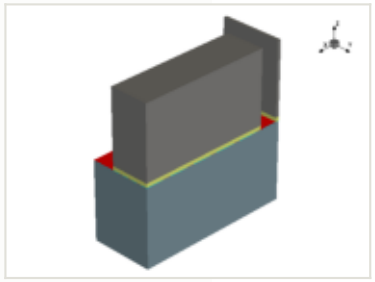
Step #650: RIE

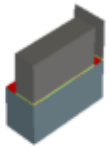
etch_anisotropic · RIE

1.04	61.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

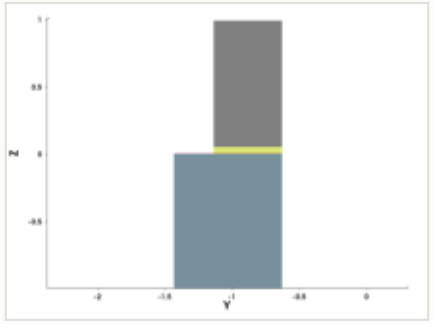
3D structure after this step

zoomed to fit

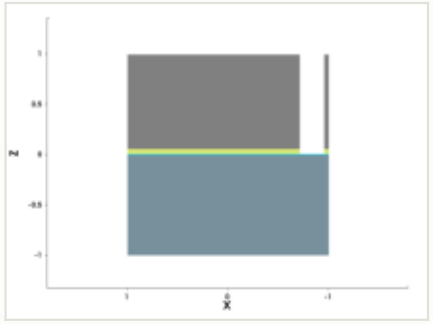




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

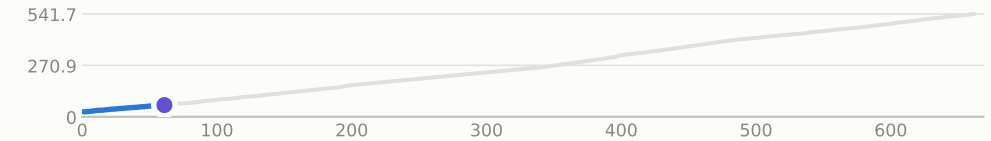
carbon so far, of the whole flow (kgCO2e)

EPW energy — 33.5 of 380.6 (this step +0.801)

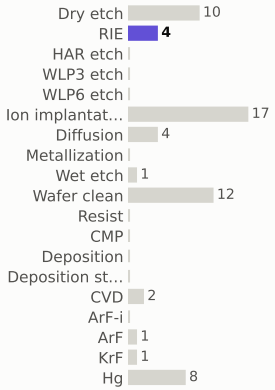
MPW materials — 26.3 of 42.4 (this step +0.0571)

GPW gases — 1.99 of 118.8 (this step +0.187)

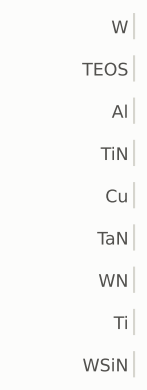
embodied carbon up to each step (kgCO2e per wafer)



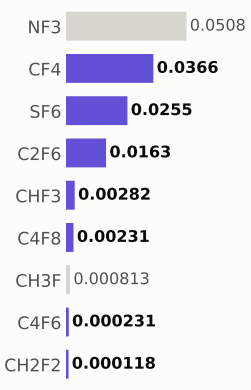
steps by process area



materials deposited (g)



gases emitted (g)



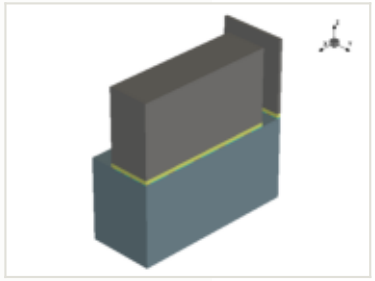
Step #660: RIE

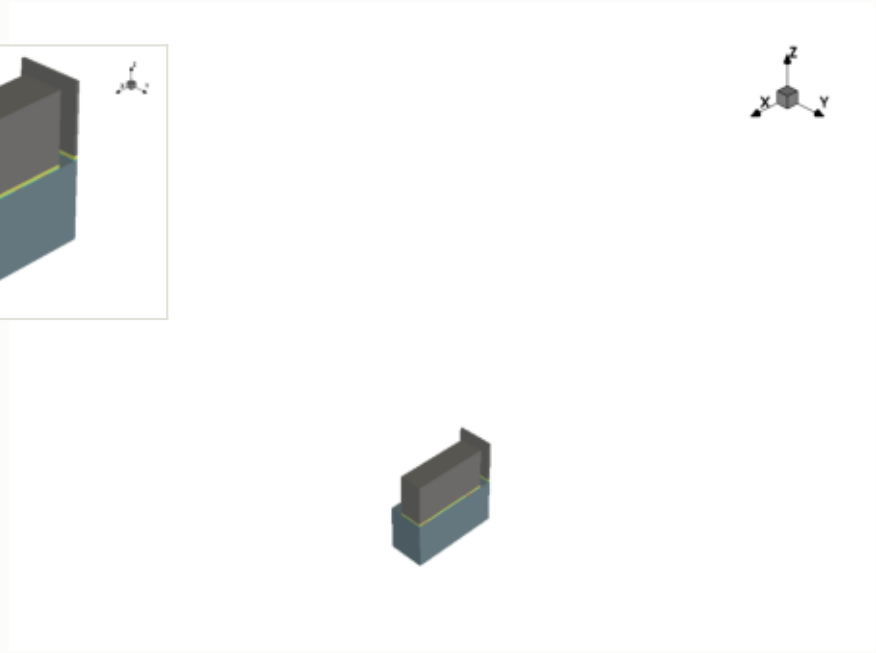
etch_anisotropic · RIE

1.04	62.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



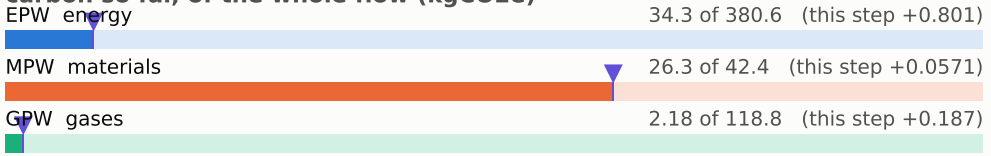


this step

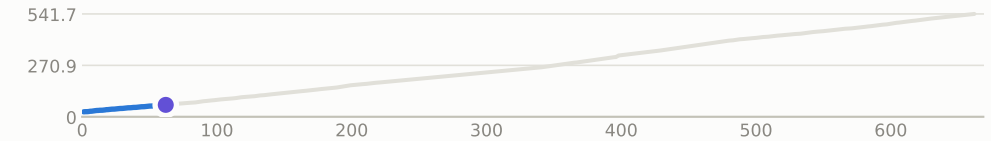
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

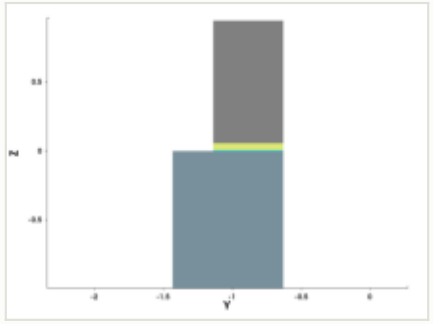
carbon so far, of the whole flow (kgCO2e)



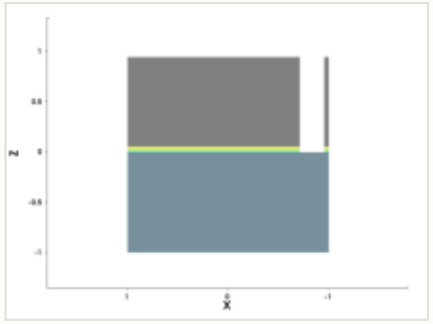
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



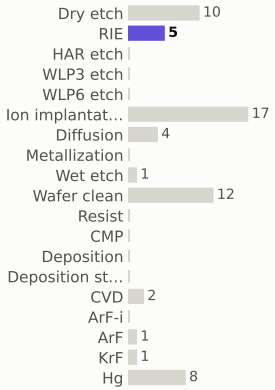
y cut at -1.032 µm



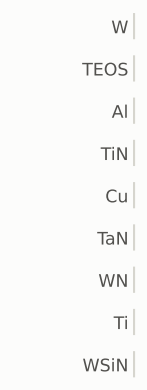
mask for this step

no mask for this step

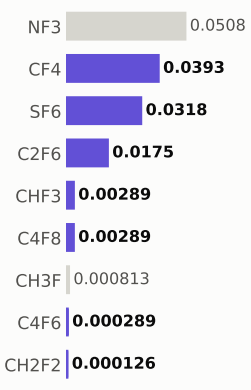
steps by process area



materials deposited (g)



gases emitted (g)



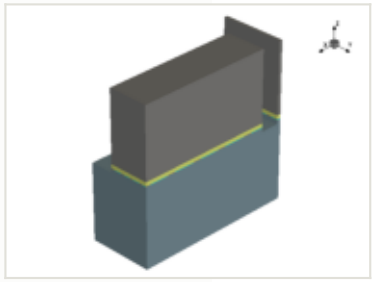
Step #670: RIE

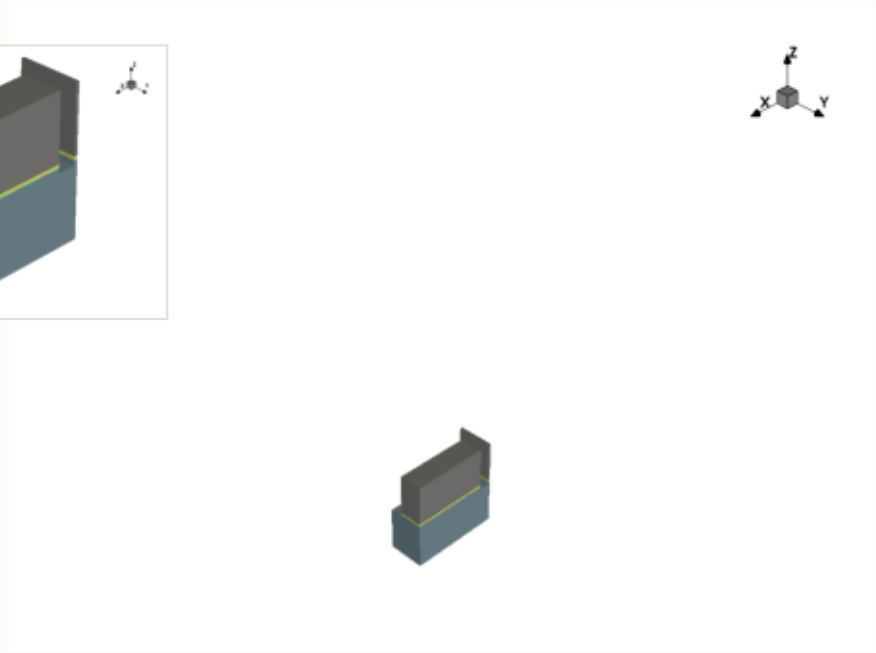
etch_anisotropic · RIE

1.04	63.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



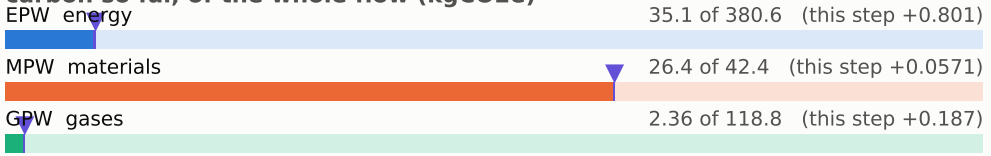


this step

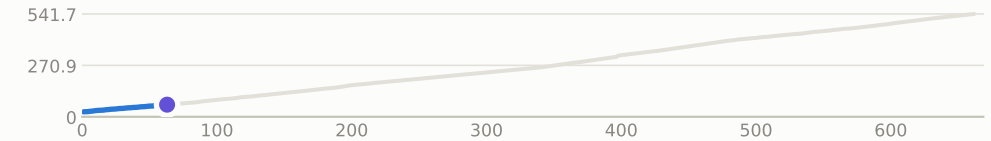
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

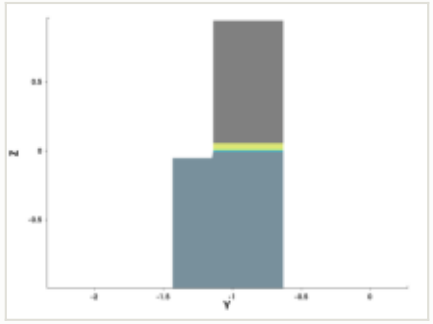
carbon so far, of the whole flow (kgCO2e)



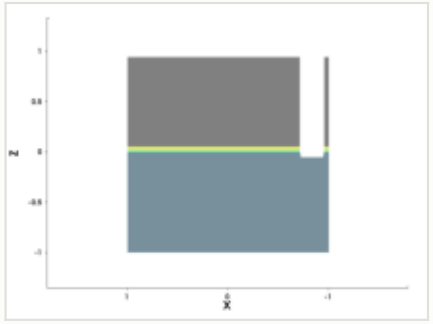
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



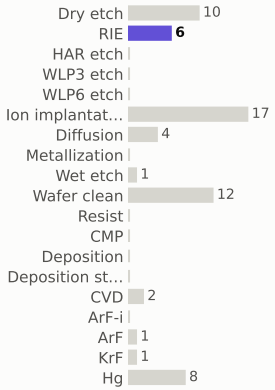
y cut at -1.032 µm



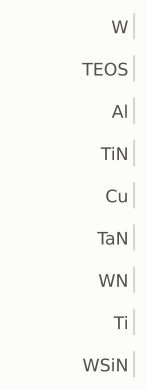
mask for this step

no mask for this step

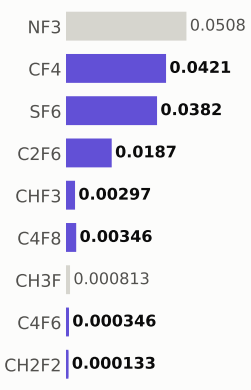
steps by process area



materials deposited (g)



gases emitted (g)



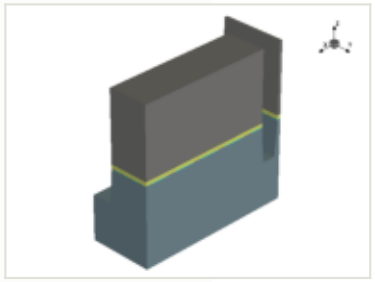
Step #680: RIE

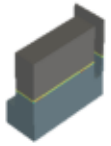
etch_anisotropic · RIE

1.04	64.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

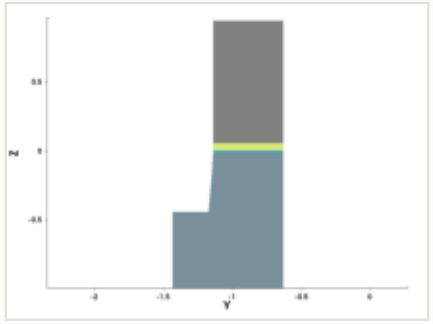
3D structure after this step

zoomed to fit

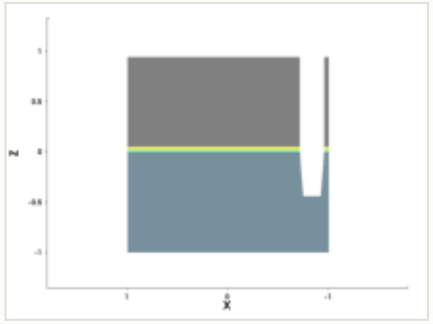




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

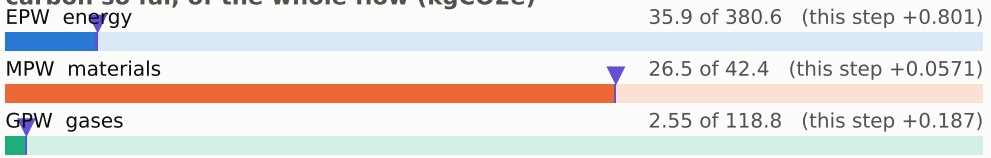
no mask for this step

this step

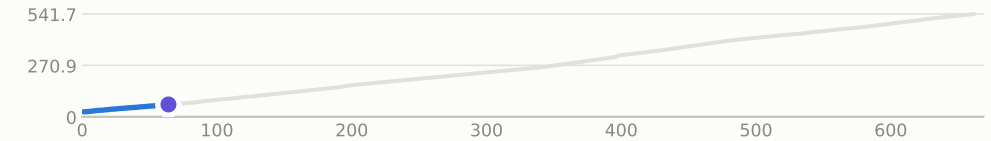
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

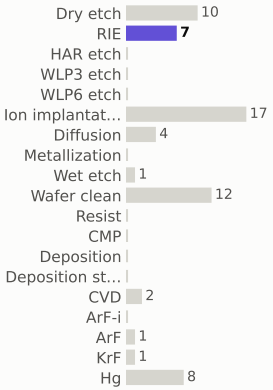
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

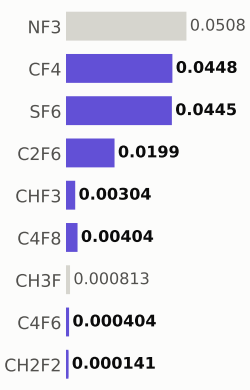


steps by process area



materials deposited (g)

gases emitted (g)



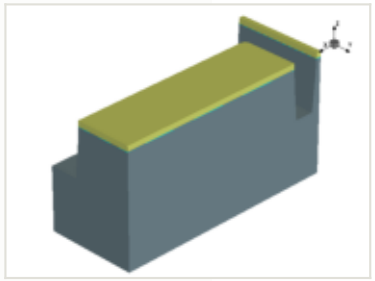
Step #690: Dry_etch

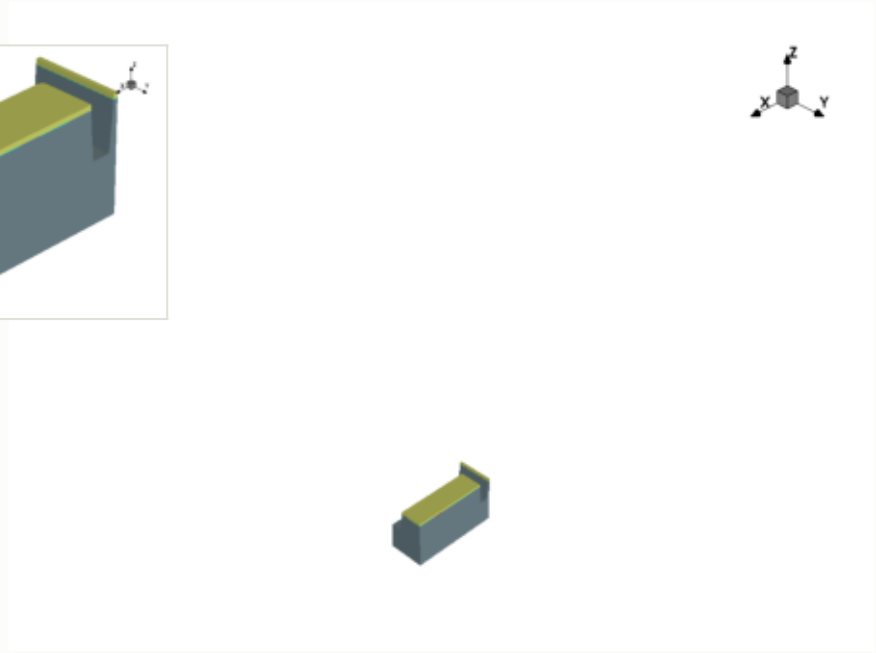
etch_strip · Dry_etch

0.801	65.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

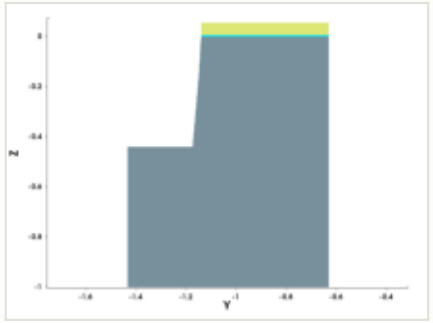
3D structure after this step

zoomed to fit

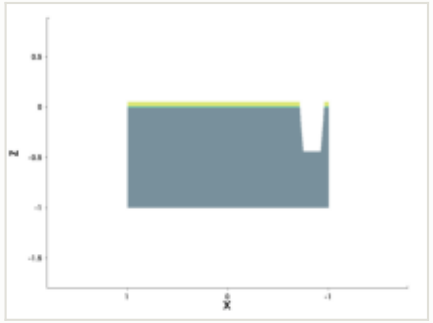




x cut at 0.000 µm



y cut at -1.032 µm



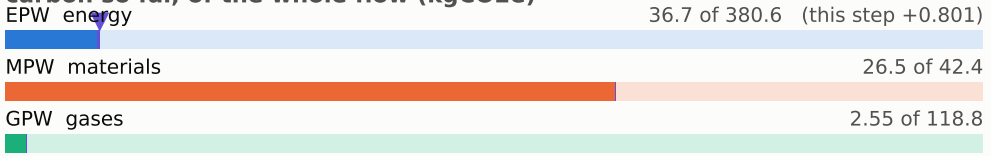
mask for this step

no mask for this step

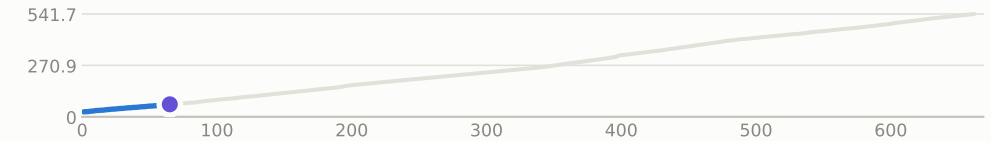
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

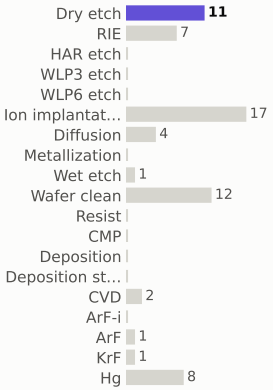
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

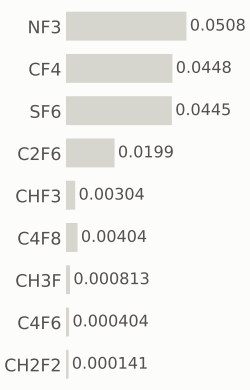


steps by process area



materials deposited (g)

gases emitted (g)



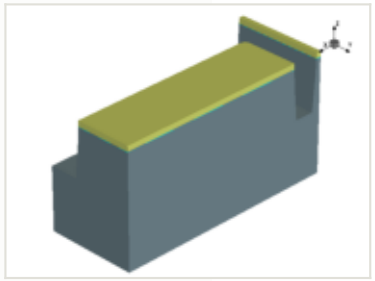
Step #700: Wafer_clean

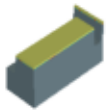
clean · Wafer_clean

0.169	65.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

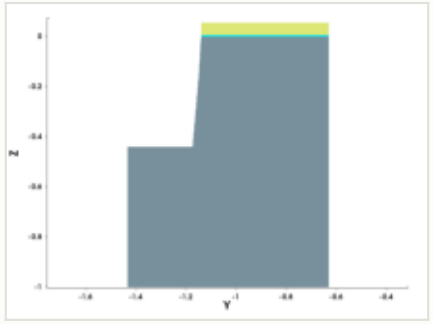
3D structure after this step

zoomed to fit

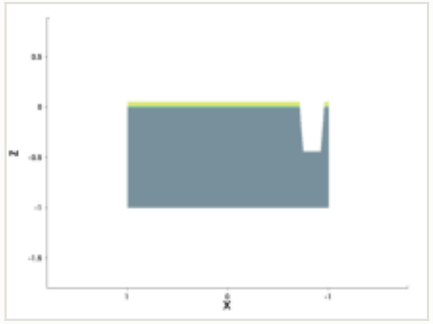




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

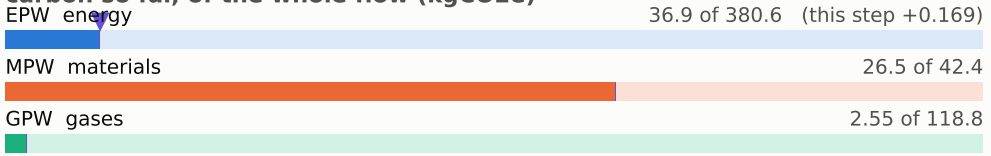
no mask for this step

this step

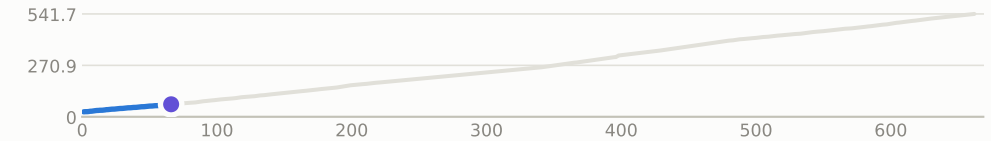
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

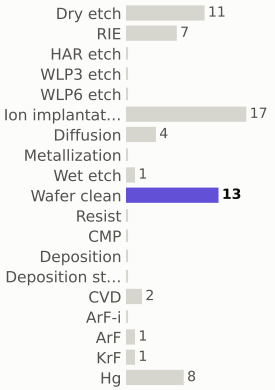
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

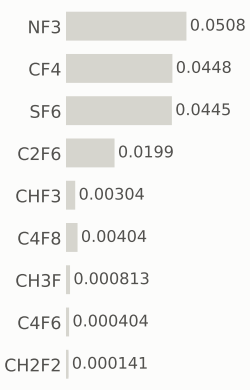


steps by process area



materials deposited (g)

gases emitted (g)



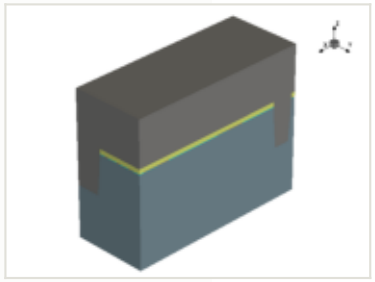
Step #710: Hg

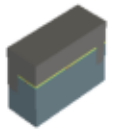
litho_patterning · Hg

0.199	66.1	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



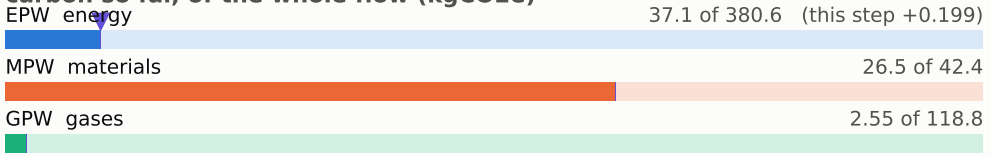


this step

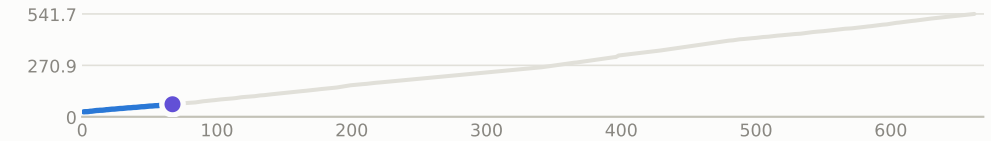
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

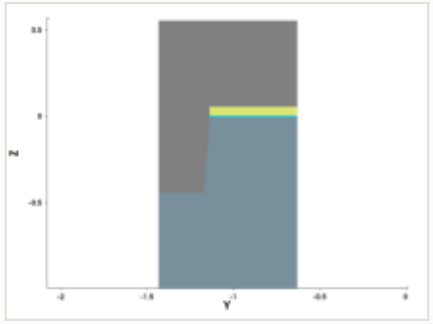
carbon so far, of the whole flow (kgCO2e)



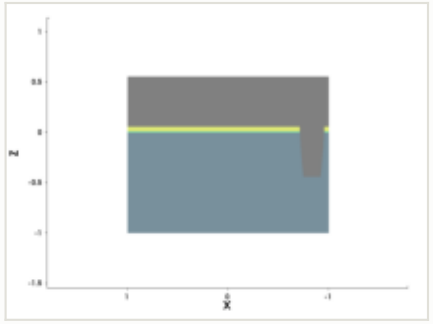
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

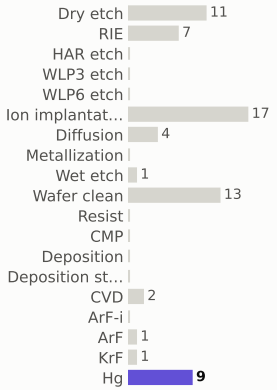


y cut at -1.032 µm

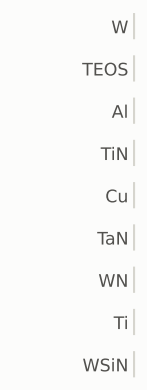


mask: layer 11:0, positive (OASIS)

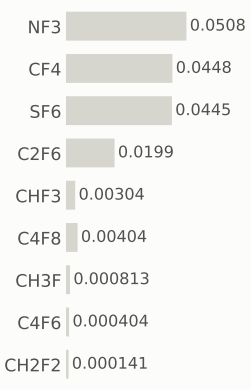
steps by process area



materials deposited (g)



gases emitted (g)



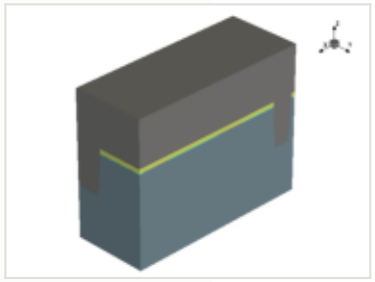
Step #720: Ion_implantation

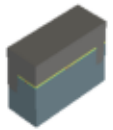
implant_ellipsoid · Ion_implantation

0.801	66.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

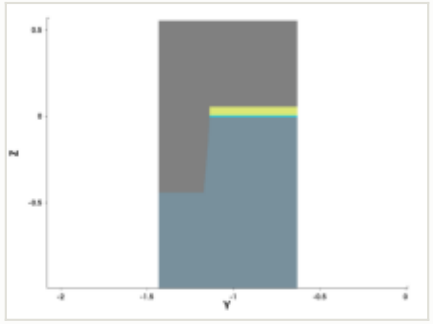
3D structure after this step

zoomed to fit

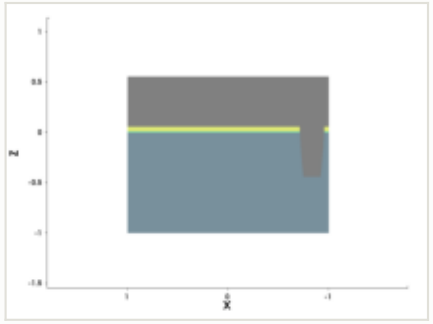




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

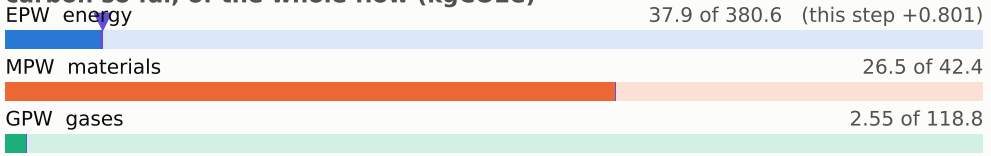
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

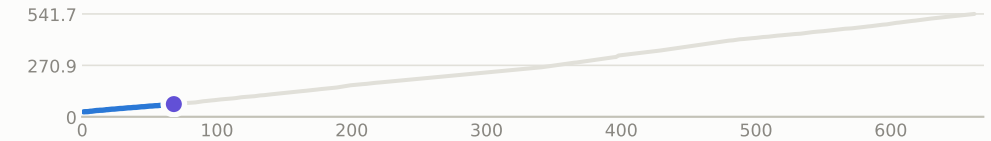
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

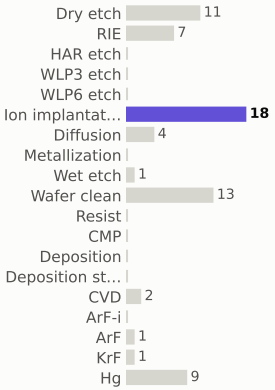
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

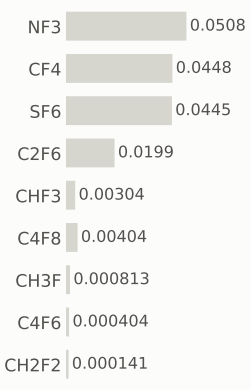


steps by process area



materials deposited (g)

gases emitted (g)



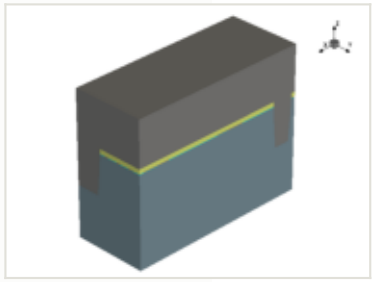
Step #730: Ion_implantation

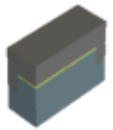
implant_ellipsoid · Ion_implantation

0.801	67.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

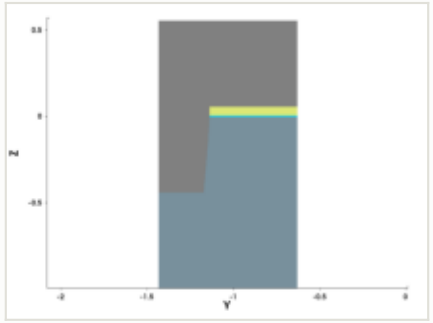
3D structure after this step

zoomed to fit

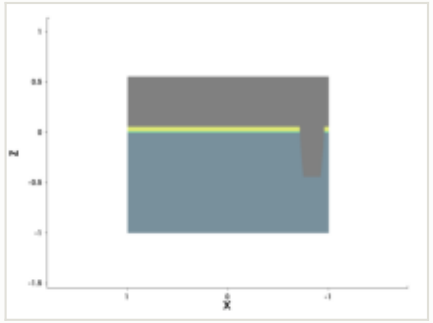




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

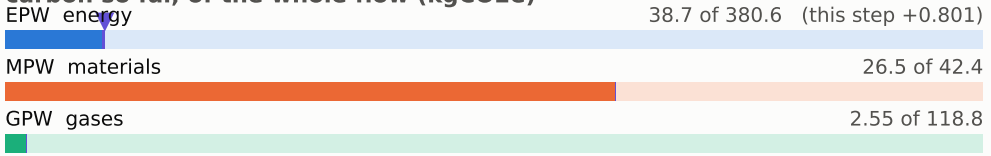
no mask for this step

this step

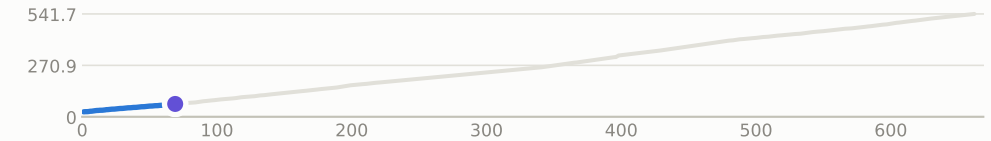
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

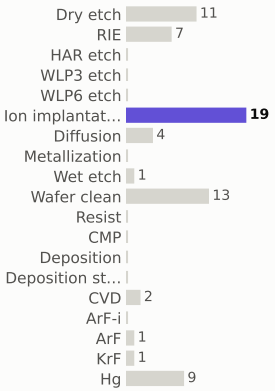
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

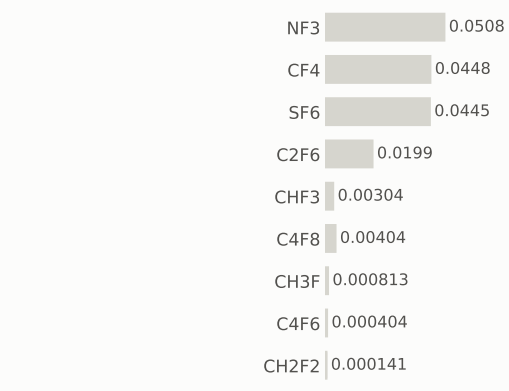


steps by process area



materials deposited (g)

gases emitted (g)



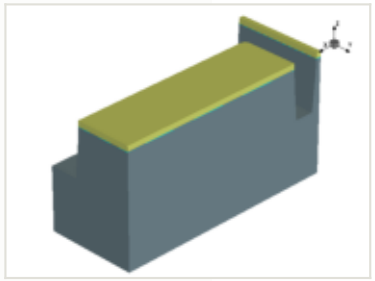
Step #740: Dry_etch

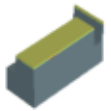
etch_strip · Dry_etch

0.801	68.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

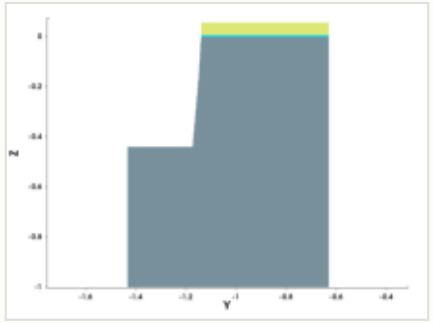
3D structure after this step

zoomed to fit

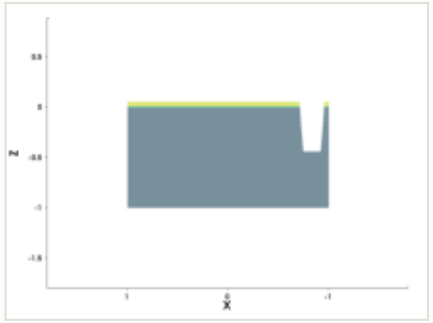




x cut at 0.000 µm



y cut at -1.032 µm



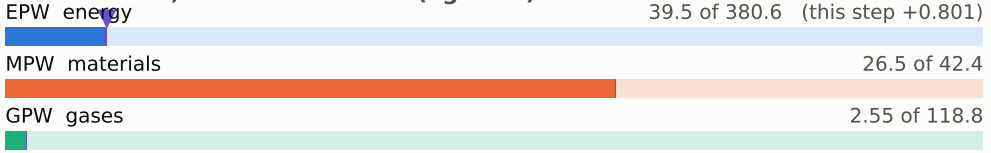
mask for this step

no mask for this step

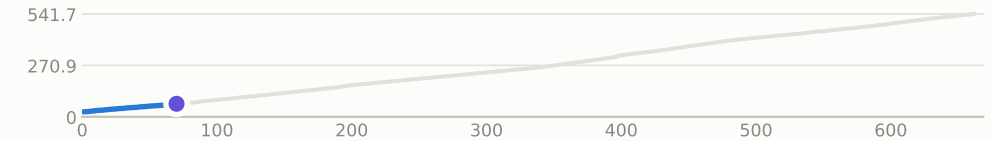
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

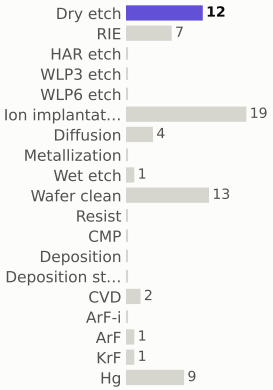
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

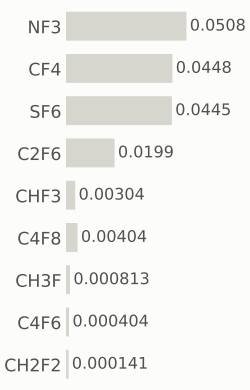


steps by process area



materials deposited (g)

gases emitted (g)



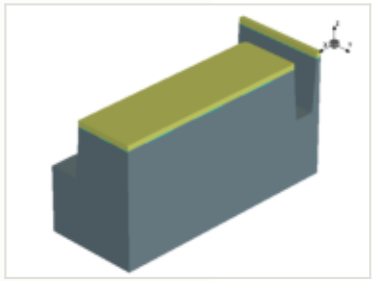
Step #750: Wafer_clean

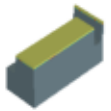
clean · Wafer_clean

0.169	68.7	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

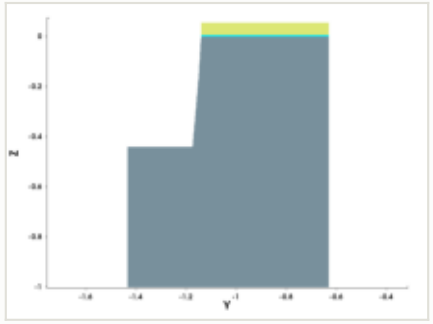
3D structure after this step

zoomed to fit

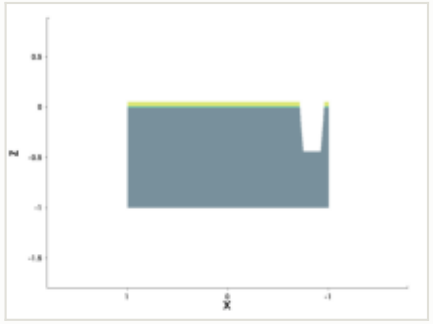




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

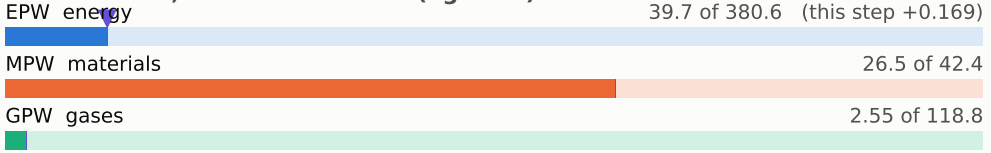
no mask for this step

this step

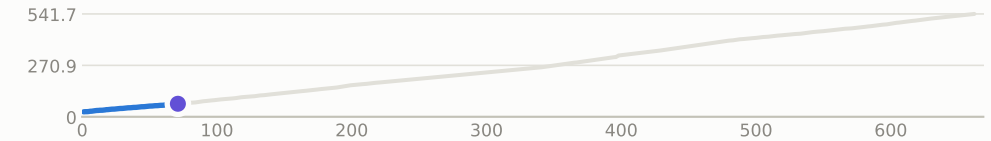
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

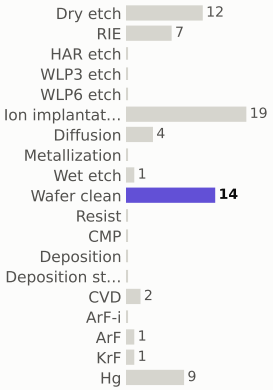
carbon so far, of the whole flow (kgCO2e)



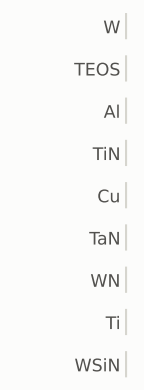
embodied carbon up to each step (kgCO2e per wafer)



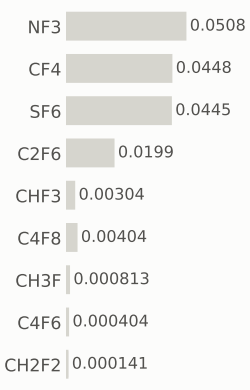
steps by process area



materials deposited (g)



gases emitted (g)



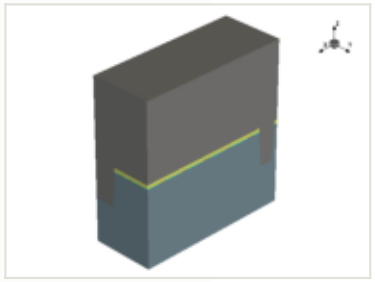
Step #760: Hg

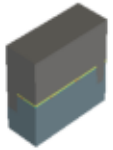
litho_patterning · Hg

0.199	68.9	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



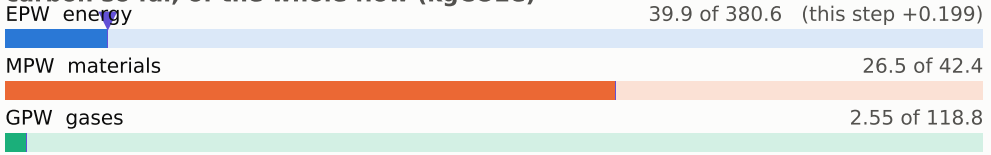


this step

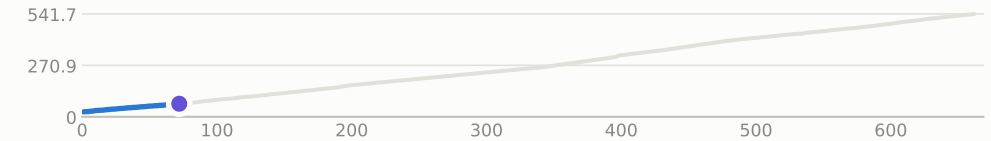
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

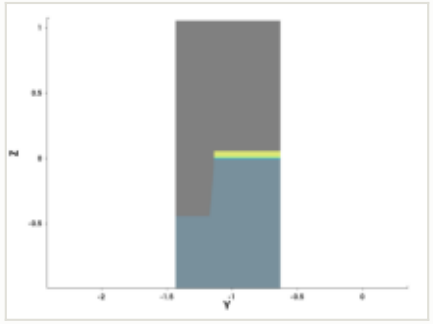
carbon so far, of the whole flow (kgCO2e)



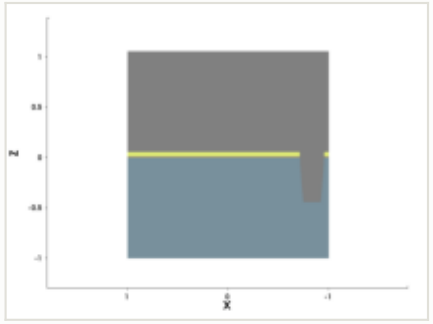
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

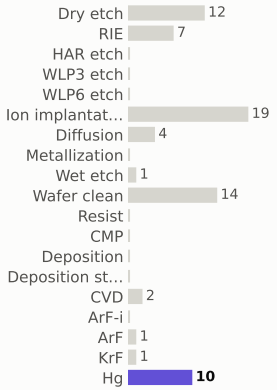


y cut at -1.032 µm

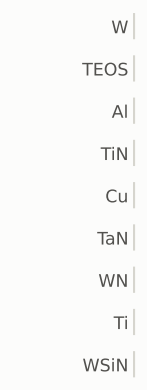


mask: layer 12:0, positive (OASIS)

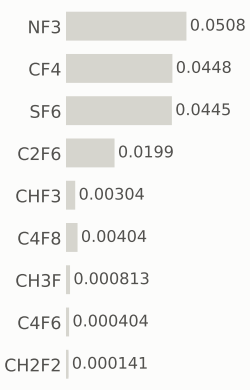
steps by process area



materials deposited (g)



gases emitted (g)



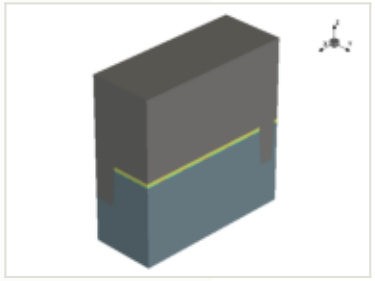
Step #770: Ion_implantation

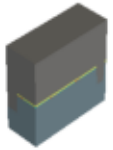
implant_ellipsoid · Ion_implantation

0.801	69.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

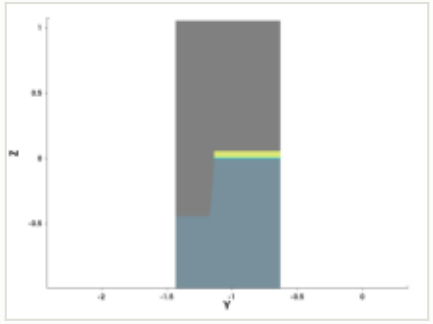
3D structure after this step

zoomed to fit

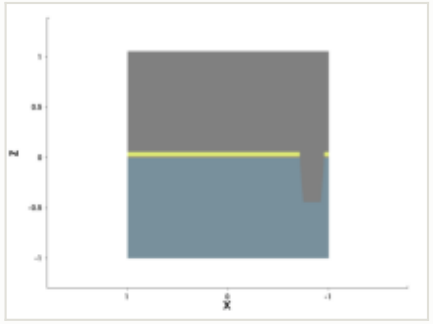




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

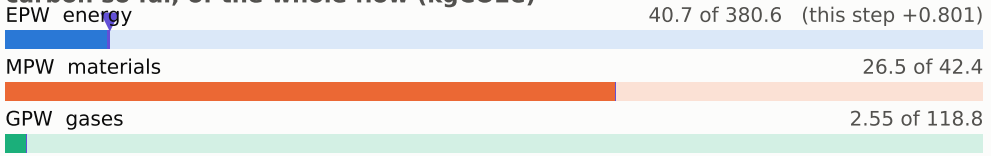
no mask for this step

this step

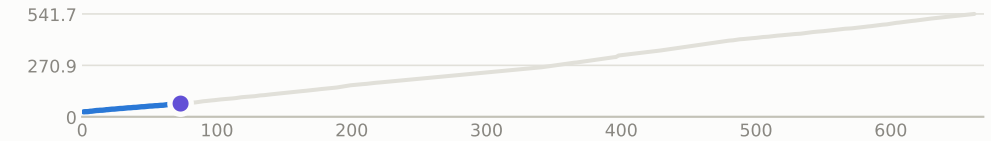
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

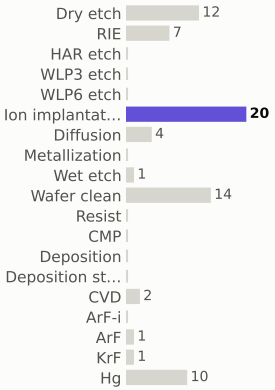
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

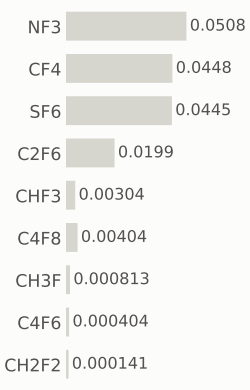


steps by process area



materials deposited (g)

gases emitted (g)



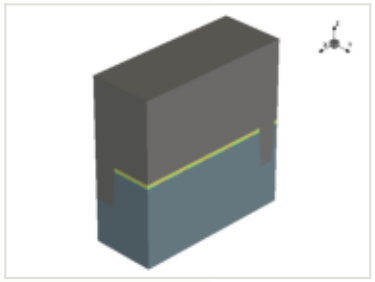
Step #780: Ion_implantation

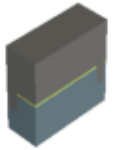
implant_ellipsoid · Ion_implantation

0.801	70.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

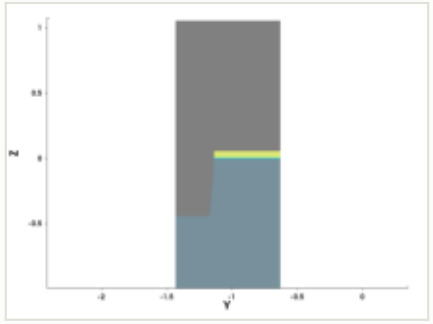
3D structure after this step

zoomed to fit

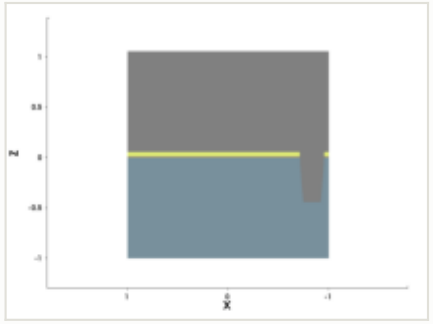




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

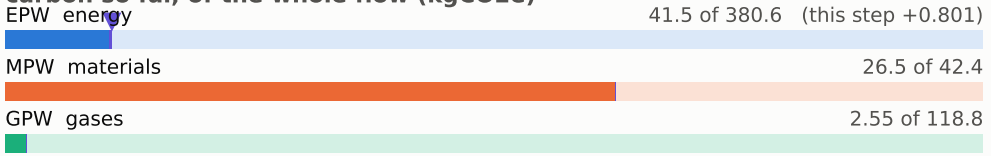
no mask for this step

this step

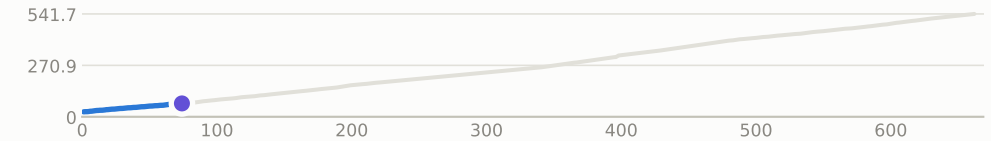
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

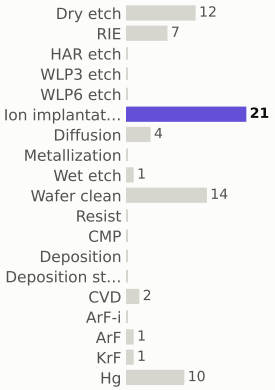
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

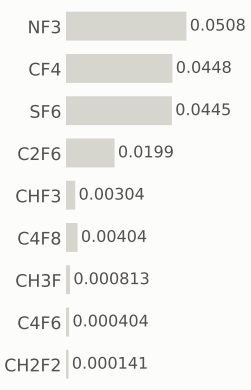


steps by process area



materials deposited (g)

gases emitted (g)



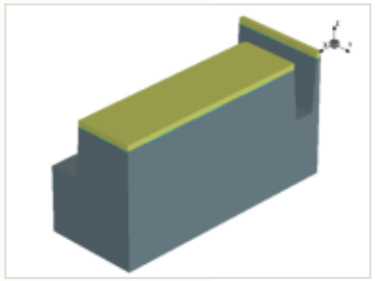
Step #790: Dry_etch

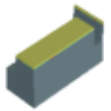
etch_strip · Dry_etch

0.801	71.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

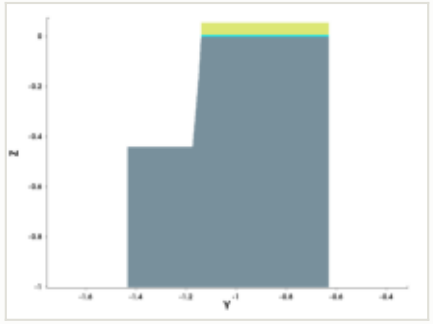
3D structure after this step

zoomed to fit

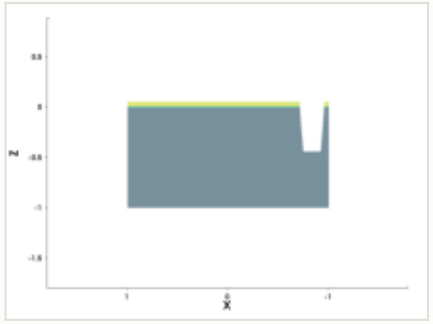




x cut at 0.000 µm



y cut at -1.032 µm



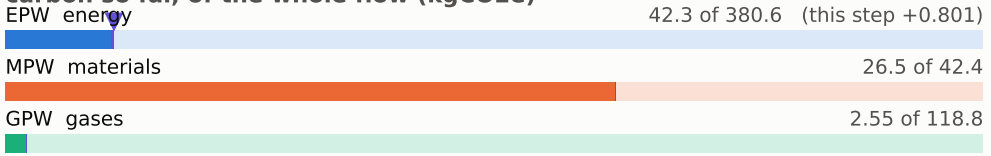
mask for this step

no mask for this step

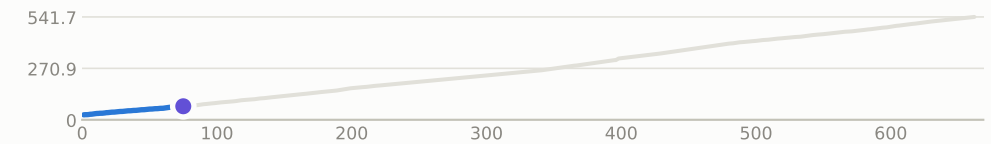
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

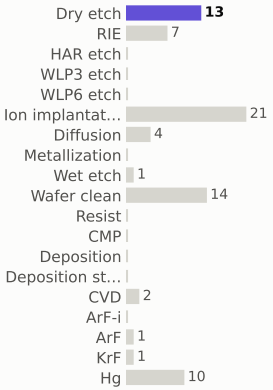
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

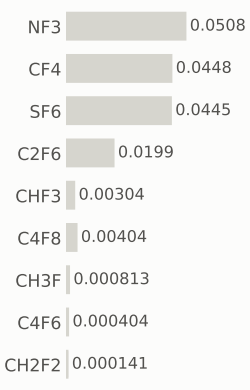


steps by process area



materials deposited (g)

gases emitted (g)



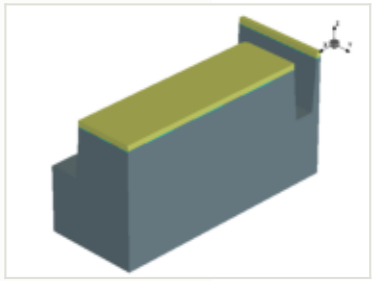
Step #800: Wafer_clean

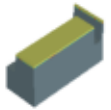
clean · Wafer_clean

0.169	71.4	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

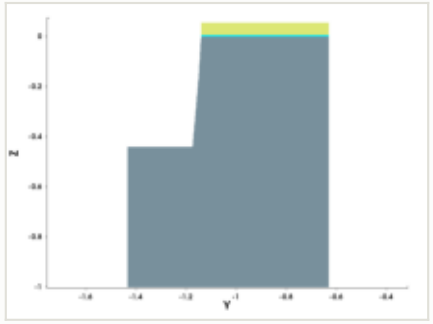
3D structure after this step

zoomed to fit

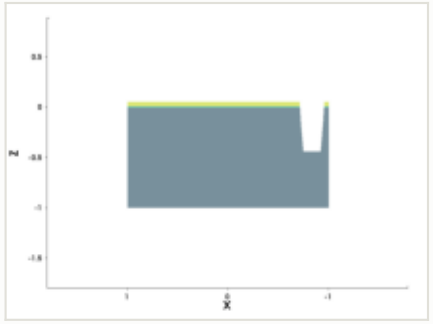




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

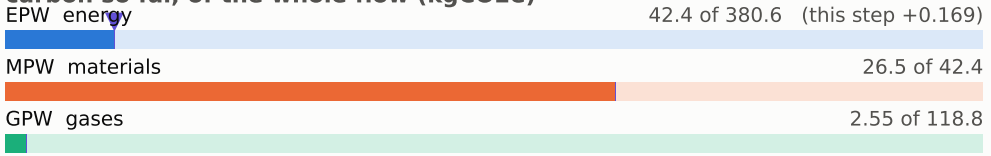
no mask for this step

this step

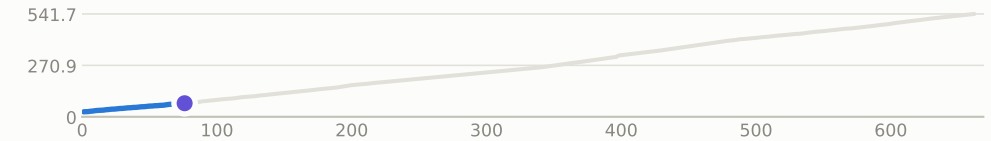
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

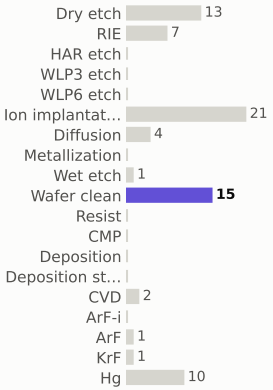
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

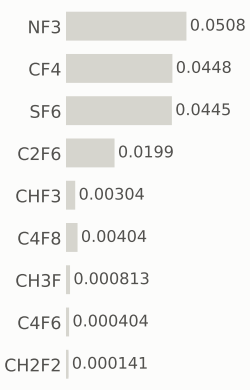


steps by process area



materials deposited (g)

gases emitted (g)



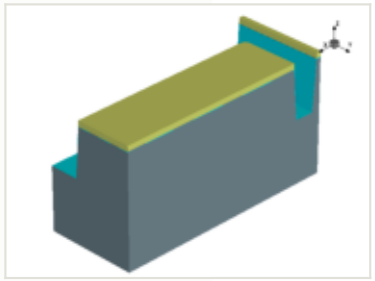
Step #810: Diffusion

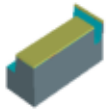
depo_isotropic · Diffusion

0.801	72.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



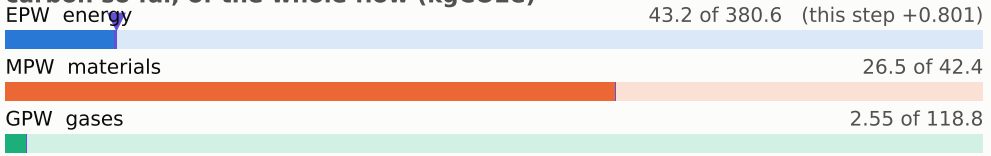


this step

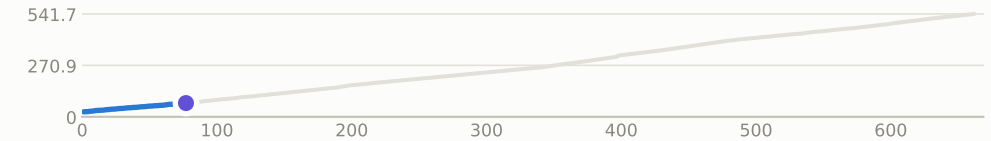
process area	Diffusion
material	—
thickness	0.002 µm
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

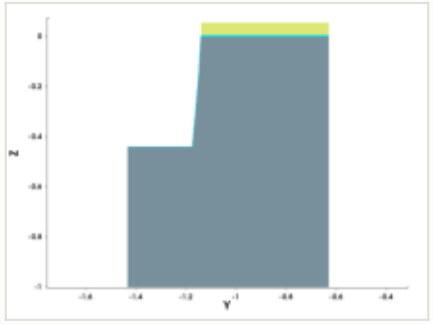
carbon so far, of the whole flow (kgCO2e)



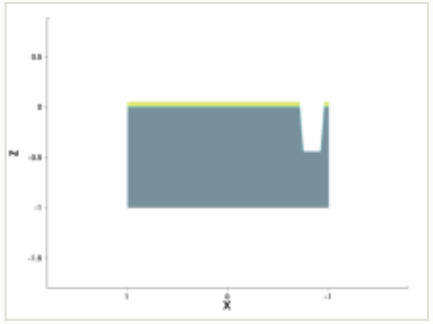
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



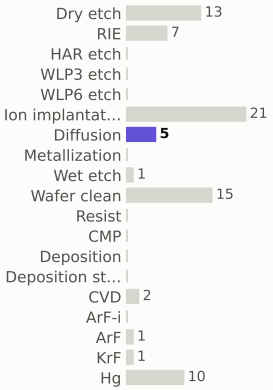
y cut at -1.032 µm



mask for this step

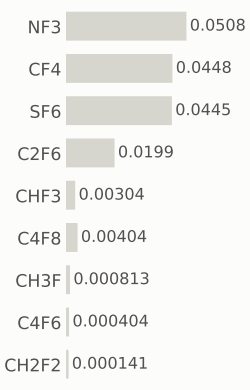
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



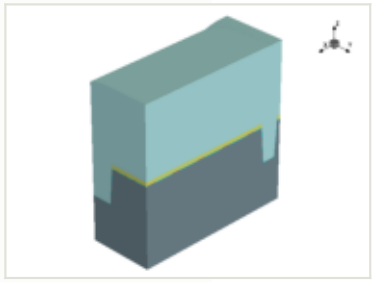
Step #820: Deposition

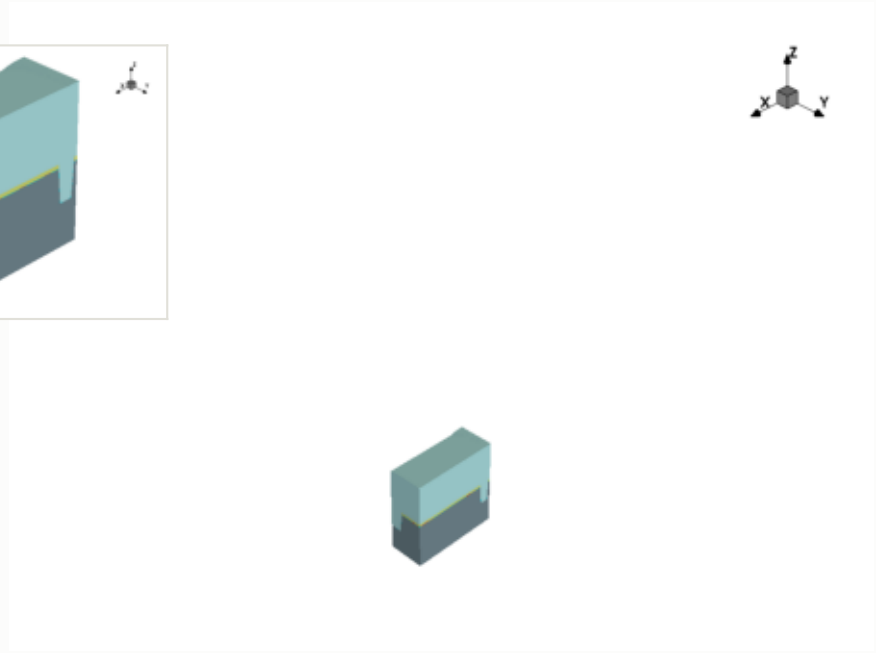
depo_isotropic · Deposition

0.656	72.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



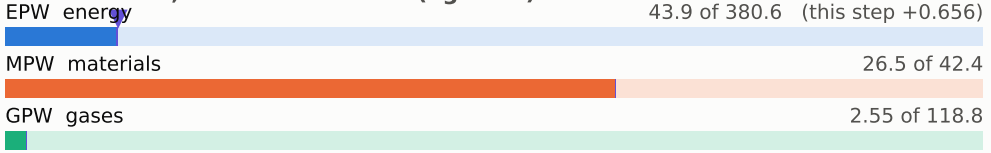


this step

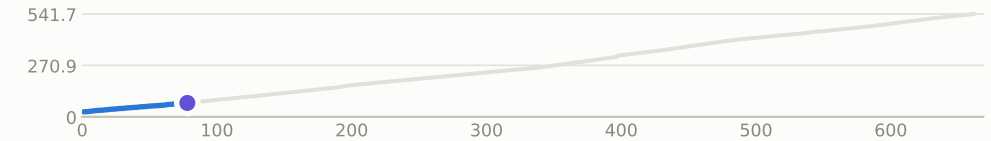
process area	Deposition
material	—
thickness	0.9 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

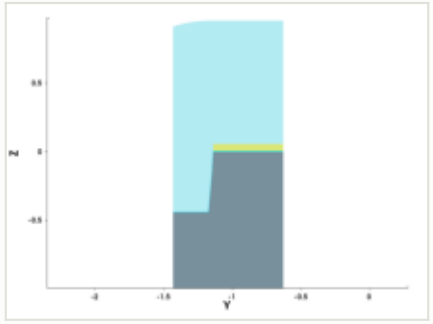
carbon so far, of the whole flow (kgCO2e)



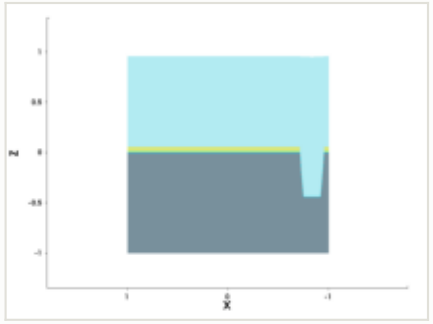
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



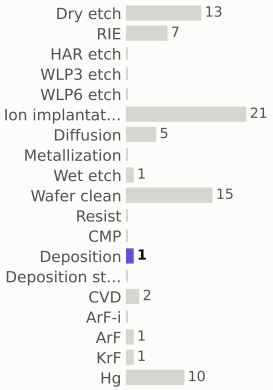
y cut at -1.032 µm



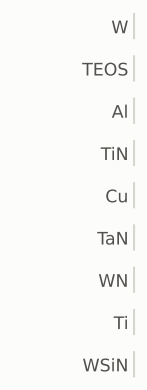
mask for this step

no mask for this step

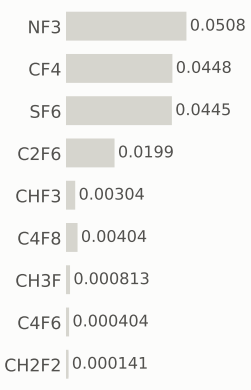
steps by process area



materials deposited (g)



gases emitted (g)



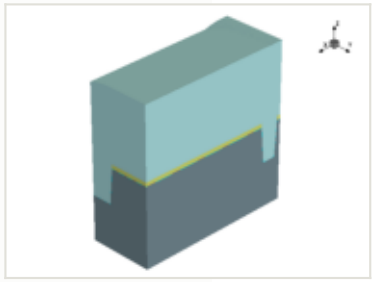
Step #830: Diffusion

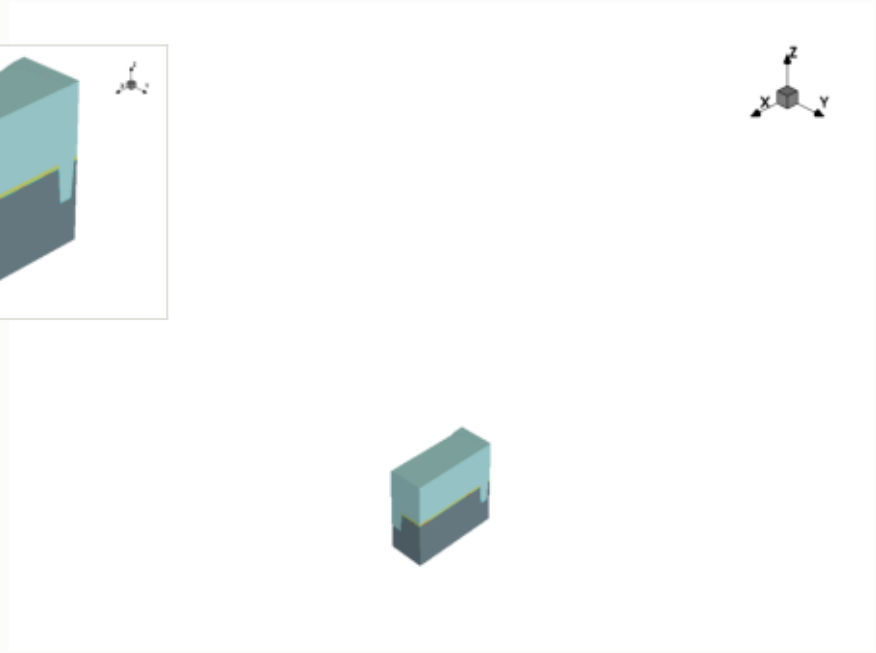
anneal · Diffusion

0.801	73.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

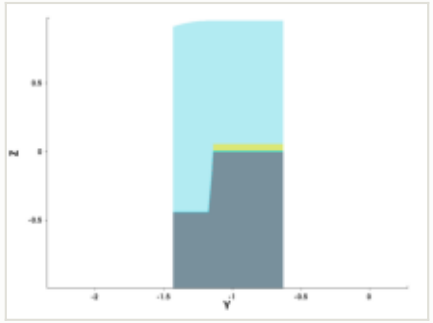
3D structure after this step

zoomed to fit

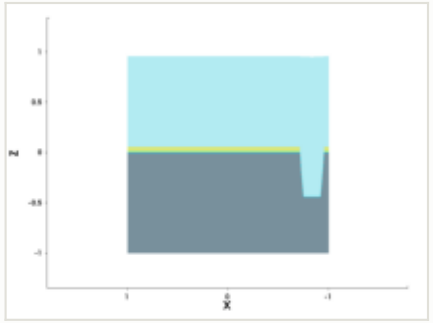




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

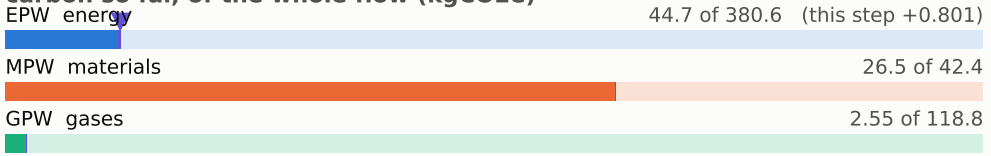
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

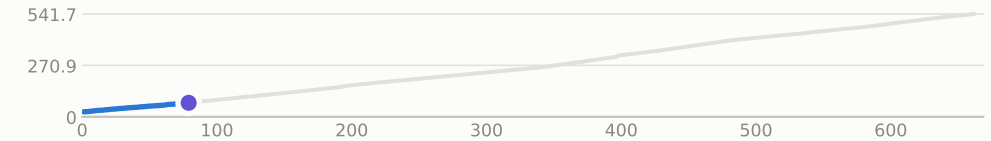
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

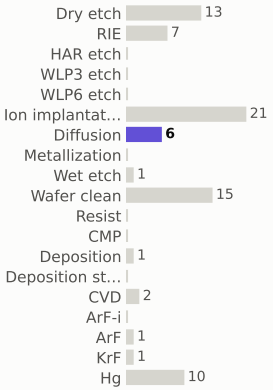
carbon so far, of the whole flow (kgCO2e)



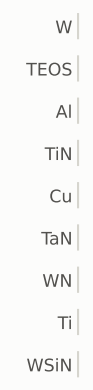
embodied carbon up to each step (kgCO2e per wafer)



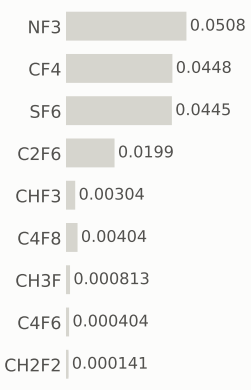
steps by process area



materials deposited (g)



gases emitted (g)



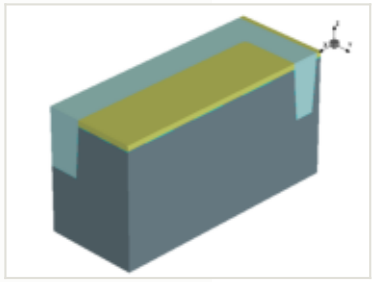
Step #840: CMP

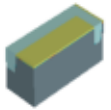
etch_cmp · CMP

0.169	73.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

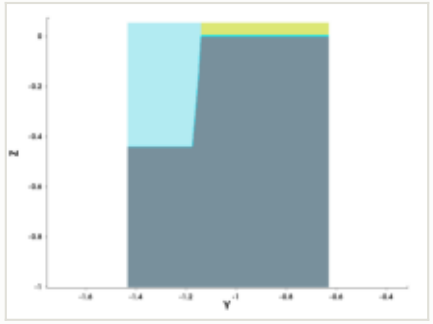
3D structure after this step

zoomed to fit

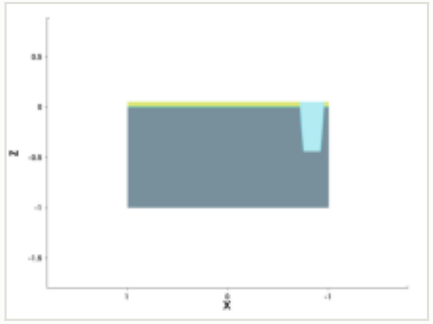




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

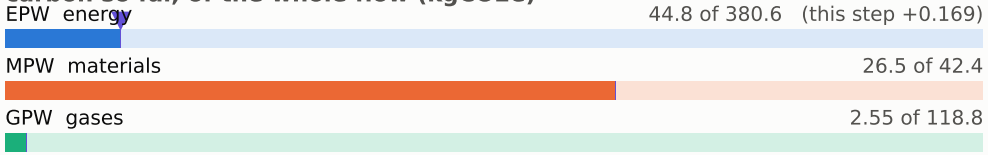
no mask for this step

this step

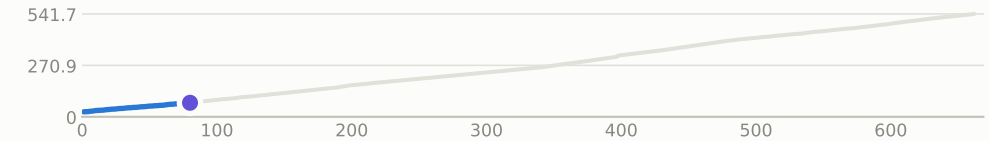
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

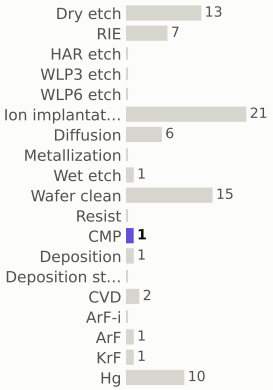
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

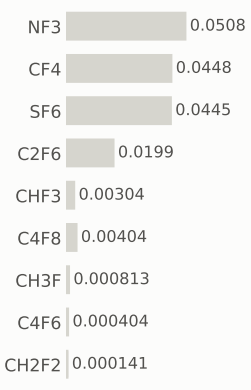


steps by process area



materials deposited (g)

gases emitted (g)



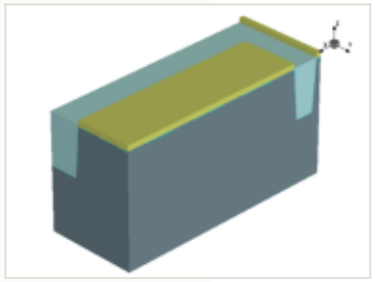
Step #850: Wet_etch

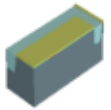
etch_anisotropic · Wet_etch

0.169	74	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



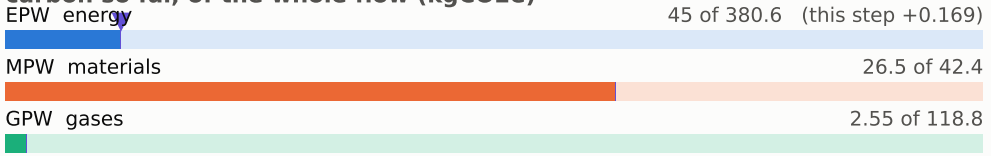


this step

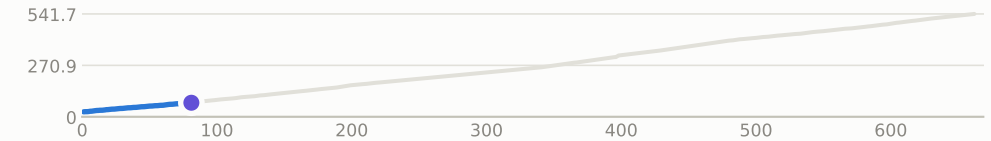
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

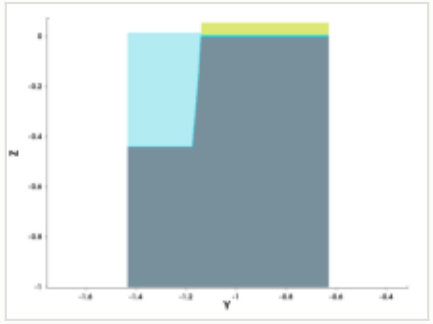
carbon so far, of the whole flow (kgCO2e)



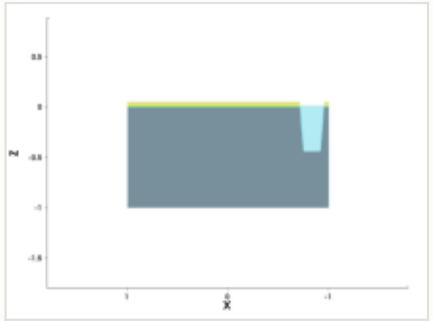
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



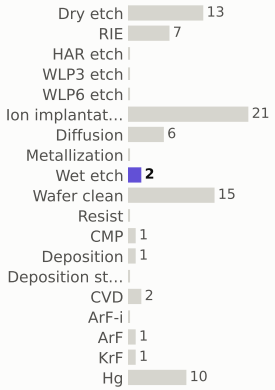
y cut at -1.032 µm



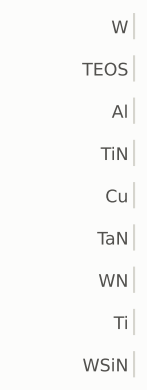
mask for this step

no mask for this step

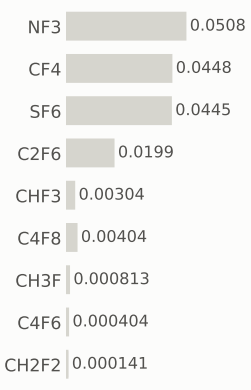
steps by process area



materials deposited (g)



gases emitted (g)



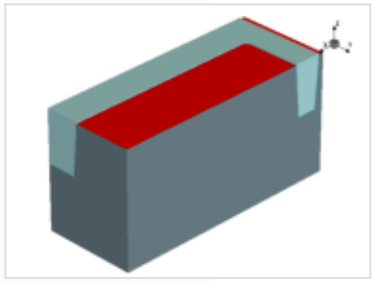
Step #860: Wet_etch

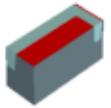
etch_isotropic · Wet_etch

0.169	74.2	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



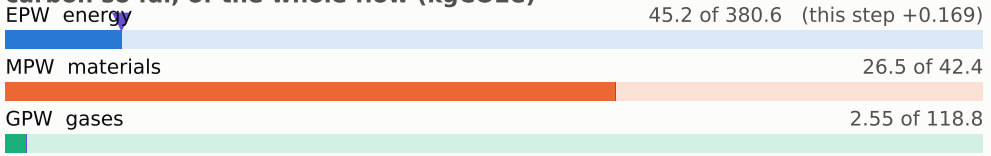


this step

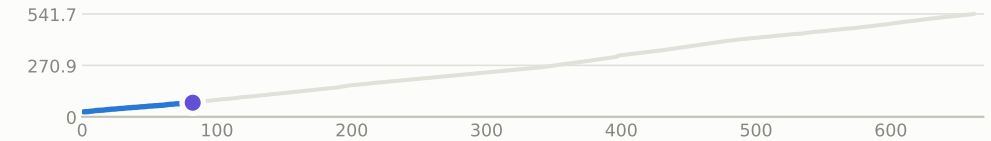
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

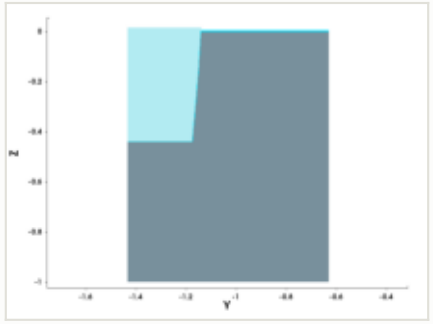
carbon so far, of the whole flow (kgCO2e)



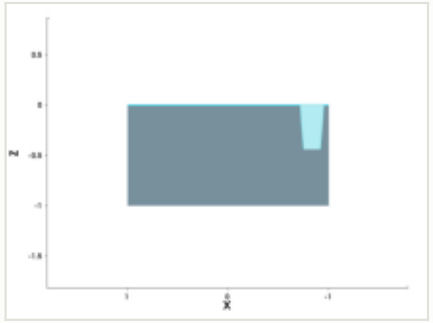
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



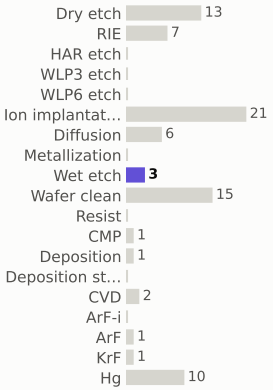
y cut at -1.032 µm



mask for this step

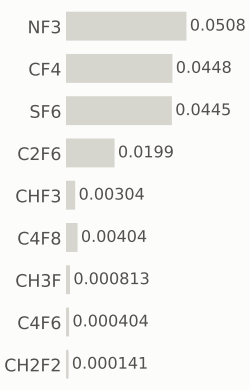
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



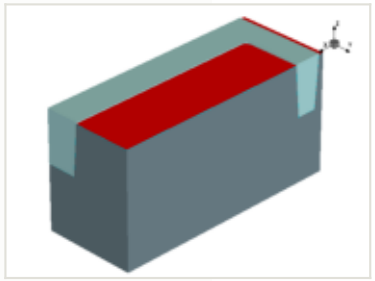
Step #870: Wafer_clean

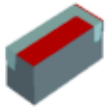
clean · Wafer_clean

0.169	74.4	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

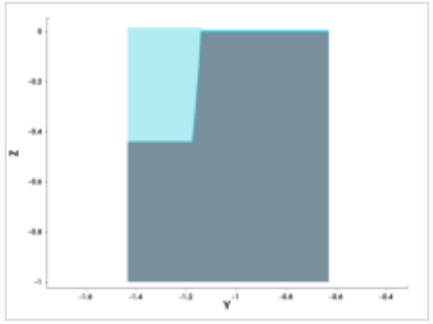
3D structure after this step

zoomed to fit

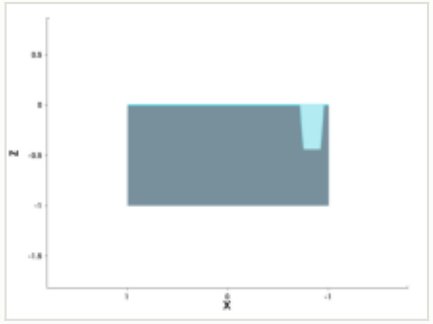




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

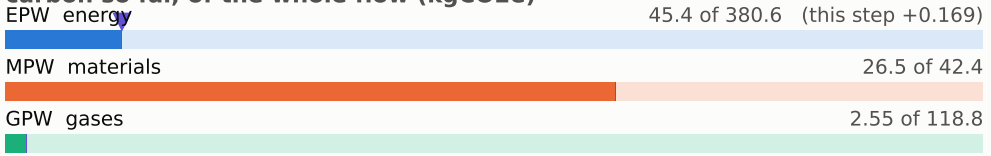
no mask for this step

this step

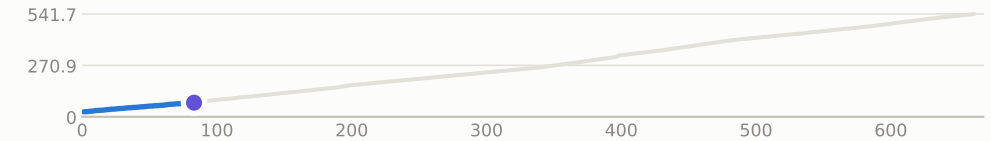
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

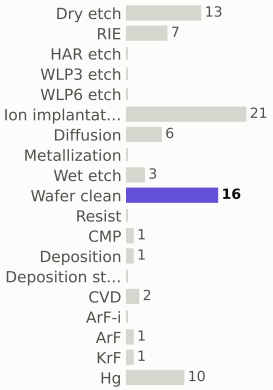
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

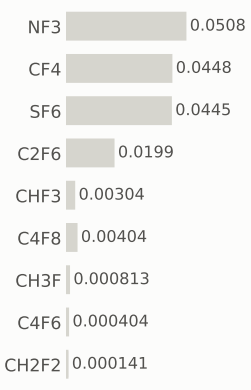


steps by process area



materials deposited (g)

gases emitted (g)



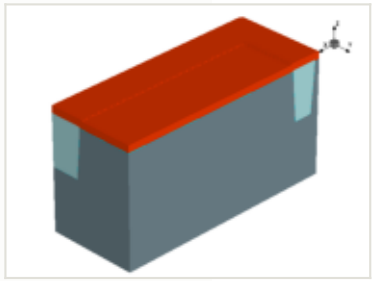
Step #890: CVD

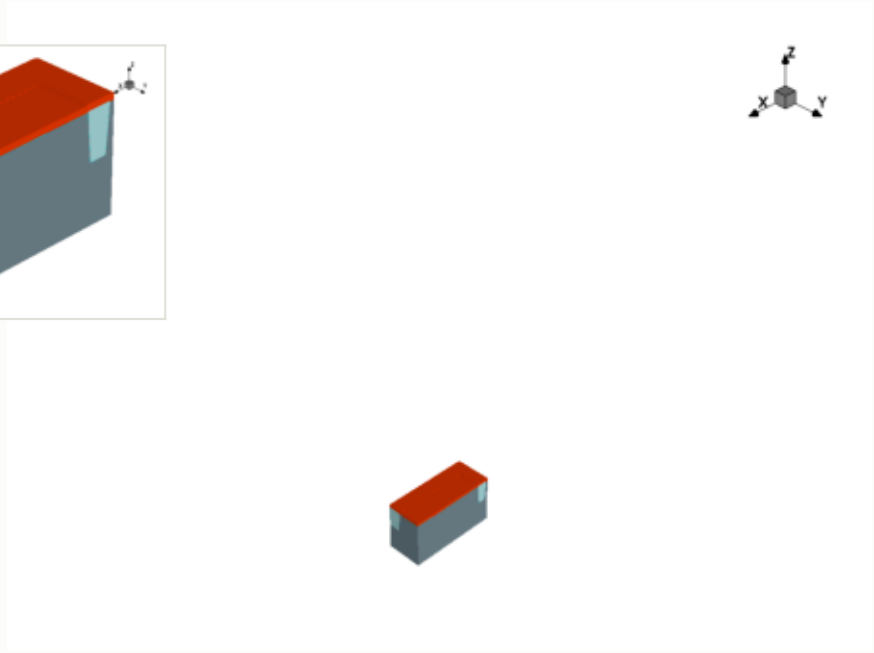
depo_isotropic · CVD

1.31	75.7	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

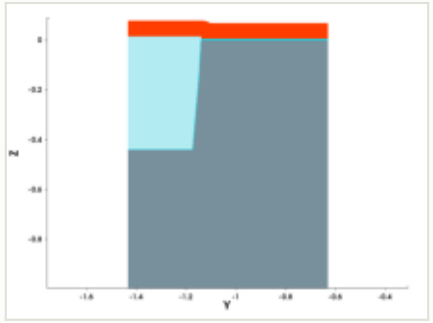
3D structure after this step

zoomed to fit

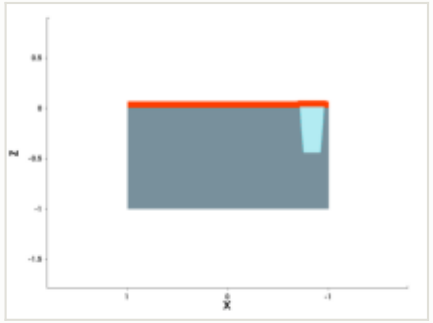




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

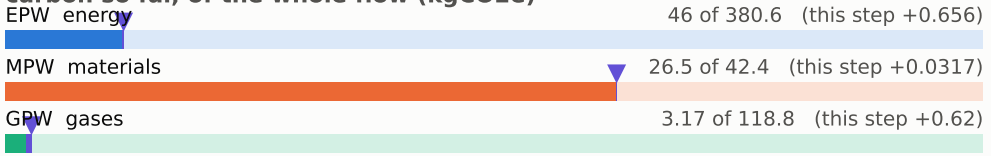
no mask for this step

this step

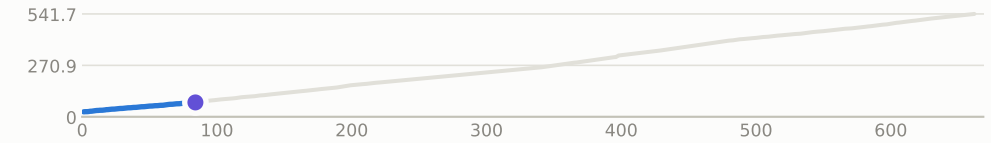
process area	CVD
material	—
thickness	0.063 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

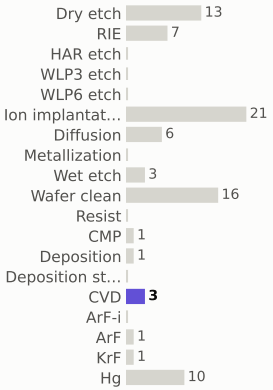
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

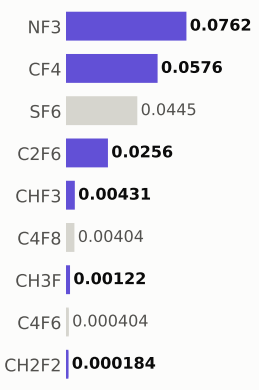


steps by process area



materials deposited (g)

gases emitted (g)



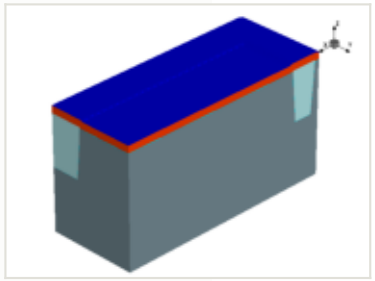
Step #900: Metallization

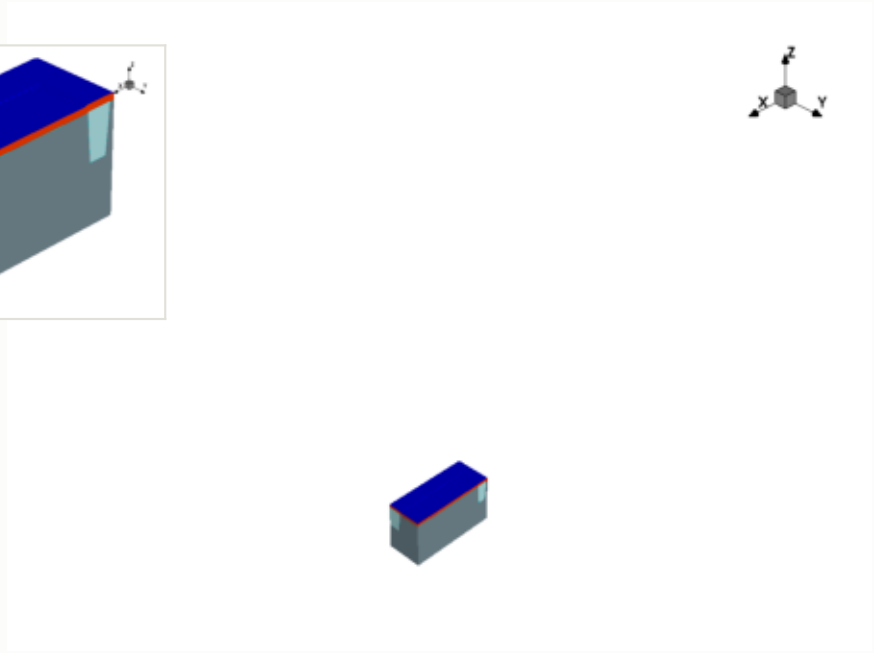
depo_isotropic · Metallization

1.2	76.9	0.22%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



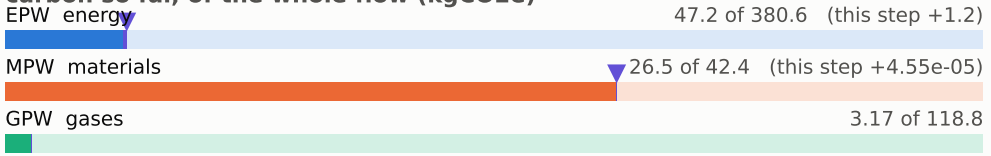


this step

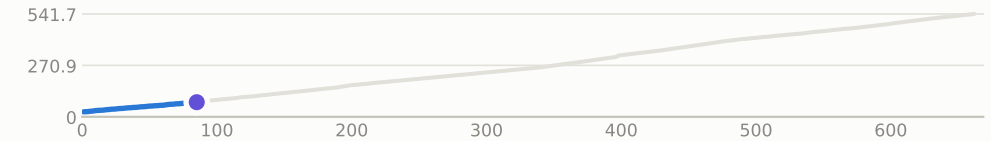
process area	Metallization
material	Ti
thickness	0.003 µm
material GWP	47.6334 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	4.55e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

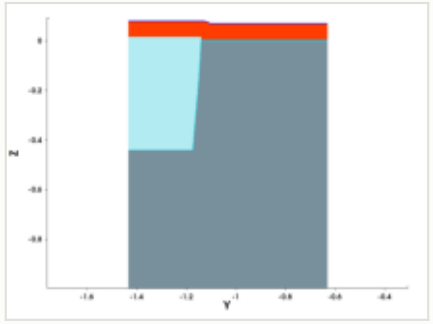
carbon so far, of the whole flow (kgCO2e)



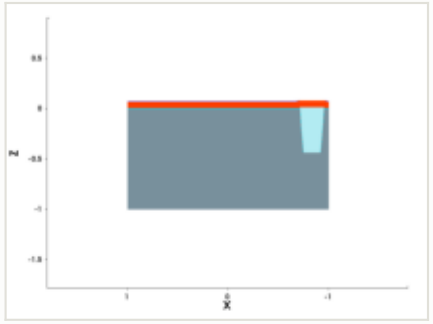
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



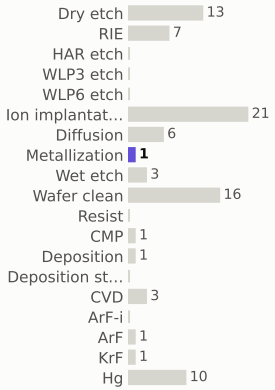
y cut at -1.032 µm



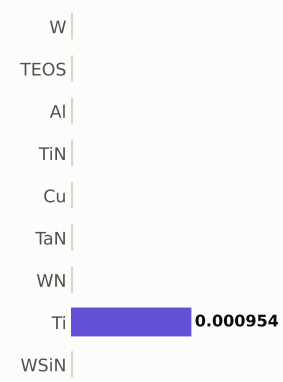
mask for this step

no mask for this step

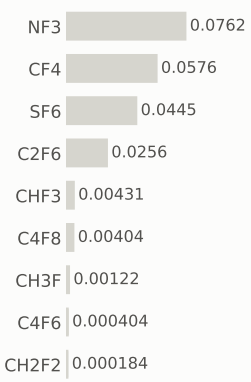
steps by process area



materials deposited (g)



gases emitted (g)



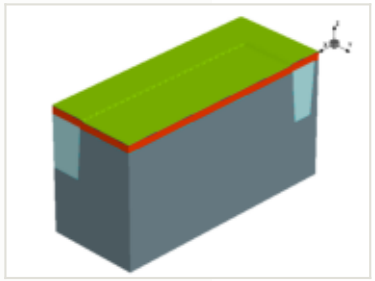
Step #910: Deposition

depo_isotropic · Deposition

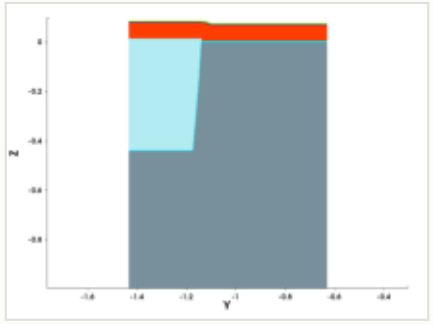
0.657	77.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

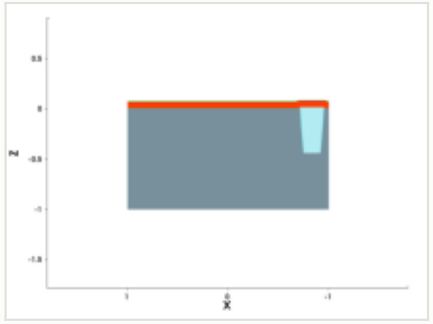
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

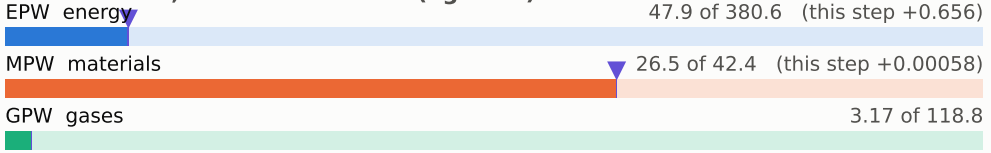
no mask for this step

this step

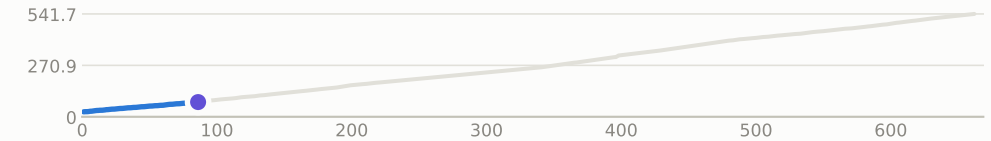
process area	Deposition
material	WN
thickness	0.006 µm
material GWP	76.8289 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.00058 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.657 kgCO2e**

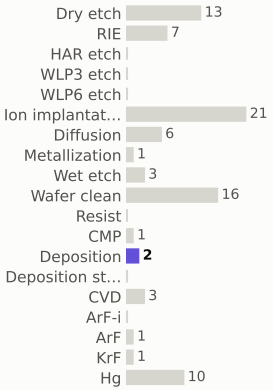
carbon so far, of the whole flow (kgCO2e)



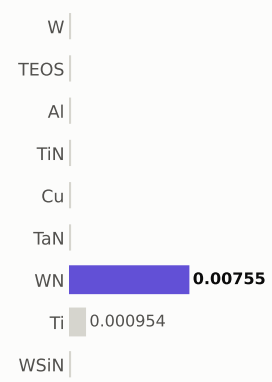
embodied carbon up to each step (kgCO2e per wafer)



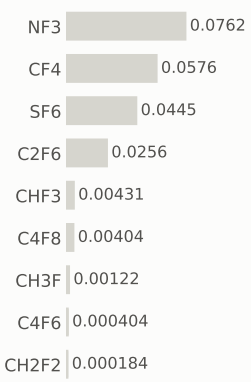
steps by process area



materials deposited (g)



gases emitted (g)



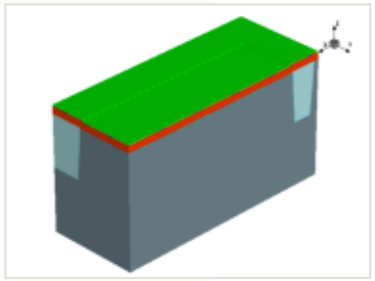
Step #920: Deposition

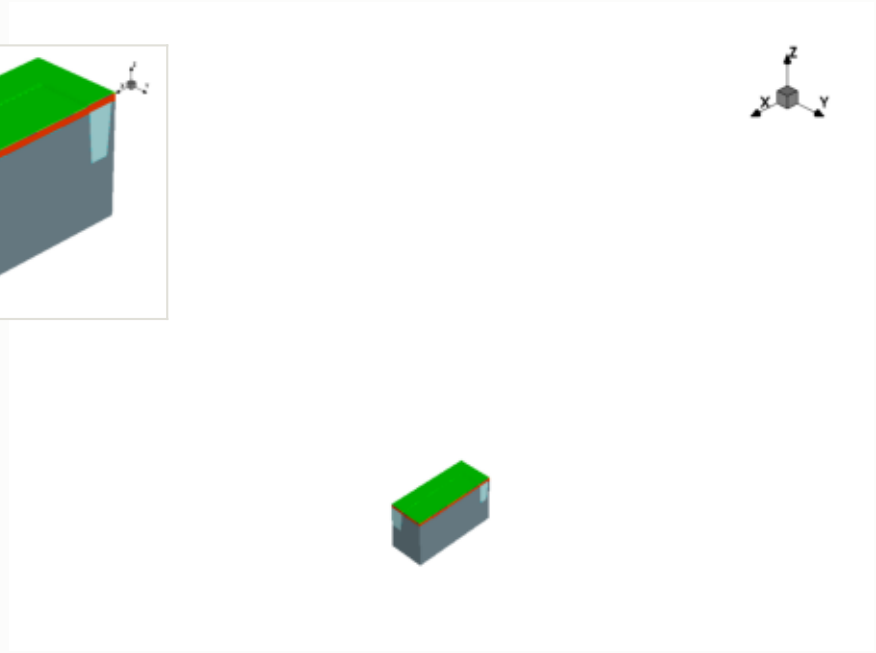
depo_isotropic · Deposition

0.656	78.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

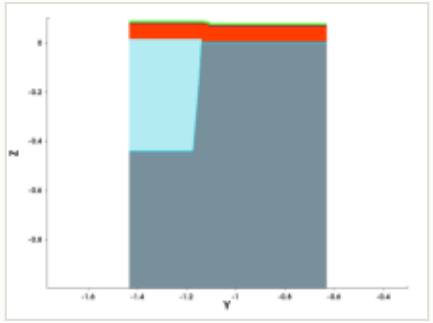
3D structure after this step

zoomed to fit

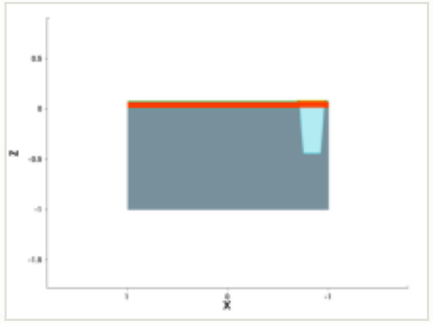




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

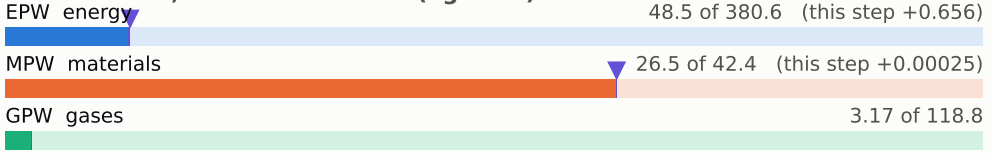
no mask for this step

this step

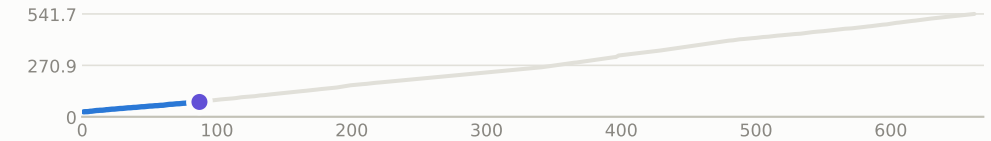
process area	Deposition
material	WSiN
thickness	0.003 µm
material GWP	73.2252 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.00025 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

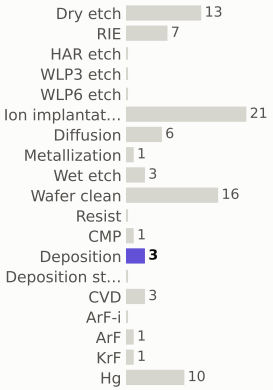
carbon so far, of the whole flow (kgCO2e)



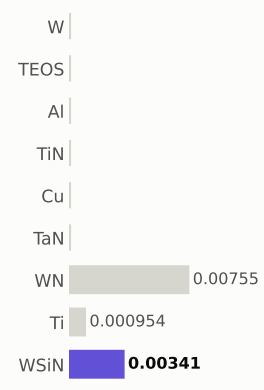
embodied carbon up to each step (kgCO2e per wafer)



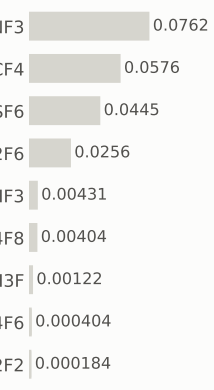
steps by process area



materials deposited (g)



gases emitted (g)



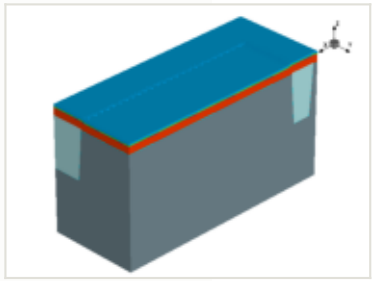
Step #930: Metallization

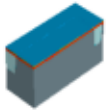
depo_isotropic · Metallization

1.21	79.4	0.22%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

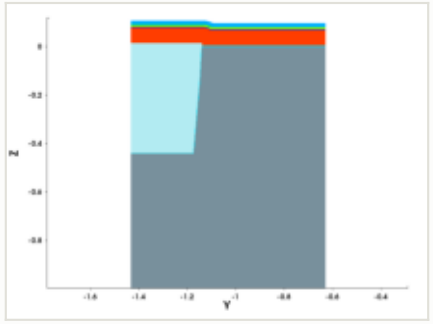
3D structure after this step

zoomed to fit

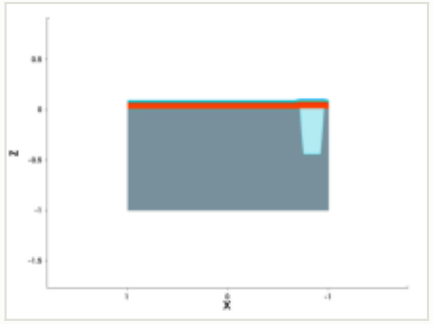




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

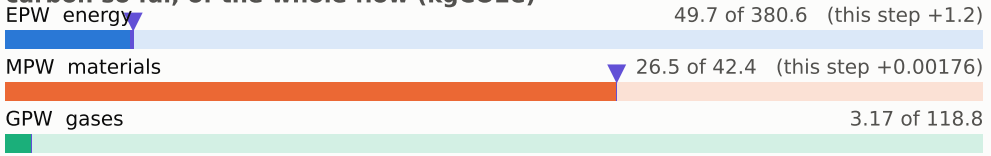
no mask for this step

this step

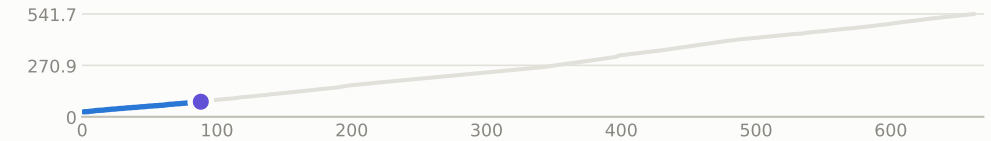
process area	Metallization
material	W
thickness	0.017 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00176 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

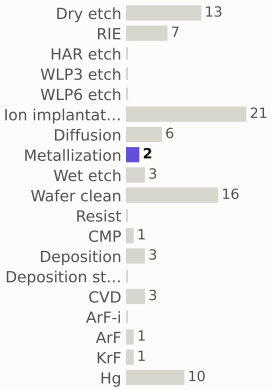
carbon so far, of the whole flow (kgCO2e)



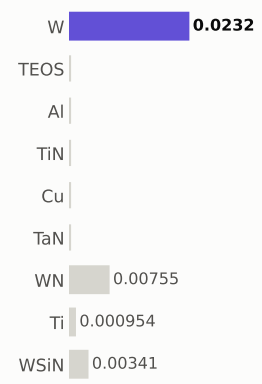
embodied carbon up to each step (kgCO2e per wafer)



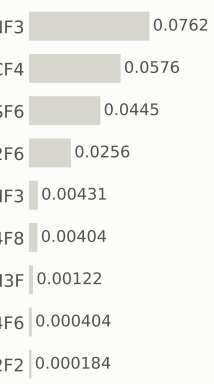
steps by process area



materials deposited (g)



gases emitted (g)



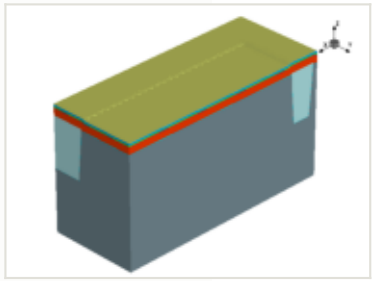
Step #940: CVD

depo_isotropic · CVD

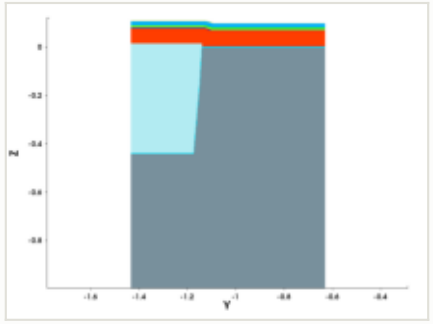
1.31	80.7	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

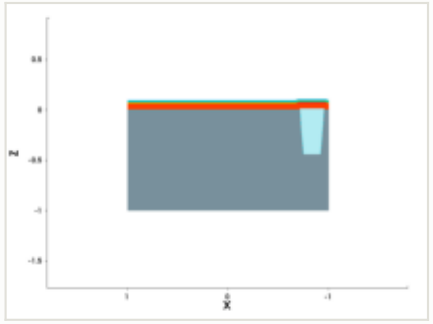
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

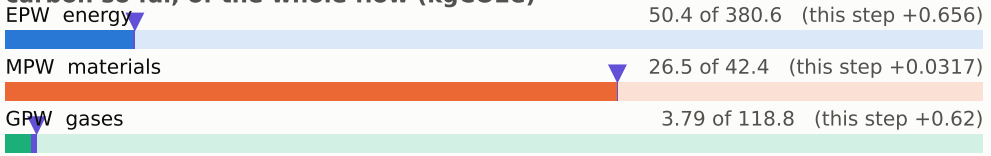
no mask for this step

this step

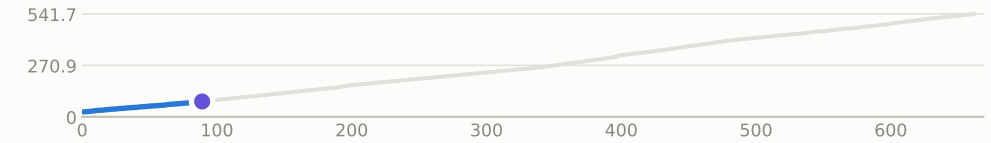
process area	CVD
material	—
thickness	0.003 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

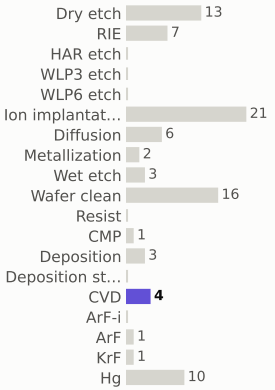
carbon so far, of the whole flow (kgCO2e)



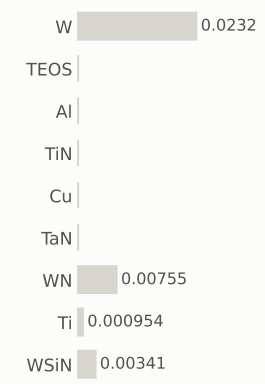
embodied carbon up to each step (kgCO2e per wafer)



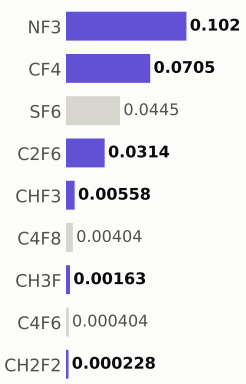
steps by process area



materials deposited (g)



gases emitted (g)



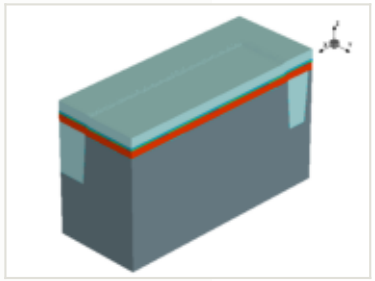
Step #950: CVD

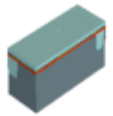
depo_isotropic · CVD

1.31	82	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

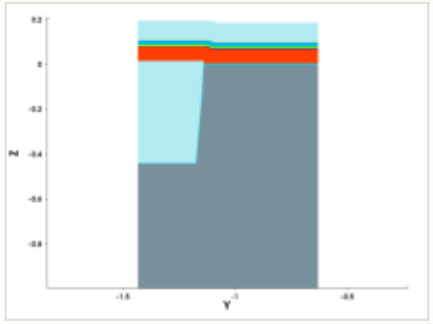
3D structure after this step

zoomed to fit

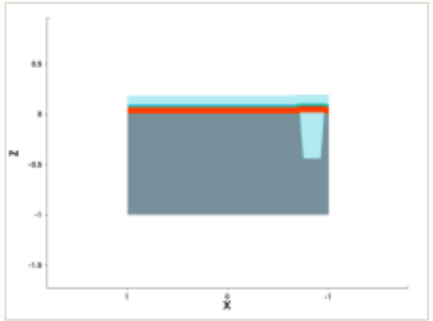




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

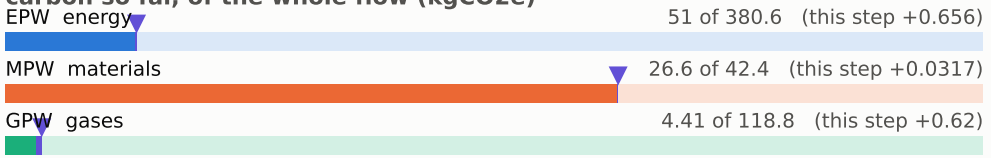
no mask for this step

this step

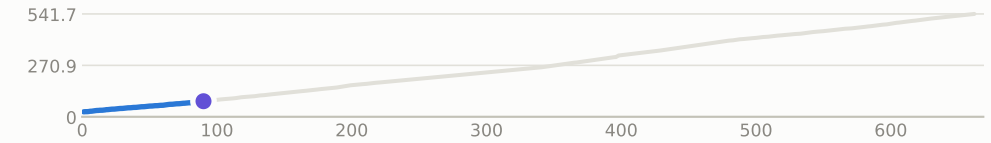
process area	CVD
material	—
thickness	0.085 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

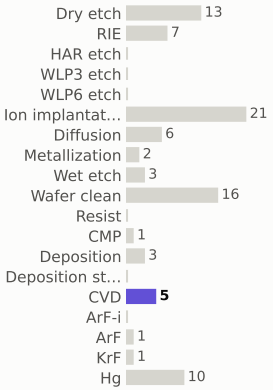
carbon so far, of the whole flow (kgCO2e)



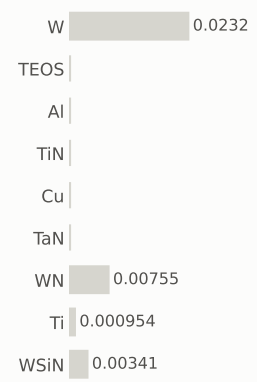
embodied carbon up to each step (kgCO2e per wafer)



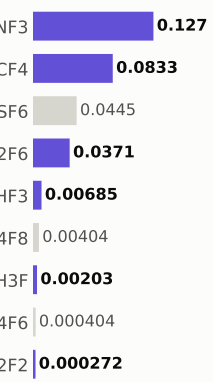
steps by process area



materials deposited (g)



gases emitted (g)



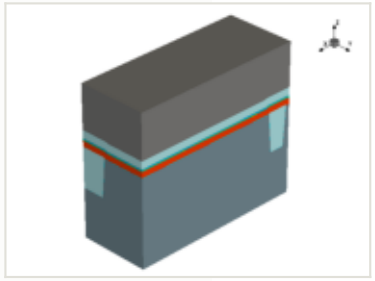
Step #960: KrF

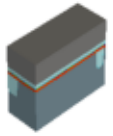
litho_patterning · KrF

0.199	82.2	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



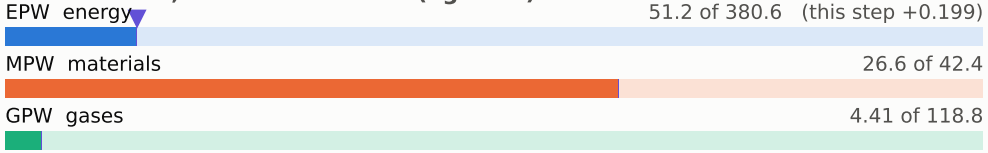


this step

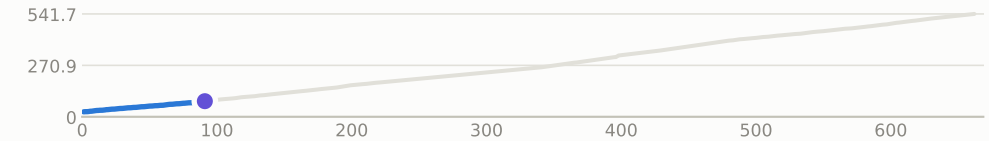
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

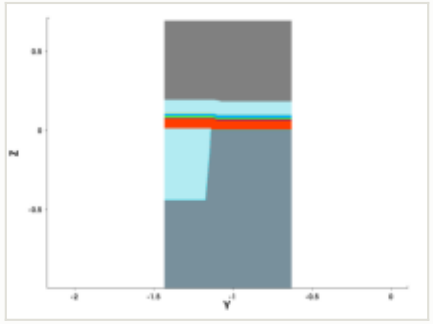
carbon so far, of the whole flow (kgCO2e)



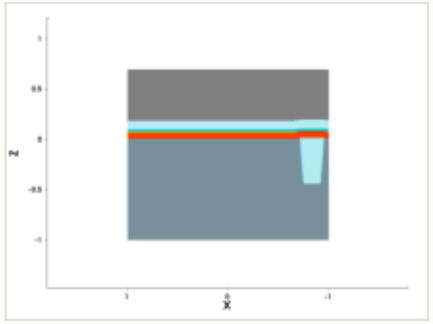
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

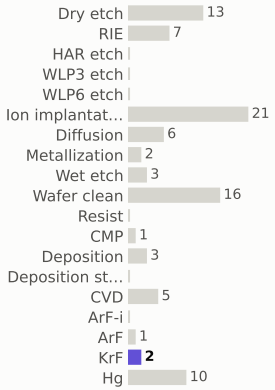


y cut at -1.032 µm

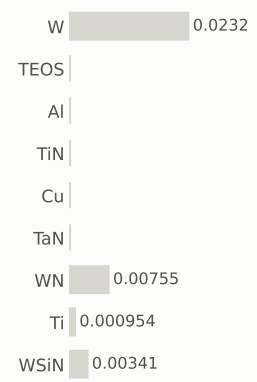


mask: layer 13:0, positive (OASIS)

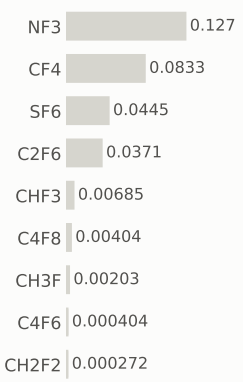
steps by process area



materials deposited (g)



gases emitted (g)



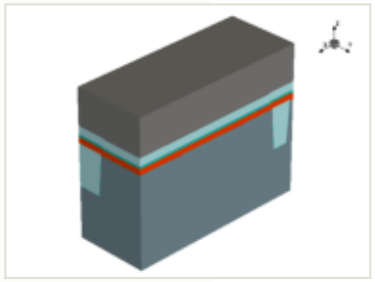
Step #970: RIE

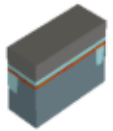
etch_anisotropic · RIE

1.04	83.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

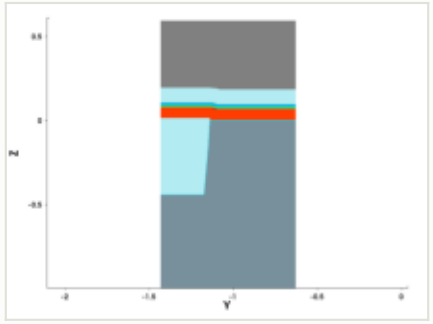
3D structure after this step

zoomed to fit

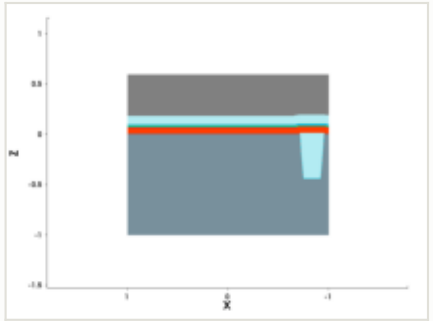




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

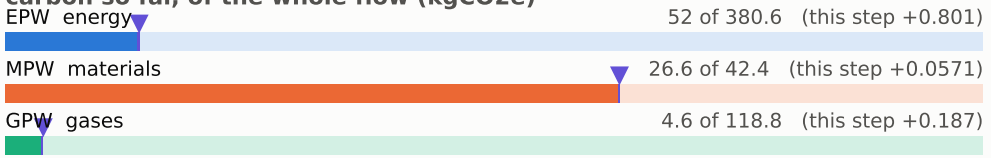
no mask for this step

this step

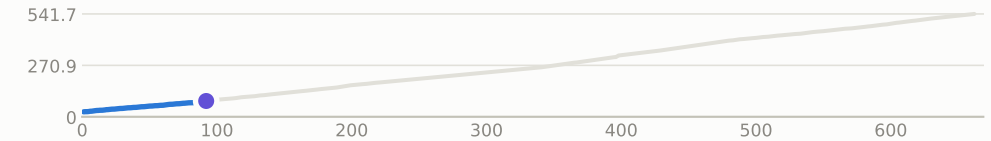
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

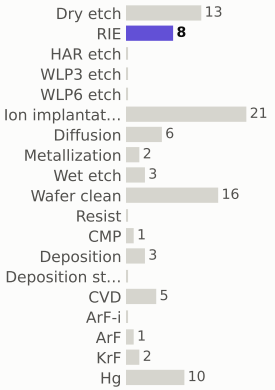
carbon so far, of the whole flow (kgCO2e)



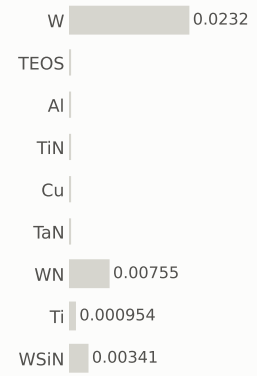
embodied carbon up to each step (kgCO2e per wafer)



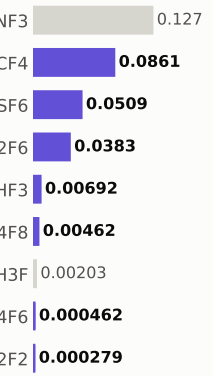
steps by process area



materials deposited (g)



gases emitted (g)



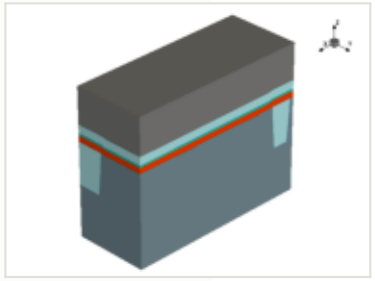
Step #980: RIE

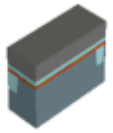
etch_anisotropic · RIE

1.04	84.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

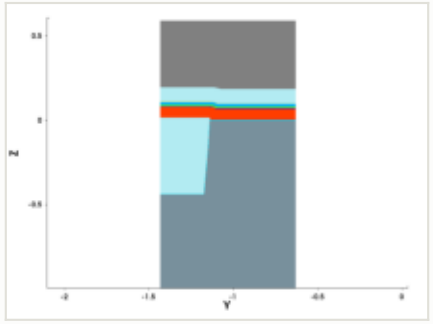
3D structure after this step

zoomed to fit

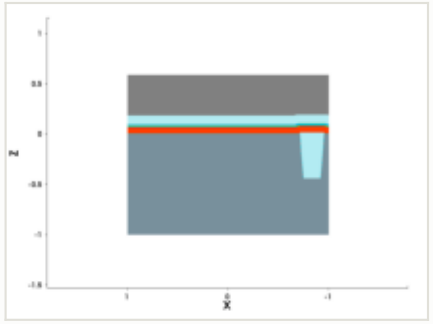




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

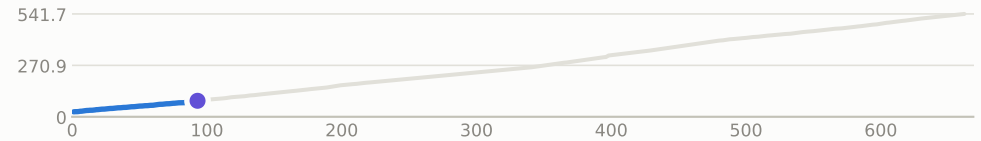
carbon so far, of the whole flow (kgCO2e)

EPW energy — 52.8 of 380.6 (this step +0.801)

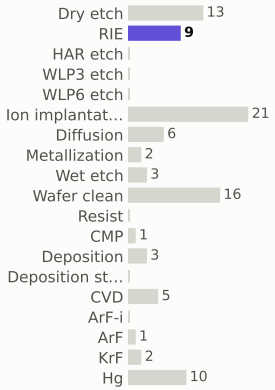
MPW materials — 26.7 of 42.4 (this step +0.0571)

GPW gases — 4.79 of 118.8 (this step +0.187)

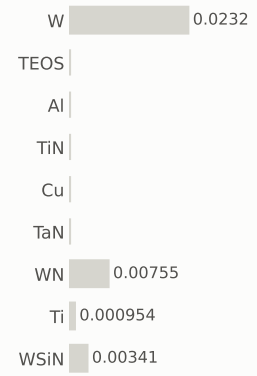
embodied carbon up to each step (kgCO2e per wafer)



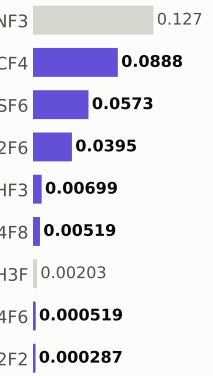
steps by process area



materials deposited (g)



gases emitted (g)



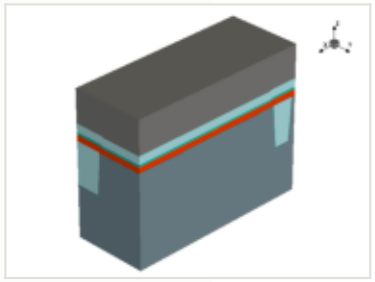
Step #990: RIE

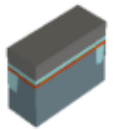
etch_anisotropic · RIE

1.04	85.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



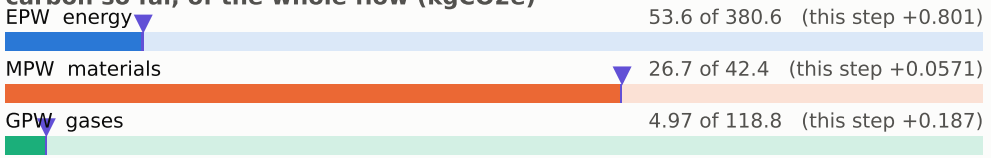


this step

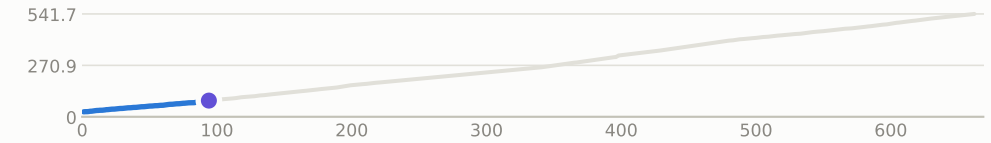
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

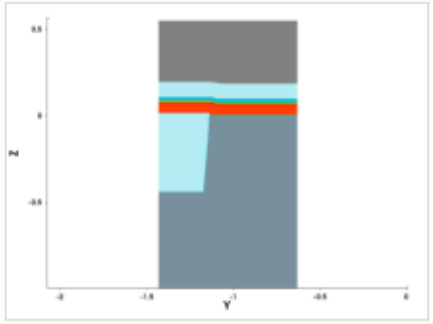
carbon so far, of the whole flow (kgCO2e)



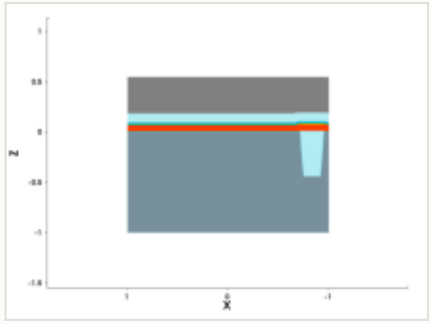
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



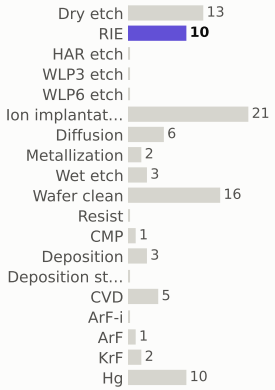
y cut at -1.032 µm



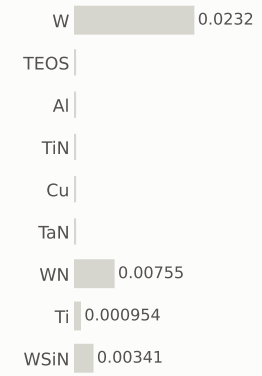
mask for this step

no mask for this step

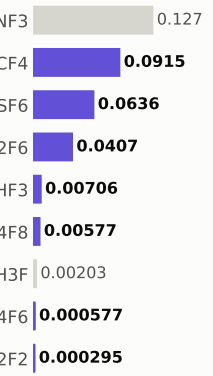
steps by process area



materials deposited (g)



gases emitted (g)



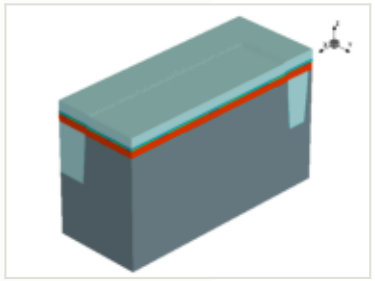
Step #1000: Dry_etch

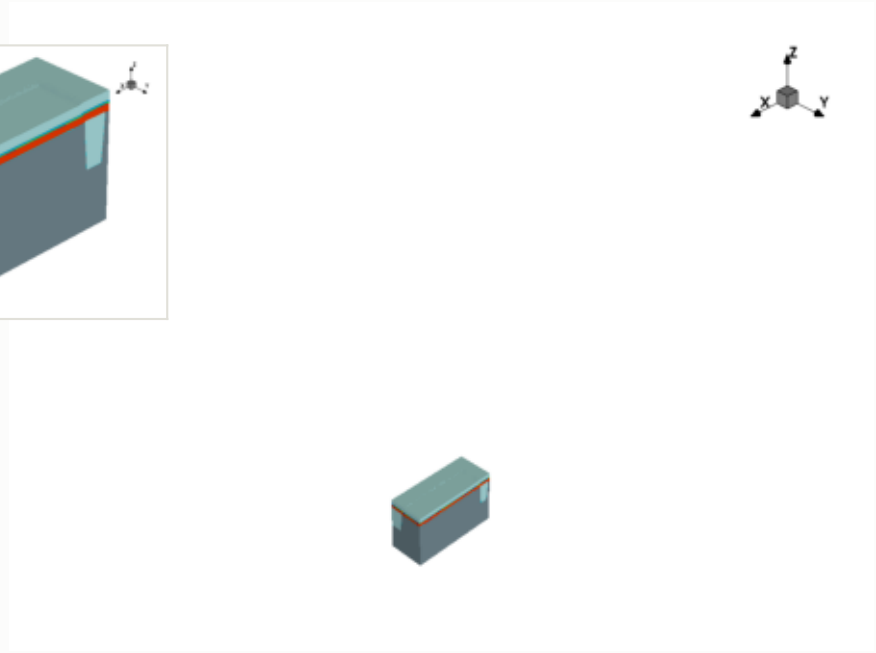
etch_strip · Dry_etch

0.801	86.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

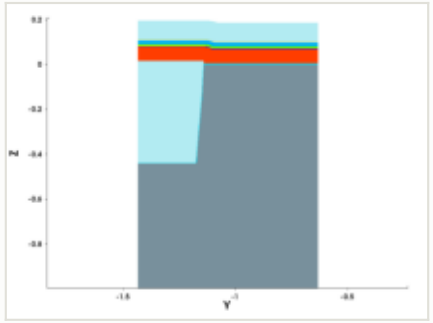
3D structure after this step

zoomed to fit

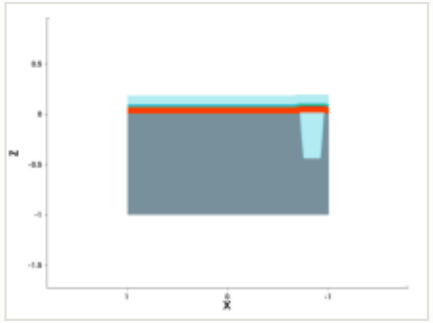




x cut at 0.000 µm



y cut at -1.032 µm



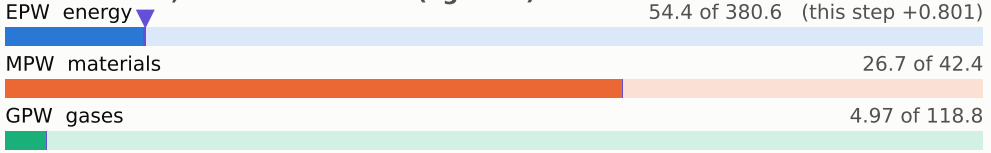
mask for this step

no mask for this step

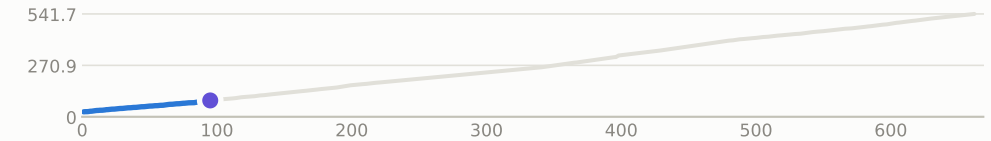
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

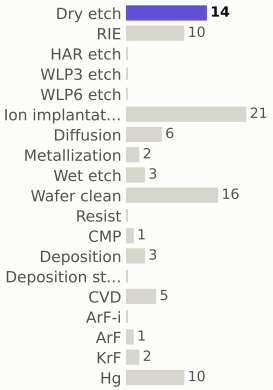
carbon so far, of the whole flow (kgCO2e)



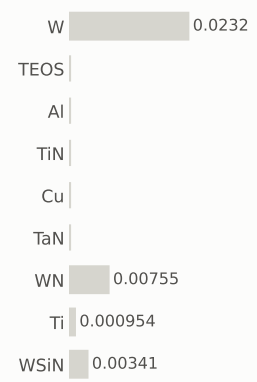
embodied carbon up to each step (kgCO2e per wafer)



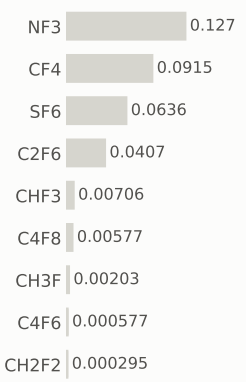
steps by process area



materials deposited (g)



gases emitted (g)



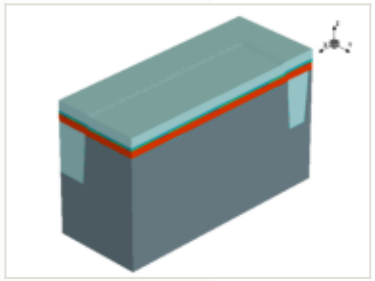
Step #1010: Wafer_clean

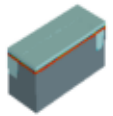
clean · Wafer_clean

0.169	86.3	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

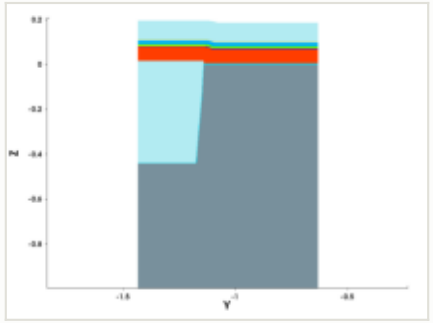
3D structure after this step

zoomed to fit

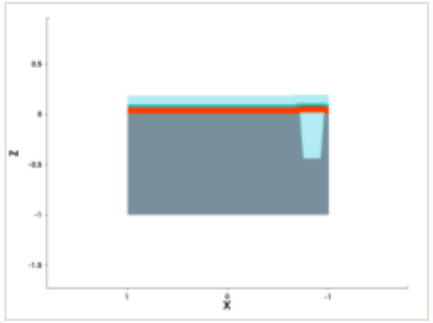




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

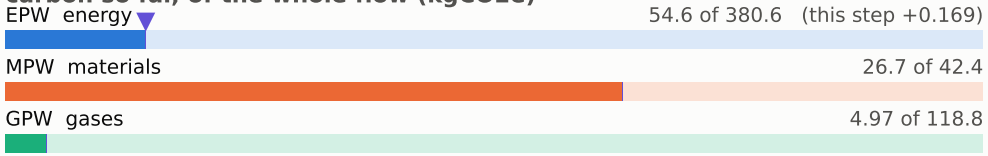
no mask for this step

this step

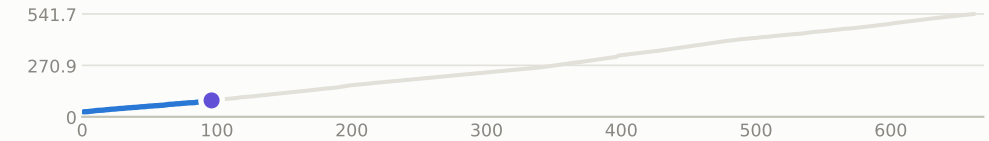
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

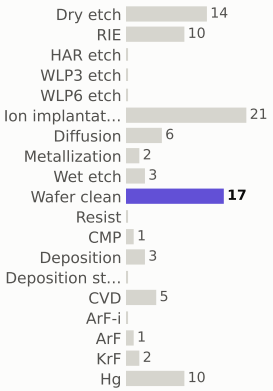
carbon so far, of the whole flow (kgCO2e)



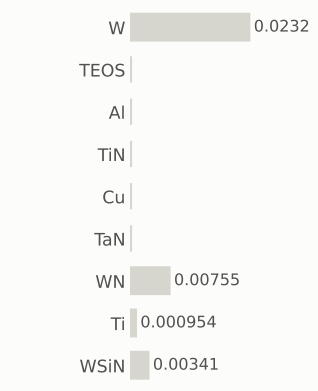
embodied carbon up to each step (kgCO2e per wafer)



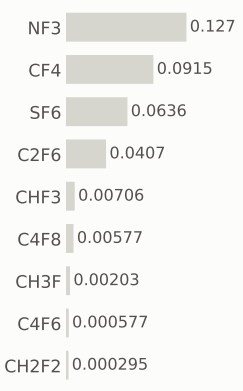
steps by process area



materials deposited (g)



gases emitted (g)



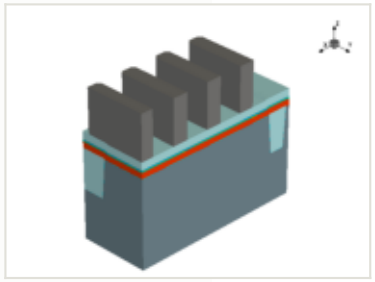
Step #1020: ArF

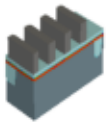
litho_patterning · ArF

0.199	86.5	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



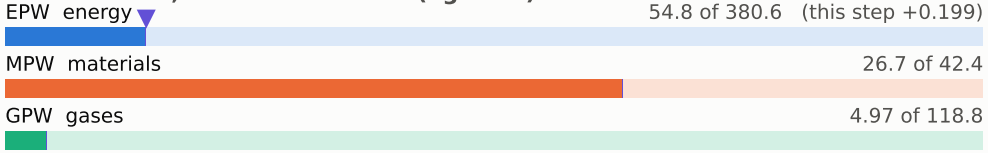


this step

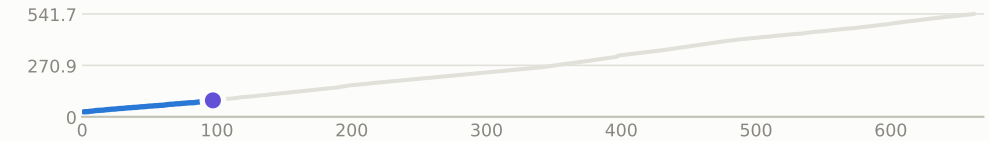
process area	ArF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

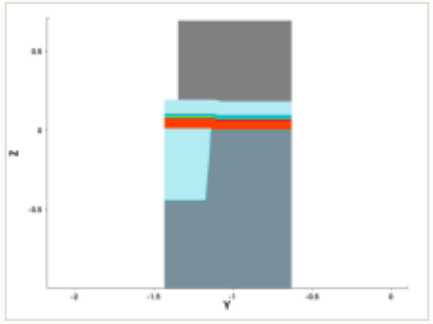
carbon so far, of the whole flow (kgCO2e)



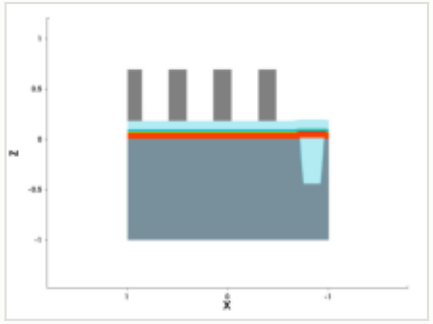
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

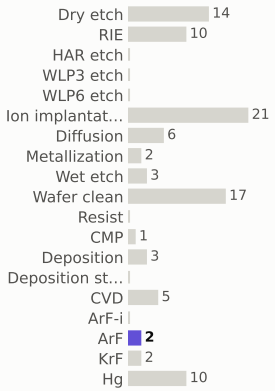


y cut at -1.032 µm

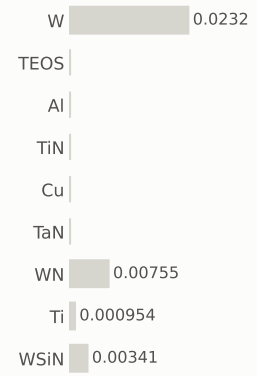


mask: layer 14:0, positive (OASIS)

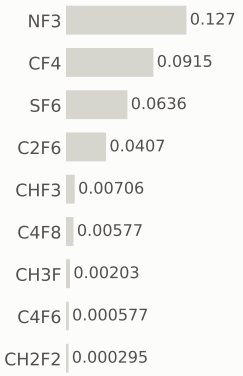
steps by process area



materials deposited (g)



gases emitted (g)



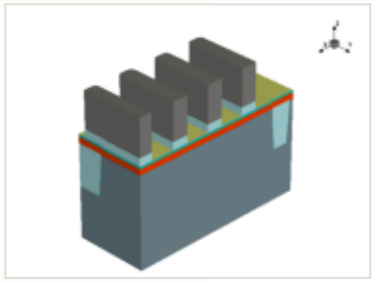
Step #1030: RIE

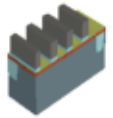
etch_anisotropic · RIE

1.04	87.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

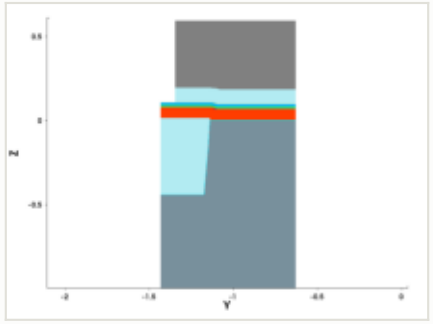
3D structure after this step

zoomed to fit

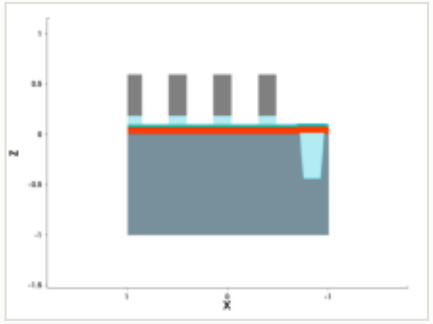




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

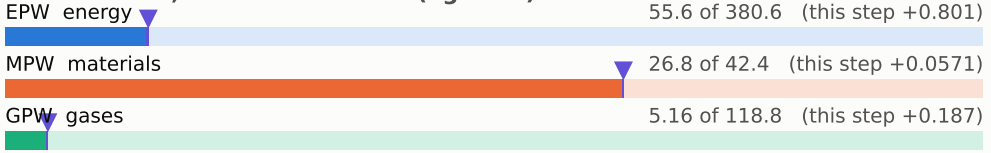
no mask for this step

this step

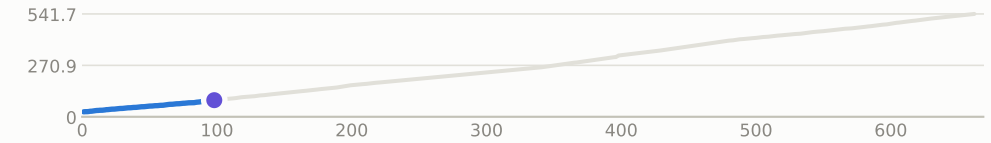
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

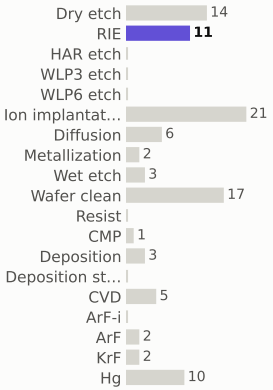
carbon so far, of the whole flow (kgCO2e)



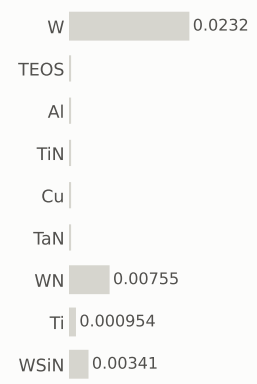
embodied carbon up to each step (kgCO2e per wafer)



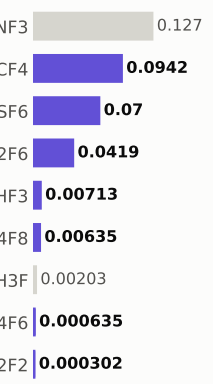
steps by process area



materials deposited (g)



gases emitted (g)



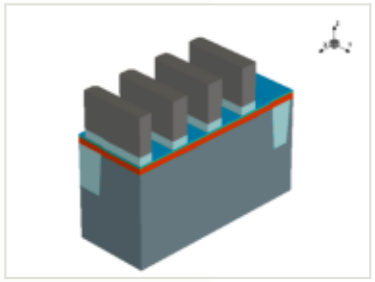
Step #1040: RIE

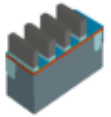
etch_anisotropic · RIE

1.04	88.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



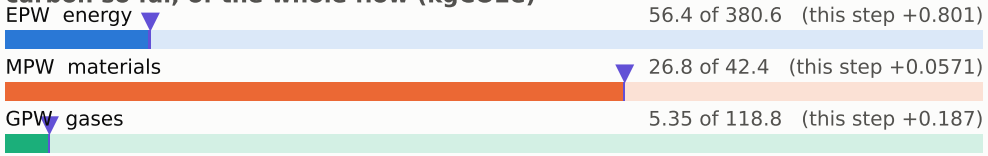


this step

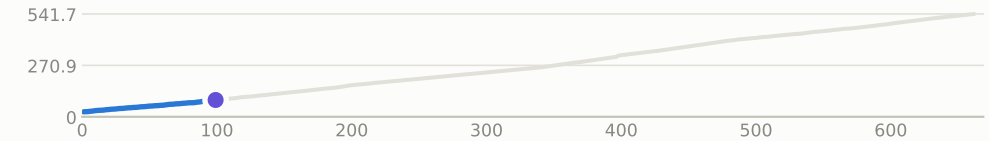
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

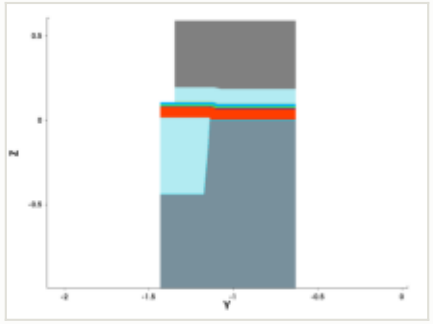
carbon so far, of the whole flow (kgCO2e)



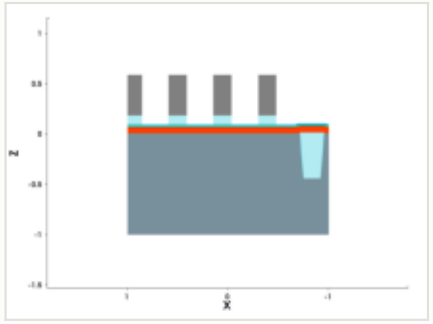
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



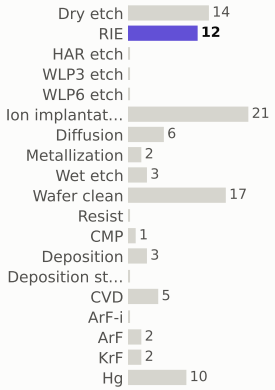
y cut at -1.032 µm



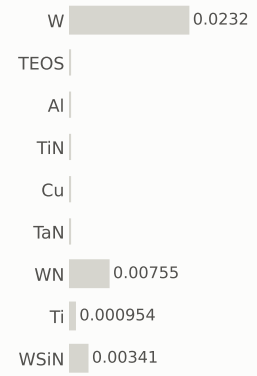
mask for this step

no mask for this step

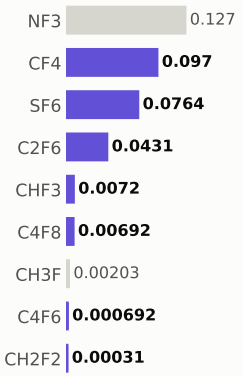
steps by process area



materials deposited (g)



gases emitted (g)



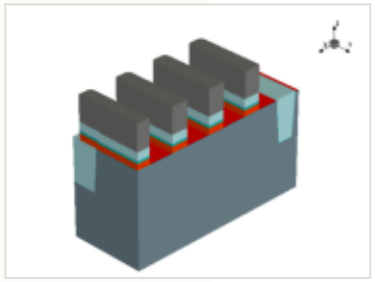
Step #1050: RIE

etch_anisotropic · RIE

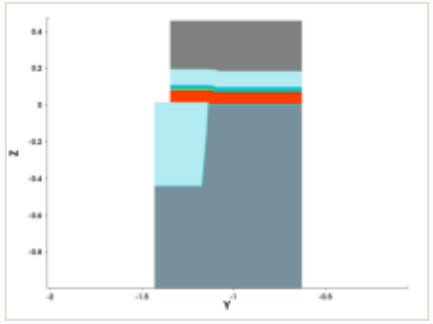
1.04	89.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

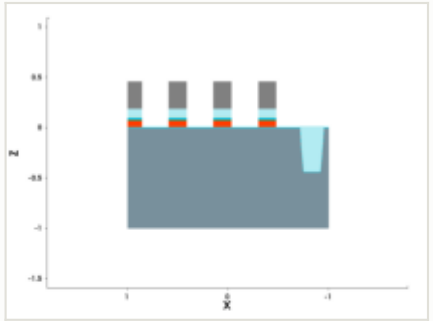
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

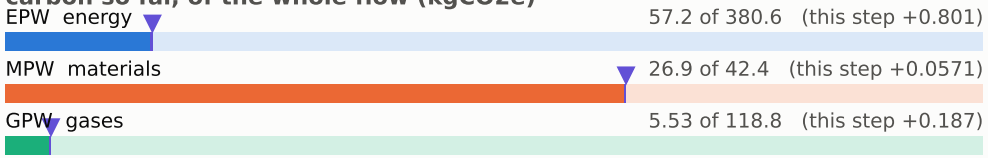
no mask for this step

this step

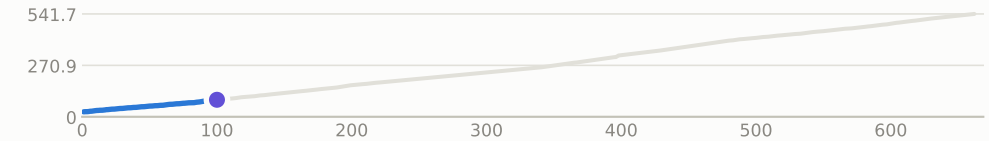
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

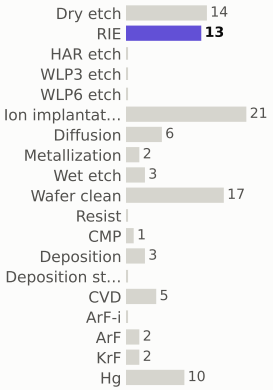
carbon so far, of the whole flow (kgCO2e)



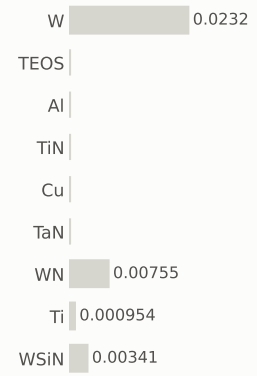
embodied carbon up to each step (kgCO2e per wafer)



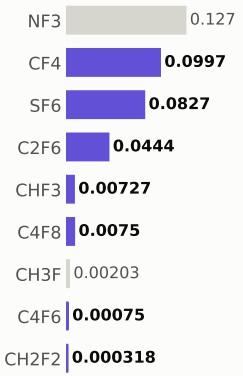
steps by process area



materials deposited (g)



gases emitted (g)



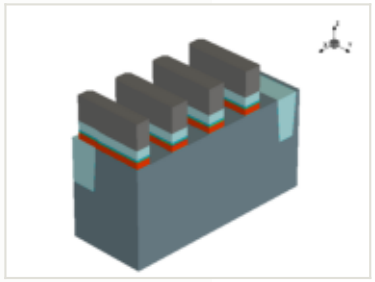
Step #1060: RIE

etch_anisotropic · RIE

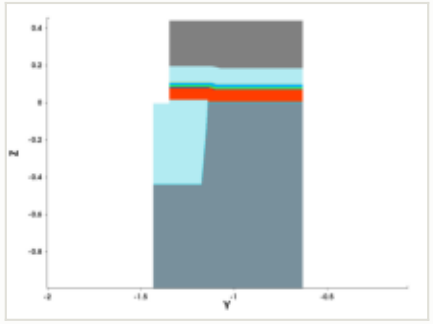
1.04	90.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

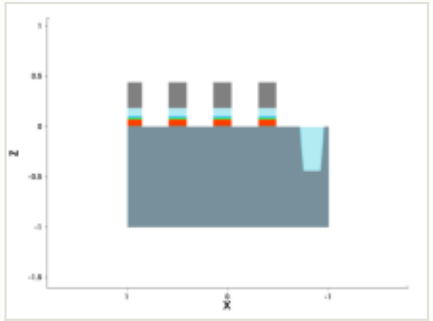
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

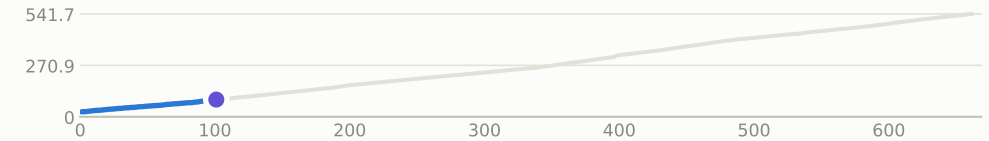
carbon so far, of the whole flow (kgCO2e)

EPW energy — 58 of 380.6 (this step +0.801)

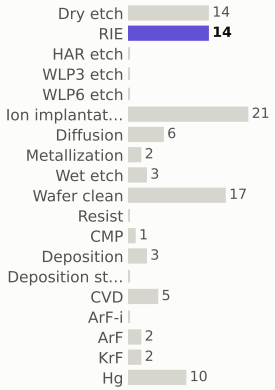
MPW materials — 27 of 42.4 (this step +0.0571)

GPW gases — 5.72 of 118.8 (this step +0.187)

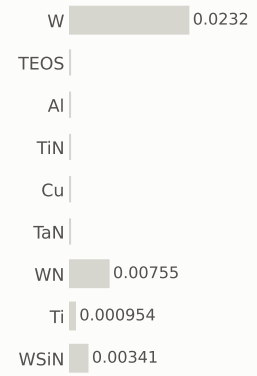
embodied carbon up to each step (kgCO2e per wafer)



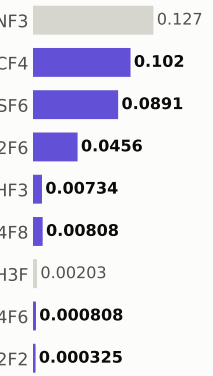
steps by process area



materials deposited (g)



gases emitted (g)



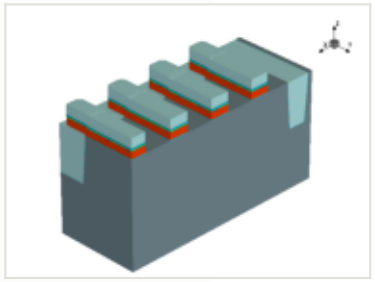
Step #1070: Dry_etch

etch_strip · Dry_etch

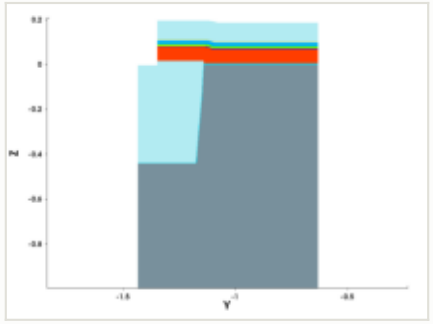
0.801	91.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

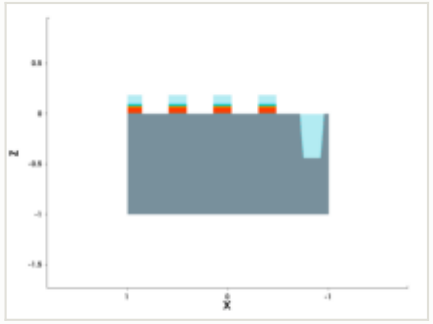
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



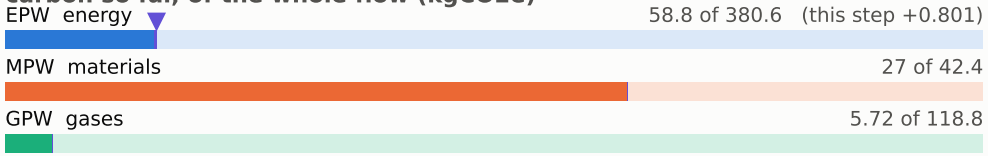
mask for this step

no mask for this step

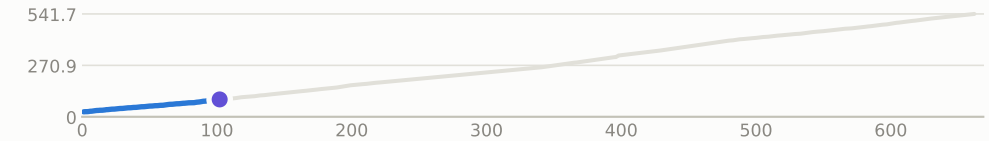
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

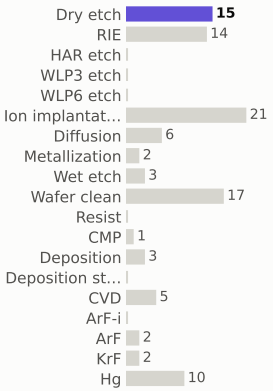
carbon so far, of the whole flow (kgCO2e)



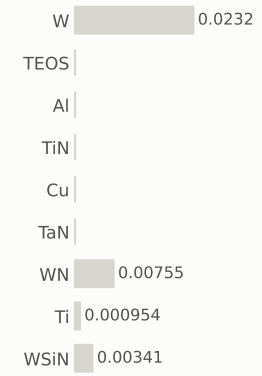
embodied carbon up to each step (kgCO2e per wafer)



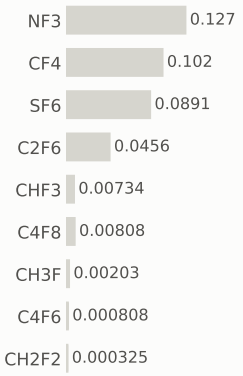
steps by process area



materials deposited (g)



gases emitted (g)



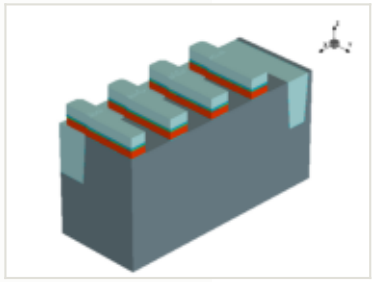
Step #1080: Wafer_clean

clean · Wafer_clean

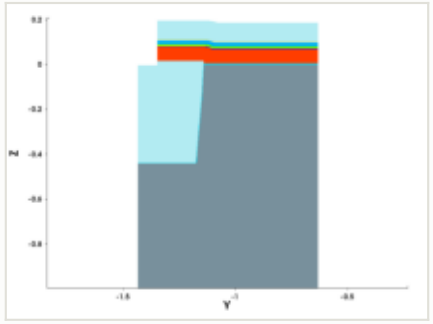
0.169	91.7	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

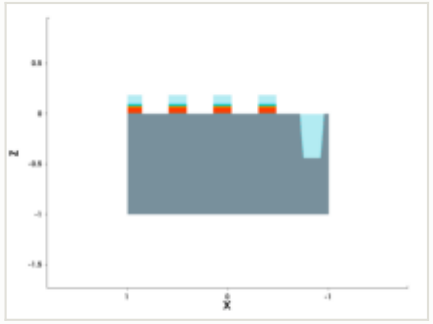
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

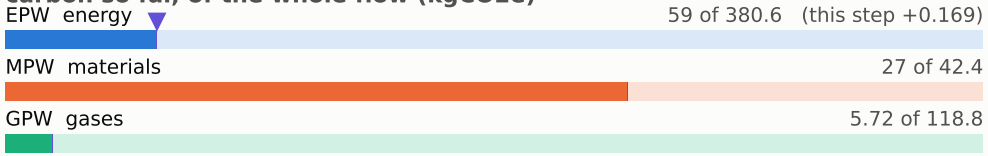
no mask for this step

this step

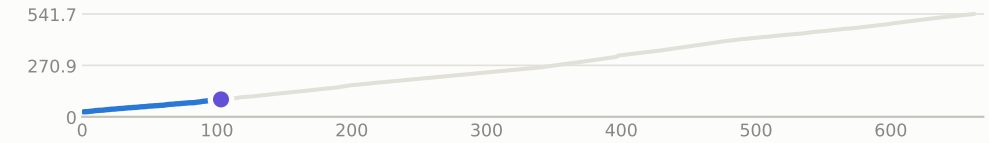
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

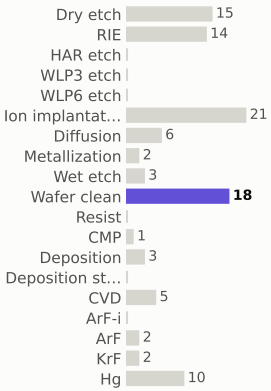
carbon so far, of the whole flow (kgCO2e)



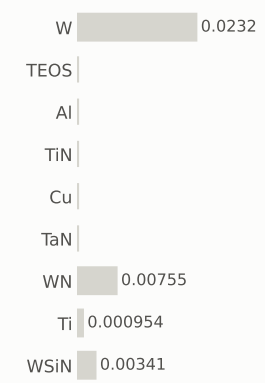
embodied carbon up to each step (kgCO2e per wafer)



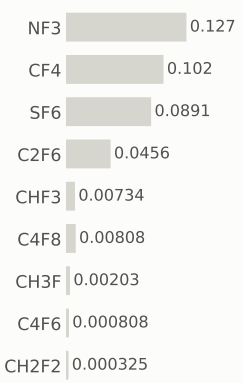
steps by process area



materials deposited (g)



gases emitted (g)



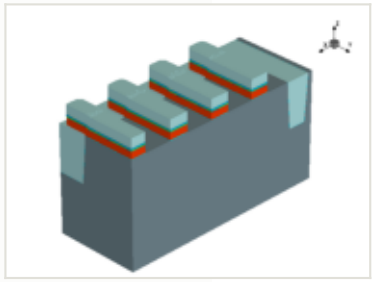
Step #1090: KrF

litho_patterning · KrF

0.199	91.9	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

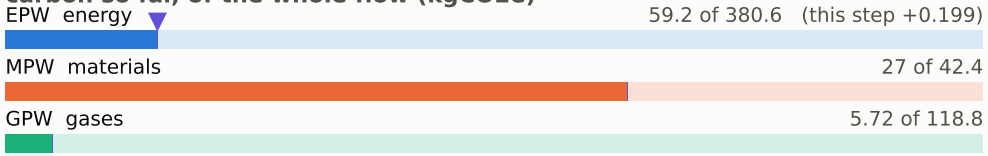


this step

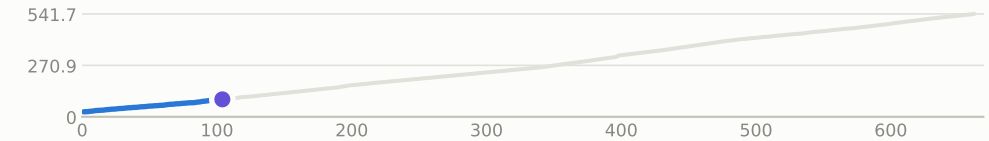
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

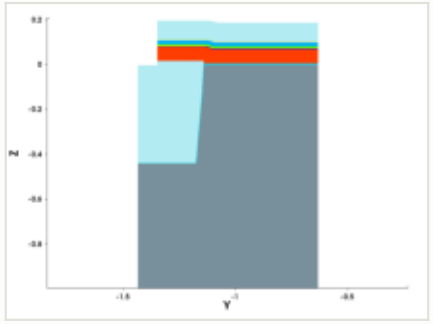
carbon so far, of the whole flow (kgCO2e)



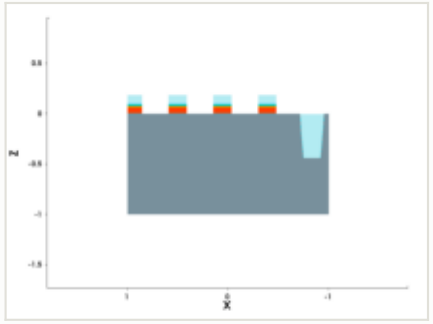
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

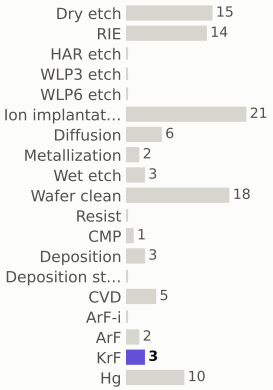


y cut at -1.032 µm

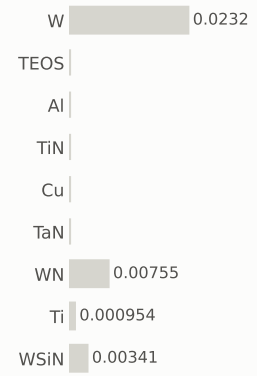


mask: layer 15:0 (OASIS)

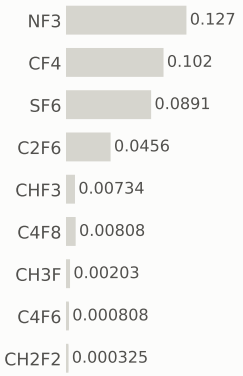
steps by process area



materials deposited (g)



gases emitted (g)



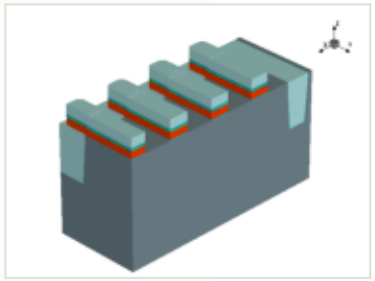
Step #1100: Ion_implantation

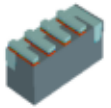
implant_ellipsoid · Ion_implantation

0.801	92.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

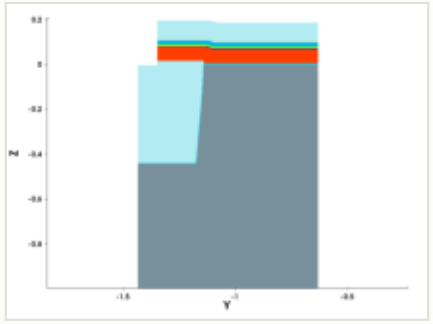
3D structure after this step

zoomed to fit

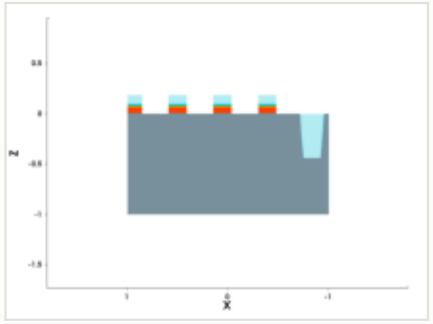




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

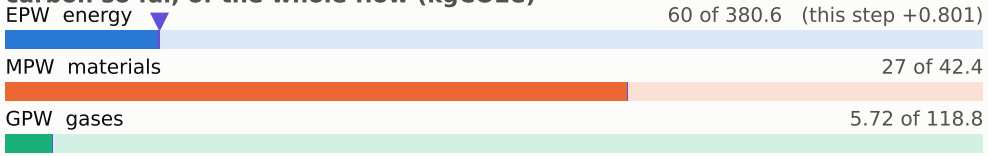
no mask for this step

this step

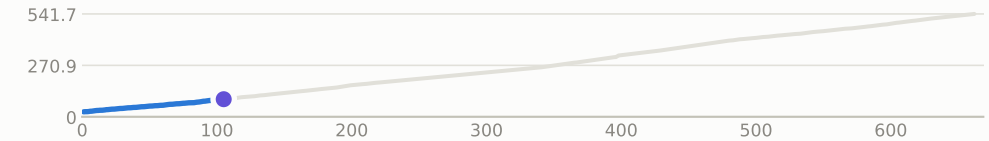
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

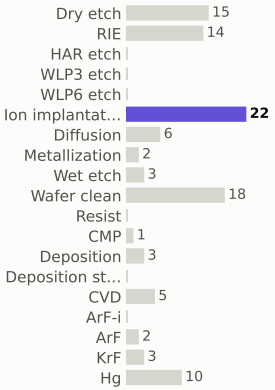
carbon so far, of the whole flow (kgCO2e)



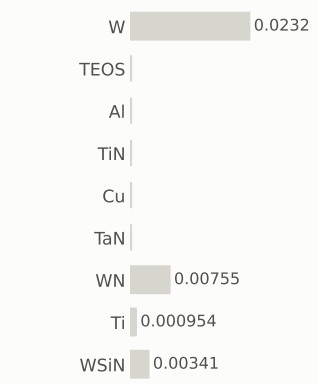
embodied carbon up to each step (kgCO2e per wafer)



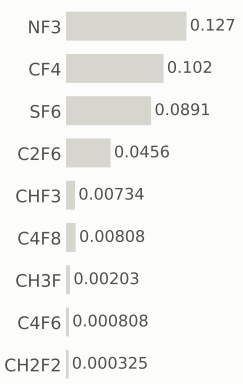
steps by process area



materials deposited (g)



gases emitted (g)



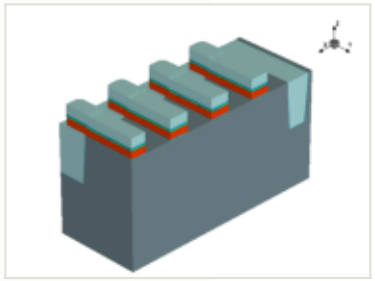
Step #1110: Ion_implantation

implant_ellipsoid · Ion_implantation

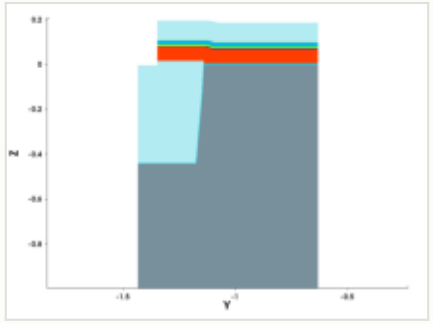
0.801	93.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

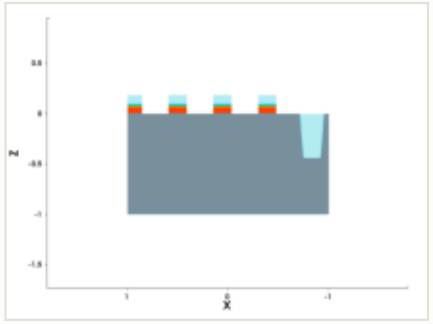
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

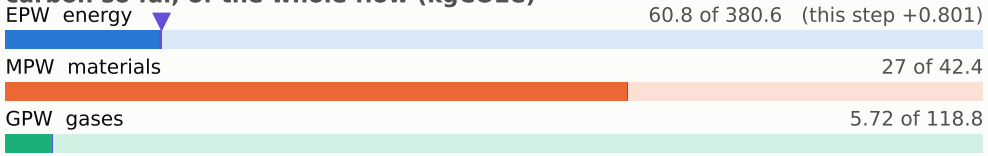
no mask for this step

this step

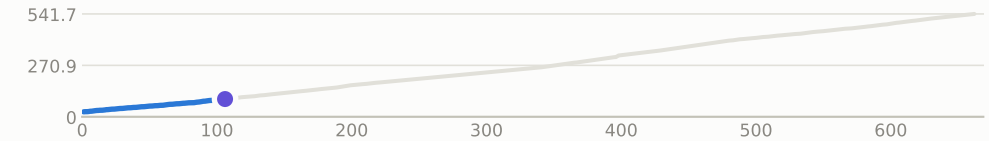
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

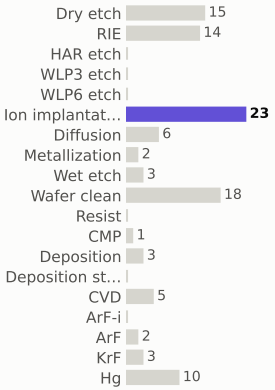
carbon so far, of the whole flow (kgCO2e)



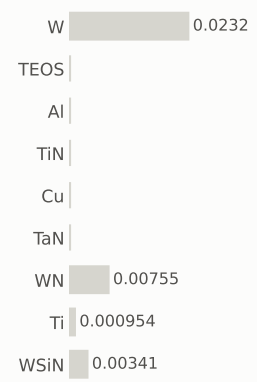
embodied carbon up to each step (kgCO2e per wafer)



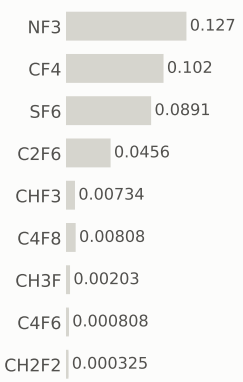
steps by process area



materials deposited (g)



gases emitted (g)



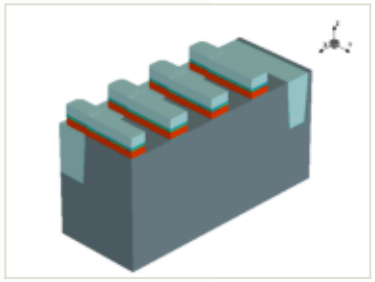
Step #1120: Dry_etch

etch_strip · Dry_etch

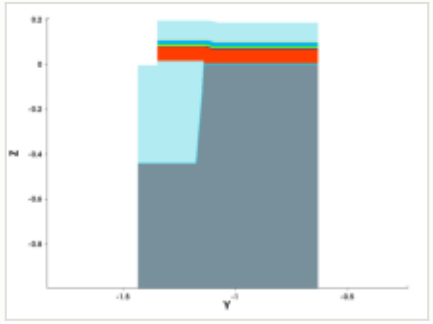
0.801	94.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

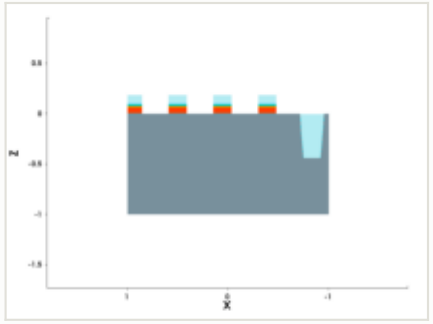
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



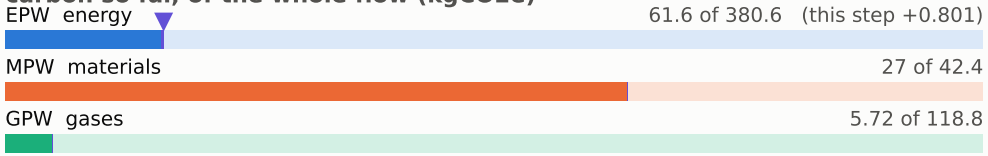
mask for this step

no mask for this step

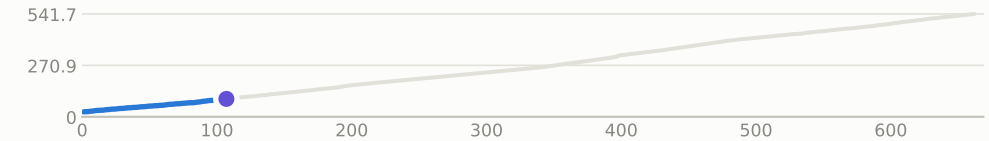
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

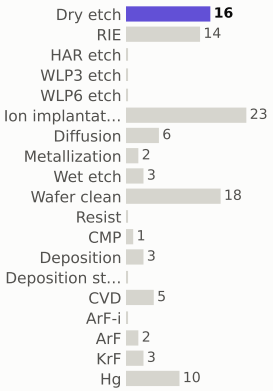
carbon so far, of the whole flow (kgCO2e)



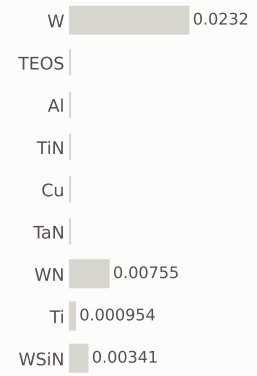
embodied carbon up to each step (kgCO2e per wafer)



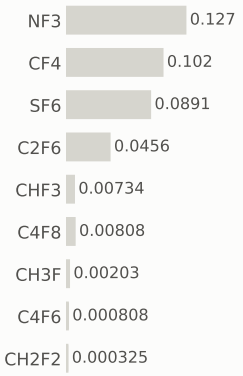
steps by process area



materials deposited (g)



gases emitted (g)



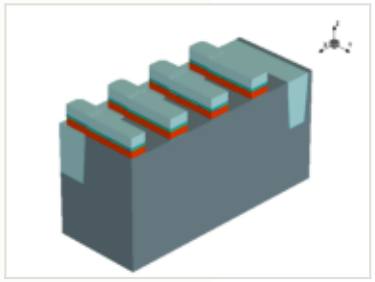
Step #1130: Wafer_clean

clean · Wafer_clean

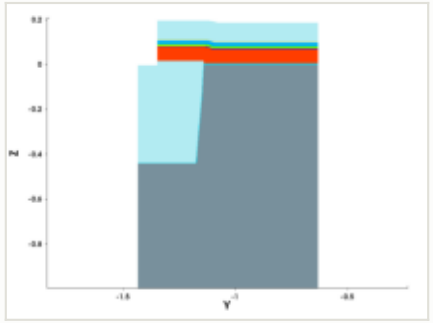
0.169	94.4	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

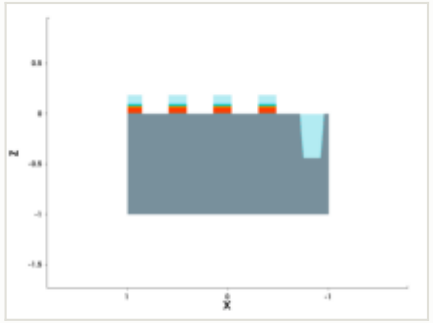
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

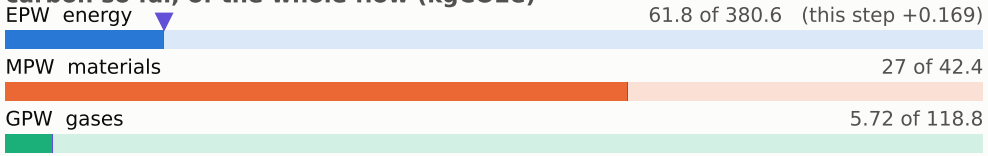
no mask for this step

this step

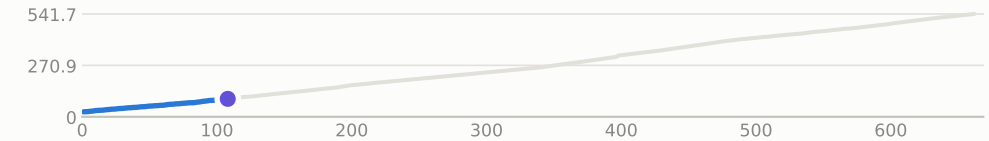
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

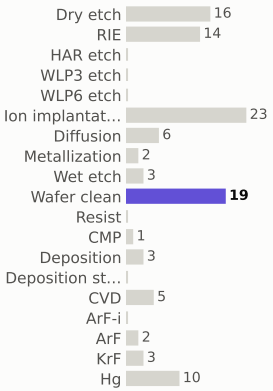
carbon so far, of the whole flow (kgCO2e)



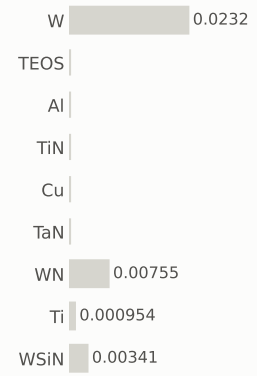
embodied carbon up to each step (kgCO2e per wafer)



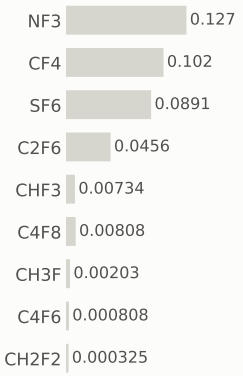
steps by process area



materials deposited (g)



gases emitted (g)



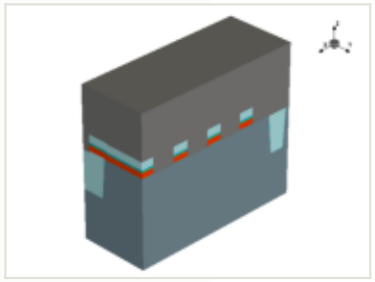
Step #1140: KrF

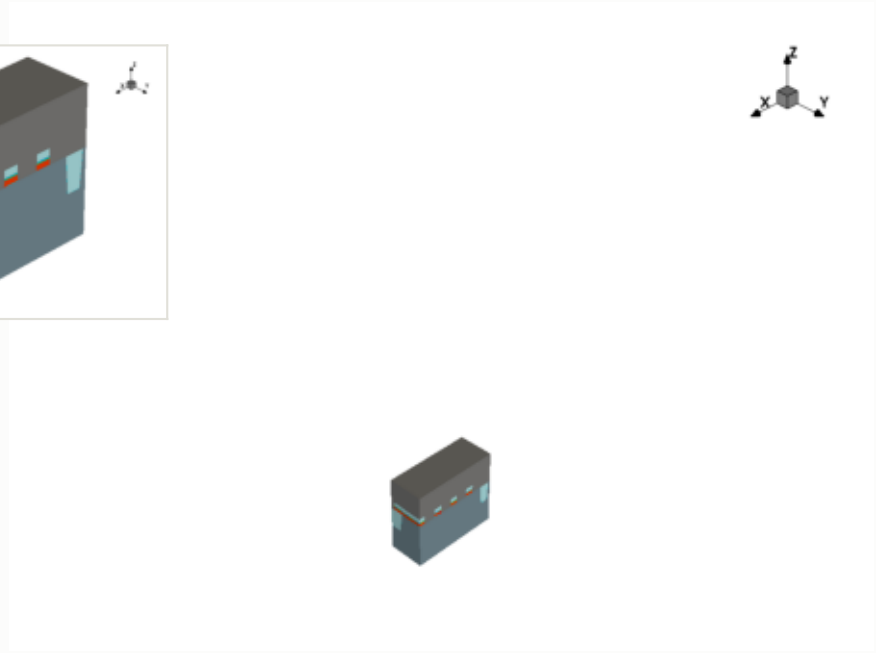
litho_patterning · KrF

0.199	94.6	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

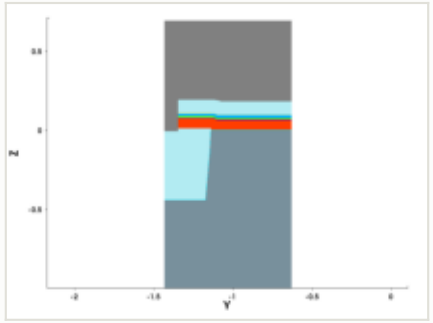
3D structure after this step

zoomed to fit

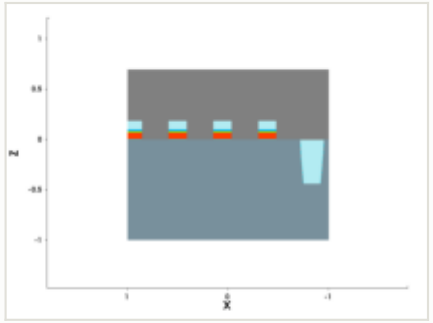




x cut at 0.000 µm



y cut at -1.032 µm



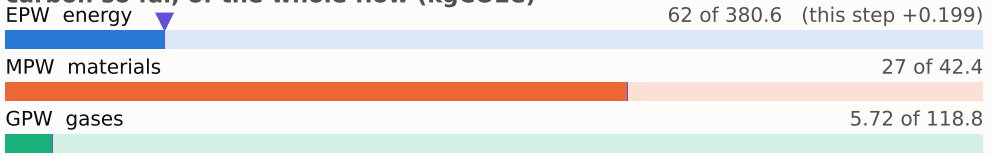
mask: layer 16:0, positive (OASIS)

this step

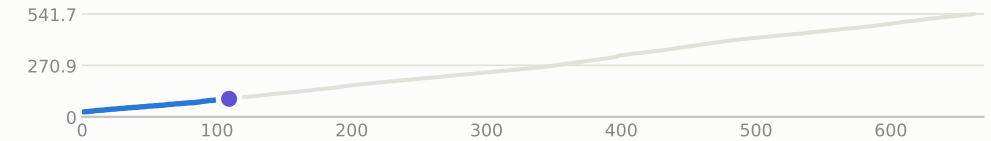
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

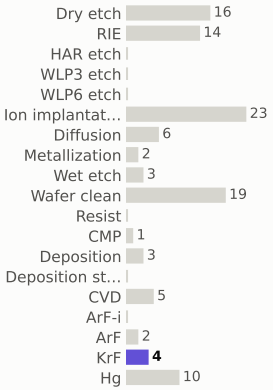
carbon so far, of the whole flow (kgCO2e)



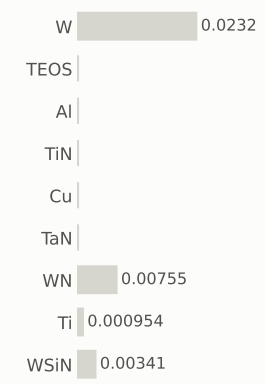
embodied carbon up to each step (kgCO2e per wafer)



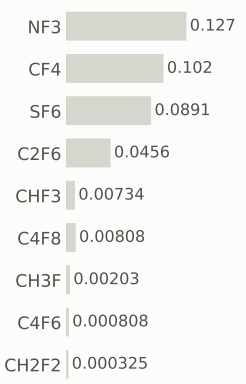
steps by process area



materials deposited (g)



gases emitted (g)



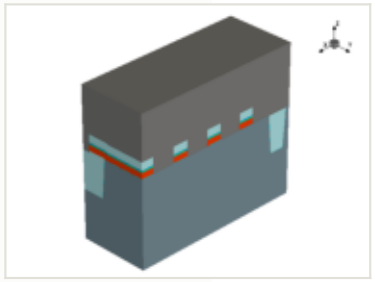
Step #1150: Ion_implantation

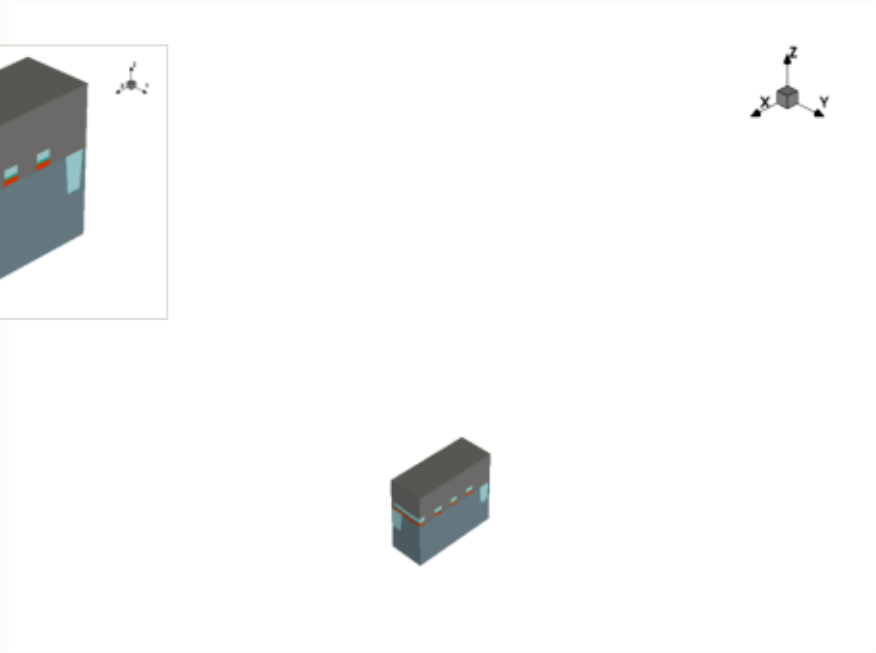
implant_ellipsoid · Ion_implantation

0.801	95.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

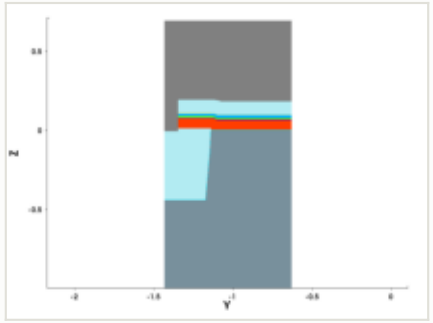
3D structure after this step

zoomed to fit

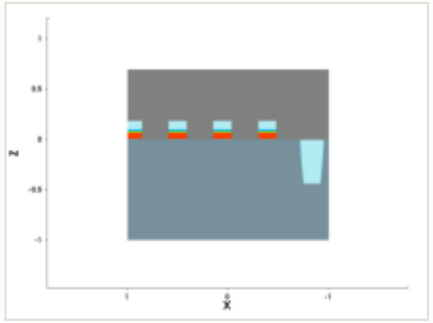




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

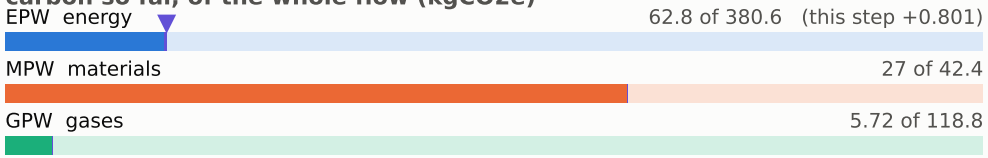
this step

process area	Ion_implantation
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

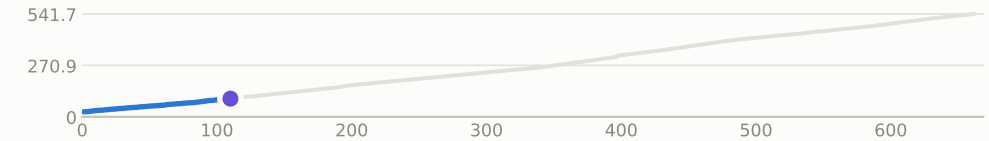
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

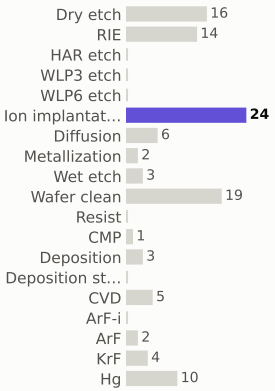
carbon so far, of the whole flow (kgCO2e)



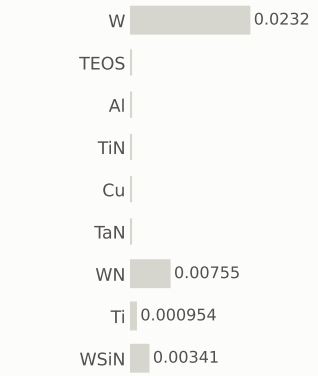
embodied carbon up to each step (kgCO2e per wafer)



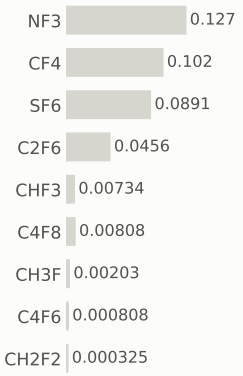
steps by process area



materials deposited (g)



gases emitted (g)



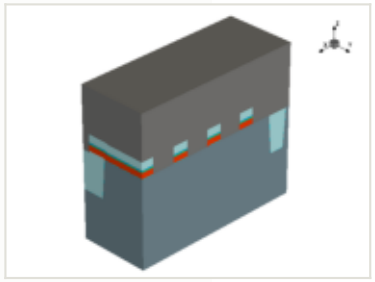
Step #1160: Ion_implantation

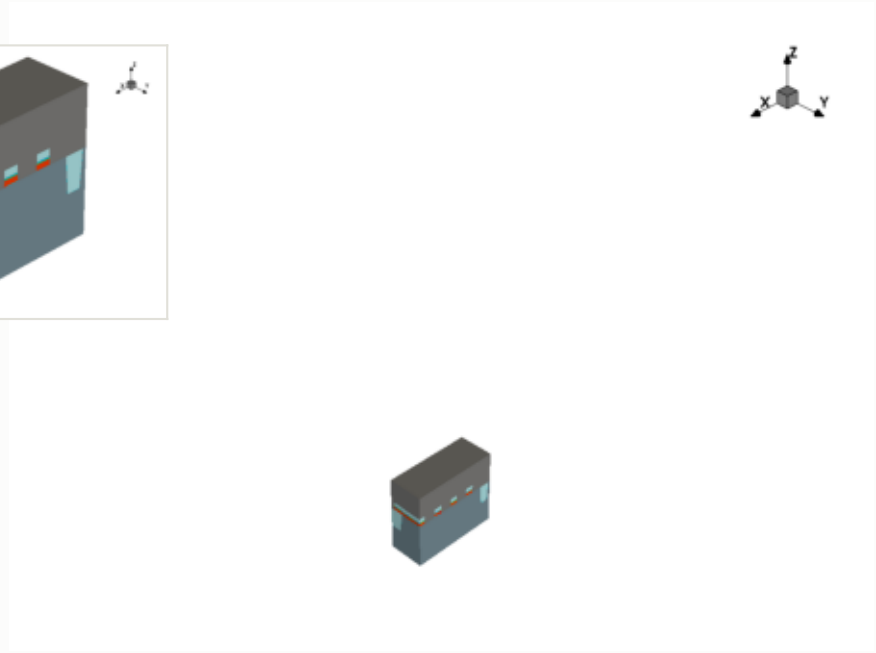
implant_ellipsoid · Ion_implantation

0.801	96.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

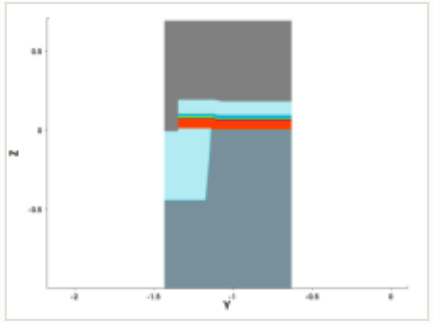
3D structure after this step

zoomed to fit

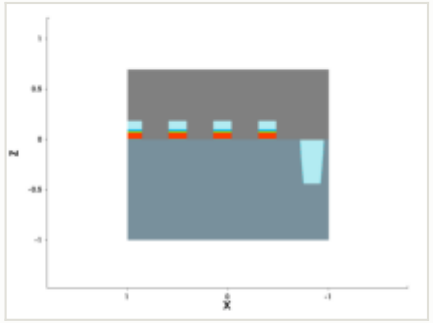




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

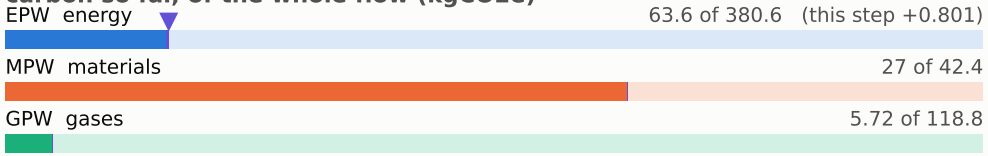
no mask for this step

this step

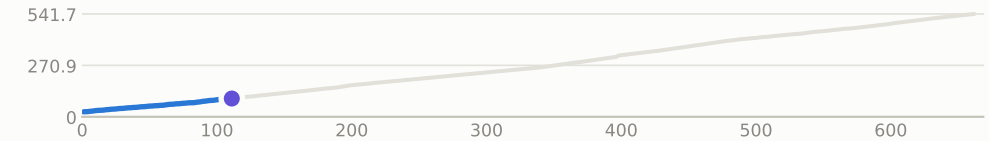
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

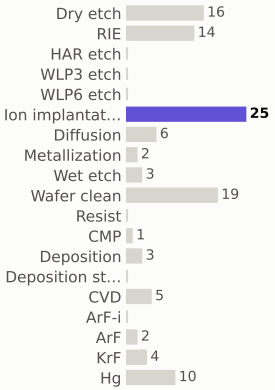
carbon so far, of the whole flow (kgCO2e)



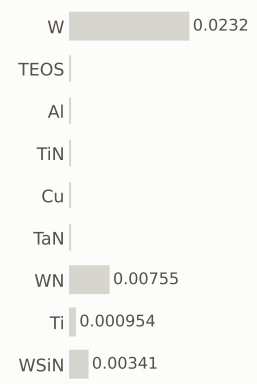
embodied carbon up to each step (kgCO2e per wafer)



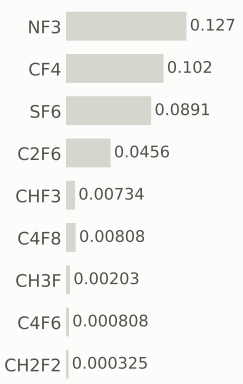
steps by process area



materials deposited (g)



gases emitted (g)



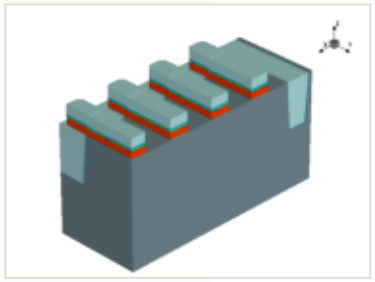
Step #1170: Dry_etch

etch_strip · Dry_etch

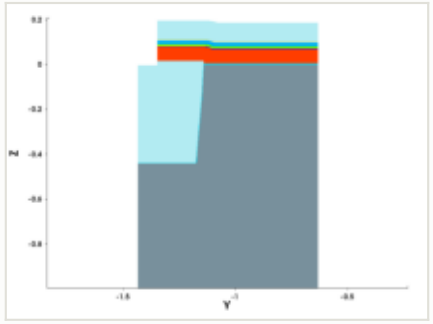
0.801	97	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

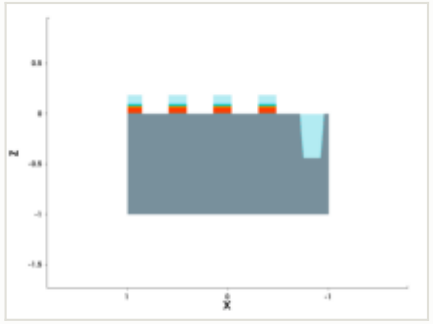
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



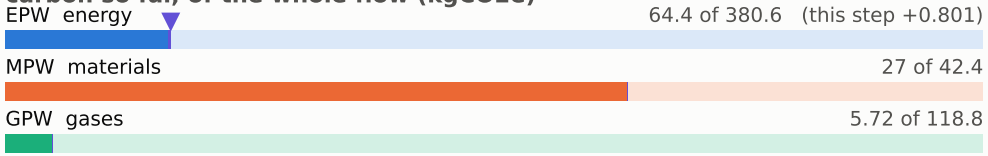
mask for this step

no mask for this step

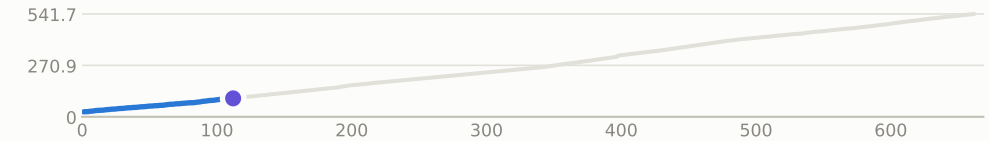
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

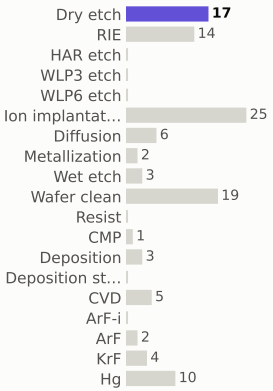
carbon so far, of the whole flow (kgCO2e)



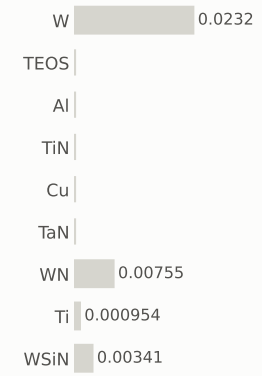
embodied carbon up to each step (kgCO2e per wafer)



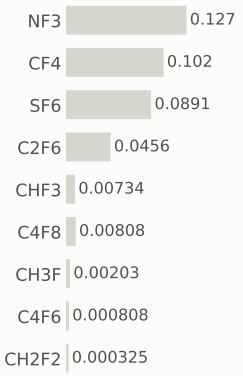
steps by process area



materials deposited (g)



gases emitted (g)



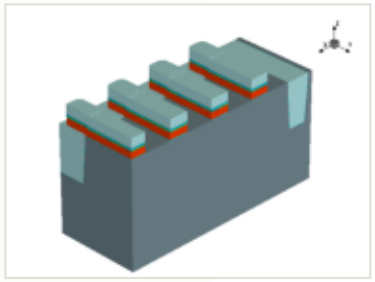
Step #1180: Wafer_clean

clean · Wafer_clean

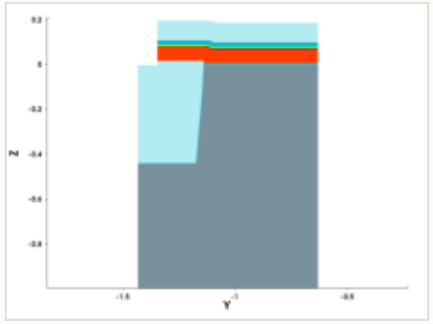
0.169	97.2	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

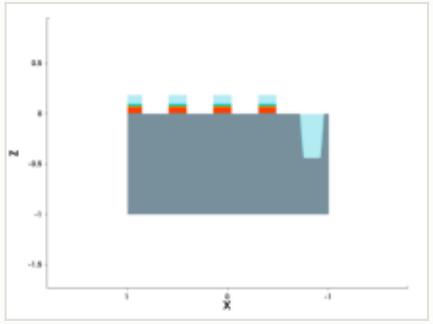
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

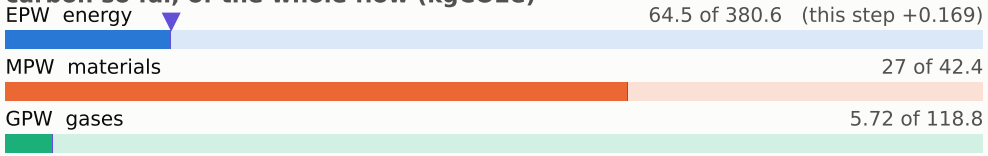
no mask for this step

this step

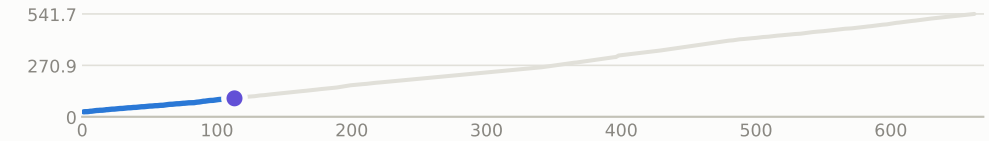
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

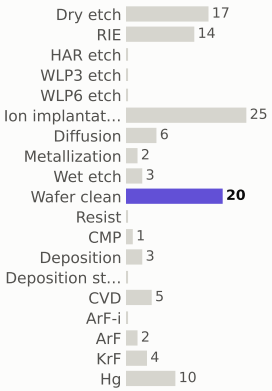
carbon so far, of the whole flow (kgCO2e)



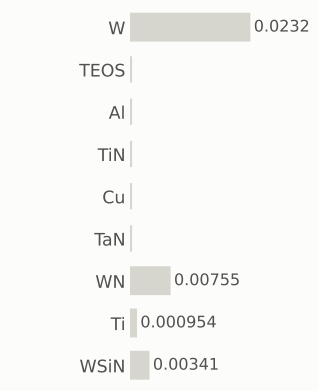
embodied carbon up to each step (kgCO2e per wafer)



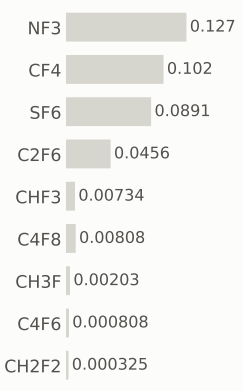
steps by process area



materials deposited (g)



gases emitted (g)



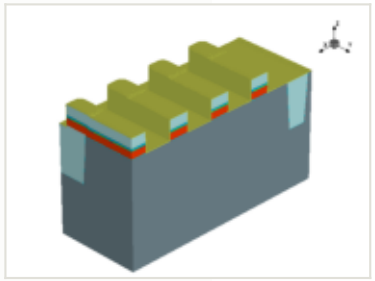
Step #1190: CVD

depo_isotropic · CVD

1.31	98.5	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

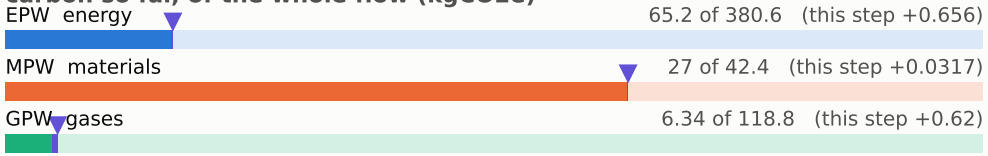


this step

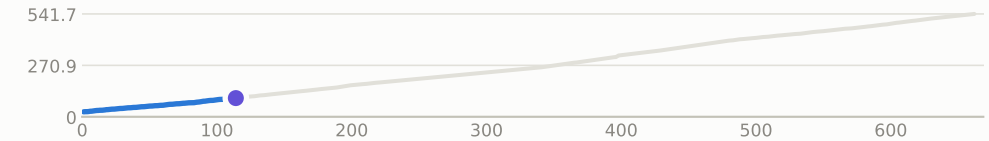
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

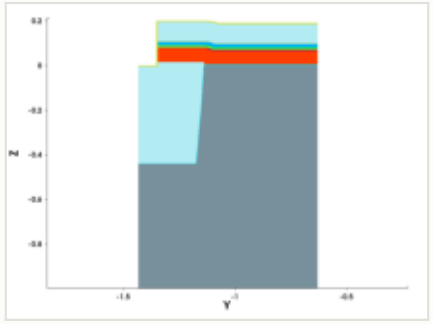
carbon so far, of the whole flow (kgCO2e)



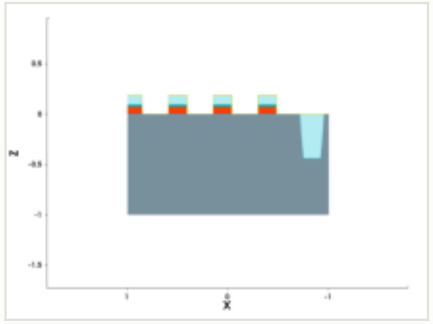
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



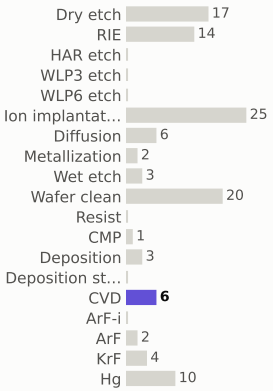
y cut at -1.032 µm



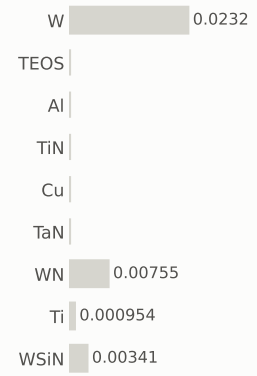
mask for this step

no mask for this step

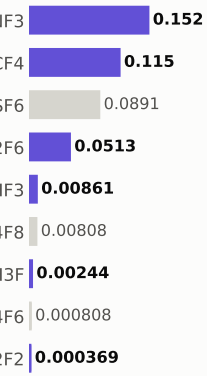
steps by process area



materials deposited (g)



gases emitted (g)



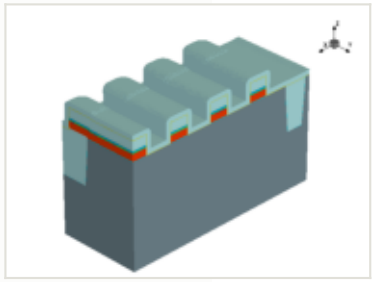
Step #1200: CVD

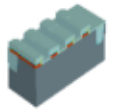
depo_isotropic · CVD

1.31	99.8	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

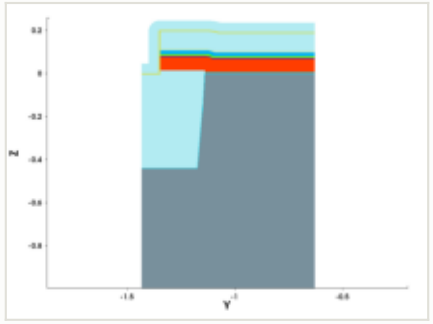
3D structure after this step

zoomed to fit

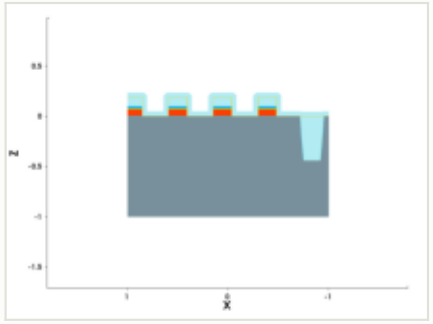




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

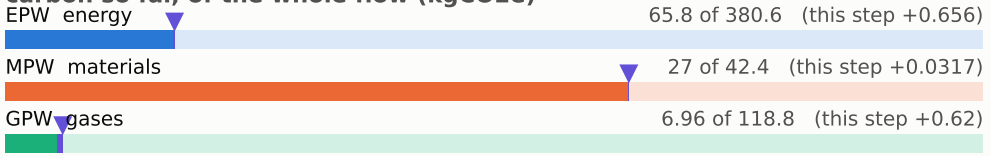
no mask for this step

this step

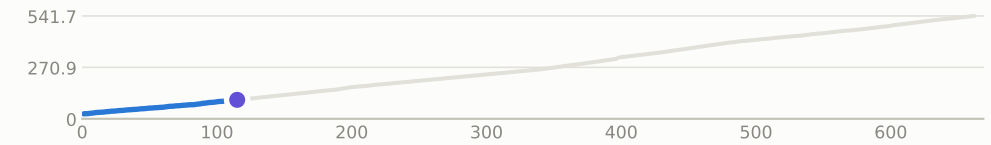
process area	CVD
material	—
thickness	0.044 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

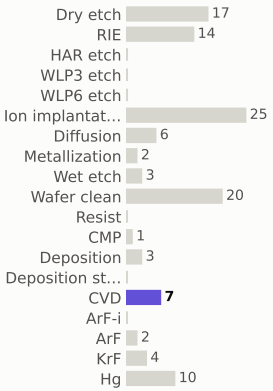
carbon so far, of the whole flow (kgCO2e)



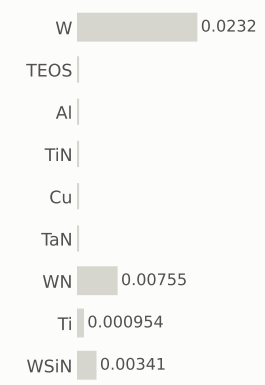
embodied carbon up to each step (kgCO2e per wafer)



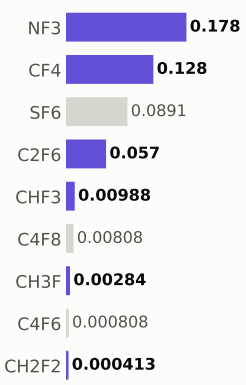
steps by process area



materials deposited (g)



gases emitted (g)



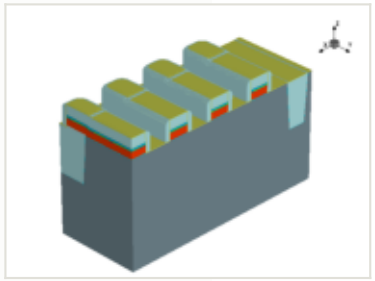
Step #1210: RIE

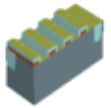
etch_anisotropic · RIE

1.04	100.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



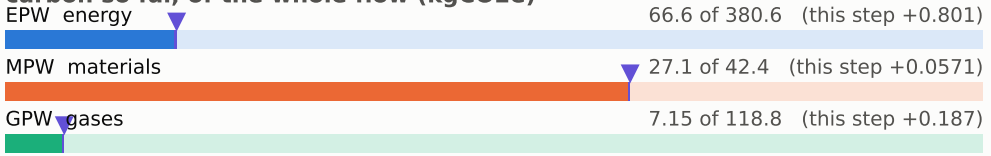


this step

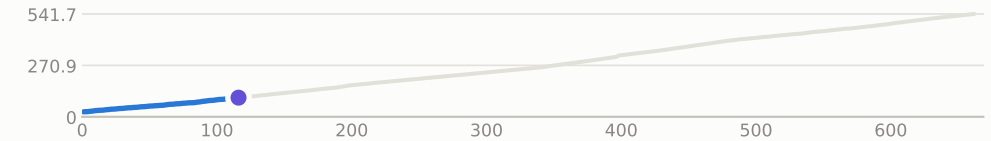
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

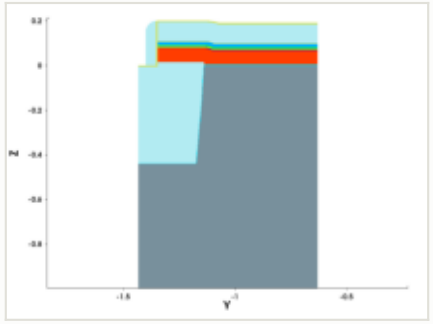
carbon so far, of the whole flow (kgCO2e)



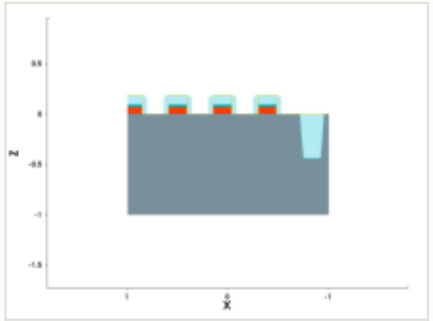
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



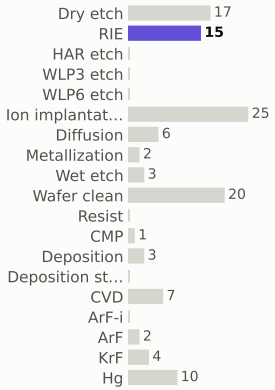
y cut at -1.032 µm



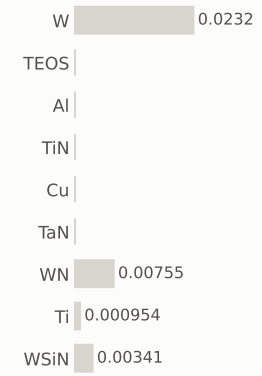
mask for this step

no mask for this step

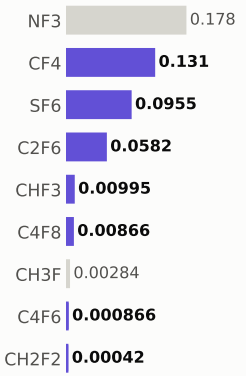
steps by process area



materials deposited (g)



gases emitted (g)



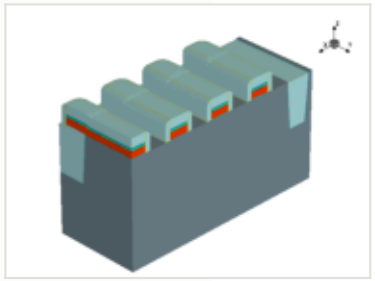
Step #1220: RIE

etch_anisotropic · RIE

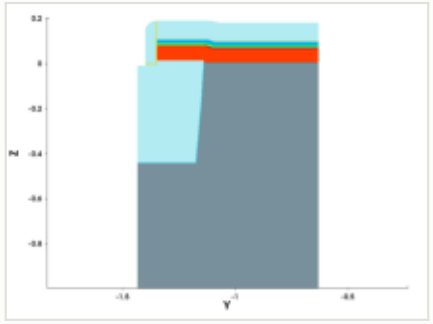
1.04	101.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

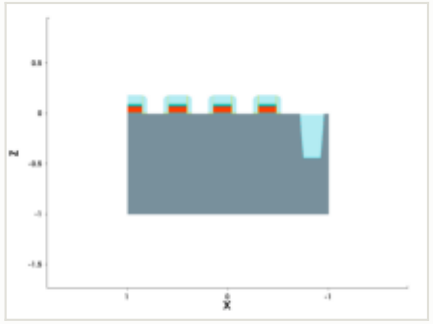
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

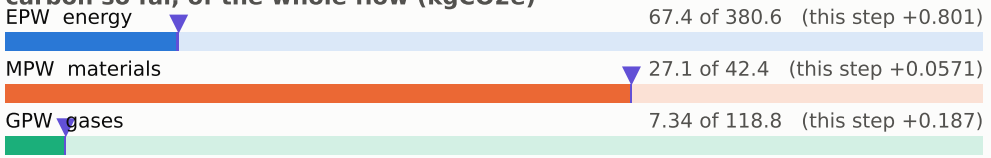
no mask for this step

this step

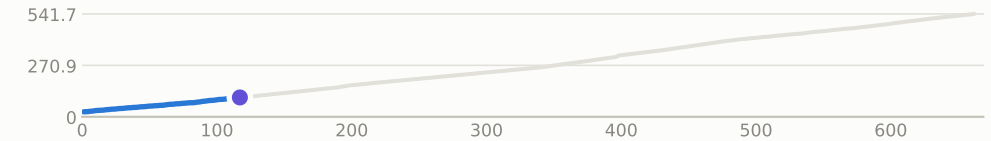
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

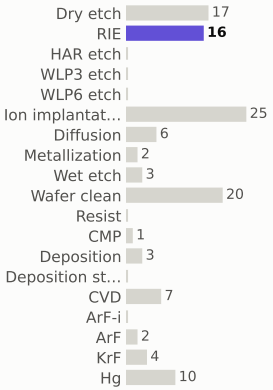
carbon so far, of the whole flow (kgCO2e)



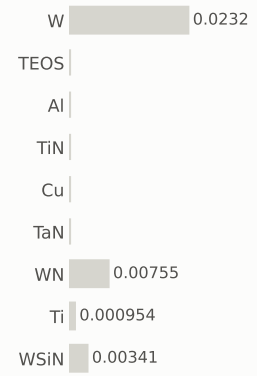
embodied carbon up to each step (kgCO2e per wafer)



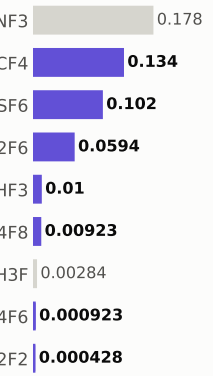
steps by process area



materials deposited (g)



gases emitted (g)



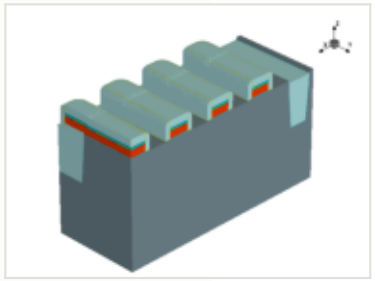
Step #1230: RIE

etch_anisotropic · RIE

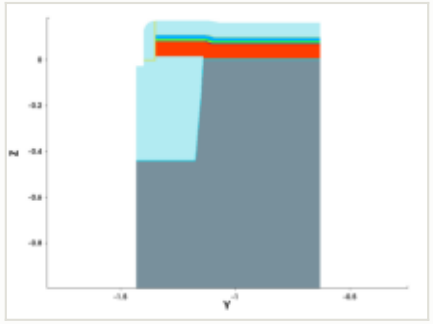
1.04	103.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

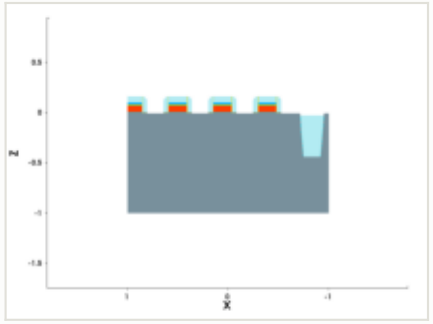
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

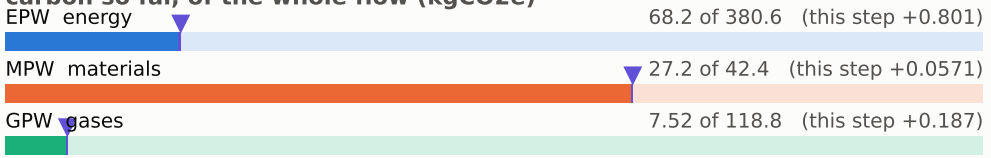
no mask for this step

this step

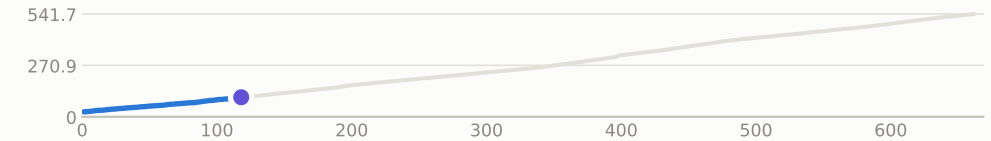
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

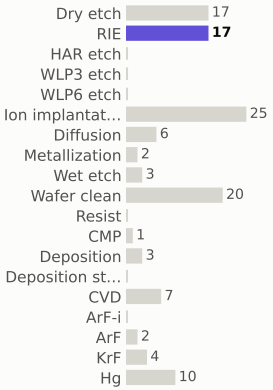
carbon so far, of the whole flow (kgCO2e)



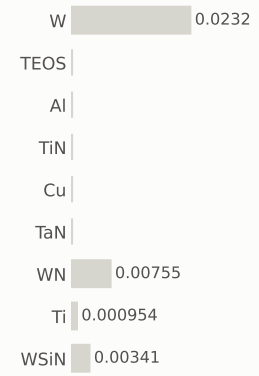
embodied carbon up to each step (kgCO2e per wafer)



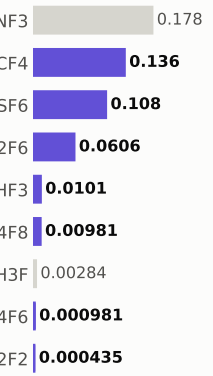
steps by process area



materials deposited (g)



gases emitted (g)



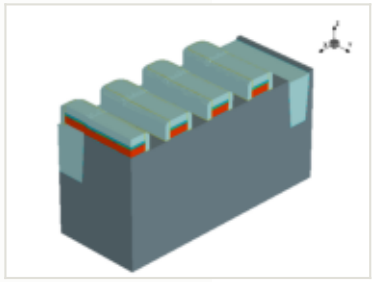
Step #1240: KrF

litho_patterning · KrF

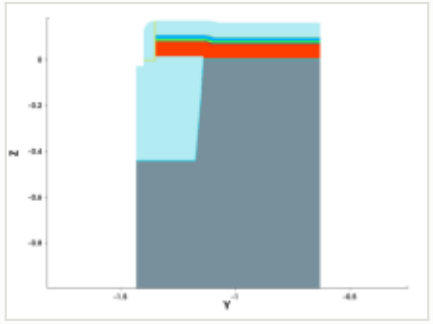
0.199	103.2	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

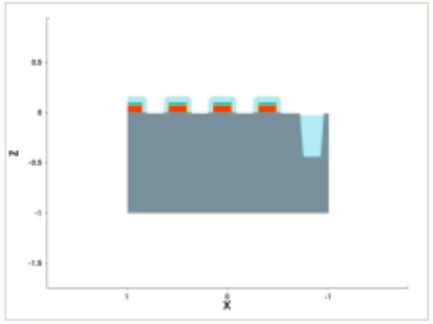
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



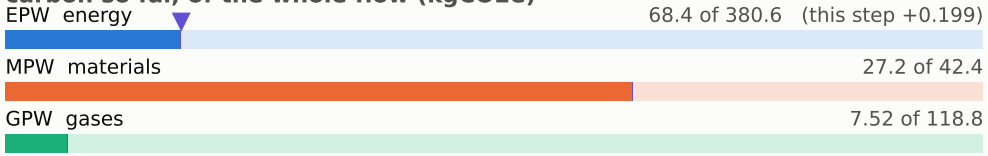
mask: layer 17:0 (OASIS)

this step

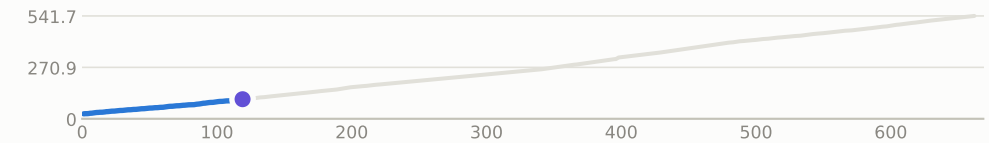
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

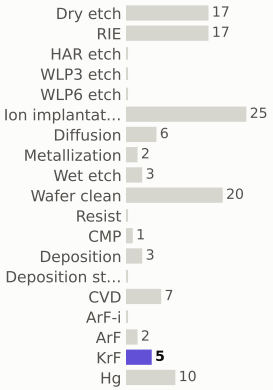
carbon so far, of the whole flow (kgCO2e)



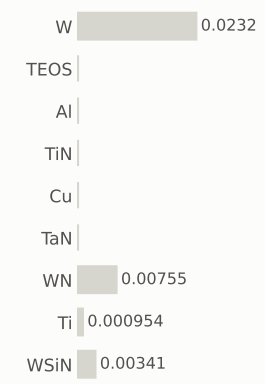
embodied carbon up to each step (kgCO2e per wafer)



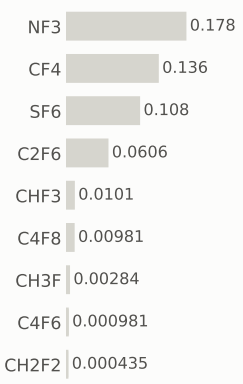
steps by process area



materials deposited (g)



gases emitted (g)



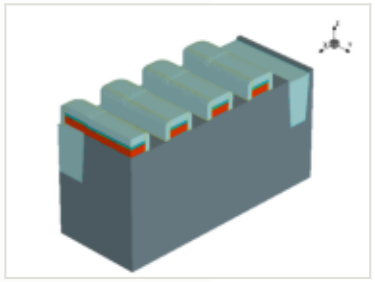
Step #1250: Ion_implantation

implant_ellipsoid · Ion_implantation

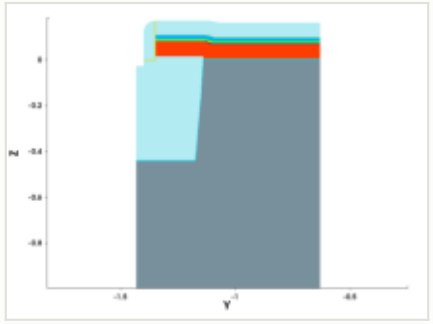
0.801	104.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

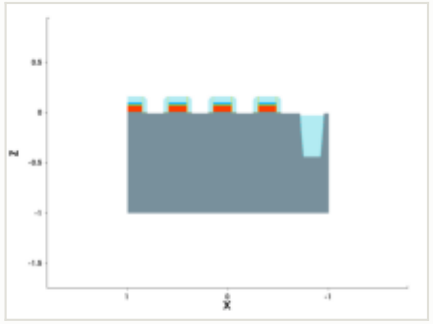
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

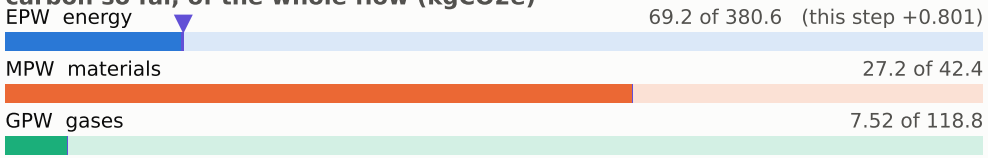
no mask for this step

this step

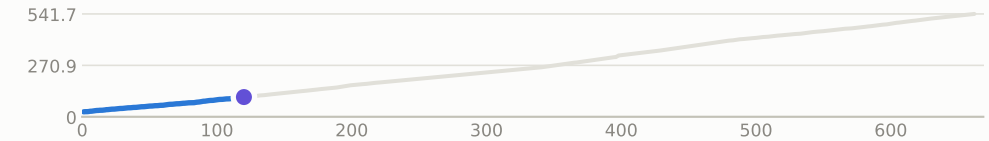
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

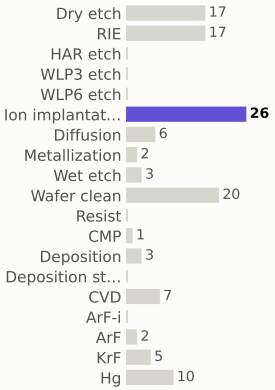
carbon so far, of the whole flow (kgCO2e)



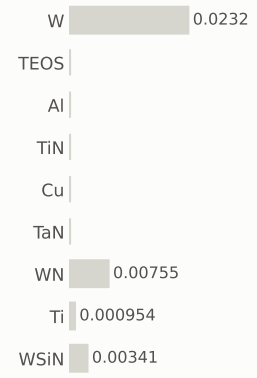
embodied carbon up to each step (kgCO2e per wafer)



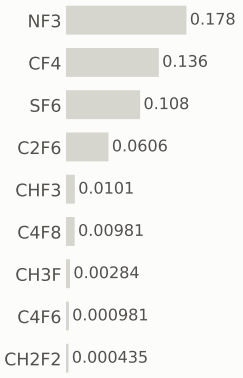
steps by process area



materials deposited (g)



gases emitted (g)



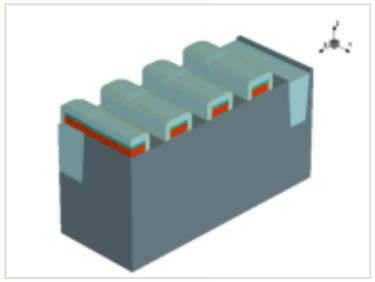
Step #1260: Ion_implantation

implant_ellipsoid · Ion_implantation

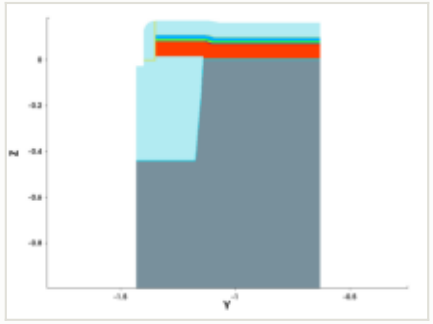
0.801	104.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

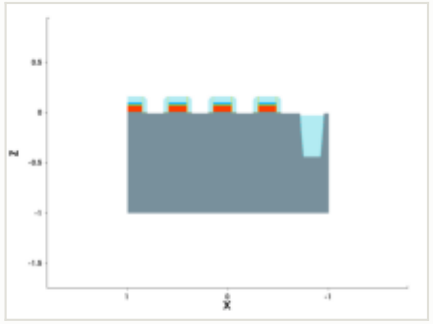
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

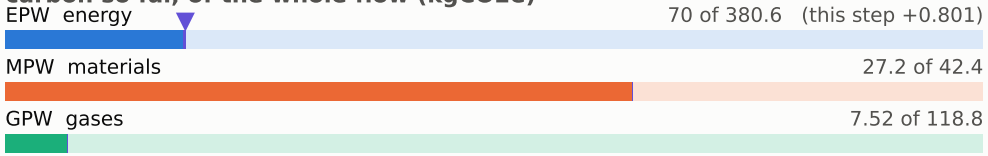
no mask for this step

this step

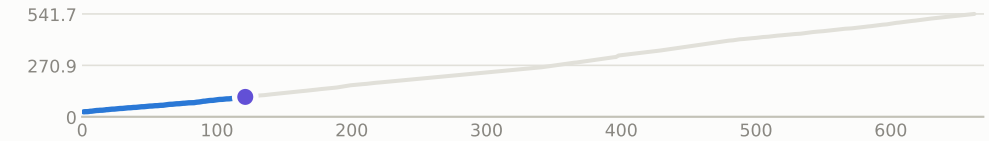
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

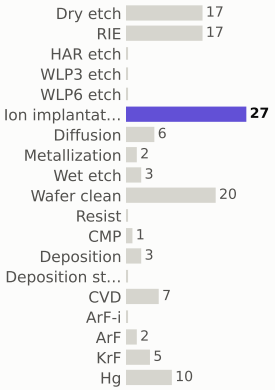
carbon so far, of the whole flow (kgCO2e)



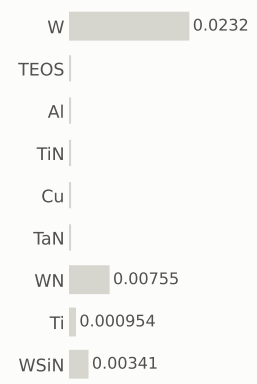
embodied carbon up to each step (kgCO2e per wafer)



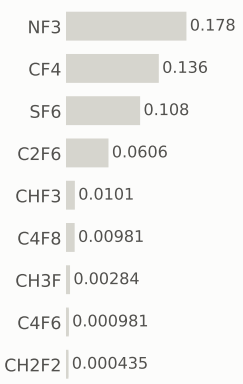
steps by process area



materials deposited (g)



gases emitted (g)



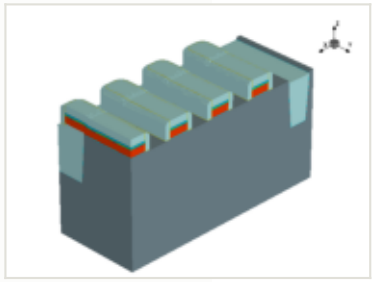
Step #1270: Dry_etch

etch_strip · Dry_etch

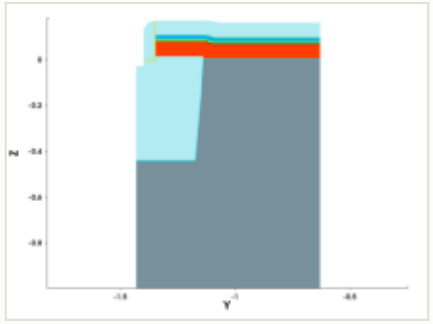
0.801	105.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

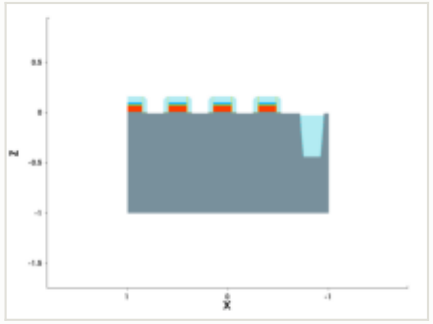
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

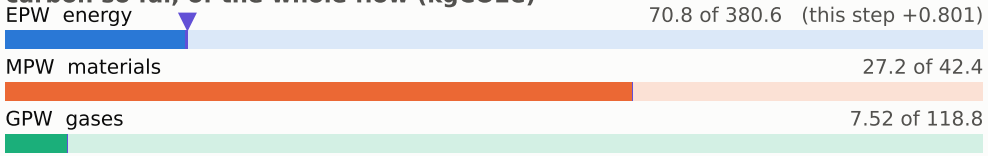
no mask for this step

this step

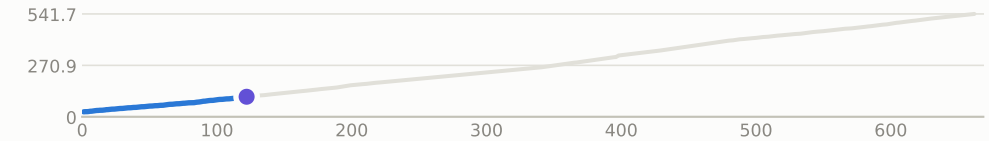
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

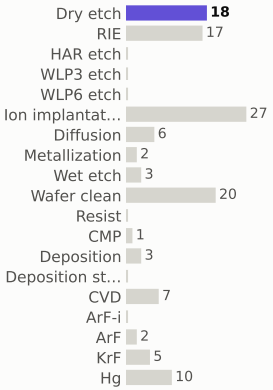
carbon so far, of the whole flow (kgCO2e)



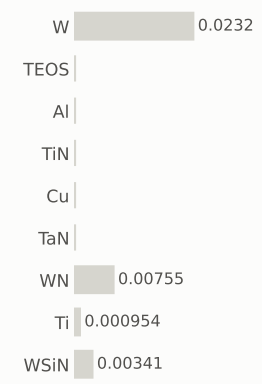
embodied carbon up to each step (kgCO2e per wafer)



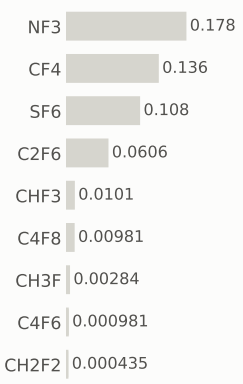
steps by process area



materials deposited (g)



gases emitted (g)



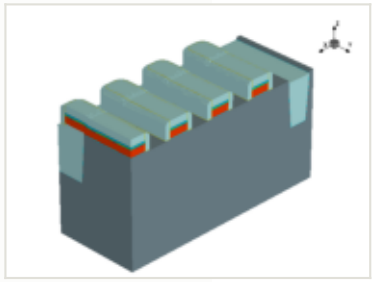
Step #1280: Wafer_clean

clean · Wafer_clean

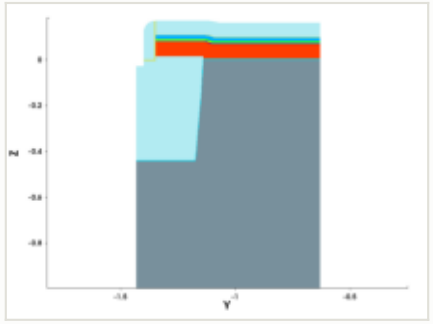
0.169	105.7	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

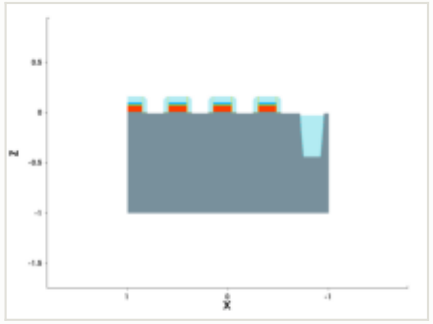
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

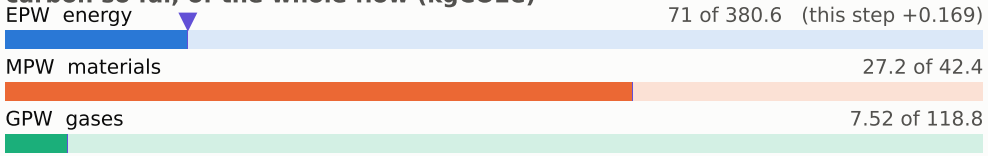
no mask for this step

this step

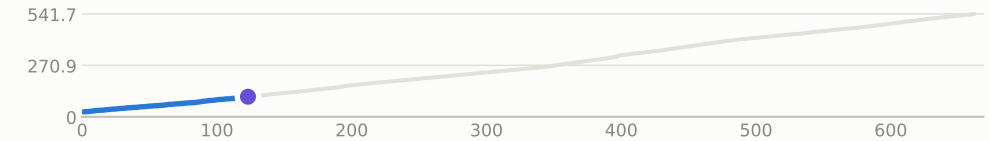
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

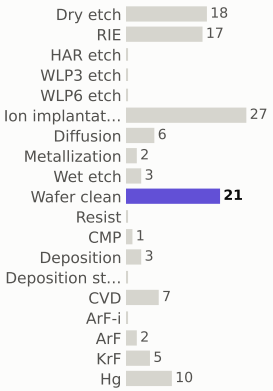
carbon so far, of the whole flow (kgCO2e)



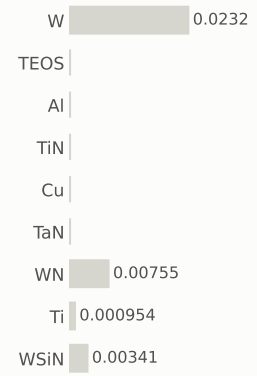
embodied carbon up to each step (kgCO2e per wafer)



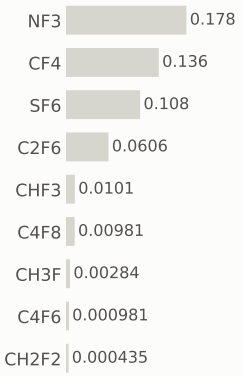
steps by process area



materials deposited (g)



gases emitted (g)



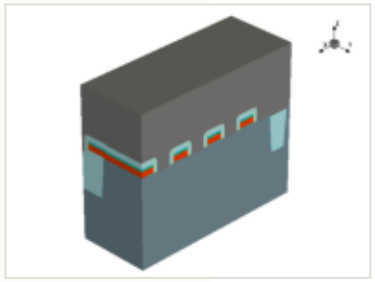
Step #1290: KrF

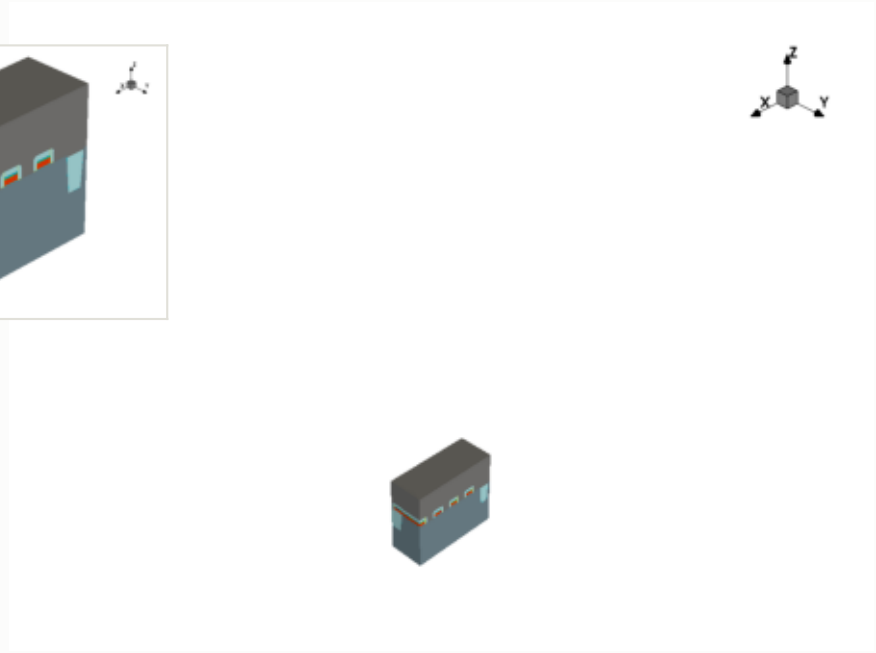
litho_patterning · KrF

0.199	105.9	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

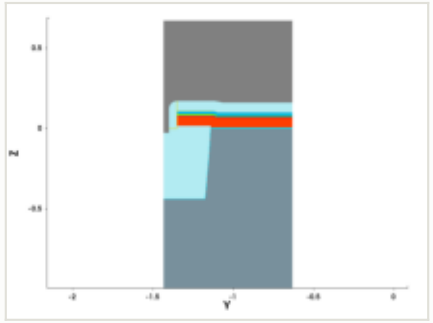
3D structure after this step

zoomed to fit

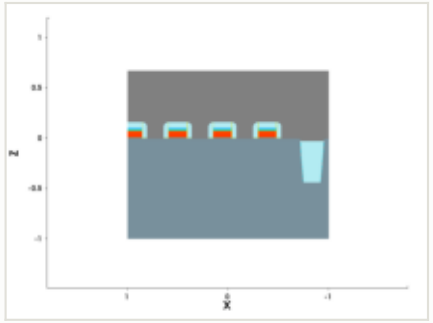




x cut at 0.000 µm



y cut at -1.032 µm

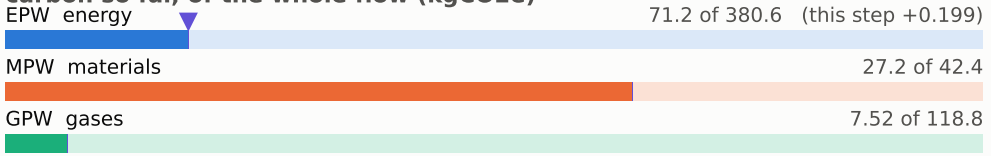


mask: layer 18:0, positive (OASIS)

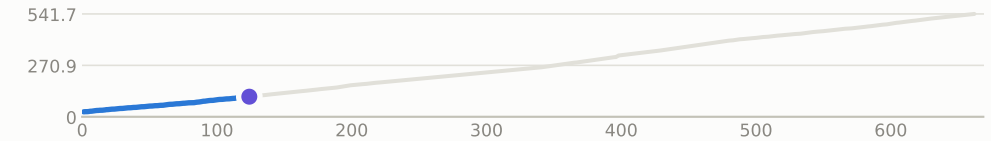
this step

process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

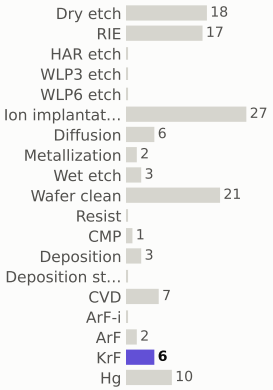
carbon so far, of the whole flow (kgCO2e)



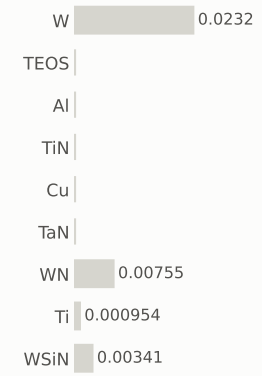
embodied carbon up to each step (kgCO2e per wafer)



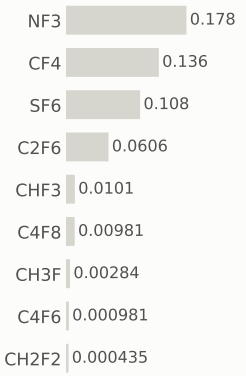
steps by process area



materials deposited (g)



gases emitted (g)



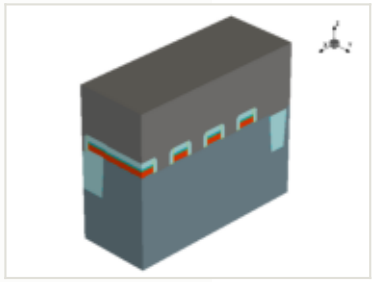
Step #1300: Ion_implantation

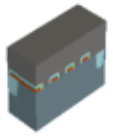
implant_ellipsoid · Ion_implantation

0.801	106.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

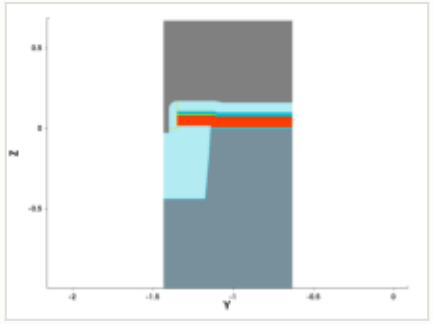
3D structure after this step

zoomed to fit

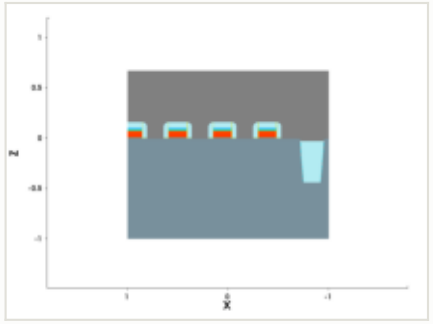




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

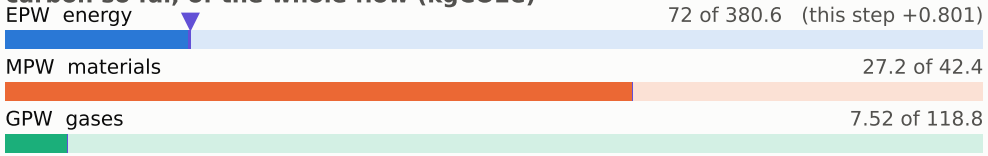
no mask for this step

this step

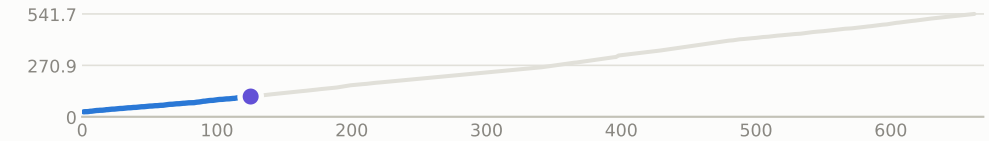
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

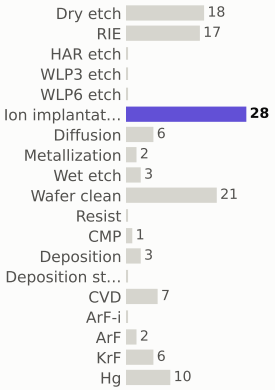
carbon so far, of the whole flow (kgCO2e)



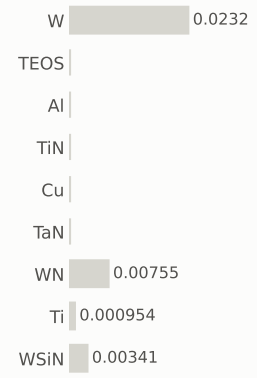
embodied carbon up to each step (kgCO2e per wafer)



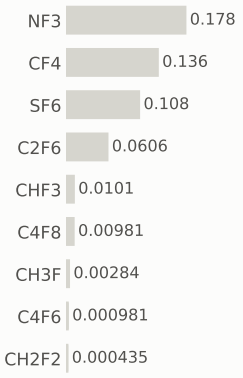
steps by process area



materials deposited (g)



gases emitted (g)



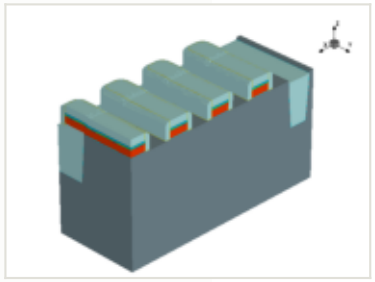
Step #1310: Dry_etch

etch_strip · Dry_etch

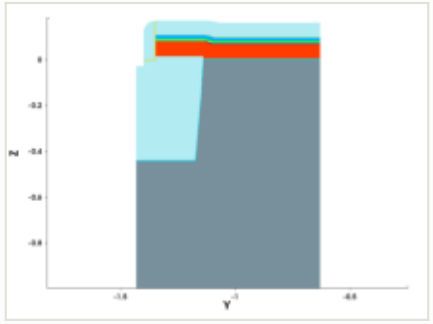
0.801	107.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

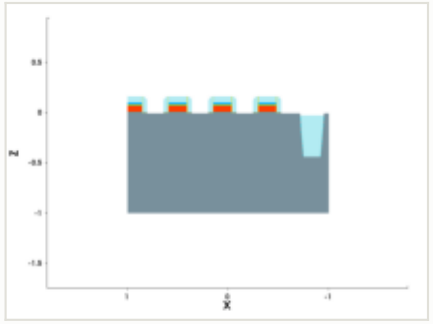
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



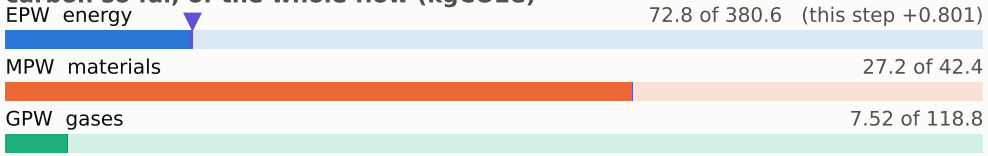
mask for this step

no mask for this step

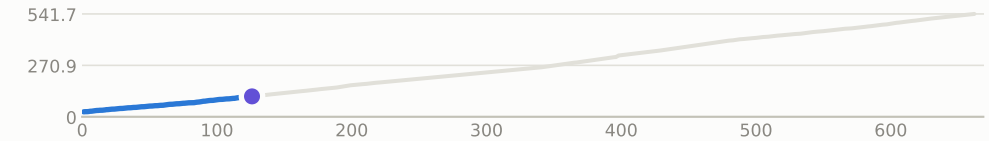
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

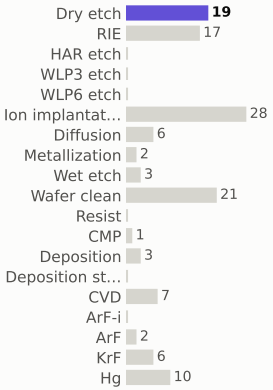
carbon so far, of the whole flow (kgCO2e)



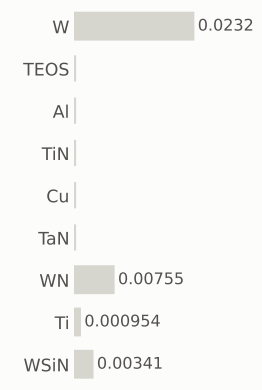
embodied carbon up to each step (kgCO2e per wafer)



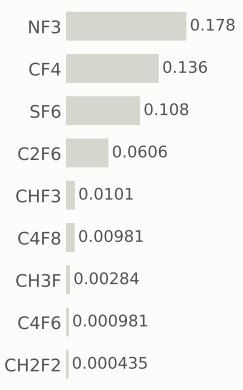
steps by process area



materials deposited (g)



gases emitted (g)



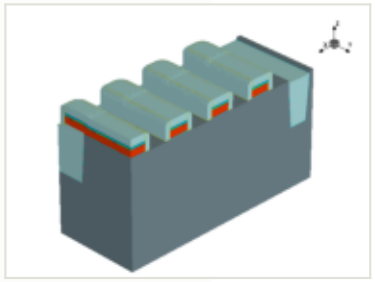
Step #1320: Wafer_clean

clean · Wafer_clean

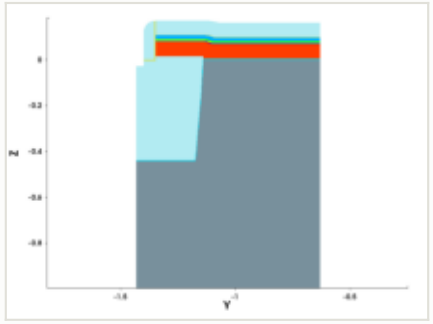
0.169	107.7	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

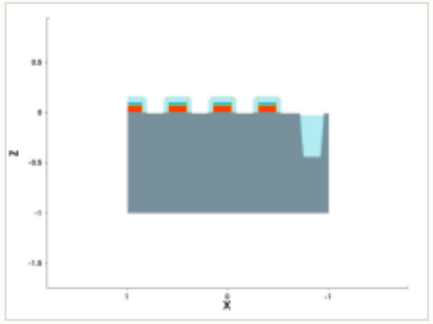
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

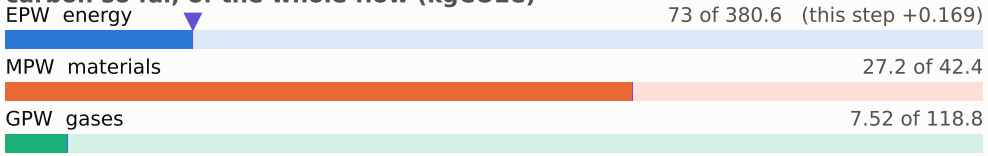
no mask for this step

this step

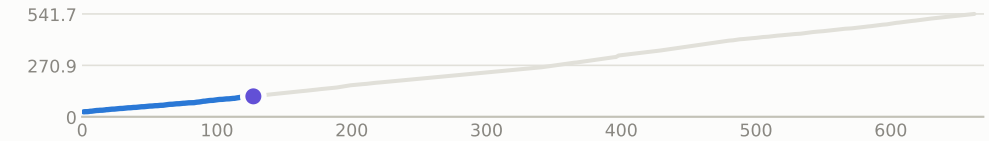
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

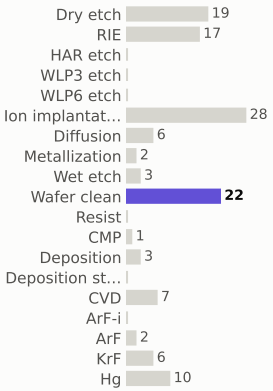
carbon so far, of the whole flow (kgCO2e)



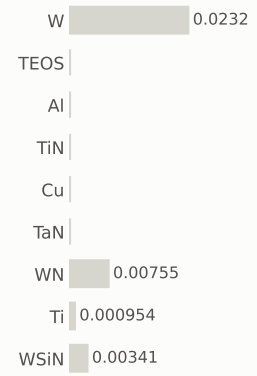
embodied carbon up to each step (kgCO2e per wafer)



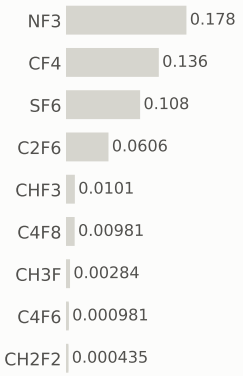
steps by process area



materials deposited (g)



gases emitted (g)



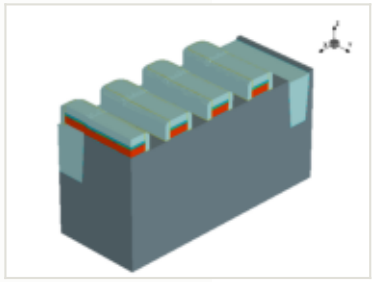
Step #1330: Diffusion

anneal · Diffusion

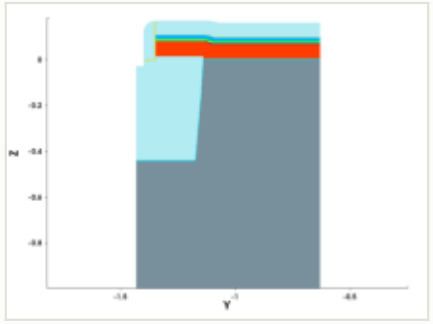
0.801	108.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

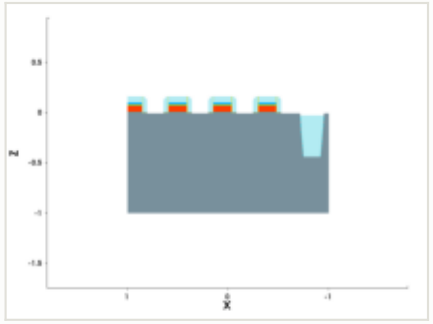
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

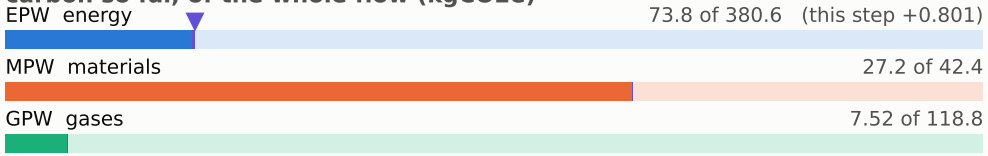
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

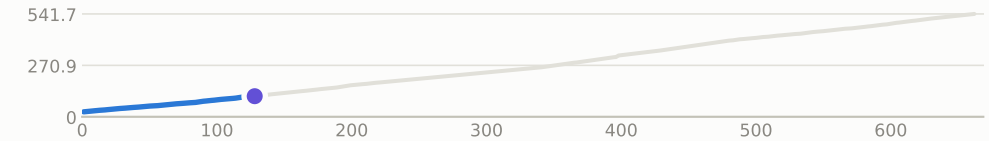
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

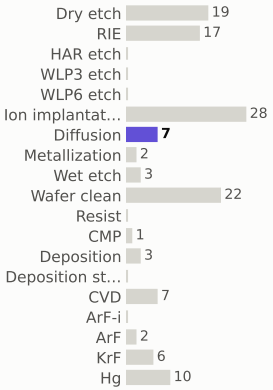
carbon so far, of the whole flow (kgCO2e)



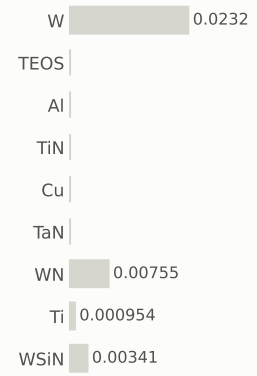
embodied carbon up to each step (kgCO2e per wafer)



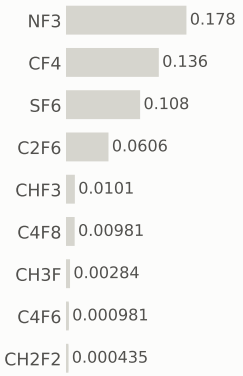
steps by process area



materials deposited (g)



gases emitted (g)



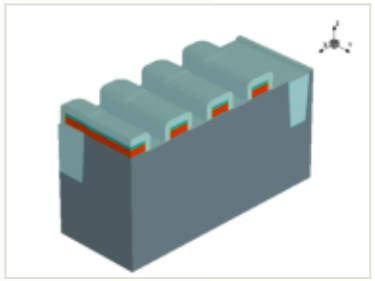
Step #1350: CVD

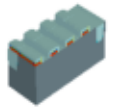
depo_isotropic · CVD

1.31	109.8	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

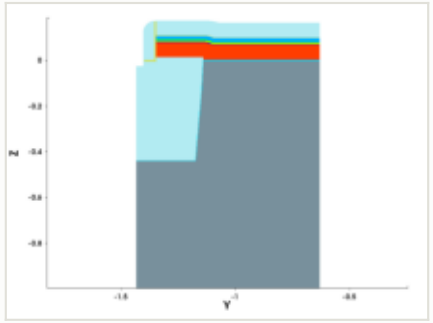
3D structure after this step

zoomed to fit

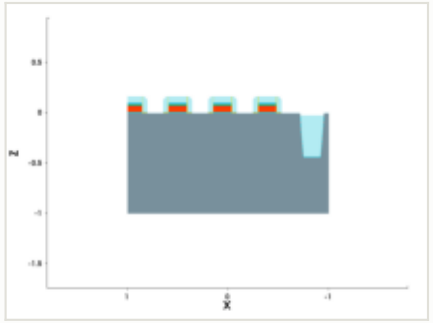




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

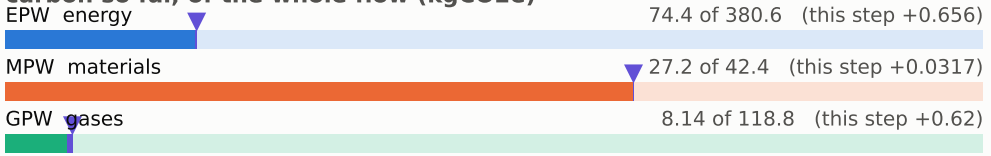
no mask for this step

this step

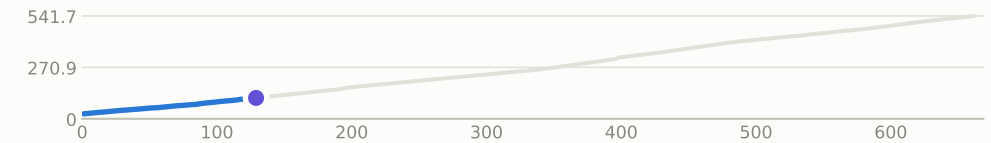
process area	CVD
material	—
thickness	0.003 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

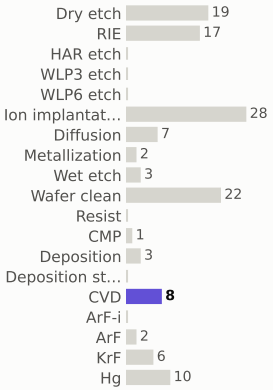
carbon so far, of the whole flow (kgCO2e)



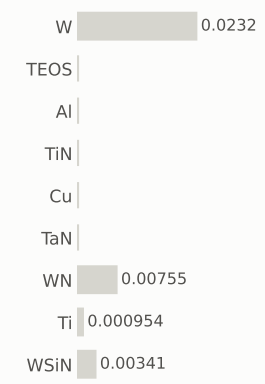
embodied carbon up to each step (kgCO2e per wafer)



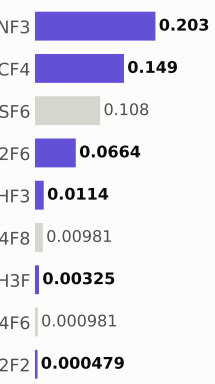
steps by process area



materials deposited (g)



gases emitted (g)



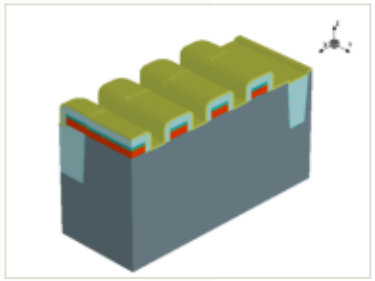
Step #1360: CVD

depo_isotropic · CVD

1.31	111.1	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

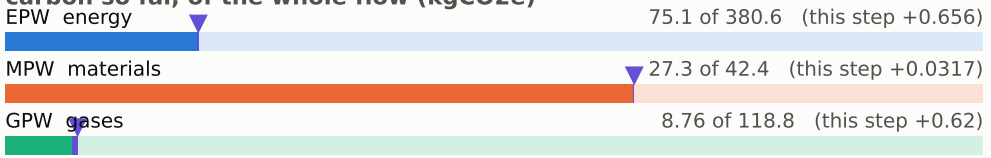


this step

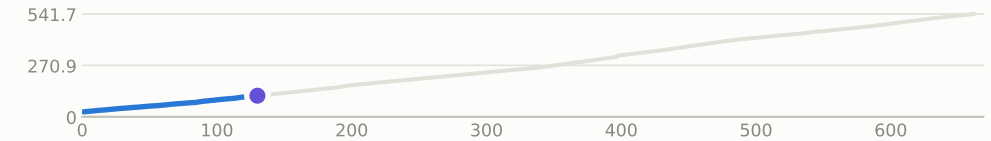
process area	CVD
material	—
thickness	0.017 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

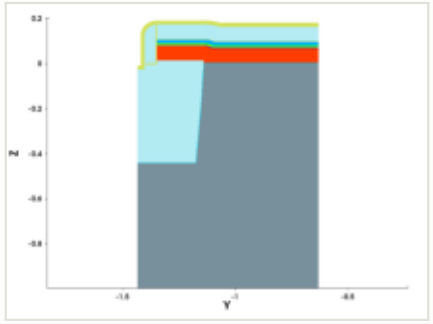
carbon so far, of the whole flow (kgCO2e)



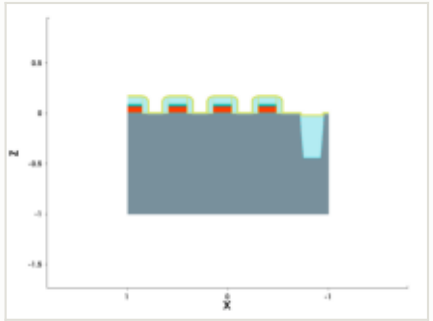
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



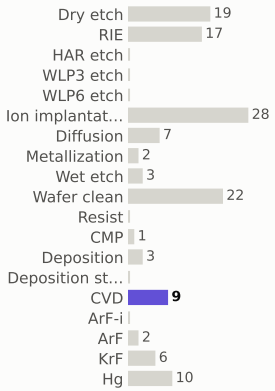
y cut at -1.032 µm



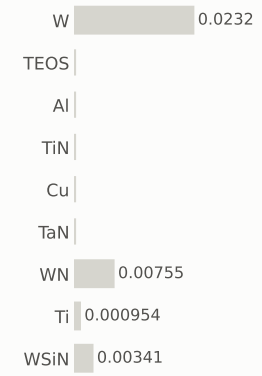
mask for this step

no mask for this step

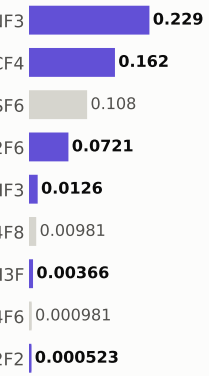
steps by process area



materials deposited (g)



gases emitted (g)



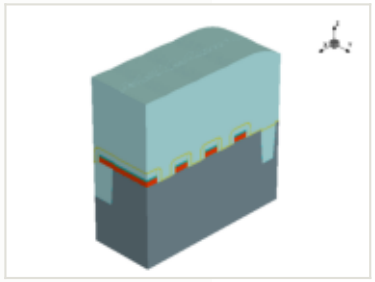
Step #1370: CVD

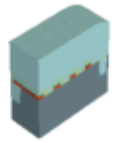
depo_isotropic · CVD

1.31	112.4	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

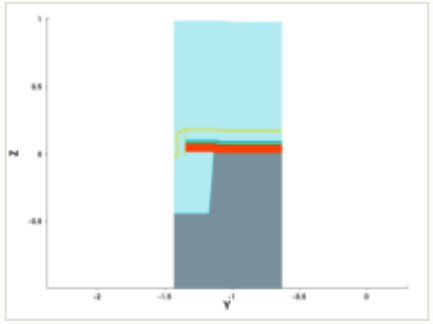
3D structure after this step

zoomed to fit

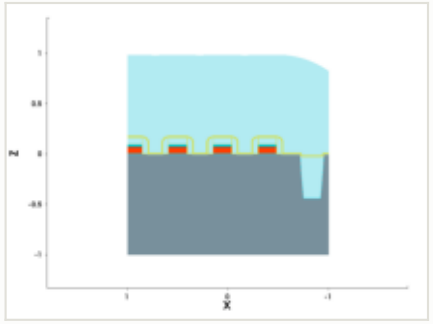




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

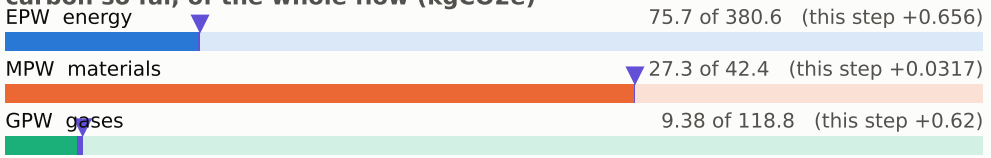
no mask for this step

this step

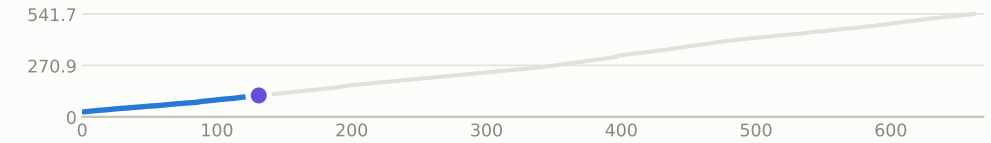
process area	CVD
material	—
thickness	0.8 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

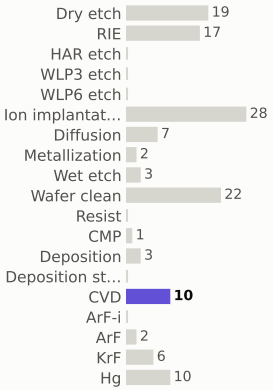
carbon so far, of the whole flow (kgCO2e)



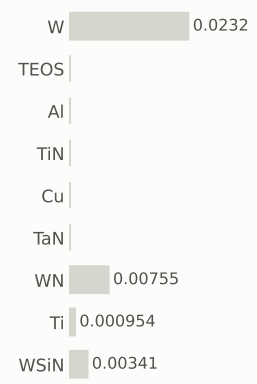
embodied carbon up to each step (kgCO2e per wafer)



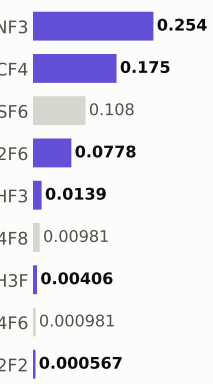
steps by process area



materials deposited (g)



gases emitted (g)



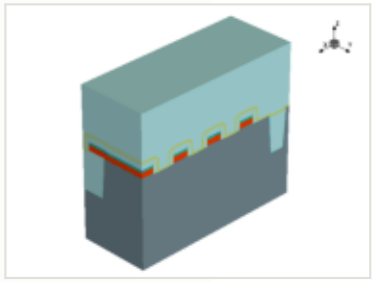
Step #1380: CMP

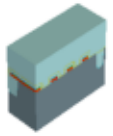
etch_cmp · CMP

0.169	112.6	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

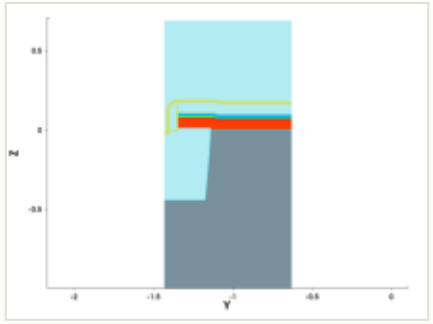
3D structure after this step

zoomed to fit

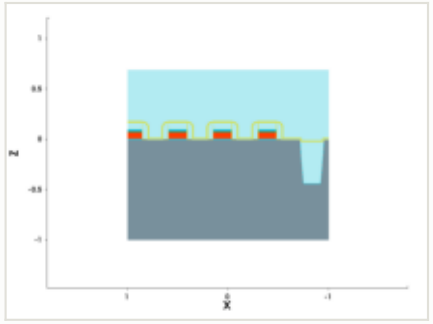




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

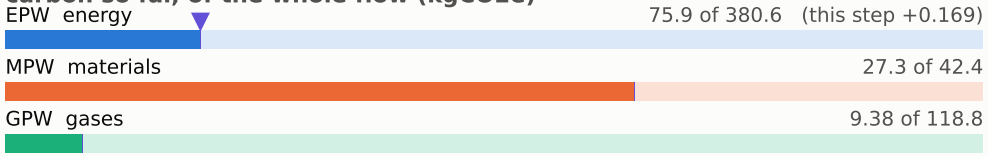
no mask for this step

this step

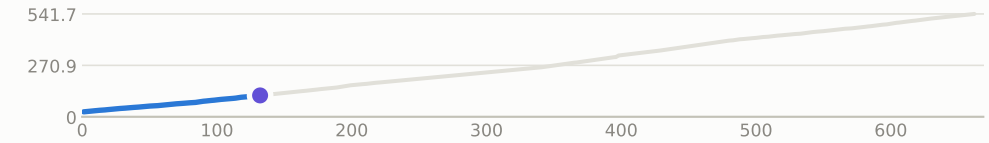
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

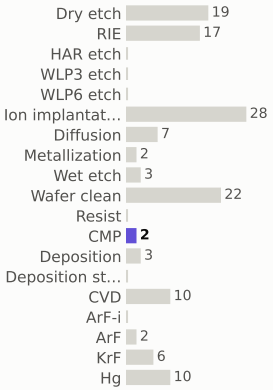
carbon so far, of the whole flow (kgCO2e)



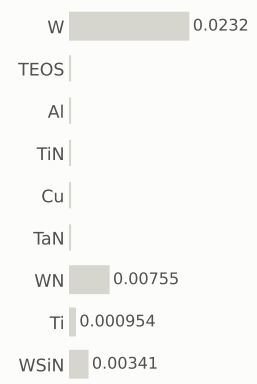
embodied carbon up to each step (kgCO2e per wafer)



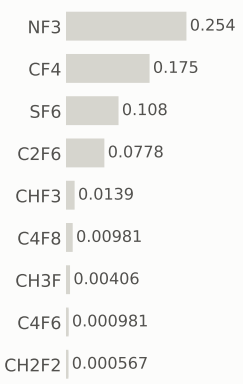
steps by process area



materials deposited (g)



gases emitted (g)



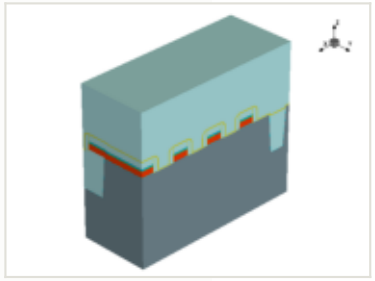
Step #1390: Wafer_clean

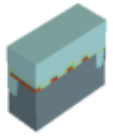
clean · Wafer_clean

0.169	112.8	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

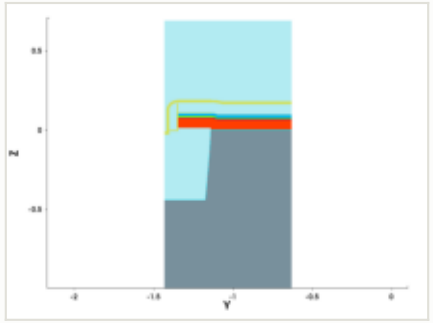
3D structure after this step

zoomed to fit

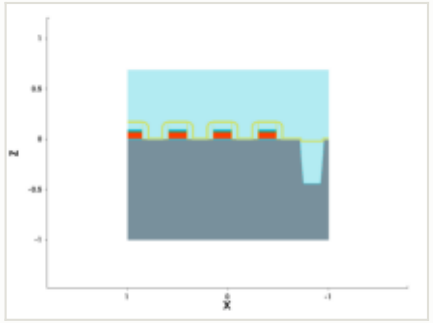




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

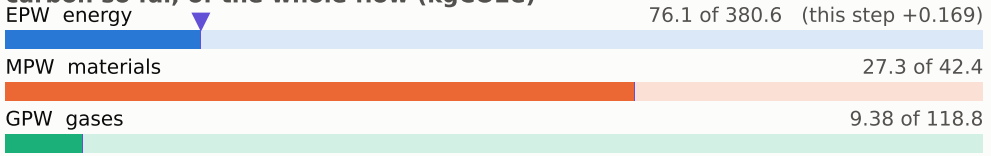
no mask for this step

this step

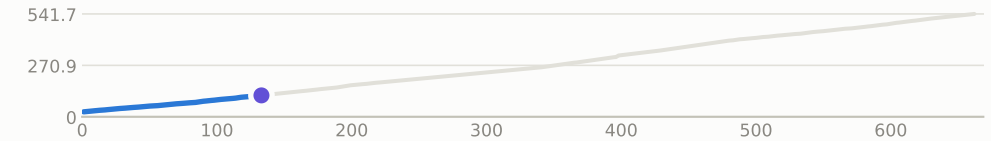
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

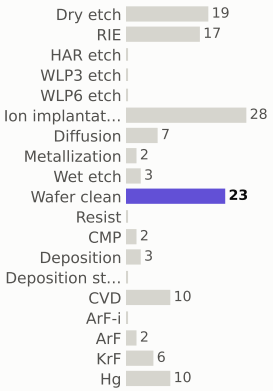
carbon so far, of the whole flow (kgCO2e)



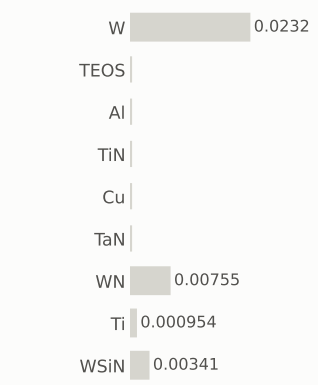
embodied carbon up to each step (kgCO2e per wafer)



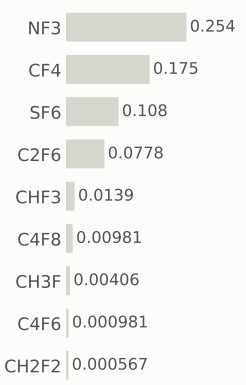
steps by process area



materials deposited (g)



gases emitted (g)



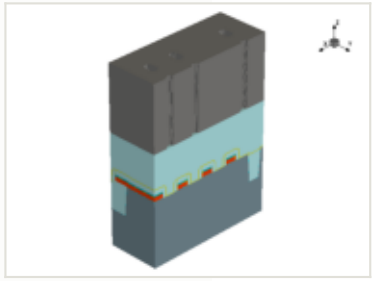
Step #1400: ArF

litho_patterning · ArF

0.199	113.0	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

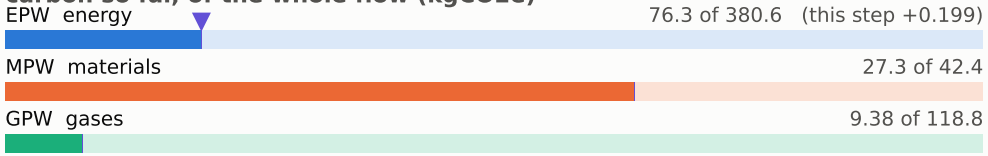
zoomed to fit



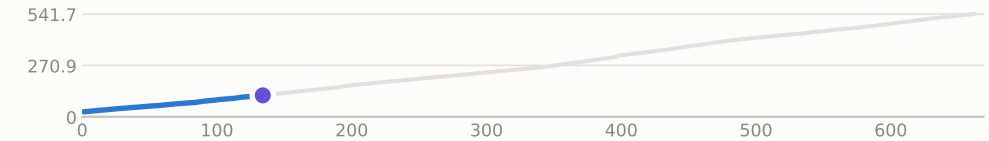
this step

process area	ArF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

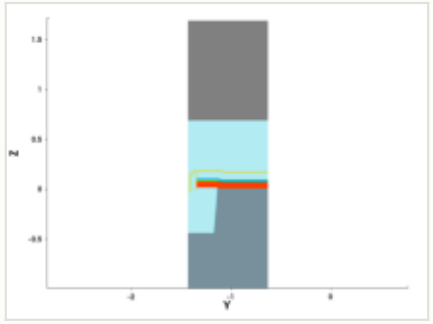
carbon so far, of the whole flow (kgCO2e)



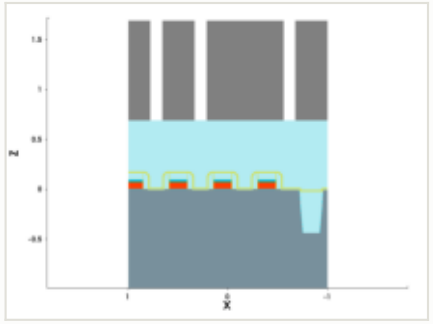
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

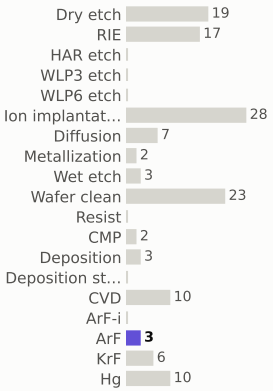


y cut at -1.032 µm

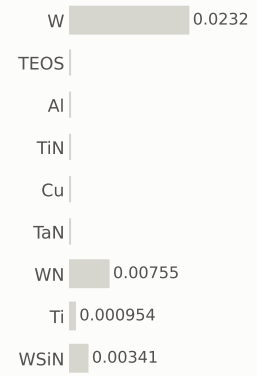


mask: layer 19:0 (OASIS)

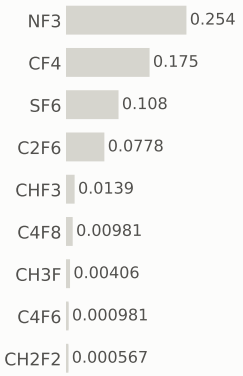
steps by process area



materials deposited (g)



gases emitted (g)



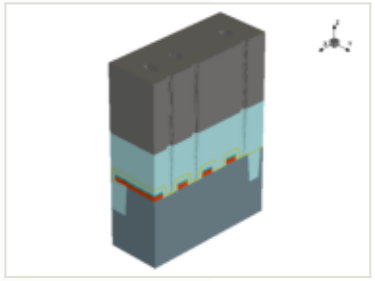
Step #1410: RIE

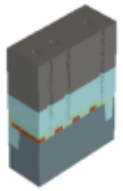
etch_anisotropic · RIE

1.04	114.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



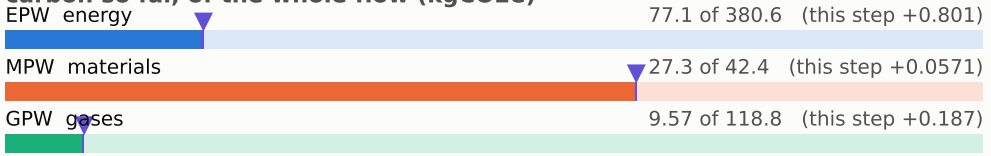


this step

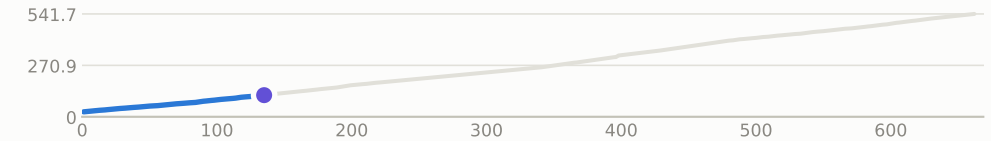
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

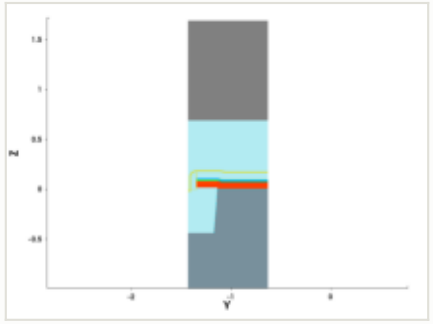
carbon so far, of the whole flow (kgCO2e)



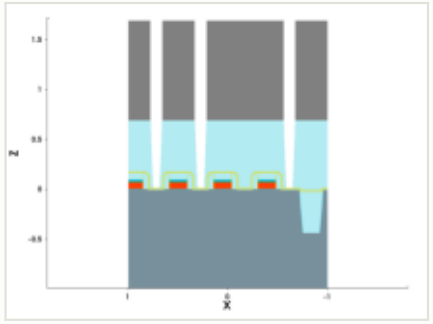
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



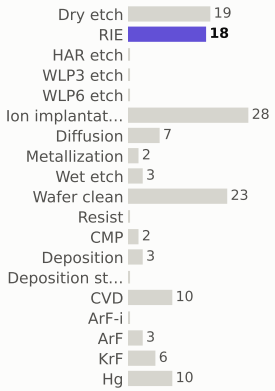
y cut at -1.032 µm



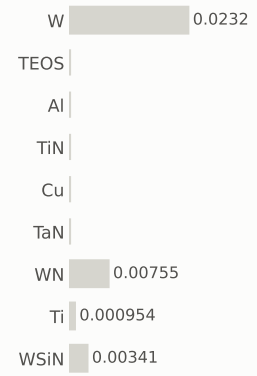
mask for this step

no mask for this step

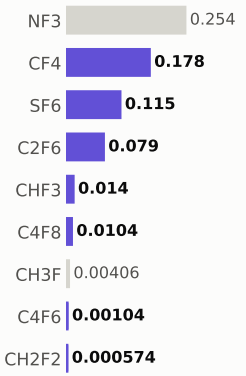
steps by process area



materials deposited (g)



gases emitted (g)



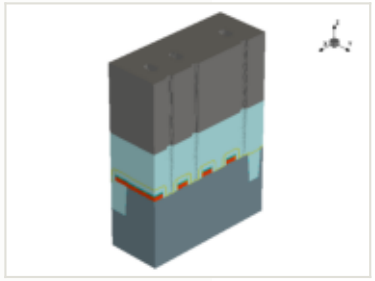
Step #1420: RIE

etch_anisotropic · RIE

1.04	115.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

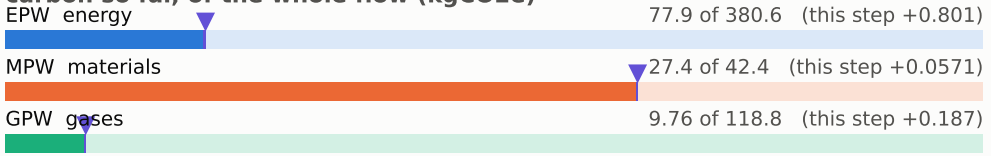


this step

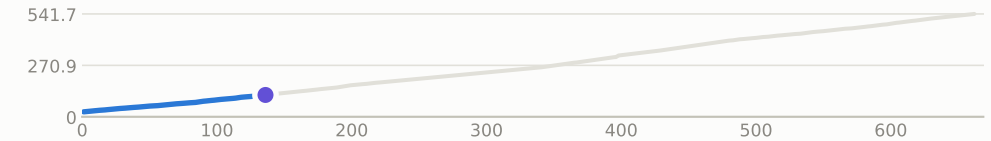
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

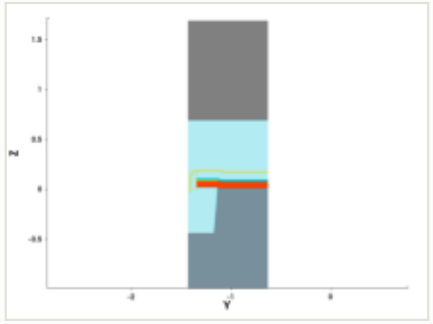
carbon so far, of the whole flow (kgCO2e)



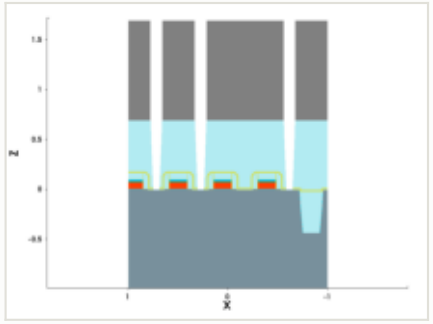
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



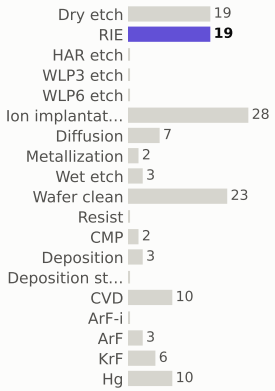
y cut at -1.032 µm



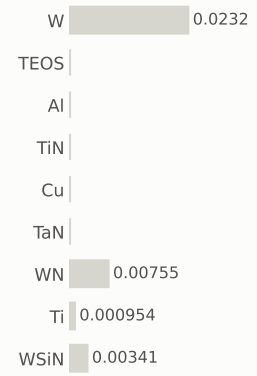
mask for this step

no mask for this step

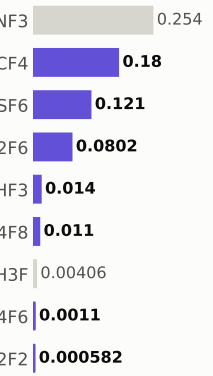
steps by process area



materials deposited (g)



gases emitted (g)



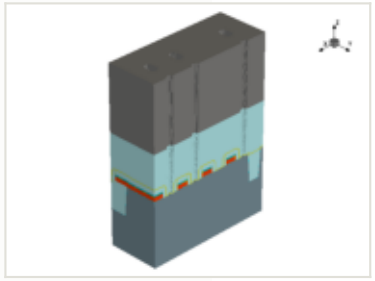
Step #1430: RIE

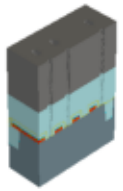
etch_anisotropic · RIE

1.04	116.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

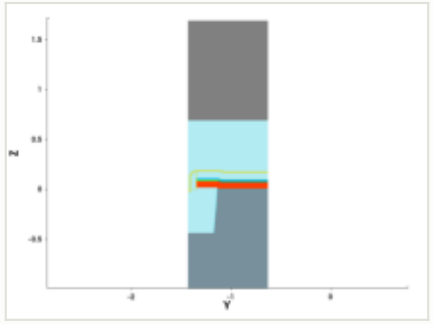
3D structure after this step

zoomed to fit

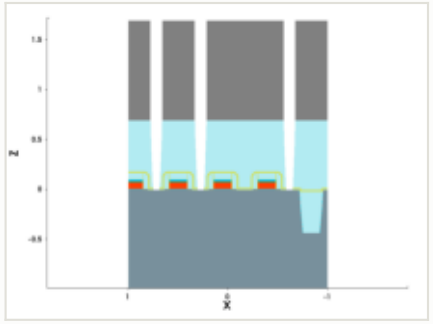




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

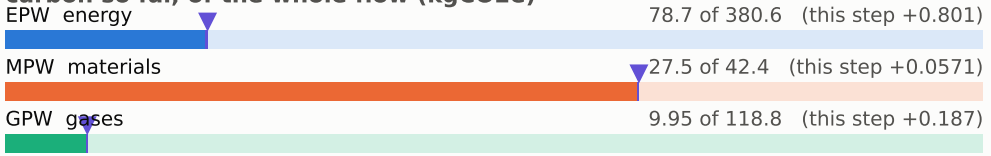
no mask for this step

this step

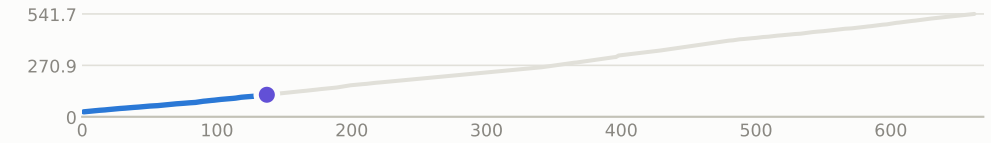
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

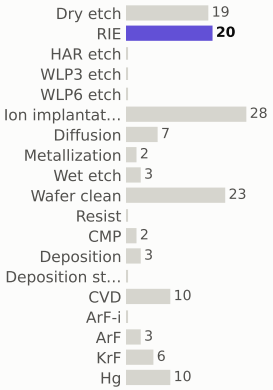
carbon so far, of the whole flow (kgCO2e)



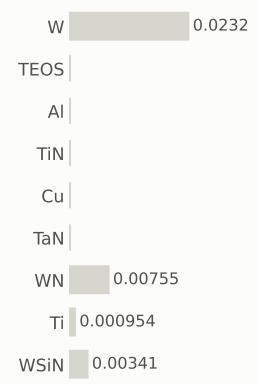
embodied carbon up to each step (kgCO2e per wafer)



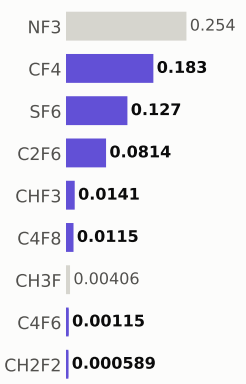
steps by process area



materials deposited (g)



gases emitted (g)



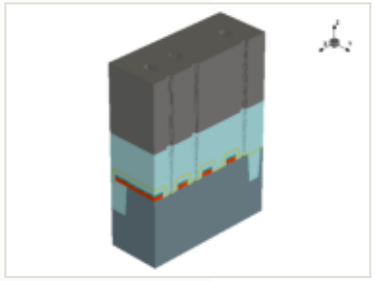
Step #1440: RIE

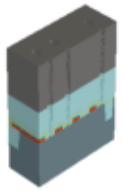
etch_anisotropic · RIE

1.04	117.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

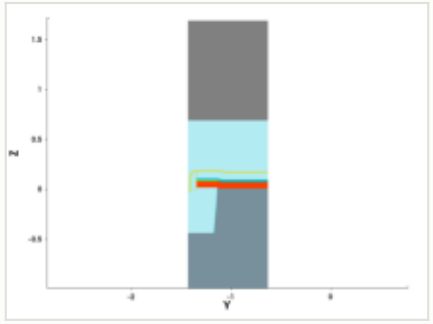
3D structure after this step

zoomed to fit

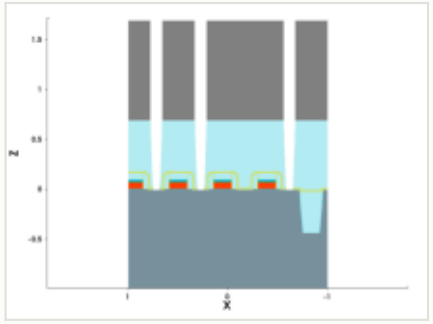




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

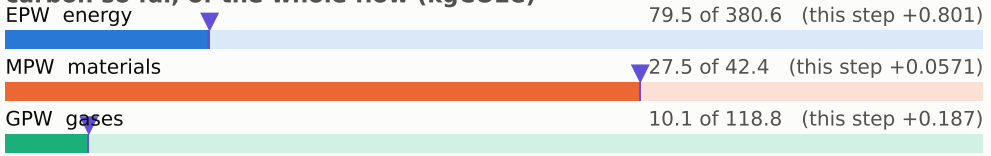
no mask for this step

this step

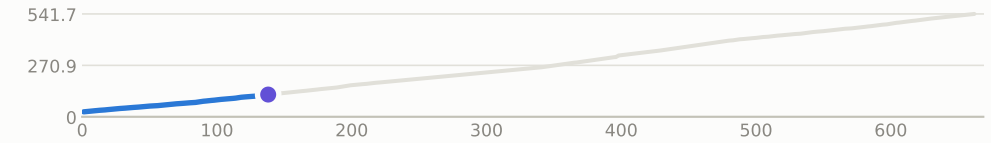
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

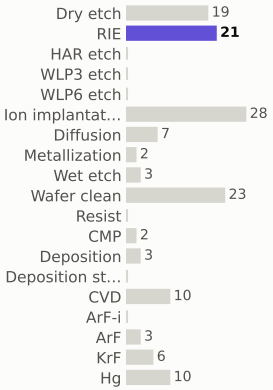
carbon so far, of the whole flow (kgCO2e)



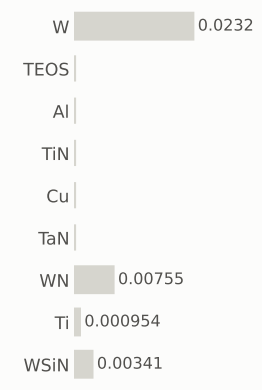
embodied carbon up to each step (kgCO2e per wafer)



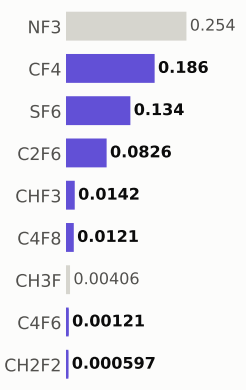
steps by process area



materials deposited (g)



gases emitted (g)



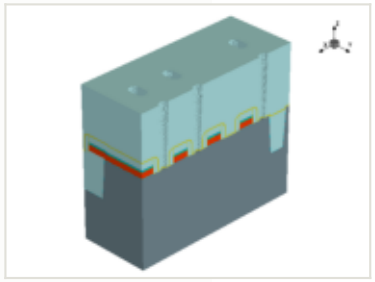
Step #1450: Dry_etch

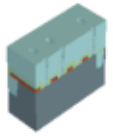
etch_strip · Dry_etch

0.801	117.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

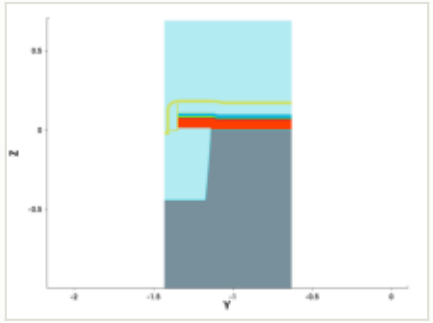
3D structure after this step

zoomed to fit

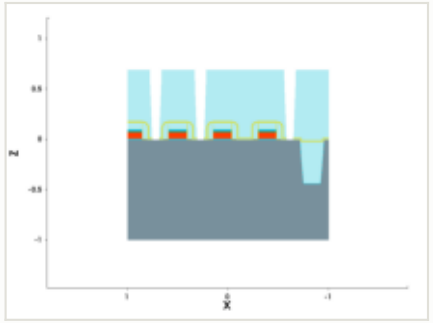




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

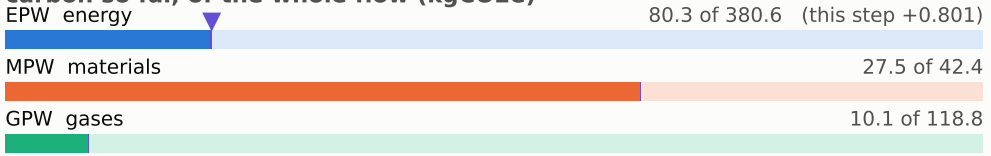
no mask for this step

this step

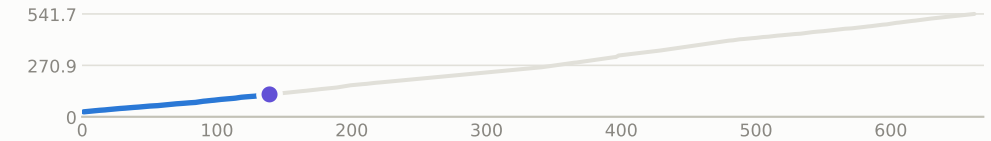
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

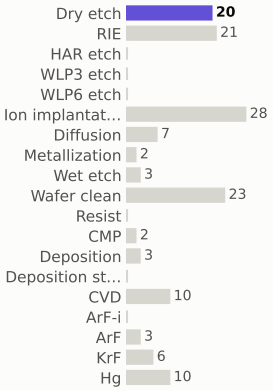
carbon so far, of the whole flow (kgCO2e)



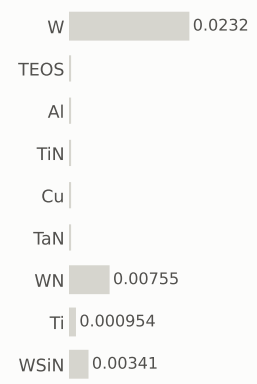
embodied carbon up to each step (kgCO2e per wafer)



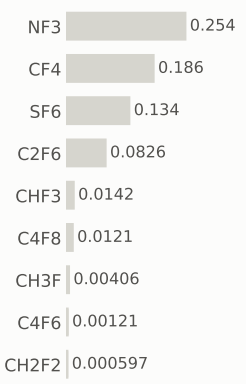
steps by process area



materials deposited (g)



gases emitted (g)



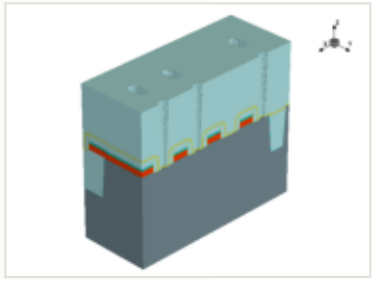
Step #1460: Wafer_clean

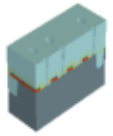
clean · Wafer_clean

0.169	118.1	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

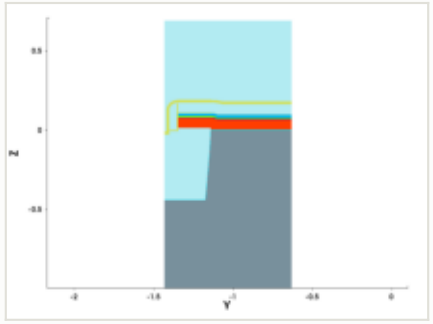
3D structure after this step

zoomed to fit

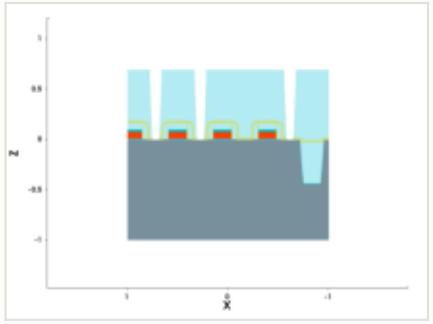




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

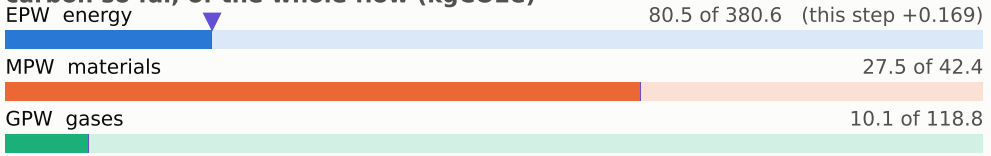
no mask for this step

this step

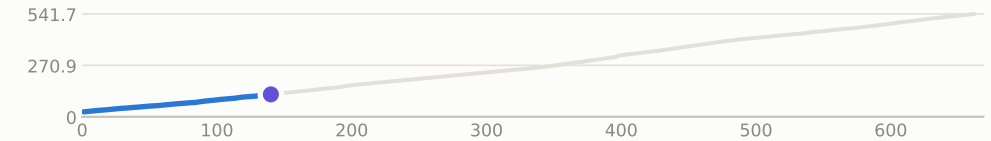
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

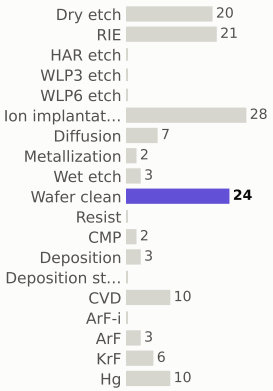
carbon so far, of the whole flow (kgCO2e)



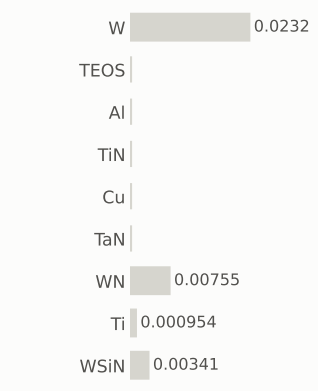
embodied carbon up to each step (kgCO2e per wafer)



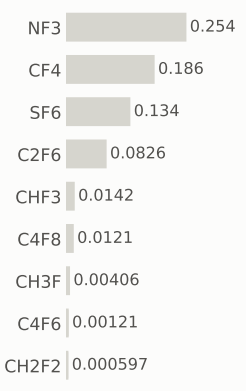
steps by process area



materials deposited (g)



gases emitted (g)



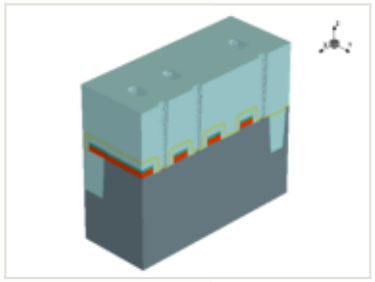
Step #1470: Diffusion

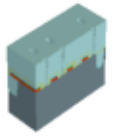
anneal · Diffusion

0.801	118.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

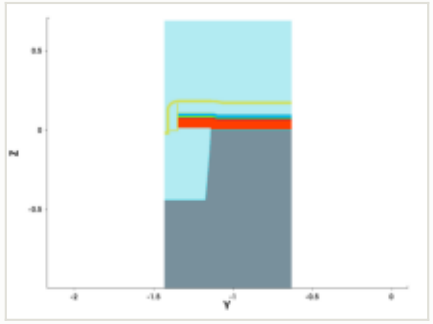
3D structure after this step

zoomed to fit

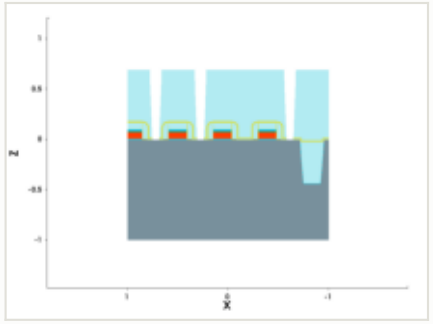




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

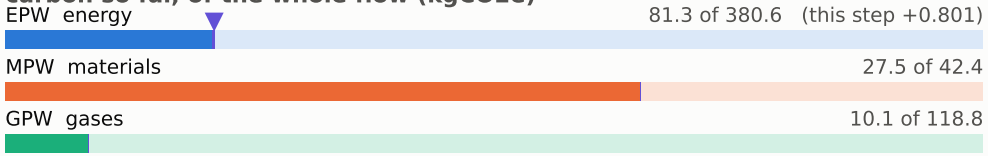
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

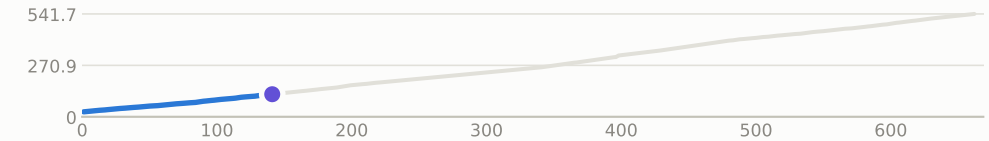
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

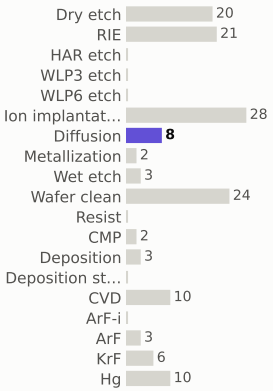
carbon so far, of the whole flow (kgCO2e)



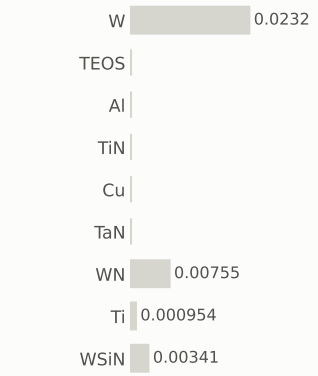
embodied carbon up to each step (kgCO2e per wafer)



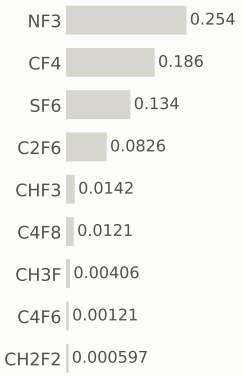
steps by process area



materials deposited (g)



gases emitted (g)



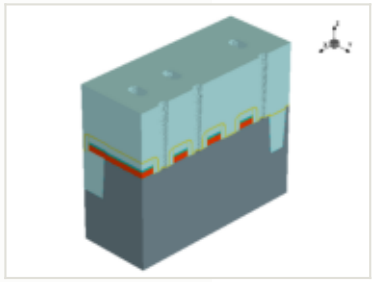
Step #1480: Diffusion

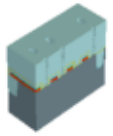
anneal · Diffusion

0.801	119.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

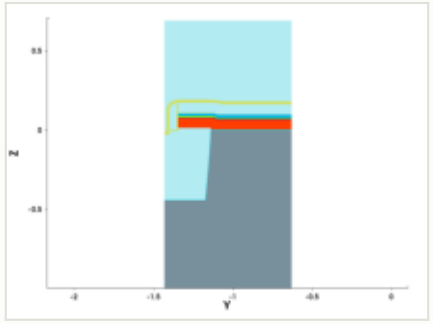
3D structure after this step

zoomed to fit

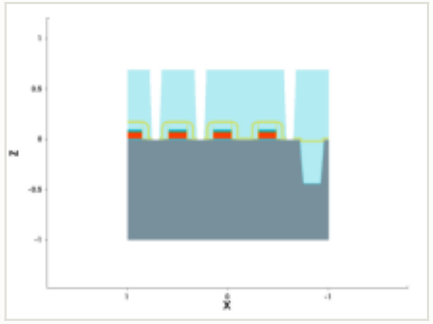




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

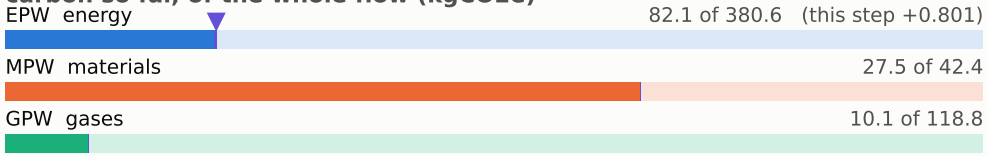
no mask for this step

this step

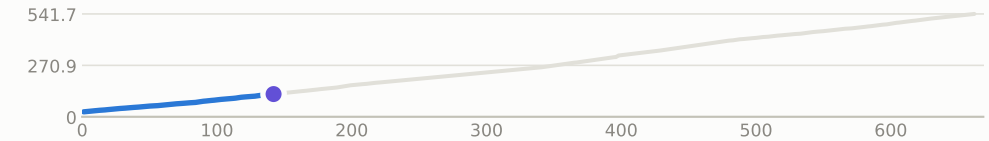
process area	Diffusion	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

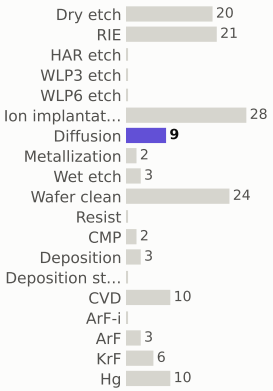
carbon so far, of the whole flow (kgCO2e)



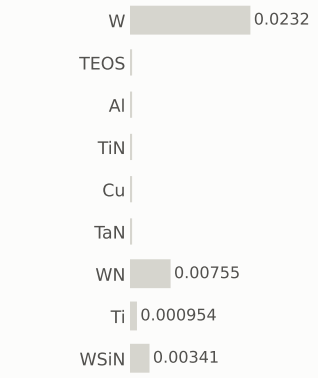
embodied carbon up to each step (kgCO2e per wafer)



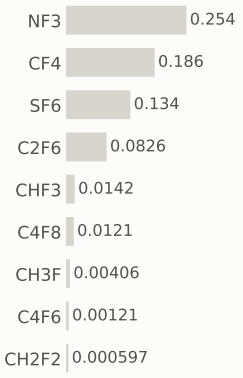
steps by process area



materials deposited (g)



gases emitted (g)



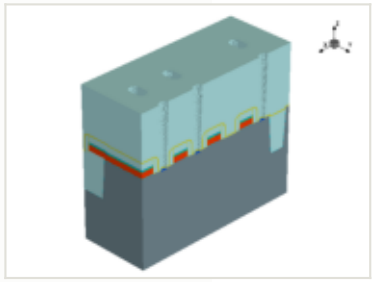
Step #1490: Wet_etch

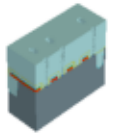
implant_ellipsoid · Wet_etch

0.169	119.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

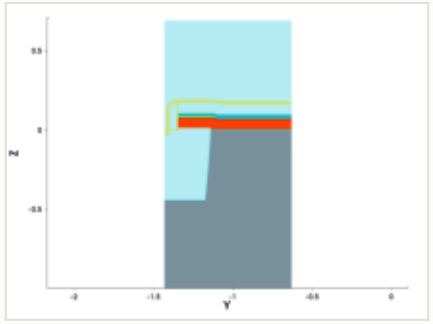
3D structure after this step

zoomed to fit

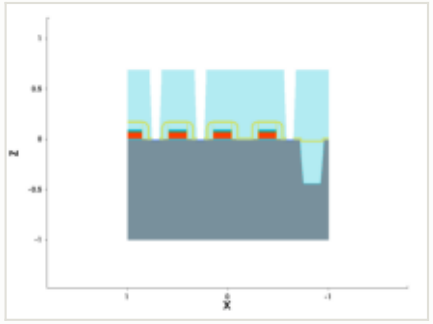




x cut at 0.000 µm



y cut at -1.032 µm



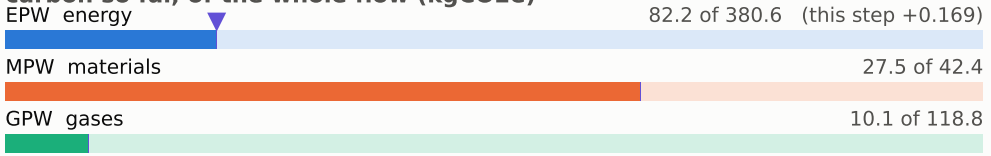
mask for this step

no mask for this step

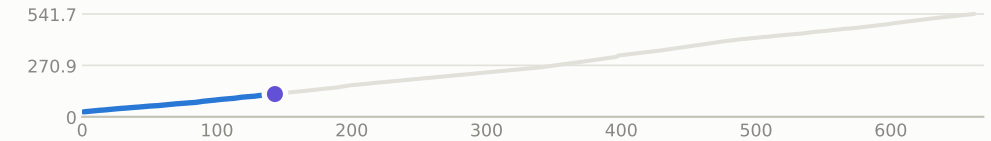
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

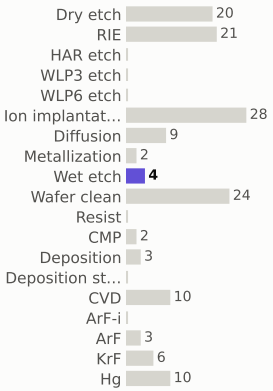
carbon so far, of the whole flow (kgCO2e)



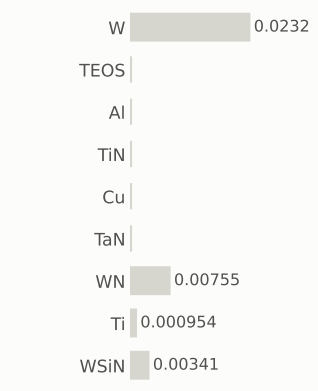
embodied carbon up to each step (kgCO2e per wafer)



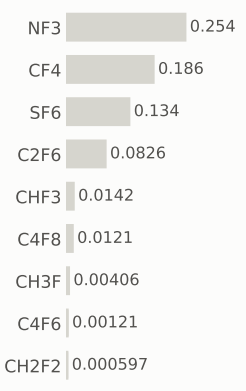
steps by process area



materials deposited (g)



gases emitted (g)



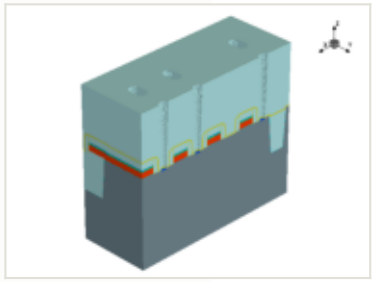
Step #1500: Wafer_clean

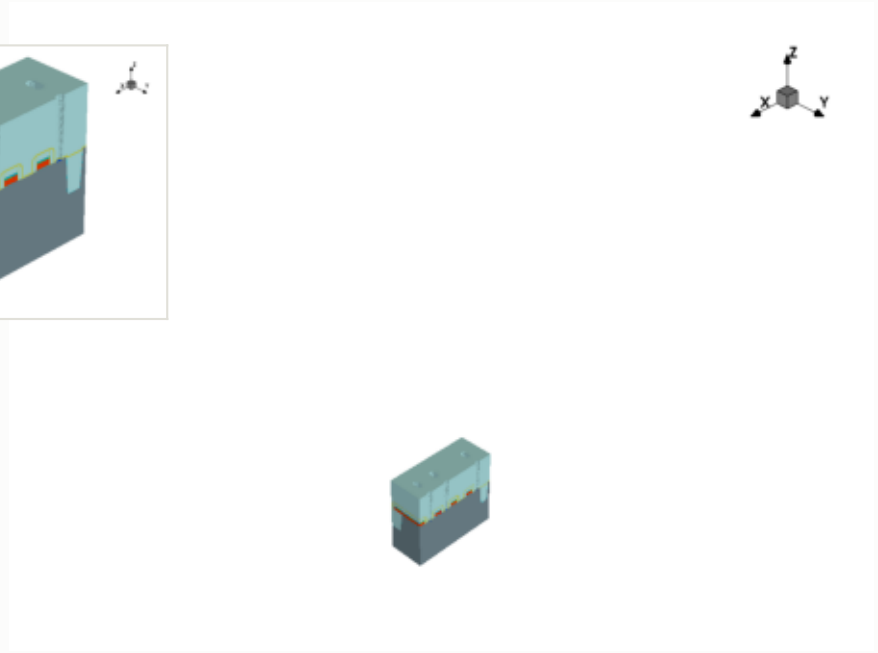
clean · Wafer_clean

0.169	120.0	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

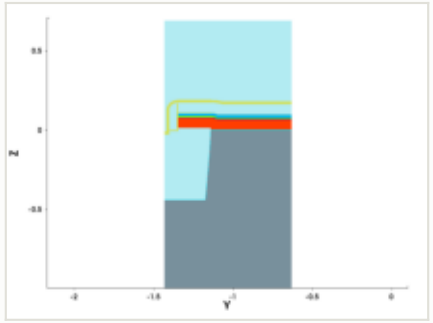
3D structure after this step

zoomed to fit

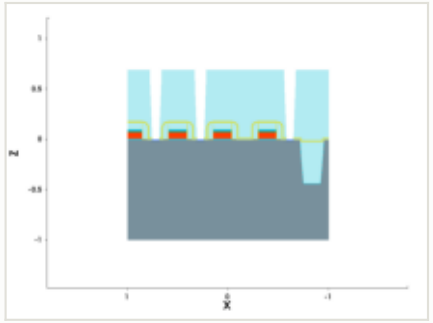




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

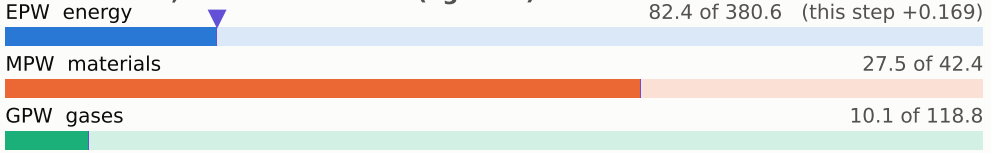
no mask for this step

this step

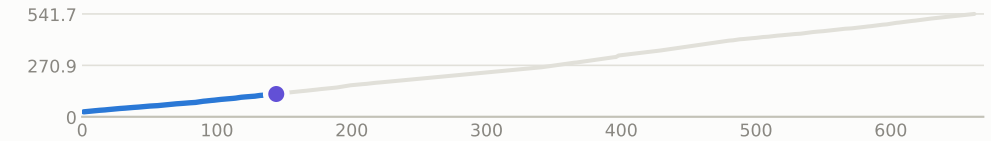
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

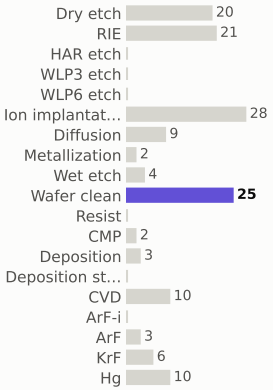
carbon so far, of the whole flow (kgCO2e)



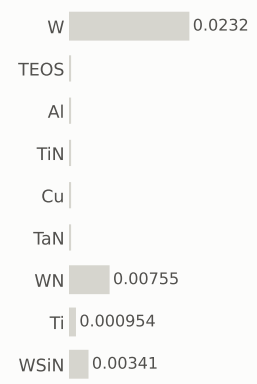
embodied carbon up to each step (kgCO2e per wafer)



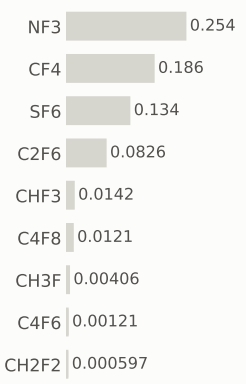
steps by process area



materials deposited (g)



gases emitted (g)



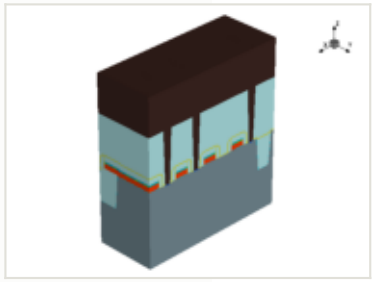
Step #1510: CVD

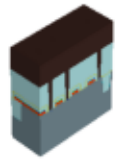
depo_isotropic · CVD

1.31	121.3	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

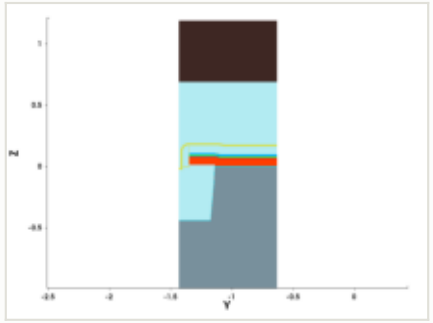
3D structure after this step

zoomed to fit

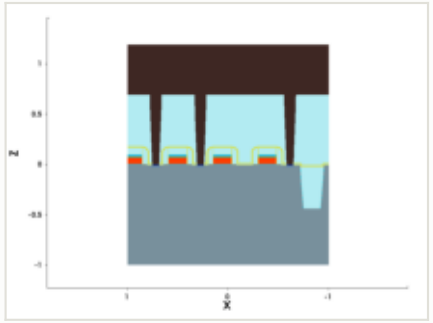




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

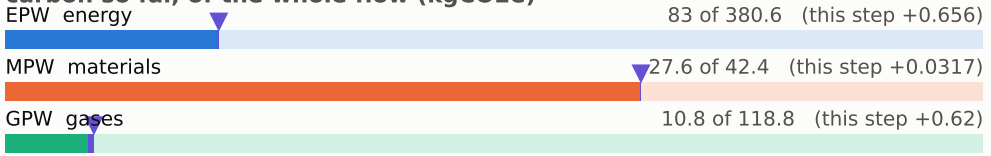
no mask for this step

this step

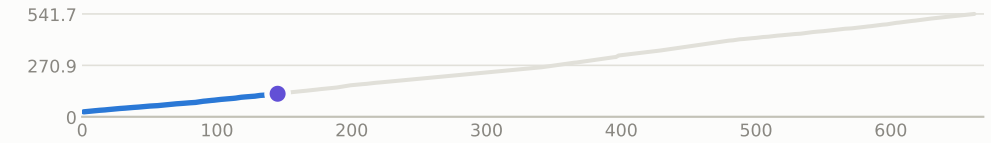
process area	CVD
material	—
thickness	0.5 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

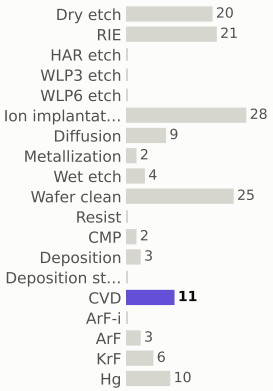
carbon so far, of the whole flow (kgCO2e)



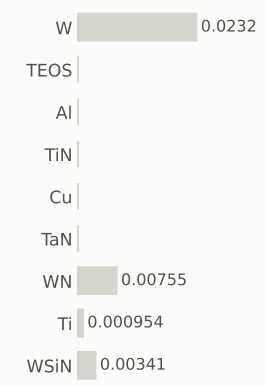
embodied carbon up to each step (kgCO2e per wafer)



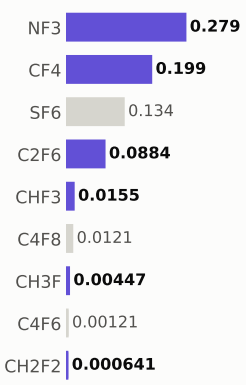
steps by process area



materials deposited (g)



gases emitted (g)



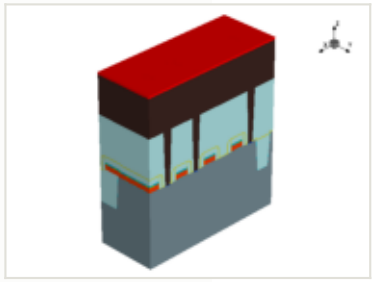
Step #1520: CVD

depo_isotropic · CVD

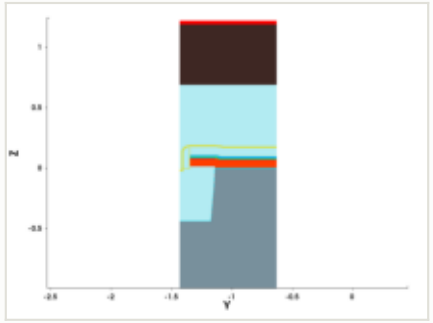
1.31	122.7	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

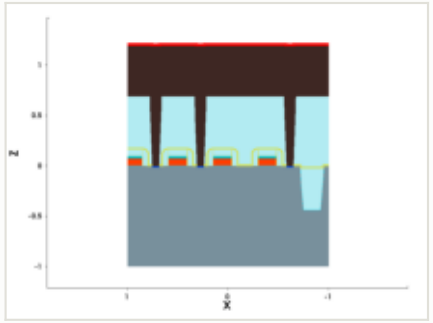
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

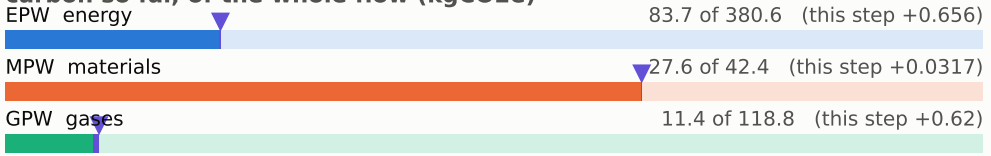
no mask for this step

this step

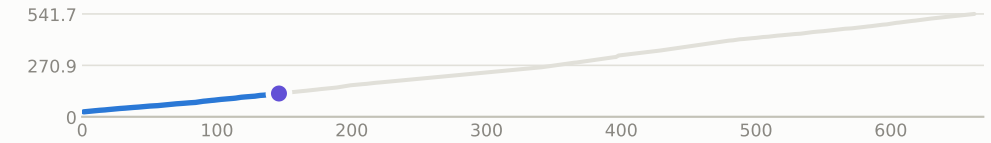
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

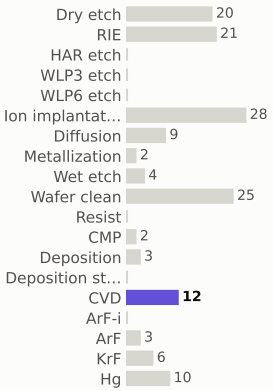
carbon so far, of the whole flow (kgCO2e)



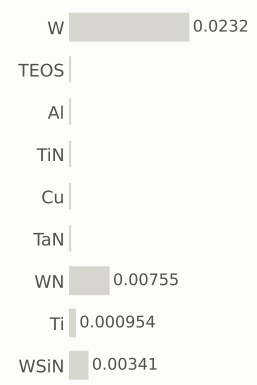
embodied carbon up to each step (kgCO2e per wafer)



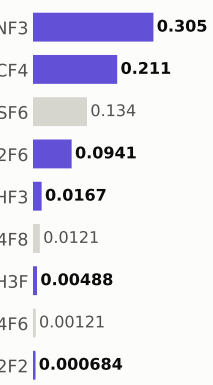
steps by process area



materials deposited (g)



gases emitted (g)



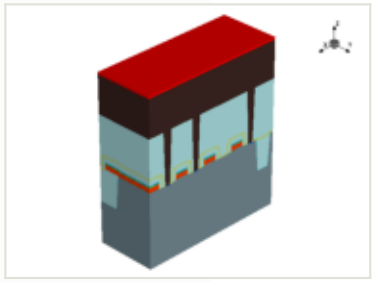
Step #1530: Wafer_clean

clean · Wafer_clean

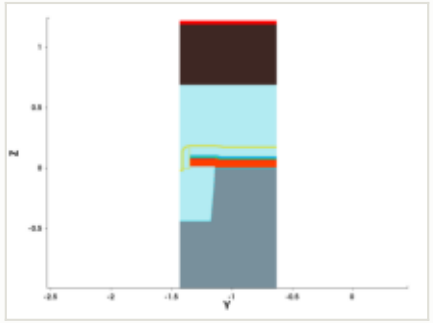
0.169	122.8	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

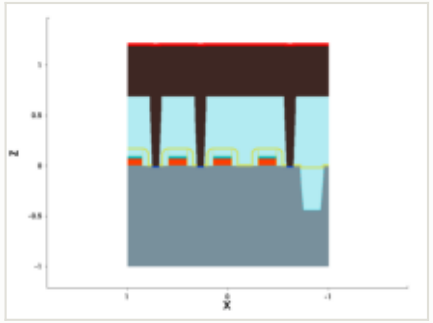
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

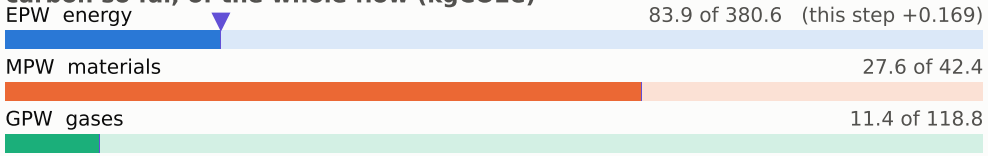
no mask for this step

this step

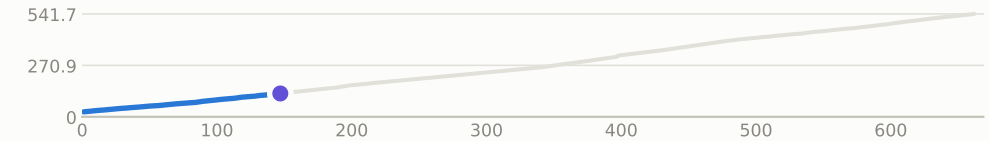
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

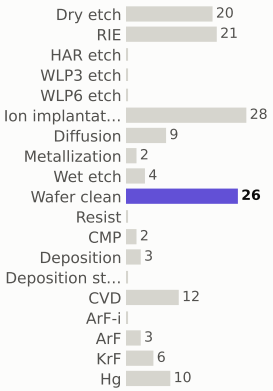
carbon so far, of the whole flow (kgCO2e)



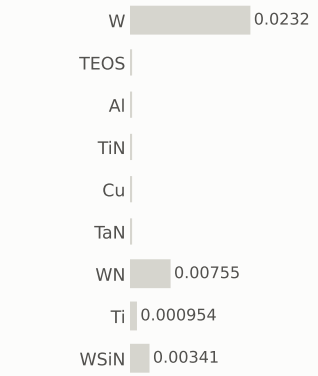
embodied carbon up to each step (kgCO2e per wafer)



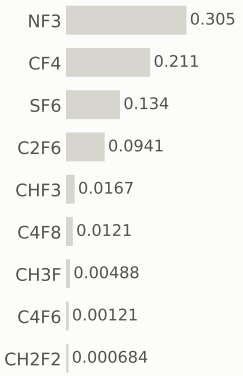
steps by process area



materials deposited (g)



gases emitted (g)



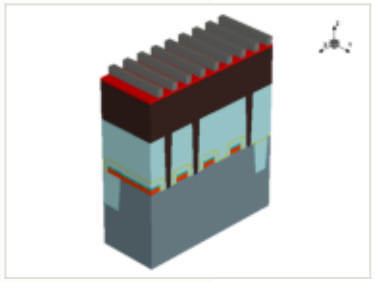
Step #1540: ArF-i

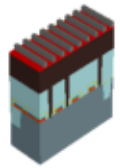
litho_patterning · ArF-i

0.217	123.0	0.04%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

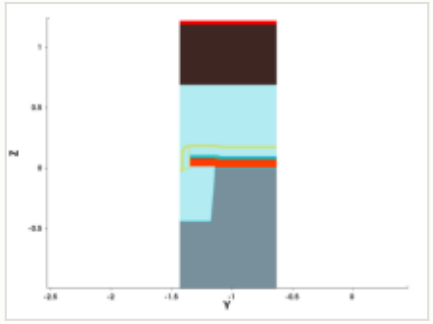
3D structure after this step

zoomed to fit

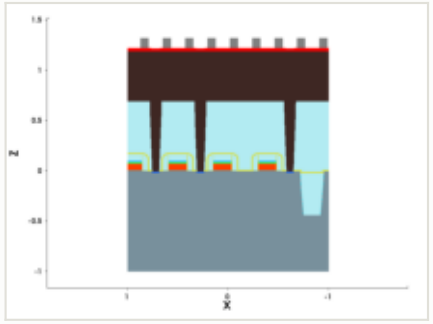




x cut at 0.000 µm



y cut at -1.032 µm



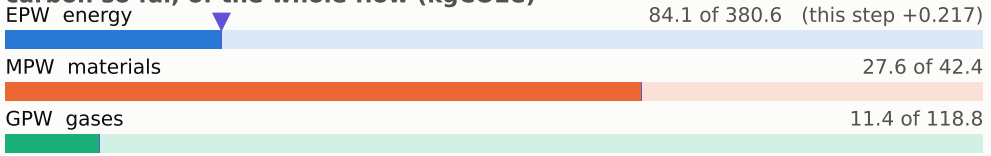
mask: layer 20:0 (OASIS)

this step

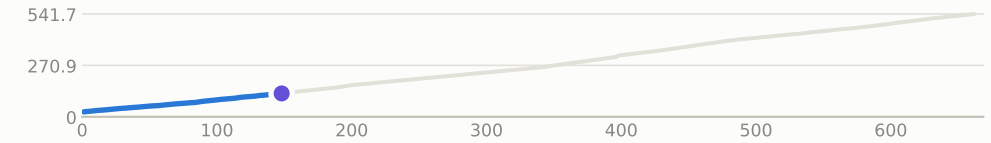
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

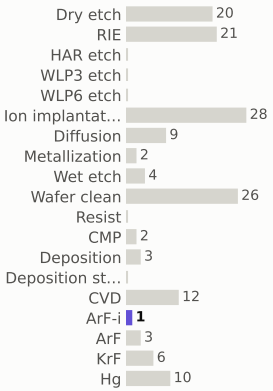
carbon so far, of the whole flow (kgCO2e)



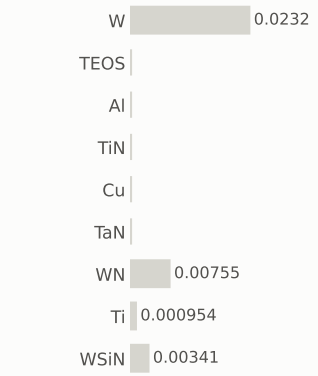
embodied carbon up to each step (kgCO2e per wafer)



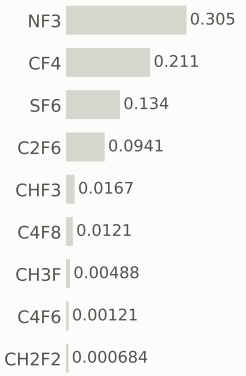
steps by process area



materials deposited (g)



gases emitted (g)



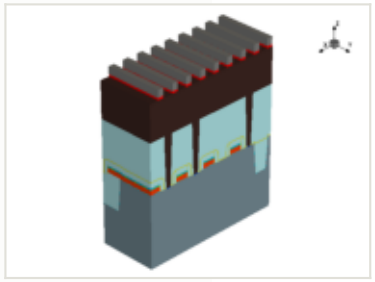
Step #1550: RIE

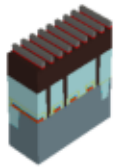
etch_anisotropic · RIE

1.04	124.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



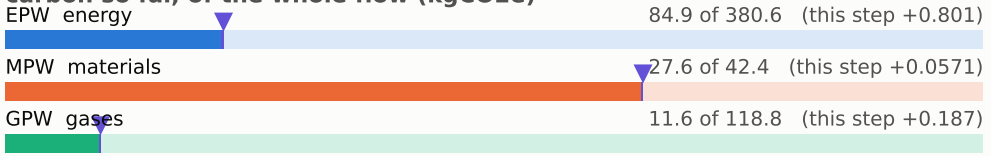


this step

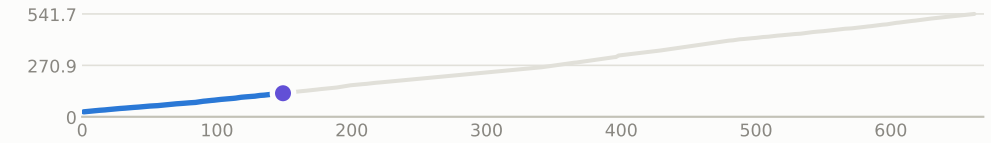
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

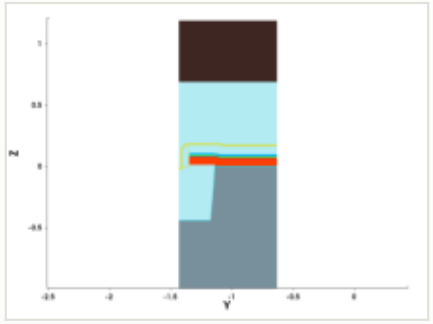
carbon so far, of the whole flow (kgCO2e)



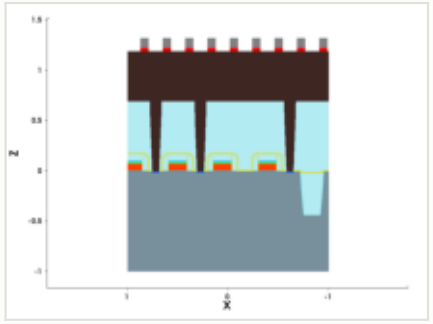
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



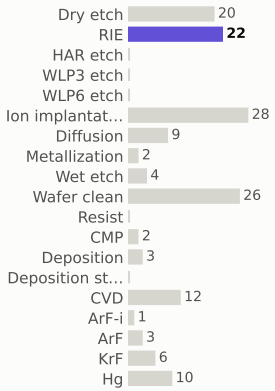
y cut at -1.032 µm



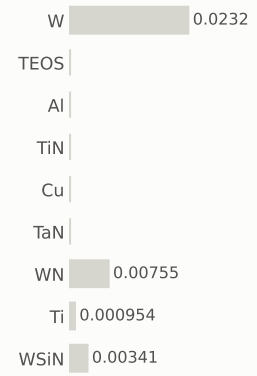
mask for this step

no mask for this step

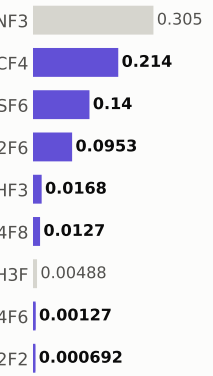
steps by process area



materials deposited (g)



gases emitted (g)



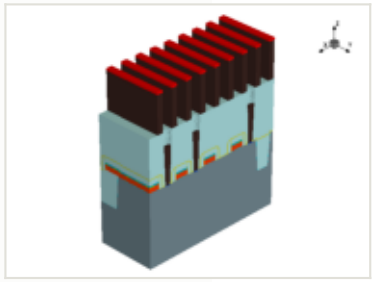
Step #1560: RIE

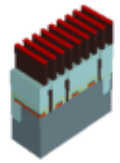
etch_anisotropic · RIE

1.04	125.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

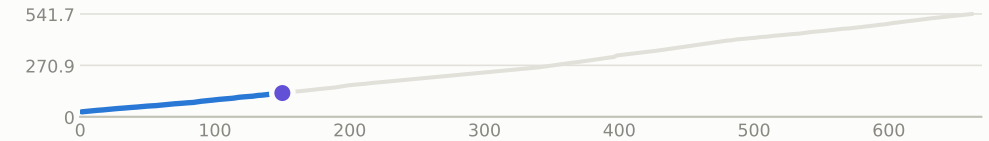
carbon so far, of the whole flow (kgCO2e)

EPW energy — 85.7 of 380.6 (this step +0.801)

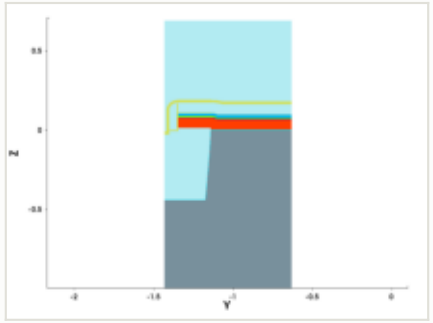
MPW materials — 27.7 of 42.4 (this step +0.0571)

GPW gases — 11.7 of 118.8 (this step +0.187)

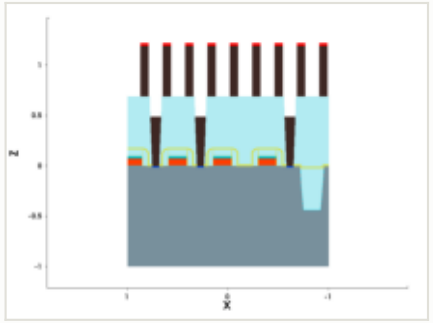
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



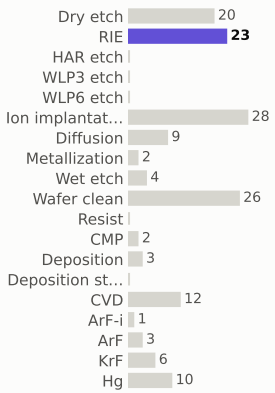
y cut at -1.032 µm



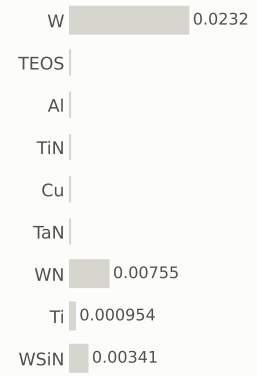
mask for this step

no mask for this step

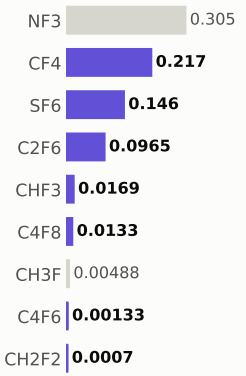
steps by process area



materials deposited (g)



gases emitted (g)



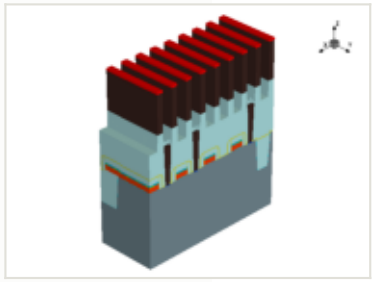
Step #1570: RIE

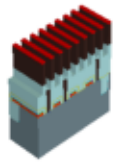
etch_anisotropic · RIE

1.04	126.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



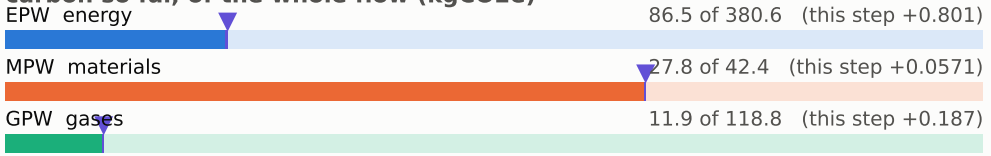


this step

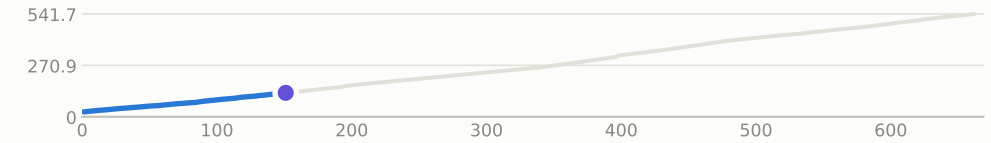
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

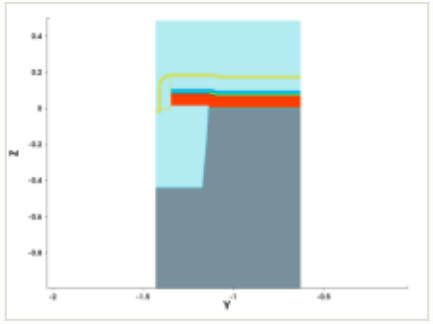
carbon so far, of the whole flow (kgCO2e)



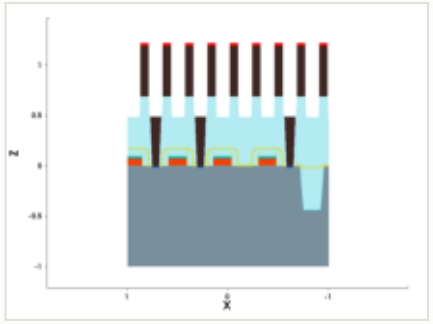
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



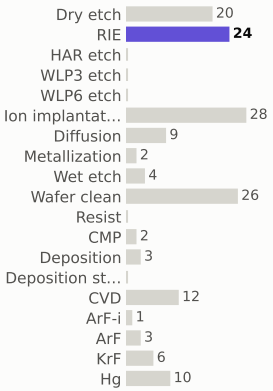
y cut at -1.032 µm



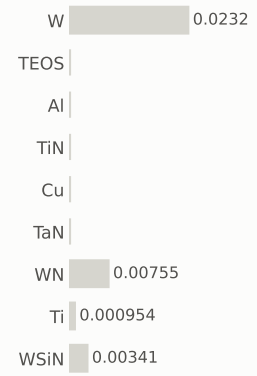
mask for this step

no mask for this step

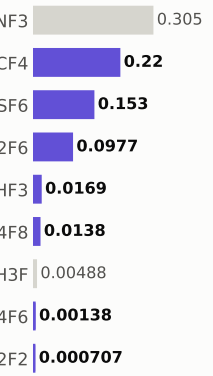
steps by process area



materials deposited (g)



gases emitted (g)



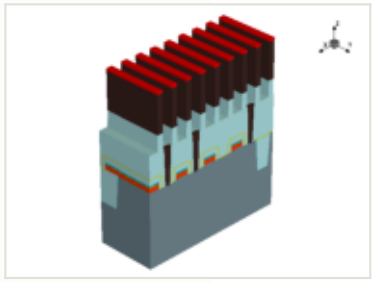
Step #1580: RIE

etch_isotropic · RIE

1.04	127.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

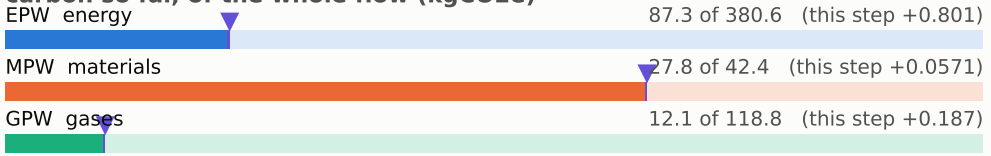


this step

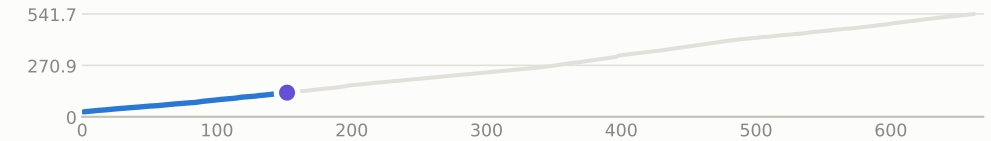
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

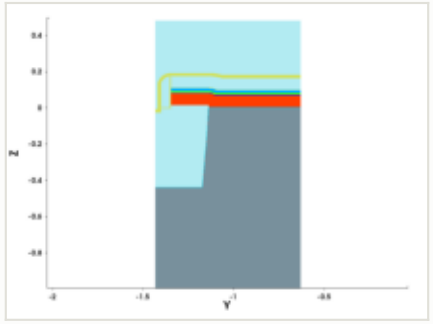
carbon so far, of the whole flow (kgCO2e)



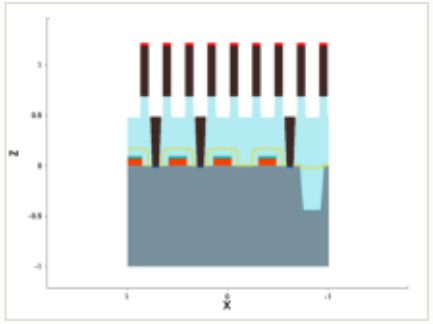
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



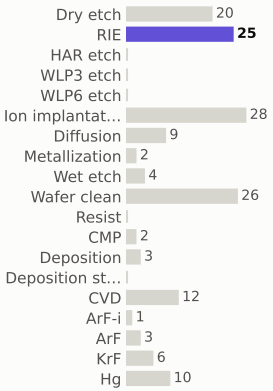
y cut at -1.032 µm



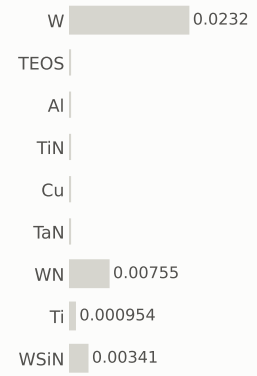
mask for this step

no mask for this step

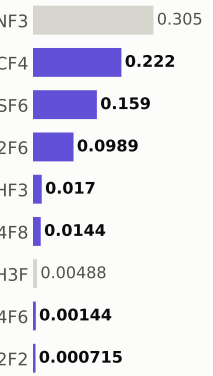
steps by process area



materials deposited (g)



gases emitted (g)



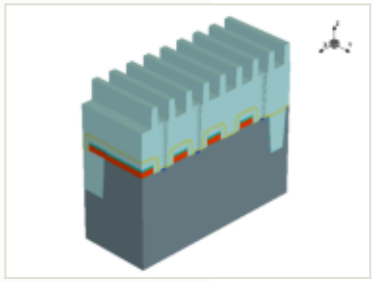
Step #1590: Dry_etch

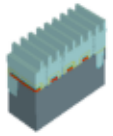
etch_strip · Dry_etch

0.801	128.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

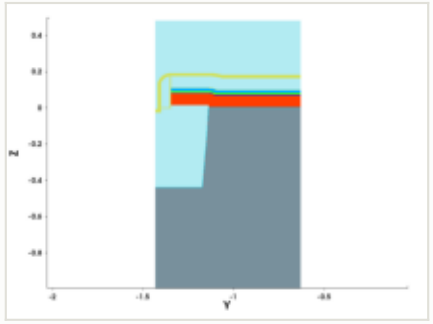
3D structure after this step

zoomed to fit

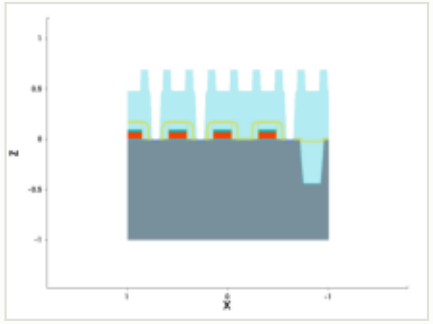




x cut at 0.000 µm



y cut at -1.032 µm



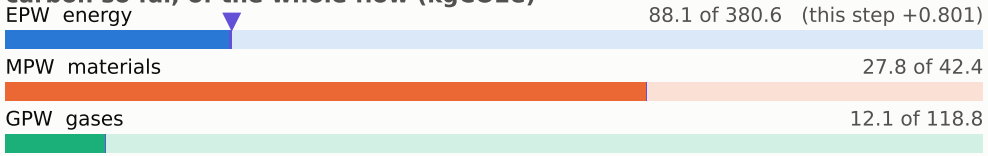
mask for this step

no mask for this step

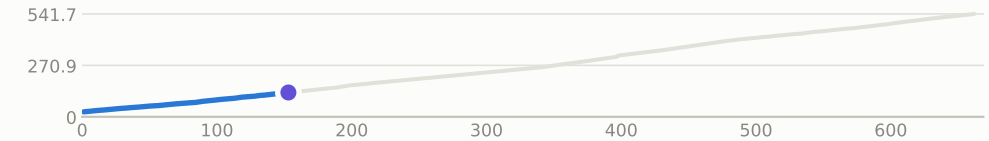
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

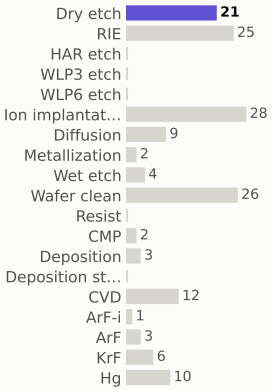
carbon so far, of the whole flow (kgCO2e)



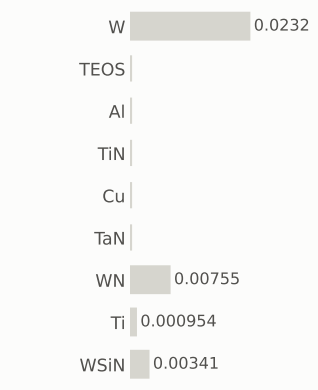
embodied carbon up to each step (kgCO2e per wafer)



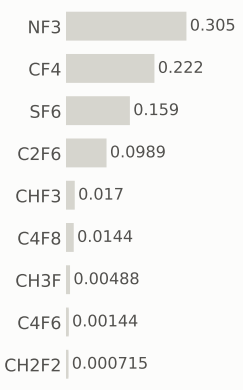
steps by process area



materials deposited (g)



gases emitted (g)



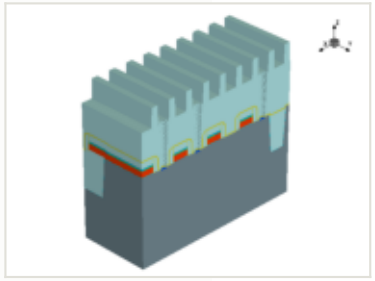
Step #1600: Wafer_clean

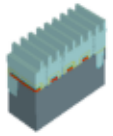
clean · Wafer_clean

0.169	128.2	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

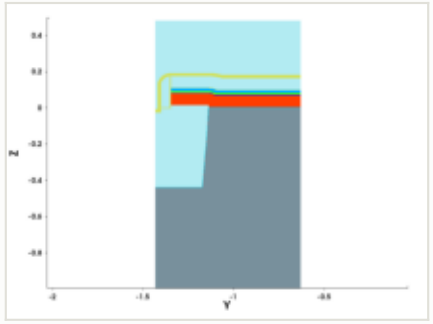
3D structure after this step

zoomed to fit

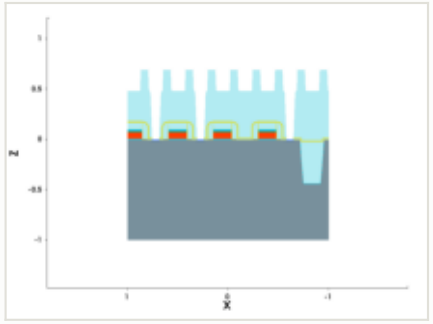




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

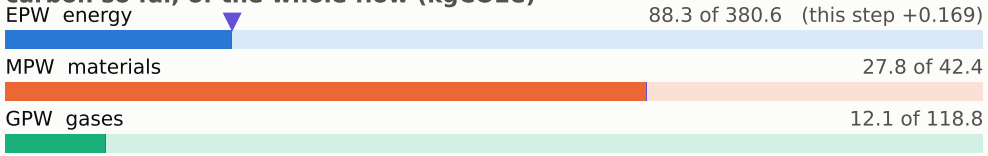
no mask for this step

this step

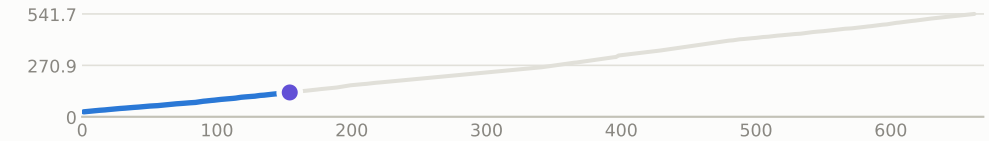
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

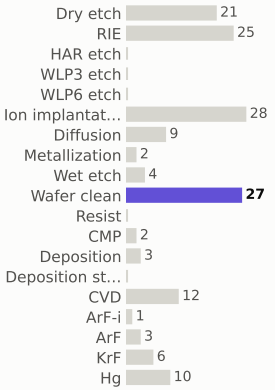
carbon so far, of the whole flow (kgCO2e)



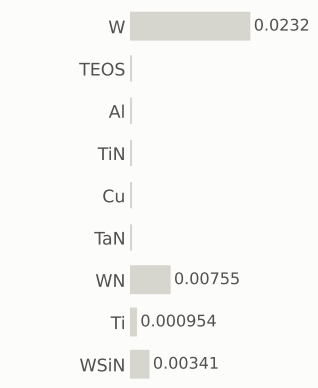
embodied carbon up to each step (kgCO2e per wafer)



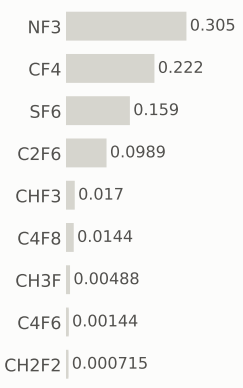
steps by process area



materials deposited (g)



gases emitted (g)



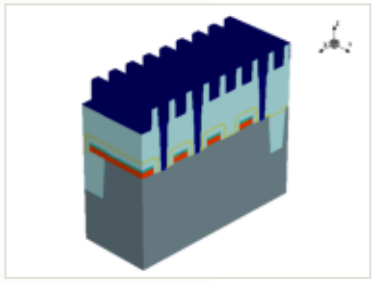
Step #1610: CVD

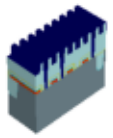
depo_isotropic · CVD

1.31	129.5	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

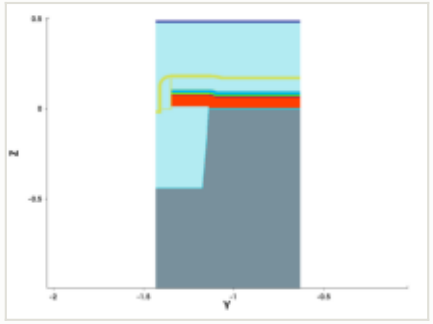
3D structure after this step

zoomed to fit

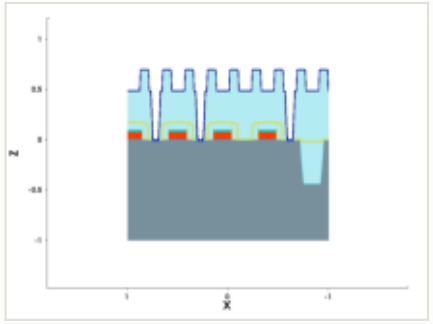




x cut at 0.000 µm



y cut at -1.032 µm



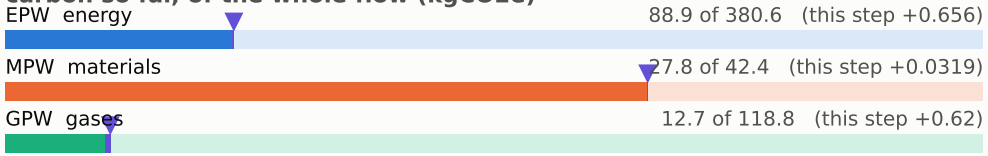
mask for this step

no mask for this step

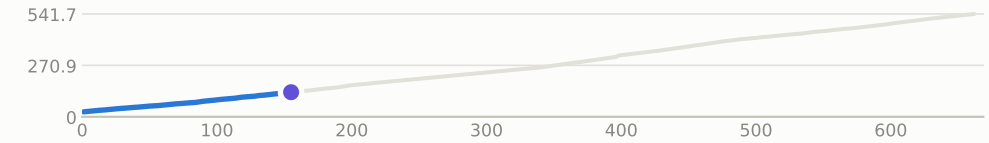
this step

process area	CVD	EPW	0.656 kgCO2e
material	TiN	EPW tool	0.469 kgCO2e
thickness	0.01 µm	EPW overhead	0.187 kgCO2e
material GWP	43.9778 gCO2e/g	MPW	0.0319 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

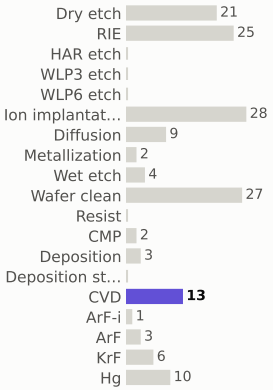
carbon so far, of the whole flow (kgCO2e)



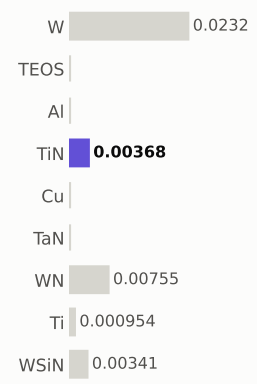
embodied carbon up to each step (kgCO2e per wafer)



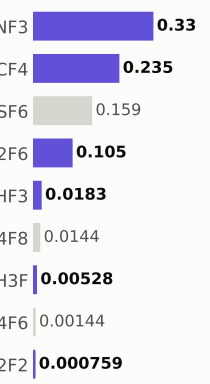
steps by process area



materials deposited (g)



gases emitted (g)



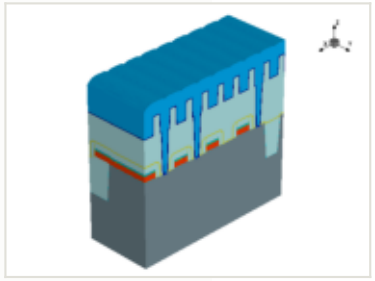
Step #1620: Metallization

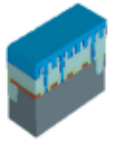
depo_isotropic · Metallization

1.22	130.7	0.23%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

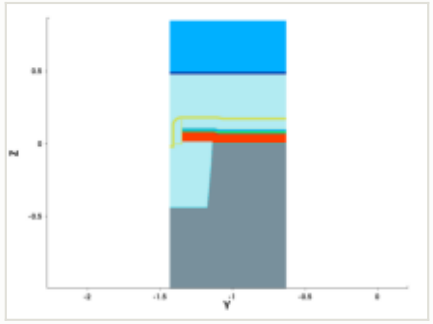
3D structure after this step

zoomed to fit

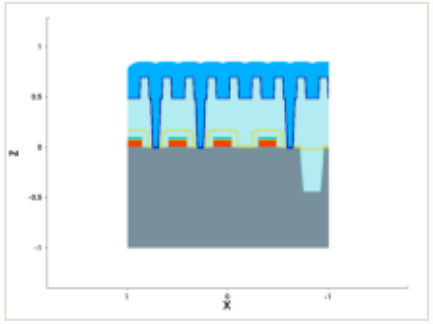




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

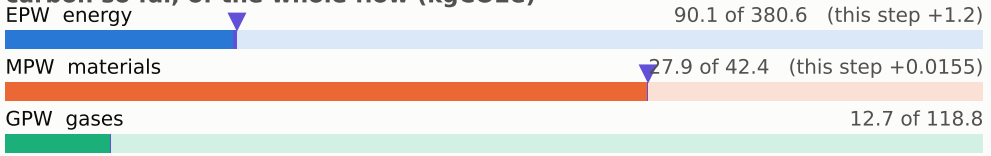
no mask for this step

this step

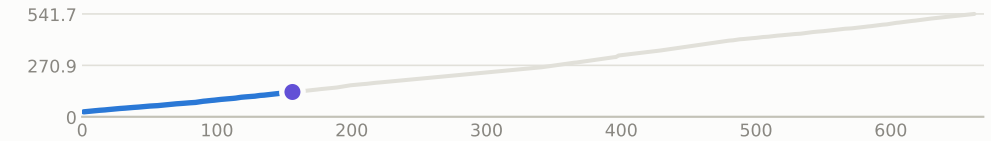
process area	Metallization
material	W
thickness	0.15 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0155 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.22 kgCO2e**

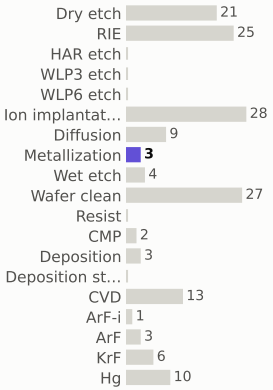
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

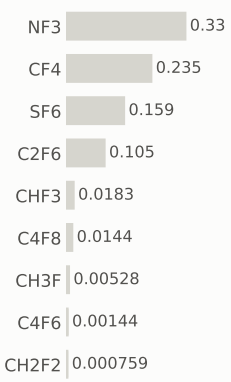


steps by process area



materials deposited (g)

gases emitted (g)



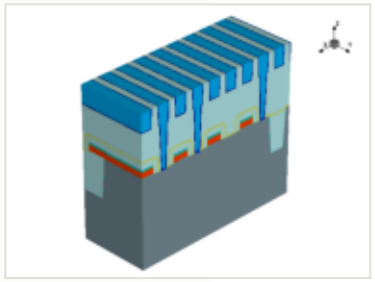
Step #1630: CMP

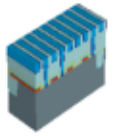
etch_cmp · CMP

0.169	130.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

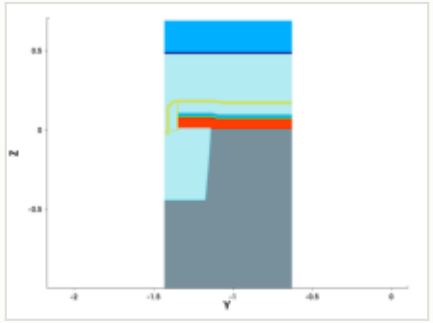
3D structure after this step

zoomed to fit

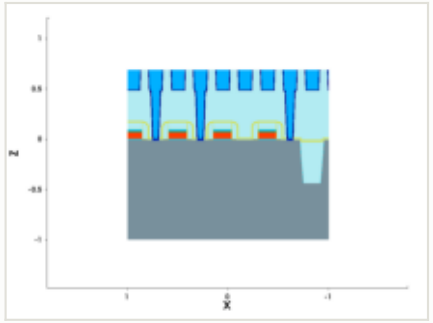




x cut at 0.000 µm



y cut at -1.032 µm



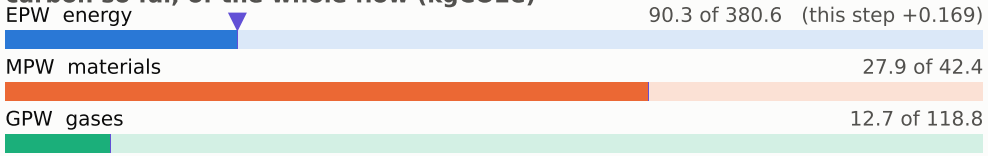
mask for this step

no mask for this step

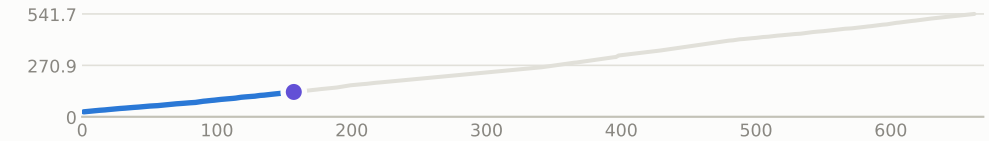
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

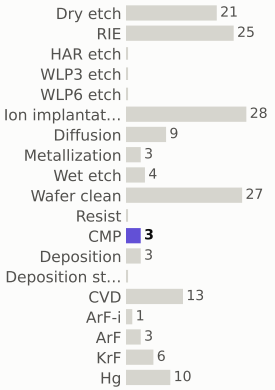
carbon so far, of the whole flow (kgCO2e)



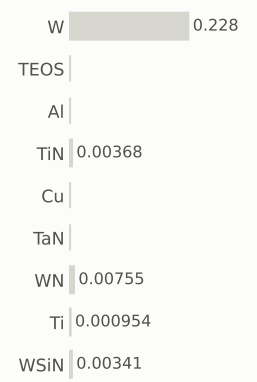
embodied carbon up to each step (kgCO2e per wafer)



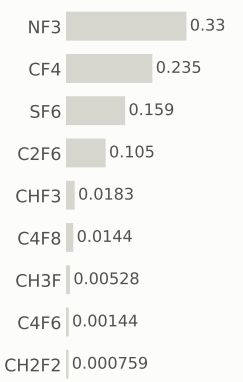
steps by process area



materials deposited (g)



gases emitted (g)



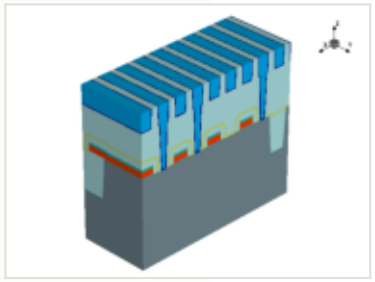
Step #1640: Wafer_clean

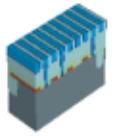
clean · Wafer_clean

0.169	131.1	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

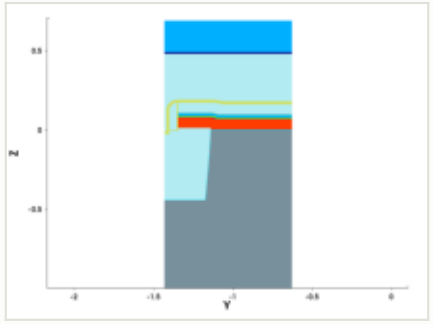
3D structure after this step

zoomed to fit

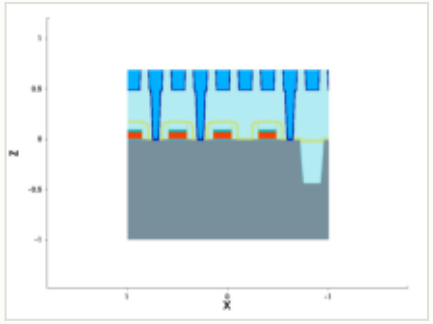




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

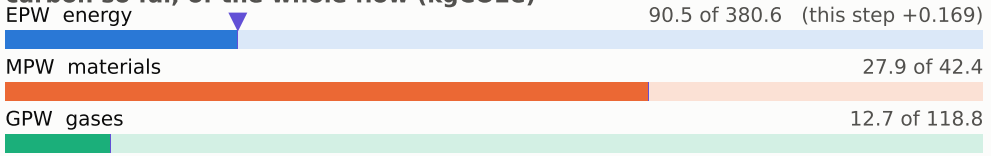
no mask for this step

this step

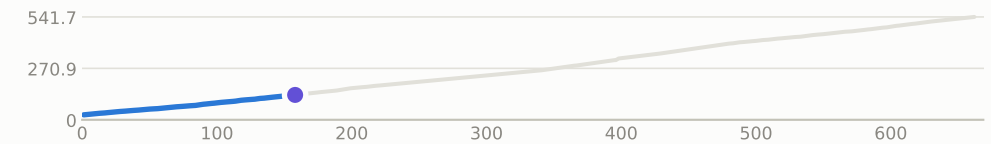
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

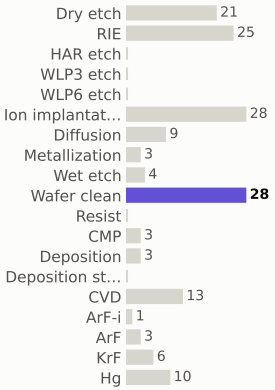
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

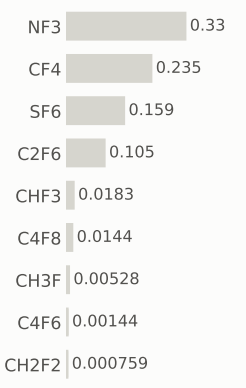


steps by process area



materials deposited (g)

gases emitted (g)



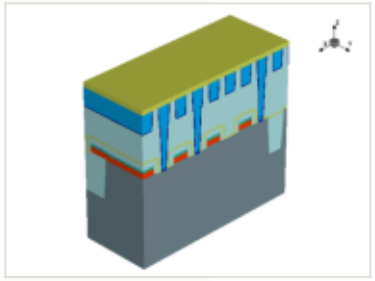
Step #1660: CVD

depo_isotropic · CVD

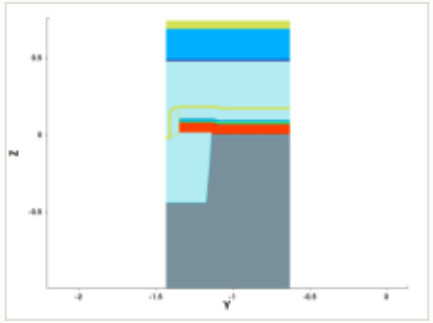
1.31	132.4	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

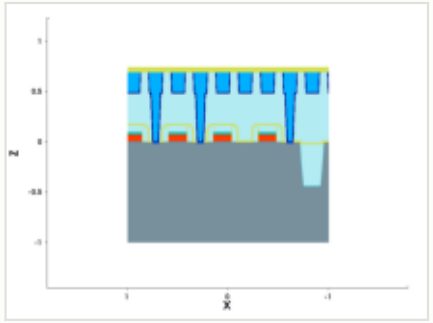
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

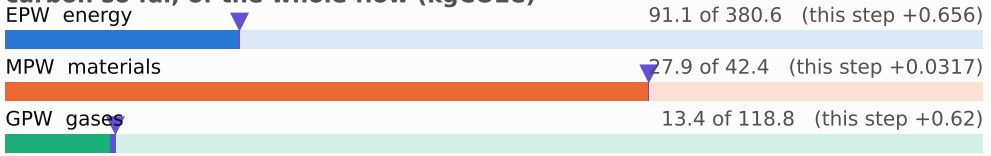
no mask for this step

this step

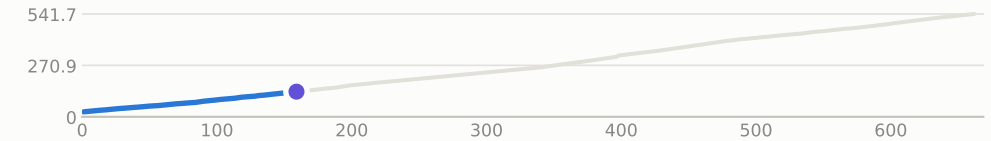
process area	CVD
material	—
thickness	0.053 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

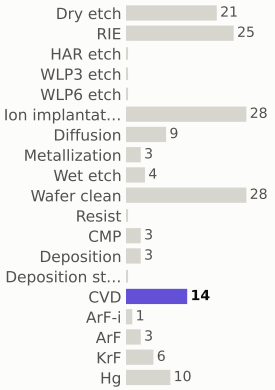
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

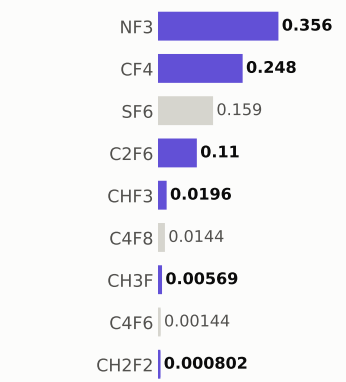


steps by process area



materials deposited (g)

gases emitted (g)



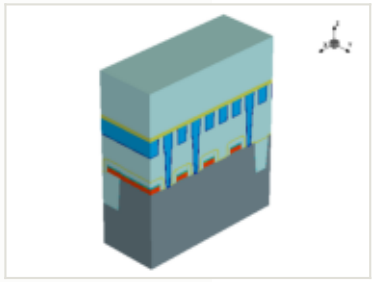
Step #1670: CVD

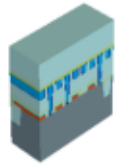
depo_isotropic · CVD

1.31	133.7	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



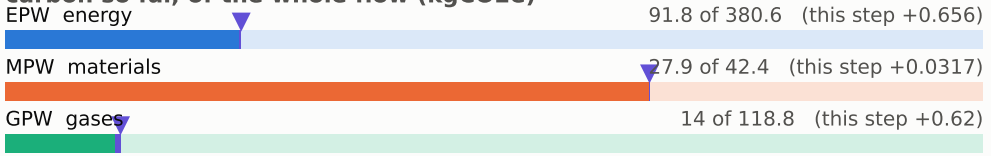


this step

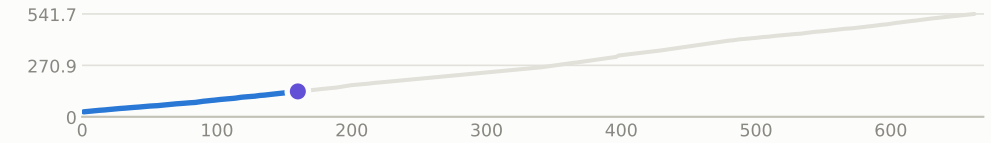
process area	CVD
material	—
thickness	0.56 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

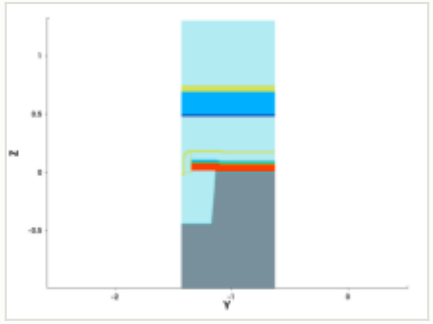
carbon so far, of the whole flow (kgCO2e)



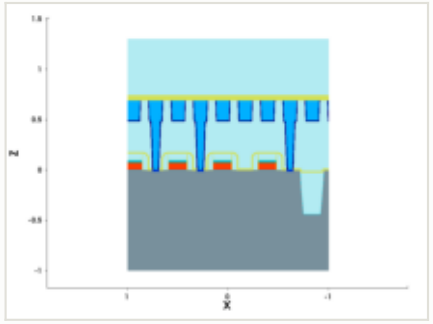
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



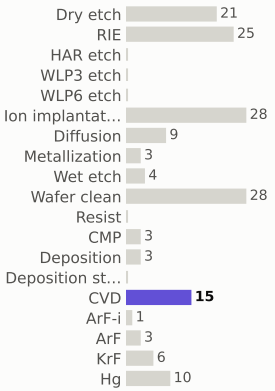
y cut at -1.032 µm



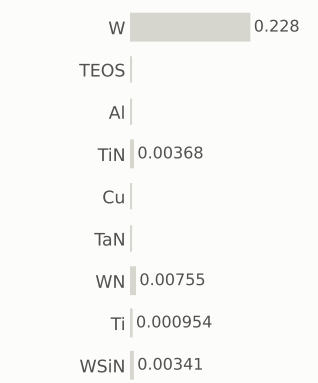
mask for this step

no mask for this step

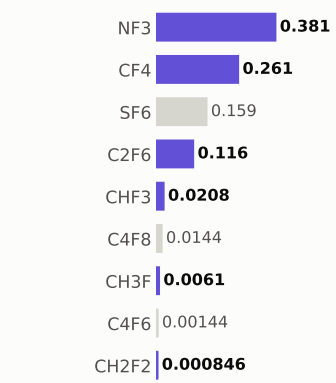
steps by process area



materials deposited (g)



gases emitted (g)



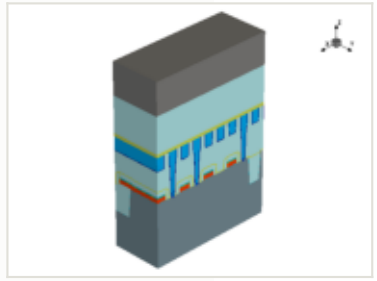
Step #1680: ArF

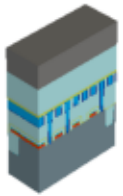
litho_patterning · ArF

0.199	133.9	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

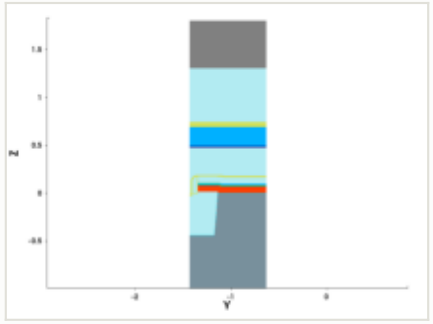
3D structure after this step

zoomed to fit

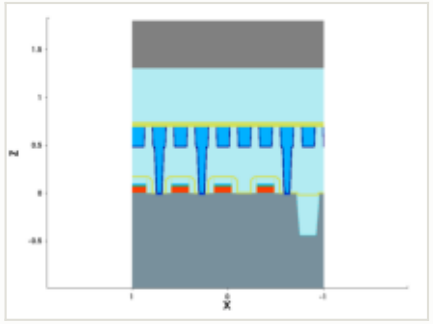




x cut at 0.000 µm



y cut at -1.032 µm

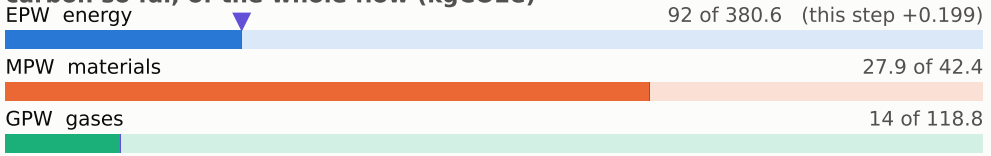


mask: layer 21:0, positive (OASIS)

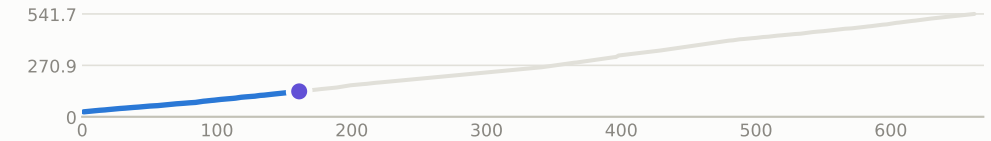
this step

process area	ArF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

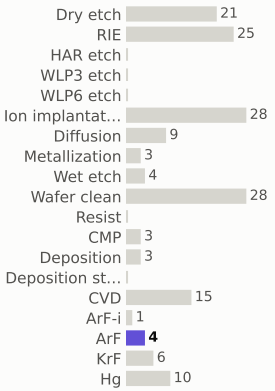
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

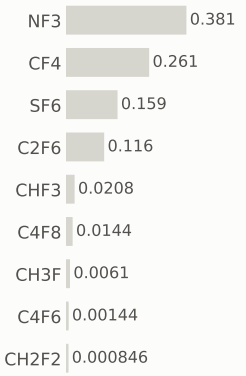


steps by process area



materials deposited (g)

gases emitted (g)



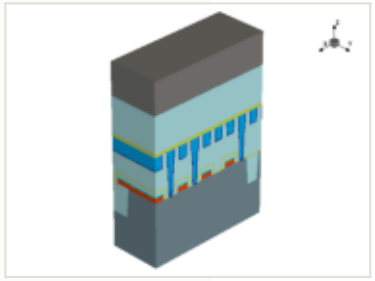
Step #1690: RIE

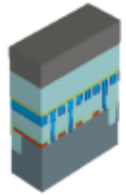
etch_anisotropic · RIE

1.04	134.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



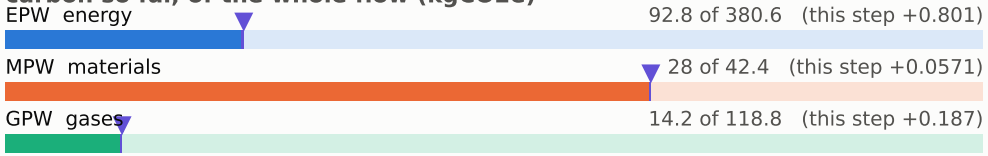


this step

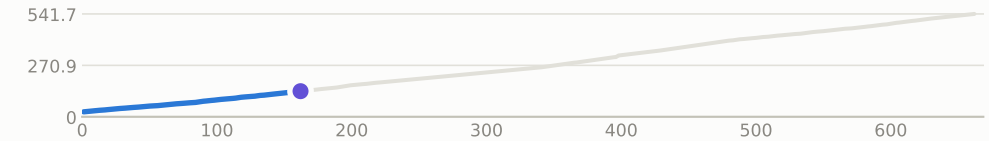
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

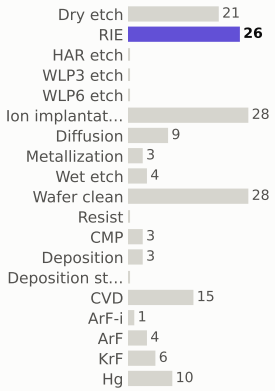
carbon so far, of the whole flow (kgCO2e)



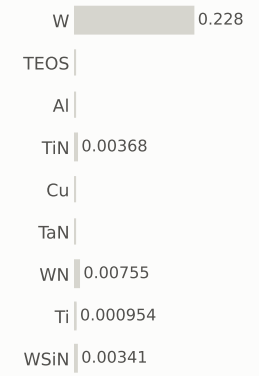
embodied carbon up to each step (kgCO2e per wafer)



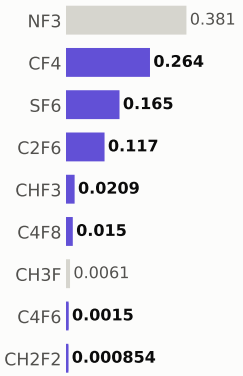
steps by process area



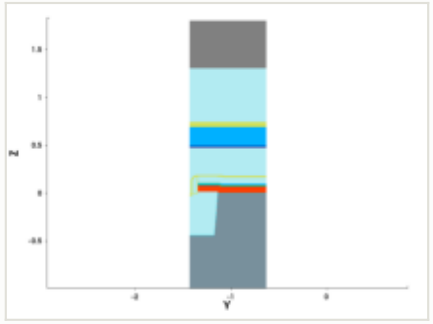
materials deposited (g)



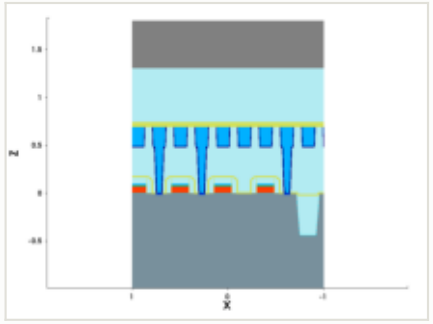
gases emitted (g)



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

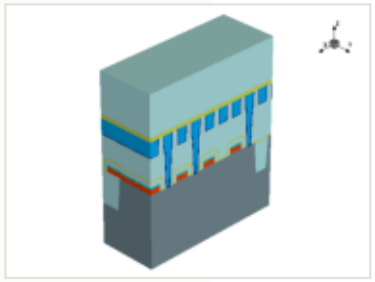
Step #1700: Dry_etch

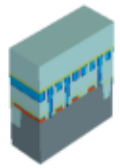
etch_strip · Dry_etch

0.801	135.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

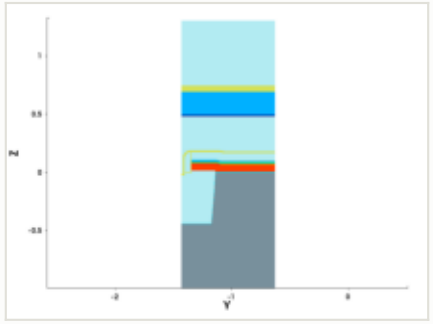
3D structure after this step

zoomed to fit

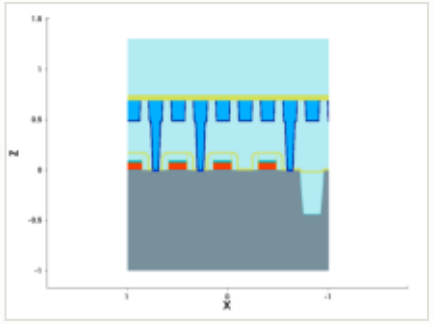




x cut at 0.000 µm



y cut at -1.032 µm



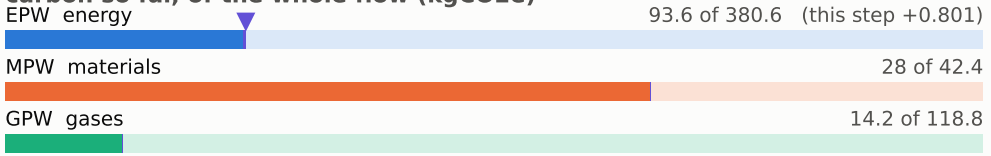
mask for this step

no mask for this step

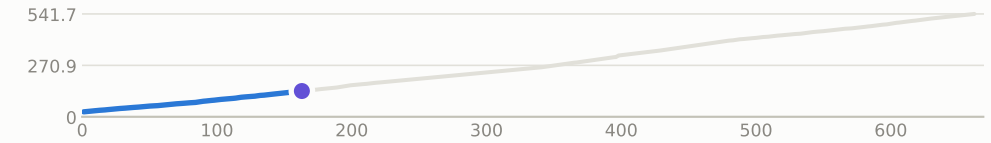
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

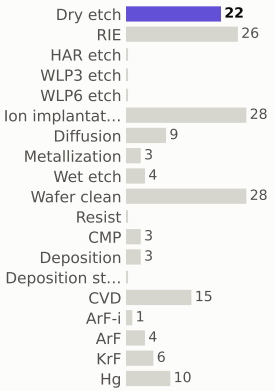
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

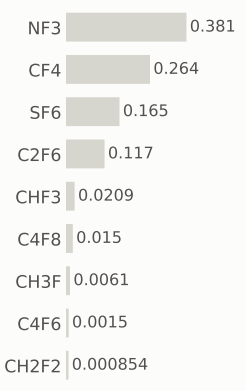


steps by process area



materials deposited (g)

gases emitted (g)



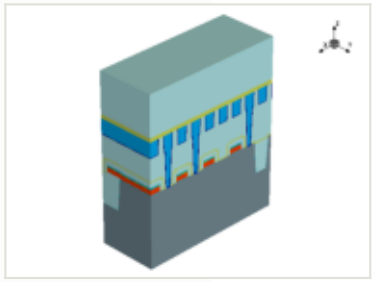
Step #1710: Wafer_clean

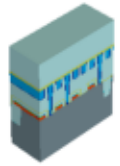
clean · Wafer_clean

0.169	135.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

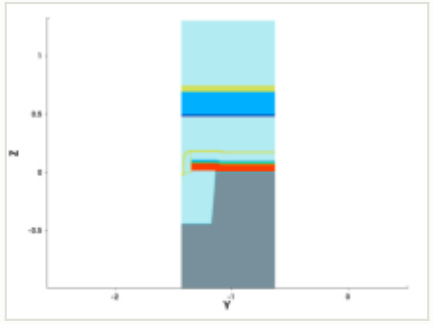
3D structure after this step

zoomed to fit

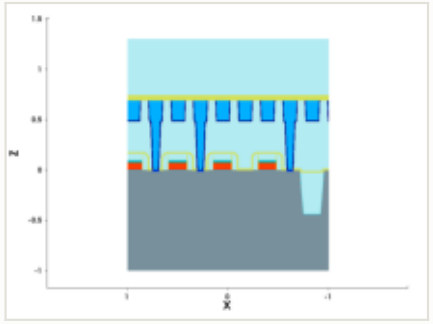




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

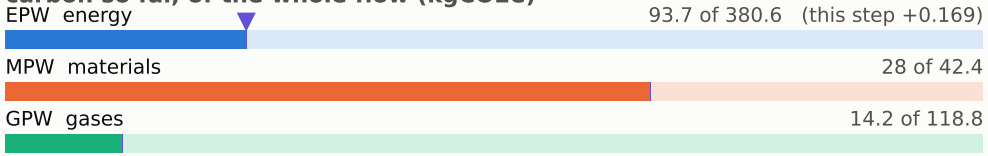
no mask for this step

this step

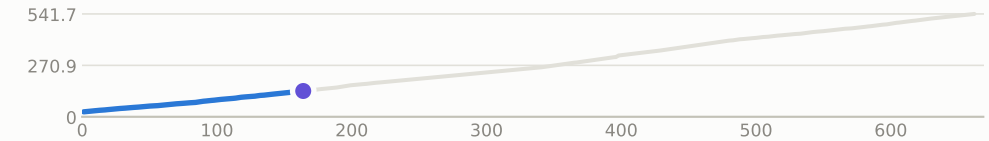
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

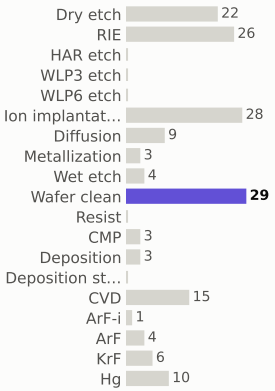
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

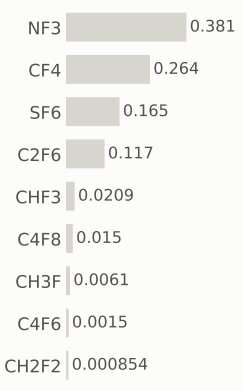


steps by process area



materials deposited (g)

gases emitted (g)



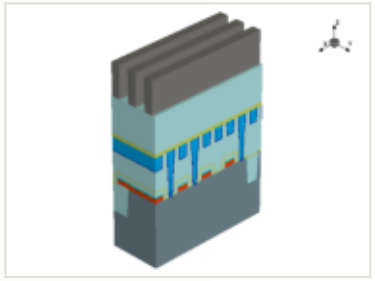
Step #1720: KrF

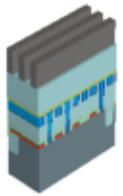
litho_patterning · KrF

0.199	136.1	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

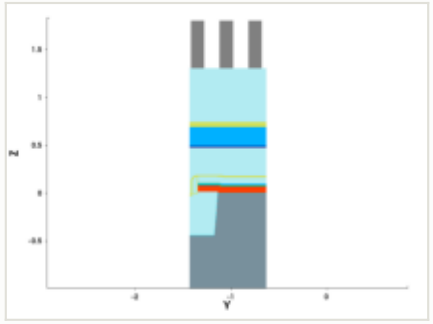
3D structure after this step

zoomed to fit

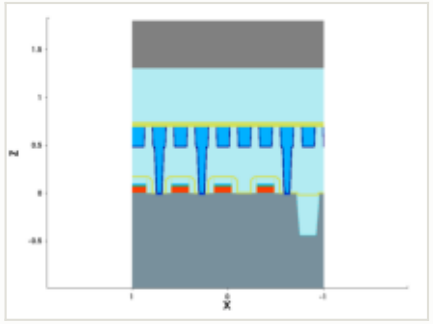




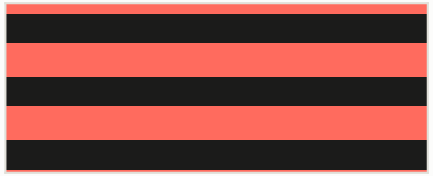
x cut at 0.000 µm



y cut at -1.032 µm



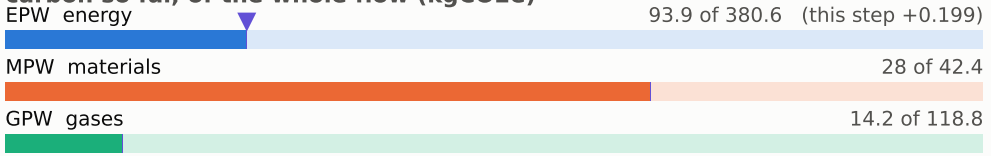
mask: layer 22:0 (OASIS)



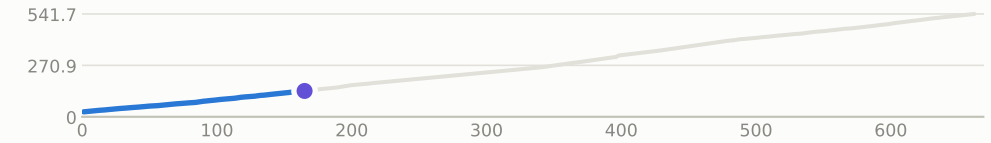
this step

process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

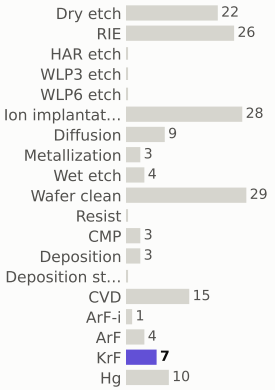
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

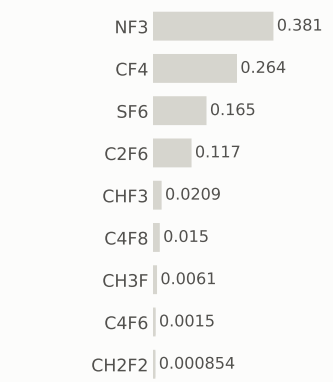


steps by process area



materials deposited (g)

gases emitted (g)



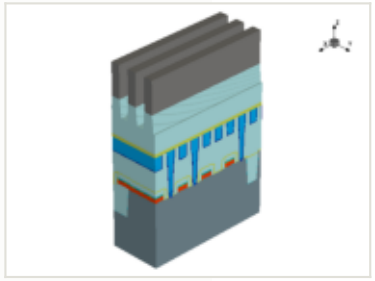
Step #1730: RIE

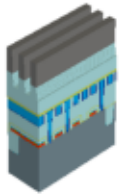
etch_anisotropic · RIE

1.04	137.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

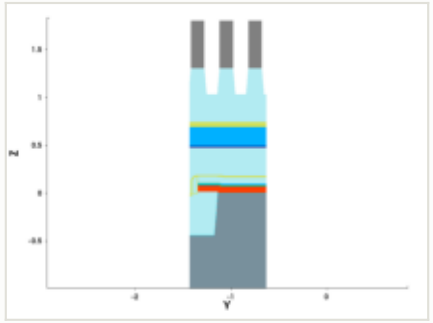
3D structure after this step

zoomed to fit

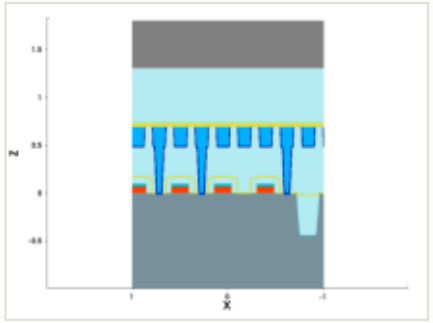




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

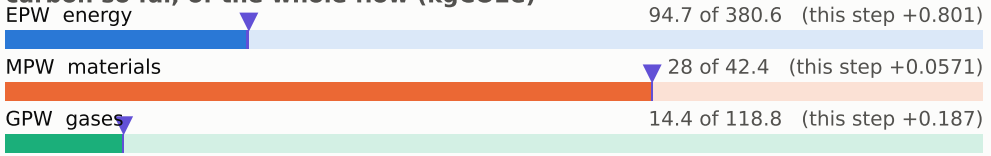
no mask for this step

this step

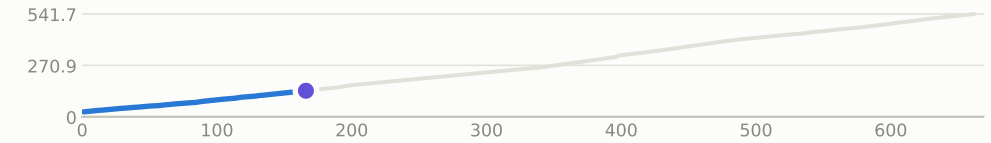
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

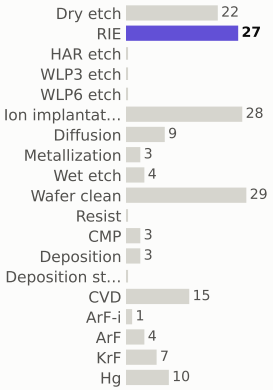
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

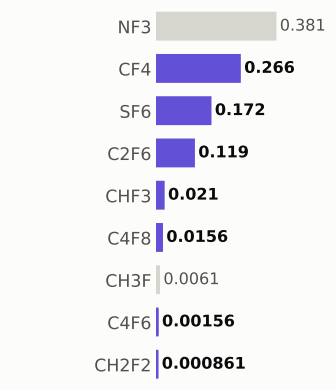


steps by process area



materials deposited (g)

gases emitted (g)



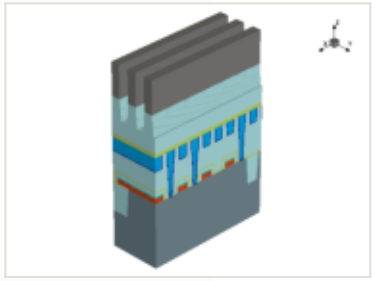
Step #1740: RIE

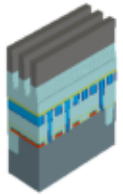
etch_anisotropic · RIE

1.04	138.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

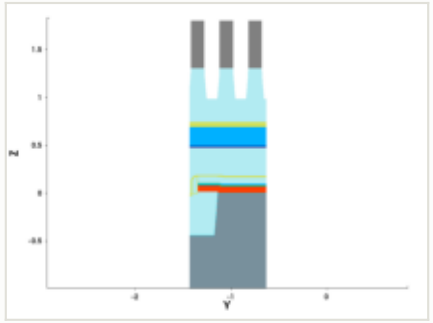
3D structure after this step

zoomed to fit

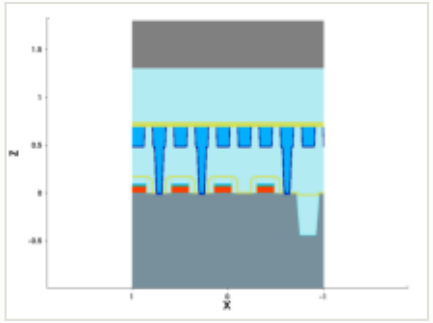




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

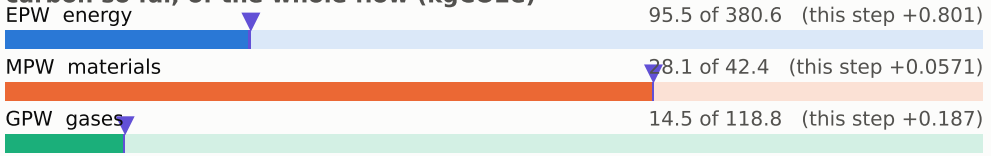
no mask for this step

this step

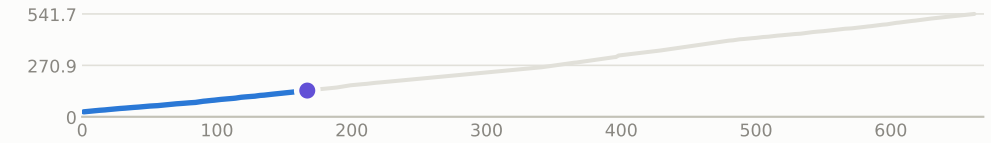
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

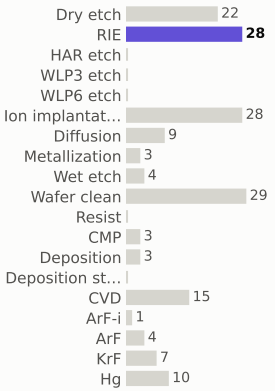
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

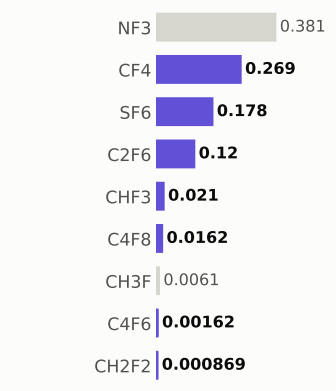


steps by process area



materials deposited (g)

gases emitted (g)



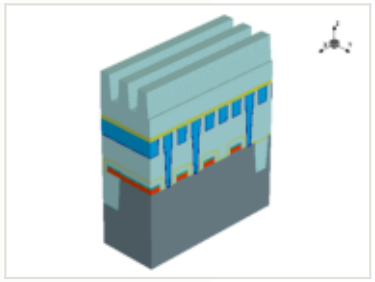
Step #1750: Dry_etch

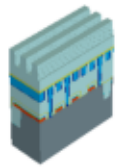
etch_strip · Dry_etch

0.801	139.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

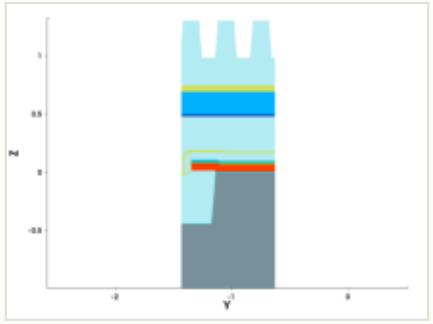
3D structure after this step

zoomed to fit

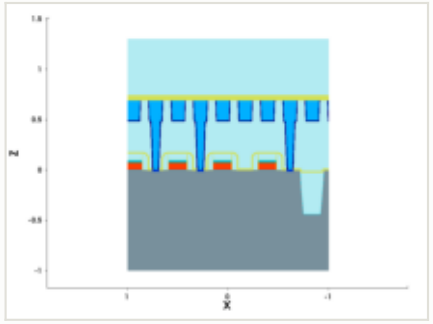




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

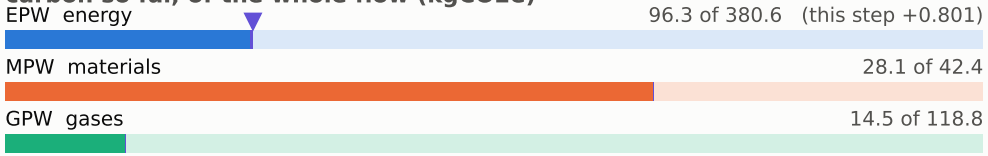
no mask for this step

this step

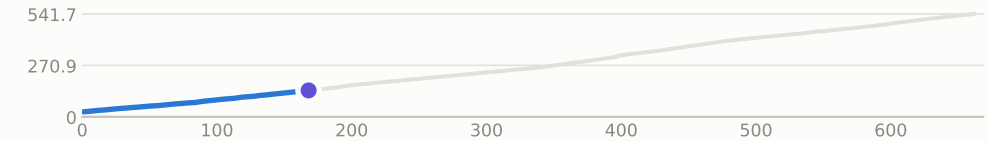
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

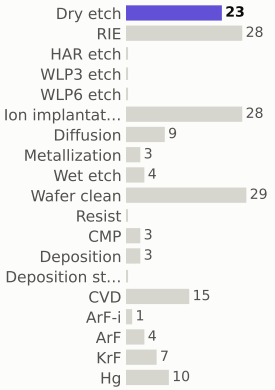
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

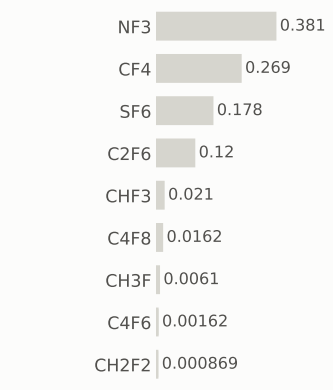


steps by process area



materials deposited (g)

gases emitted (g)



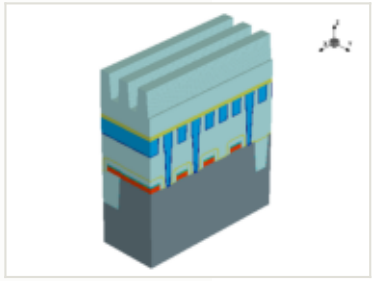
Step #1760: Wafer_clean

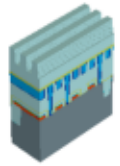
clean · Wafer_clean

0.169	139.1	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

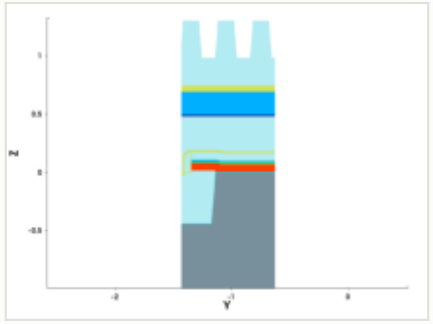
3D structure after this step

zoomed to fit

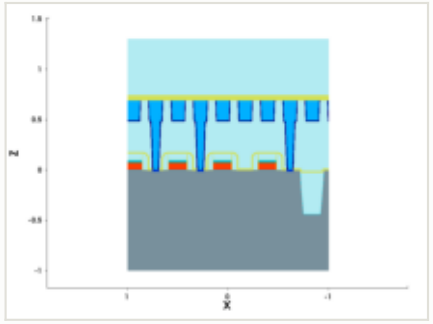




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

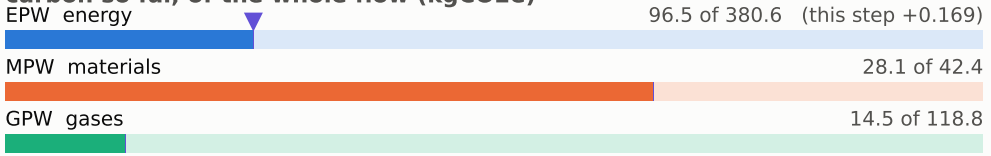
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

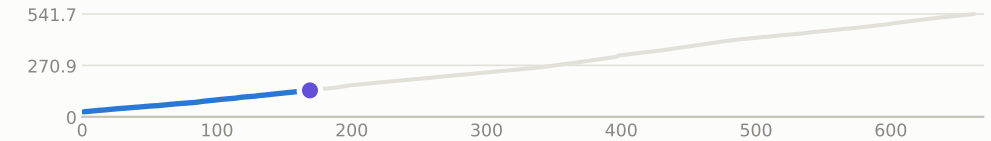
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

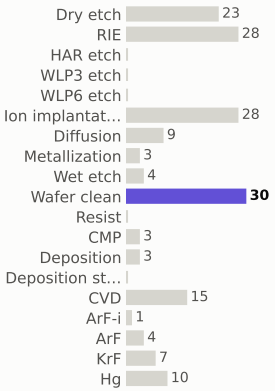
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

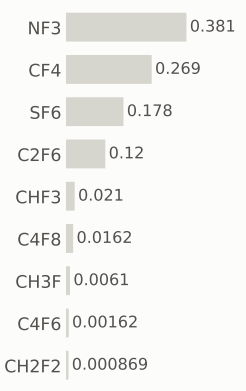


steps by process area



materials deposited (g)

gases emitted (g)



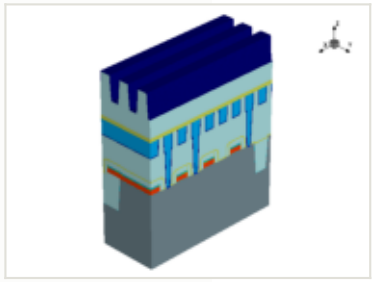
Step #1770: CVD

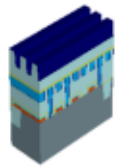
depo_isotropic · CVD

1.31	140.5	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

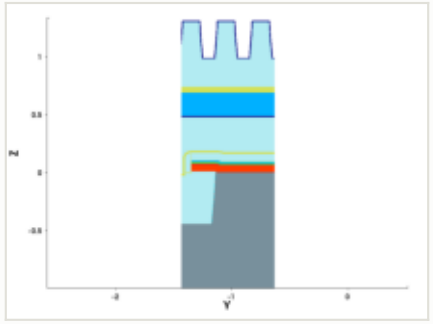
3D structure after this step

zoomed to fit

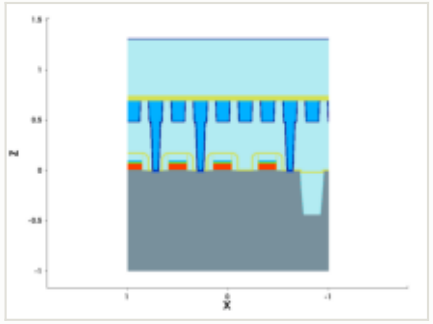




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

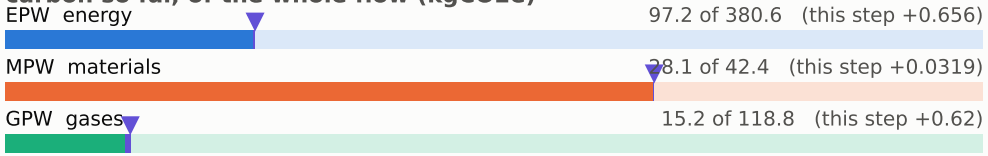
no mask for this step

this step

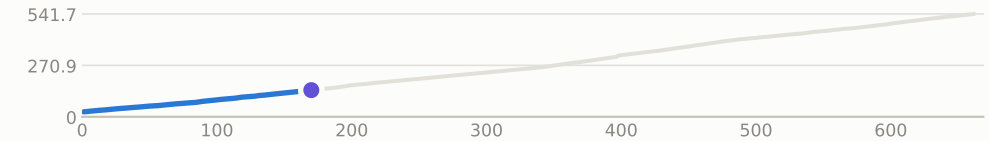
process area	CVD
material	TiN
thickness	0.01 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0319 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

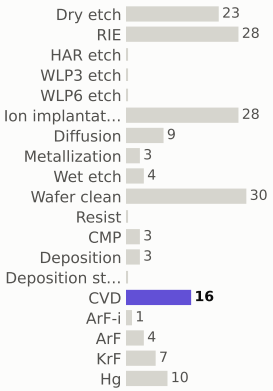
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

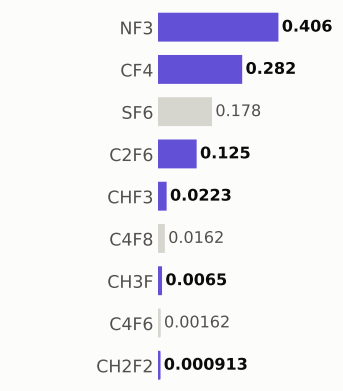


steps by process area



materials deposited (g)

gases emitted (g)



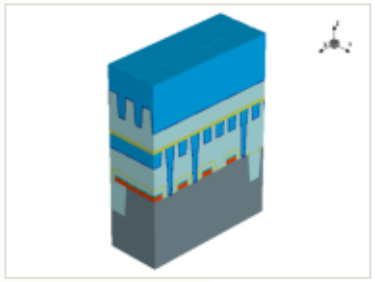
Step #1780: Metallization

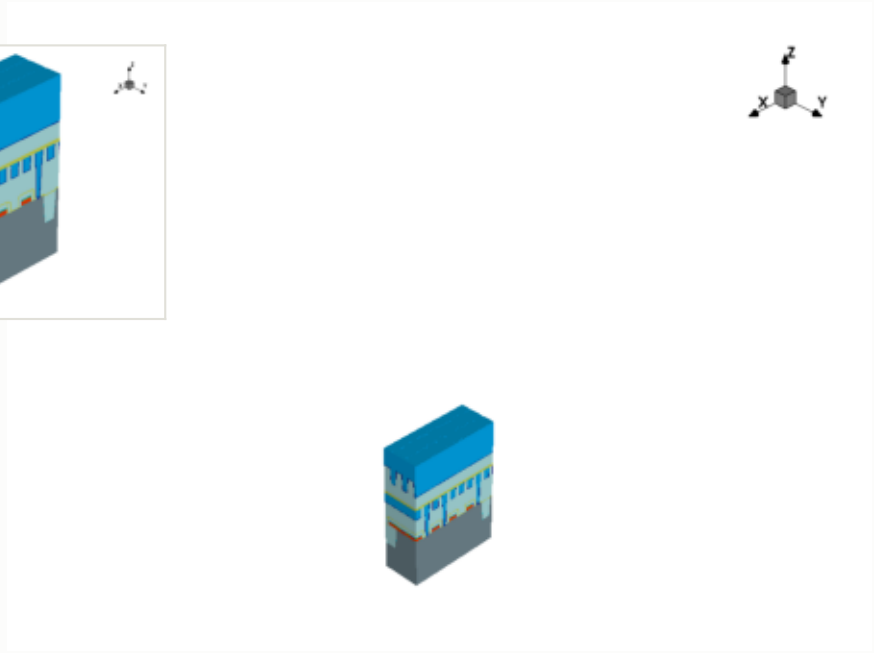
depo_isotropic · Metallization

1.24	141.7	0.23%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



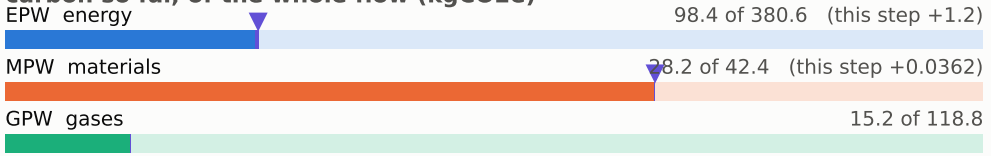


this step

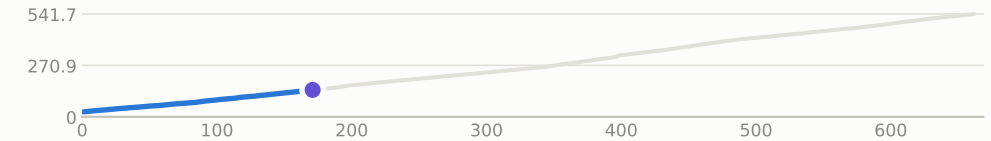
process area	Metallization
material	W
thickness	0.35 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0362 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.24 kgCO2e**

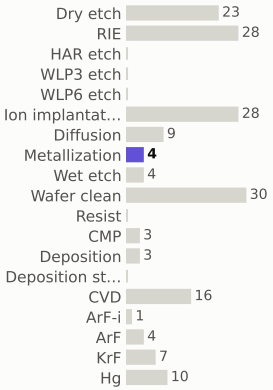
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

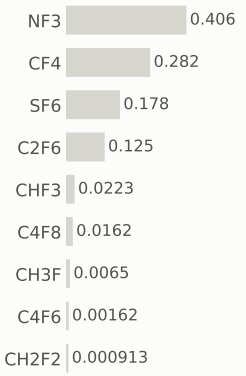


steps by process area

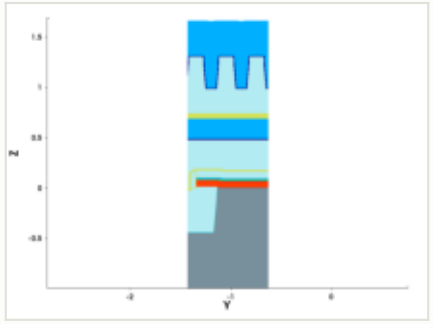


materials deposited (g)

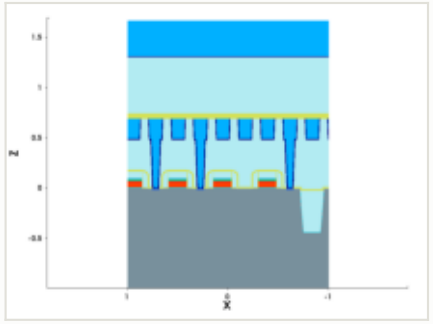
gases emitted (g)



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

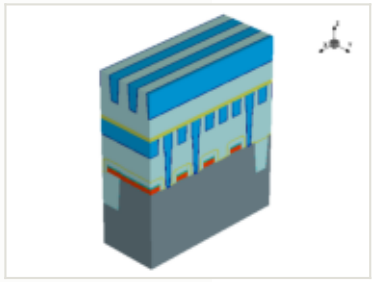
Step #1790: CMP

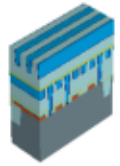
etch_cmp · CMP

0.169	141.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

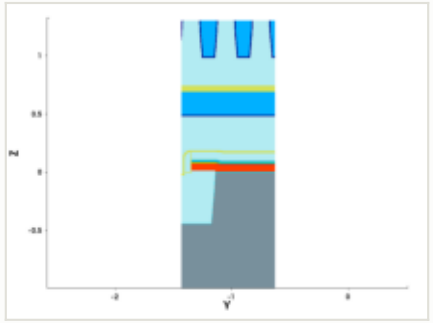
3D structure after this step

zoomed to fit

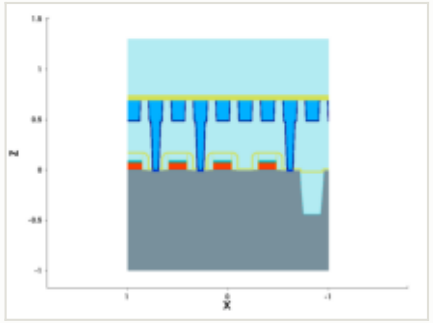




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

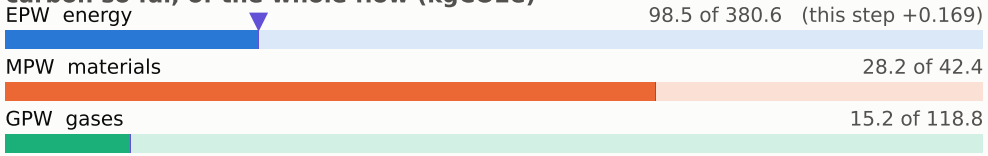
no mask for this step

this step

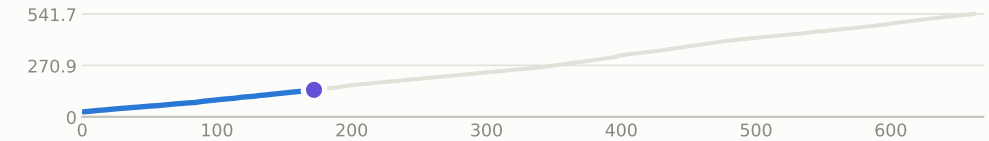
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

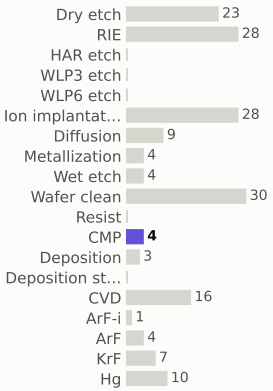
carbon so far, of the whole flow (kgCO2e)



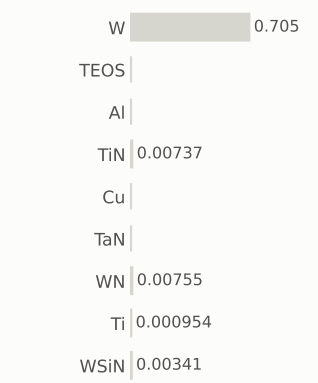
embodied carbon up to each step (kgCO2e per wafer)



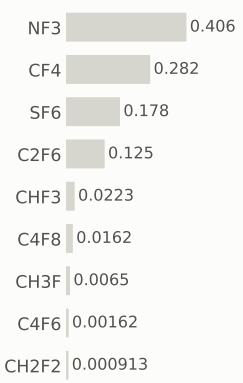
steps by process area



materials deposited (g)



gases emitted (g)



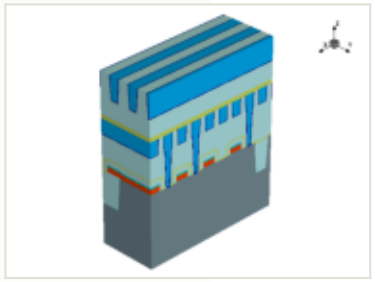
Step #1800: Wafer_clean

clean · Wafer_clean

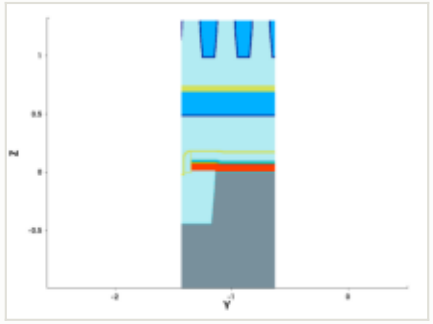
0.169	142.0	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

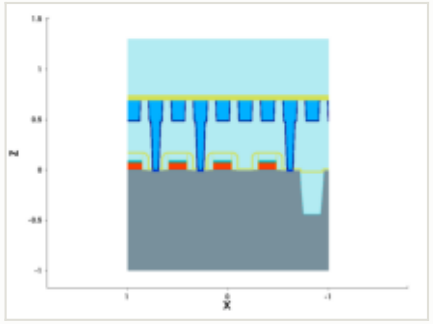
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

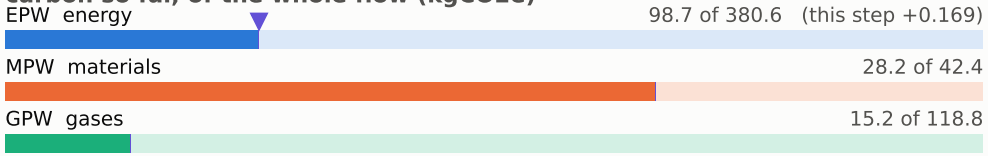
no mask for this step

this step

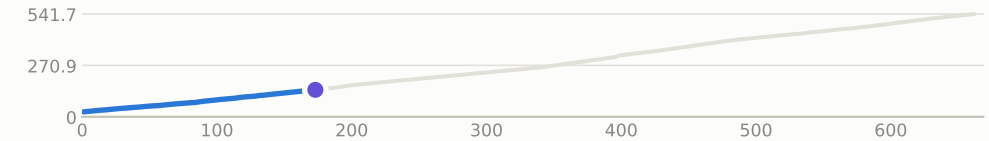
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

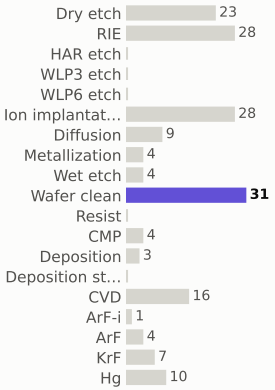
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

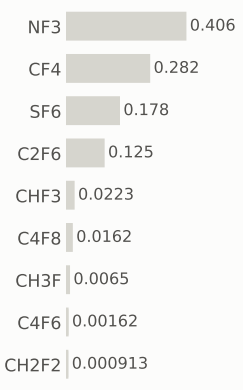


steps by process area



materials deposited (g)

gases emitted (g)



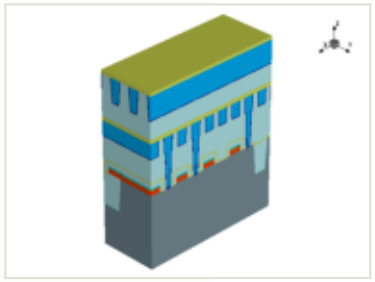
Step #1820: CVD

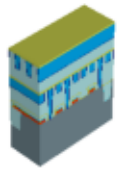
depo_isotropic · CVD

1.31	143.3	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

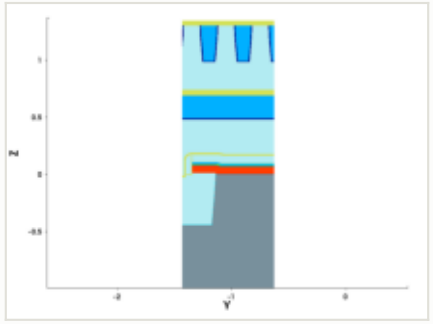
3D structure after this step

zoomed to fit

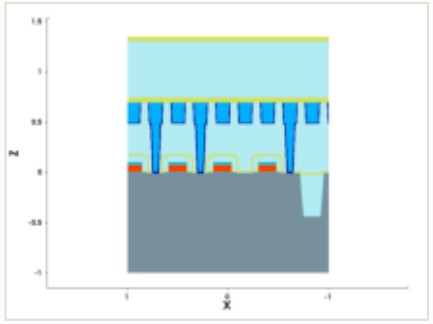




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

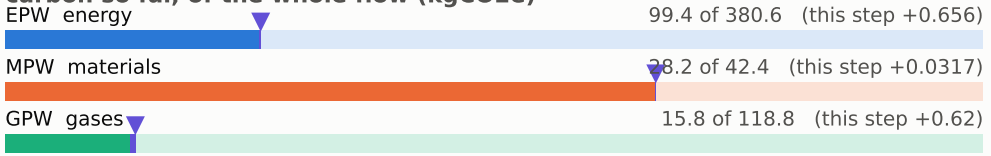
no mask for this step

this step

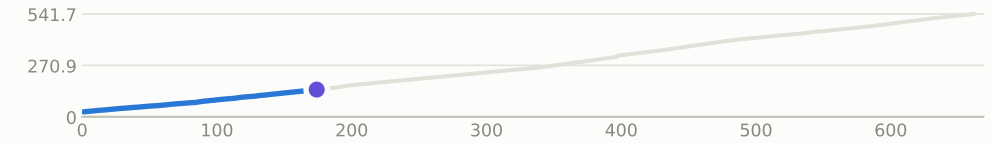
process area	CVD
material	—
thickness	0.045 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

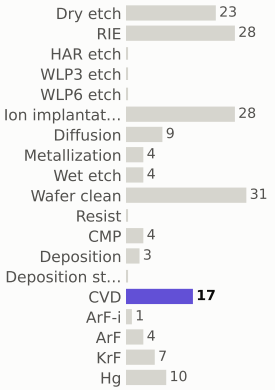
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

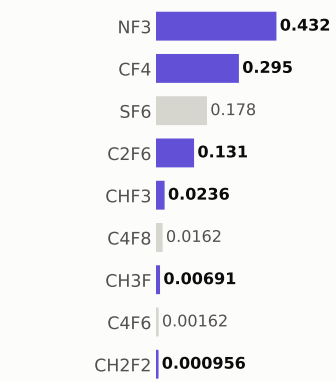


steps by process area



materials deposited (g)

gases emitted (g)



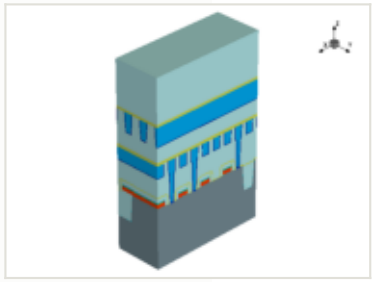
Step #1830: CVD

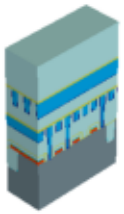
depo_isotropic · CVD

1.31	144.6	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

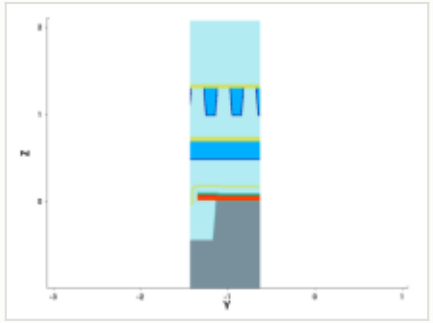
3D structure after this step

zoomed to fit

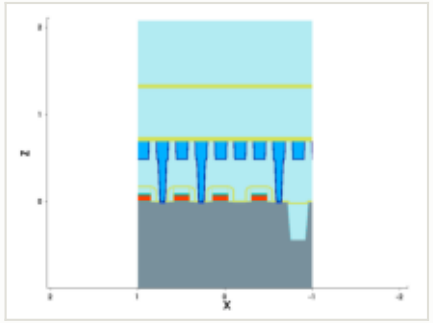




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

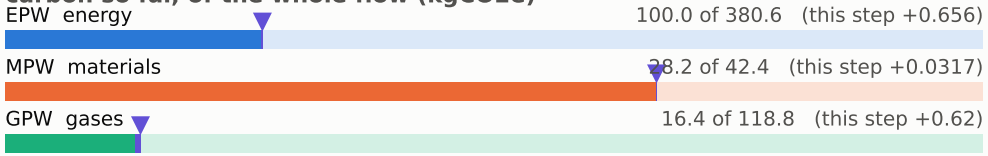
no mask for this step

this step

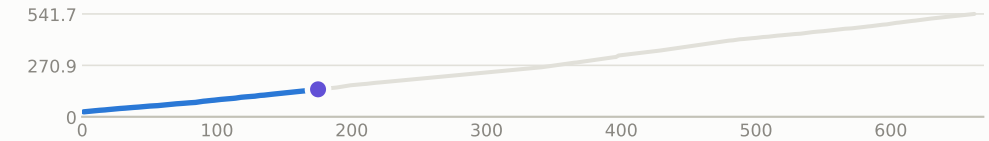
process area	CVD
material	—
thickness	0.73 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

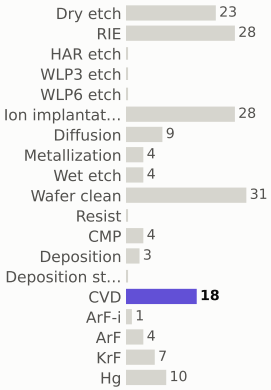
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

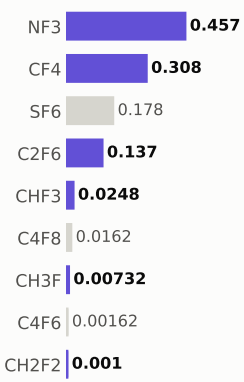


steps by process area



materials deposited (g)

gases emitted (g)



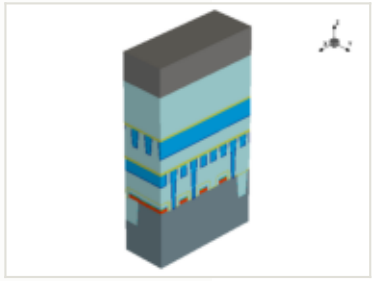
Step #1840: ArF

litho_patterning · ArF

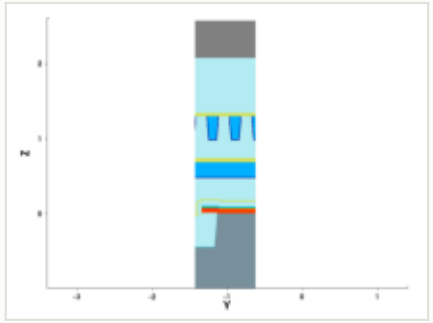
0.199	144.8	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

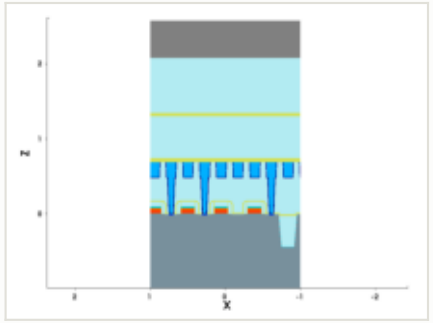
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



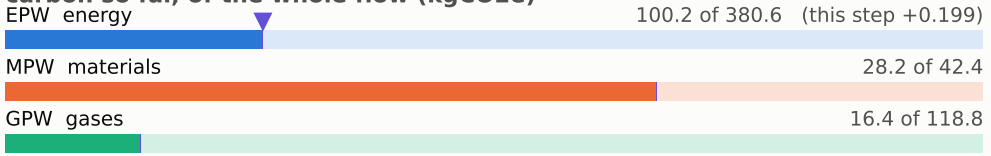
mask: layer 23:0, positive (OASIS)

this step

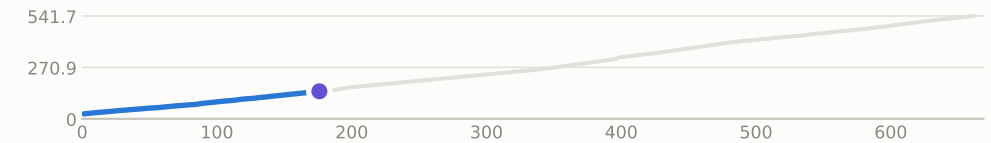
process area	ArF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

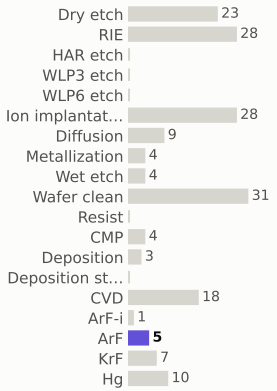
carbon so far, of the whole flow (kgCO2e)



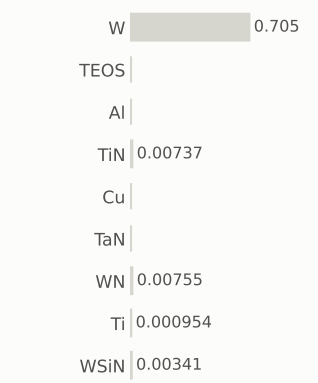
embodied carbon up to each step (kgCO2e per wafer)



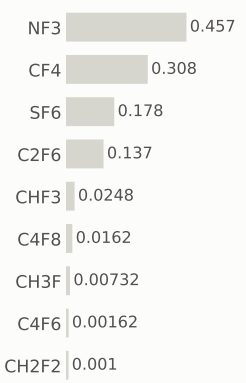
steps by process area



materials deposited (g)



gases emitted (g)



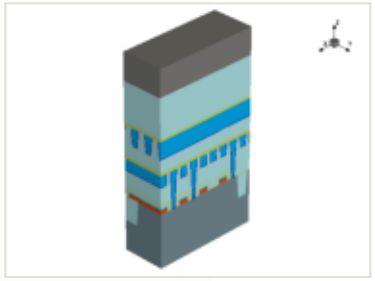
Step #1850: RIE

etch_anisotropic · RIE

1.04	145.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

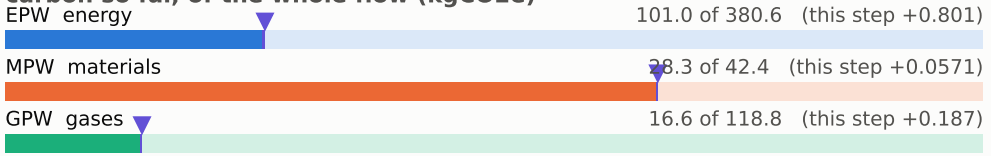


this step

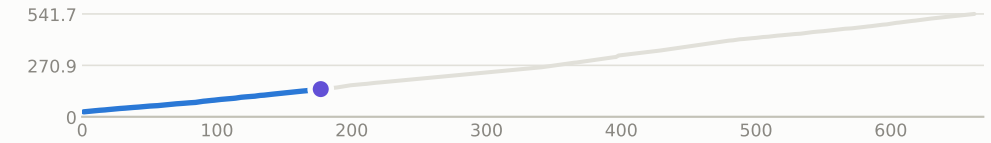
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

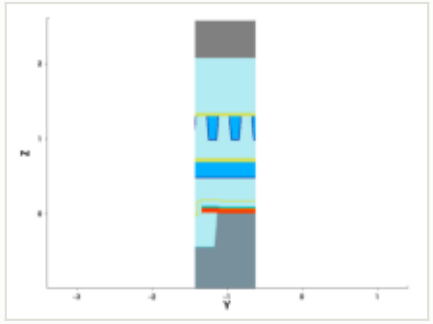
carbon so far, of the whole flow (kgCO2e)



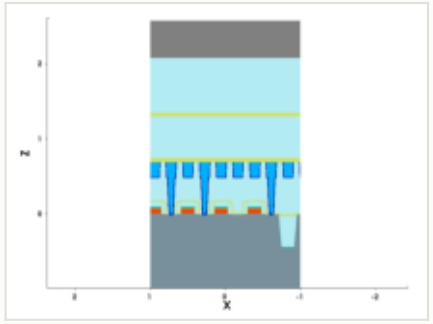
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



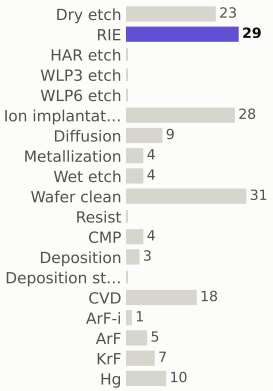
y cut at -1.032 µm



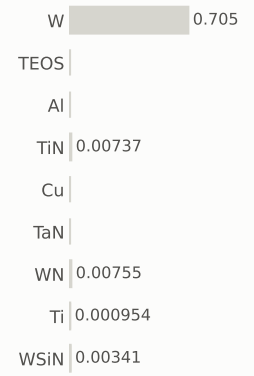
mask for this step

no mask for this step

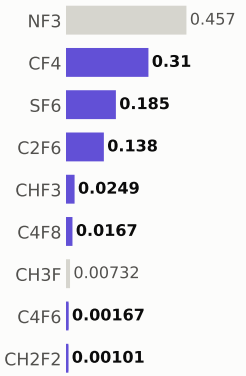
steps by process area



materials deposited (g)



gases emitted (g)



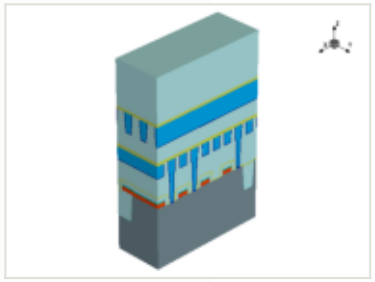
Step #1860: Dry_etch

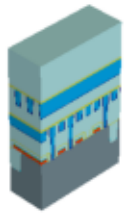
etch_strip · Dry_etch

0.801	146.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

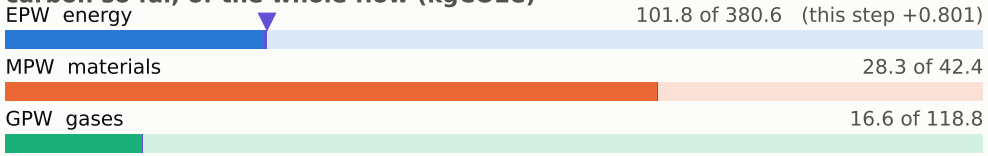




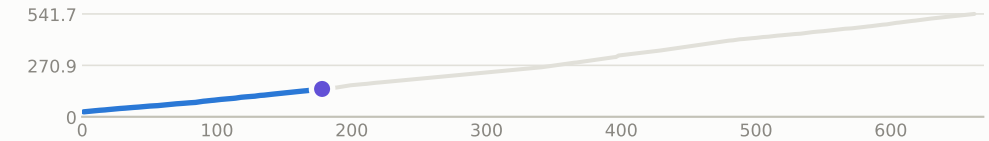
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

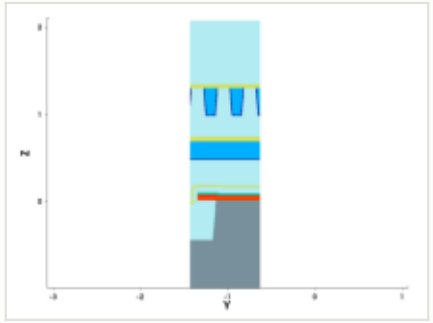
carbon so far, of the whole flow (kgCO2e)



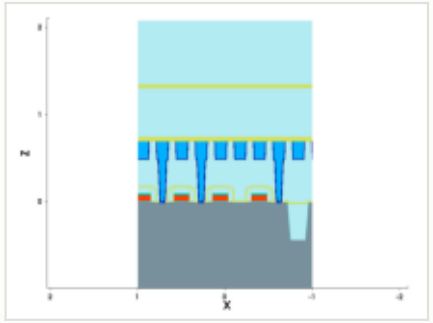
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



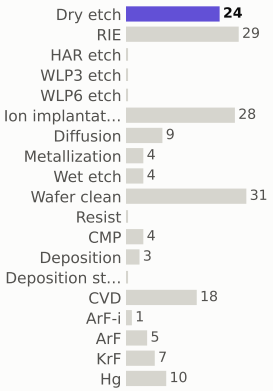
y cut at -1.032 µm



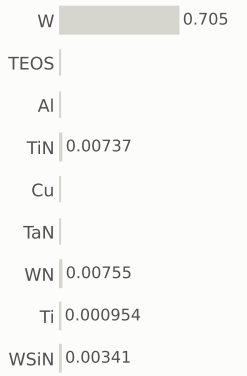
mask for this step

no mask for this step

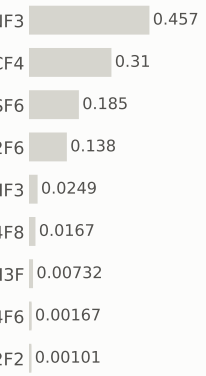
steps by process area



materials deposited (g)



gases emitted (g)



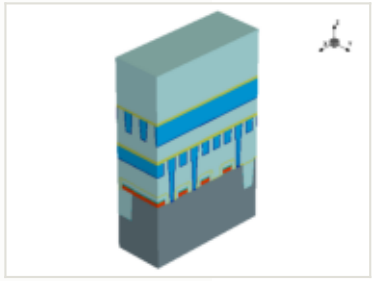
Step #1870: Wafer_clean

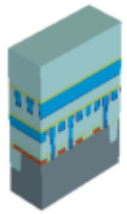
clean · Wafer_clean

0.169	146.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

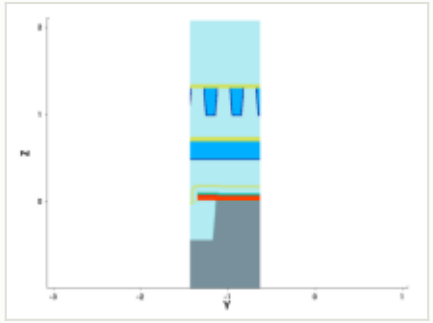
3D structure after this step

zoomed to fit

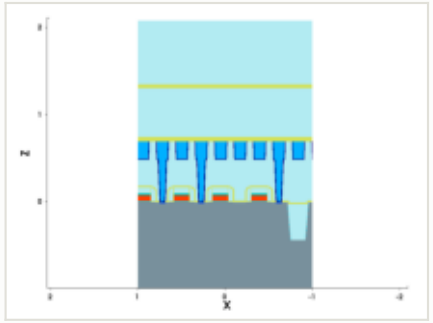




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

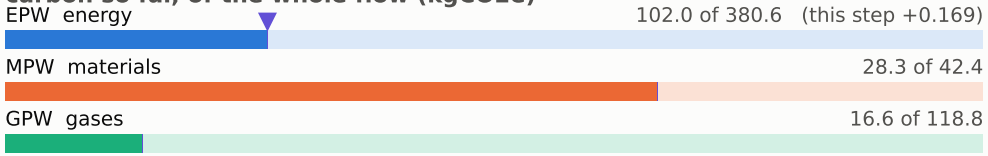
no mask for this step

this step

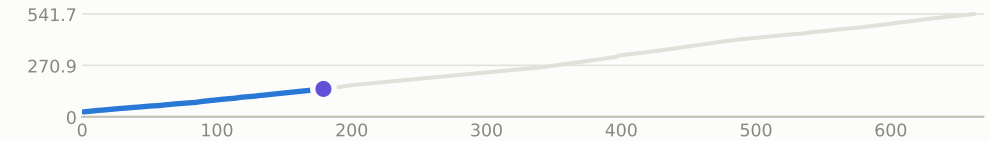
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

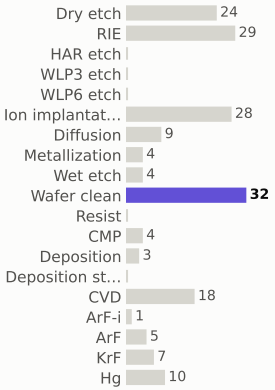
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

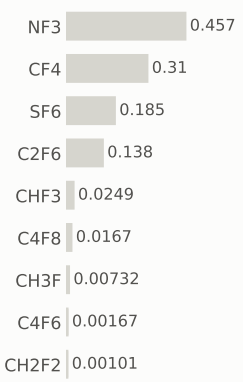


steps by process area



materials deposited (g)

gases emitted (g)



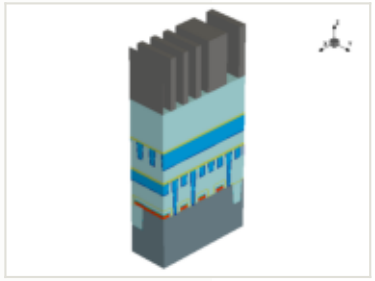
Step #1880: KrF

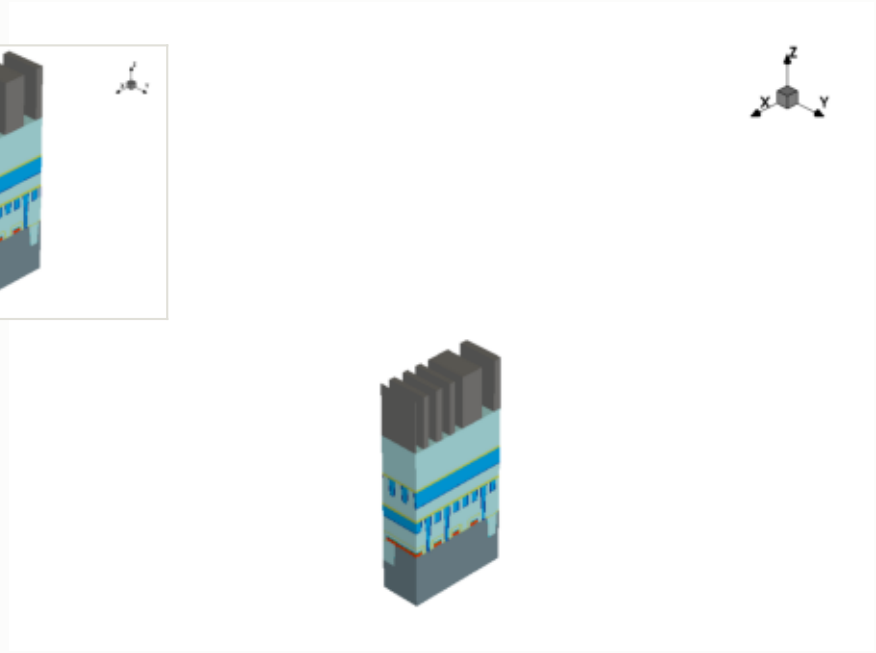
litho_patterning · KrF

0.199	147.1	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

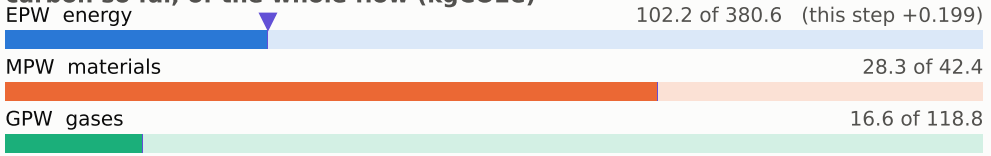




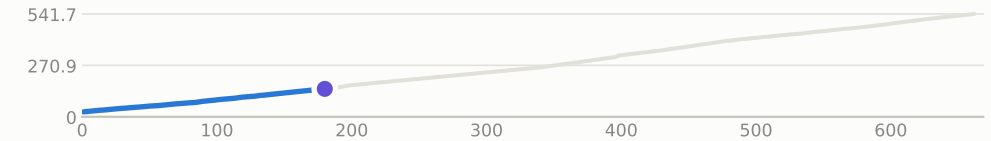
this step

process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

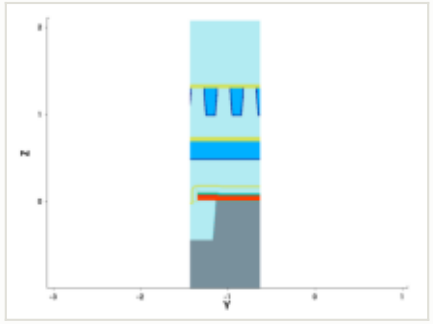
carbon so far, of the whole flow (kgCO2e)



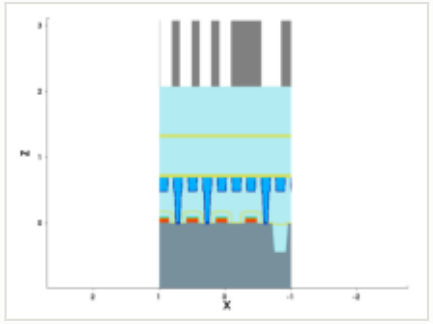
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

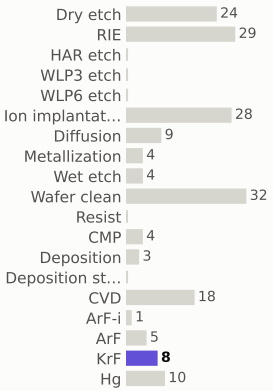


y cut at -1.032 µm

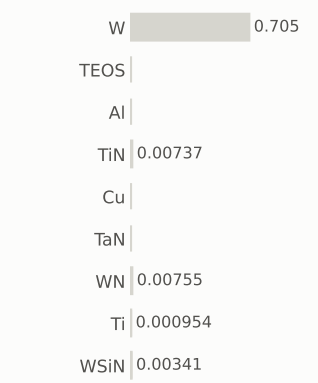


mask: layer 24:0 (OASIS)

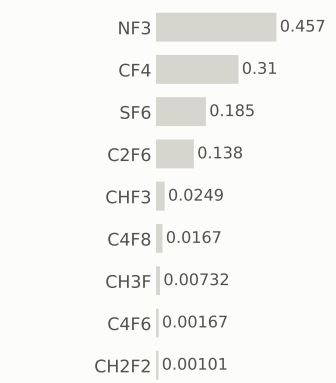
steps by process area



materials deposited (g)



gases emitted (g)



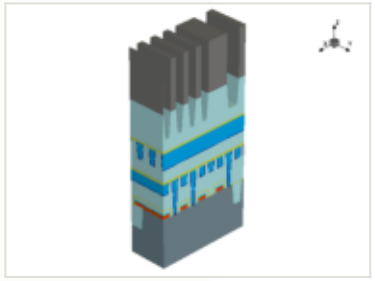
Step #1890: RIE

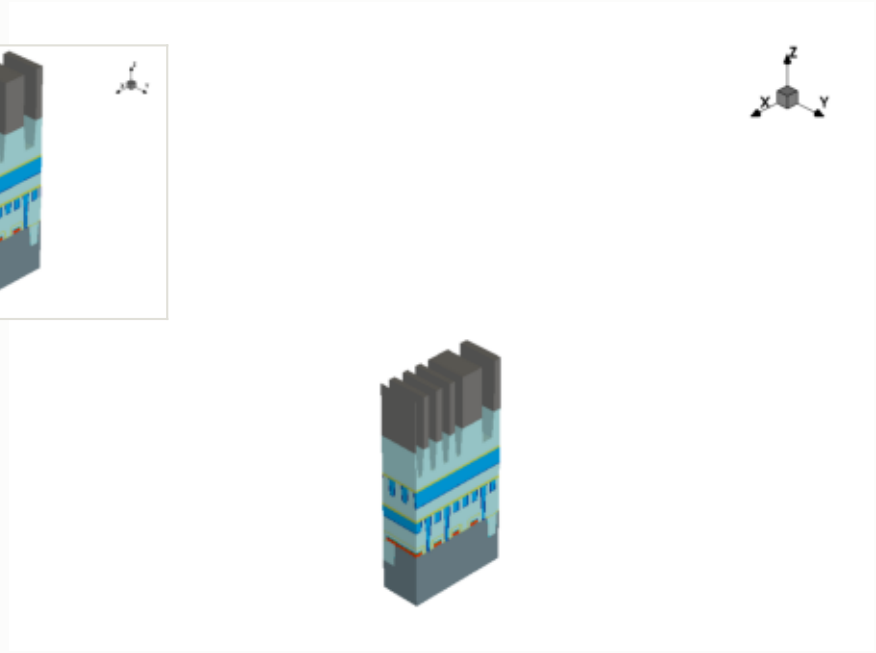
etch_anisotropic · RIE

1.04	148.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



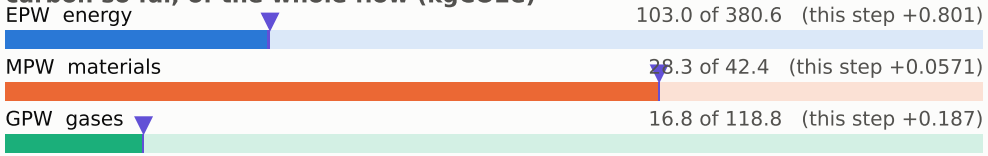


this step

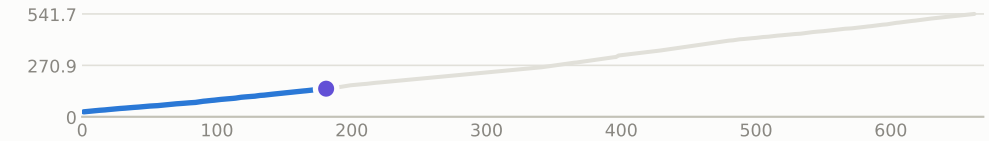
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

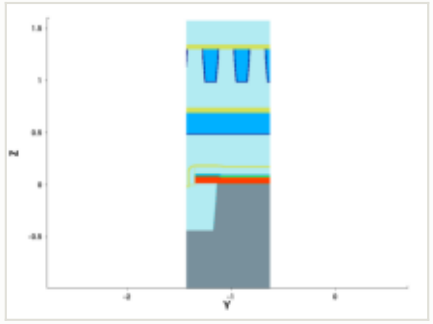
carbon so far, of the whole flow (kgCO2e)



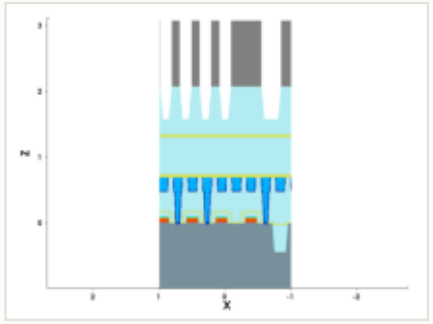
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



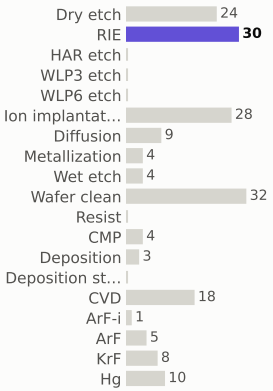
y cut at -1.032 µm



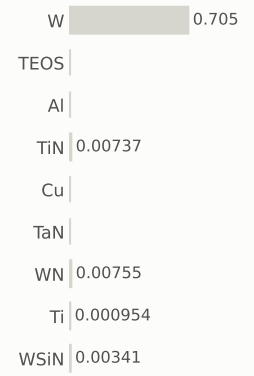
mask for this step

no mask for this step

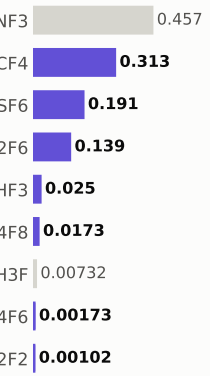
steps by process area



materials deposited (g)



gases emitted (g)



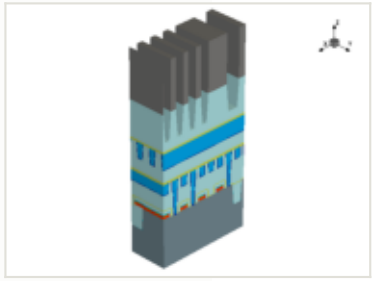
Step #1900: RIE

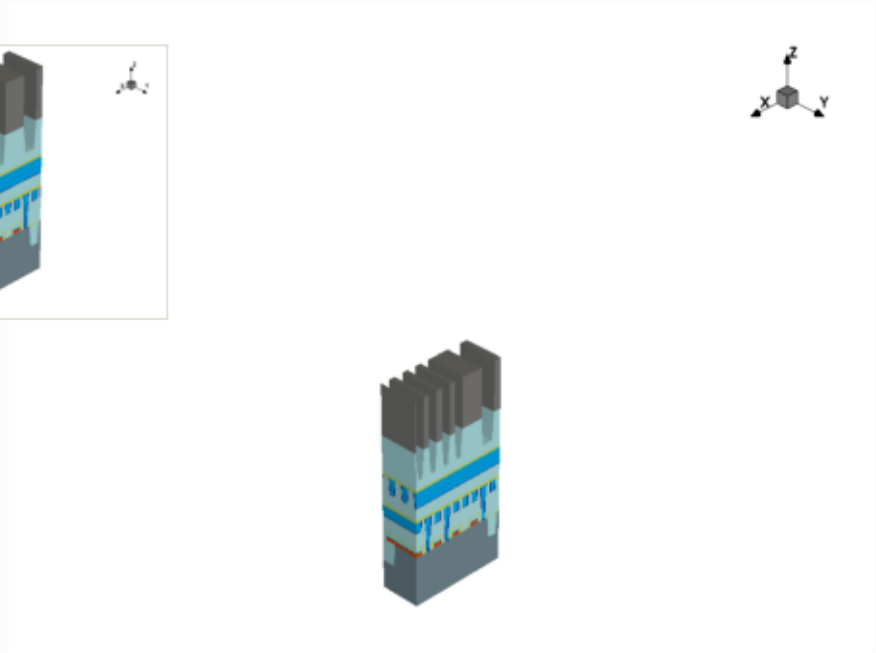
etch_anisotropic · RIE

1.04	149.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



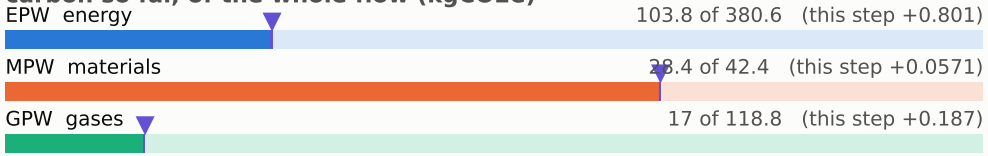


this step

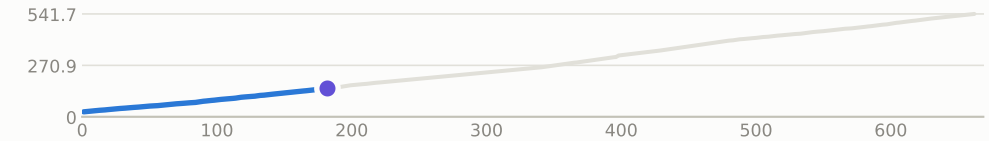
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

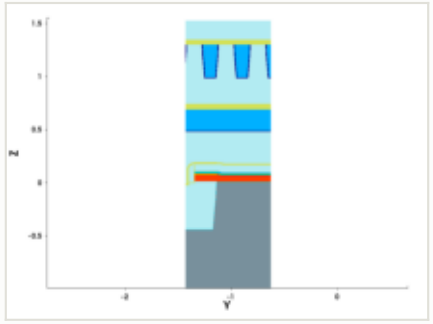
carbon so far, of the whole flow (kgCO2e)



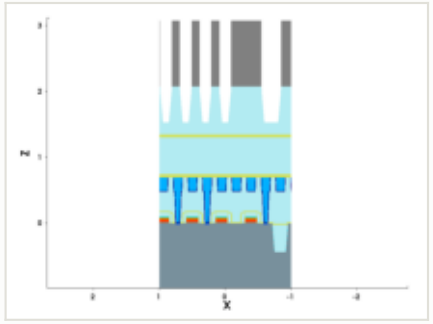
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



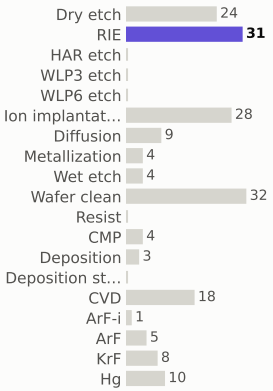
y cut at -1.032 µm



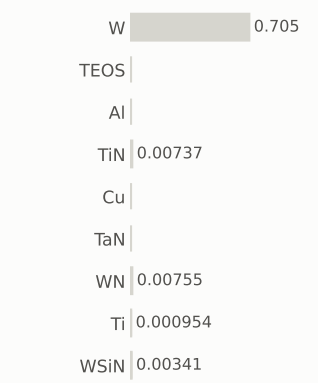
mask for this step

no mask for this step

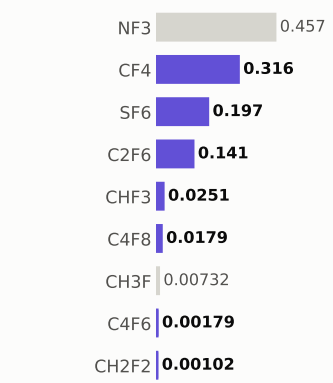
steps by process area



materials deposited (g)



gases emitted (g)



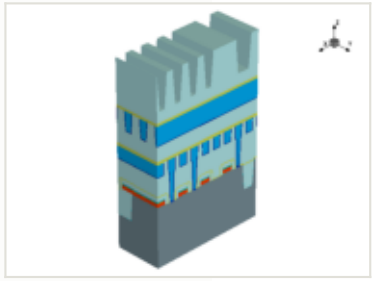
Step #1910: Dry_etch

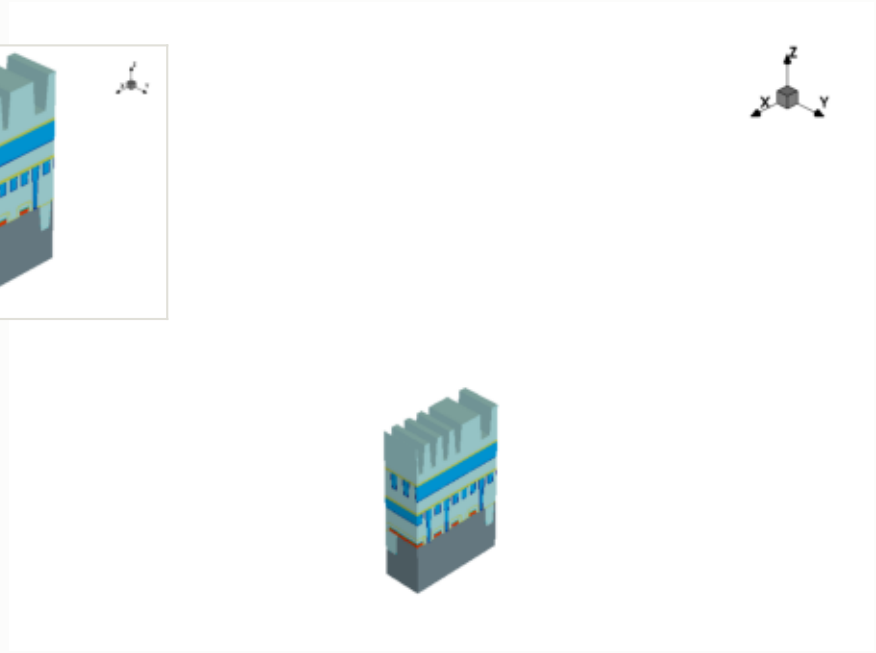
etch_strip · Dry_etch

0.801	149.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

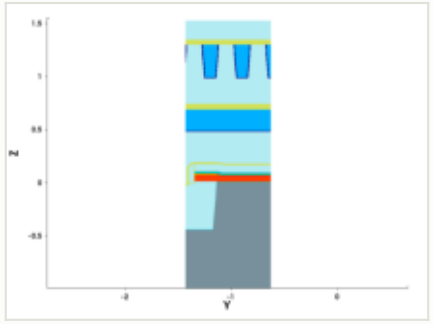
3D structure after this step

zoomed to fit

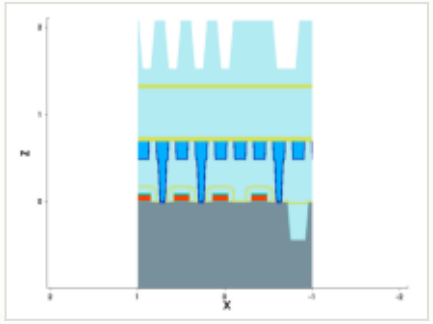




x cut at 0.000 µm



y cut at -1.032 µm



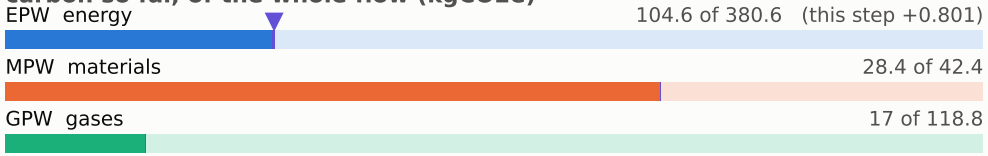
mask for this step

no mask for this step

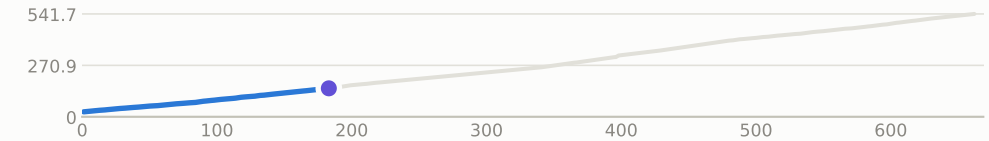
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

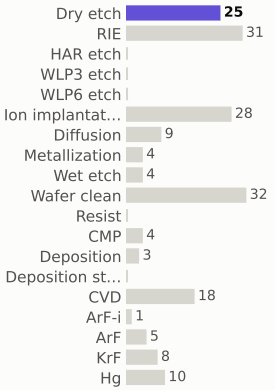
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

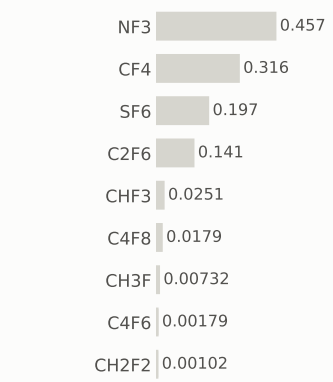


steps by process area



materials deposited (g)

gases emitted (g)



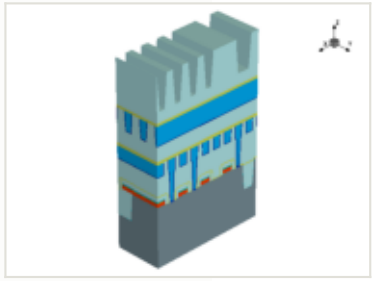
Step #1920: Wafer_clean

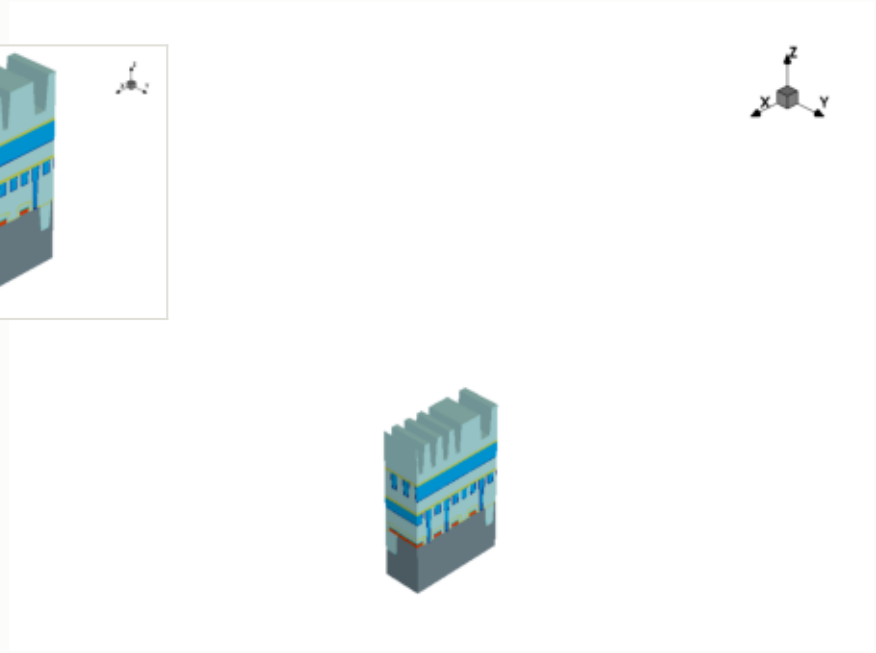
clean · Wafer_clean

0.169	150.1	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

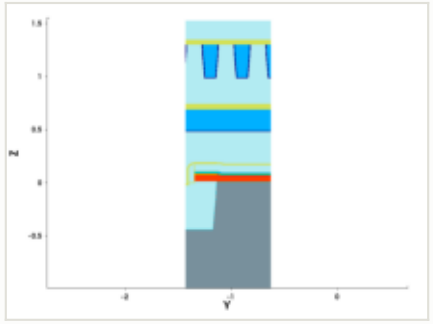
3D structure after this step

zoomed to fit

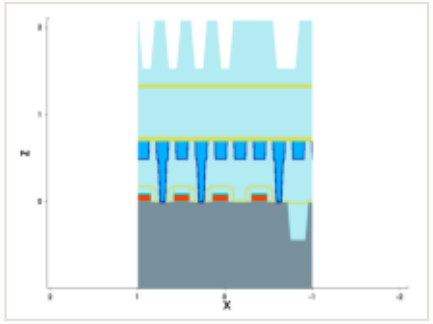




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

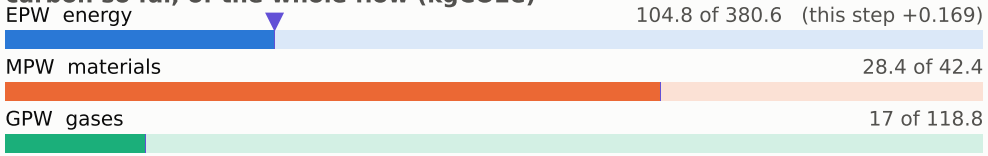
no mask for this step

this step

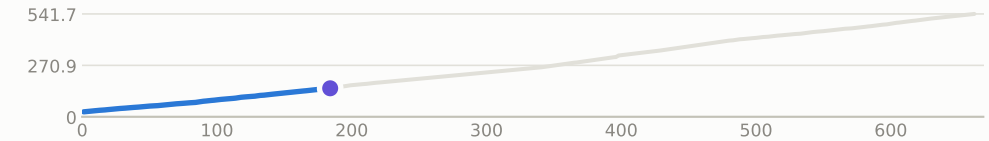
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

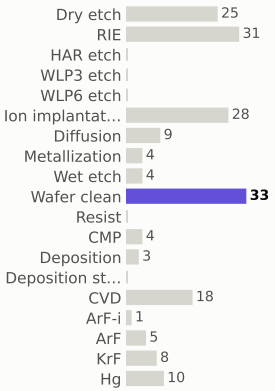
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

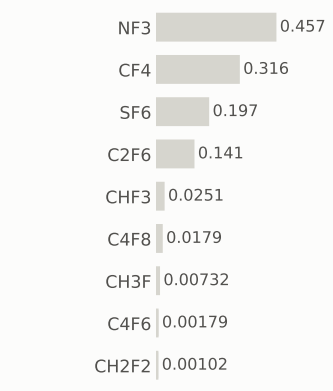


steps by process area



materials deposited (g)

gases emitted (g)



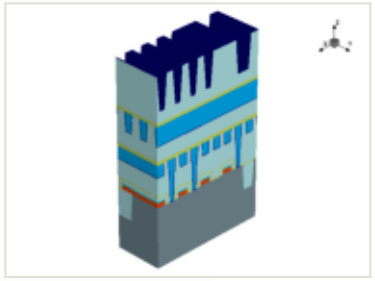
Step #1930: CVD

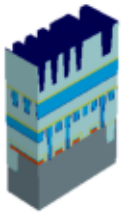
depo_isotropic · CVD

1.31	151.4	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

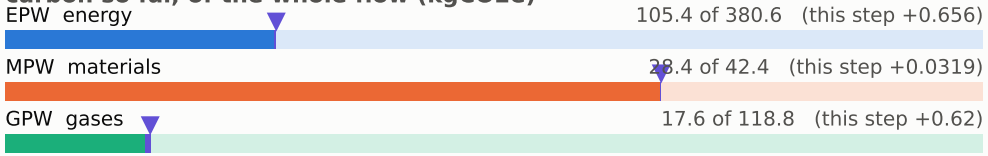




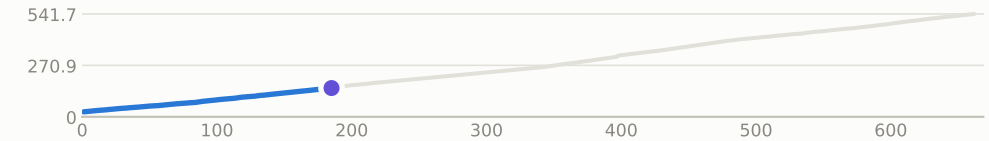
this step

process area	CVD	EPW	0.656 kgCO2e
material	TiN	EPW tool	0.469 kgCO2e
thickness	0.01 µm	EPW overhead	0.187 kgCO2e
material GWP	43.9778 gCO2e/g	MPW	0.0319 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

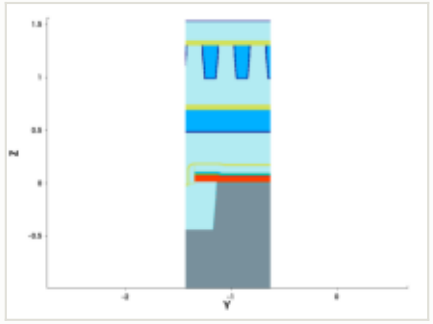
carbon so far, of the whole flow (kgCO2e)



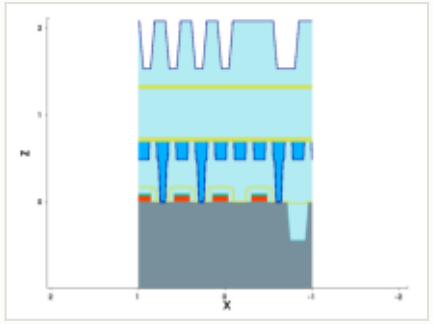
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



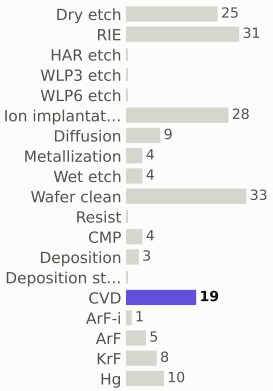
y cut at -1.032 µm



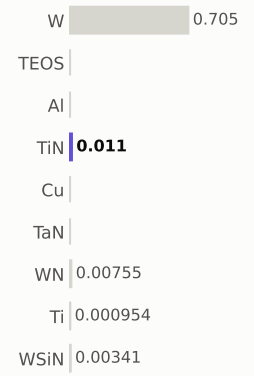
mask for this step

no mask for this step

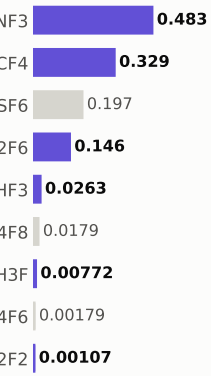
steps by process area



materials deposited (g)



gases emitted (g)



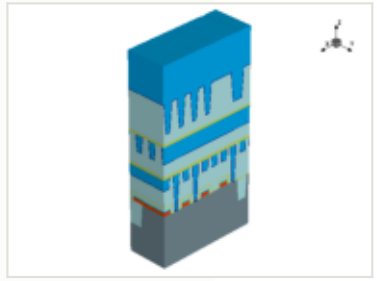
Step #1940: Metallization

depo_isotropic · Metallization

1.28	152.7	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

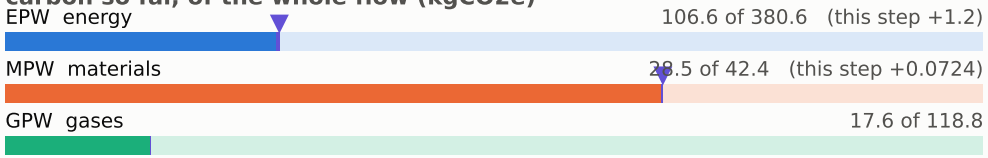


this step

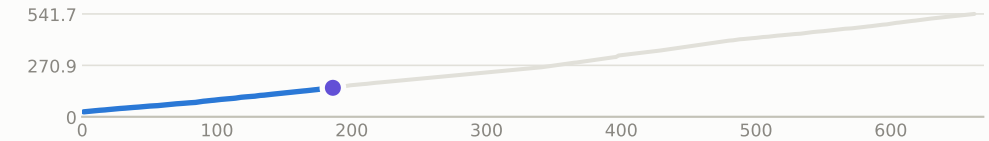
process area	Metallization
material	W
thickness	0.7 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0724 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.28 kgCO2e**

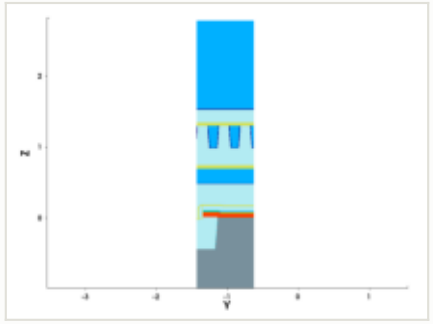
carbon so far, of the whole flow (kgCO2e)



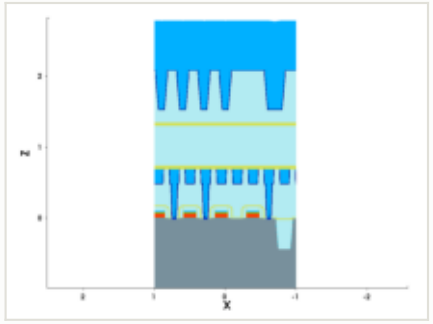
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



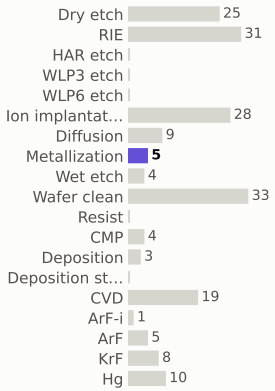
y cut at -1.032 µm



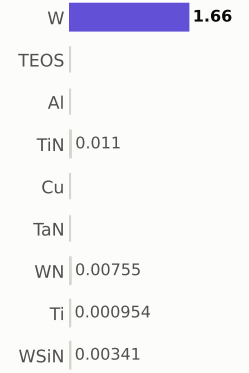
mask for this step

no mask for this step

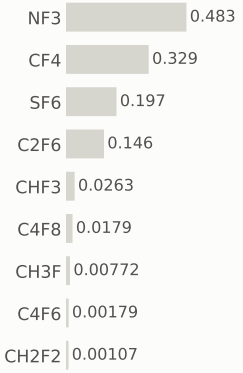
steps by process area



materials deposited (g)



gases emitted (g)



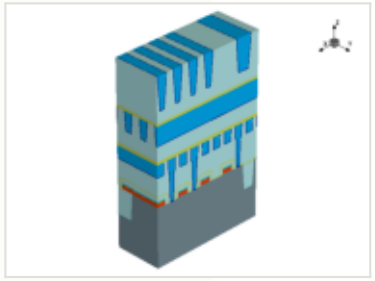
Step #1950: CMP

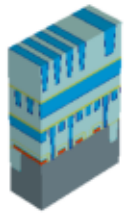
etch_cmp · CMP

0.169	152.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

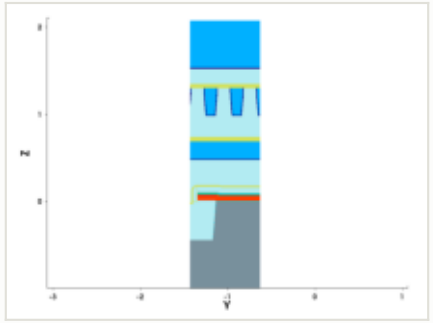
3D structure after this step

zoomed to fit

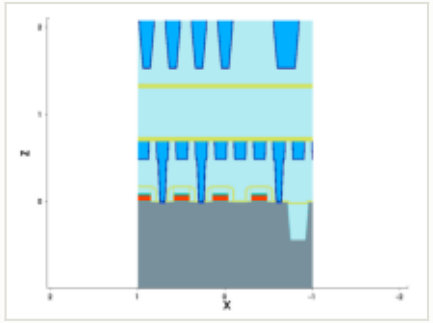




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

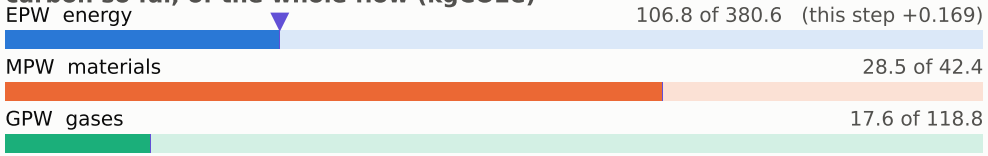
no mask for this step

this step

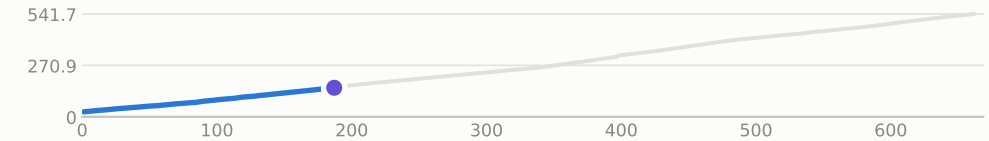
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

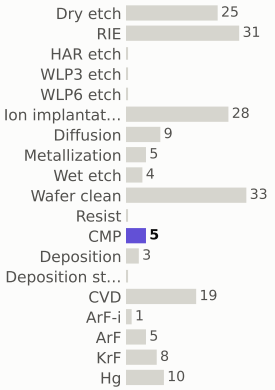
carbon so far, of the whole flow (kgCO2e)



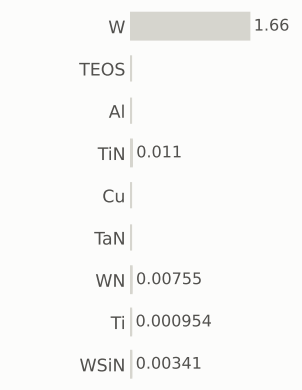
embodied carbon up to each step (kgCO2e per wafer)



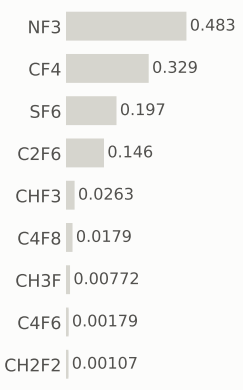
steps by process area



materials deposited (g)



gases emitted (g)



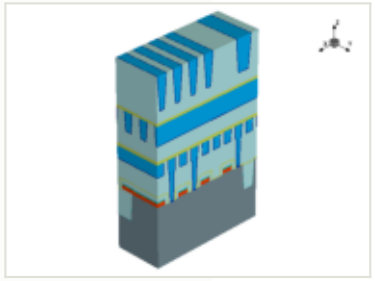
Step #1960: Wafer_clean

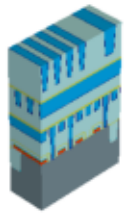
clean · Wafer_clean

0.169	153.0	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

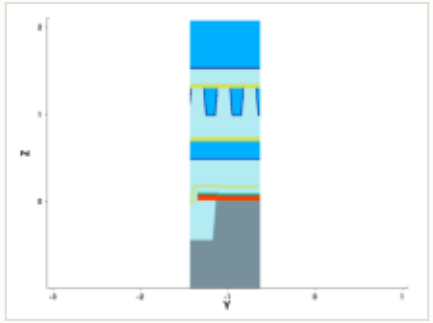
3D structure after this step

zoomed to fit

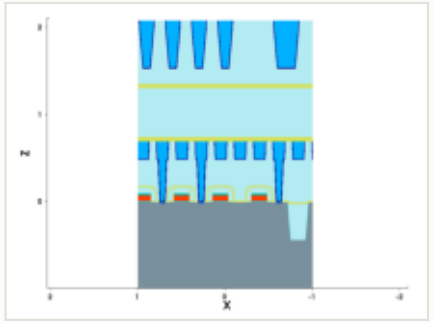




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

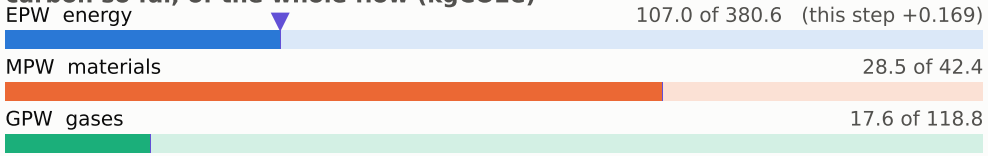
no mask for this step

this step

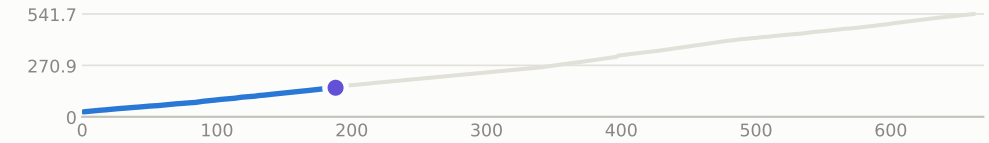
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

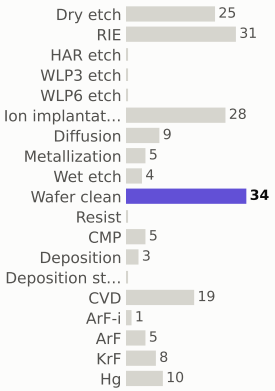
carbon so far, of the whole flow (kgCO2e)



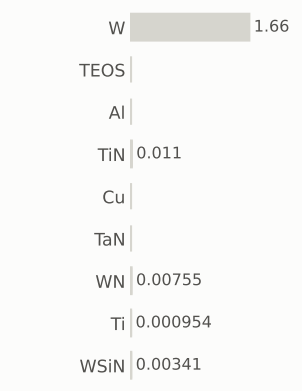
embodied carbon up to each step (kgCO2e per wafer)



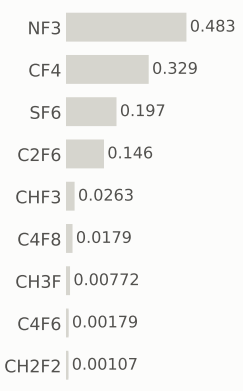
steps by process area



materials deposited (g)



gases emitted (g)



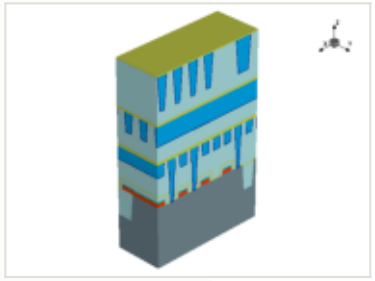
Step #1970: CVD

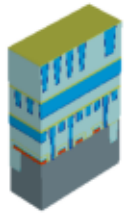
depo_isotropic · CVD

1.31	154.3	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



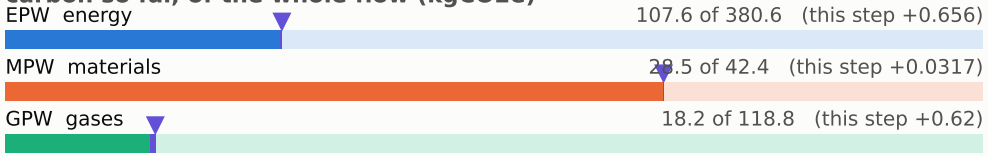


this step

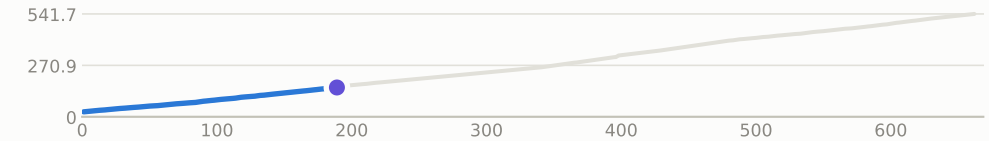
process area	CVD
material	—
thickness	0.019 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

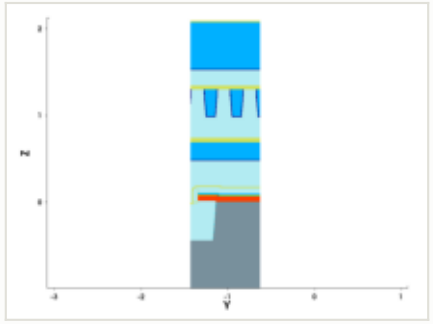
carbon so far, of the whole flow (kgCO2e)



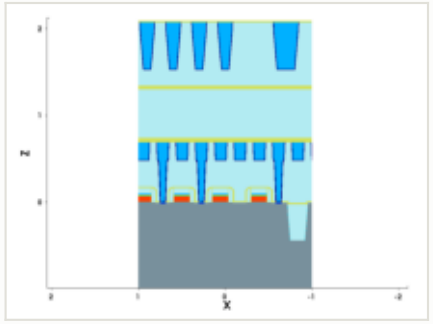
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



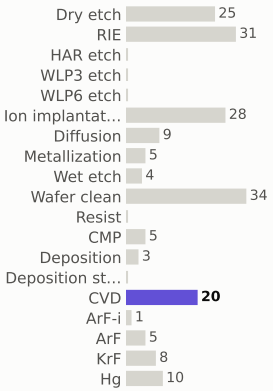
y cut at -1.032 µm



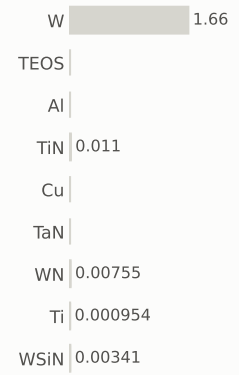
mask for this step

no mask for this step

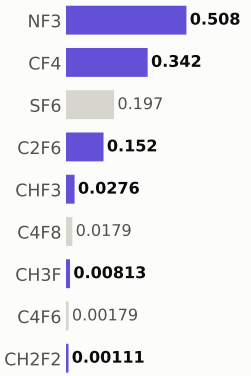
steps by process area



materials deposited (g)



gases emitted (g)



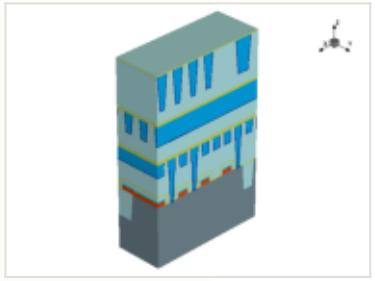
Step #1980: CVD

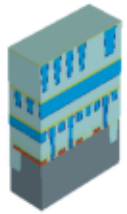
depo_isotropic · CVD

1.31	155.7	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

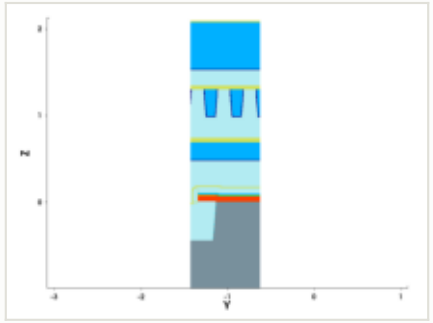
3D structure after this step

zoomed to fit

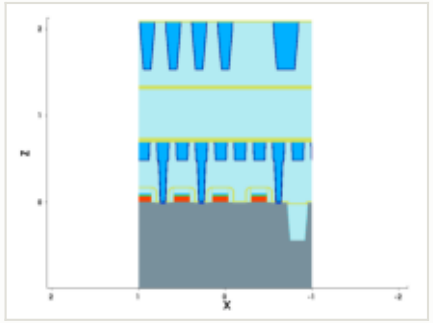




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

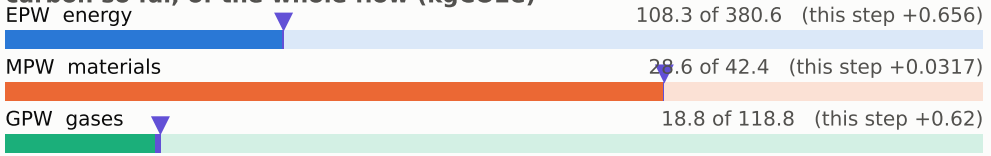
no mask for this step

this step

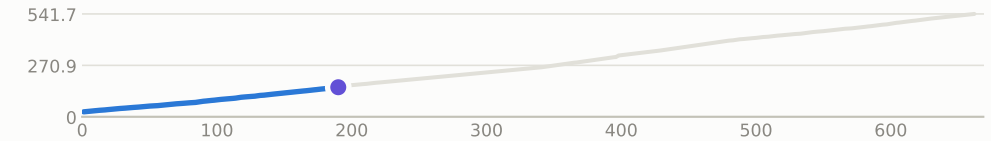
process area	CVD
material	—
thickness	0.002 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

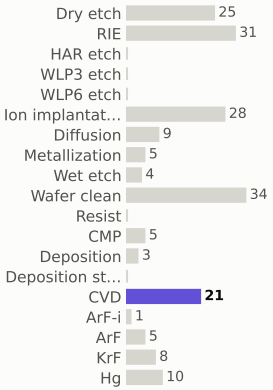
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

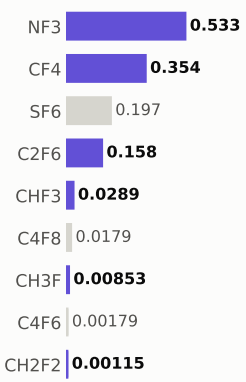


steps by process area



materials deposited (g)

gases emitted (g)



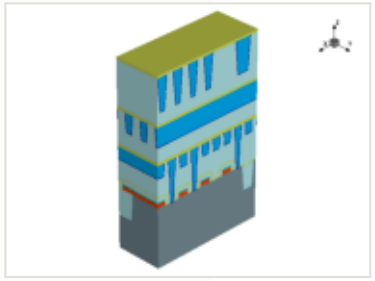
Step #1990: CVD

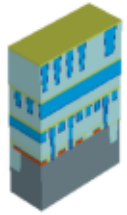
depo_isotropic · CVD

1.31	157.0	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



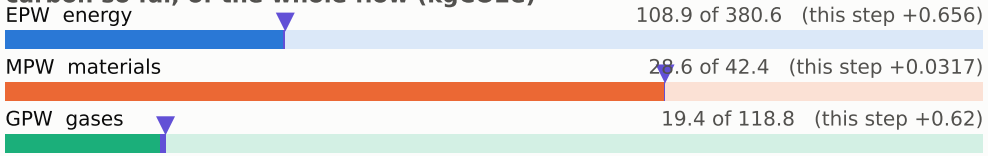


this step

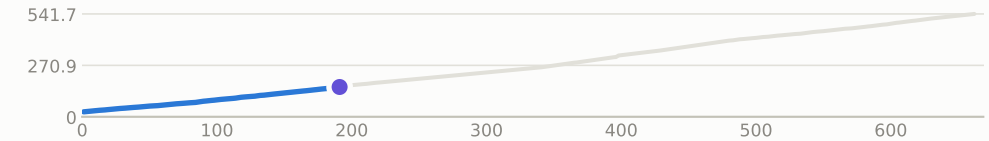
process area	CVD
material	—
thickness	0.019 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

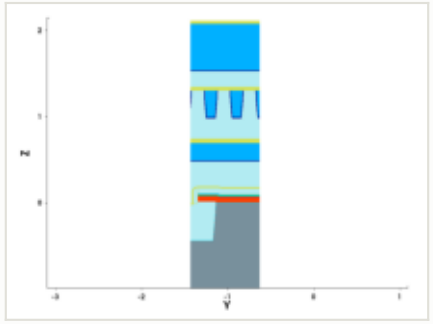
carbon so far, of the whole flow (kgCO2e)



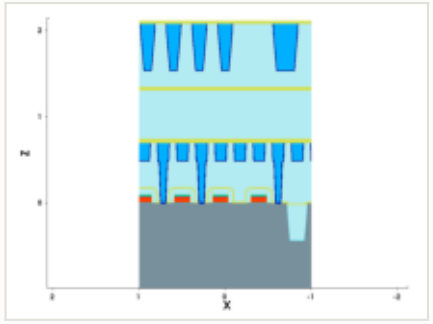
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



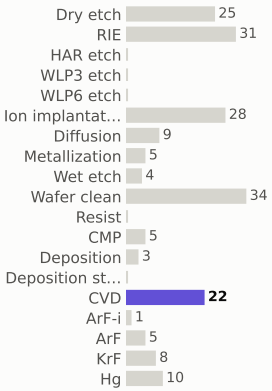
y cut at -1.032 µm



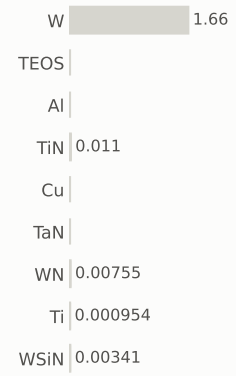
mask for this step

no mask for this step

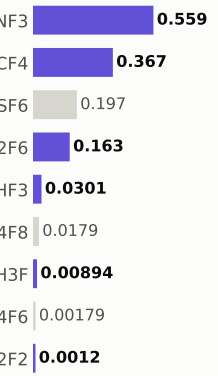
steps by process area



materials deposited (g)



gases emitted (g)



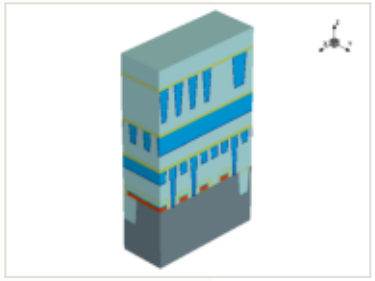
Step #2000: CVD

depo_isotropic · CVD

1.31	158.3	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

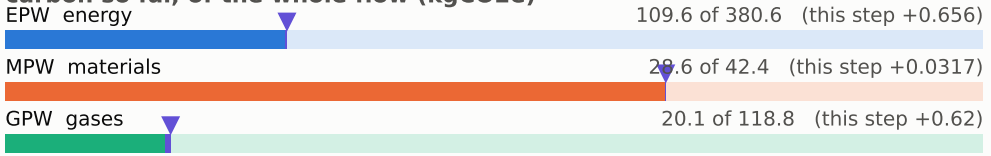


this step

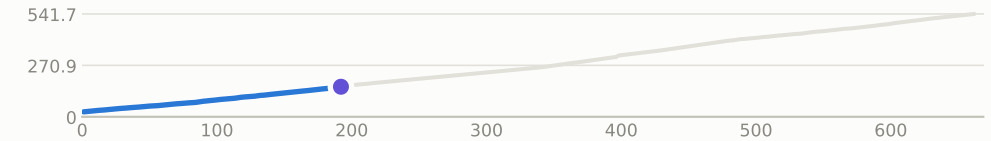
process area	CVD
material	—
thickness	0.31 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

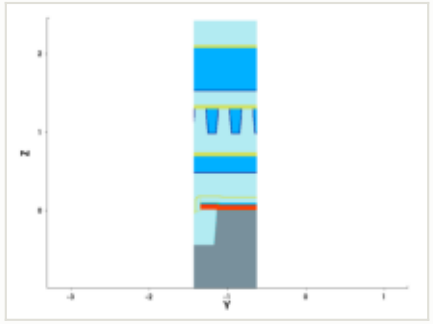
carbon so far, of the whole flow (kgCO2e)



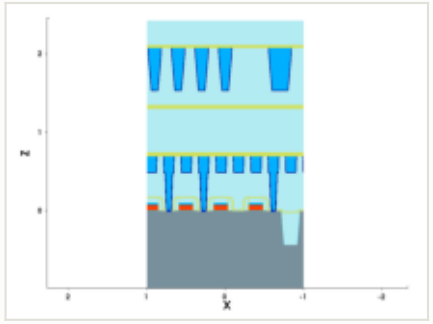
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



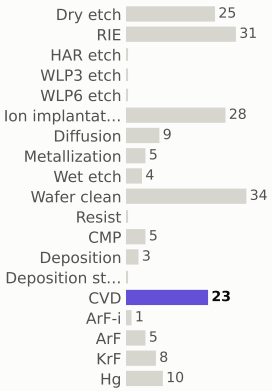
y cut at -1.032 µm



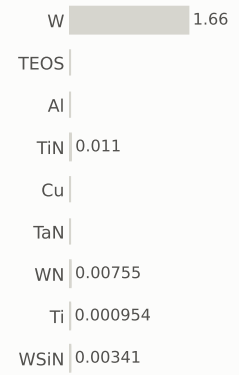
mask for this step

no mask for this step

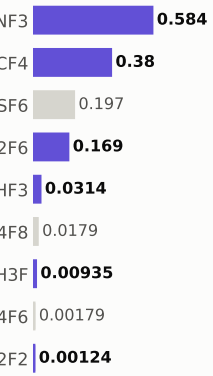
steps by process area



materials deposited (g)



gases emitted (g)



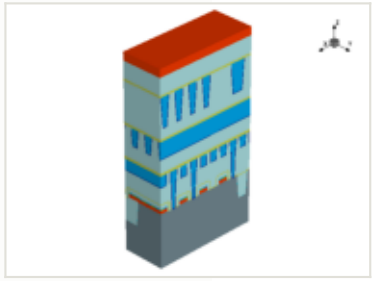
Step #2020: CVD

depo_isotropic · CVD

1.31	159.6	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

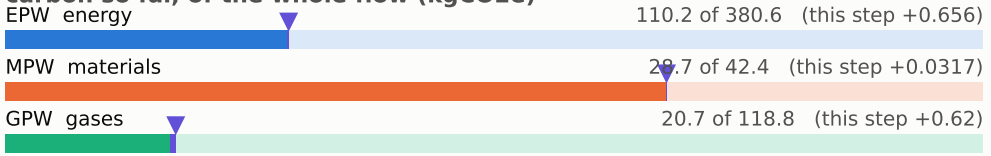


this step

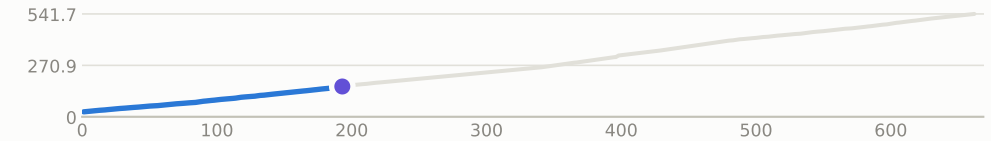
process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

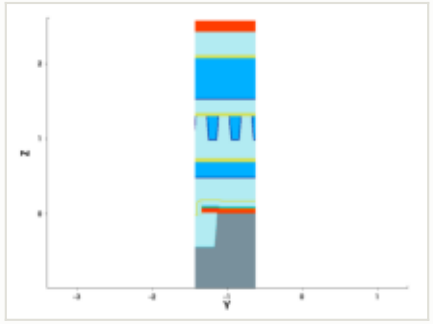
carbon so far, of the whole flow (kgCO2e)



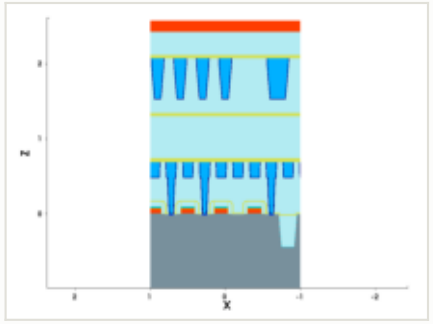
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



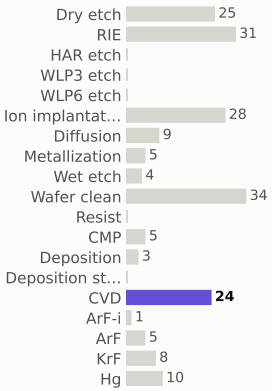
y cut at -1.032 µm



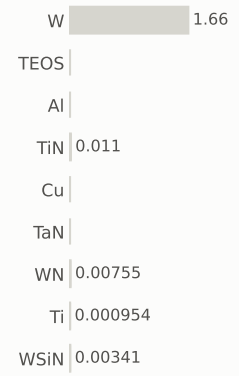
mask for this step

no mask for this step

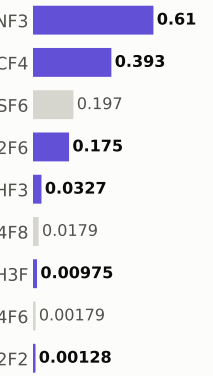
steps by process area



materials deposited (g)



gases emitted (g)



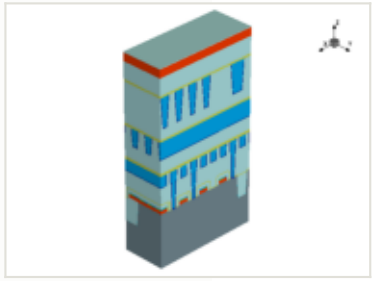
Step #2030: CVD

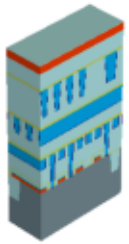
depo_isotropic · CVD

1.31	160.9	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

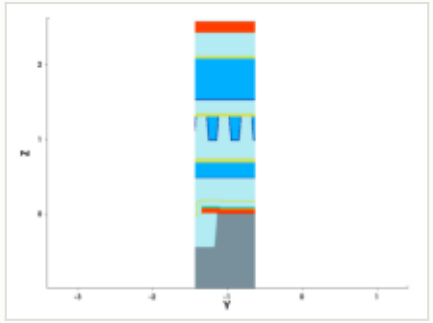
3D structure after this step

zoomed to fit

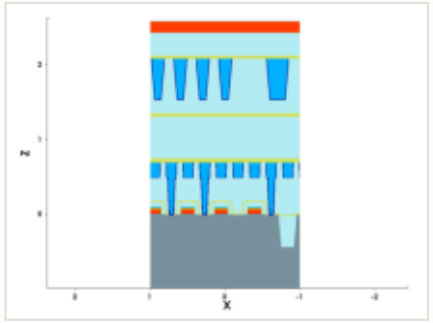




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

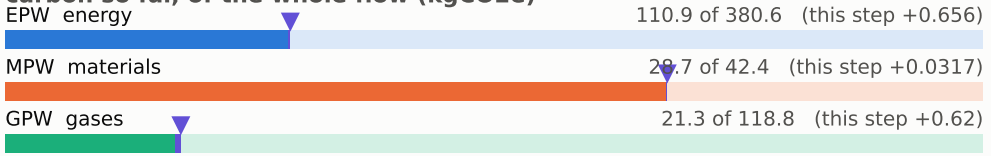
no mask for this step

this step

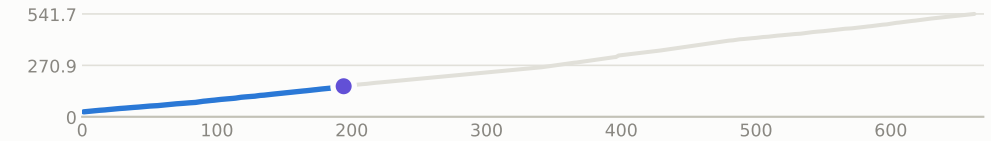
process area	CVD
material	—
thickness	0.012 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

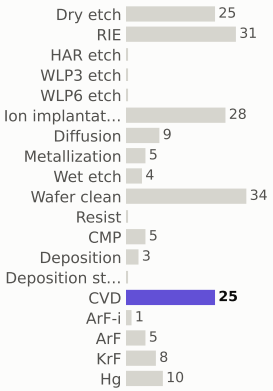
carbon so far, of the whole flow (kgCO2e)



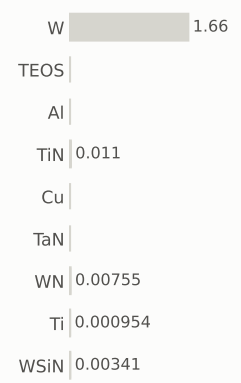
embodied carbon up to each step (kgCO2e per wafer)



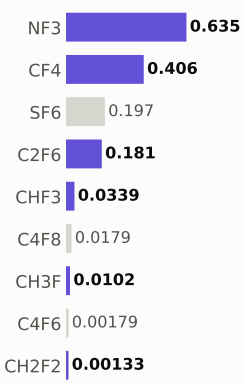
steps by process area



materials deposited (g)



gases emitted (g)



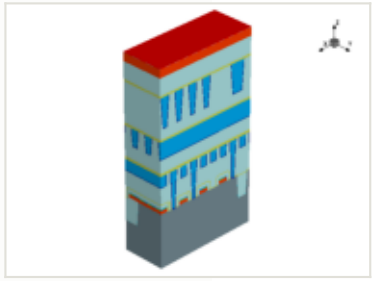
Step #2040: Diffusion

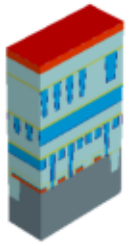
implant_ellipsoid · Diffusion

0.801	161.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

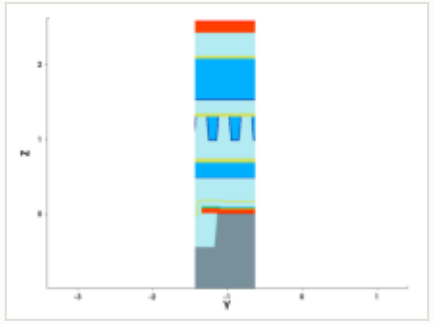
3D structure after this step

zoomed to fit

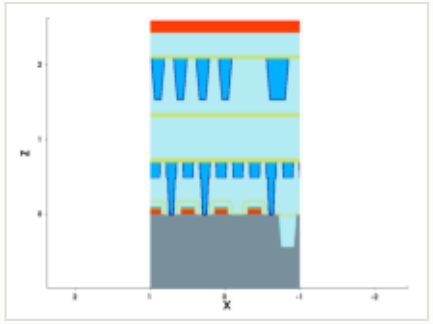




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

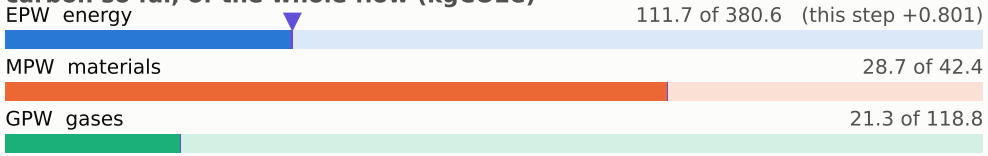
no mask for this step

this step

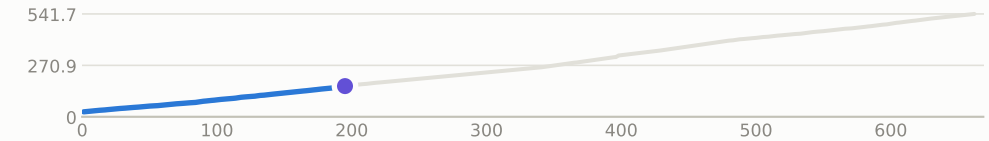
process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

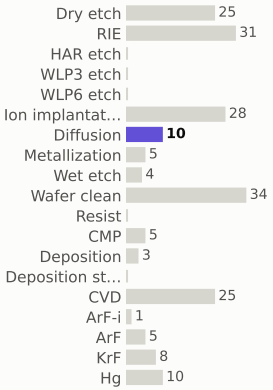
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

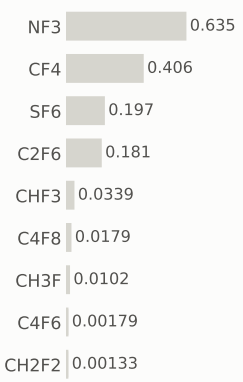


steps by process area



materials deposited (g)

gases emitted (g)



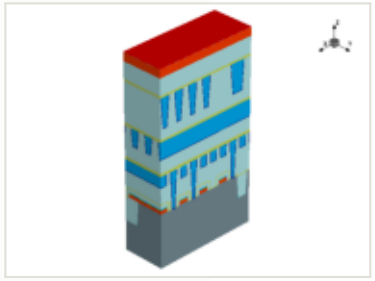
Step #2050: Diffusion

anneal · Diffusion

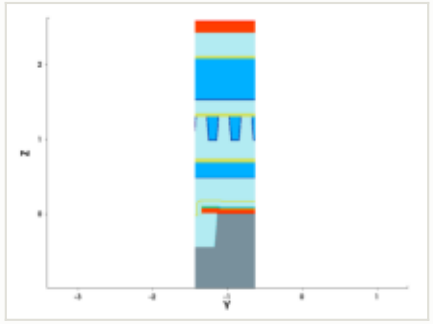
0.801	162.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

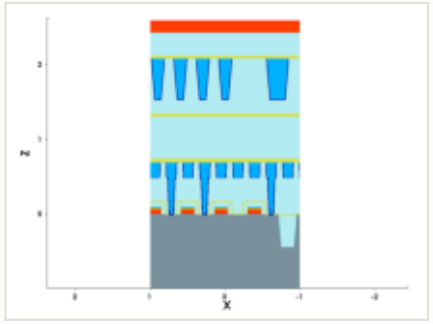
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

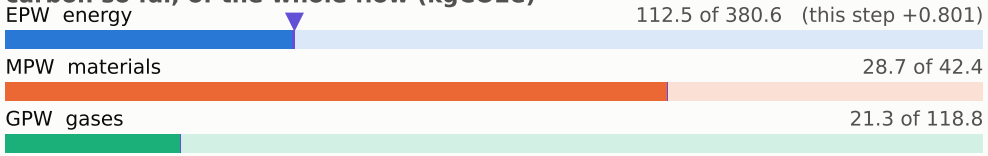
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

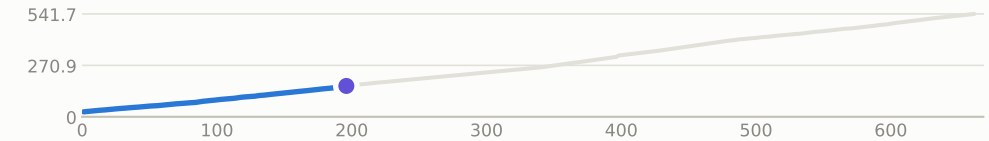
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

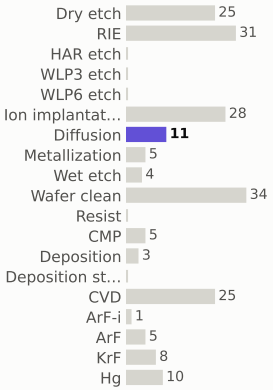
carbon so far, of the whole flow (kgCO2e)



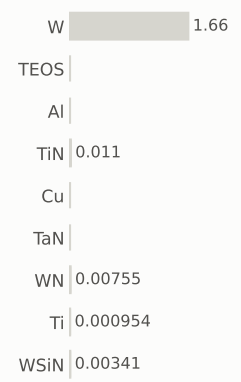
embodied carbon up to each step (kgCO2e per wafer)



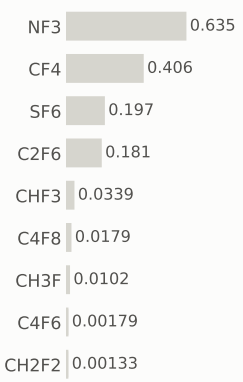
steps by process area



materials deposited (g)



gases emitted (g)



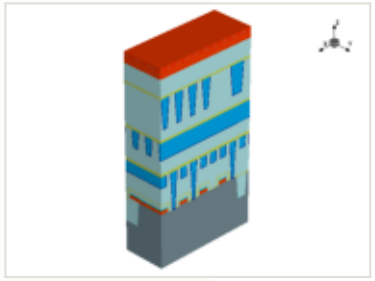
Step #2060: CVD

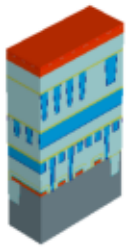
depo_isotropic · CVD

1.31	163.8	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



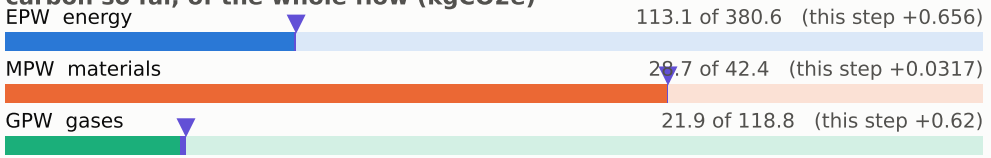


this step

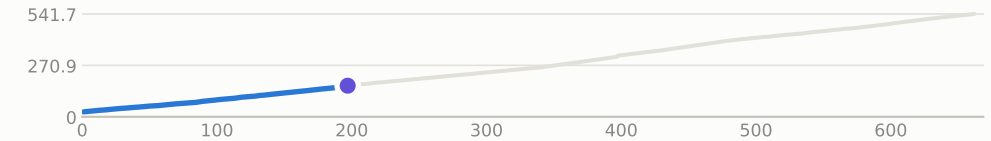
process area	CVD
material	—
thickness	0.036 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

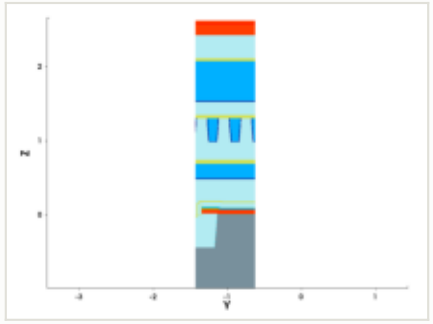
carbon so far, of the whole flow (kgCO2e)



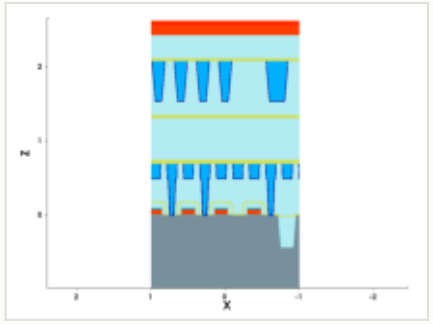
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



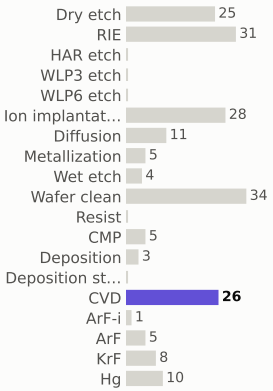
y cut at -1.032 µm



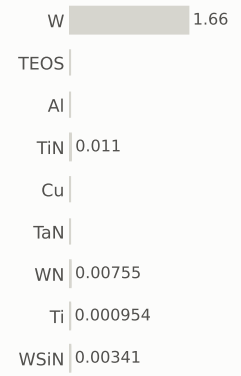
mask for this step

no mask for this step

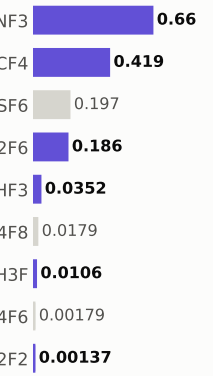
steps by process area



materials deposited (g)



gases emitted (g)



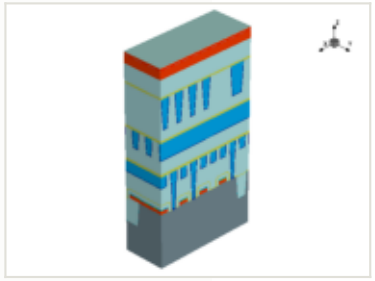
Step #2070: CVD

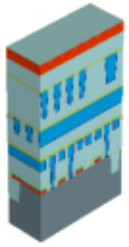
depo_isotropic · CVD

1.31	165.1	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

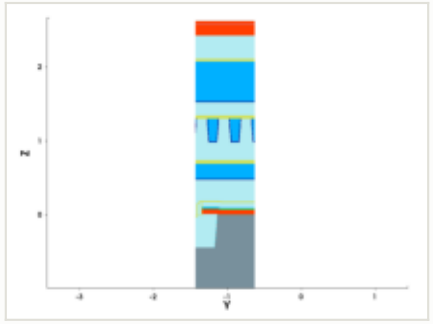
3D structure after this step

zoomed to fit

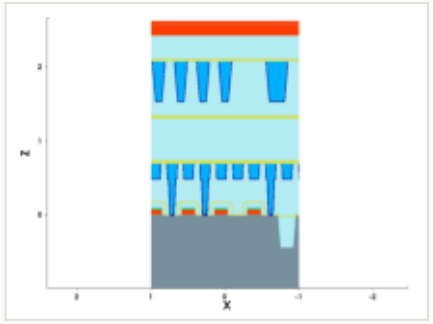




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

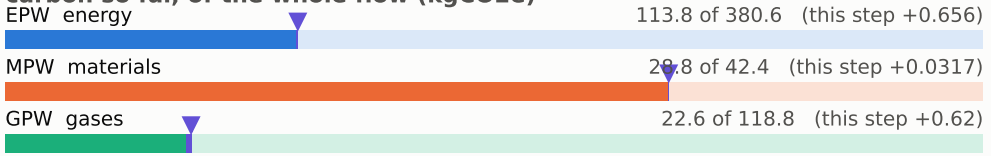
no mask for this step

this step

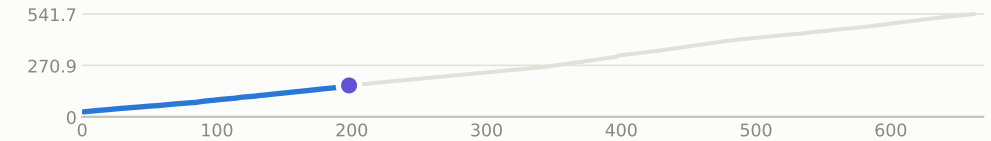
process area	CVD
material	—
thickness	0.007 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

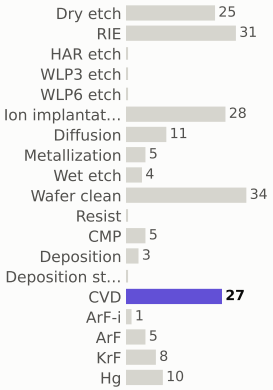
carbon so far, of the whole flow (kgCO2e)



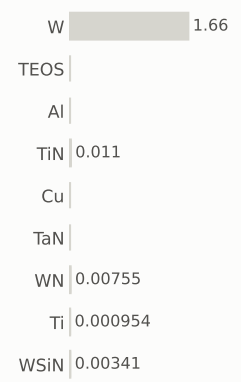
embodied carbon up to each step (kgCO2e per wafer)



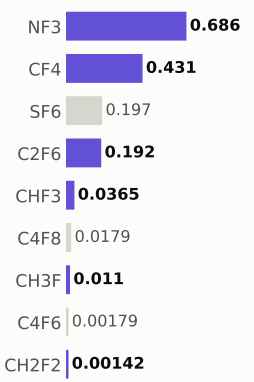
steps by process area



materials deposited (g)



gases emitted (g)



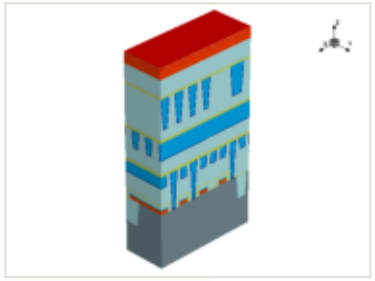
Step #2080: Diffusion

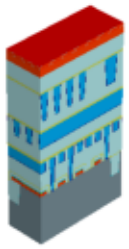
implant_ellipsoid · Diffusion

0.801	165.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

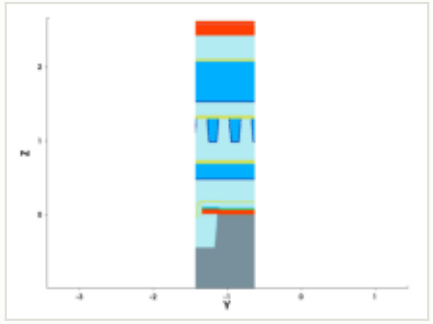
3D structure after this step

zoomed to fit

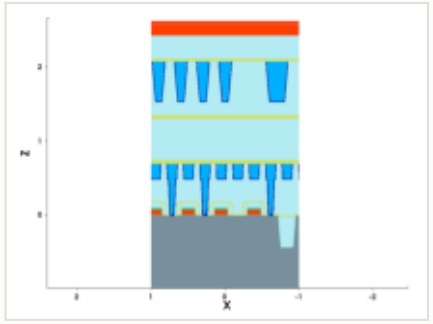




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

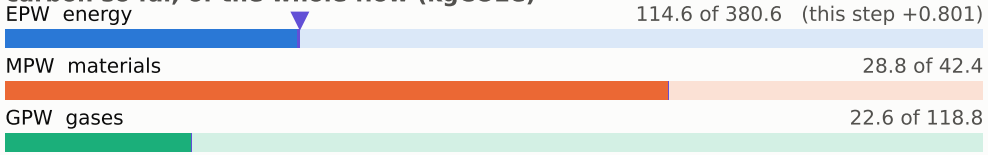
no mask for this step

this step

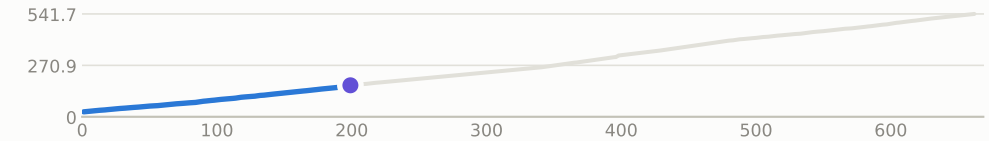
process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

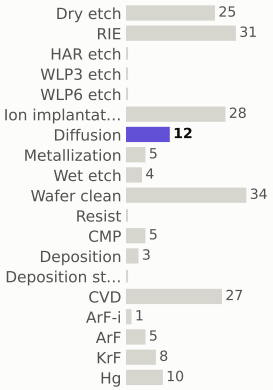
carbon so far, of the whole flow (kgCO2e)



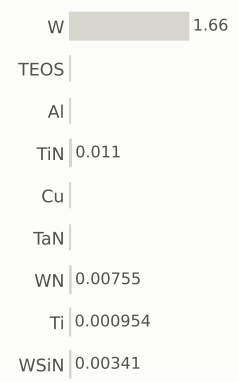
embodied carbon up to each step (kgCO2e per wafer)



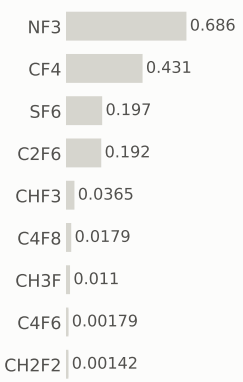
steps by process area



materials deposited (g)



gases emitted (g)



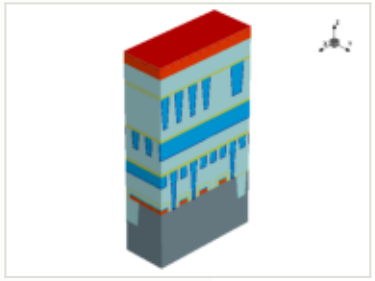
Step #2090: Diffusion

anneal · Diffusion

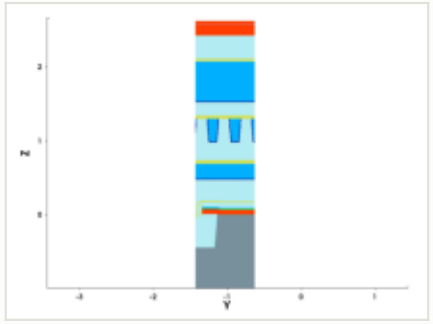
0.801	166.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

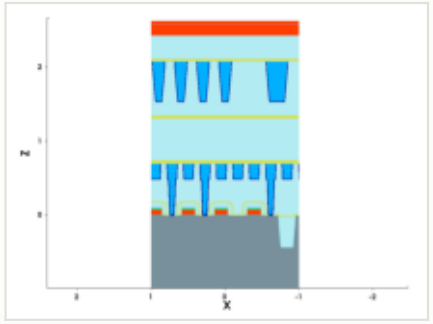
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

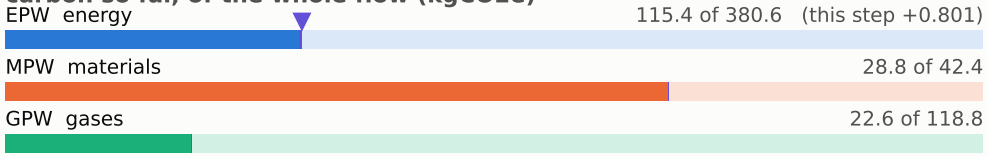
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

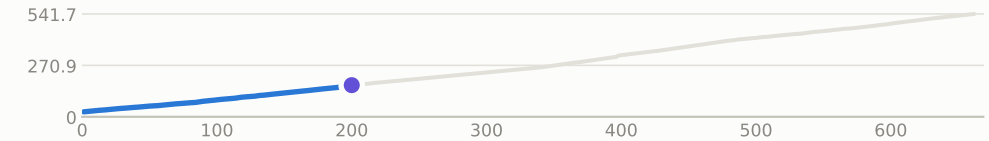
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

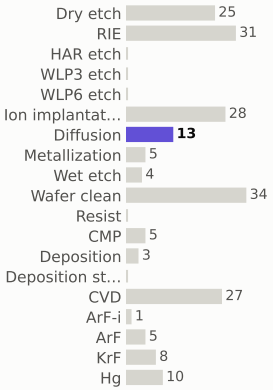
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

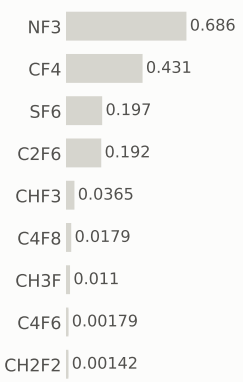


steps by process area



materials deposited (g)

gases emitted (g)



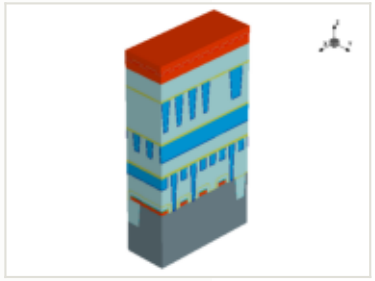
Step #2100: CVD

depo_isotropic · CVD

1.31	168.0	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

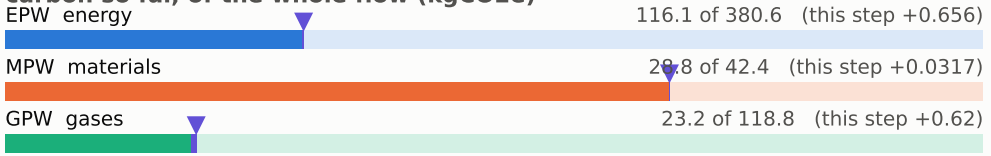


this step

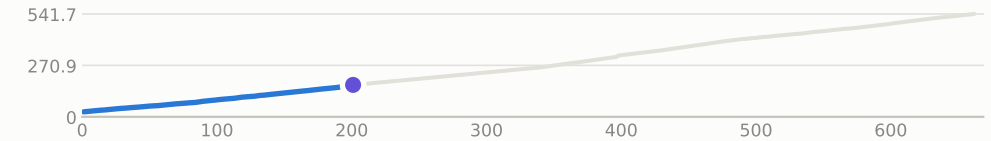
process area	CVD
material	—
thickness	0.105 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

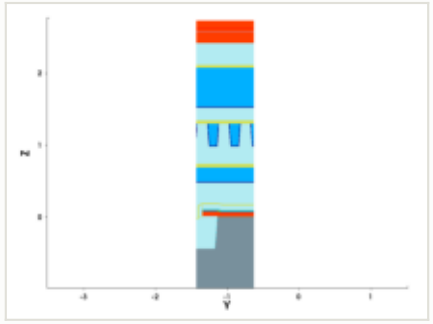
carbon so far, of the whole flow (kgCO2e)



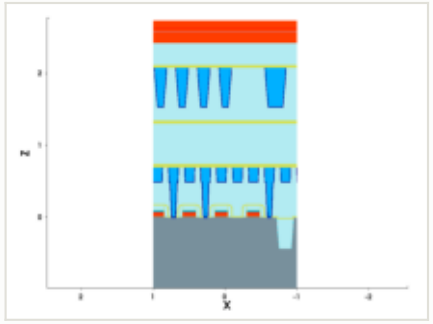
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



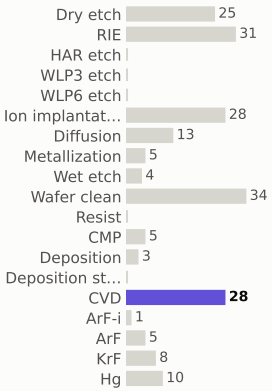
y cut at -1.032 µm



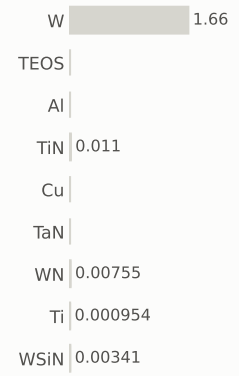
mask for this step

no mask for this step

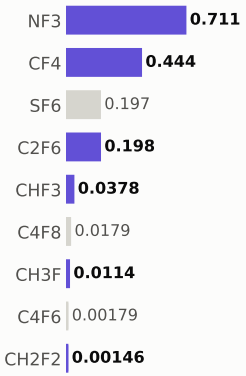
steps by process area



materials deposited (g)



gases emitted (g)



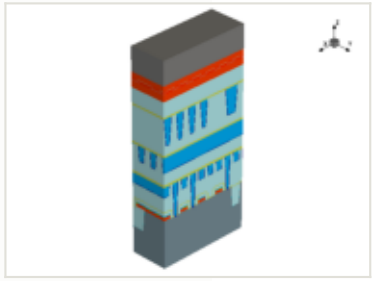
Step #2110: KrF

litho_patterning · KrF

0.199	168.2	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

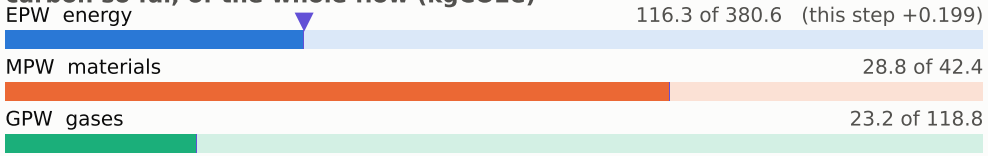
zoomed to fit



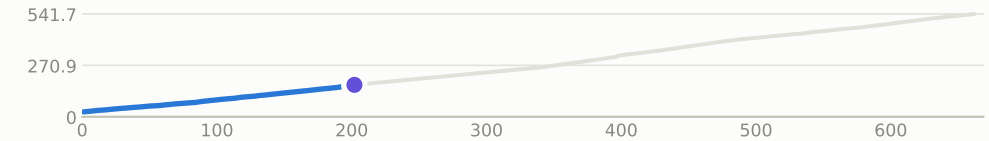
this step

process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

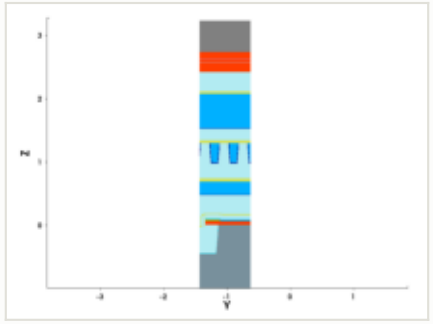
carbon so far, of the whole flow (kgCO2e)



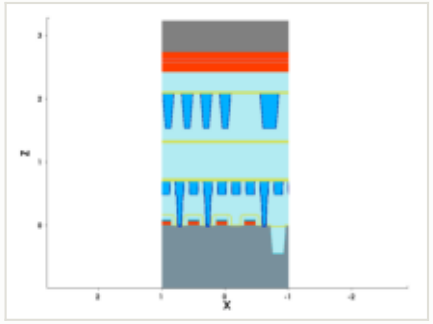
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

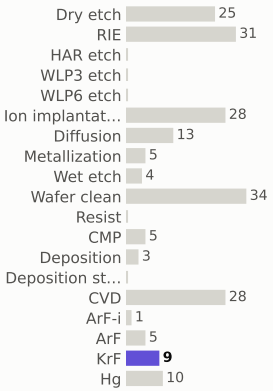


y cut at -1.032 µm

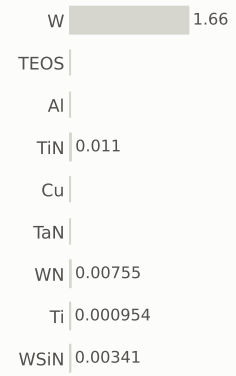


mask: layer 25:0, positive (OASIS)

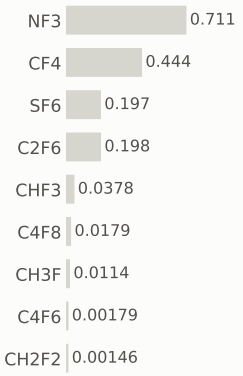
steps by process area



materials deposited (g)



gases emitted (g)



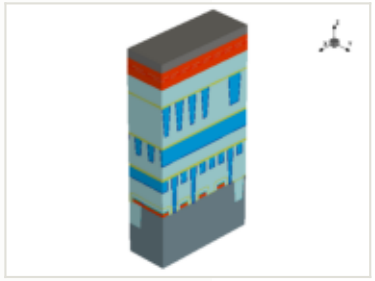
Step #2120: RIE

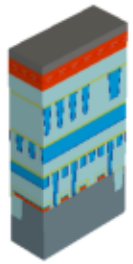
etch_anisotropic · RIE

1.04	169.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



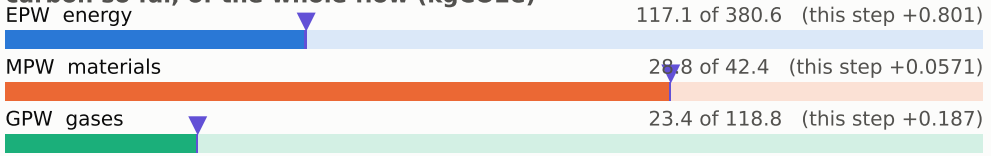


this step

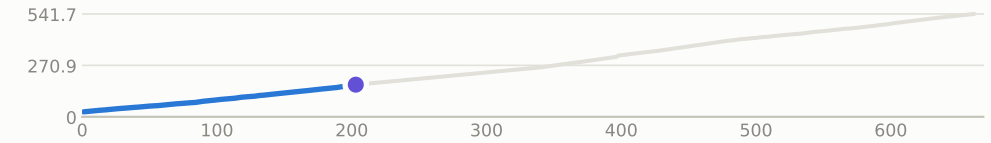
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

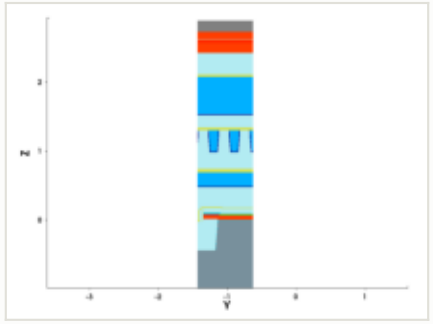
carbon so far, of the whole flow (kgCO2e)



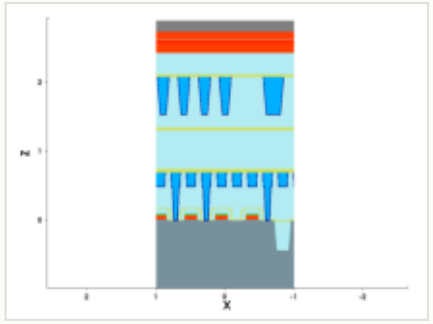
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



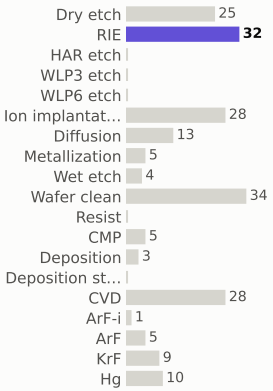
y cut at -1.032 µm



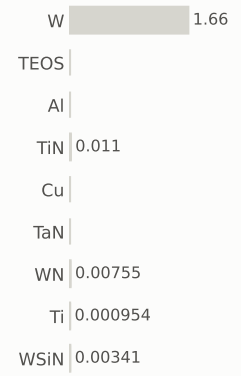
mask for this step

no mask for this step

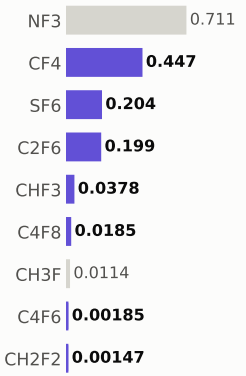
steps by process area



materials deposited (g)



gases emitted (g)



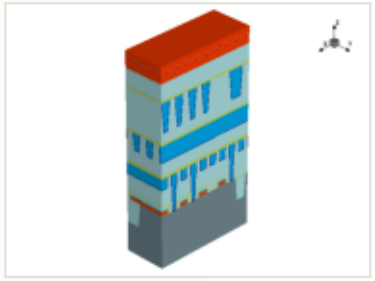
Step #2130: Dry_etch

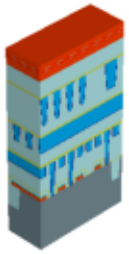
etch_strip · Dry_etch

0.801	170.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

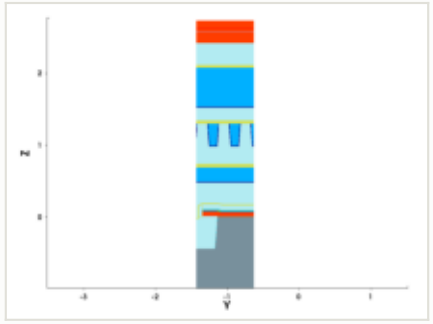
3D structure after this step

zoomed to fit

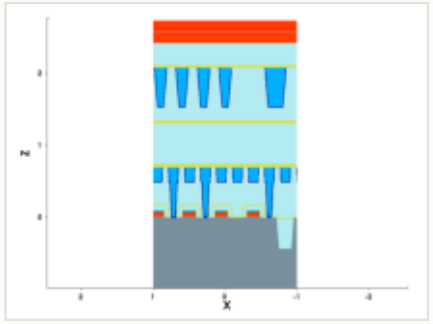




x cut at 0.000 µm



y cut at -1.032 µm



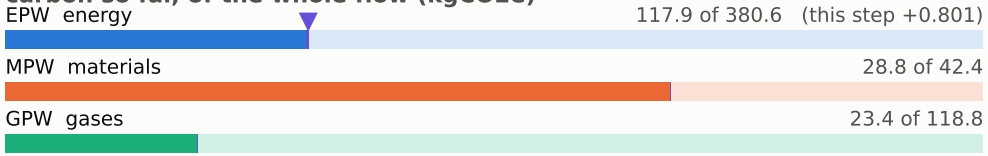
mask for this step

no mask for this step

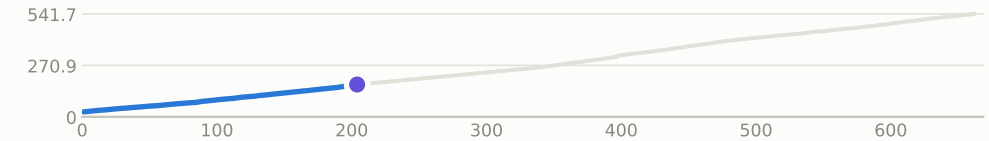
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

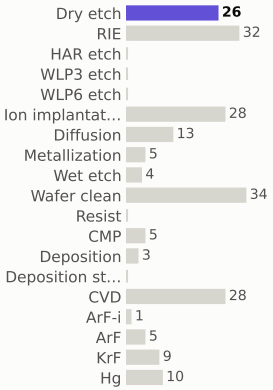
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

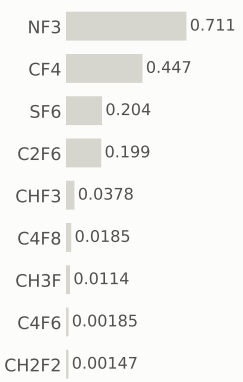


steps by process area



materials deposited (g)

gases emitted (g)



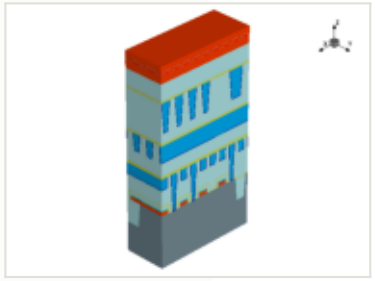
Step #2140: Wafer_clean

clean · Wafer_clean

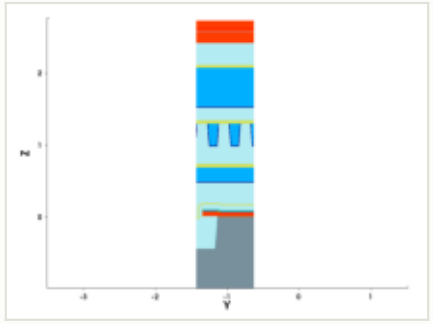
0.169	170.2	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

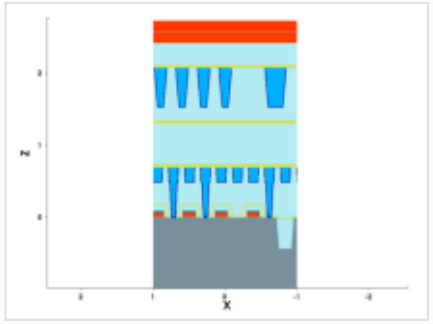
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

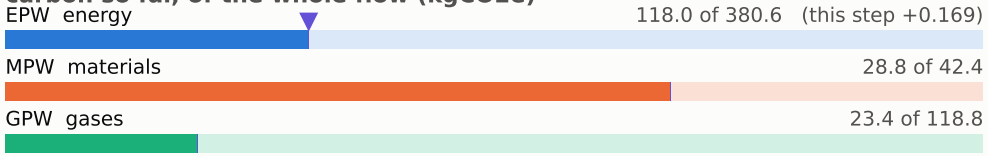
no mask for this step

this step

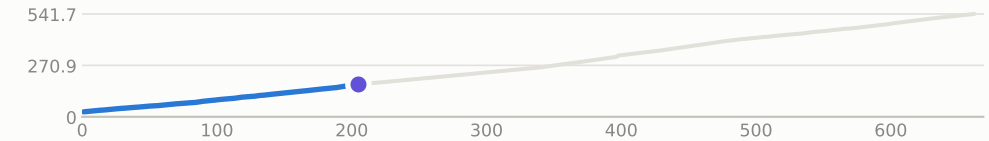
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

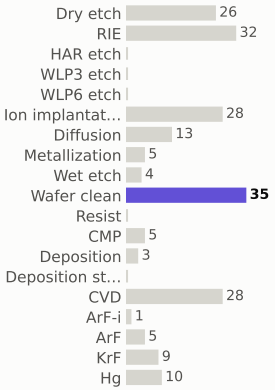
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

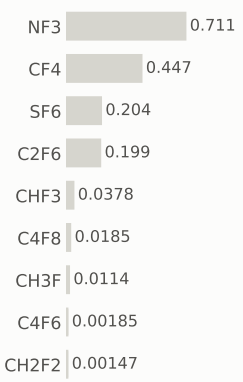


steps by process area



materials deposited (g)

gases emitted (g)



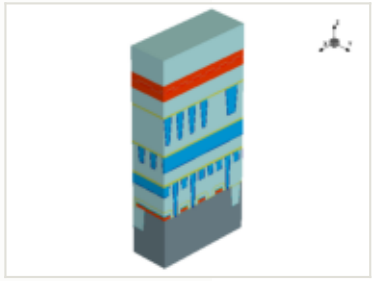
Step #2150: CVD

depo_isotropic · CVD

1.31	171.5	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

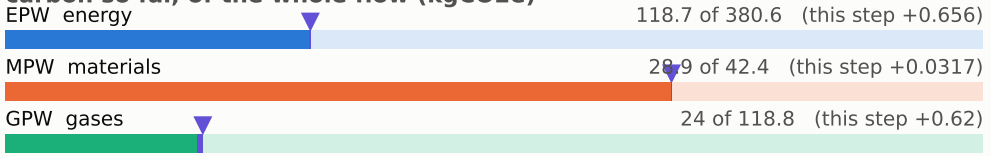


this step

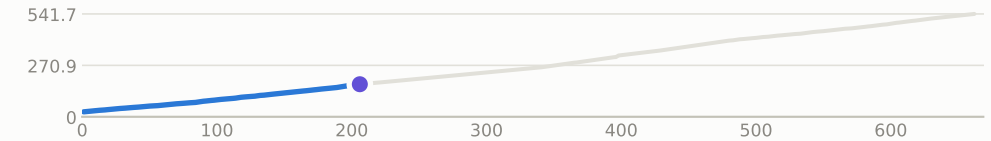
process area	CVD
material	—
thickness	0.5 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

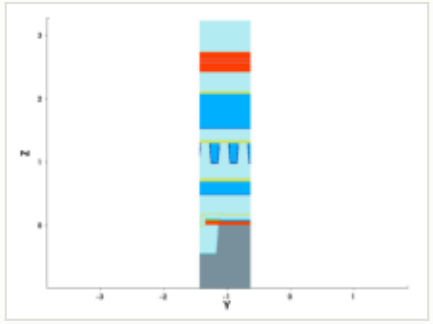
carbon so far, of the whole flow (kgCO2e)



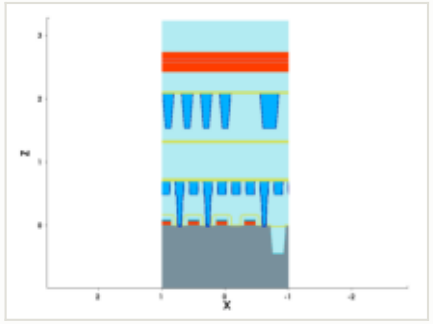
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



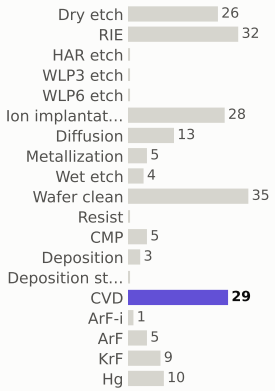
y cut at -1.032 µm



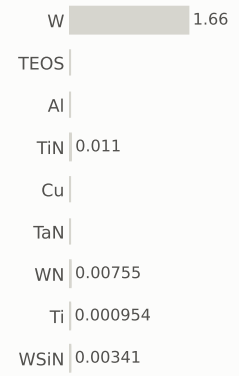
mask for this step

no mask for this step

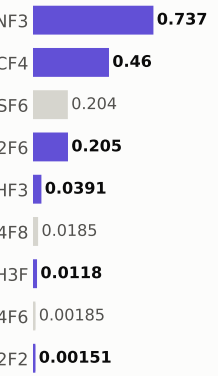
steps by process area



materials deposited (g)



gases emitted (g)



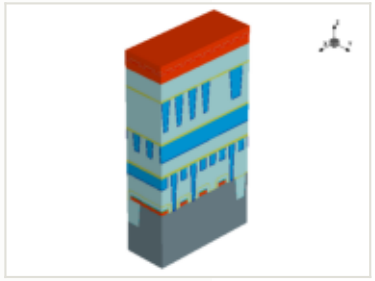
Step #2160: CMP

etch_cmp · CMP

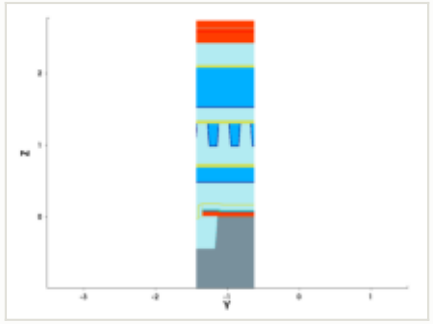
0.169	171.7	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

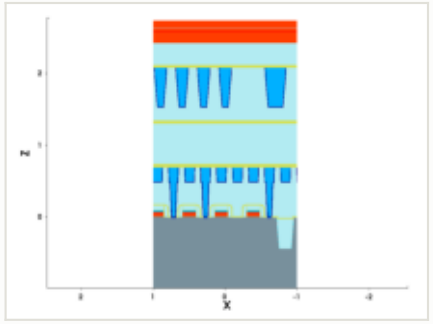
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

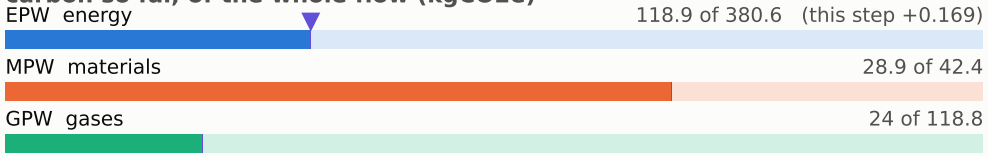
no mask for this step

this step

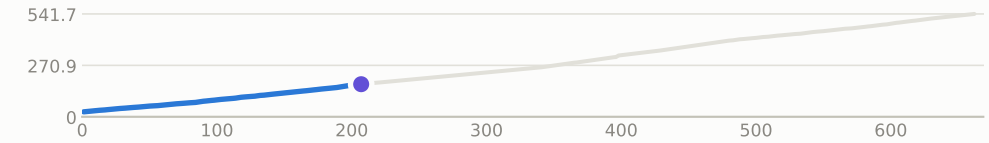
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

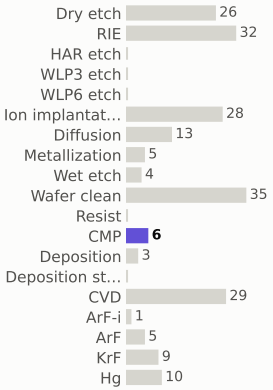
carbon so far, of the whole flow (kgCO2e)



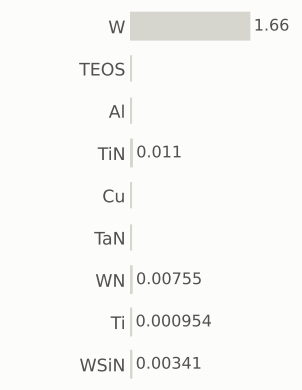
embodied carbon up to each step (kgCO2e per wafer)



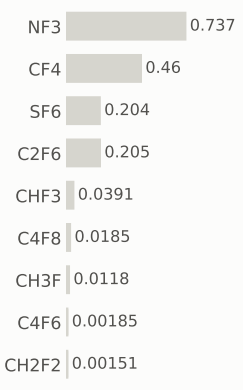
steps by process area



materials deposited (g)



gases emitted (g)



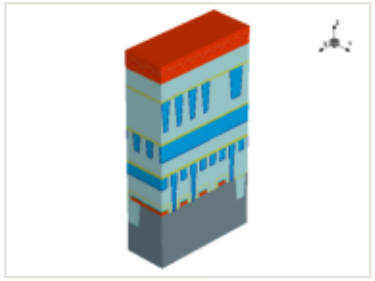
Step #2170: Wafer_clean

clean · Wafer_clean

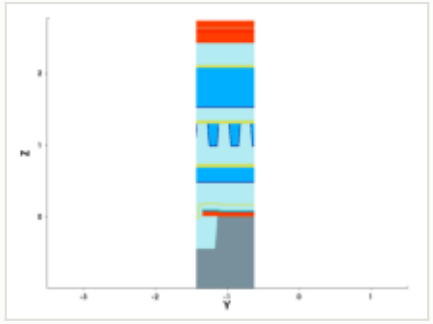
0.169	171.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

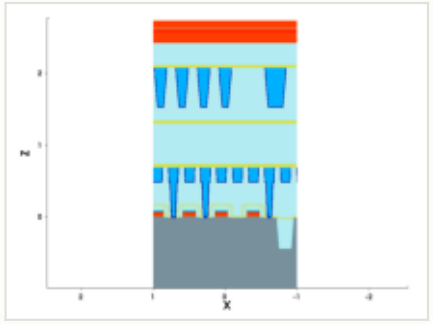
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

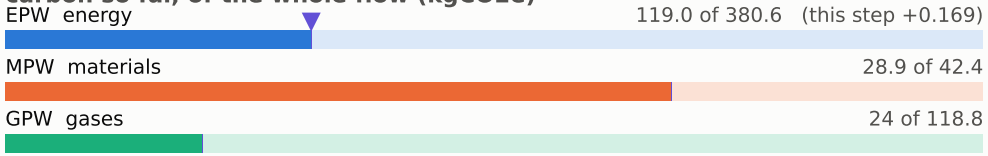
no mask for this step

this step

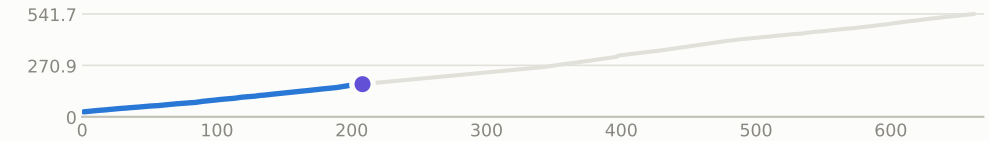
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

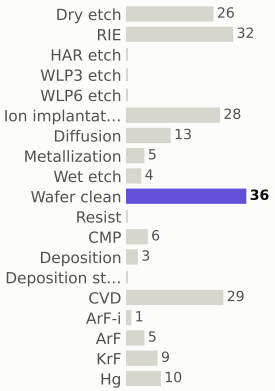
carbon so far, of the whole flow (kgCO2e)



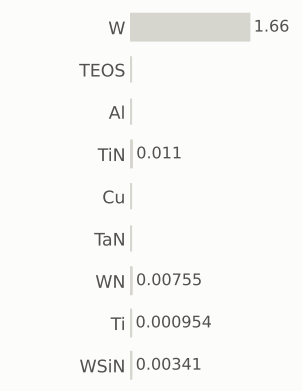
embodied carbon up to each step (kgCO2e per wafer)



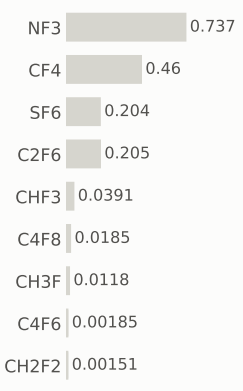
steps by process area



materials deposited (g)



gases emitted (g)



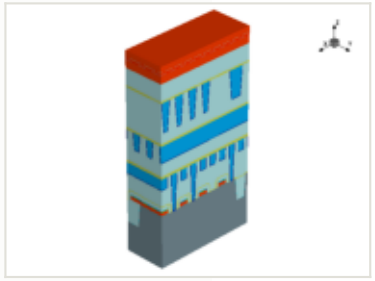
Step #2180: Wafer_clean

clean · Wafer_clean

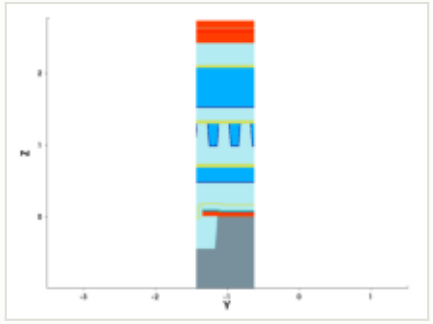
0.169	172.0	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

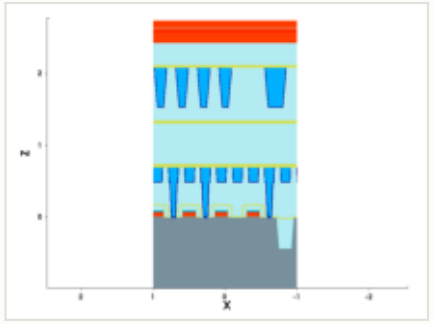
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

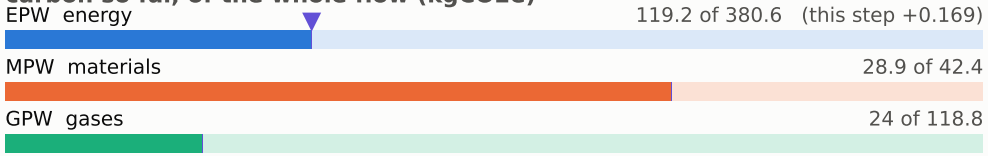
no mask for this step

this step

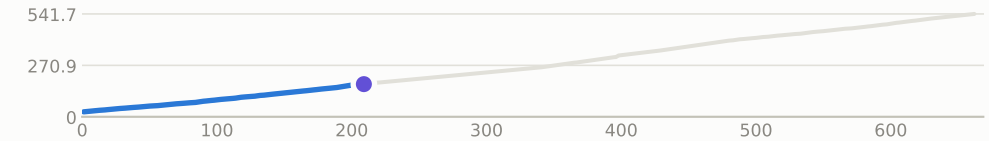
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

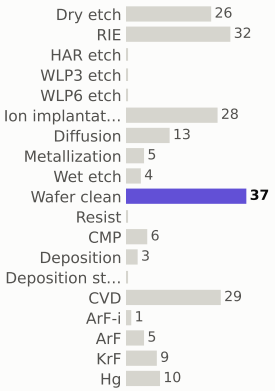
carbon so far, of the whole flow (kgCO2e)



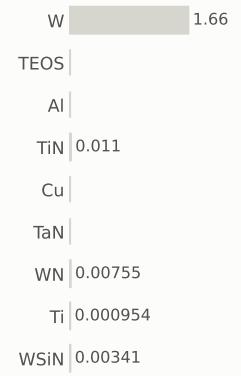
embodied carbon up to each step (kgCO2e per wafer)



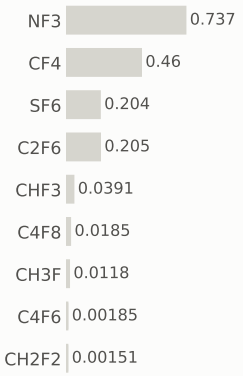
steps by process area



materials deposited (g)



gases emitted (g)



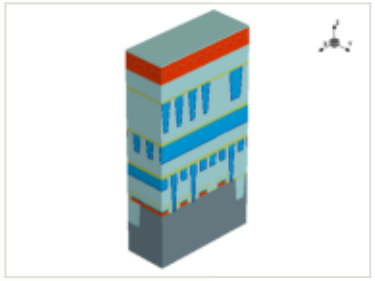
Step #2190: CVD

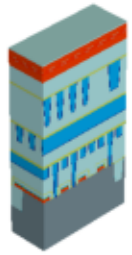
depo_isotropic · CVD

1.31	173.4	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



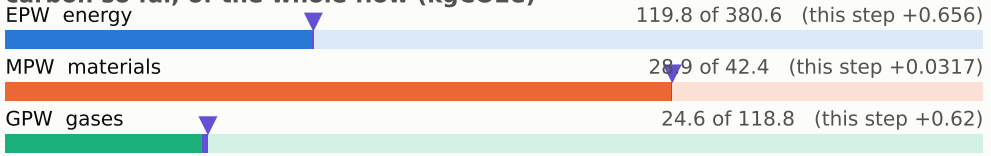


this step

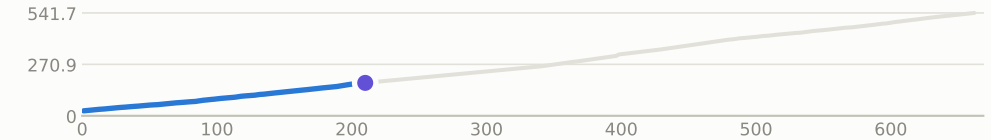
process area	CVD
material	—
thickness	0.025 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

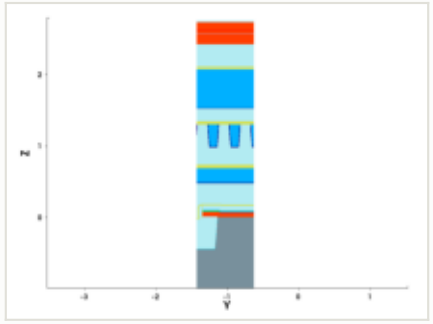
carbon so far, of the whole flow (kgCO2e)



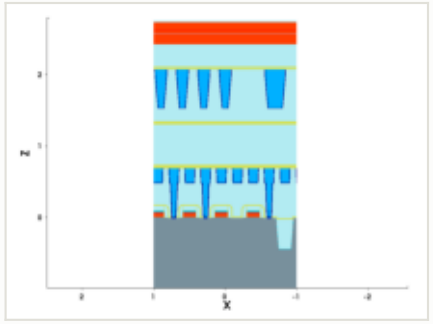
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



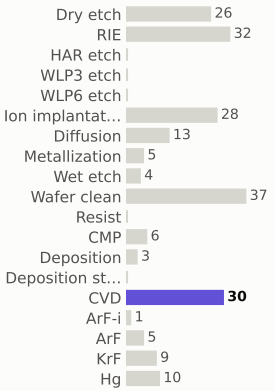
y cut at -1.032 µm



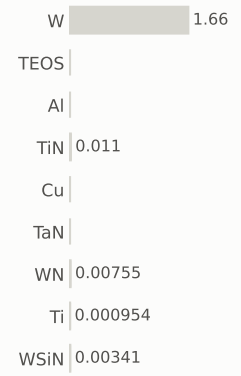
mask for this step

no mask for this step

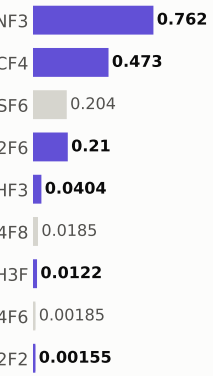
steps by process area



materials deposited (g)



gases emitted (g)



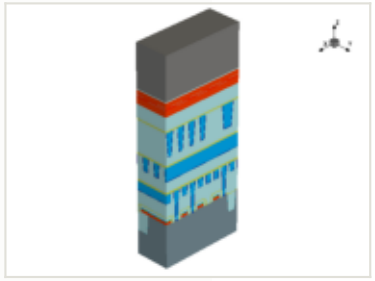
Step #2200: ArF

litho_patterning · ArF

0.199	173.5	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

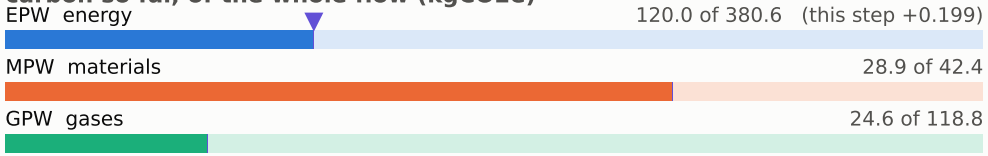
zoomed to fit



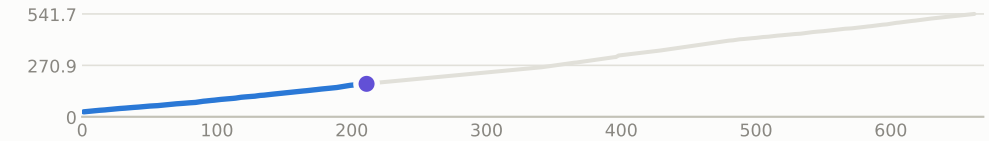
this step

process area	ArF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

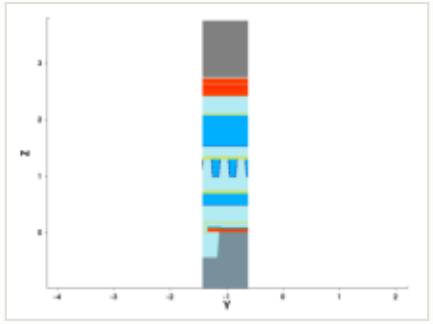
carbon so far, of the whole flow (kgCO2e)



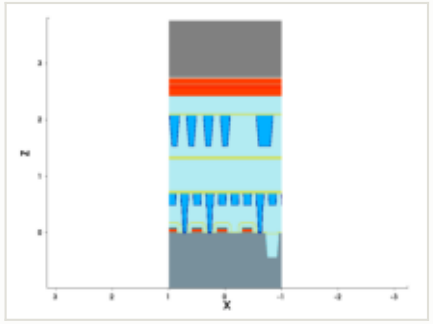
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



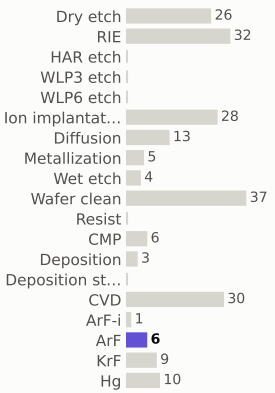
y cut at -1.032 µm



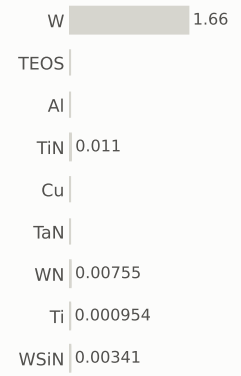
mask: layer 26:0 (OASIS)

nothing in the clip window

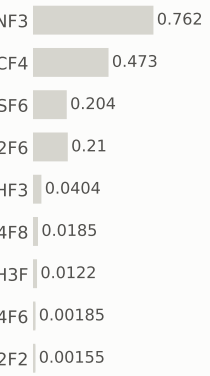
steps by process area



materials deposited (g)



gases emitted (g)



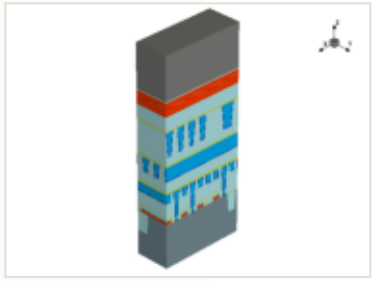
Step #2210: RIE

etch_anisotropic · RIE

1.04	174.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

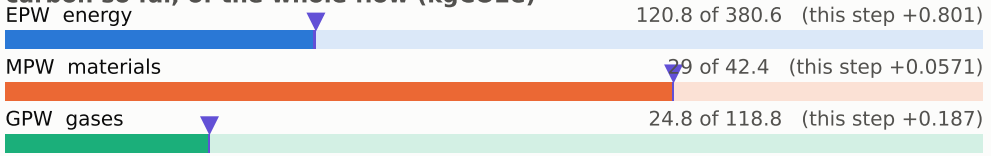


this step

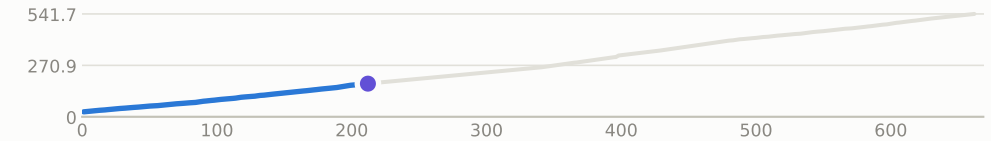
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

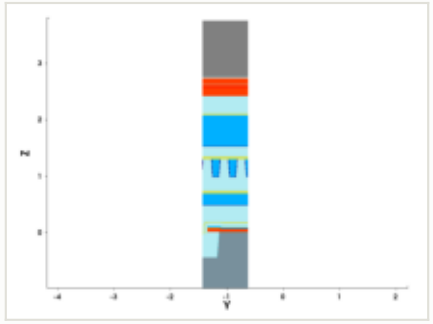
carbon so far, of the whole flow (kgCO2e)



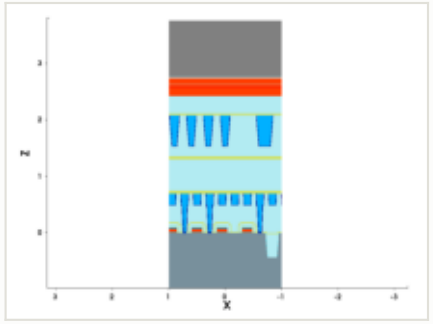
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



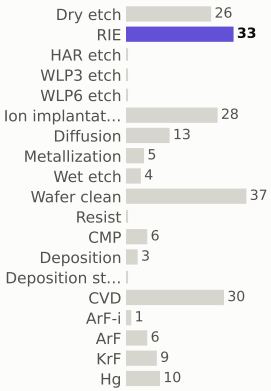
y cut at -1.032 µm



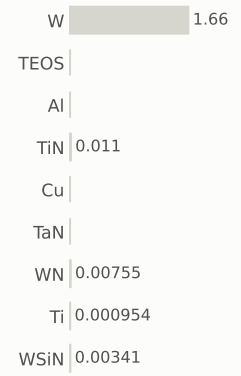
mask for this step

no mask for this step

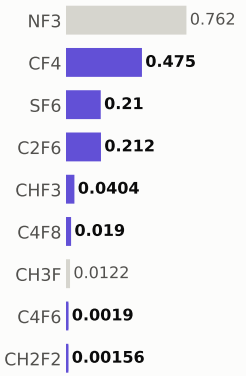
steps by process area



materials deposited (g)



gases emitted (g)



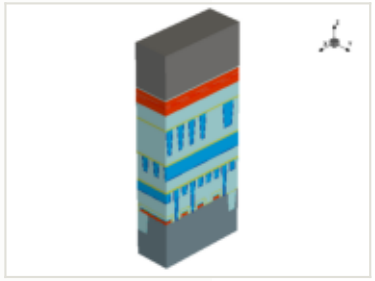
Step #2220: RIE

etch_anisotropic · RIE

1.04	175.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

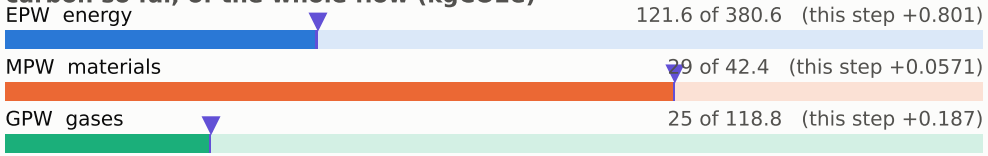


this step

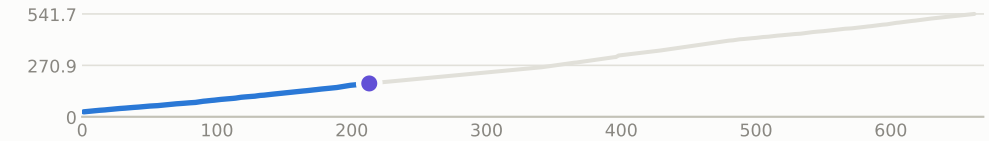
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

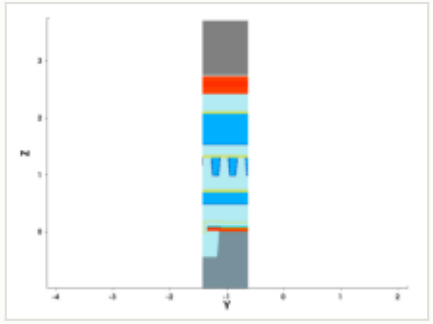
carbon so far, of the whole flow (kgCO2e)



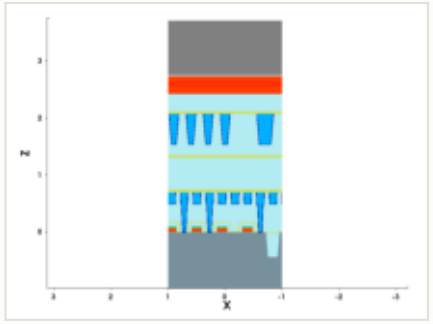
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



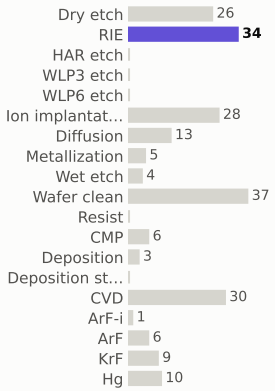
y cut at -1.032 µm



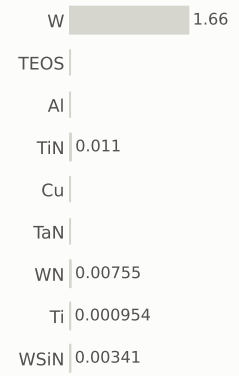
mask for this step

no mask for this step

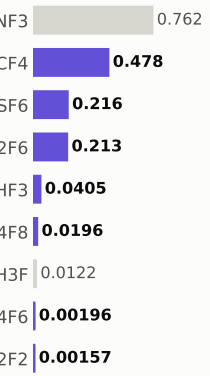
steps by process area



materials deposited (g)



gases emitted (g)



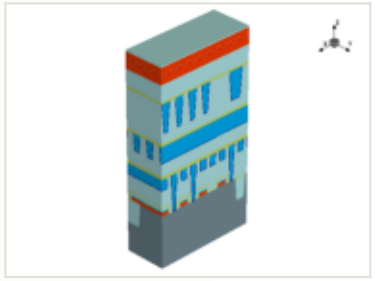
Step #2230: Dry_etch

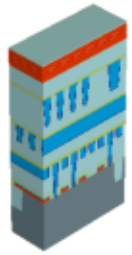
etch_strip · Dry_etch

0.801	176.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

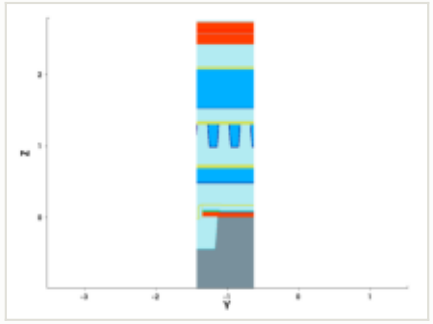
3D structure after this step

zoomed to fit

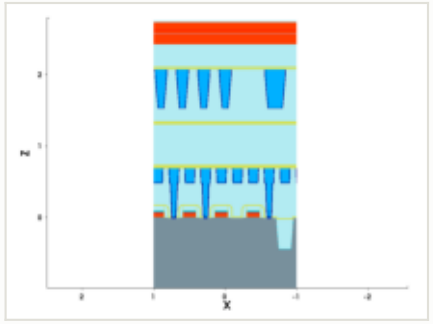




x cut at 0.000 µm



y cut at -1.032 µm



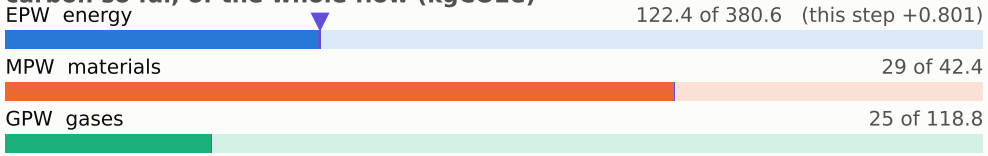
mask for this step

no mask for this step

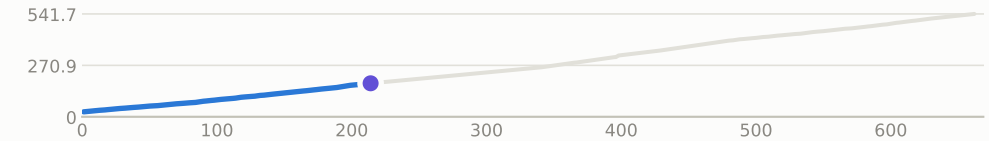
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

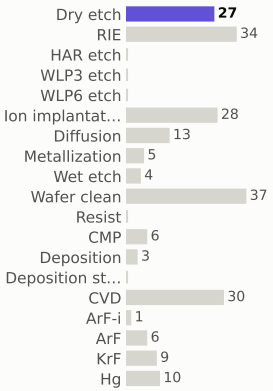
carbon so far, of the whole flow (kgCO2e)



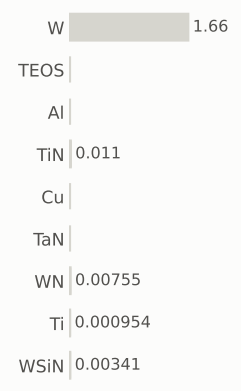
embodied carbon up to each step (kgCO2e per wafer)



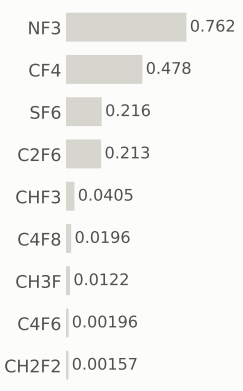
steps by process area



materials deposited (g)



gases emitted (g)



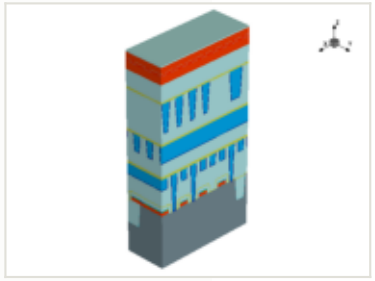
Step #2240: Wafer_clean

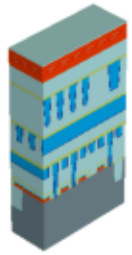
clean · Wafer_clean

0.169	176.6	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

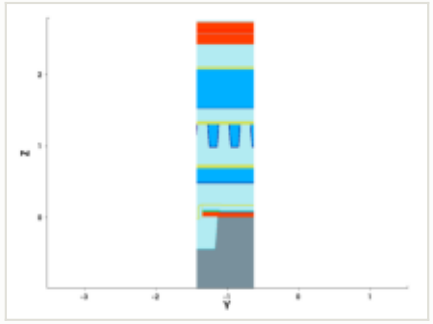
3D structure after this step

zoomed to fit

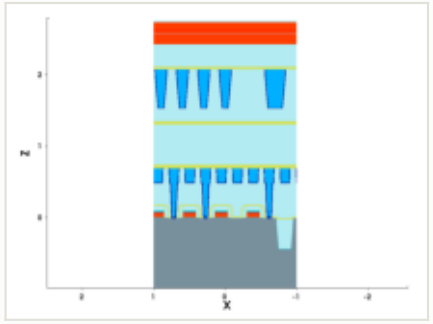




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

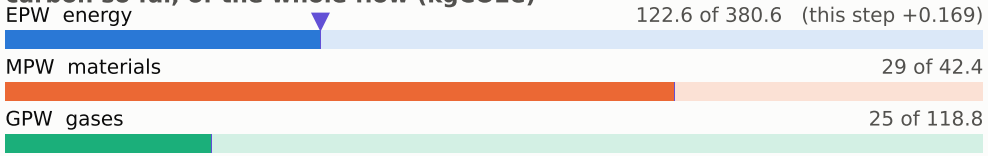
no mask for this step

this step

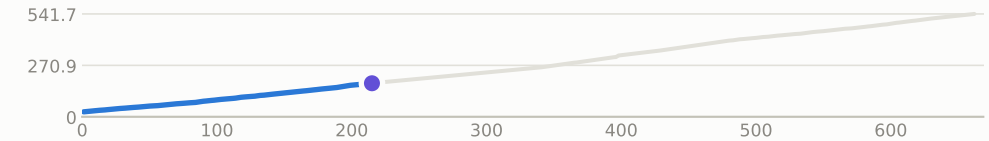
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

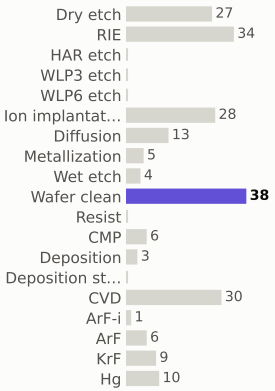
carbon so far, of the whole flow (kgCO2e)



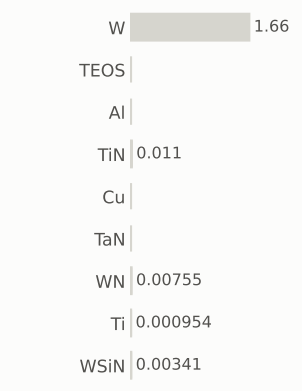
embodied carbon up to each step (kgCO2e per wafer)



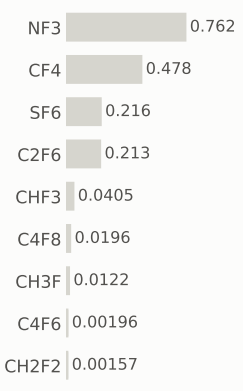
steps by process area



materials deposited (g)



gases emitted (g)



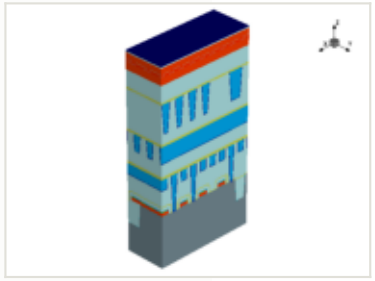
Step #2250: CVD

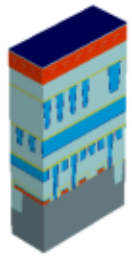
depo_isotropic · CVD

1.31	177.9	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



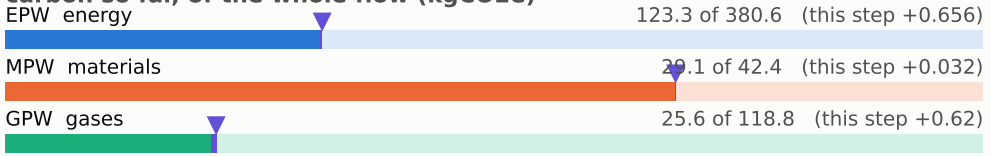


this step

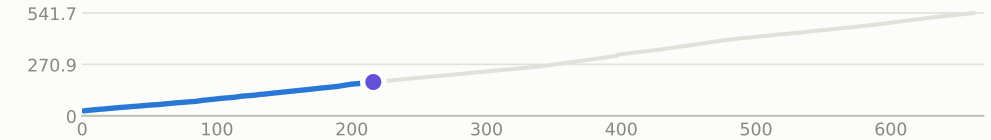
process area	CVD
material	TiN
thickness	0.015 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.032 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

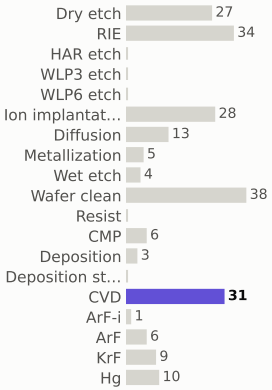
carbon so far, of the whole flow (kgCO2e)



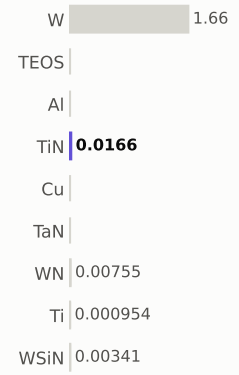
embodied carbon up to each step (kgCO2e per wafer)



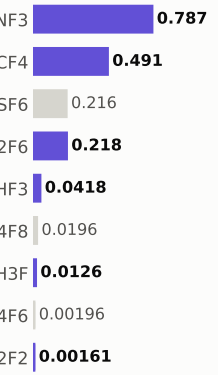
steps by process area



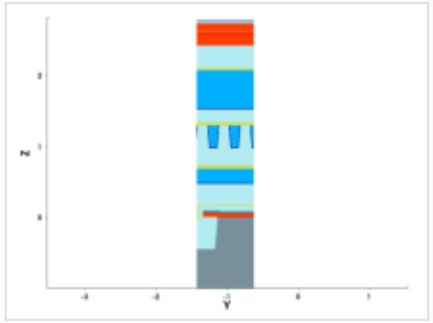
materials deposited (g)



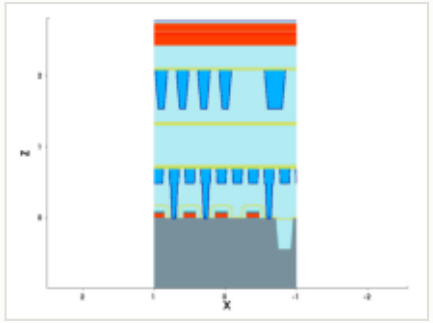
gases emitted (g)



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

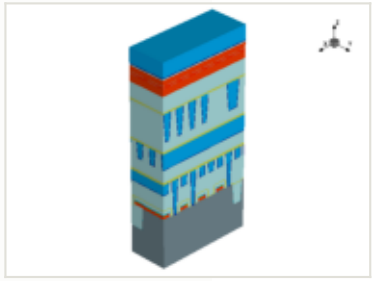
Step #2260: Metallization

depo_isotropic · Metallization

1.24	179.2	0.23%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

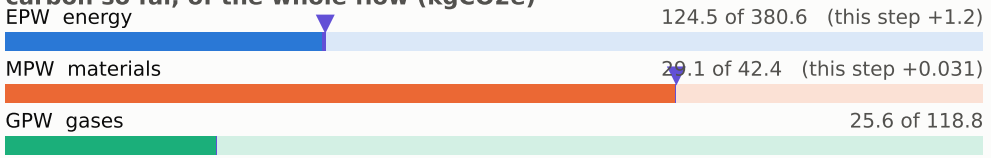


this step

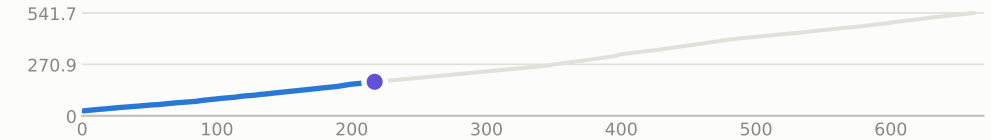
process area	Metallization
material	W
thickness	0.3 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.031 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.24 kgCO2e**

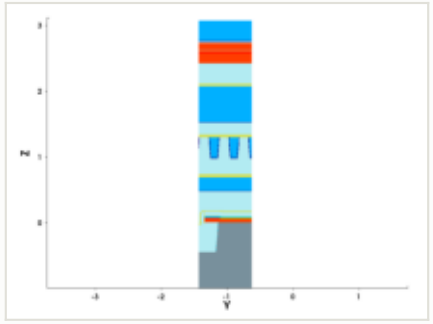
carbon so far, of the whole flow (kgCO2e)



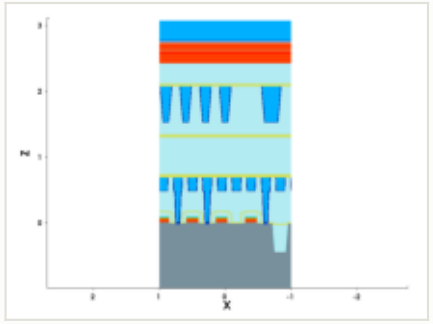
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



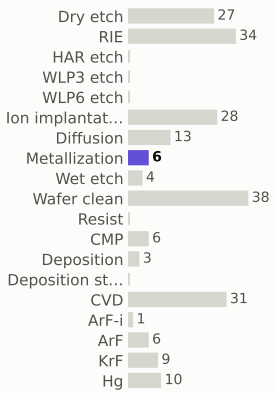
y cut at -1.032 µm



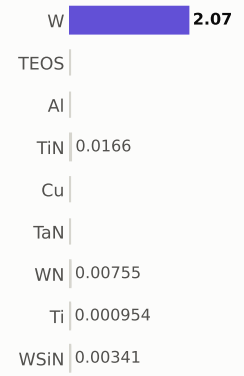
mask for this step

no mask for this step

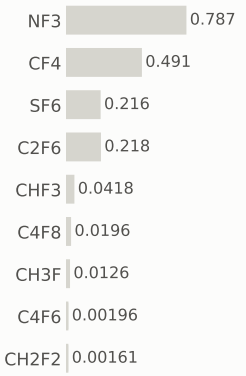
steps by process area



materials deposited (g)



gases emitted (g)



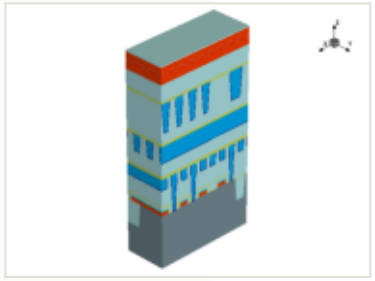
Step #2270: CMP

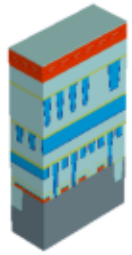
etch_cmp · CMP

0.169	179.3	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

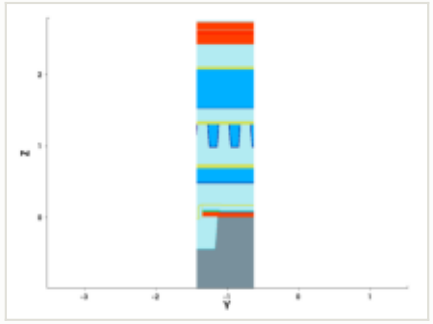
3D structure after this step

zoomed to fit

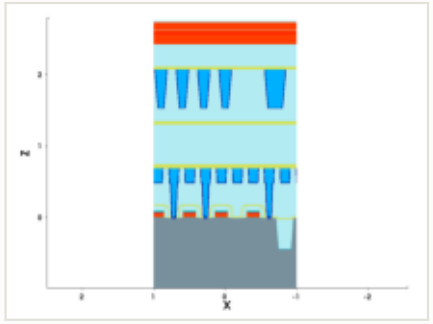




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

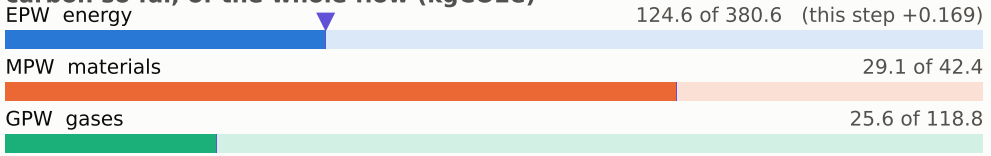
no mask for this step

this step

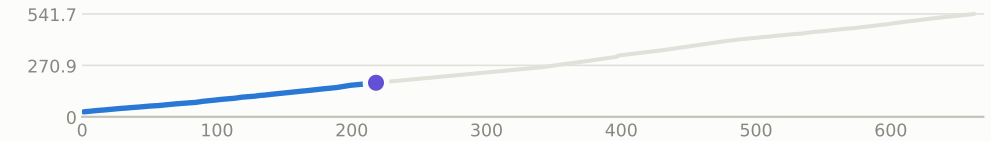
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

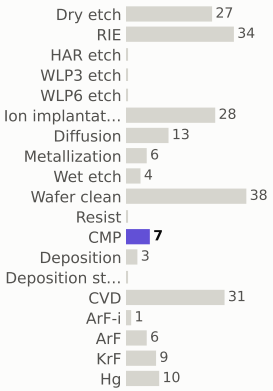
carbon so far, of the whole flow (kgCO2e)



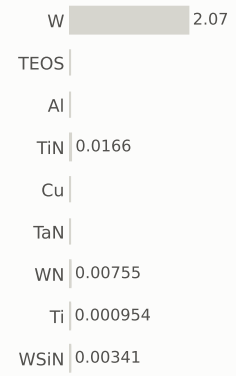
embodied carbon up to each step (kgCO2e per wafer)



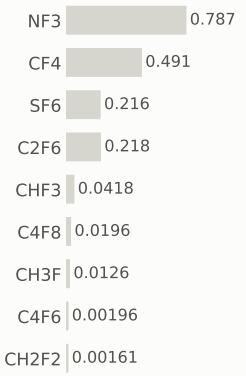
steps by process area



materials deposited (g)



gases emitted (g)



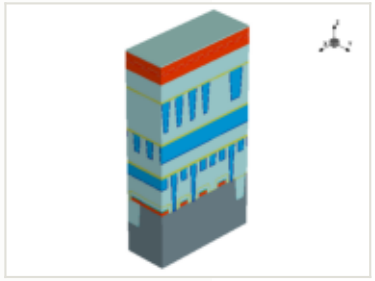
Step #2280: Wafer_clean

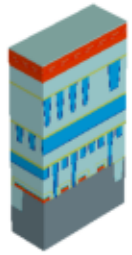
clean · Wafer_clean

0.169	179.5	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

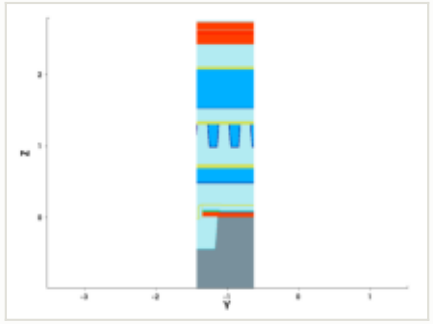
3D structure after this step

zoomed to fit

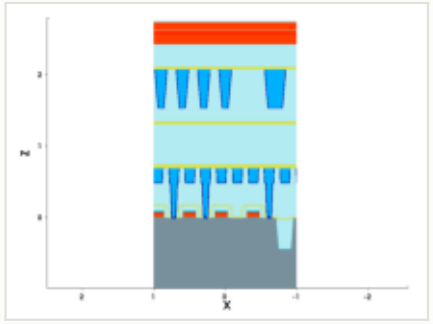




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

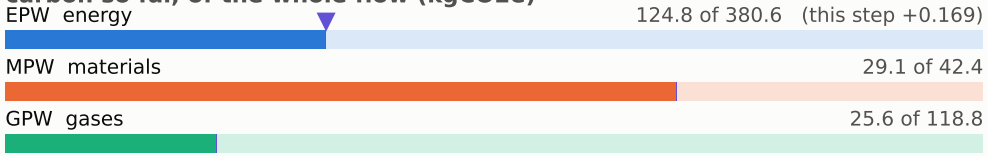
no mask for this step

this step

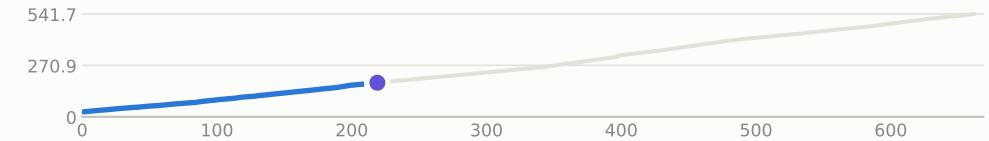
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

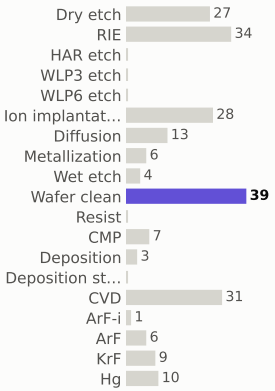
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

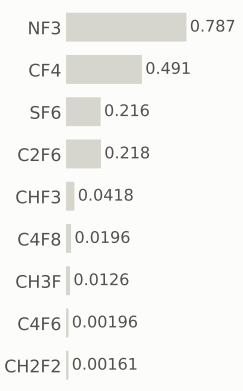


steps by process area



materials deposited (g)

gases emitted (g)



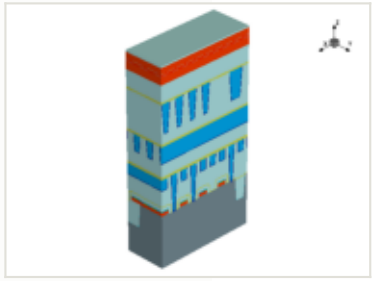
Step #2290: CVD

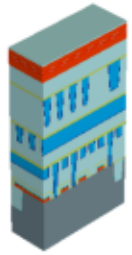
depo_isotropic · CVD

1.31	180.8	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

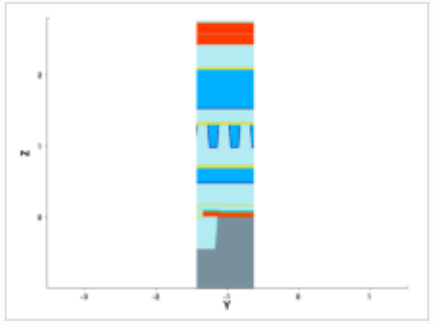
3D structure after this step

zoomed to fit

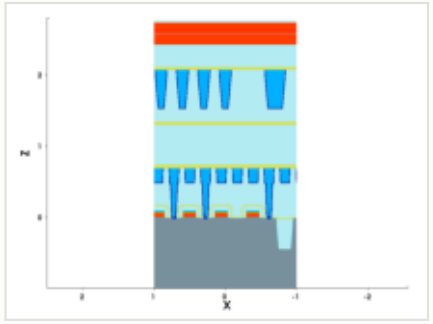




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

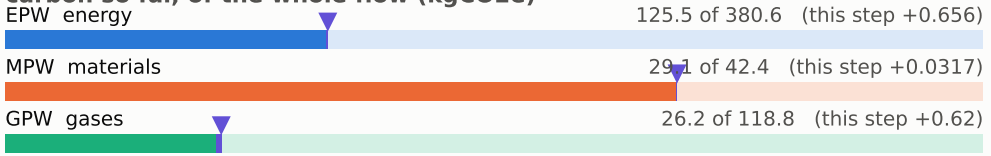
no mask for this step

this step

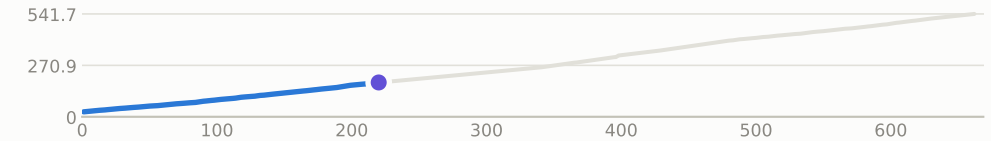
process area	CVD
material	—
thickness	0.007 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

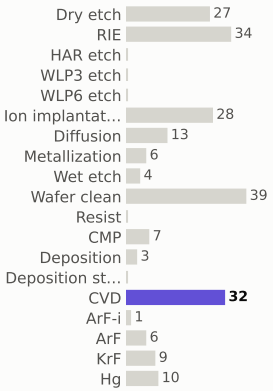
carbon so far, of the whole flow (kgCO2e)



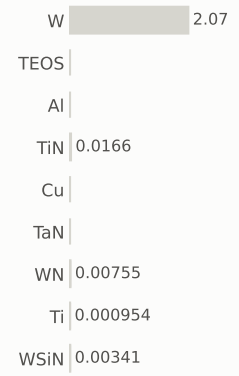
embodied carbon up to each step (kgCO2e per wafer)



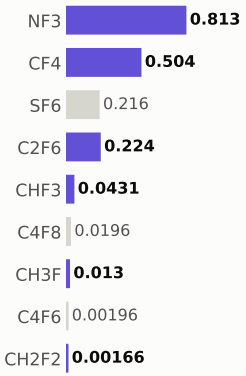
steps by process area



materials deposited (g)



gases emitted (g)



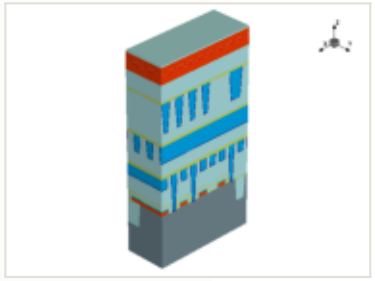
Step #2310: Wafer_clean

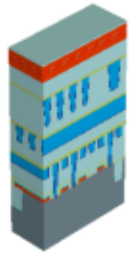
clean · Wafer_clean

0.169	181.0	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

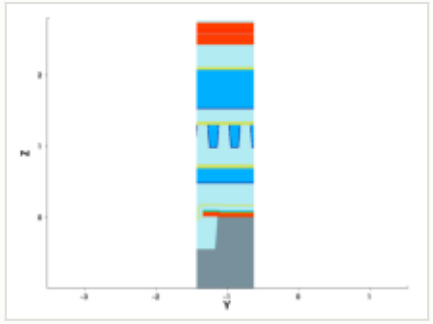
3D structure after this step

zoomed to fit

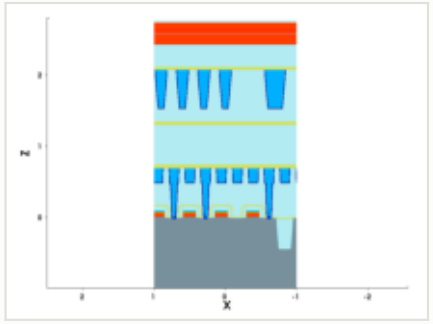




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

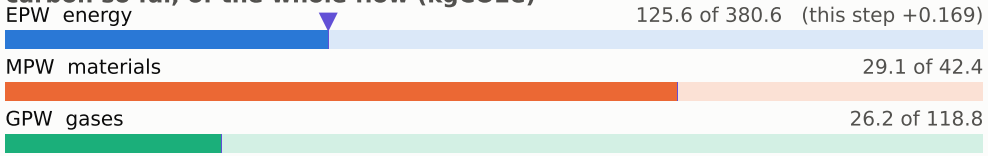
no mask for this step

this step

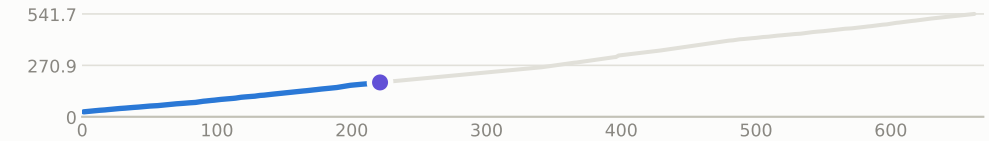
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

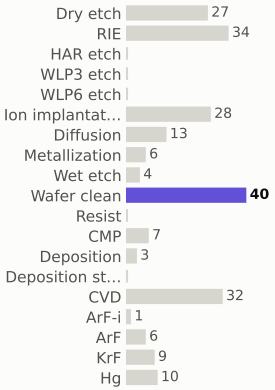
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

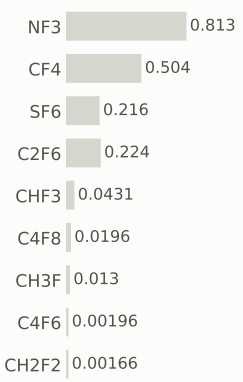


steps by process area



materials deposited (g)

gases emitted (g)



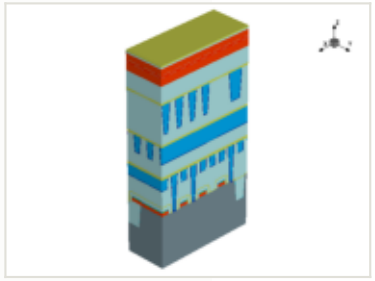
Step #2320: Deposition_stack

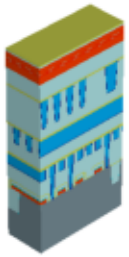
depo_anisotropic · Deposition_stack

0.654	181.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

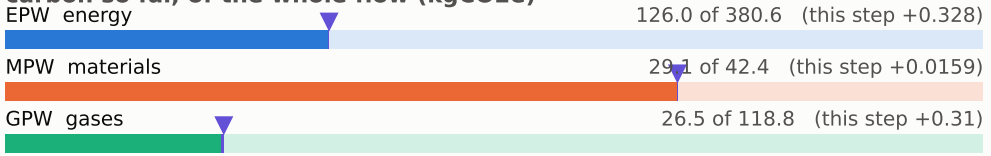




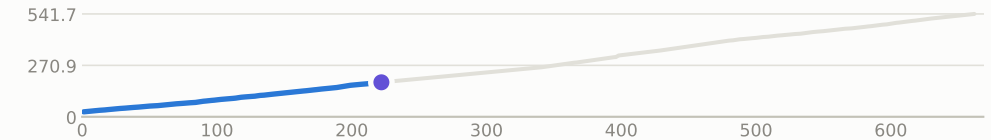
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

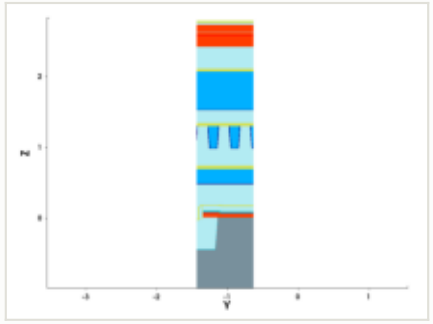
carbon so far, of the whole flow (kgCO2e)



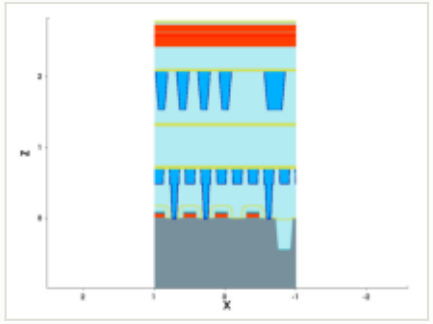
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



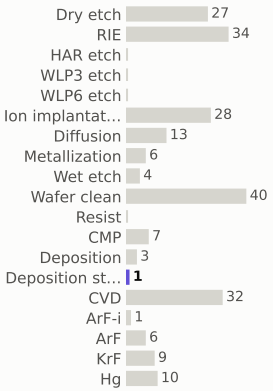
y cut at -1.032 µm



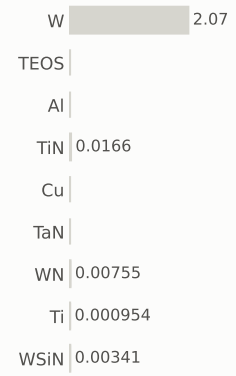
mask for this step

no mask for this step

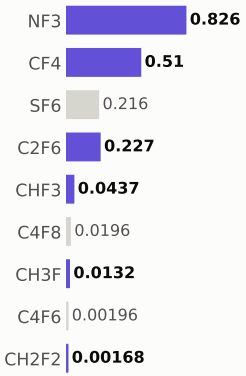
steps by process area



materials deposited (g)



gases emitted (g)



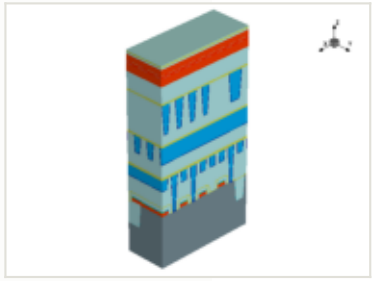
Step #2330: Deposition_stack

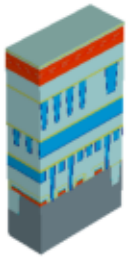
depo_anisotropic · Deposition_stack

0.654	182.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

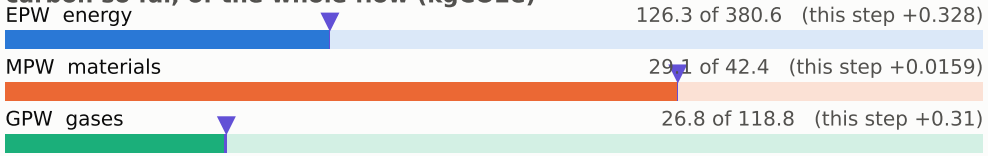




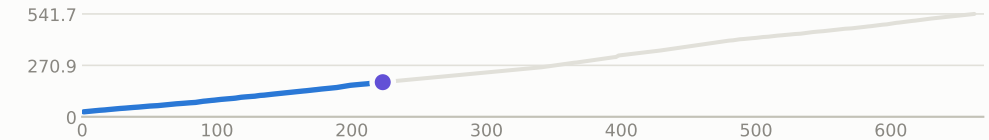
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

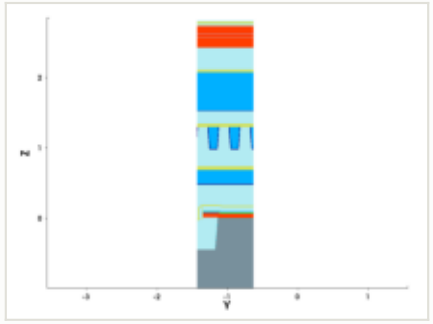
carbon so far, of the whole flow (kgCO2e)



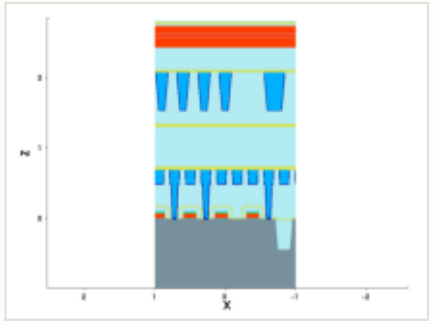
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



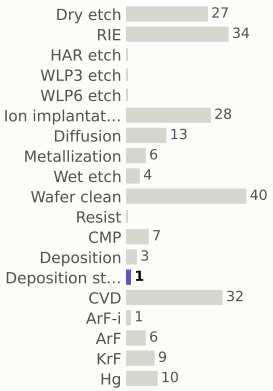
y cut at -1.032 µm



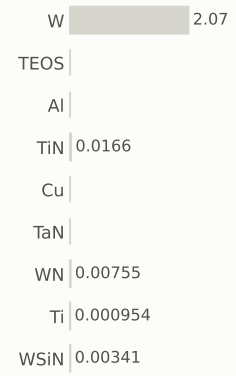
mask for this step

no mask for this step

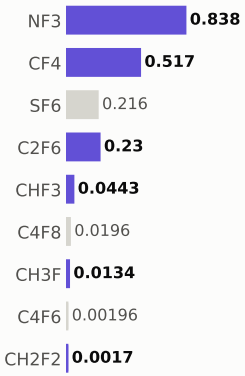
steps by process area



materials deposited (g)



gases emitted (g)



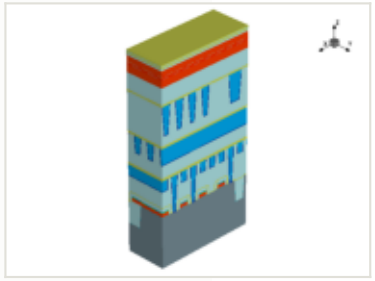
Step #2340: Deposition_stack

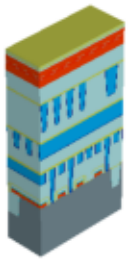
depo_anisotropic · Deposition_stack

0.654	182.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

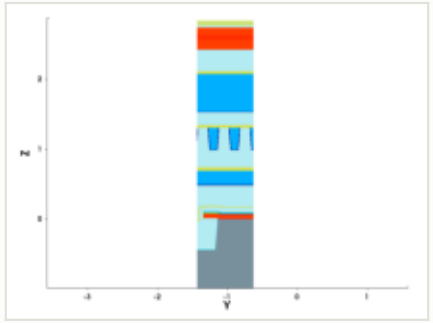
3D structure after this step

zoomed to fit

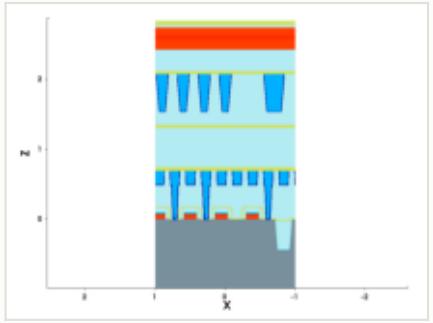




x cut at 0.000 µm



y cut at -1.032 µm



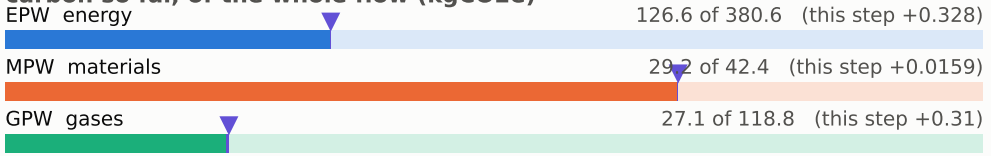
mask for this step

no mask for this step

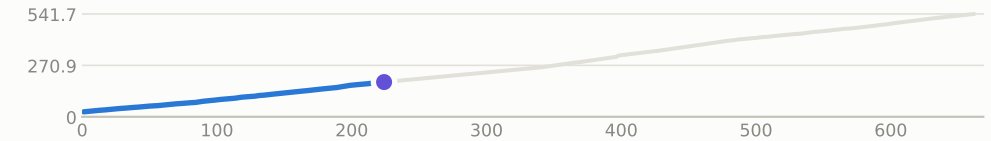
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

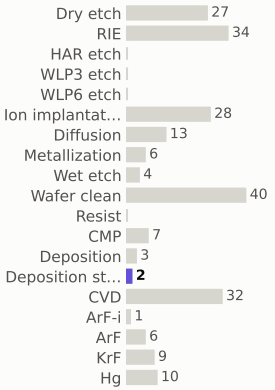
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

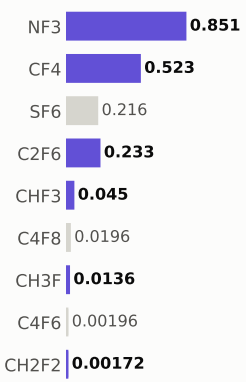


steps by process area



materials deposited (g)

gases emitted (g)



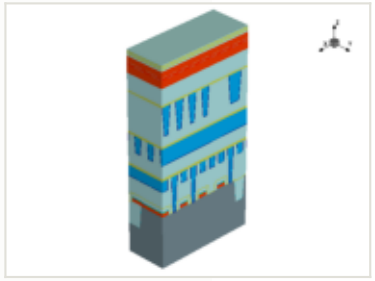
Step #2350: Deposition_stack

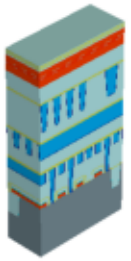
depo_anisotropic · Deposition_stack

0.654	183.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

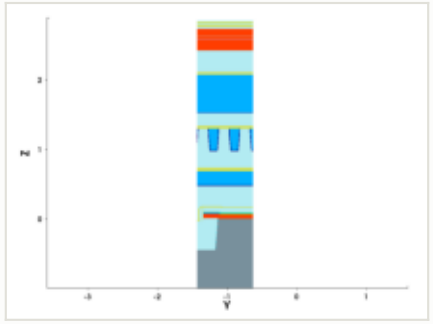
3D structure after this step

zoomed to fit

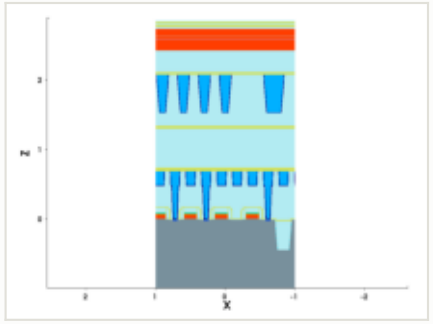




x cut at 0.000 µm



y cut at -1.032 µm



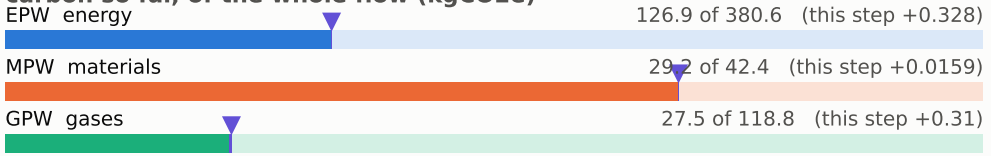
mask for this step

no mask for this step

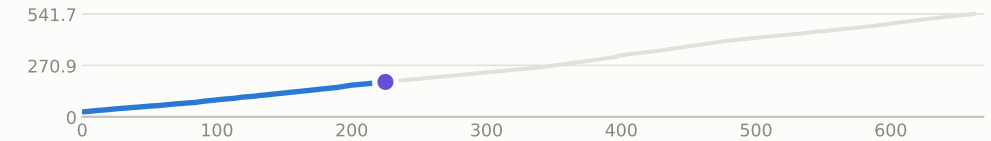
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

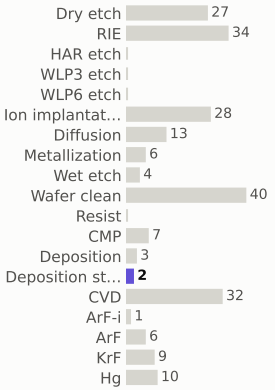
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

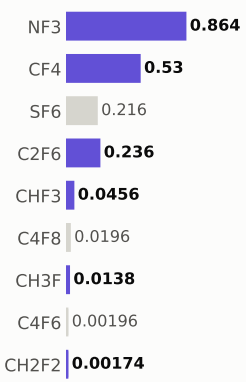


steps by process area



materials deposited (g)

gases emitted (g)



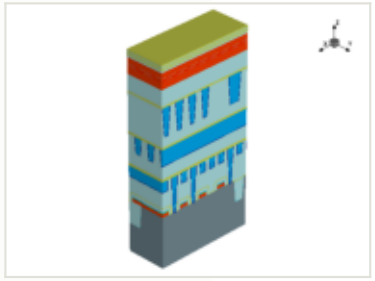
Step #2360: Deposition_stack

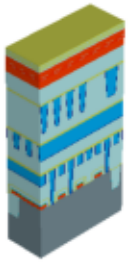
depo_anisotropic · Deposition_stack

0.654	184.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

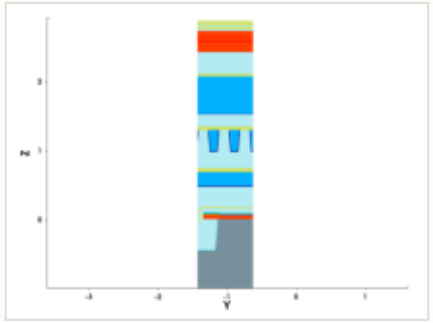
3D structure after this step

zoomed to fit

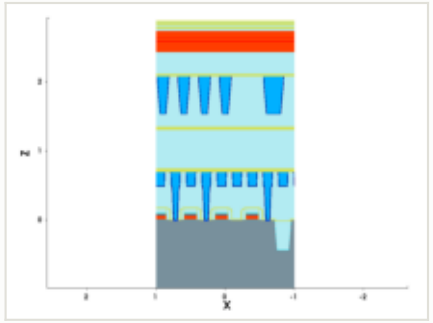




x cut at 0.000 µm



y cut at -1.032 µm



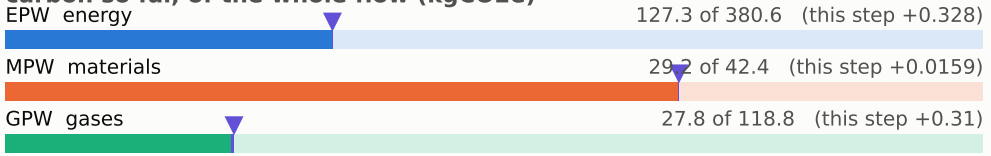
mask for this step

no mask for this step

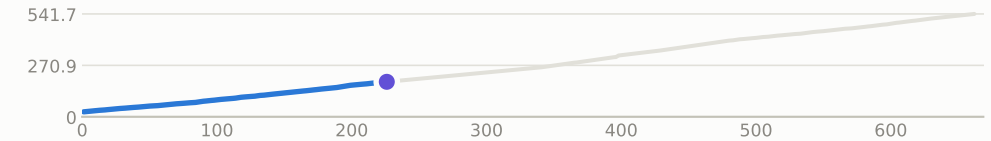
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

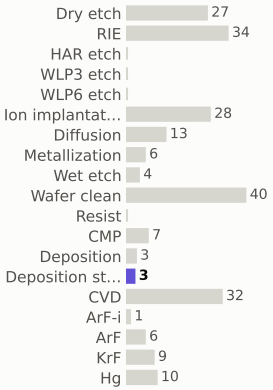
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

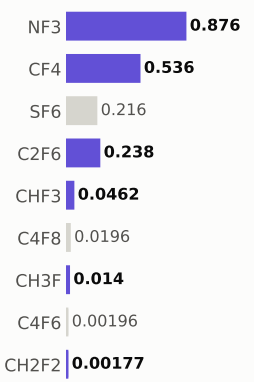


steps by process area



materials deposited (g)

gases emitted (g)



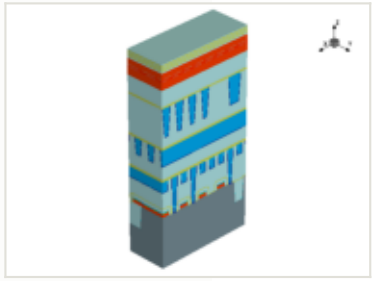
Step #2370: Deposition_stack

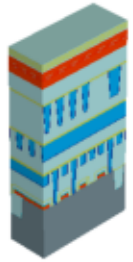
depo_anisotropic · Deposition_stack

0.654	184.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



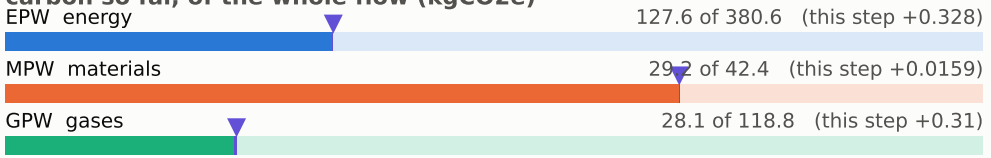


this step

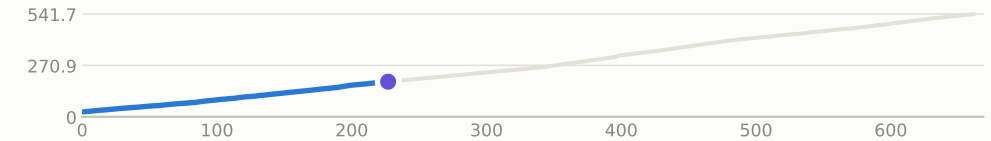
process area	Deposition_stack
material	—
thickness	0.017 µm
material GWP	—
input gases	NF3 1.27 g
energy	0.545 kWh + 0.218 overhead

EPW	0.328 kgCO2e
EPW tool	0.234 kgCO2e
EPW overhead	0.0937 kgCO2e
MPW	0.0159 kgCO2e (gas 0.0159)
GPW	0.31 kgCO2e
embodied carbon	**0.654 kgCO2e**

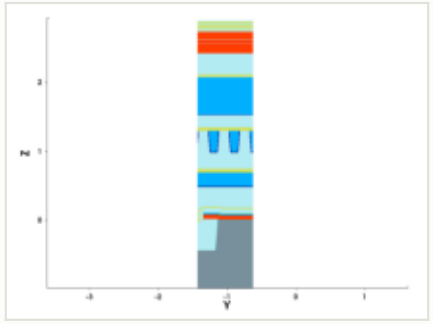
carbon so far, of the whole flow (kgCO2e)



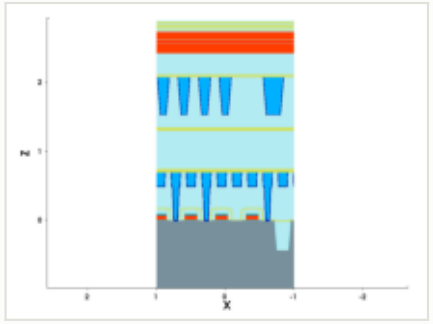
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



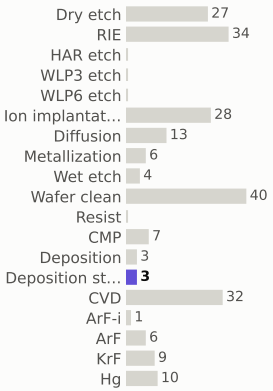
y cut at -1.032 µm



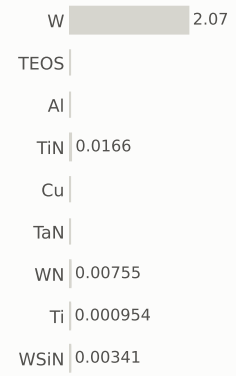
mask for this step

no mask for this step

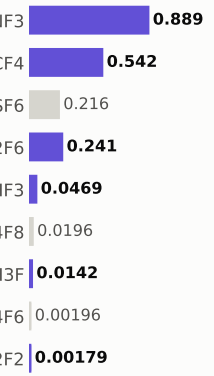
steps by process area



materials deposited (g)



gases emitted (g)



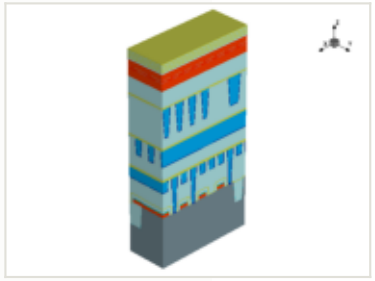
Step #2380: Deposition_stack

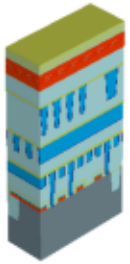
depo_anisotropic · Deposition_stack

0.654	185.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

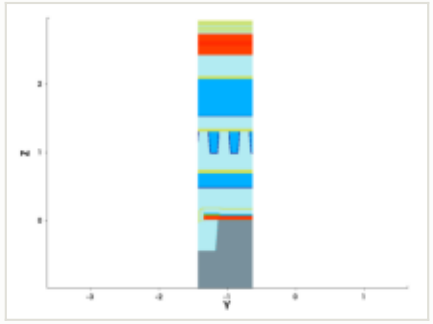
3D structure after this step

zoomed to fit

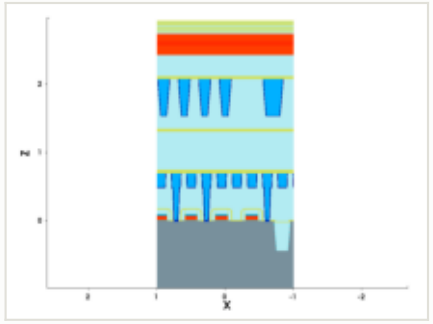




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

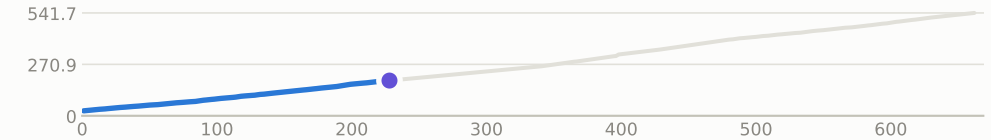
carbon so far, of the whole flow (kgCO2e)

EPW energy — 127.9 of 380.6 (this step +0.328)

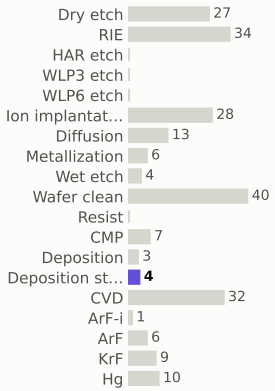
MPW materials — 29.2 of 42.4 (this step +0.0159)

GPW gases — 28.4 of 118.8 (this step +0.31)

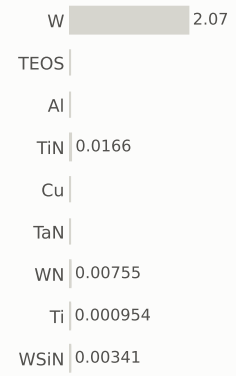
embodied carbon up to each step (kgCO2e per wafer)



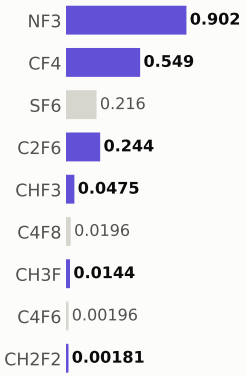
steps by process area



materials deposited (g)



gases emitted (g)



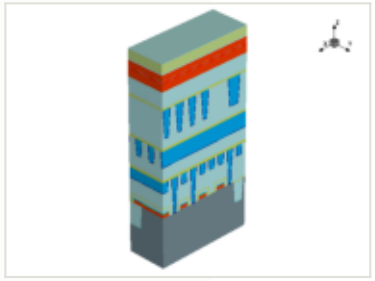
Step #2390: Deposition_stack

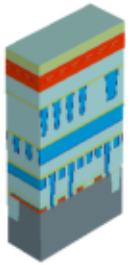
depo_anisotropic · Deposition_stack

0.654	186.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

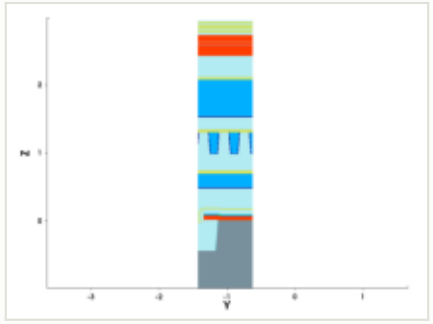
3D structure after this step

zoomed to fit

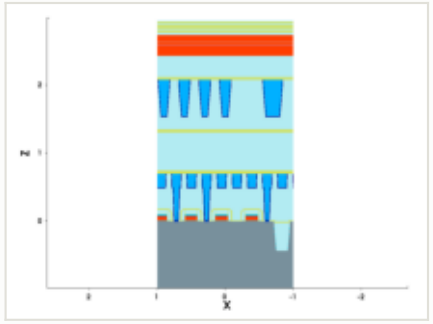




x cut at 0.000 µm



y cut at -1.032 µm



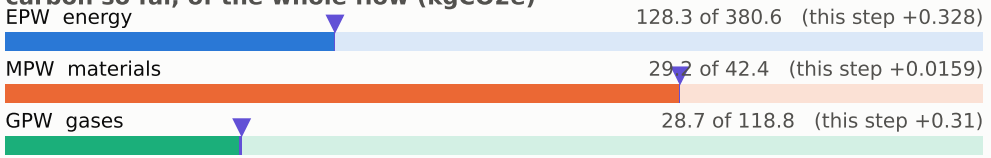
mask for this step

no mask for this step

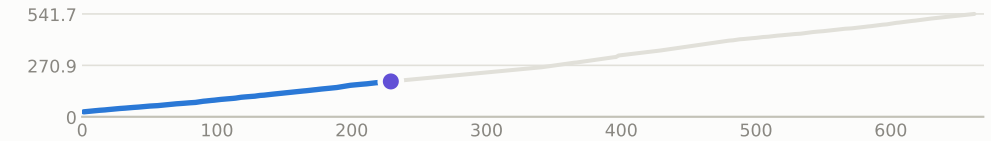
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

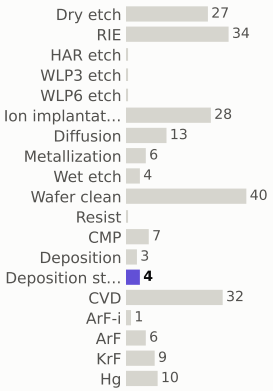
carbon so far, of the whole flow (kgCO2e)



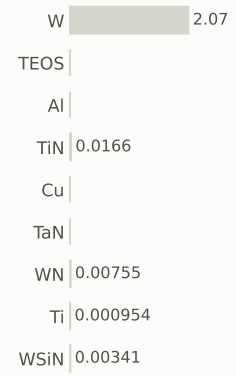
embodied carbon up to each step (kgCO2e per wafer)



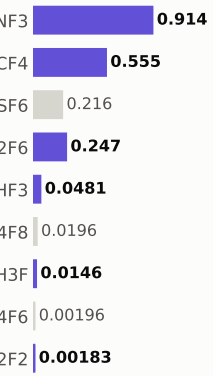
steps by process area



materials deposited (g)



gases emitted (g)



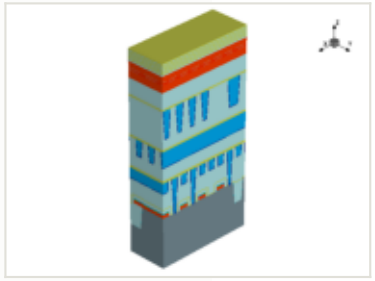
Step #2400: Deposition_stack

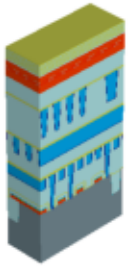
depo_anisotropic · Deposition_stack

0.654	186.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

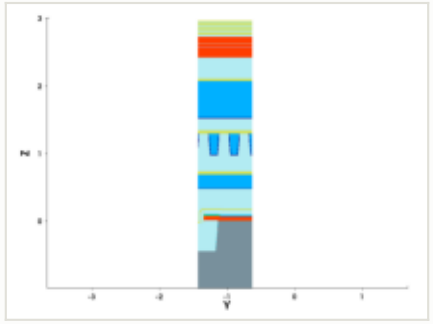
3D structure after this step

zoomed to fit

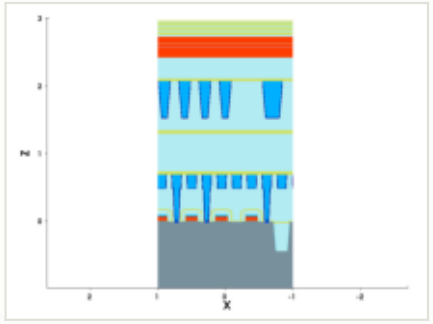




x cut at 0.000 µm



y cut at -1.032 µm



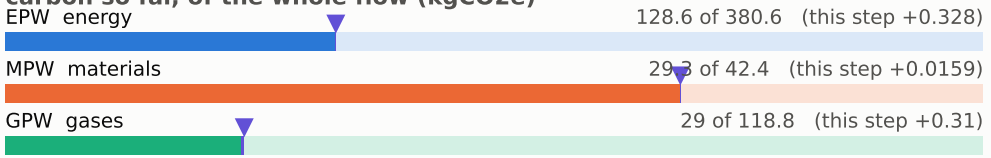
mask for this step

no mask for this step

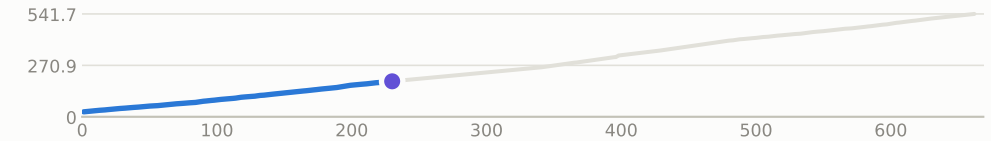
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

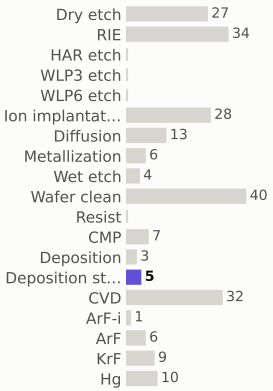
carbon so far, of the whole flow (kgCO2e)



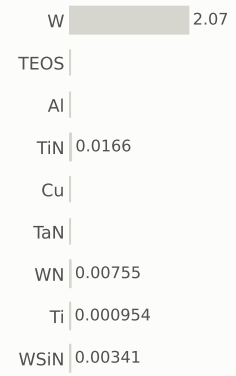
embodied carbon up to each step (kgCO2e per wafer)



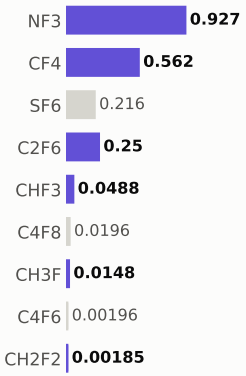
steps by process area



materials deposited (g)



gases emitted (g)



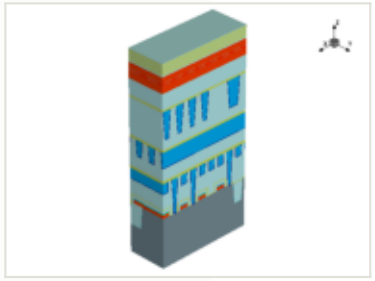
Step #2410: Deposition_stack

depo_anisotropic · Deposition_stack

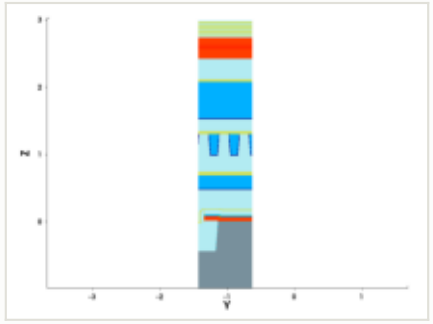
0.654	187.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

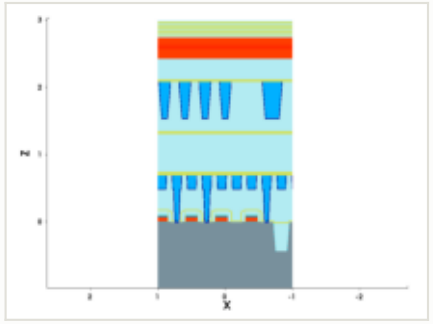
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

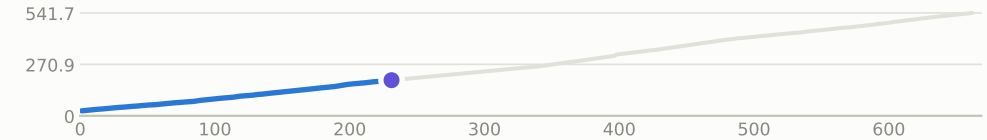
carbon so far, of the whole flow (kgCO2e)

EPW energy — 128.9 of 380.6 (this step +0.328)

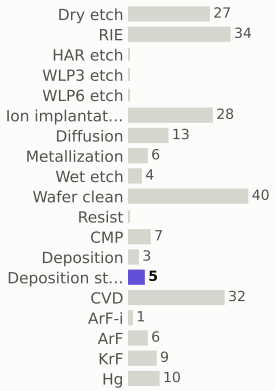
MPW materials — 29.3 of 42.4 (this step +0.0159)

GPW gases — 29.3 of 118.8 (this step +0.31)

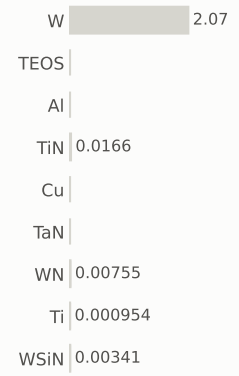
embodied carbon up to each step (kgCO2e per wafer)



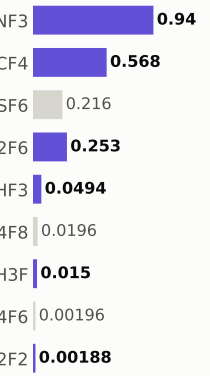
steps by process area



materials deposited (g)



gases emitted (g)



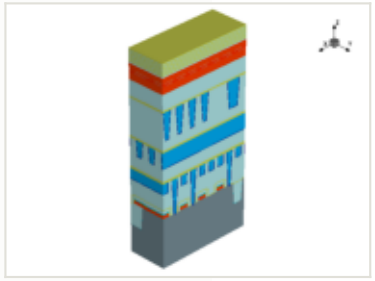
Step #2420: Deposition_stack

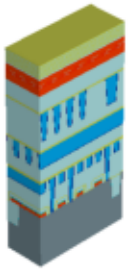
depo_anisotropic · Deposition_stack

0.654	188.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

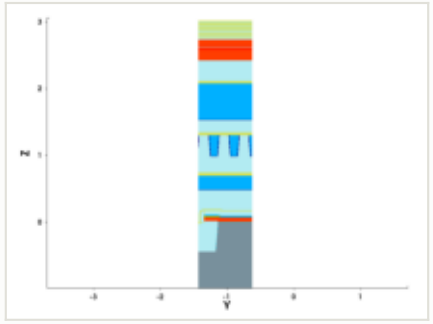
3D structure after this step

zoomed to fit

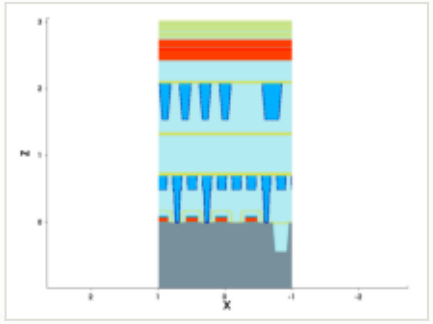




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

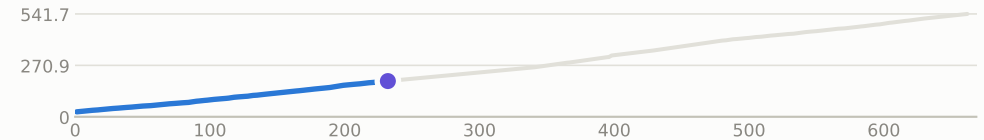
carbon so far, of the whole flow (kgCO2e)

EPW energy — 129.2 of 380.6 (this step +0.328)

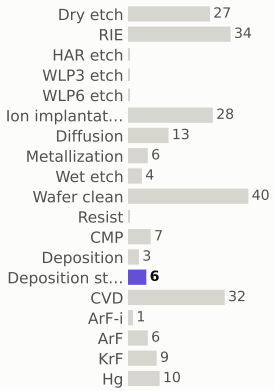
MPW materials — 29.3 of 42.4 (this step +0.0159)

GPW gases — 29.6 of 118.8 (this step +0.31)

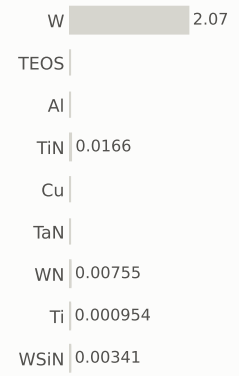
embodied carbon up to each step (kgCO2e per wafer)



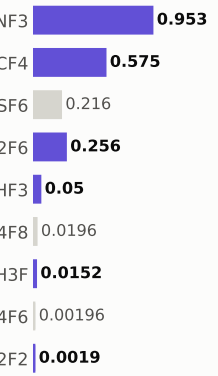
steps by process area



materials deposited (g)



gases emitted (g)



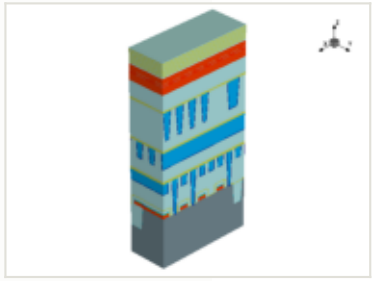
Step #2430: Deposition_stack

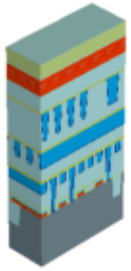
depo_anisotropic · Deposition_stack

0.654	188.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

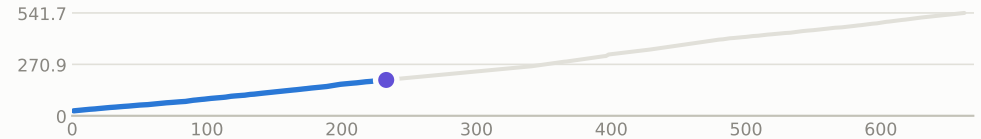
carbon so far, of the whole flow (kgCO2e)

EPW energy — 129.6 of 380.6 (this step +0.328)

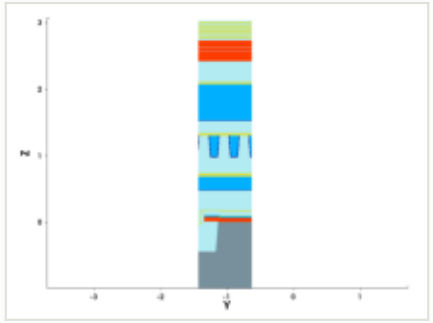
MPW materials — 29.3 of 42.4 (this step +0.0159)

GPW gases — 29.9 of 118.8 (this step +0.31)

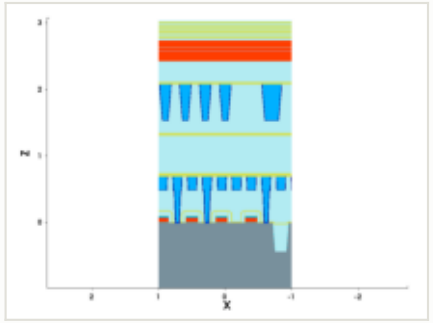
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



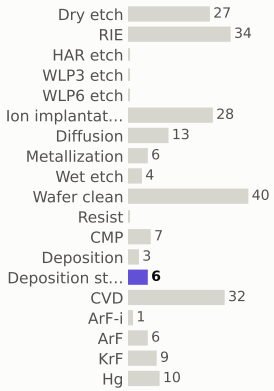
y cut at -1.032 µm



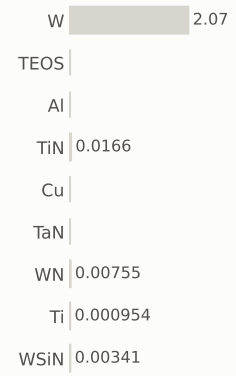
mask for this step

no mask for this step

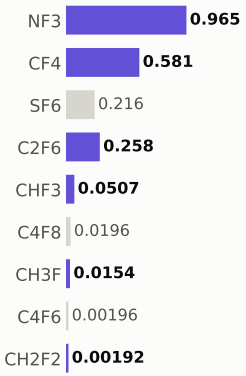
steps by process area



materials deposited (g)



gases emitted (g)



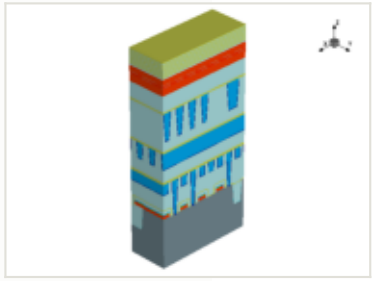
Step #2440: Deposition_stack

depo_anisotropic · Deposition_stack

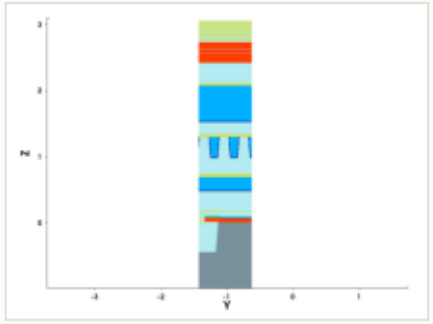
0.654	189.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

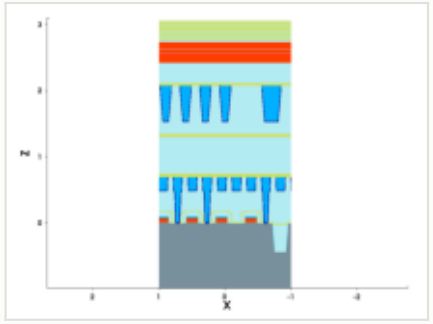
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



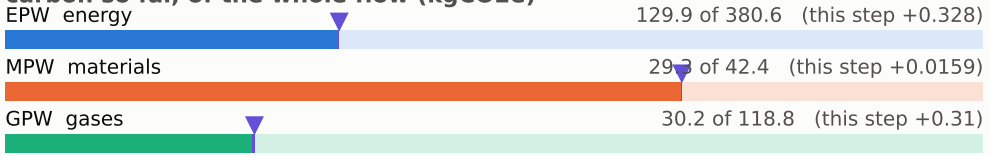
mask for this step

no mask for this step

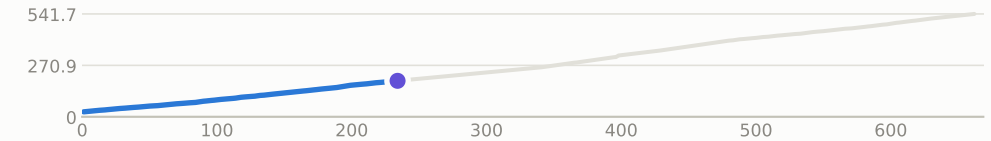
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

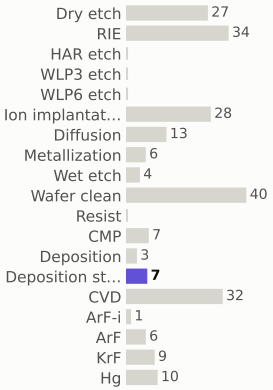
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

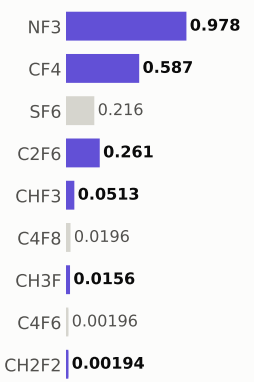


steps by process area



materials deposited (g)

gases emitted (g)



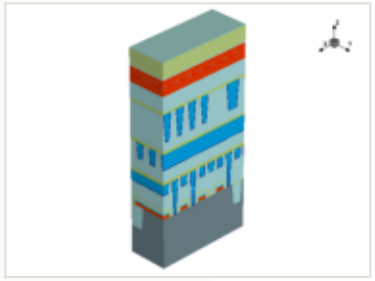
Step #2450: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	190.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

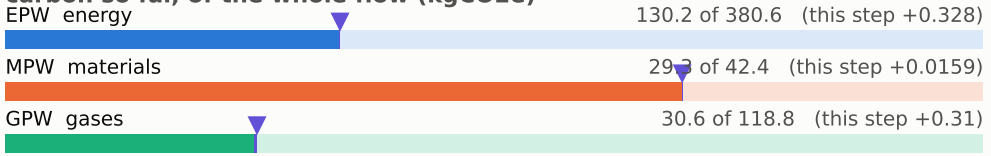
zoomed to fit



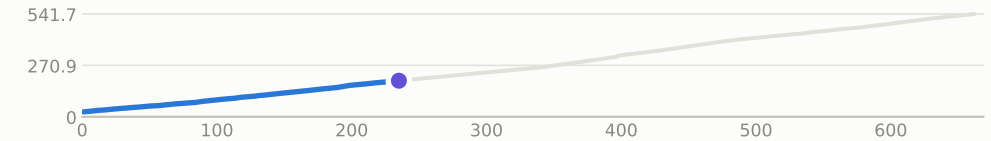
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

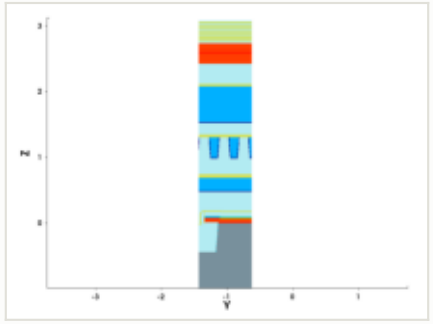
carbon so far, of the whole flow (kgCO2e)



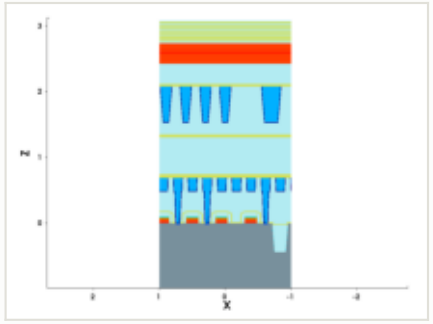
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



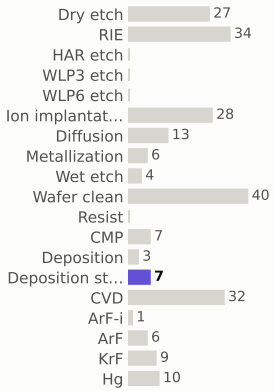
y cut at -1.032 µm



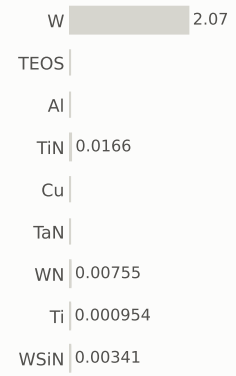
mask for this step

no mask for this step

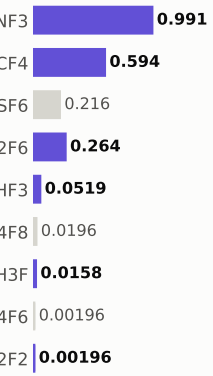
steps by process area



materials deposited (g)



gases emitted (g)



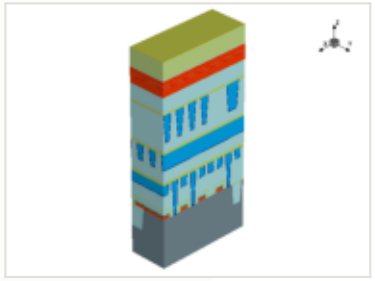
Step #2460: Deposition_stack

depo_anisotropic · Deposition_stack

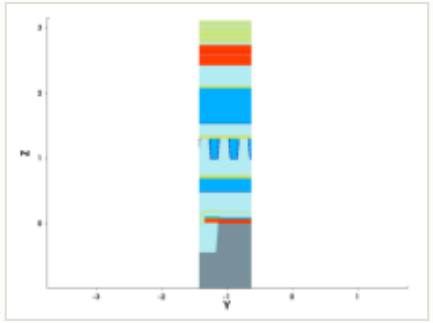
0.654	190.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

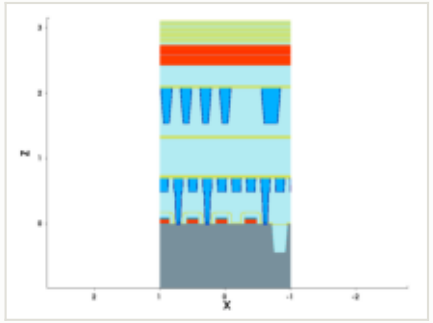
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



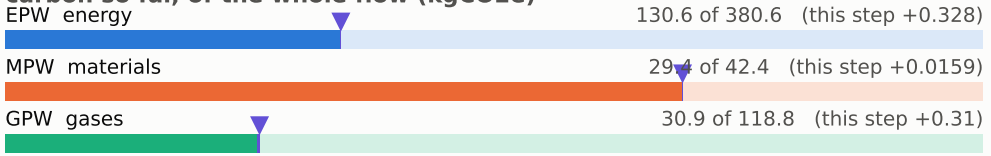
mask for this step

no mask for this step

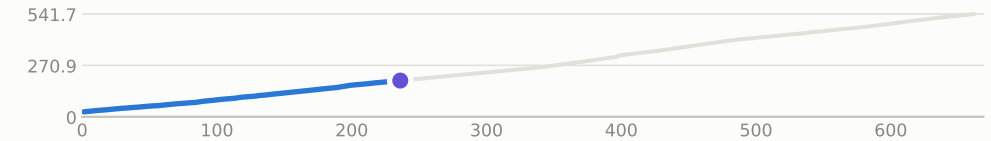
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

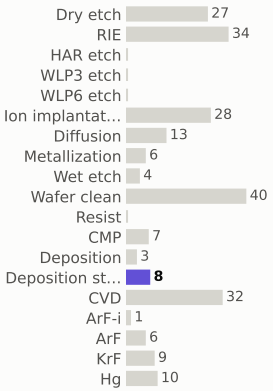
carbon so far, of the whole flow (kgCO2e)



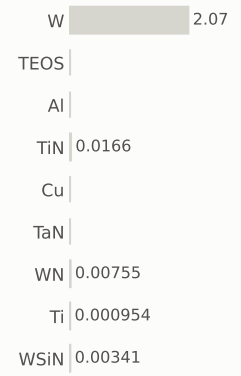
embodied carbon up to each step (kgCO2e per wafer)



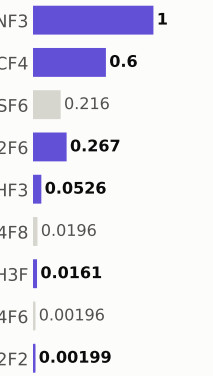
steps by process area



materials deposited (g)



gases emitted (g)



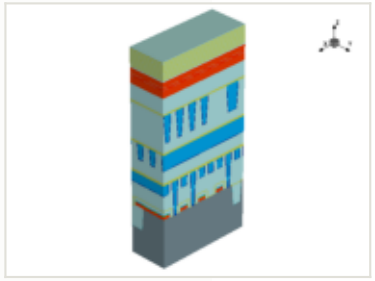
Step #2470: Deposition_stack

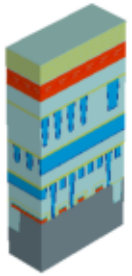
depo_anisotropic · Deposition_stack

0.654	191.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

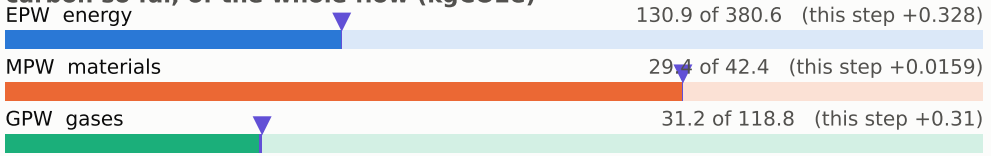




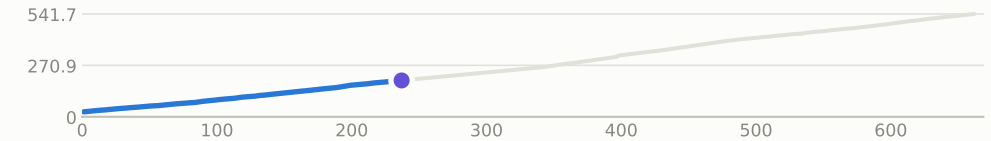
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

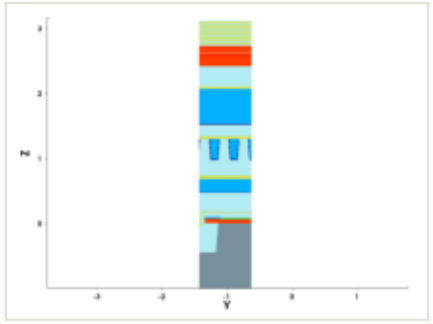
carbon so far, of the whole flow (kgCO2e)



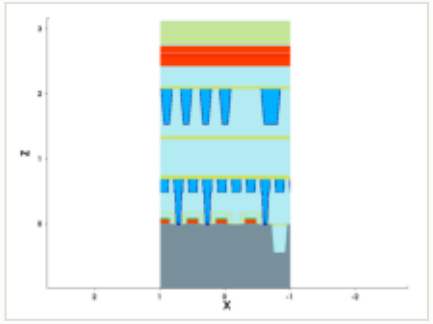
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



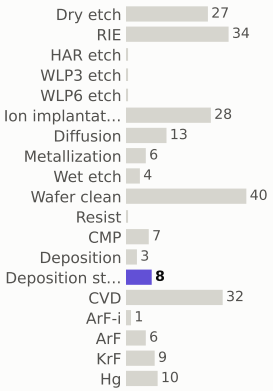
y cut at -1.032 µm



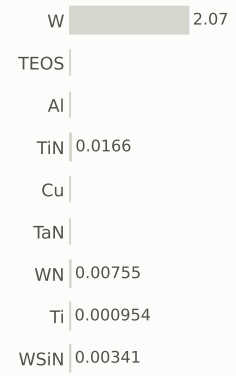
mask for this step

no mask for this step

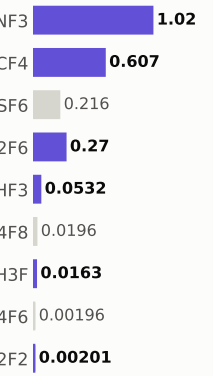
steps by process area



materials deposited (g)



gases emitted (g)



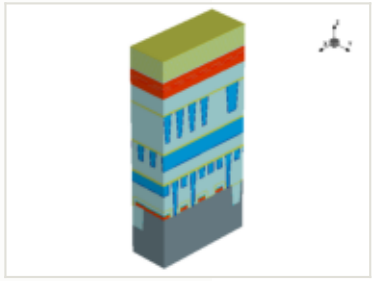
Step #2480: Deposition_stack

depo_anisotropic · Deposition_stack

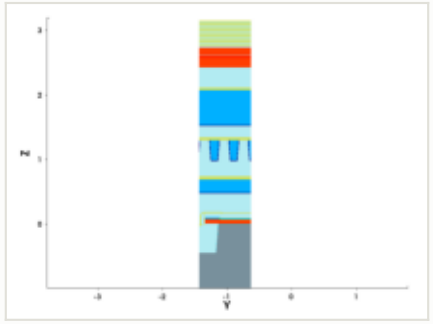
0.654	192.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

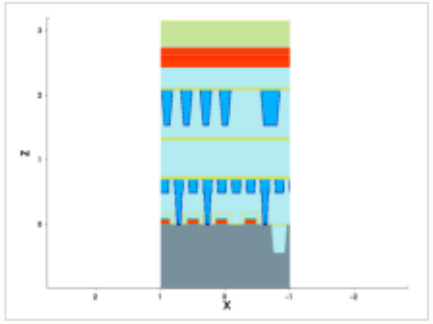
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



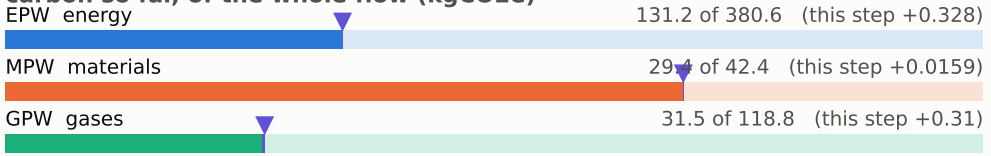
mask for this step

no mask for this step

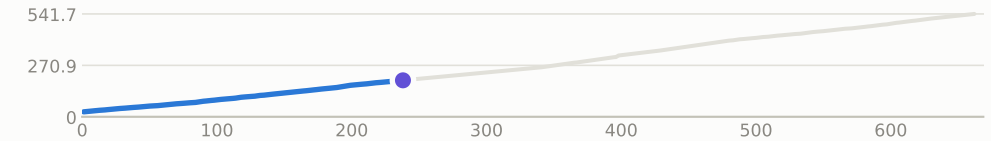
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

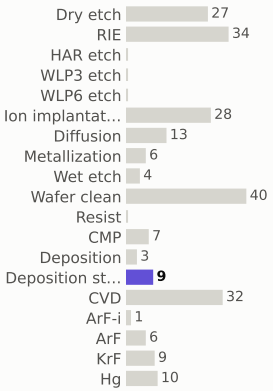
carbon so far, of the whole flow (kgCO2e)



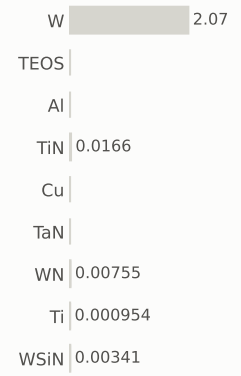
embodied carbon up to each step (kgCO2e per wafer)



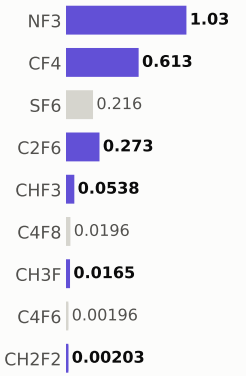
steps by process area



materials deposited (g)



gases emitted (g)



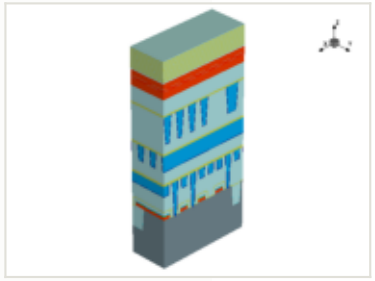
Step #2490: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	192.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

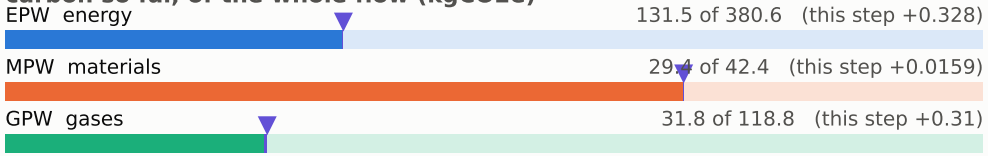
zoomed to fit



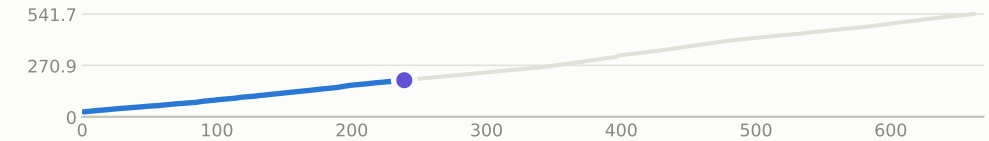
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

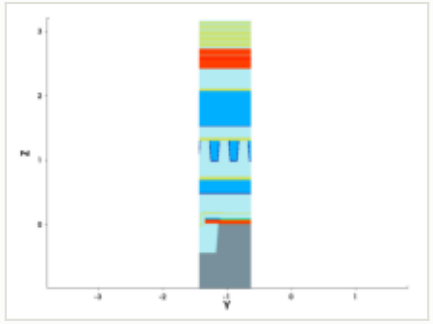
carbon so far, of the whole flow (kgCO2e)



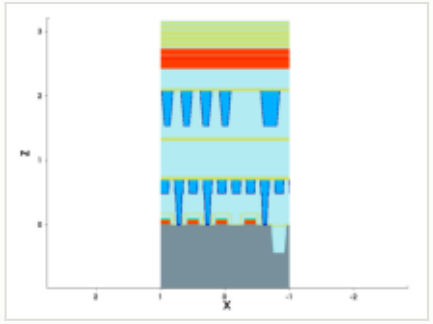
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



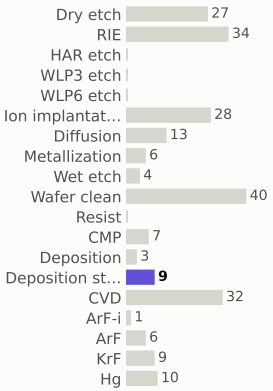
y cut at -1.032 µm



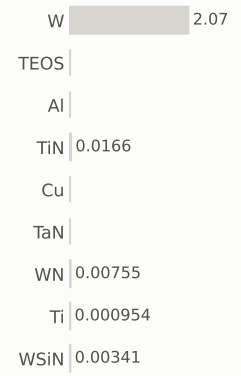
mask for this step

no mask for this step

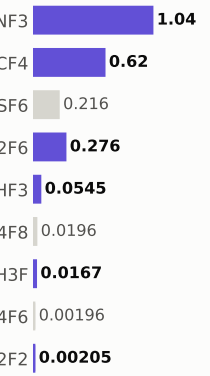
steps by process area



materials deposited (g)



gases emitted (g)



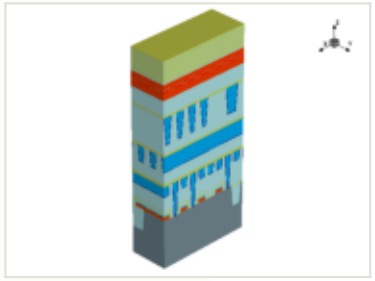
Step #2500: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	193.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

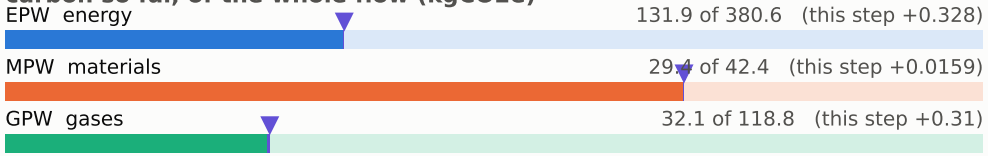
zoomed to fit



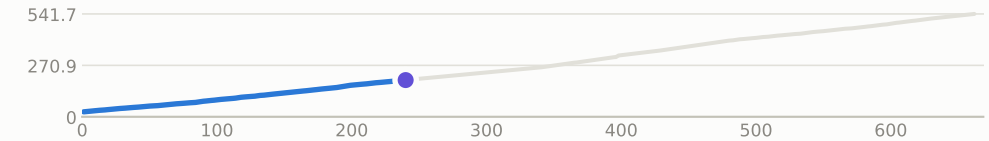
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

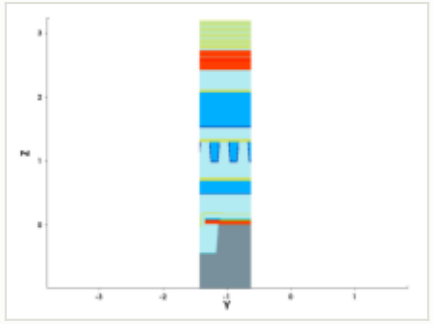
carbon so far, of the whole flow (kgCO2e)



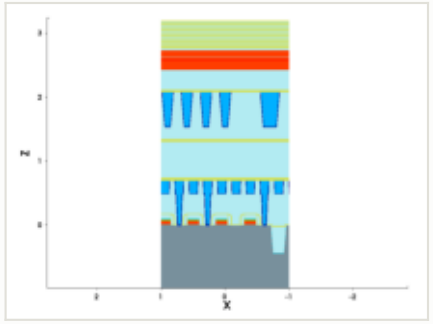
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



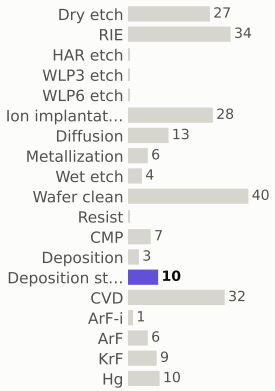
y cut at -1.032 µm



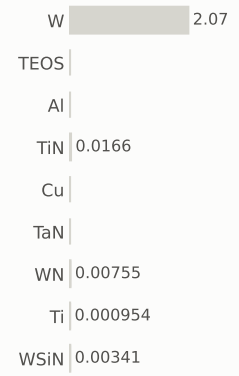
mask for this step

no mask for this step

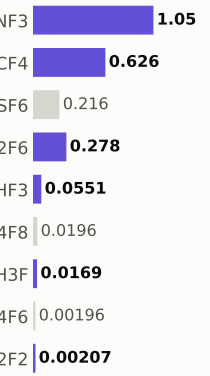
steps by process area



materials deposited (g)



gases emitted (g)



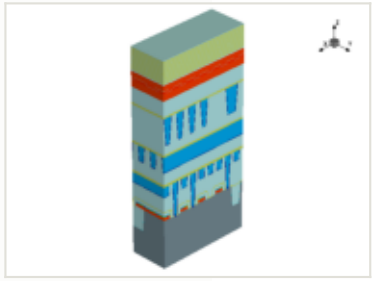
Step #2510: Deposition_stack

depo_anisotropic · Deposition_stack

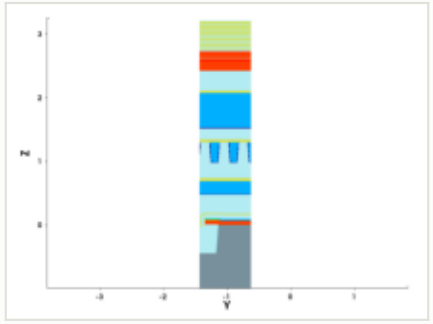
0.654	194.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

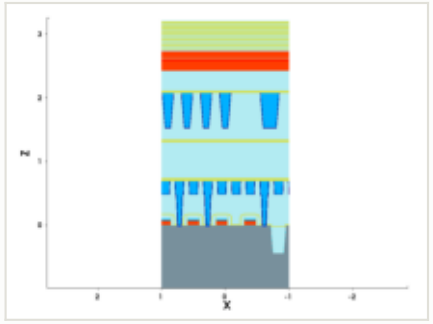
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



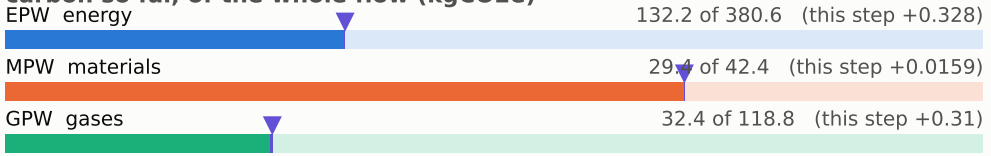
mask for this step

no mask for this step

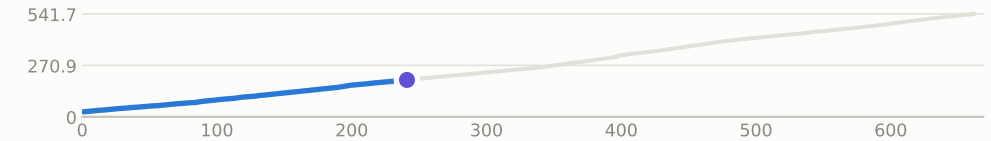
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

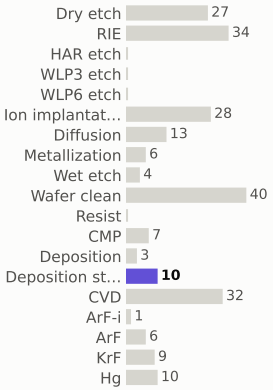
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

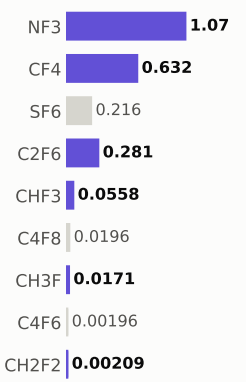


steps by process area



materials deposited (g)

gases emitted (g)



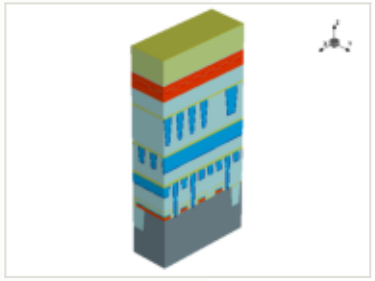
Step #2520: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	194.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

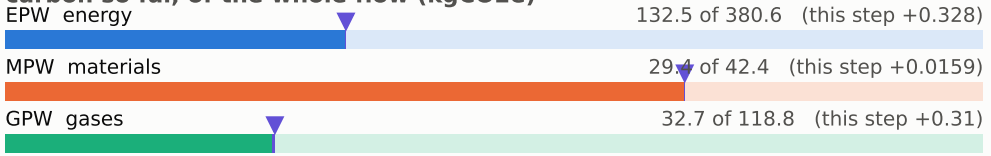
zoomed to fit



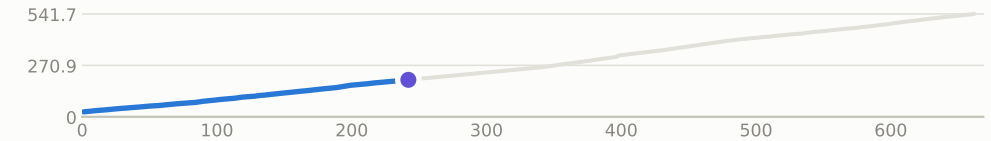
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

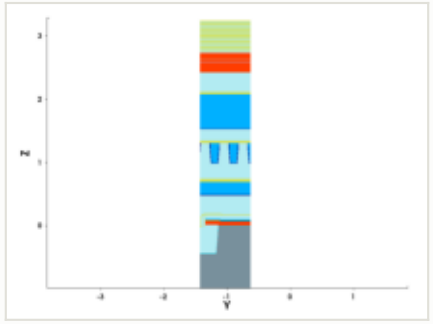
carbon so far, of the whole flow (kgCO2e)



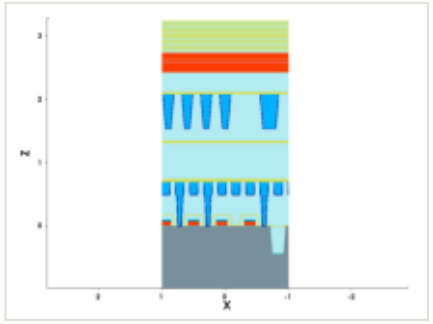
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



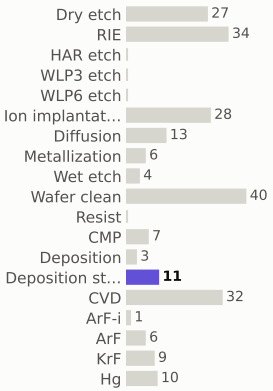
y cut at -1.032 µm



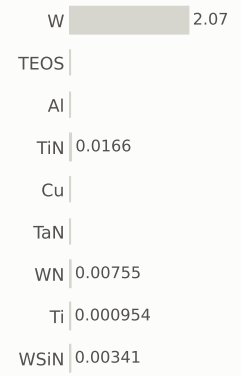
mask for this step

no mask for this step

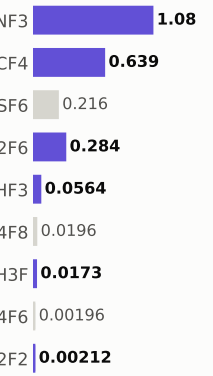
steps by process area



materials deposited (g)



gases emitted (g)



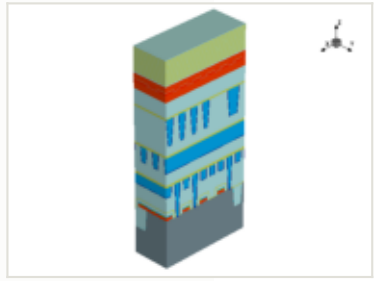
Step #2530: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	195.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

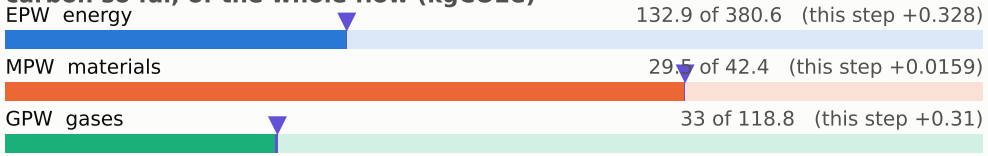
zoomed to fit



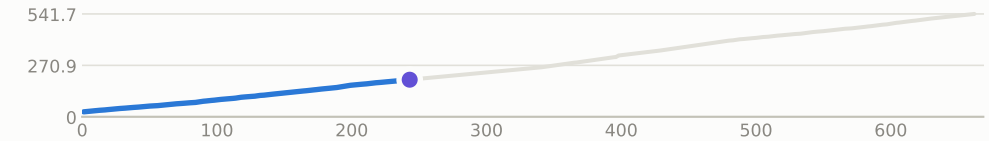
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

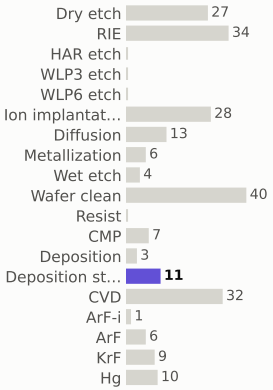
carbon so far, of the whole flow (kgCO2e)



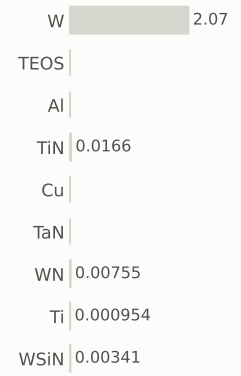
embodied carbon up to each step (kgCO2e per wafer)



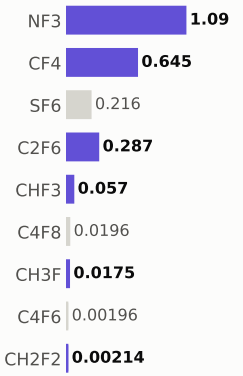
steps by process area



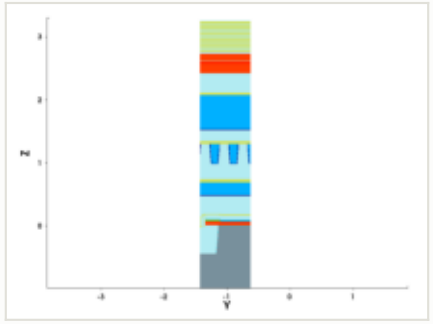
materials deposited (g)



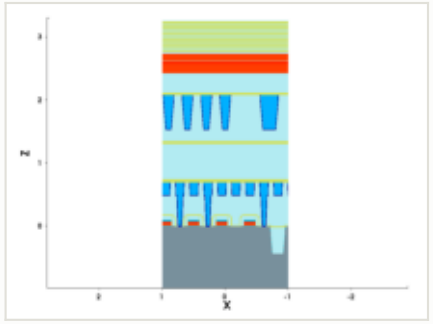
gases emitted (g)



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

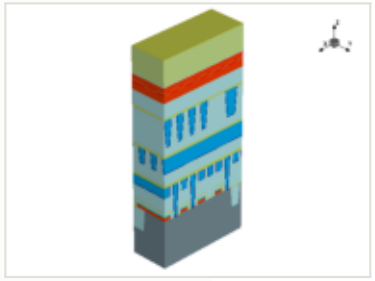
Step #2540: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	196.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

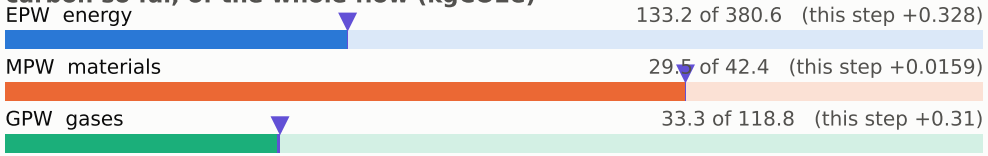
zoomed to fit



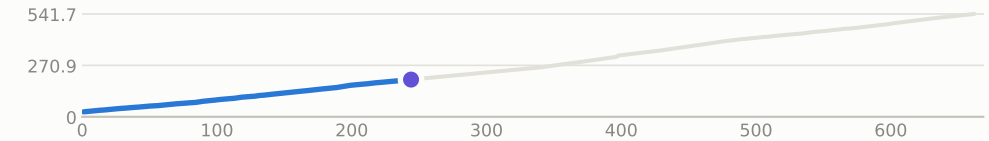
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

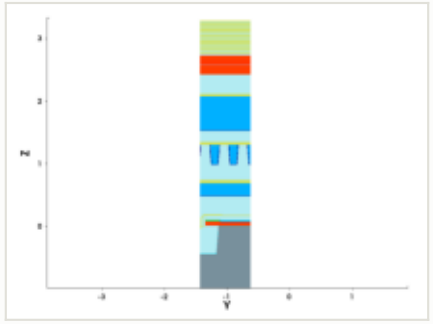
carbon so far, of the whole flow (kgCO2e)



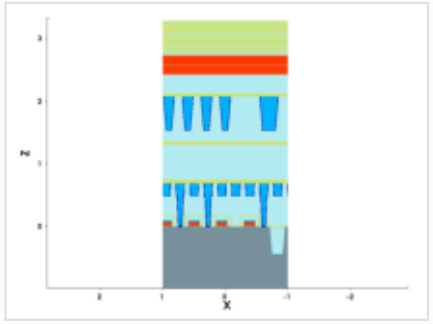
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



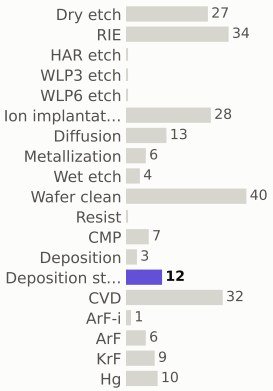
y cut at -1.032 µm



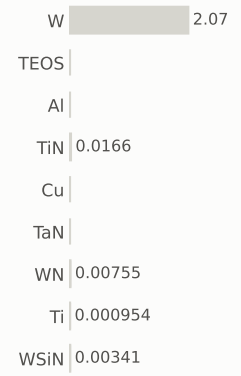
mask for this step

no mask for this step

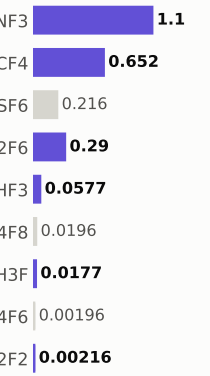
steps by process area



materials deposited (g)



gases emitted (g)



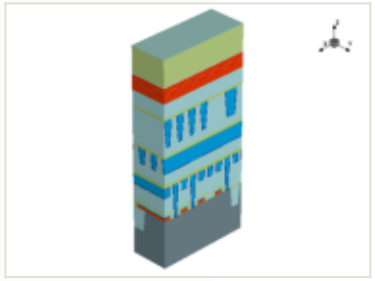
Step #2550: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	196.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

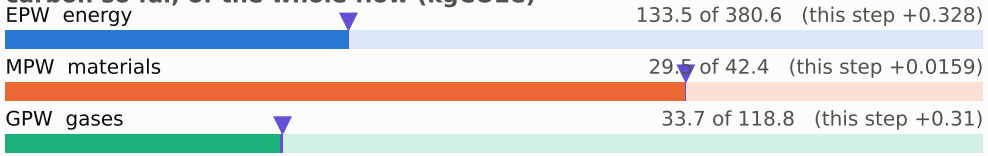
zoomed to fit



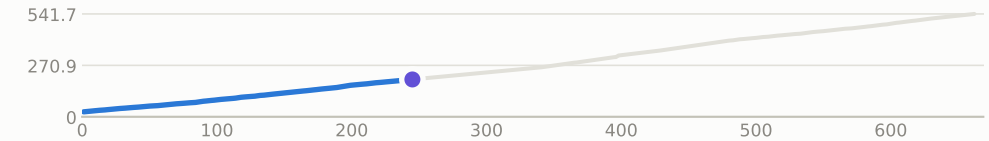
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

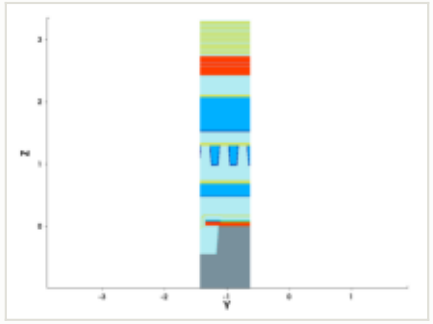
carbon so far, of the whole flow (kgCO2e)



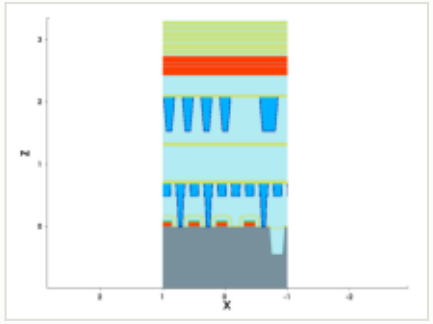
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



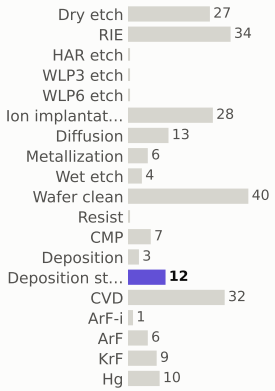
y cut at -1.032 µm



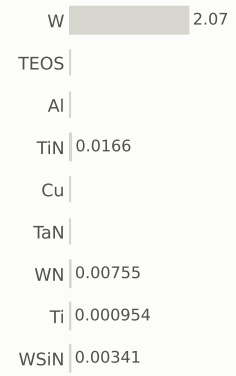
mask for this step

no mask for this step

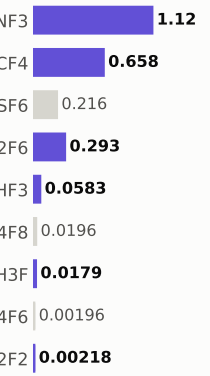
steps by process area



materials deposited (g)



gases emitted (g)



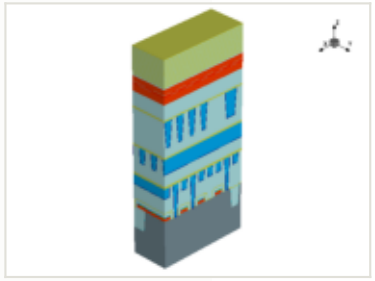
Step #2560: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	197.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

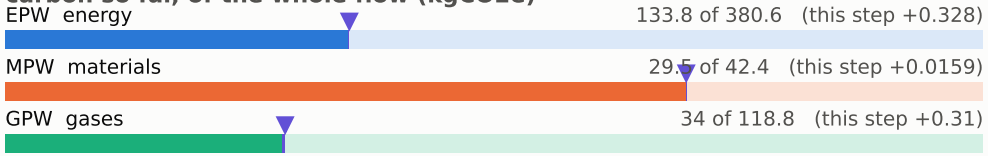
zoomed to fit



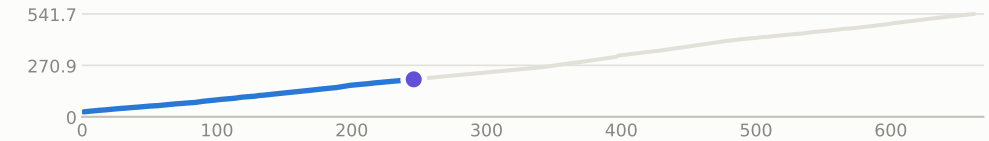
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

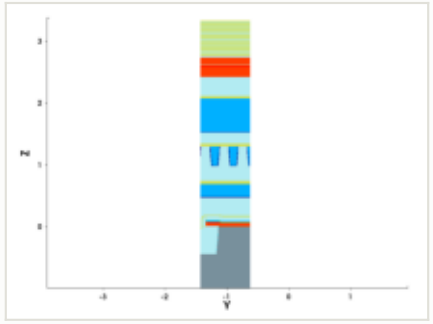
carbon so far, of the whole flow (kgCO2e)



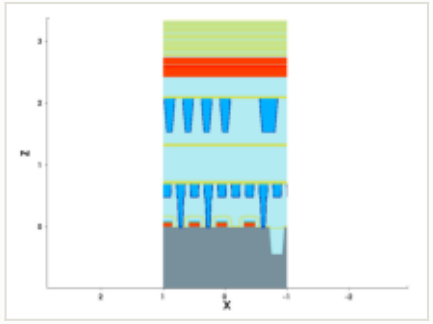
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



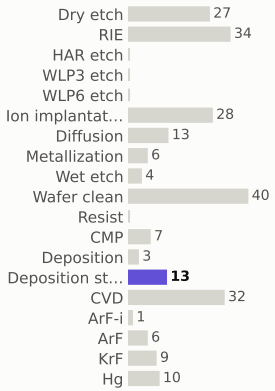
y cut at -1.032 µm



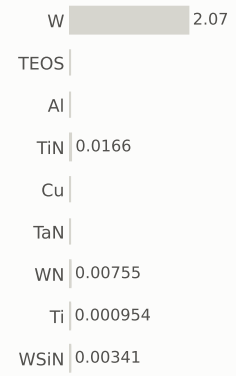
mask for this step

no mask for this step

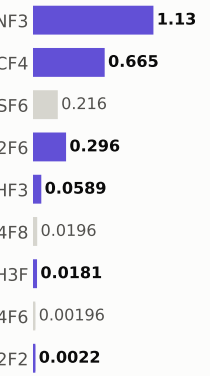
steps by process area



materials deposited (g)



gases emitted (g)



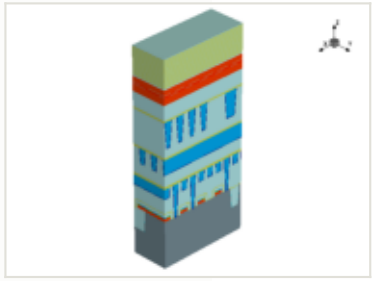
Step #2570: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	198.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

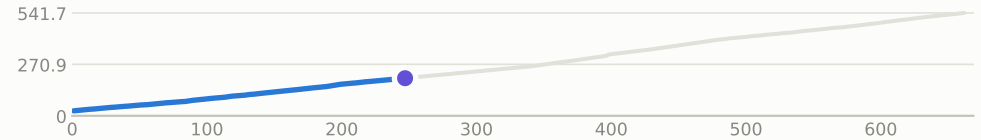
carbon so far, of the whole flow (kgCO2e)

EPW energy — 134.2 of 380.6 (this step +0.328)

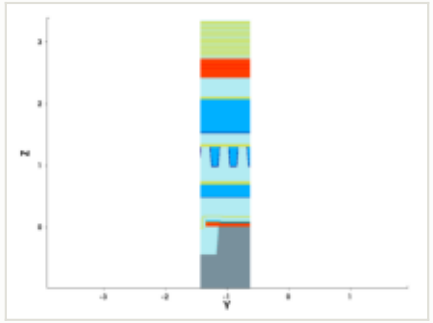
MPW materials — 29.5 of 42.4 (this step +0.0159)

GPW gases — 34.3 of 118.8 (this step +0.31)

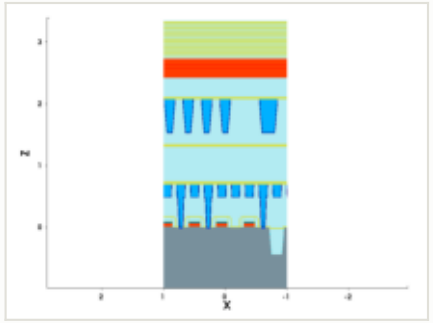
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



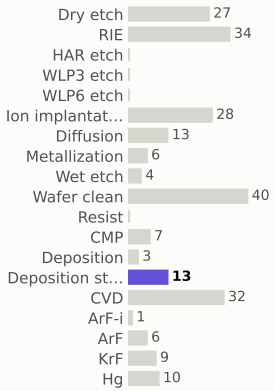
y cut at -1.032 µm



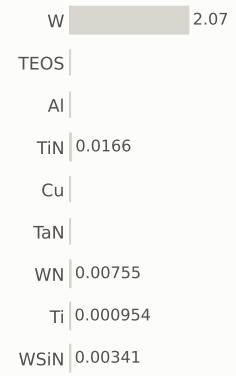
mask for this step

no mask for this step

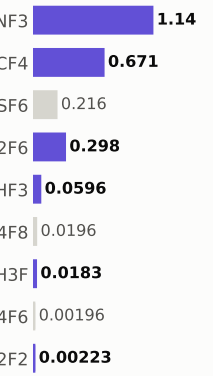
steps by process area



materials deposited (g)



gases emitted (g)



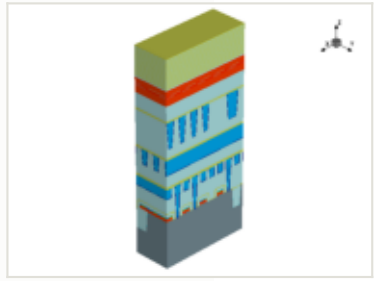
Step #2580: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	198.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

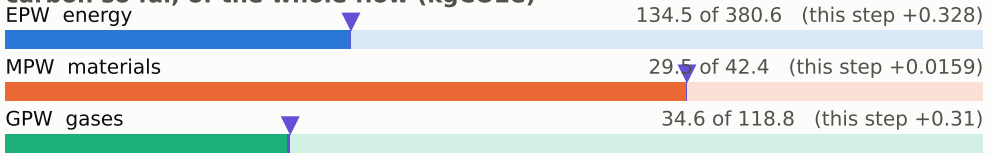
zoomed to fit



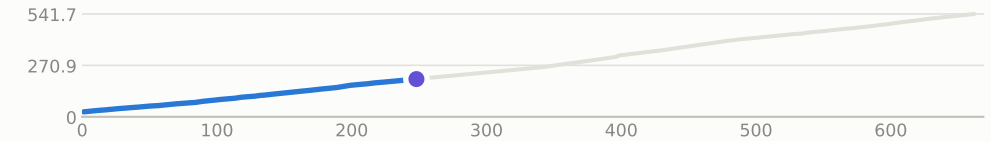
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

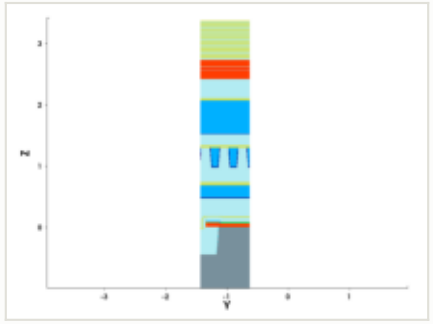
carbon so far, of the whole flow (kgCO2e)



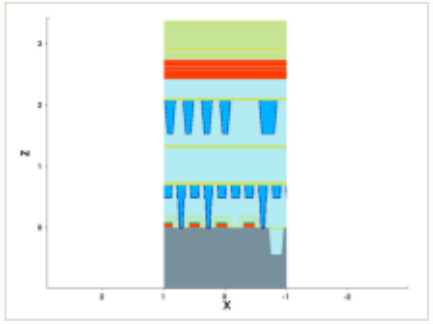
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



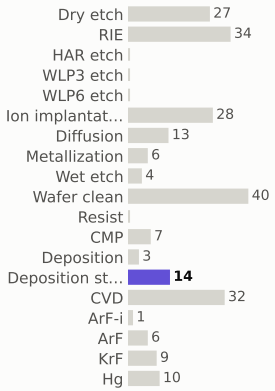
y cut at -1.032 µm



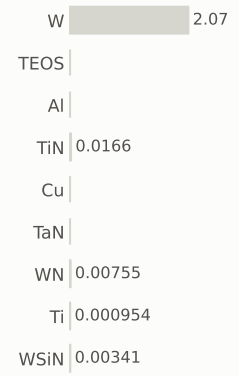
mask for this step

no mask for this step

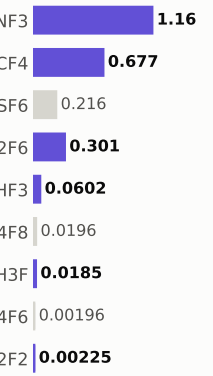
steps by process area



materials deposited (g)



gases emitted (g)



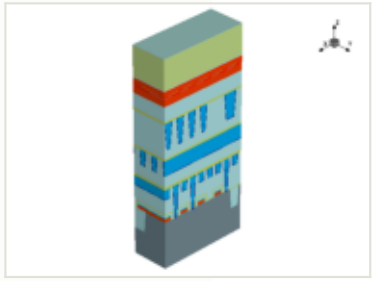
Step #2590: Deposition_stack

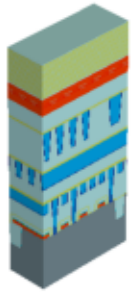
depo_anisotropic · Deposition_stack

0.654	199.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

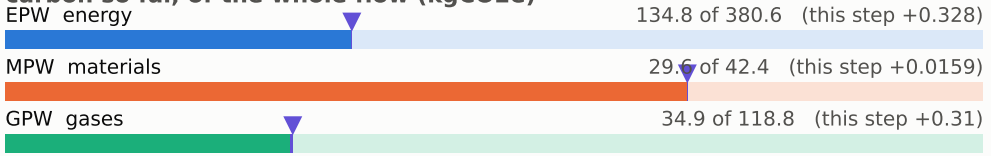




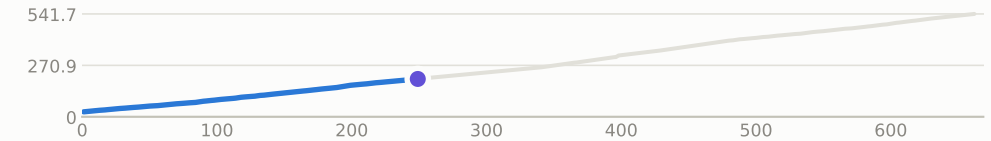
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

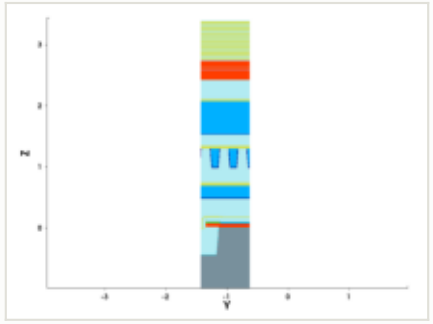
carbon so far, of the whole flow (kgCO2e)



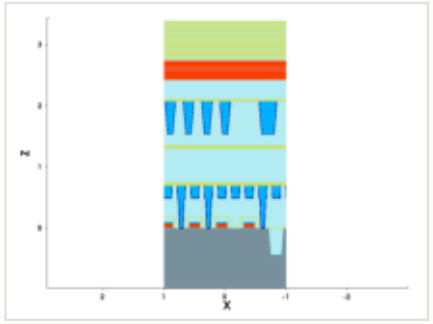
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



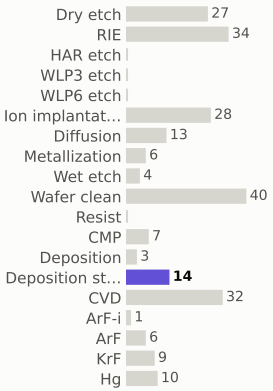
y cut at -1.032 µm



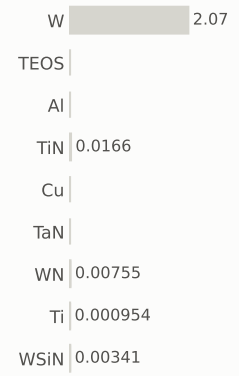
mask for this step

no mask for this step

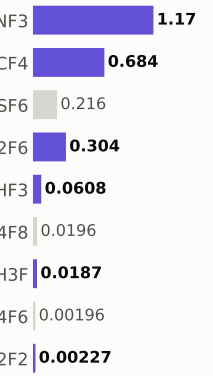
steps by process area



materials deposited (g)



gases emitted (g)



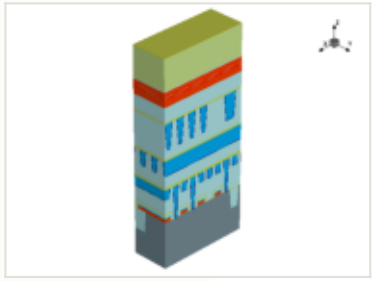
Step #2600: Deposition_stack

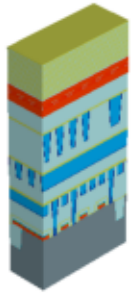
depo_anisotropic · Deposition_stack

0.654	199.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

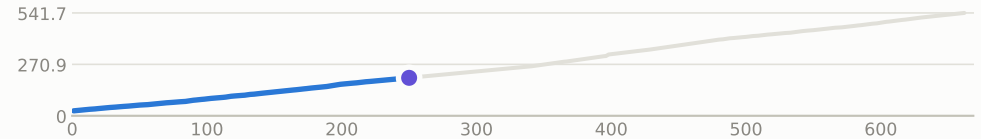
carbon so far, of the whole flow (kgCO2e)

EPW energy — 135.2 of 380.6 (this step +0.328)

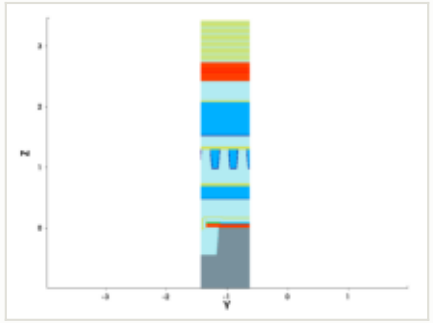
MPW materials — 29.6 of 42.4 (this step +0.0159)

GPW gases — 35.2 of 118.8 (this step +0.31)

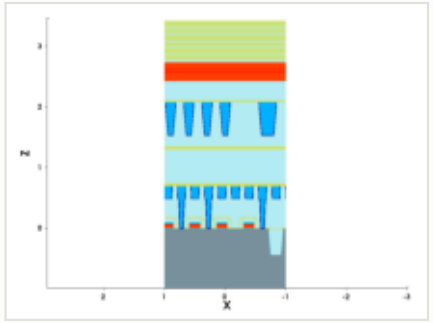
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



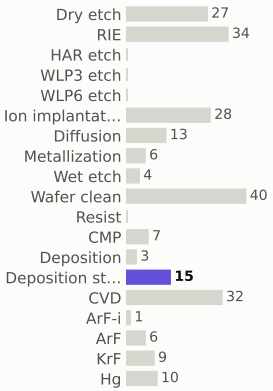
y cut at -1.032 µm



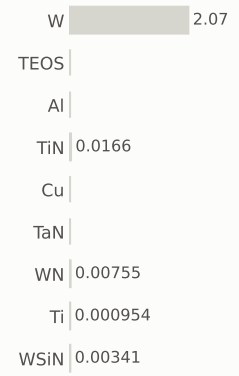
mask for this step

no mask for this step

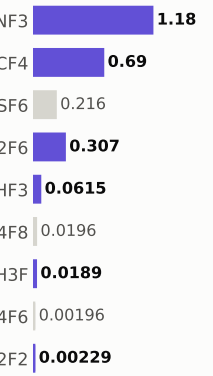
steps by process area



materials deposited (g)



gases emitted (g)



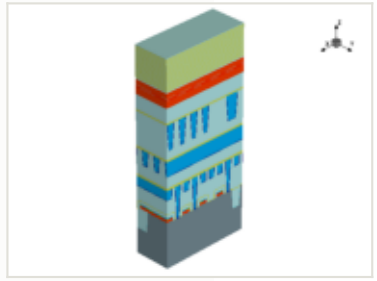
Step #2610: Deposition_stack

depo_anisotropic · Deposition_stack

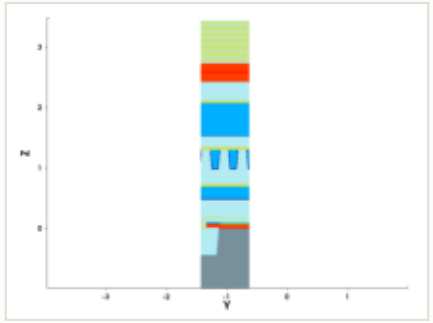
0.654	200.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

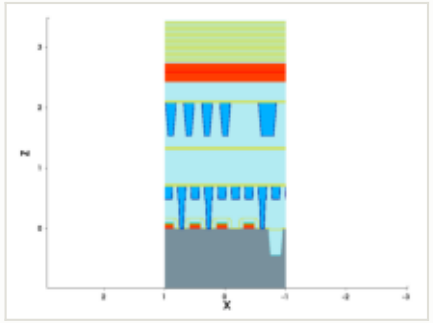
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



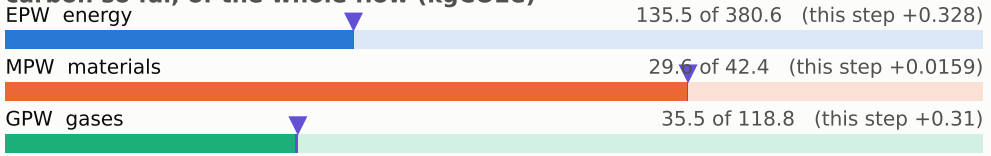
mask for this step

no mask for this step

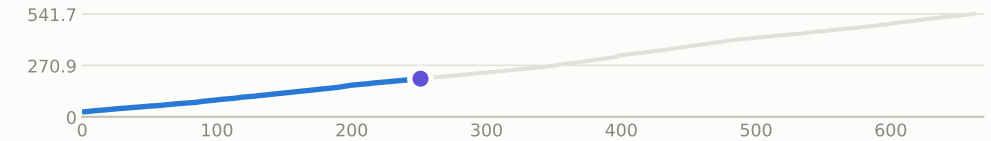
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

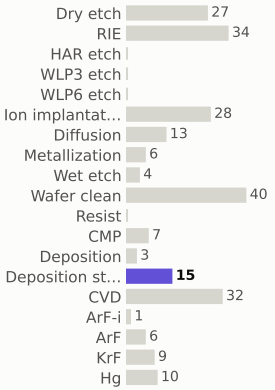
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

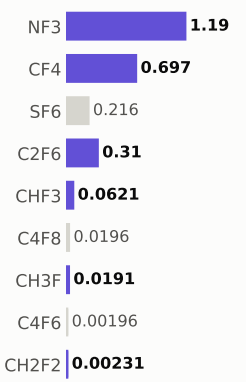


steps by process area



materials deposited (g)

gases emitted (g)



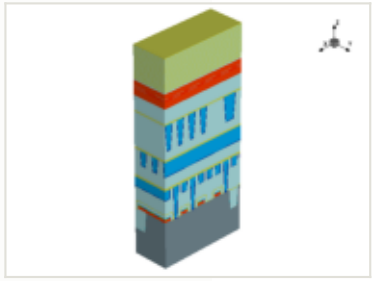
Step #2620: Deposition_stack

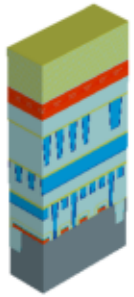
depo_anisotropic · Deposition_stack

0.654	201.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



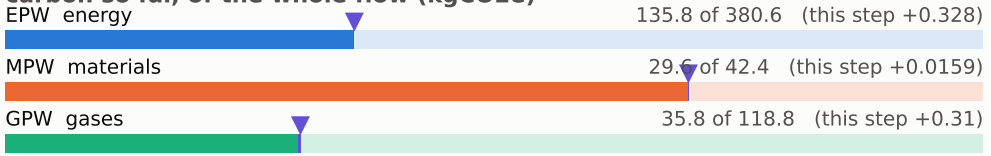


this step

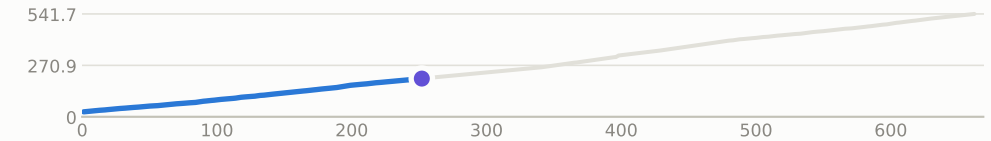
process area	Deposition_stack
material	—
thickness	0.028 µm
material GWP	—
input gases	NF3 1.27 g
energy	0.545 kWh + 0.218 overhead

EPW	0.328 kgCO2e
EPW tool	0.234 kgCO2e
EPW overhead	0.0937 kgCO2e
MPW	0.0159 kgCO2e (gas 0.0159)
GPW	0.31 kgCO2e
embodied carbon	**0.654 kgCO2e**

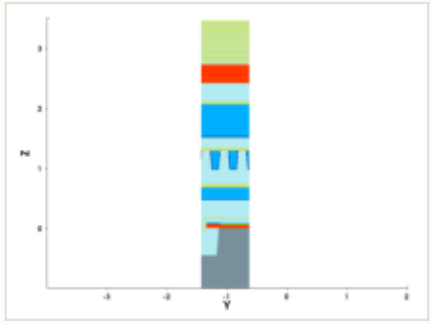
carbon so far, of the whole flow (kgCO2e)



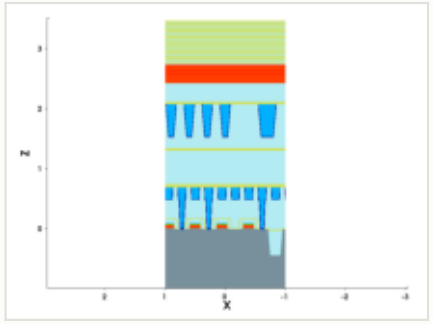
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



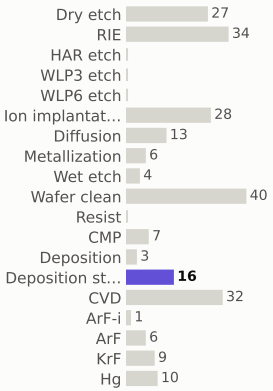
y cut at -1.032 µm



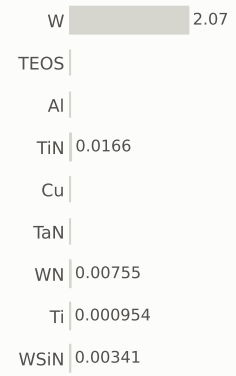
mask for this step

no mask for this step

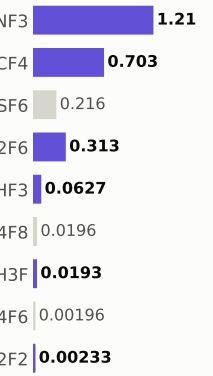
steps by process area



materials deposited (g)



gases emitted (g)



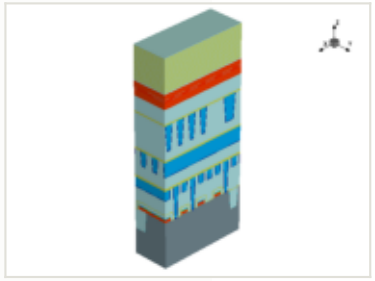
Step #2630: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	201.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

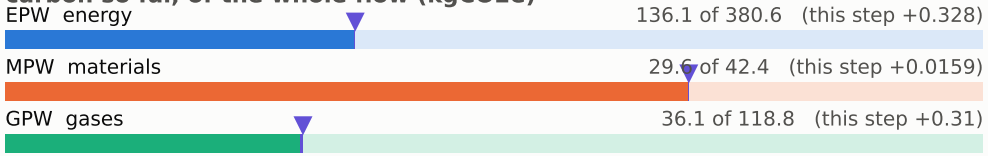
zoomed to fit



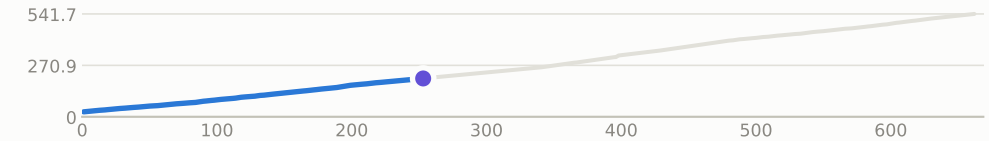
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

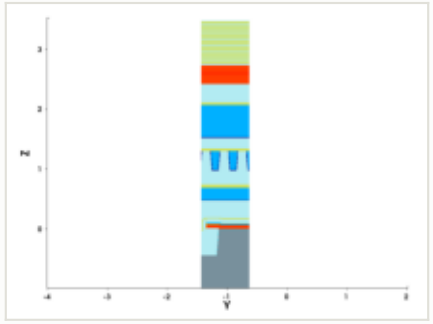
carbon so far, of the whole flow (kgCO2e)



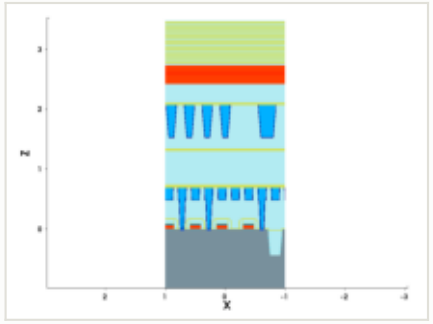
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



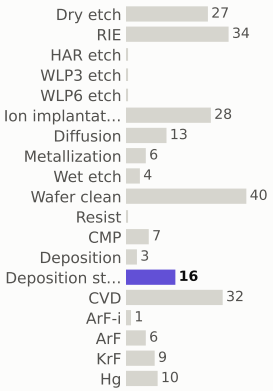
y cut at -1.032 µm



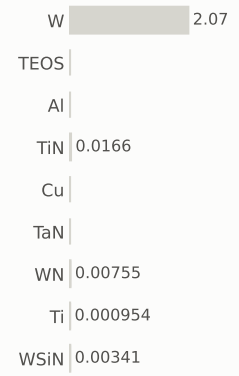
mask for this step

no mask for this step

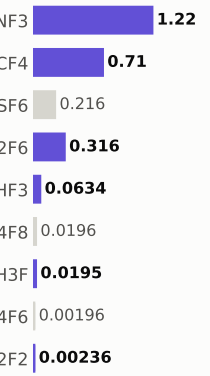
steps by process area



materials deposited (g)



gases emitted (g)



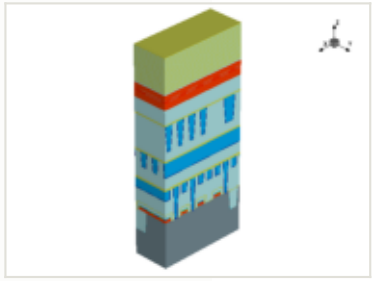
Step #2640: Deposition_stack

depo_anisotropic · Deposition_stack

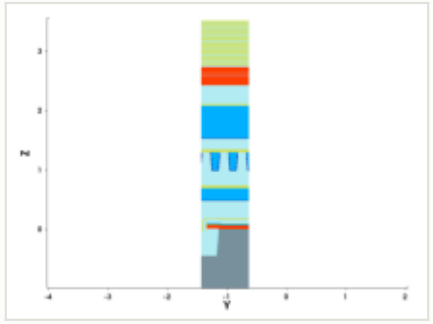
0.654	202.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

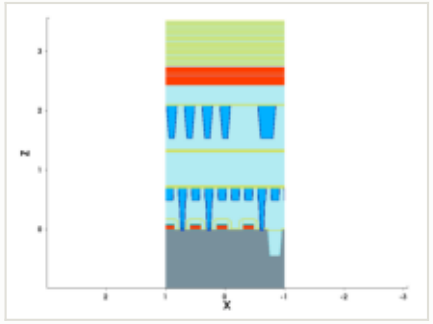
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



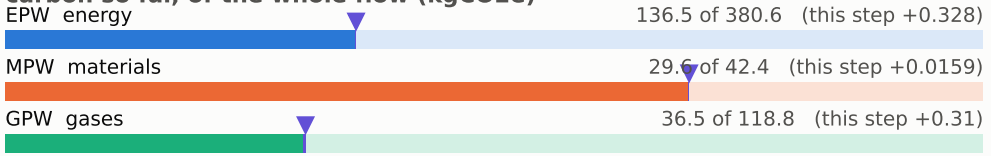
mask for this step

no mask for this step

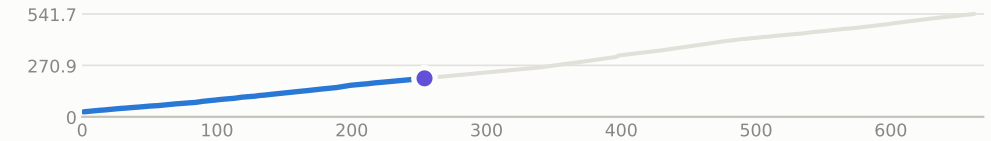
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

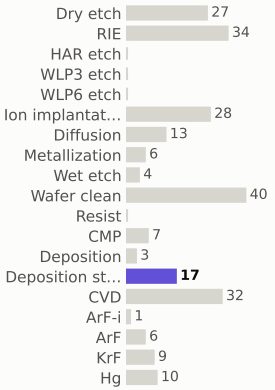
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

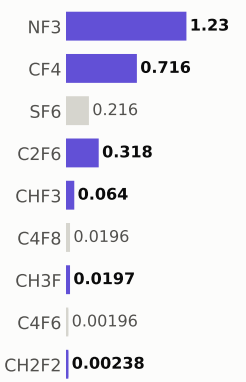


steps by process area



materials deposited (g)

gases emitted (g)



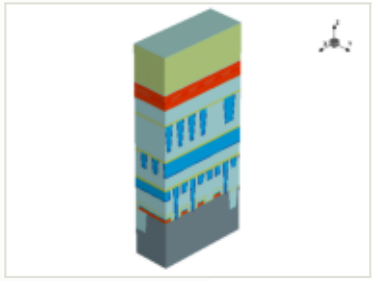
Step #2650: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	203.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

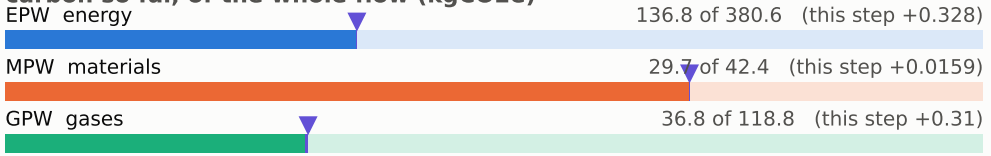
zoomed to fit



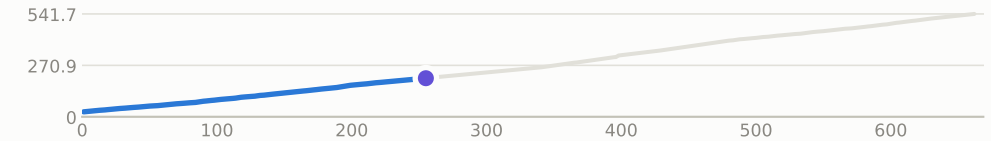
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

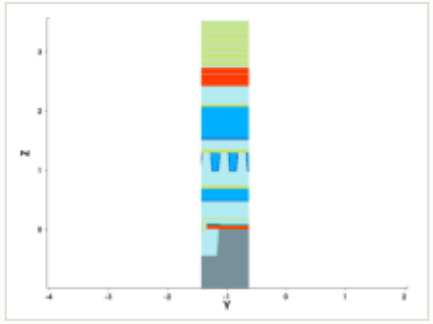
carbon so far, of the whole flow (kgCO2e)



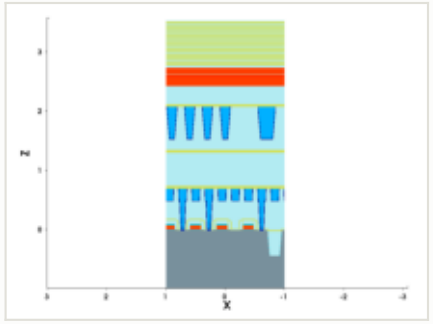
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



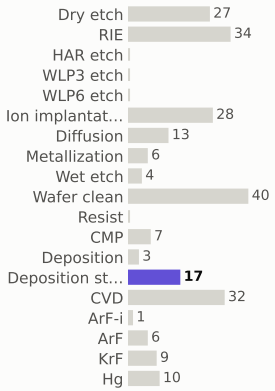
y cut at -1.032 µm



mask for this step

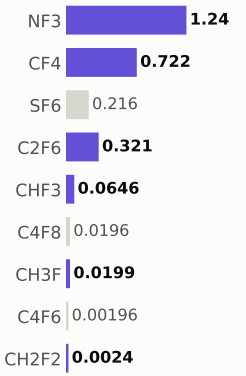
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



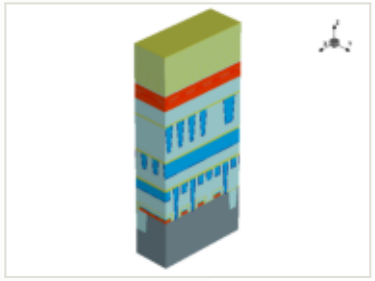
Step #2660: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	203.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

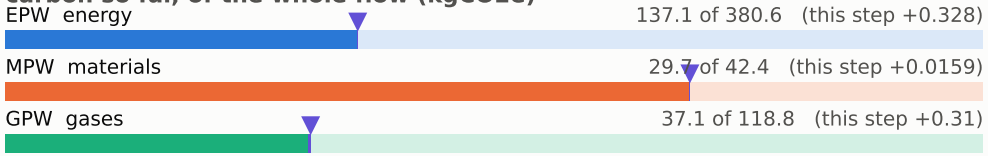
zoomed to fit



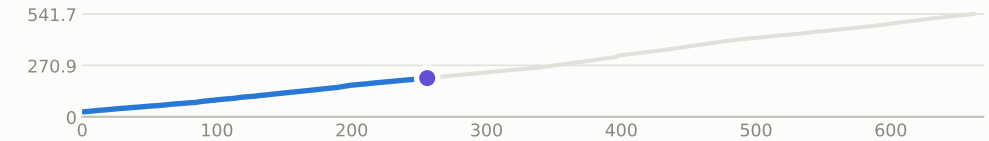
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

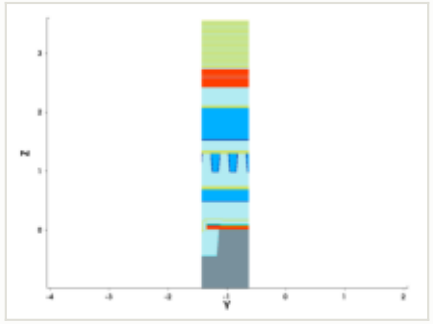
carbon so far, of the whole flow (kgCO2e)



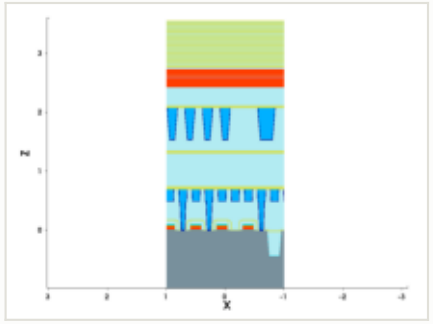
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



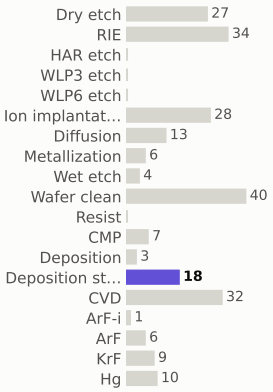
y cut at -1.032 µm



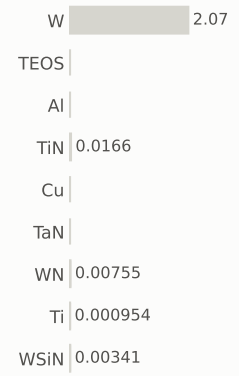
mask for this step

no mask for this step

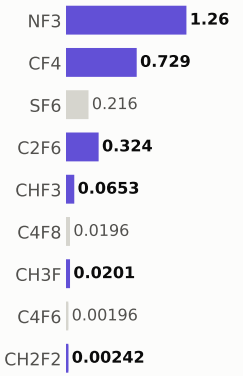
steps by process area



materials deposited (g)



gases emitted (g)



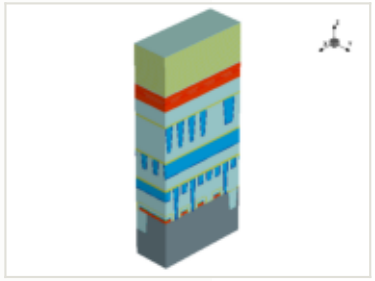
Step #2670: Deposition_stack

depo_anisotropic · Deposition_stack

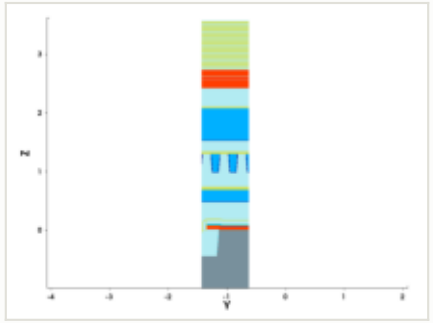
0.654	204.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

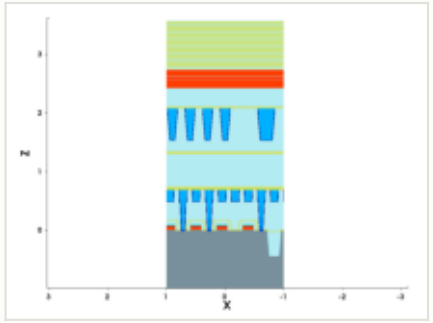
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



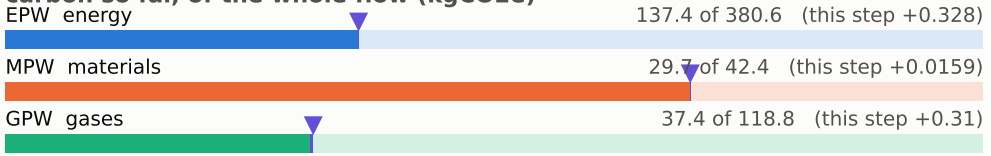
mask for this step

no mask for this step

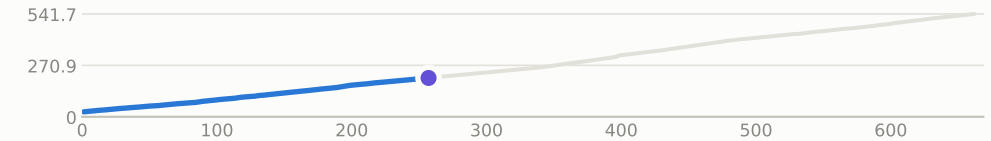
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

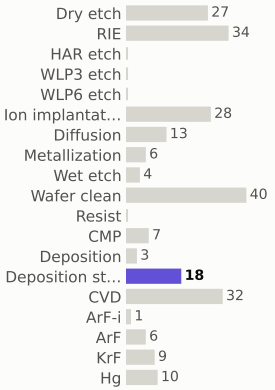
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

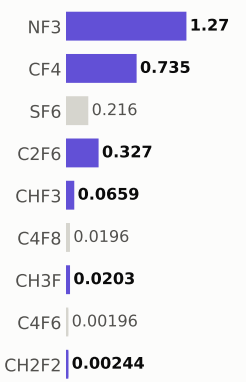


steps by process area



materials deposited (g)

gases emitted (g)



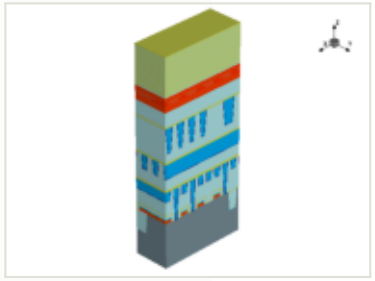
Step #2680: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	205.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

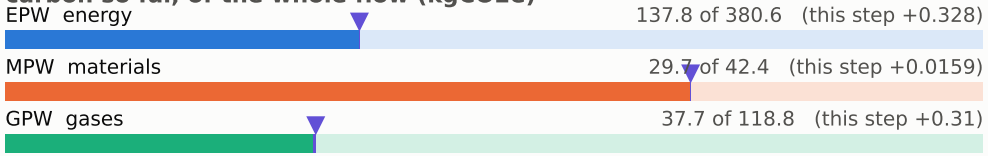
zoomed to fit



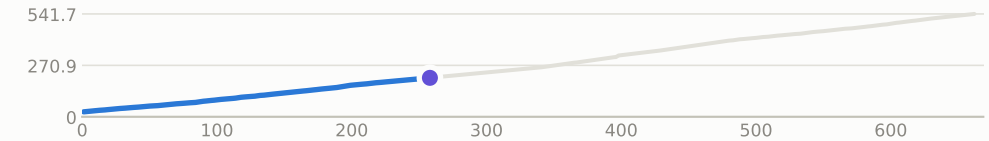
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

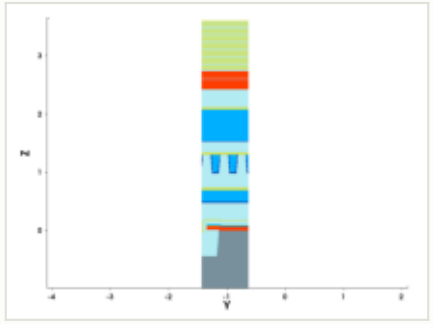
carbon so far, of the whole flow (kgCO2e)



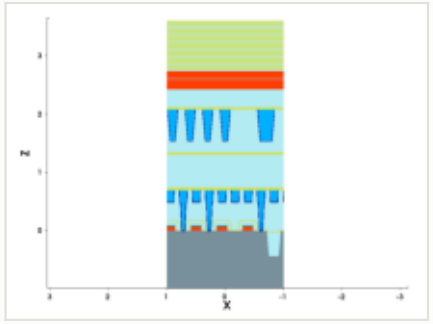
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



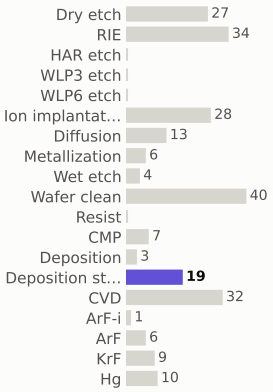
y cut at -1.032 µm



mask for this step

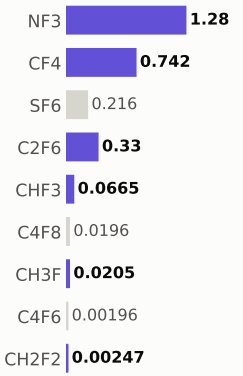
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



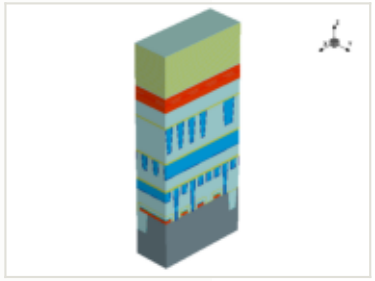
Step #2690: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	205.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

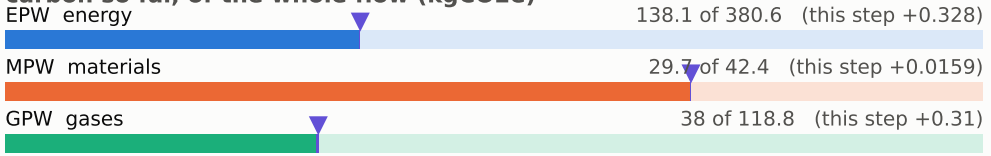
zoomed to fit



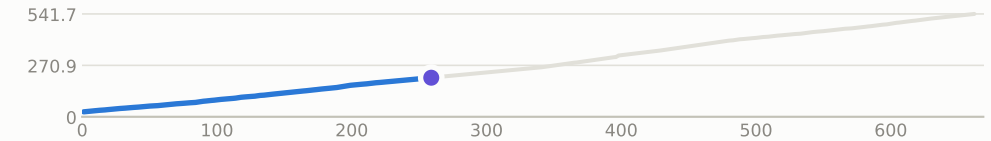
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

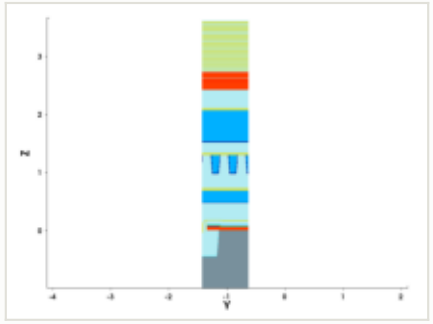
carbon so far, of the whole flow (kgCO2e)



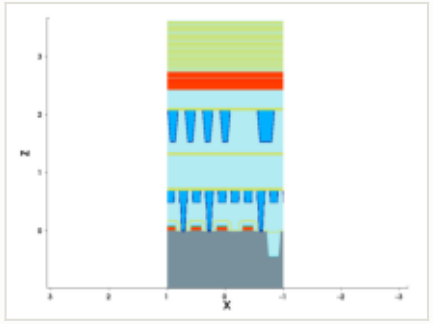
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



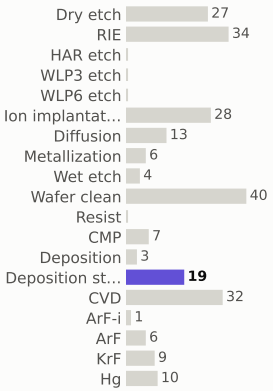
y cut at -1.032 µm



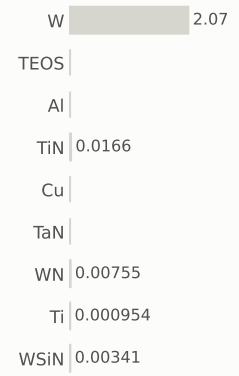
mask for this step

no mask for this step

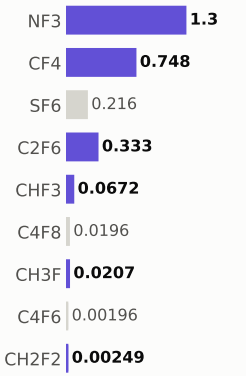
steps by process area



materials deposited (g)



gases emitted (g)



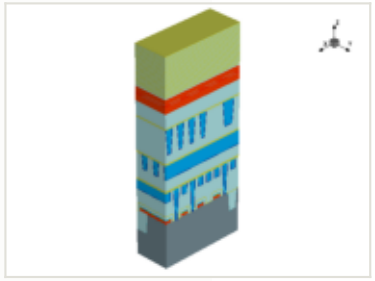
Step #2700: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	206.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

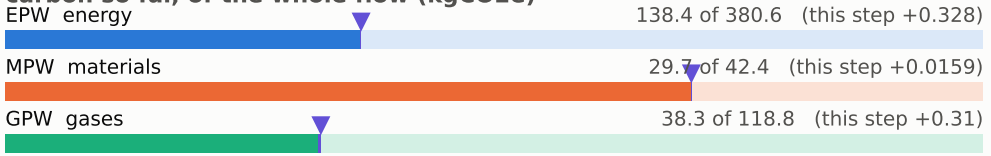
zoomed to fit



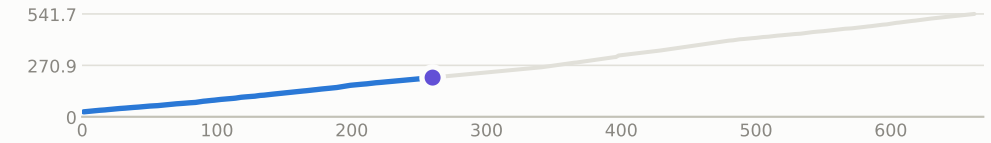
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

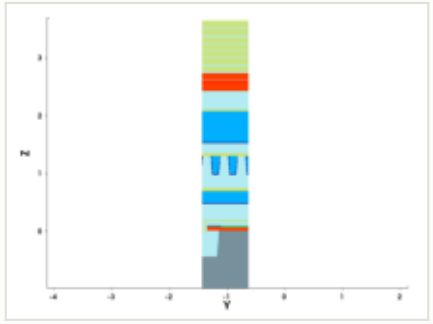
carbon so far, of the whole flow (kgCO2e)



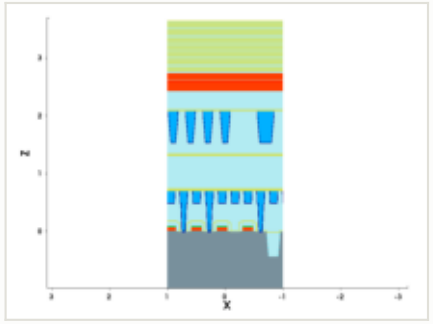
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



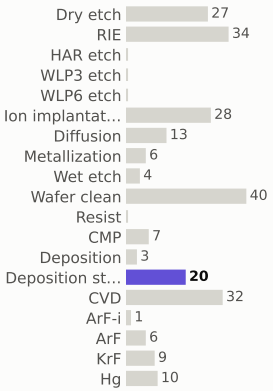
y cut at -1.032 µm



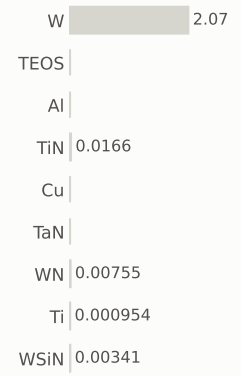
mask for this step

no mask for this step

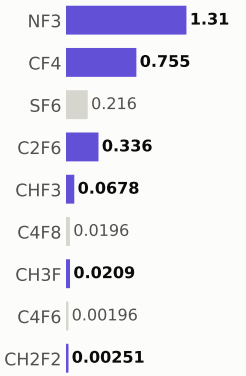
steps by process area



materials deposited (g)



gases emitted (g)



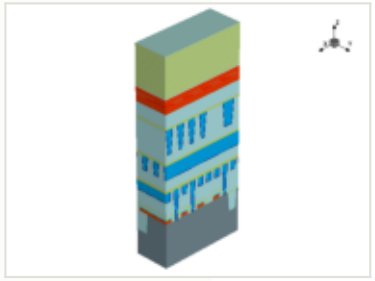
Step #2710: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	207.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

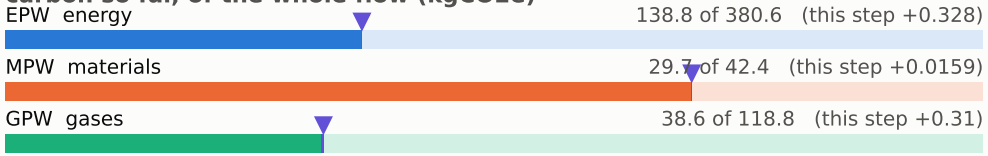
zoomed to fit



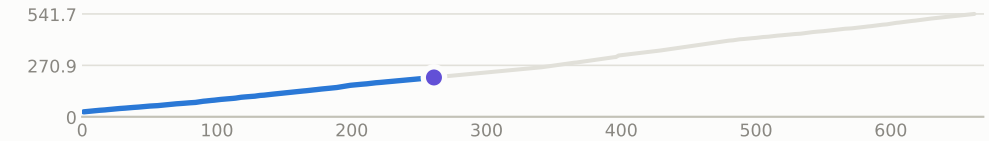
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

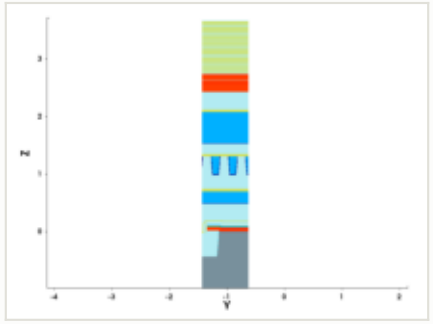
carbon so far, of the whole flow (kgCO2e)



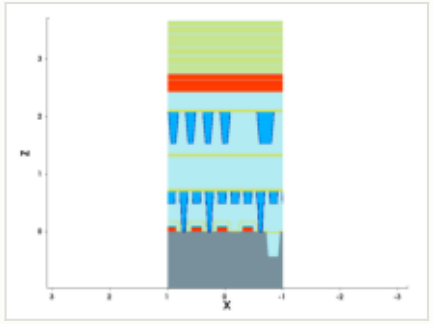
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



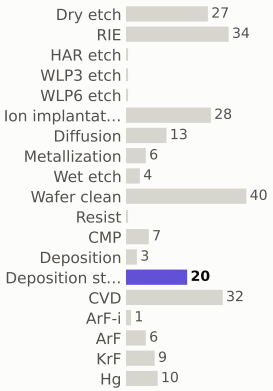
y cut at -1.032 µm



mask for this step

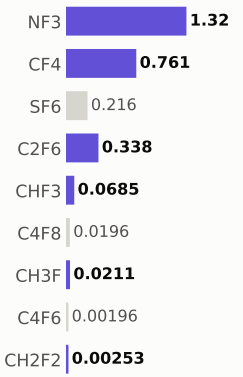
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



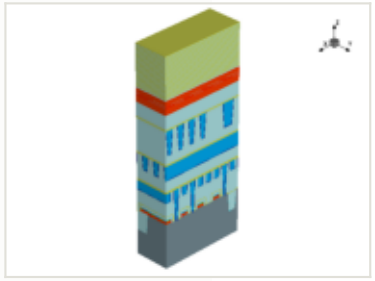
Step #2720: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	207.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

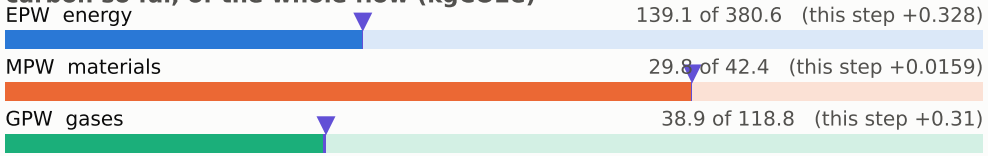
zoomed to fit



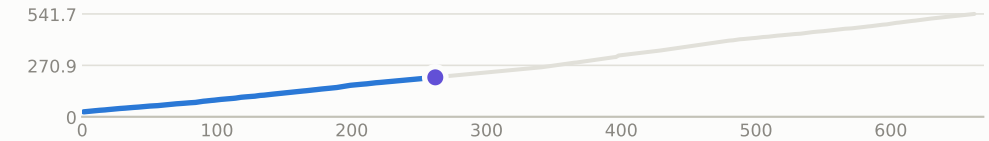
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

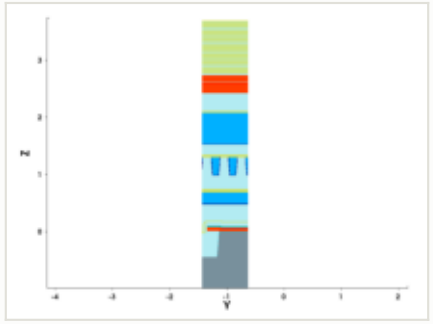
carbon so far, of the whole flow (kgCO2e)



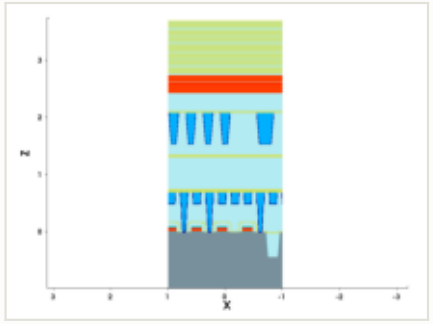
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



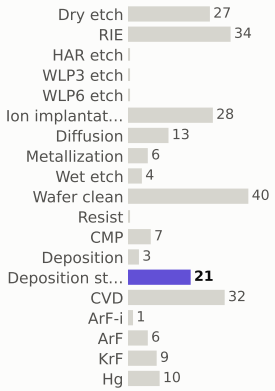
y cut at -1.032 µm



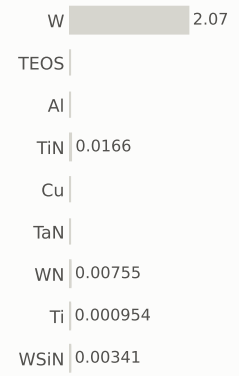
mask for this step

no mask for this step

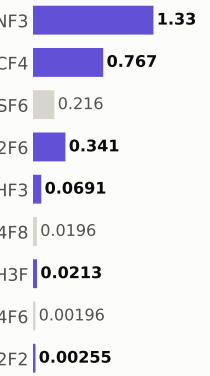
steps by process area



materials deposited (g)



gases emitted (g)



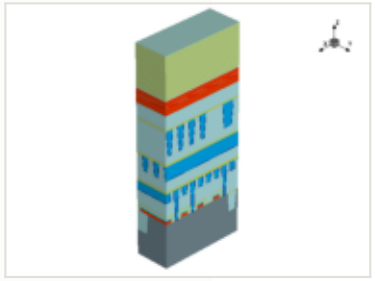
Step #2730: Deposition_stack

depo_anisotropic · Deposition_stack

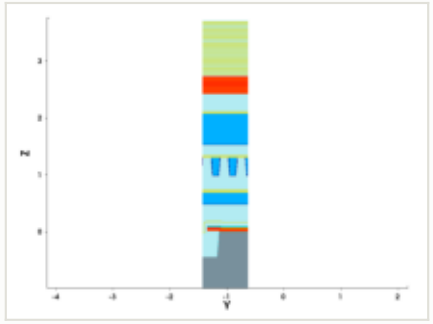
0.654	208.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

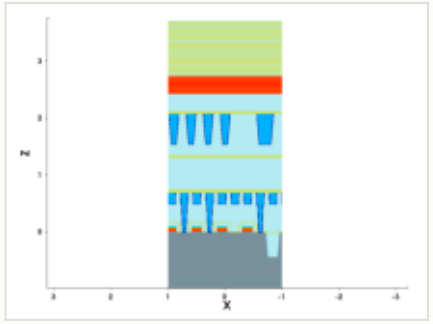
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



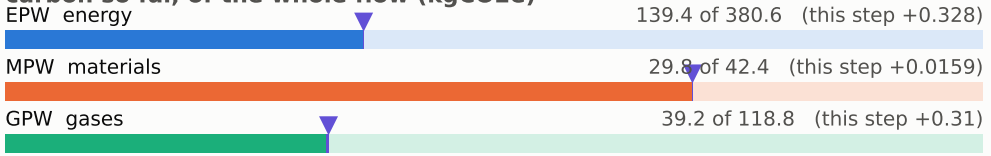
mask for this step

no mask for this step

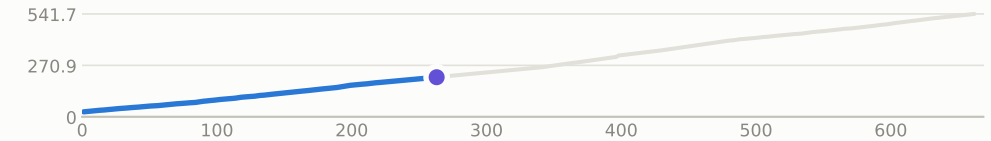
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

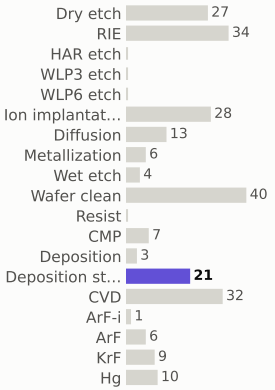
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

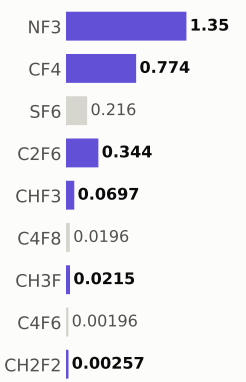


steps by process area



materials deposited (g)

gases emitted (g)



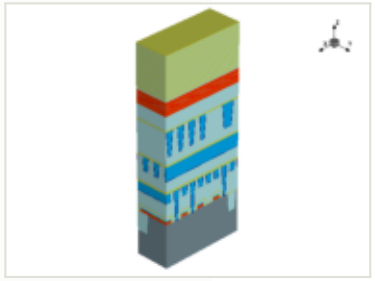
Step #2740: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	209.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

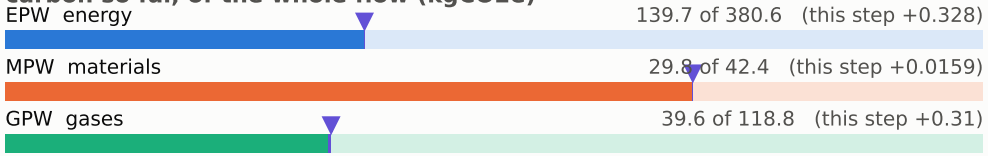
zoomed to fit



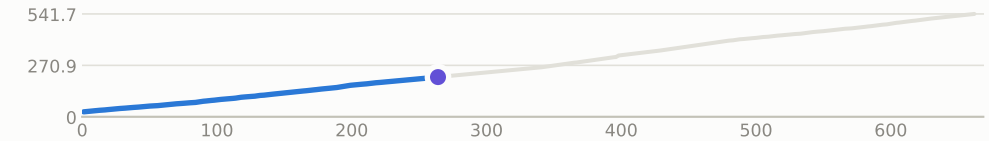
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

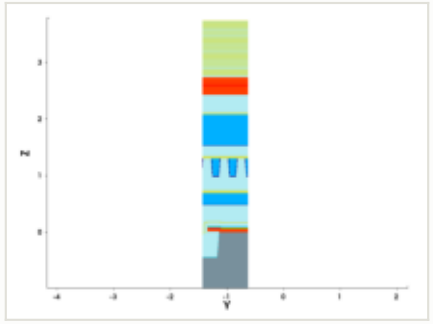
carbon so far, of the whole flow (kgCO2e)



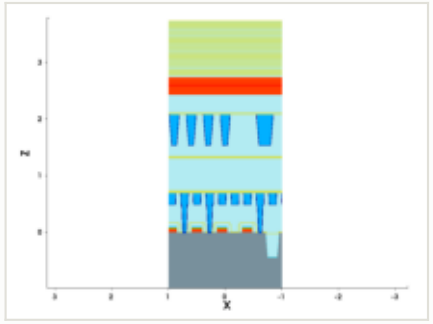
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



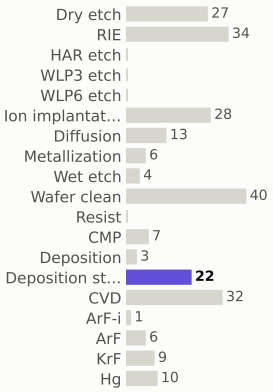
y cut at -1.032 µm



mask for this step

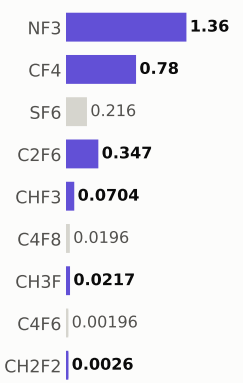
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



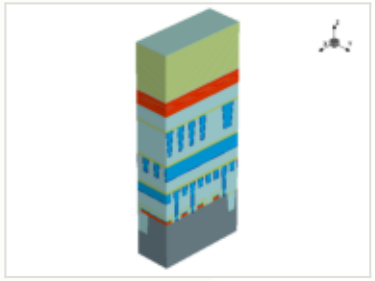
Step #2750: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	209.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

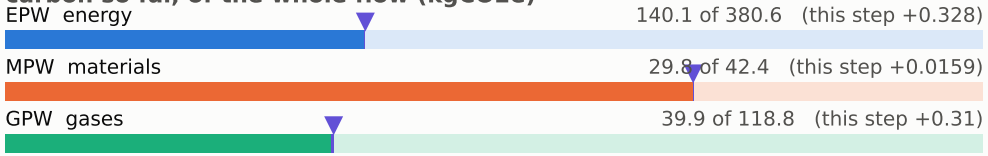
zoomed to fit



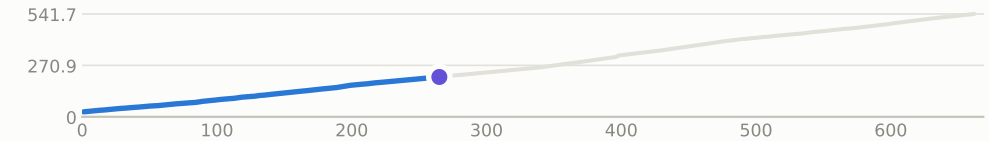
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

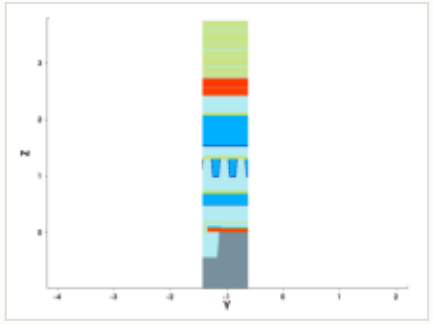
carbon so far, of the whole flow (kgCO2e)



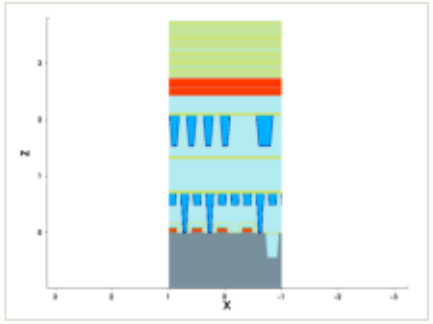
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



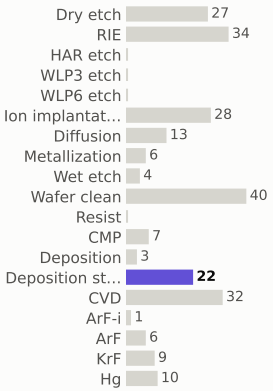
y cut at -1.032 µm



mask for this step

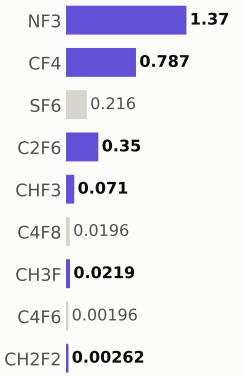
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



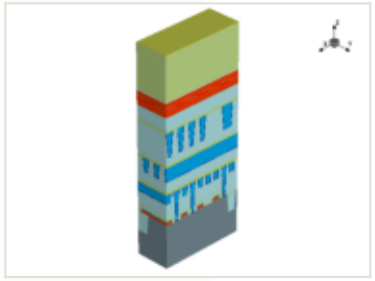
Step #2760: Deposition_stack

depo_anisotropic · Deposition_stack

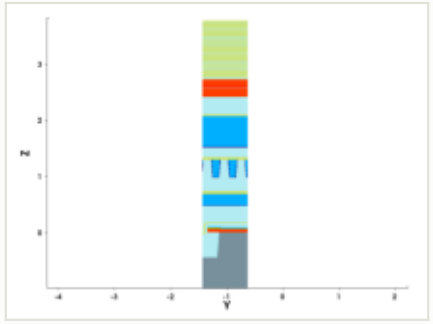
0.654	210.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

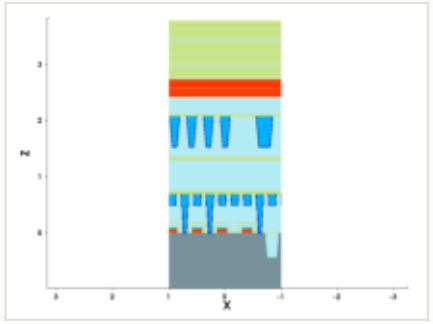
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



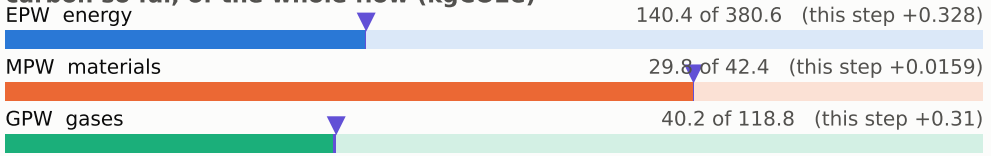
mask for this step

no mask for this step

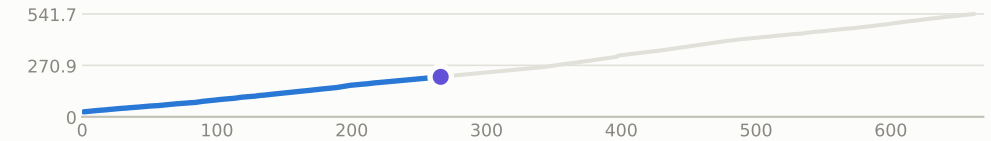
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

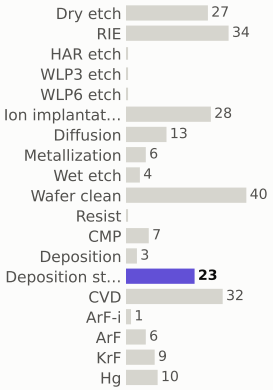
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

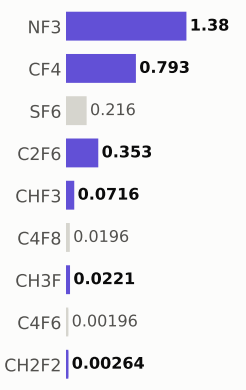


steps by process area



materials deposited (g)

gases emitted (g)



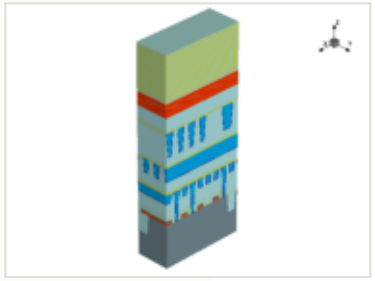
Step #2770: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	211.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

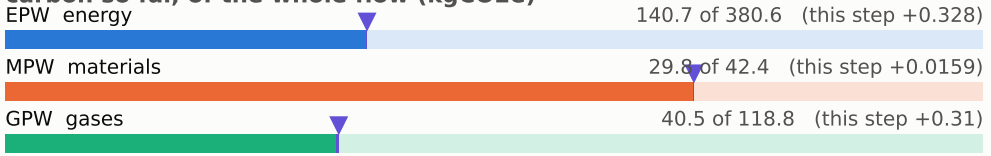
zoomed to fit



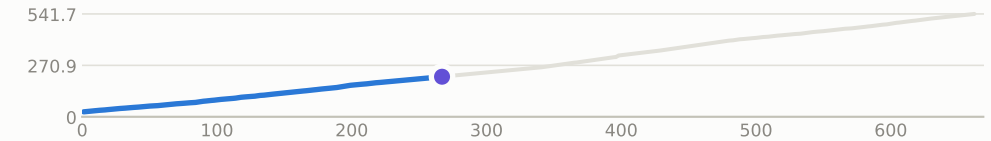
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

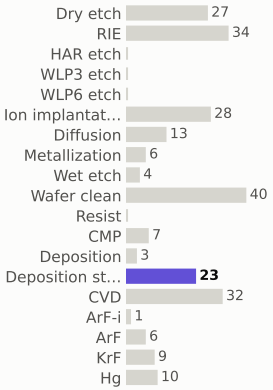
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

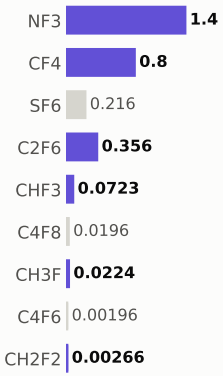


steps by process area

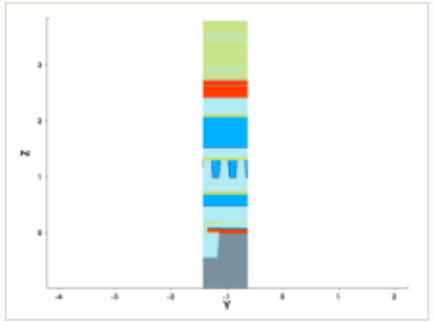


materials deposited (g)

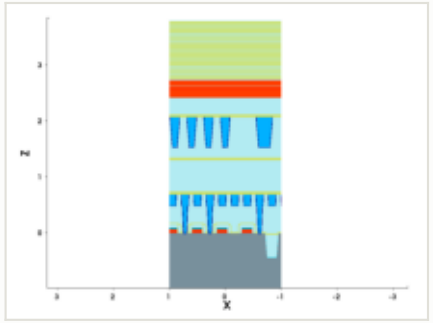
gases emitted (g)



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

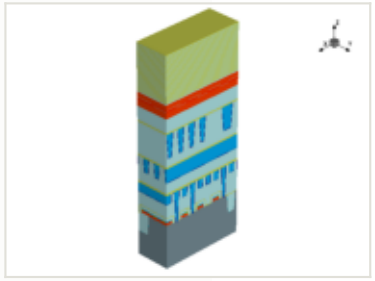
Step #2780: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	211.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

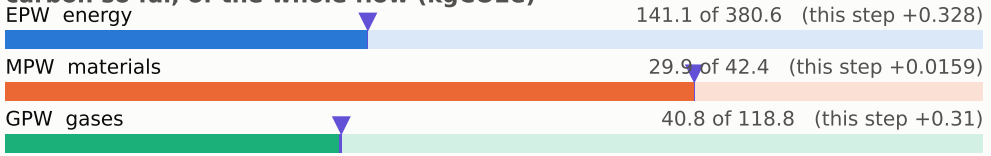
zoomed to fit



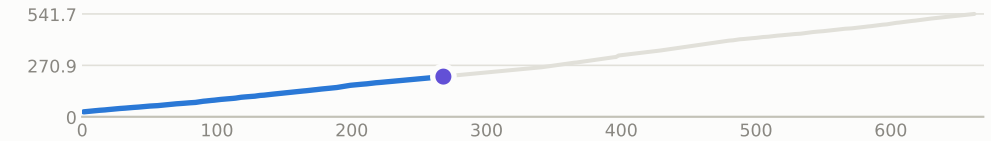
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

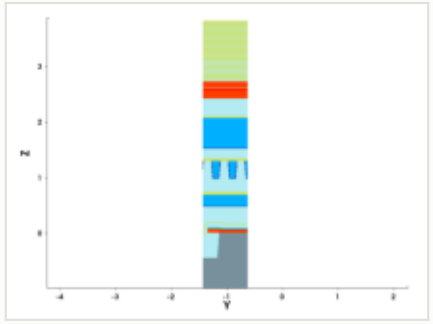
carbon so far, of the whole flow (kgCO2e)



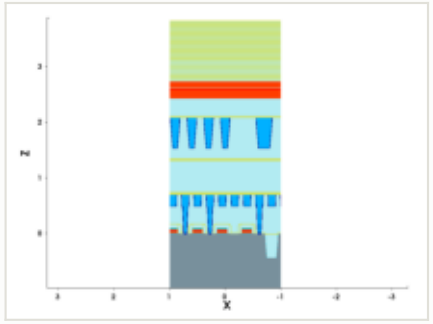
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



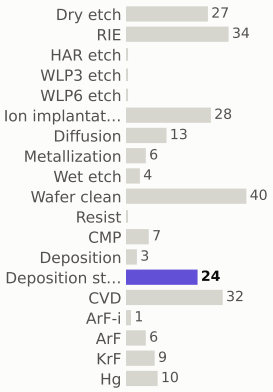
y cut at -1.032 µm



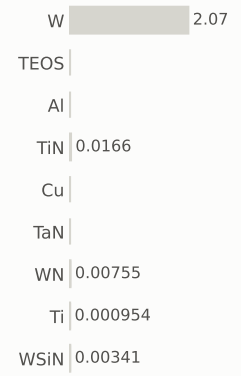
mask for this step

no mask for this step

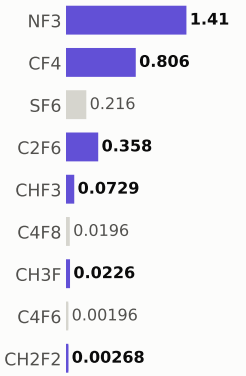
steps by process area



materials deposited (g)



gases emitted (g)



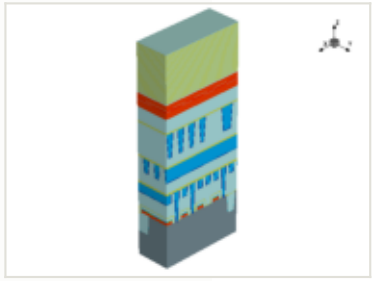
Step #2790: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	212.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

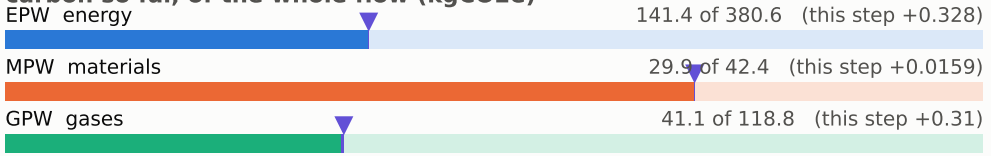
zoomed to fit



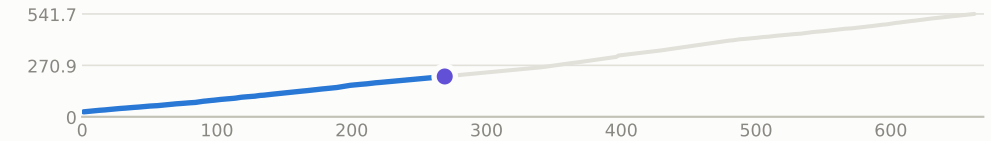
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

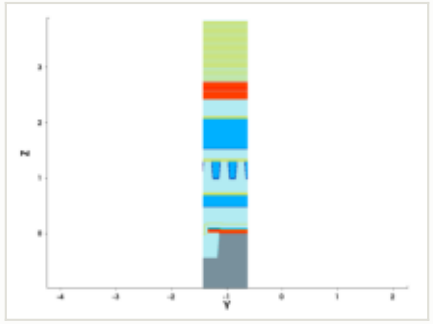
carbon so far, of the whole flow (kgCO2e)



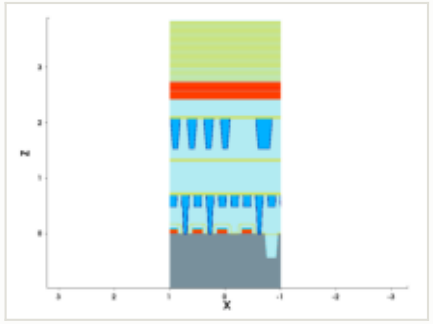
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



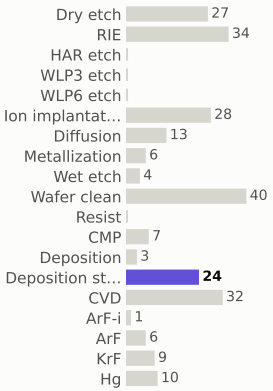
y cut at -1.032 µm



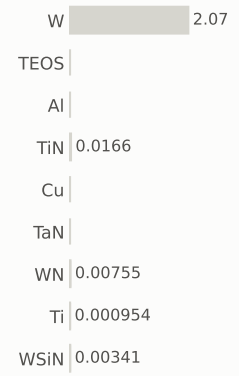
mask for this step

no mask for this step

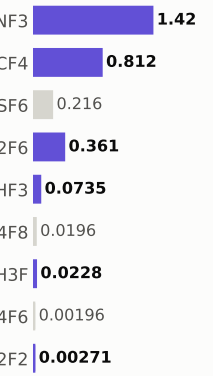
steps by process area



materials deposited (g)



gases emitted (g)



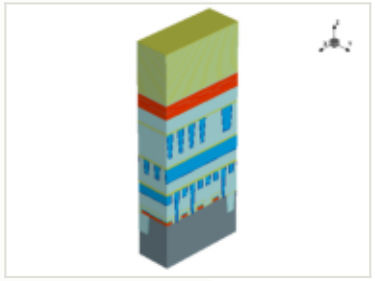
Step #2800: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	213.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

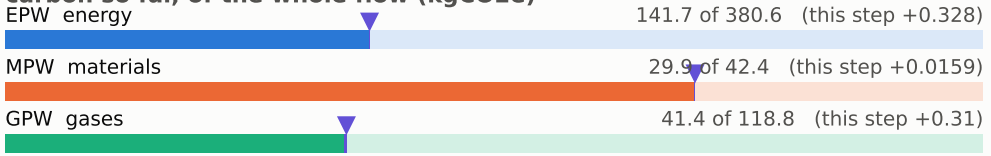
zoomed to fit



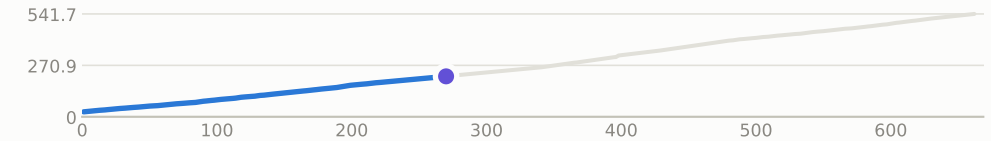
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

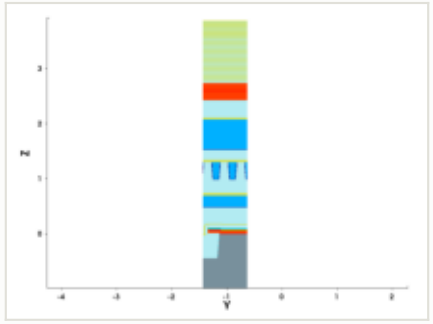
carbon so far, of the whole flow (kgCO2e)



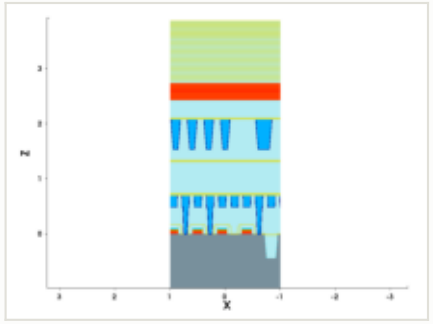
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



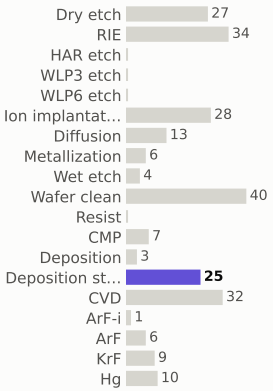
y cut at -1.032 µm



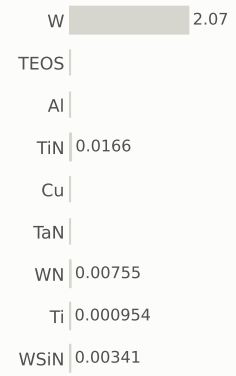
mask for this step

no mask for this step

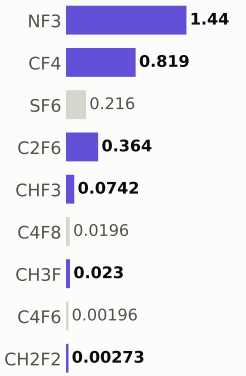
steps by process area



materials deposited (g)



gases emitted (g)



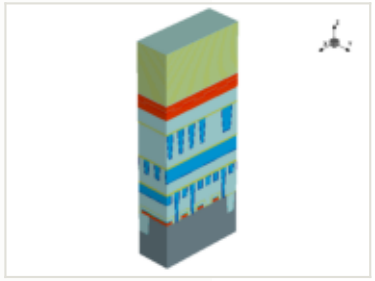
Step #2810: Deposition_stack

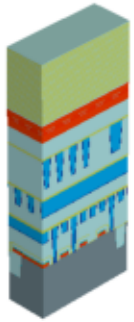
depo_anisotropic · Deposition_stack

0.654	213.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

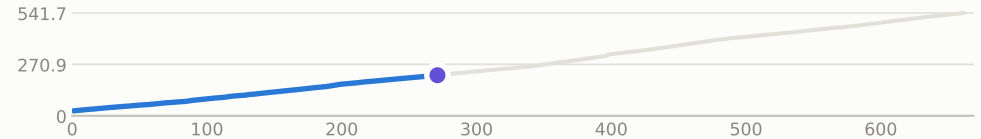
carbon so far, of the whole flow (kgCO2e)

EPW energy — 142.0 of 380.6 (this step +0.328)

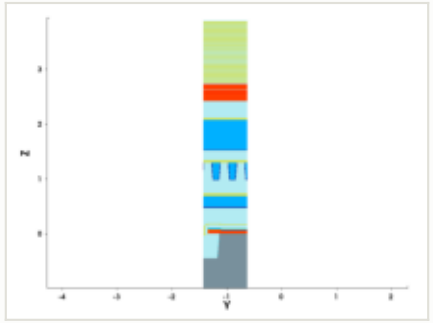
MPW materials — 29.9 of 42.4 (this step +0.0159)

GPW gases — 41.7 of 118.8 (this step +0.31)

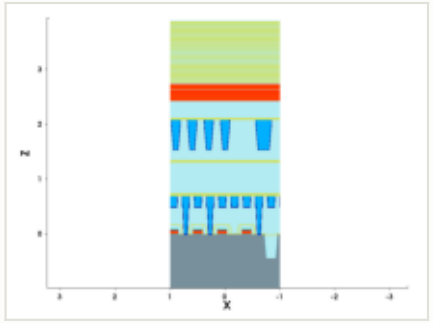
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



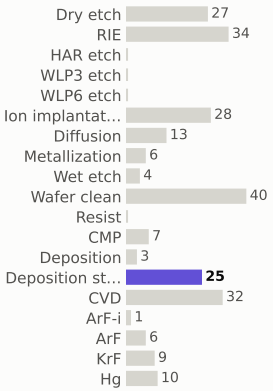
y cut at -1.032 µm



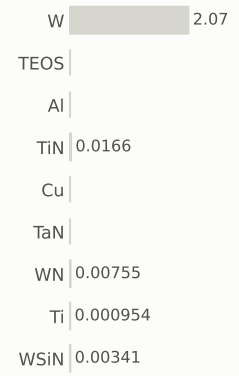
mask for this step

no mask for this step

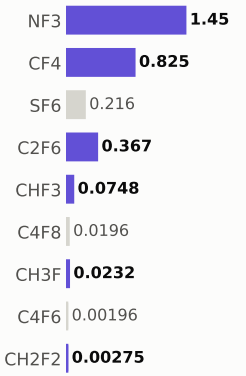
steps by process area



materials deposited (g)



gases emitted (g)



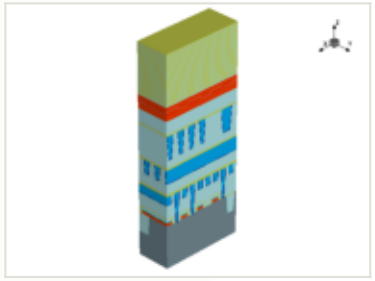
Step #2820: Deposition_stack

depo_anisotropic · Deposition_stack

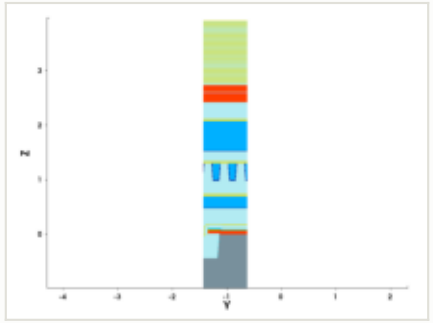
0.654	214.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

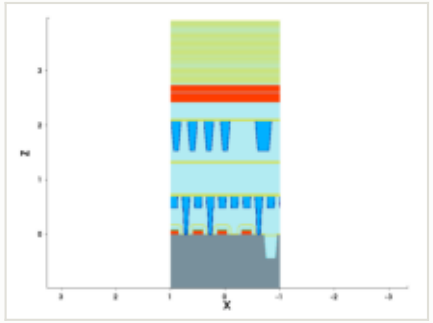
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



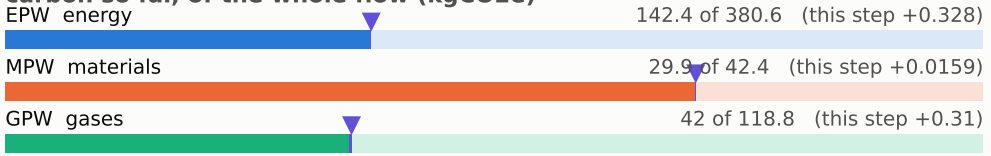
mask for this step

no mask for this step

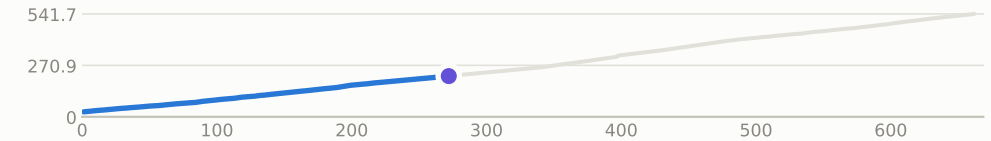
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

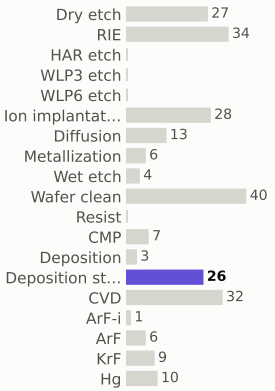
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

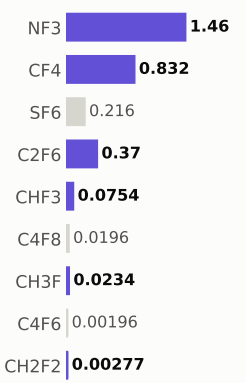


steps by process area



materials deposited (g)

gases emitted (g)



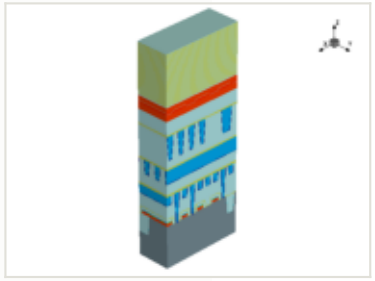
Step #2830: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	215.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

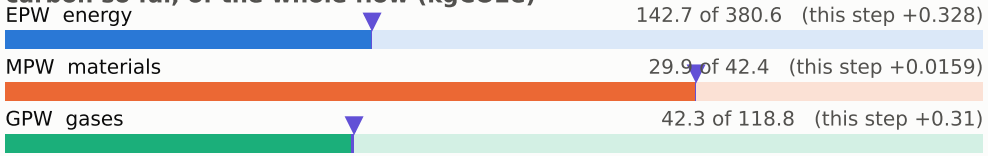
zoomed to fit



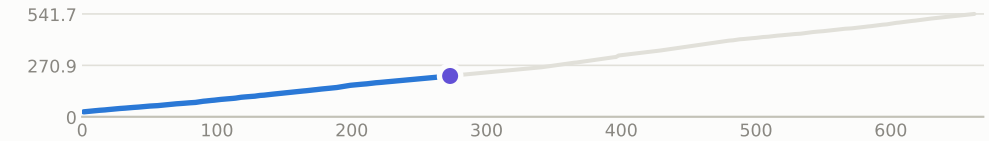
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

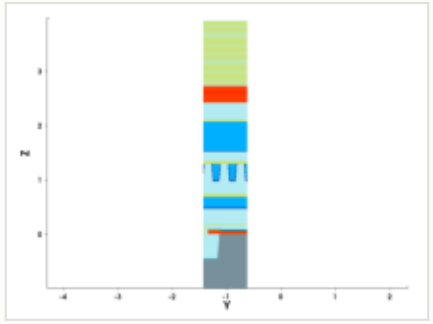
carbon so far, of the whole flow (kgCO2e)



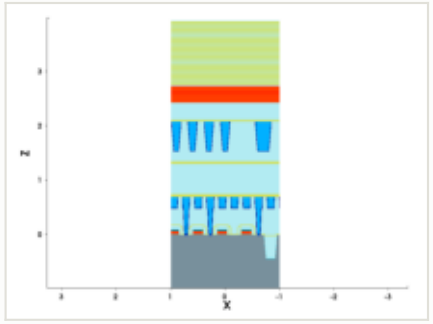
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



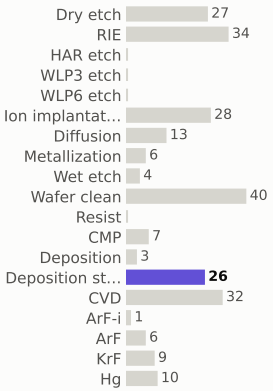
y cut at -1.032 µm



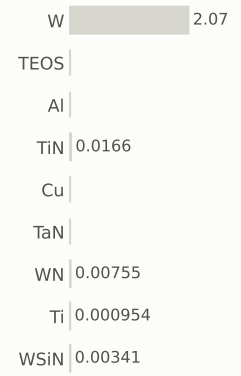
mask for this step

no mask for this step

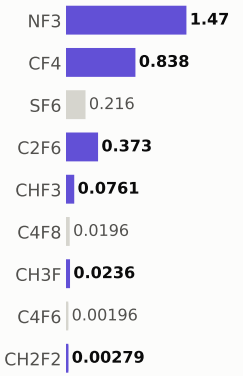
steps by process area



materials deposited (g)



gases emitted (g)



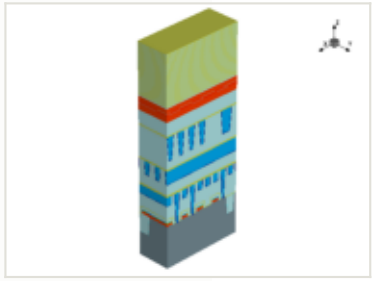
Step #2840: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	215.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

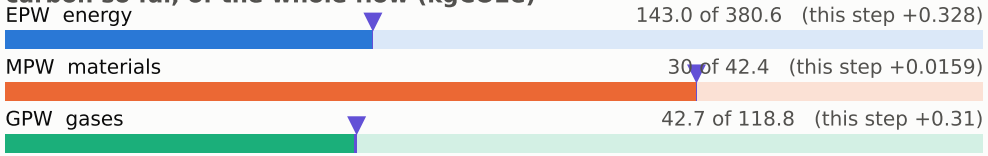
zoomed to fit



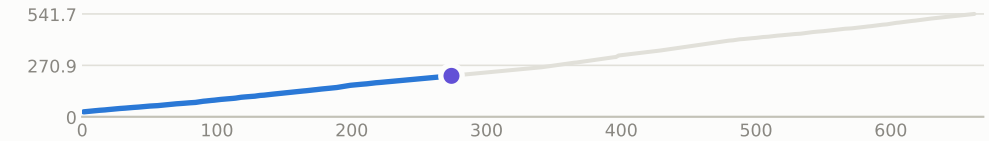
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

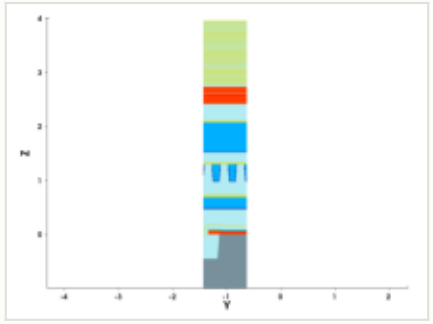
carbon so far, of the whole flow (kgCO2e)



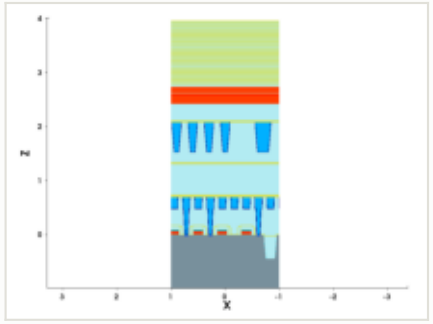
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



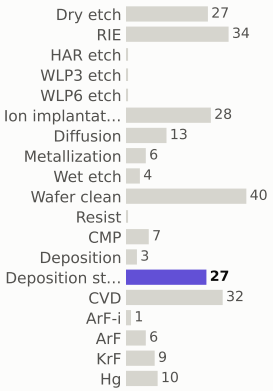
y cut at -1.032 µm



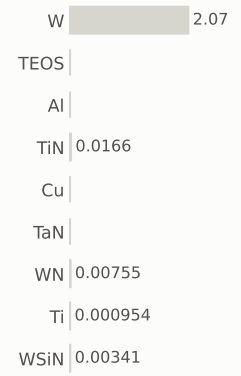
mask for this step

no mask for this step

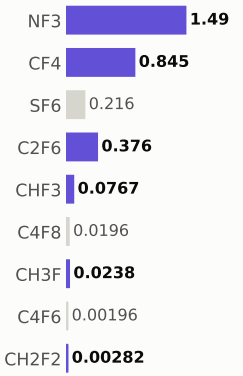
steps by process area



materials deposited (g)



gases emitted (g)



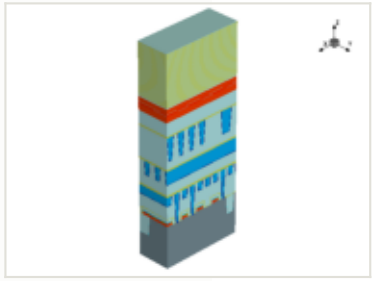
Step #2850: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	216.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

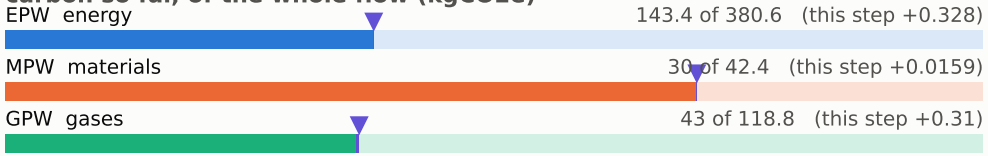
zoomed to fit



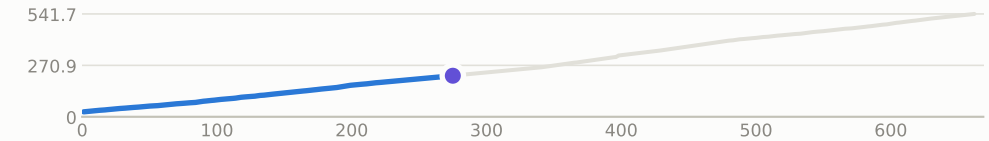
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

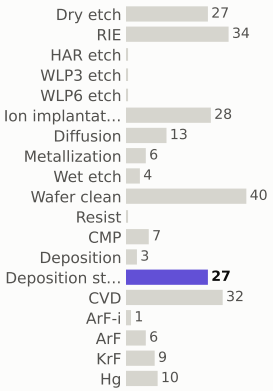
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

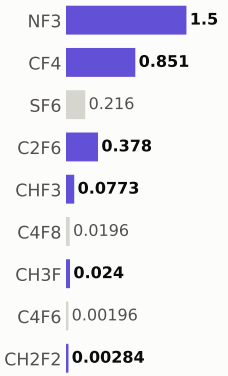


steps by process area

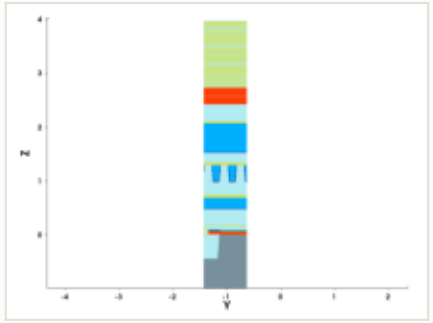


materials deposited (g)

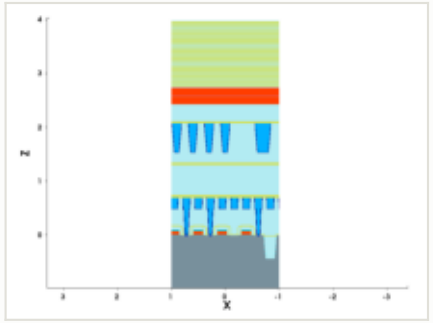
gases emitted (g)



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

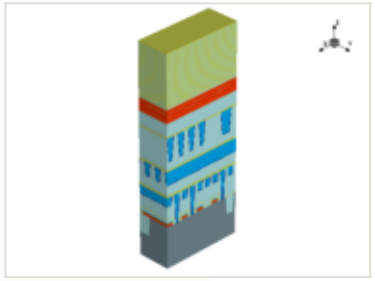
Step #2860: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	216.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

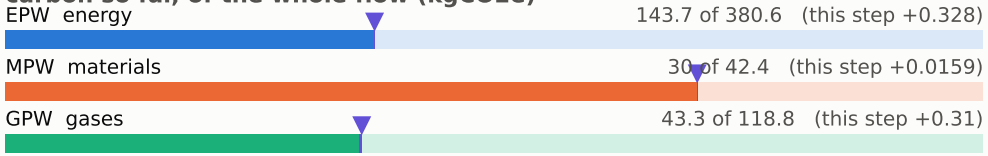
zoomed to fit



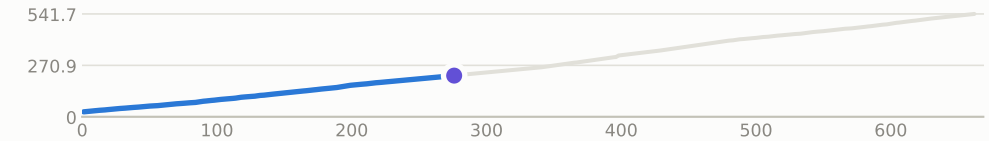
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

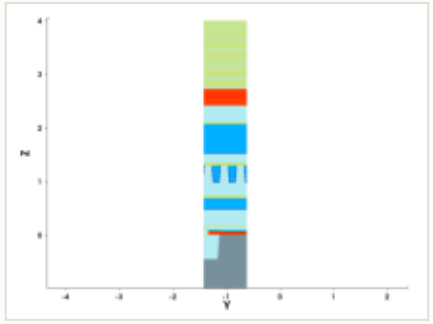
carbon so far, of the whole flow (kgCO2e)



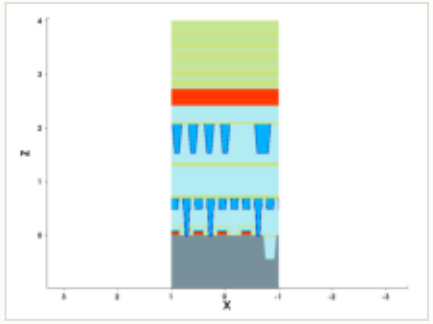
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



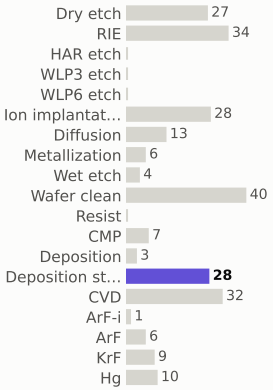
y cut at -1.032 µm



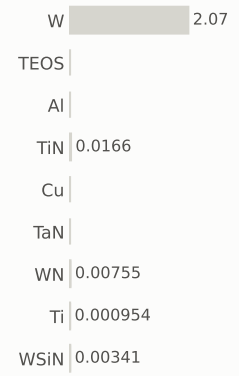
mask for this step

no mask for this step

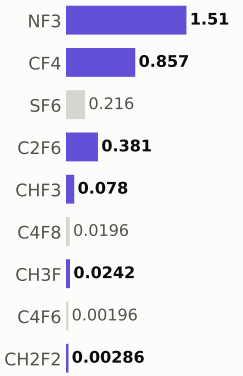
steps by process area



materials deposited (g)



gases emitted (g)



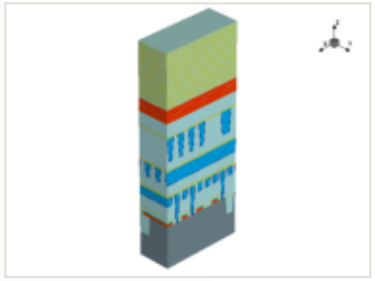
Step #2870: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	217.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

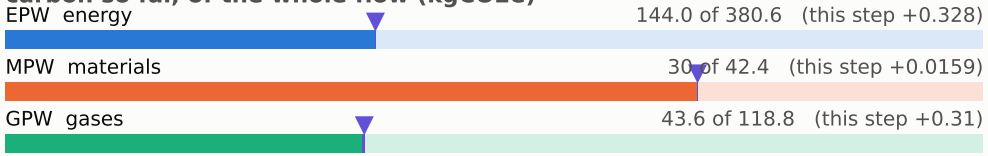
zoomed to fit



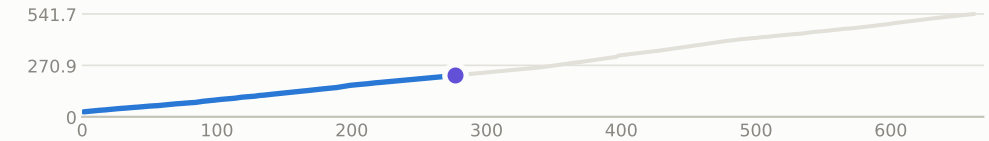
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

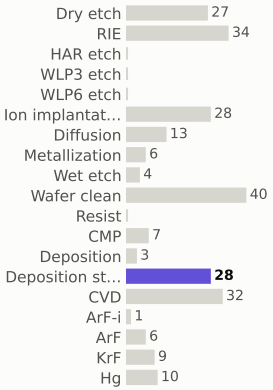
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

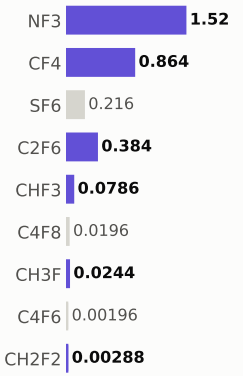


steps by process area

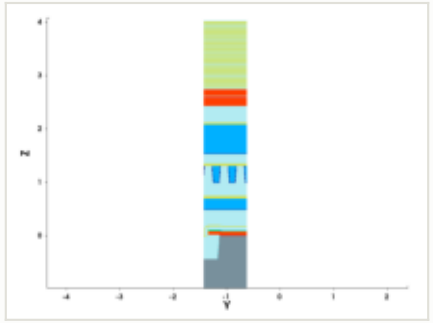


materials deposited (g)

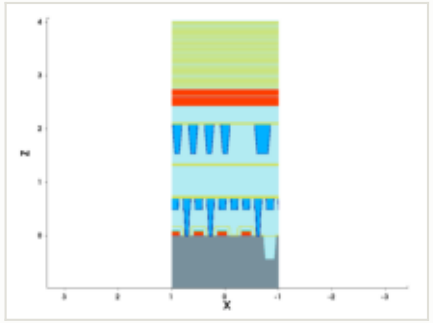
gases emitted (g)



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

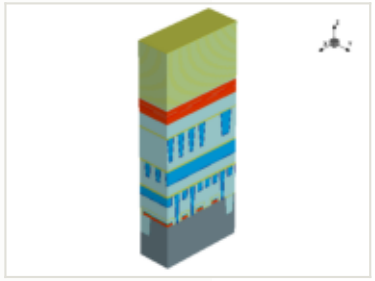
Step #2880: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	218.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

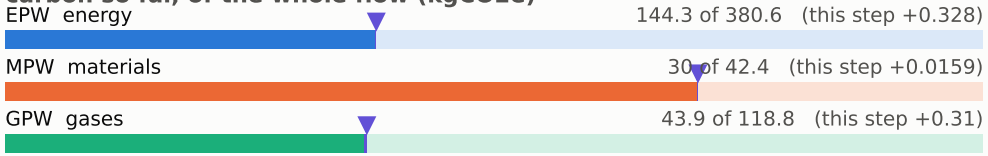
zoomed to fit



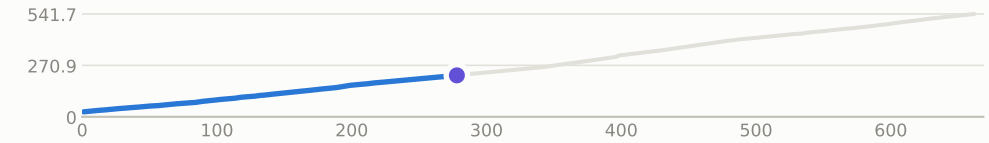
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

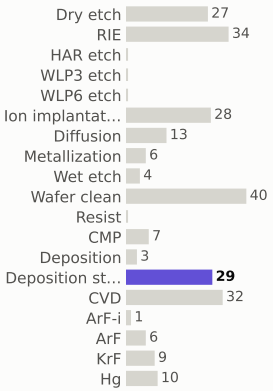
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

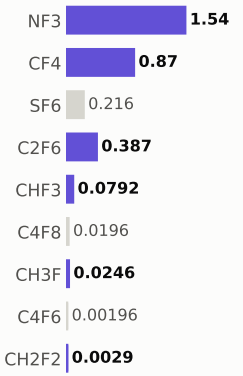


steps by process area

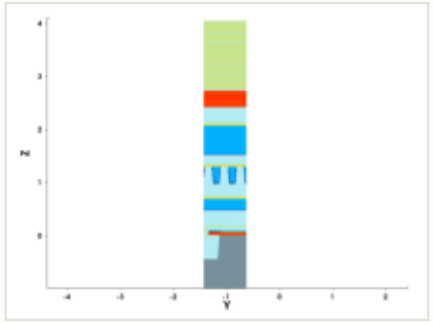


materials deposited (g)

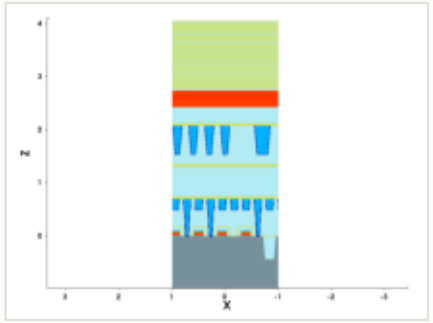
gases emitted (g)



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

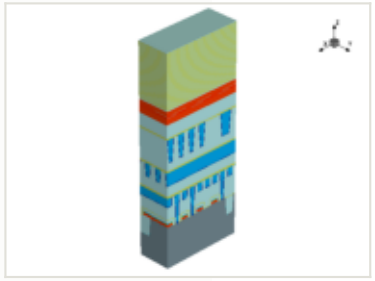
Step #2890: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	218.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

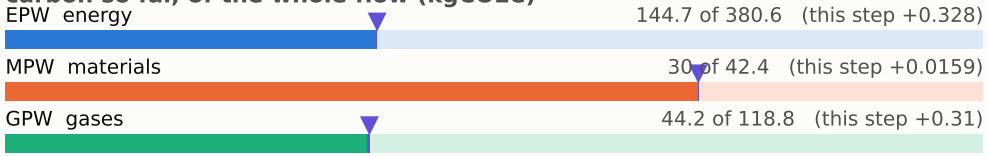
zoomed to fit



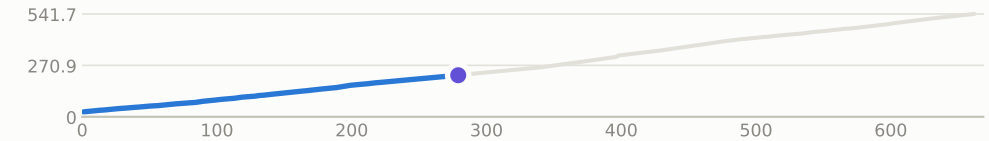
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

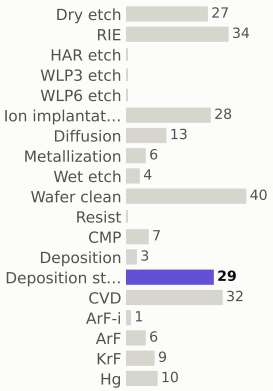
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

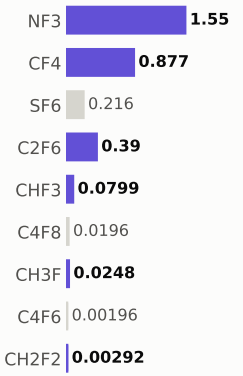


steps by process area

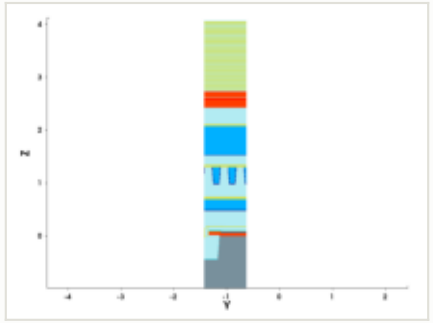


materials deposited (g)

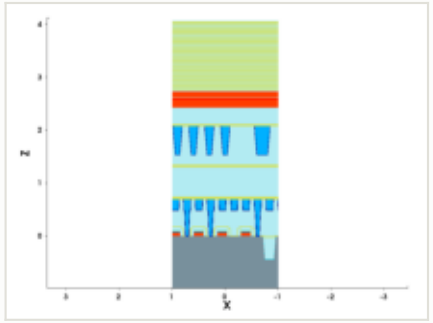
gases emitted (g)



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

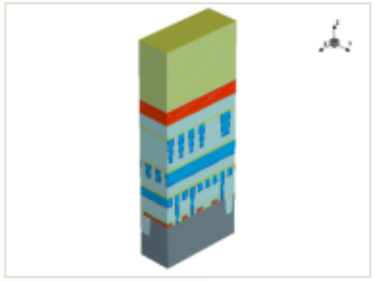
Step #2900: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	219.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

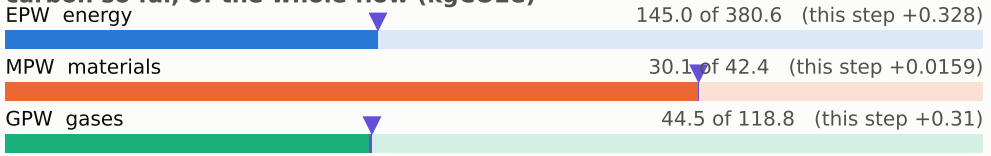
zoomed to fit



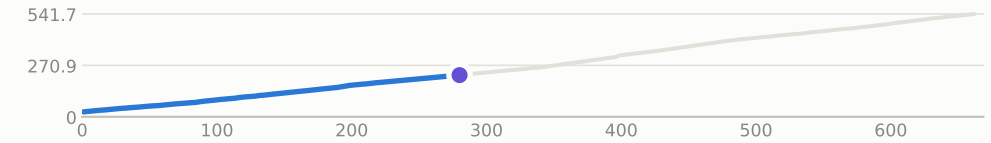
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

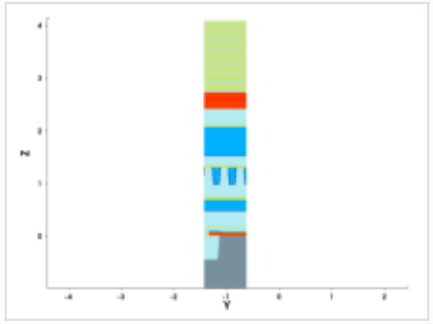
carbon so far, of the whole flow (kgCO2e)



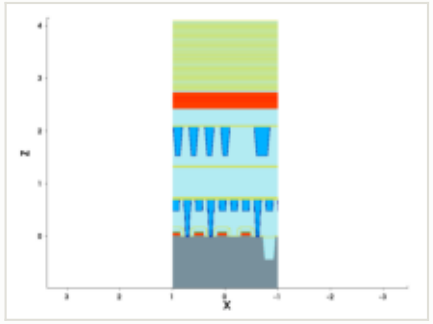
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



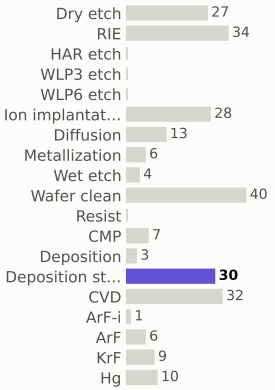
y cut at -1.032 µm



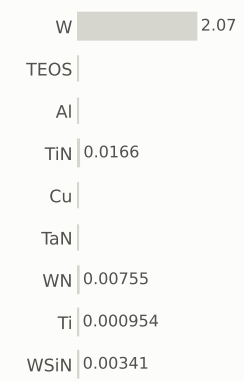
mask for this step

no mask for this step

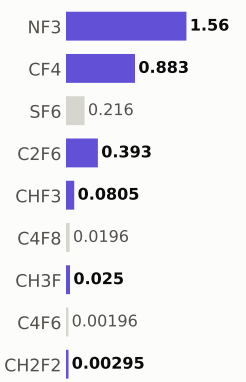
steps by process area



materials deposited (g)



gases emitted (g)



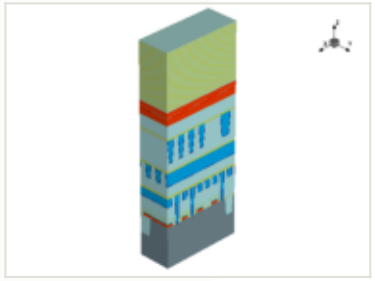
Step #2910: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	220.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

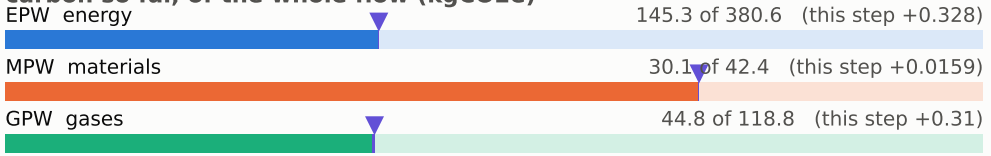
zoomed to fit



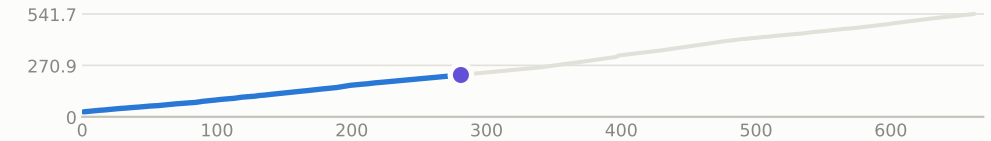
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

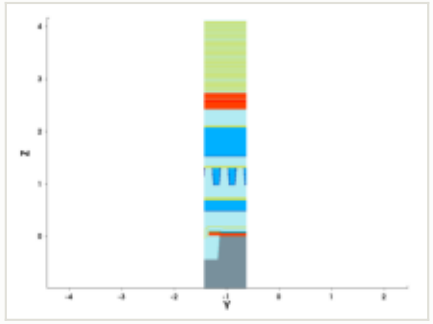
carbon so far, of the whole flow (kgCO2e)



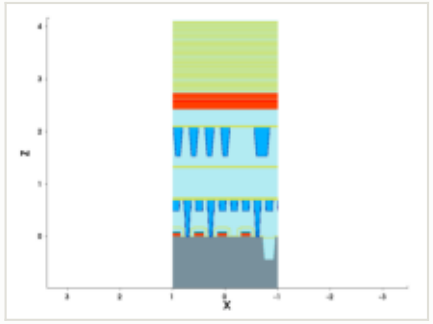
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



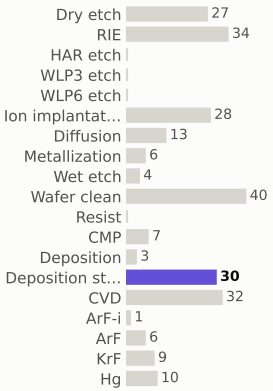
y cut at -1.032 µm



mask for this step

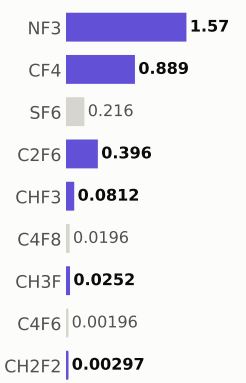
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



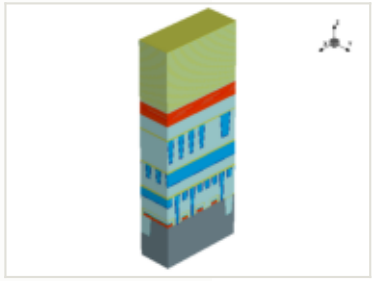
Step #2920: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	220.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

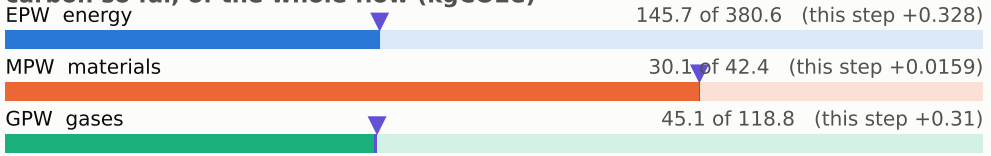
zoomed to fit



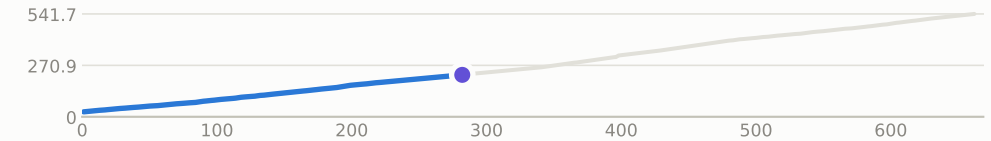
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

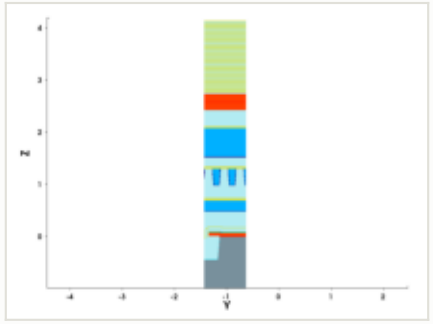
carbon so far, of the whole flow (kgCO2e)



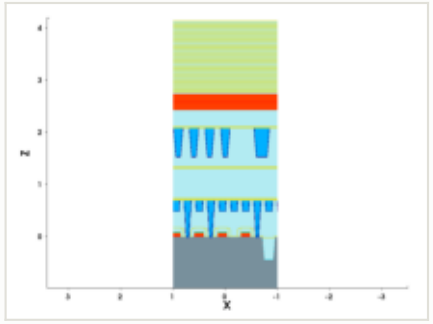
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



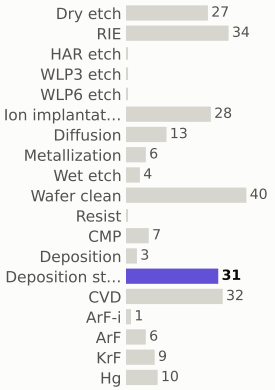
y cut at -1.032 µm



mask for this step

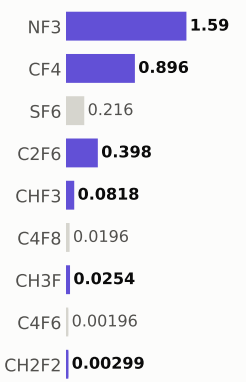
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



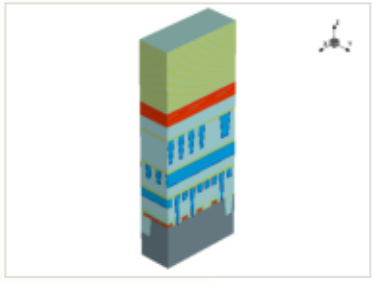
Step #2930: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	221.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

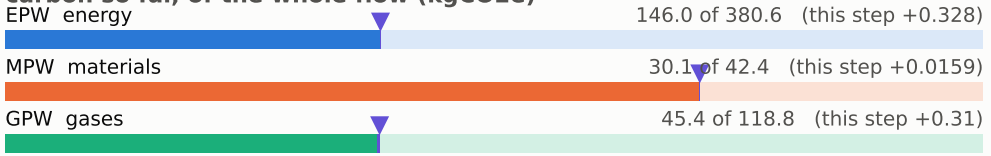
zoomed to fit



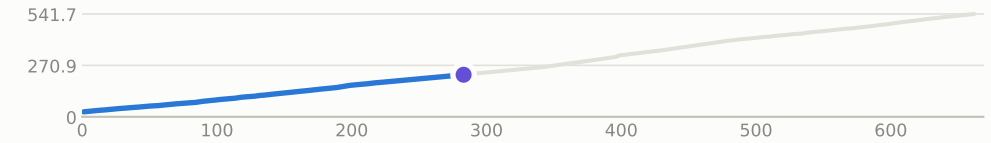
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

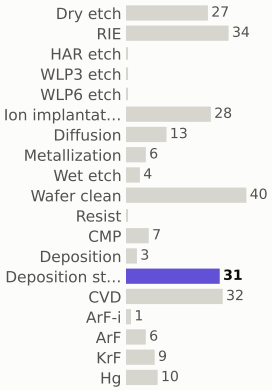
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

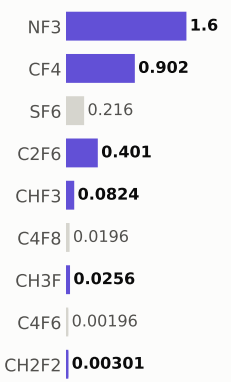


steps by process area

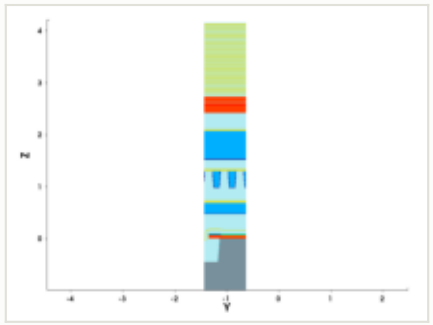


materials deposited (g)

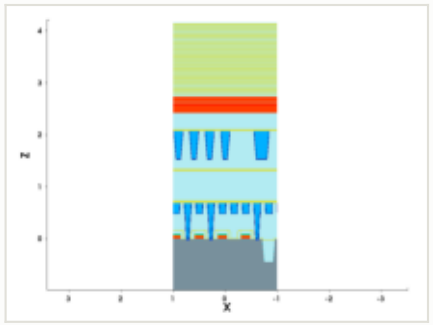
gases emitted (g)



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

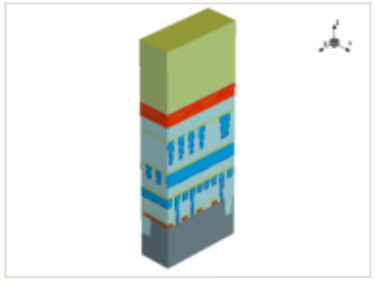
Step #2940: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	222.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

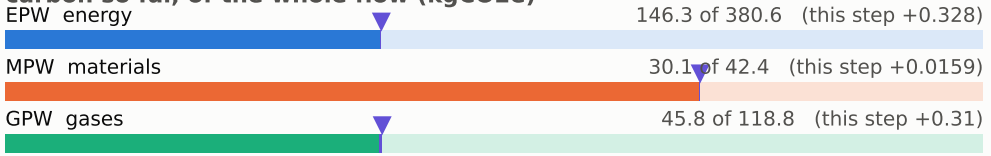
zoomed to fit



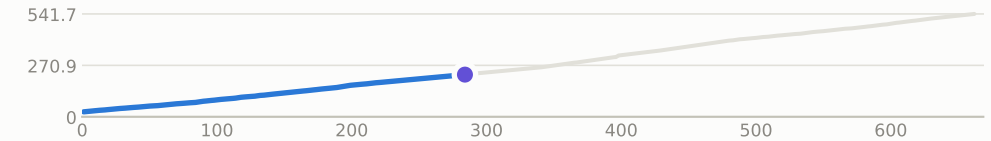
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

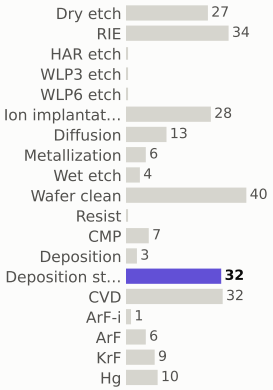
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

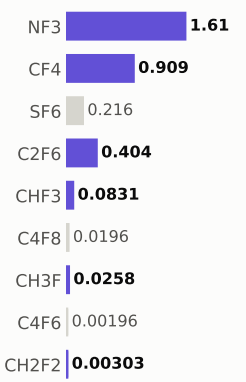


steps by process area

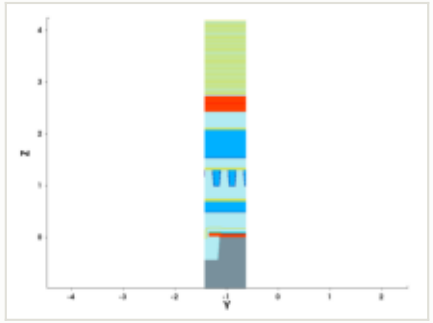


materials deposited (g)

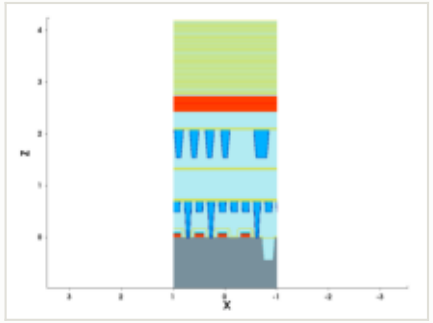
gases emitted (g)



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

Step #2950: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	222.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

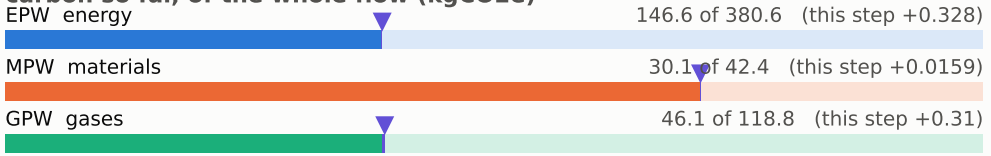
3D structure after this step

zoomed to fit

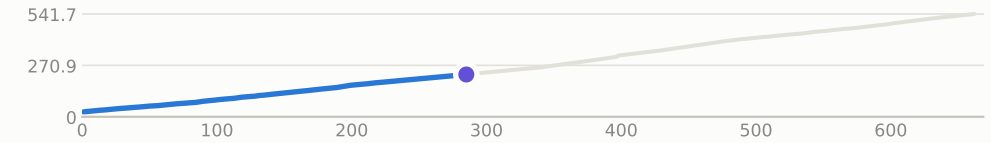
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

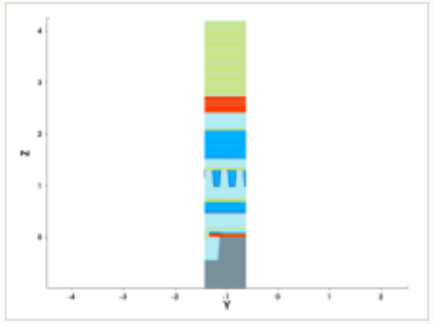
carbon so far, of the whole flow (kgCO2e)



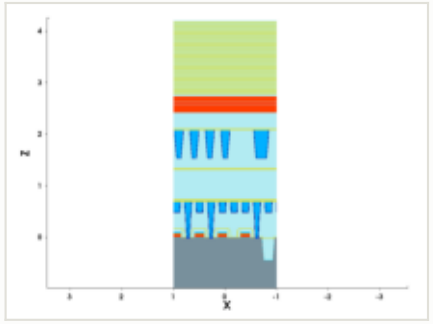
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



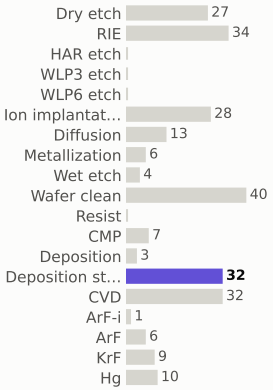
y cut at -1.032 µm



mask for this step

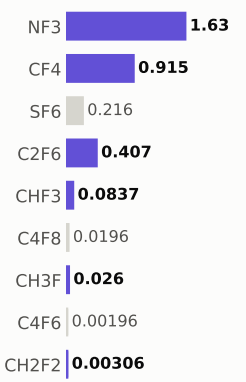
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



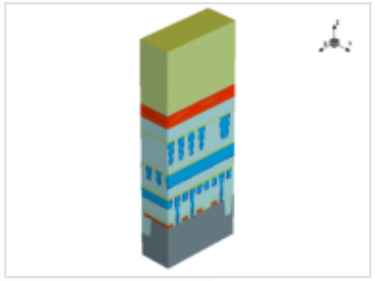
Step #2960: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	223.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

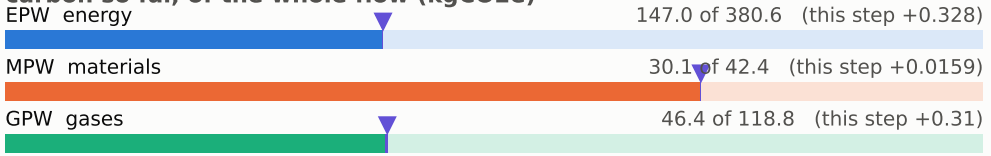
zoomed to fit



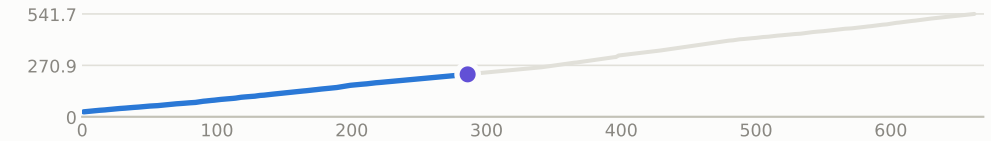
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

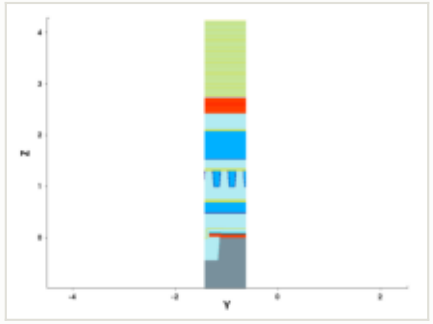
carbon so far, of the whole flow (kgCO2e)



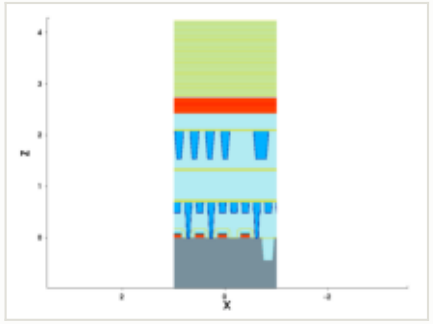
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



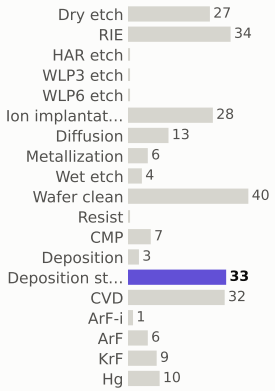
y cut at -1.032 µm



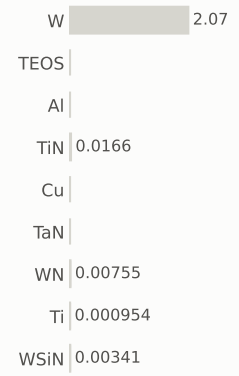
mask for this step

no mask for this step

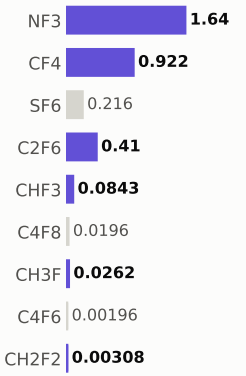
steps by process area



materials deposited (g)



gases emitted (g)



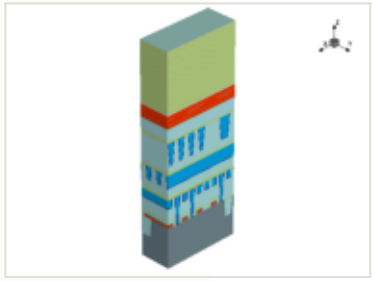
Step #2970: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	224.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

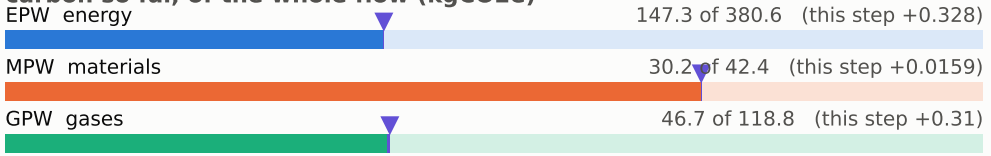
zoomed to fit



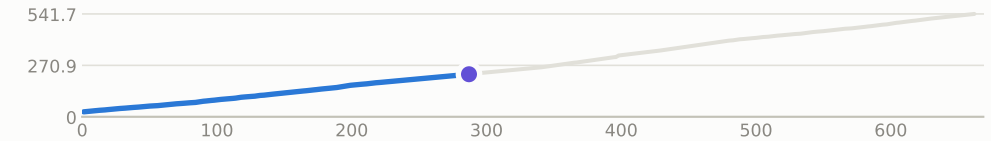
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

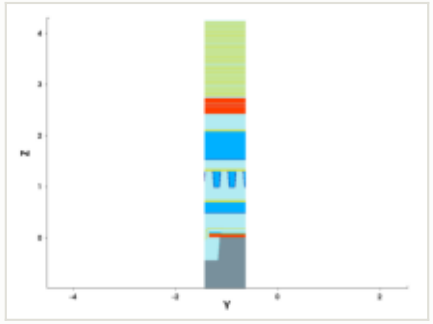
carbon so far, of the whole flow (kgCO2e)



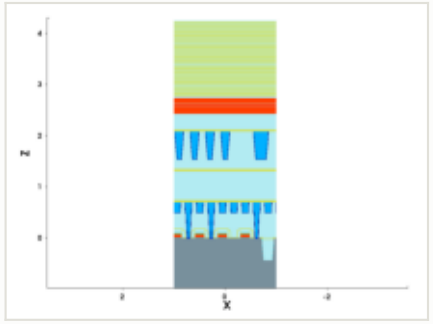
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



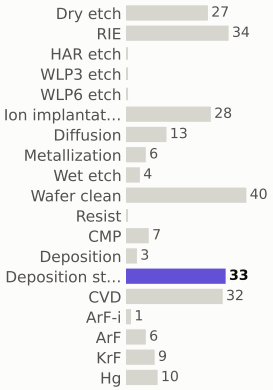
y cut at -1.032 µm



mask for this step

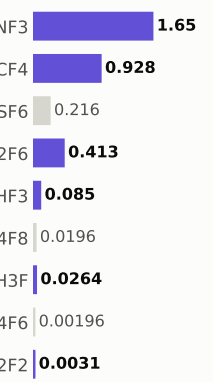
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



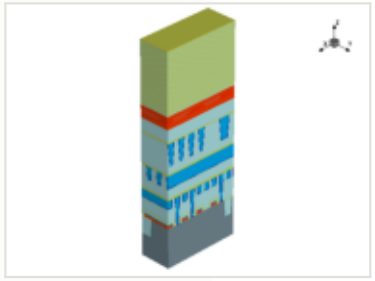
Step #2980: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	224.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

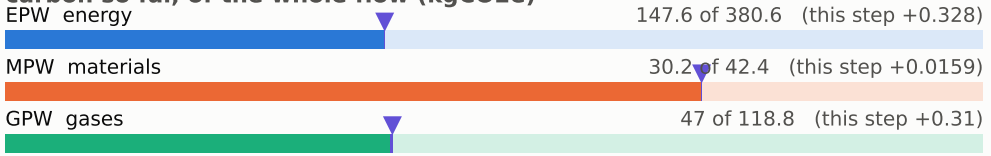
zoomed to fit



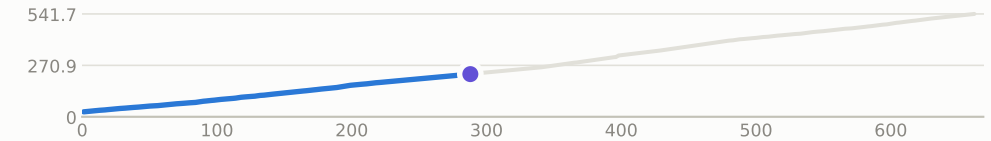
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

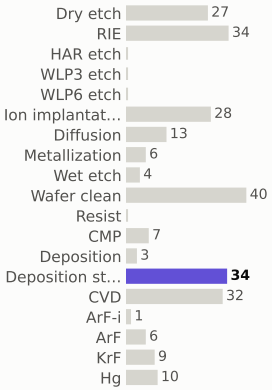
carbon so far, of the whole flow (kgCO2e)



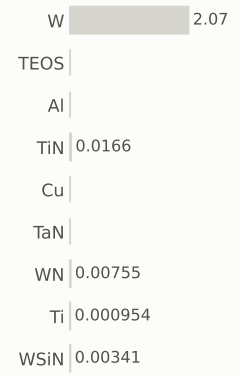
embodied carbon up to each step (kgCO2e per wafer)



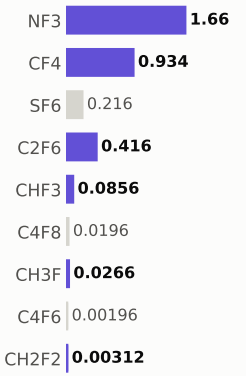
steps by process area



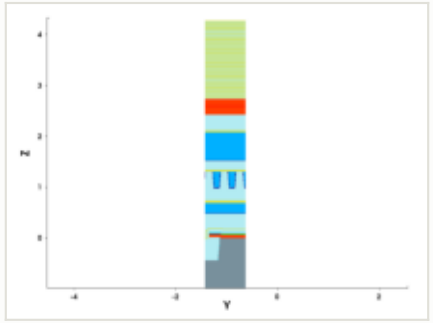
materials deposited (g)



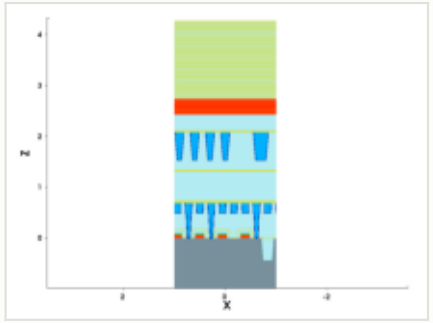
gases emitted (g)



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

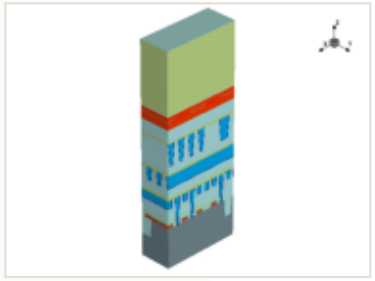
Step #2990: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	225.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

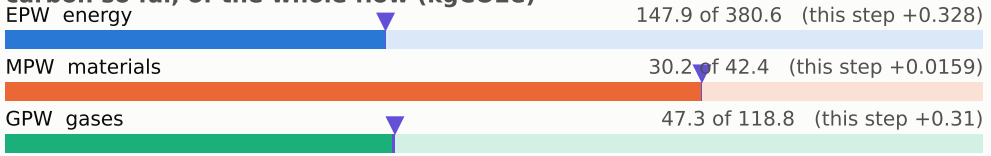
zoomed to fit



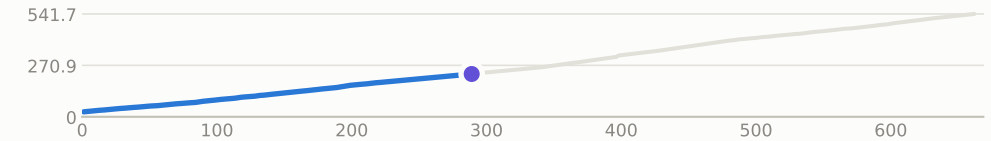
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

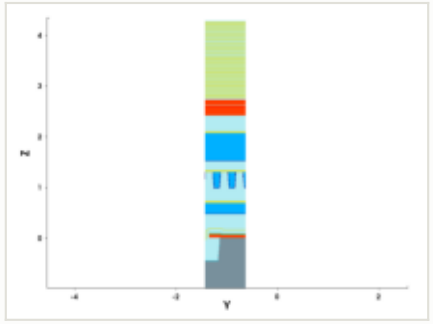
carbon so far, of the whole flow (kgCO2e)



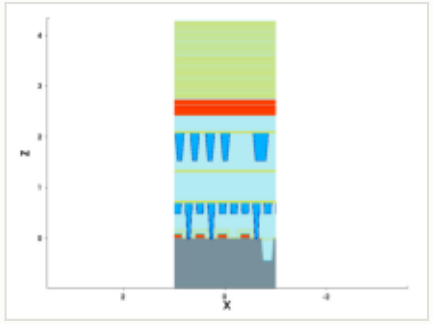
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



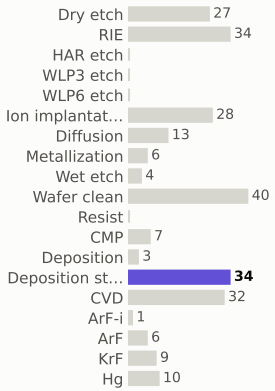
y cut at -1.032 µm



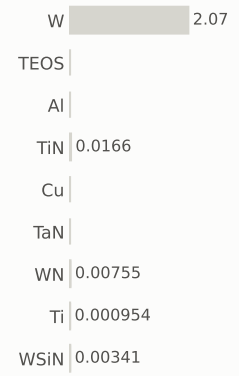
mask for this step

no mask for this step

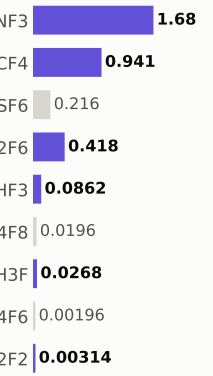
steps by process area



materials deposited (g)



gases emitted (g)



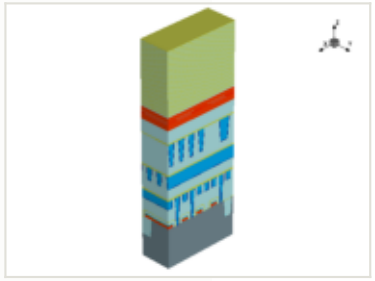
Step #3000: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	226.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

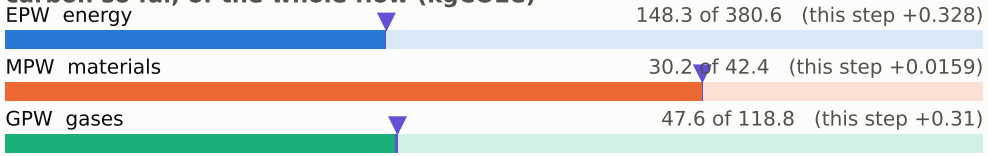
zoomed to fit



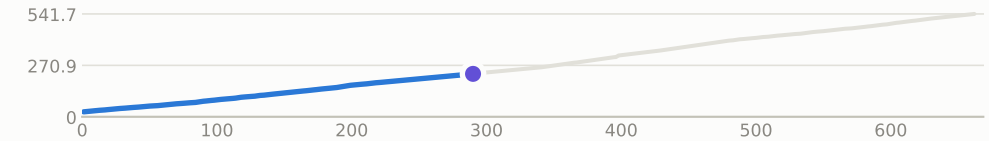
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

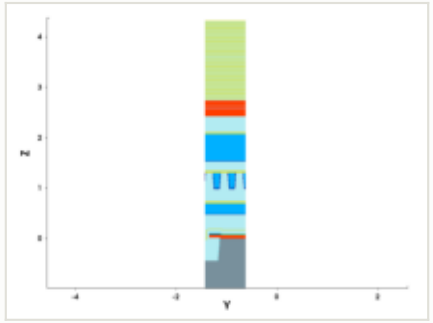
carbon so far, of the whole flow (kgCO2e)



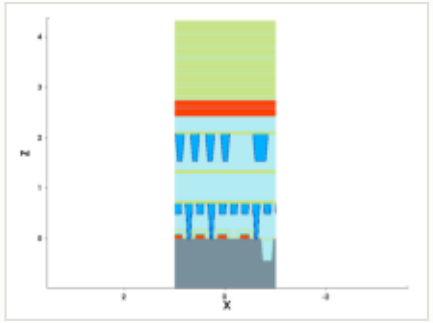
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



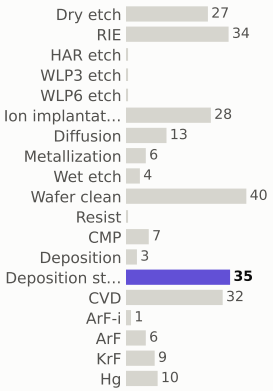
y cut at -1.032 µm



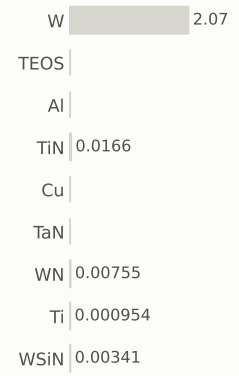
mask for this step

no mask for this step

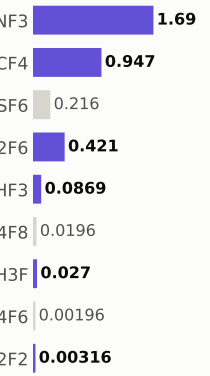
steps by process area



materials deposited (g)



gases emitted (g)



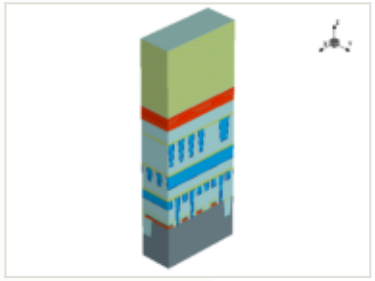
Step #3010: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	226.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

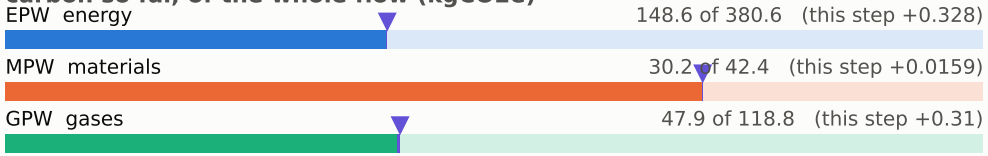
zoomed to fit



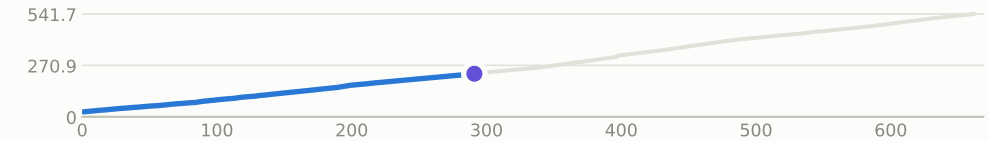
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

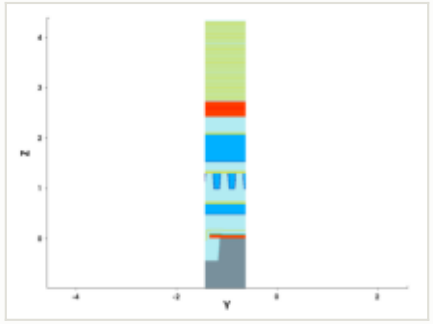
carbon so far, of the whole flow (kgCO2e)



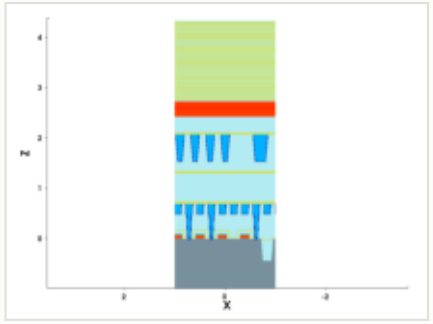
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



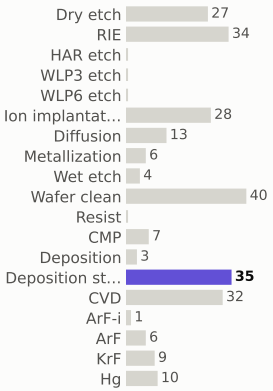
y cut at -1.032 µm



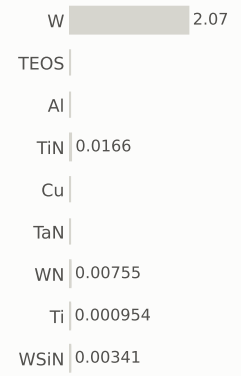
mask for this step

no mask for this step

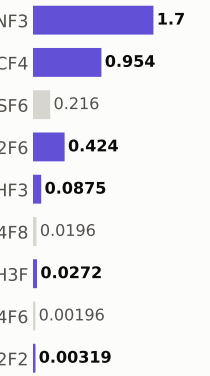
steps by process area



materials deposited (g)



gases emitted (g)



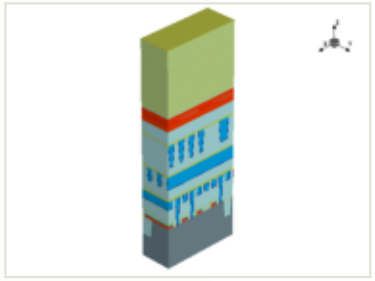
Step #3020: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	227.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

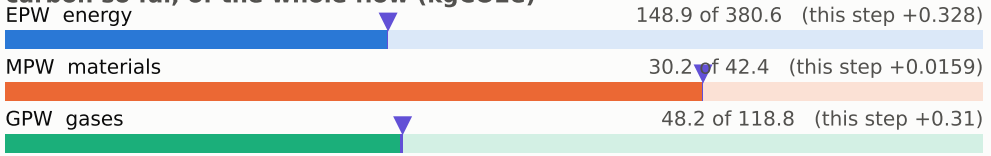
zoomed to fit



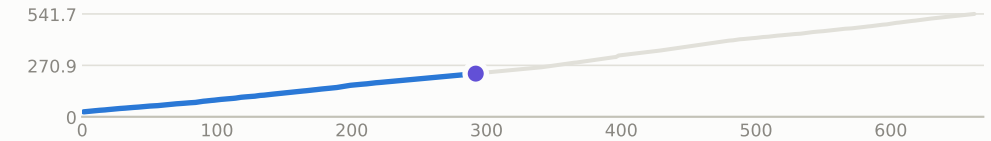
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

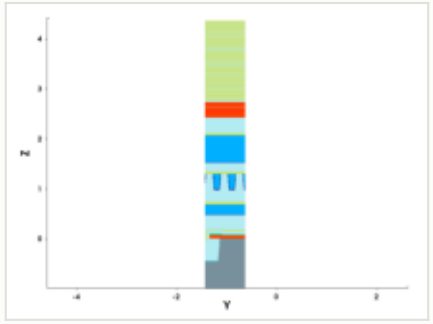
carbon so far, of the whole flow (kgCO2e)



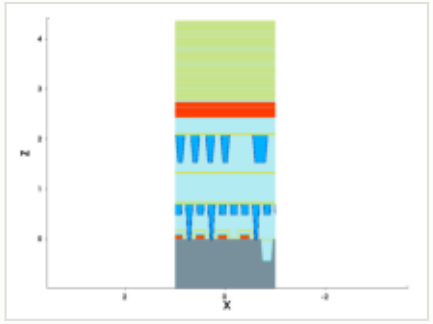
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



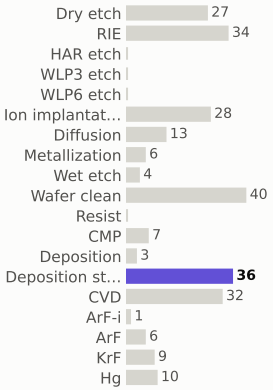
y cut at -1.032 µm



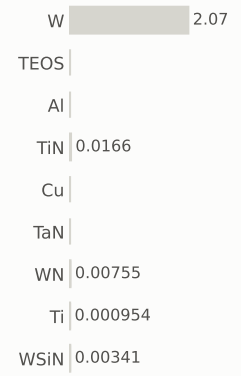
mask for this step

no mask for this step

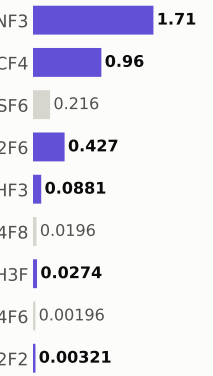
steps by process area



materials deposited (g)



gases emitted (g)



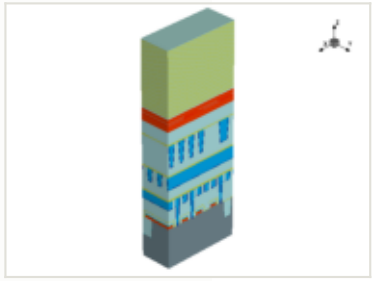
Step #3030: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	228.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

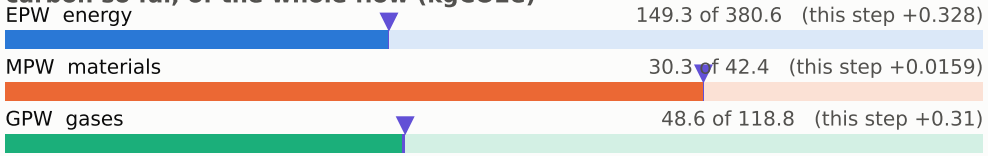
zoomed to fit



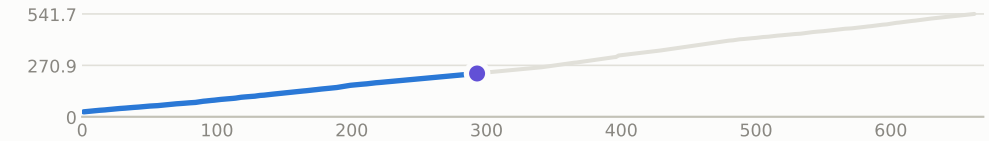
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

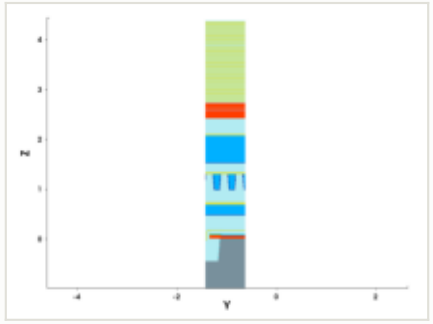
carbon so far, of the whole flow (kgCO2e)



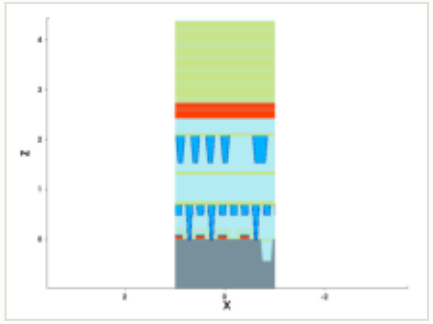
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



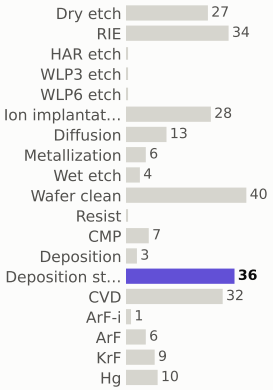
y cut at -1.032 µm



mask for this step

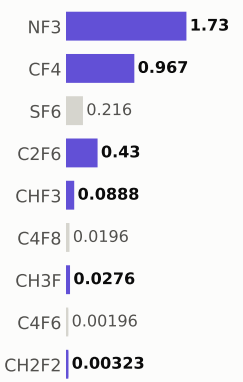
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



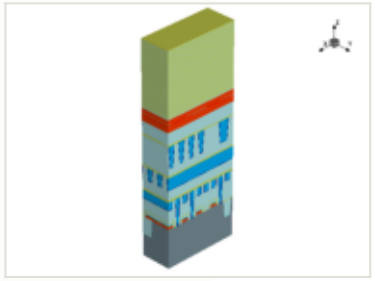
Step #3040: Deposition_stack

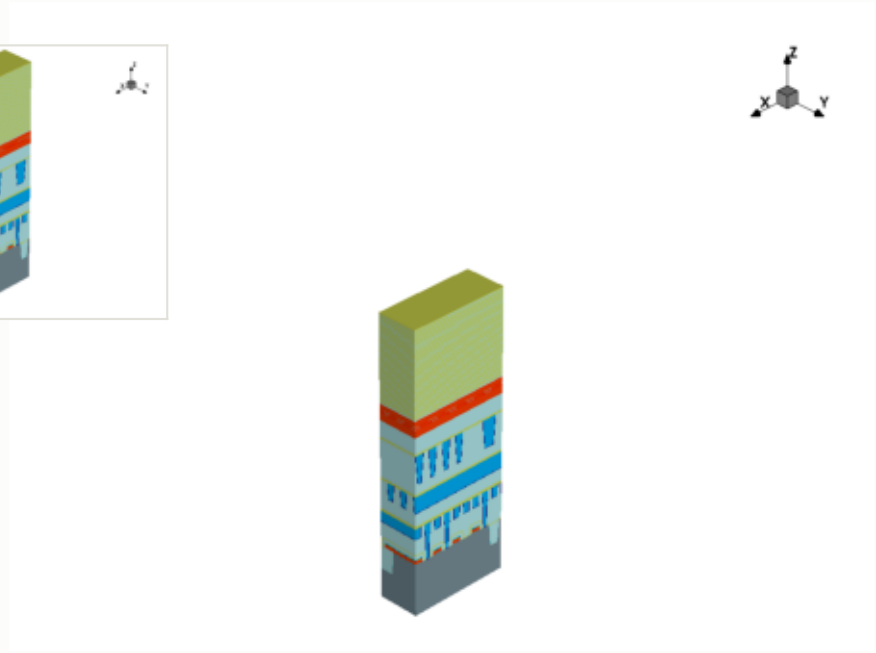
depo_anisotropic · Deposition_stack

0.654	228.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

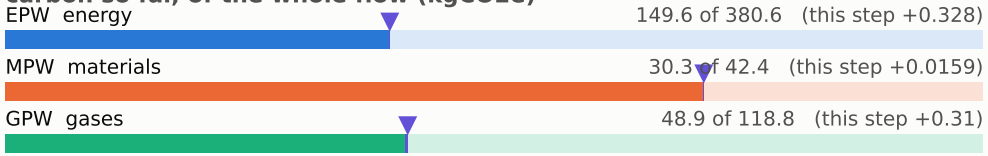




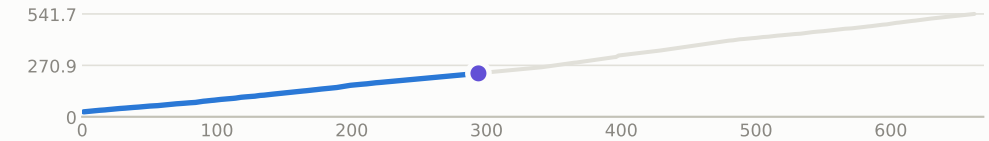
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

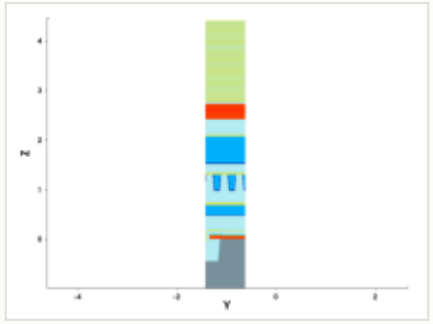
carbon so far, of the whole flow (kgCO2e)



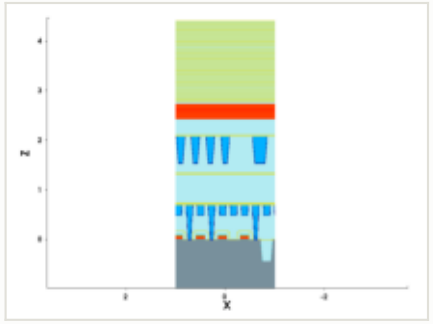
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



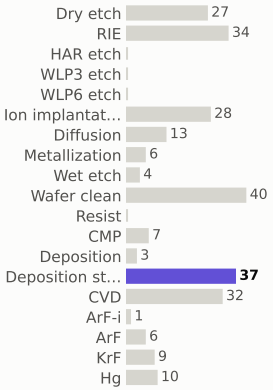
y cut at -1.032 µm



mask for this step

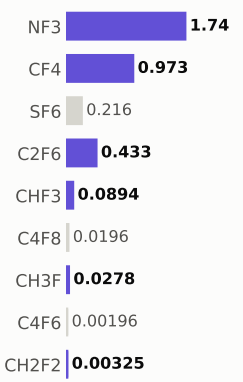
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



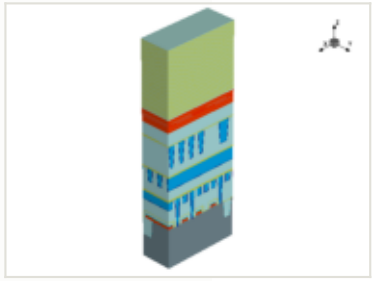
Step #3050: Deposition_stack

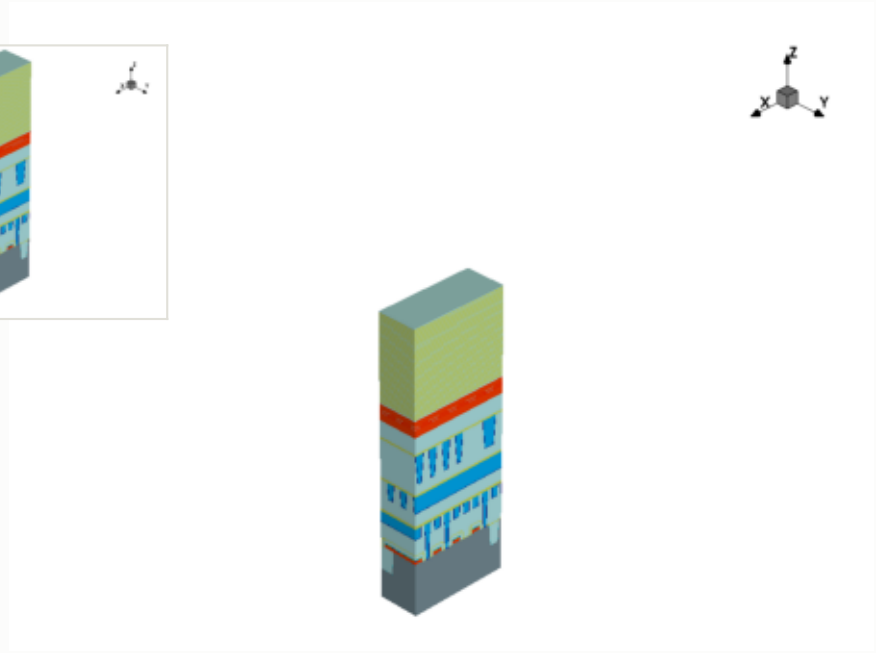
depo_anisotropic · Deposition_stack

0.654	229.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

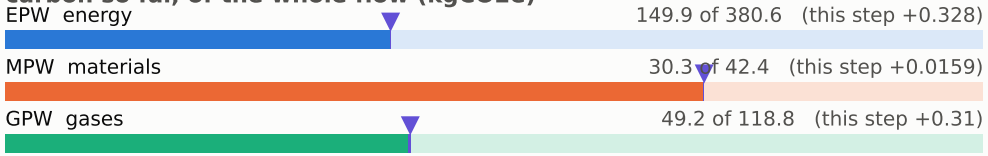




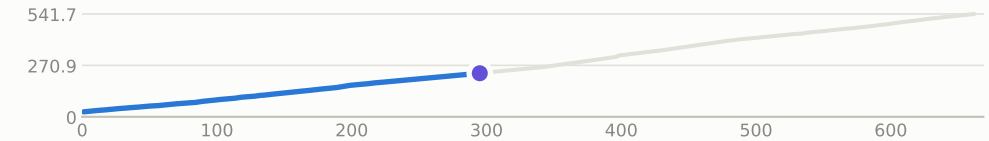
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

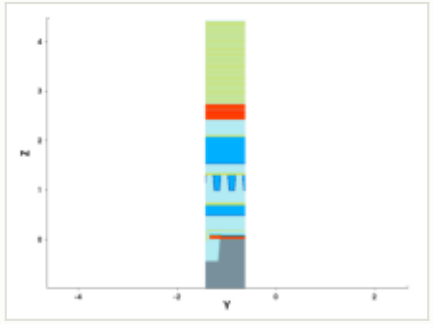
carbon so far, of the whole flow (kgCO2e)



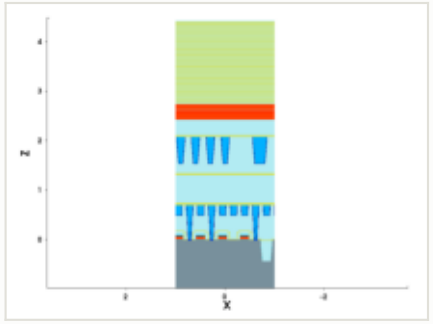
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



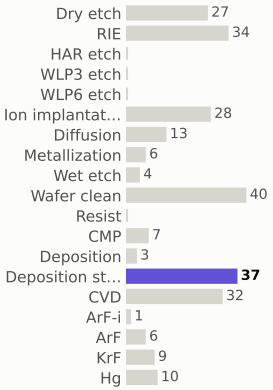
y cut at -1.032 µm



mask for this step

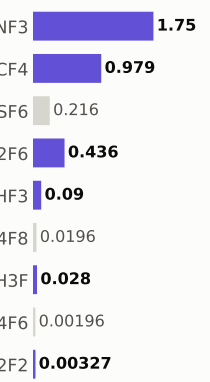
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



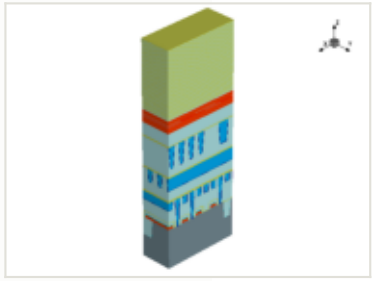
Step #3060: Deposition_stack

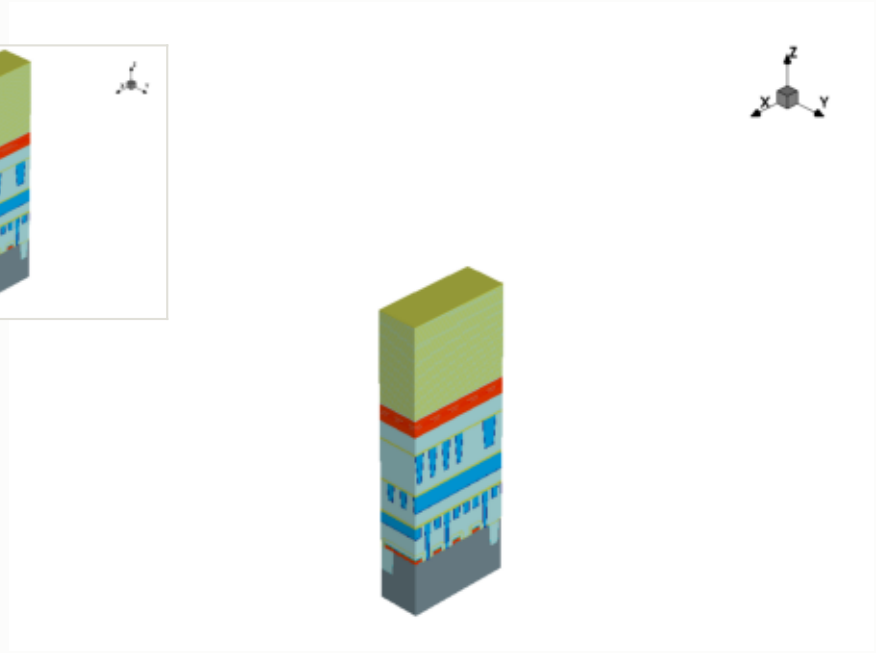
depo_anisotropic · Deposition_stack

0.654	230.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

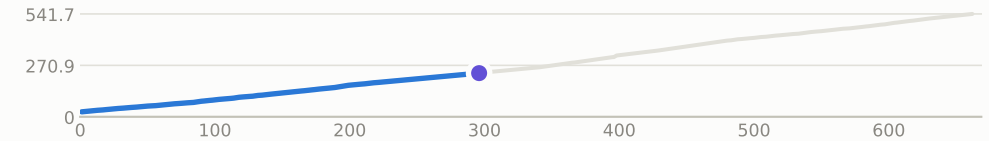
carbon so far, of the whole flow (kgCO2e)

EPW energy — 150.2 of 380.6 (this step +0.328)

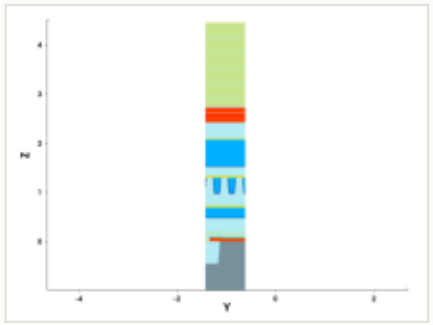
MPW materials — 30.3 of 42.4 (this step +0.0159)

GPW gases — 49.5 of 118.8 (this step +0.31)

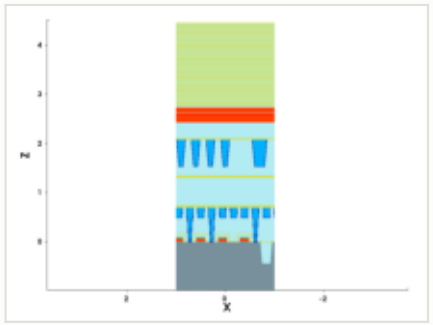
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



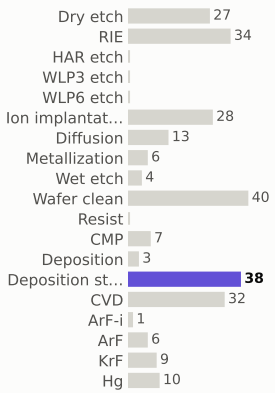
y cut at -1.032 µm



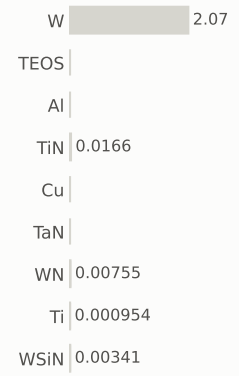
mask for this step

no mask for this step

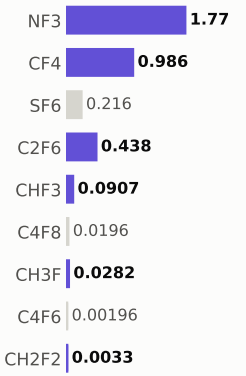
steps by process area



materials deposited (g)



gases emitted (g)



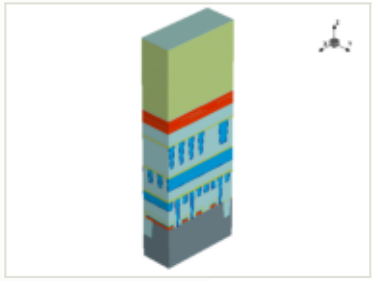
Step #3070: Deposition_stack

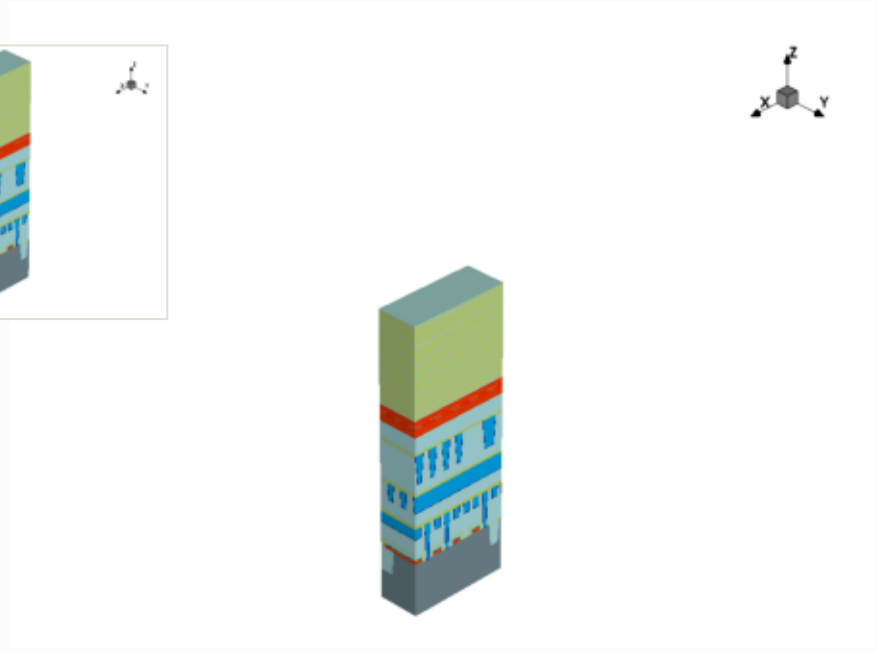
depo_anisotropic · Deposition_stack

0.654	230.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

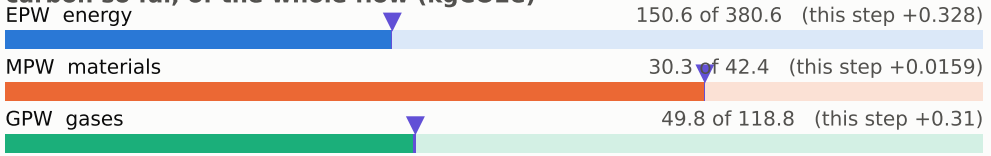




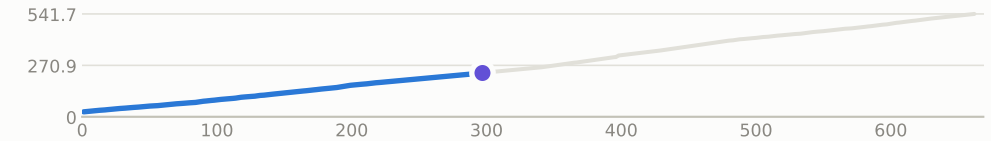
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

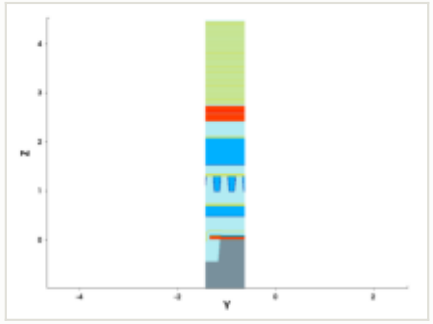
carbon so far, of the whole flow (kgCO2e)



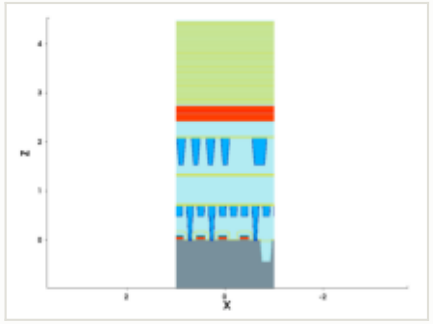
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



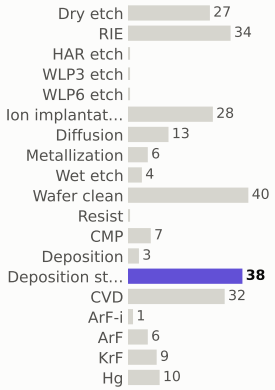
y cut at -1.032 µm



mask for this step

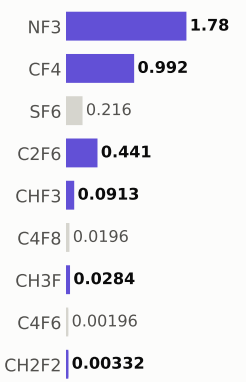
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



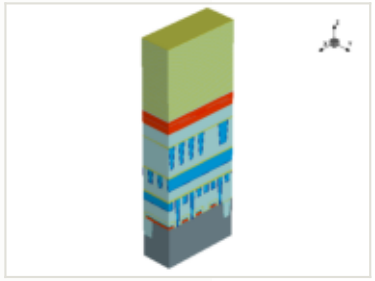
Step #3080: Deposition_stack

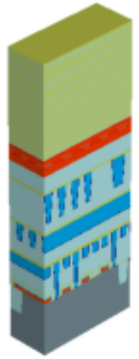
depo_anisotropic · Deposition_stack

0.654	231.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

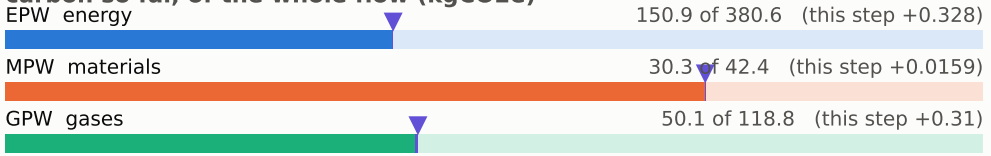




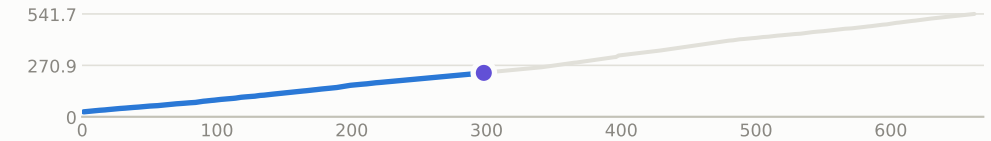
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

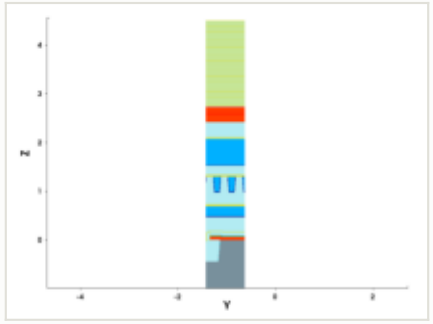
carbon so far, of the whole flow (kgCO2e)



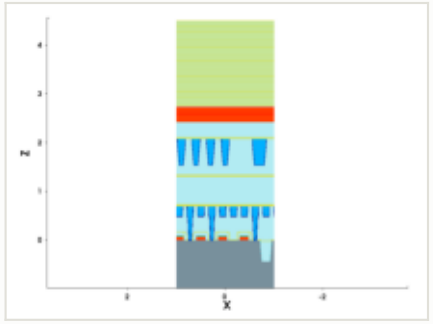
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



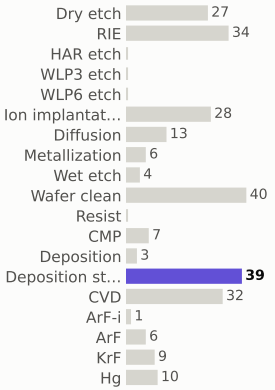
y cut at -1.032 µm



mask for this step

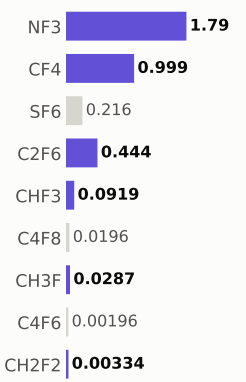
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



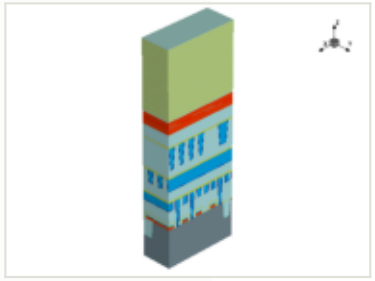
Step #3090: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	232.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

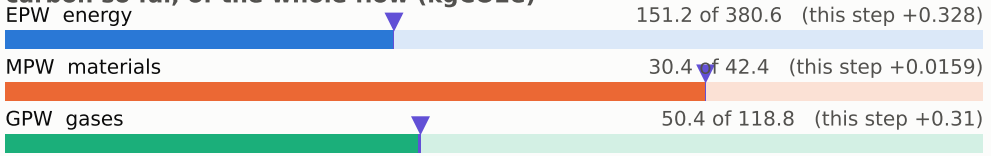
zoomed to fit



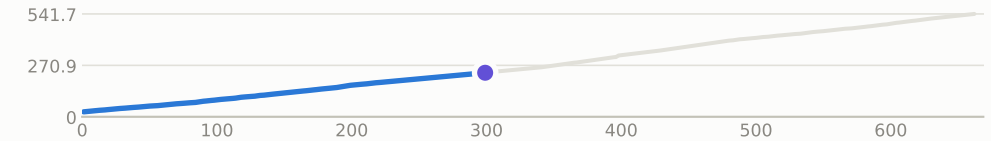
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

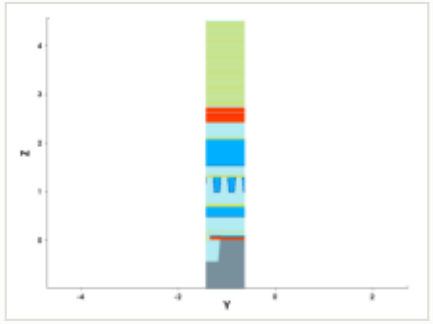
carbon so far, of the whole flow (kgCO2e)



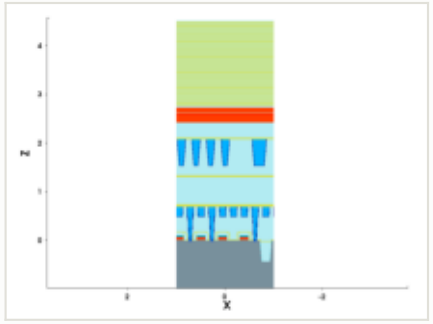
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



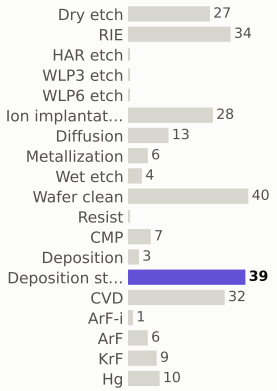
y cut at -1.032 µm



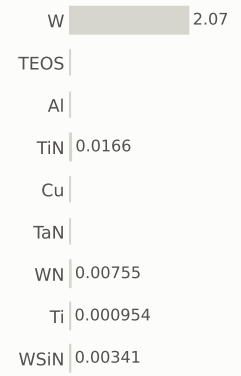
mask for this step

no mask for this step

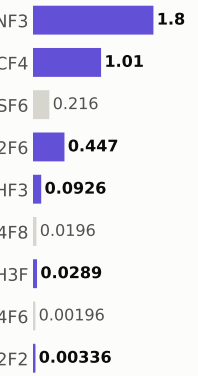
steps by process area



materials deposited (g)



gases emitted (g)



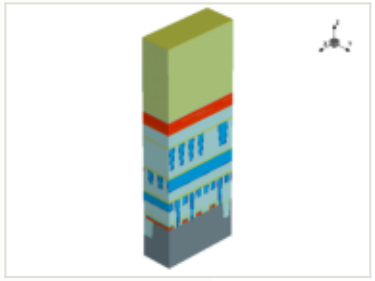
Step #3100: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	232.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

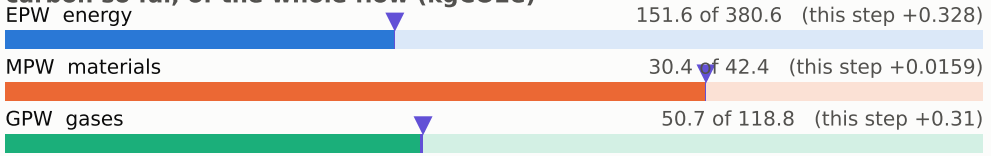
zoomed to fit



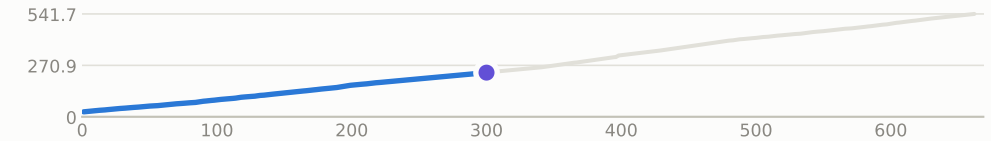
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

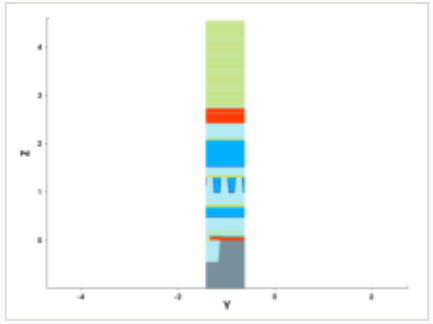
carbon so far, of the whole flow (kgCO2e)



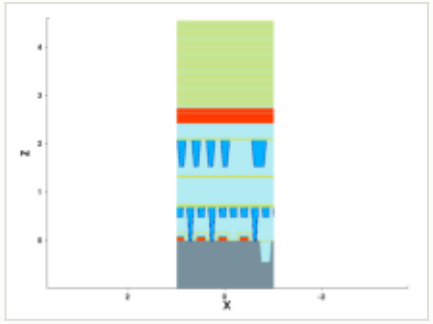
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



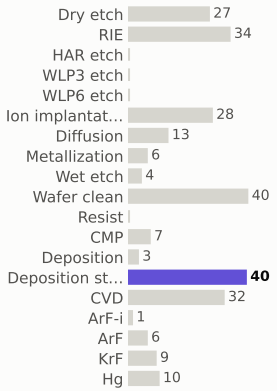
y cut at -1.032 µm



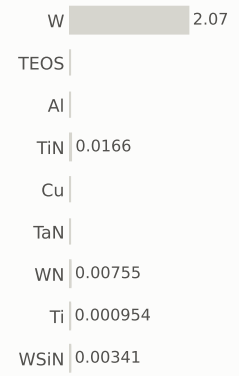
mask for this step

no mask for this step

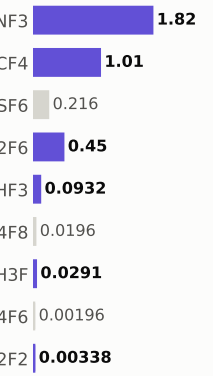
steps by process area



materials deposited (g)



gases emitted (g)



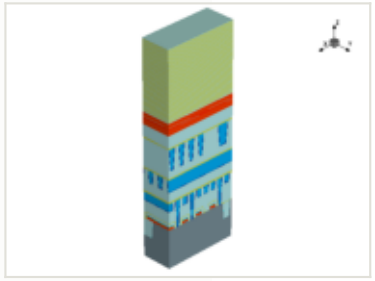
Step #3110: Deposition_stack

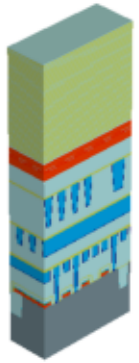
depo_anisotropic · Deposition_stack

0.654	233.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

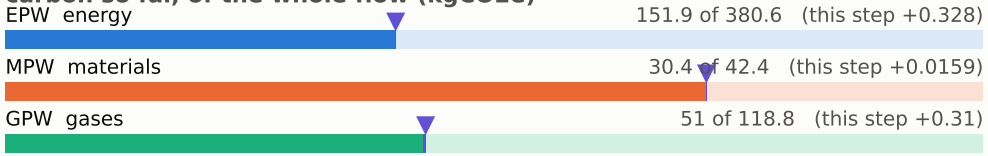




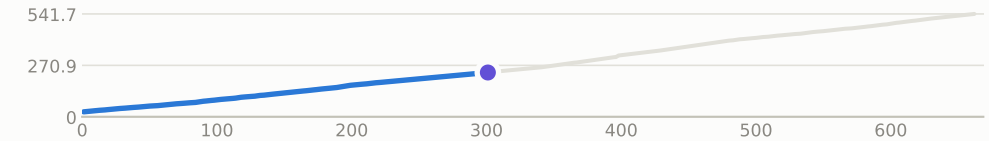
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

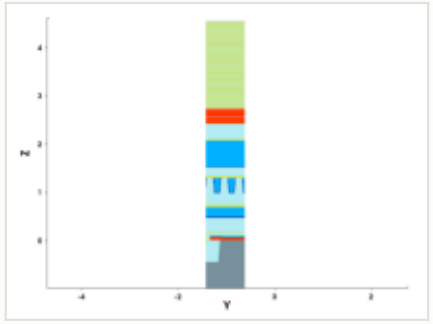
carbon so far, of the whole flow (kgCO2e)



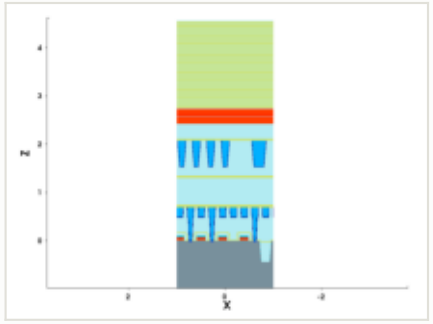
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



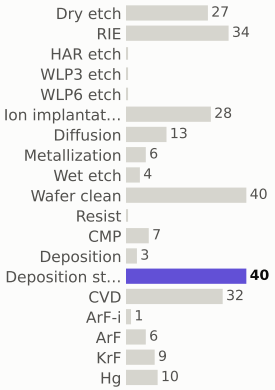
y cut at -1.032 µm



mask for this step

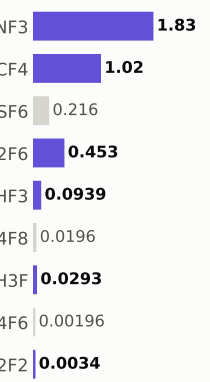
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



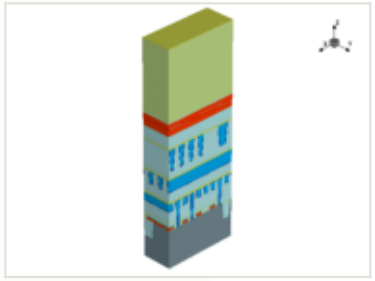
Step #3120: Deposition_stack

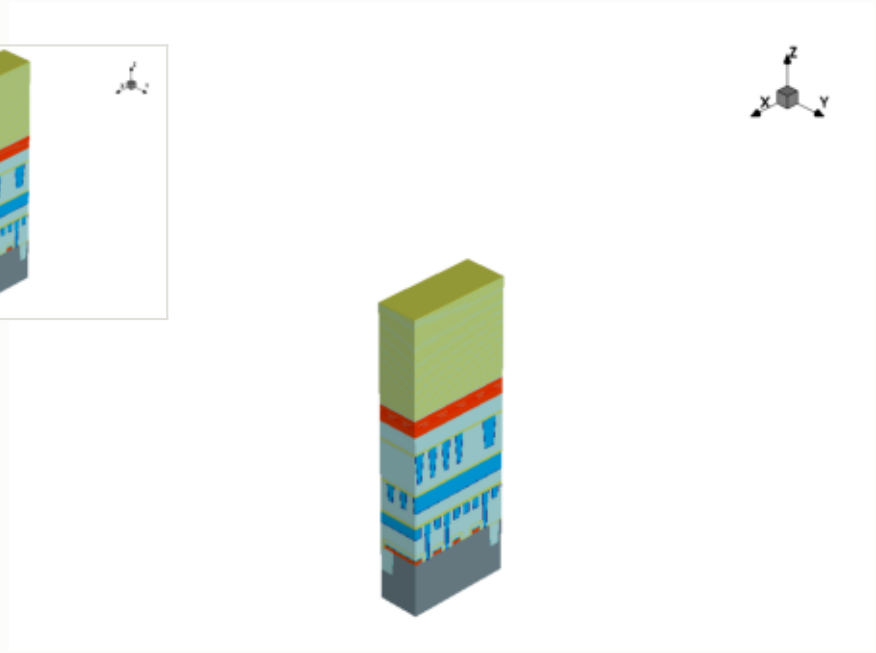
depo_anisotropic · Deposition_stack

0.654	234.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

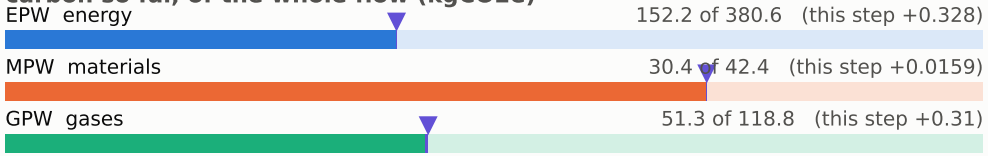




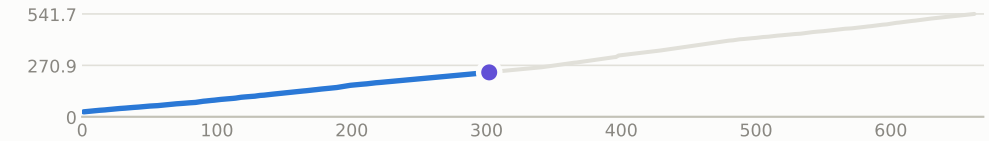
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

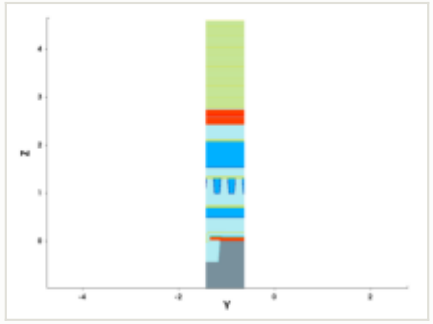
carbon so far, of the whole flow (kgCO2e)



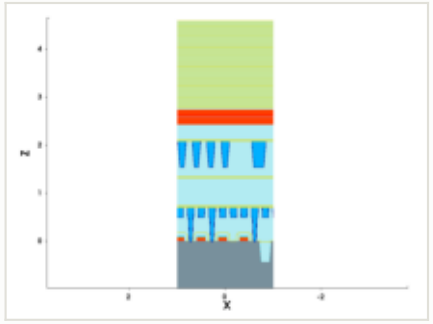
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



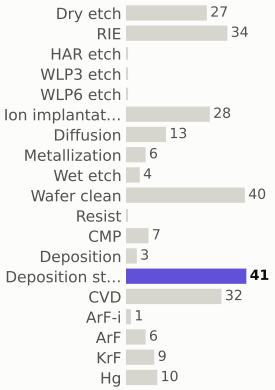
y cut at -1.032 µm



mask for this step

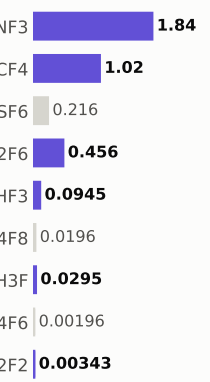
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



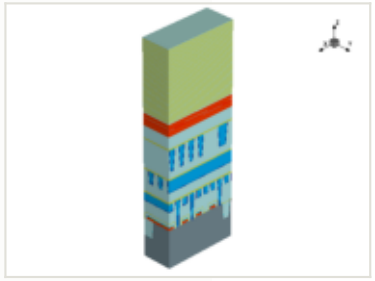
Step #3130: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	234.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

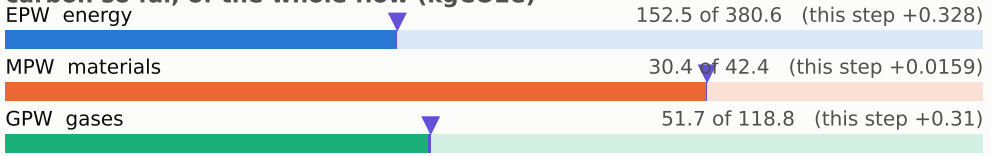
zoomed to fit



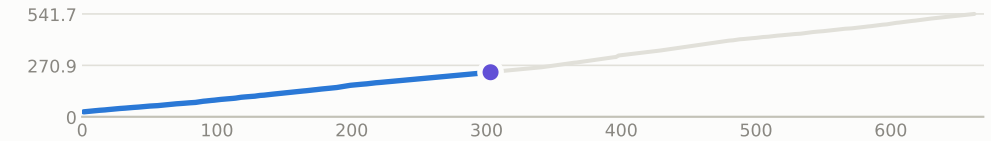
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

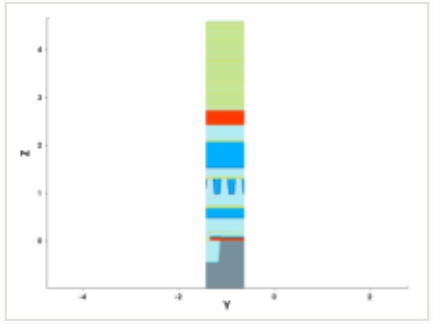
carbon so far, of the whole flow (kgCO2e)



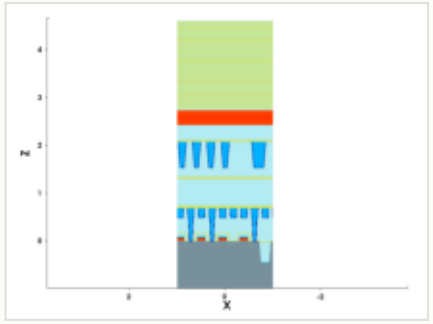
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



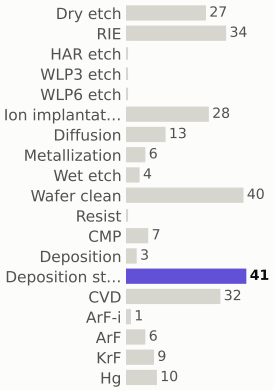
y cut at -1.032 µm



mask for this step

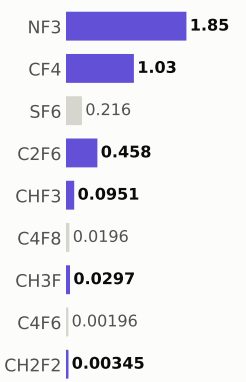
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



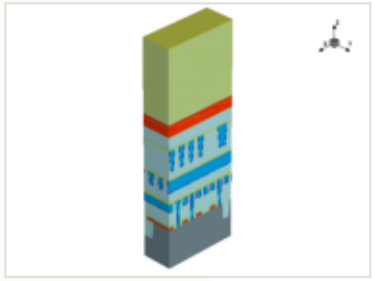
Step #3140: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	235.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

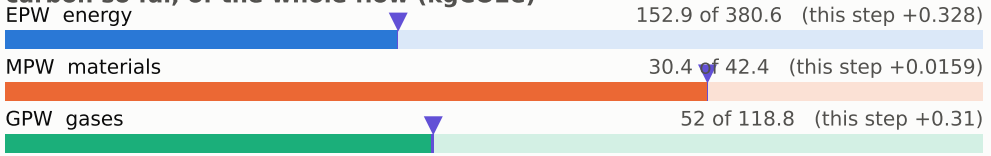
zoomed to fit



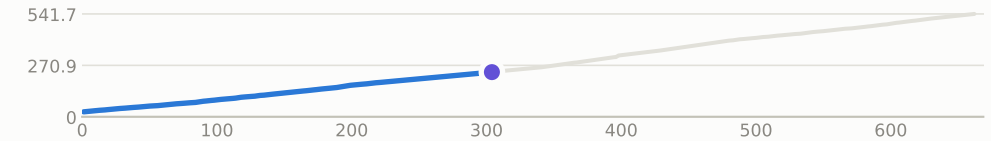
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

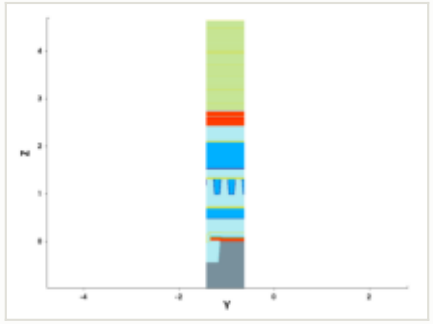
carbon so far, of the whole flow (kgCO2e)



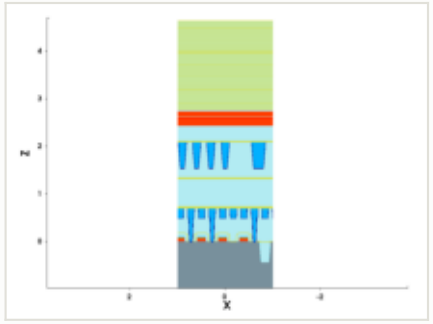
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



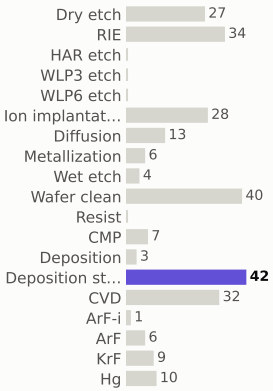
y cut at -1.032 µm



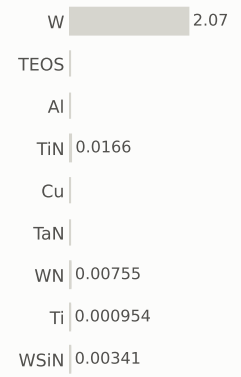
mask for this step

no mask for this step

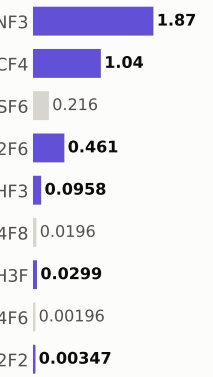
steps by process area



materials deposited (g)



gases emitted (g)



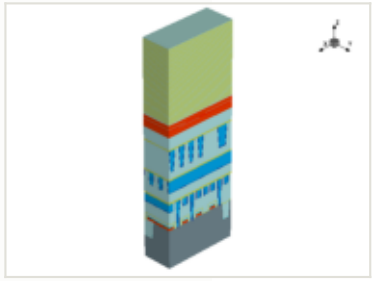
Step #3150: Deposition_stack

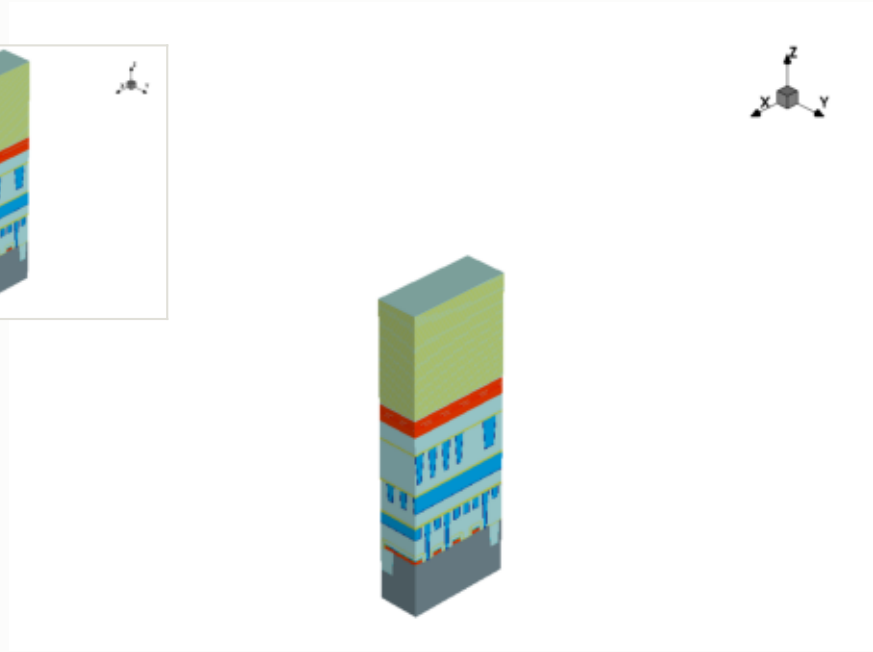
depo_anisotropic · Deposition_stack

0.654	235.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

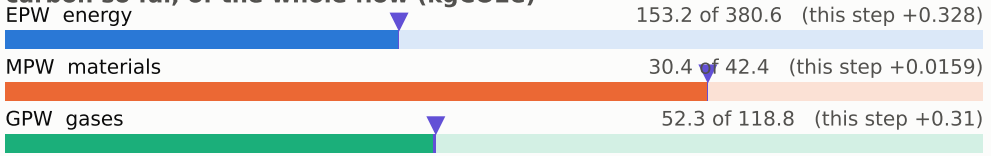




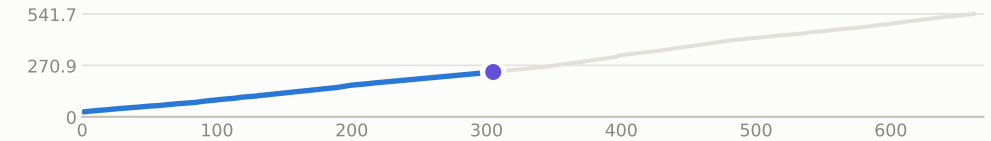
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

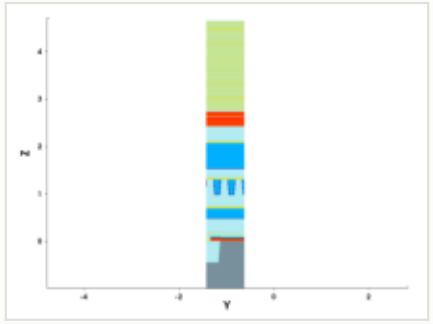
carbon so far, of the whole flow (kgCO2e)



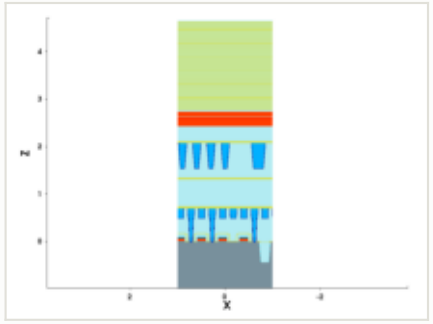
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



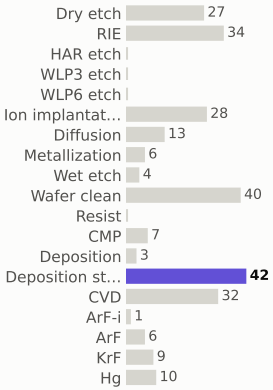
y cut at -1.032 µm



mask for this step

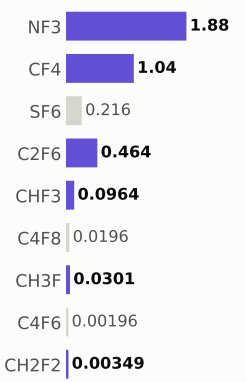
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



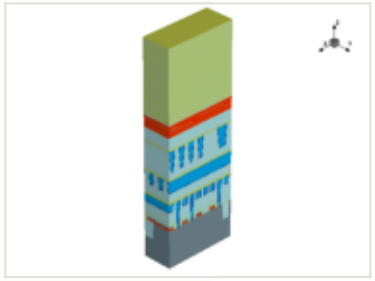
Step #3160: Deposition_stack

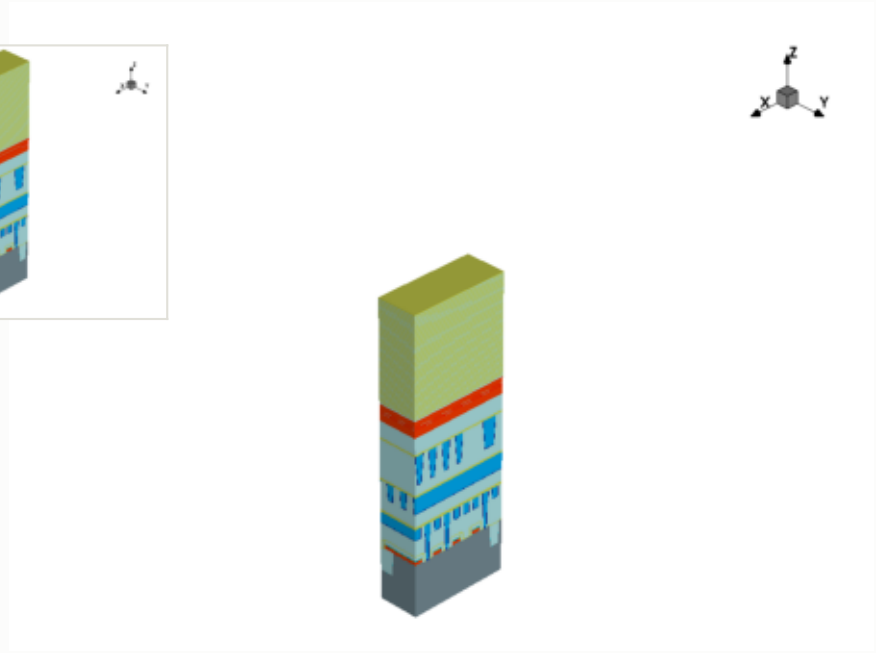
depo_anisotropic · Deposition_stack

0.654	236.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

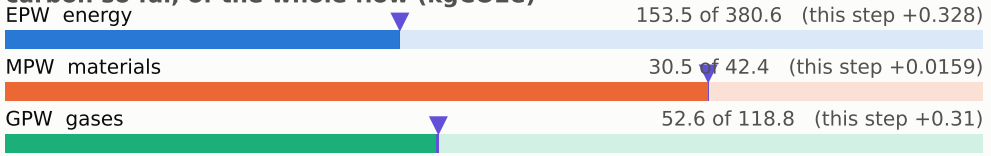




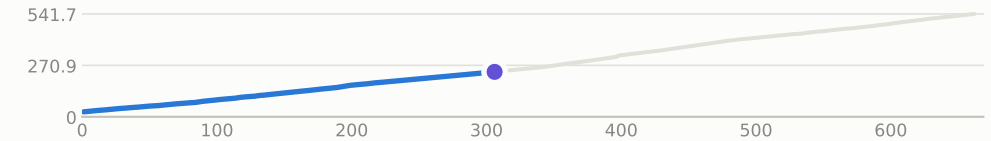
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

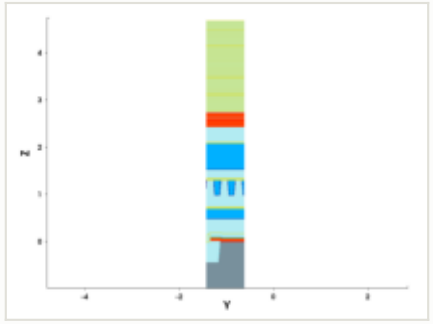
carbon so far, of the whole flow (kgCO2e)



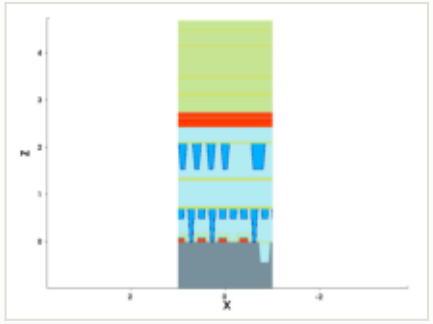
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



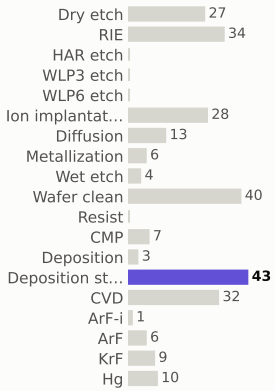
y cut at -1.032 µm



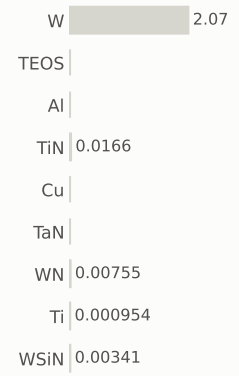
mask for this step

no mask for this step

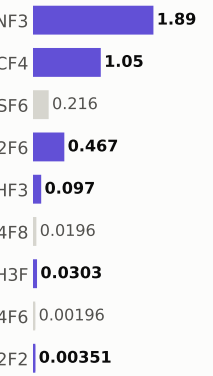
steps by process area



materials deposited (g)



gases emitted (g)



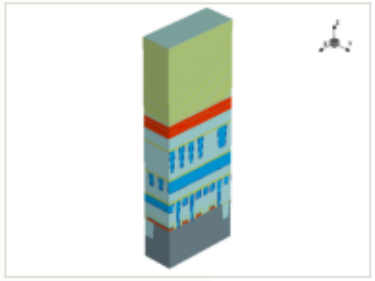
Step #3170: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	237.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

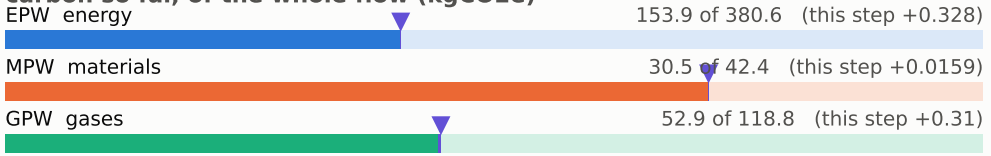
zoomed to fit



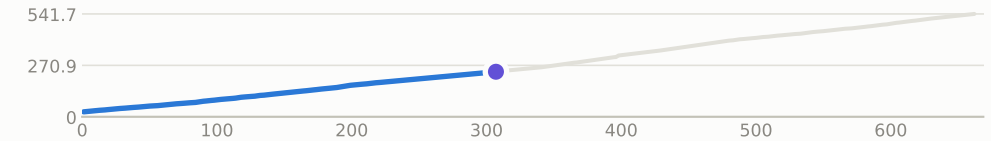
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

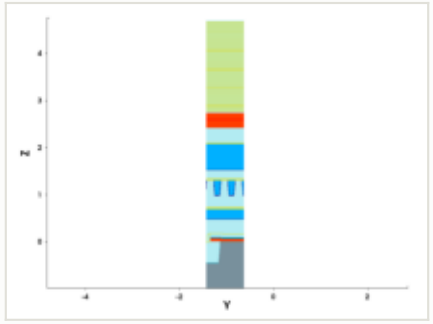
carbon so far, of the whole flow (kgCO2e)



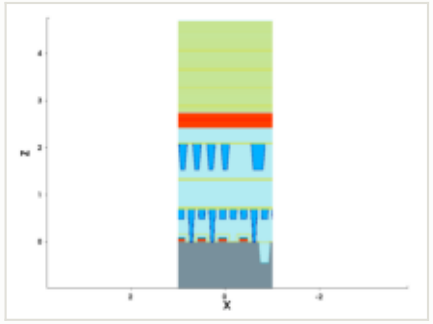
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



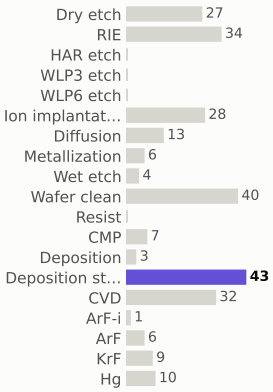
y cut at -1.032 µm



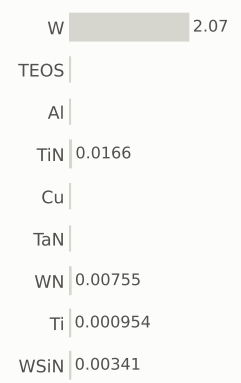
mask for this step

no mask for this step

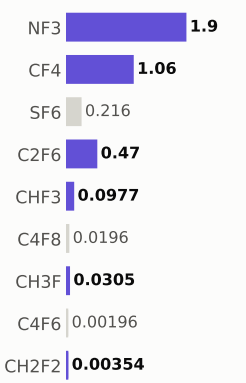
steps by process area



materials deposited (g)



gases emitted (g)



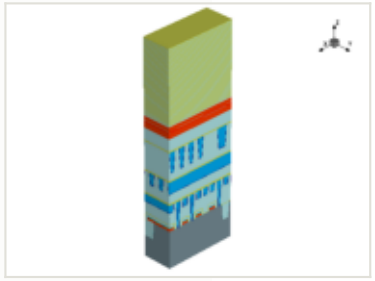
Step #3180: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	237.9	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

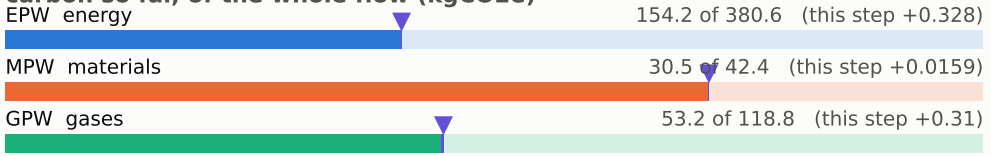
zoomed to fit



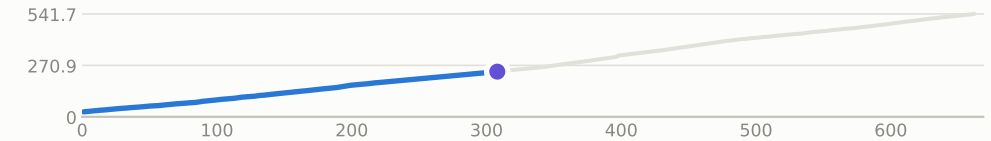
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

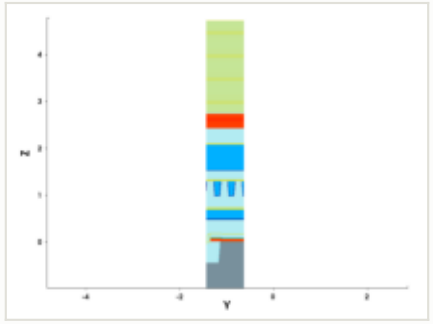
carbon so far, of the whole flow (kgCO2e)



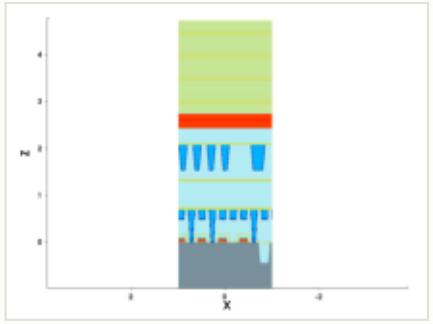
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



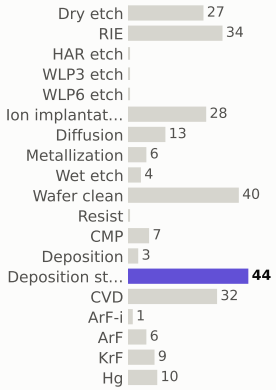
y cut at -1.032 µm



mask for this step

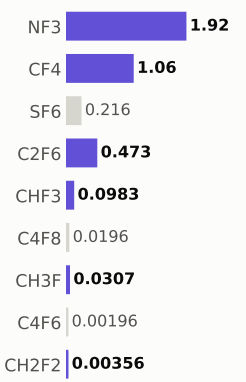
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



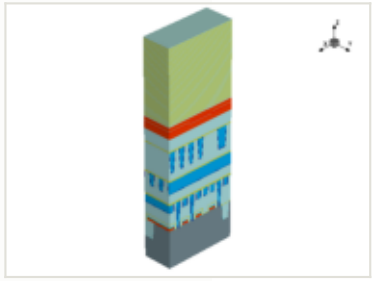
Step #3190: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	238.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

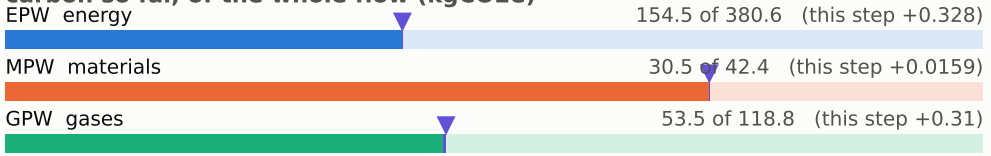
zoomed to fit



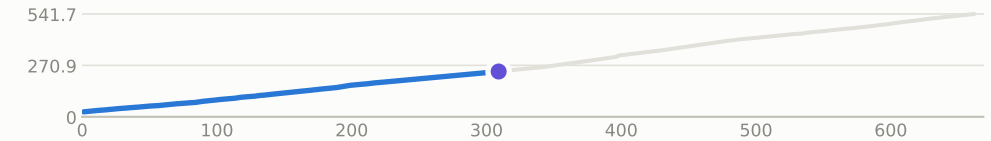
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

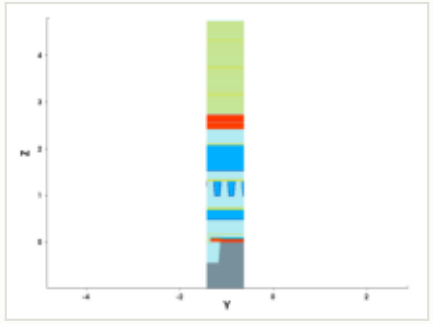
carbon so far, of the whole flow (kgCO2e)



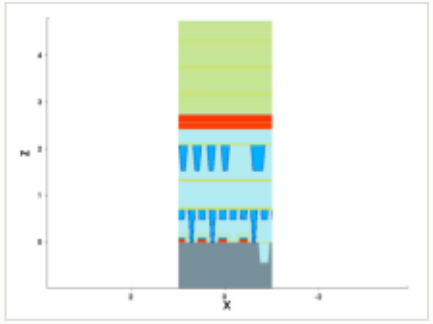
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



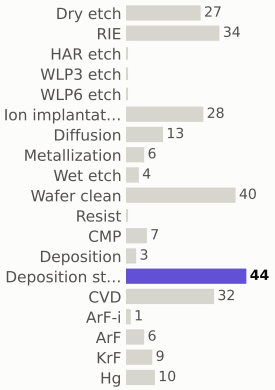
y cut at -1.032 µm



mask for this step

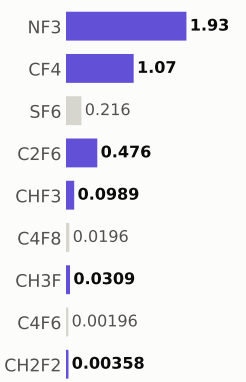
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



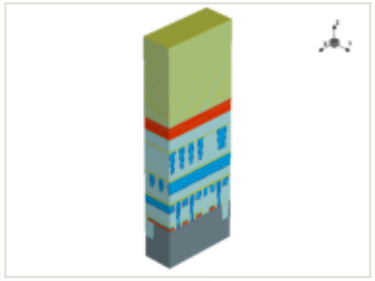
Step #3200: Deposition_stack

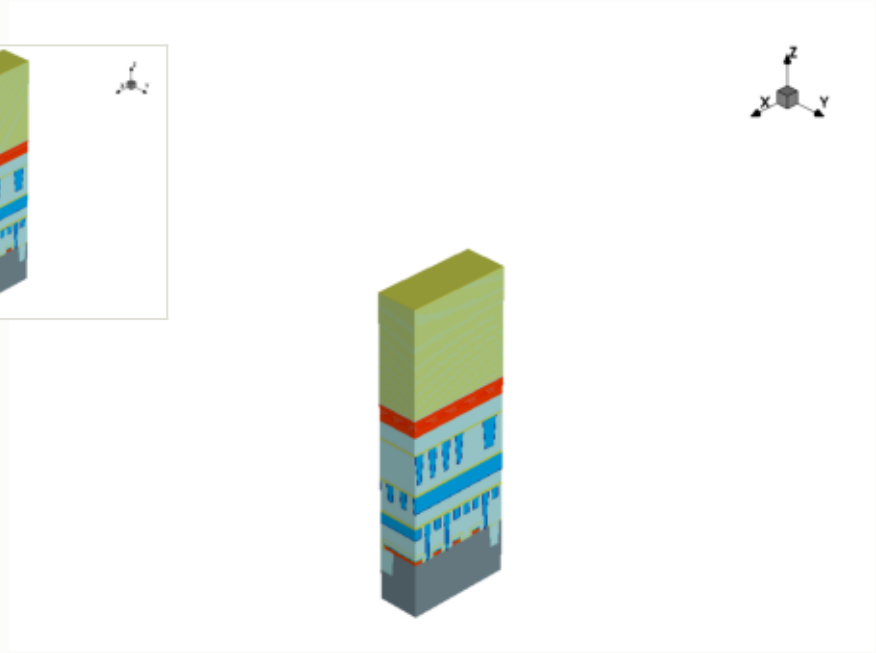
depo_anisotropic · Deposition_stack

0.654	239.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

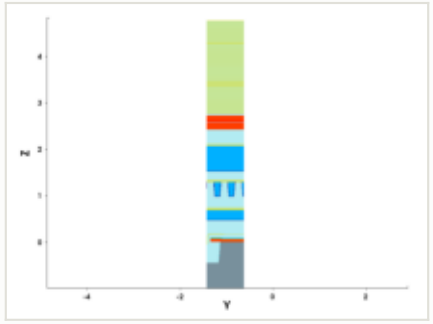
3D structure after this step

zoomed to fit

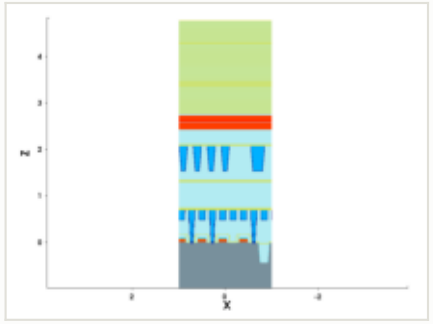




x cut at 0.000 µm



y cut at -1.032 µm



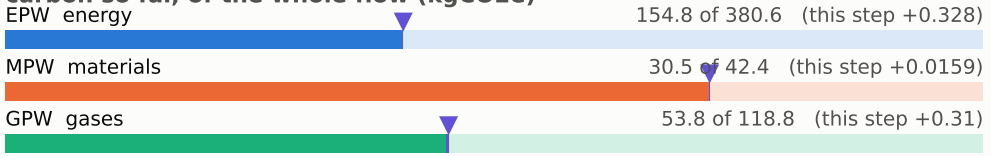
mask for this step

no mask for this step

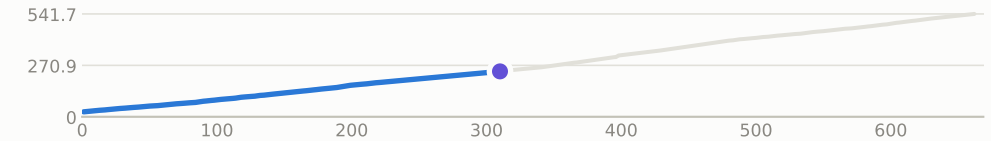
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

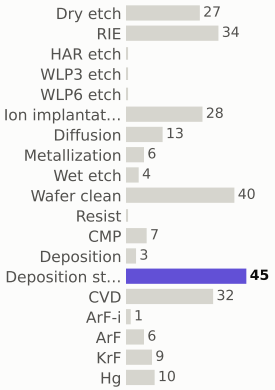
carbon so far, of the whole flow (kgCO2e)



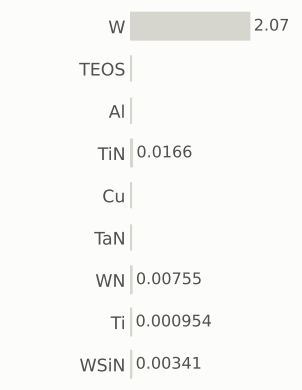
embodied carbon up to each step (kgCO2e per wafer)



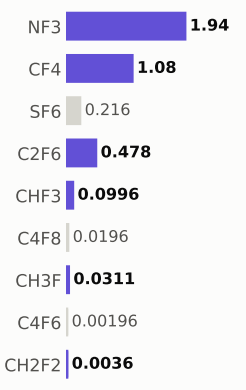
steps by process area



materials deposited (g)



gases emitted (g)



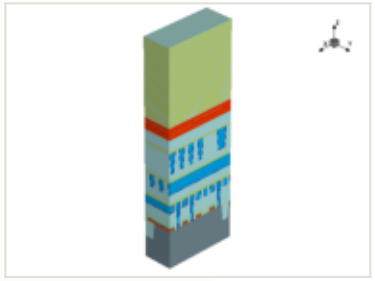
Step #3210: Deposition_stack

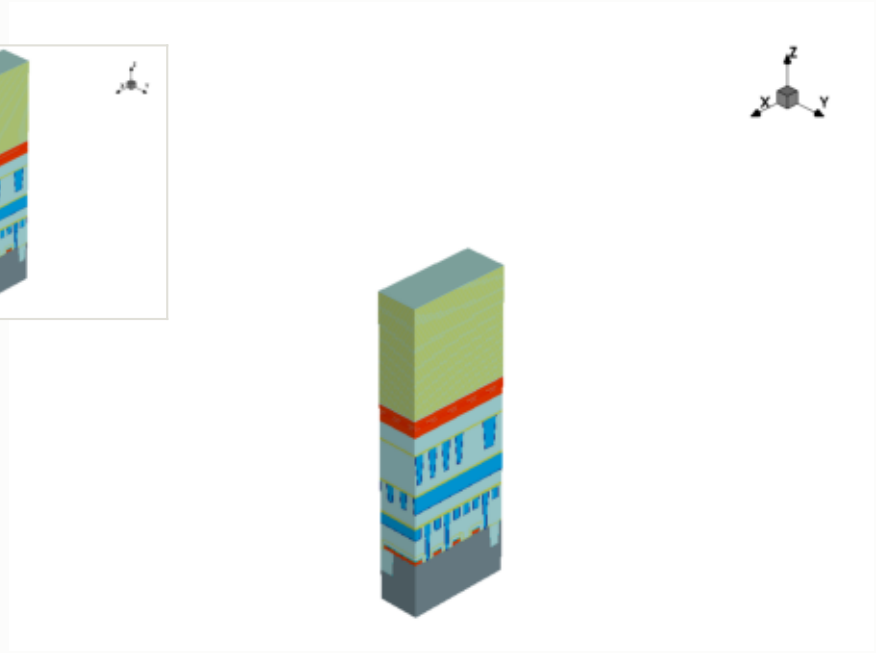
depo_anisotropic · Deposition_stack

0.654	239.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

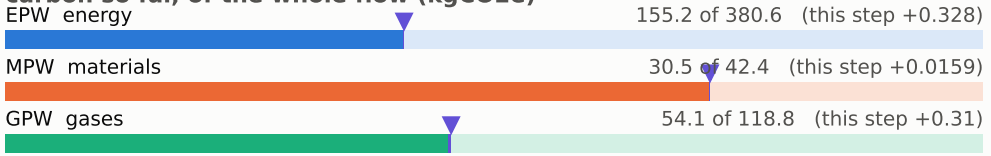




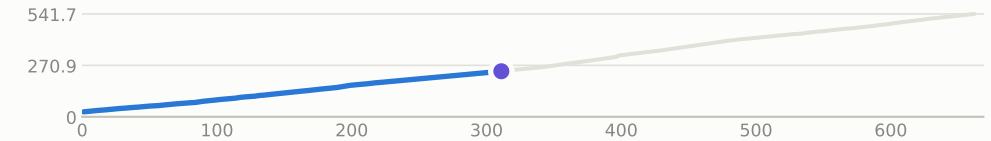
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

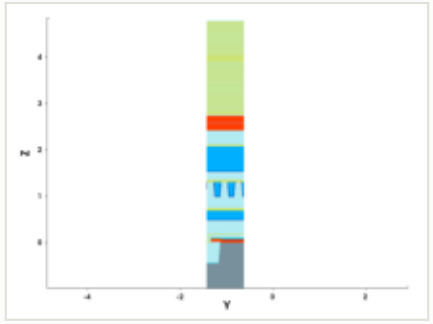
carbon so far, of the whole flow (kgCO2e)



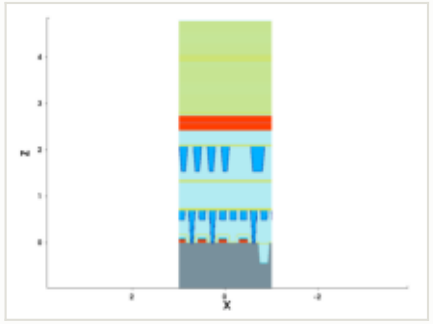
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



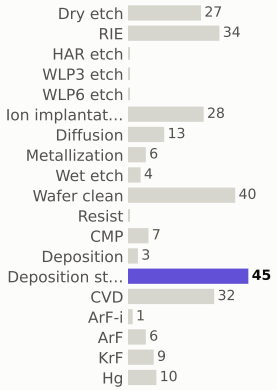
y cut at -1.032 µm



mask for this step

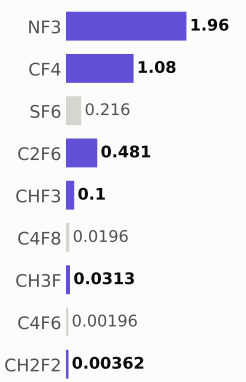
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



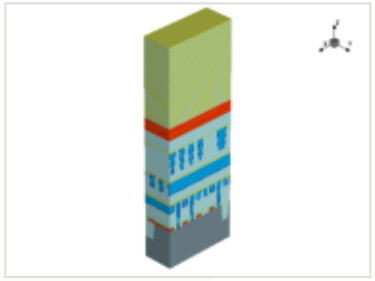
Step #3220: Deposition_stack

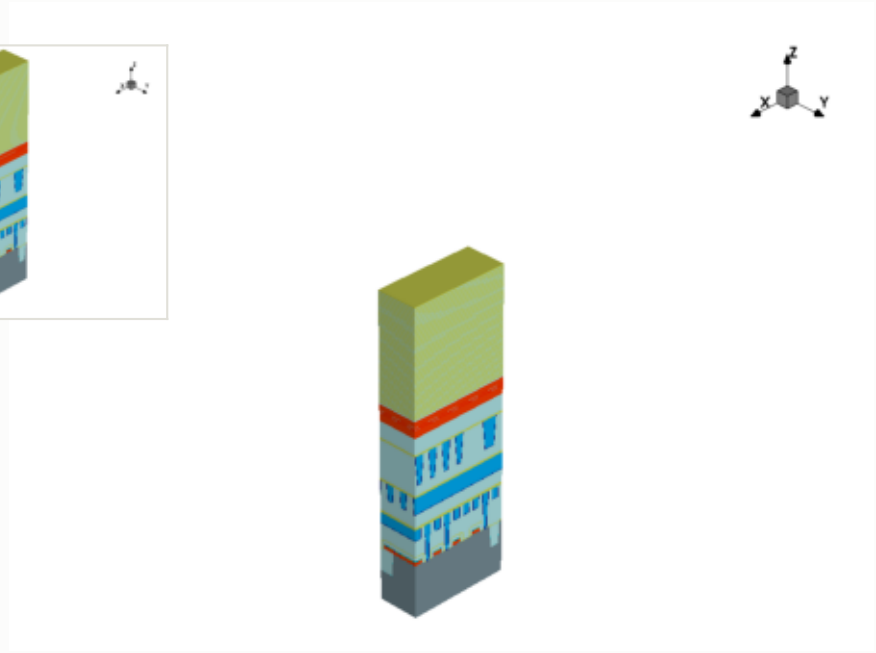
depo_anisotropic · Deposition_stack

0.654	240.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

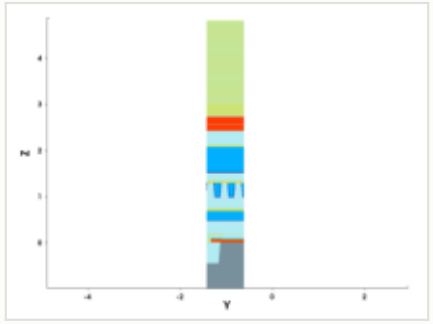
3D structure after this step

zoomed to fit

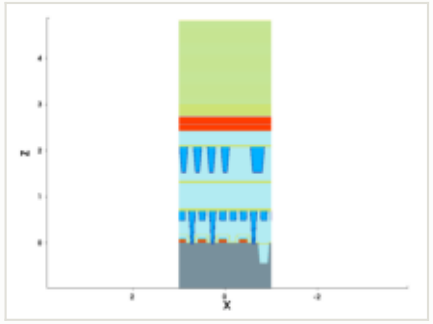




x cut at 0.000 µm



y cut at -1.032 µm



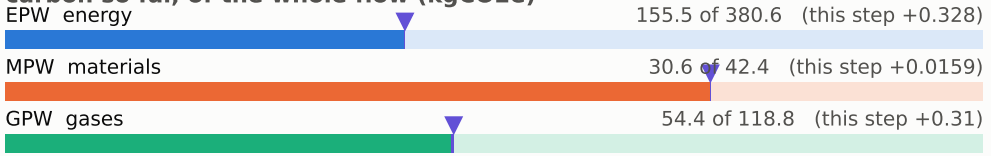
mask for this step

no mask for this step

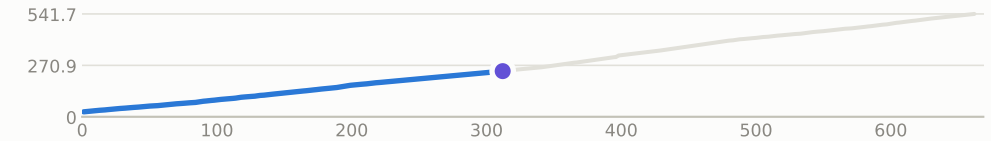
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

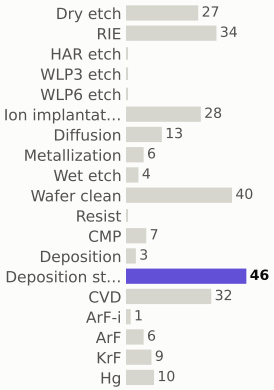
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

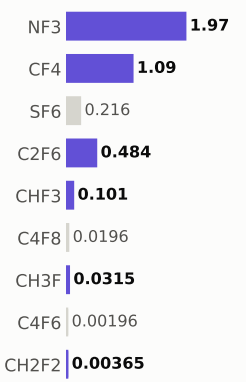


steps by process area



materials deposited (g)

gases emitted (g)



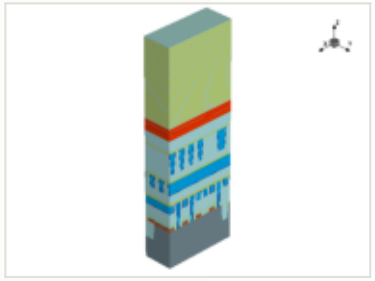
Step #3230: Deposition_stack

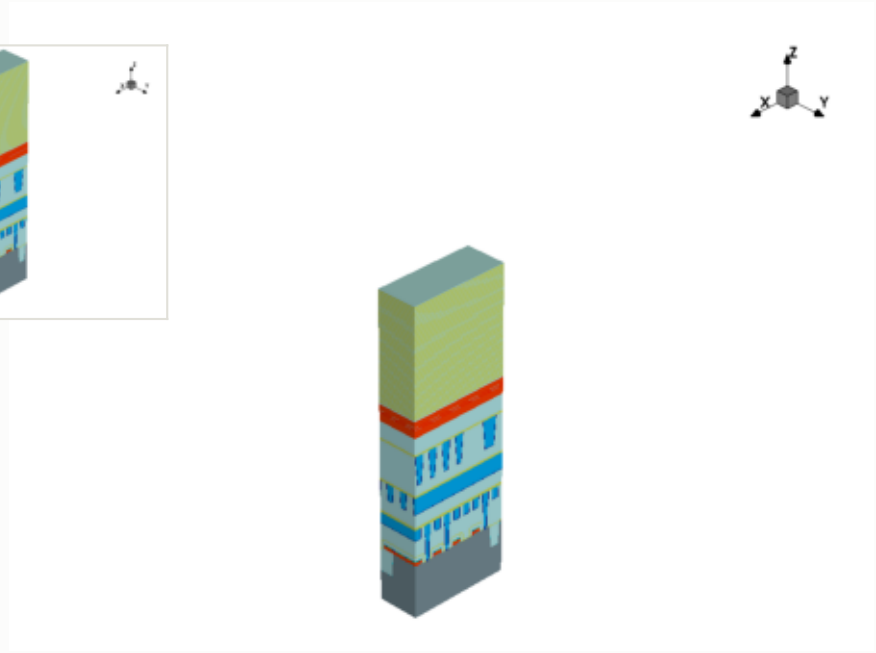
depo_anisotropic · Deposition_stack

0.654	241.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

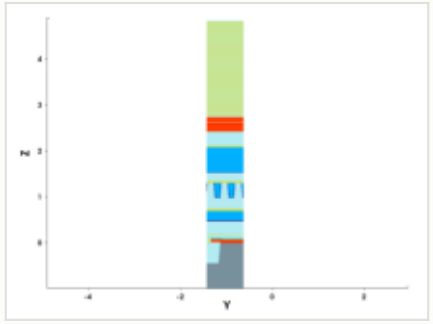
3D structure after this step

zoomed to fit

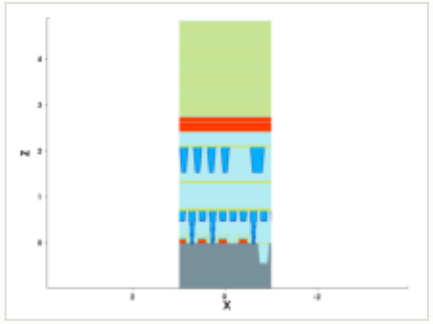




x cut at 0.000 µm



y cut at -1.032 µm



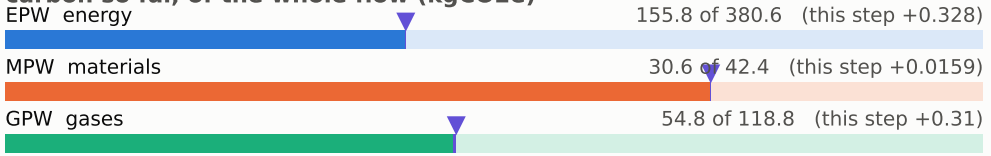
mask for this step

no mask for this step

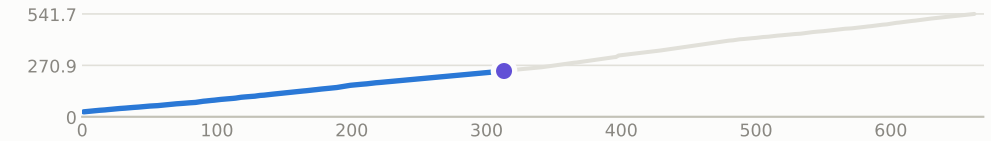
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

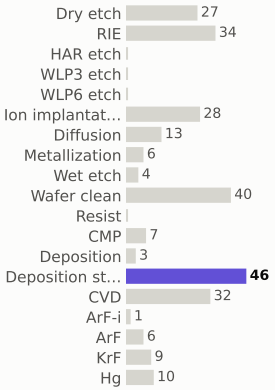
carbon so far, of the whole flow (kgCO2e)



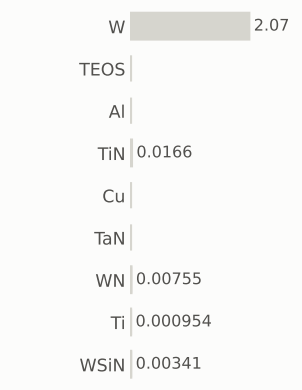
embodied carbon up to each step (kgCO2e per wafer)



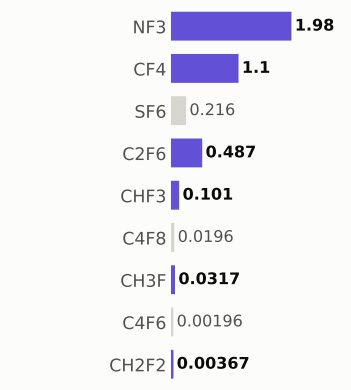
steps by process area



materials deposited (g)



gases emitted (g)



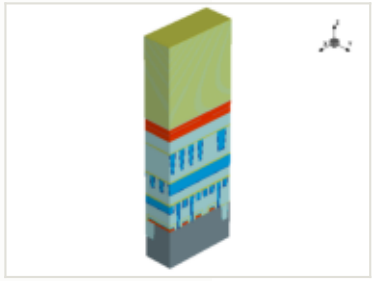
Step #3240: Deposition_stack

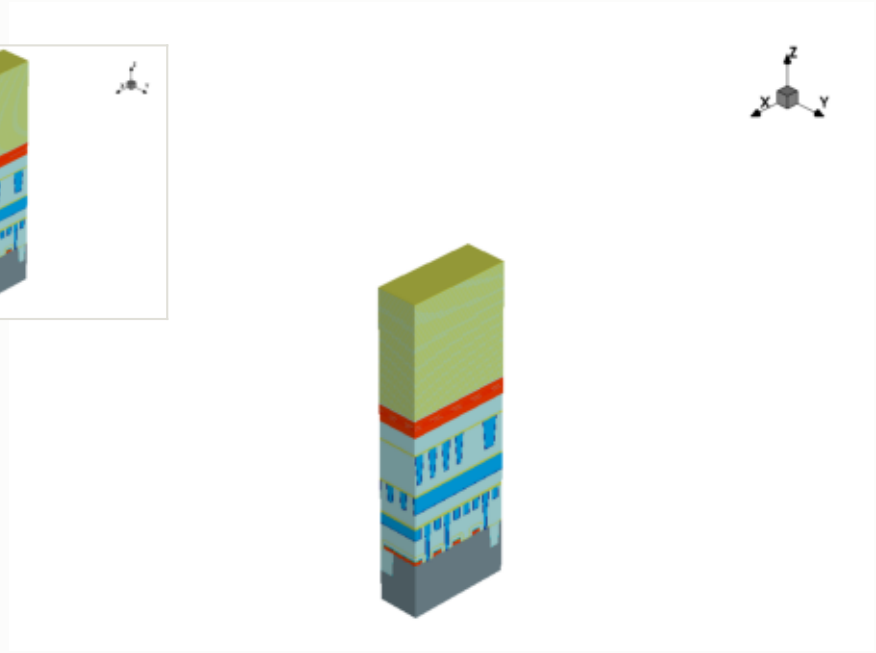
depo_anisotropic · Deposition_stack

0.654	241.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

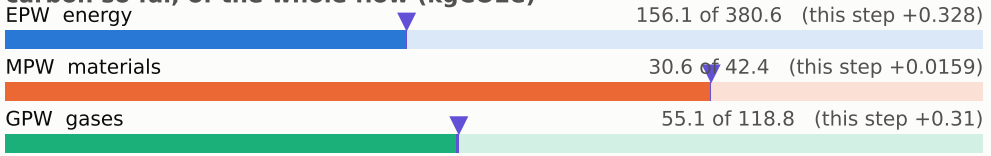




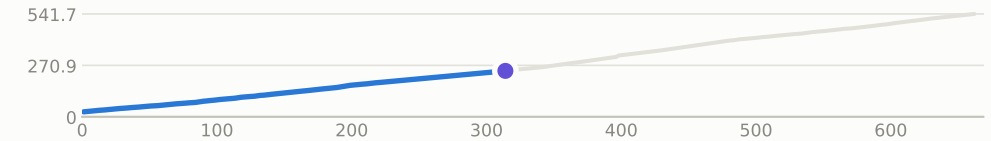
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

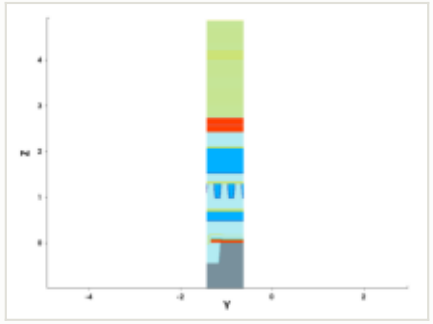
carbon so far, of the whole flow (kgCO2e)



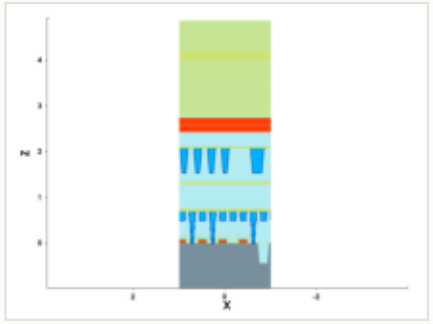
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



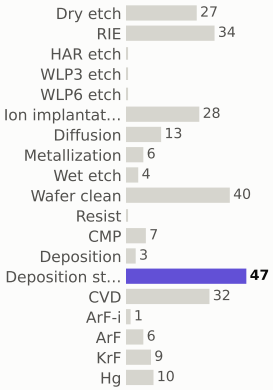
y cut at -1.032 µm



mask for this step

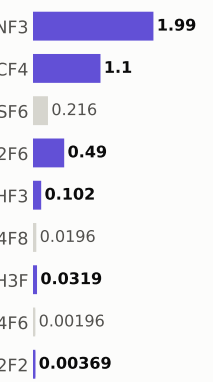
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



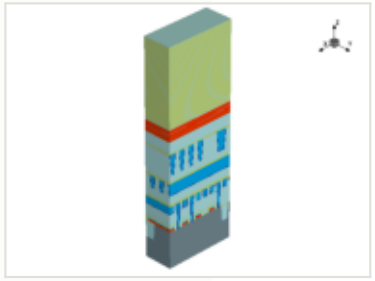
Step #3250: Deposition_stack

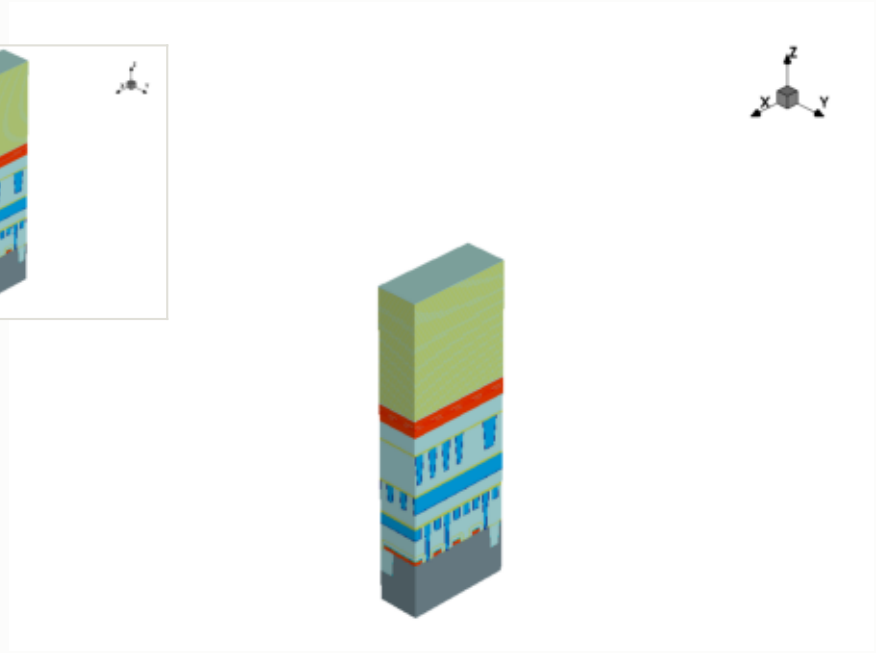
depo_anisotropic · Deposition_stack

0.654	242.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

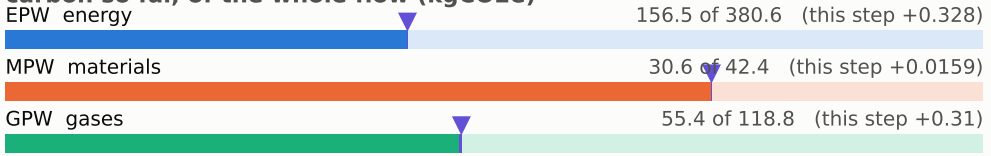




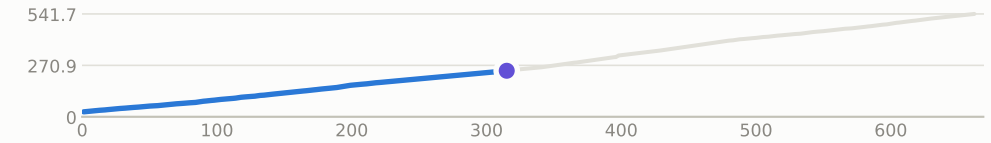
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

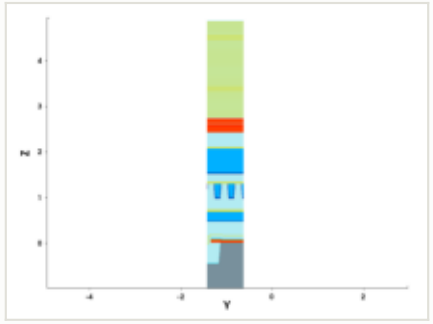
carbon so far, of the whole flow (kgCO2e)



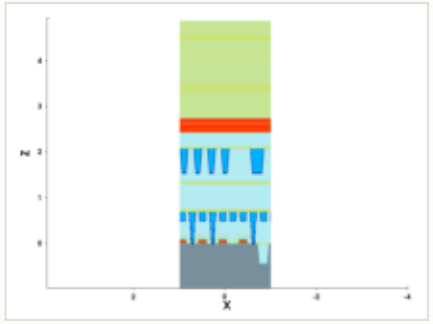
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



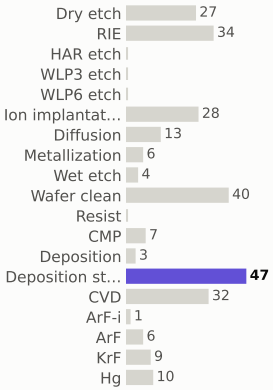
y cut at -1.032 µm



mask for this step

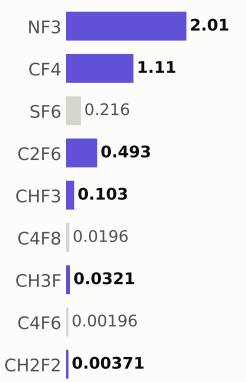
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



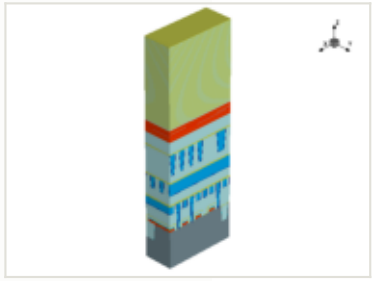
Step #3260: Deposition_stack

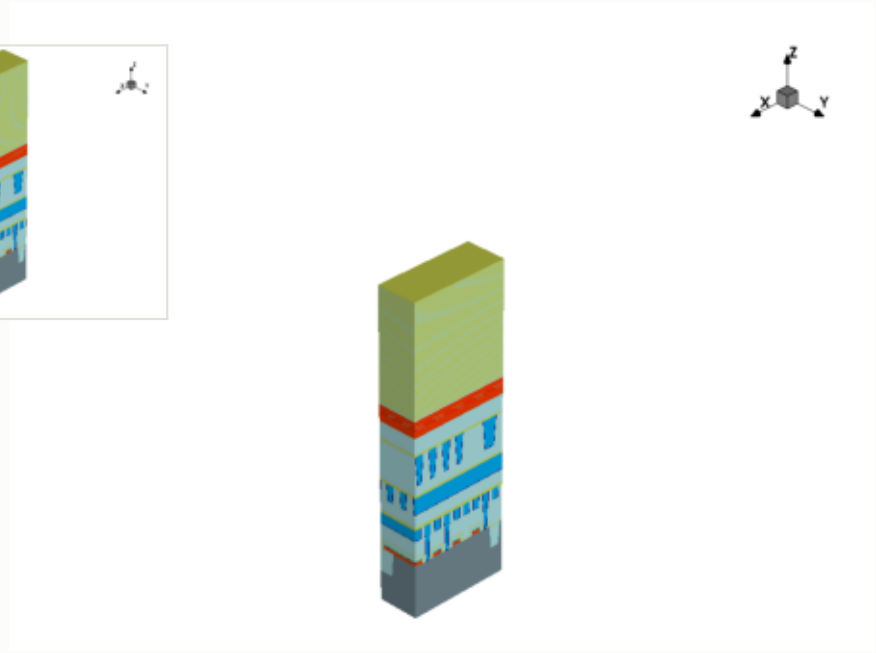
depo_anisotropic · Deposition_stack

0.654	243.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

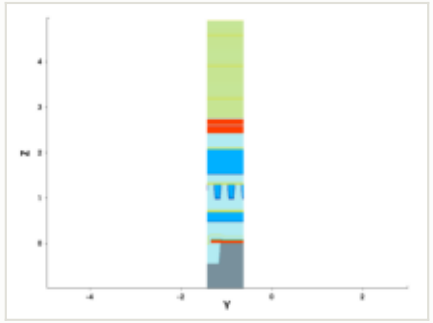
3D structure after this step

zoomed to fit

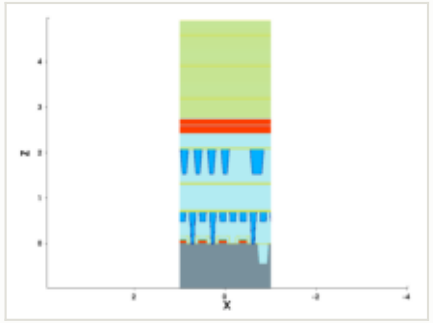




x cut at 0.000 µm



y cut at -1.032 µm



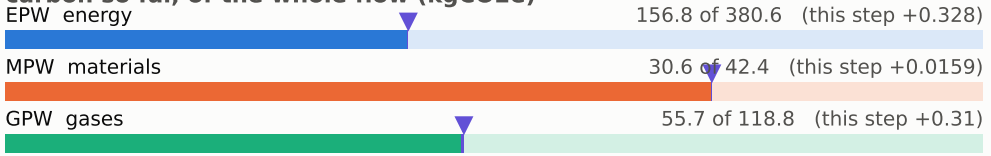
mask for this step

no mask for this step

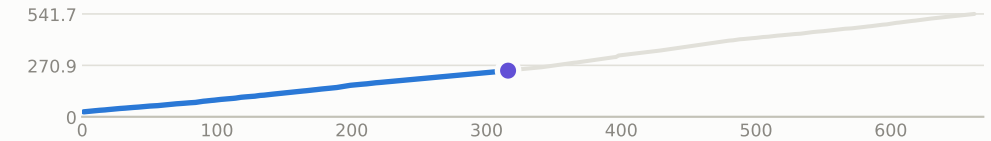
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

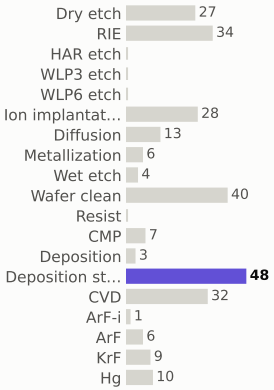
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

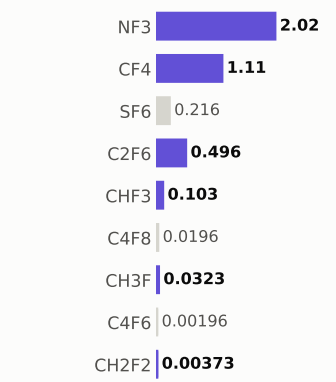


steps by process area



materials deposited (g)

gases emitted (g)



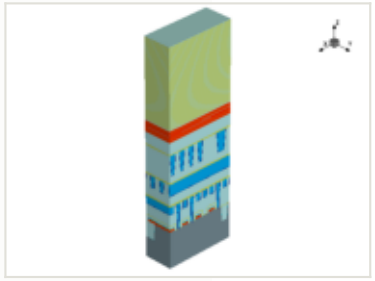
Step #3270: Deposition_stack

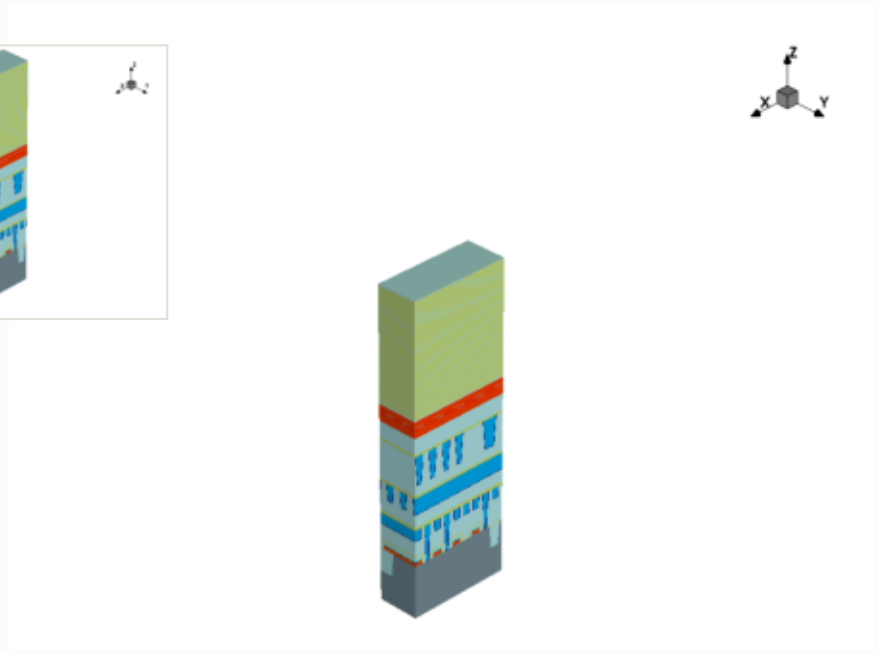
depo_anisotropic · Deposition_stack

0.654	243.8	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

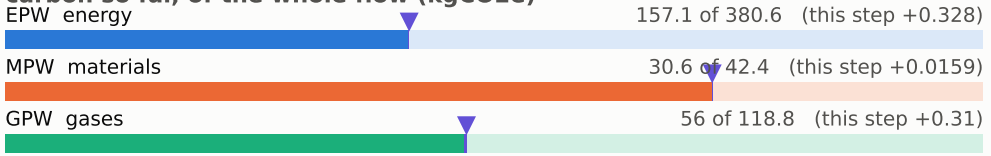




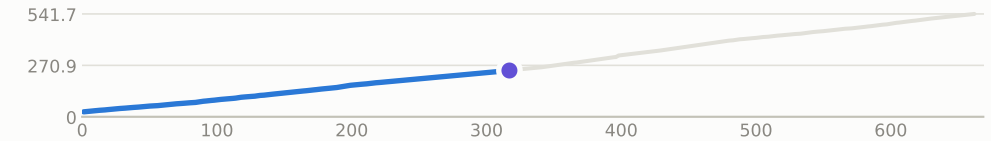
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

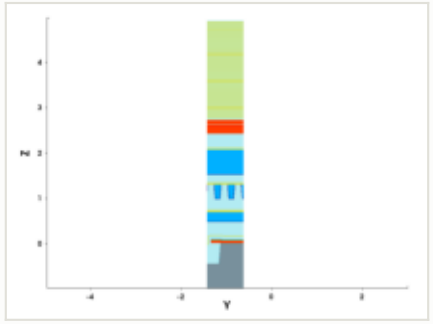
carbon so far, of the whole flow (kgCO2e)



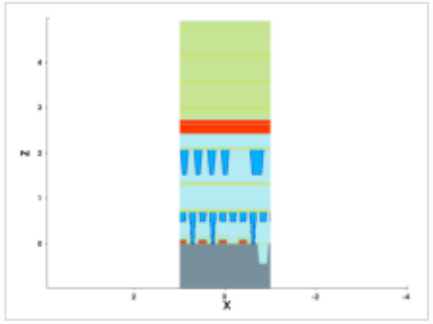
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



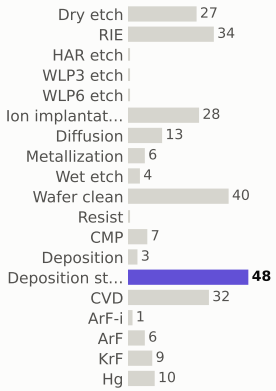
y cut at -1.032 µm



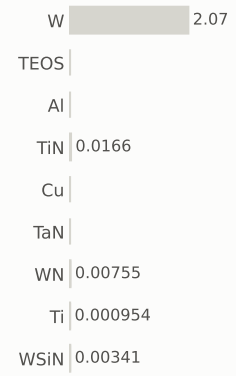
mask for this step

no mask for this step

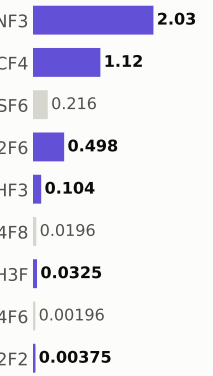
steps by process area



materials deposited (g)



gases emitted (g)



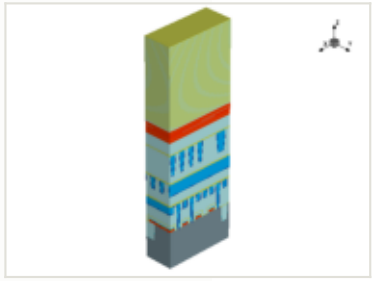
Step #3280: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	244.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

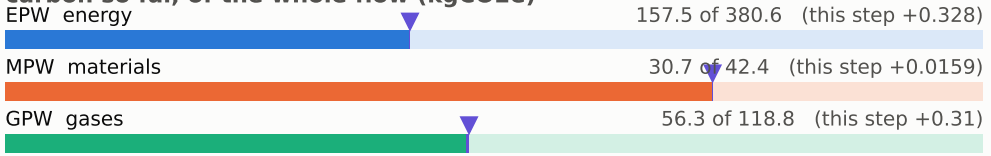
zoomed to fit



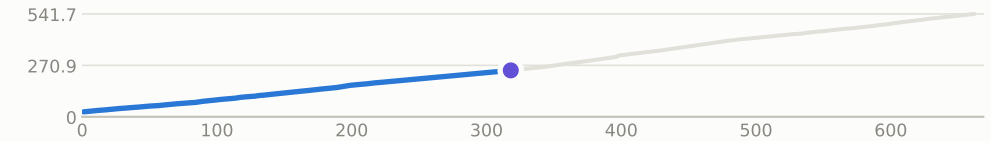
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

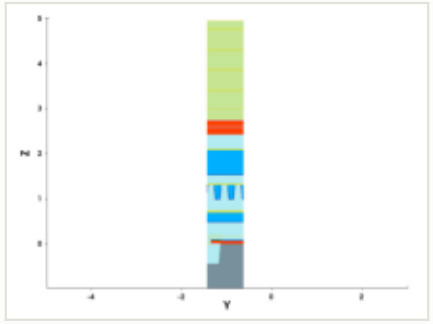
carbon so far, of the whole flow (kgCO2e)



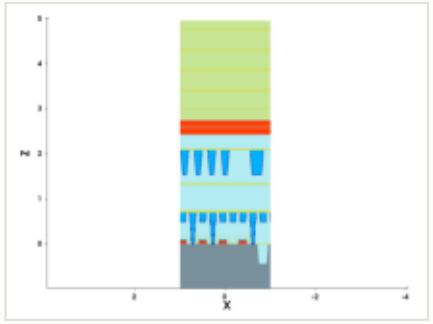
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



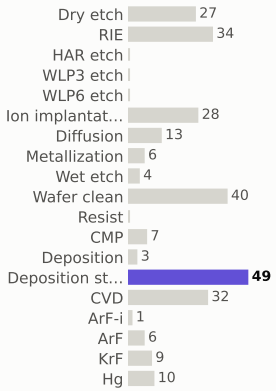
y cut at -1.032 µm



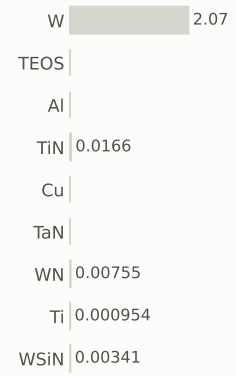
mask for this step

no mask for this step

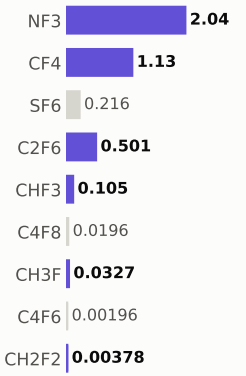
steps by process area



materials deposited (g)



gases emitted (g)



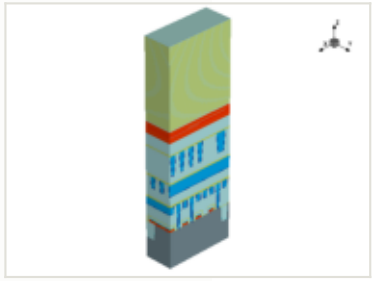
Step #3290: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	245.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

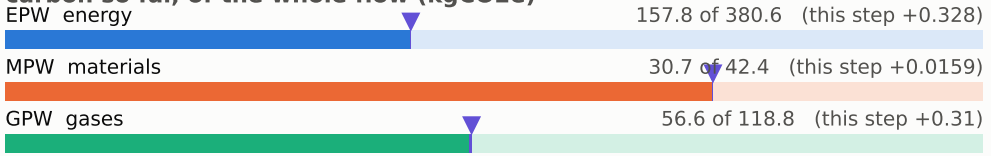
zoomed to fit



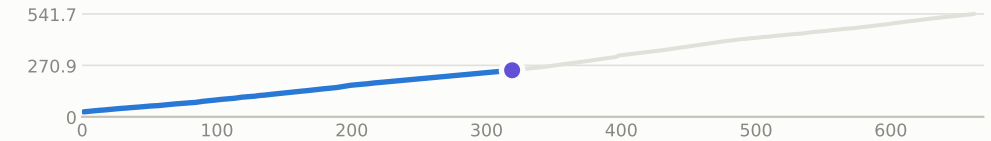
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

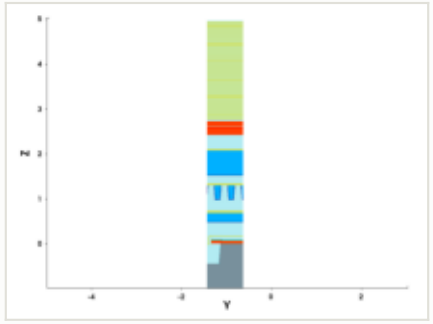
carbon so far, of the whole flow (kgCO2e)



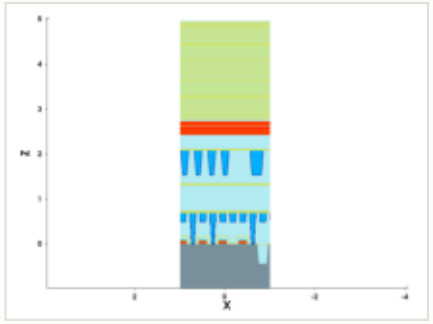
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



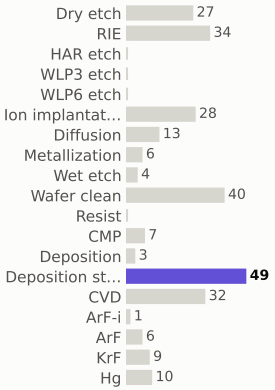
y cut at -1.032 µm



mask for this step

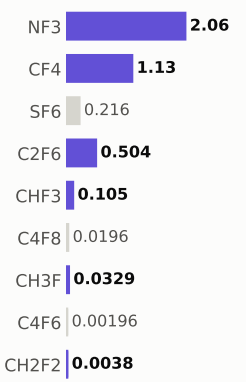
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



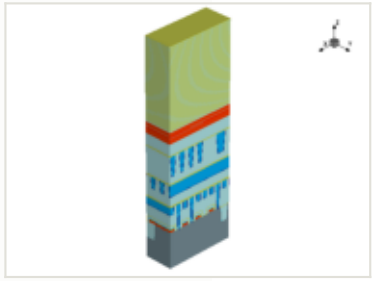
Step #3300: Deposition_stack

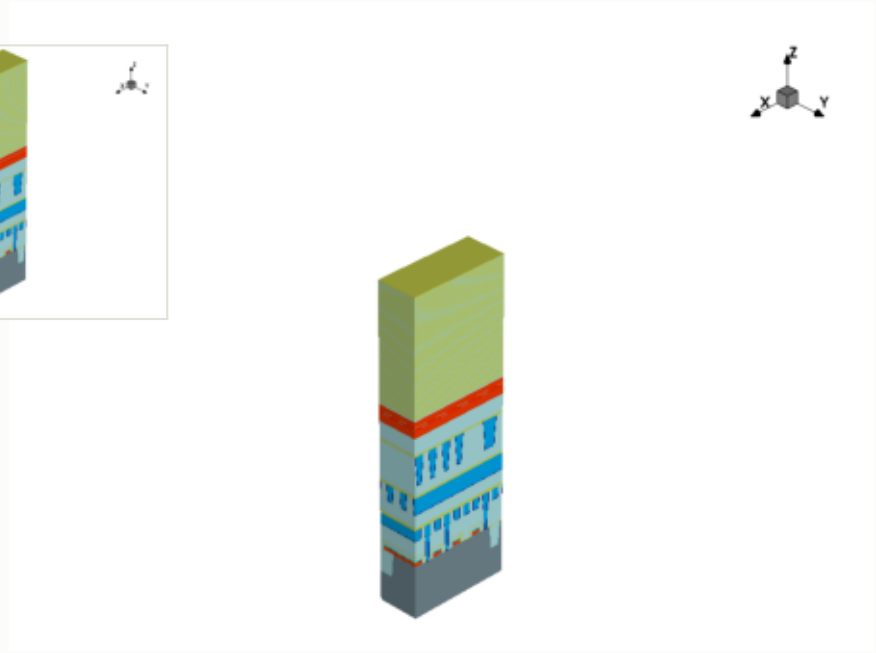
depo_anisotropic · Deposition_stack

0.654	245.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

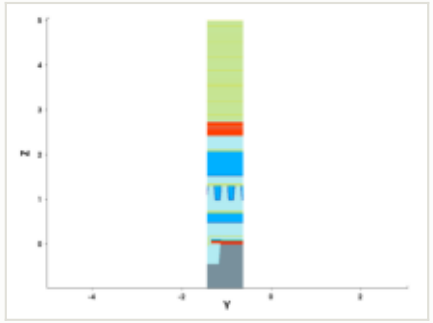
3D structure after this step

zoomed to fit

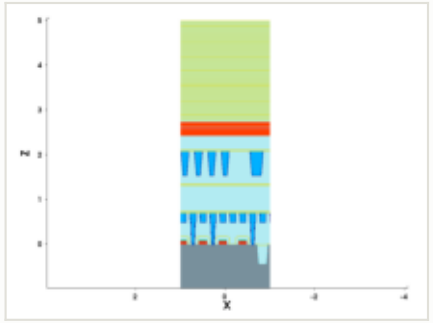




x cut at 0.000 µm



y cut at -1.032 µm



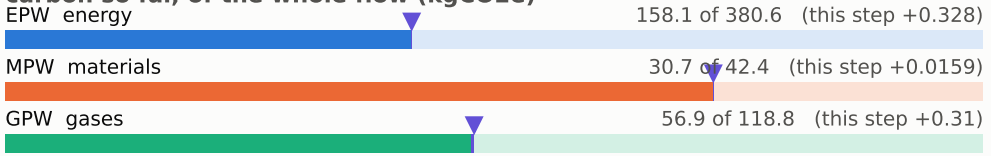
mask for this step

no mask for this step

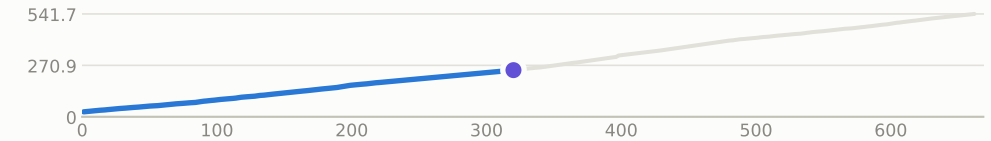
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

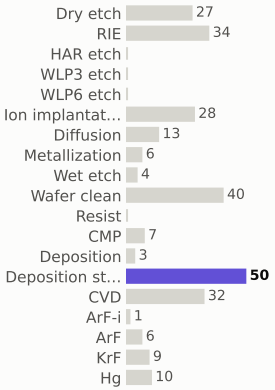
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

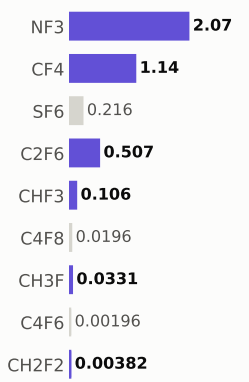


steps by process area



materials deposited (g)

gases emitted (g)



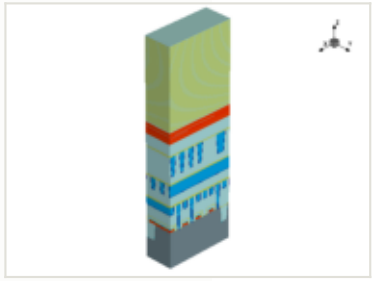
Step #3310: Deposition_stack

depo_anisotropic · Deposition_stack

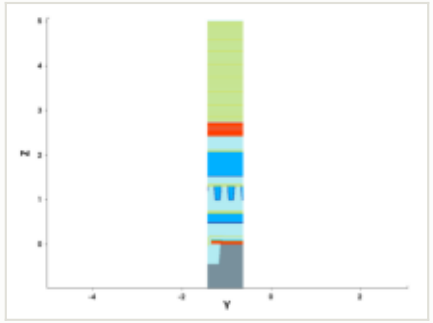
0.654	246.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

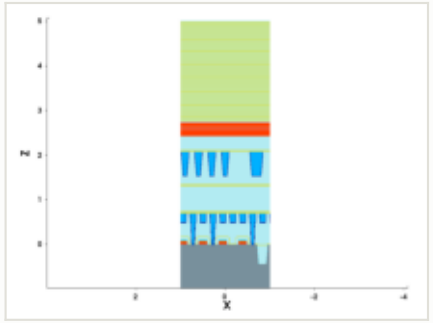
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



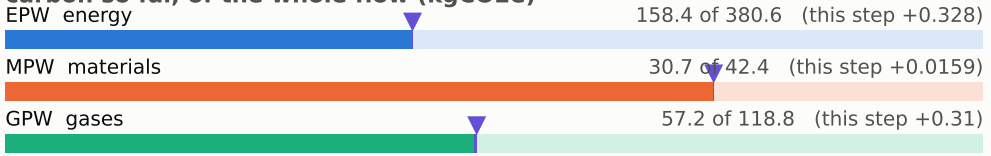
mask for this step

no mask for this step

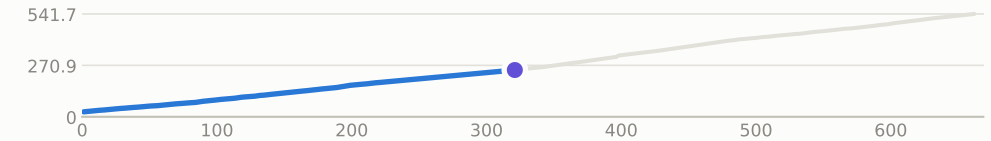
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

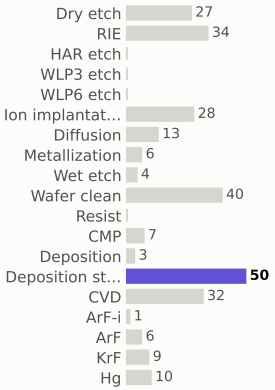
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

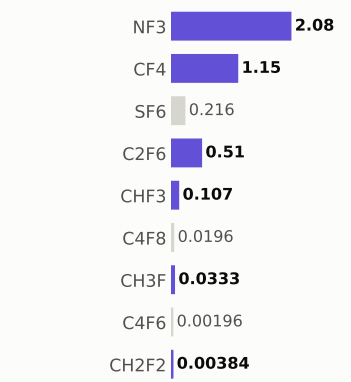


steps by process area



materials deposited (g)

gases emitted (g)



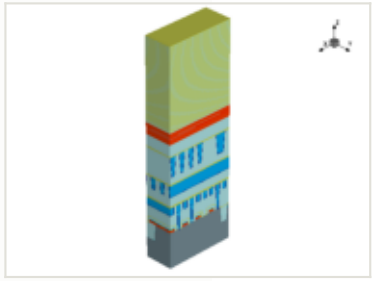
Step #3320: Deposition_stack

depo_anisotropic · Deposition_stack

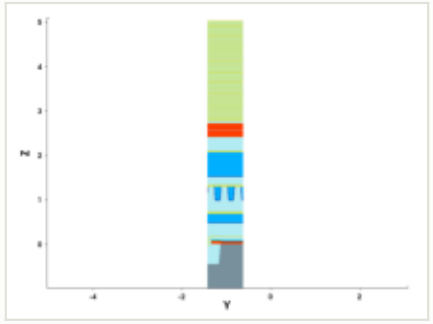
0.654	247.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

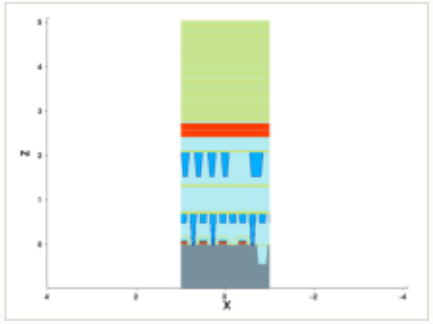
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



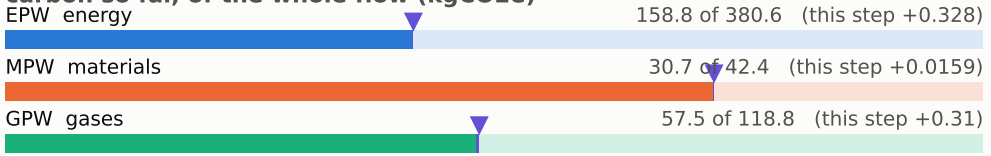
mask for this step

no mask for this step

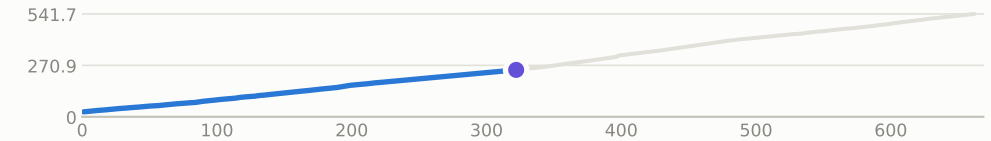
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

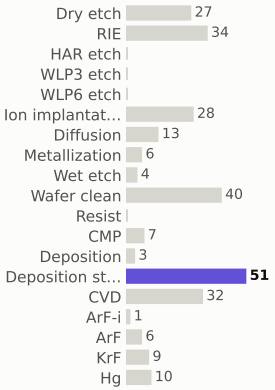
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

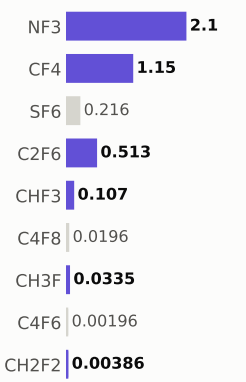


steps by process area



materials deposited (g)

gases emitted (g)



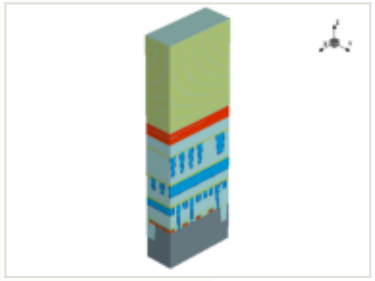
Step #3330: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	247.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

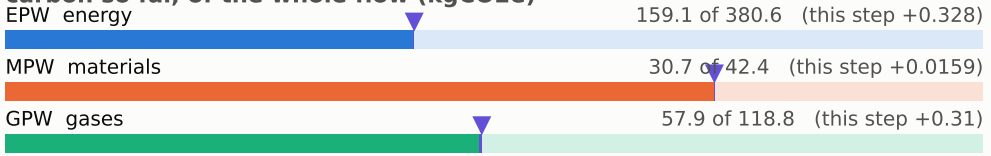
zoomed to fit



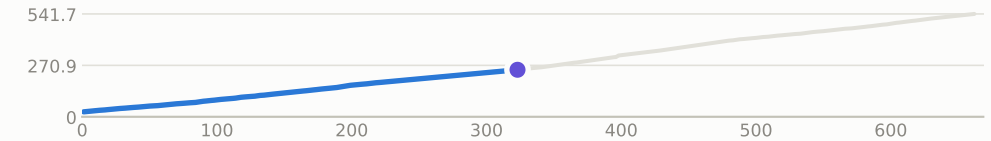
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

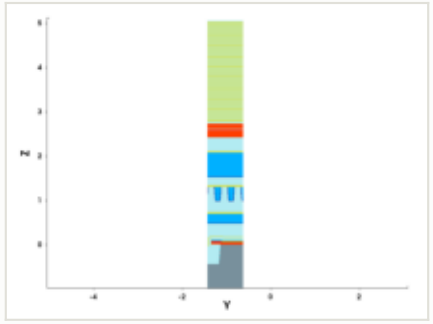
carbon so far, of the whole flow (kgCO2e)



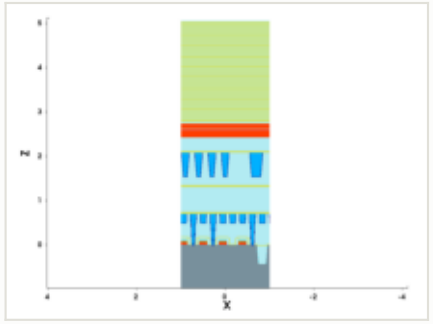
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



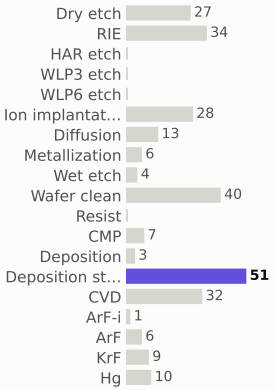
y cut at -1.032 µm



mask for this step

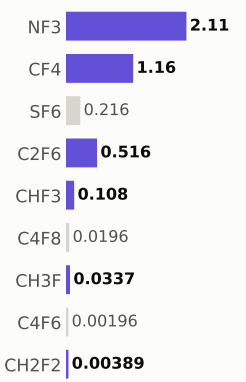
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



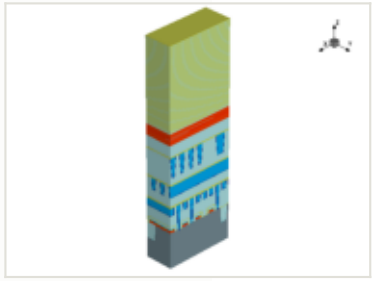
Step #3340: Deposition_stack

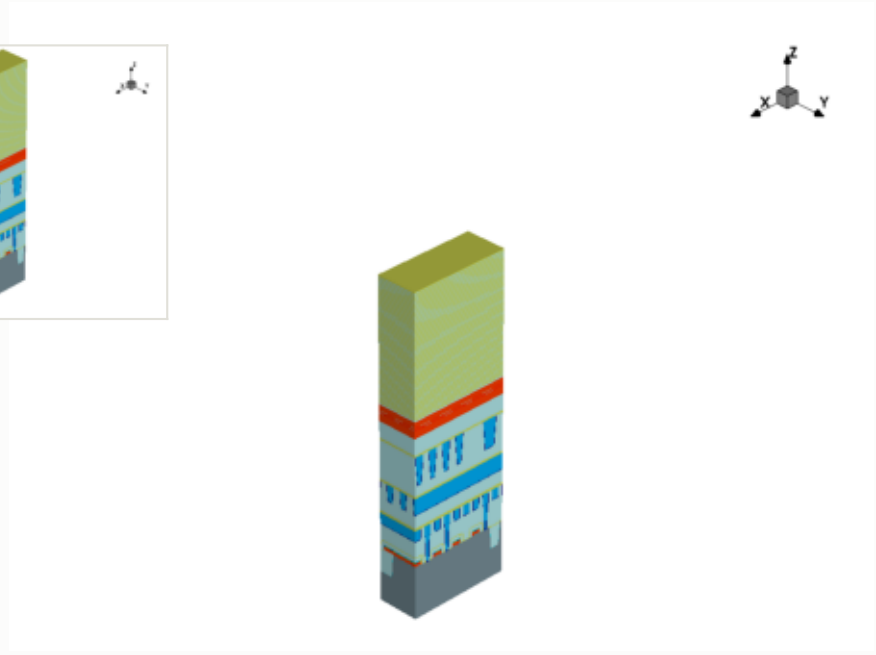
depo_anisotropic · Deposition_stack

0.654	248.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

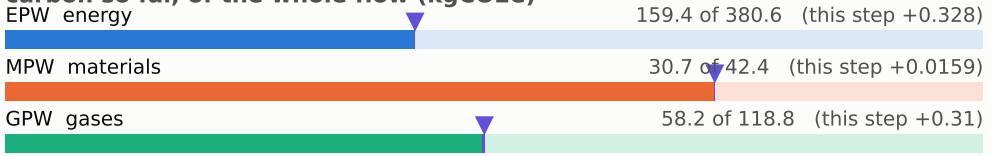




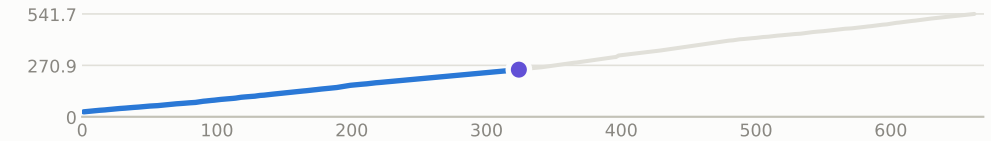
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

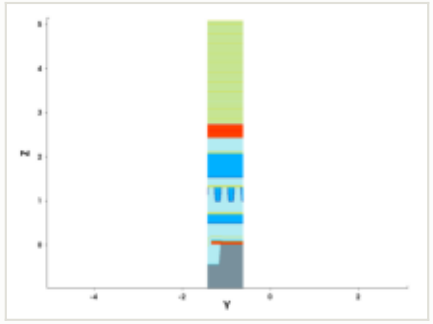
carbon so far, of the whole flow (kgCO2e)



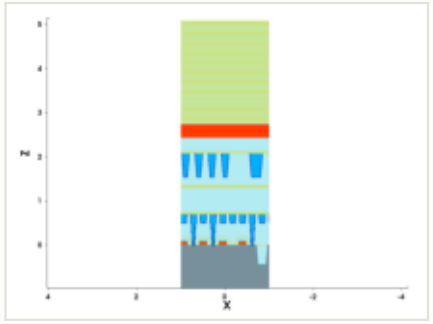
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



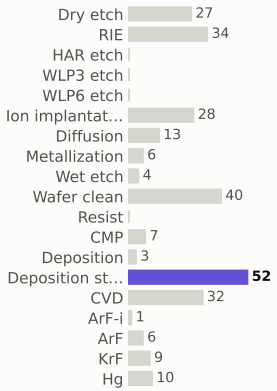
y cut at -1.032 µm



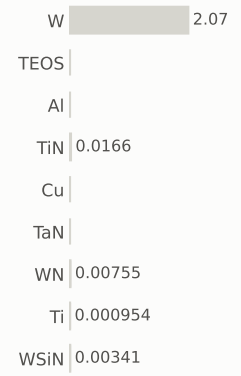
mask for this step

no mask for this step

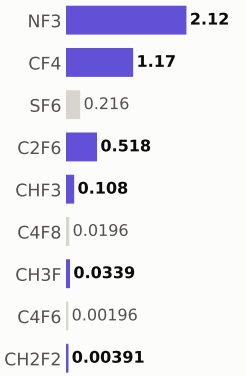
steps by process area



materials deposited (g)



gases emitted (g)



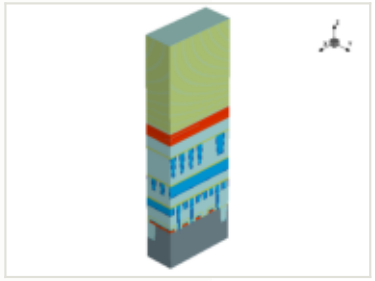
Step #3350: Deposition_stack

depo_anisotropic · Deposition_stack

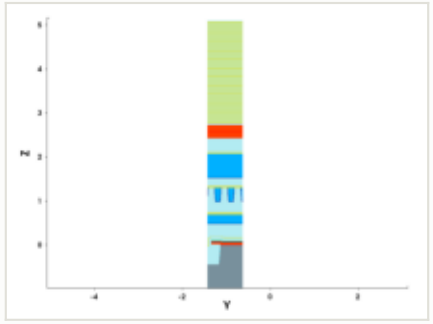
0.654	249.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

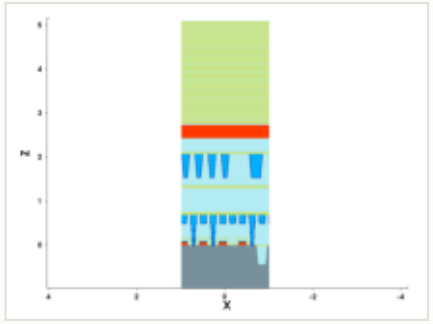
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



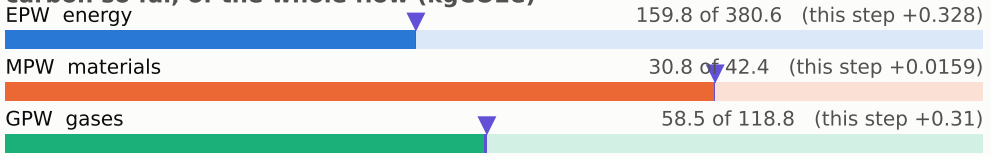
mask for this step

no mask for this step

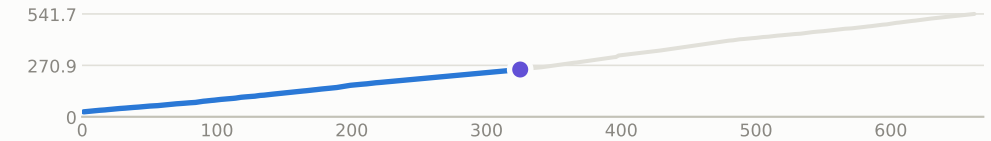
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

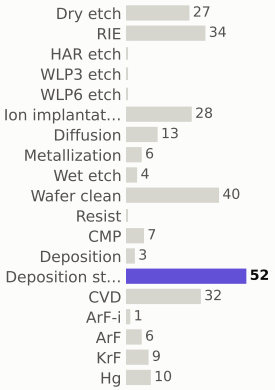
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

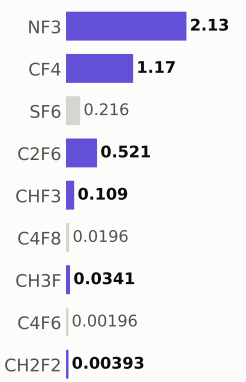


steps by process area



materials deposited (g)

gases emitted (g)



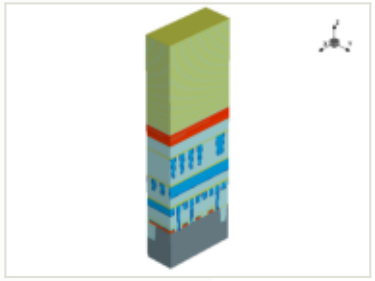
Step #3360: Deposition_stack

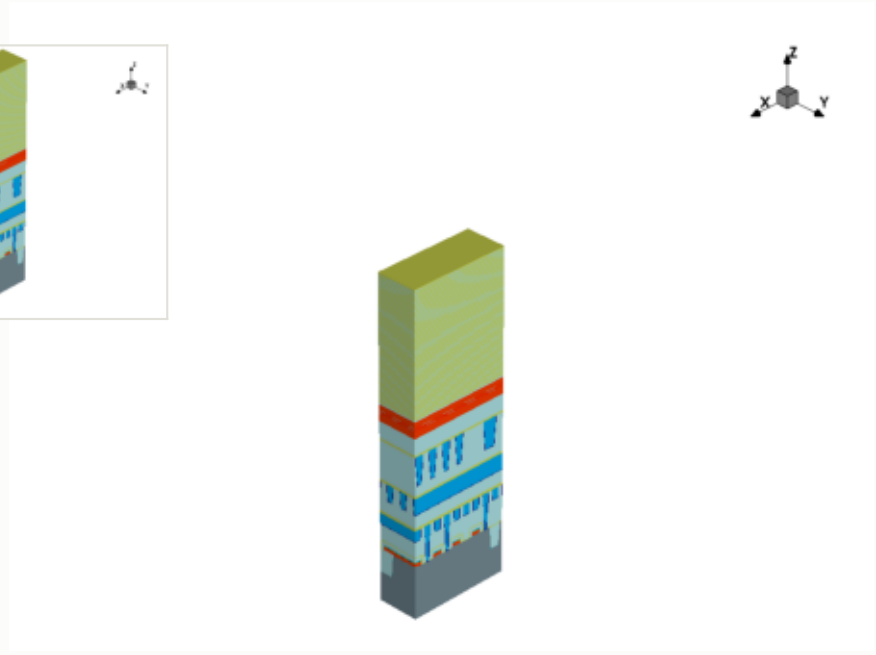
depo_anisotropic · Deposition_stack

0.654	249.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

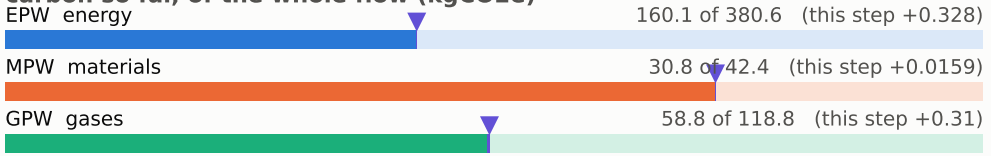




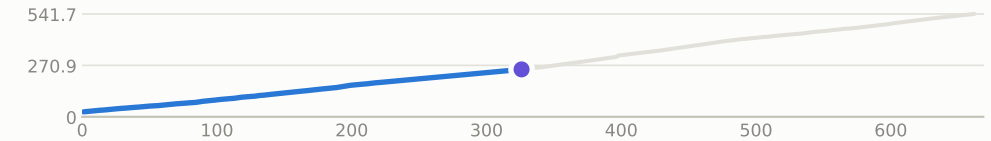
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

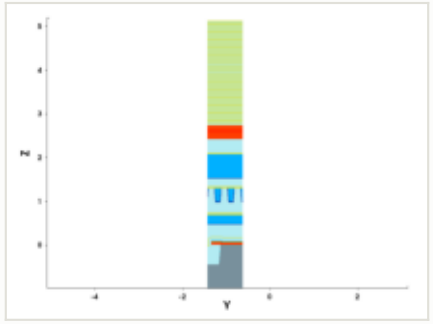
carbon so far, of the whole flow (kgCO2e)



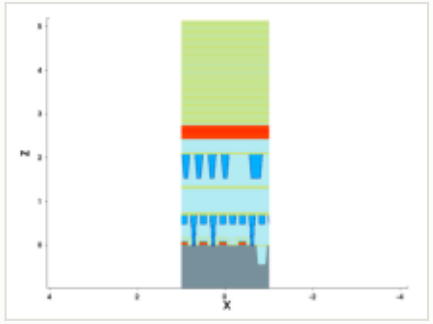
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



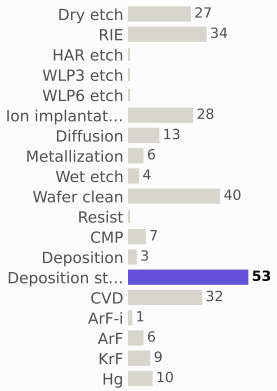
y cut at -1.032 µm



mask for this step

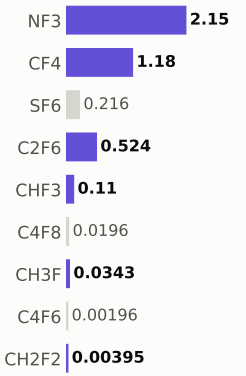
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



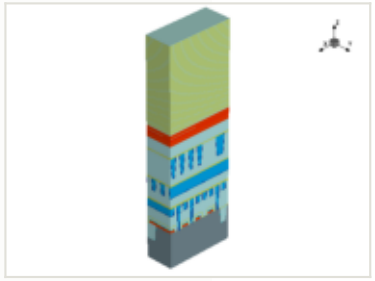
Step #3370: Deposition_stack

depo_anisotropic · Deposition_stack

0.654	250.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

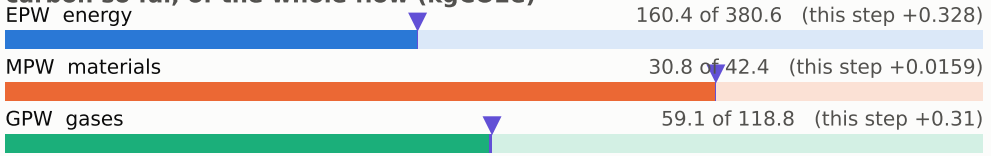
zoomed to fit



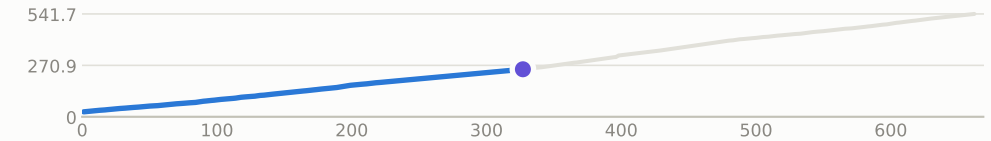
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

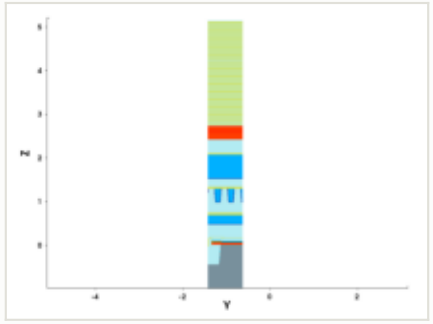
carbon so far, of the whole flow (kgCO2e)



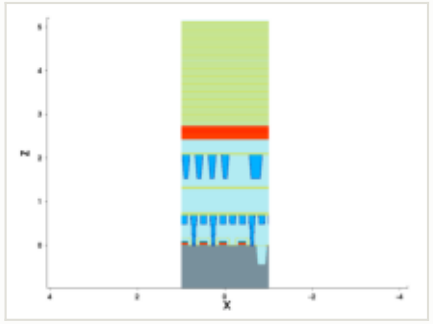
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



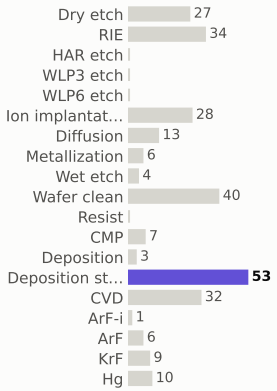
y cut at -1.032 µm



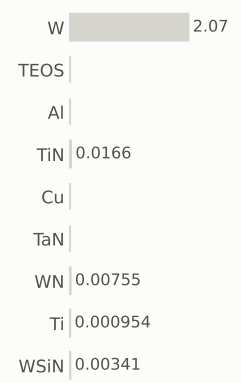
mask for this step

no mask for this step

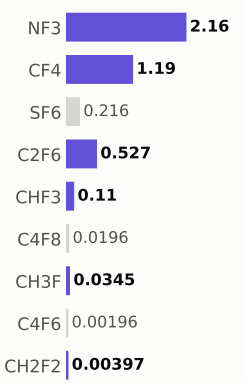
steps by process area



materials deposited (g)



gases emitted (g)



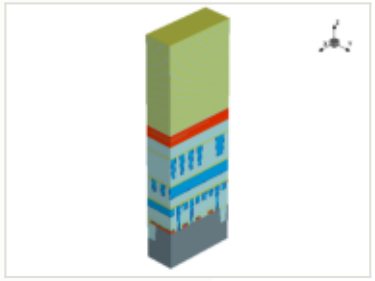
Step #3380: Deposition_stack

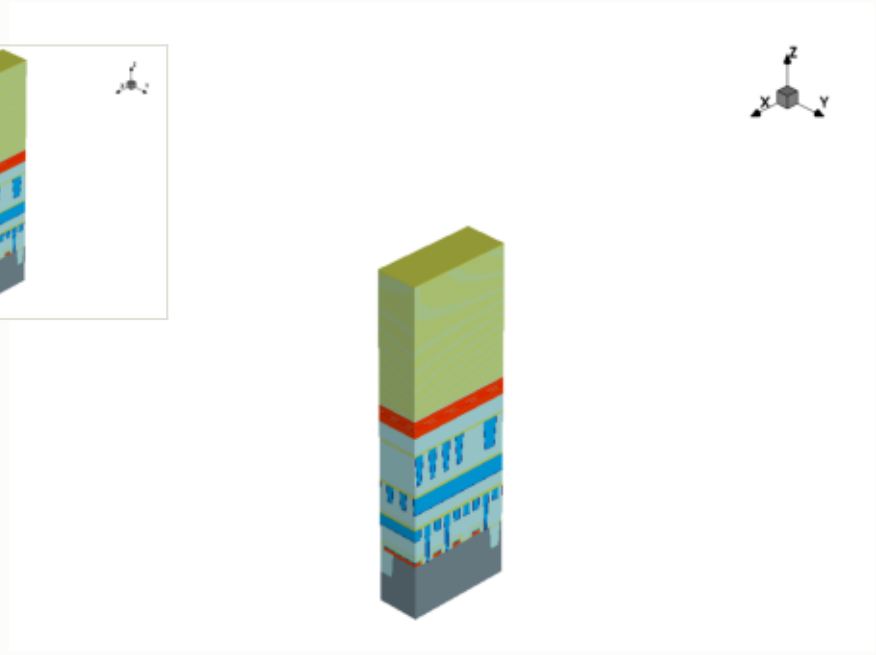
depo_anisotropic · Deposition_stack

0.654	251.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

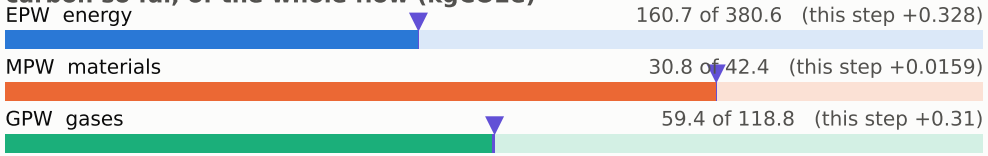




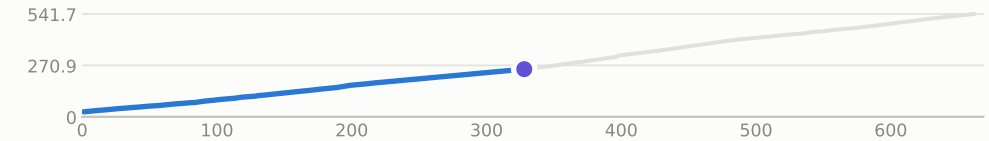
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.028 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

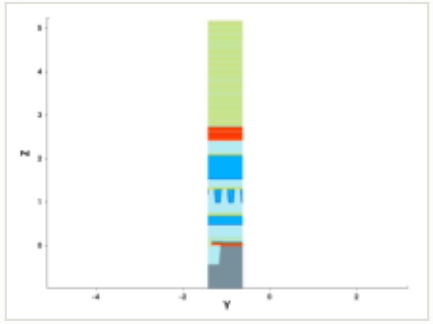
carbon so far, of the whole flow (kgCO2e)



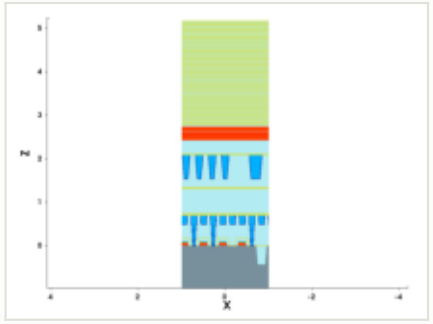
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



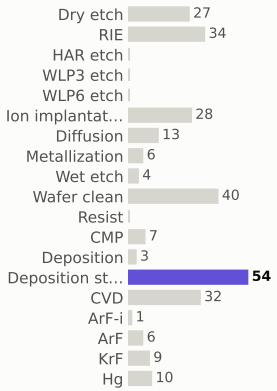
y cut at -1.032 µm



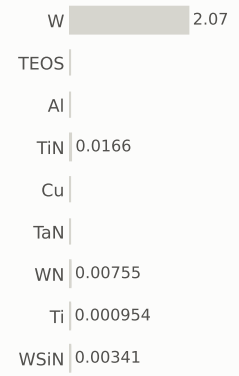
mask for this step

no mask for this step

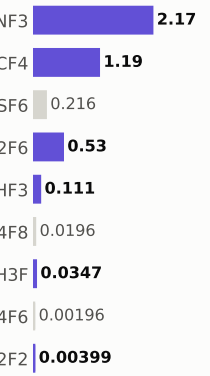
steps by process area



materials deposited (g)



gases emitted (g)



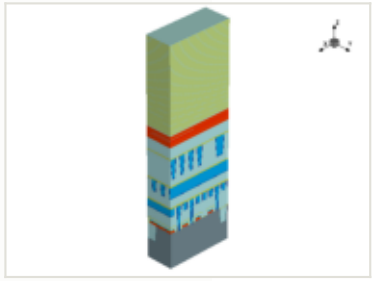
Step #3390: Deposition_stack

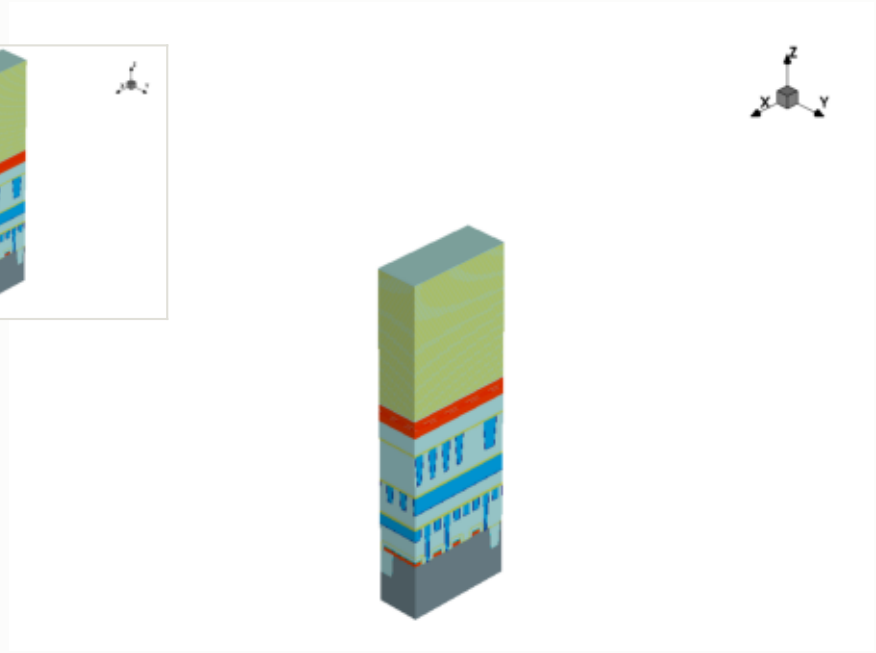
depo_anisotropic · Deposition_stack

0.654	251.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

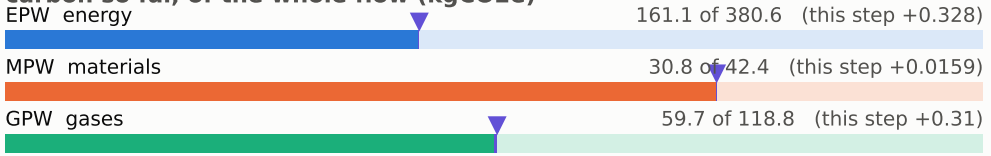




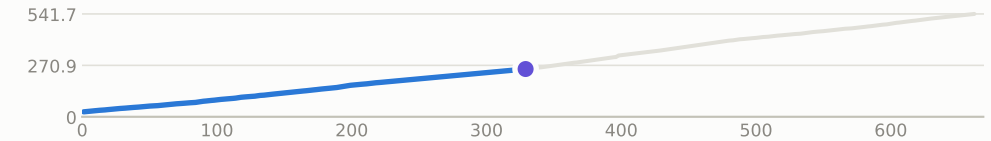
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.017 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

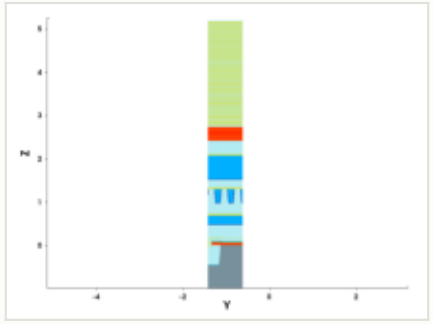
carbon so far, of the whole flow (kgCO2e)



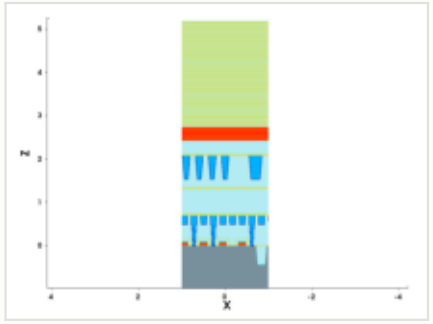
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



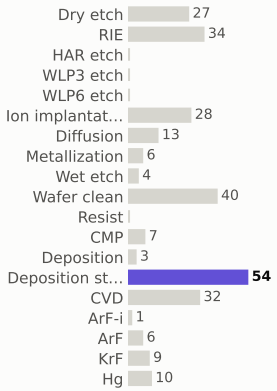
y cut at -1.032 µm



mask for this step

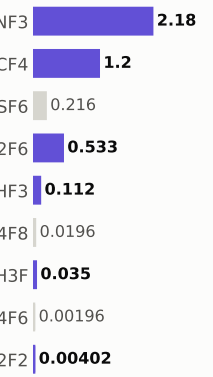
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



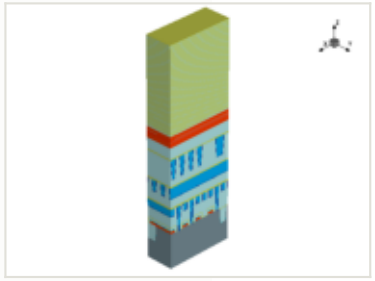
Step #3400: Deposition_stack

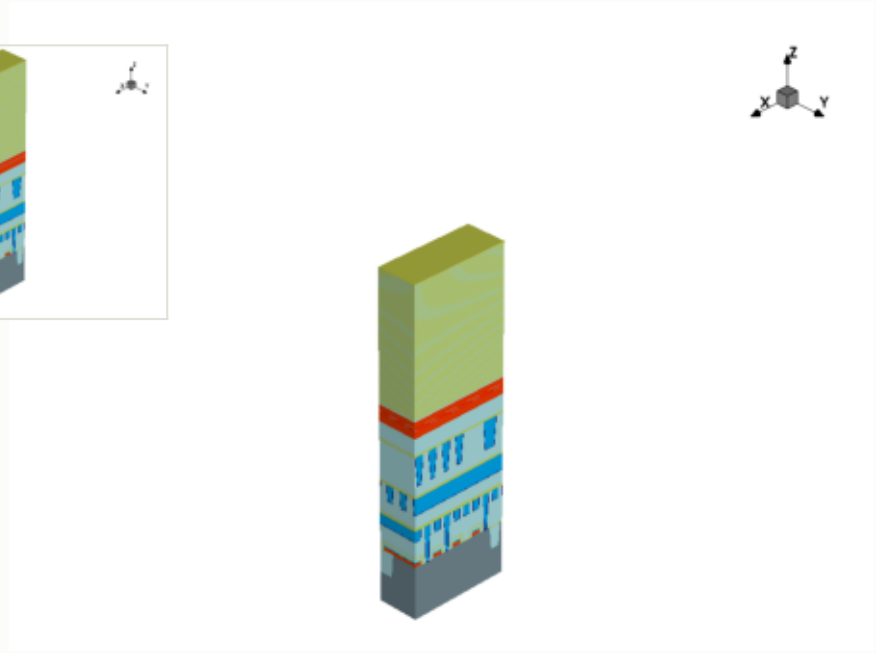
depo_anisotropic · Deposition_stack

0.654	252.3	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

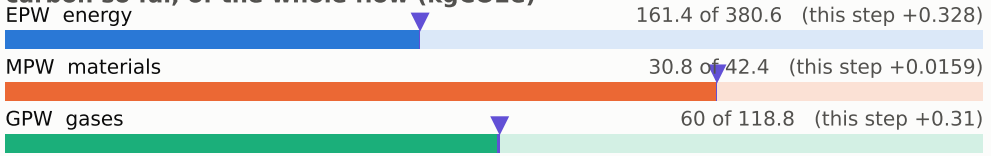




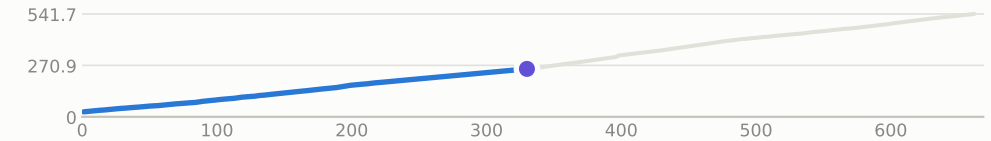
this step

process area	Deposition_stack	EPW	0.328 kgCO2e
material	—	EPW tool	0.234 kgCO2e
thickness	0.025 µm	EPW overhead	0.0937 kgCO2e
material GWP	—	MPW	0.0159 kgCO2e (gas 0.0159)
input gases	NF3 1.27 g	GPW	0.31 kgCO2e
energy	0.545 kWh + 0.218 overhead	embodied carbon	**0.654 kgCO2e**

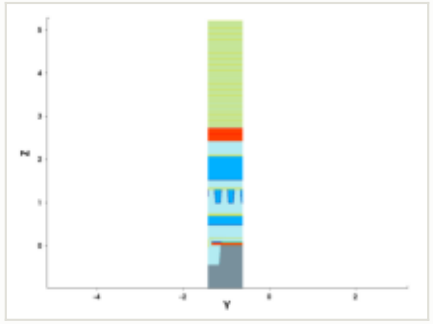
carbon so far, of the whole flow (kgCO2e)



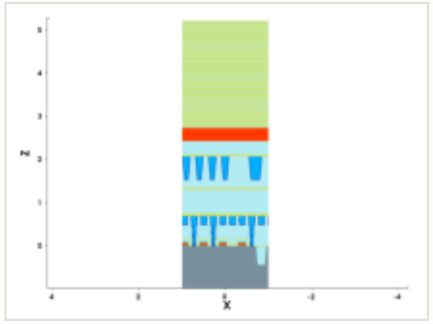
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



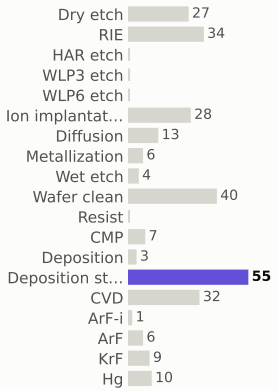
y cut at -1.032 µm



mask for this step

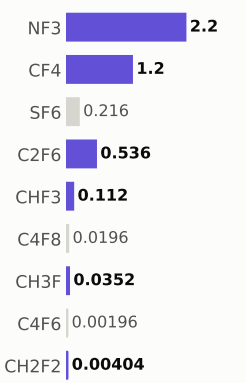
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



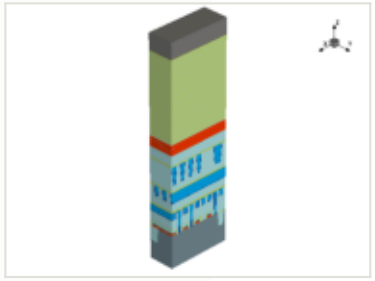
Step #3420: KrF

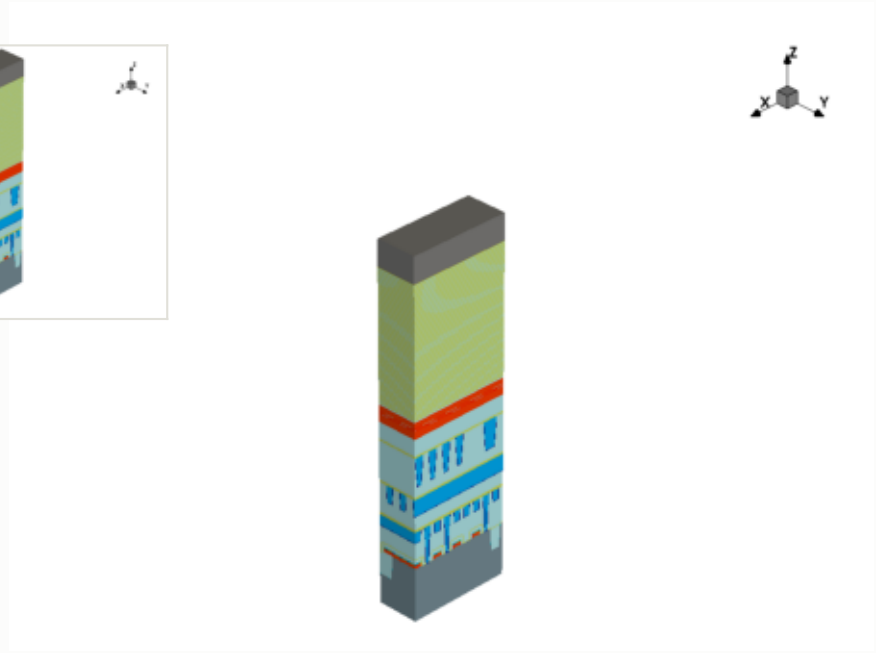
litho_patterning · KrF

0.199	252.5	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



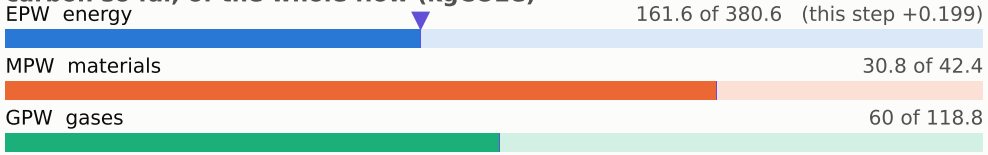


this step

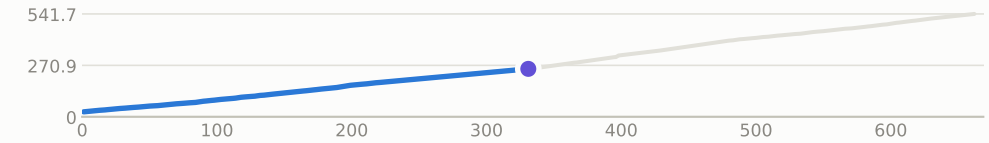
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

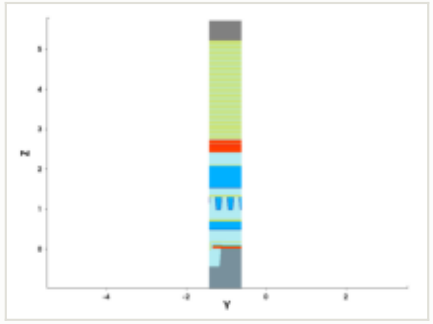
carbon so far, of the whole flow (kgCO2e)



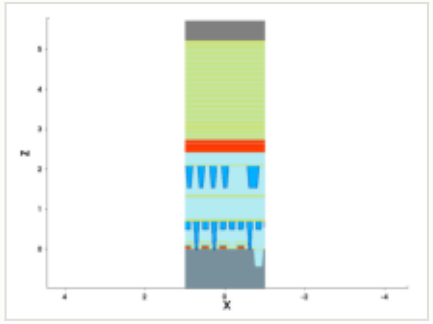
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

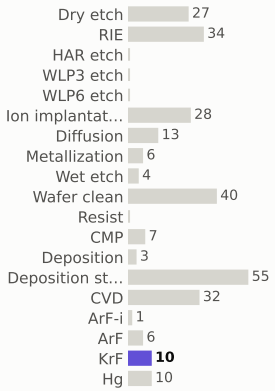


y cut at -1.032 µm



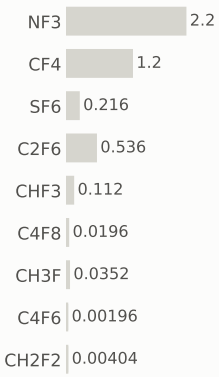
mask: layer 28:0, positive (OASIS)

steps by process area



materials deposited (g)

gases emitted (g)



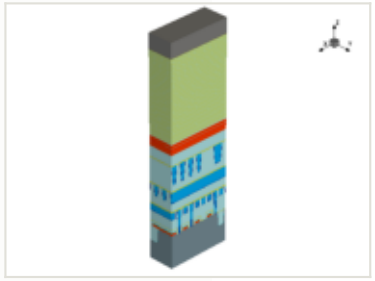
Step #3430: RIE

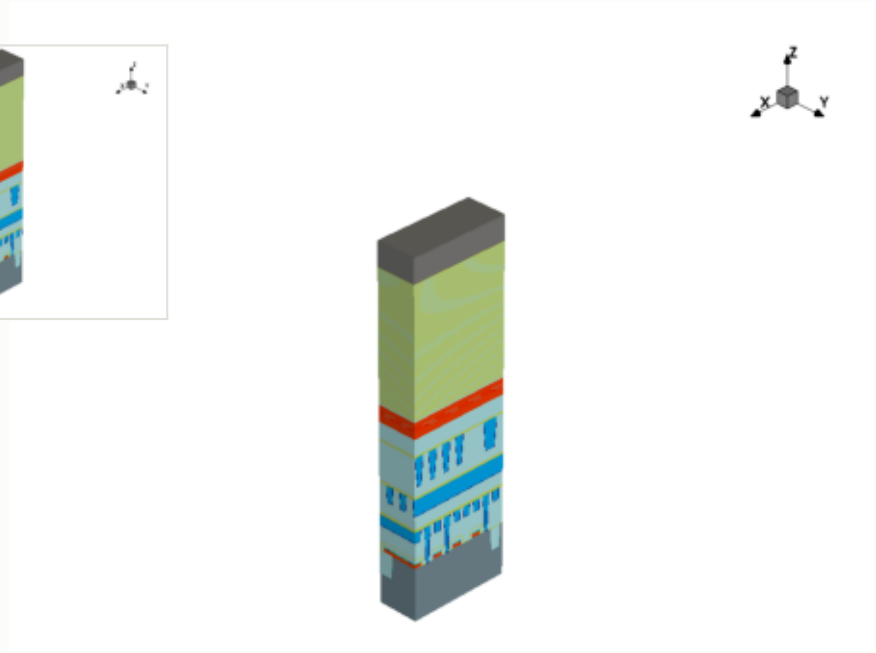
etch_anisotropic · RIE

1.04	253.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



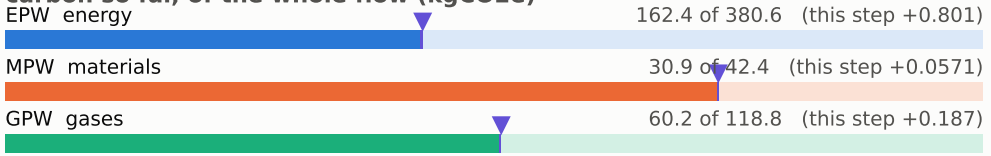


this step

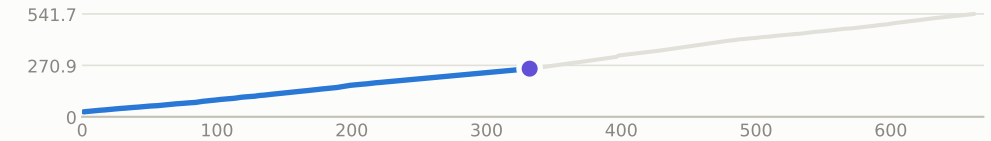
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

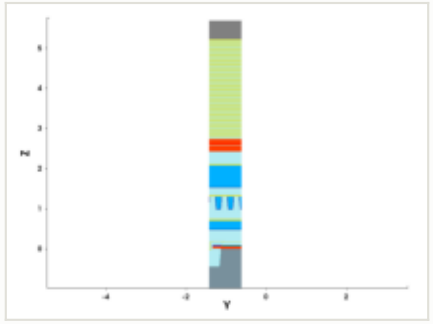
carbon so far, of the whole flow (kgCO2e)



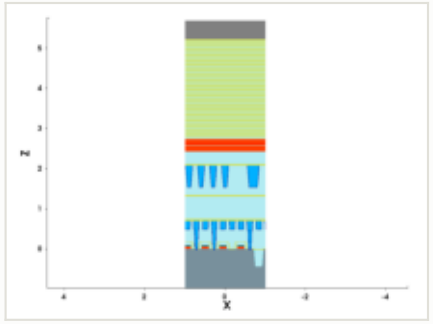
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



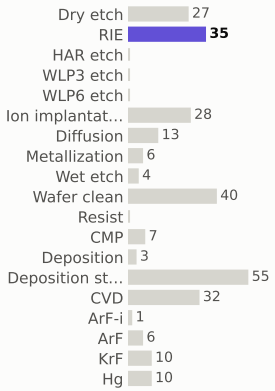
y cut at -1.032 µm



mask for this step

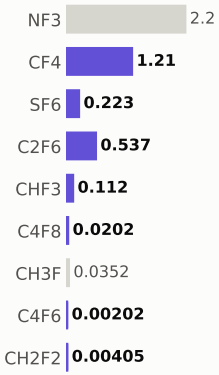
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



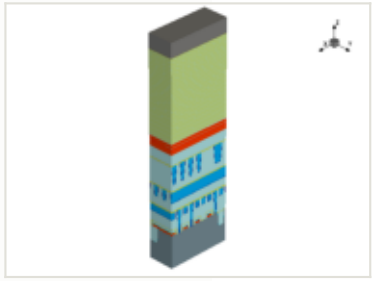
Step #3440: RIE

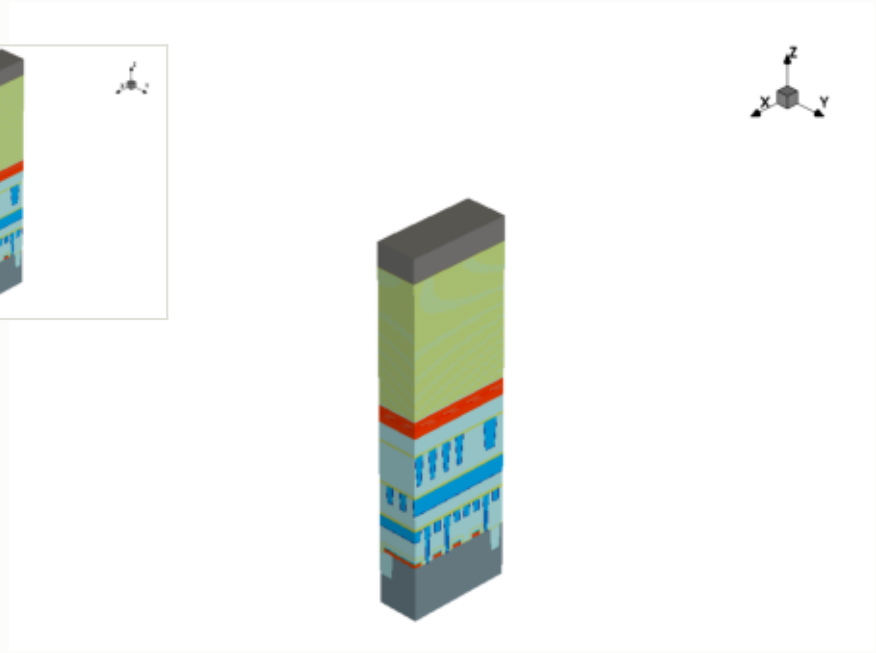
etch_anisotropic · RIE

1.04	254.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



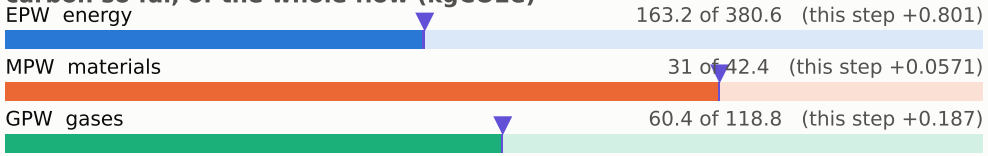


this step

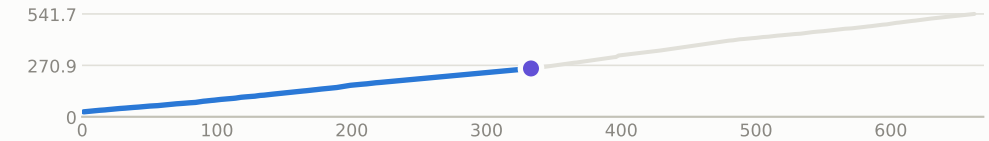
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

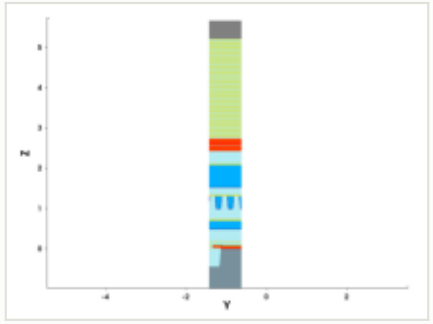
carbon so far, of the whole flow (kgCO2e)



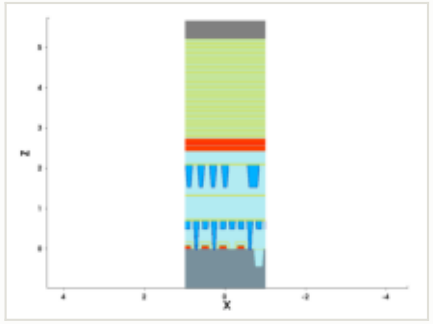
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



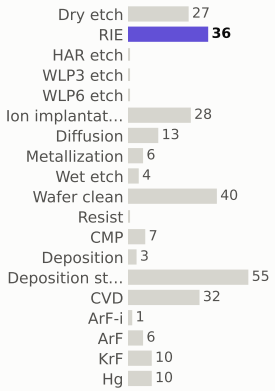
y cut at -1.032 µm



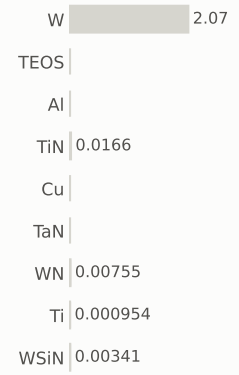
mask for this step

no mask for this step

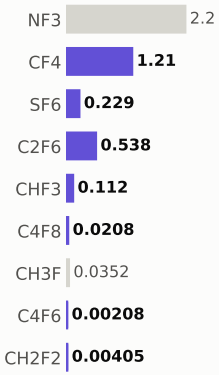
steps by process area



materials deposited (g)



gases emitted (g)



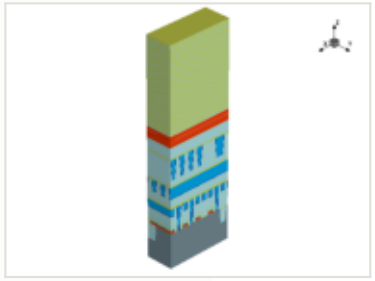
Step #3450: Dry_etch

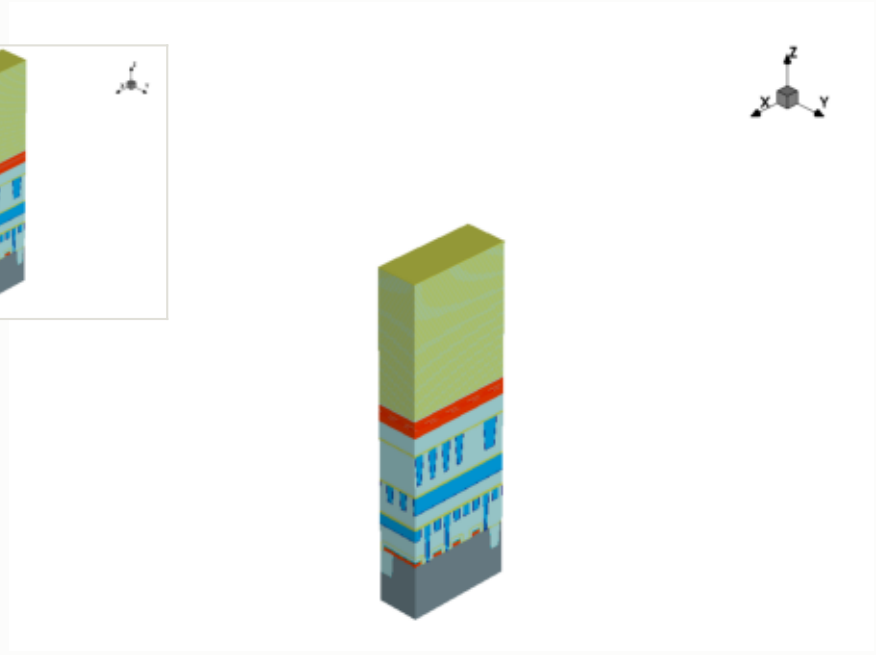
etch_strip · Dry_etch

0.801	255.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

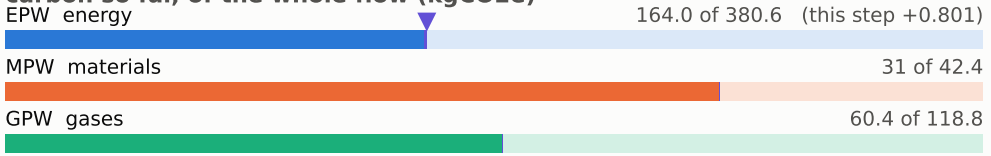




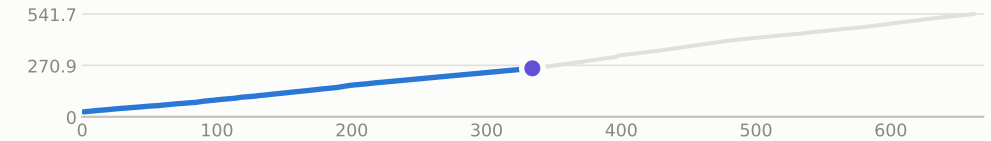
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

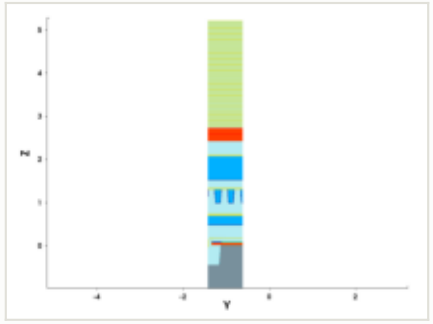
carbon so far, of the whole flow (kgCO2e)



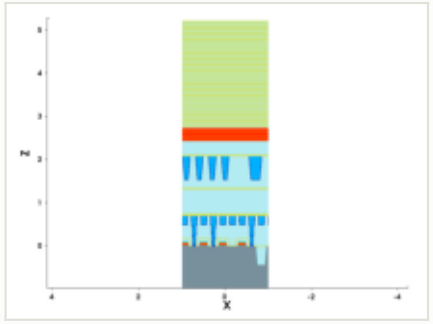
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



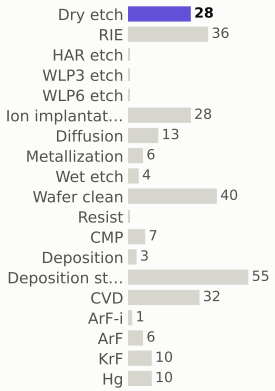
y cut at -1.032 µm



mask for this step

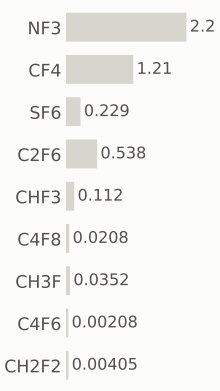
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



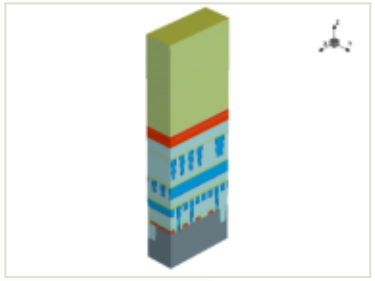
Step #3460: Wafer_clean

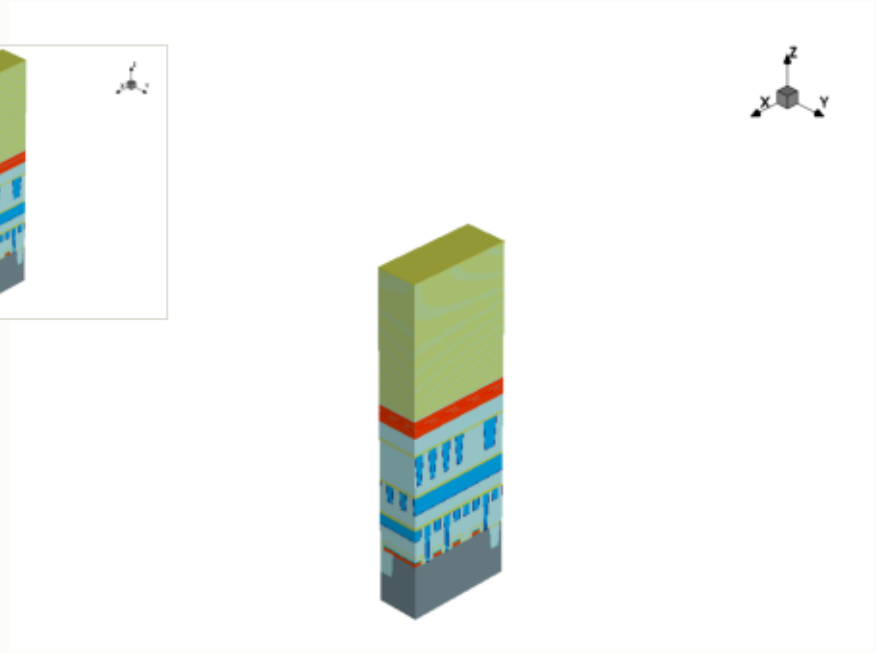
clean · Wafer_clean

0.169	255.5	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

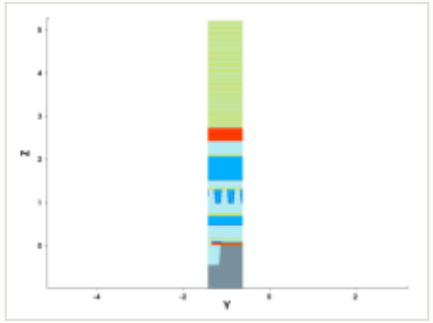
3D structure after this step

zoomed to fit

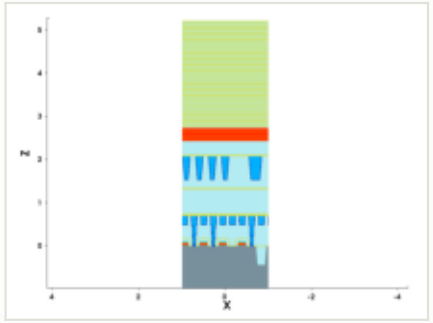




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

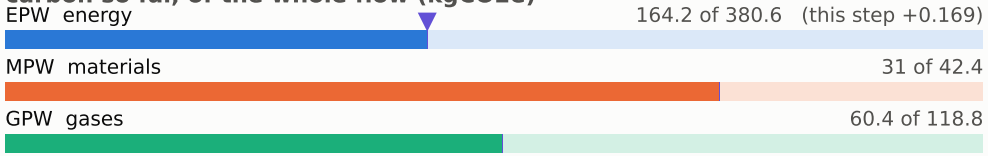
no mask for this step

this step

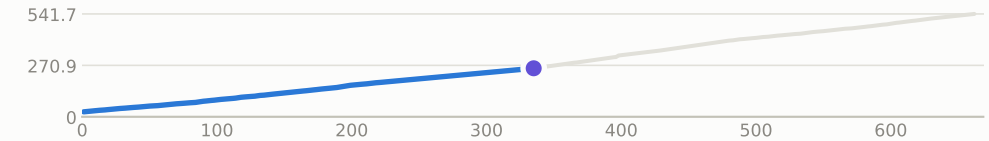
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

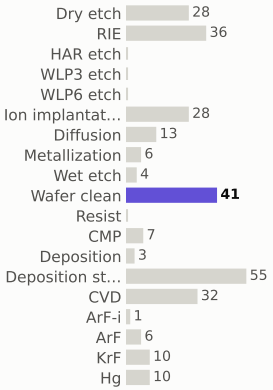
carbon so far, of the whole flow (kgCO2e)



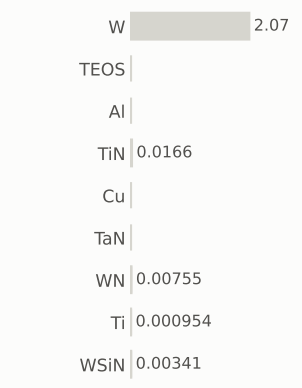
embodied carbon up to each step (kgCO2e per wafer)



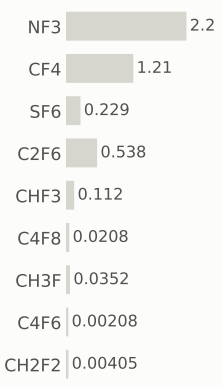
steps by process area



materials deposited (g)



gases emitted (g)



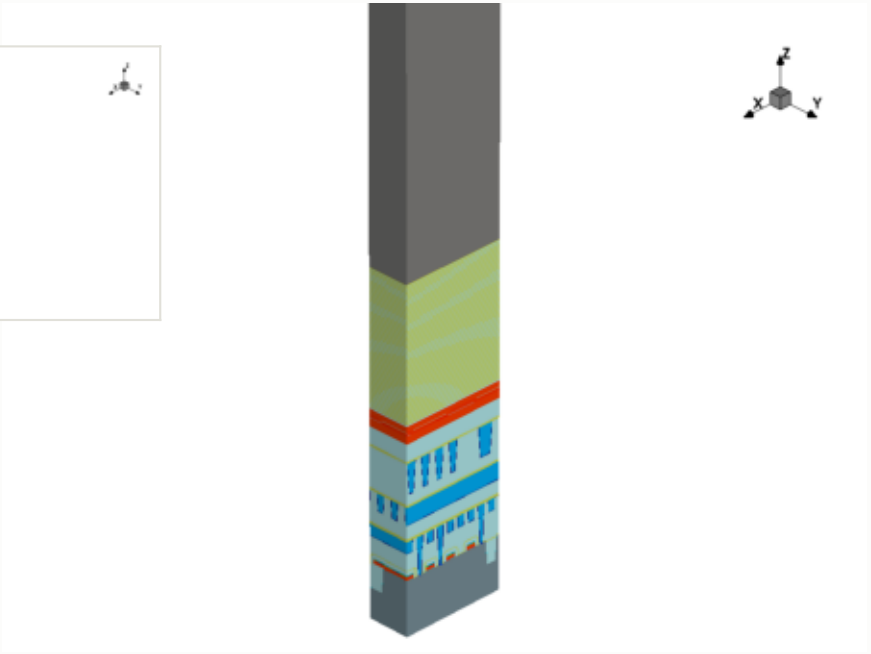
Step #3470: KrF

litho_patterning · KrF

0.199	255.7	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

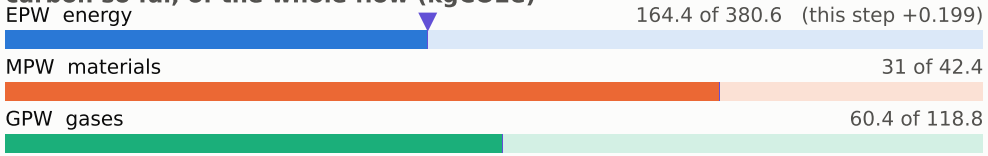


this step

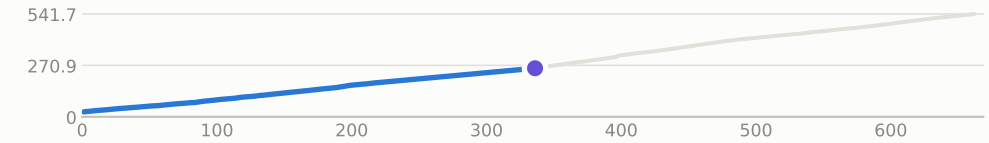
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

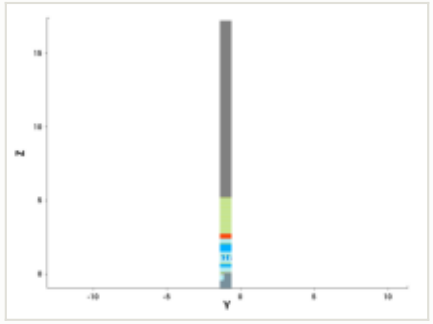
carbon so far, of the whole flow (kgCO2e)



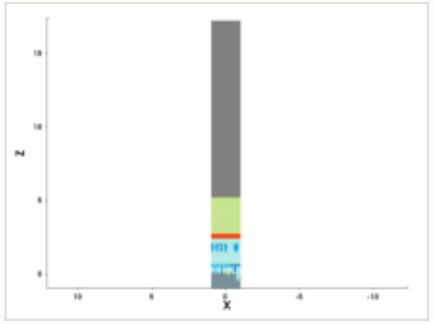
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

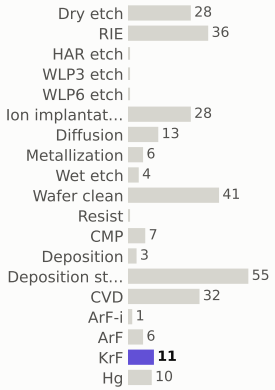


y cut at -1.032 µm

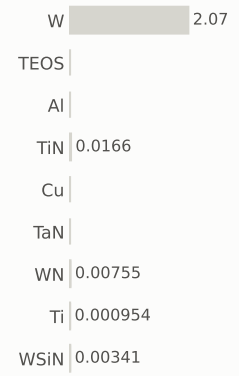


mask: layer 29:0, positive (OASIS)

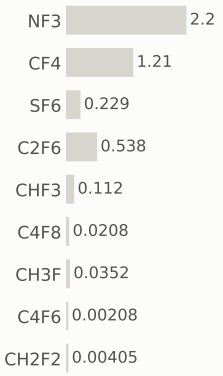
steps by process area



materials deposited (g)



gases emitted (g)



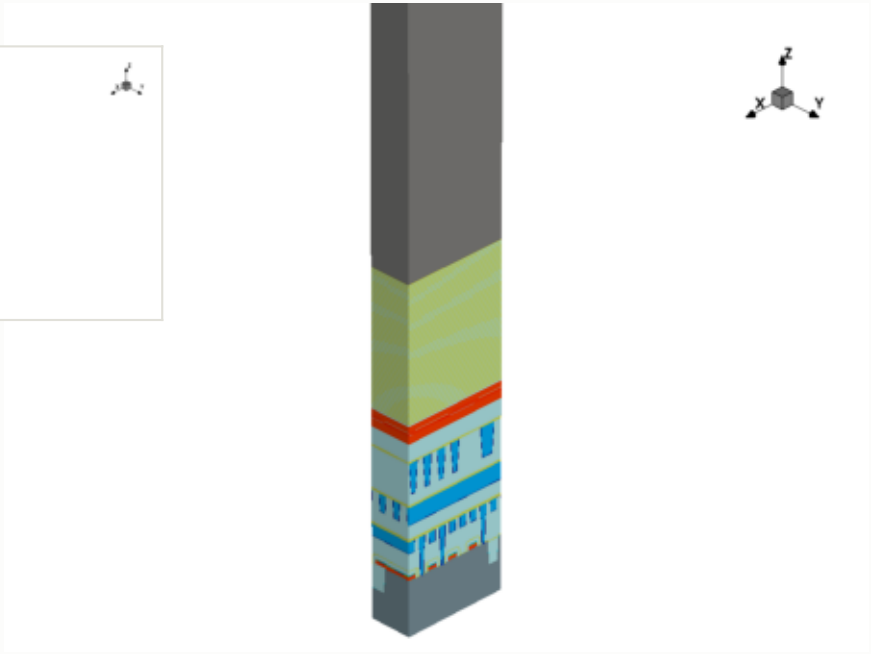
Step #3480: RIE

etch_anisotropic · RIE

1.04	256.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

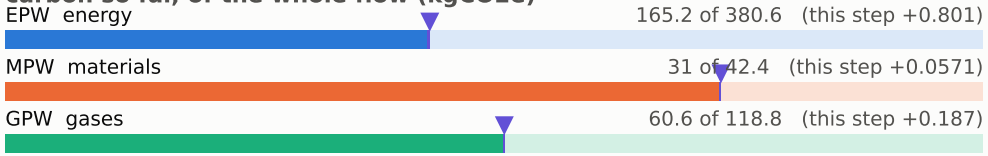


this step

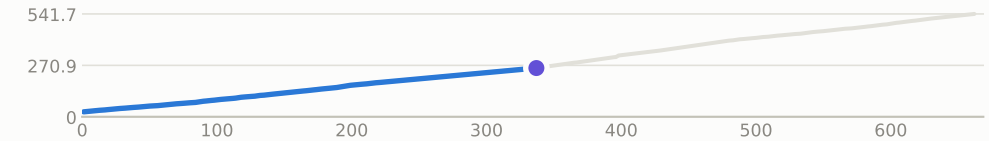
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

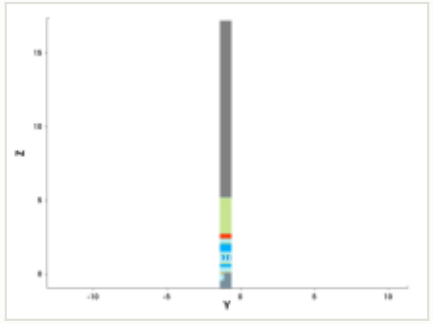
carbon so far, of the whole flow (kgCO2e)



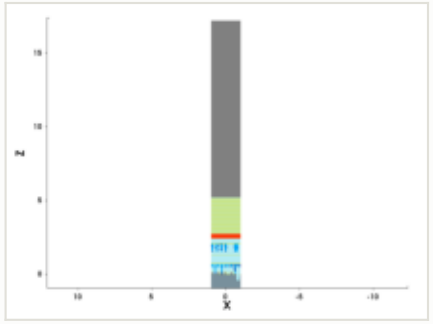
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



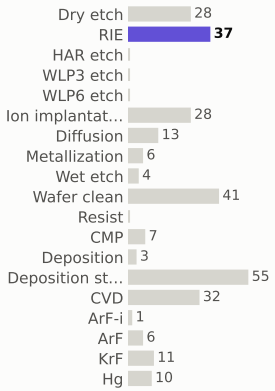
y cut at -1.032 µm



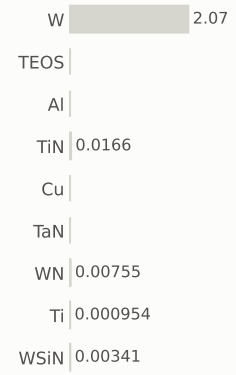
mask for this step

no mask for this step

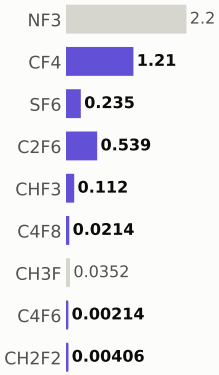
steps by process area



materials deposited (g)



gases emitted (g)



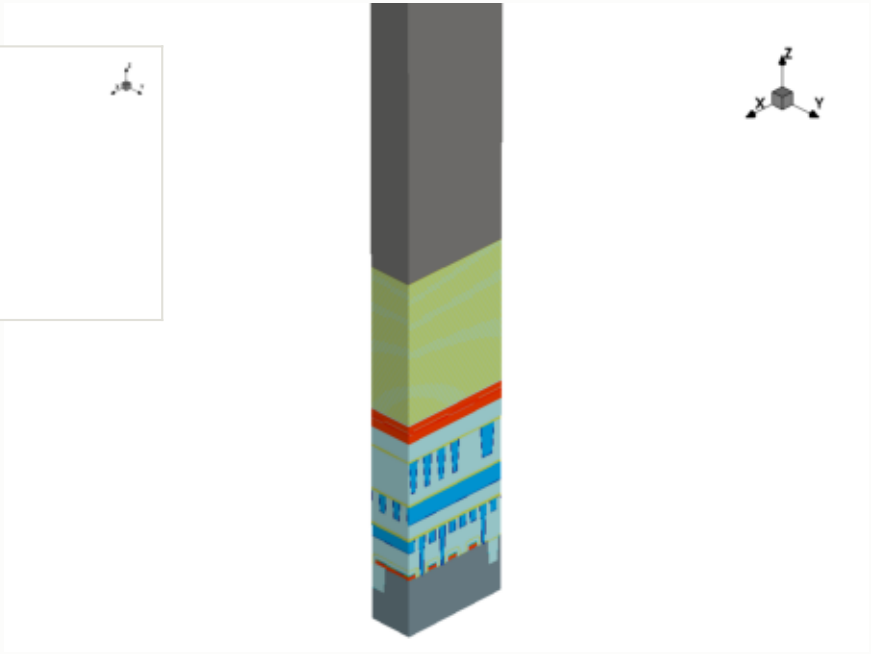
Step #3490: RIE

etch_anisotropic · RIE

1.04	257.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

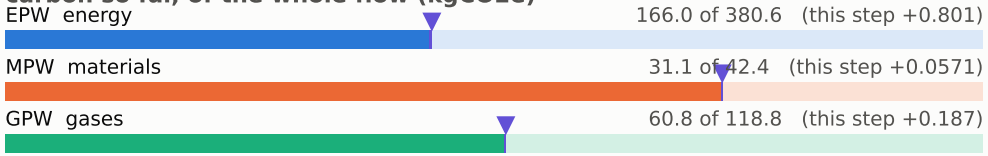


this step

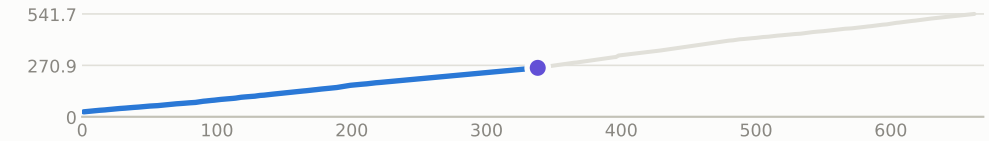
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

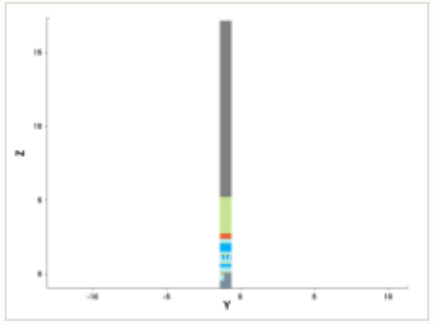
carbon so far, of the whole flow (kgCO2e)



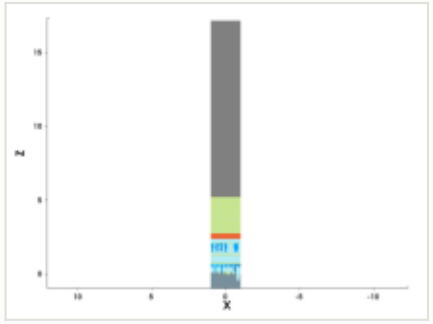
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



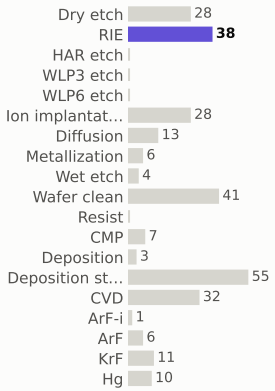
y cut at -1.032 µm



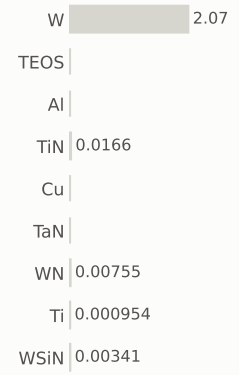
mask for this step

no mask for this step

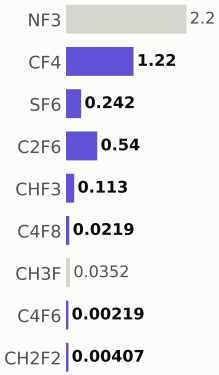
steps by process area



materials deposited (g)



gases emitted (g)



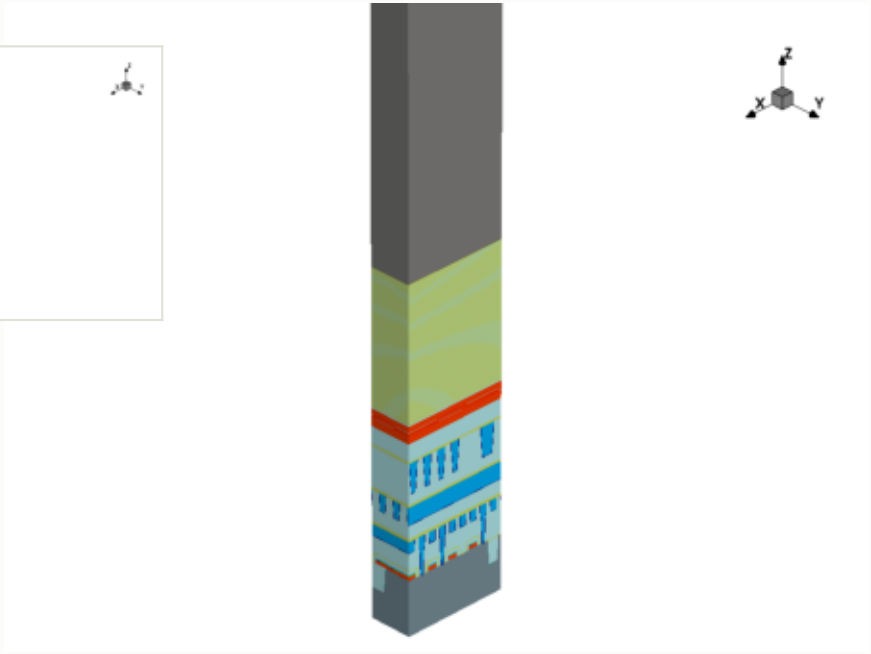
Step #3500: RIE

etch_anisotropic · RIE

1.04	258.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

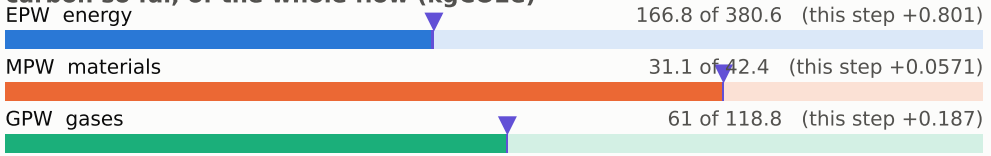


this step

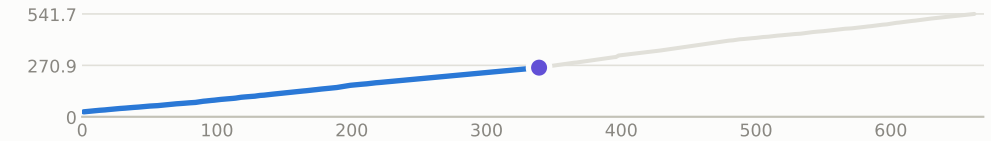
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

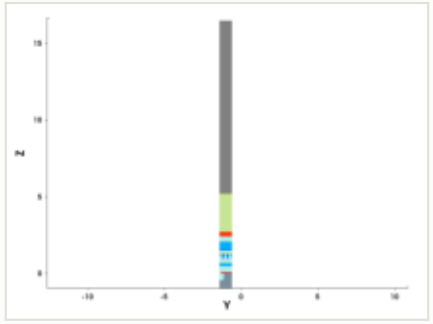
carbon so far, of the whole flow (kgCO2e)



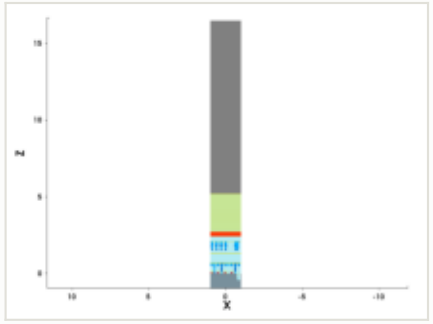
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



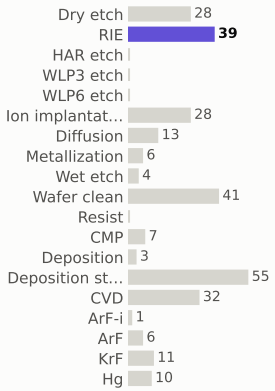
y cut at -1.032 µm



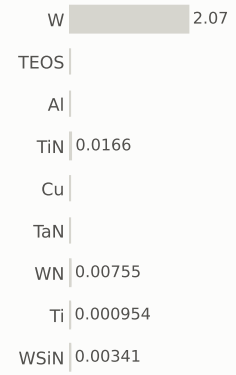
mask for this step

no mask for this step

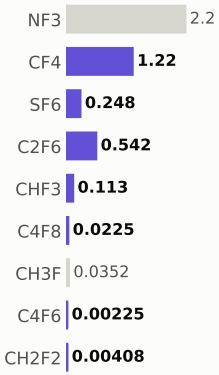
steps by process area



materials deposited (g)



gases emitted (g)



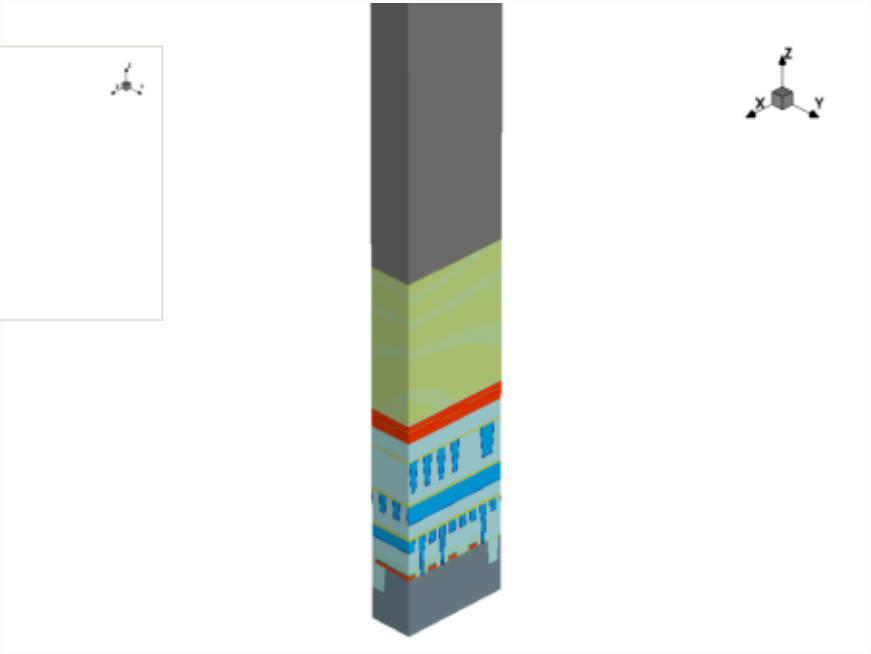
Step #3510: RIE

etch_anisotropic · RIE

1.04	259.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

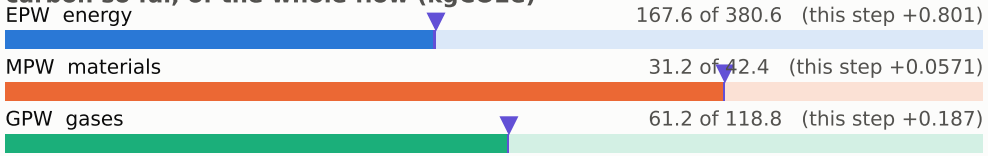


this step

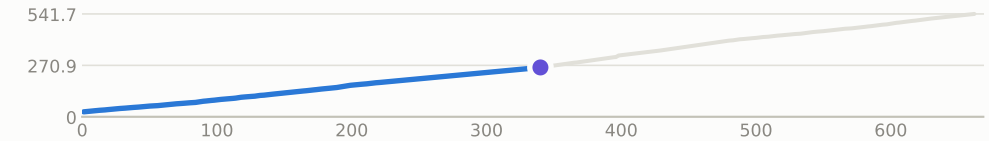
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

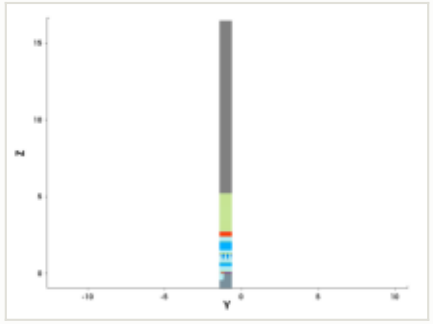
carbon so far, of the whole flow (kgCO2e)



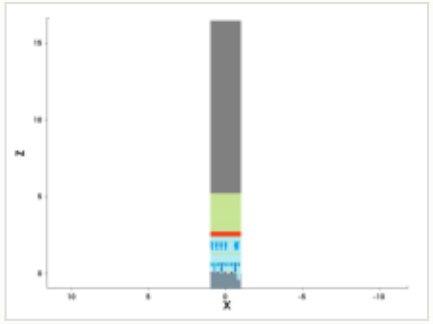
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



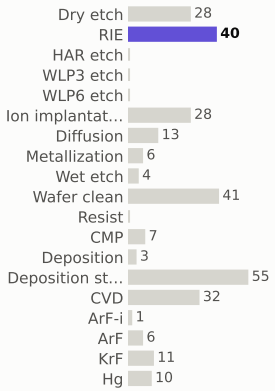
y cut at -1.032 µm



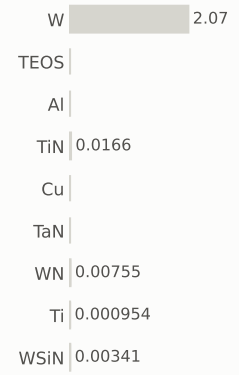
mask for this step

no mask for this step

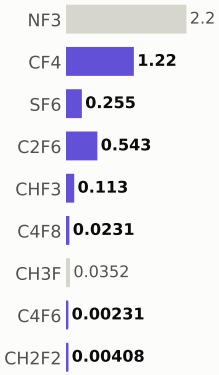
steps by process area



materials deposited (g)



gases emitted (g)



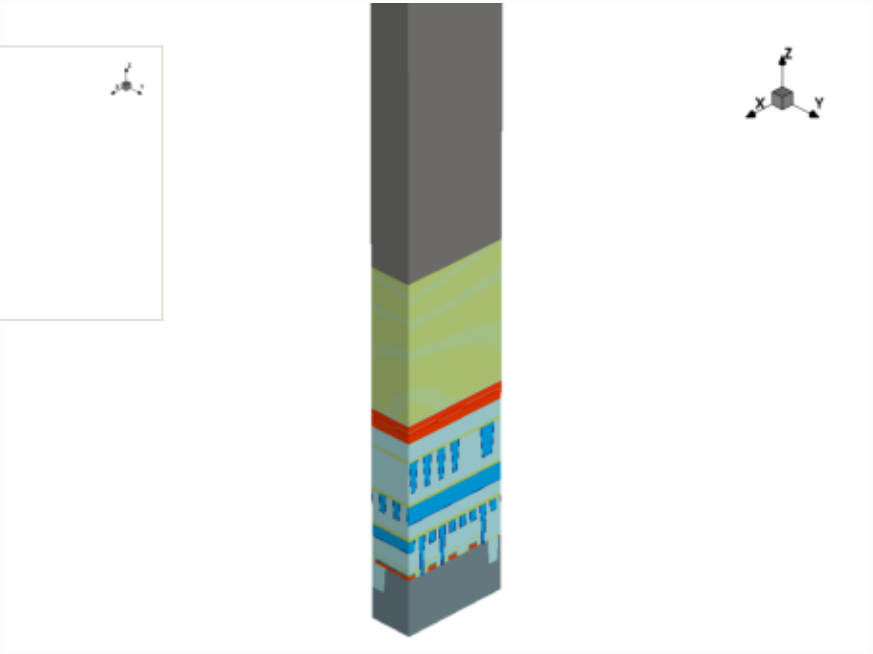
Step #3520: RIE

etch_anisotropic · RIE

1.04	261.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

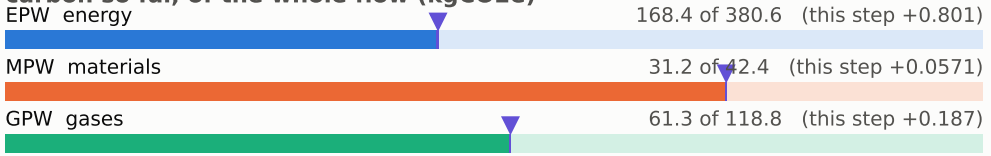


this step

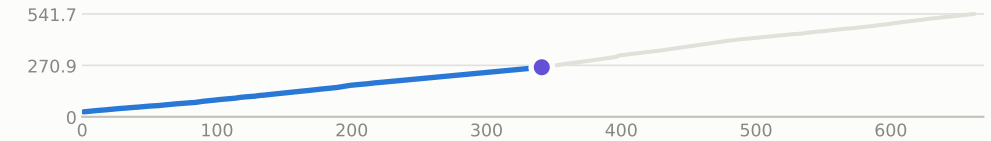
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

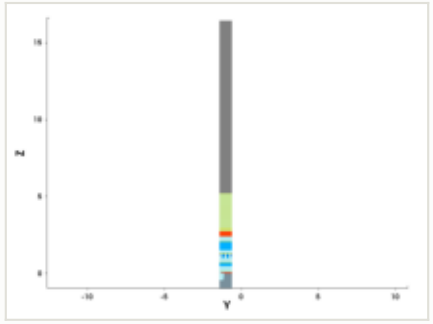
carbon so far, of the whole flow (kgCO2e)



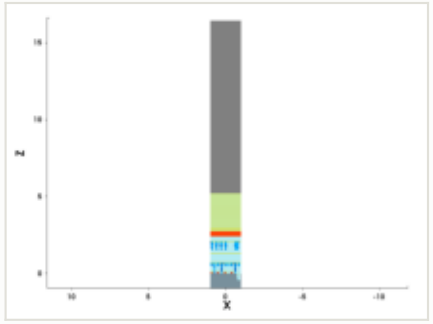
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



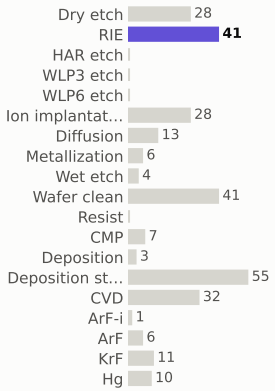
y cut at -1.032 µm



mask for this step

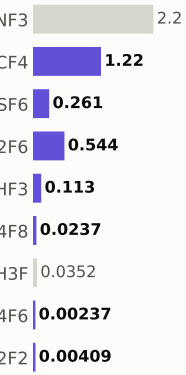
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



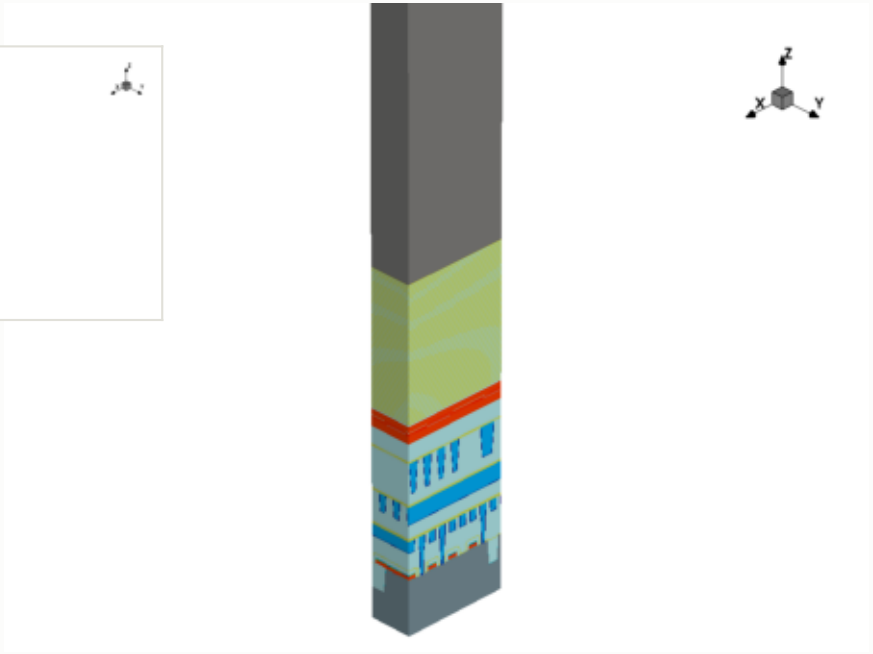
Step #3530: RIE

etch_anisotropic · RIE

1.04	262.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

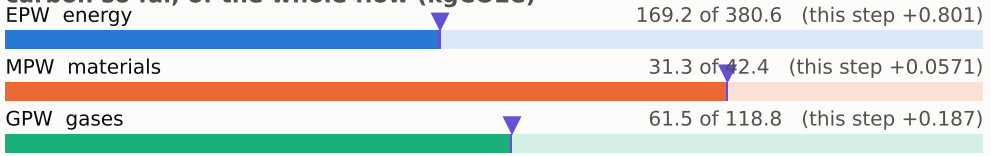


this step

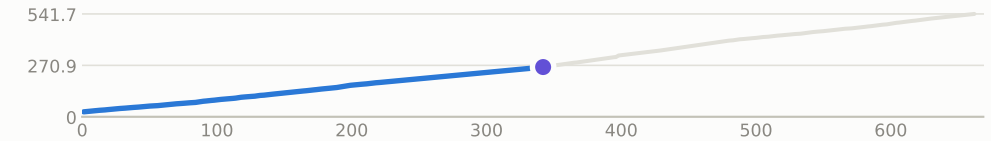
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

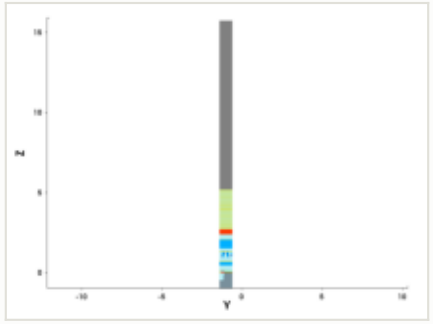
carbon so far, of the whole flow (kgCO2e)



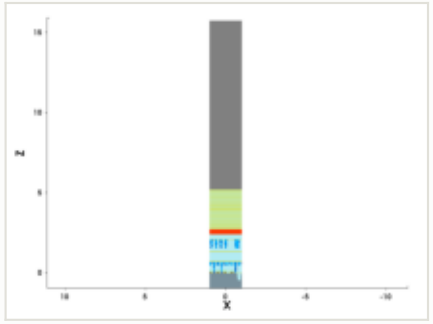
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



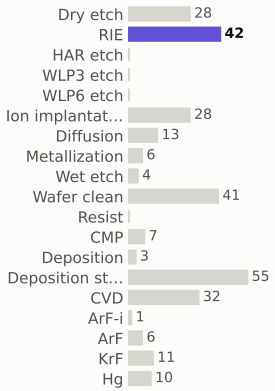
y cut at -1.032 µm



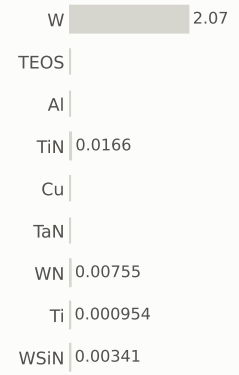
mask for this step

no mask for this step

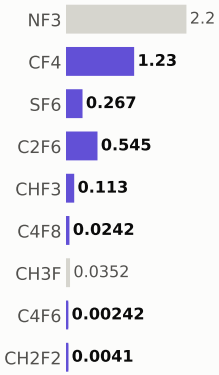
steps by process area



materials deposited (g)



gases emitted (g)



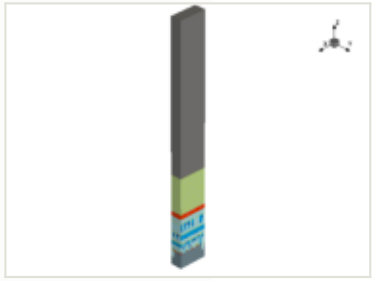
Step #3540: RIE

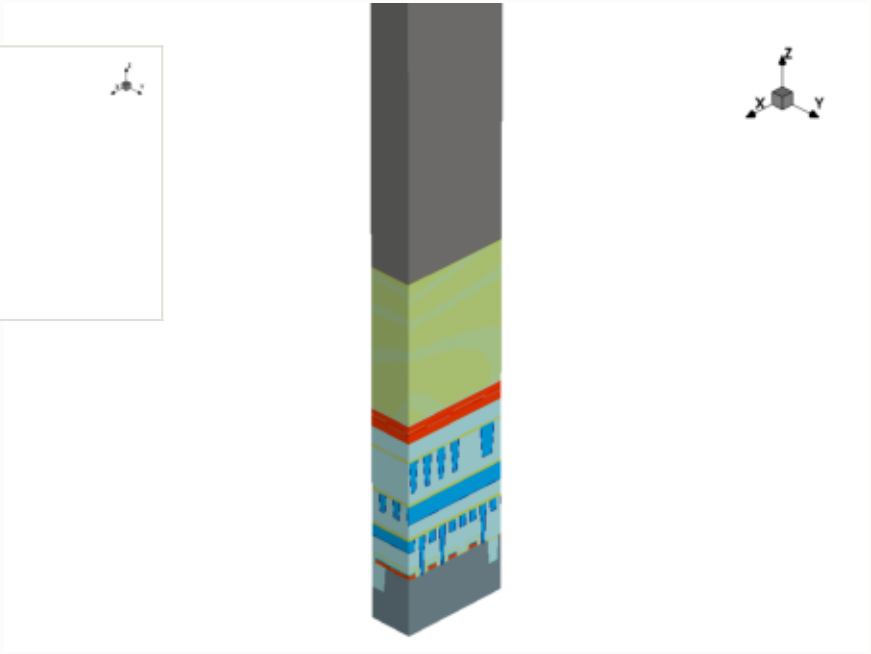
etch_anisotropic · RIE

1.04	263.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



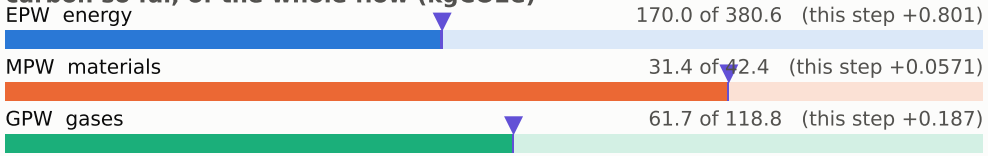


this step

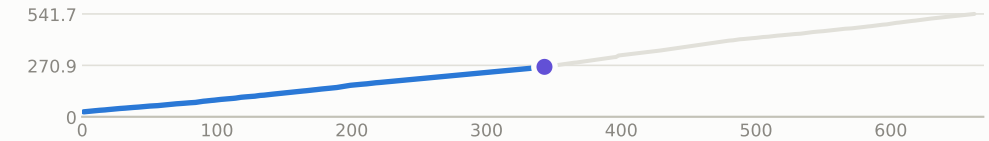
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

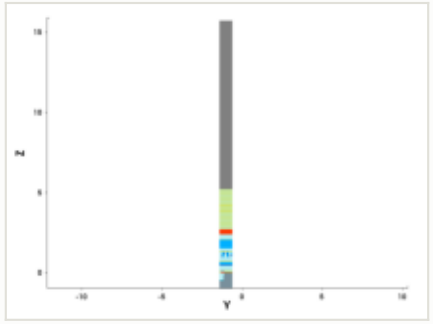
carbon so far, of the whole flow (kgCO2e)



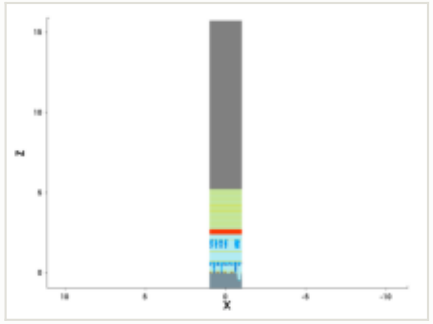
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



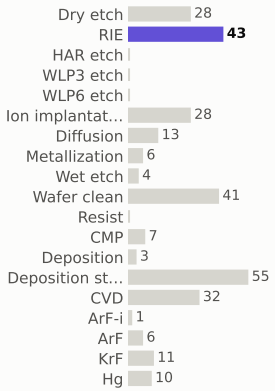
y cut at -1.032 µm



mask for this step

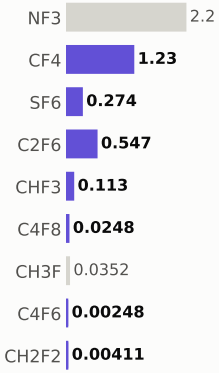
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



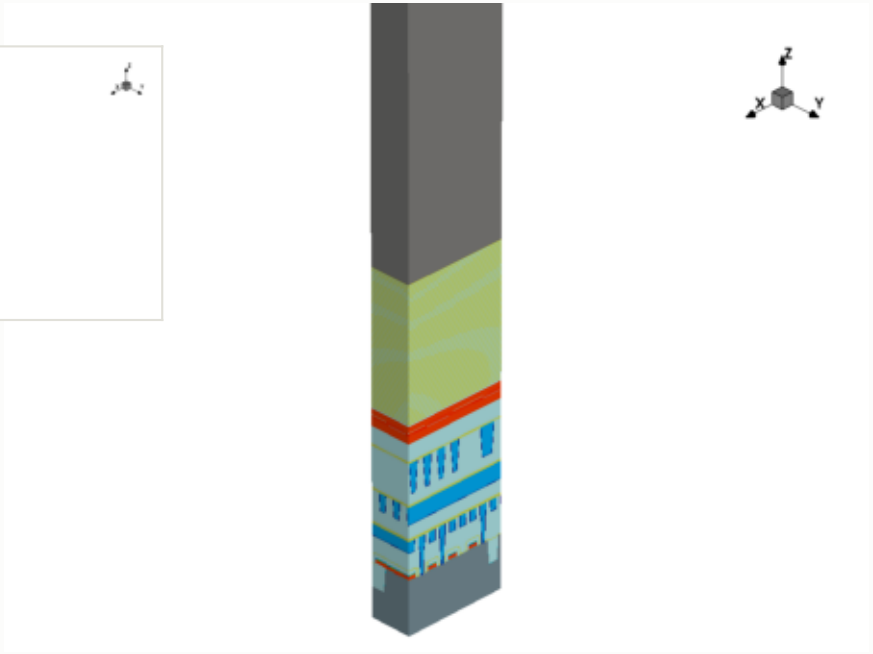
Step #3550: RIE

etch_anisotropic · RIE

1.04	264.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

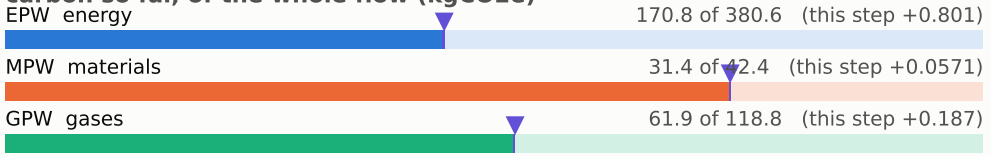


this step

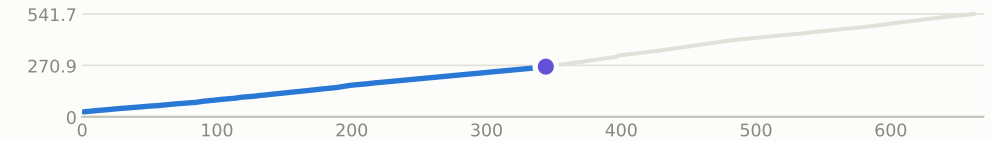
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

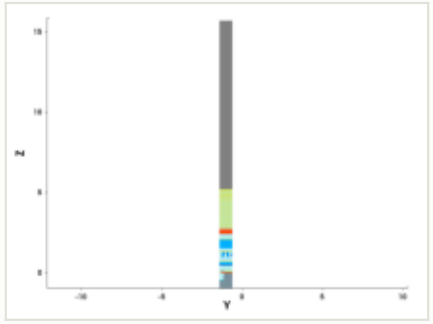
carbon so far, of the whole flow (kgCO2e)



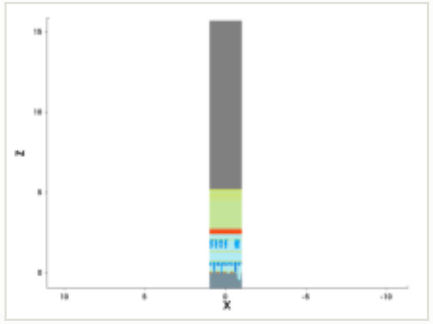
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



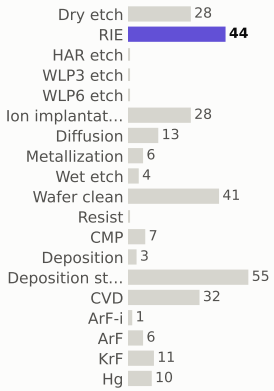
y cut at -1.032 µm



mask for this step

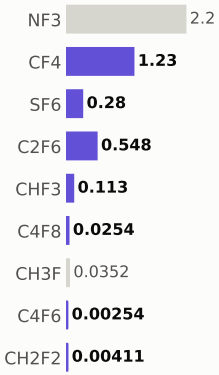
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



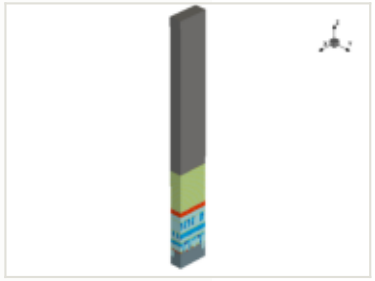
Step #3560: RIE

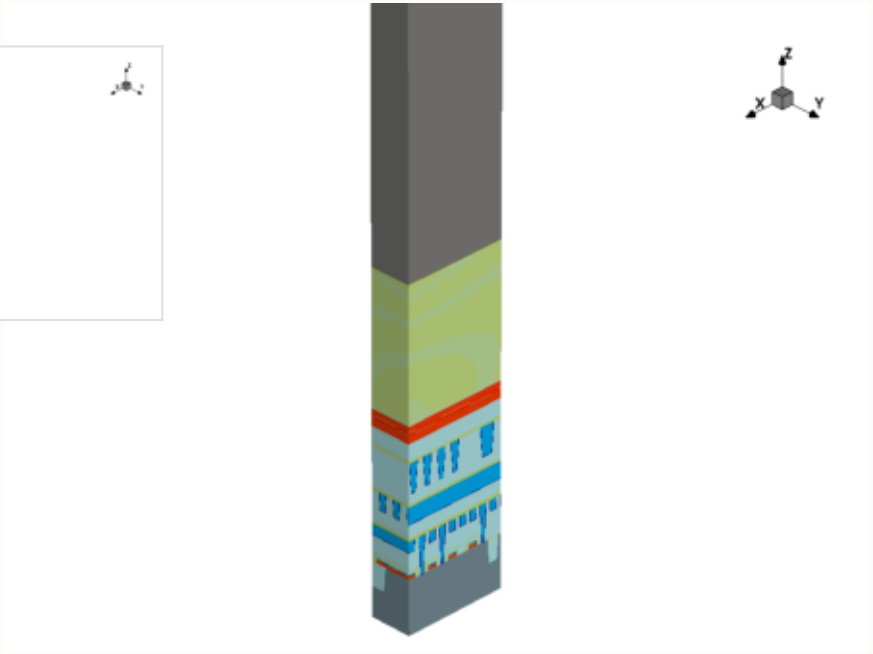
etch_anisotropic · RIE

1.04	265.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



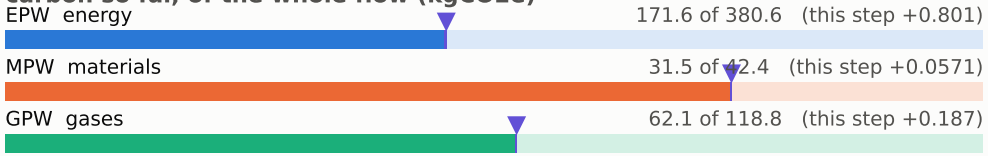


this step

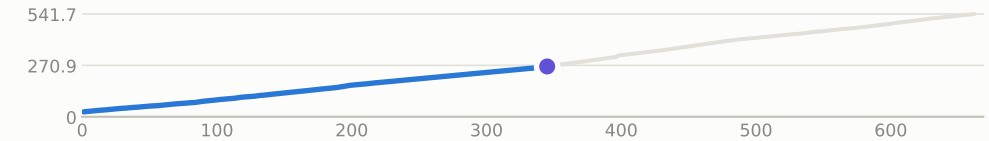
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

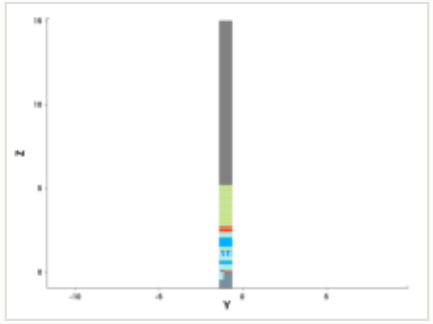
carbon so far, of the whole flow (kgCO2e)



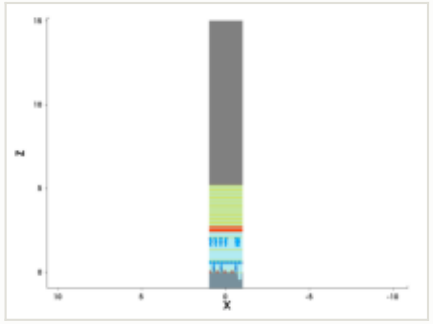
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



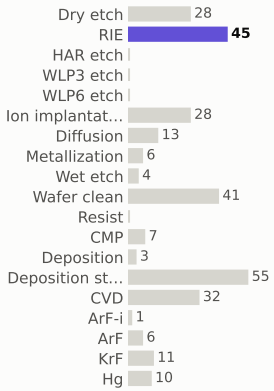
y cut at -1.032 µm



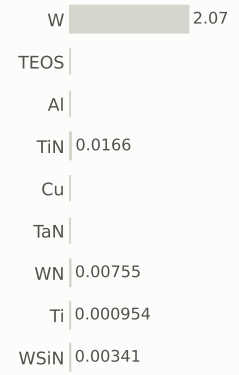
mask for this step

no mask for this step

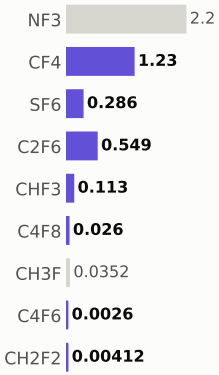
steps by process area



materials deposited (g)



gases emitted (g)



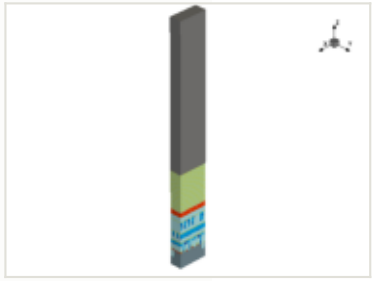
Step #3570: RIE

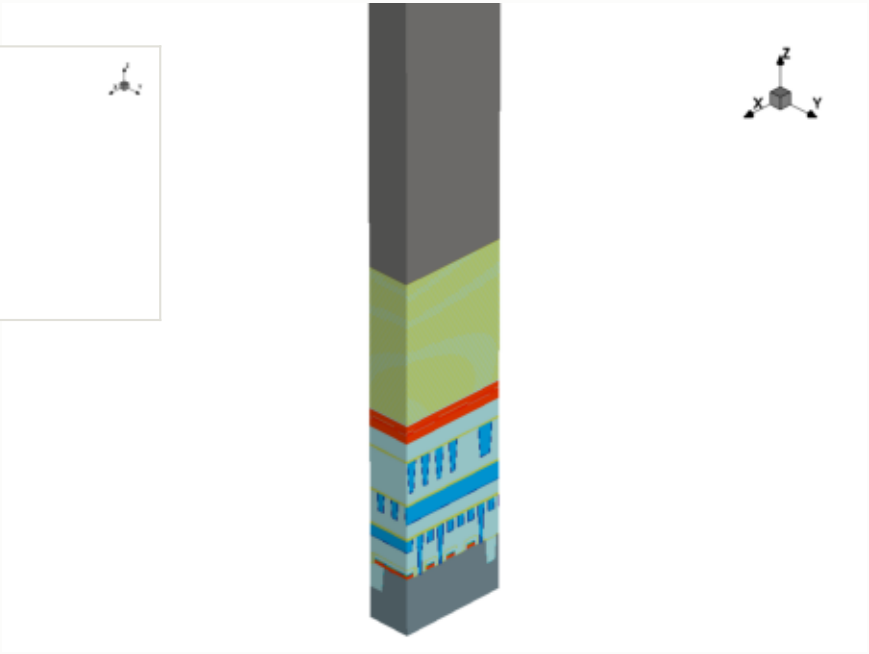
etch_anisotropic · RIE

1.04	266.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



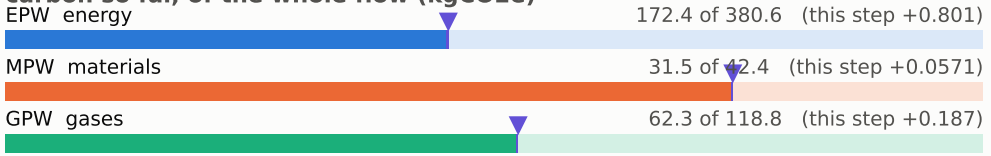


this step

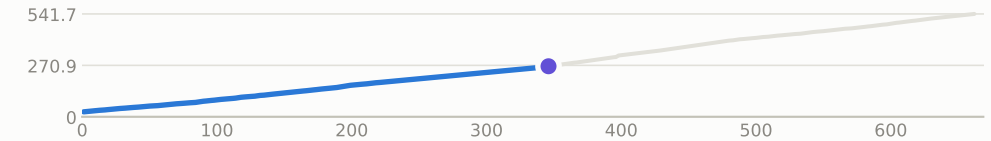
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

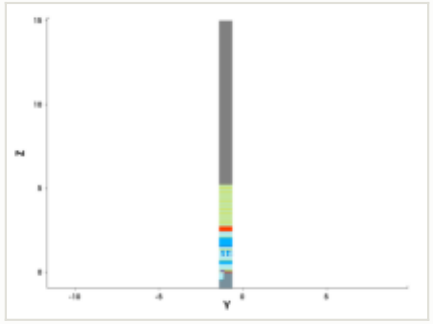
carbon so far, of the whole flow (kgCO2e)



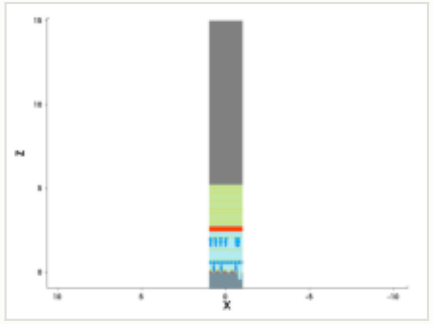
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



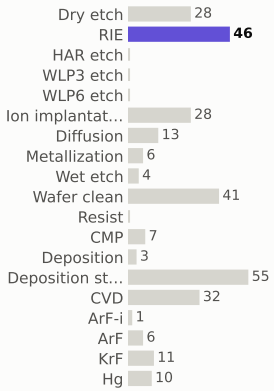
y cut at -1.032 µm



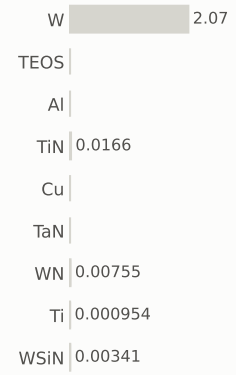
mask for this step

no mask for this step

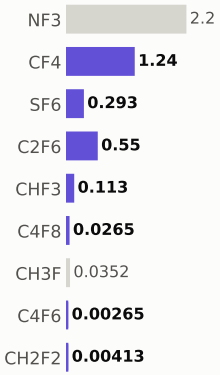
steps by process area



materials deposited (g)



gases emitted (g)



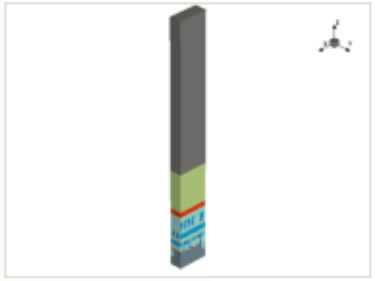
Step #3580: RIE

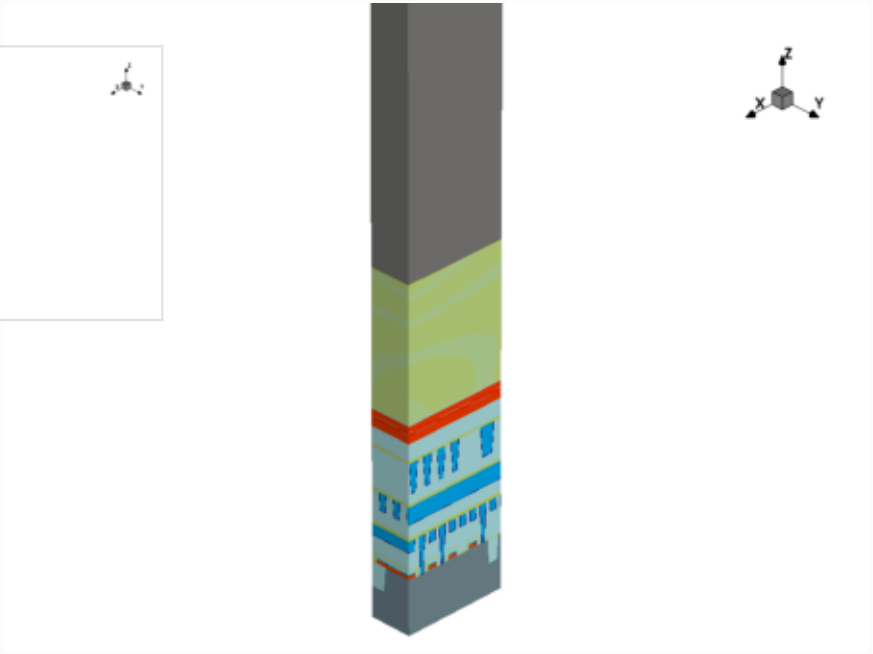
etch_anisotropic · RIE

1.04	267.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



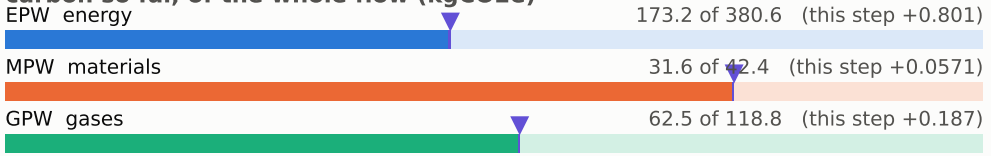


this step

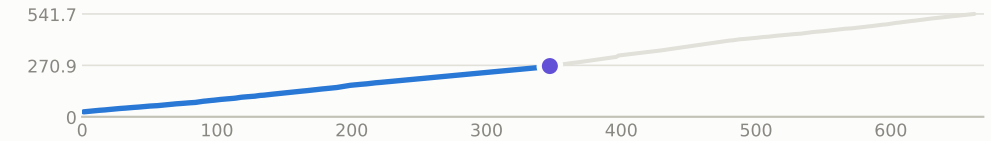
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

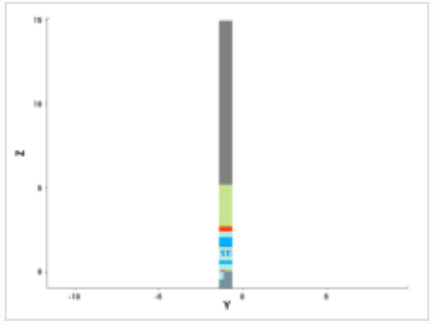
carbon so far, of the whole flow (kgCO2e)



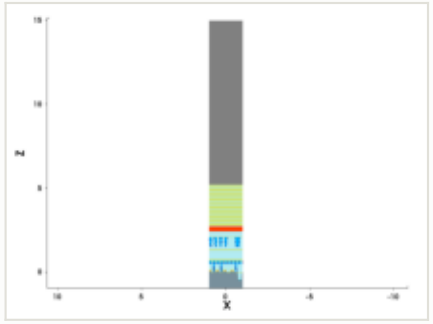
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



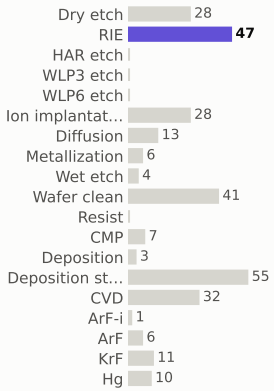
y cut at -1.032 µm



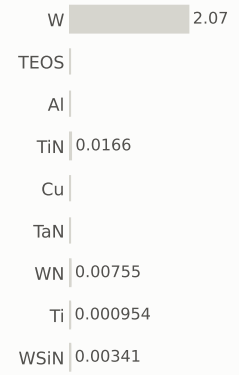
mask for this step

no mask for this step

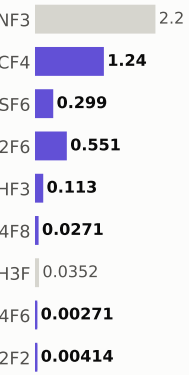
steps by process area



materials deposited (g)



gases emitted (g)



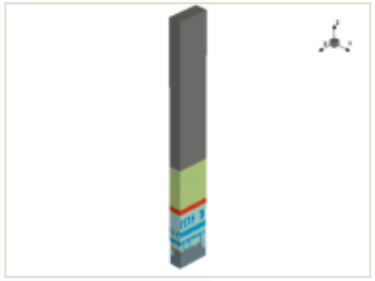
Step #3590: RIE

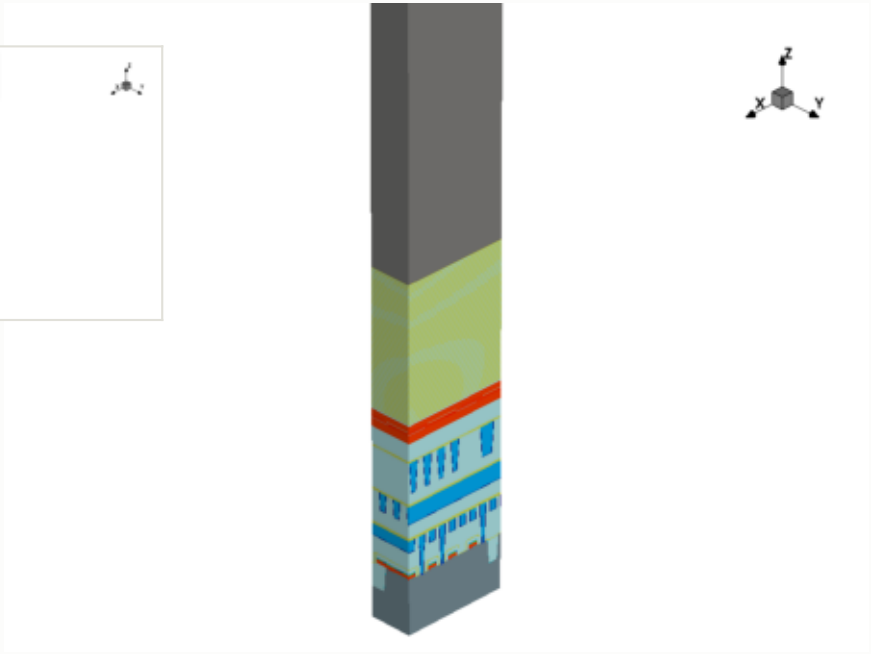
etch_anisotropic · RIE

1.04	268.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



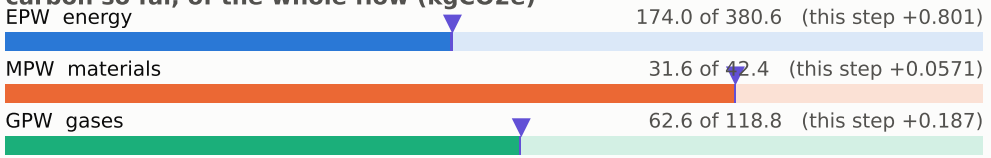


this step

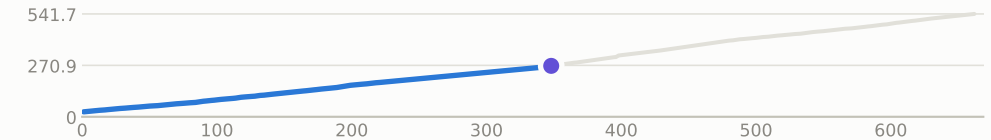
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

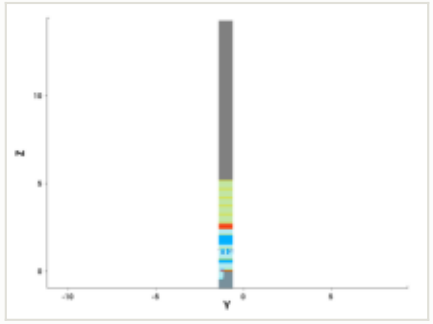
carbon so far, of the whole flow (kgCO2e)



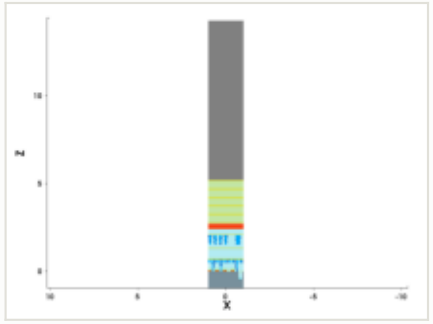
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



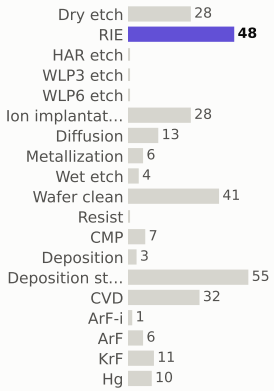
y cut at -1.032 µm



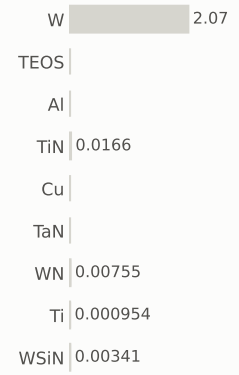
mask for this step

no mask for this step

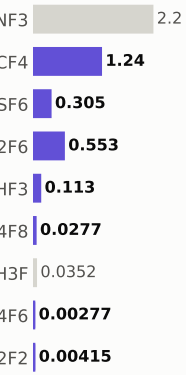
steps by process area



materials deposited (g)



gases emitted (g)



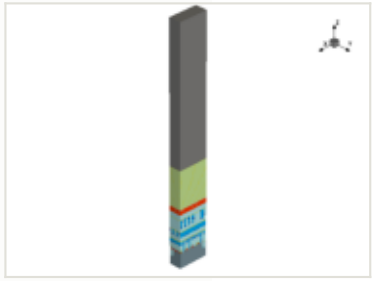
Step #3600: RIE

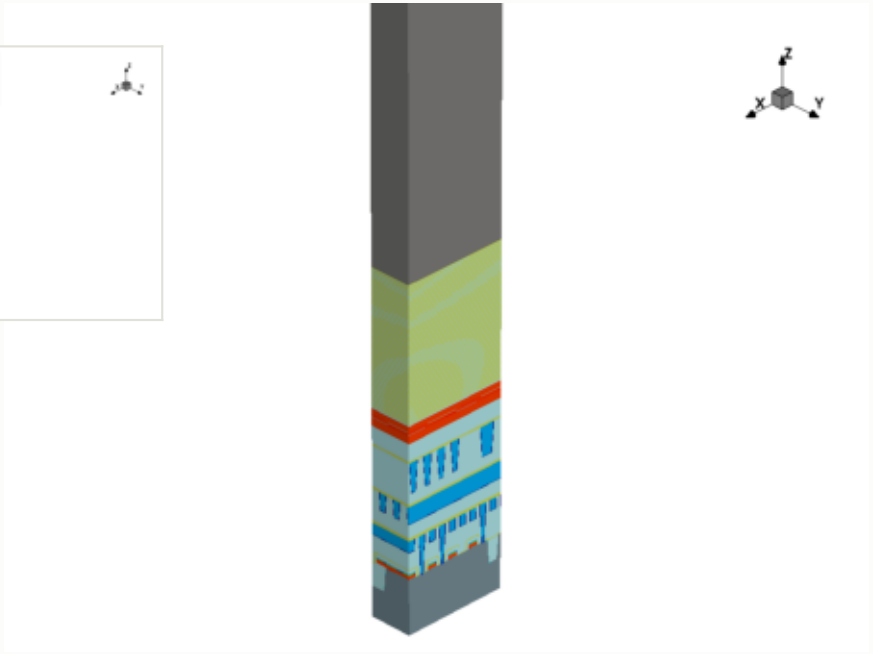
etch_anisotropic · RIE

1.04	269.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



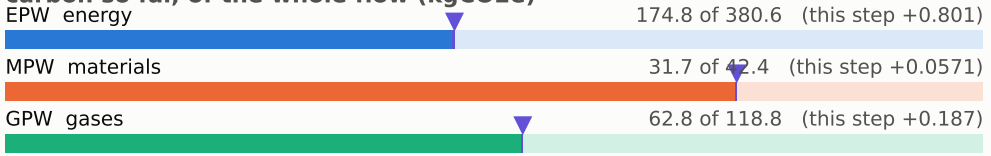


this step

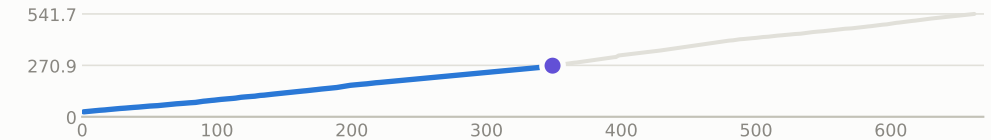
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

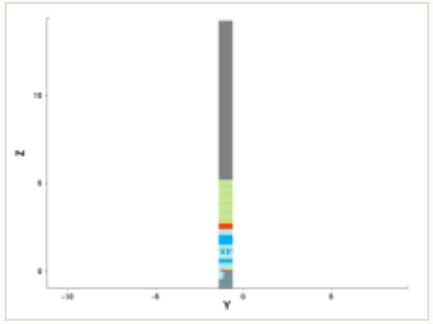
carbon so far, of the whole flow (kgCO2e)



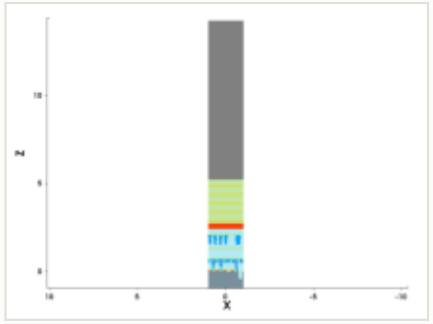
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



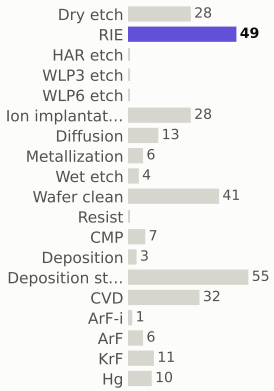
y cut at -1.032 µm



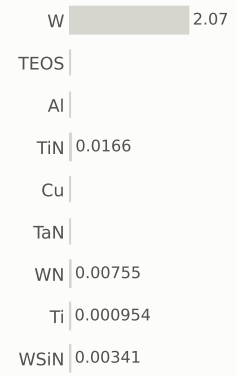
mask for this step

no mask for this step

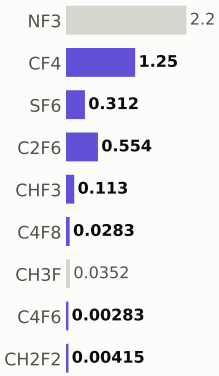
steps by process area



materials deposited (g)



gases emitted (g)



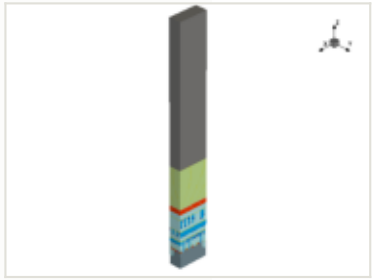
Step #3610: RIE

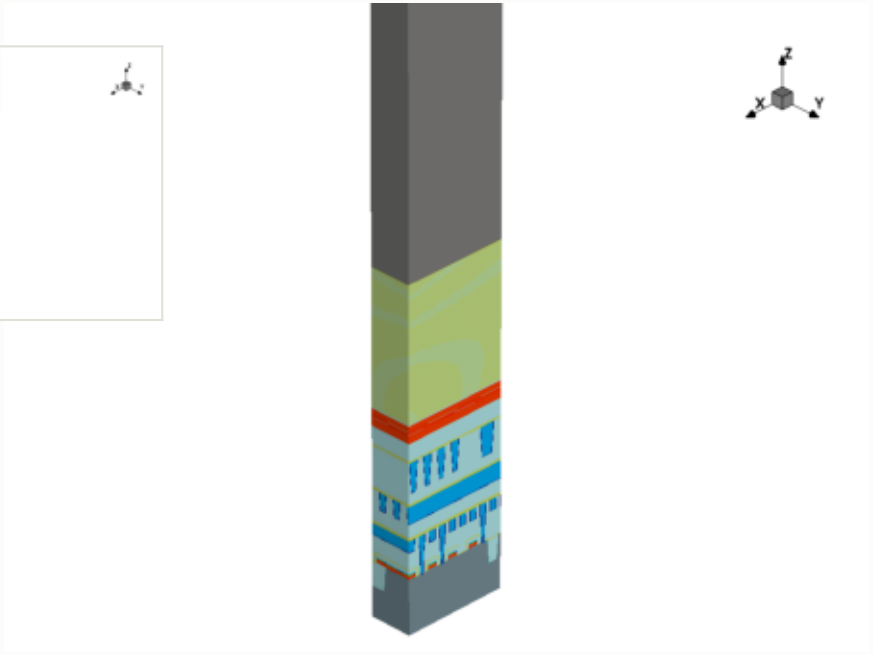
etch_anisotropic · RIE

1.04	270.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



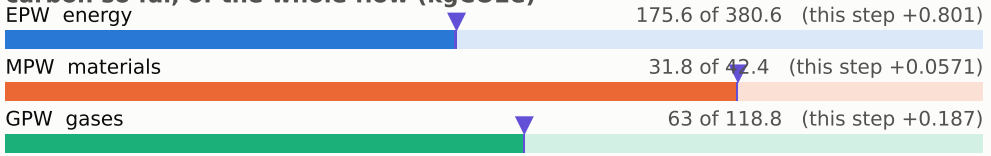


this step

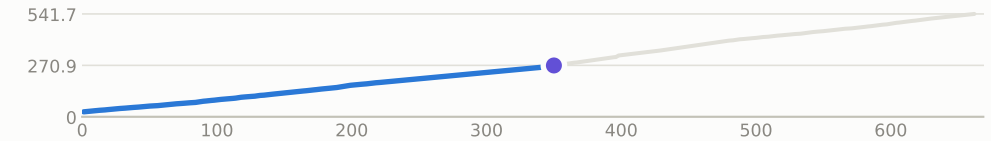
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

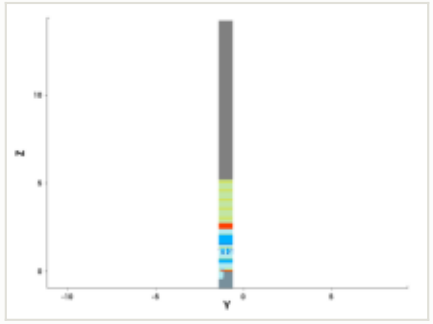
carbon so far, of the whole flow (kgCO2e)



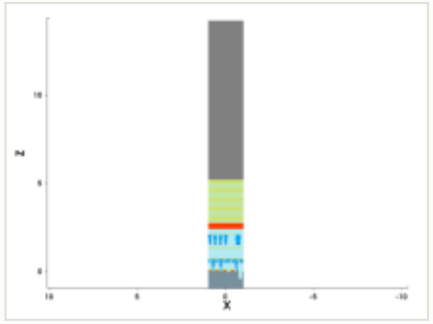
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



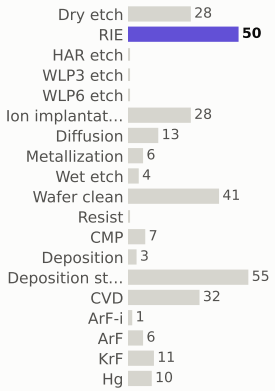
y cut at -1.032 µm



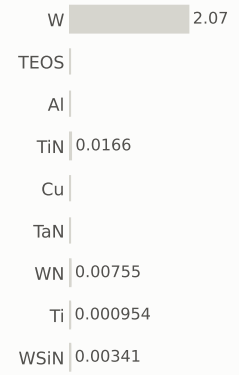
mask for this step

no mask for this step

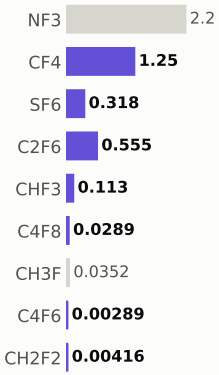
steps by process area



materials deposited (g)



gases emitted (g)



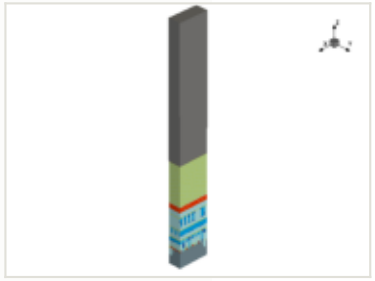
Step #3620: RIE

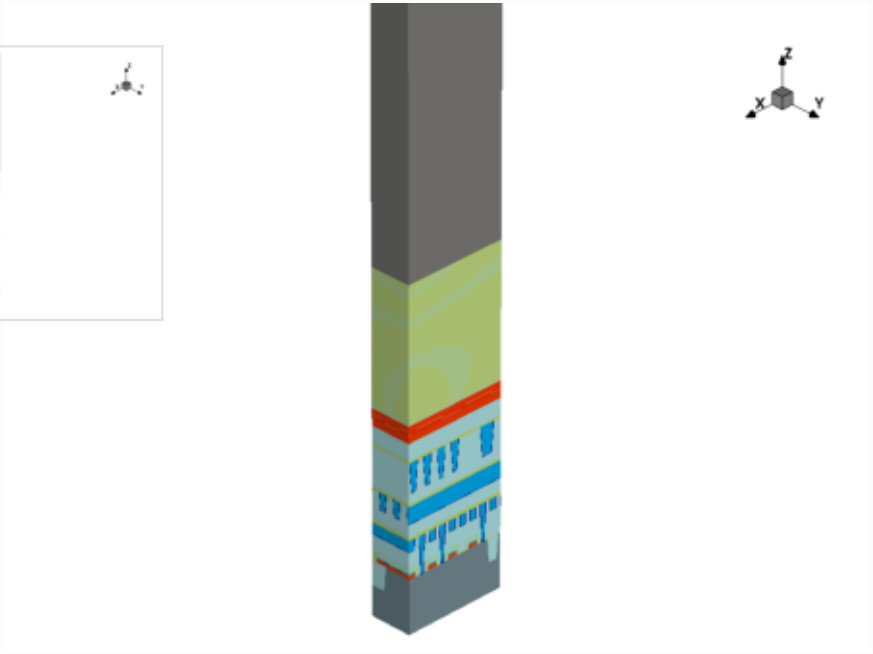
etch_anisotropic · RIE

1.04	271.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



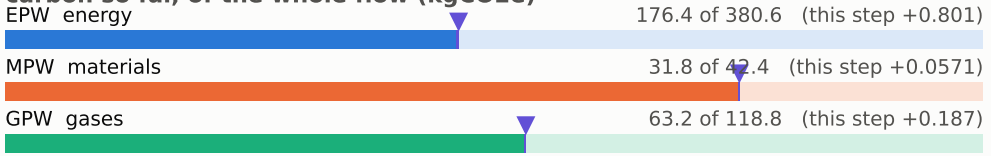


this step

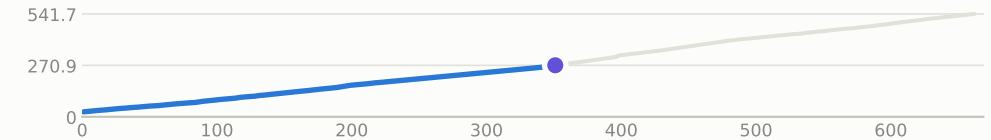
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

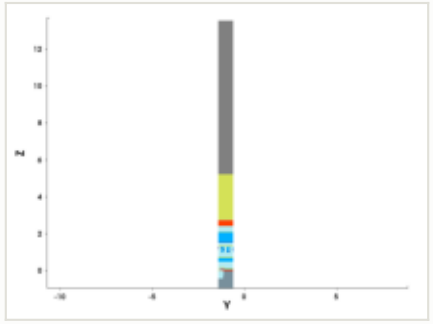
carbon so far, of the whole flow (kgCO2e)



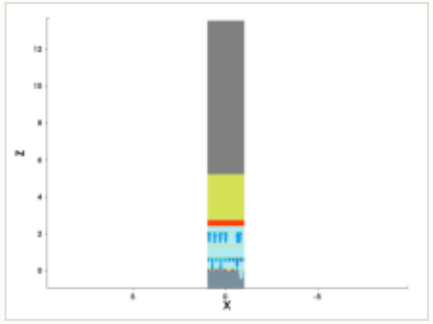
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



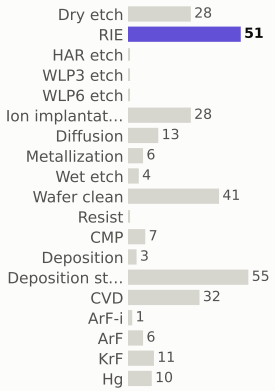
y cut at -1.032 µm



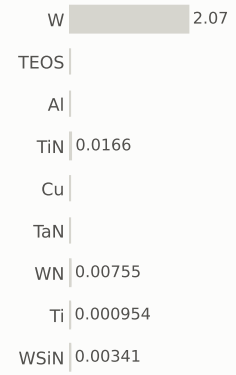
mask for this step

no mask for this step

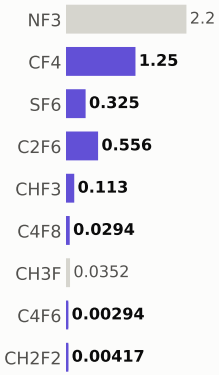
steps by process area



materials deposited (g)



gases emitted (g)



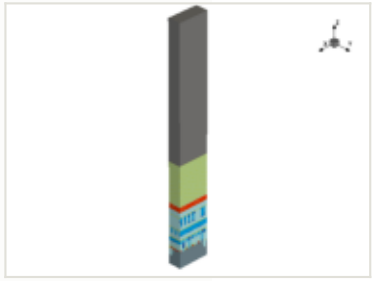
Step #3630: RIE

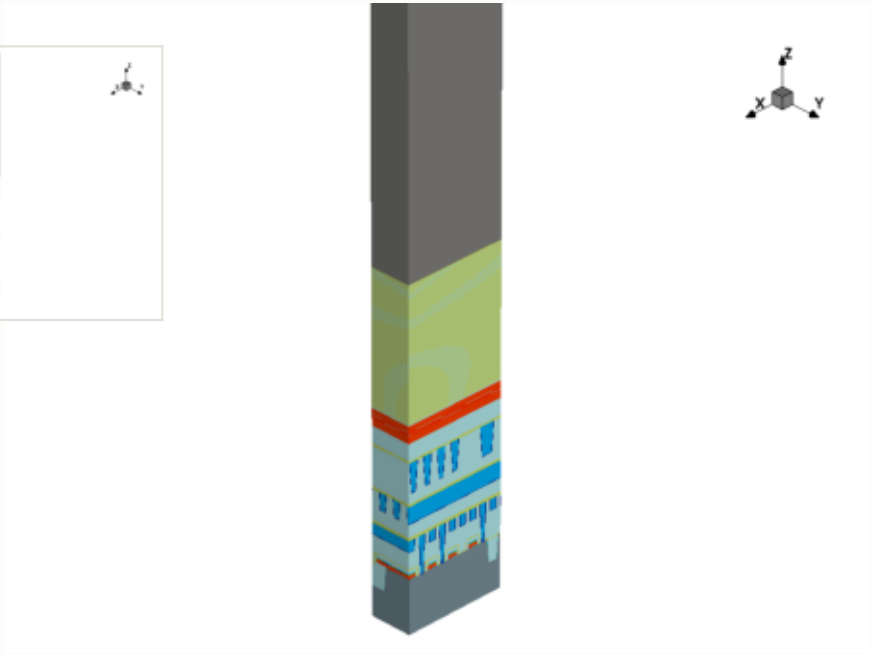
etch_anisotropic · RIE

1.04	272.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



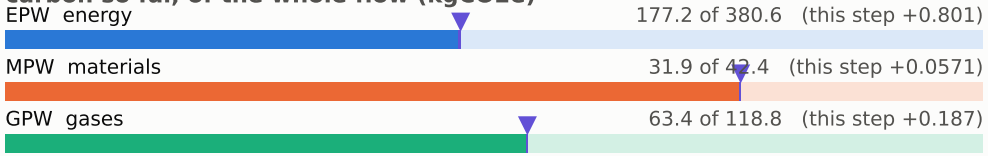


this step

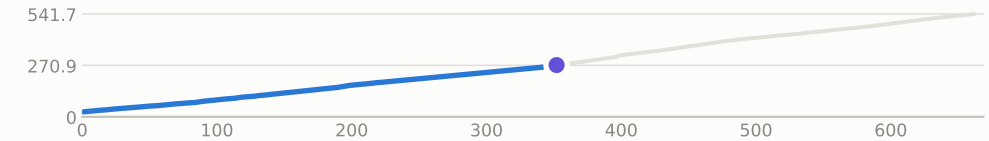
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

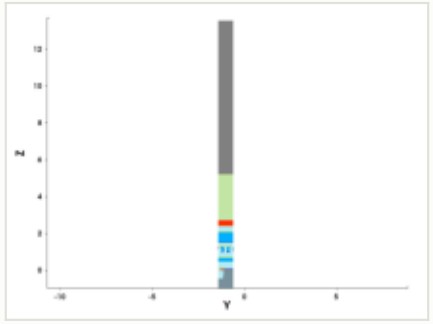
carbon so far, of the whole flow (kgCO2e)



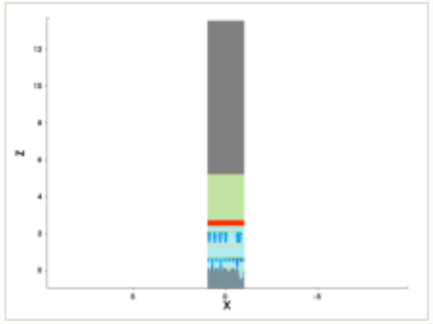
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



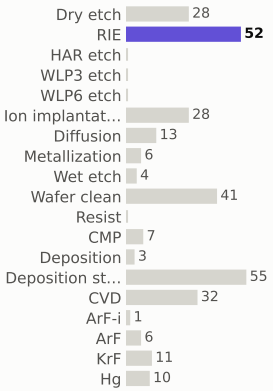
y cut at -1.032 µm



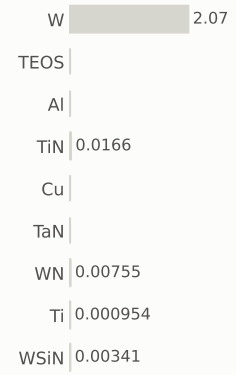
mask for this step

no mask for this step

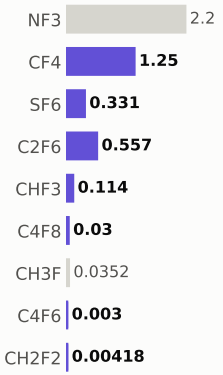
steps by process area



materials deposited (g)



gases emitted (g)



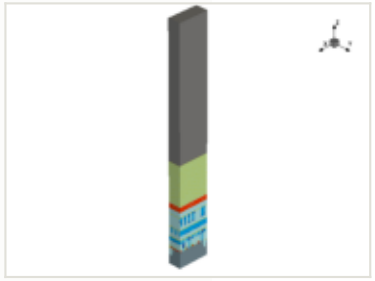
Step #3640: RIE

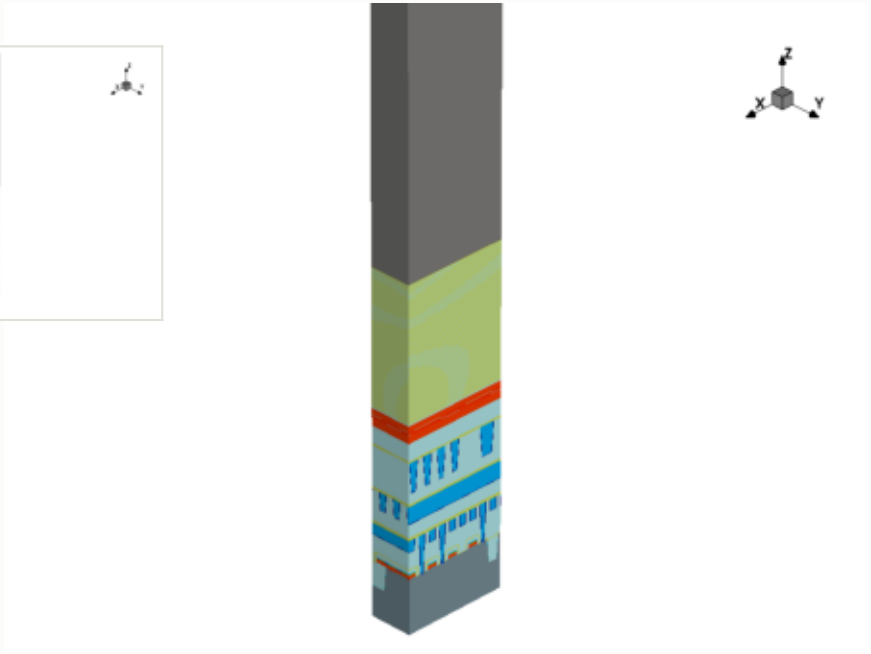
etch_anisotropic · RIE

1.04	273.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



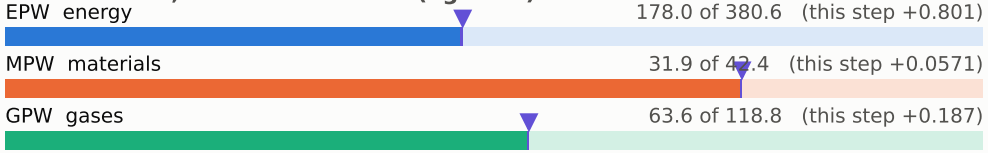


this step

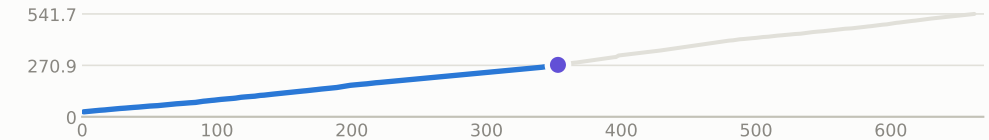
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

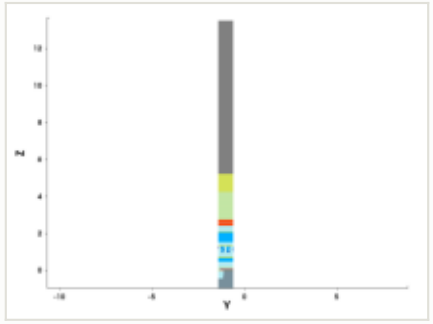
carbon so far, of the whole flow (kgCO2e)



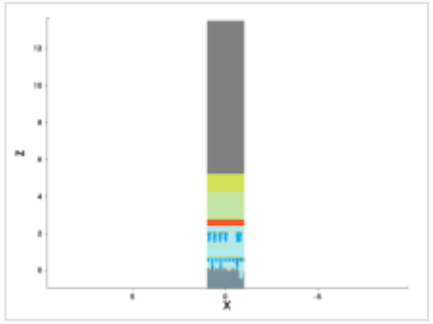
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



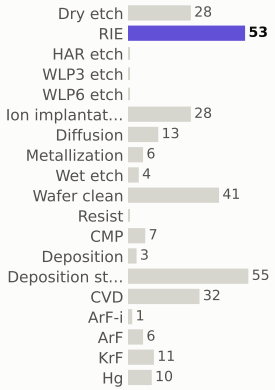
y cut at -1.032 µm



mask for this step

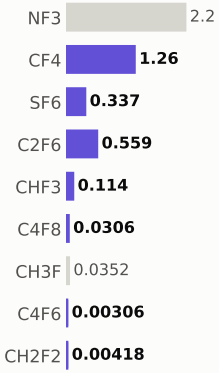
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



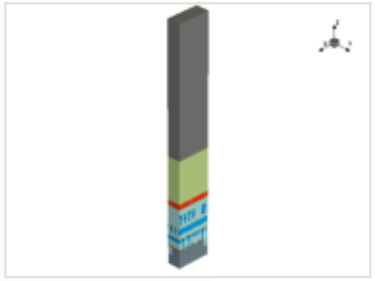
Step #3650: RIE

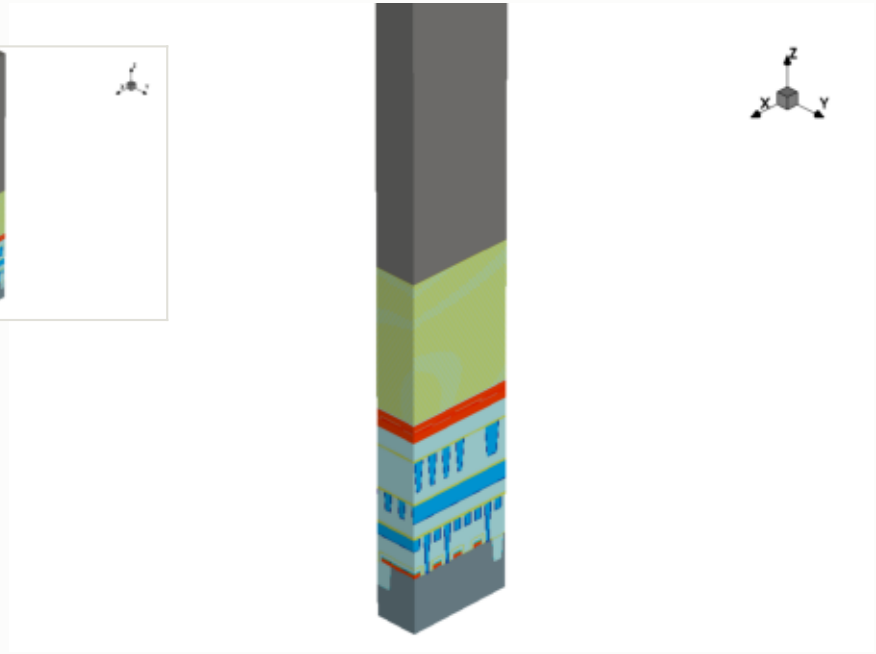
etch_anisotropic · RIE

1.04	274.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



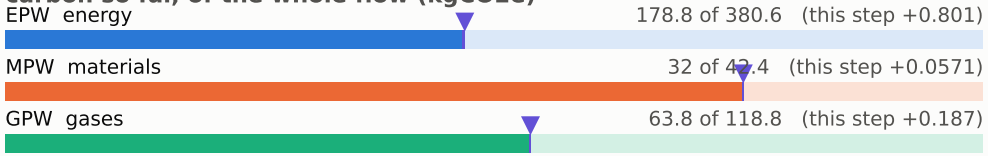


this step

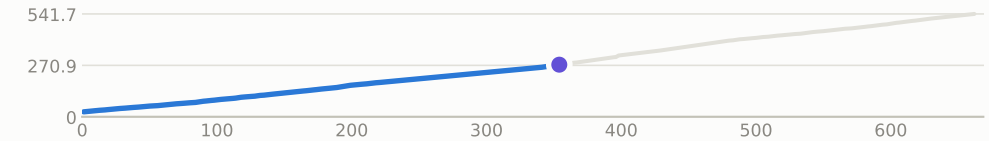
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

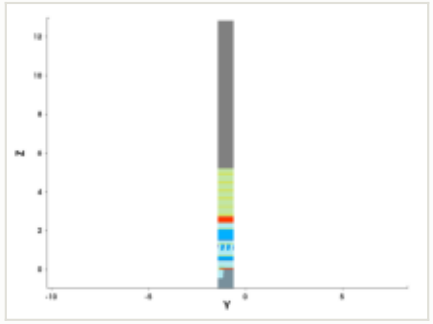
carbon so far, of the whole flow (kgCO2e)



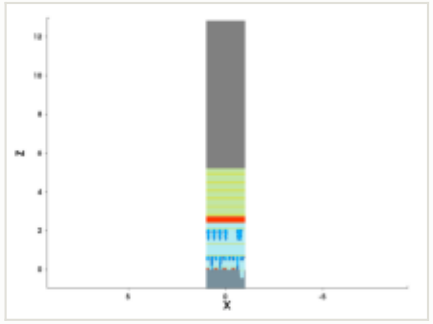
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



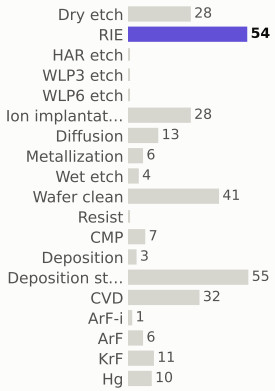
y cut at -1.032 µm



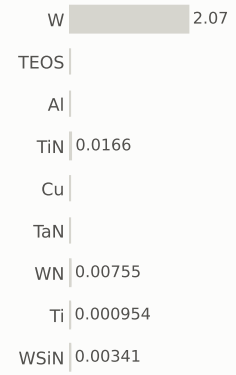
mask for this step

no mask for this step

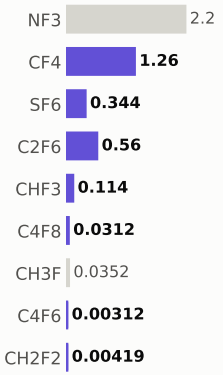
steps by process area



materials deposited (g)



gases emitted (g)



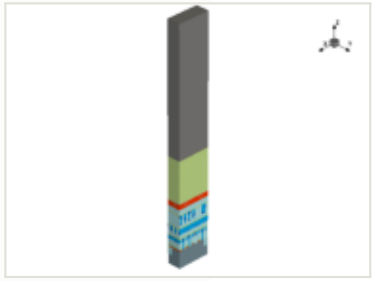
Step #3660: RIE

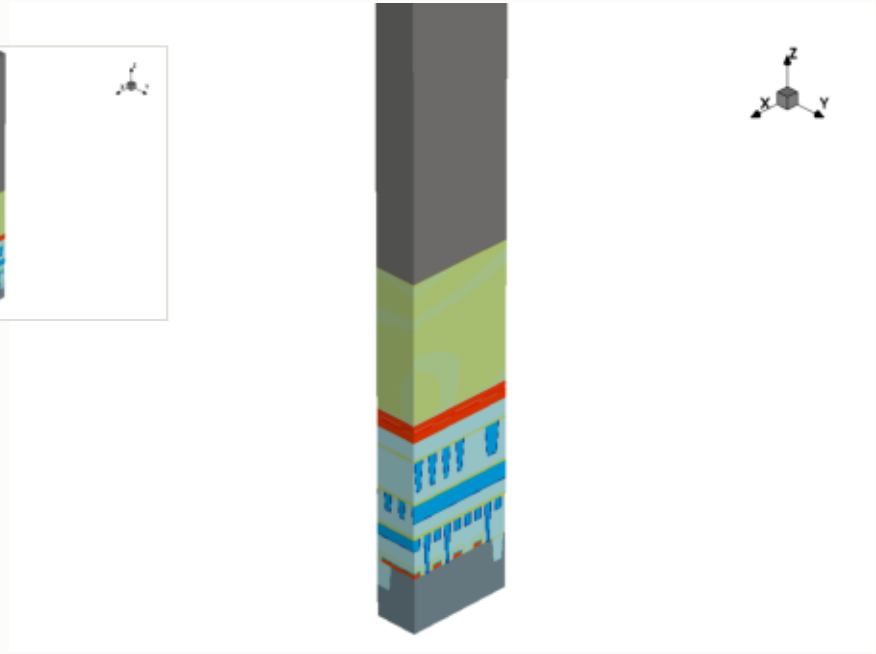
etch_anisotropic · RIE

1.04	275.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



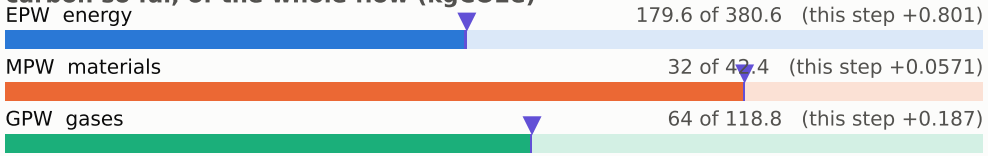


this step

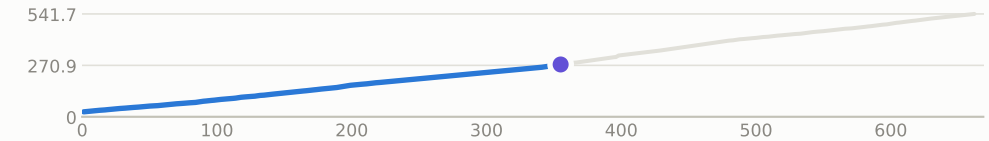
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

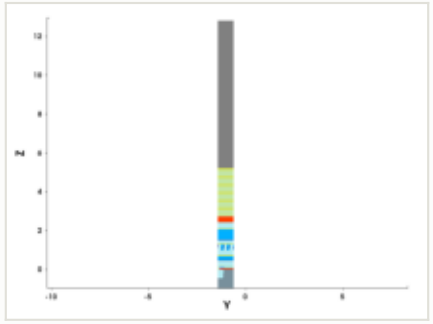
carbon so far, of the whole flow (kgCO2e)



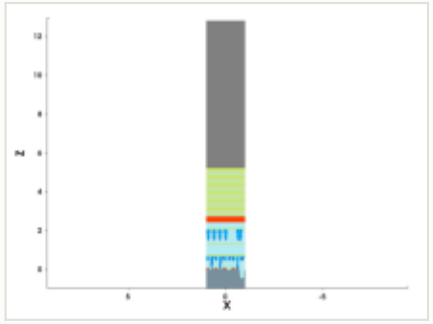
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



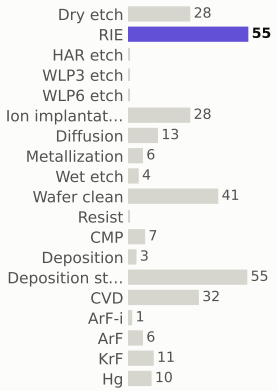
y cut at -1.032 µm



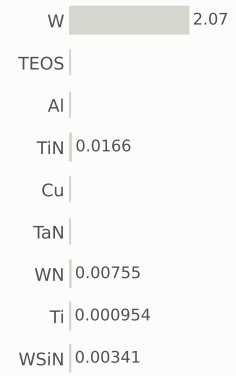
mask for this step

no mask for this step

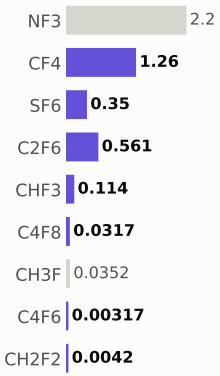
steps by process area



materials deposited (g)



gases emitted (g)



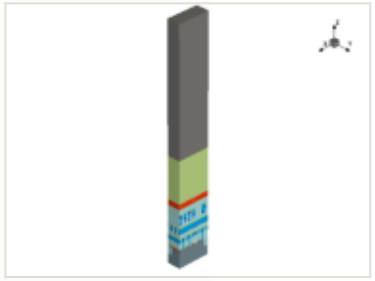
Step #3670: RIE

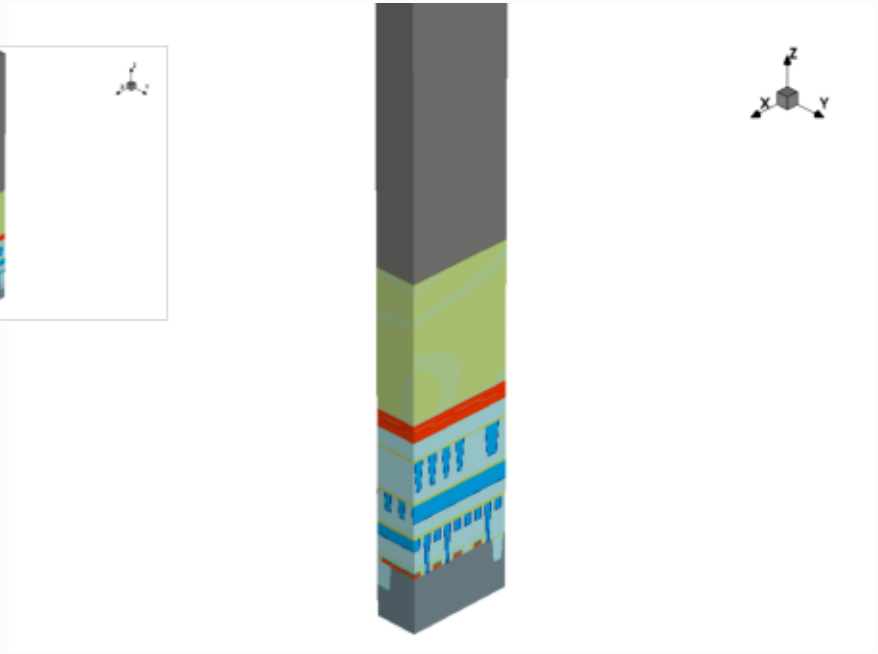
etch_anisotropic · RIE

1.04	276.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



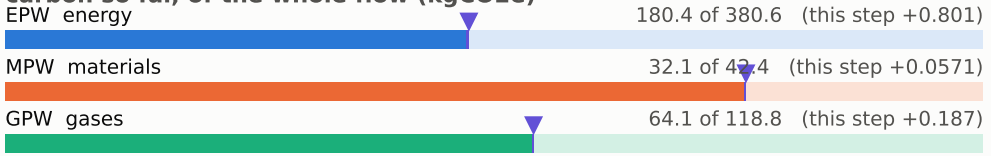


this step

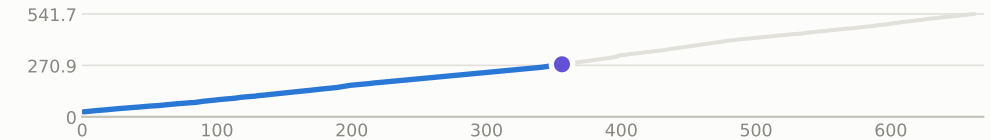
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

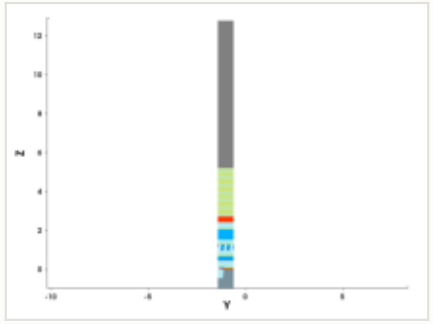
carbon so far, of the whole flow (kgCO2e)



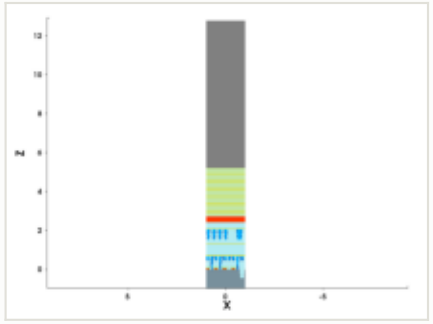
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



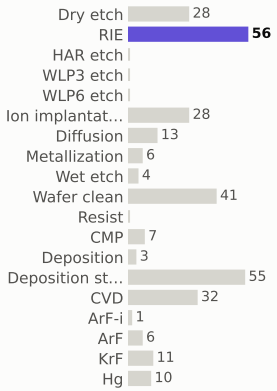
y cut at -1.032 µm



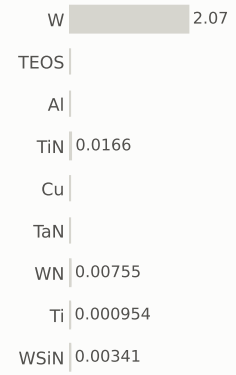
mask for this step

no mask for this step

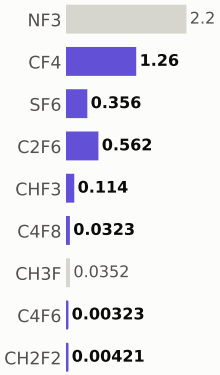
steps by process area



materials deposited (g)



gases emitted (g)



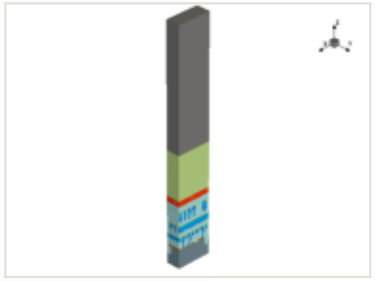
Step #3680: RIE

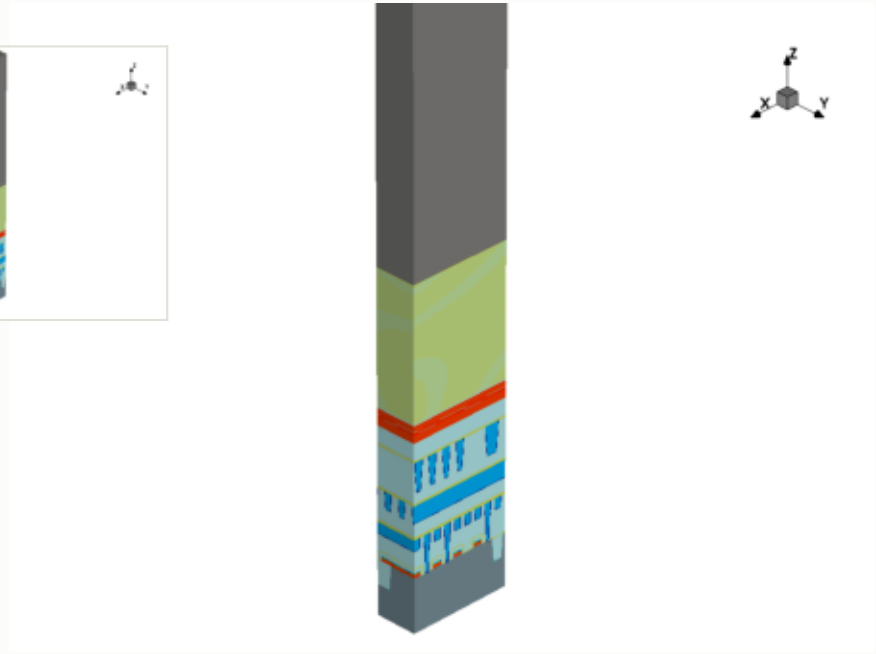
etch_anisotropic · RIE

1.04	277.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



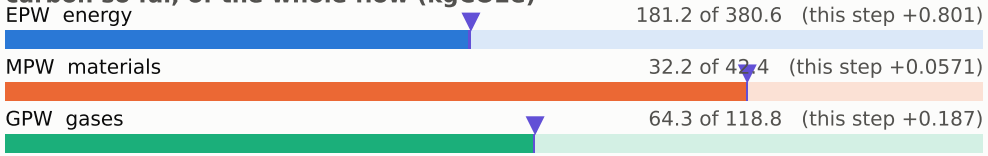


this step

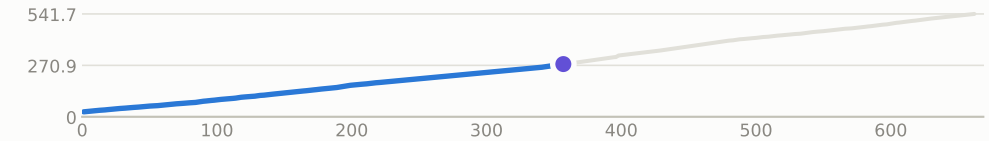
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

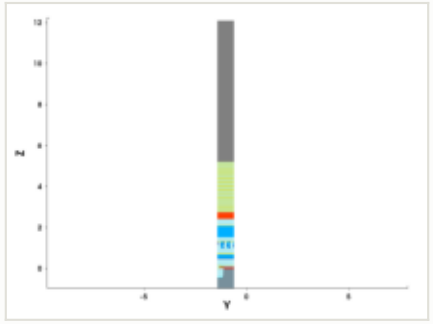
carbon so far, of the whole flow (kgCO2e)



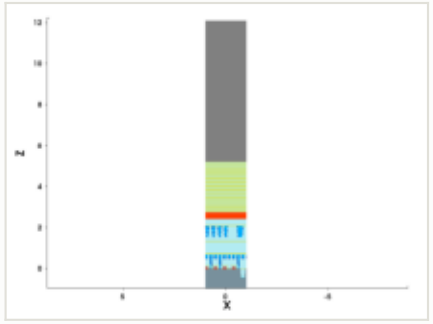
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



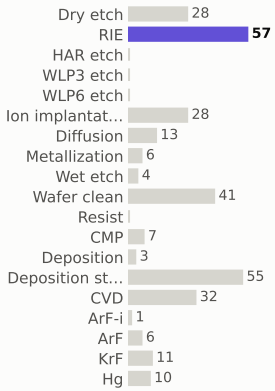
y cut at -1.032 µm



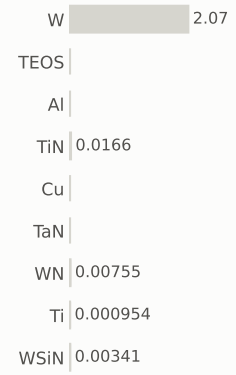
mask for this step

no mask for this step

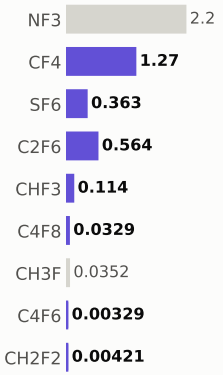
steps by process area



materials deposited (g)



gases emitted (g)



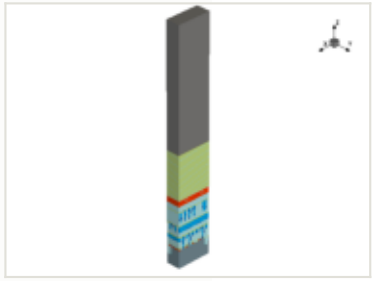
Step #3690: RIE

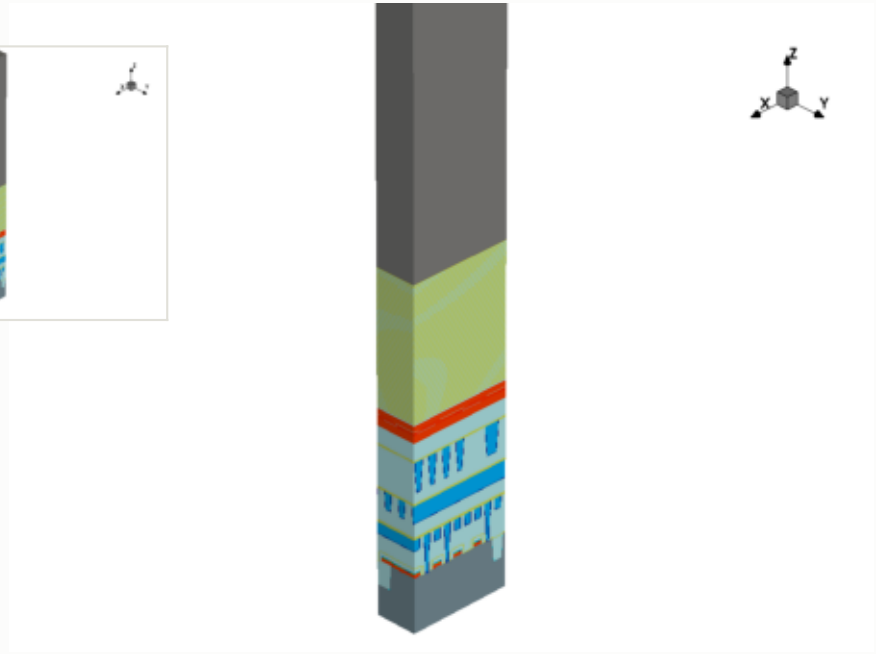
etch_anisotropic · RIE

1.04	278.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

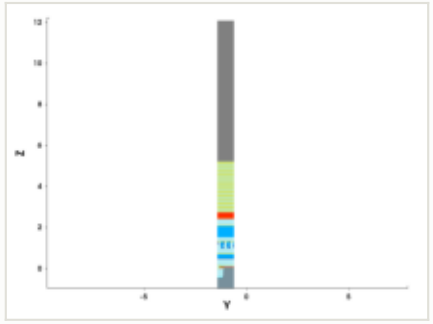
3D structure after this step

zoomed to fit

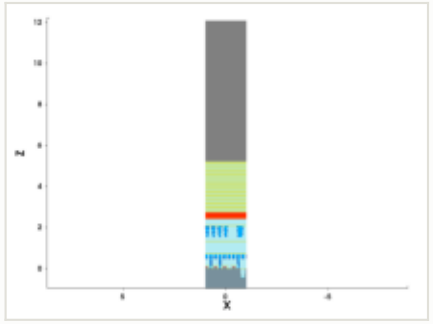




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

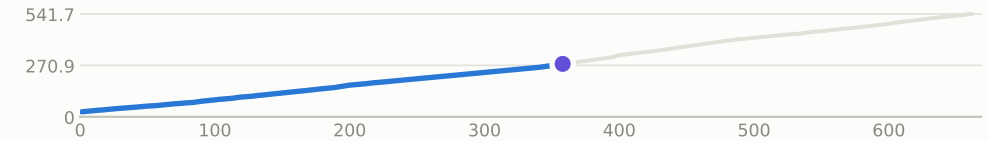
carbon so far, of the whole flow (kgCO2e)

EPW energy — 182.0 of 380.6 (this step +0.801)

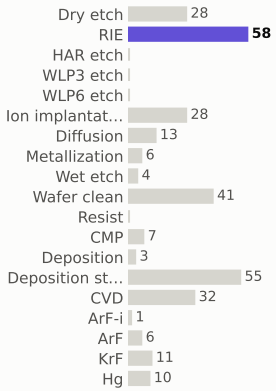
MPW materials — 32.2 of 43.4 (this step +0.0571)

GPW gases — 64.5 of 118.8 (this step +0.187)

embodied carbon up to each step (kgCO2e per wafer)

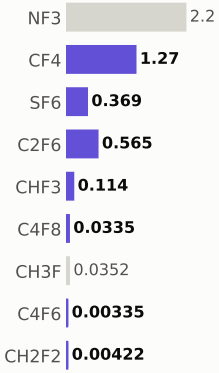


steps by process area



materials deposited (g)

gases emitted (g)



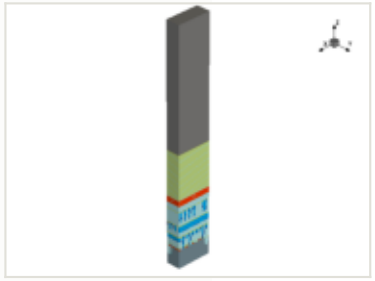
Step #3700: RIE

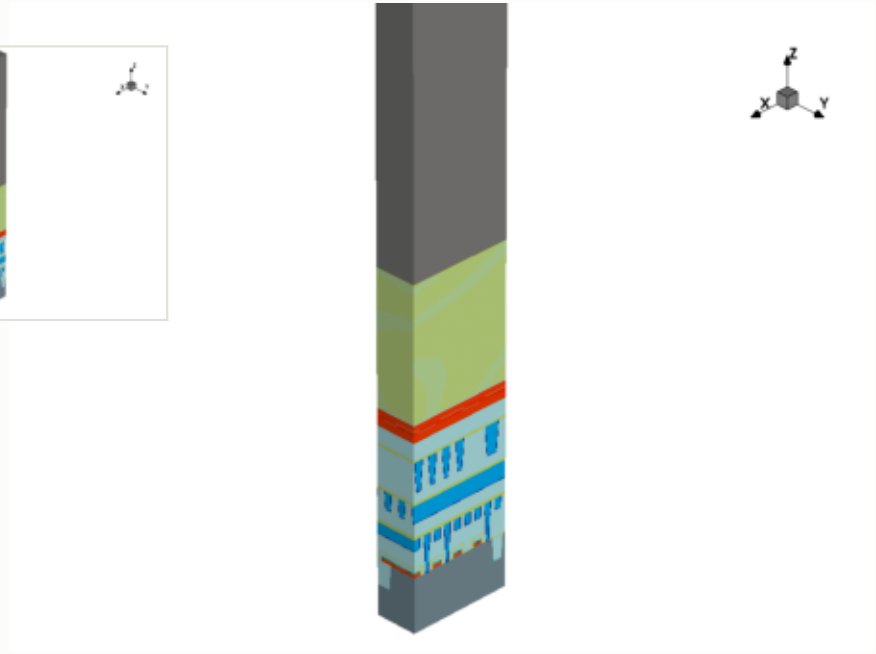
etch_anisotropic · RIE

1.04	279.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



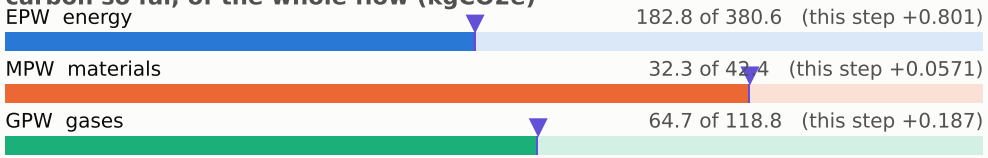


this step

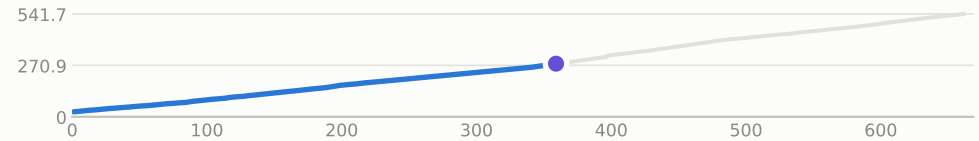
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

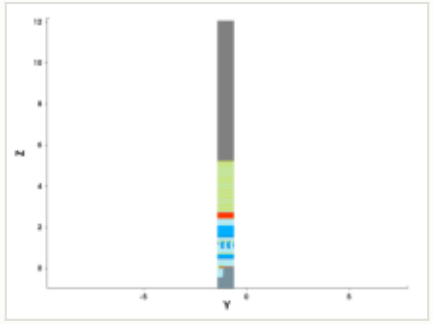
carbon so far, of the whole flow (kgCO2e)



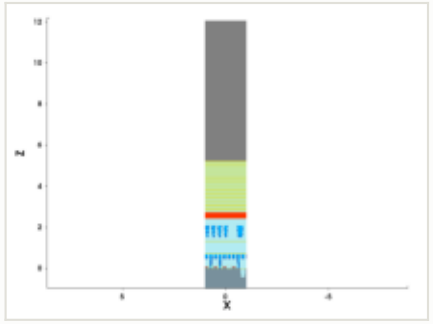
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



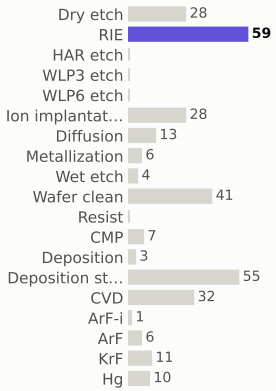
y cut at -1.032 µm



mask for this step

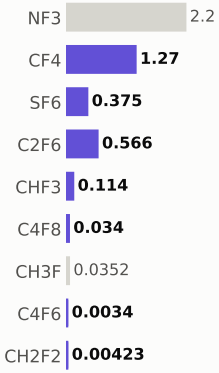
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



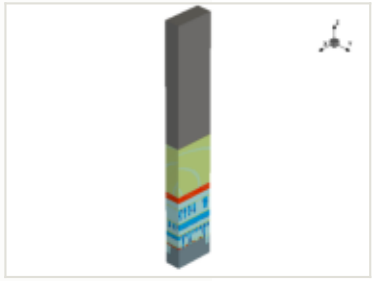
Step #3710: RIE

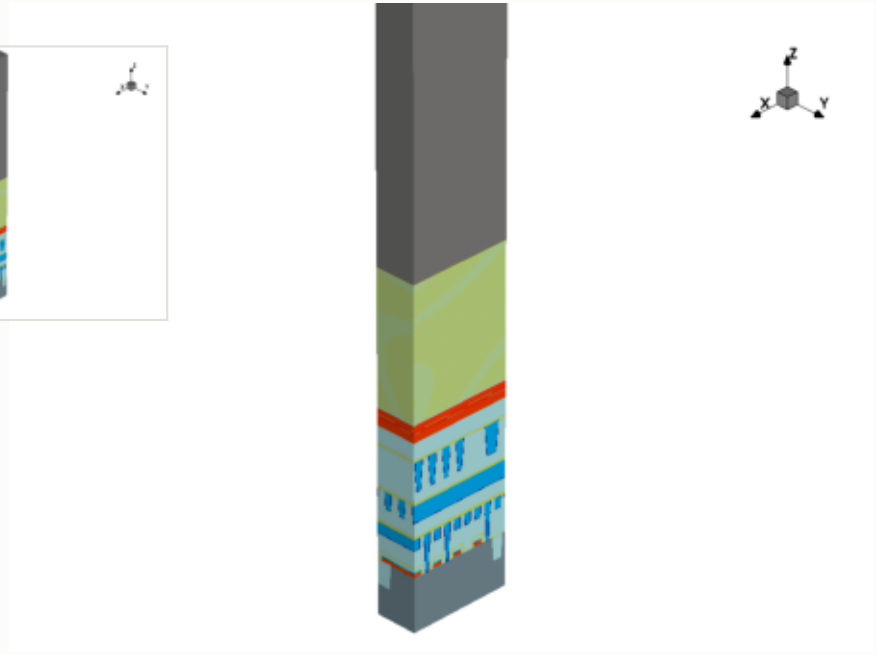
etch_anisotropic · RIE

1.04	280.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



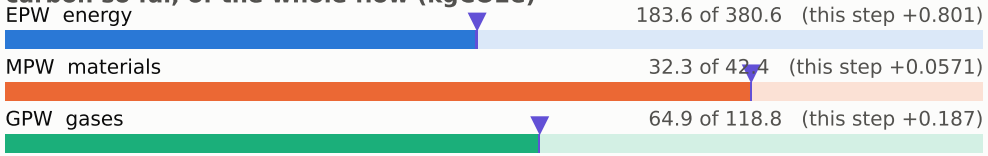


this step

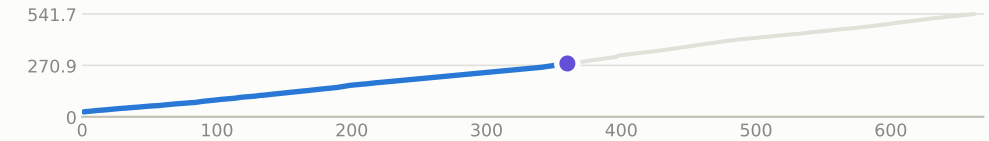
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

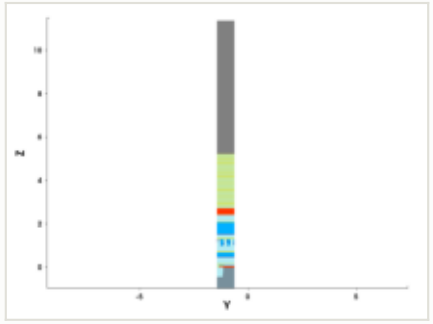
carbon so far, of the whole flow (kgCO2e)



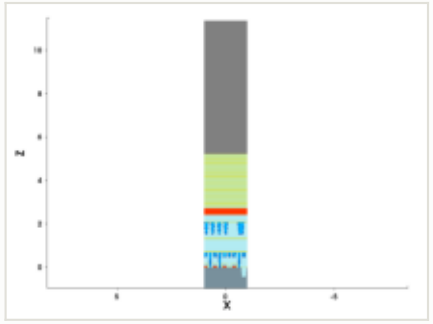
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



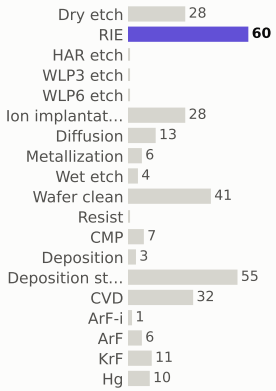
y cut at -1.032 µm



mask for this step

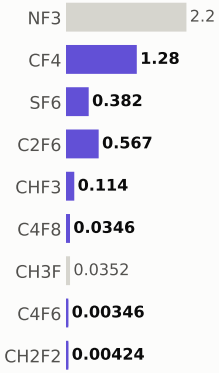
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



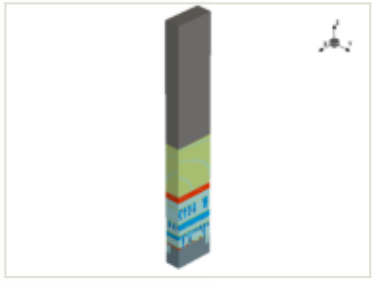
Step #3720: RIE

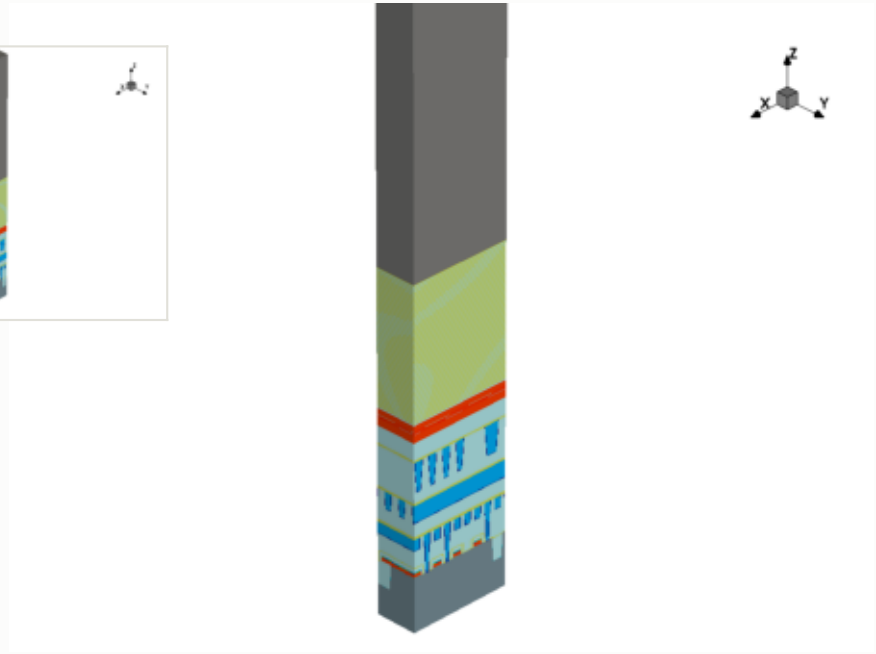
etch_anisotropic · RIE

1.04	281.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



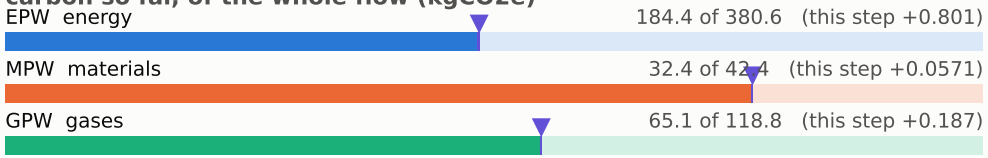


this step

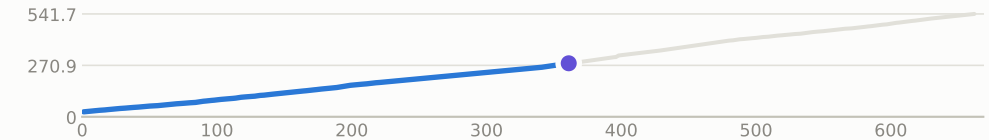
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

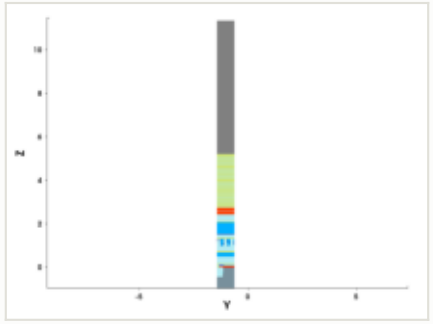
carbon so far, of the whole flow (kgCO2e)



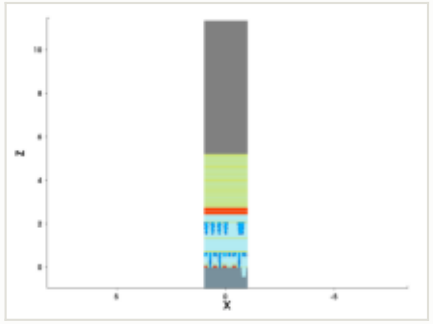
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



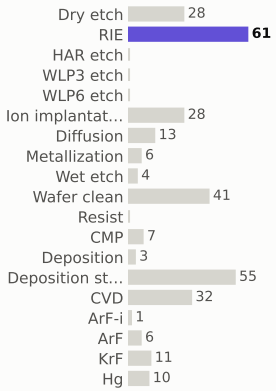
y cut at -1.032 µm



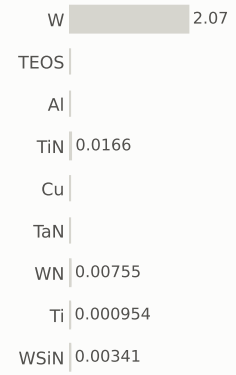
mask for this step

no mask for this step

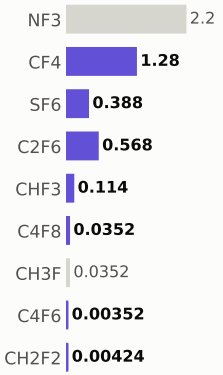
steps by process area



materials deposited (g)



gases emitted (g)



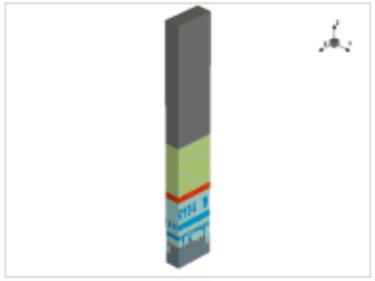
Step #3730: RIE

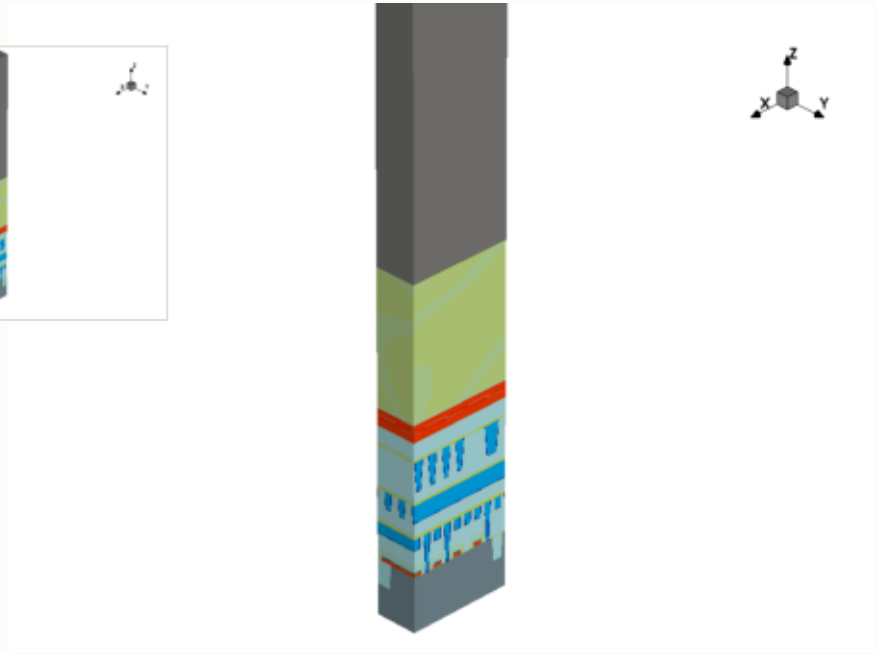
etch_anisotropic · RIE

1.04	282.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



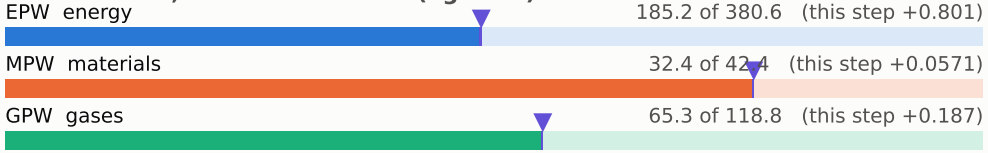


this step

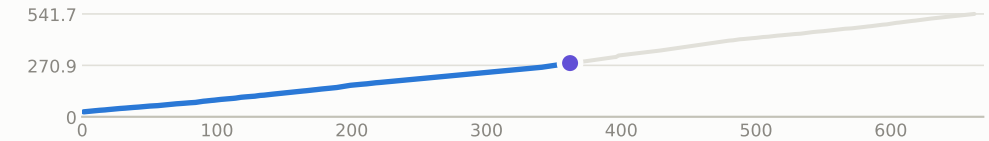
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

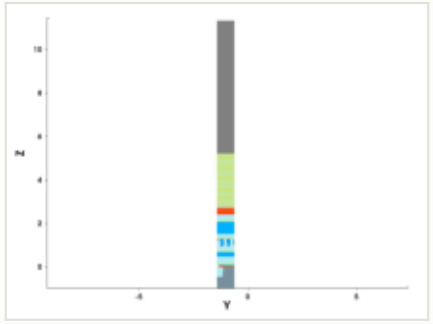
carbon so far, of the whole flow (kgCO2e)



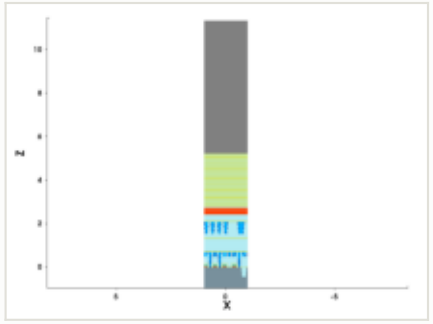
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



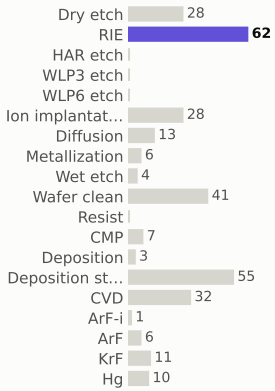
y cut at -1.032 µm



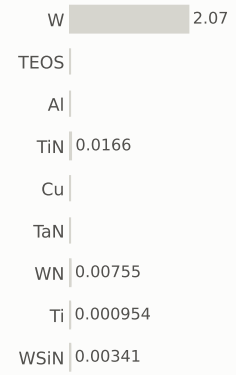
mask for this step

no mask for this step

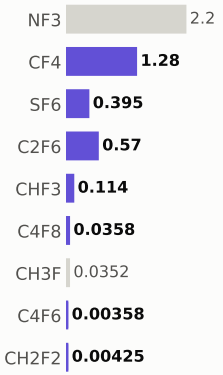
steps by process area



materials deposited (g)



gases emitted (g)



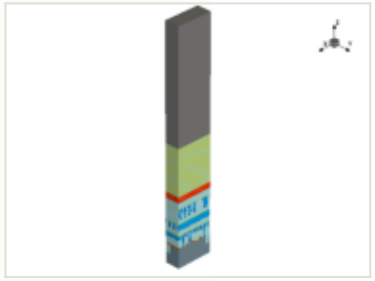
Step #3740: RIE

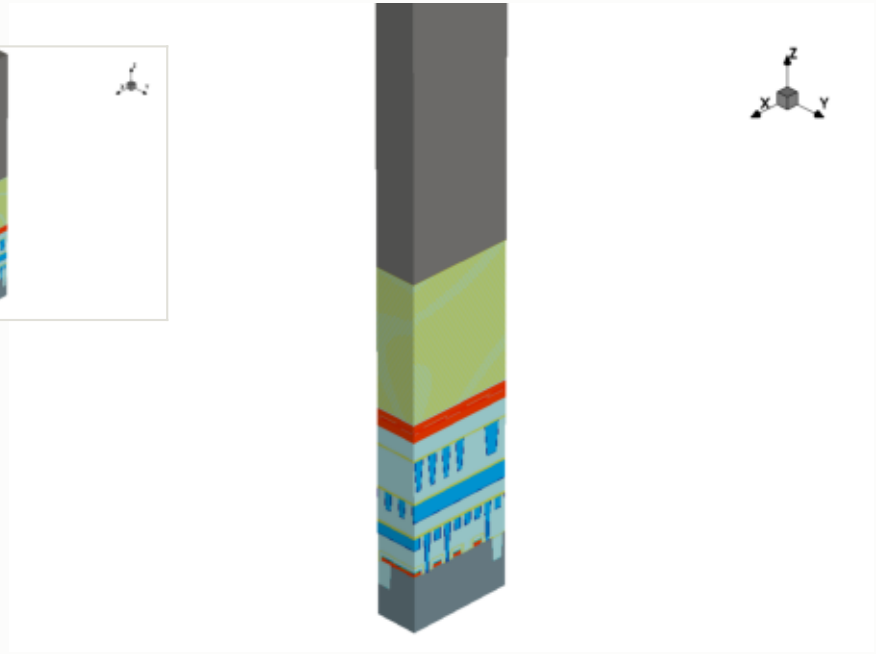
etch_anisotropic · RIE

1.04	283.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



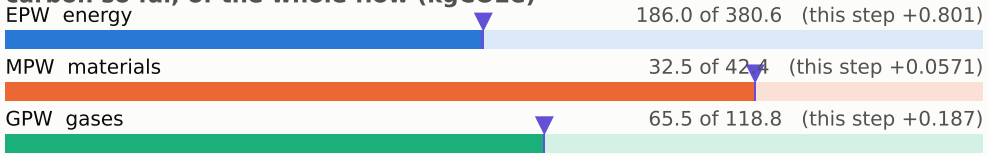


this step

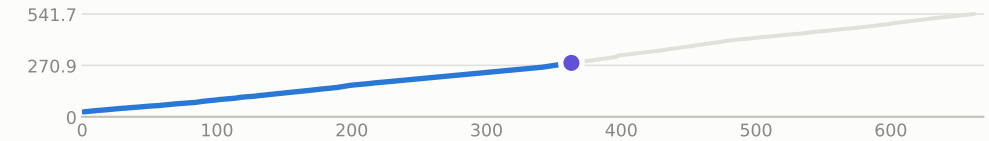
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

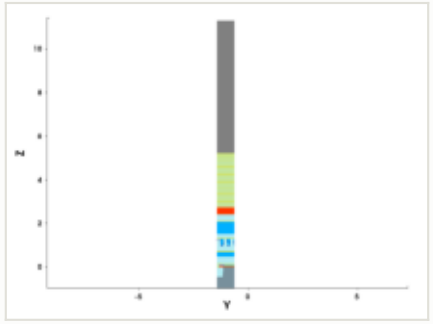
carbon so far, of the whole flow (kgCO2e)



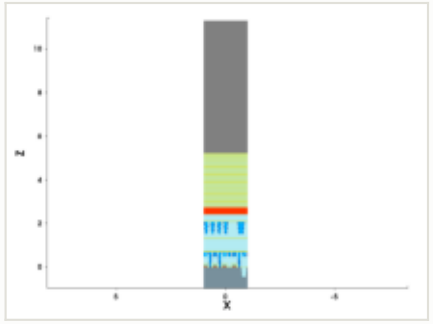
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



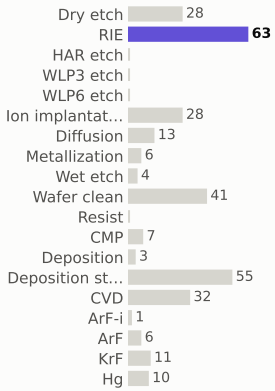
y cut at -1.032 µm



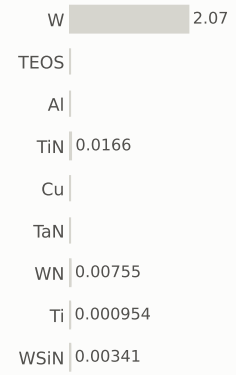
mask for this step

no mask for this step

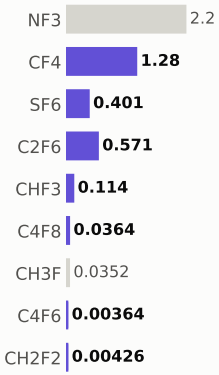
steps by process area



materials deposited (g)



gases emitted (g)



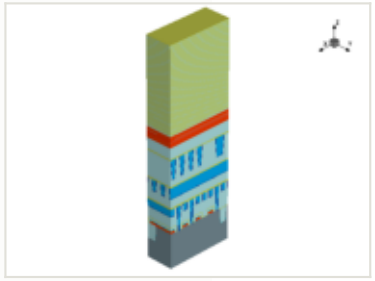
Step #3750: Dry_etch

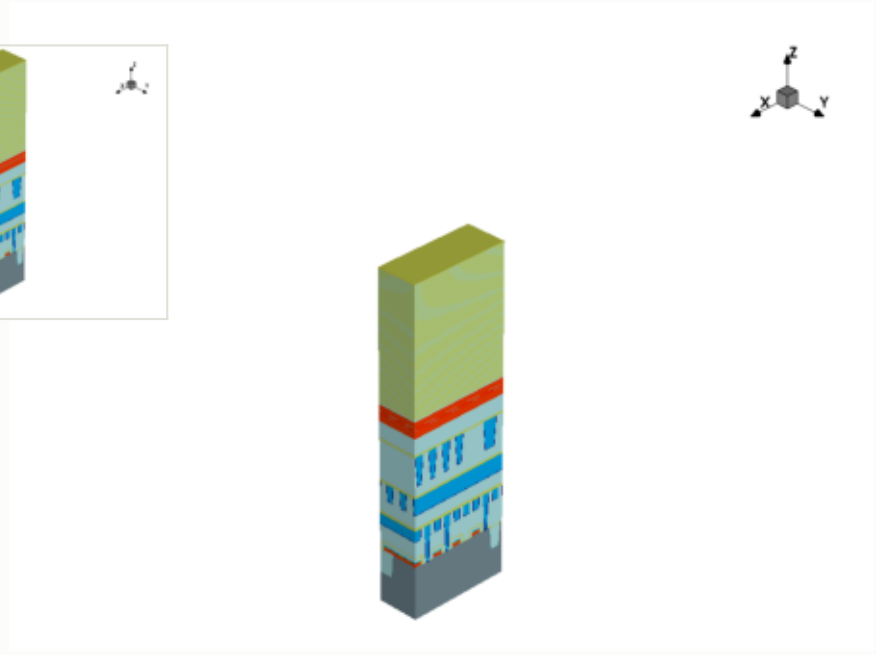
etch_strip · Dry_etch

0.801	284.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

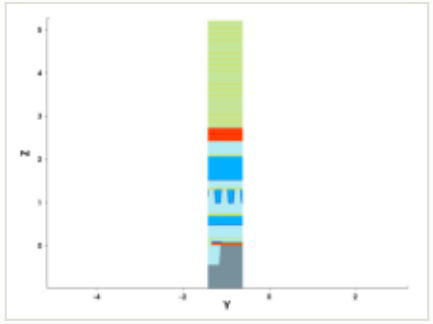
3D structure after this step

zoomed to fit

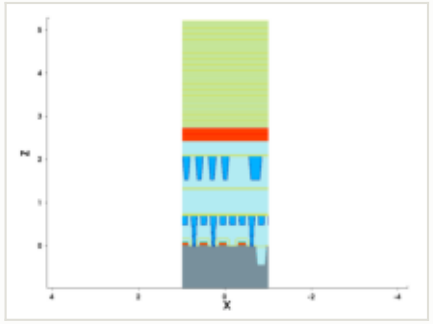




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

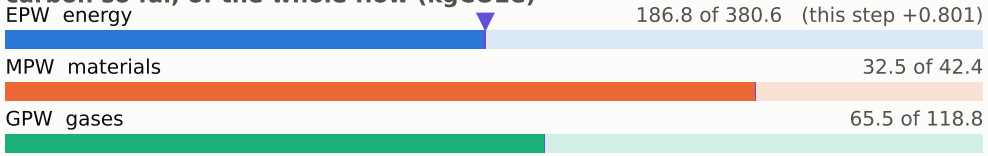
no mask for this step

this step

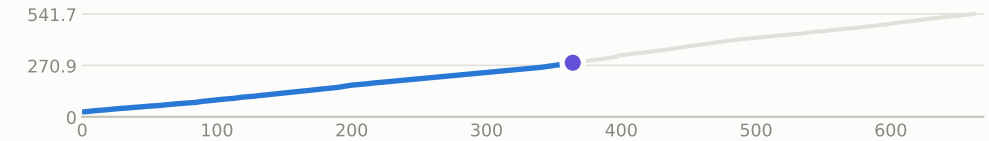
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

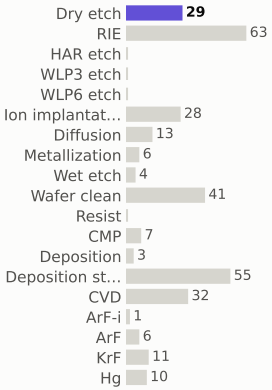
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

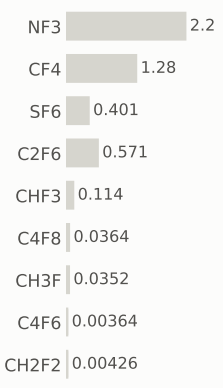


steps by process area



materials deposited (g)

gases emitted (g)



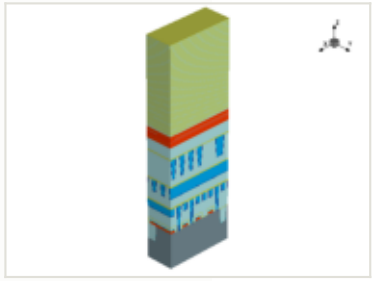
Step #3760: Wafer_clean

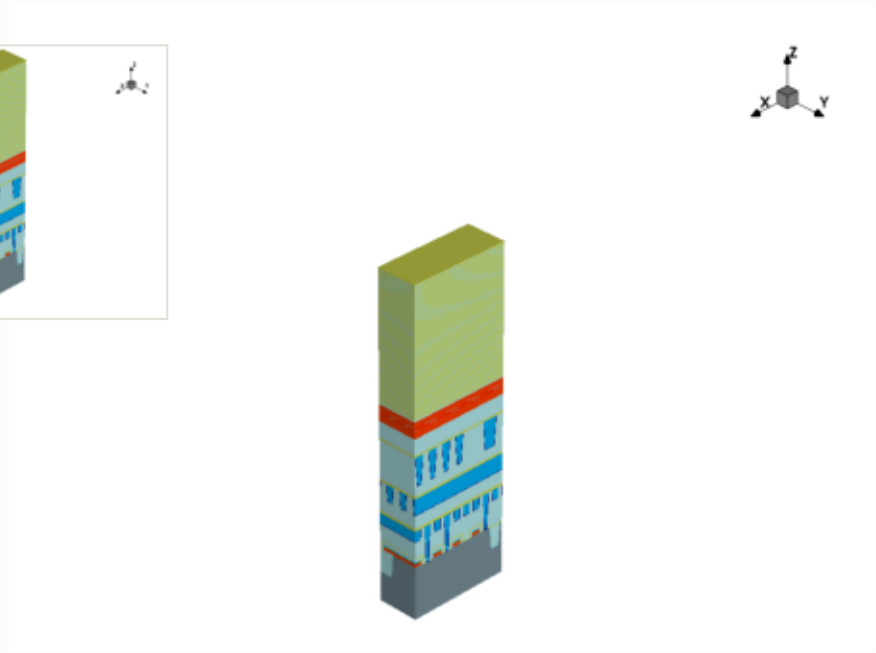
clean · Wafer_clean

0.169	284.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

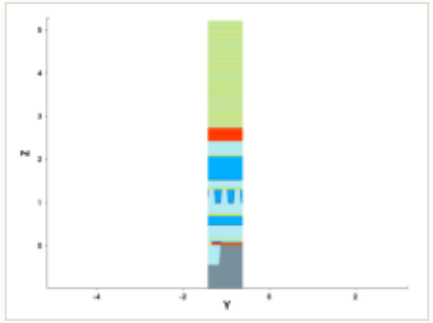
3D structure after this step

zoomed to fit

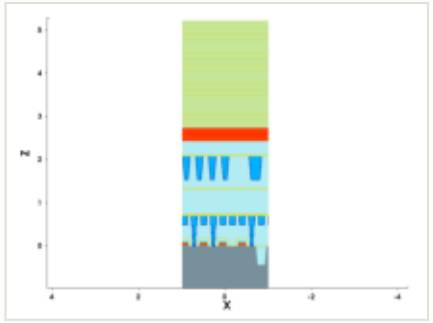




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

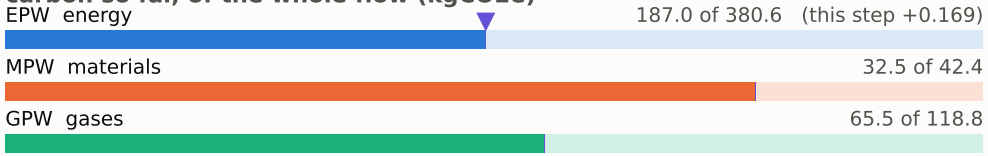
no mask for this step

this step

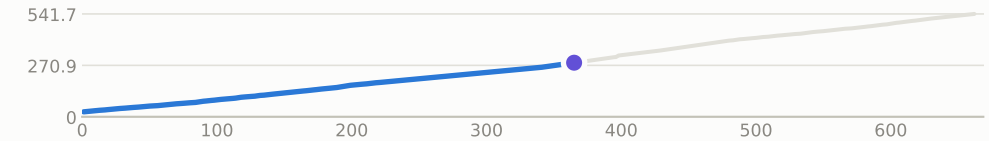
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

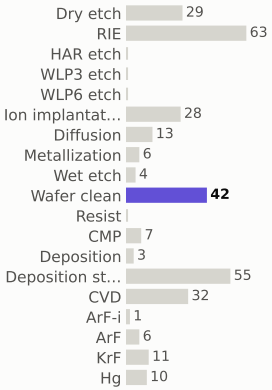
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

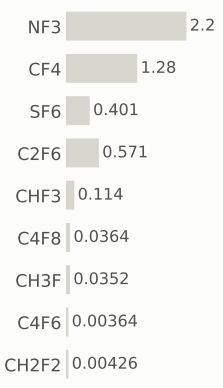


steps by process area



materials deposited (g)

gases emitted (g)



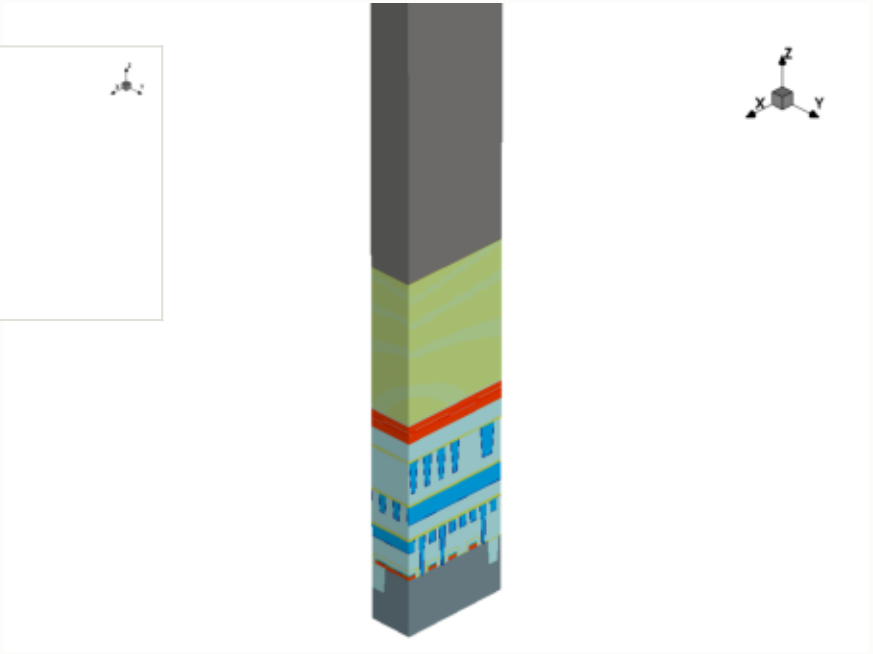
Step #3770: KrF

litho_patterning · KrF

0.199	285.1	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

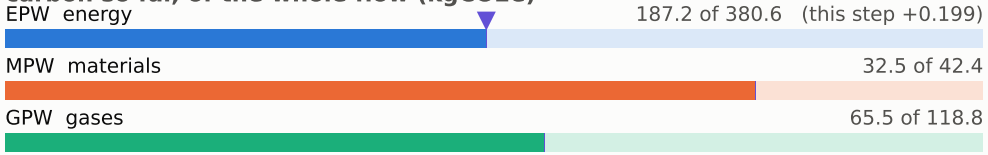


this step

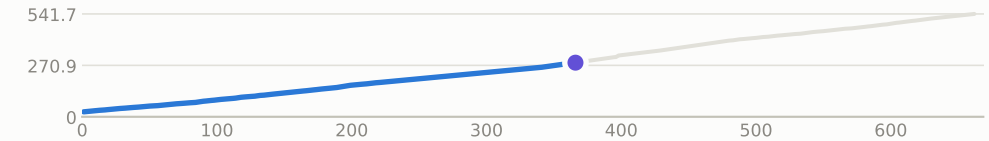
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

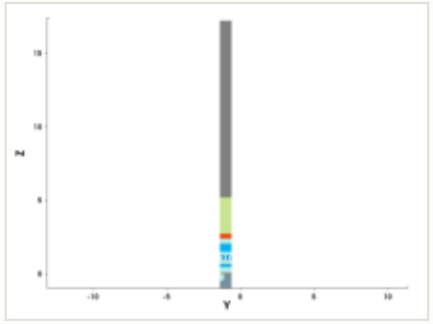
carbon so far, of the whole flow (kgCO2e)



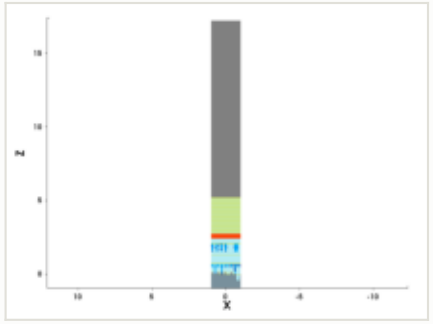
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

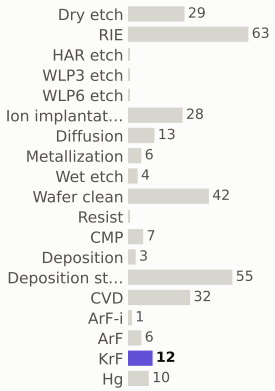


y cut at -1.032 µm



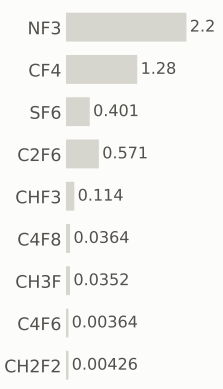
mask: layer 30:0, positive (OASIS)

steps by process area



materials deposited (g)

gases emitted (g)



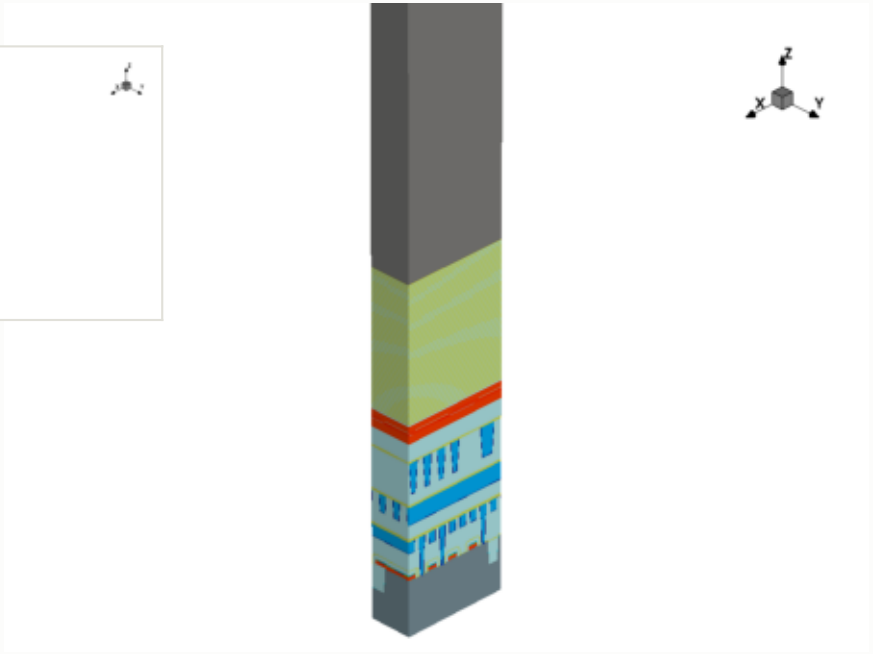
Step #3780: RIE

etch_anisotropic · RIE

1.04	286.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

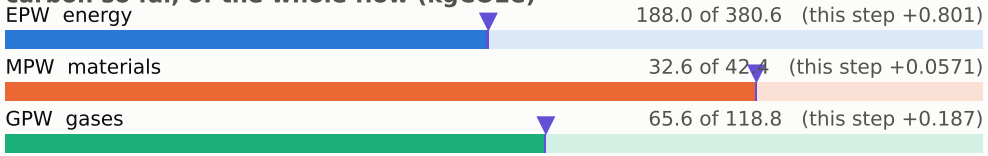


this step

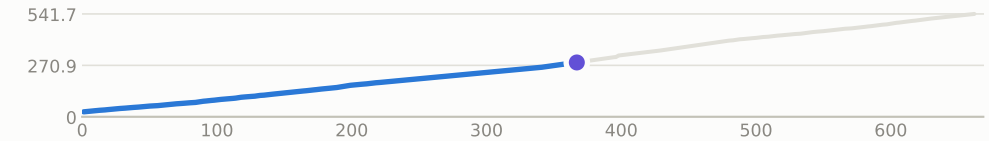
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

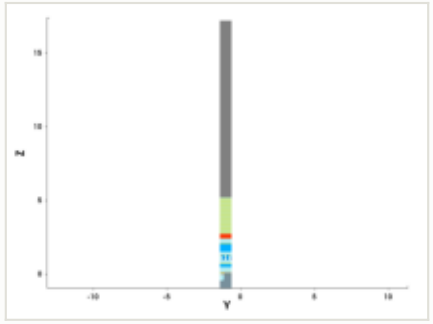
carbon so far, of the whole flow (kgCO2e)



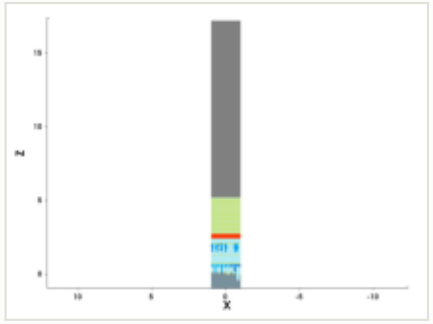
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



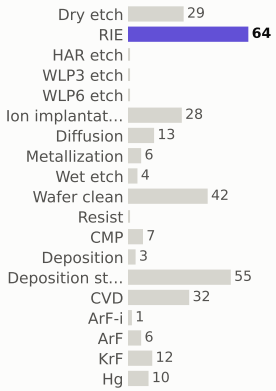
y cut at -1.032 µm



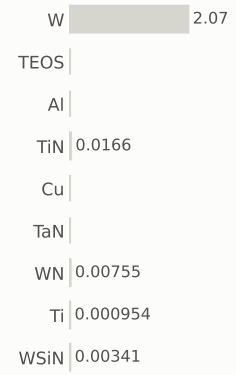
mask for this step

no mask for this step

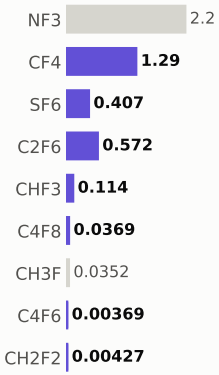
steps by process area



materials deposited (g)



gases emitted (g)



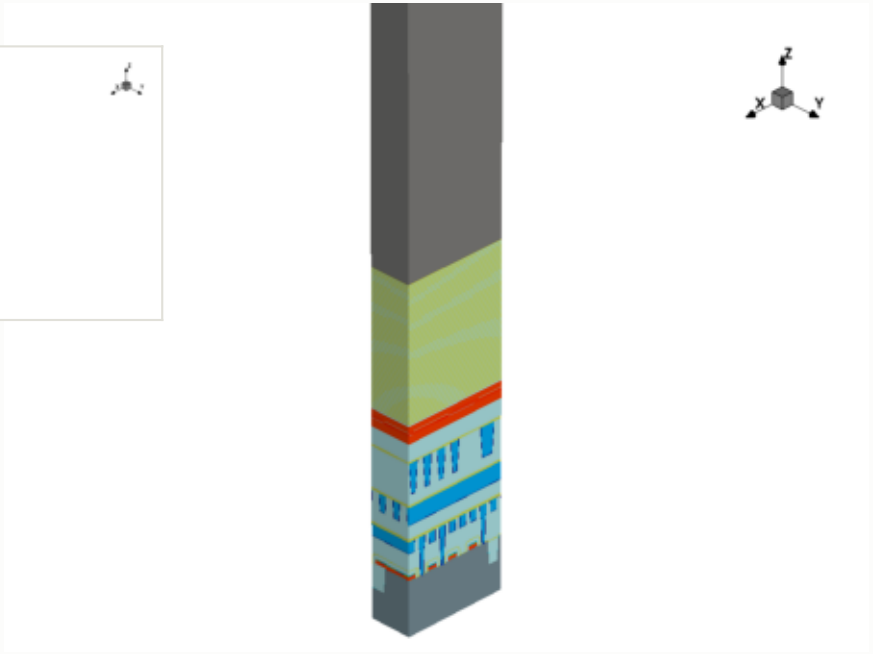
Step #3790: RIE

etch_anisotropic · RIE

1.04	287.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

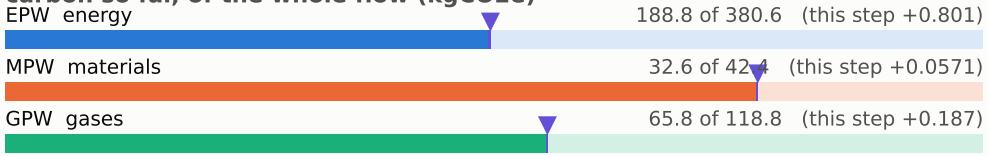


this step

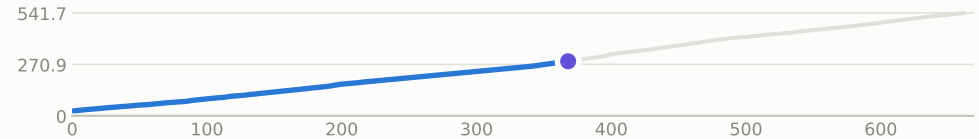
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

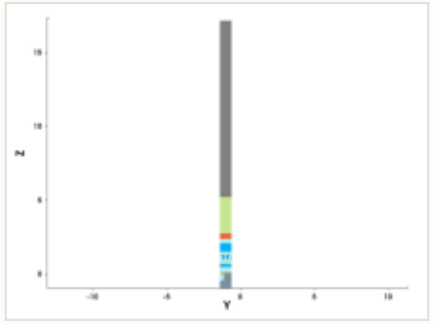
carbon so far, of the whole flow (kgCO2e)



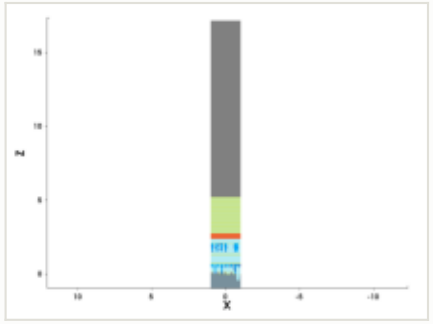
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



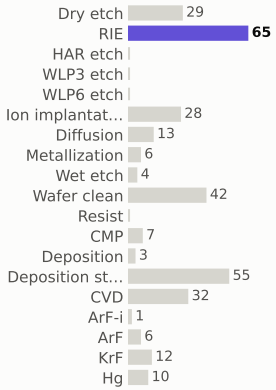
y cut at -1.032 µm



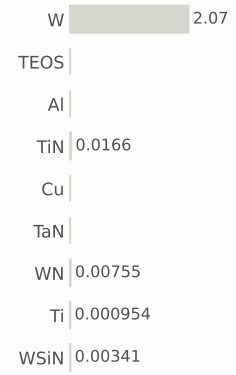
mask for this step

no mask for this step

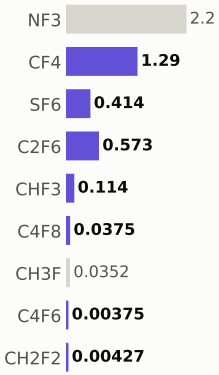
steps by process area



materials deposited (g)



gases emitted (g)



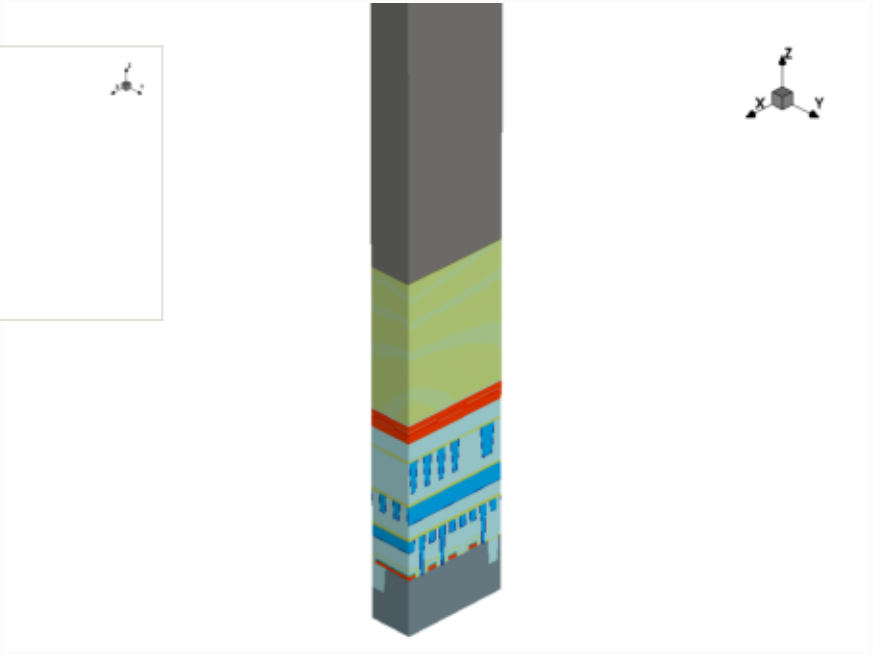
Step #3800: RIE

etch_anisotropic · RIE

1.04	288.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

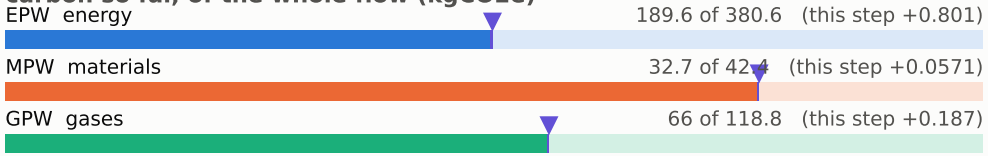


this step

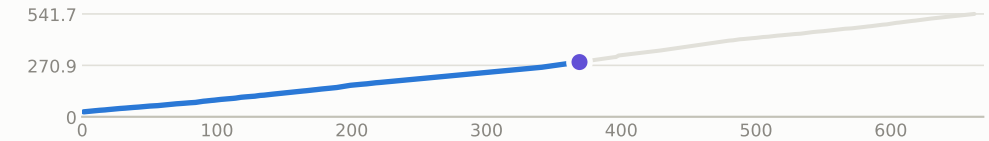
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

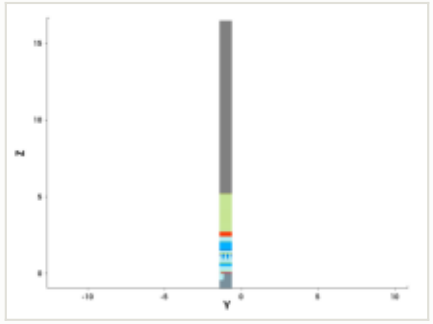
carbon so far, of the whole flow (kgCO2e)



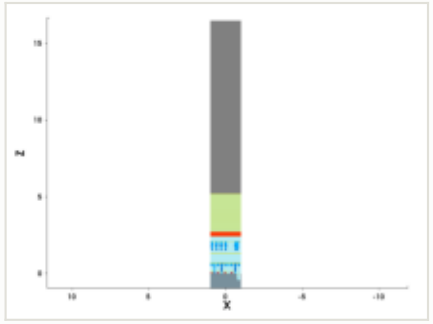
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



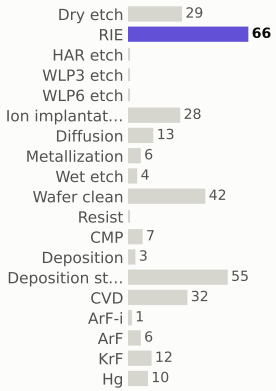
y cut at -1.032 µm



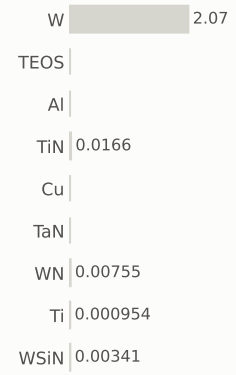
mask for this step

no mask for this step

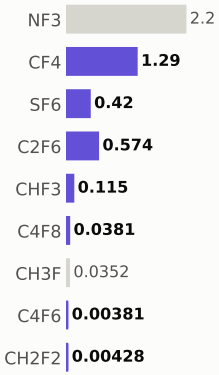
steps by process area



materials deposited (g)



gases emitted (g)



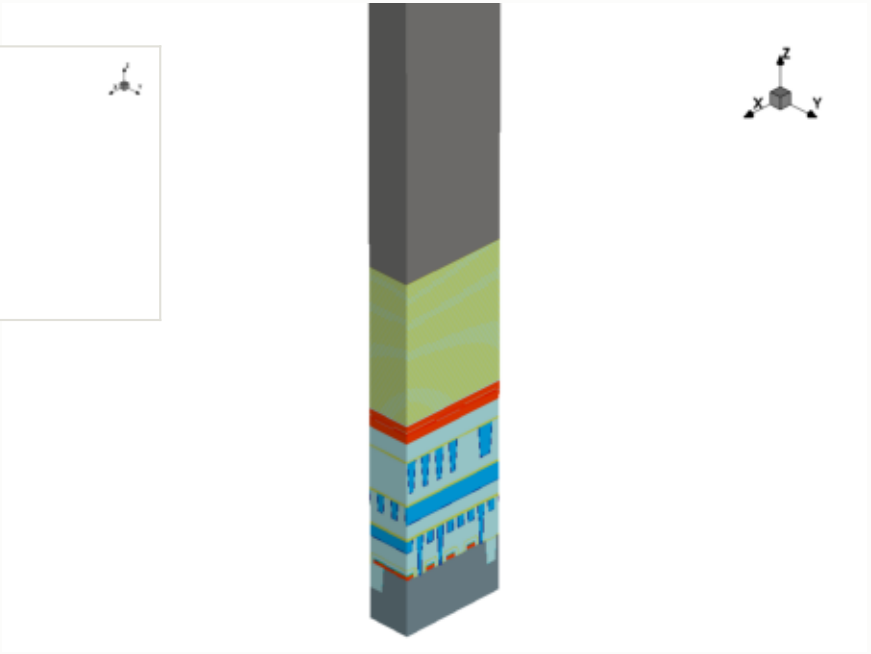
Step #3810: RIE

etch_anisotropic · RIE

1.04	289.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

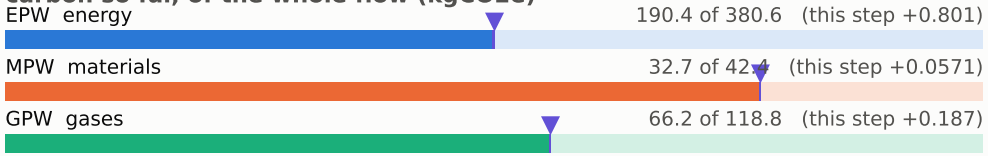


this step

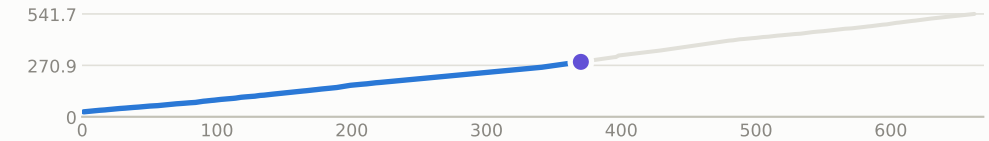
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

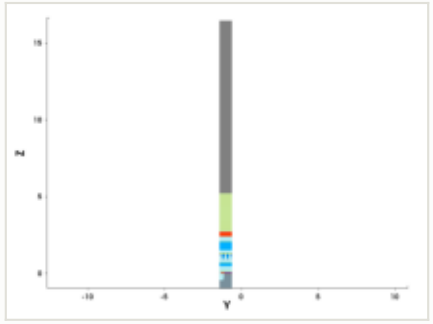
carbon so far, of the whole flow (kgCO2e)



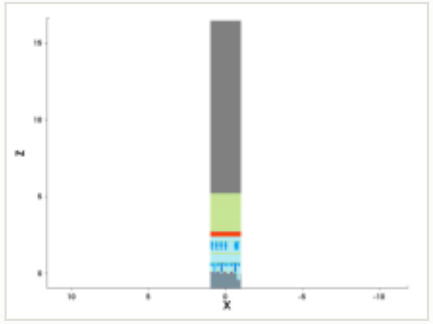
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



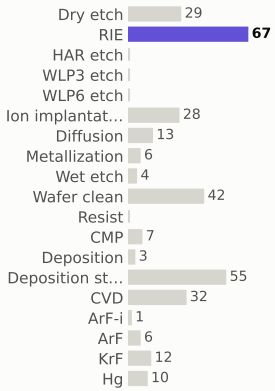
y cut at -1.032 µm



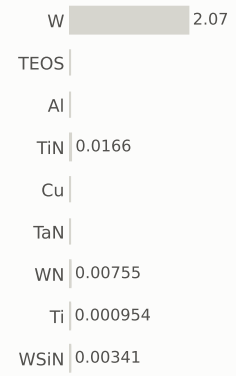
mask for this step

no mask for this step

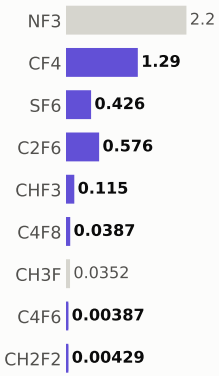
steps by process area



materials deposited (g)



gases emitted (g)



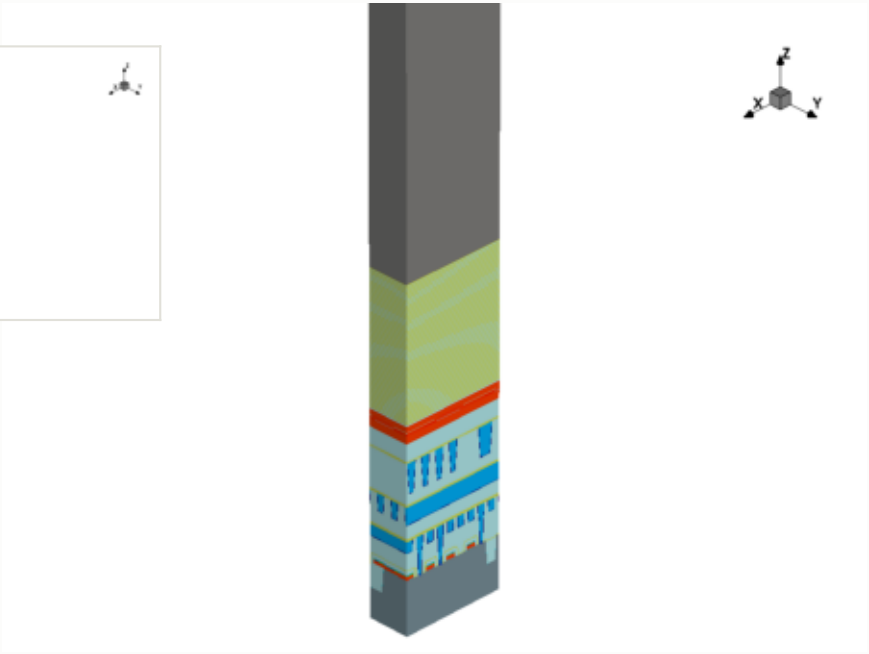
Step #3820: RIE

etch_anisotropic · RIE

1.04	290.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

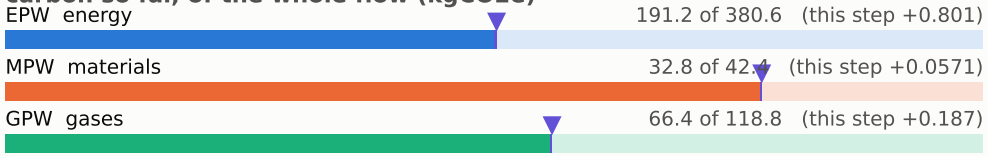


this step

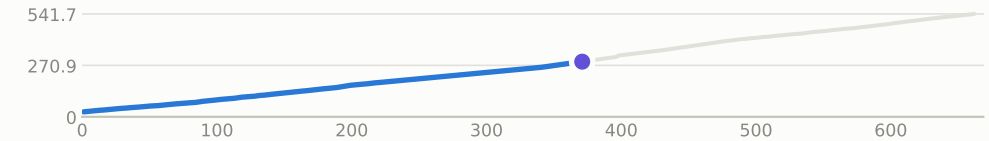
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

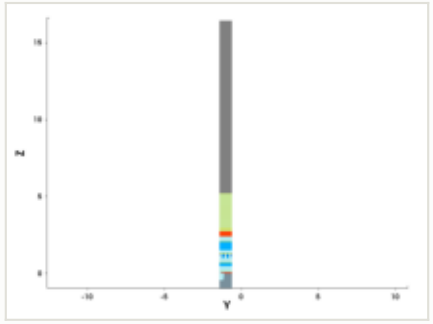
carbon so far, of the whole flow (kgCO2e)



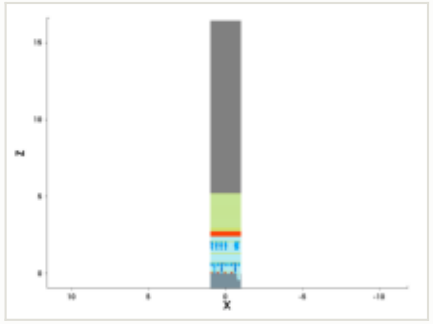
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



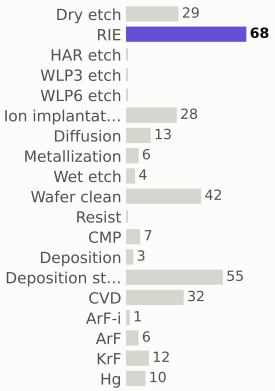
y cut at -1.032 µm



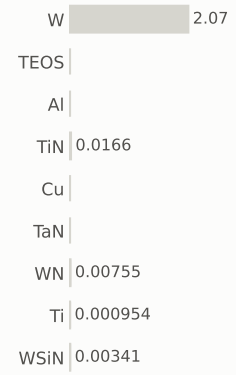
mask for this step

no mask for this step

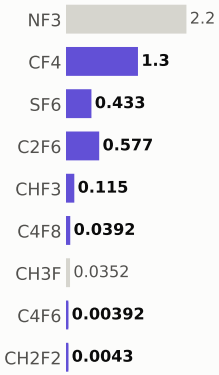
steps by process area



materials deposited (g)



gases emitted (g)



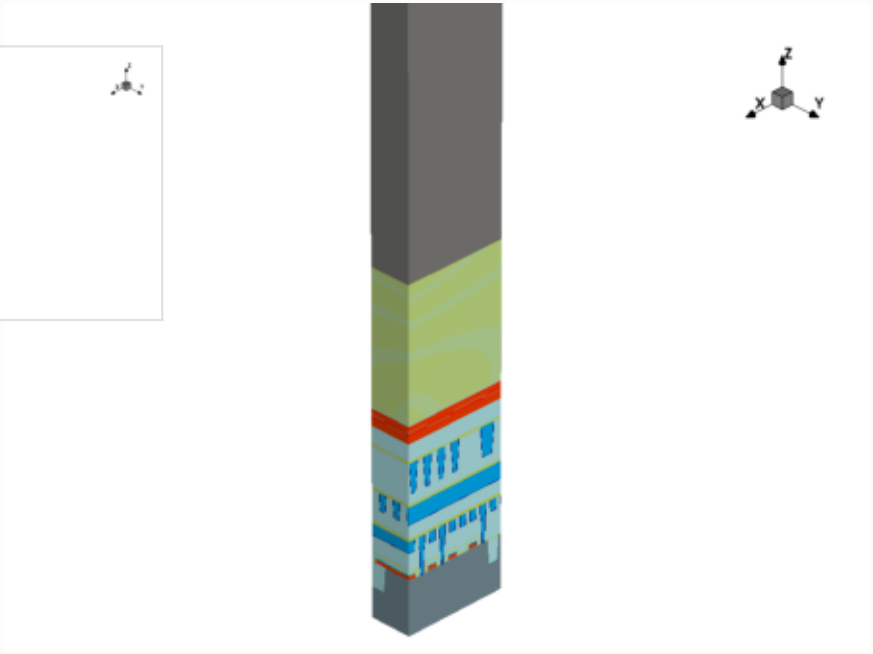
Step #3830: RIE

etch_anisotropic · RIE

1.04	291.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

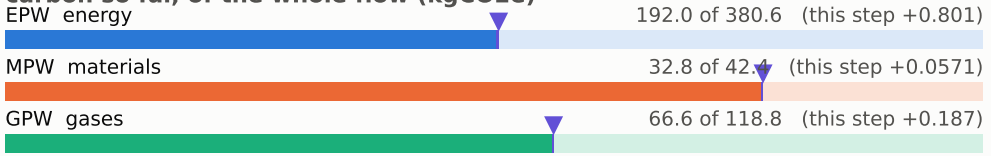


this step

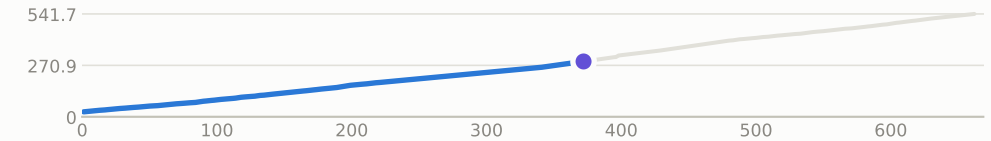
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

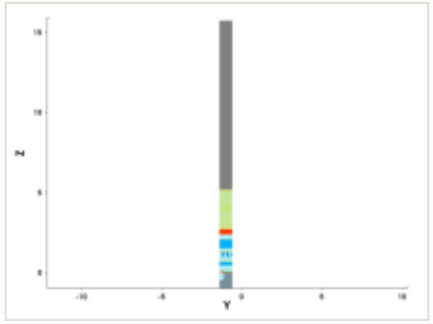
carbon so far, of the whole flow (kgCO2e)



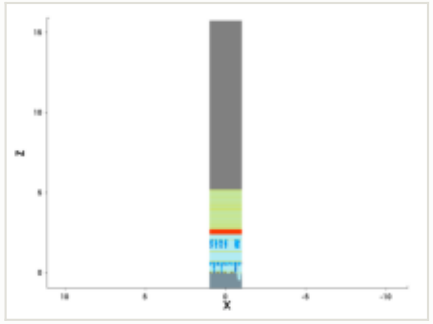
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



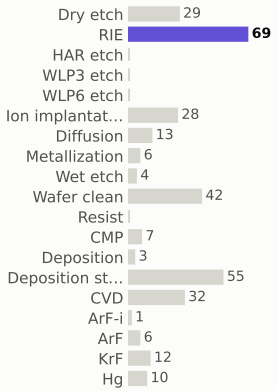
y cut at -1.032 µm



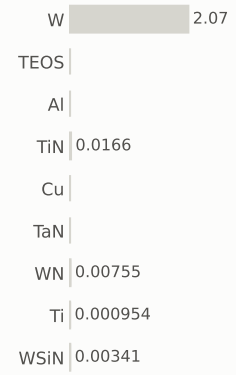
mask for this step

no mask for this step

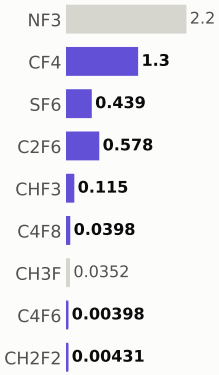
steps by process area



materials deposited (g)



gases emitted (g)



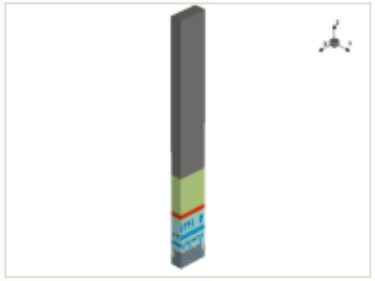
Step #3840: RIE

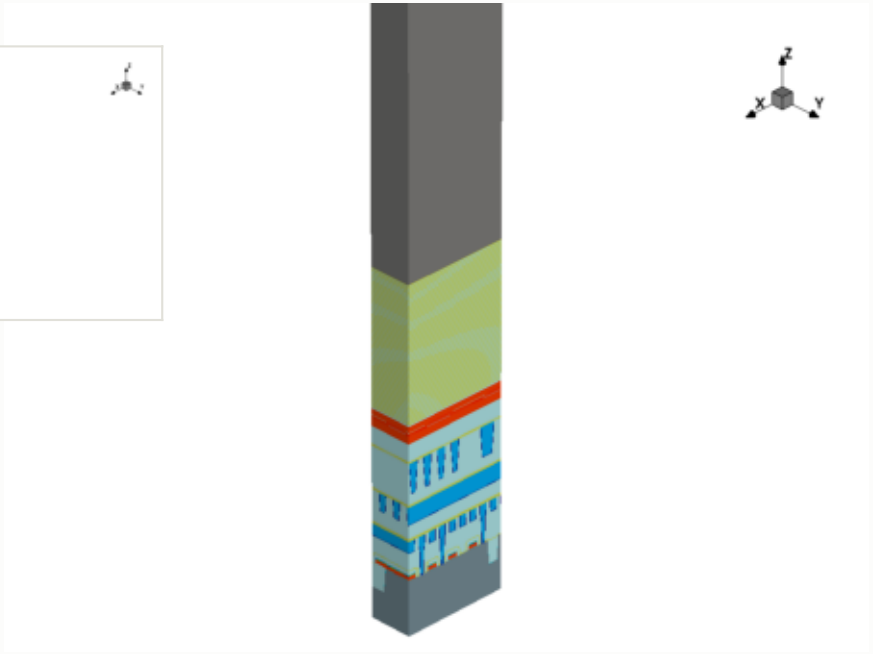
etch_anisotropic · RIE

1.04	292.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



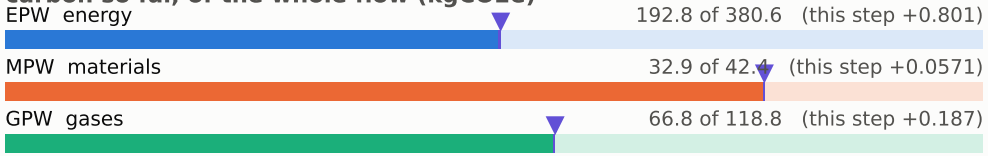


this step

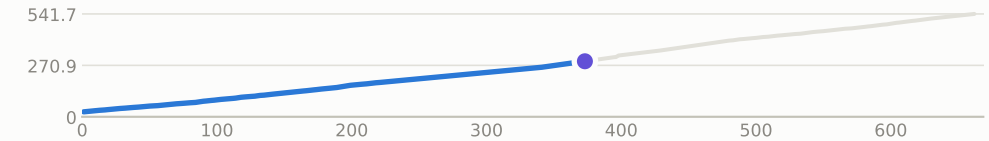
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

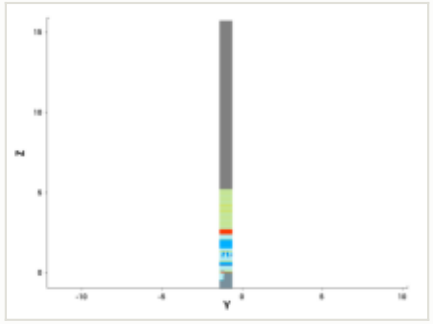
carbon so far, of the whole flow (kgCO2e)



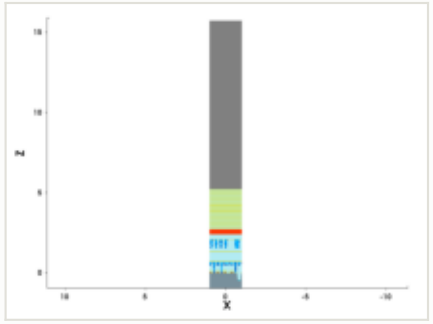
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



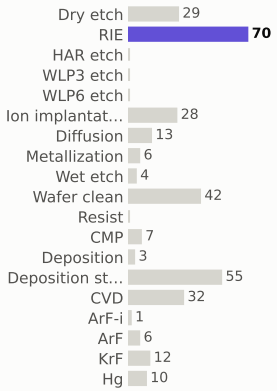
y cut at -1.032 µm



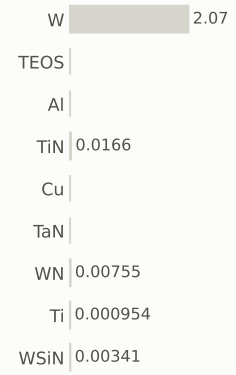
mask for this step

no mask for this step

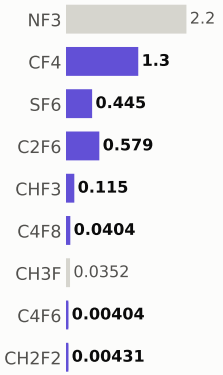
steps by process area



materials deposited (g)



gases emitted (g)



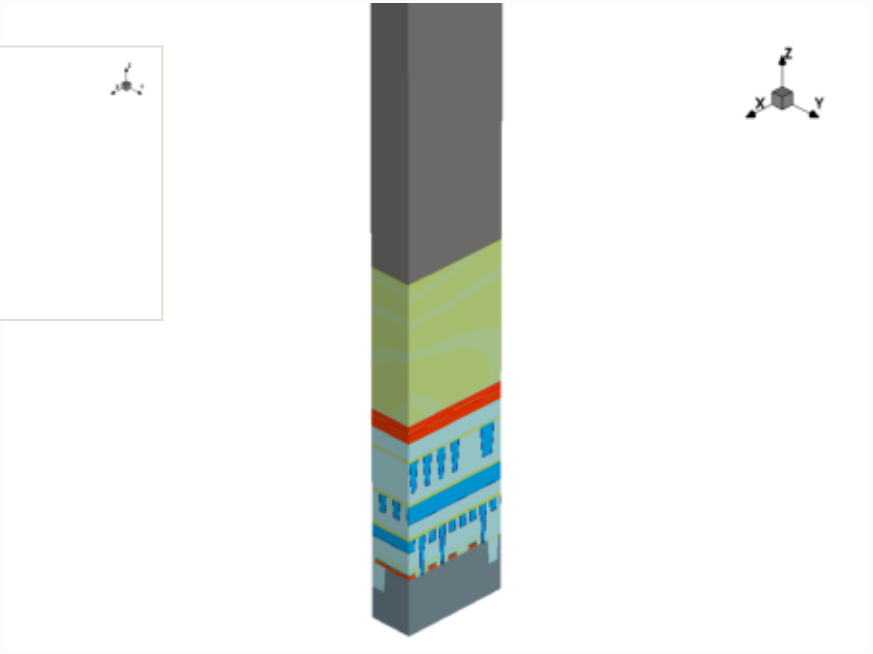
Step #3850: RIE

etch_anisotropic · RIE

1.04	293.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

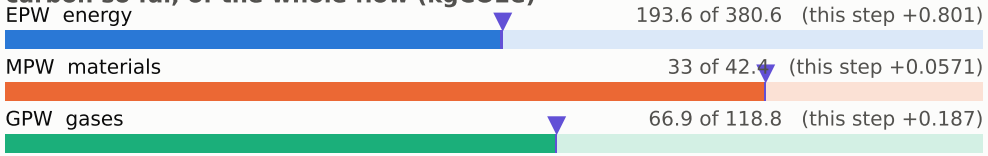


this step

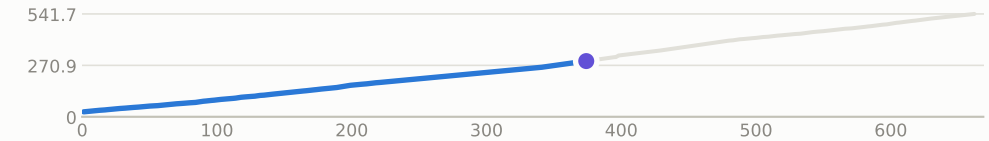
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

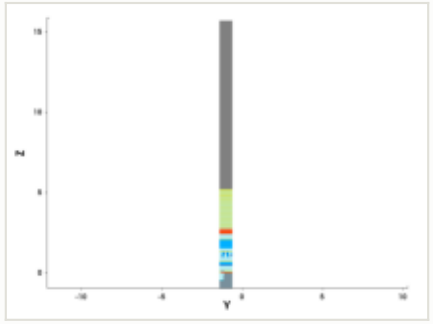
carbon so far, of the whole flow (kgCO2e)



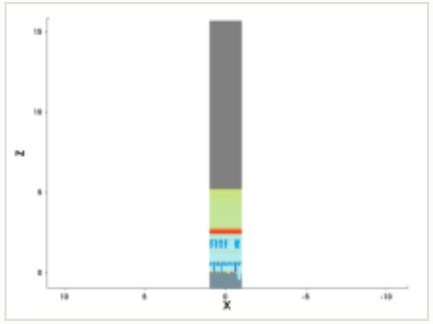
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



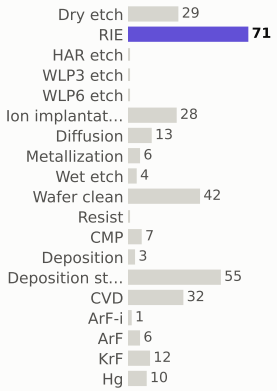
y cut at -1.032 µm



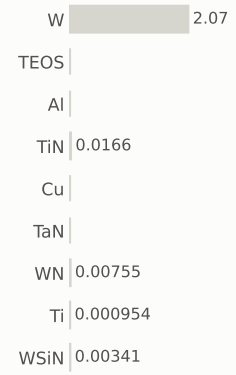
mask for this step

no mask for this step

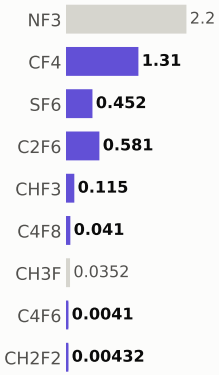
steps by process area



materials deposited (g)



gases emitted (g)



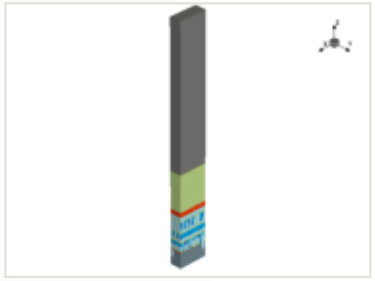
Step #3860: RIE

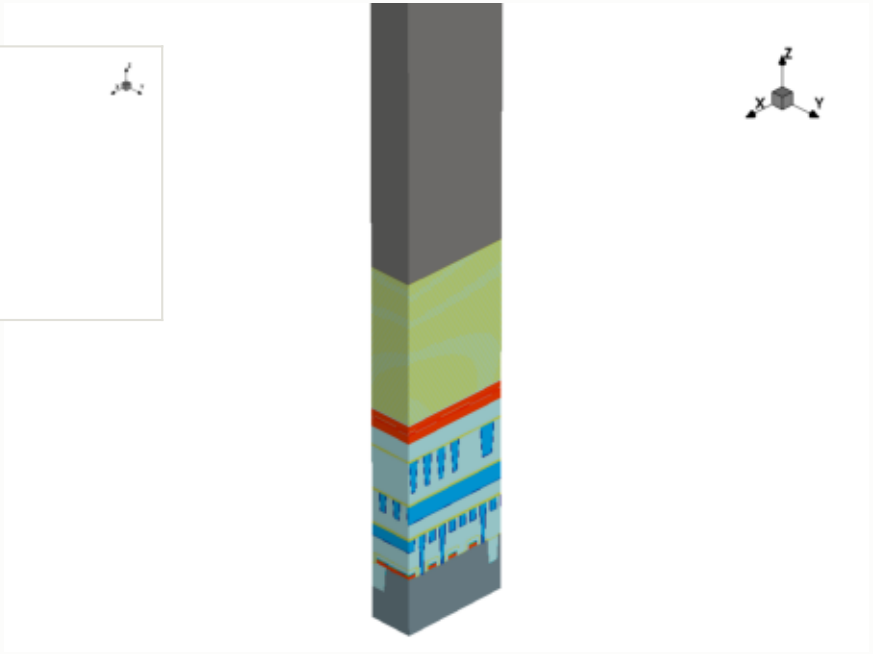
etch_anisotropic · RIE

1.04	294.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



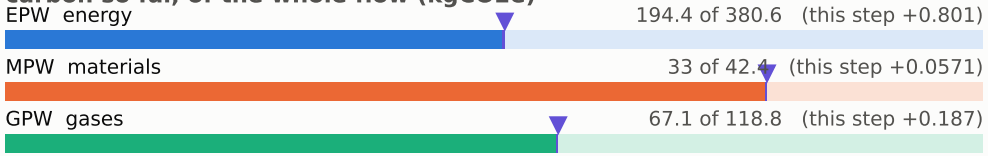


this step

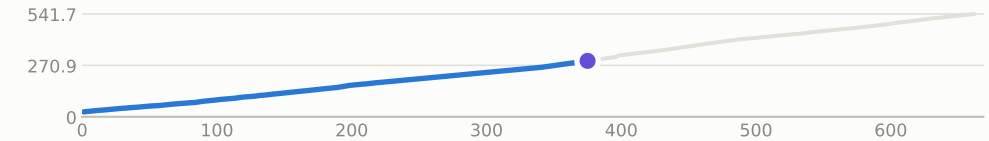
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

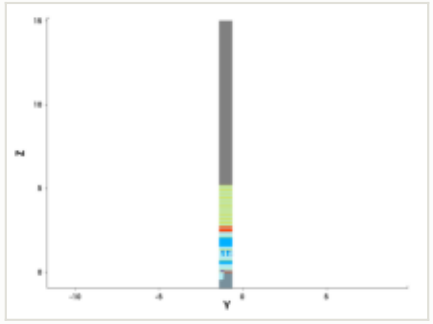
carbon so far, of the whole flow (kgCO2e)



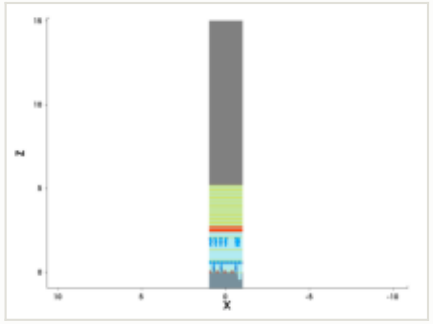
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



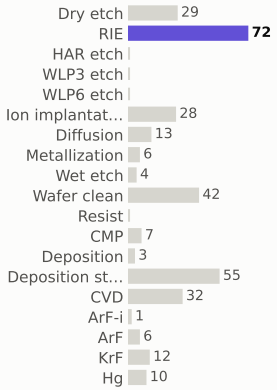
y cut at -1.032 µm



mask for this step

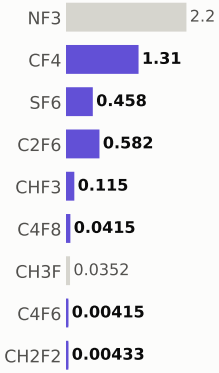
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



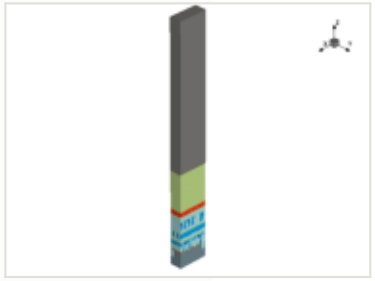
Step #3870: RIE

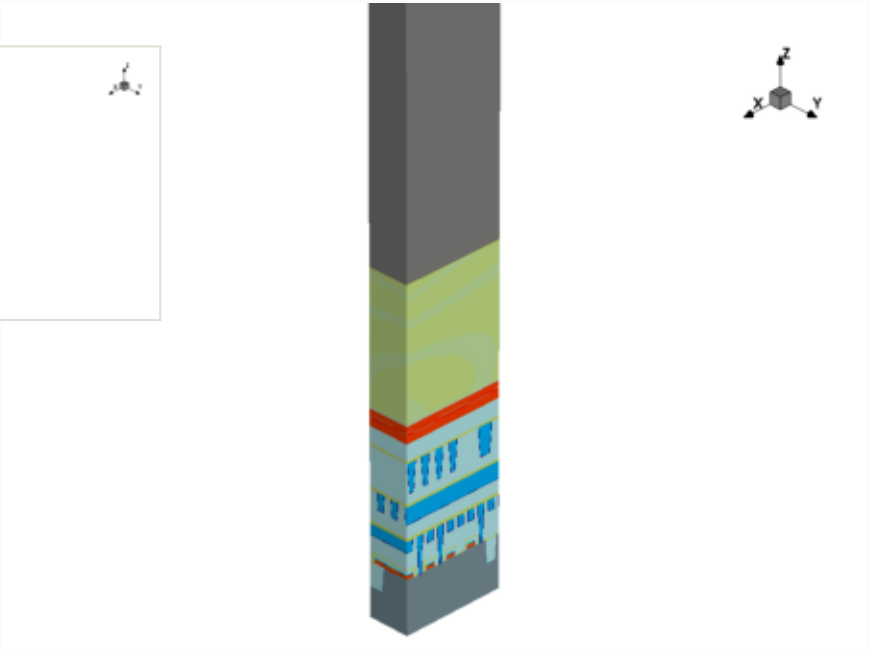
etch_anisotropic · RIE

1.04	295.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





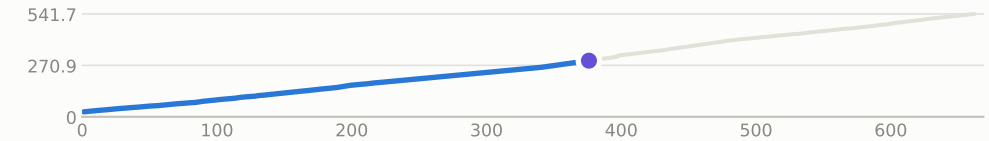
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

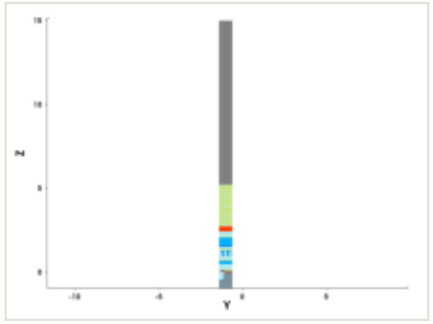
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

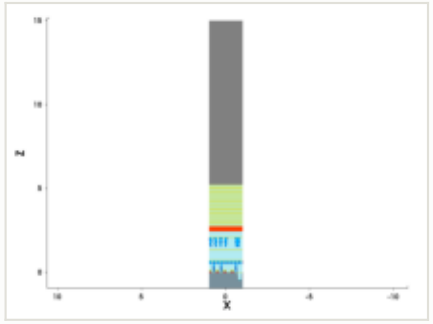
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



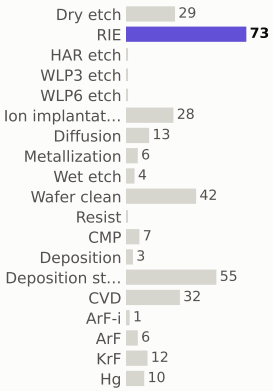
y cut at -1.032 µm



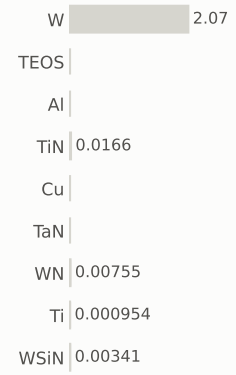
mask for this step

no mask for this step

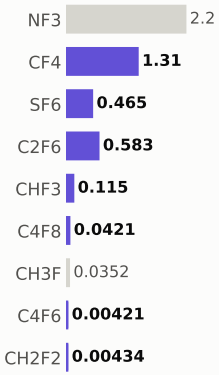
steps by process area



materials deposited (g)



gases emitted (g)



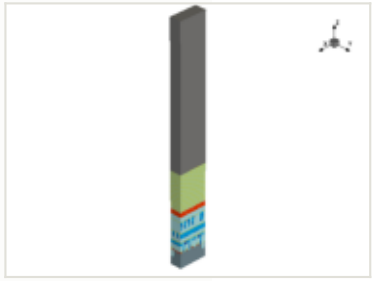
Step #3880: RIE

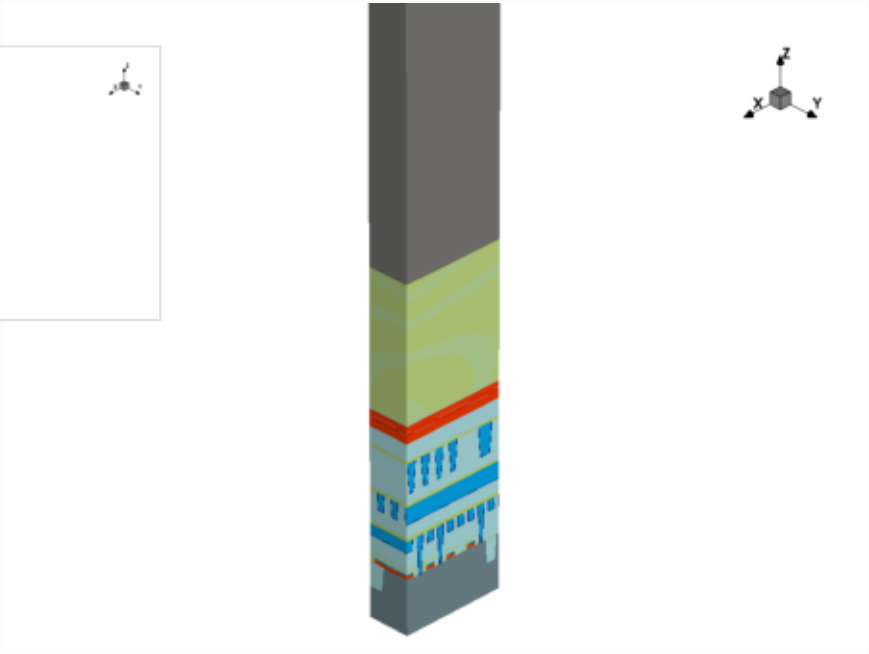
etch_anisotropic · RIE

1.04	296.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



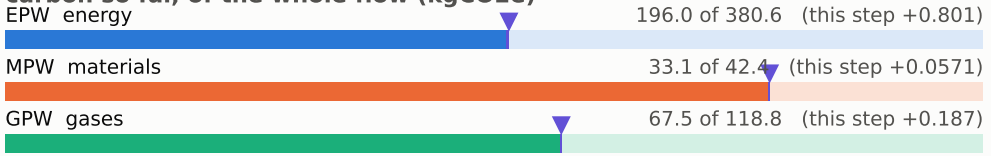


this step

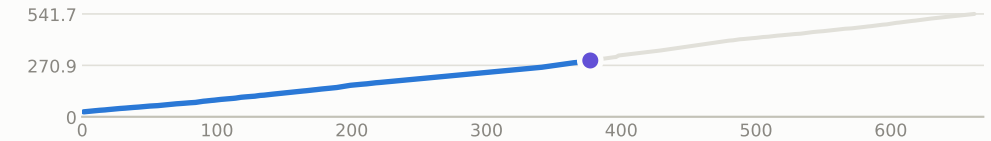
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

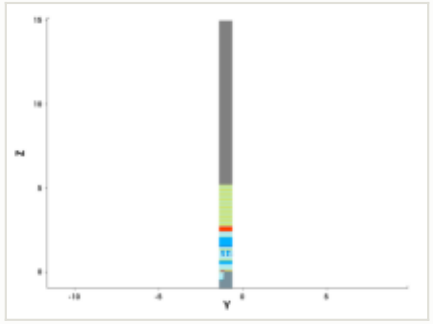
carbon so far, of the whole flow (kgCO2e)



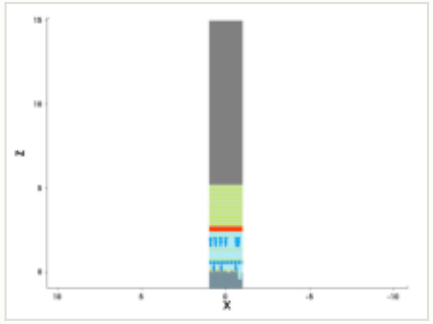
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



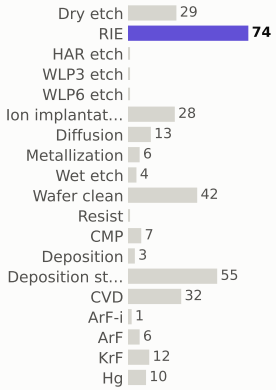
y cut at -1.032 µm



mask for this step

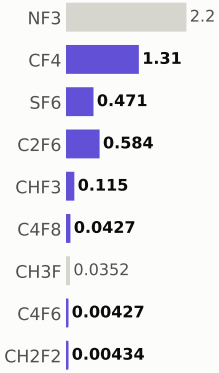
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



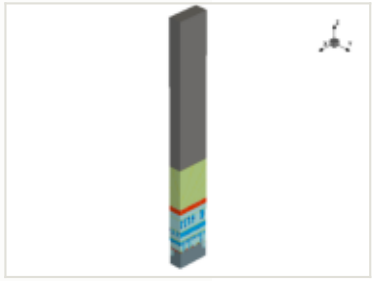
Step #3890: RIE

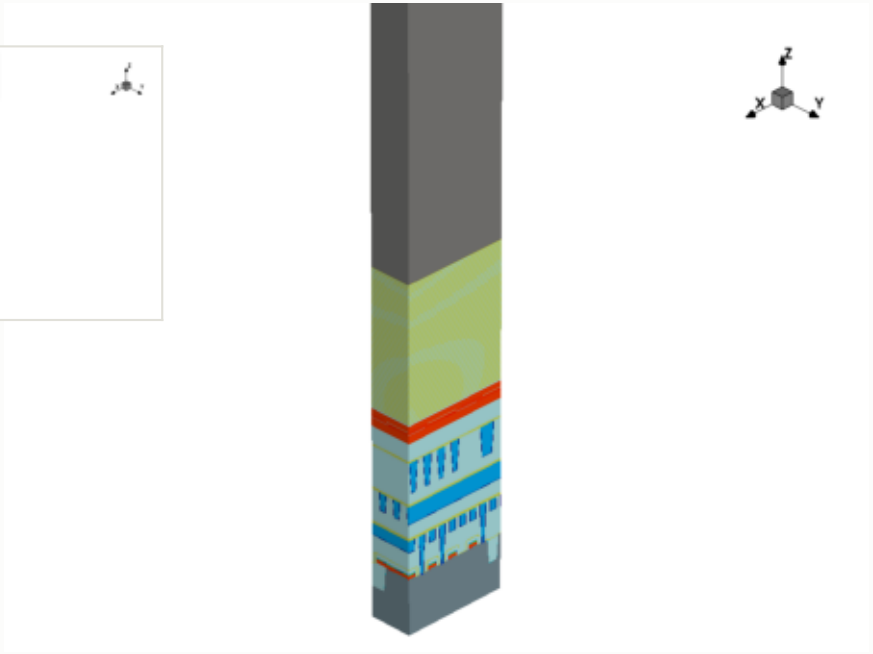
etch_anisotropic · RIE

1.04	297.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



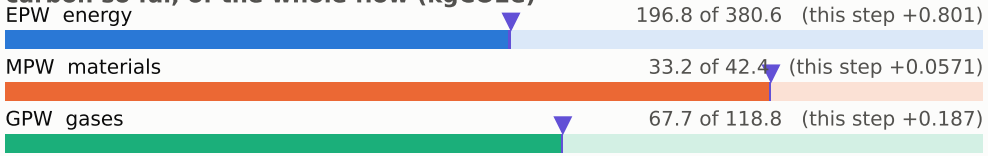


this step

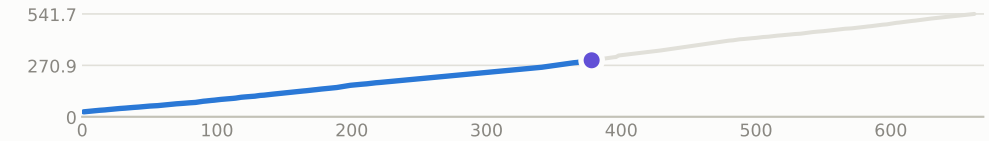
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

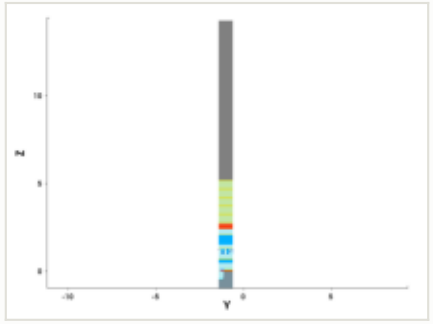
carbon so far, of the whole flow (kgCO2e)



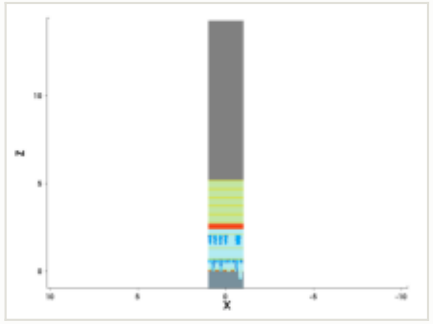
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



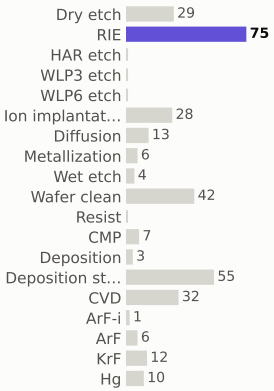
y cut at -1.032 µm



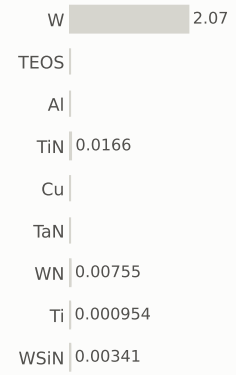
mask for this step

no mask for this step

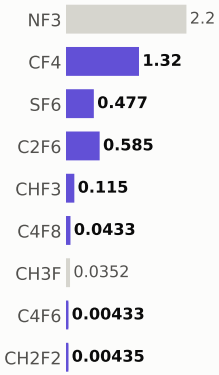
steps by process area



materials deposited (g)



gases emitted (g)



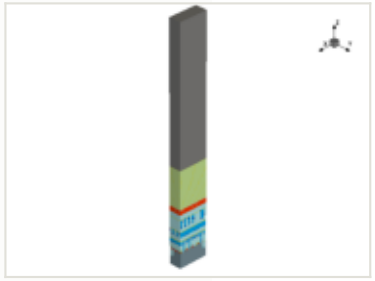
Step #3900: RIE

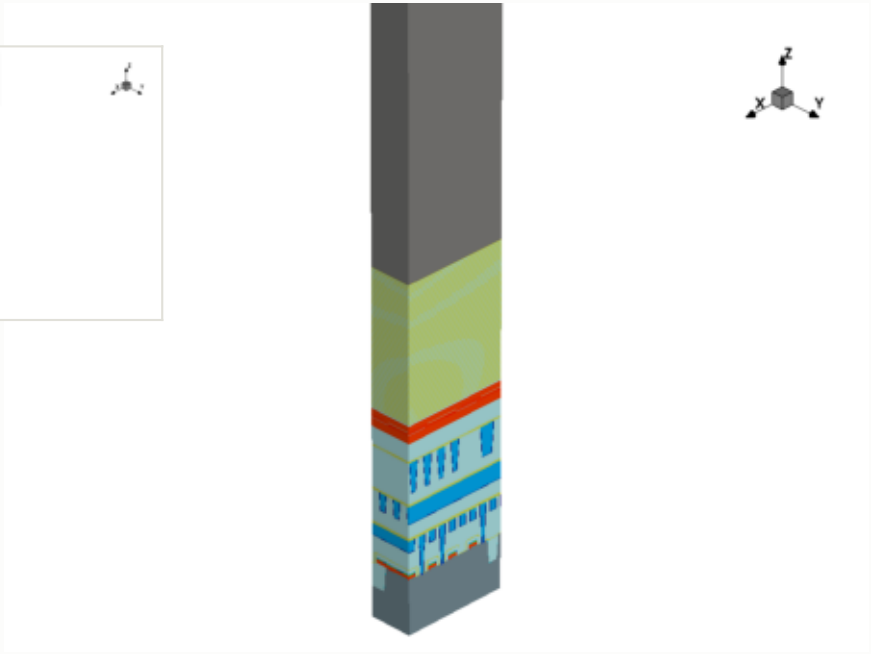
etch_anisotropic · RIE

1.04	298.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



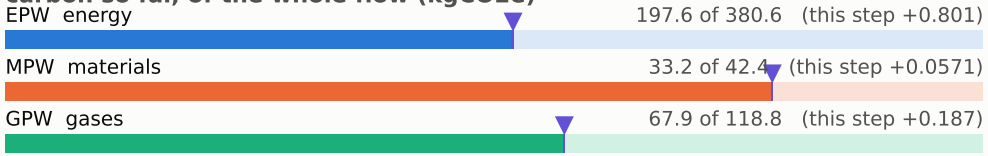


this step

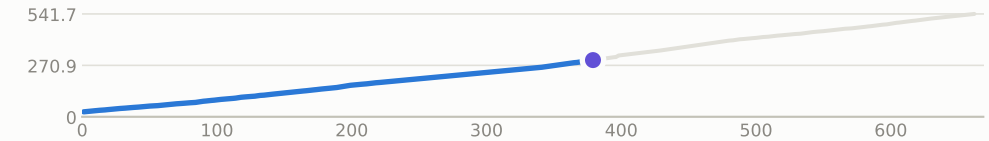
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

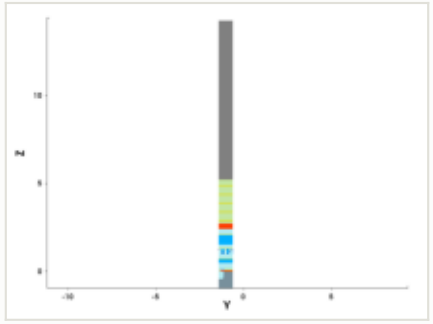
carbon so far, of the whole flow (kgCO2e)



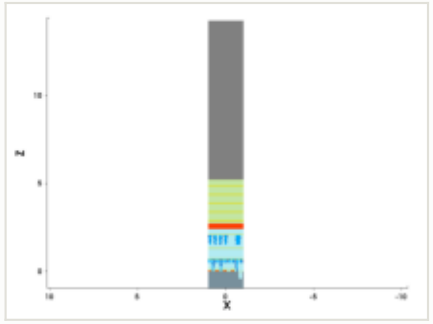
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



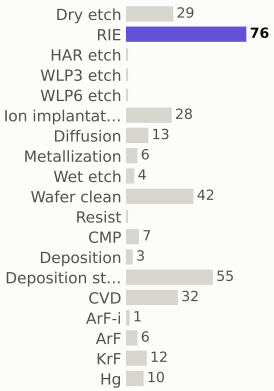
y cut at -1.032 µm



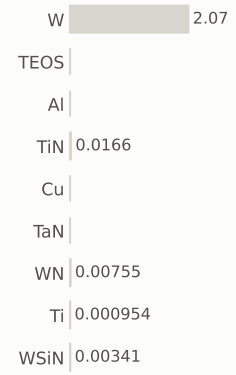
mask for this step

no mask for this step

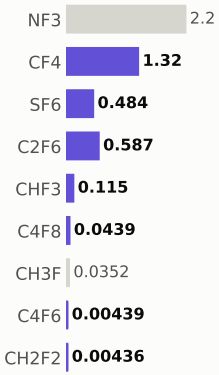
steps by process area



materials deposited (g)



gases emitted (g)



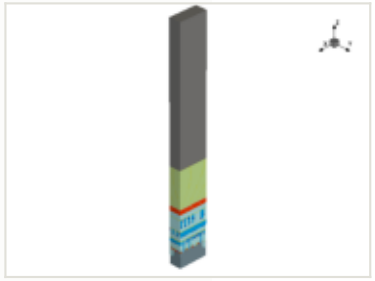
Step #3910: RIE

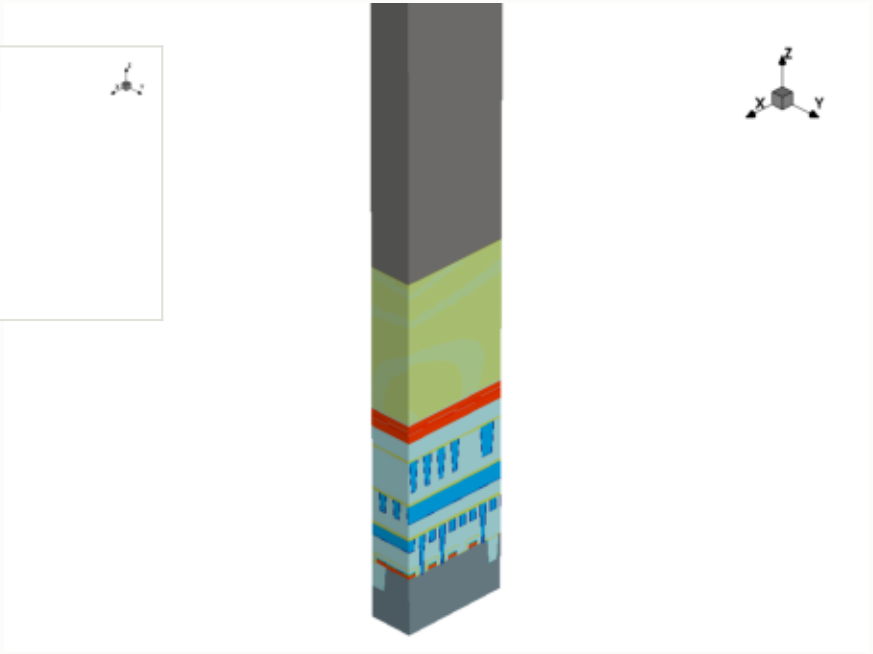
etch_anisotropic · RIE

1.04	299.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





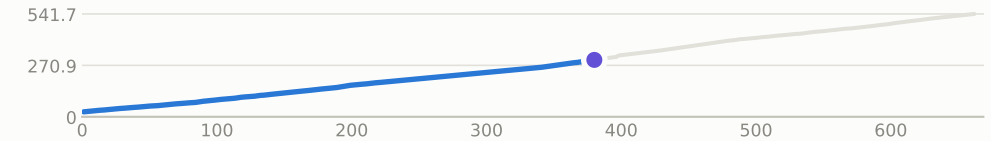
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

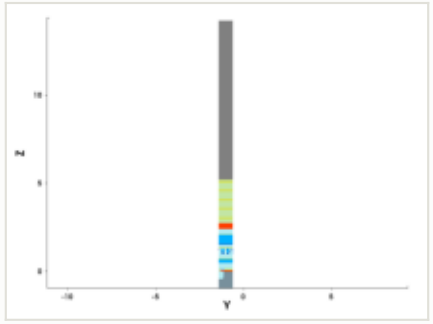
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

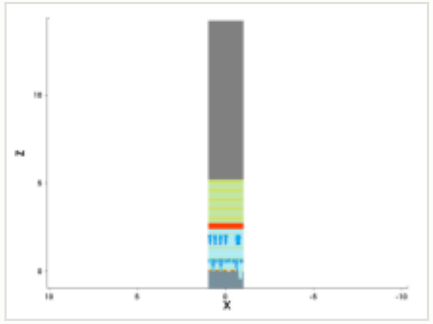
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



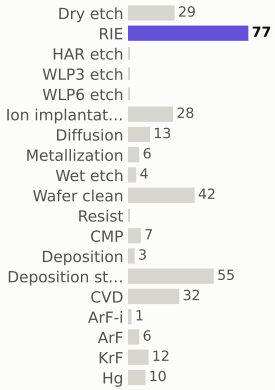
y cut at -1.032 µm



mask for this step

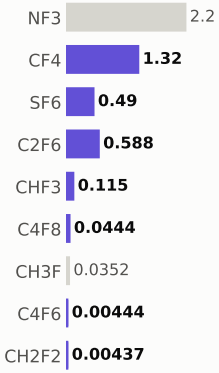
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



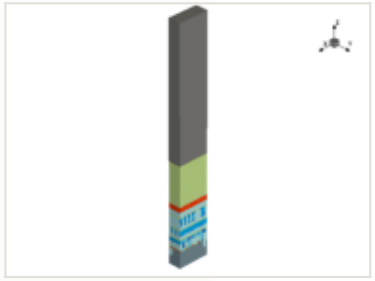
Step #3920: RIE

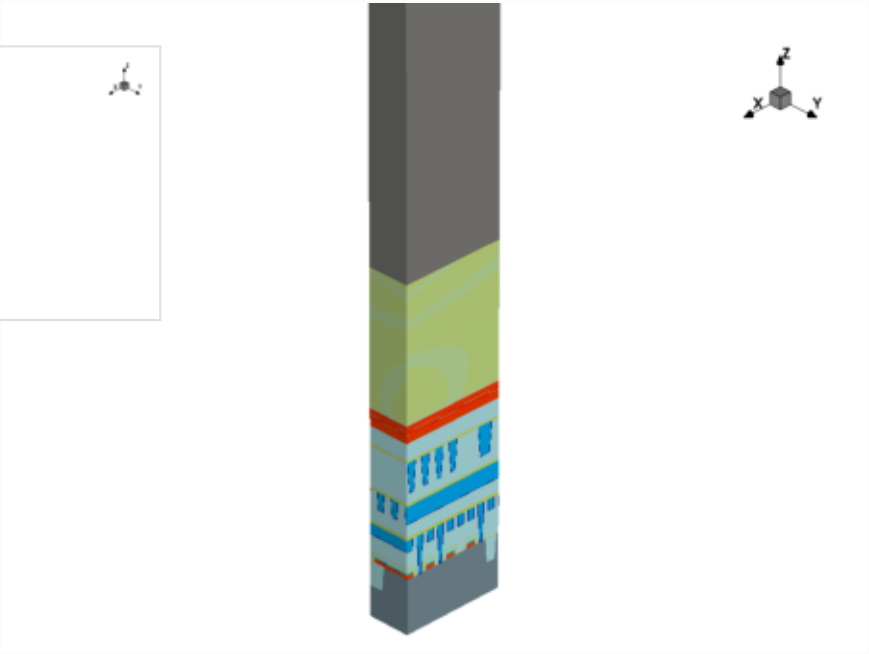
etch_anisotropic · RIE

1.04	300.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



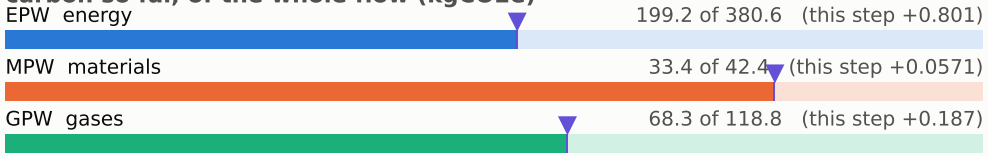


this step

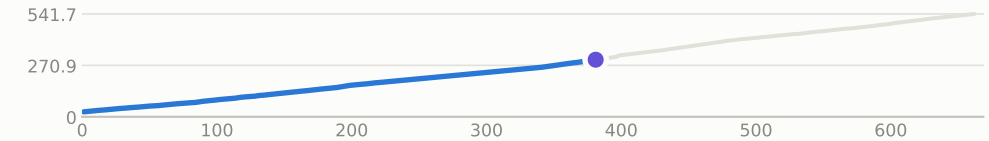
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

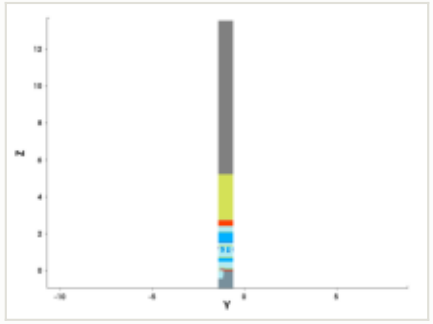
carbon so far, of the whole flow (kgCO2e)



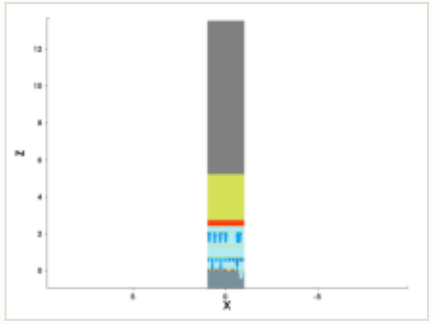
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



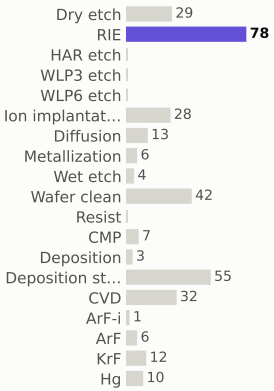
y cut at -1.032 µm



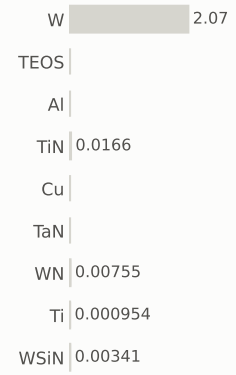
mask for this step

no mask for this step

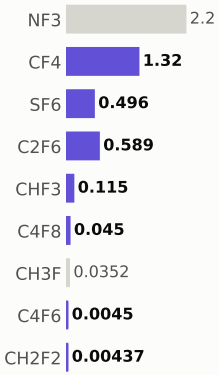
steps by process area



materials deposited (g)



gases emitted (g)



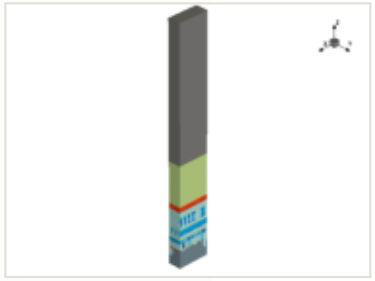
Step #3930: RIE

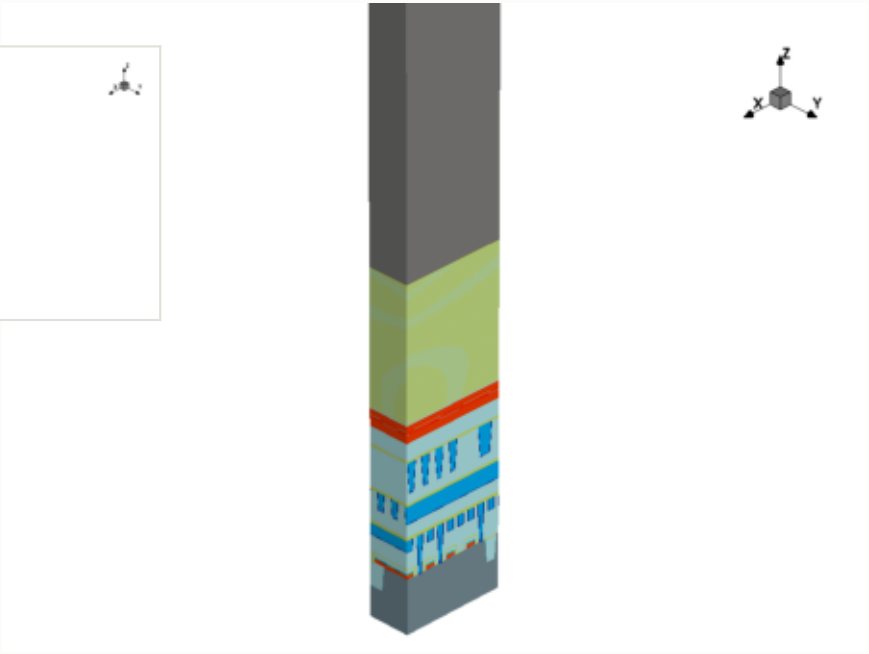
etch_anisotropic · RIE

1.04	301.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





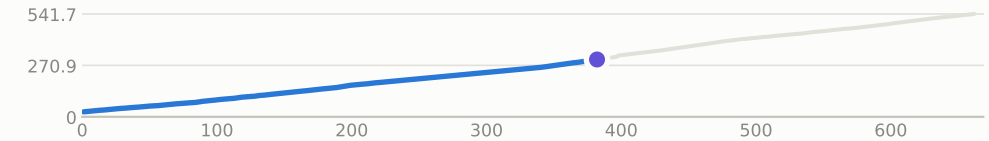
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

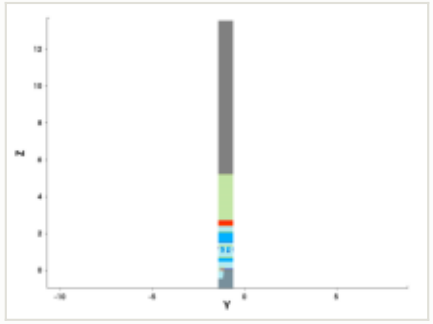
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

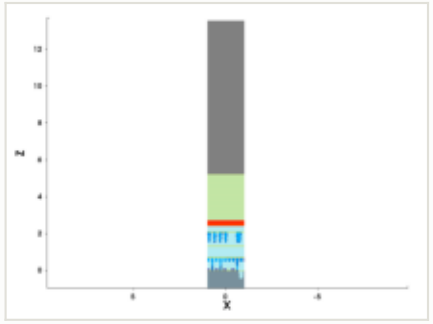
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



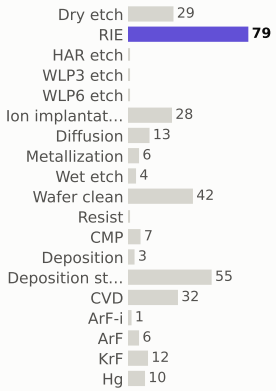
y cut at -1.032 µm



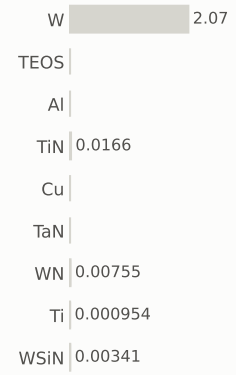
mask for this step

no mask for this step

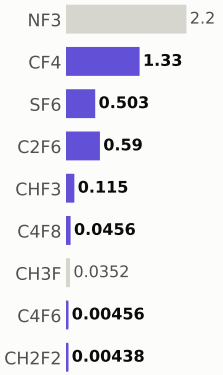
steps by process area



materials deposited (g)



gases emitted (g)



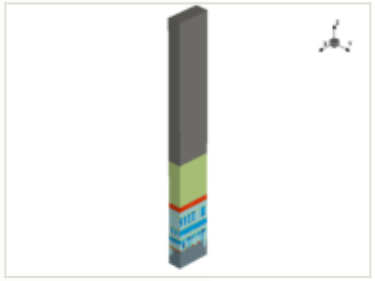
Step #3940: RIE

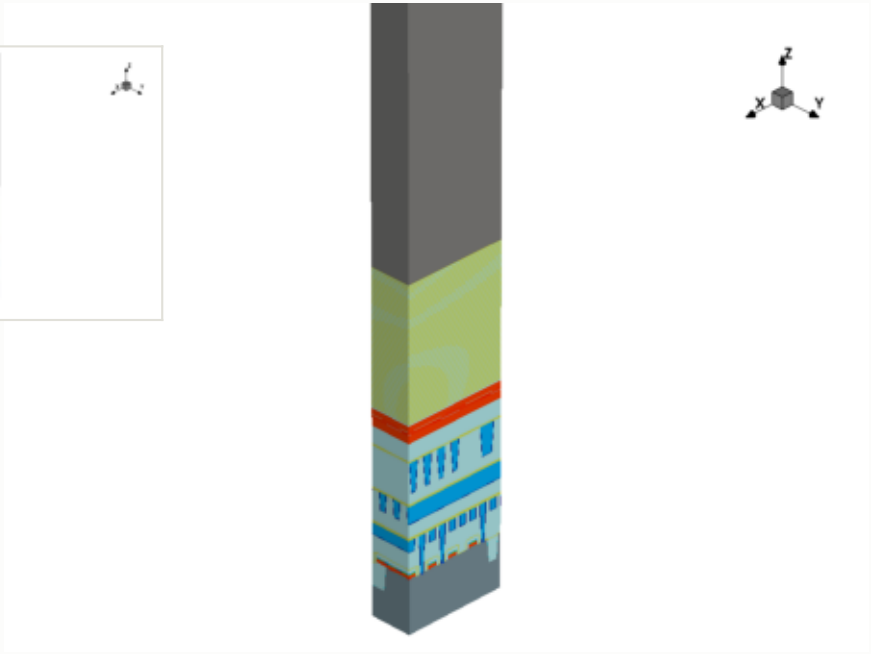
etch_anisotropic · RIE

1.04	302.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



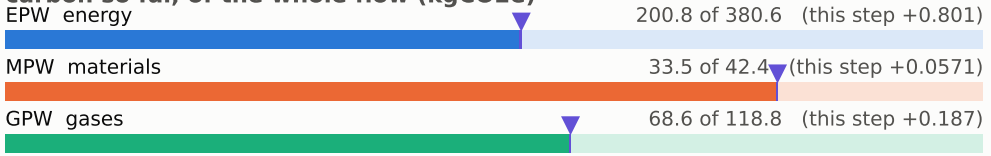


this step

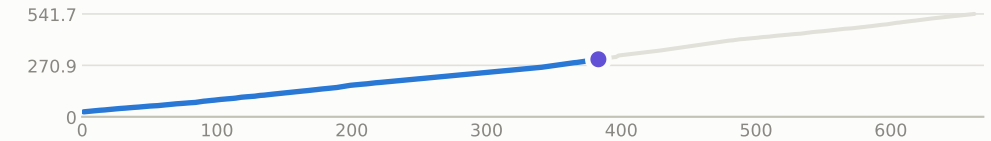
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

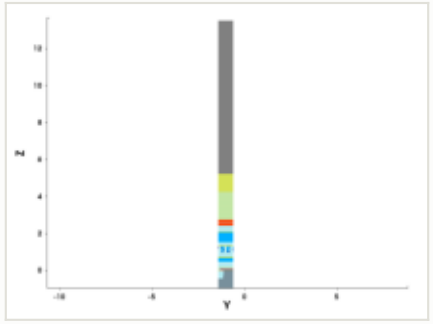
carbon so far, of the whole flow (kgCO2e)



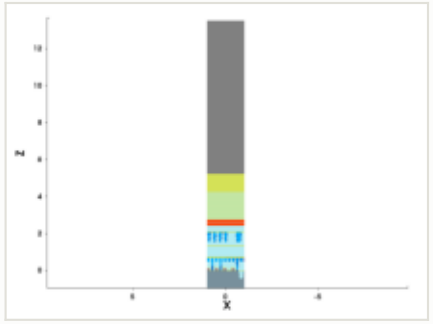
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



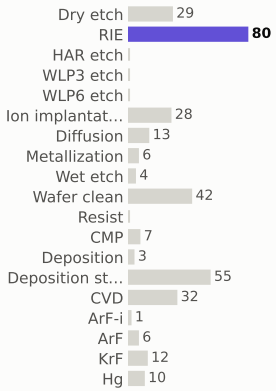
y cut at -1.032 µm



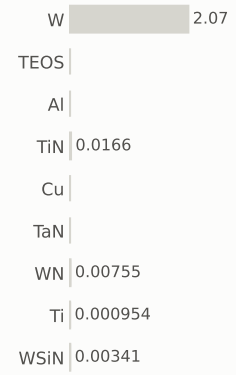
mask for this step

no mask for this step

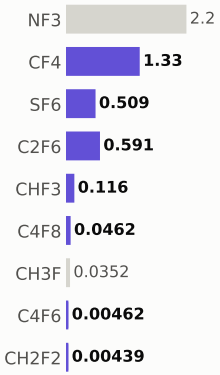
steps by process area



materials deposited (g)



gases emitted (g)



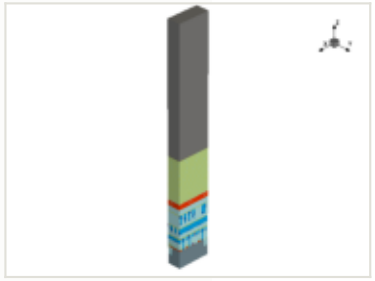
Step #3950: RIE

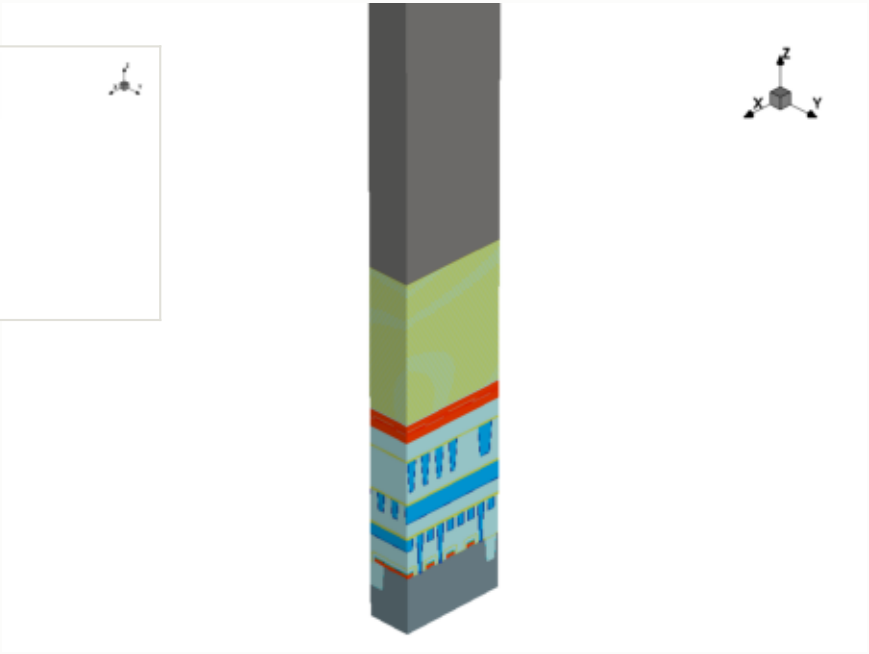
etch_anisotropic · RIE

1.04	303.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





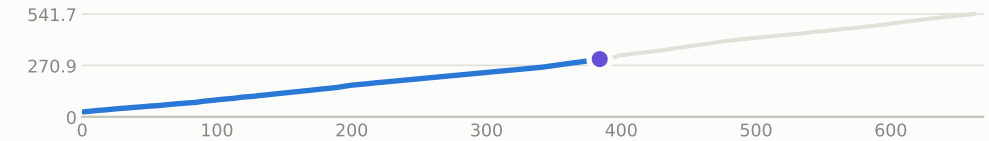
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

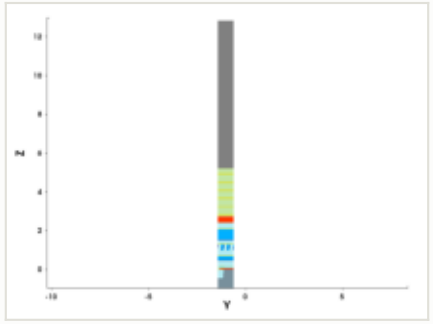
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

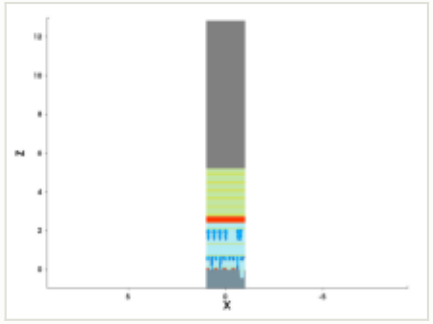
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



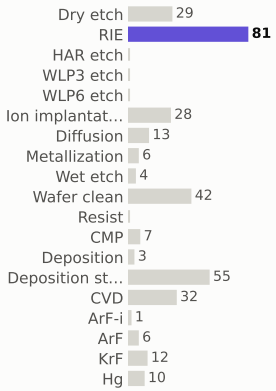
y cut at -1.032 µm



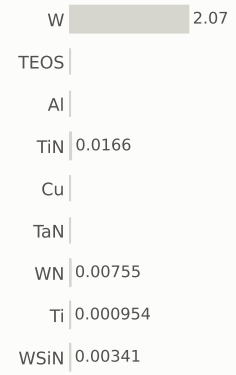
mask for this step

no mask for this step

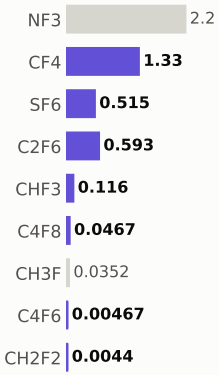
steps by process area



materials deposited (g)



gases emitted (g)



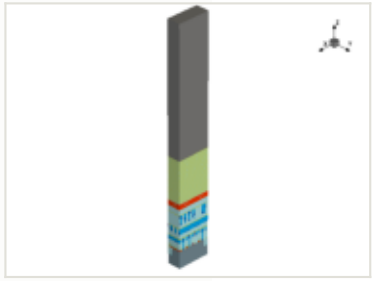
Step #3960: RIE

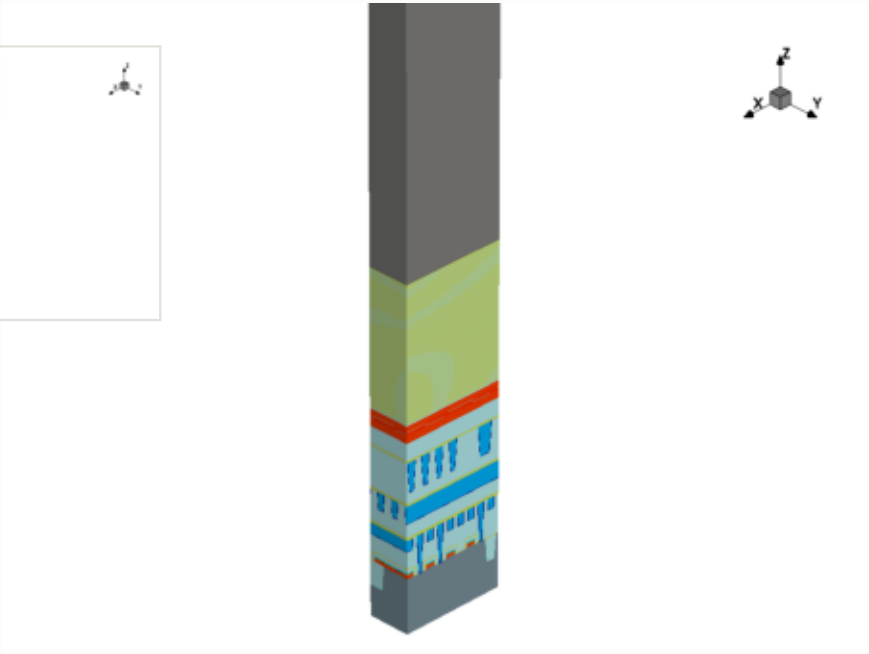
etch_anisotropic · RIE

1.04	305.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





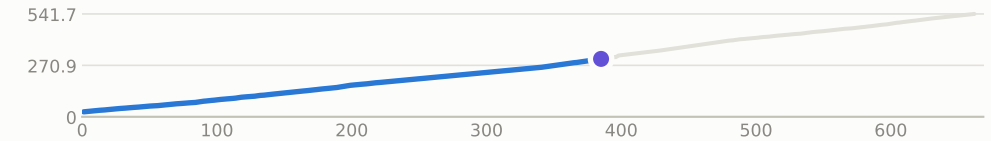
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

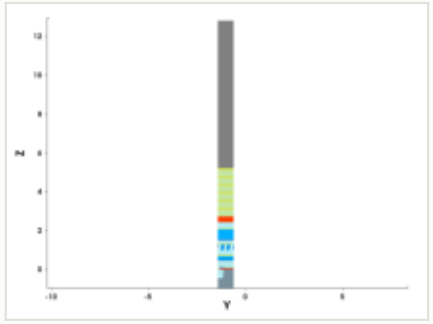
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

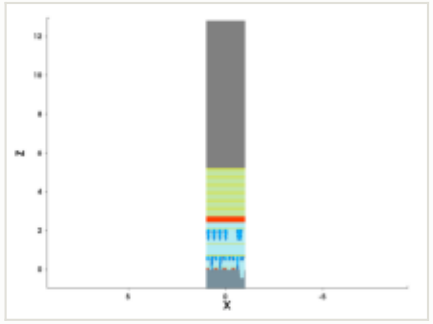
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



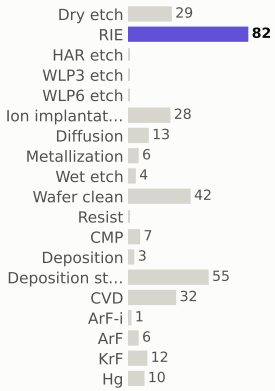
y cut at -1.032 µm



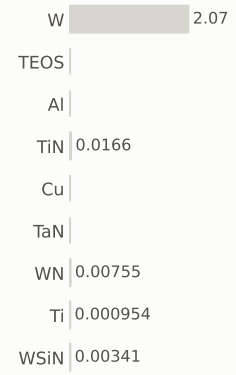
mask for this step

no mask for this step

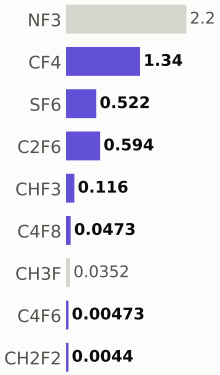
steps by process area



materials deposited (g)



gases emitted (g)



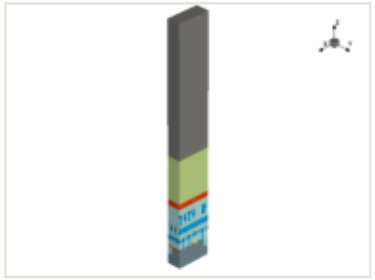
Step #3970: RIE

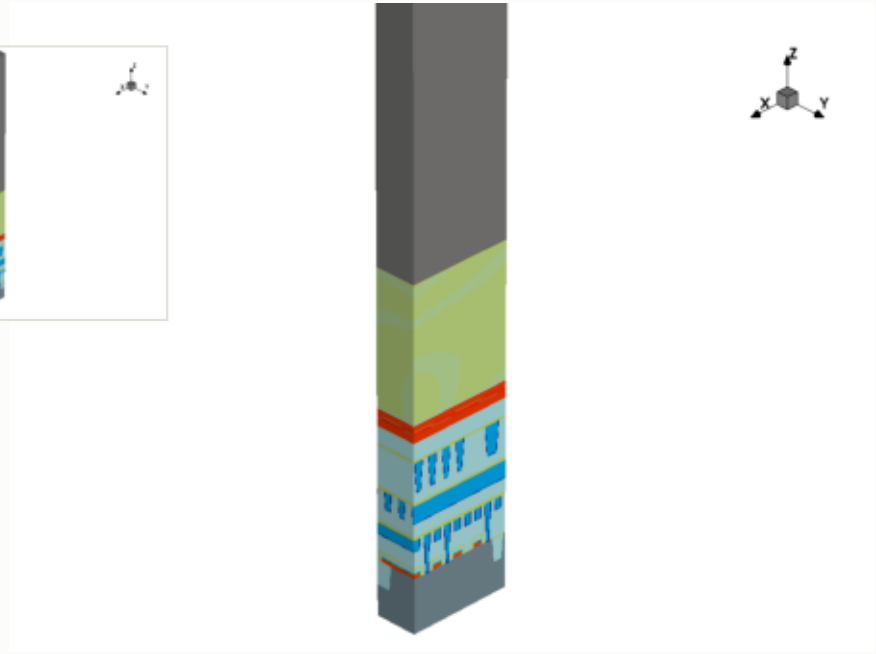
etch_anisotropic · RIE

1.04	306.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





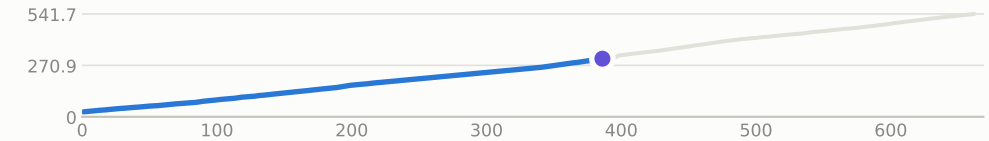
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

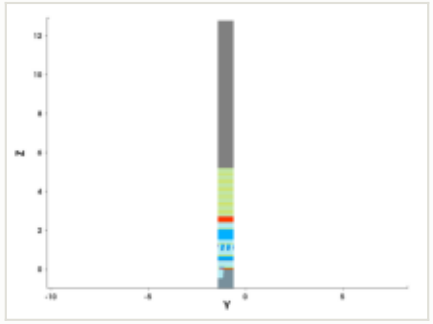
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

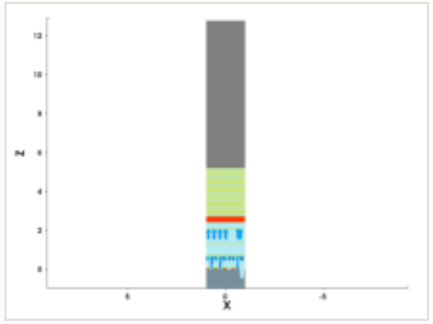
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



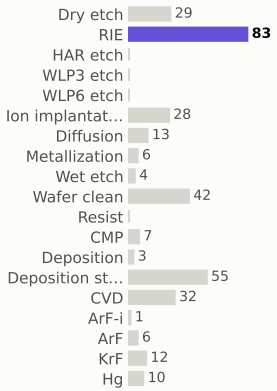
y cut at -1.032 µm



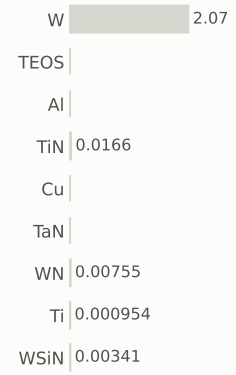
mask for this step

no mask for this step

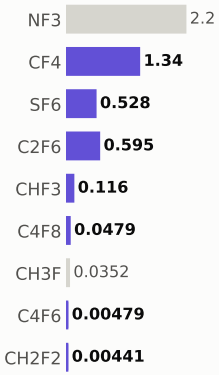
steps by process area



materials deposited (g)



gases emitted (g)



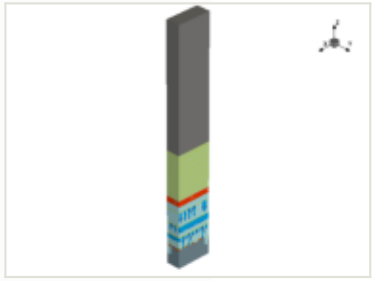
Step #3980: RIE

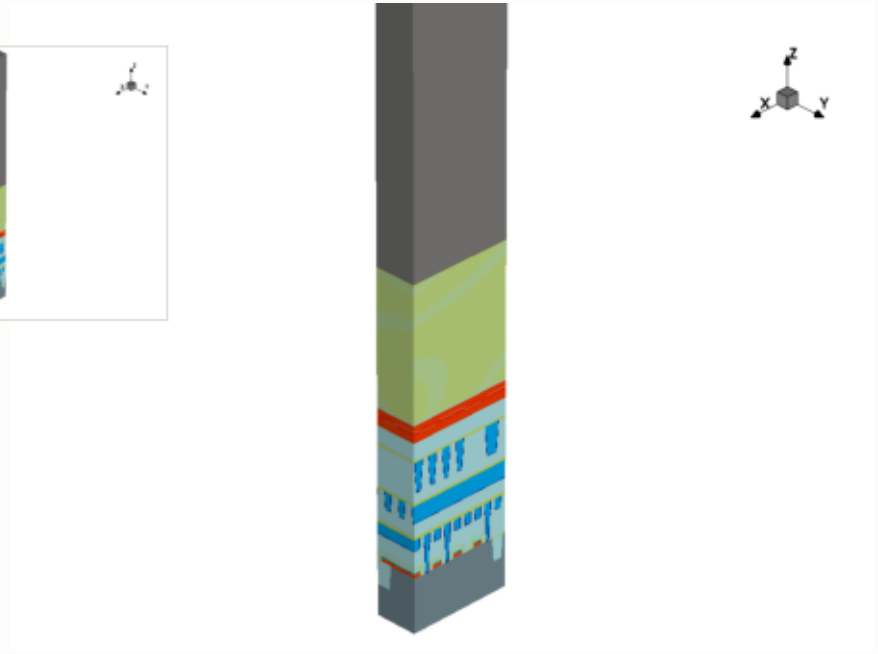
etch_anisotropic · RIE

1.04	307.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





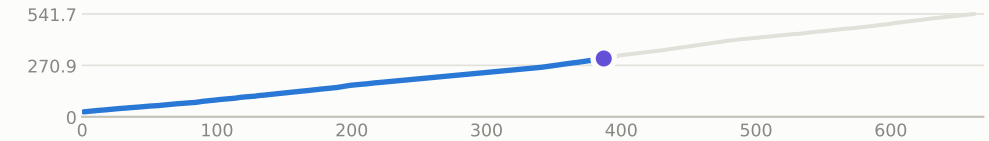
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

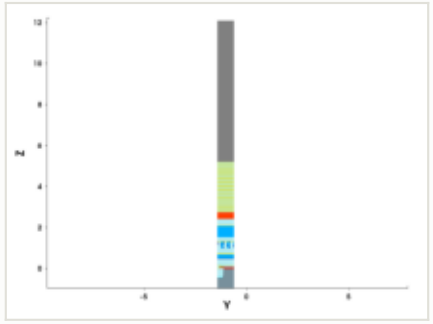
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

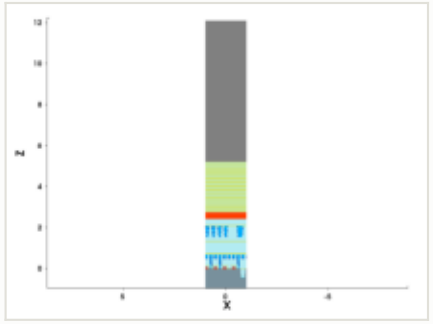
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



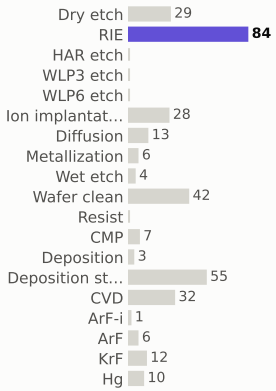
y cut at -1.032 µm



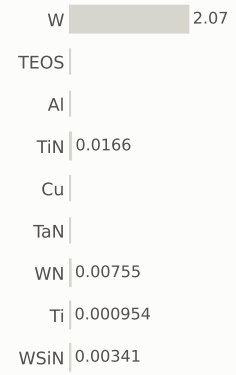
mask for this step

no mask for this step

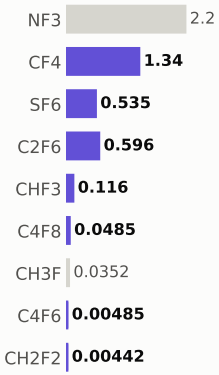
steps by process area



materials deposited (g)



gases emitted (g)



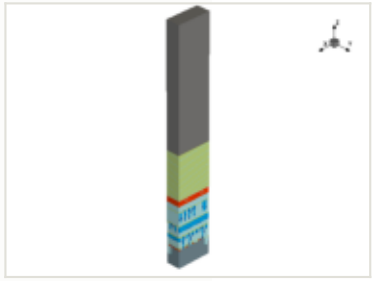
Step #3990: RIE

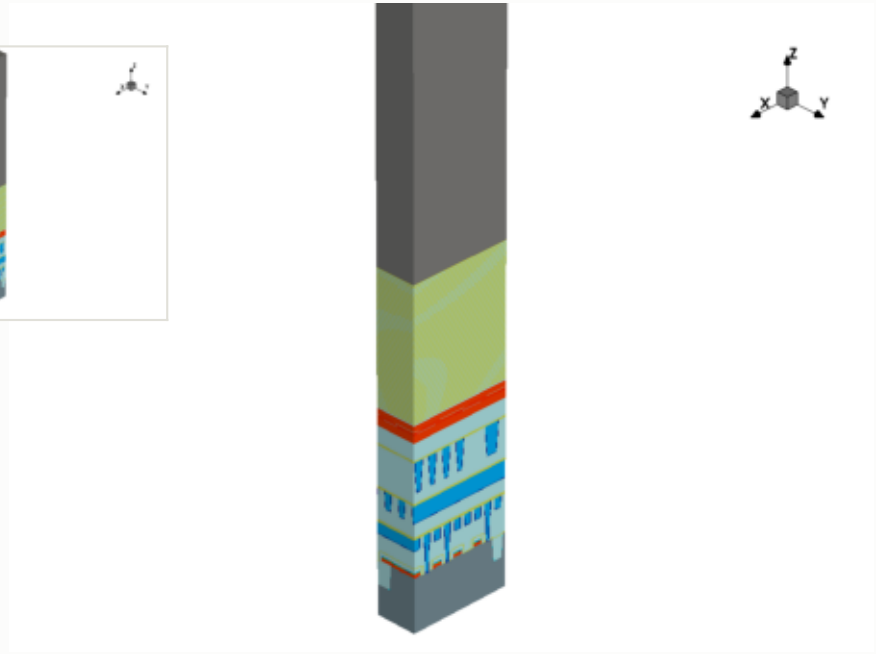
etch_anisotropic · RIE

1.04	308.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





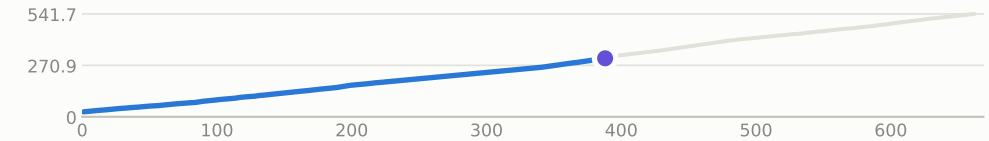
this step

process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

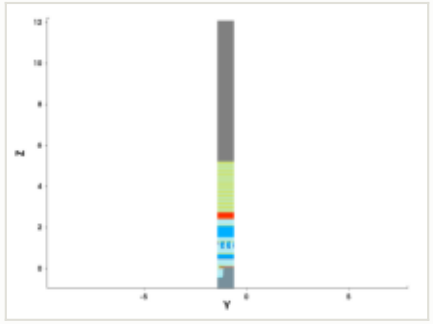
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon so far, of the whole flow (kgCO2e)

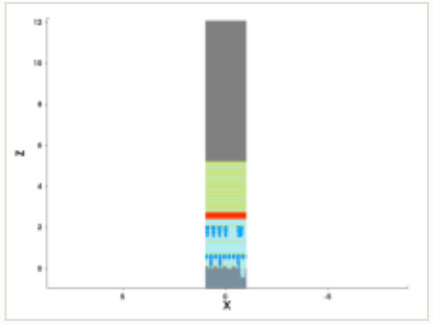
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



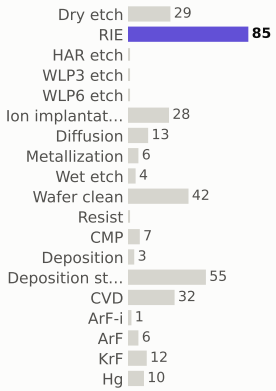
y cut at -1.032 µm



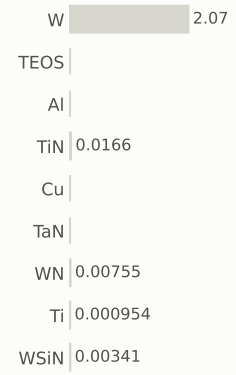
mask for this step

no mask for this step

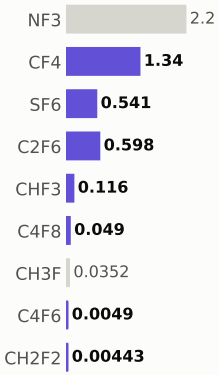
steps by process area



materials deposited (g)



gases emitted (g)



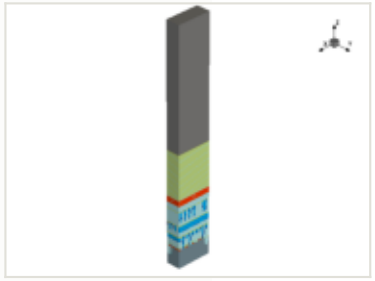
Step #4000: RIE

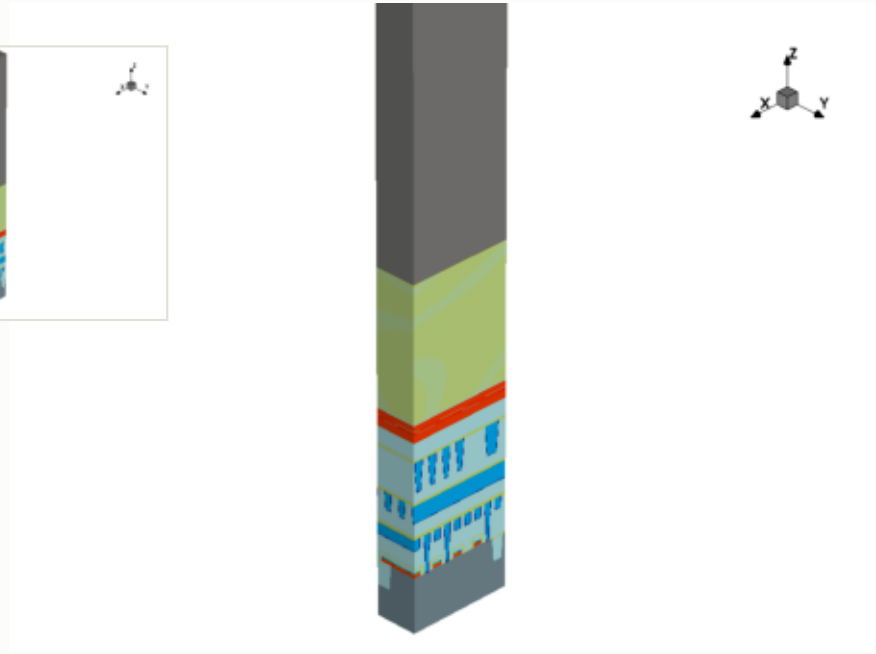
etch_anisotropic · RIE

1.04	309.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



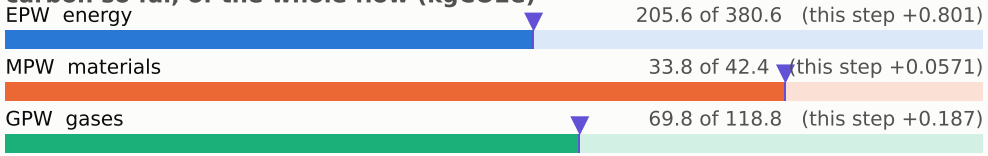


this step

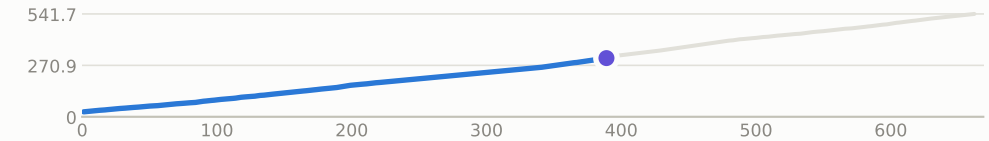
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

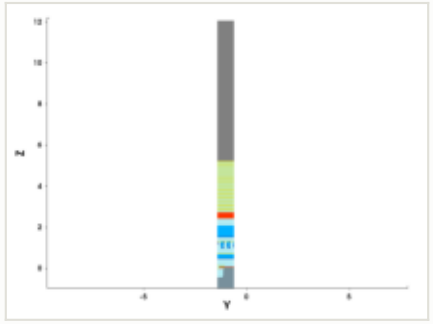
carbon so far, of the whole flow (kgCO2e)



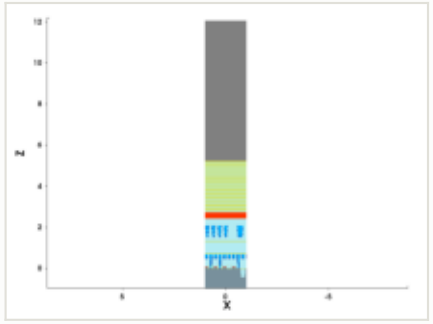
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



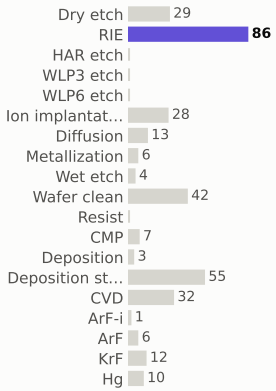
y cut at -1.032 µm



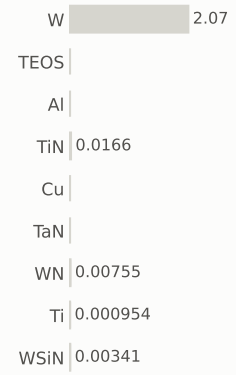
mask for this step

no mask for this step

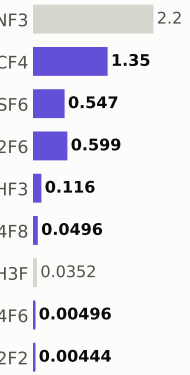
steps by process area



materials deposited (g)



gases emitted (g)



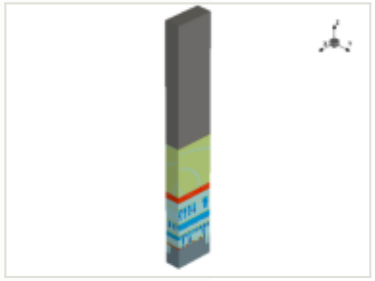
Step #4010: RIE

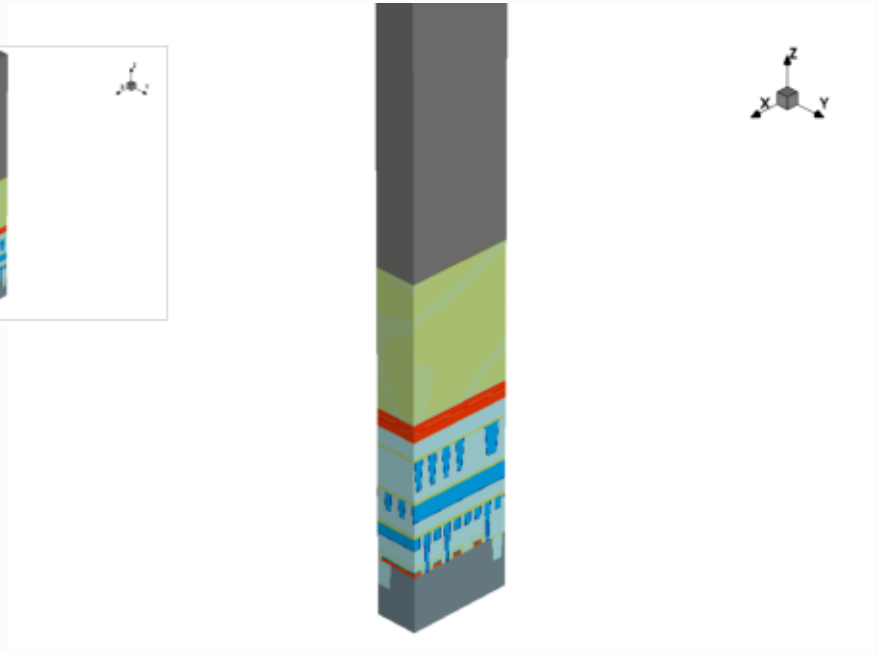
etch_anisotropic · RIE

1.04	310.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



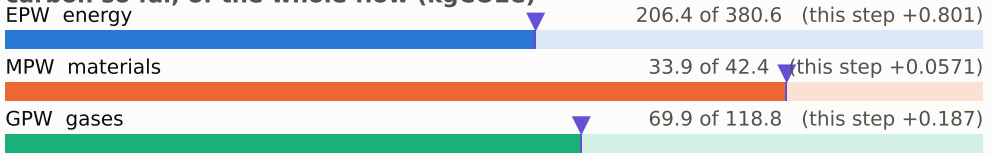


this step

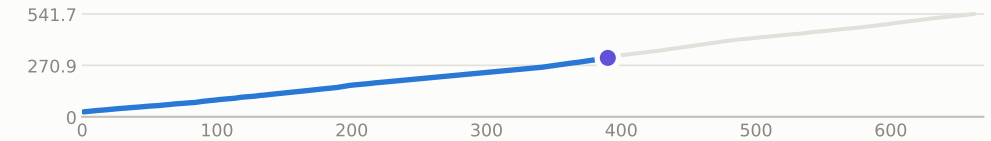
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

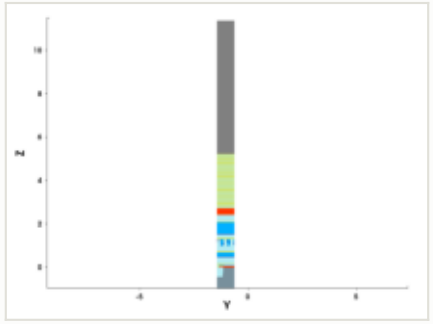
carbon so far, of the whole flow (kgCO2e)



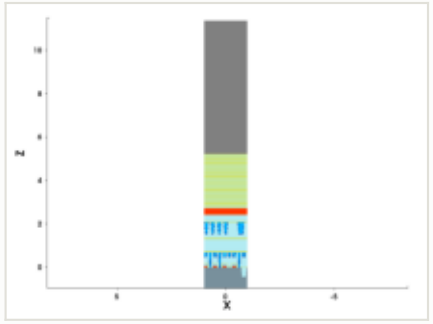
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



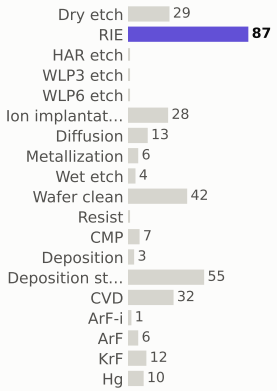
y cut at -1.032 µm



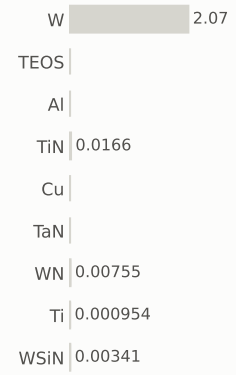
mask for this step

no mask for this step

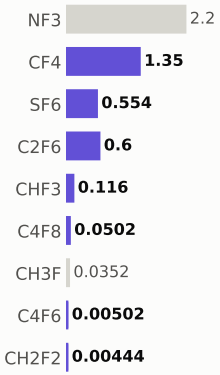
steps by process area



materials deposited (g)



gases emitted (g)



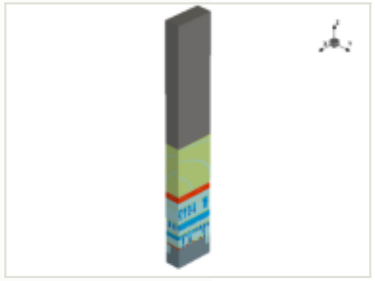
Step #4020: RIE

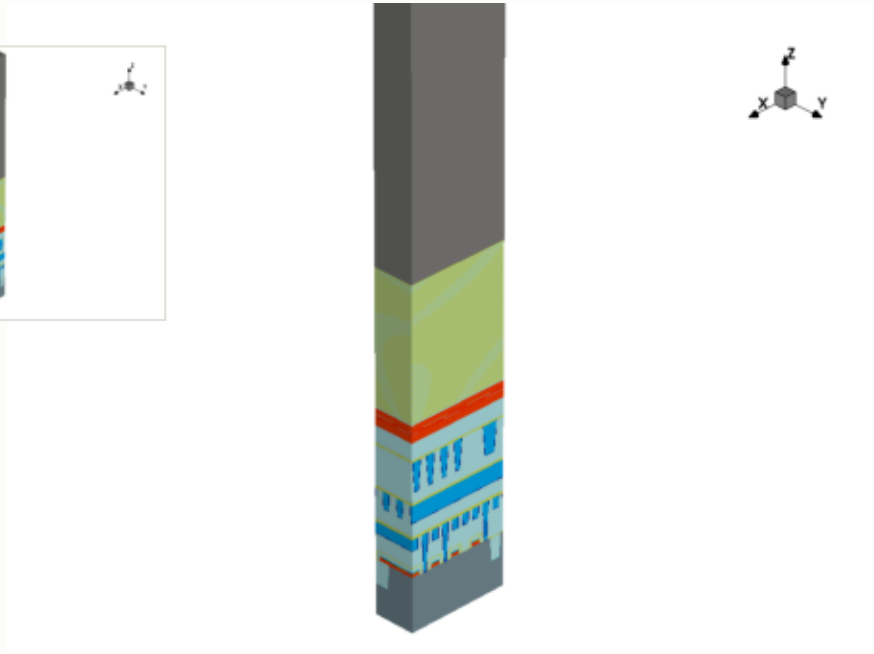
etch_anisotropic · RIE

1.04	311.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



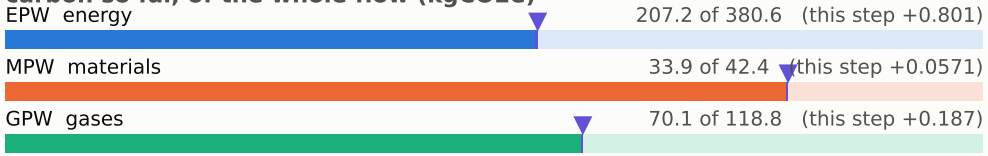


this step

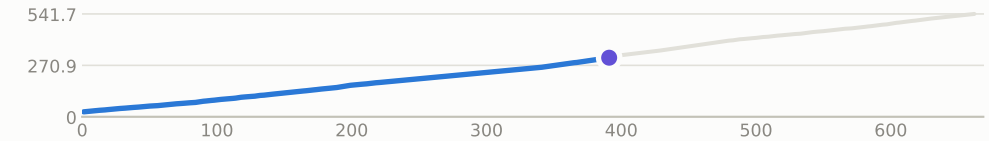
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

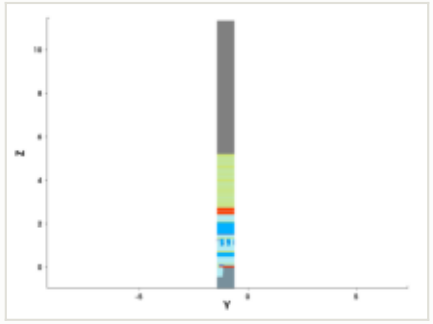
carbon so far, of the whole flow (kgCO2e)



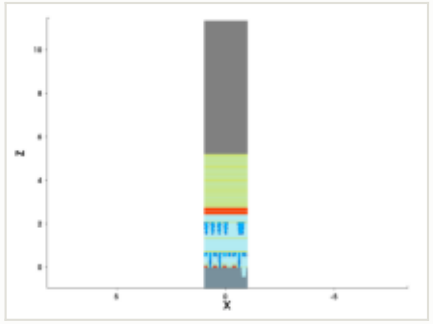
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



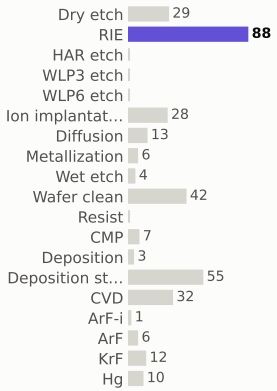
y cut at -1.032 µm



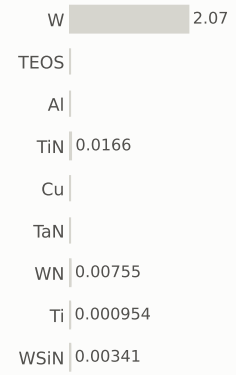
mask for this step

no mask for this step

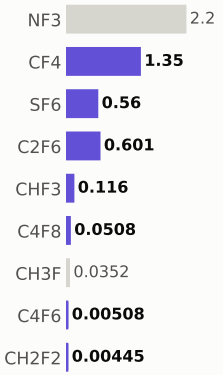
steps by process area



materials deposited (g)



gases emitted (g)



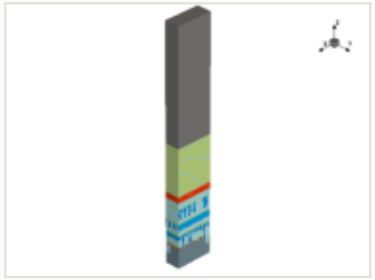
Step #4030: RIE

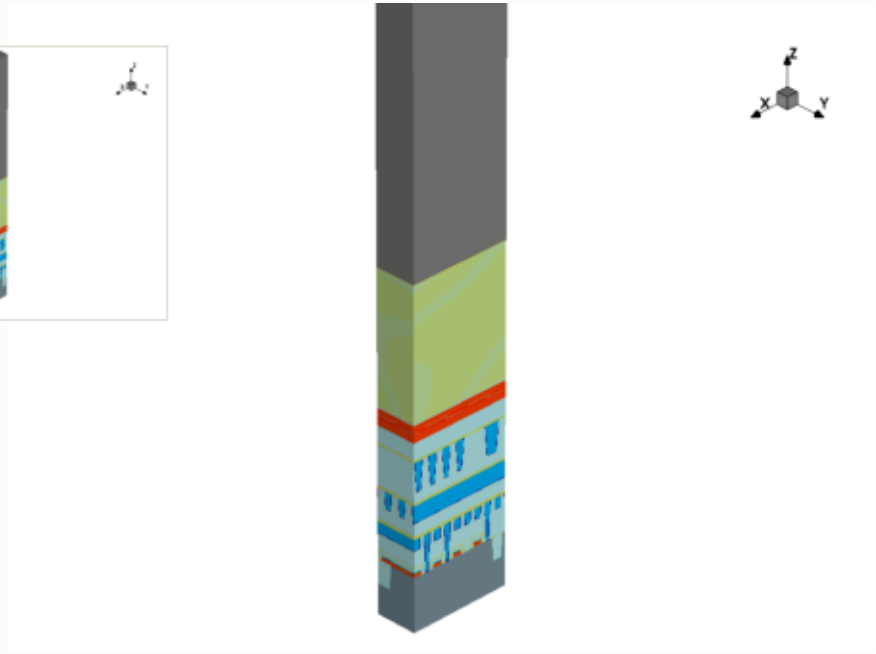
etch_anisotropic · RIE

1.04	312.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



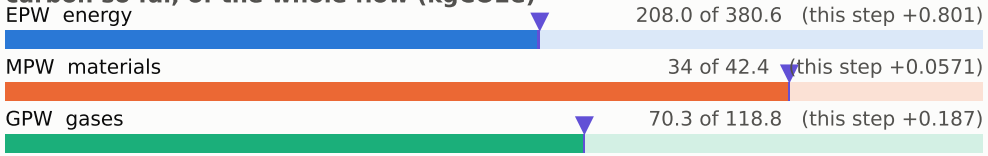


this step

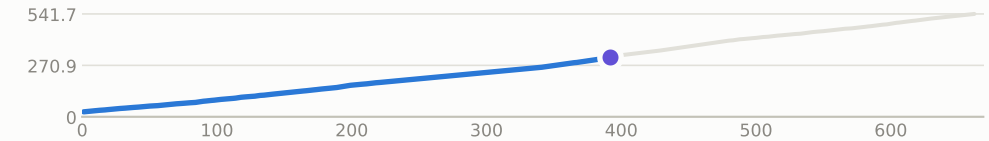
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

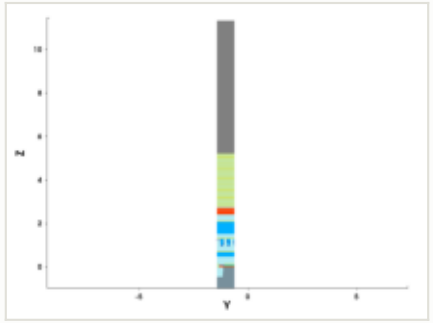
carbon so far, of the whole flow (kgCO2e)



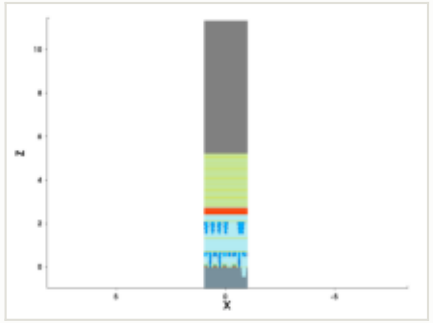
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



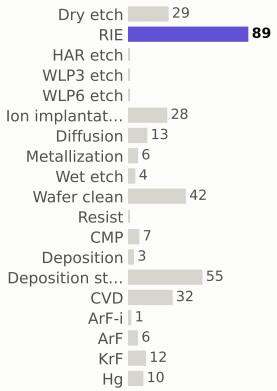
y cut at -1.032 µm



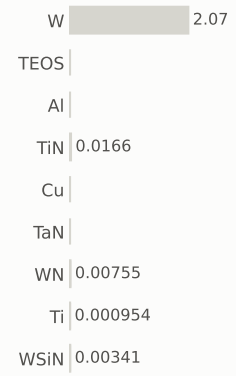
mask for this step

no mask for this step

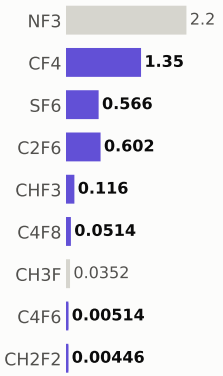
steps by process area



materials deposited (g)



gases emitted (g)



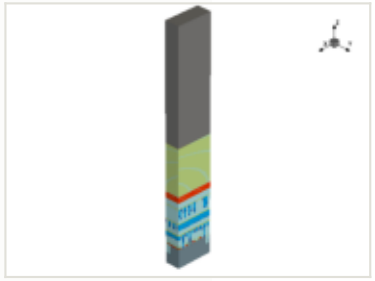
Step #4040: RIE

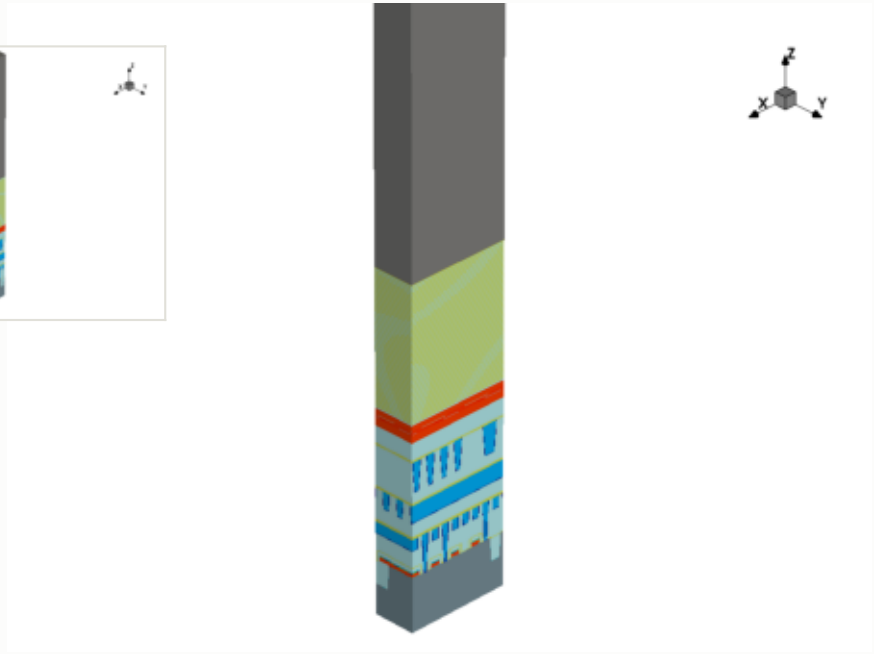
etch_anisotropic · RIE

1.04	313.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



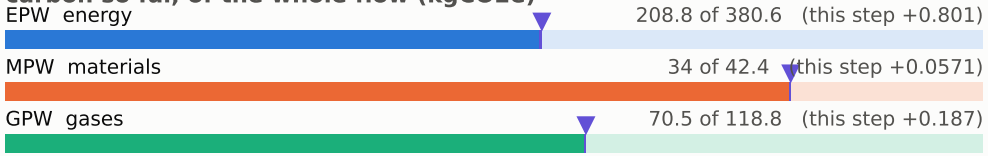


this step

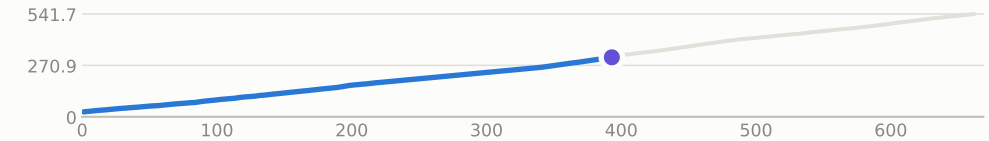
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

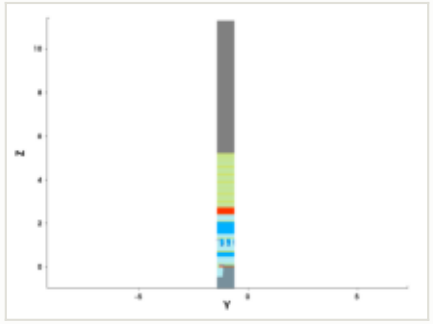
carbon so far, of the whole flow (kgCO2e)



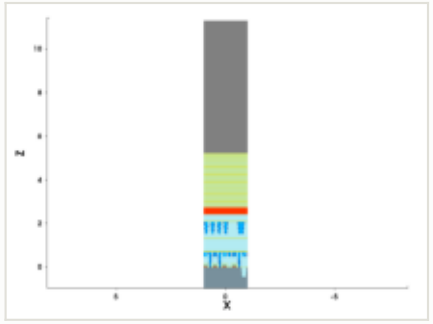
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



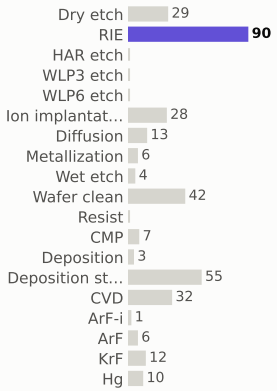
y cut at -1.032 µm



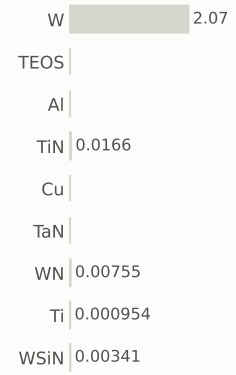
mask for this step

no mask for this step

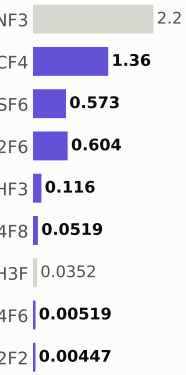
steps by process area



materials deposited (g)



gases emitted (g)



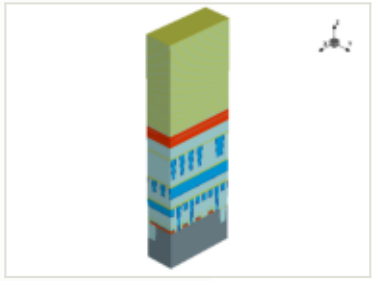
Step #4050: Dry_etch

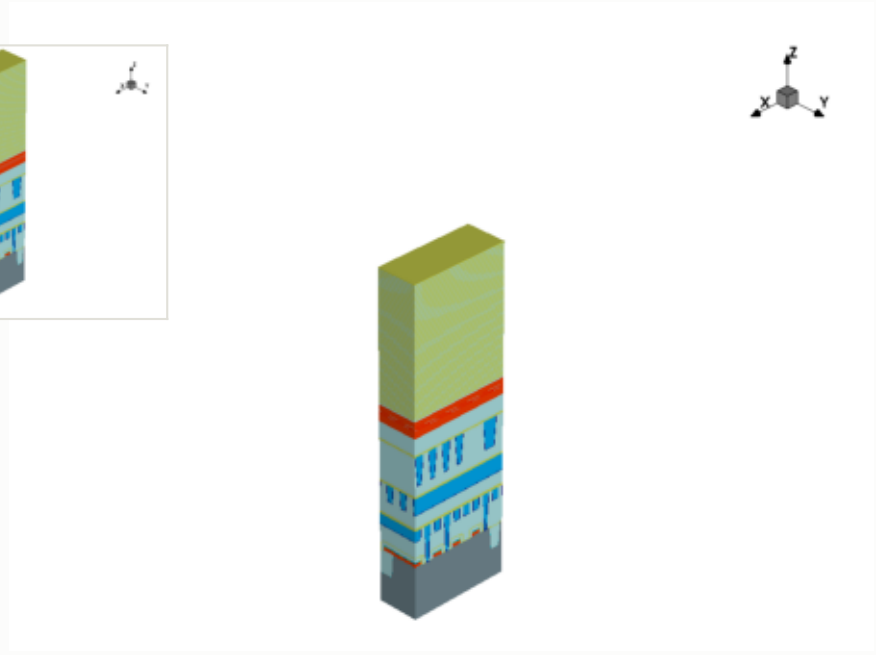
etch_strip · Dry_etch

0.801	314.1	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

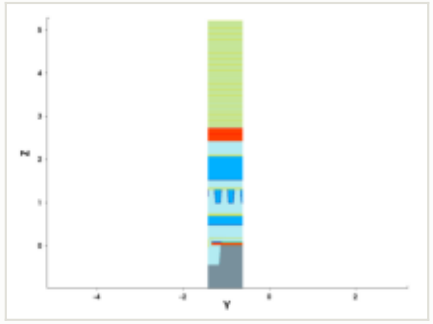
3D structure after this step

zoomed to fit

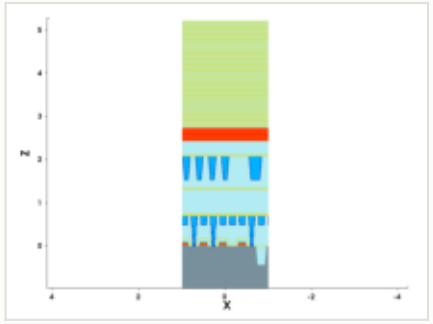




x cut at 0.000 µm



y cut at -1.032 µm



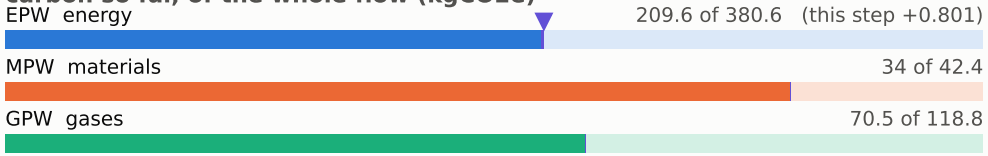
mask for this step

no mask for this step

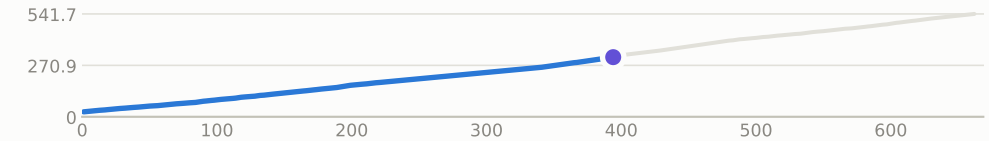
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

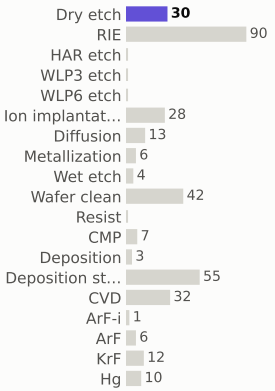
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

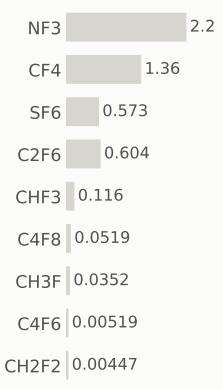


steps by process area



materials deposited (g)

gases emitted (g)



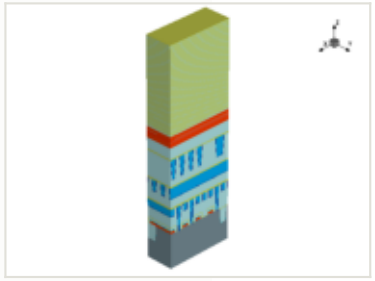
Step #4060: Wafer_clean

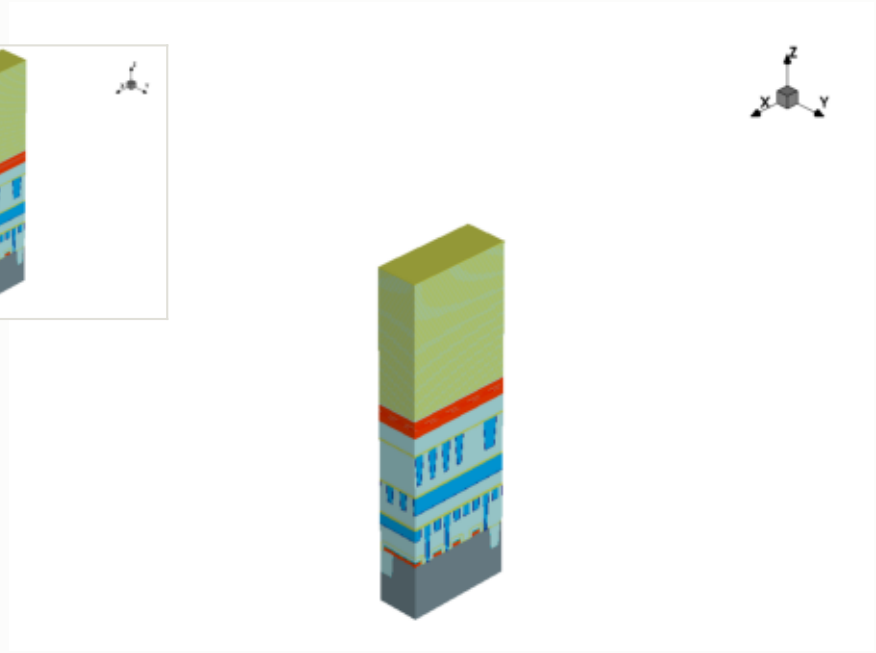
clean · Wafer_clean

0.169	314.3	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

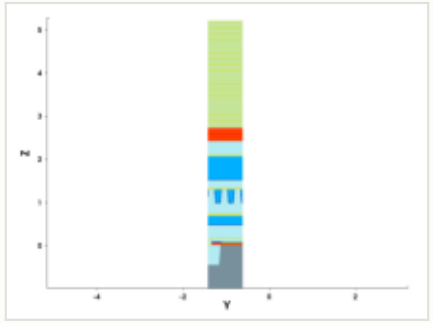
3D structure after this step

zoomed to fit

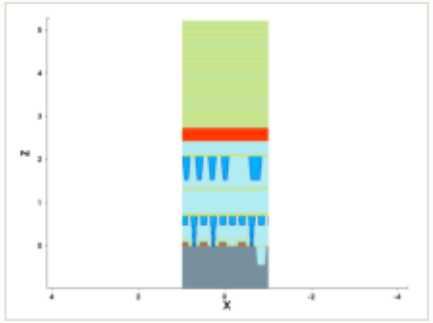




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

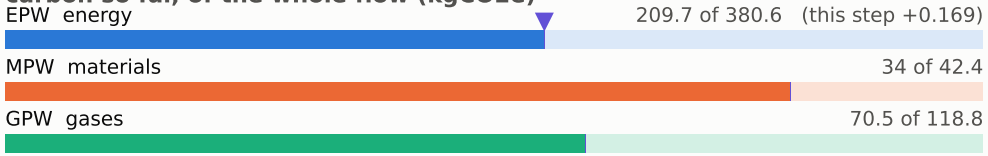
no mask for this step

this step

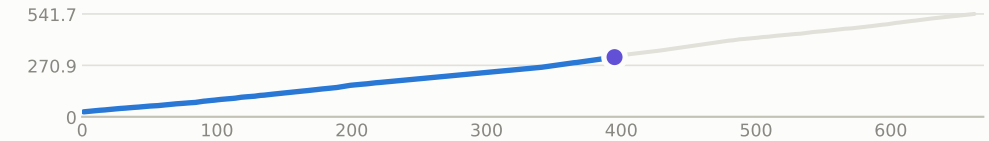
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

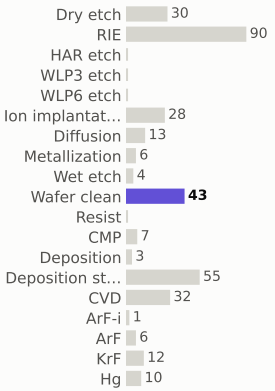
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

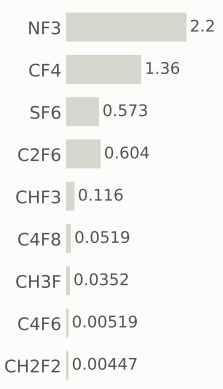


steps by process area



materials deposited (g)

gases emitted (g)



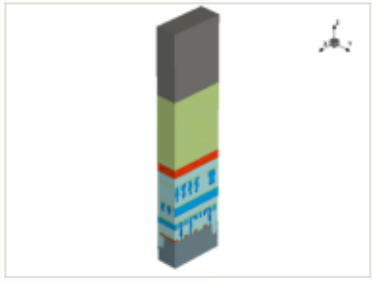
Step #4070: KrF

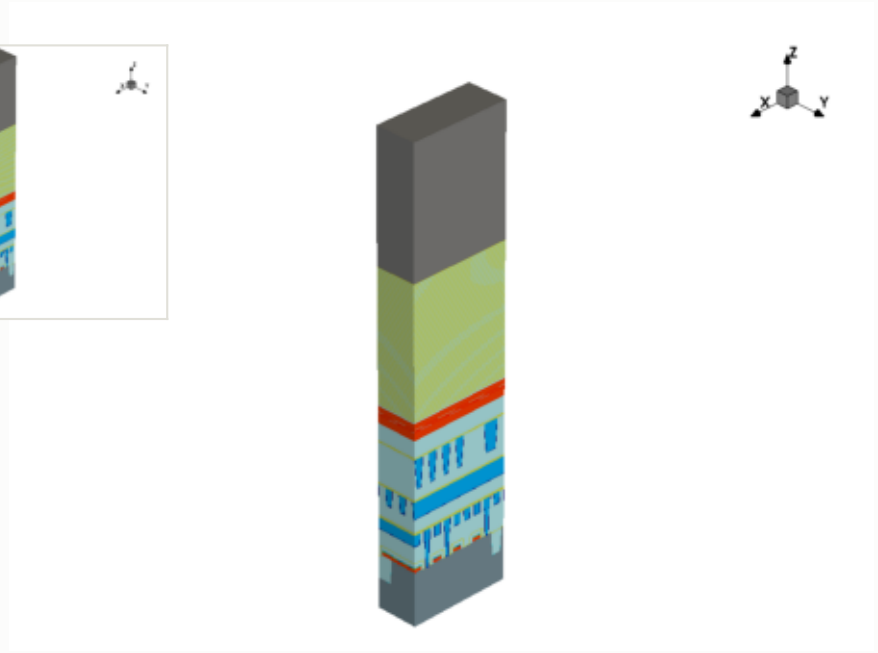
litho_patterning · KrF

0.199	314.5	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

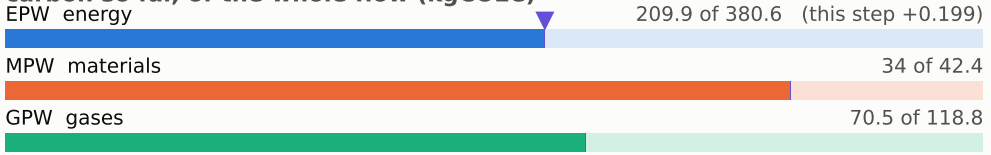




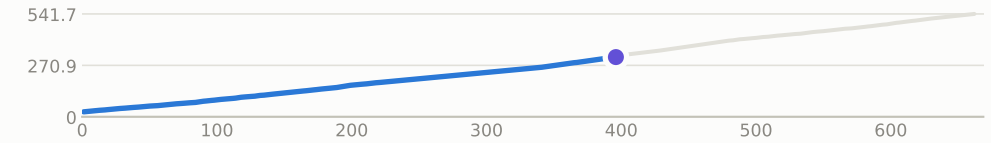
this step

process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

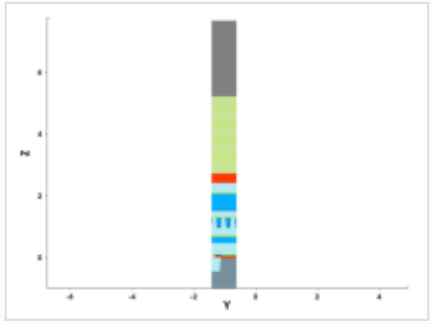
carbon so far, of the whole flow (kgCO2e)



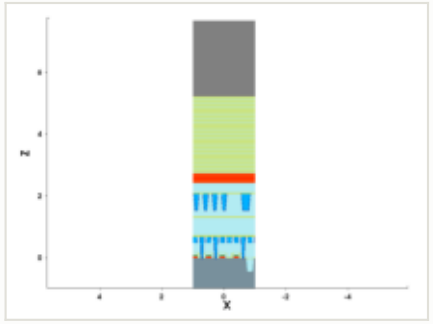
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

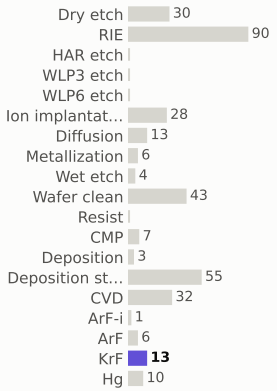


y cut at -1.032 µm

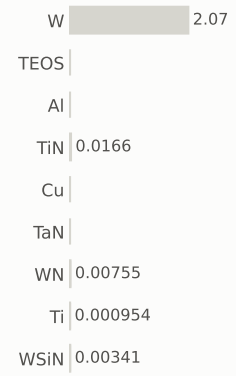


mask: layer 31:0, positive (OASIS)

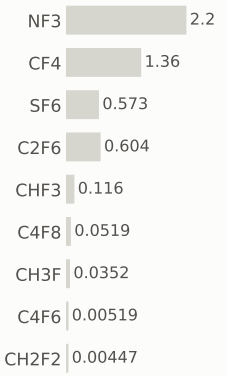
steps by process area



materials deposited (g)



gases emitted (g)



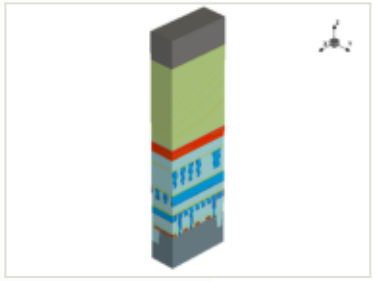
Step #4080: WLP3_etch

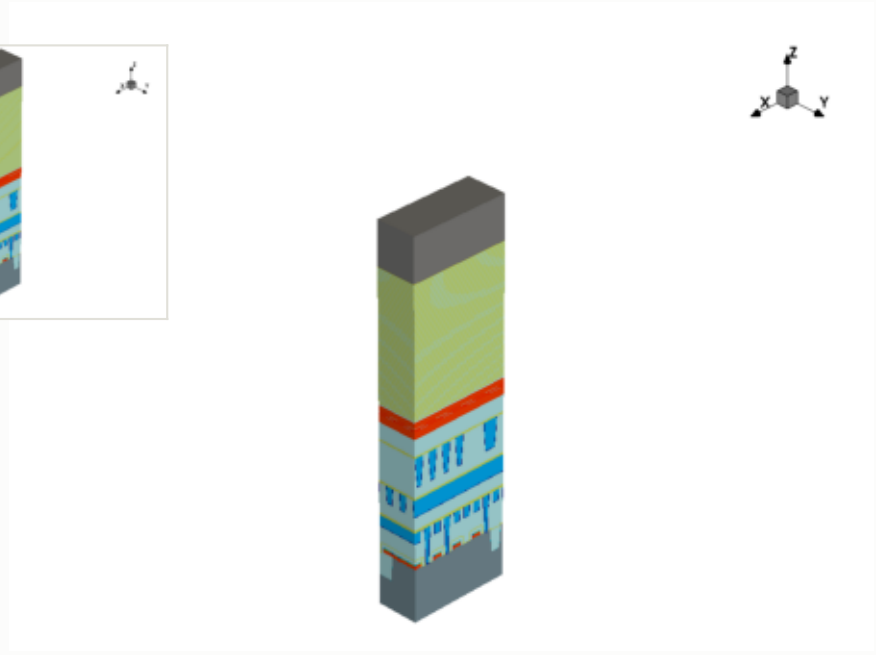
etch_anisotropic · WLP3_etch

4.19	318.7	0.77%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

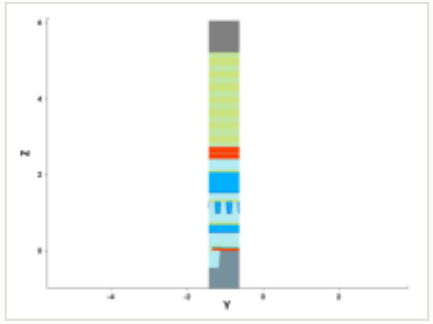
3D structure after this step

zoomed to fit

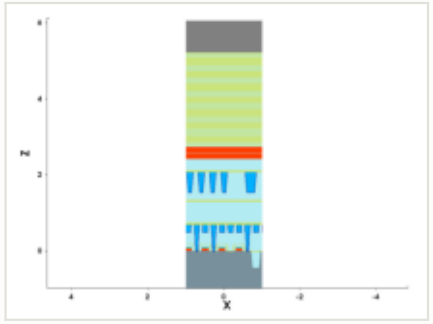




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

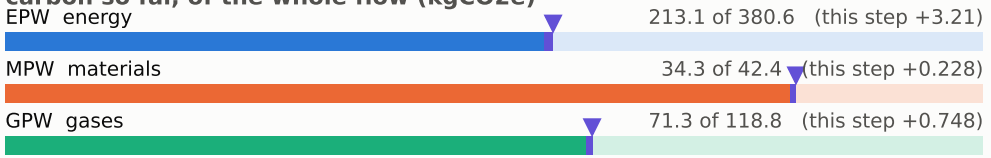
no mask for this step

this step

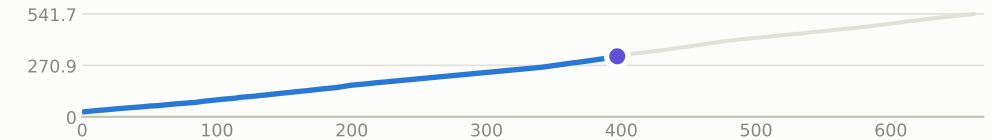
process area	WLP3_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.641 g, SF6 1.76 g
energy	5.33 kWh + 2.13 overhead

EPW	3.21 kgCO2e
EPW tool	2.29 kgCO2e
EPW overhead	0.917 kgCO2e
MPW	0.228 kgCO2e (gas 0.228)
GPW	0.748 kgCO2e
embodied carbon	**4.19 kgCO2e**

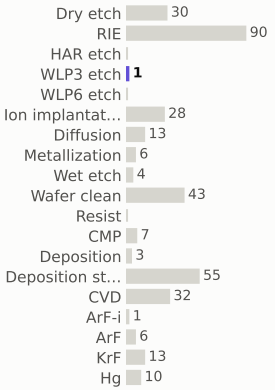
carbon so far, of the whole flow (kgCO2e)



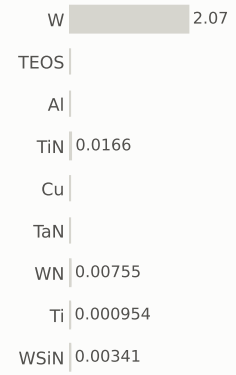
embodied carbon up to each step (kgCO2e per wafer)



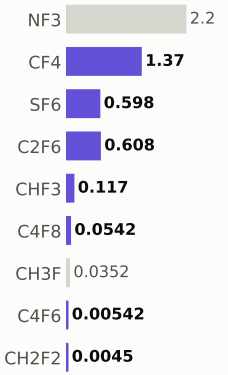
steps by process area



materials deposited (g)



gases emitted (g)



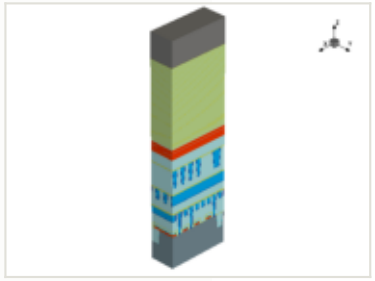
Step #4090: WLP3_etch

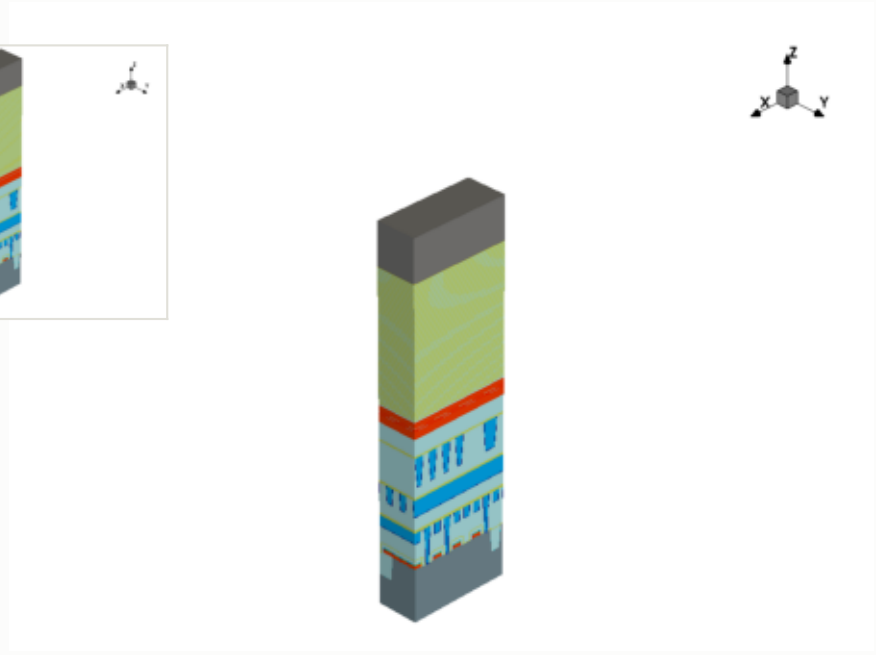
etch_anisotropic · WLP3_etch

4.19	322.9	0.77%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



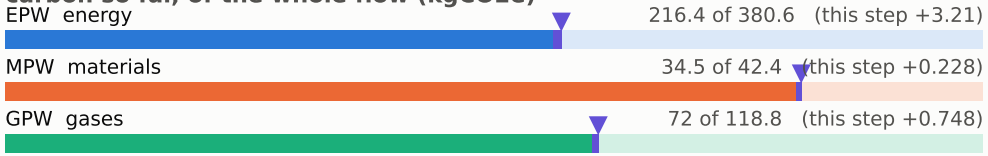


this step

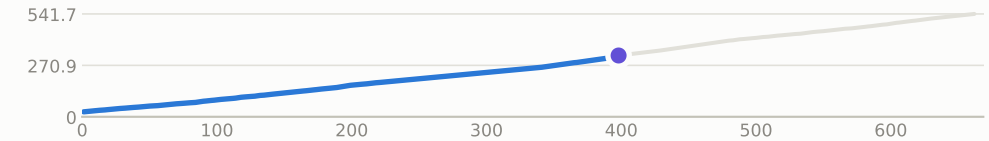
process area	WLP3_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.641 g, SF6 1.76 g
energy	5.33 kWh + 2.13 overhead

EPW	3.21 kgCO2e
EPW tool	2.29 kgCO2e
EPW overhead	0.917 kgCO2e
MPW	0.228 kgCO2e (gas 0.228)
GPW	0.748 kgCO2e
embodied carbon	**4.19 kgCO2e**

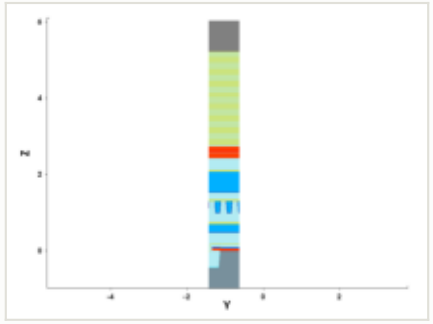
carbon so far, of the whole flow (kgCO2e)



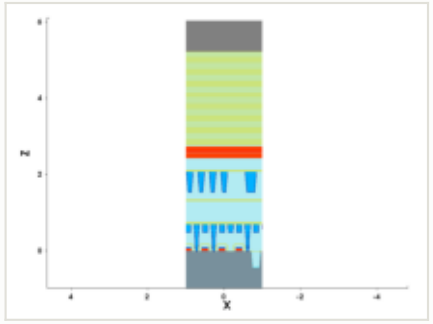
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



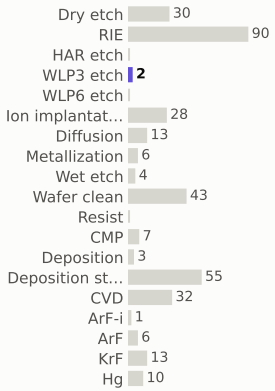
y cut at -1.032 µm



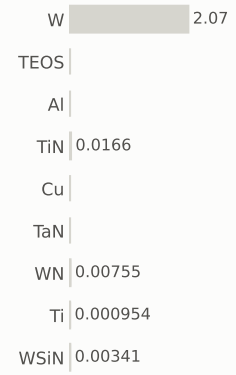
mask for this step

no mask for this step

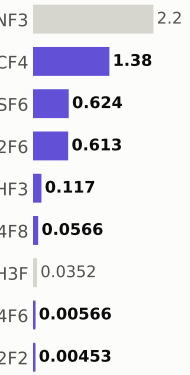
steps by process area



materials deposited (g)



gases emitted (g)



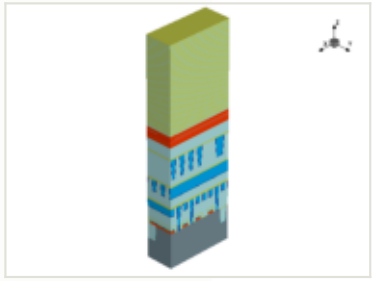
Step #4100: Dry_etch

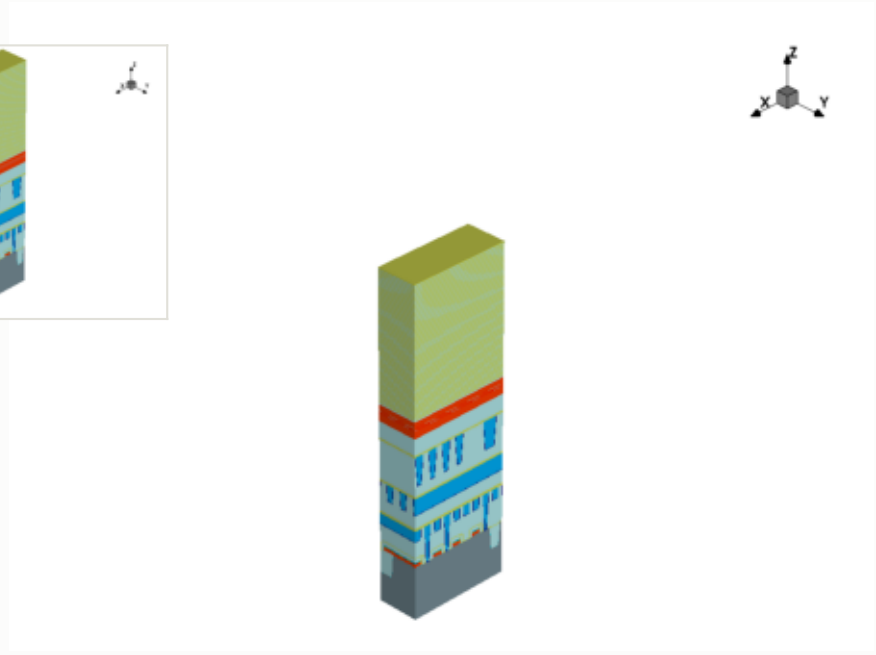
etch_strip · Dry_etch

0.801	323.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

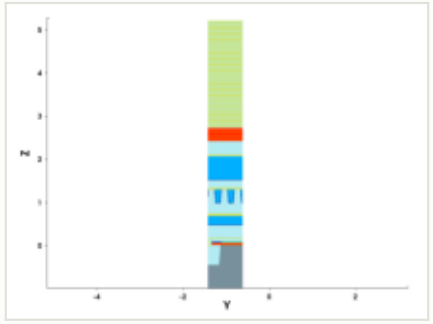
3D structure after this step

zoomed to fit

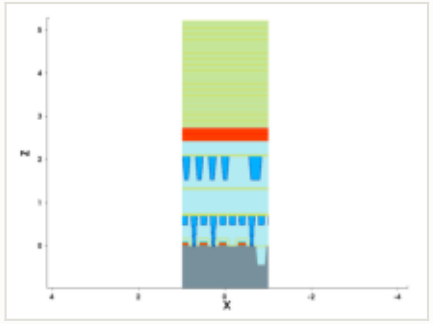




x cut at 0.000 µm



y cut at -1.032 µm



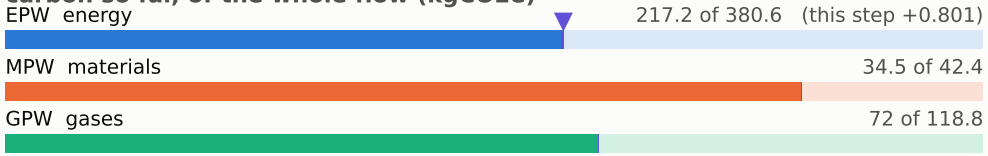
mask for this step

no mask for this step

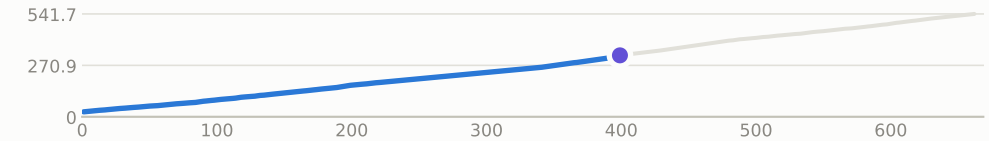
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

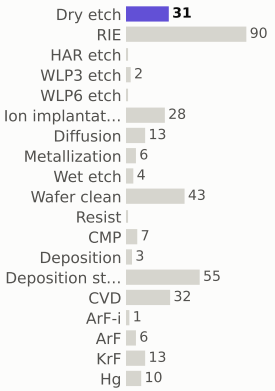
carbon so far, of the whole flow (kgCO2e)



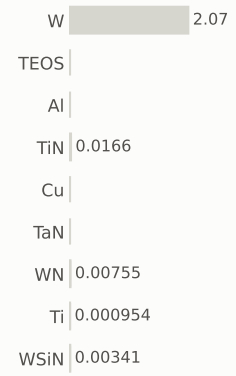
embodied carbon up to each step (kgCO2e per wafer)



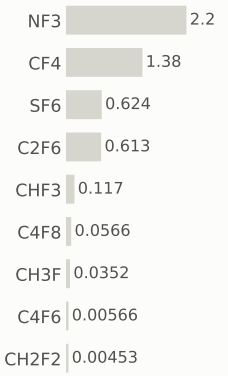
steps by process area



materials deposited (g)



gases emitted (g)



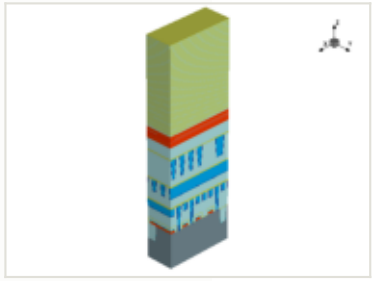
Step #4110: Wafer_clean

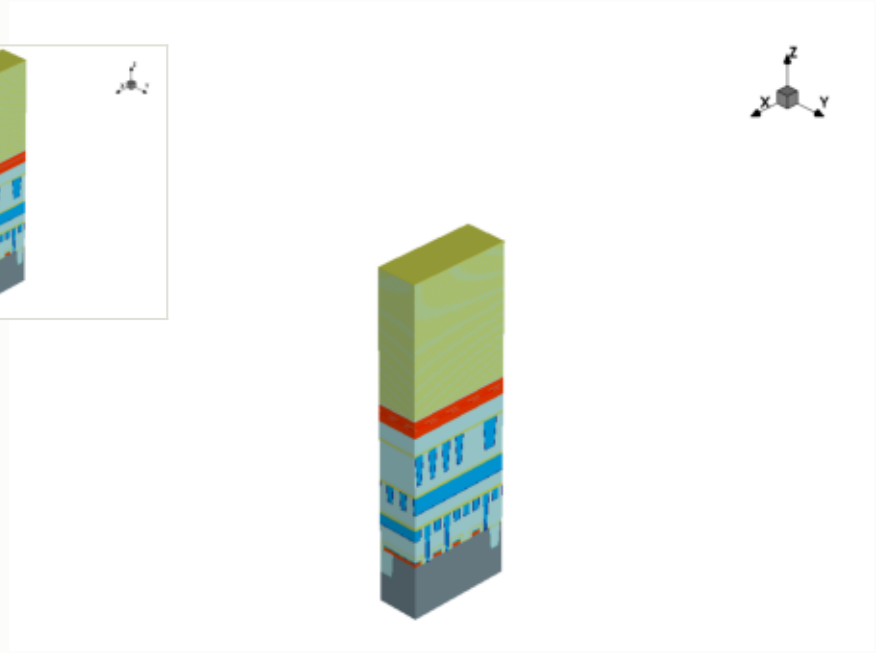
clean · Wafer_clean

0.169	323.8	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

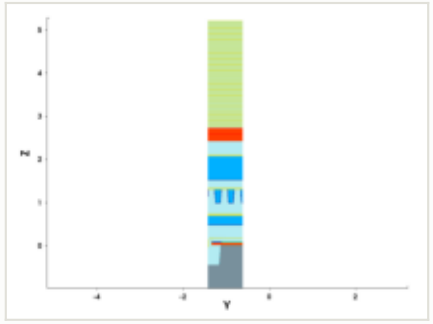
3D structure after this step

zoomed to fit

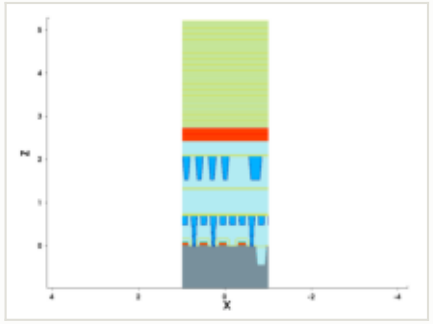




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

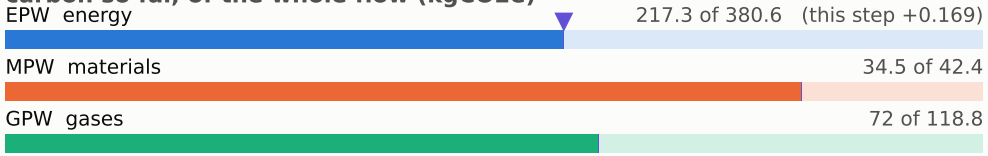
no mask for this step

this step

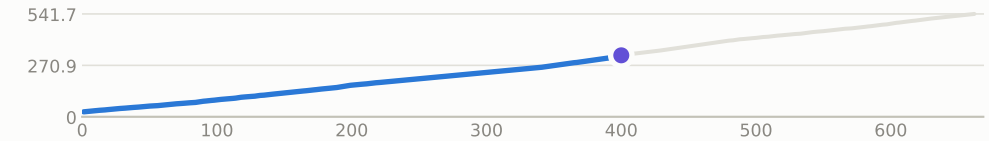
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

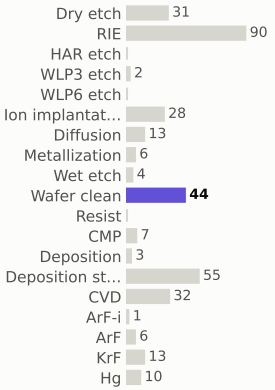
carbon so far, of the whole flow (kgCO2e)



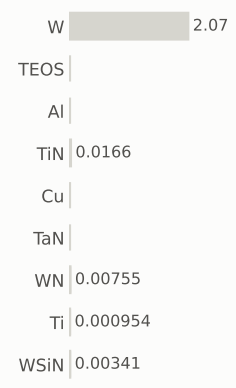
embodied carbon up to each step (kgCO2e per wafer)



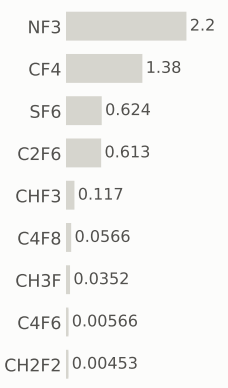
steps by process area



materials deposited (g)



gases emitted (g)



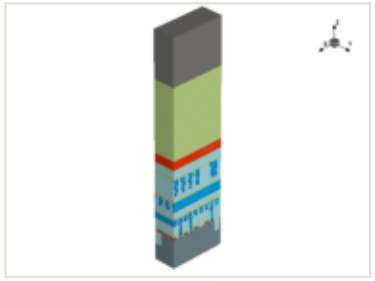
Step #4120: KrF

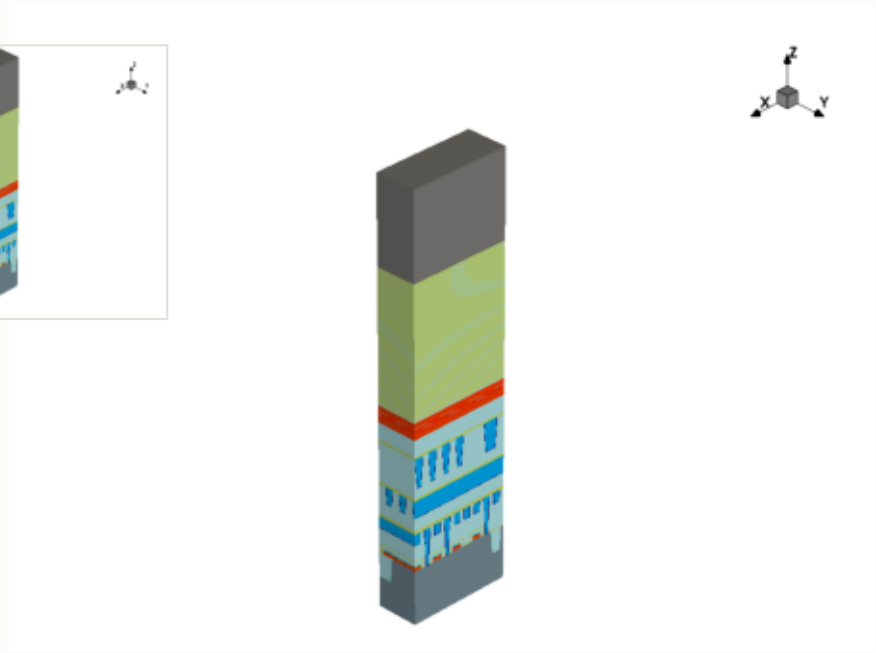
litho_patterning · KrF

0.199	324.0	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

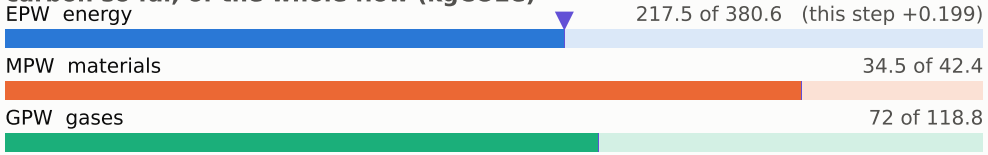




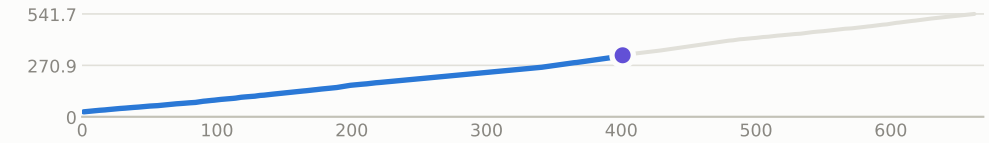
this step

process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

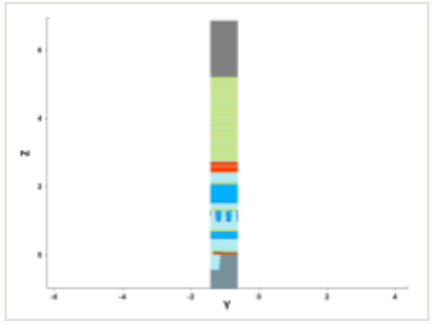
carbon so far, of the whole flow (kgCO2e)



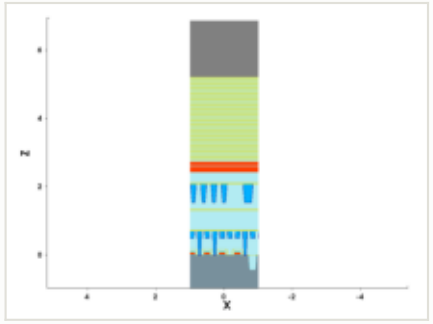
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

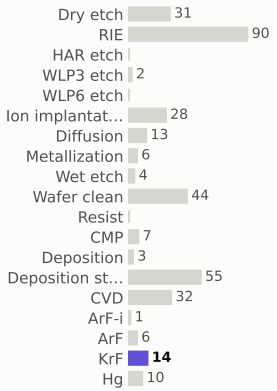


y cut at -1.032 µm



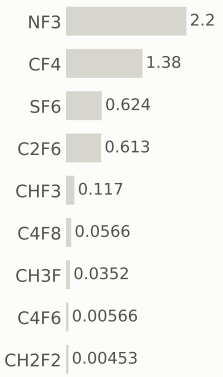
mask: layer 34:0, positive (OASIS)

steps by process area



materials deposited (g)

gases emitted (g)



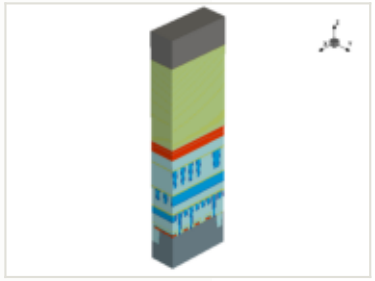
Step #4130: WLP6_etch

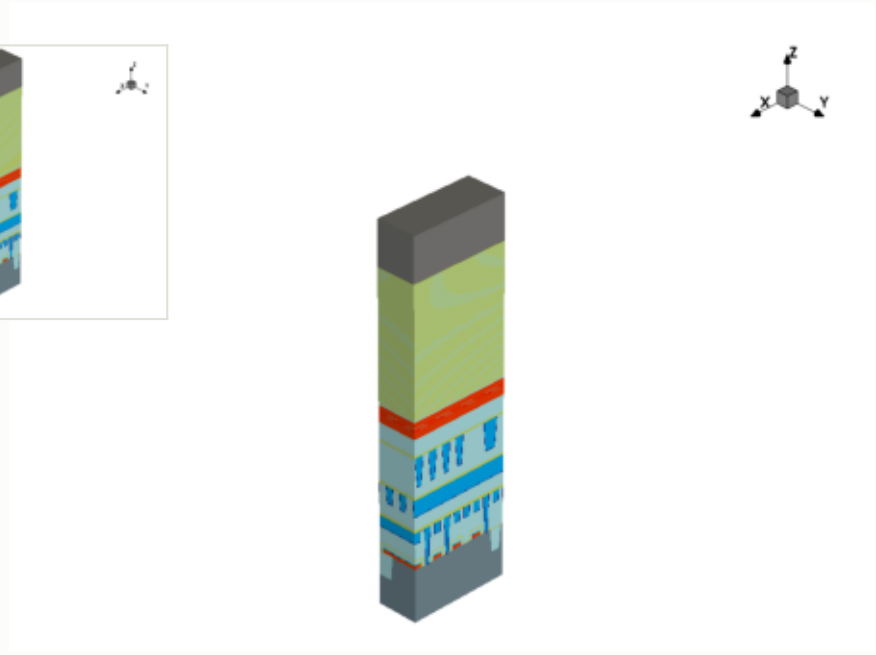
etch_anisotropic · WLP6_etch

1.04	325.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

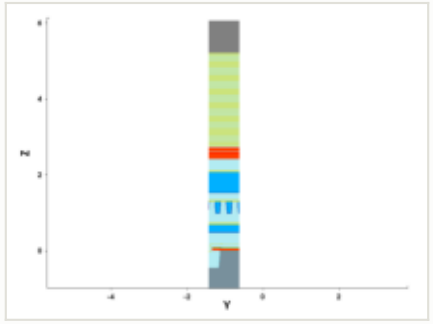
3D structure after this step

zoomed to fit

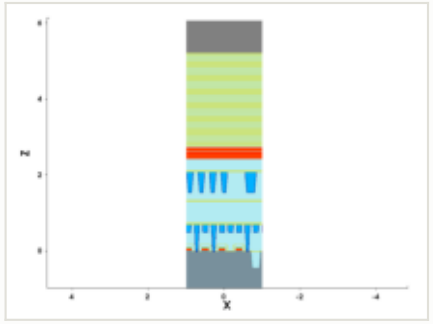




x cut at 0.000 µm



y cut at -1.032 µm



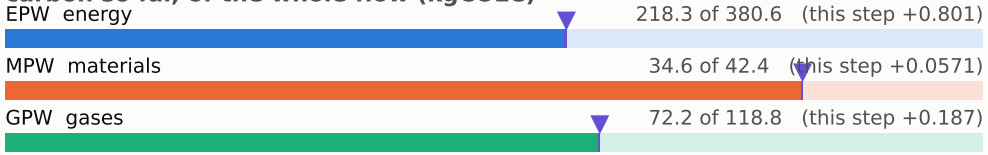
mask for this step

no mask for this step

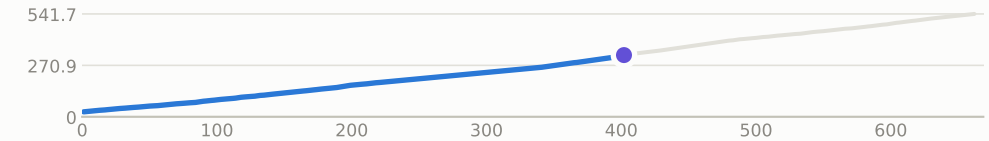
this step

process area	WLP6_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

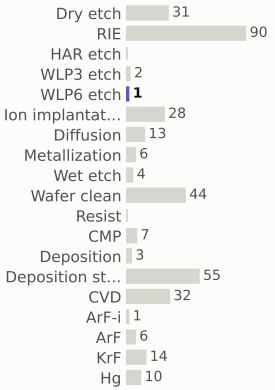
carbon so far, of the whole flow (kgCO2e)



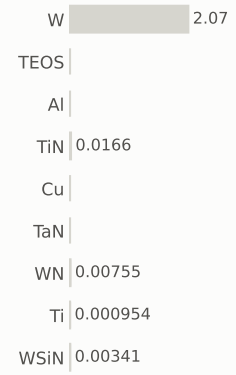
embodied carbon up to each step (kgCO2e per wafer)



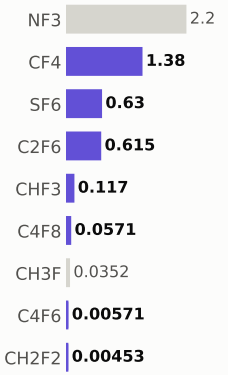
steps by process area



materials deposited (g)



gases emitted (g)



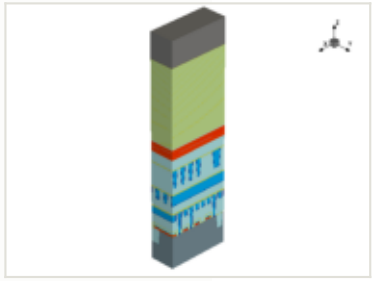
Step #4140: WLP6_etch

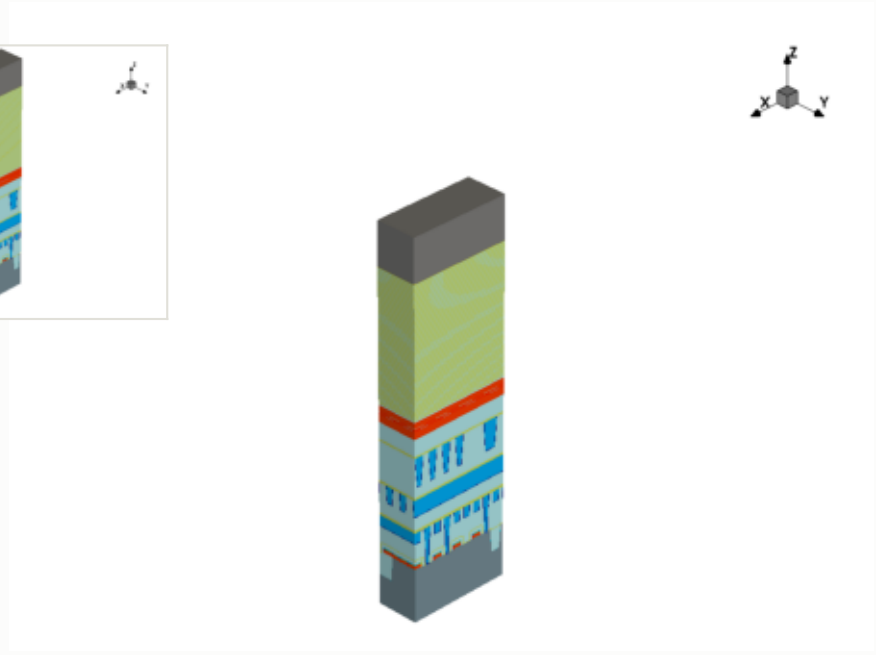
etch_anisotropic · WLP6_etch

1.04	326.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



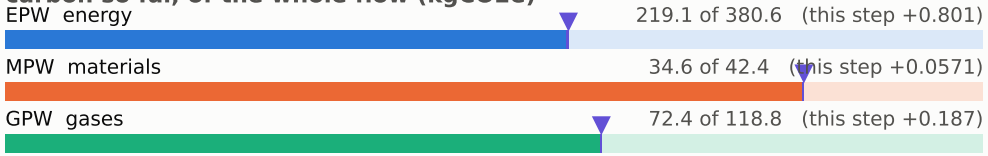


this step

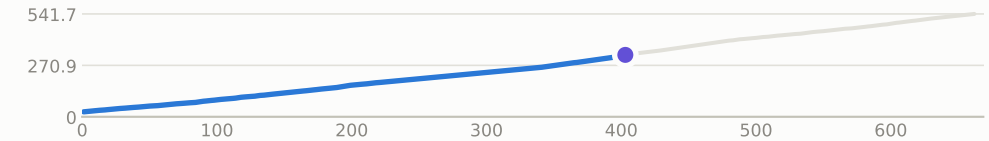
process area	WLP6_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

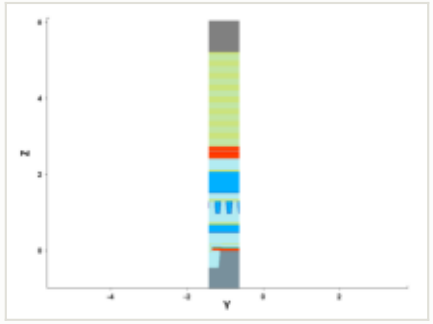
carbon so far, of the whole flow (kgCO2e)



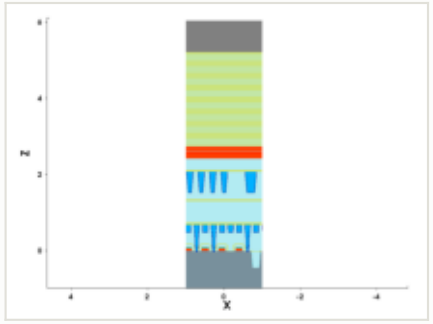
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



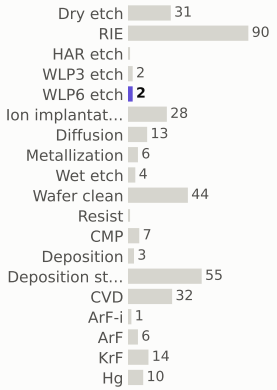
y cut at -1.032 µm



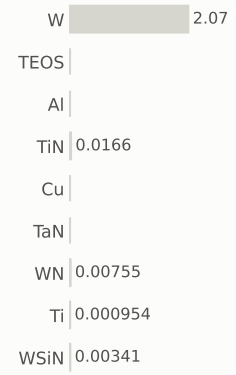
mask for this step

no mask for this step

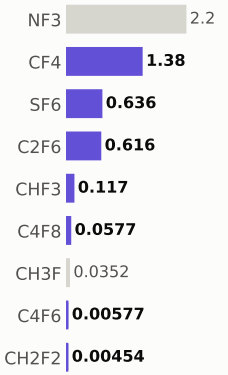
steps by process area



materials deposited (g)



gases emitted (g)



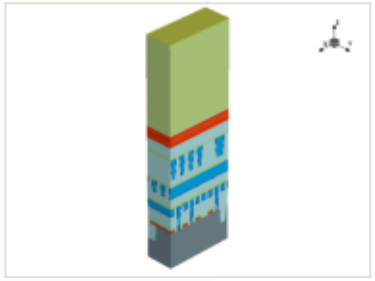
Step #4150: Dry_etch

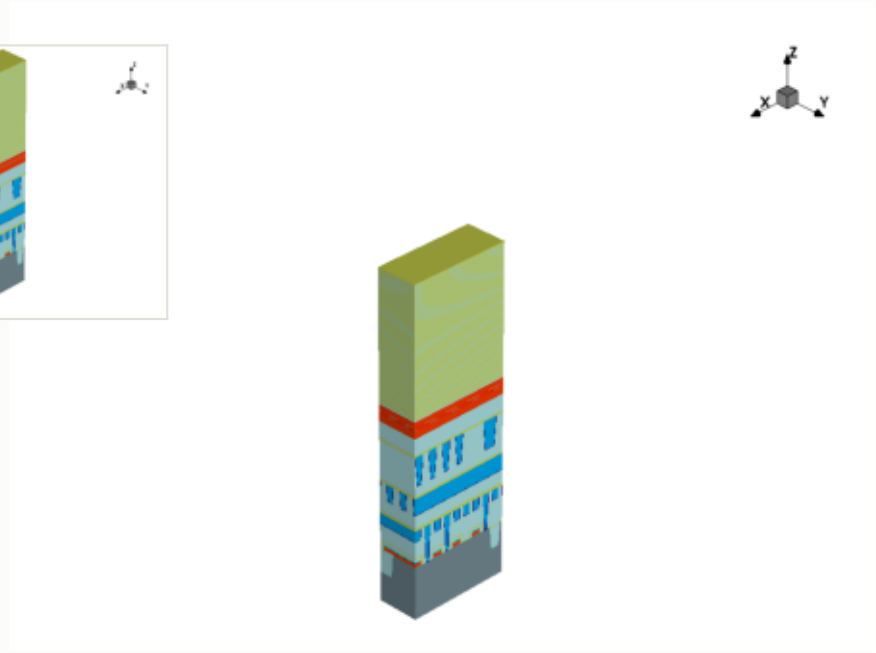
etch_strip · Dry_etch

0.801	326.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

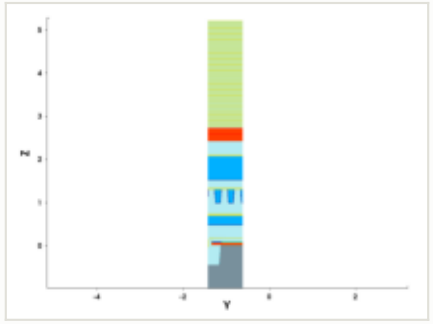
3D structure after this step

zoomed to fit

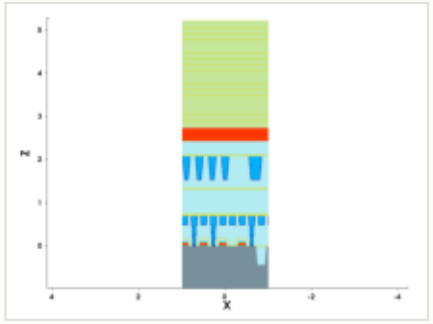




x cut at 0.000 µm



y cut at -1.032 µm



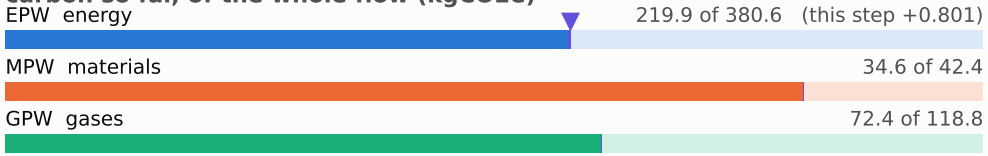
mask for this step

no mask for this step

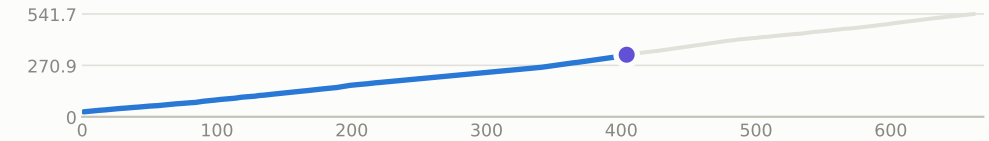
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

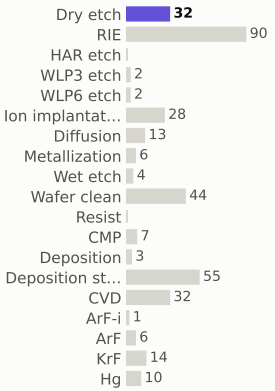
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

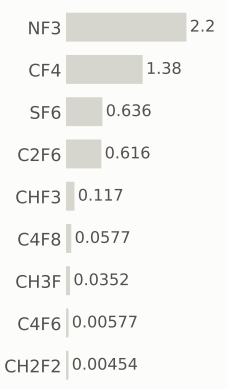


steps by process area



materials deposited (g)

gases emitted (g)



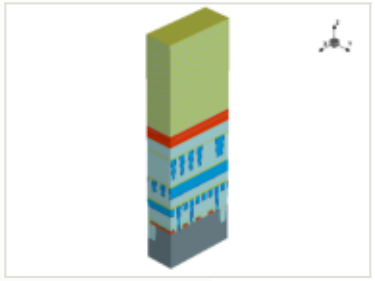
Step #4160: Wafer_clean

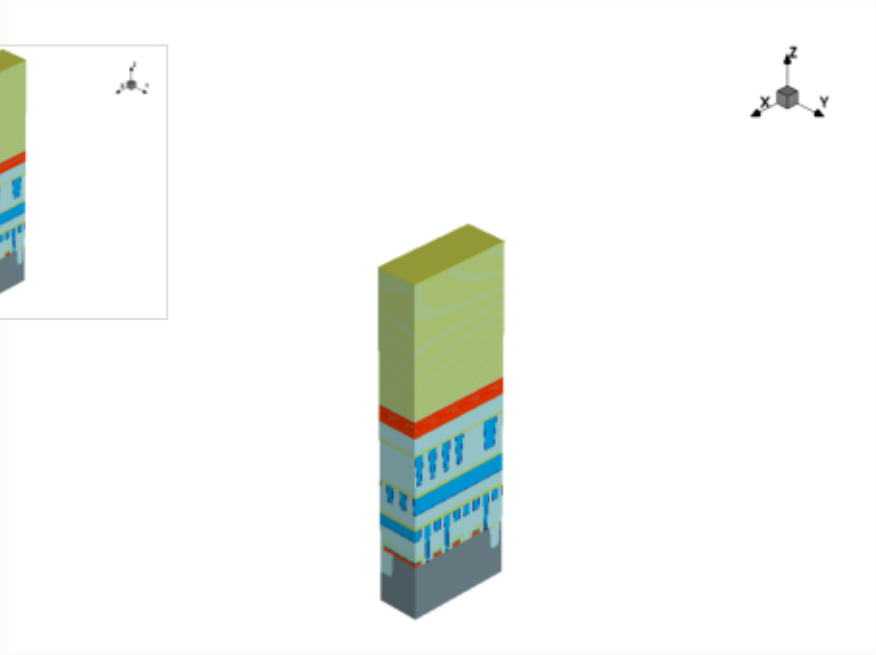
clean · Wafer_clean

0.169	327.1	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

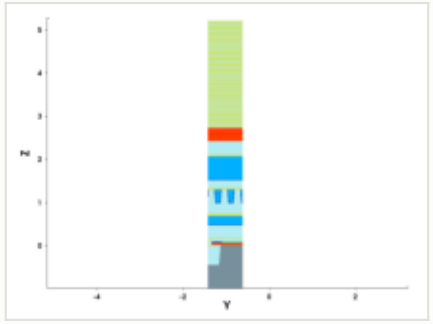
3D structure after this step

zoomed to fit

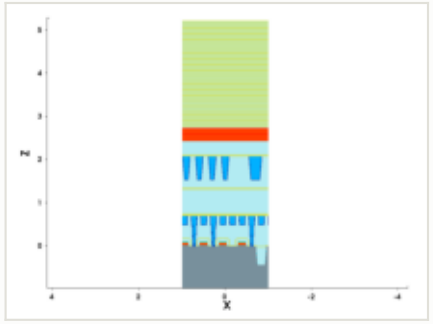




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

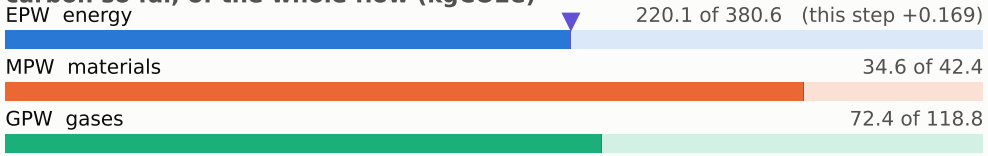
no mask for this step

this step

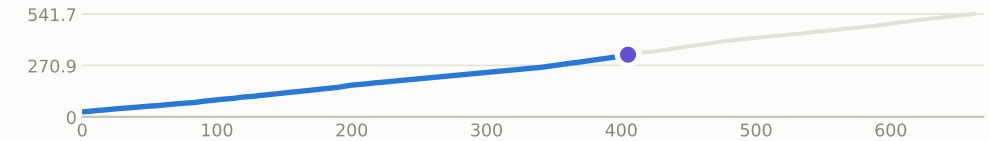
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

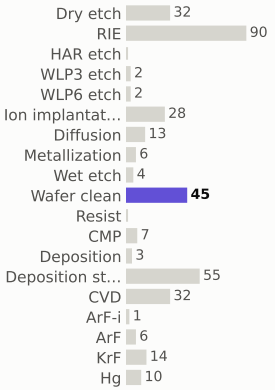
carbon so far, of the whole flow (kgCO2e)



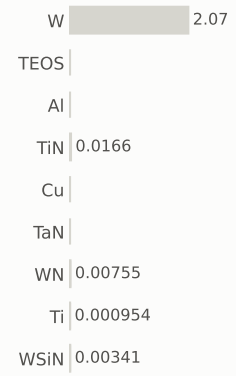
embodied carbon up to each step (kgCO2e per wafer)



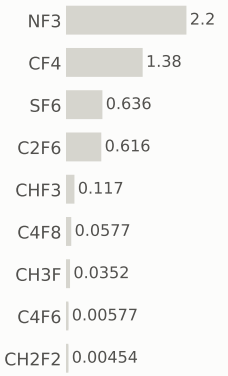
steps by process area



materials deposited (g)



gases emitted (g)



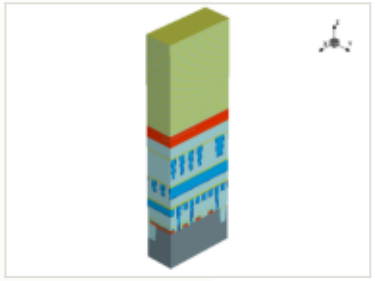
Step #4170: RIE

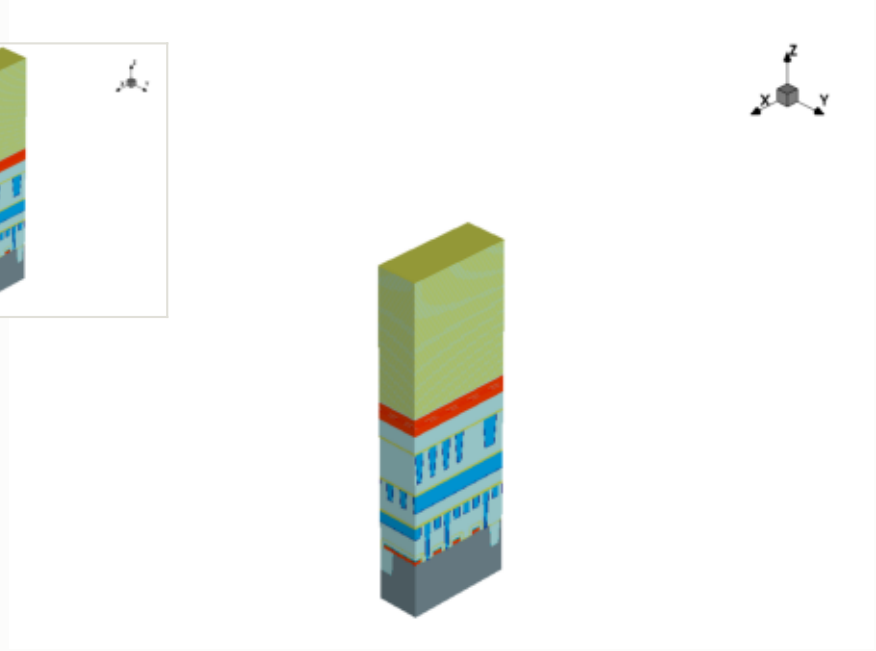
etch_anisotropic · RIE

1.04	328.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

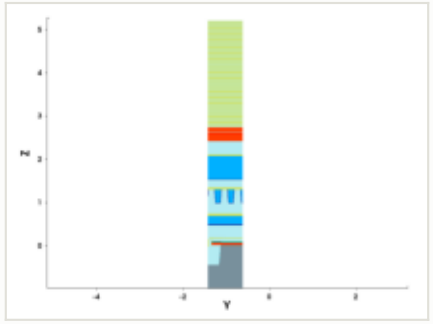
3D structure after this step

zoomed to fit

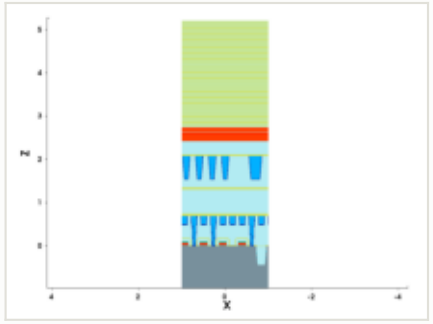




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

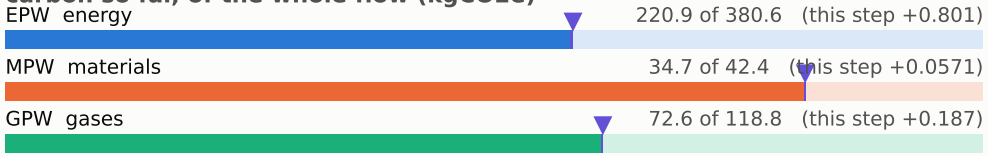
no mask for this step

this step

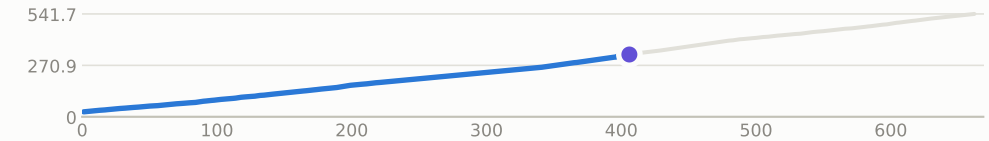
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

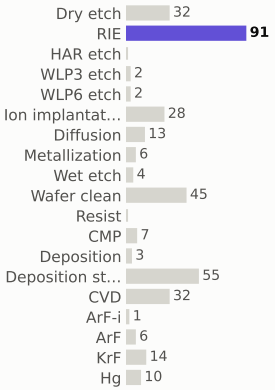
carbon so far, of the whole flow (kgCO2e)



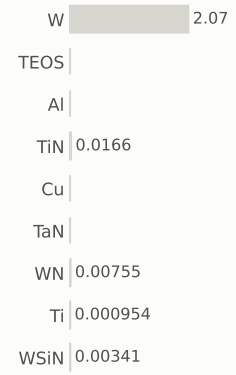
embodied carbon up to each step (kgCO2e per wafer)



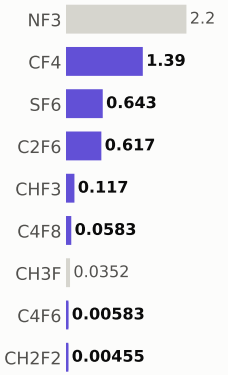
steps by process area



materials deposited (g)



gases emitted (g)



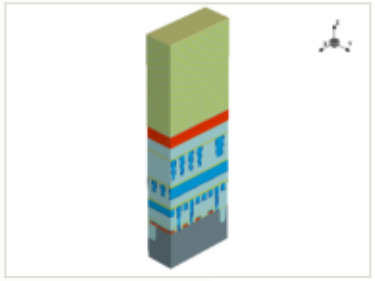
Step #4180: CVD

depo_isotropic · CVD

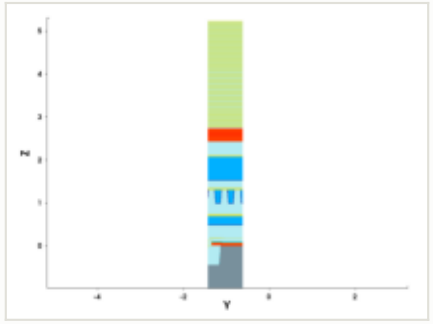
1.31	329.4	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

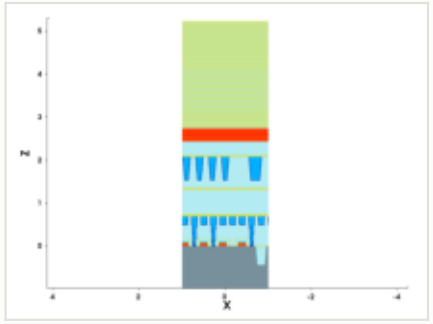
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

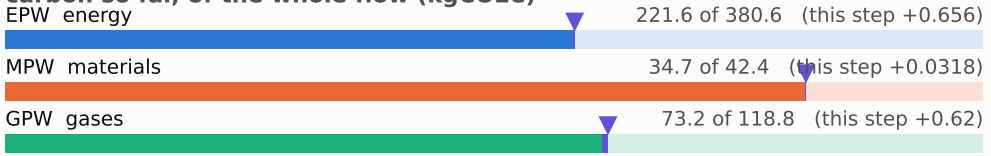
no mask for this step

this step

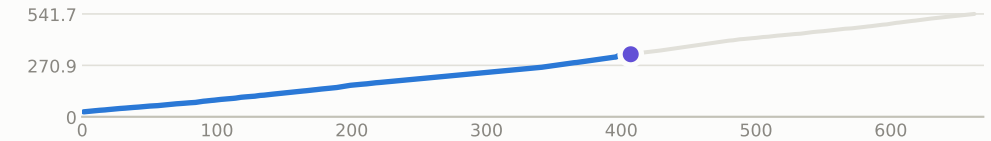
process area	CVD
material	—
thickness	0.034 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

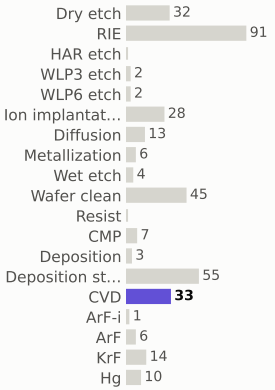
carbon so far, of the whole flow (kgCO2e)



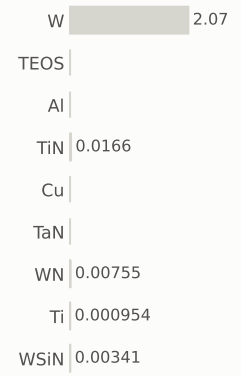
embodied carbon up to each step (kgCO2e per wafer)



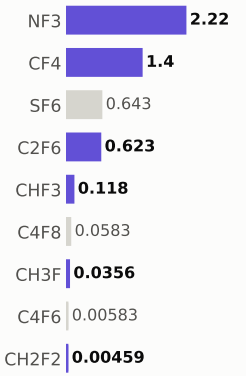
steps by process area



materials deposited (g)



gases emitted (g)



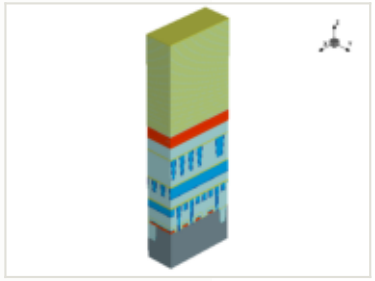
Step #4190: Ion_implantation

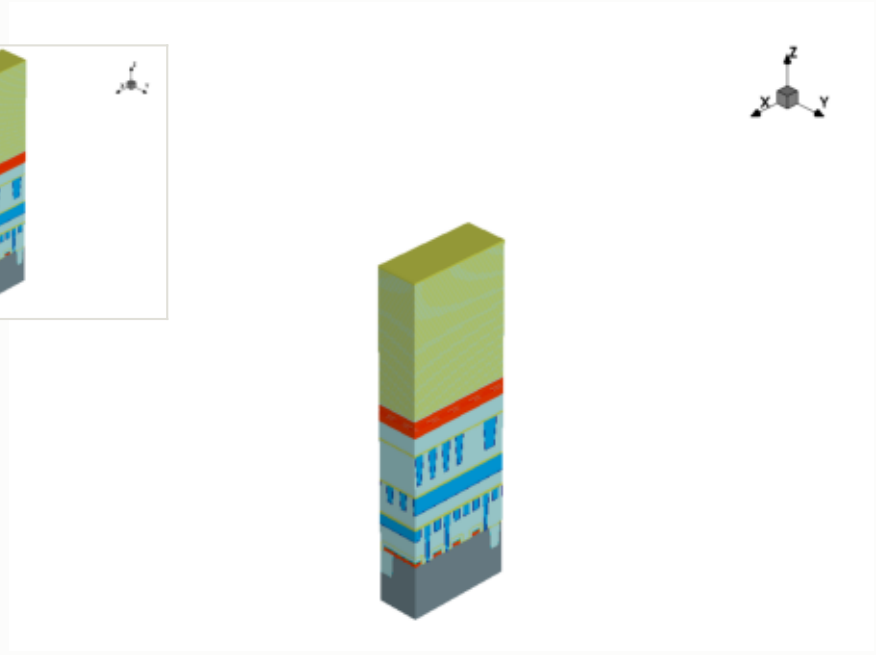
implant_ellipsoid · Ion_implantation

0.801	330.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

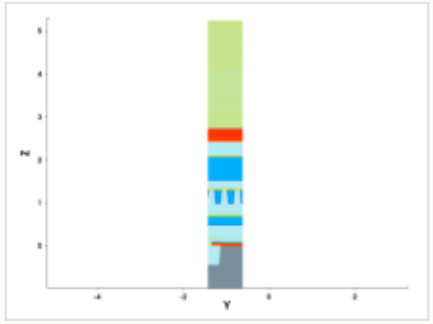
3D structure after this step

zoomed to fit

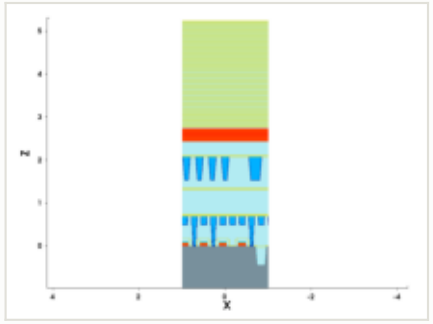




x cut at 0.000 µm



y cut at -1.032 µm



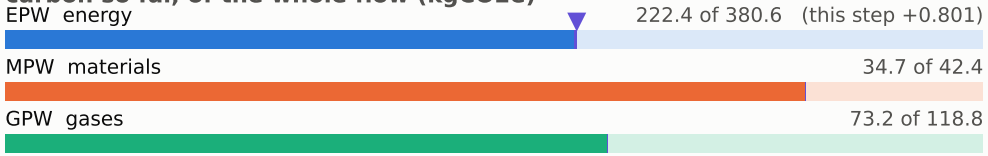
mask for this step

no mask for this step

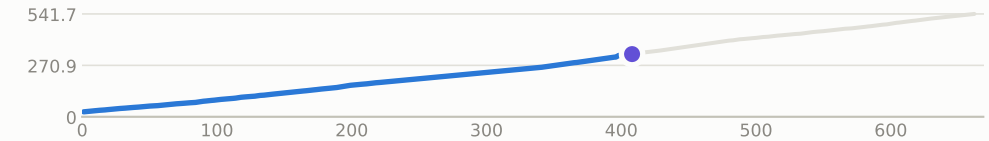
this step

process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

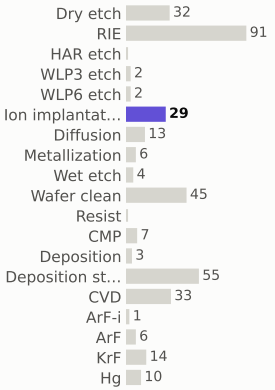
carbon so far, of the whole flow (kgCO2e)



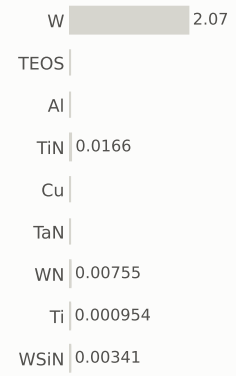
embodied carbon up to each step (kgCO2e per wafer)



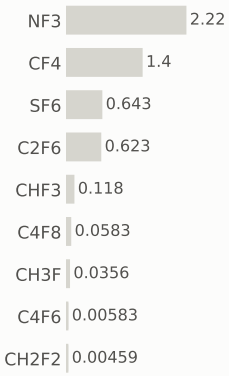
steps by process area



materials deposited (g)



gases emitted (g)



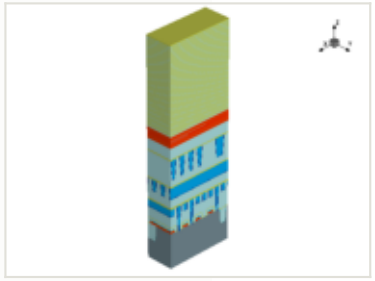
Step #4200: Wet_etch

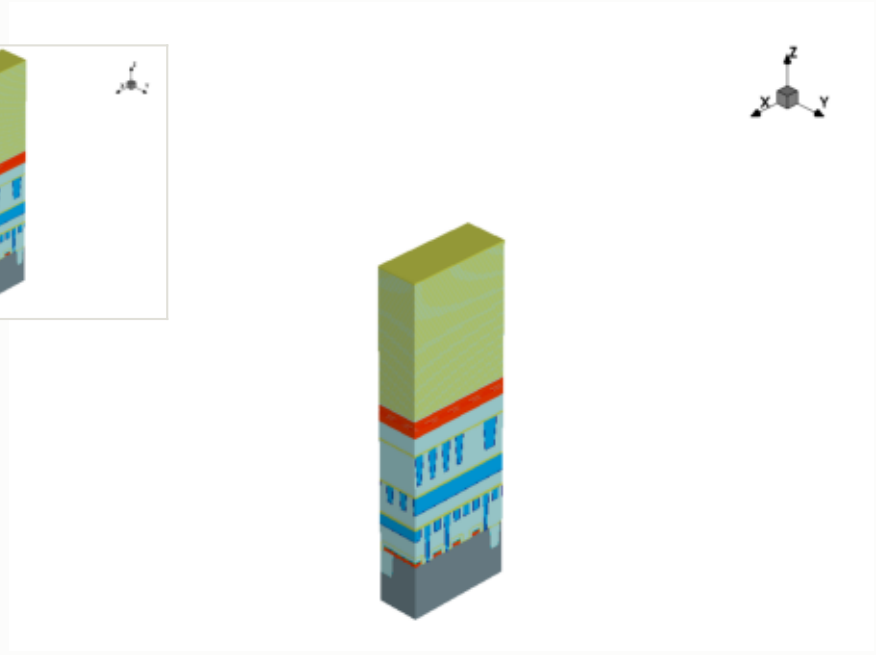
etch_isotropic · Wet_etch

0.169	330.4	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

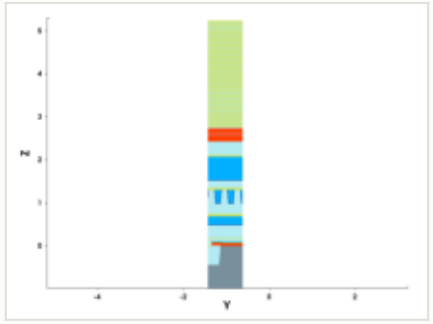
3D structure after this step

zoomed to fit

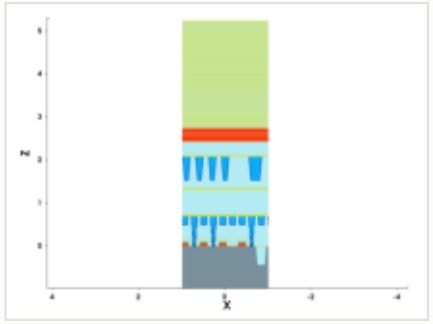




x cut at 0.000 µm



y cut at -1.032 µm



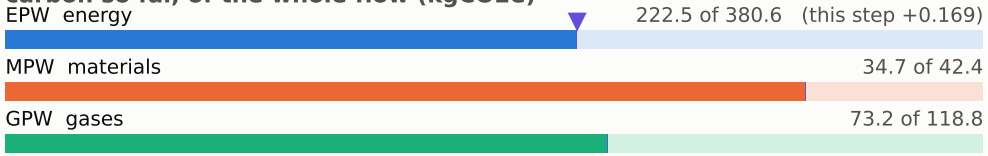
mask for this step

no mask for this step

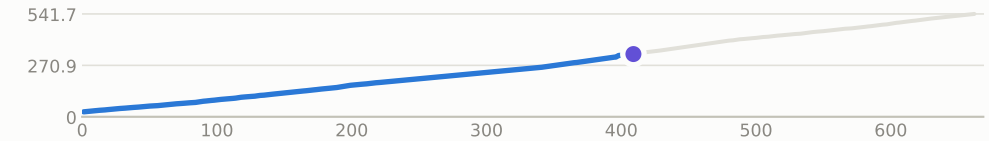
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

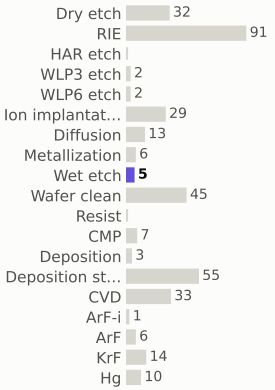
carbon so far, of the whole flow (kgCO2e)



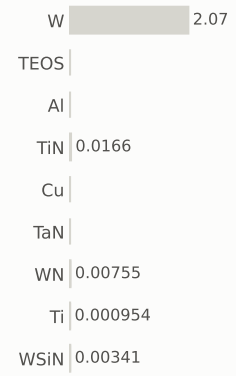
embodied carbon up to each step (kgCO2e per wafer)



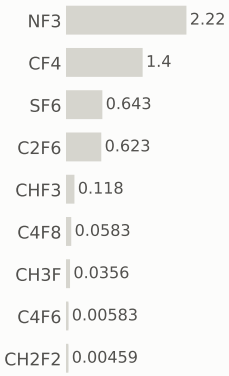
steps by process area



materials deposited (g)



gases emitted (g)



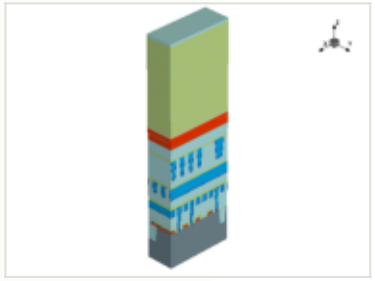
Step #4210: CVD

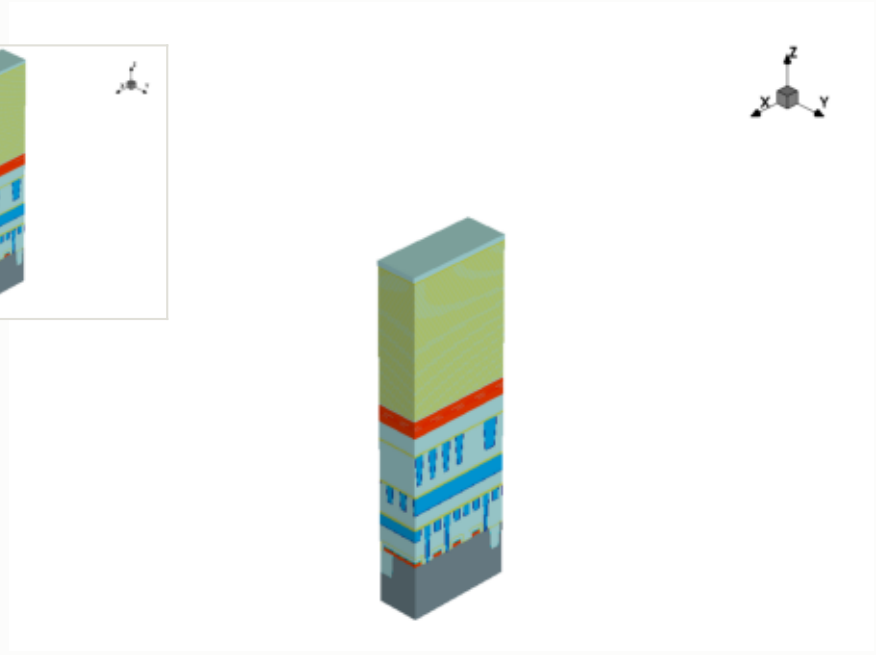
depo_isotropic · CVD

1.31	331.7	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

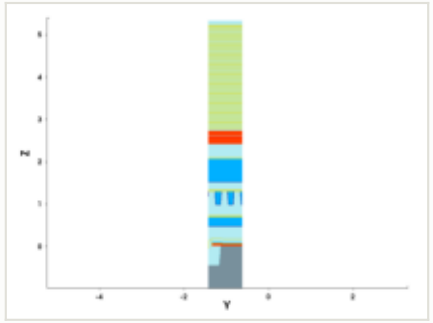
3D structure after this step

zoomed to fit

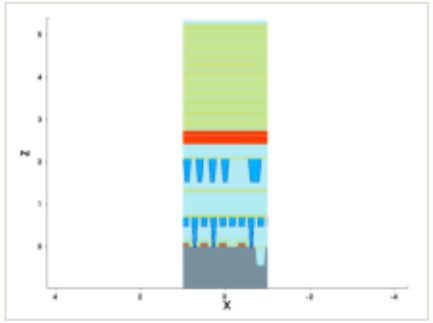




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

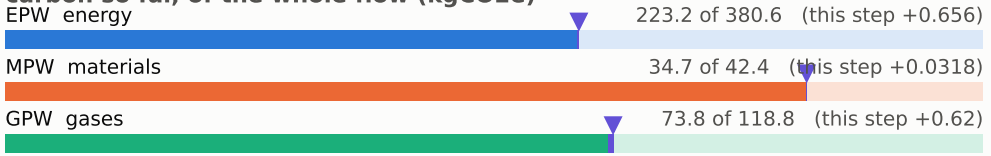
no mask for this step

this step

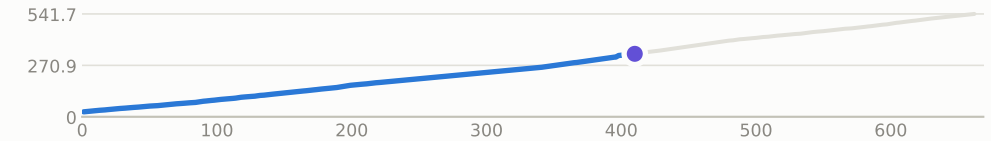
process area	CVD
material	—
thickness	0.1 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

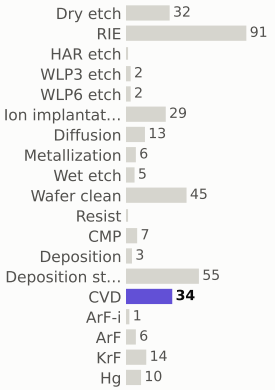
carbon so far, of the whole flow (kgCO2e)



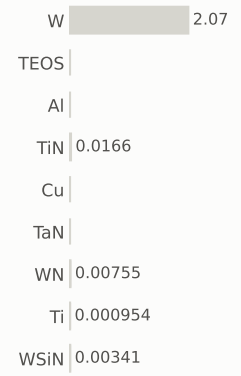
embodied carbon up to each step (kgCO2e per wafer)



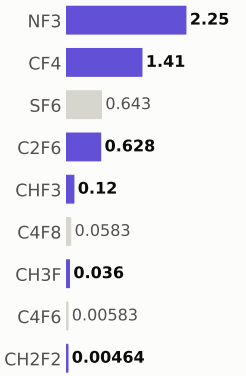
steps by process area



materials deposited (g)



gases emitted (g)



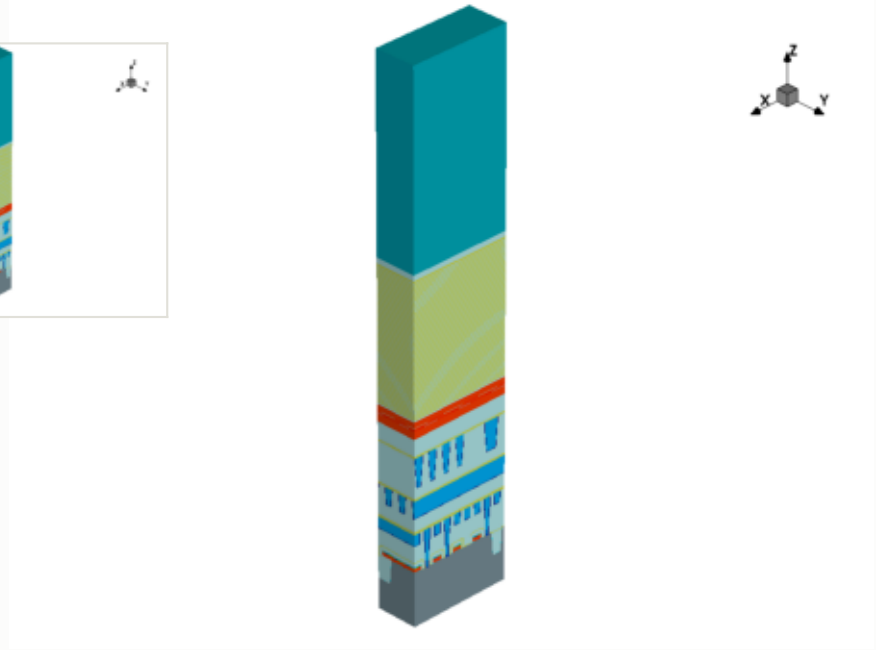
Step #4220: CVD

depo_isotropic · CVD

1.31	333.0	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

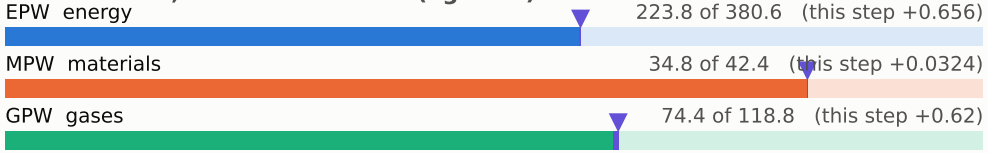


this step

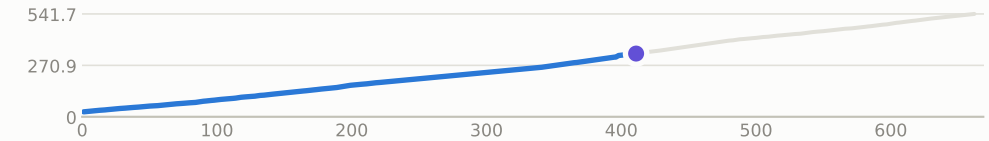
process area	CVD
material	TEOS
thickness	3.6 µm
material GWP	2.85952 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0324 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

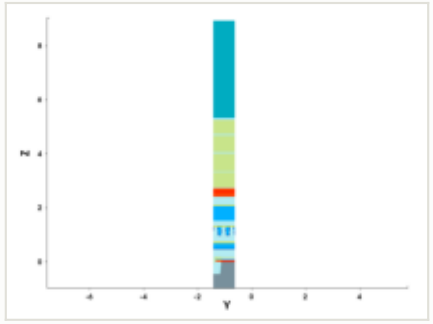
carbon so far, of the whole flow (kgCO2e)



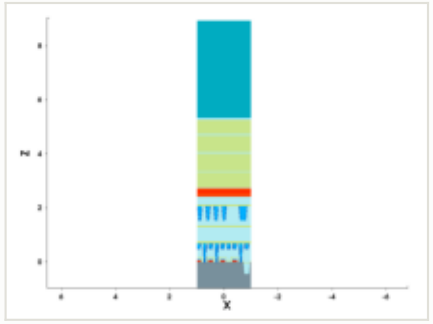
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



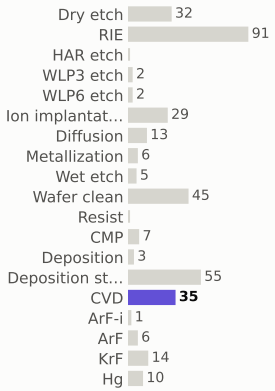
y cut at -1.032 µm



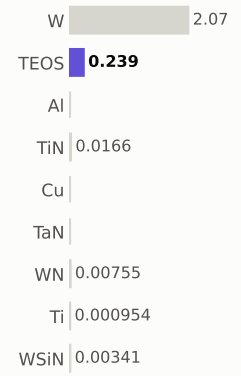
mask for this step

no mask for this step

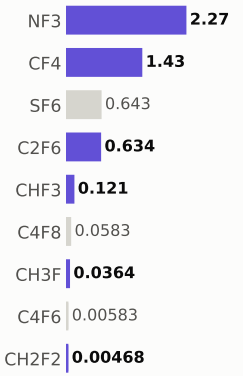
steps by process area



materials deposited (g)



gases emitted (g)



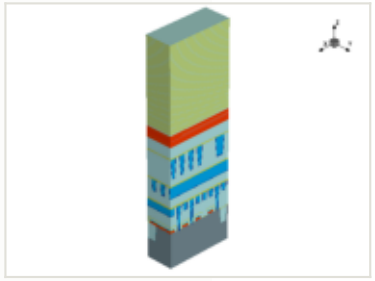
Step #4230: CMP

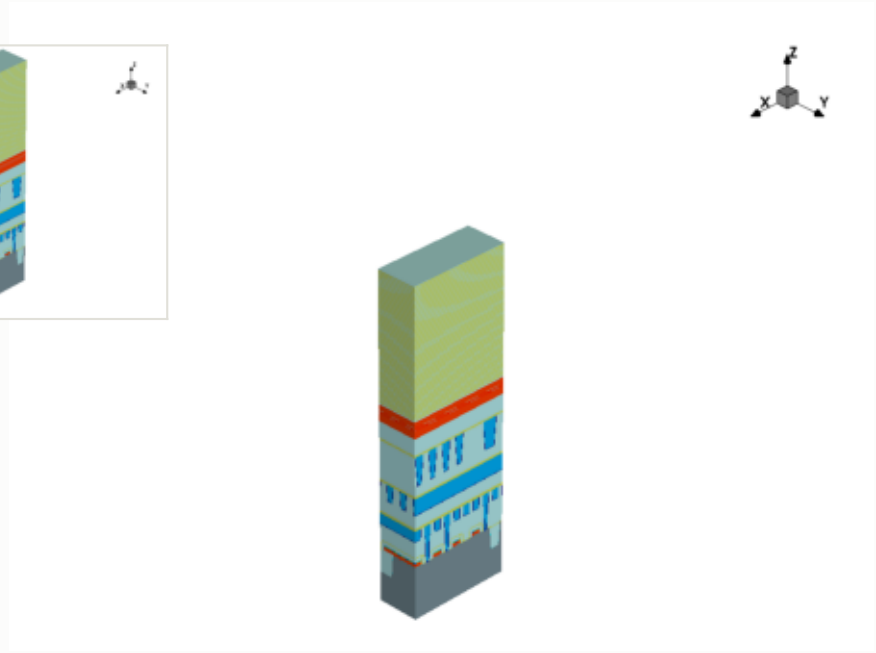
etch_cmp · CMP

0.169	333.2	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

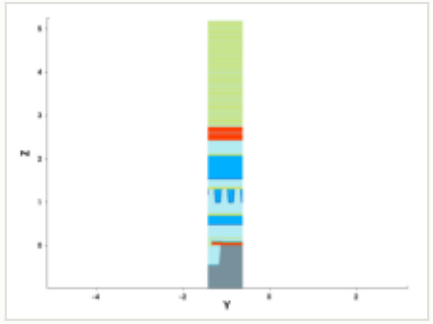
3D structure after this step

zoomed to fit

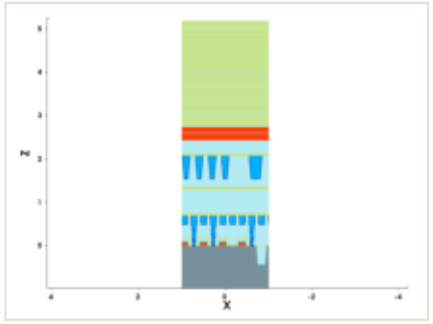




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

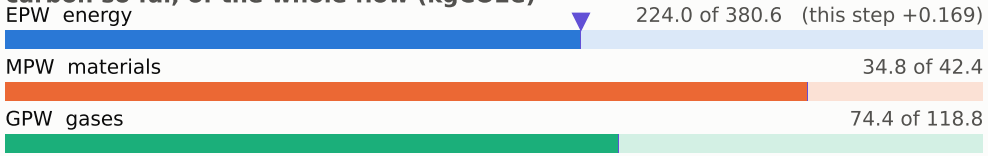
no mask for this step

this step

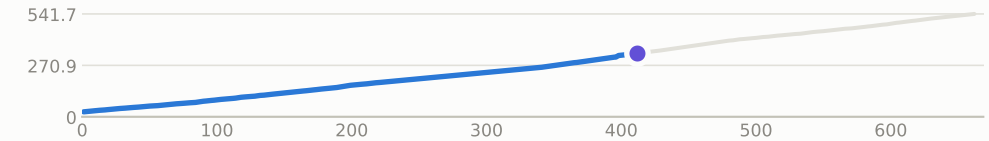
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

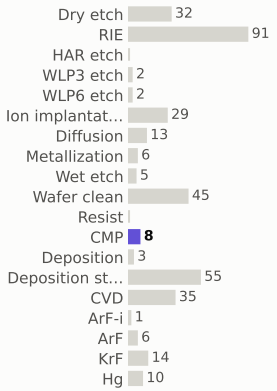
carbon so far, of the whole flow (kgCO2e)



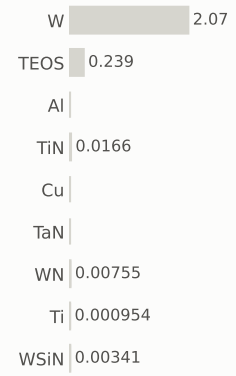
embodied carbon up to each step (kgCO2e per wafer)



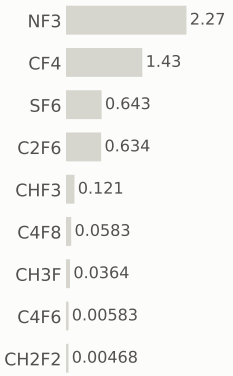
steps by process area



materials deposited (g)



gases emitted (g)



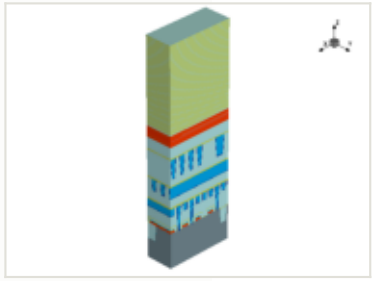
Step #4240: Wafer_clean

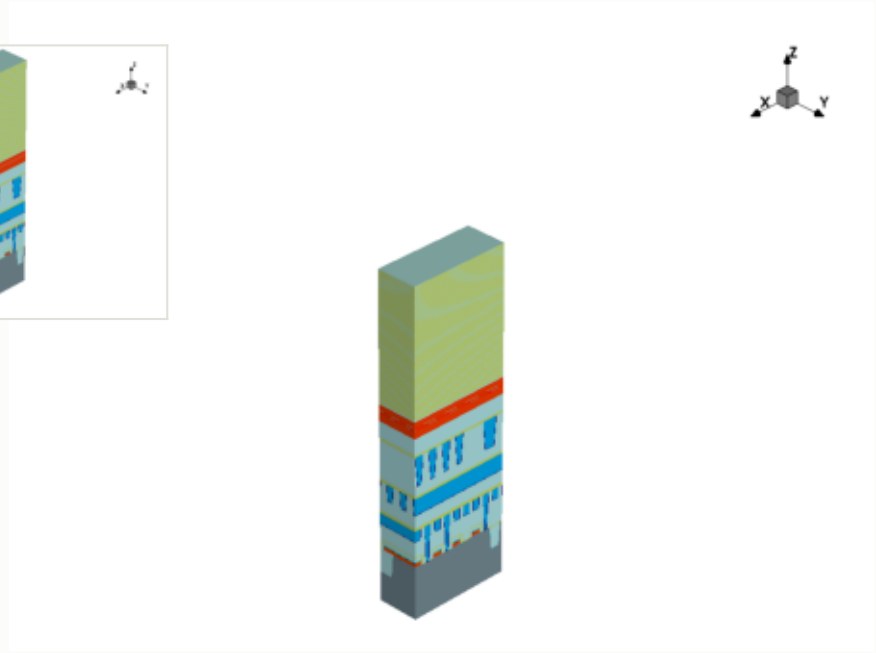
clean · Wafer_clean

0.169	333.4	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

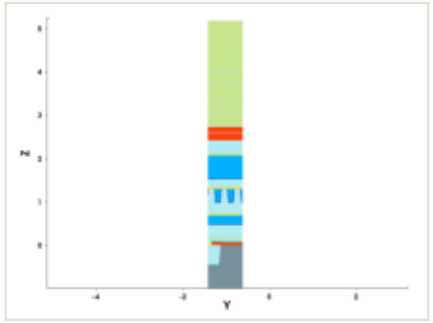
3D structure after this step

zoomed to fit

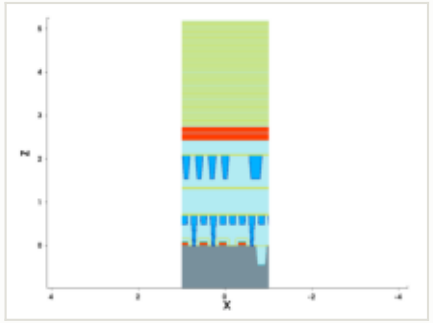




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

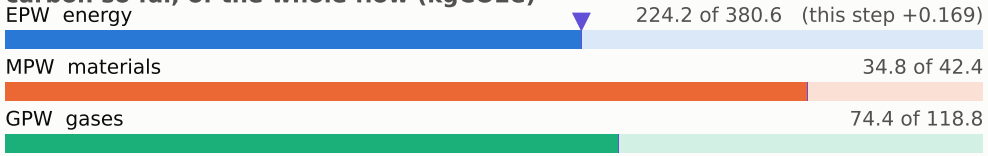
no mask for this step

this step

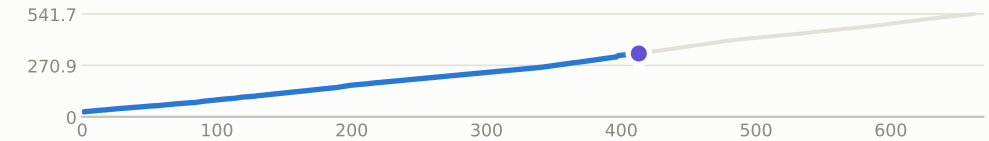
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

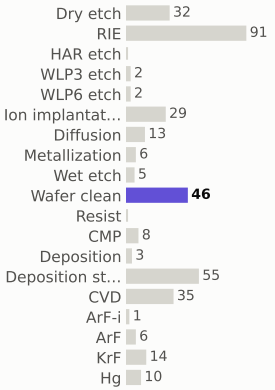
carbon so far, of the whole flow (kgCO2e)



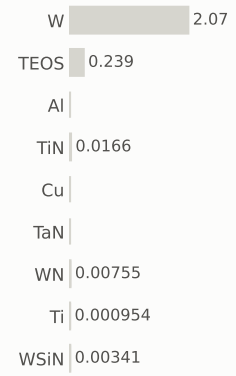
embodied carbon up to each step (kgCO2e per wafer)



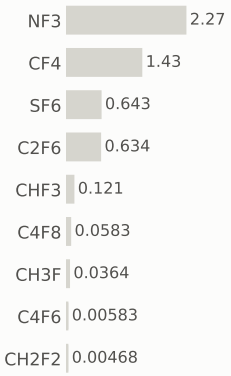
steps by process area



materials deposited (g)



gases emitted (g)



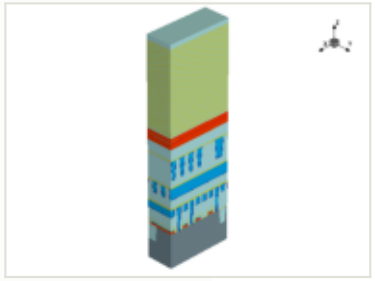
Step #4250: CVD

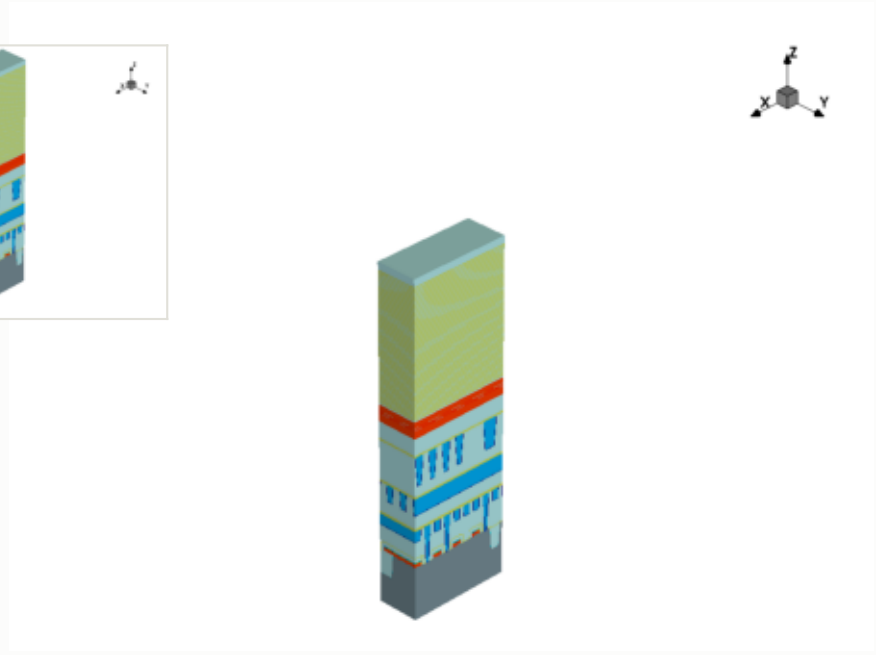
depo_isotropic · CVD

1.31	334.7	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

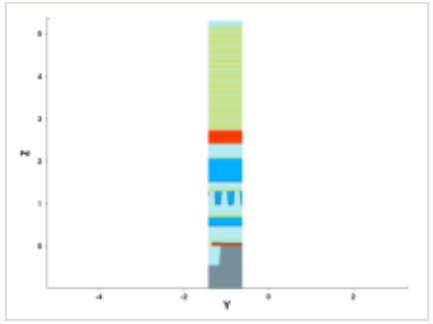
3D structure after this step

zoomed to fit

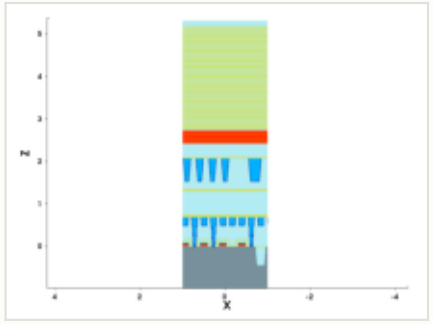




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

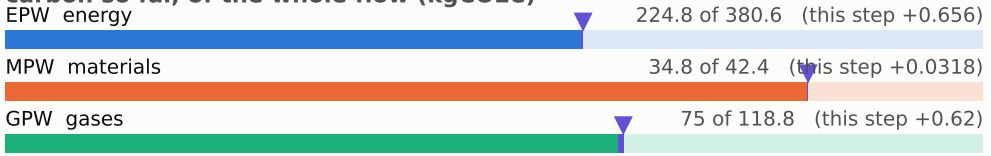
no mask for this step

this step

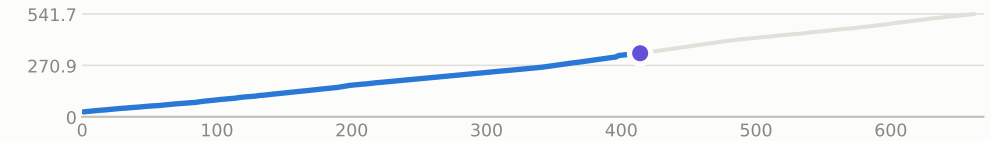
process area	CVD
material	—
thickness	0.13 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

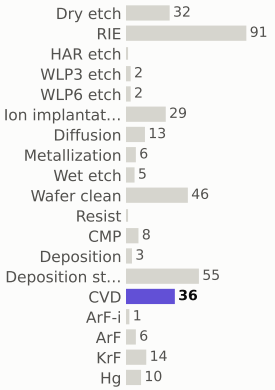
carbon so far, of the whole flow (kgCO2e)



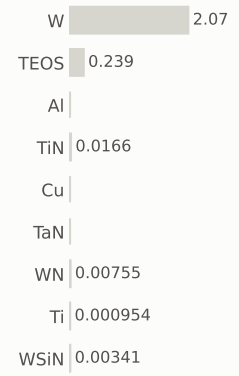
embodied carbon up to each step (kgCO2e per wafer)



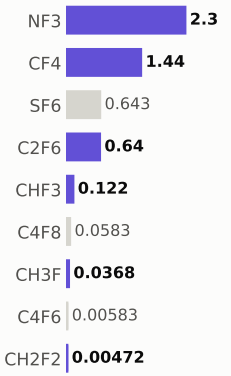
steps by process area



materials deposited (g)



gases emitted (g)



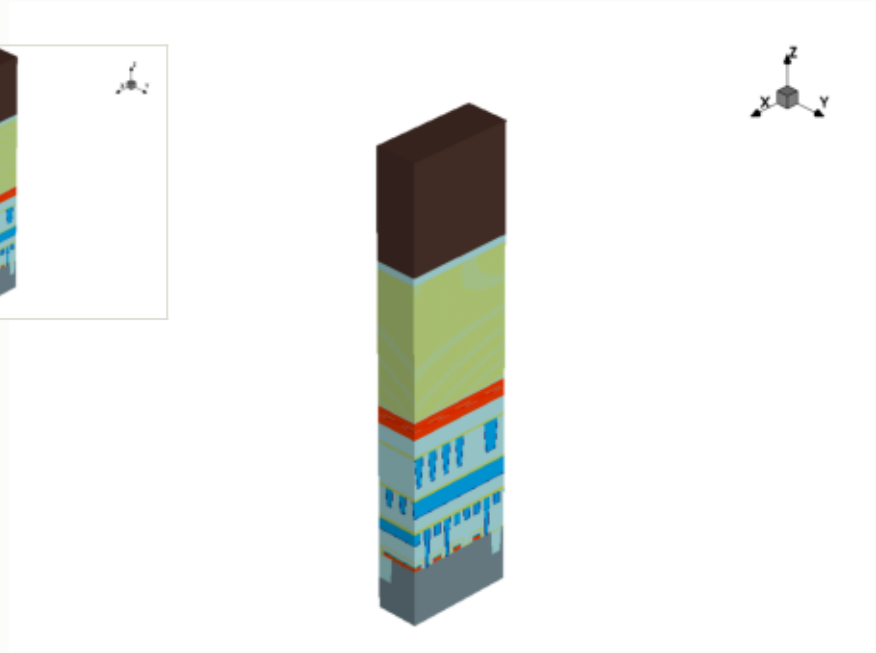
Step #4270: CVD

depo_isotropic · CVD

1.31	336.0	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

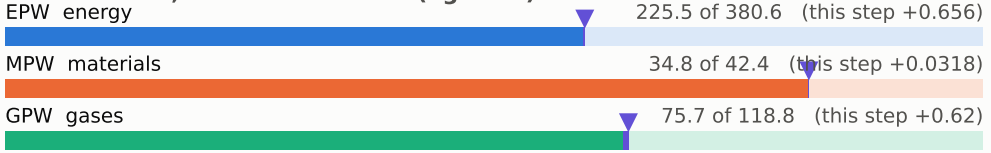


this step

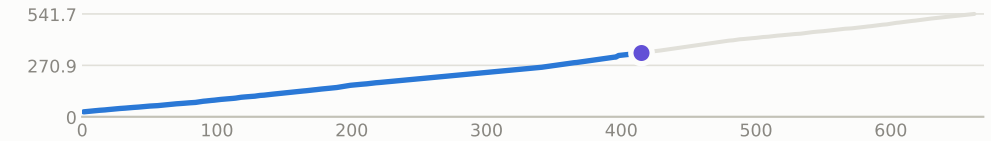
process area	CVD
material	—
thickness	2 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

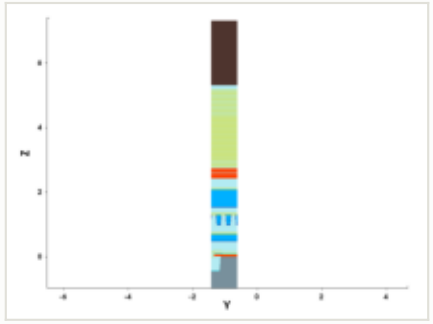
carbon so far, of the whole flow (kgCO2e)



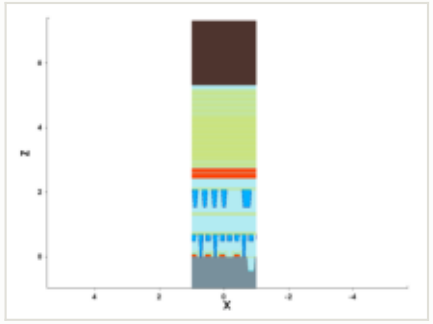
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



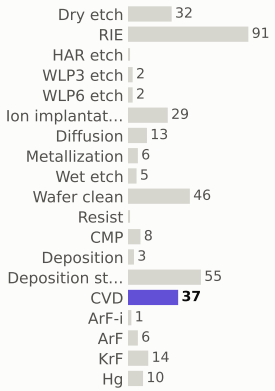
y cut at -1.032 µm



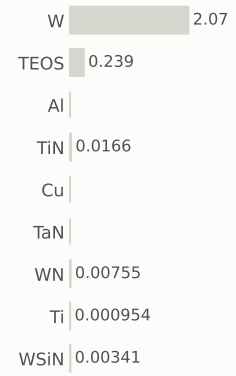
mask for this step

no mask for this step

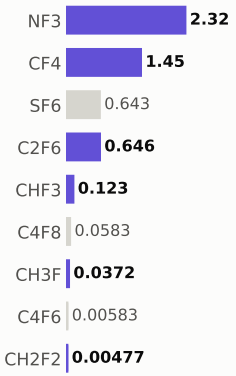
steps by process area



materials deposited (g)



gases emitted (g)



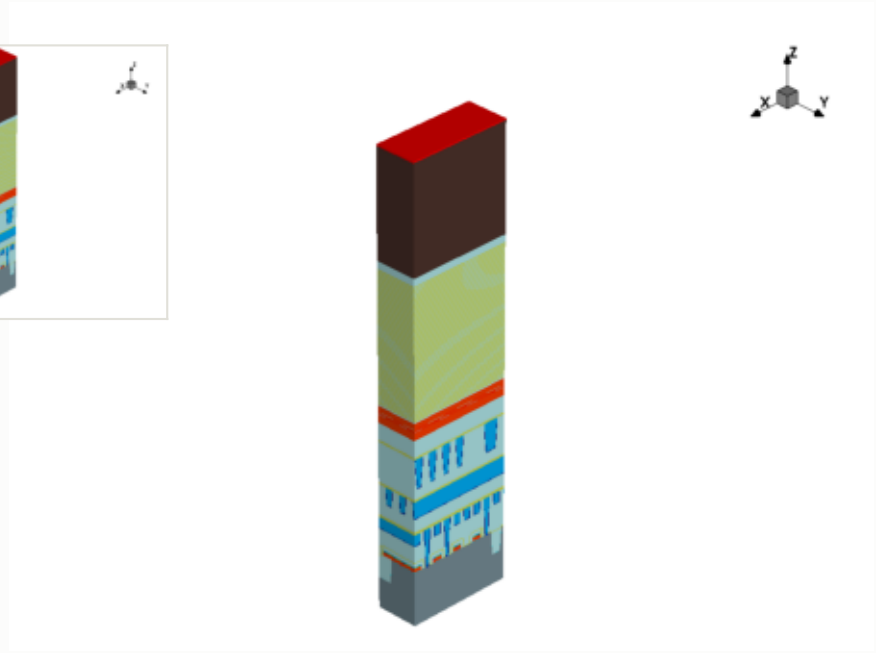
Step #4280: CVD

depo_isotropic · CVD

1.31	337.3	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

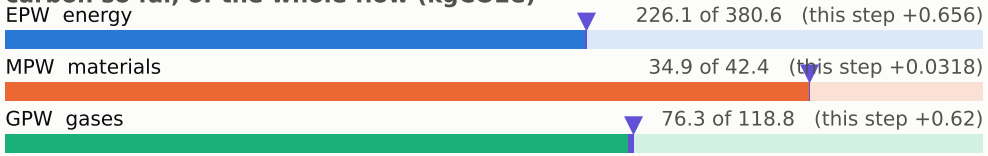


this step

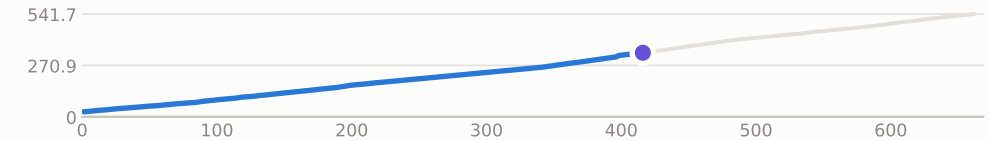
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

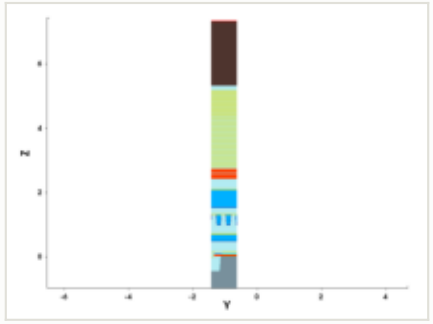
carbon so far, of the whole flow (kgCO2e)



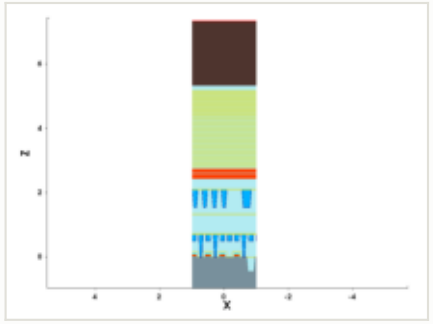
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



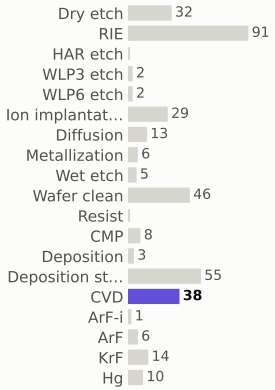
y cut at -1.032 µm



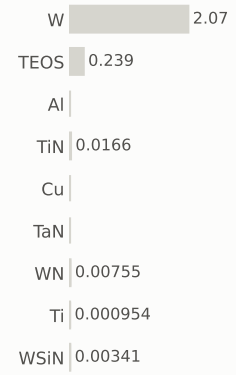
mask for this step

no mask for this step

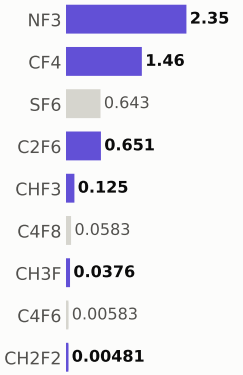
steps by process area



materials deposited (g)



gases emitted (g)



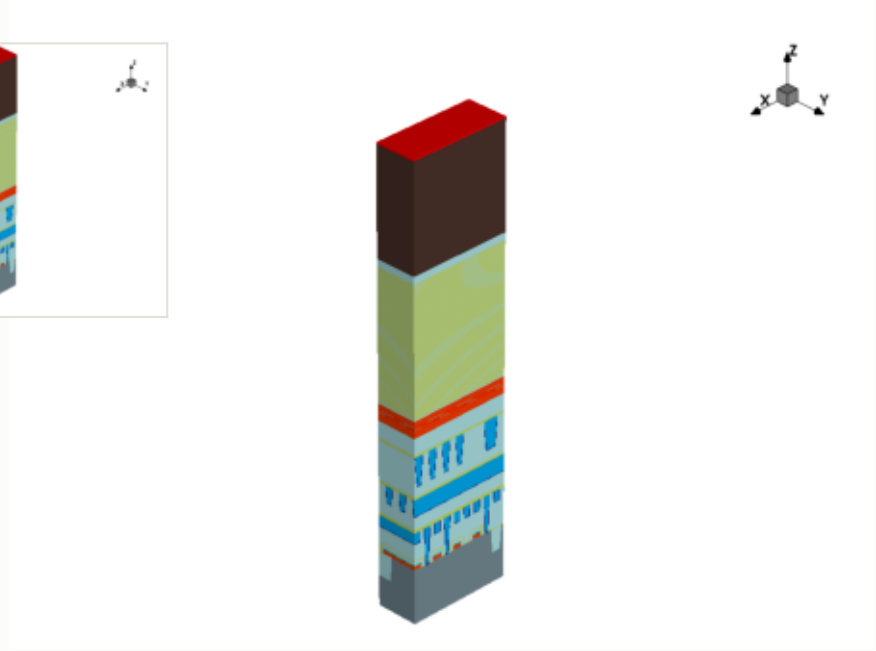
Step #4290: Wafer_clean

clean · Wafer_clean

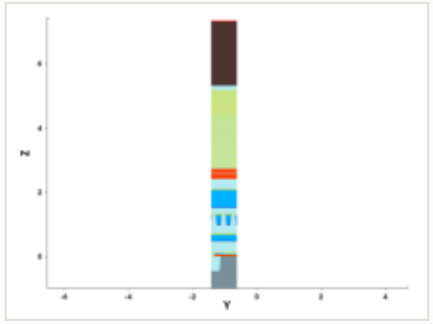
0.169	337.5	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

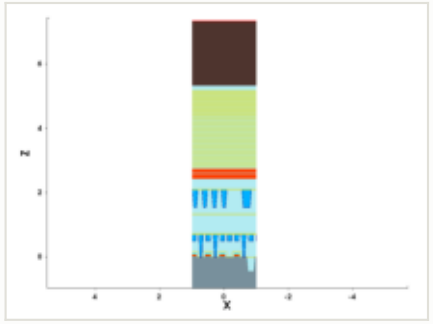
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

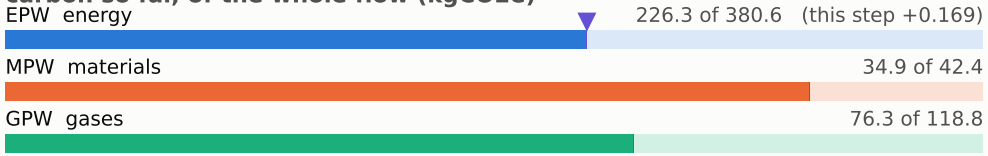
no mask for this step

this step

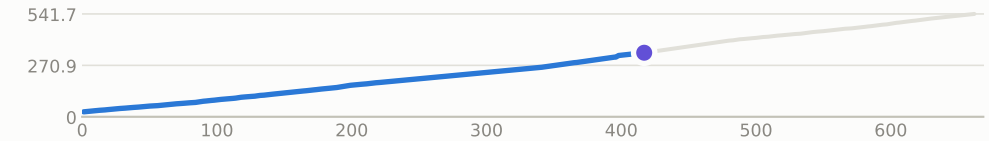
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

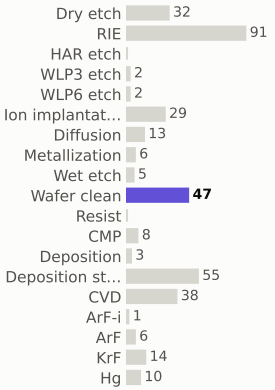
carbon so far, of the whole flow (kgCO2e)



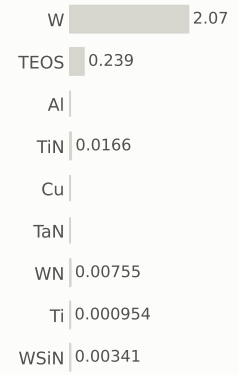
embodied carbon up to each step (kgCO2e per wafer)



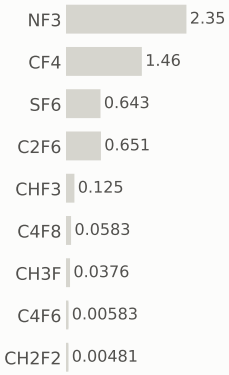
steps by process area



materials deposited (g)



gases emitted (g)



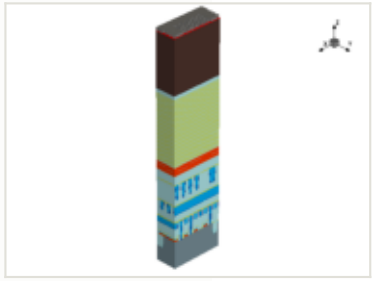
Step #4300: ArF-i

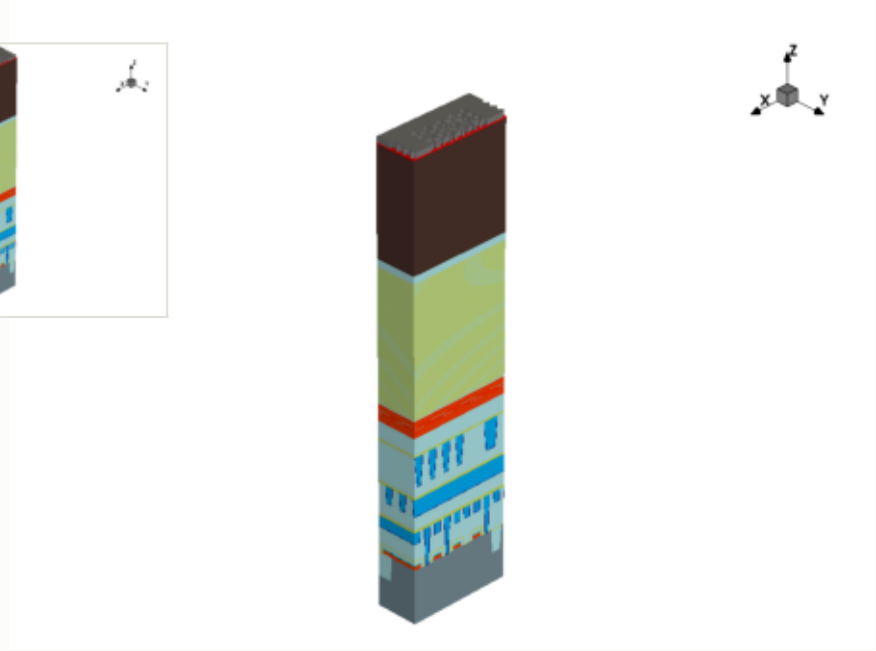
litho_patterning · ArF-i

0.217	337.7	0.04%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



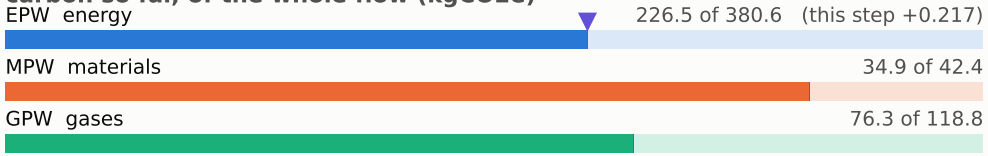


this step

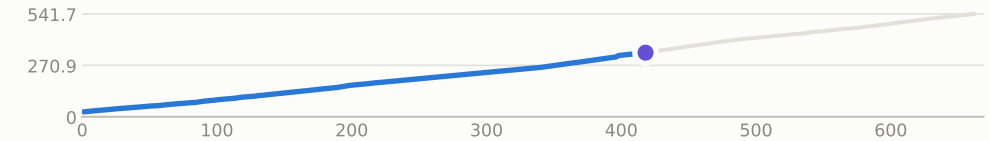
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

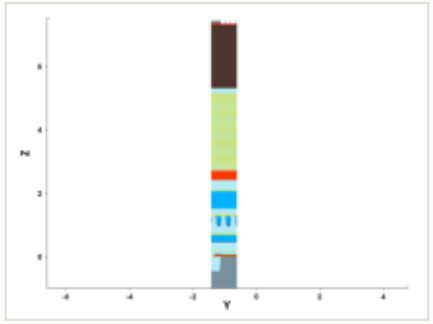
carbon so far, of the whole flow (kgCO2e)



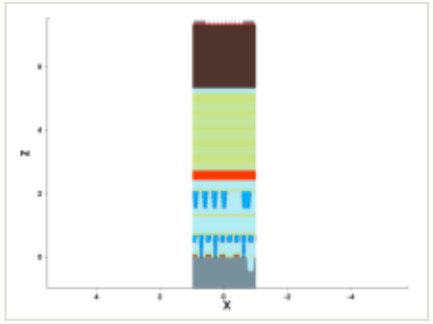
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

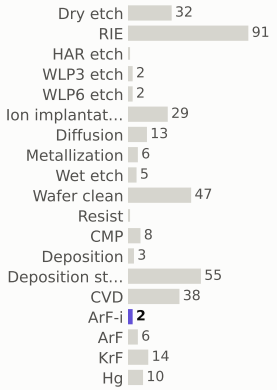


y cut at -1.032 µm

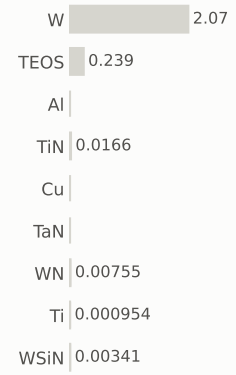


mask: layer 35:0 (OASIS)

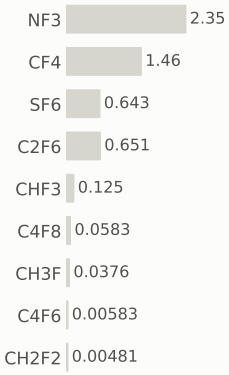
steps by process area



materials deposited (g)



gases emitted (g)



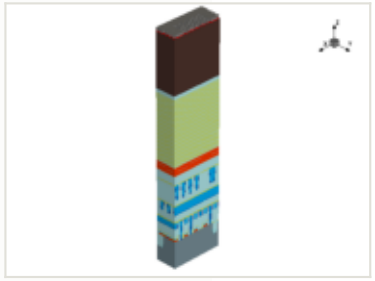
Step #4310: RIE

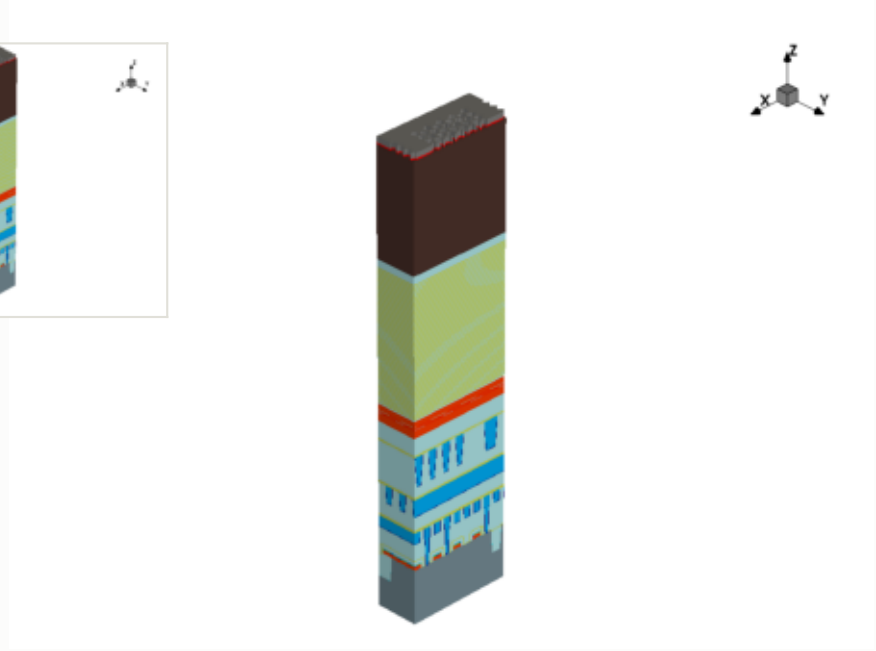
etch_anisotropic · RIE

1.04	338.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



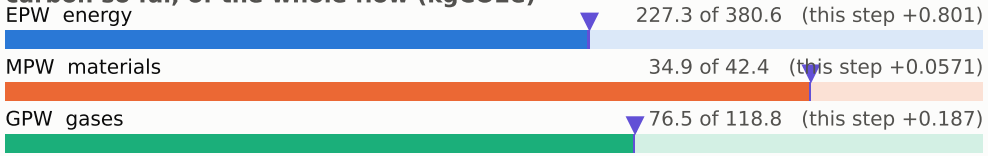


this step

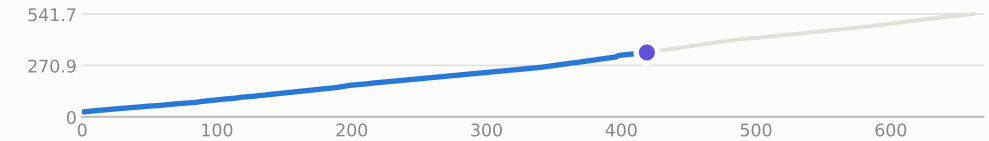
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

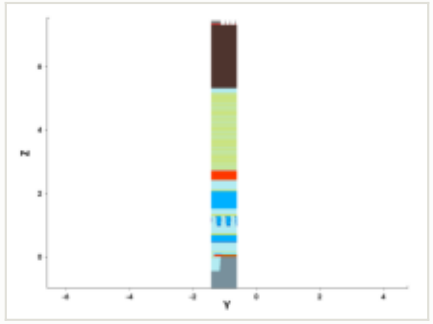
carbon so far, of the whole flow (kgCO2e)



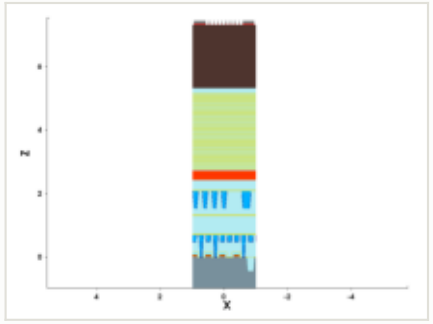
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



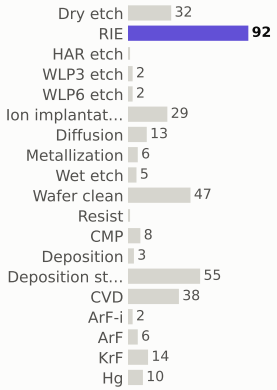
y cut at -1.032 µm



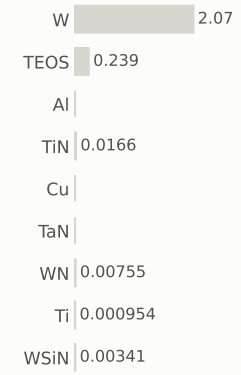
mask for this step

no mask for this step

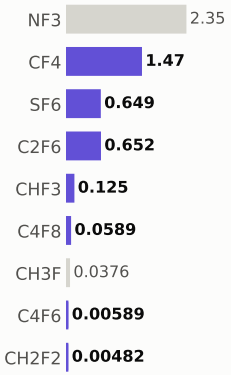
steps by process area



materials deposited (g)



gases emitted (g)



Step #4320: RIE

etch_anisotropic · RIE

1.04	339.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

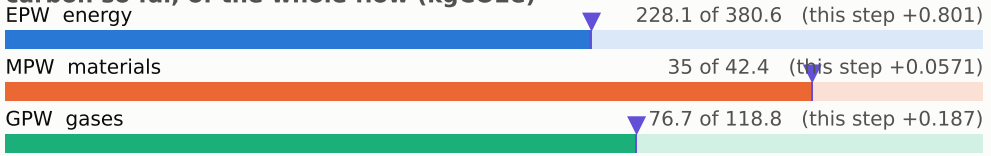
zoomed to fit

this step

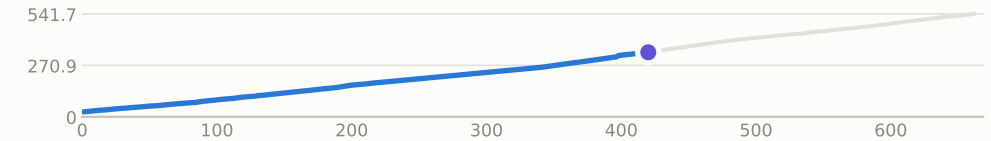
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

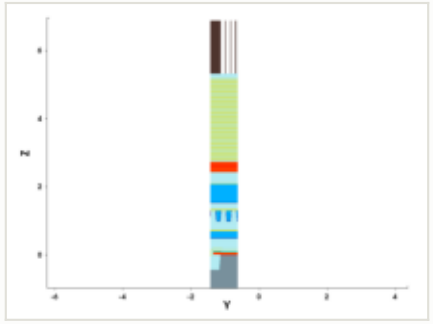
carbon so far, of the whole flow (kgCO2e)



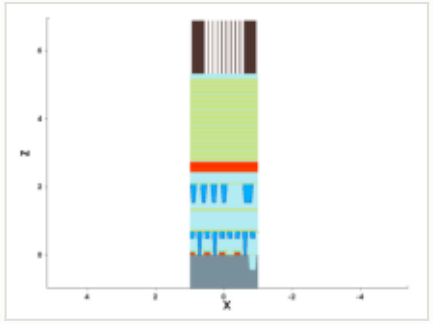
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



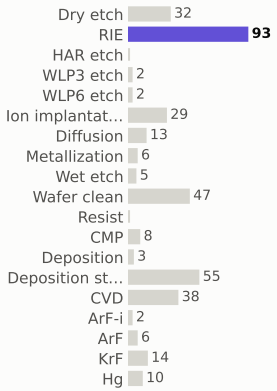
y cut at -1.032 µm



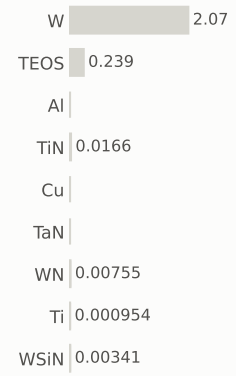
mask for this step

no mask for this step

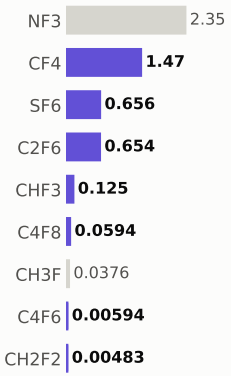
steps by process area



materials deposited (g)



gases emitted (g)



Step #4330: RIE

etch_anisotropic · RIE

1.04	340.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

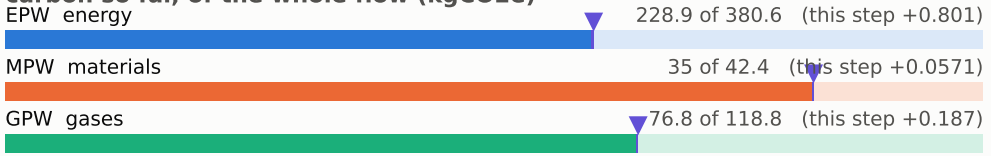
zoomed to fit

this step

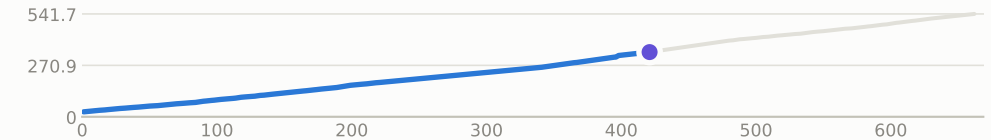
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

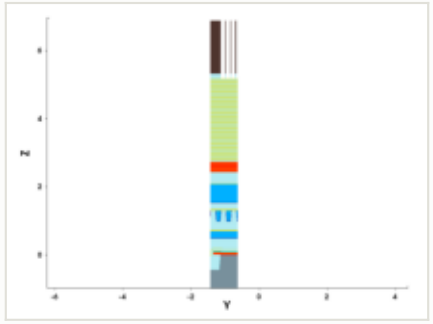
carbon so far, of the whole flow (kgCO2e)



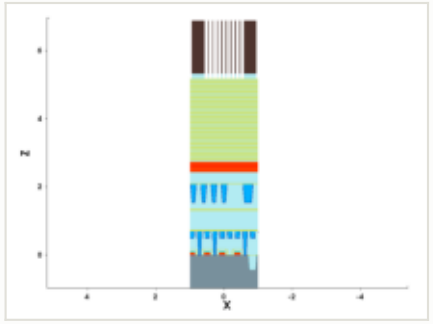
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



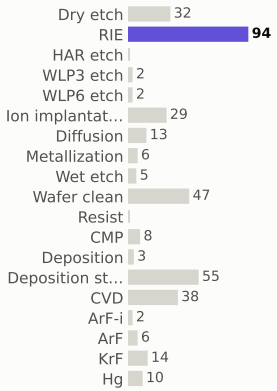
y cut at -1.032 µm



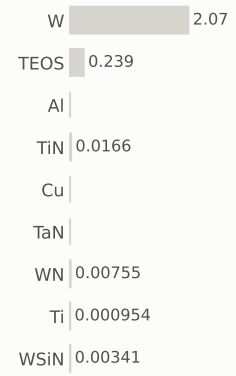
mask for this step

no mask for this step

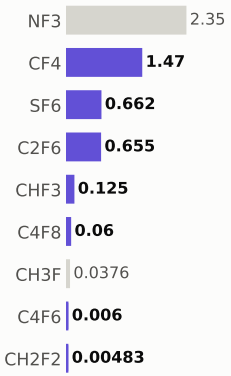
steps by process area



materials deposited (g)



gases emitted (g)



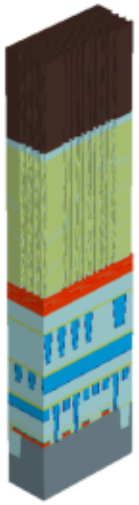
Step #4340: HAR_etch

etch_anisotropic · HAR_etch

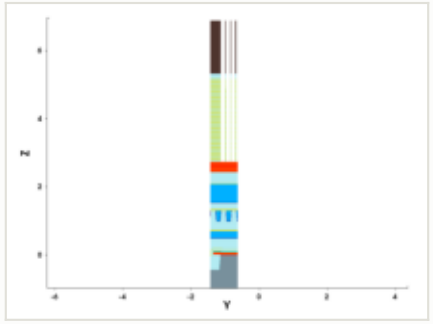
1.04	341.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

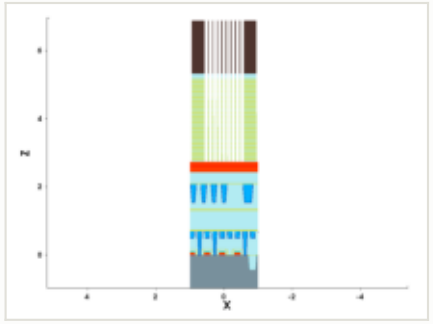
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

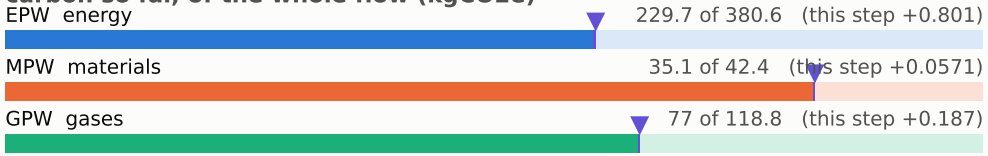
no mask for this step

this step

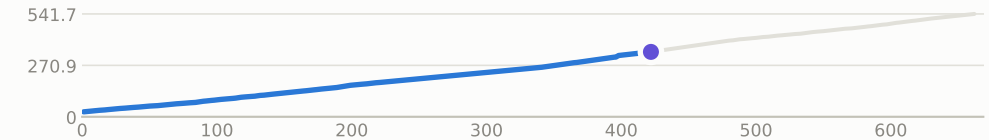
process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

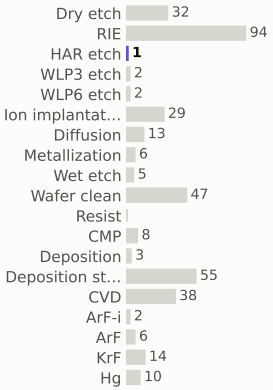
carbon so far, of the whole flow (kgCO2e)



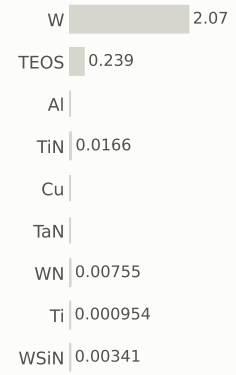
embodied carbon up to each step (kgCO2e per wafer)



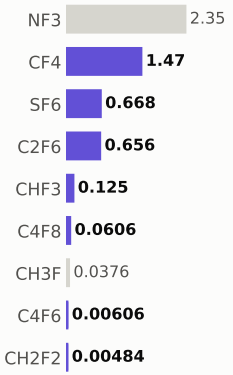
steps by process area



materials deposited (g)



gases emitted (g)



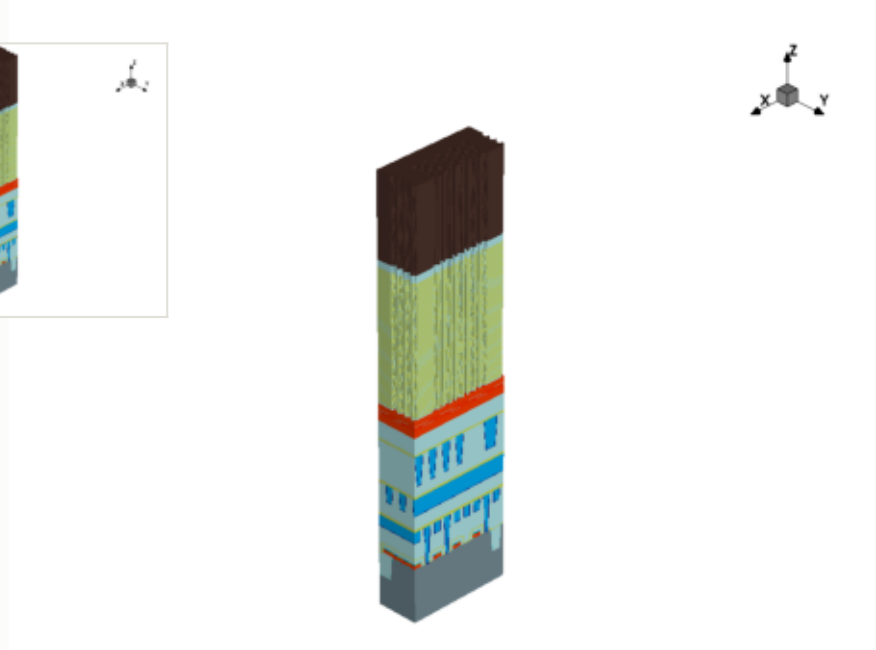
Step #4340.19: HAR_etch

etch_noop · HAR_etch

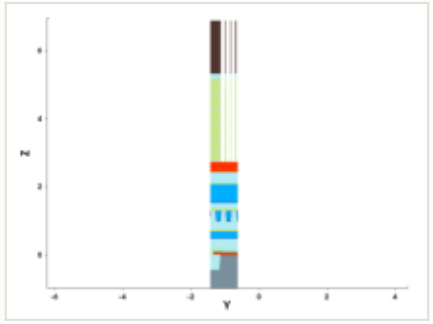
1.04	342.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

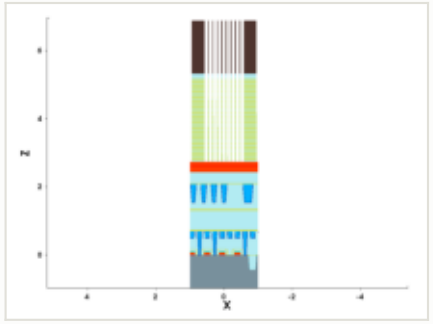
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

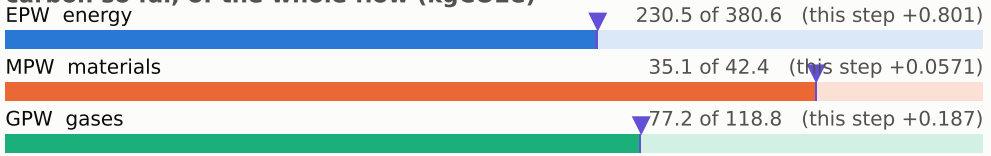
no mask for this step

this step

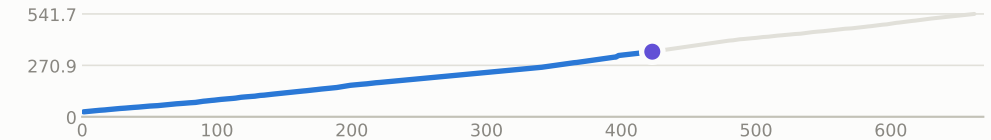
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

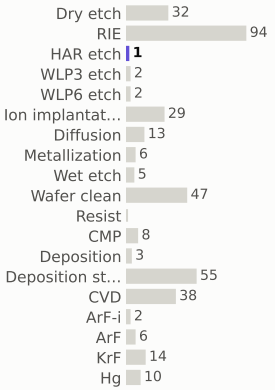
carbon so far, of the whole flow (kgCO2e)



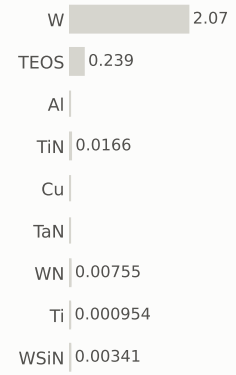
embodied carbon up to each step (kgCO2e per wafer)



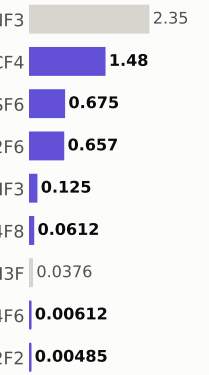
steps by process area



materials deposited (g)



gases emitted (g)



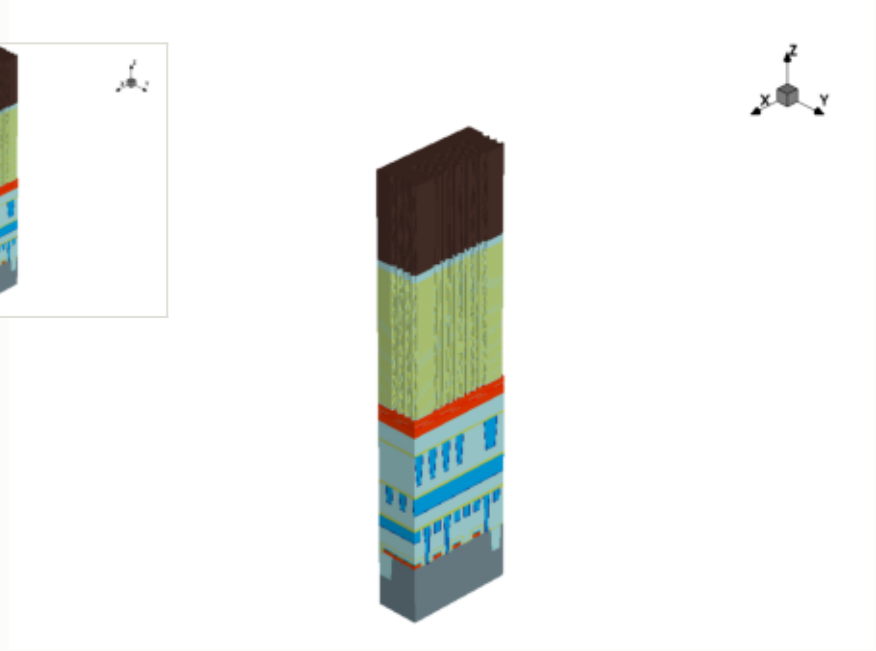
Step #4340.37: HAR_etch

etch_noop · HAR_etch

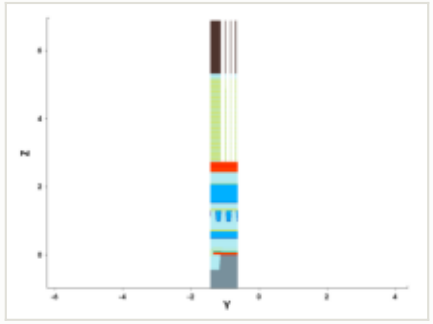
1.04	343.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

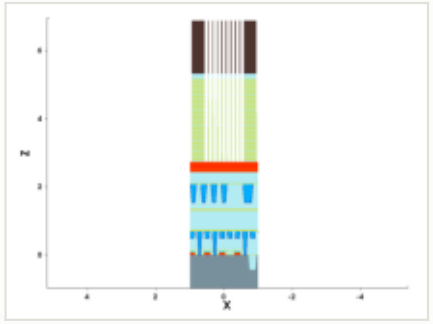
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

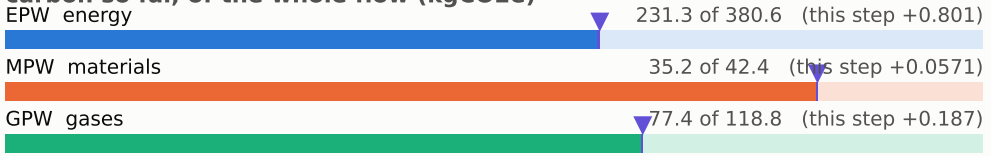
no mask for this step

this step

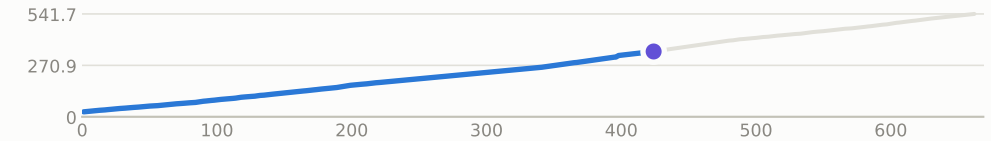
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

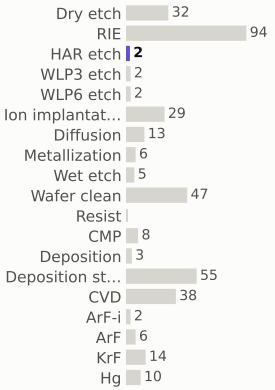
carbon so far, of the whole flow (kgCO2e)



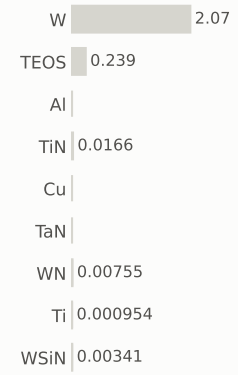
embodied carbon up to each step (kgCO2e per wafer)



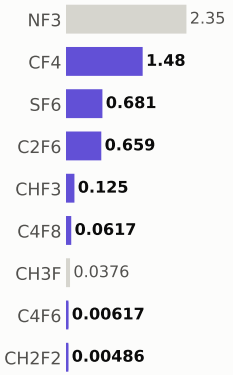
steps by process area



materials deposited (g)



gases emitted (g)



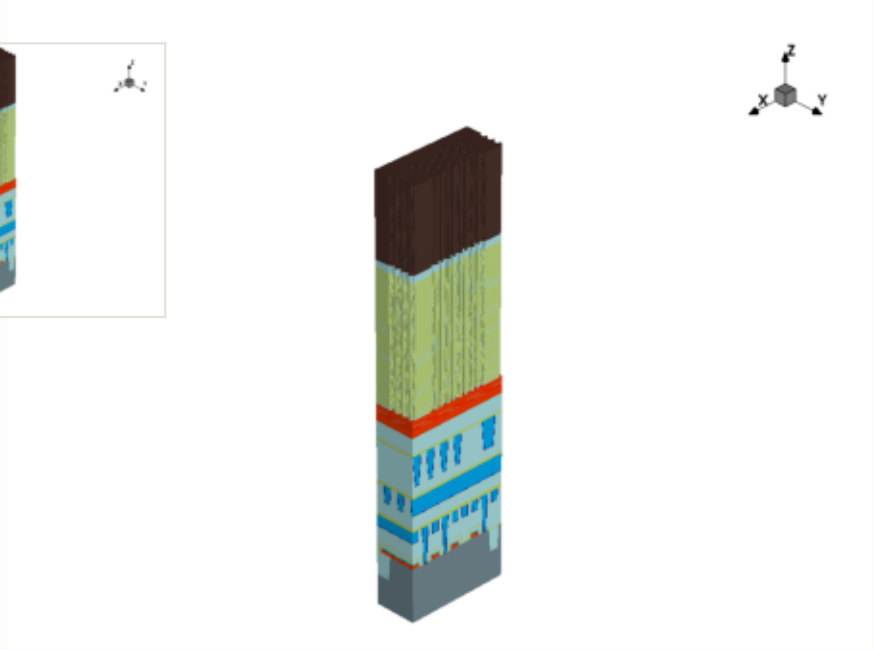
Step #4340.56: HAR_etch

etch_noop · HAR_etch

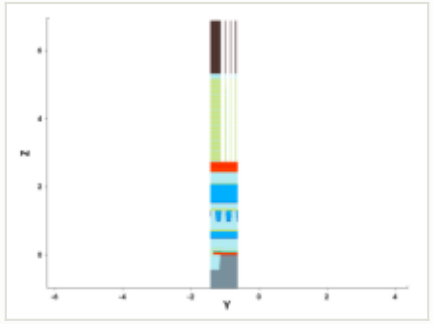
1.04	345.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

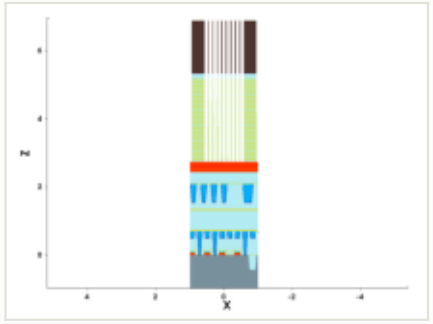
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

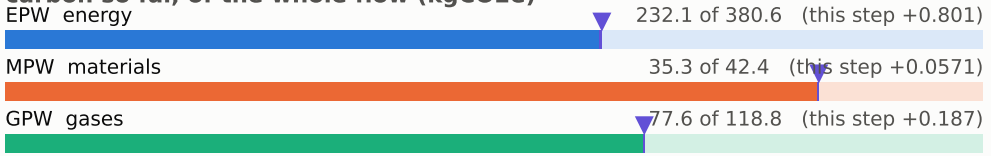
no mask for this step

this step

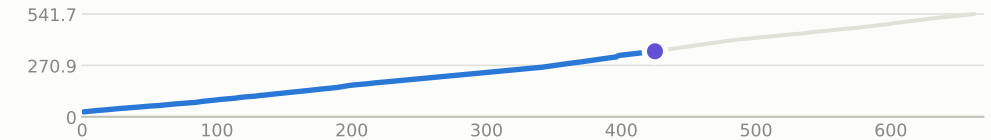
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

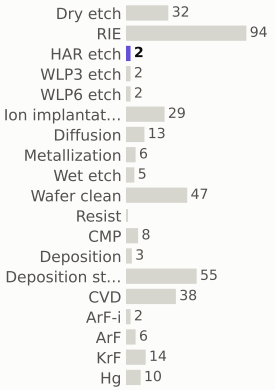
carbon so far, of the whole flow (kgCO2e)



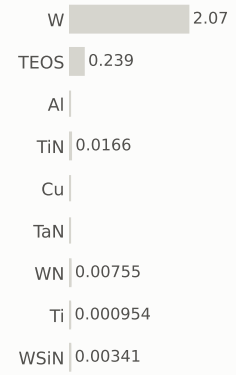
embodied carbon up to each step (kgCO2e per wafer)



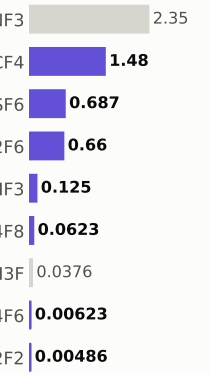
steps by process area



materials deposited (g)



gases emitted (g)



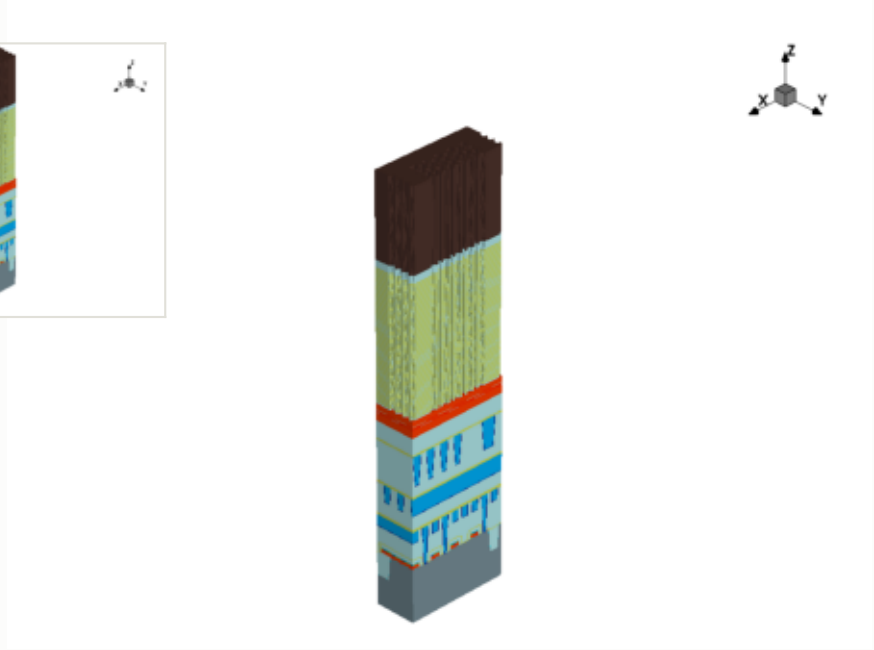
Step #4340.74: HAR_etch

etch_noop · HAR_etch

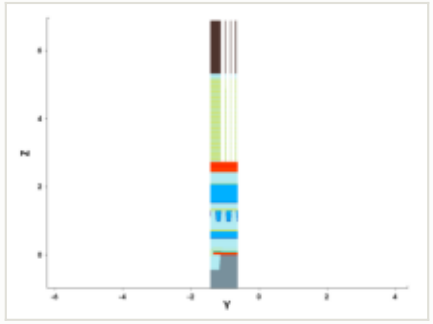
1.04	346.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

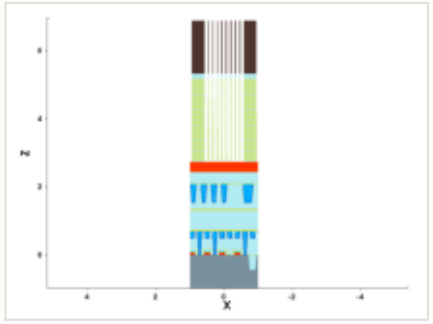
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

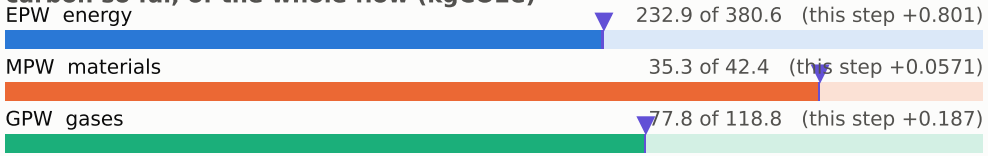
no mask for this step

this step

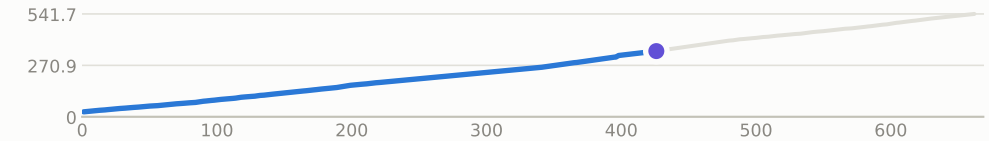
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

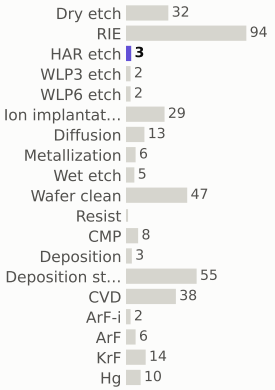
carbon so far, of the whole flow (kgCO2e)



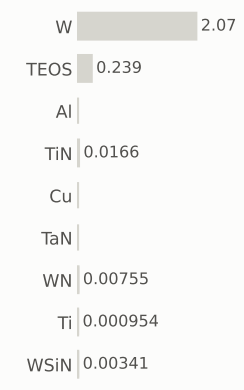
embodied carbon up to each step (kgCO2e per wafer)



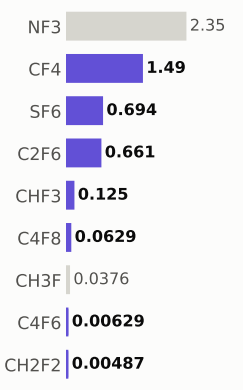
steps by process area



materials deposited (g)



gases emitted (g)



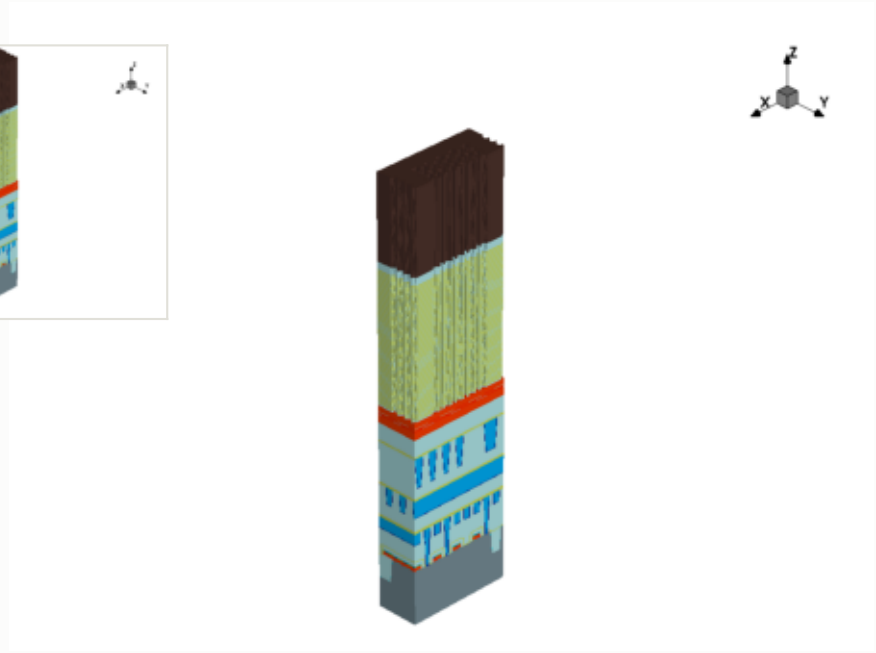
Step #4340.93: HAR_etch

etch_noop · HAR_etch

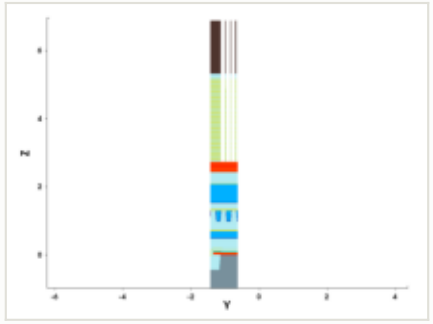
1.04	347.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

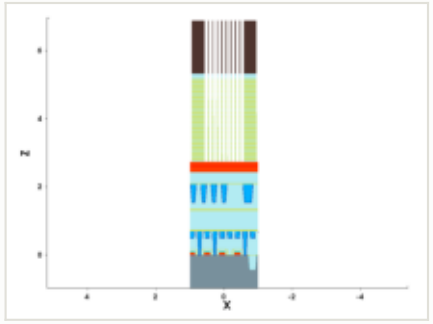
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

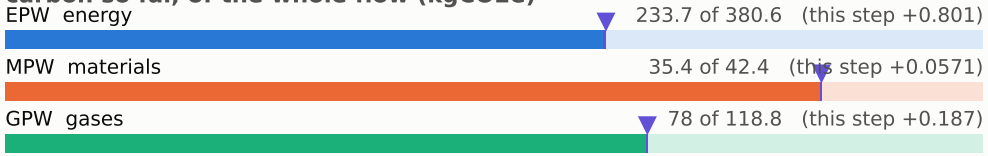
no mask for this step

this step

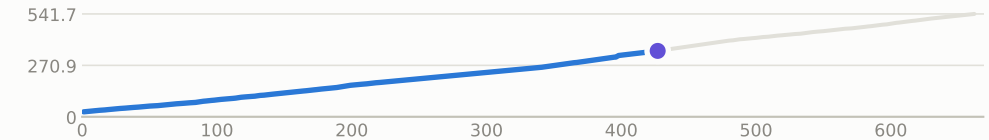
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

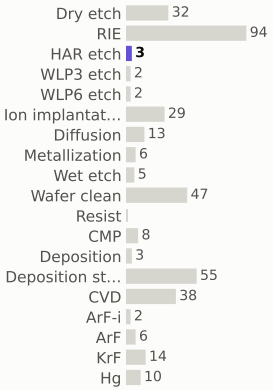
carbon so far, of the whole flow (kgCO2e)



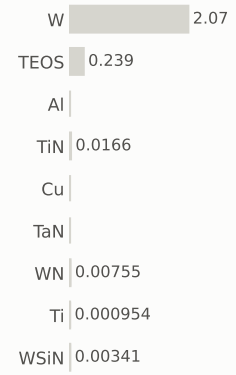
embodied carbon up to each step (kgCO2e per wafer)



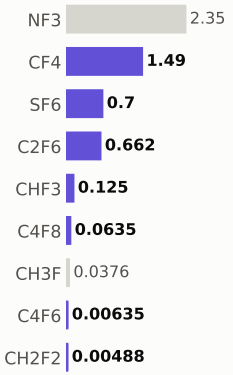
steps by process area



materials deposited (g)



gases emitted (g)



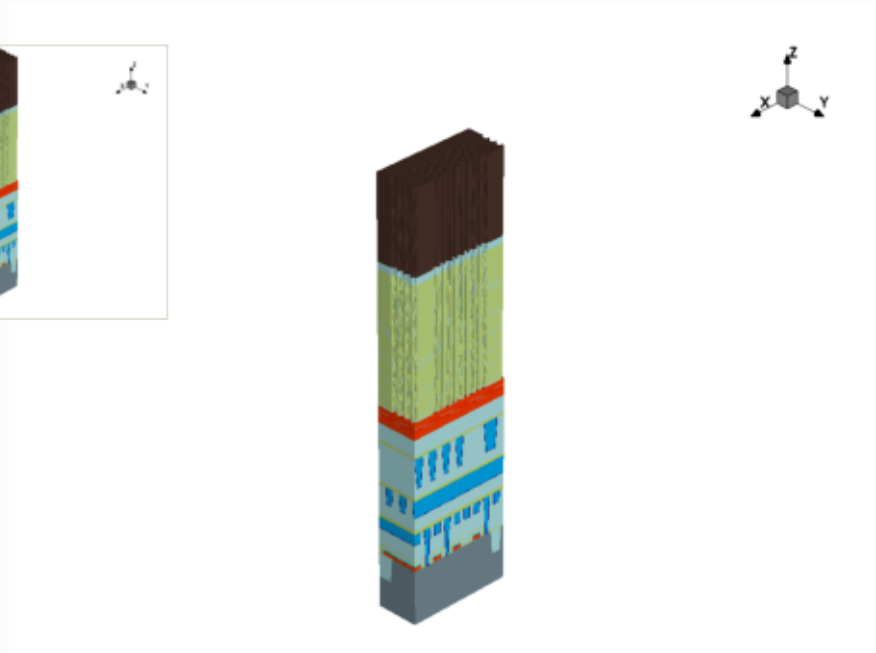
Step #4341.11: HAR_etch

etch_noop · HAR_etch

1.04	348.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

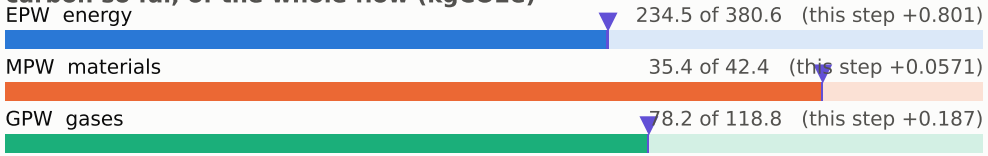


this step

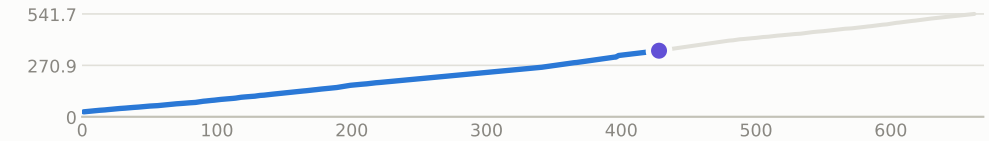
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

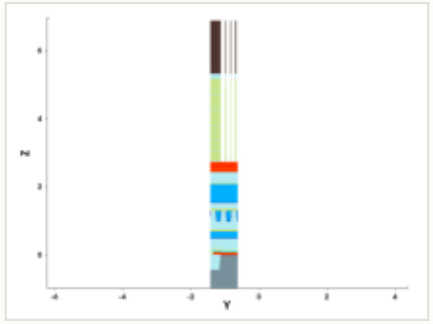
carbon so far, of the whole flow (kgCO2e)



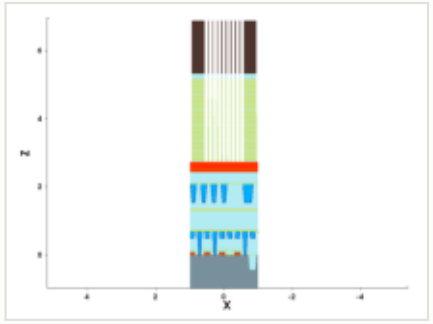
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



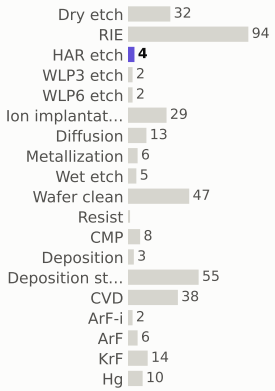
y cut at -1.032 µm



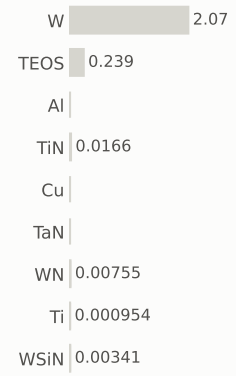
mask for this step

no mask for this step

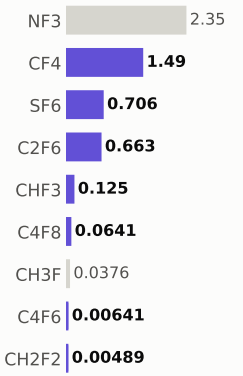
steps by process area



materials deposited (g)



gases emitted (g)



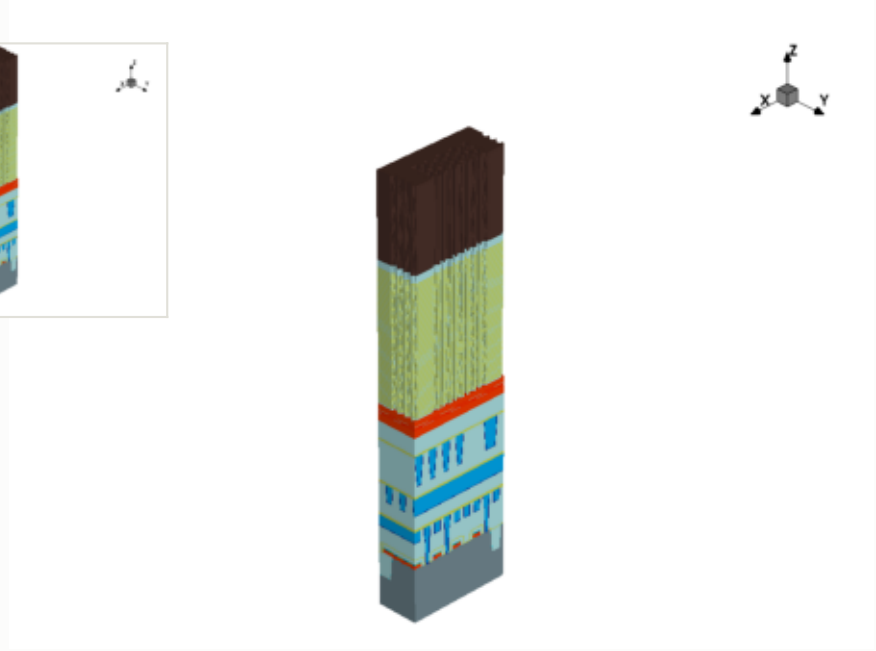
Step #4341.3: HAR_etch

etch_noop · HAR_etch

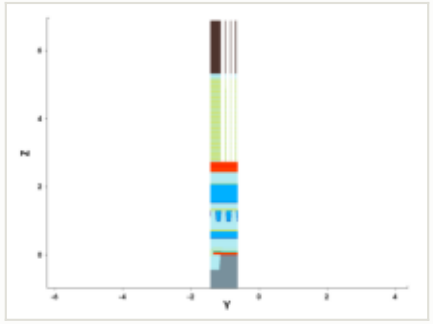
1.04	349.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

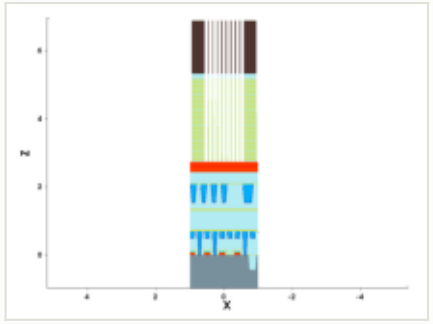
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

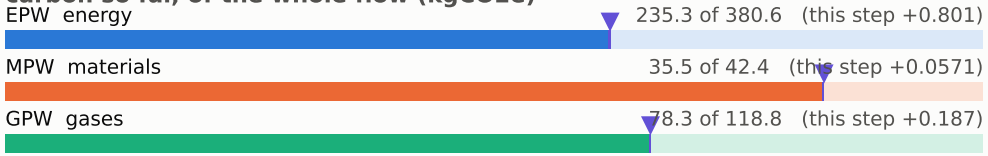
no mask for this step

this step

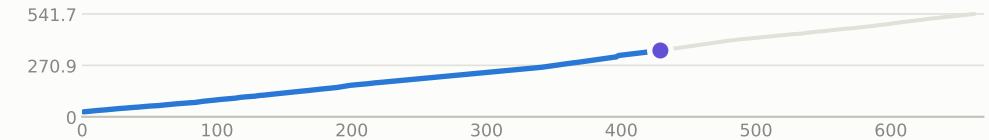
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

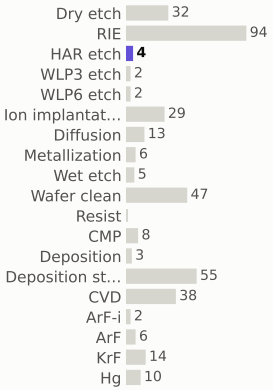
carbon so far, of the whole flow (kgCO2e)



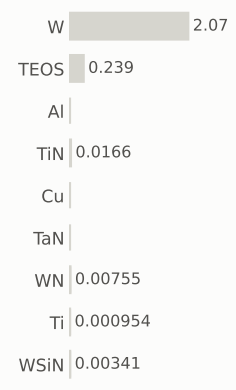
embodied carbon up to each step (kgCO2e per wafer)



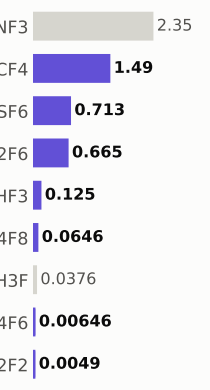
steps by process area



materials deposited (g)



gases emitted (g)



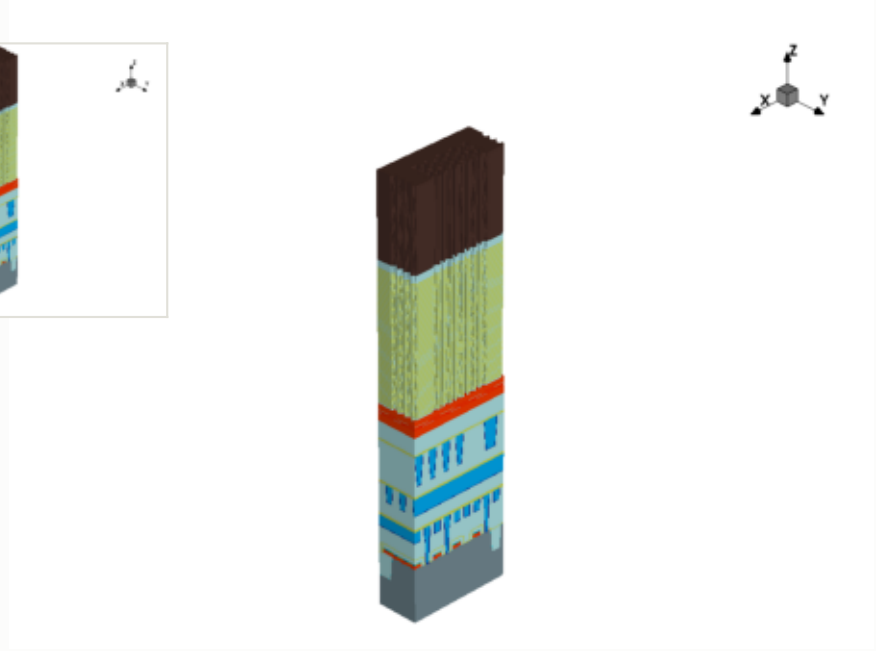
Step #4341.48: HAR_etch

etch_noop · HAR_etch

1.04	350.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



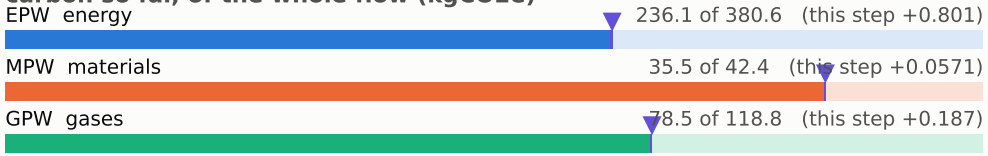
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

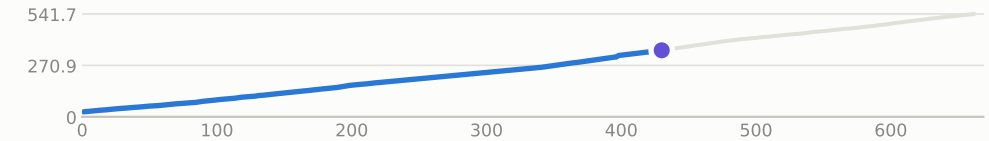
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

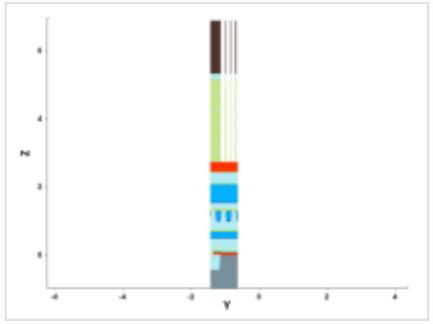
carbon so far, of the whole flow (kgCO2e)



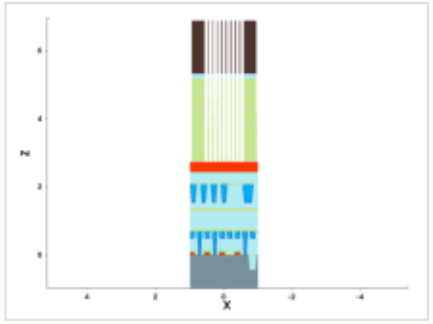
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



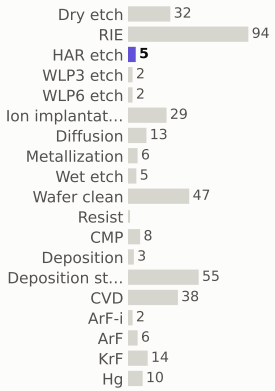
y cut at -1.032 µm



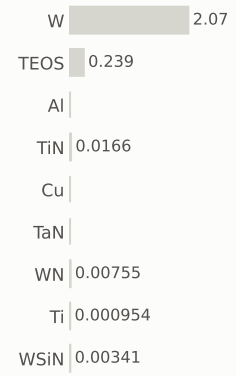
mask for this step

no mask for this step

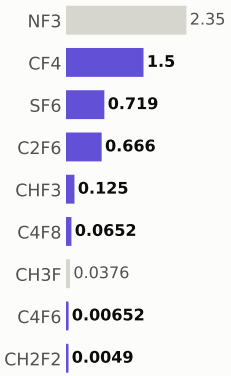
steps by process area



materials deposited (g)



gases emitted (g)



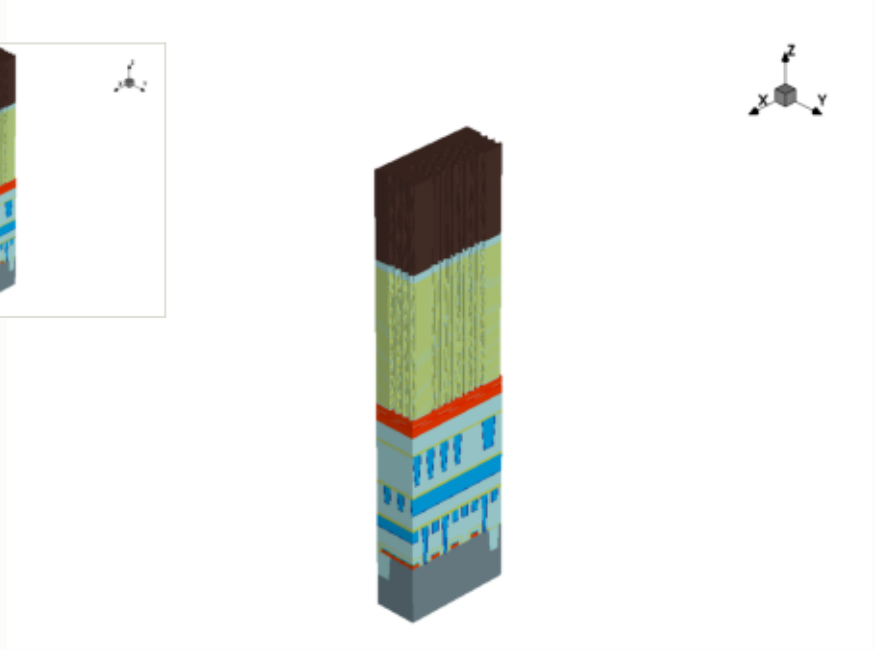
Step #4341.67: HAR_etch

etch_noop · HAR_etch

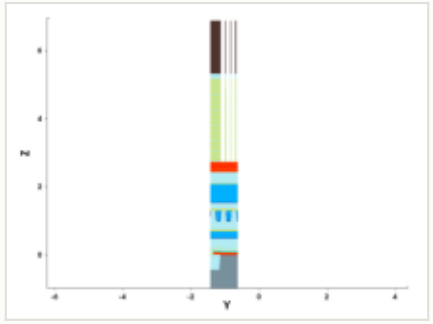
1.04	351.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

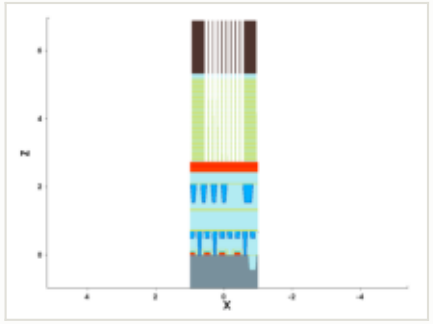
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

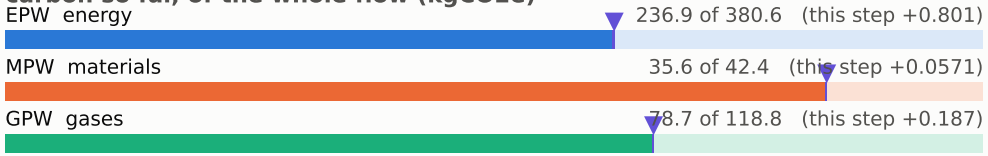
no mask for this step

this step

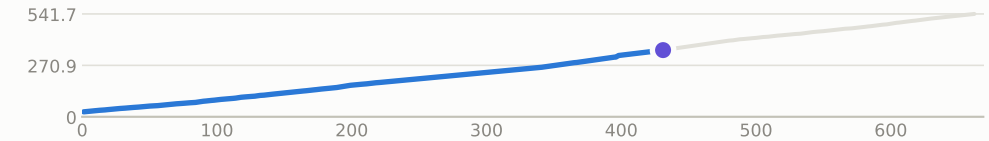
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

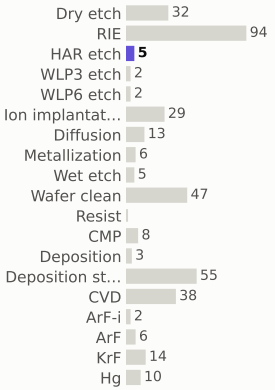
carbon so far, of the whole flow (kgCO2e)



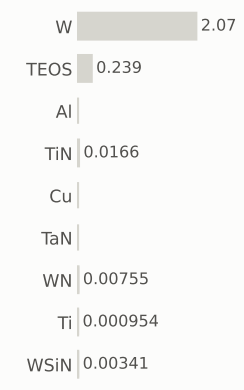
embodied carbon up to each step (kgCO2e per wafer)



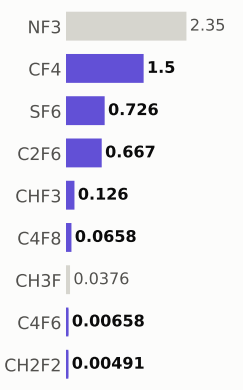
steps by process area



materials deposited (g)



gases emitted (g)



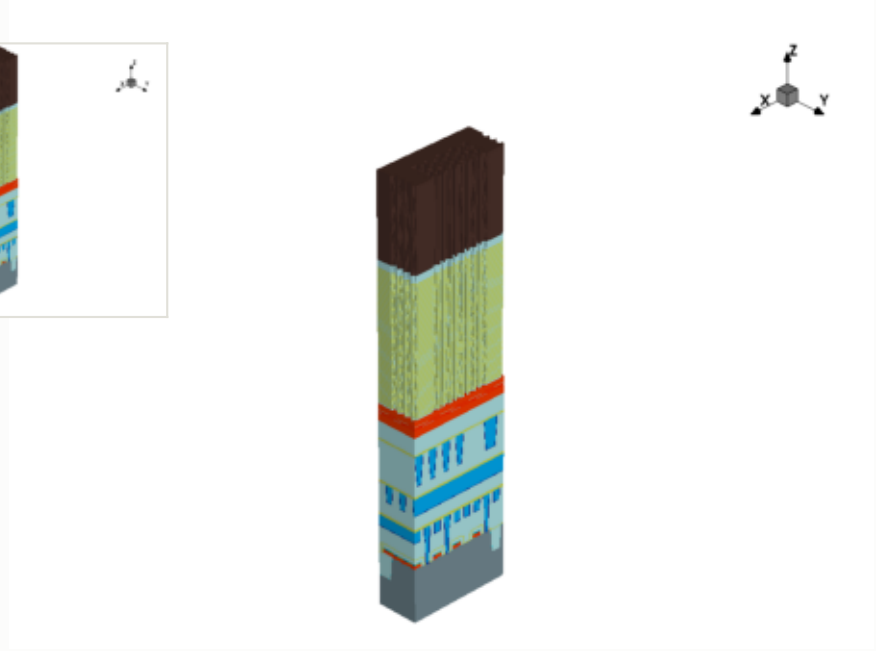
Step #4341.85: HAR_etch

etch_noop · HAR_etch

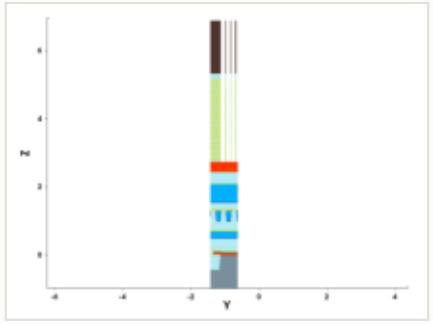
1.04	352.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

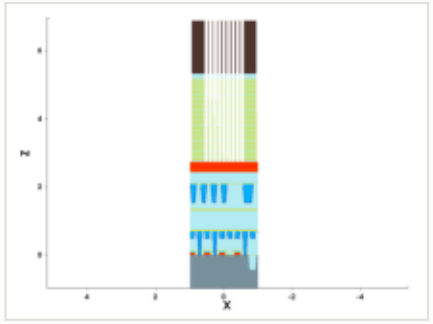
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

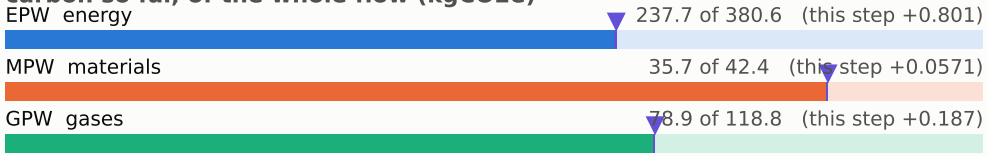
no mask for this step

this step

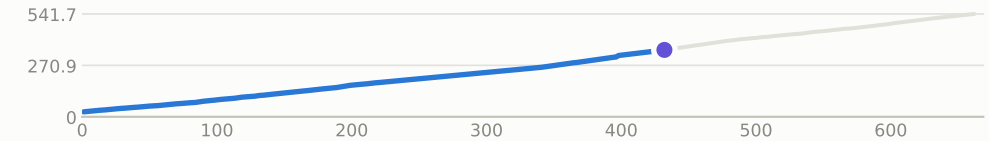
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

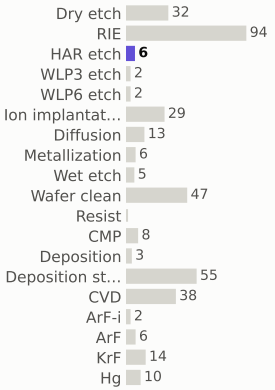
carbon so far, of the whole flow (kgCO2e)



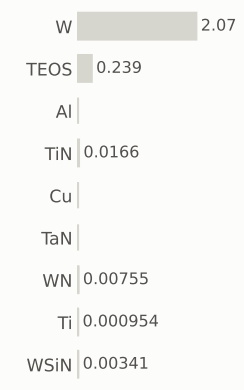
embodied carbon up to each step (kgCO2e per wafer)



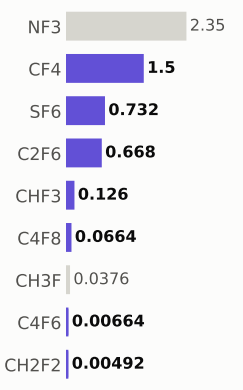
steps by process area



materials deposited (g)



gases emitted (g)



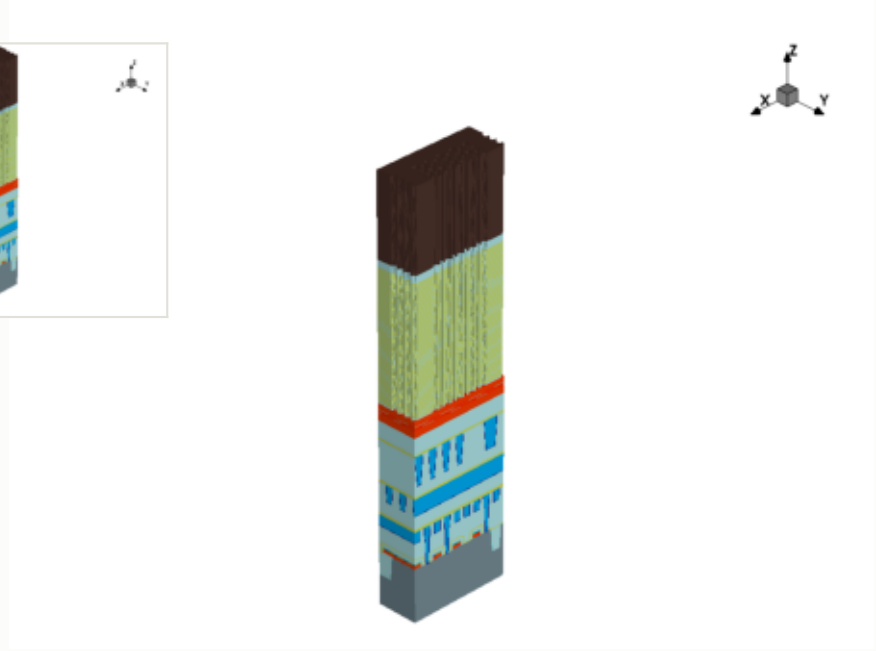
Step #4342.04: HAR_etch

etch_noop · HAR_etch

1.04	353.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

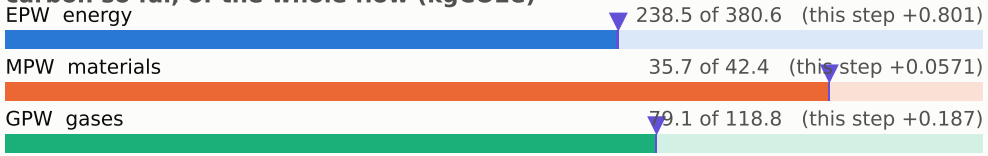


this step

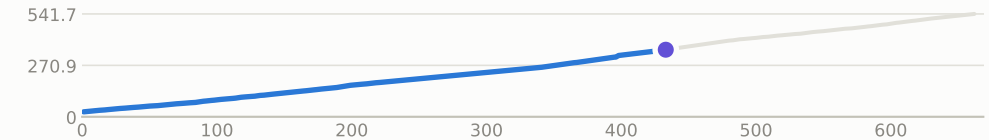
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

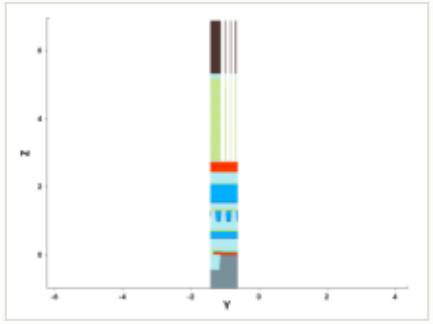
carbon so far, of the whole flow (kgCO2e)



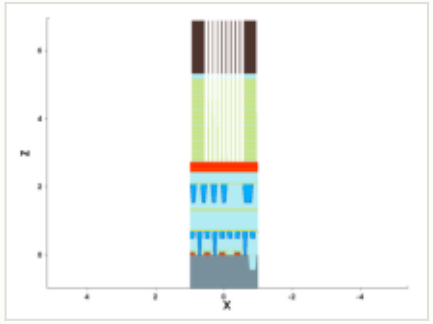
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



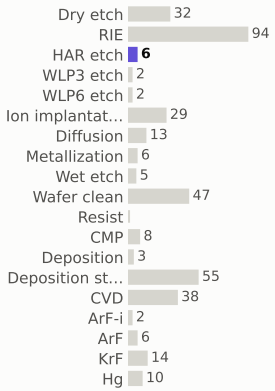
y cut at -1.032 µm



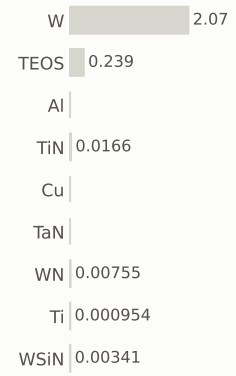
mask for this step

no mask for this step

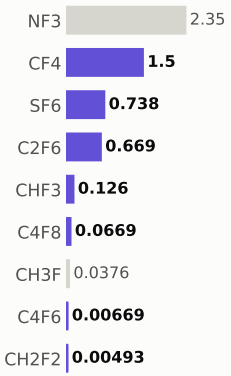
steps by process area



materials deposited (g)



gases emitted (g)



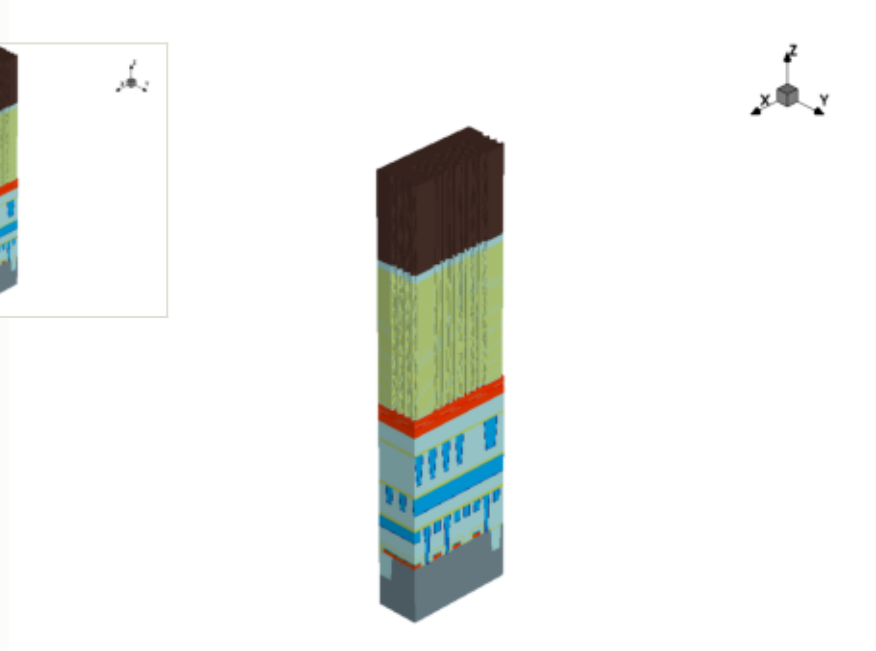
Step #4342.22: HAR_etch

etch_noop · HAR_etch

1.04	354.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

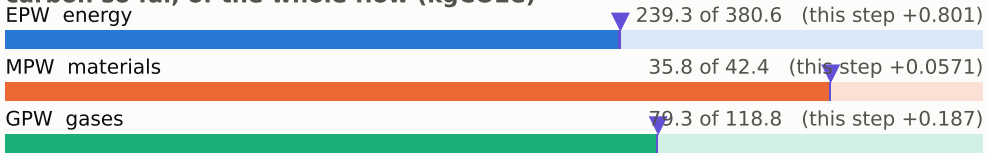


this step

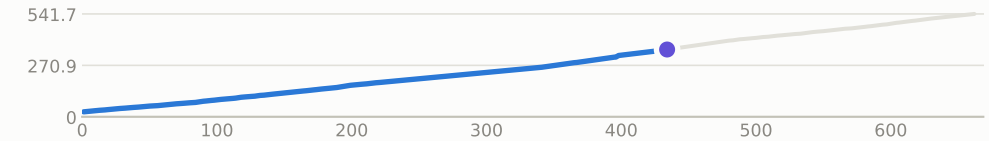
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

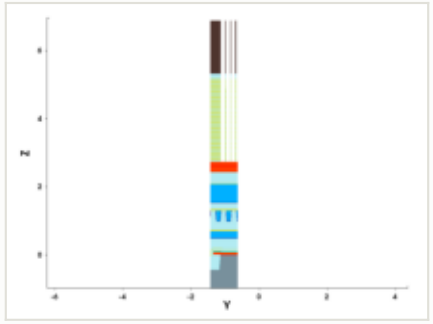
carbon so far, of the whole flow (kgCO2e)



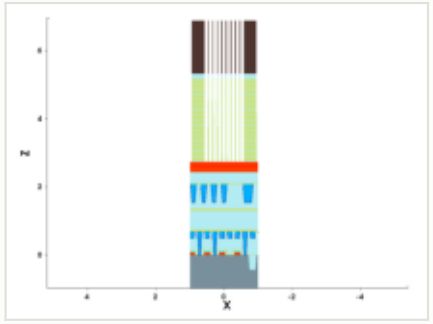
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



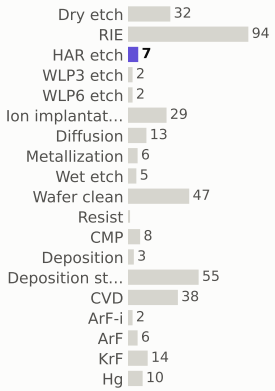
y cut at -1.032 µm



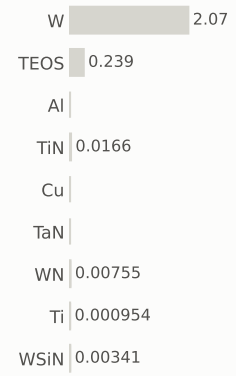
mask for this step

no mask for this step

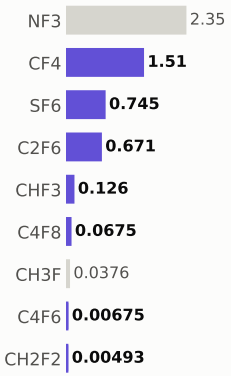
steps by process area



materials deposited (g)



gases emitted (g)



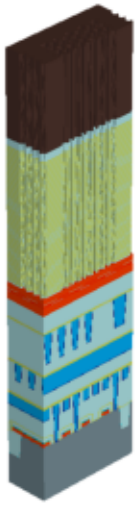
Step #4342.41: HAR_etch

etch_noop · HAR_etch

1.04	355.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



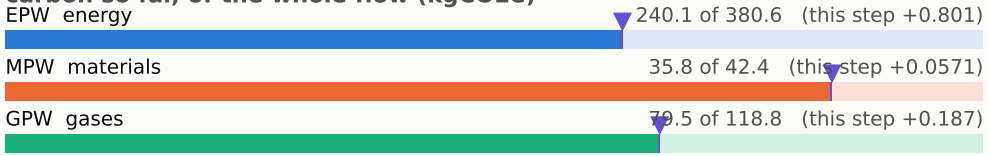
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

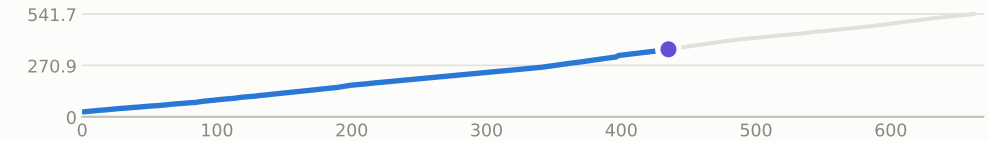
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

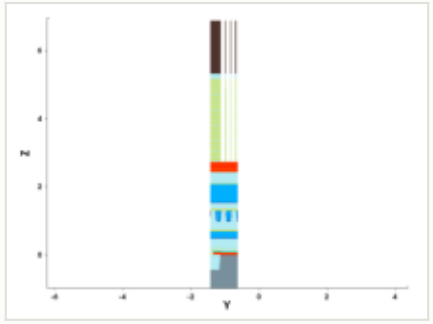
carbon so far, of the whole flow (kgCO2e)



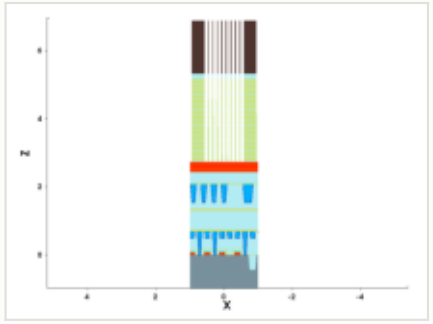
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



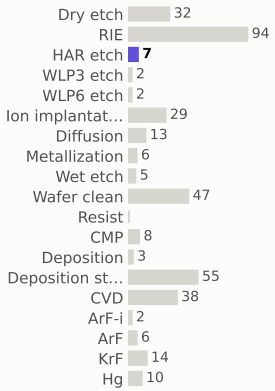
y cut at -1.032 µm



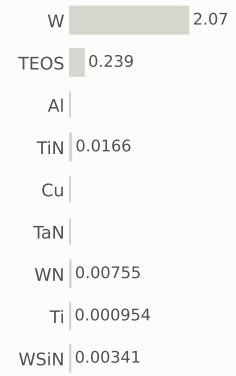
mask for this step

no mask for this step

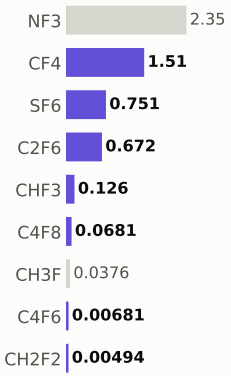
steps by process area



materials deposited (g)



gases emitted (g)



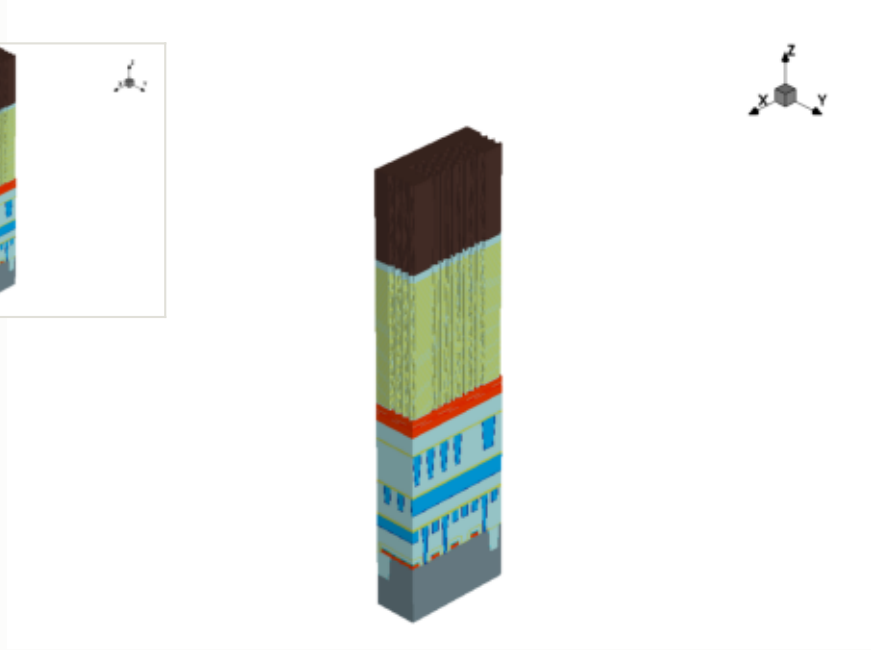
Step #4342.59: HAR_etch

etch_noop · HAR_etch

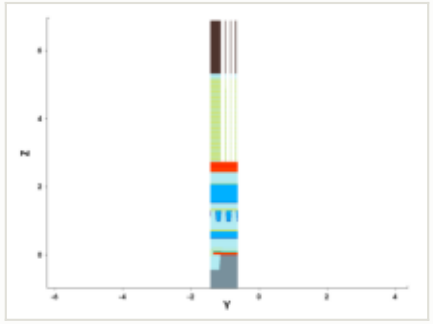
1.04	356.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

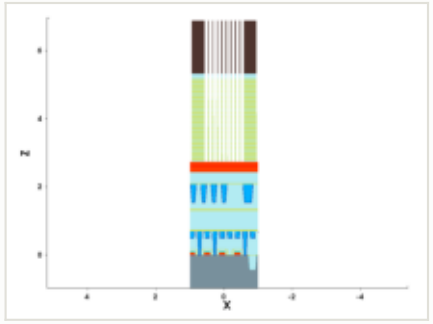
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

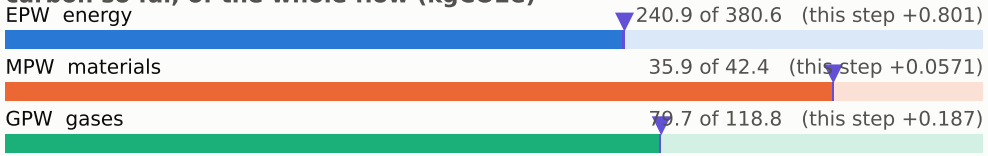
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

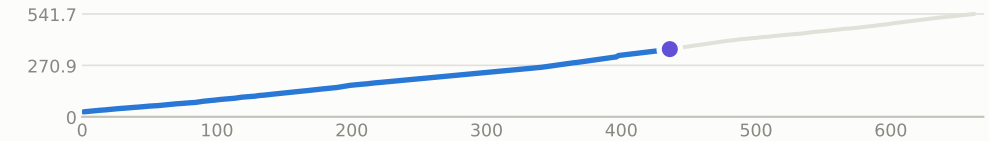
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

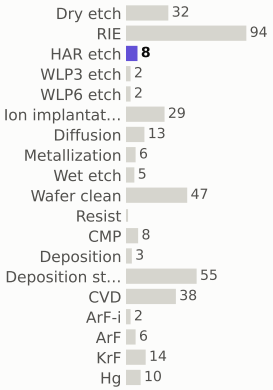
carbon so far, of the whole flow (kgCO2e)



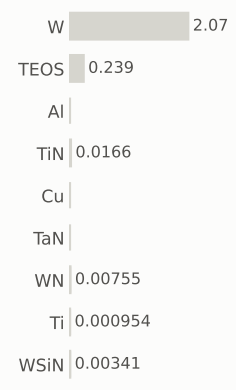
embodied carbon up to each step (kgCO2e per wafer)



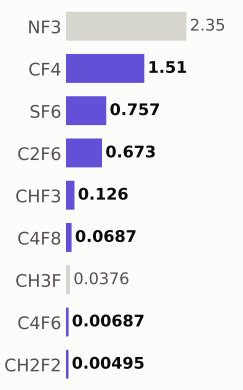
steps by process area



materials deposited (g)



gases emitted (g)



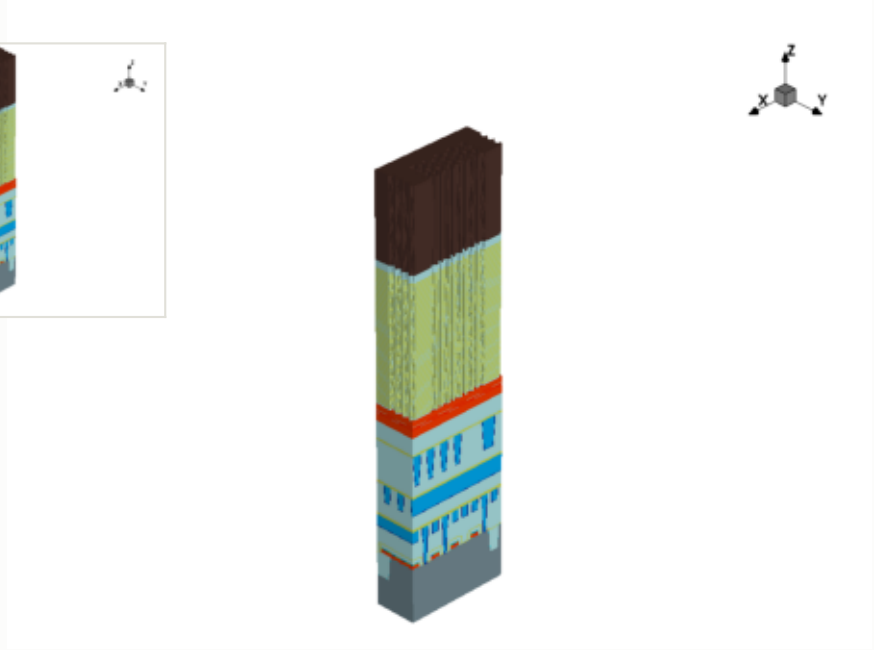
Step #4342.78: HAR_etch

etch_noop · HAR_etch

1.04	357.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



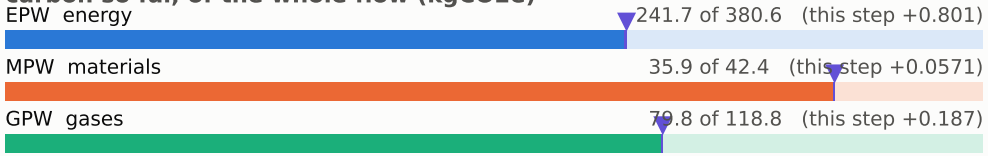
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

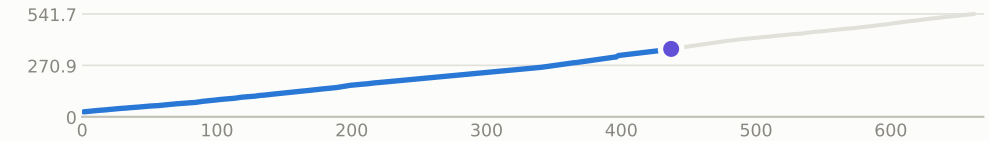
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

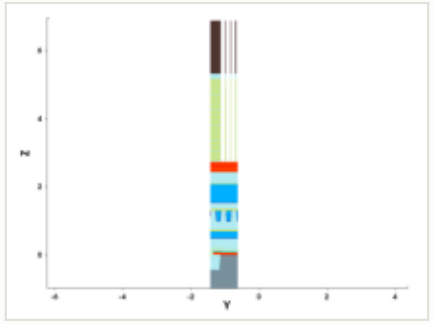
carbon so far, of the whole flow (kgCO2e)



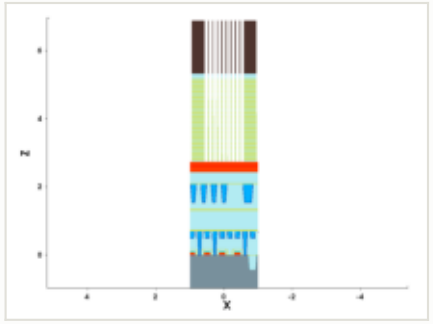
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



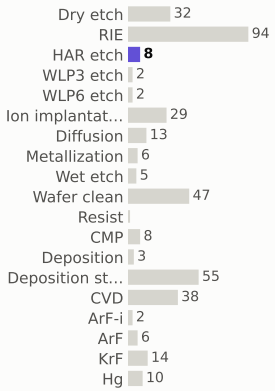
y cut at -1.032 µm



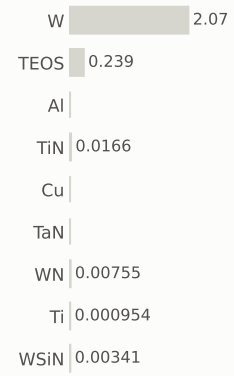
mask for this step

no mask for this step

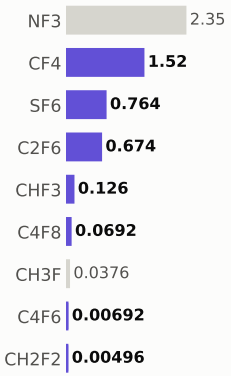
steps by process area



materials deposited (g)



gases emitted (g)



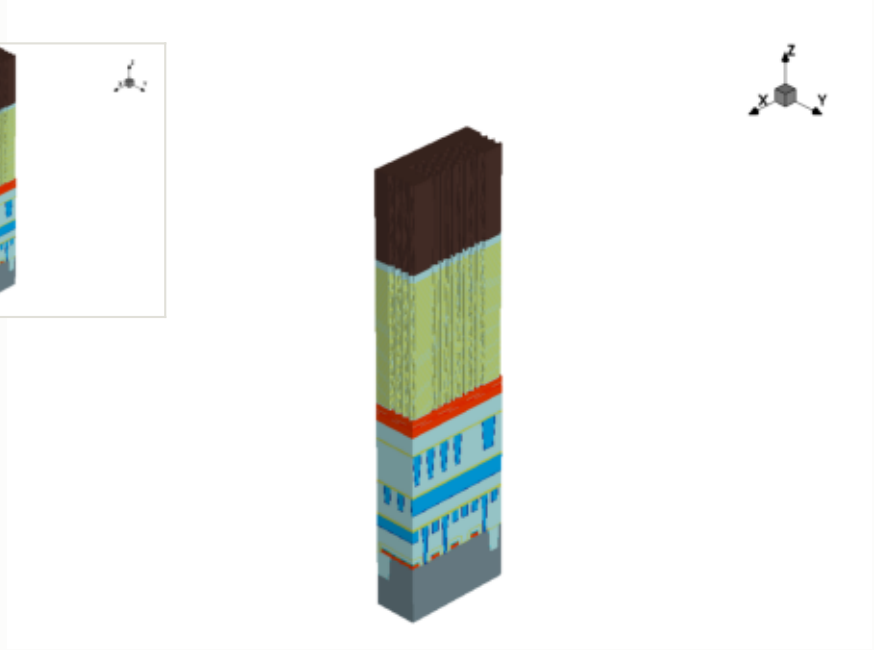
Step #4342.96: HAR_etch

etch_noop · HAR_etch

1.04	358.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



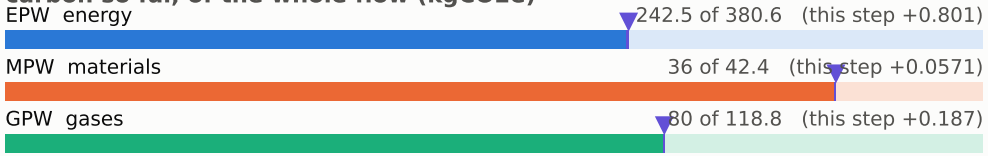
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

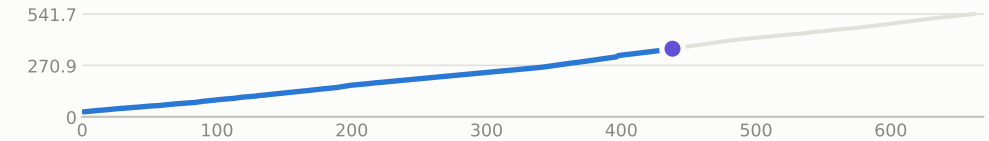
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

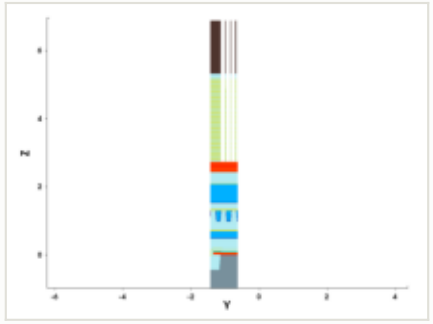
carbon so far, of the whole flow (kgCO2e)



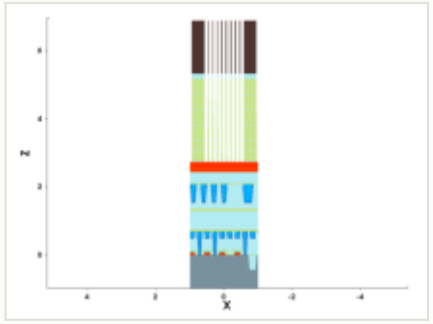
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



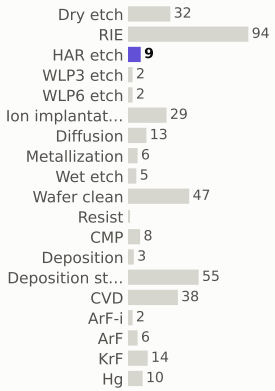
y cut at -1.032 µm



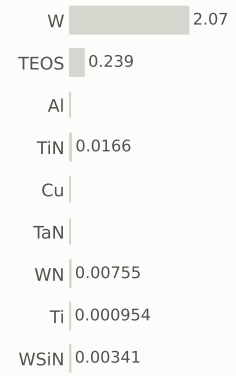
mask for this step

no mask for this step

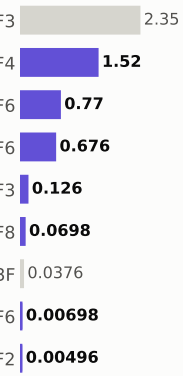
steps by process area



materials deposited (g)



gases emitted (g)



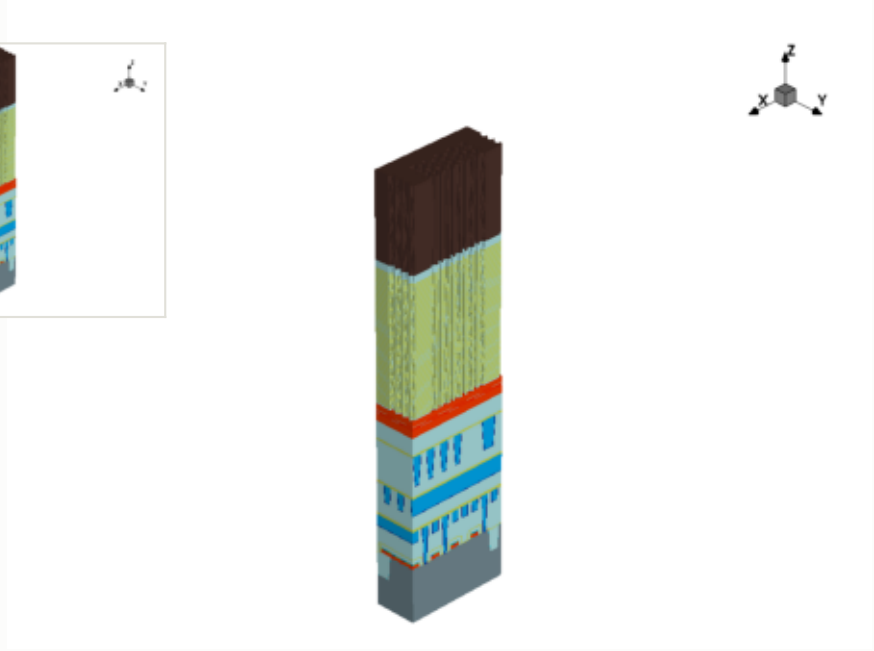
Step #4343.15: HAR_etch

etch_noop · HAR_etch

1.04	359.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



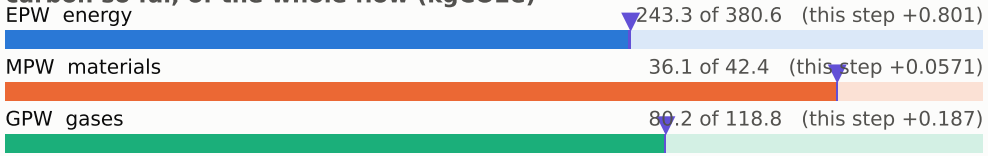
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

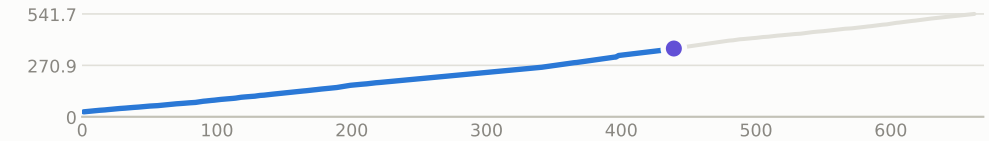
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

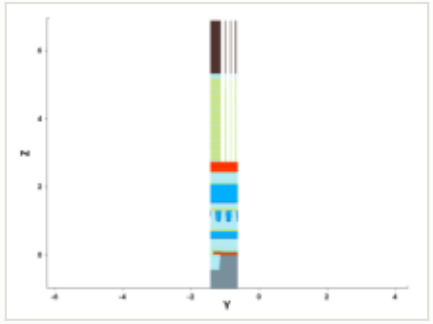
carbon so far, of the whole flow (kgCO2e)



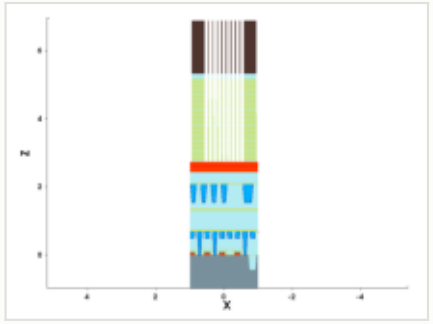
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



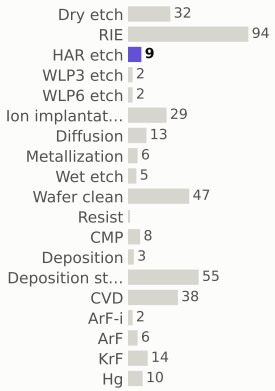
y cut at -1.032 µm



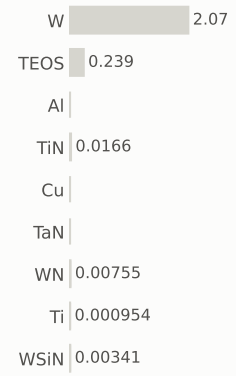
mask for this step

no mask for this step

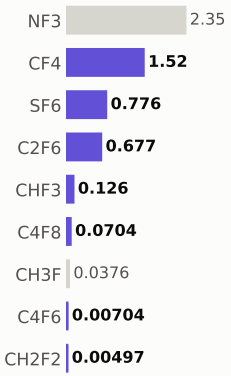
steps by process area



materials deposited (g)



gases emitted (g)



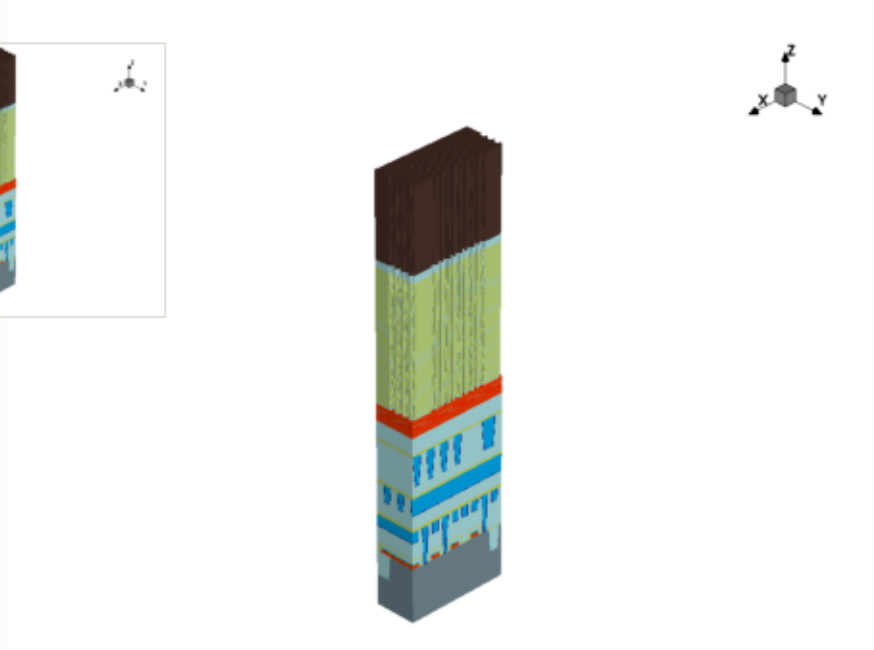
Step #4343.33: HAR_etch

etch_noop · HAR_etch

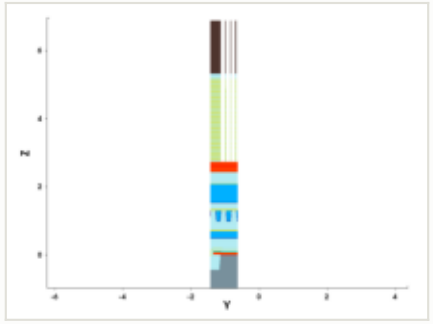
1.04	360.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

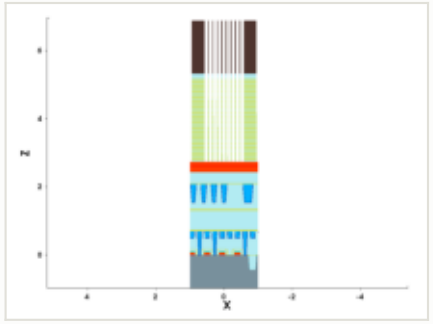
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

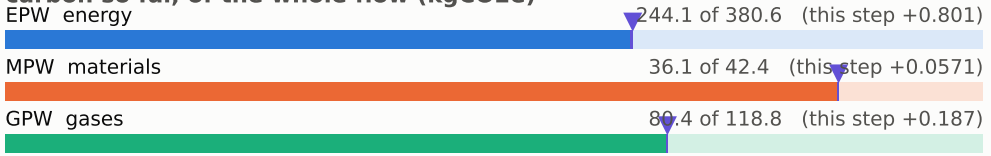
no mask for this step

this step

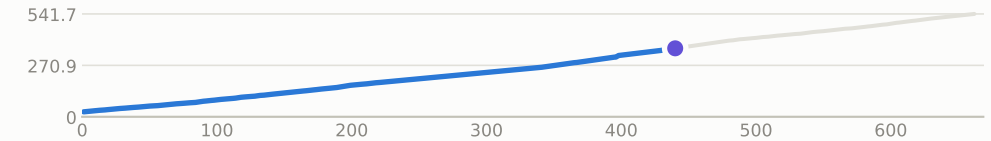
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

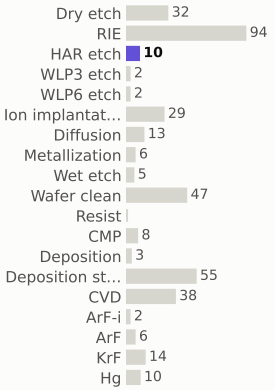
carbon so far, of the whole flow (kgCO2e)



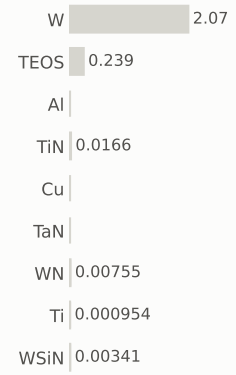
embodied carbon up to each step (kgCO2e per wafer)



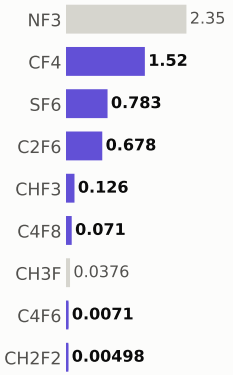
steps by process area



materials deposited (g)



gases emitted (g)



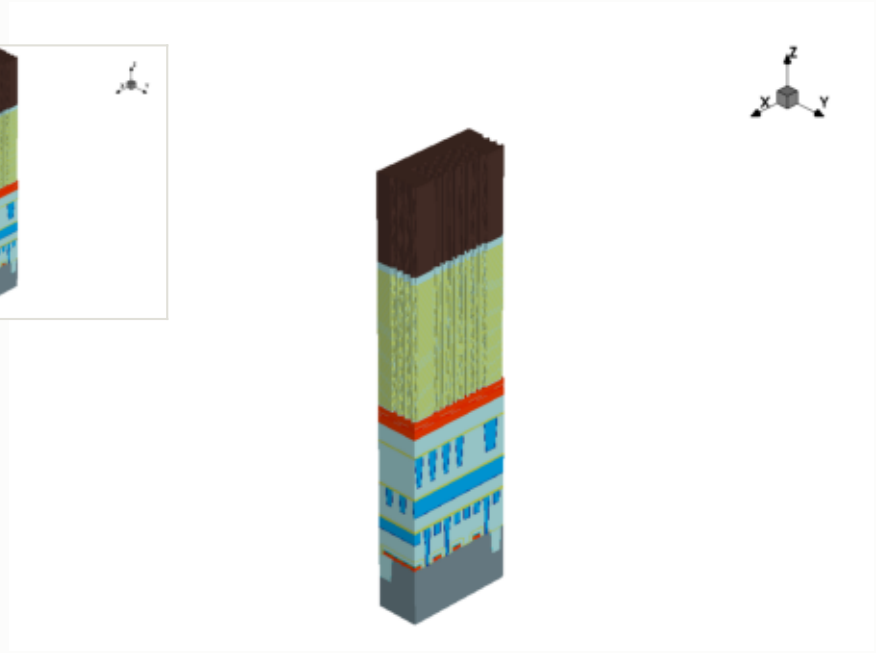
Step #4343.52: HAR_etch

etch_noop · HAR_etch

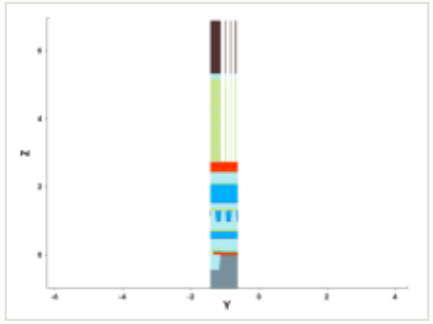
1.04	361.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

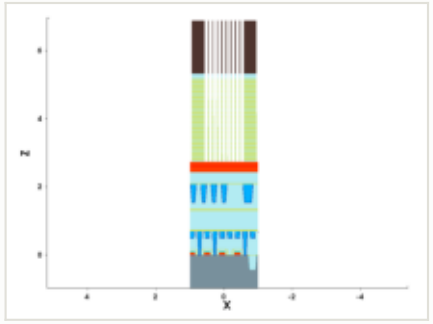
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

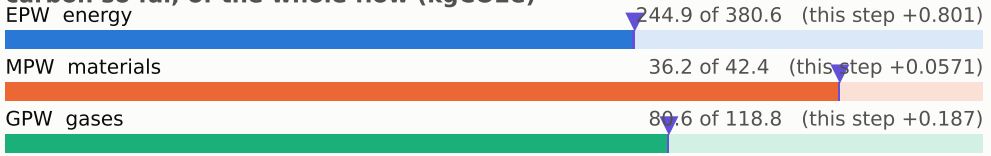
no mask for this step

this step

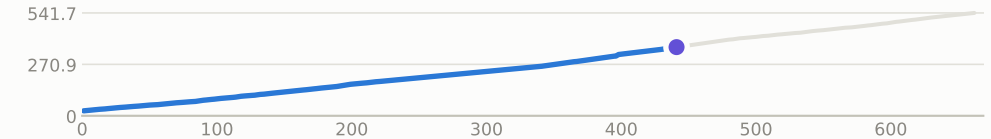
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

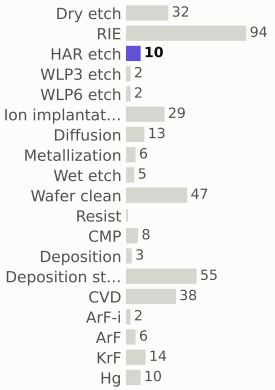
carbon so far, of the whole flow (kgCO2e)



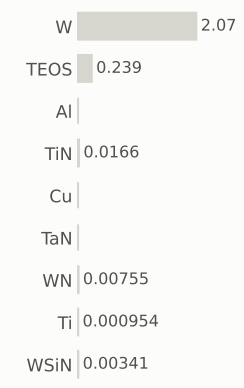
embodied carbon up to each step (kgCO2e per wafer)



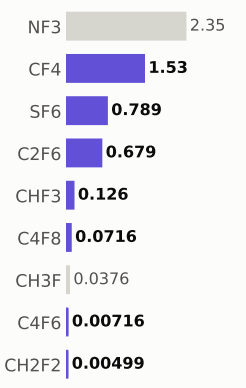
steps by process area



materials deposited (g)



gases emitted (g)



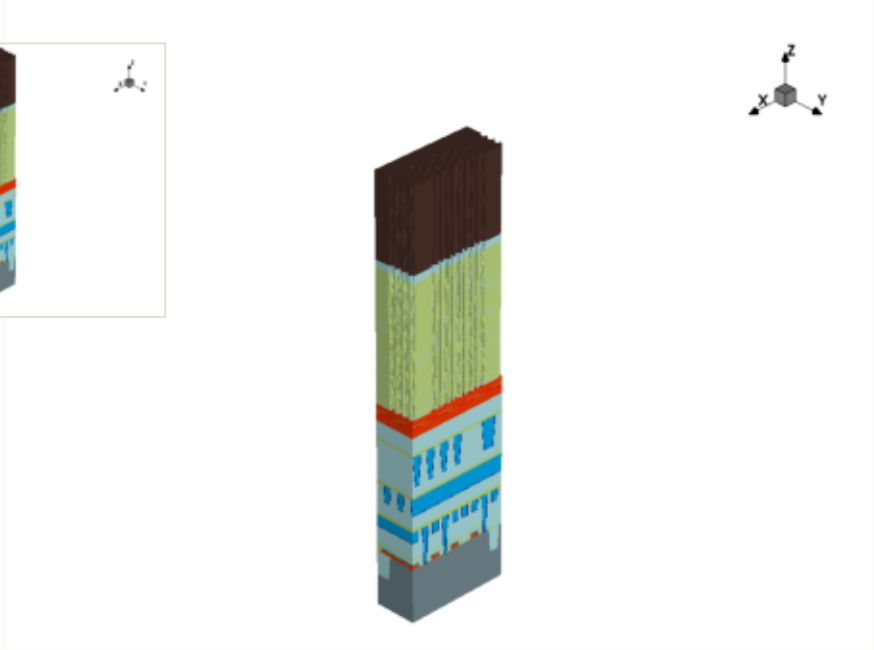
Step #4343.7: HAR_etch

etch_noop · HAR_etch

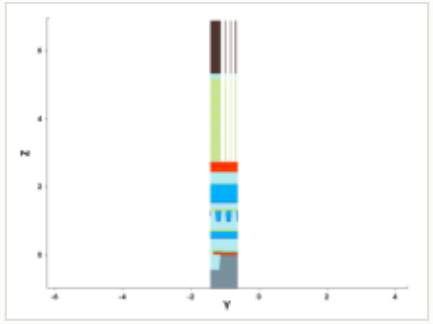
1.04	362.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

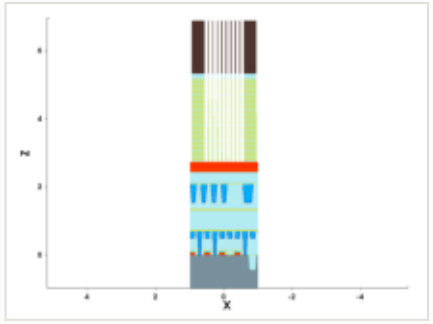
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

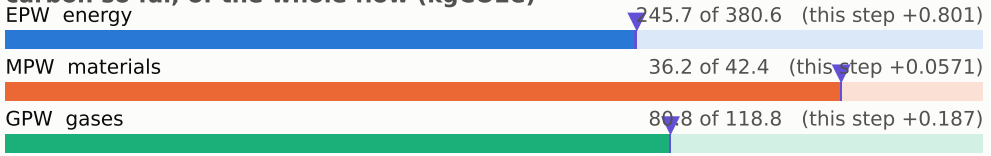
no mask for this step

this step

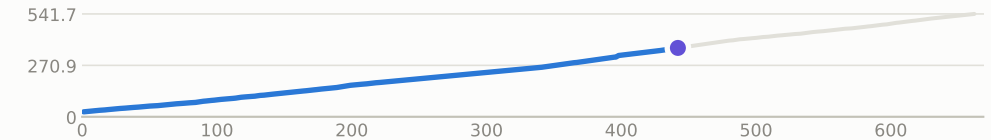
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

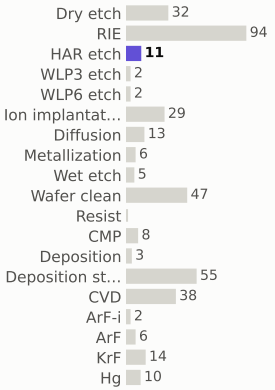
carbon so far, of the whole flow (kgCO2e)



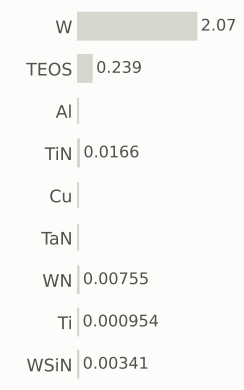
embodied carbon up to each step (kgCO2e per wafer)



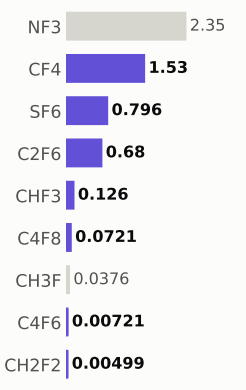
steps by process area



materials deposited (g)



gases emitted (g)



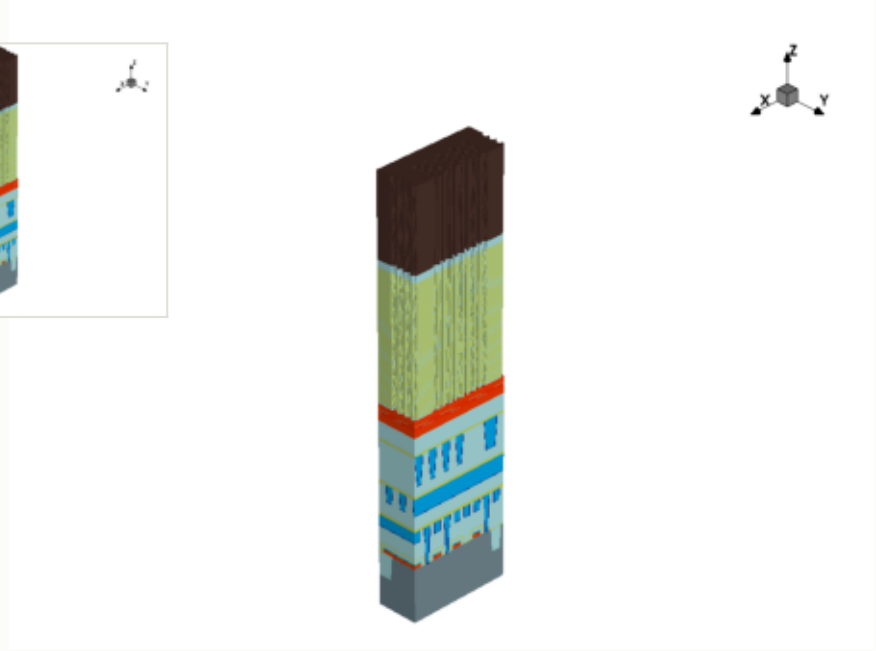
Step #4343.89: HAR_etch

etch_noop · HAR_etch

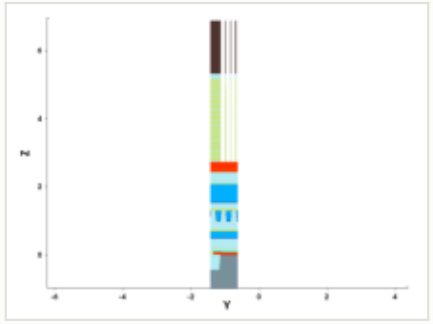
1.04	363.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

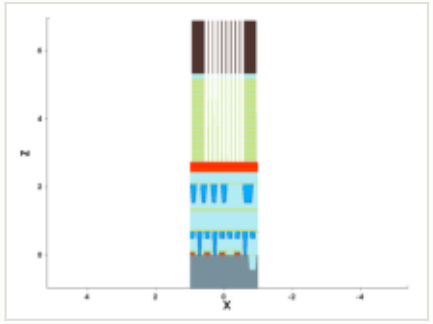
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

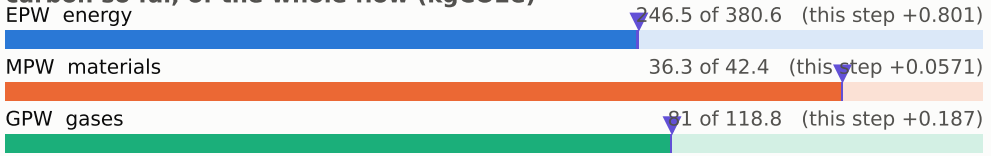
no mask for this step

this step

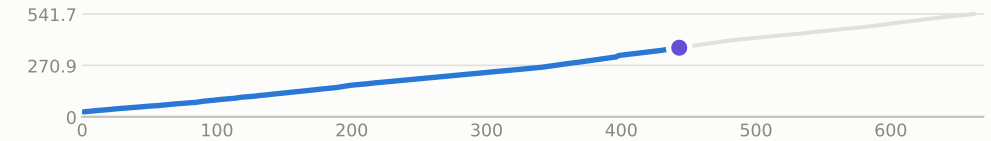
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

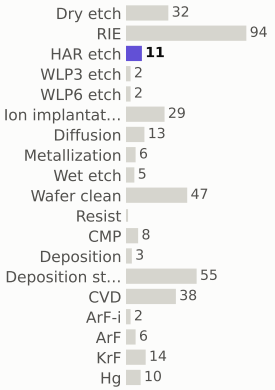
carbon so far, of the whole flow (kgCO2e)



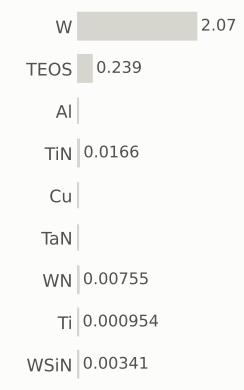
embodied carbon up to each step (kgCO2e per wafer)



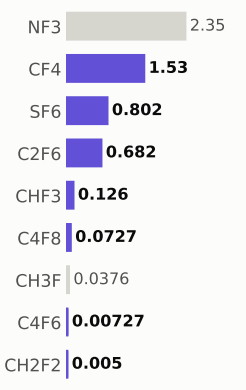
steps by process area



materials deposited (g)



gases emitted (g)



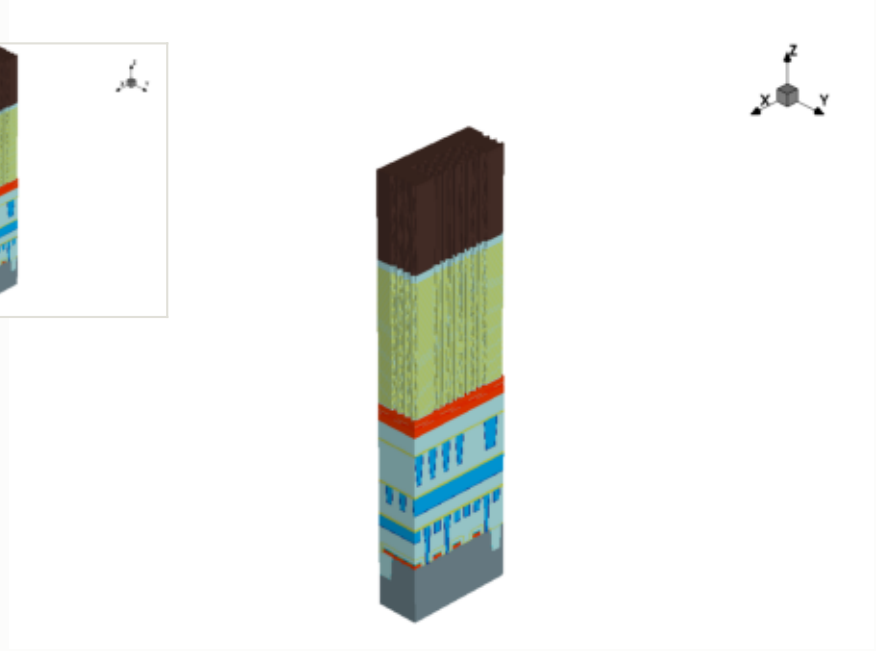
Step #4344.07: HAR_etch

etch_noop · HAR_etch

1.04	364.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



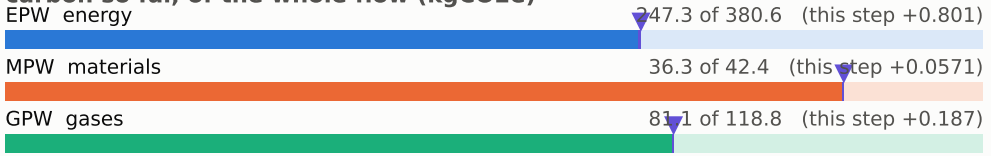
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

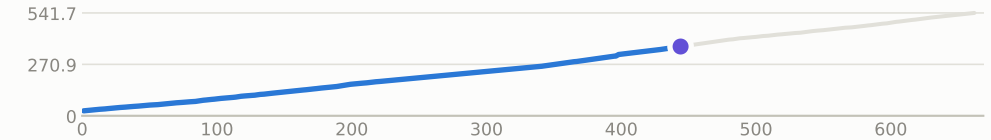
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

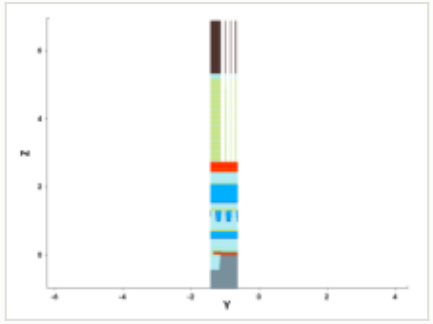
carbon so far, of the whole flow (kgCO2e)



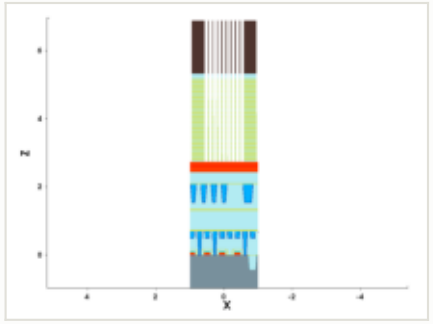
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



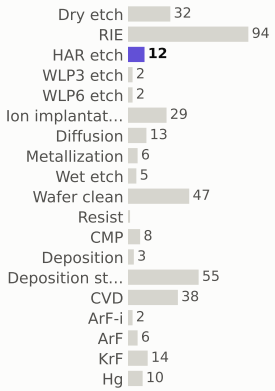
y cut at -1.032 µm



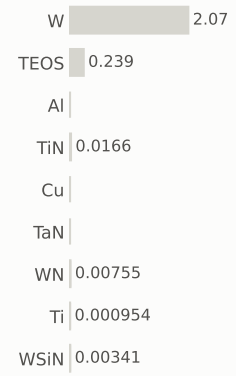
mask for this step

no mask for this step

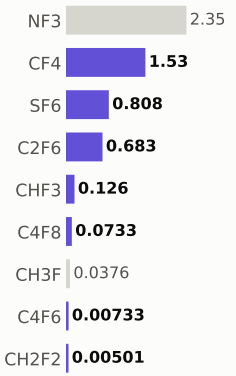
steps by process area



materials deposited (g)



gases emitted (g)



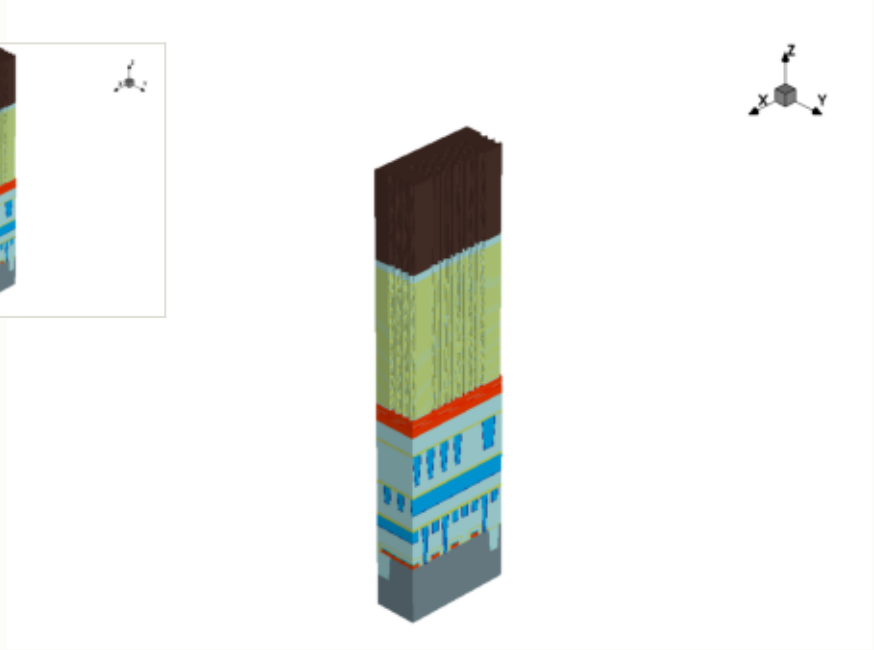
Step #4344.26: HAR_etch

etch_noop · HAR_etch

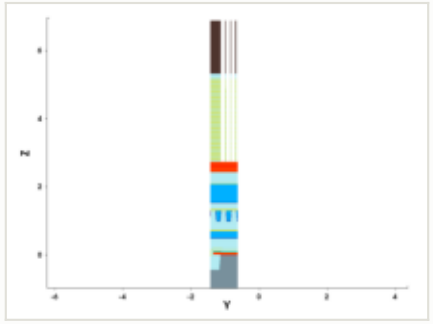
1.04	365.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

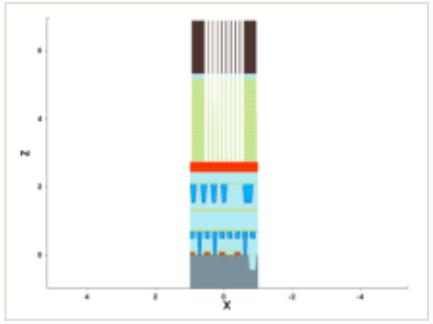
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

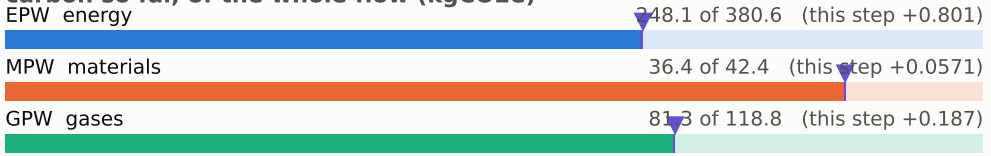
no mask for this step

this step

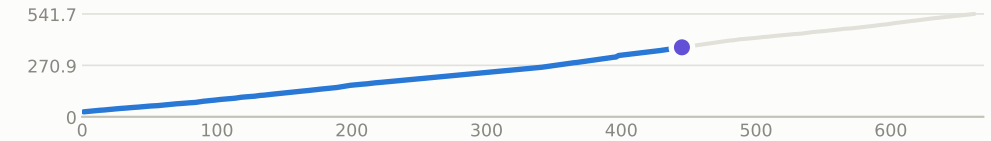
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

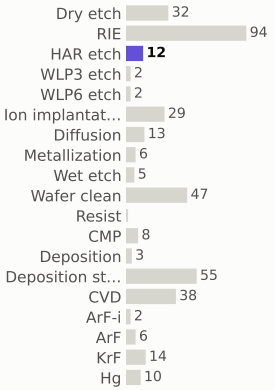
carbon so far, of the whole flow (kgCO2e)



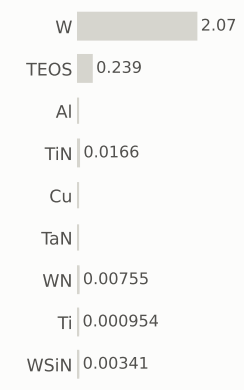
embodied carbon up to each step (kgCO2e per wafer)



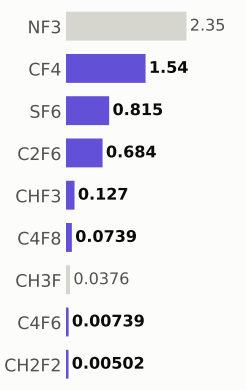
steps by process area



materials deposited (g)



gases emitted (g)



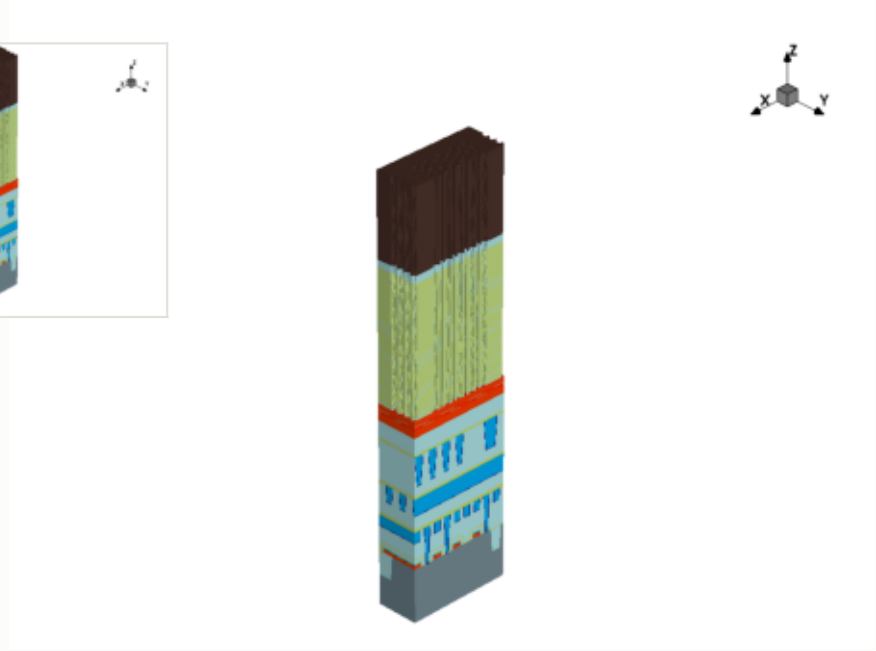
Step #4344.44: HAR_etch

etch_noop · HAR_etch

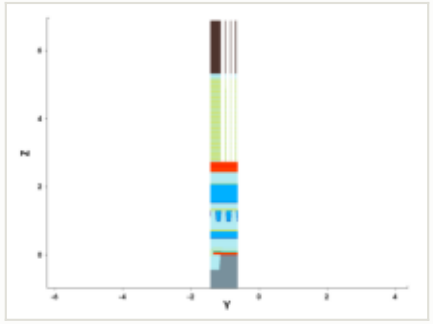
1.04	366.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

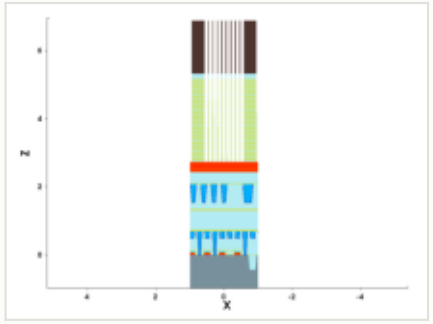
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

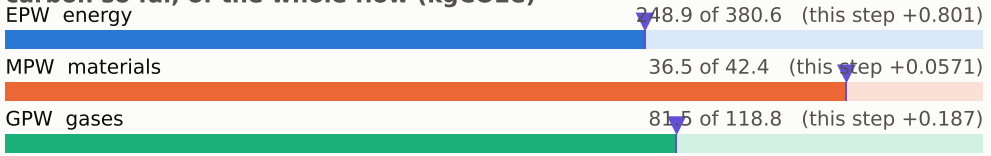
no mask for this step

this step

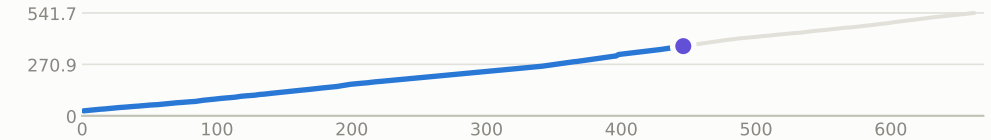
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

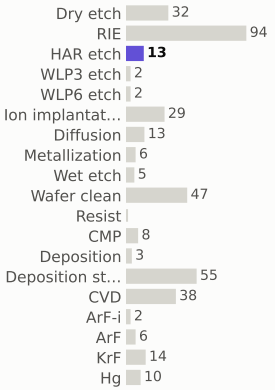
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

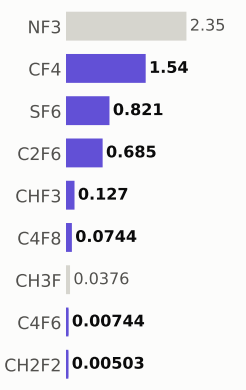


steps by process area



materials deposited (g)

gases emitted (g)



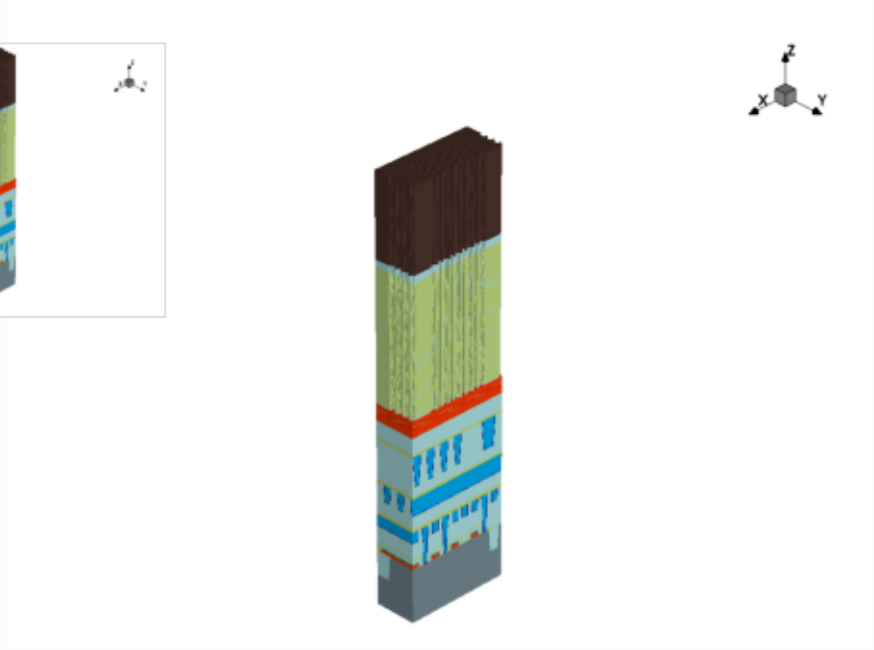
Step #4344.63: HAR_etch

etch_noop · HAR_etch

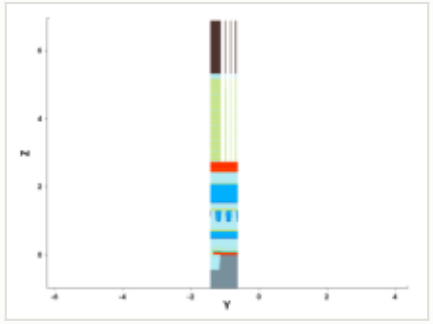
1.04	368.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

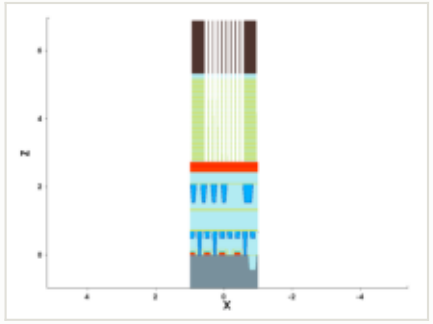
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

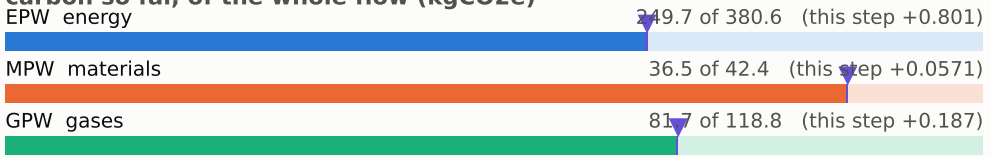
no mask for this step

this step

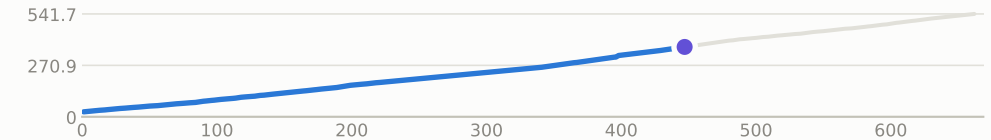
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

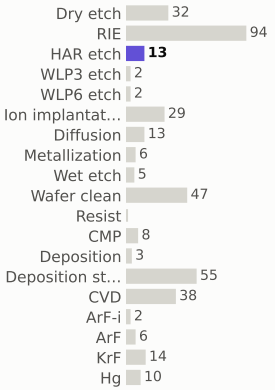
carbon so far, of the whole flow (kgCO2e)



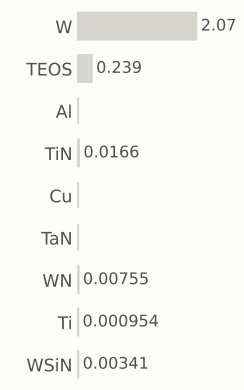
embodied carbon up to each step (kgCO2e per wafer)



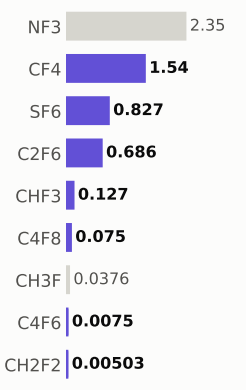
steps by process area



materials deposited (g)



gases emitted (g)



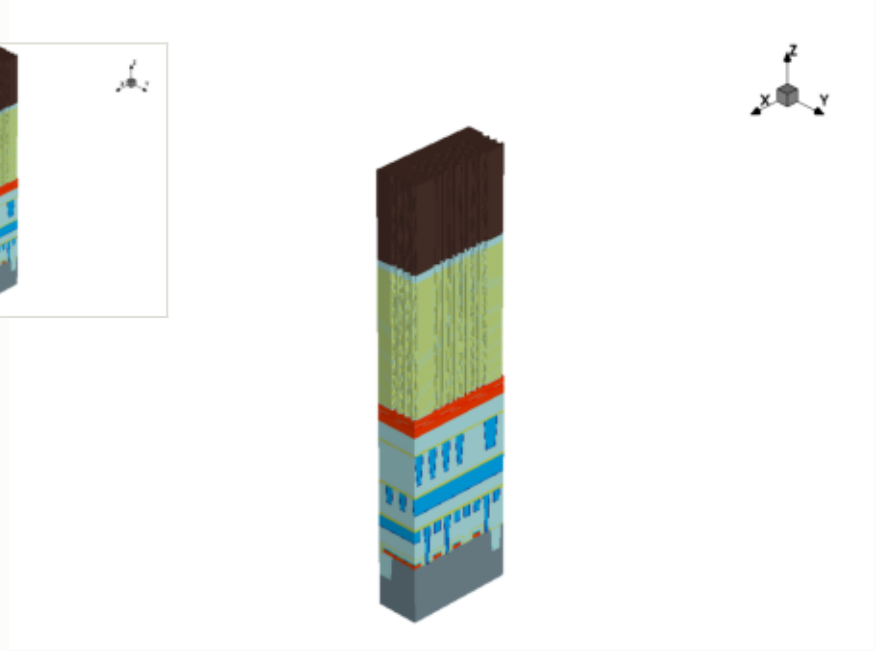
Step #4344.81: HAR_etch

etch_noop · HAR_etch

1.04	369.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



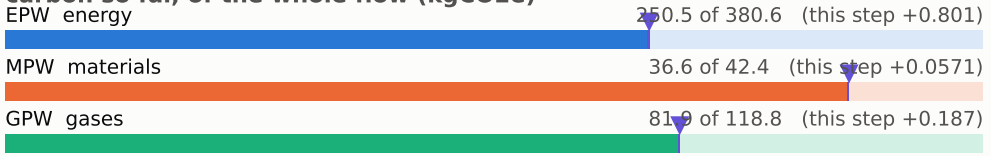
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

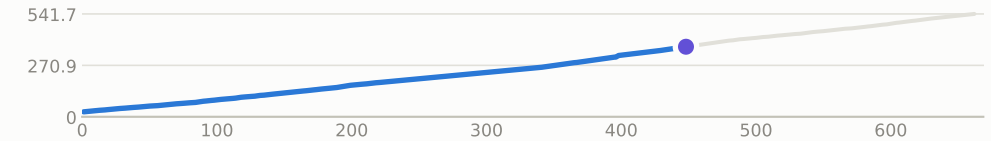
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

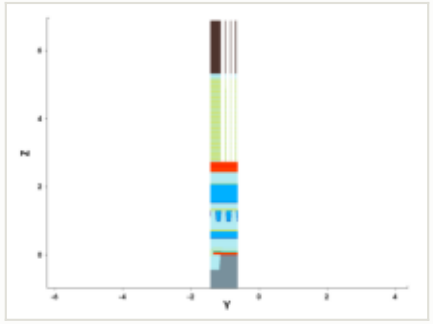
carbon so far, of the whole flow (kgCO2e)



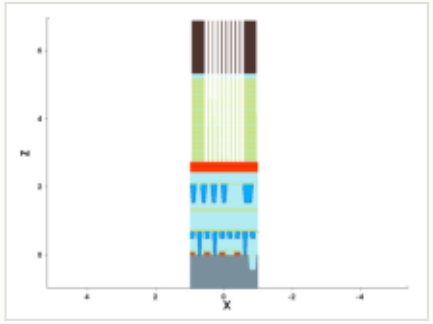
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



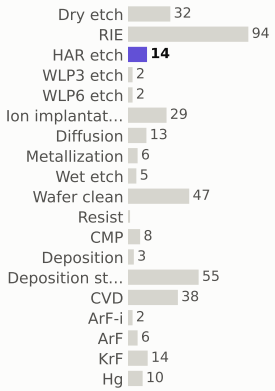
y cut at -1.032 µm



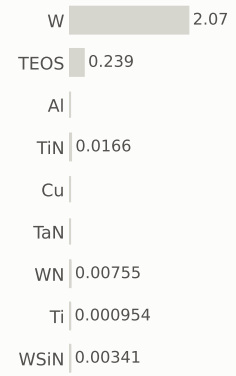
mask for this step

no mask for this step

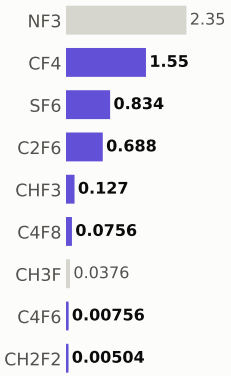
steps by process area



materials deposited (g)



gases emitted (g)



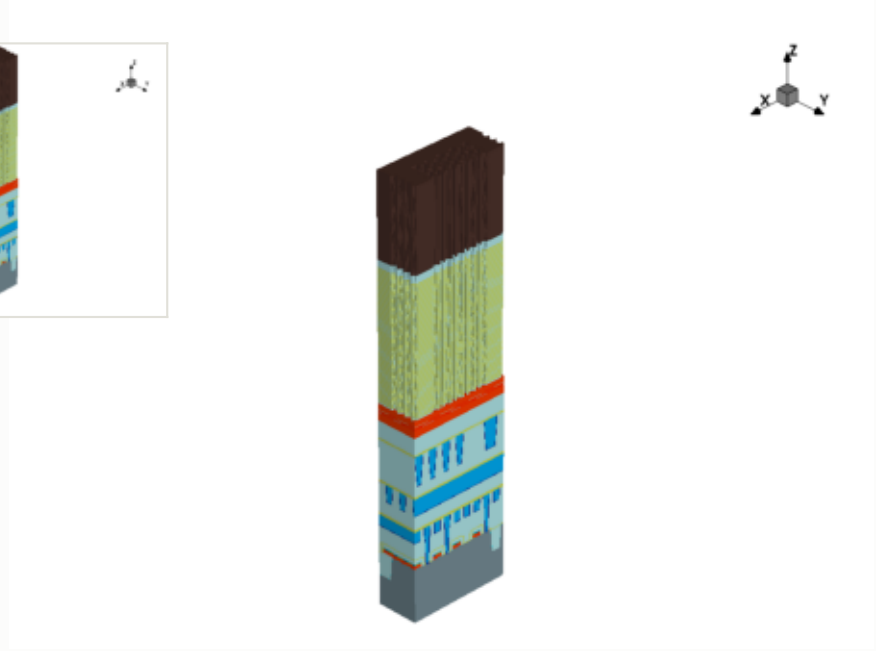
Step #4345: HAR_etch

etch_noop · HAR_etch

1.04	370.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



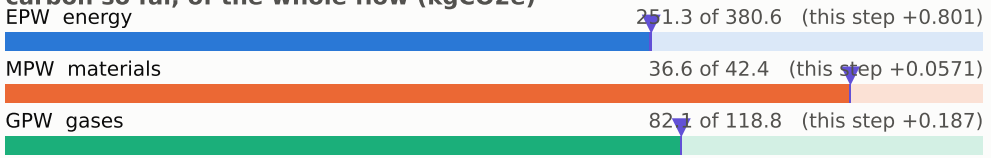
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

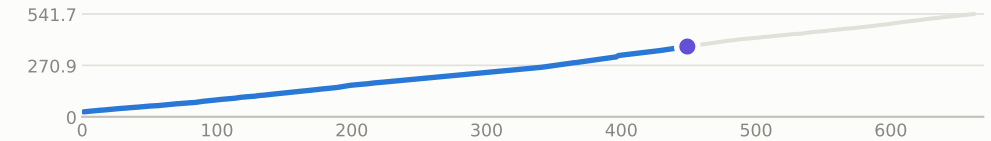
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

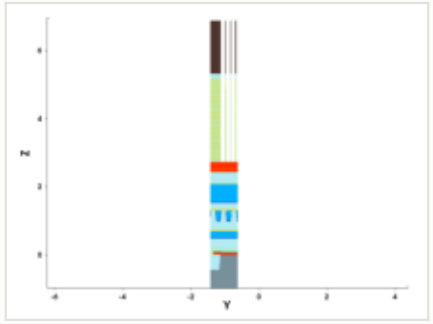
carbon so far, of the whole flow (kgCO2e)



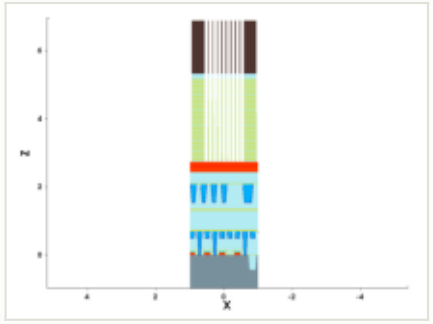
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



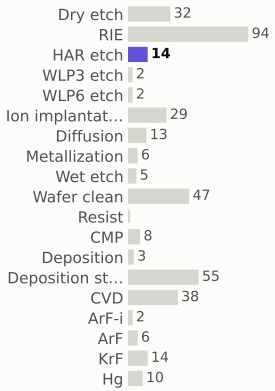
y cut at -1.032 µm



mask for this step

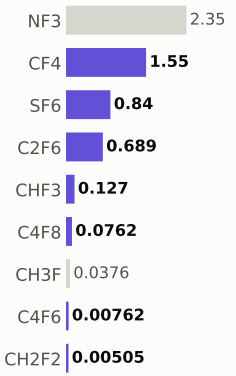
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



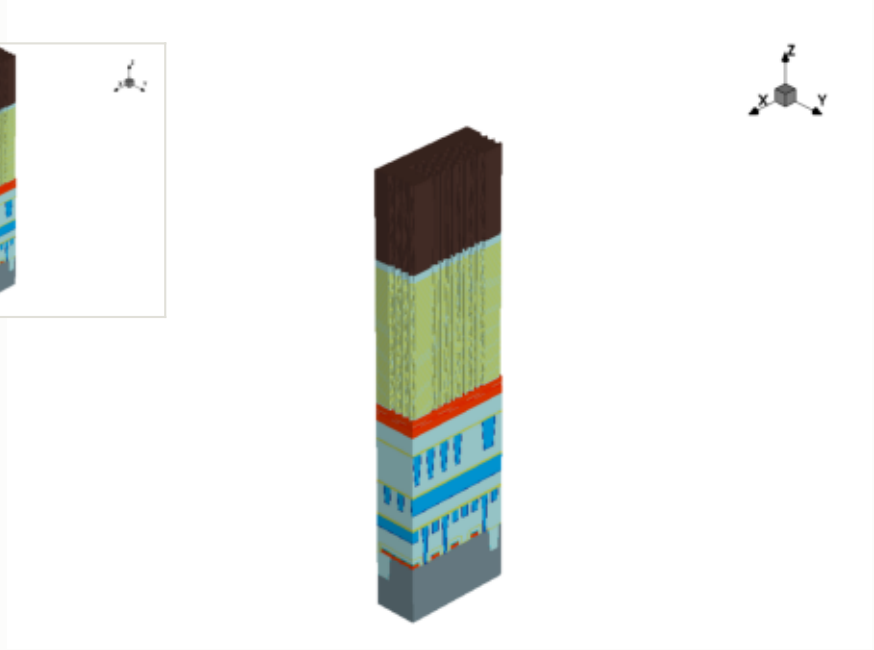
Step #4345.19: HAR_etch

etch_noop · HAR_etch

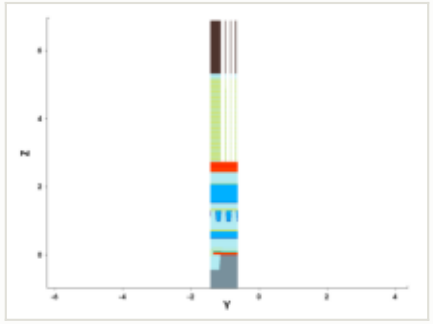
1.04	371.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

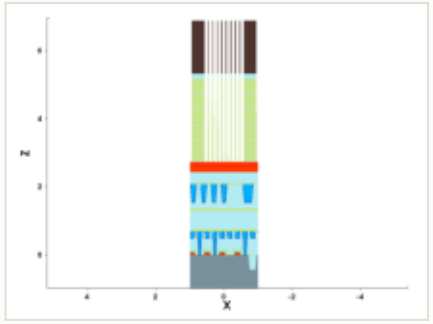
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

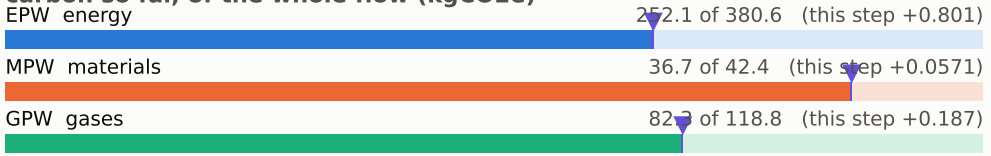
no mask for this step

this step

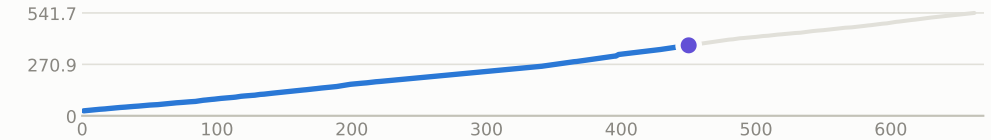
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

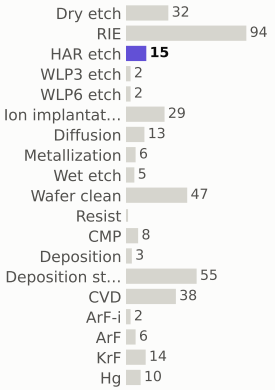
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

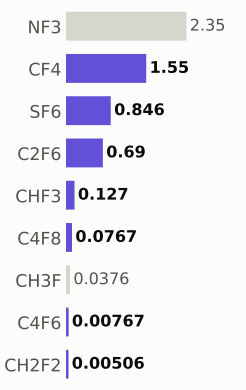


steps by process area



materials deposited (g)

gases emitted (g)



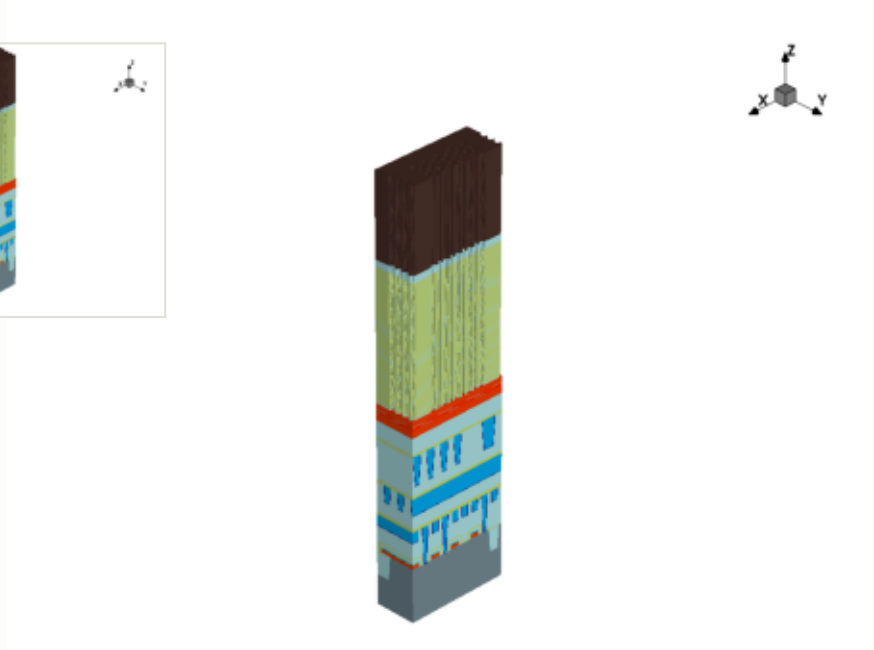
Step #4345.37: HAR_etch

etch_noop · HAR_etch

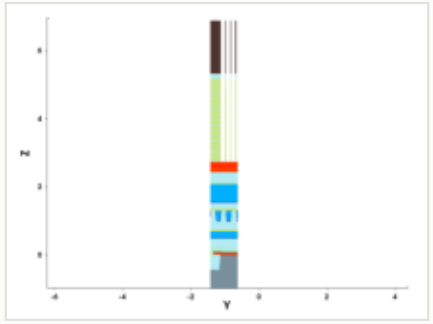
1.04	372.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

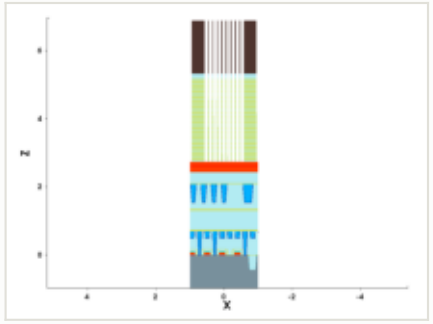
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

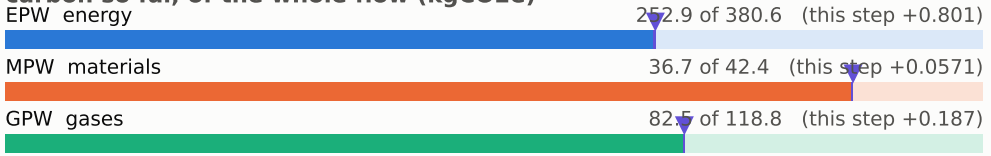
no mask for this step

this step

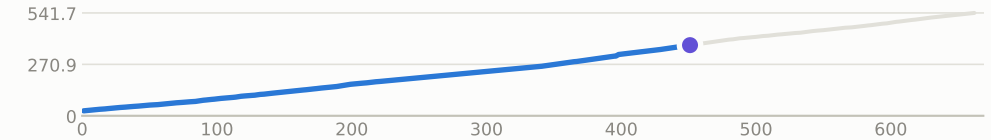
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

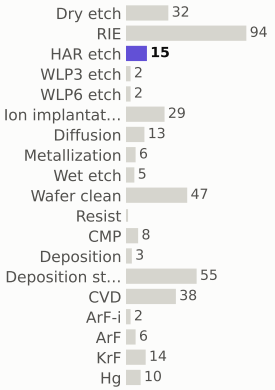
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

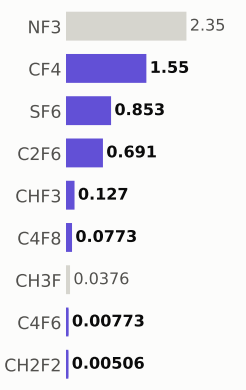


steps by process area



materials deposited (g)

gases emitted (g)



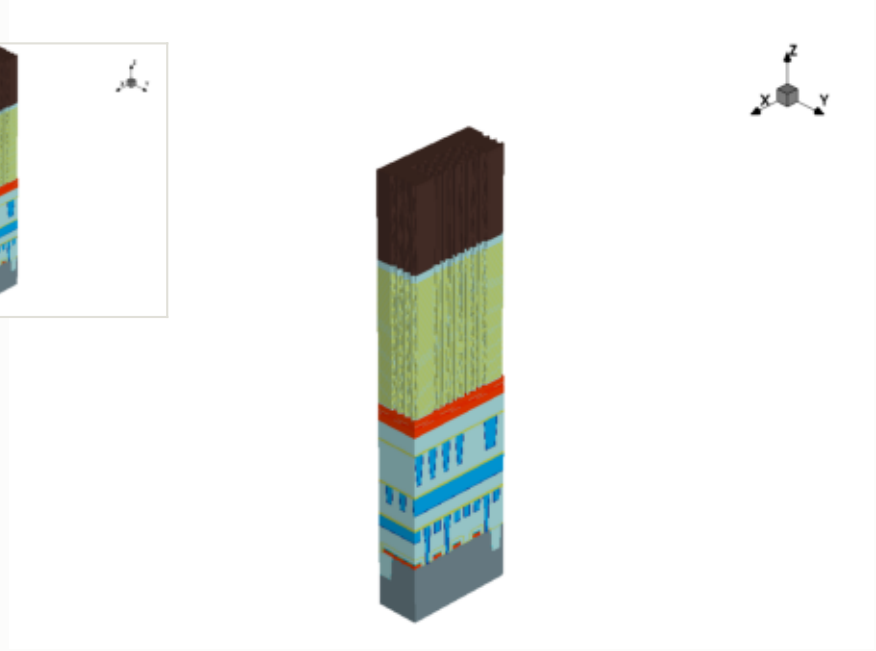
Step #4345.56: HAR_etch

etch_noop · HAR_etch

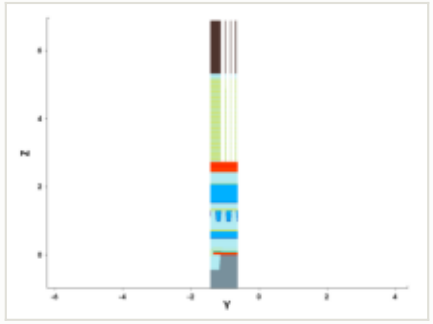
1.04	373.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

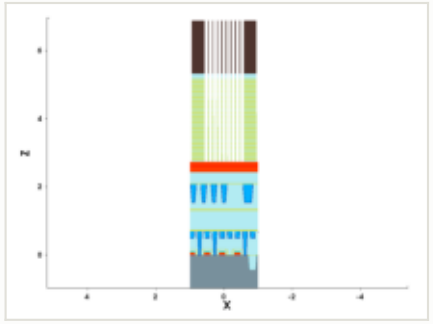
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

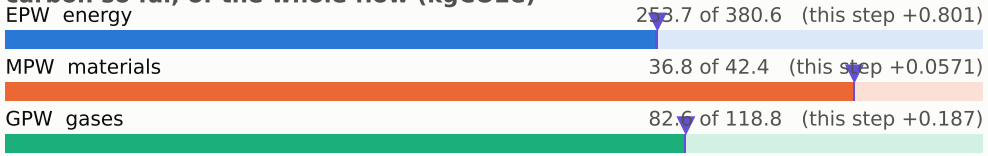
no mask for this step

this step

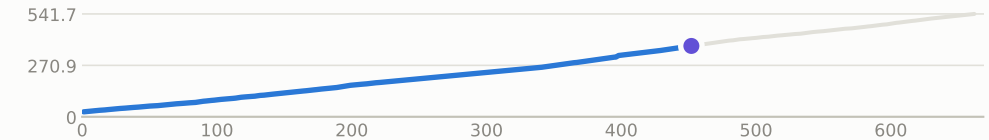
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

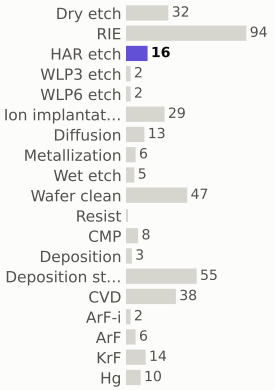
carbon so far, of the whole flow (kgCO2e)



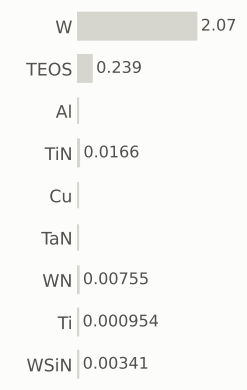
embodied carbon up to each step (kgCO2e per wafer)



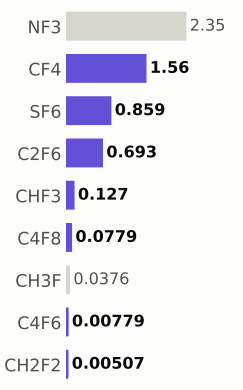
steps by process area



materials deposited (g)



gases emitted (g)



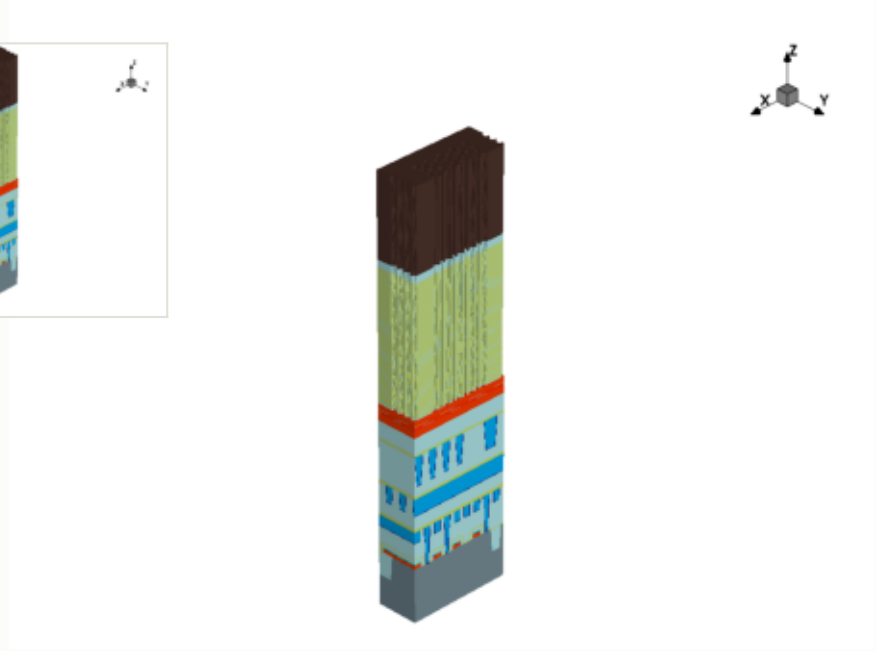
Step #4345.74: HAR_etch

etch_noop · HAR_etch

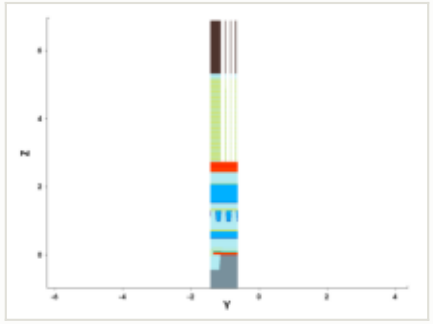
1.04	374.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

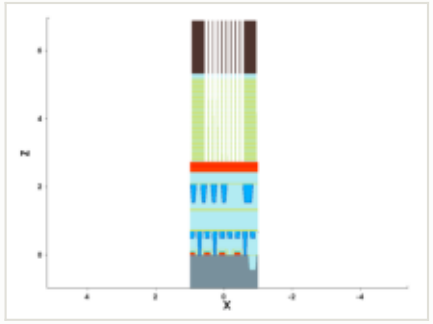
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

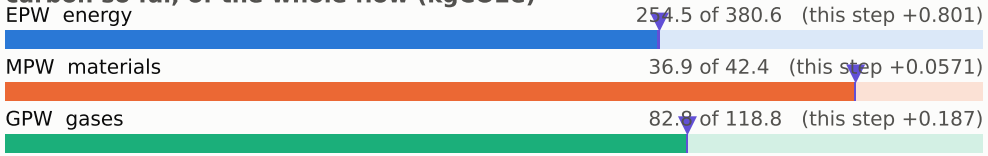
no mask for this step

this step

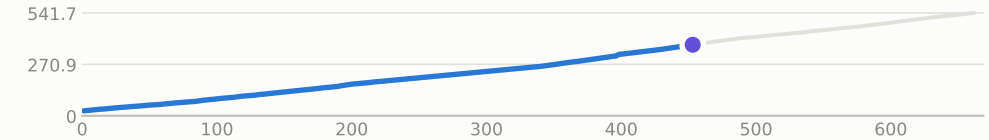
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

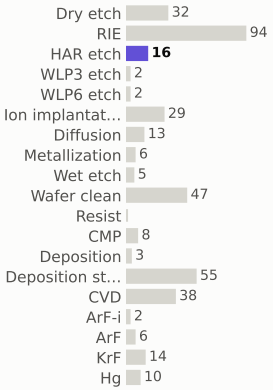
carbon so far, of the whole flow (kgCO2e)



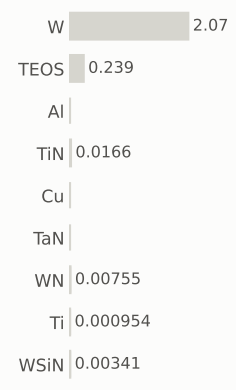
embodied carbon up to each step (kgCO2e per wafer)



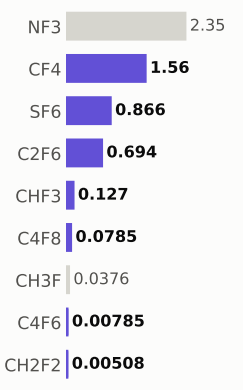
steps by process area



materials deposited (g)



gases emitted (g)



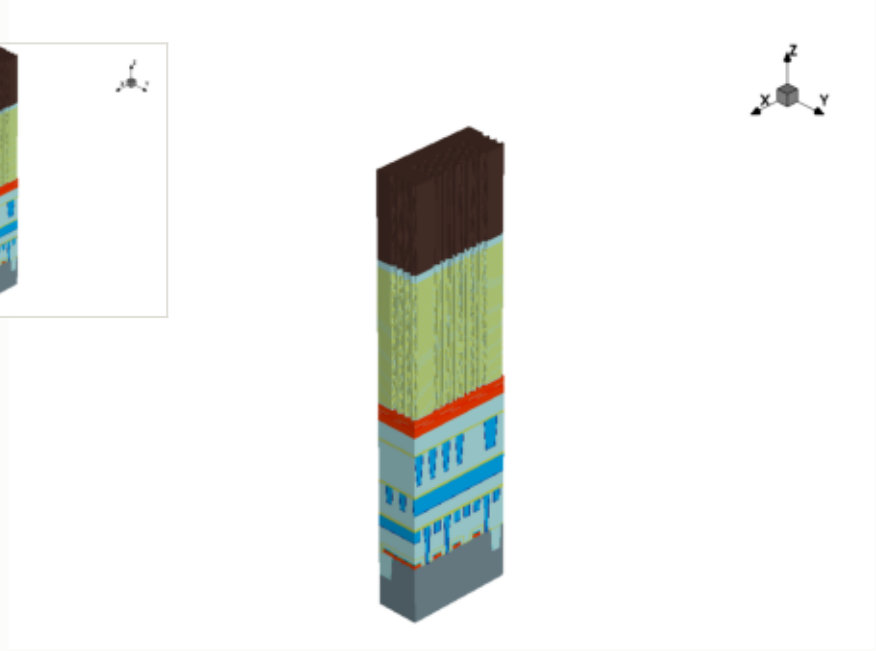
Step #4345.93: HAR_etch

etch_noop · HAR_etch

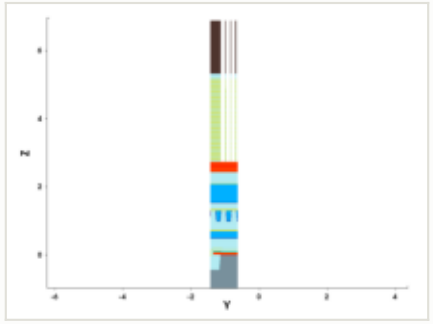
1.04	375.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

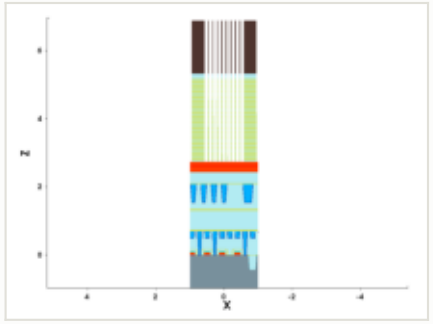
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

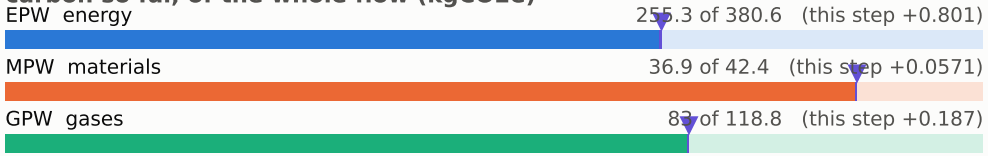
no mask for this step

this step

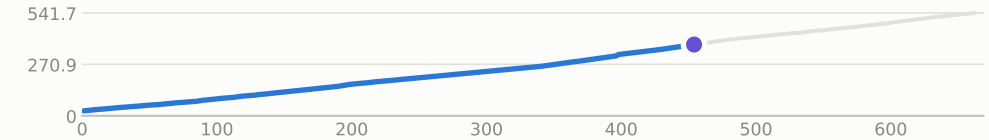
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

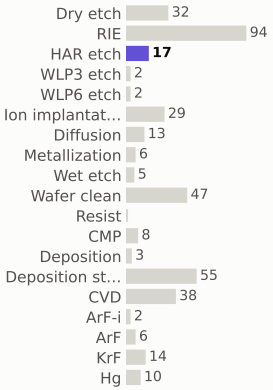
carbon so far, of the whole flow (kgCO2e)



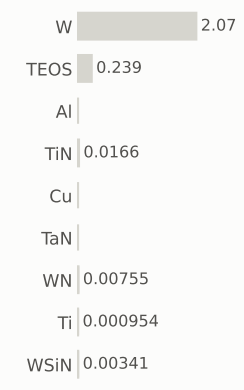
embodied carbon up to each step (kgCO2e per wafer)



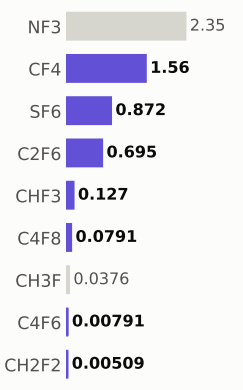
steps by process area



materials deposited (g)



gases emitted (g)



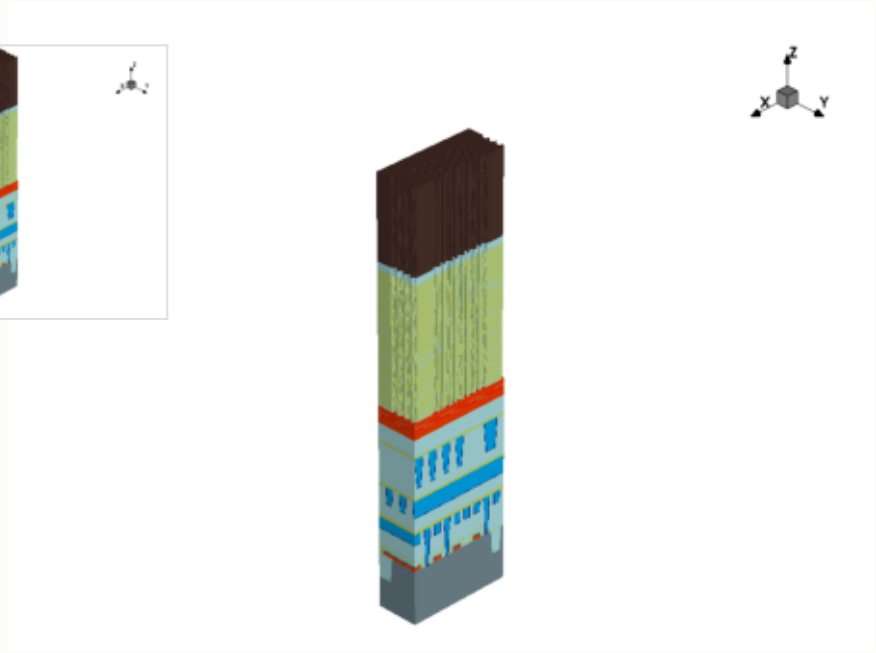
Step #4346.11: HAR_etch

etch_noop · HAR_etch

1.04	376.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



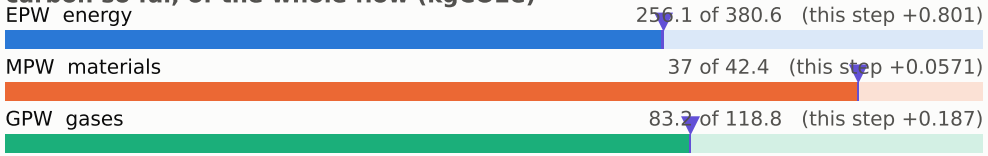
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

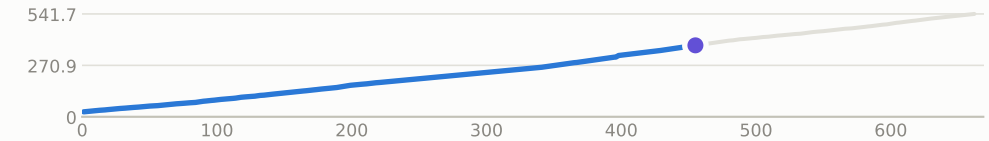
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

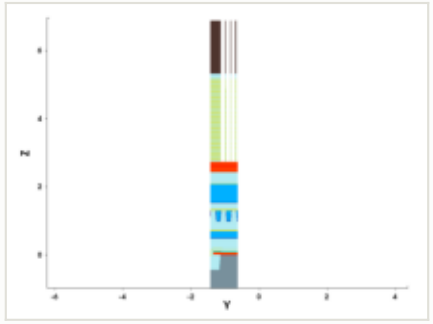
carbon so far, of the whole flow (kgCO2e)



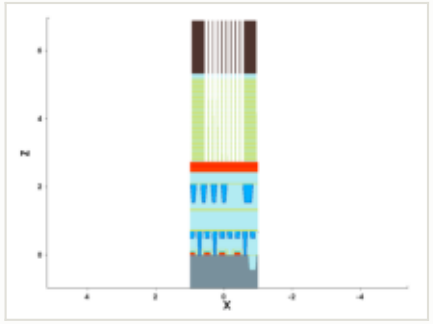
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



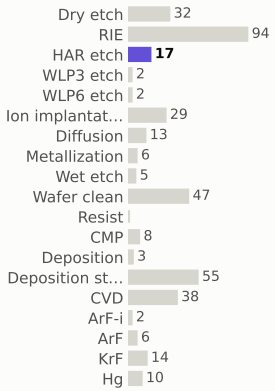
y cut at -1.032 µm



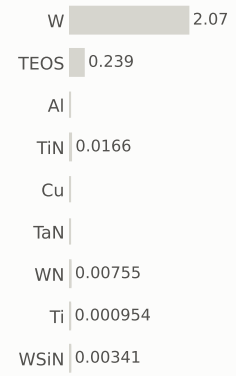
mask for this step

no mask for this step

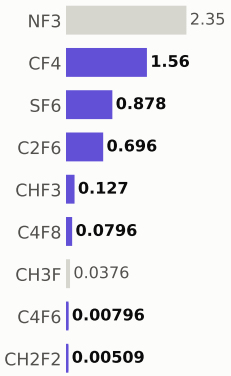
steps by process area



materials deposited (g)



gases emitted (g)



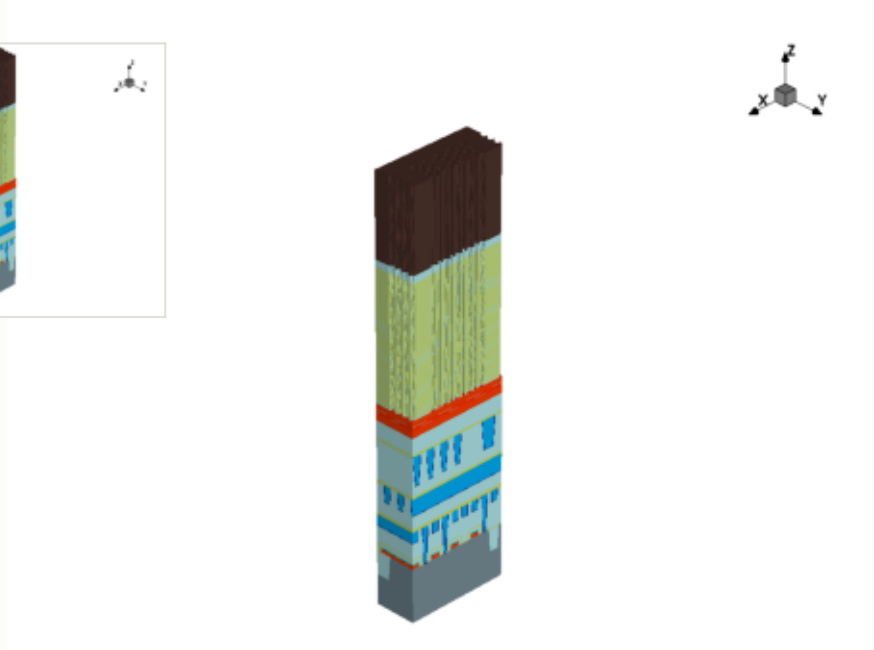
Step #4346.3: HAR_etch

etch_noop · HAR_etch

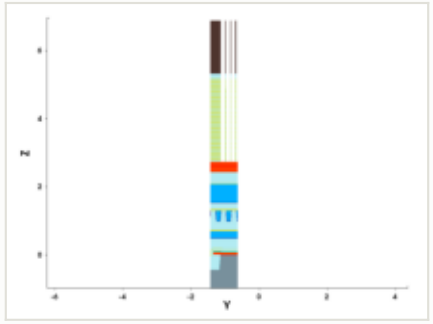
1.04	377.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

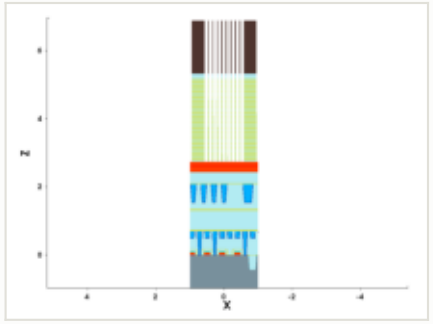
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

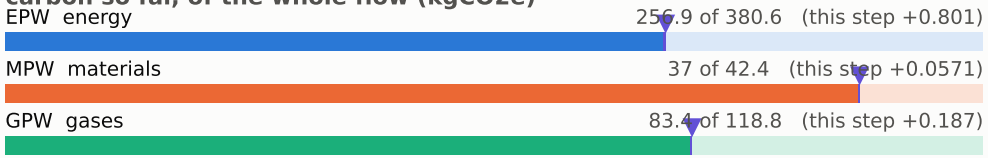
no mask for this step

this step

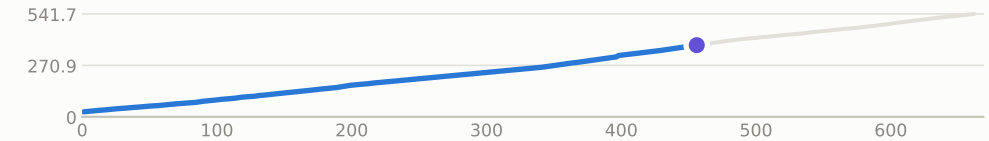
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

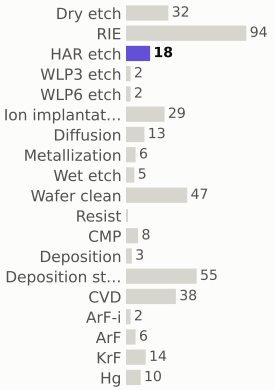
carbon so far, of the whole flow (kgCO2e)



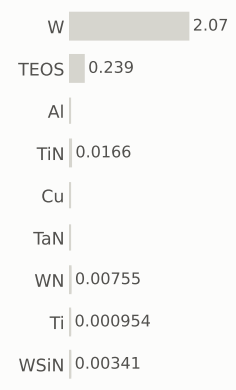
embodied carbon up to each step (kgCO2e per wafer)



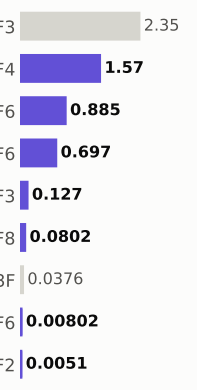
steps by process area



materials deposited (g)



gases emitted (g)



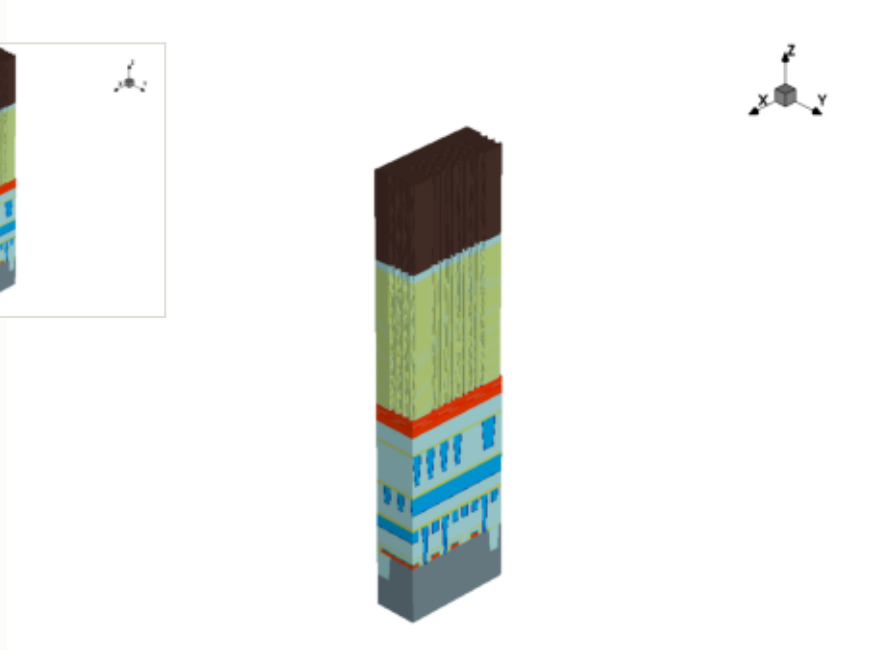
Step #4346.48: HAR_etch

etch_noop · HAR_etch

1.04	378.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



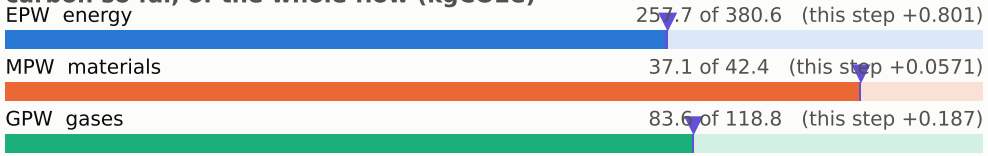
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

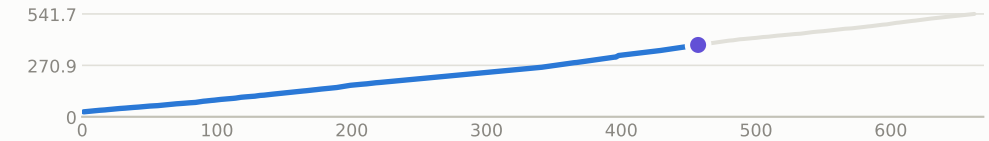
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

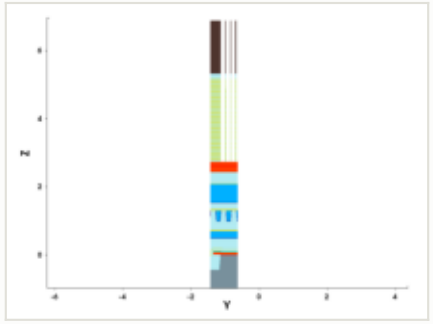
carbon so far, of the whole flow (kgCO2e)



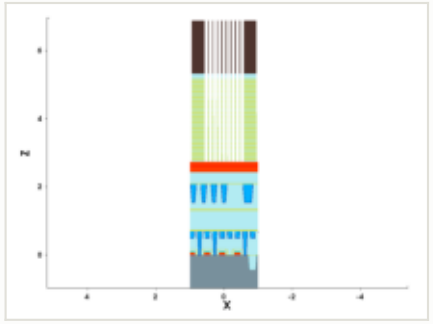
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



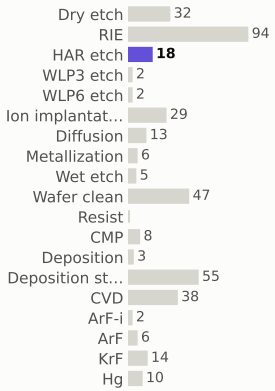
y cut at -1.032 µm



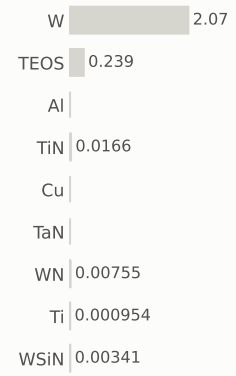
mask for this step

no mask for this step

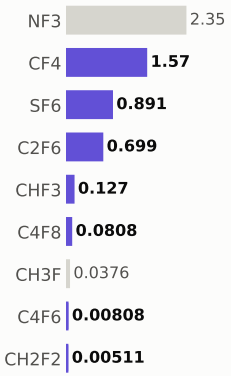
steps by process area



materials deposited (g)



gases emitted (g)



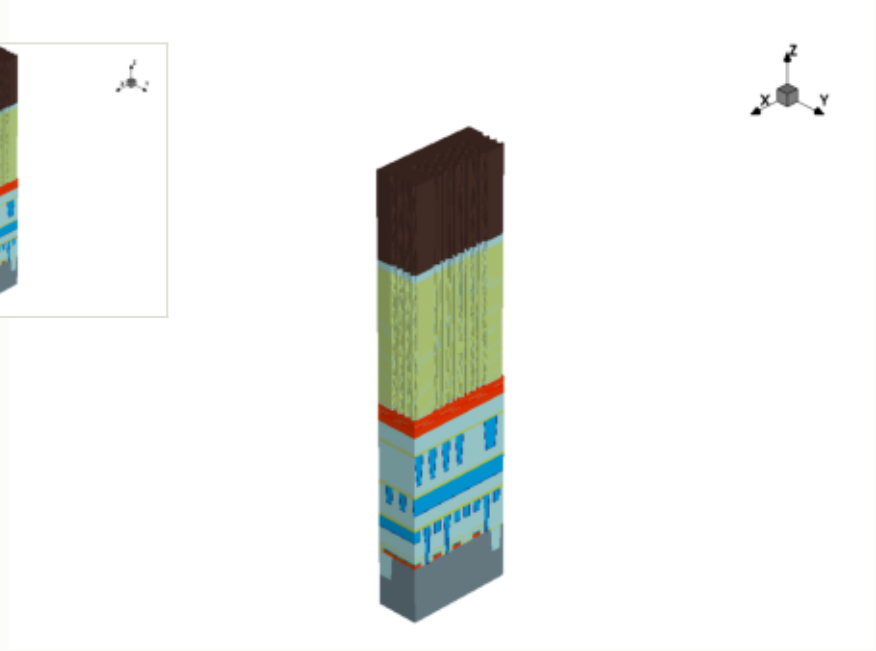
Step #4346.67: HAR_etch

etch_noop · HAR_etch

1.04	379.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



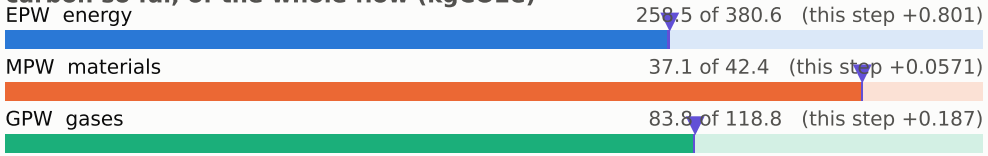
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

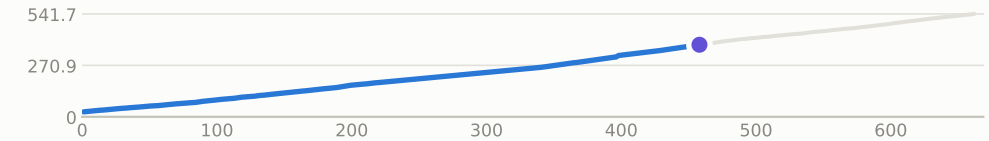
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

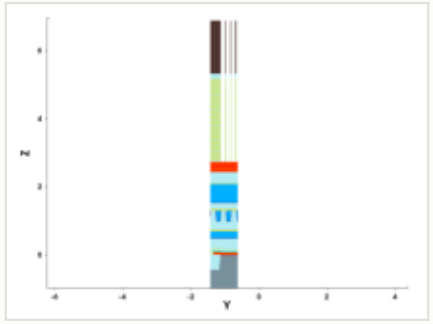
carbon so far, of the whole flow (kgCO2e)



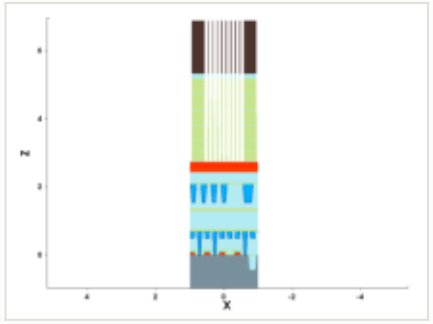
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



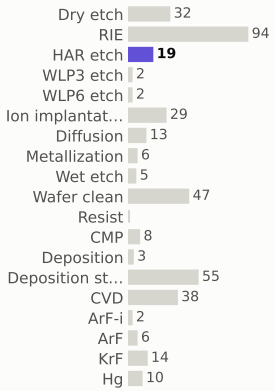
y cut at -1.032 µm



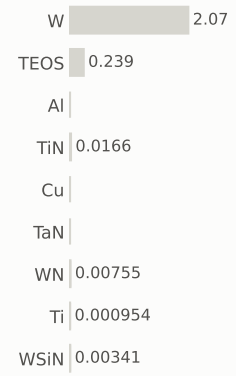
mask for this step

no mask for this step

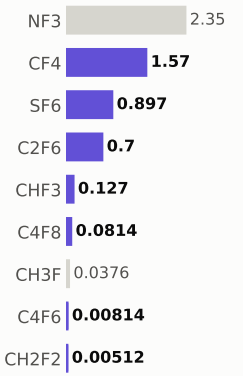
steps by process area



materials deposited (g)



gases emitted (g)



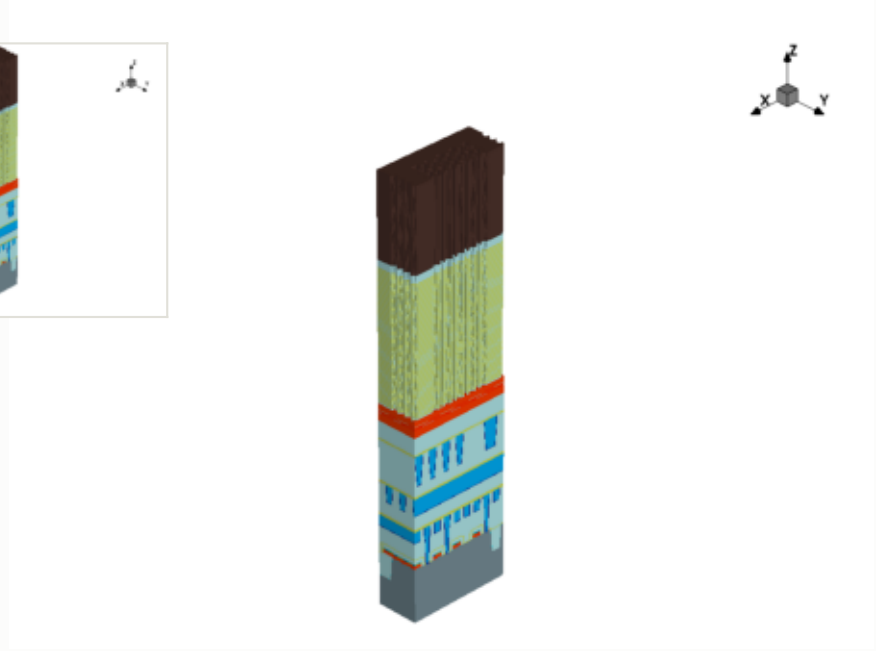
Step #4346.85: HAR_etch

etch_noop · HAR_etch

1.04	380.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



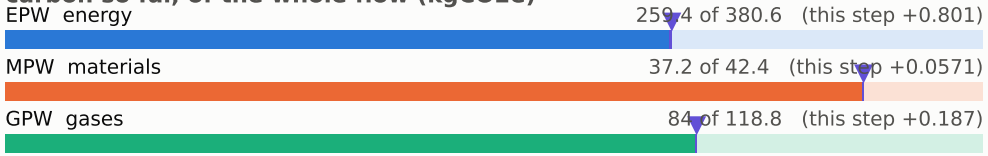
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

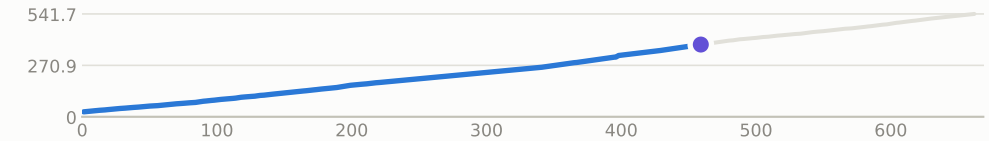
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

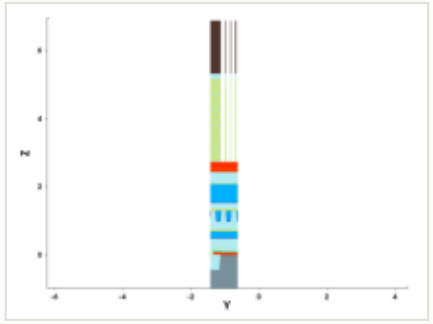
carbon so far, of the whole flow (kgCO2e)



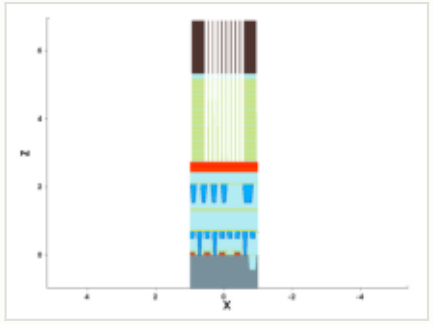
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



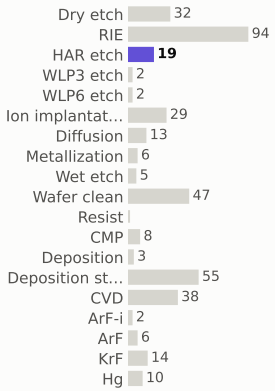
y cut at -1.032 µm



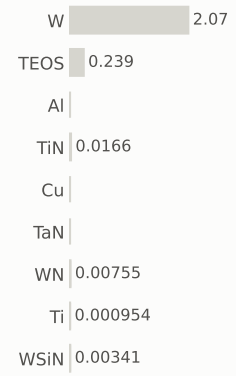
mask for this step

no mask for this step

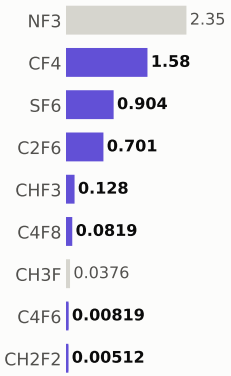
steps by process area



materials deposited (g)



gases emitted (g)



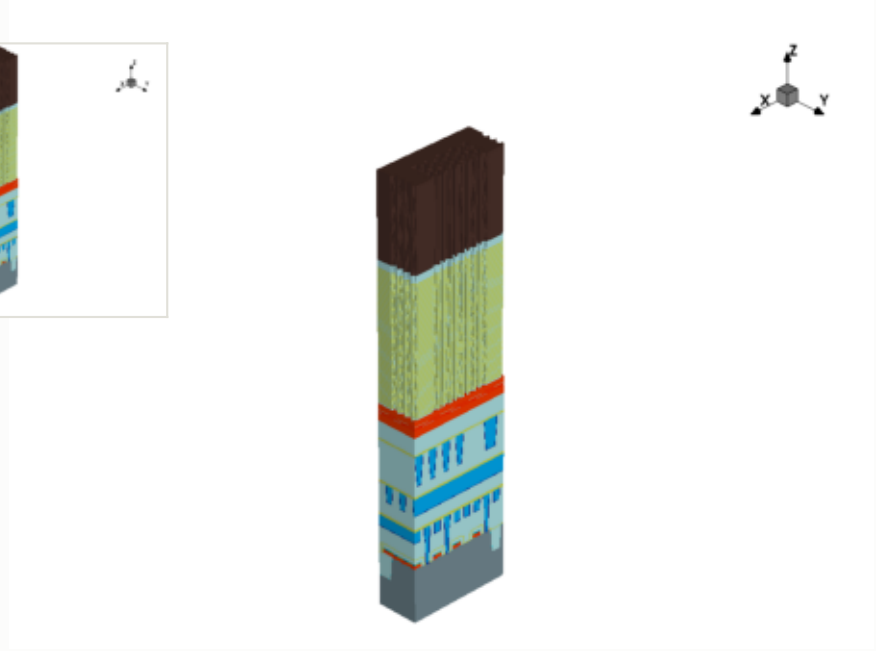
Step #4347.04: HAR_etch

etch_noop · HAR_etch

1.04	381.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



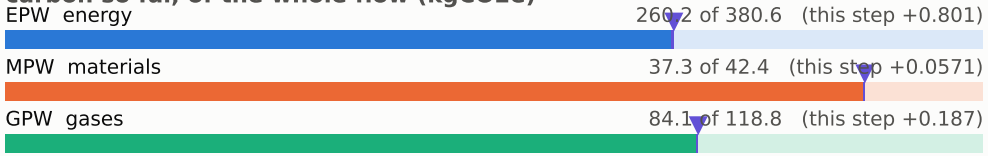
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

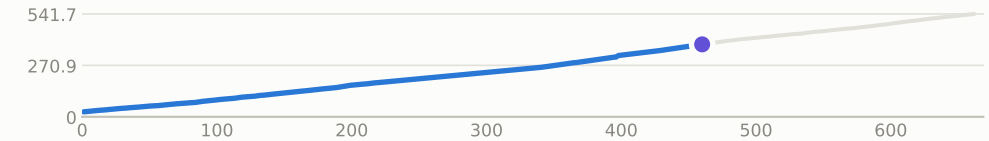
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

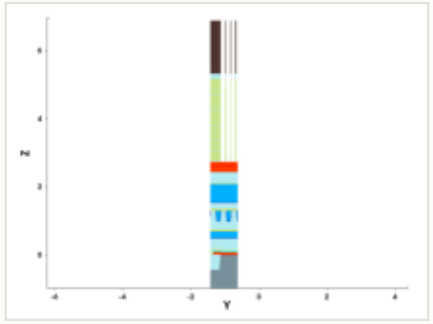
carbon so far, of the whole flow (kgCO2e)



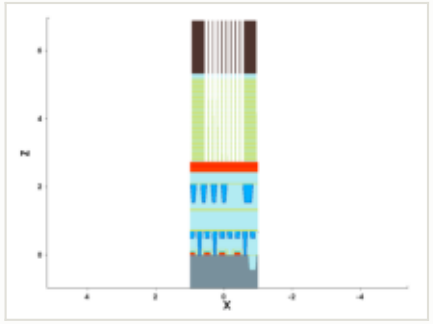
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



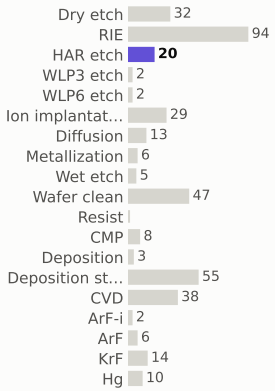
y cut at -1.032 µm



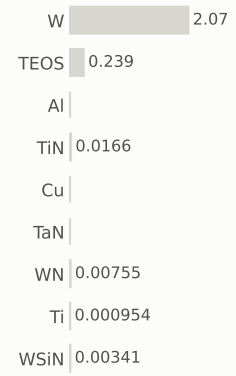
mask for this step

no mask for this step

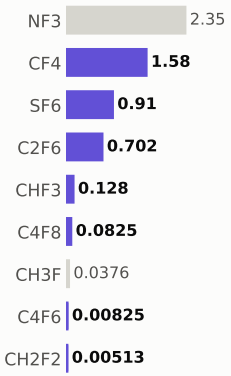
steps by process area



materials deposited (g)



gases emitted (g)



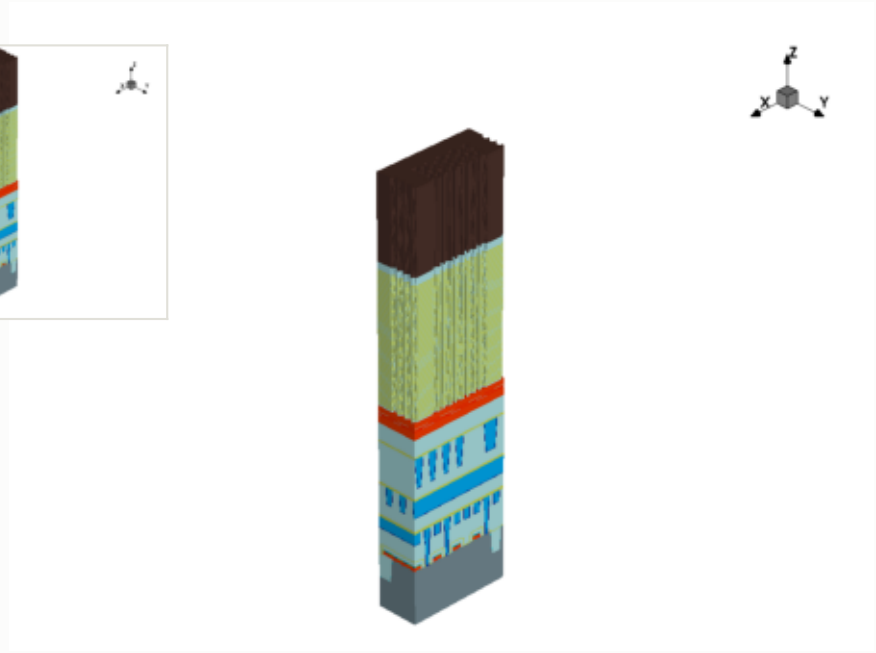
Step #4347.22: HAR_etch

etch_noop · HAR_etch

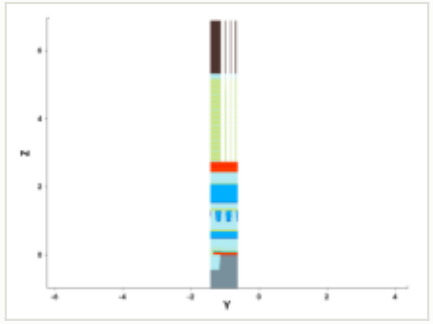
1.04	382.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

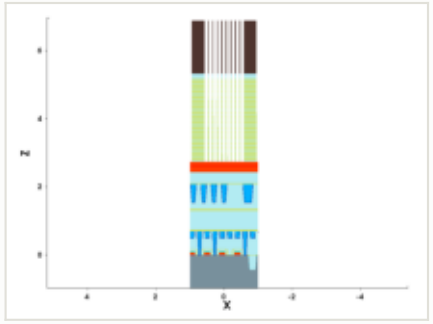
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

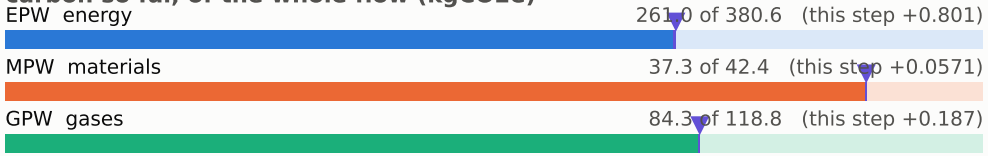
no mask for this step

this step

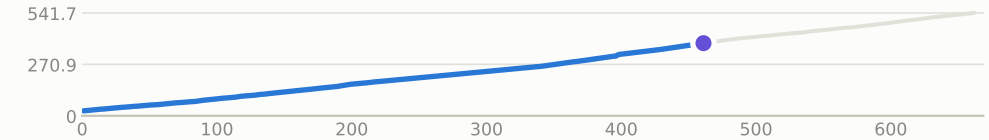
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

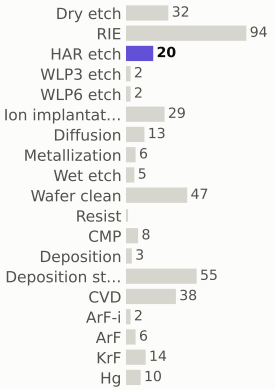
carbon so far, of the whole flow (kgCO2e)



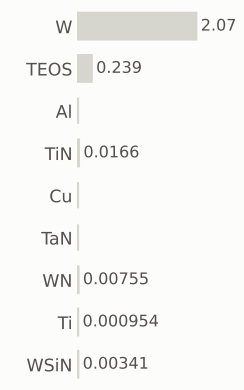
embodied carbon up to each step (kgCO2e per wafer)



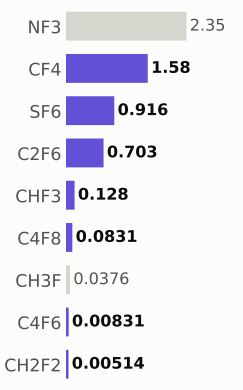
steps by process area



materials deposited (g)



gases emitted (g)



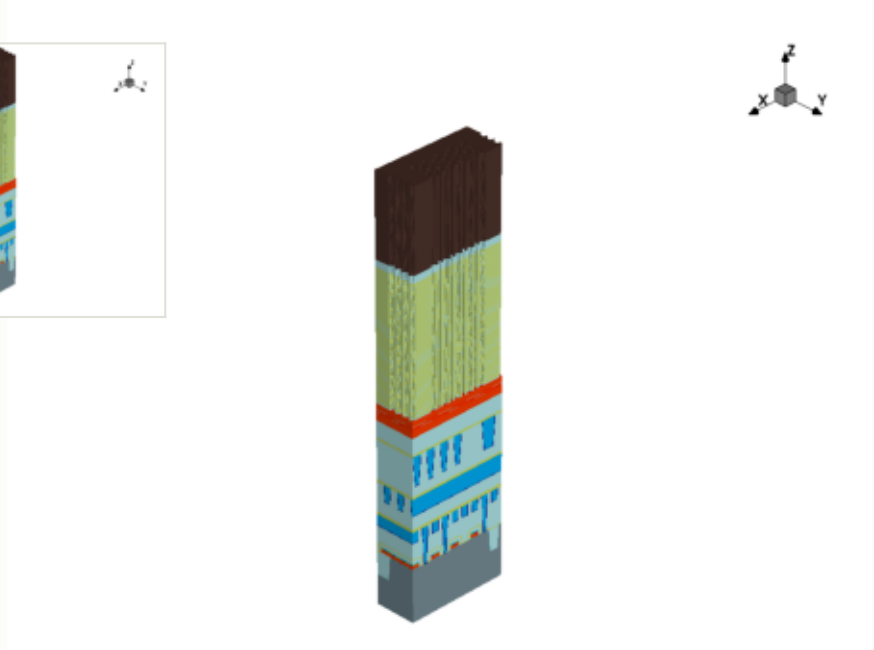
Step #4347.41: HAR_etch

etch_noop · HAR_etch

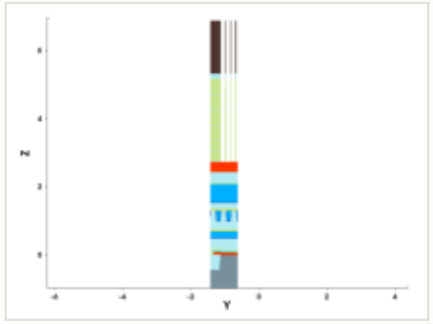
1.04	383.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

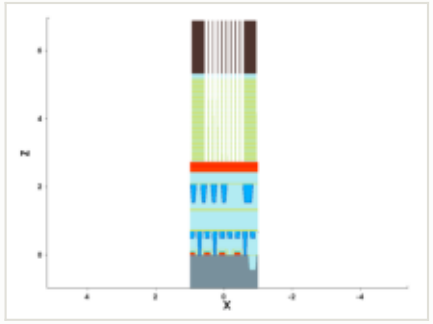
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

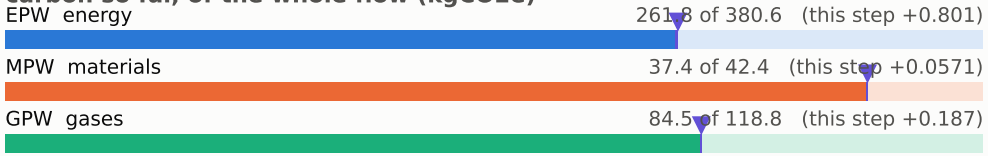
no mask for this step

this step

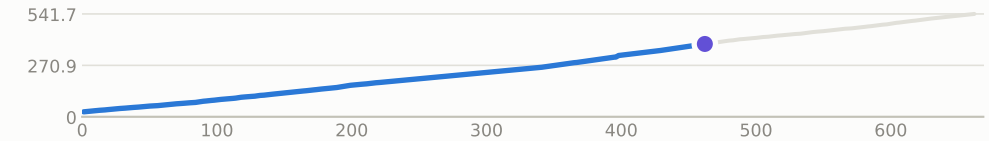
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

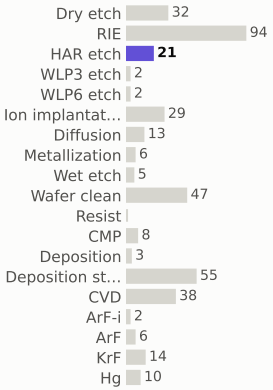
carbon so far, of the whole flow (kgCO2e)



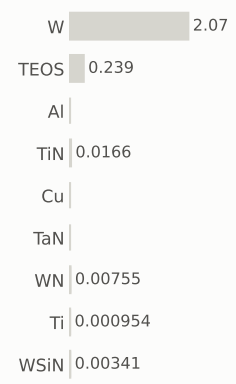
embodied carbon up to each step (kgCO2e per wafer)



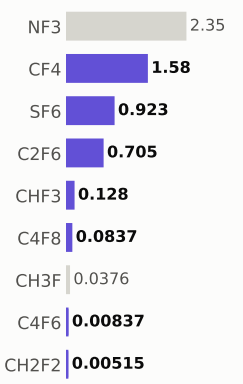
steps by process area



materials deposited (g)



gases emitted (g)



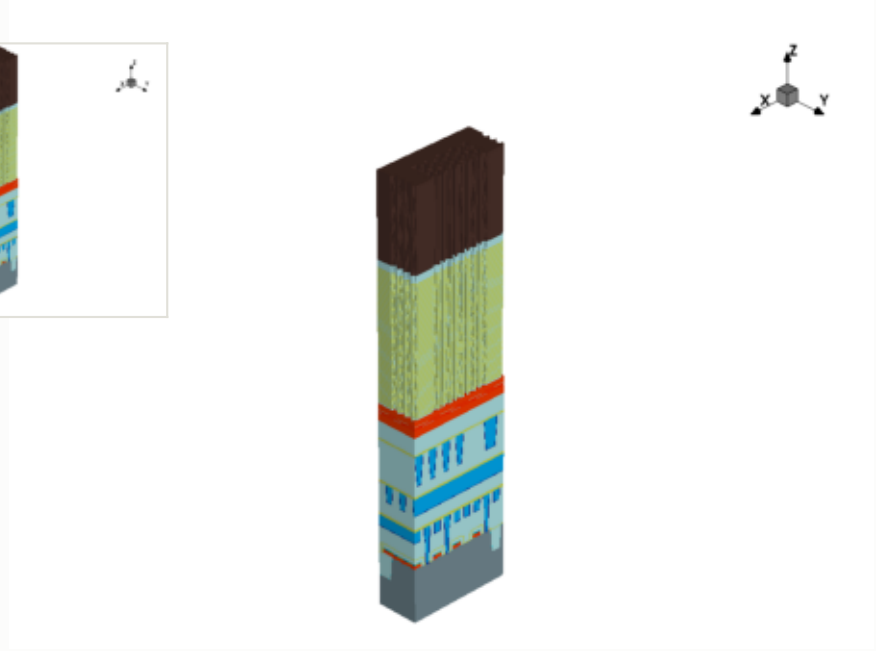
Step #4347.59: HAR_etch

etch_noop · HAR_etch

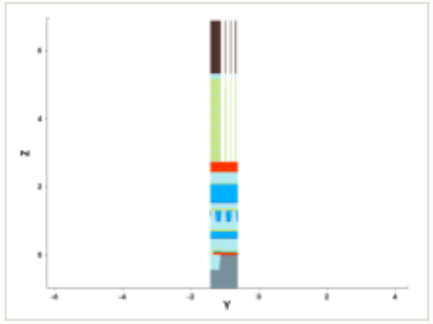
1.04	384.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

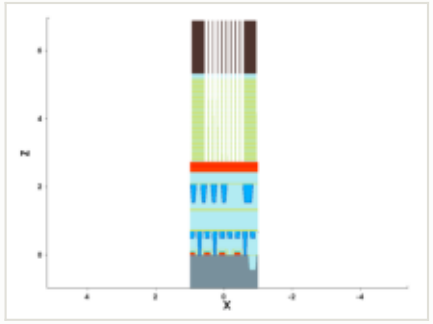
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

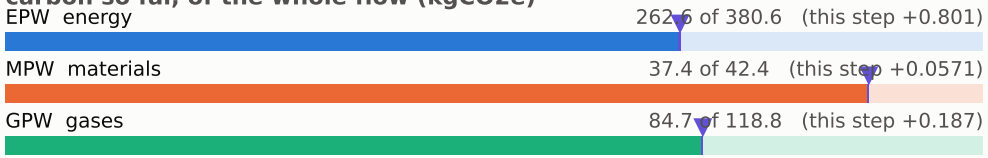
no mask for this step

this step

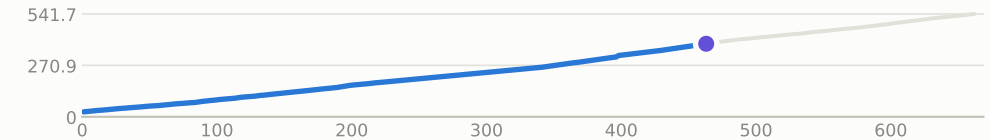
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

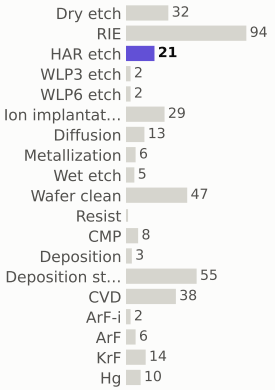
carbon so far, of the whole flow (kgCO2e)



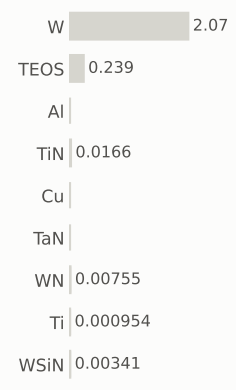
embodied carbon up to each step (kgCO2e per wafer)



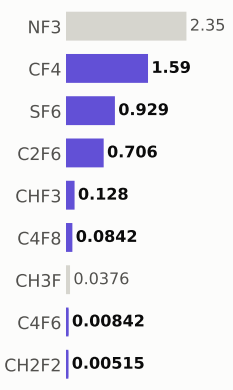
steps by process area



materials deposited (g)



gases emitted (g)



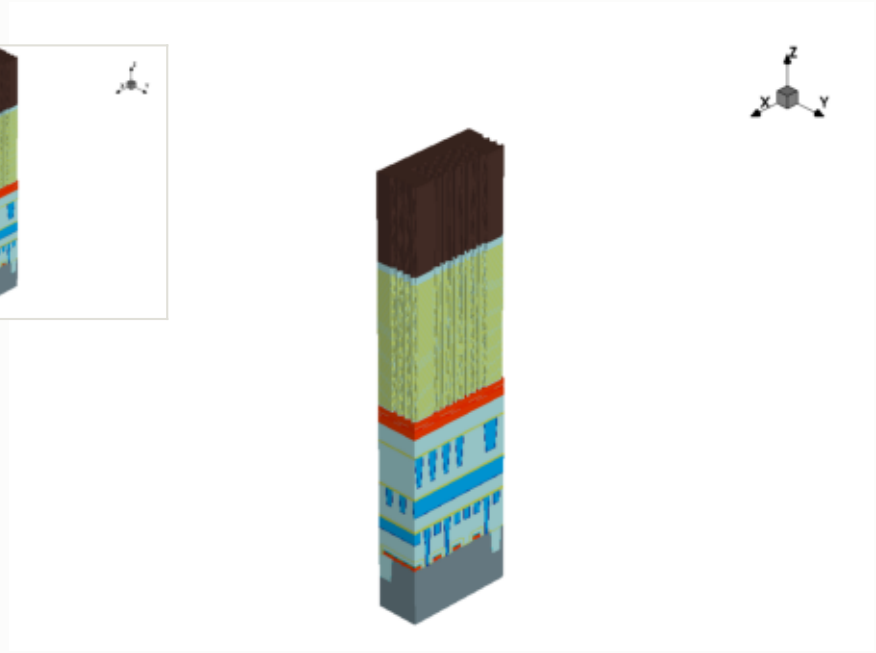
Step #4347.78: HAR_etch

etch_noop · HAR_etch

1.04	385.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



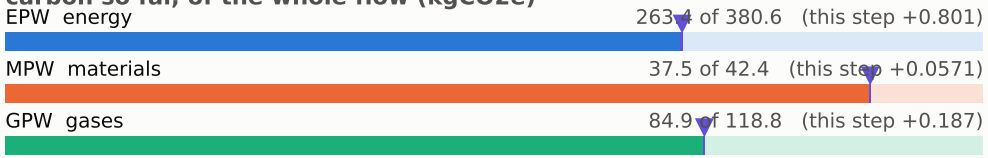
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

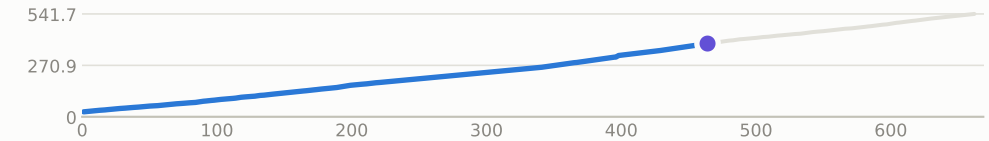
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

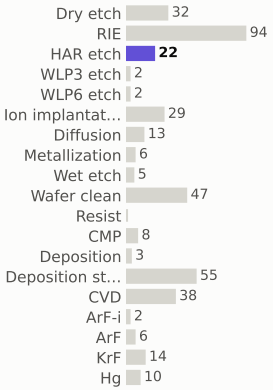
carbon so far, of the whole flow (kgCO2e)



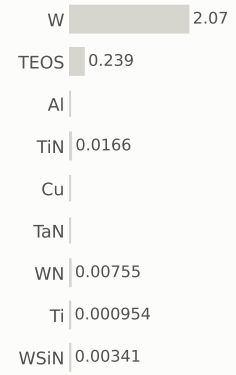
embodied carbon up to each step (kgCO2e per wafer)



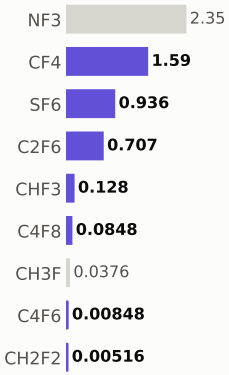
steps by process area



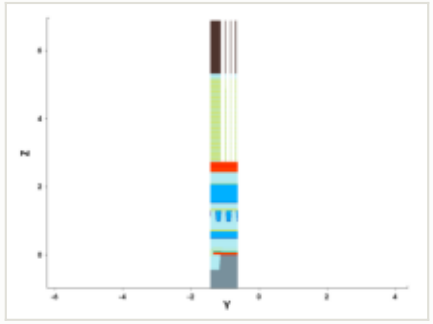
materials deposited (g)



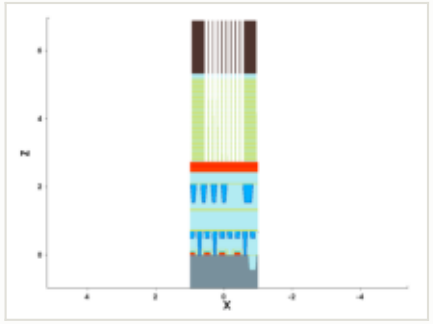
gases emitted (g)



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

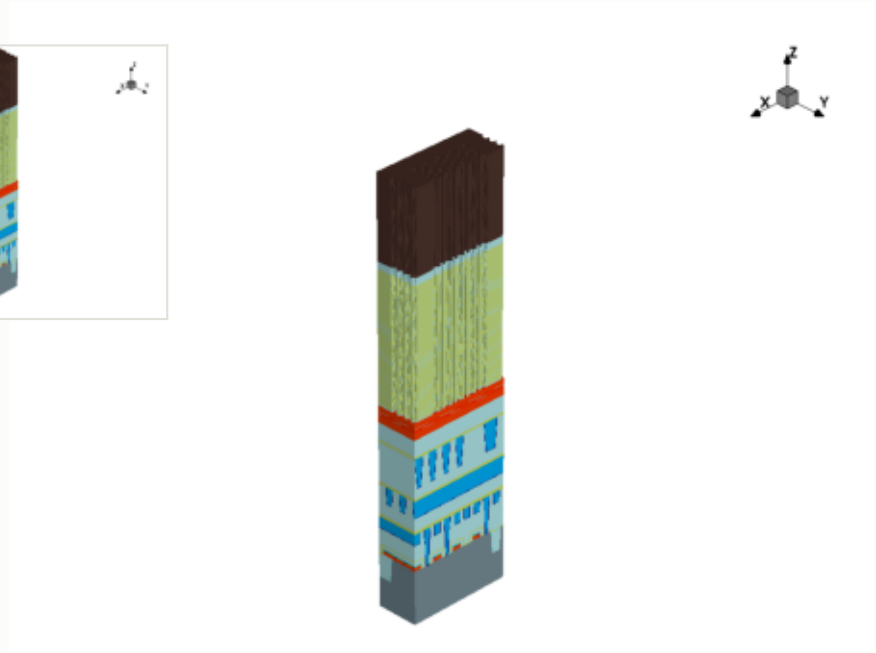
Step #4347.96: HAR_etch

etch_noop · HAR_etch

1.04	386.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



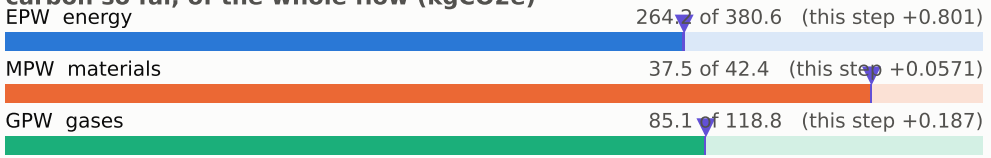
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

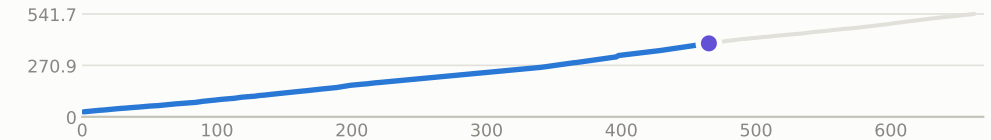
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

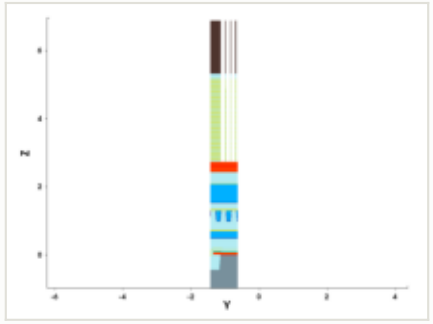
carbon so far, of the whole flow (kgCO2e)



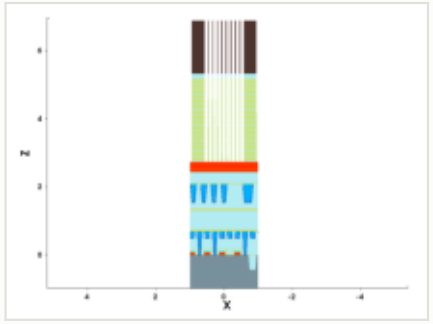
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



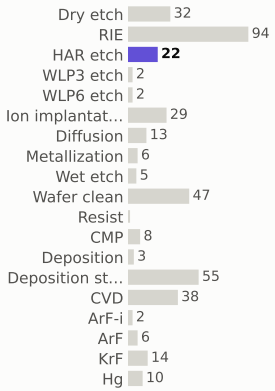
y cut at -1.032 µm



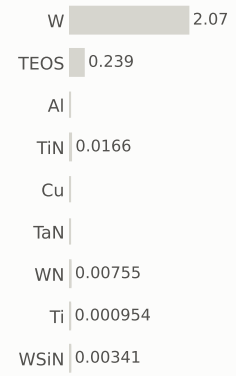
mask for this step

no mask for this step

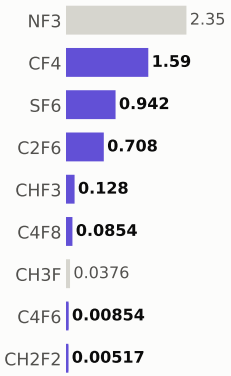
steps by process area



materials deposited (g)



gases emitted (g)



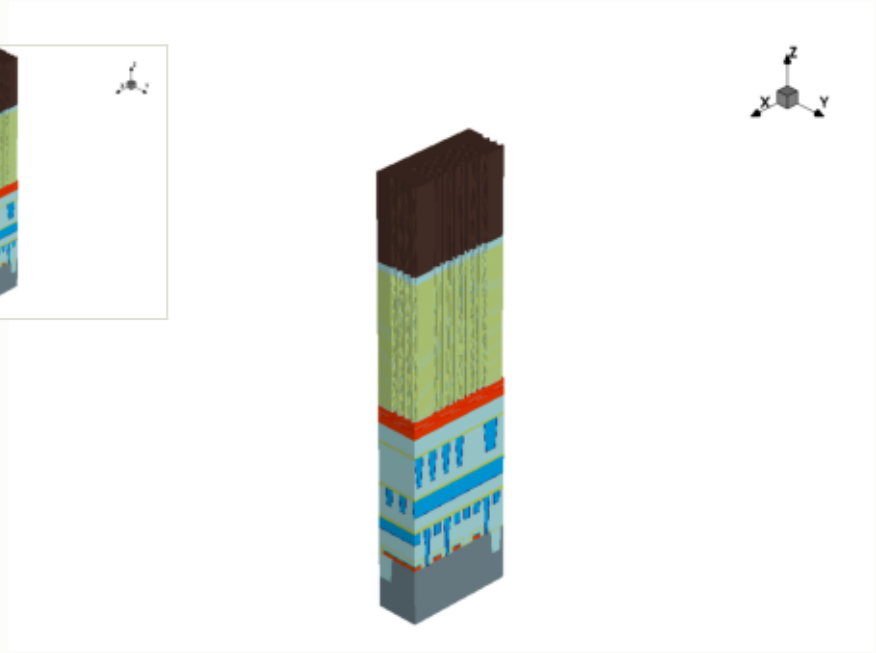
Step #4348.15: HAR_etch

etch_noop · HAR_etch

1.04	387.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

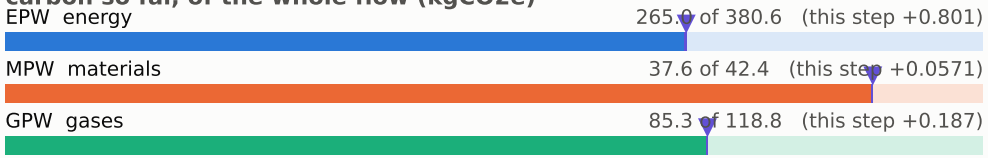


this step

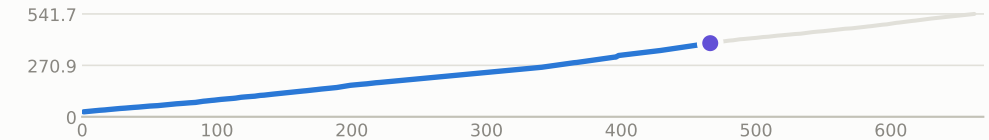
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

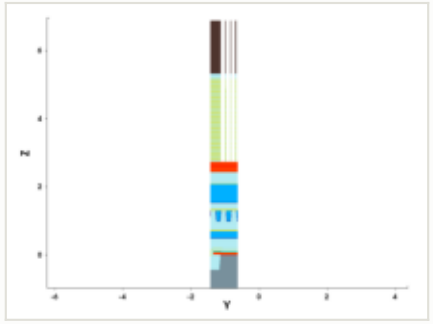
carbon so far, of the whole flow (kgCO2e)



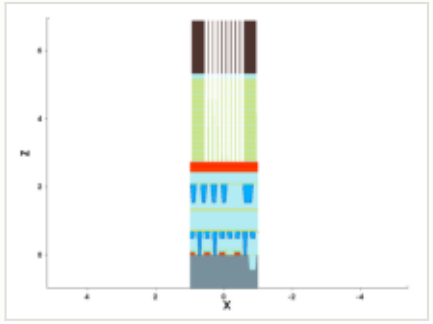
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



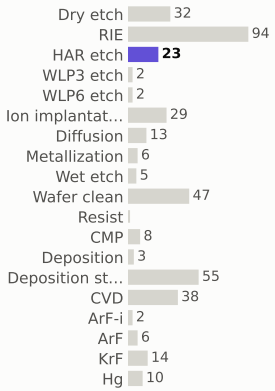
y cut at -1.032 µm



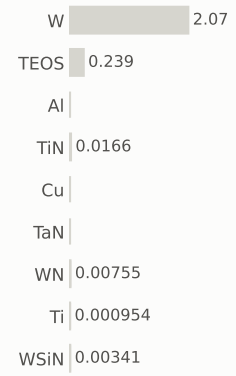
mask for this step

no mask for this step

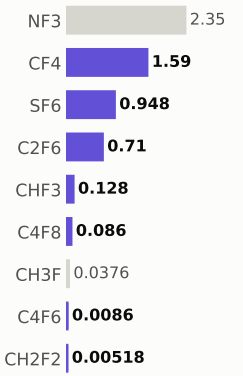
steps by process area



materials deposited (g)



gases emitted (g)



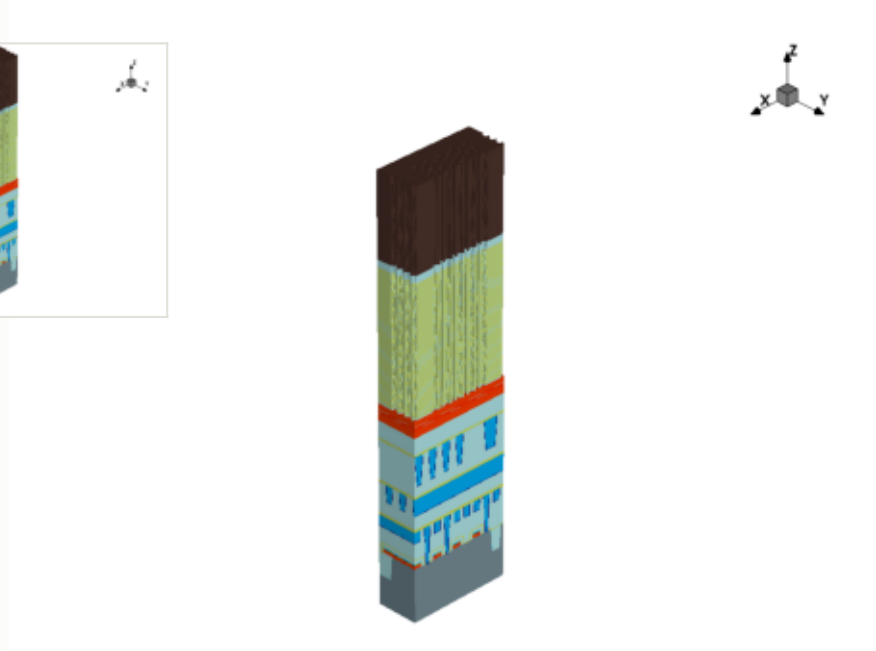
Step #4348.33: HAR_etch

etch_noop · HAR_etch

1.04	388.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

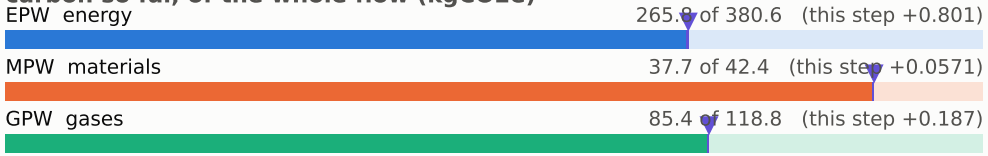


this step

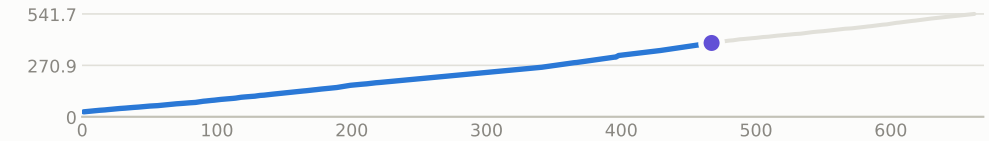
process area	HAR_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

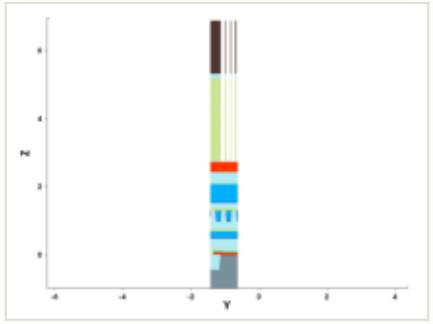
carbon so far, of the whole flow (kgCO2e)



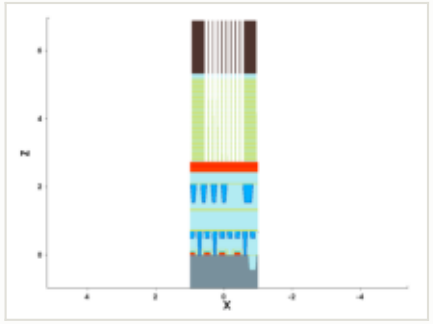
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



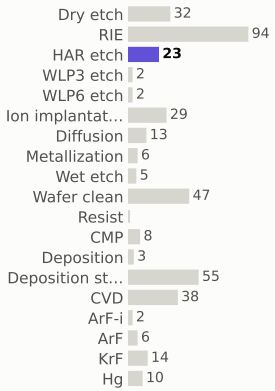
y cut at -1.032 µm



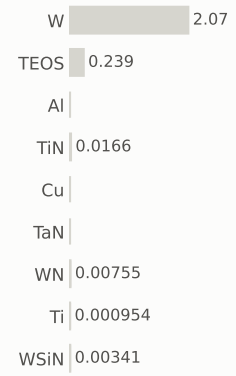
mask for this step

no mask for this step

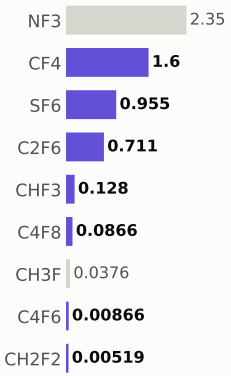
steps by process area



materials deposited (g)



gases emitted (g)



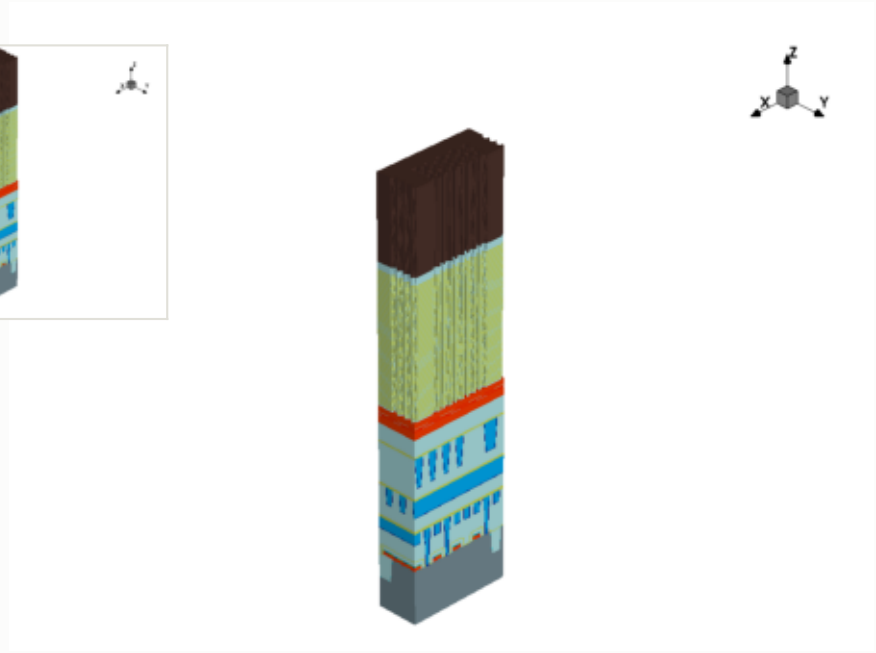
Step #4348.52: HAR_etch

etch_noop · HAR_etch

1.04	389.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



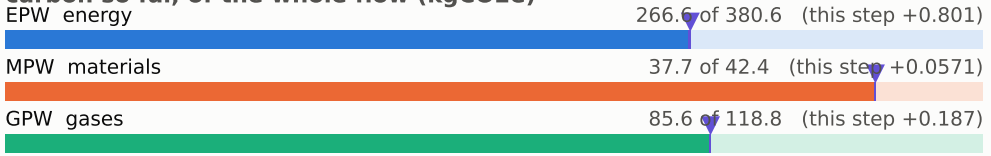
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

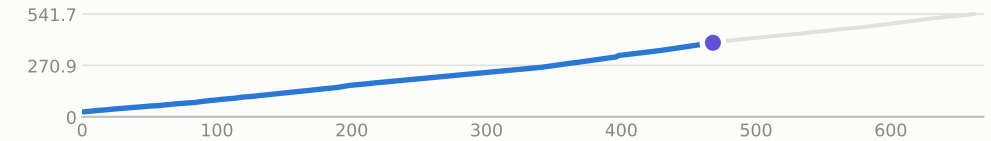
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

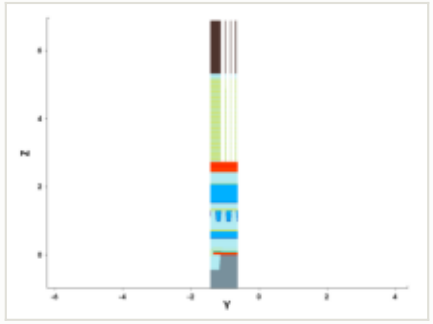
carbon so far, of the whole flow (kgCO2e)



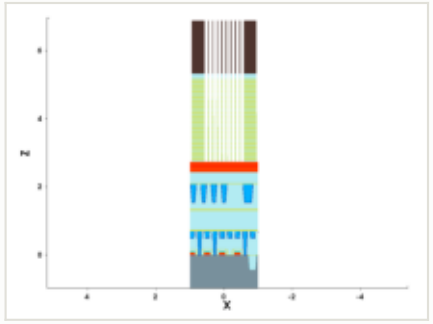
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



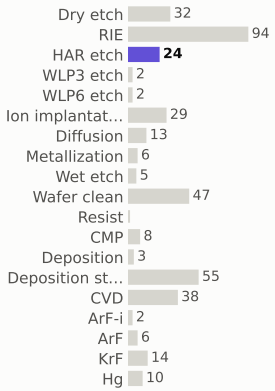
y cut at -1.032 µm



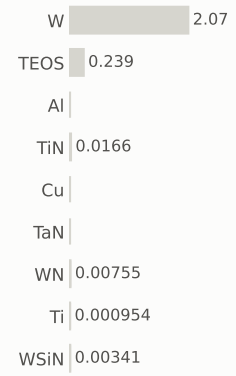
mask for this step

no mask for this step

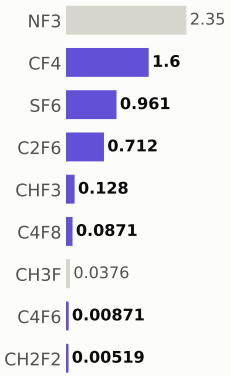
steps by process area



materials deposited (g)



gases emitted (g)



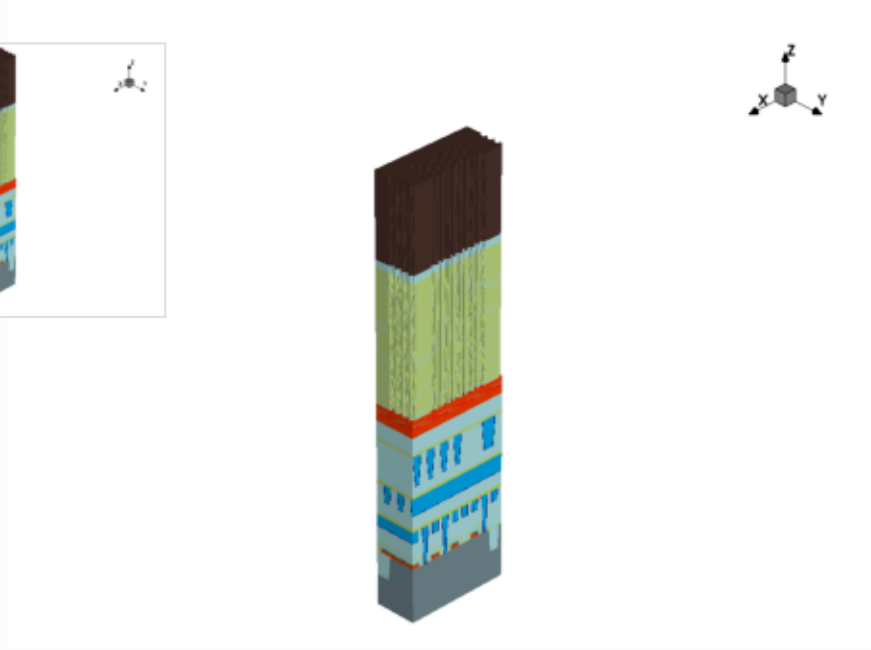
Step #4348.7: HAR_etch

etch_noop · HAR_etch

1.04	391.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



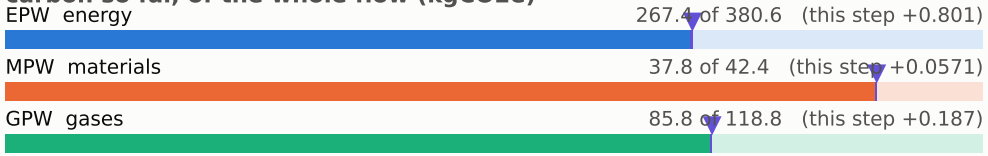
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

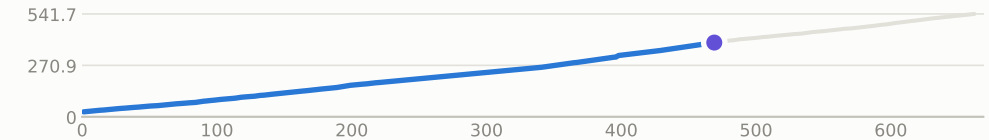
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

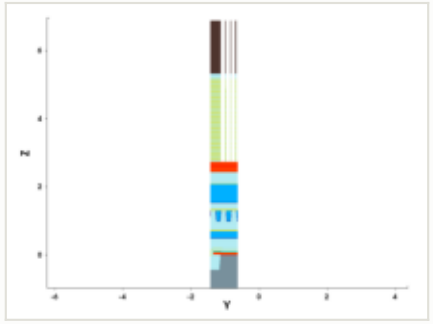
carbon so far, of the whole flow (kgCO2e)



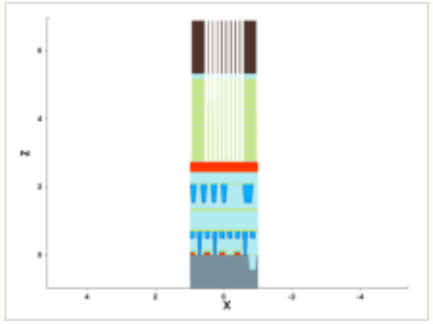
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



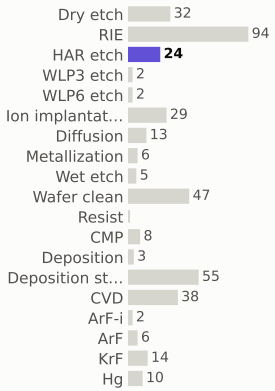
y cut at -1.032 µm



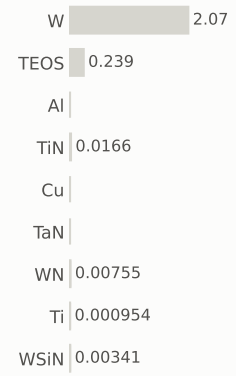
mask for this step

no mask for this step

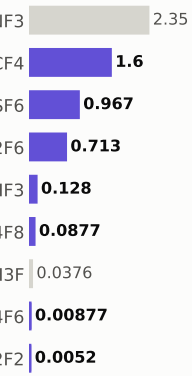
steps by process area



materials deposited (g)



gases emitted (g)



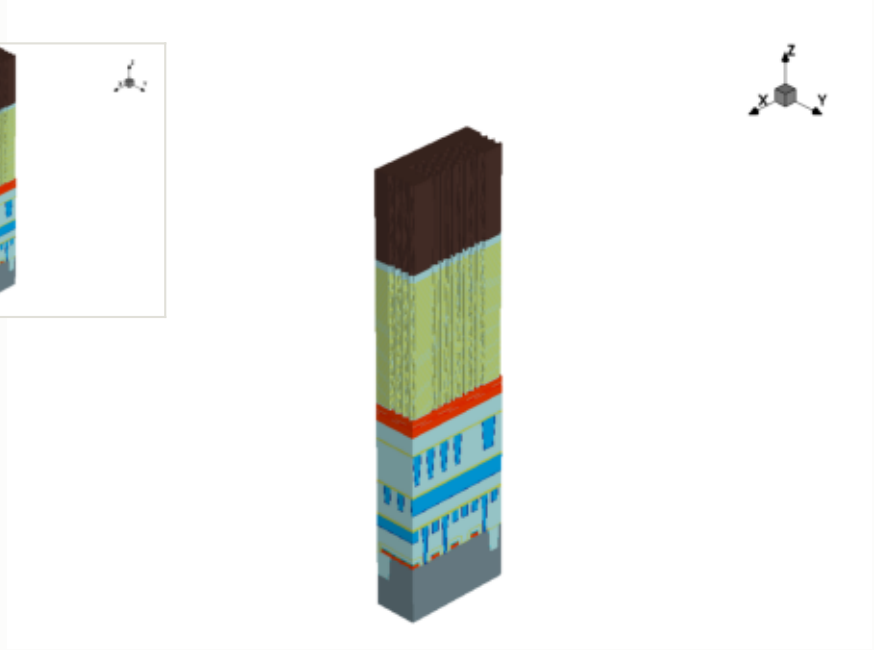
Step #4348.89: HAR_etch

etch_noop · HAR_etch

1.04	392.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



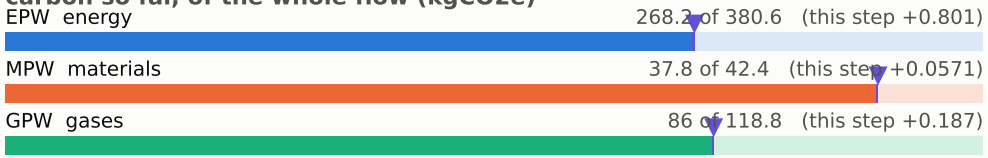
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

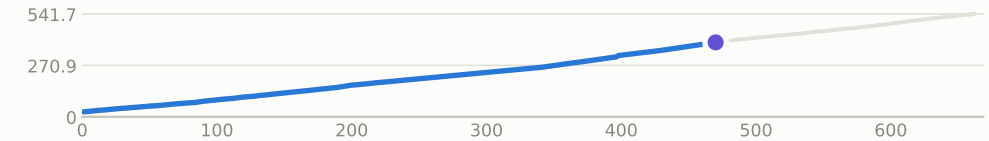
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

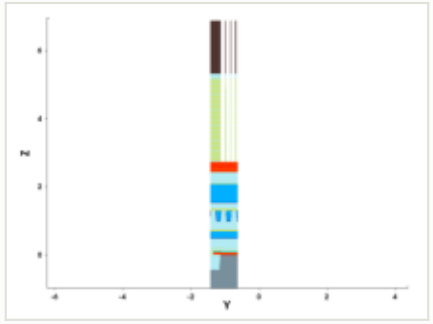
carbon so far, of the whole flow (kgCO2e)



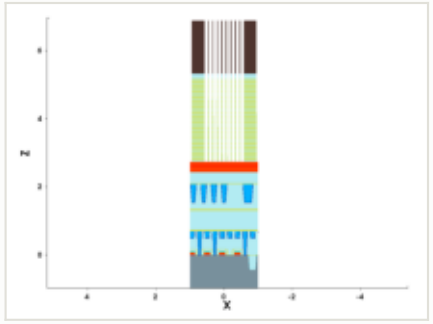
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



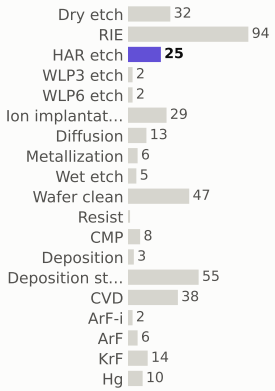
y cut at -1.032 µm



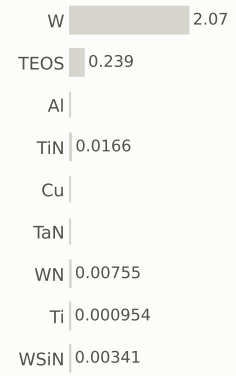
mask for this step

no mask for this step

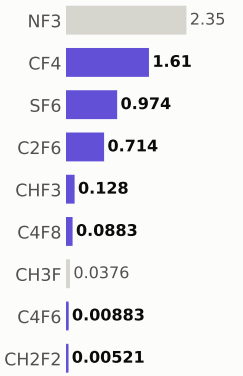
steps by process area



materials deposited (g)



gases emitted (g)



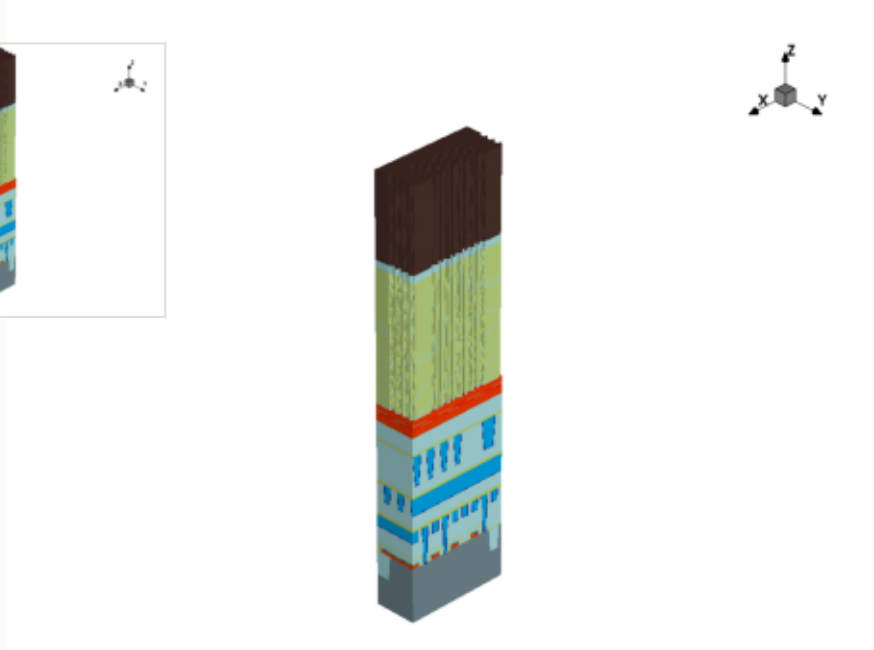
Step #4349.07: HAR_etch

etch_noop · HAR_etch

1.04	393.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



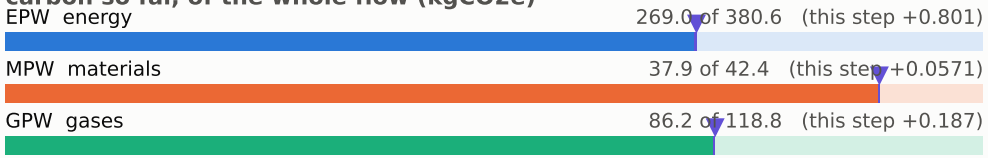
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

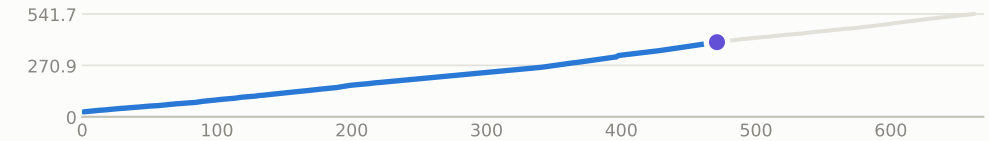
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

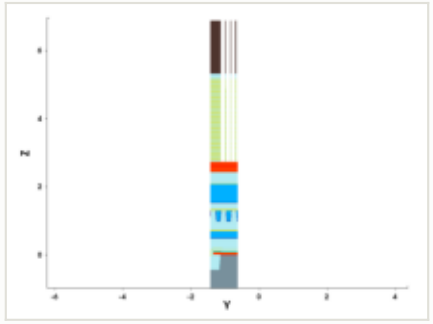
carbon so far, of the whole flow (kgCO2e)



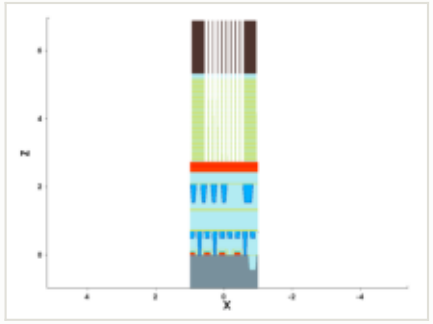
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



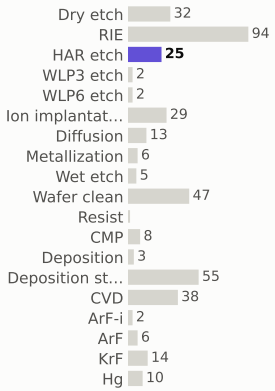
y cut at -1.032 µm



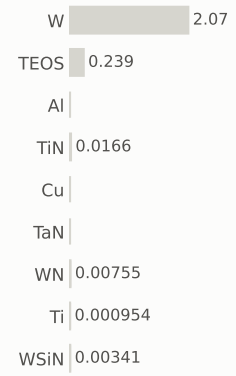
mask for this step

no mask for this step

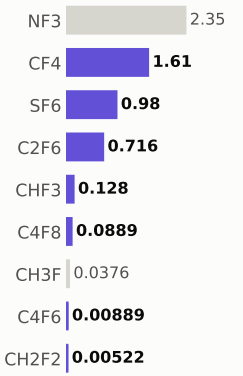
steps by process area



materials deposited (g)



gases emitted (g)



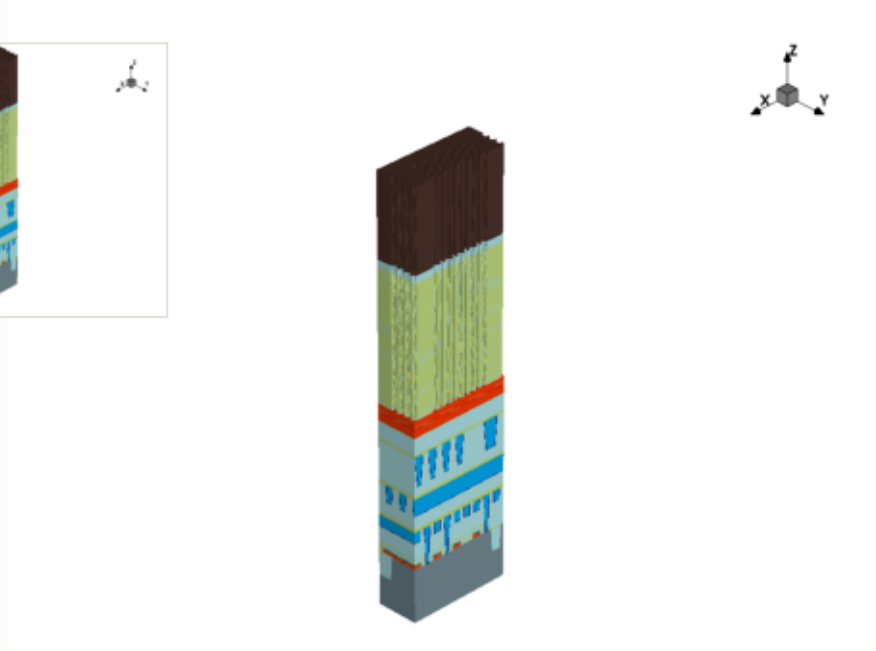
Step #4349.26: HAR_etch

etch_noop · HAR_etch

1.04	394.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



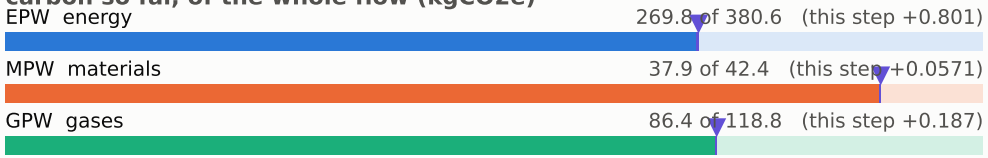
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

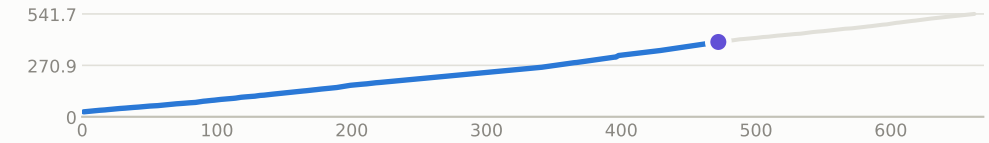
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

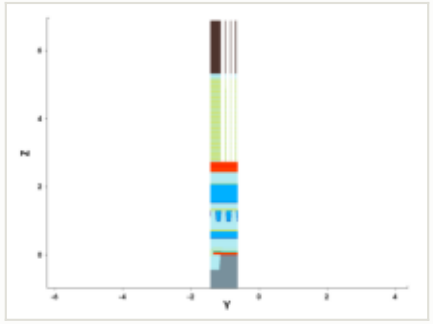
carbon so far, of the whole flow (kgCO2e)



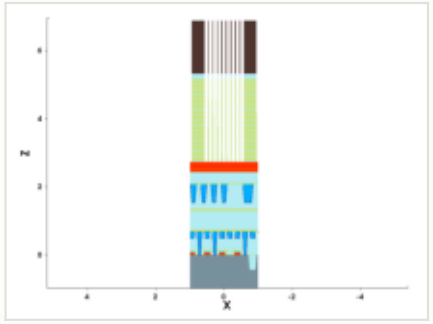
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



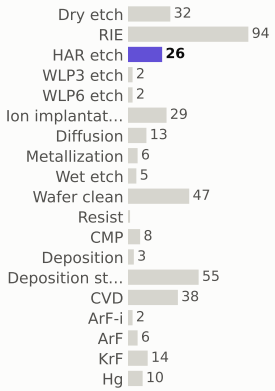
y cut at -1.032 µm



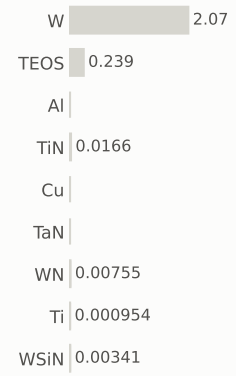
mask for this step

no mask for this step

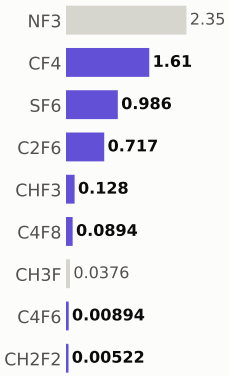
steps by process area



materials deposited (g)



gases emitted (g)



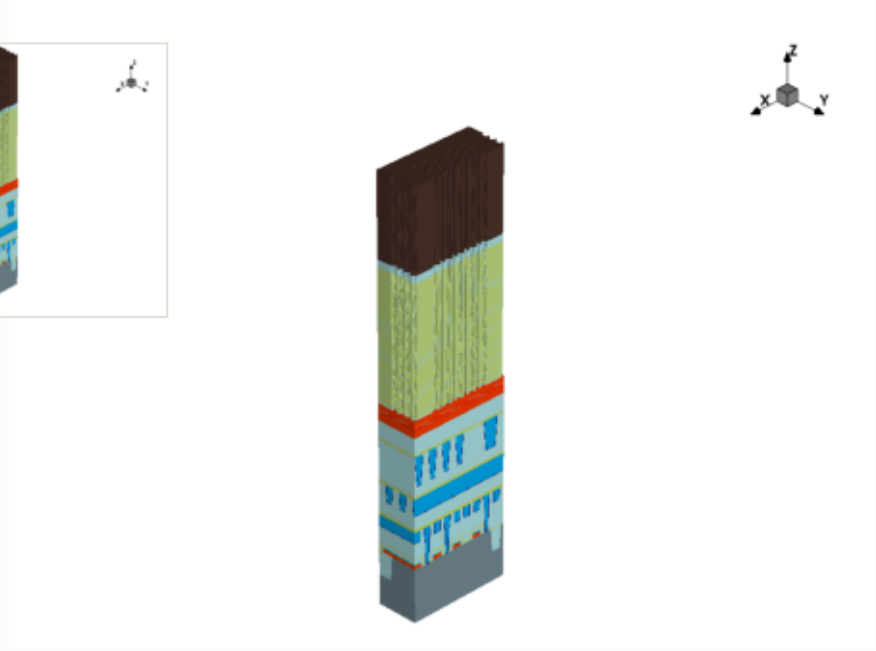
Step #4349.44: HAR_etch

etch_noop · HAR_etch

1.04	395.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



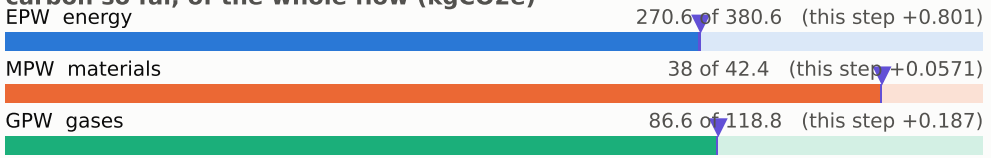
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

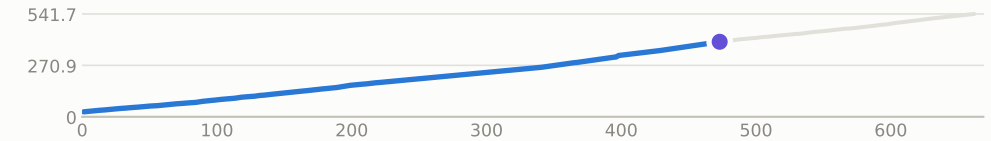
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

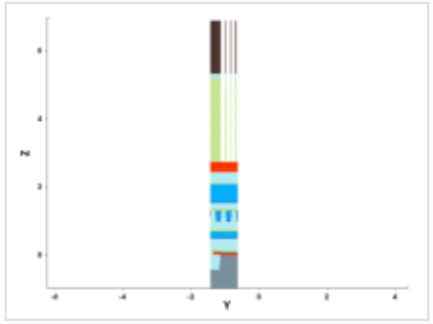
carbon so far, of the whole flow (kgCO2e)



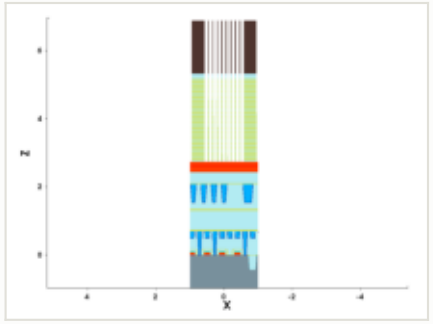
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



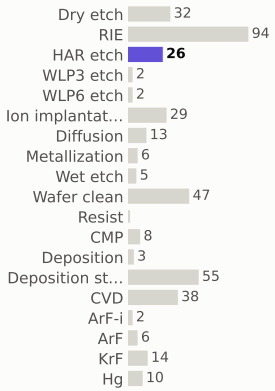
y cut at -1.032 µm



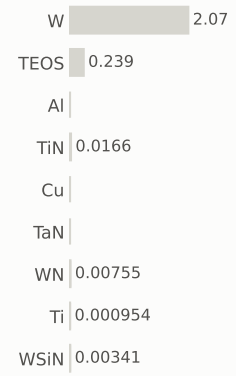
mask for this step

no mask for this step

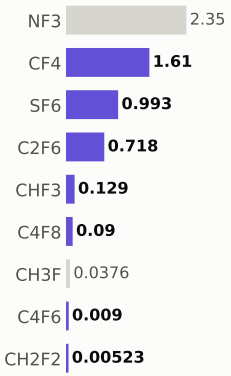
steps by process area



materials deposited (g)



gases emitted (g)



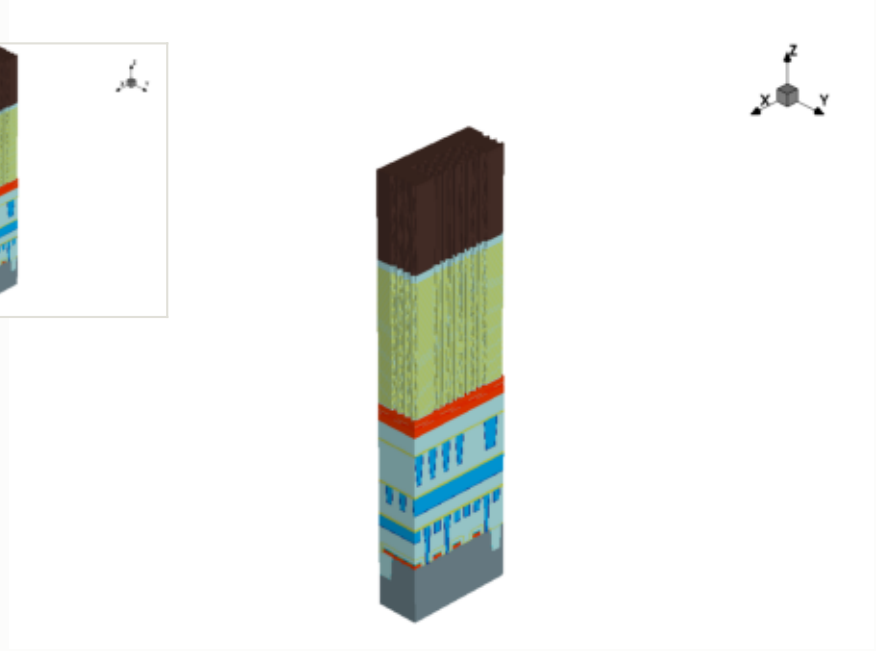
Step #4349.63: HAR_etch

etch_noop · HAR_etch

1.04	396.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



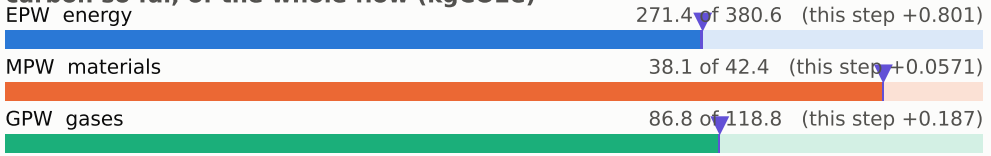
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

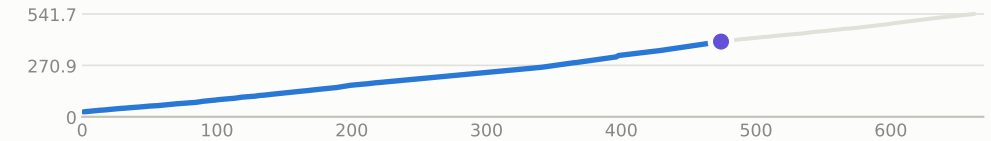
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

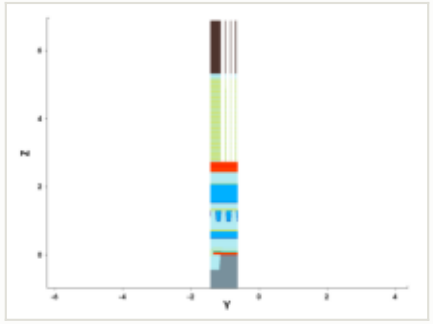
carbon so far, of the whole flow (kgCO2e)



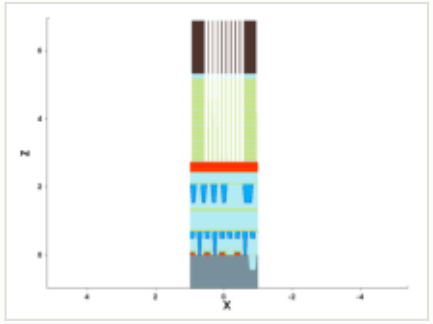
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



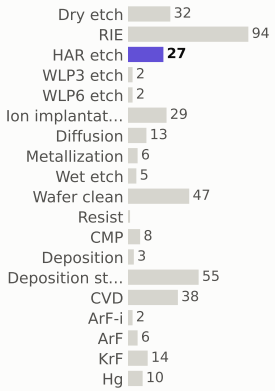
y cut at -1.032 µm



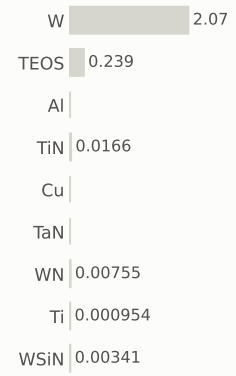
mask for this step

no mask for this step

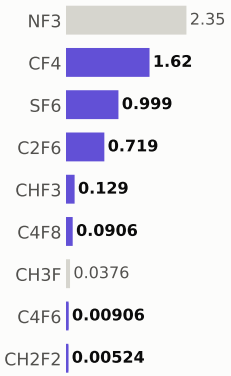
steps by process area



materials deposited (g)



gases emitted (g)



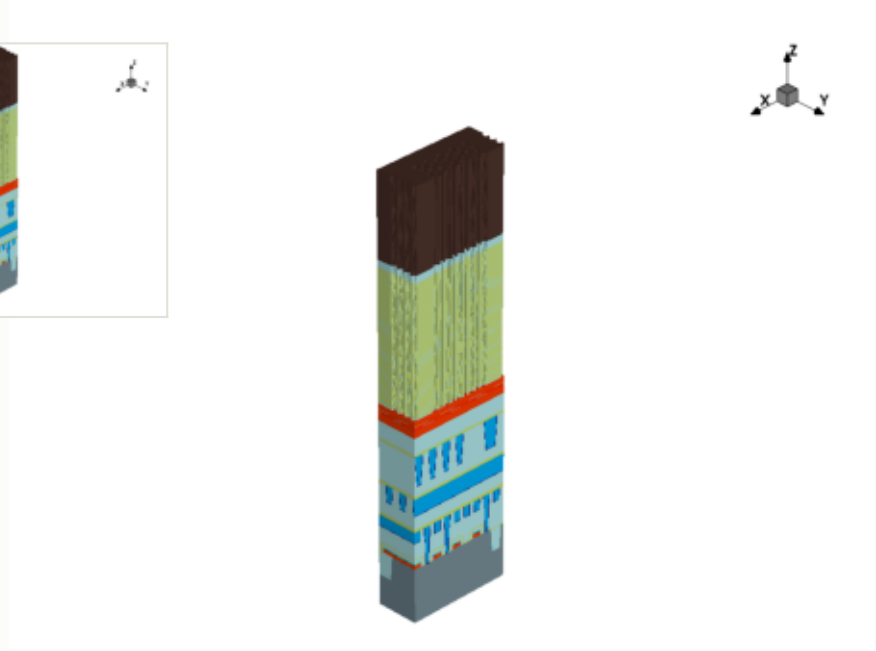
Step #4349.81: HAR_etch

etch_noop · HAR_etch

1.04	397.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



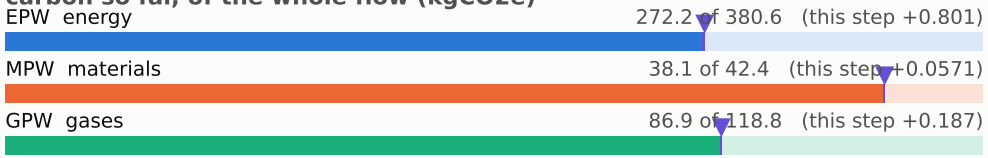
this step

process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

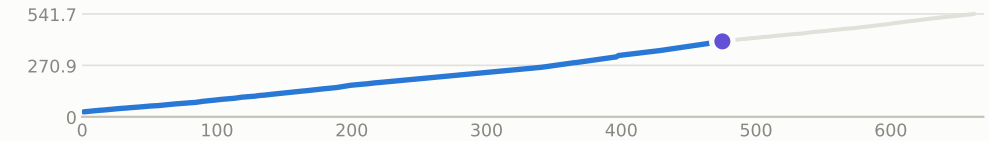
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

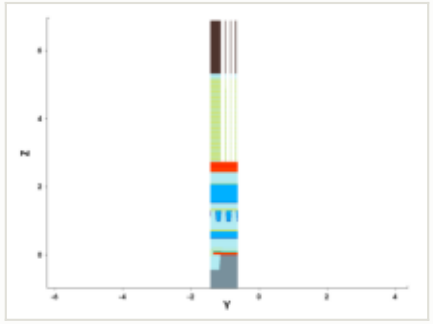
carbon so far, of the whole flow (kgCO2e)



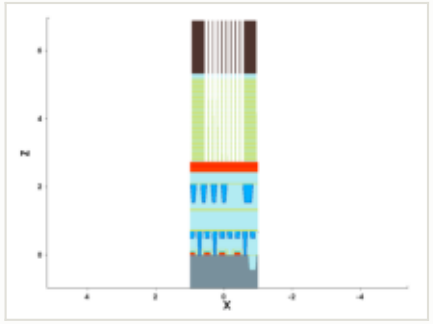
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



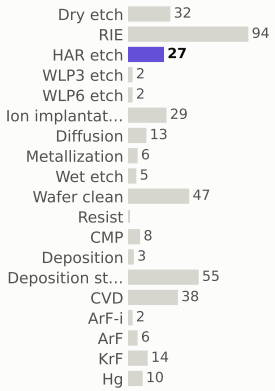
y cut at -1.032 µm



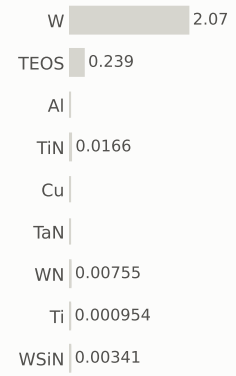
mask for this step

no mask for this step

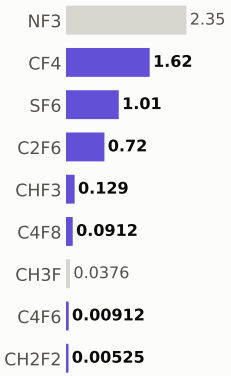
steps by process area



materials deposited (g)



gases emitted (g)



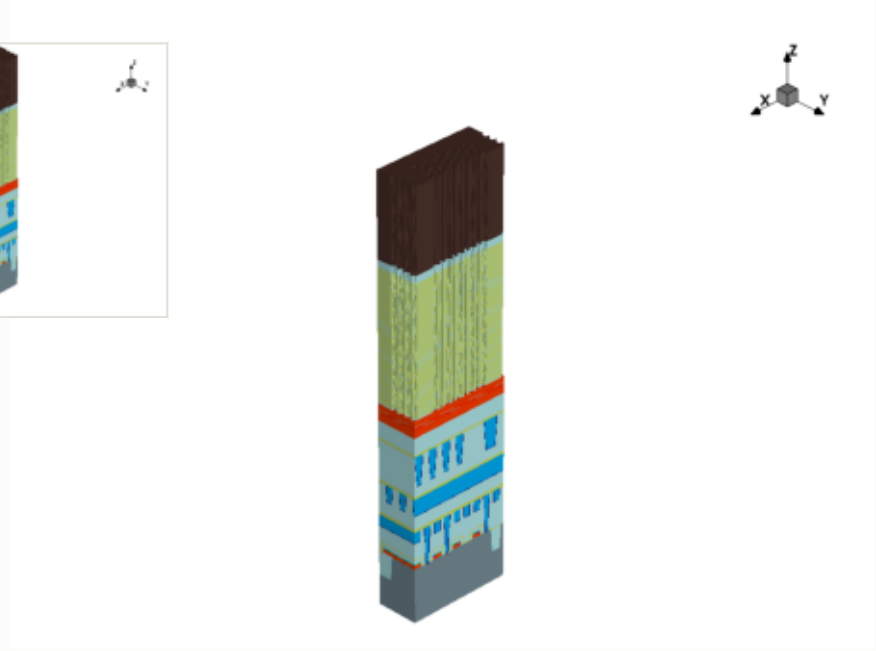
Step #4350: RIE

etch_anisotropic · RIE

1.04	398.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

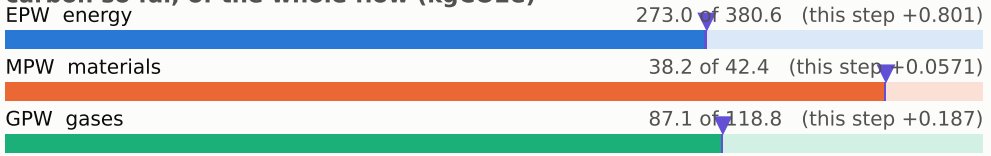


this step

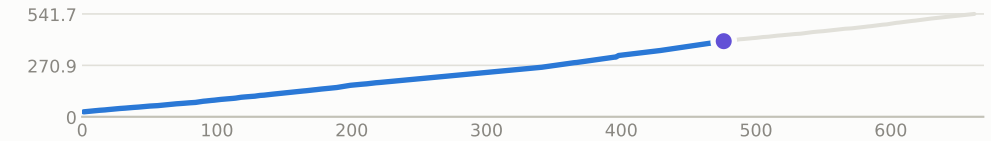
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

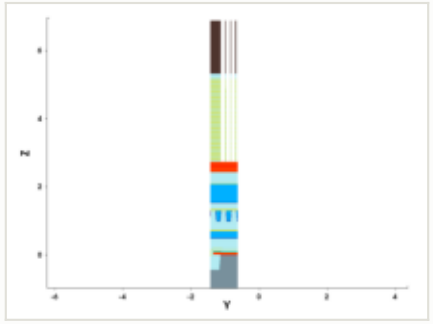
carbon so far, of the whole flow (kgCO2e)



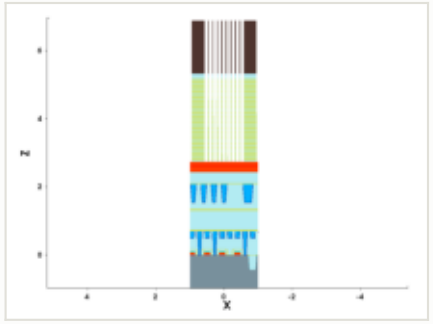
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



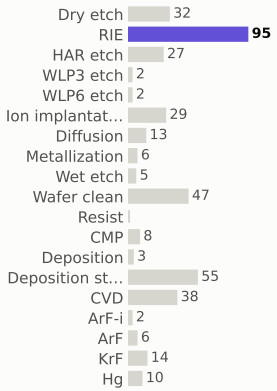
y cut at -1.032 µm



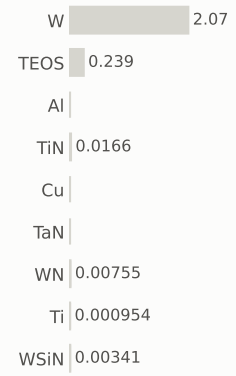
mask for this step

no mask for this step

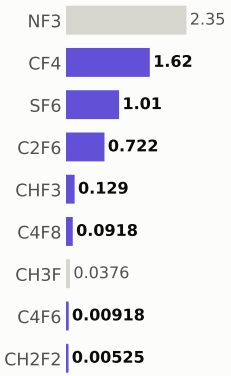
steps by process area



materials deposited (g)



gases emitted (g)



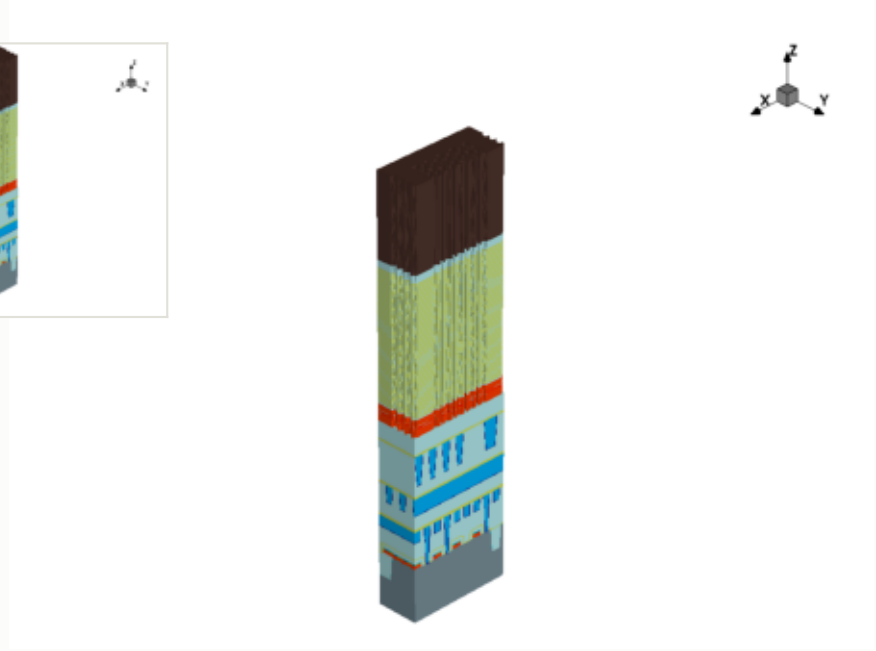
Step #4360: RIE

etch_anisotropic · RIE

1.04	399.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

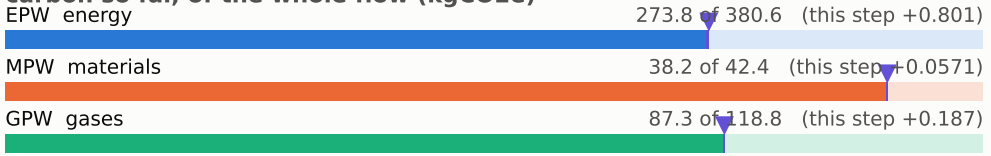


this step

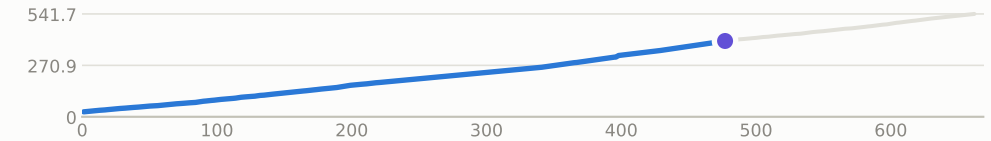
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

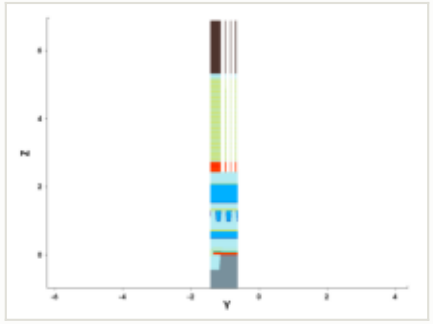
carbon so far, of the whole flow (kgCO2e)



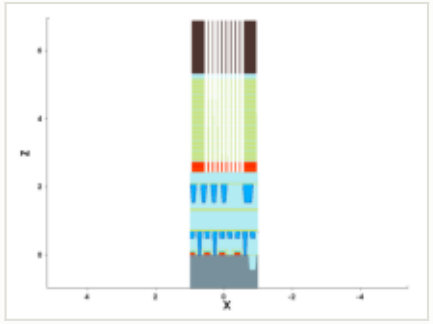
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



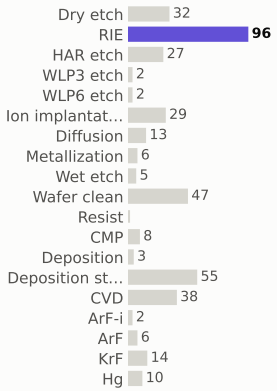
y cut at -1.032 µm



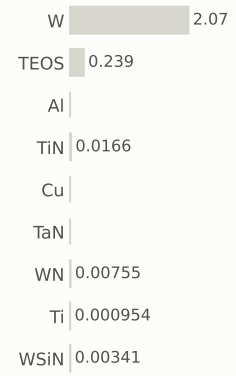
mask for this step

no mask for this step

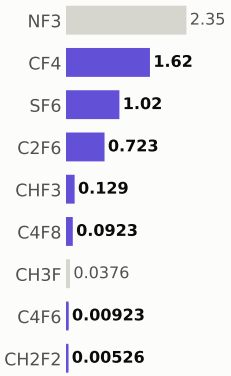
steps by process area



materials deposited (g)



gases emitted (g)



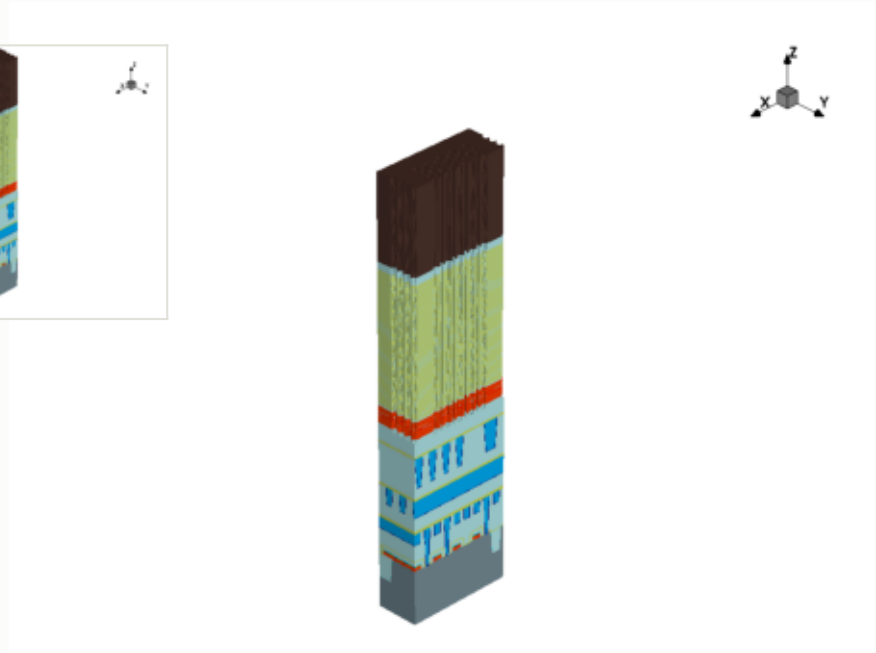
Step #4370: RIE

etch_anisotropic · RIE

1.04	400.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

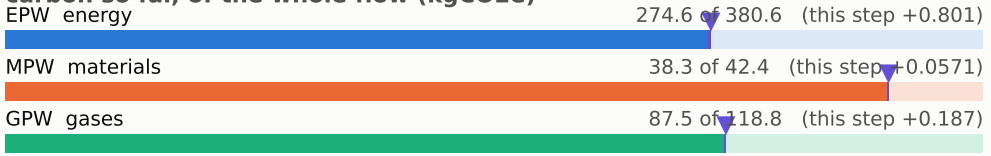


this step

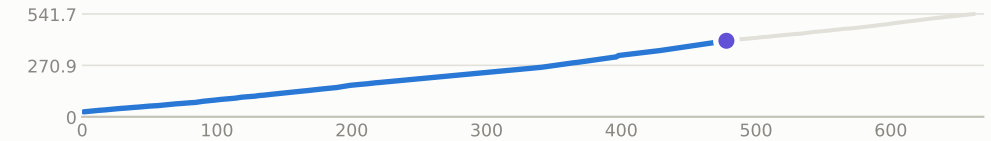
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

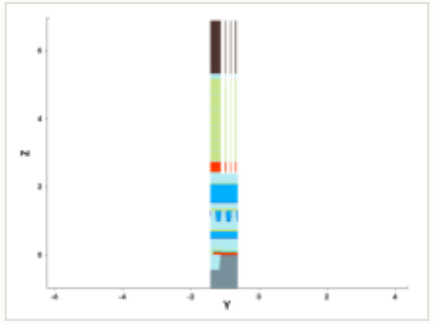
carbon so far, of the whole flow (kgCO2e)



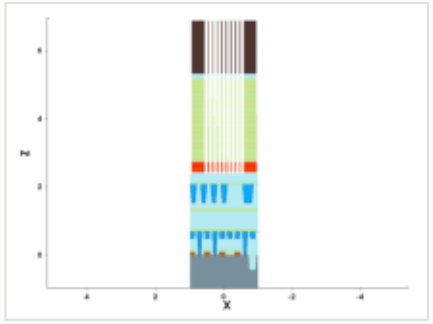
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



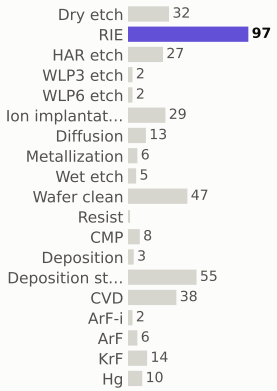
y cut at -1.032 µm



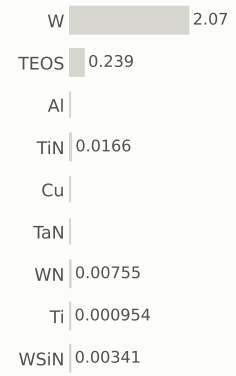
mask for this step

no mask for this step

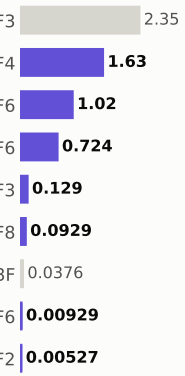
steps by process area



materials deposited (g)



gases emitted (g)



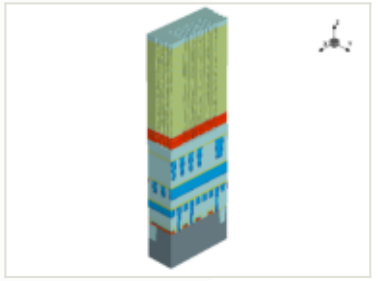
Step #4380: Dry_etch

etch_strip · Dry_etch

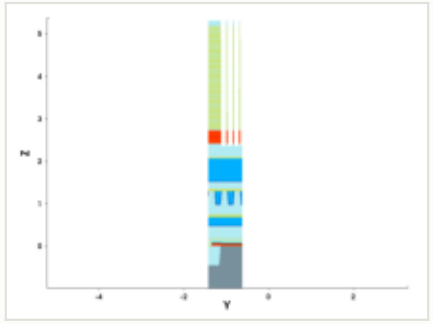
0.801	401.2	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

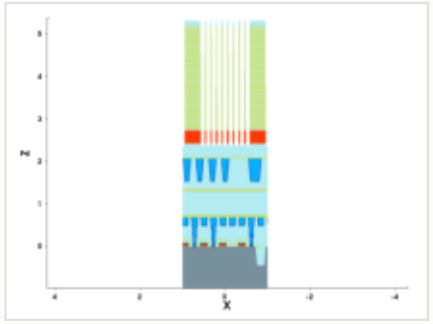
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



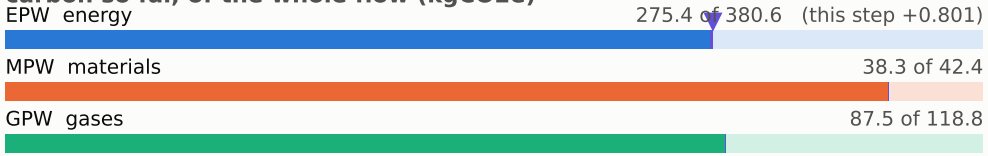
mask for this step

no mask for this step

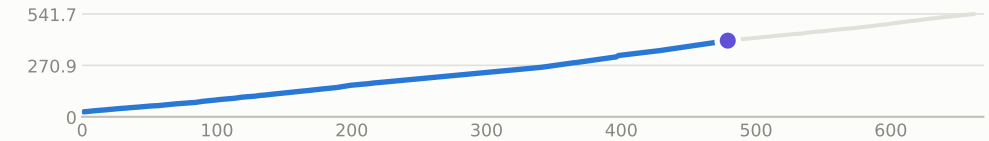
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

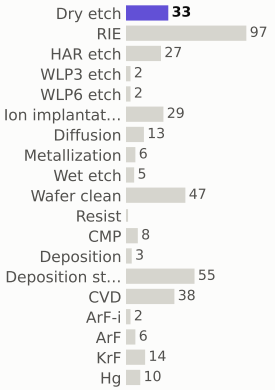
carbon so far, of the whole flow (kgCO2e)



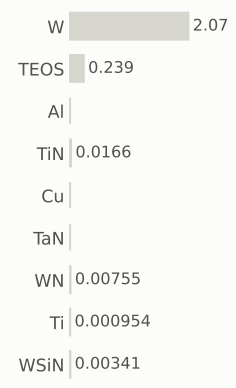
embodied carbon up to each step (kgCO2e per wafer)



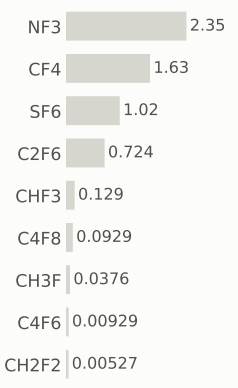
steps by process area



materials deposited (g)



gases emitted (g)



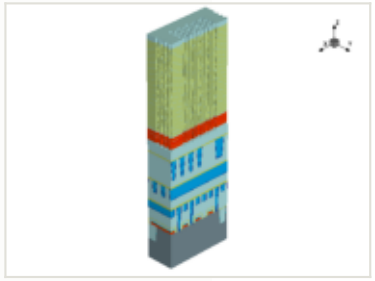
Step #4390: Wafer_clean

clean · Wafer_clean

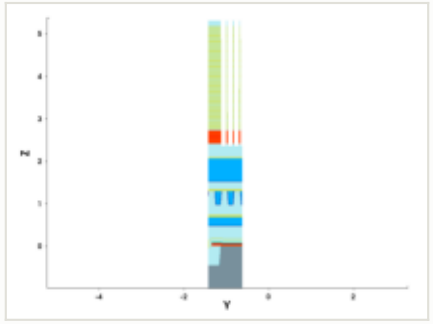
0.169	401.3	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

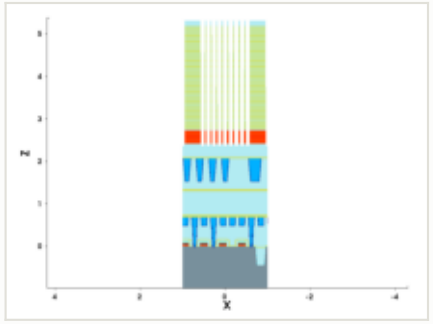
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

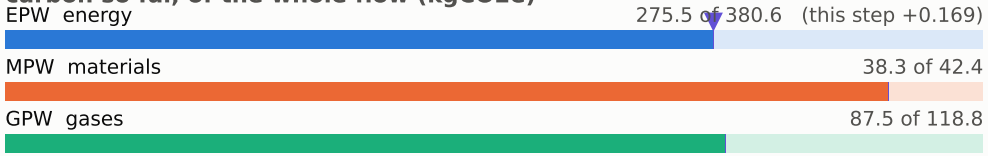
no mask for this step

this step

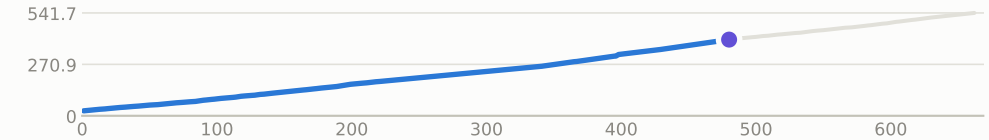
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

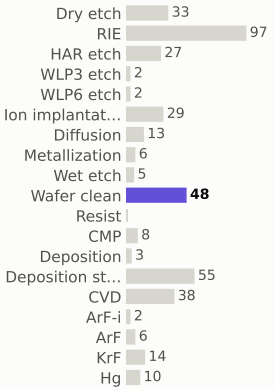
carbon so far, of the whole flow (kgCO2e)



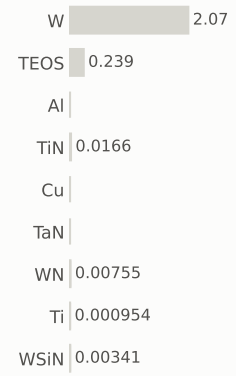
embodied carbon up to each step (kgCO2e per wafer)



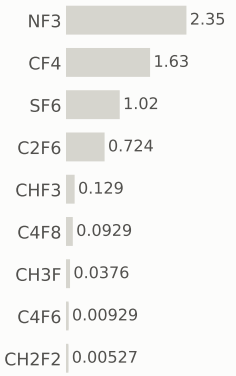
steps by process area



materials deposited (g)



gases emitted (g)



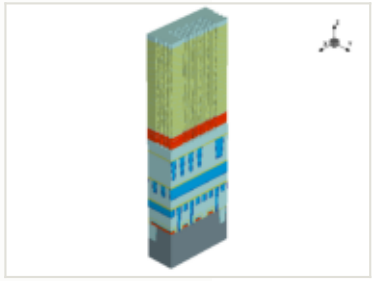
Step #4400: Wet_etch

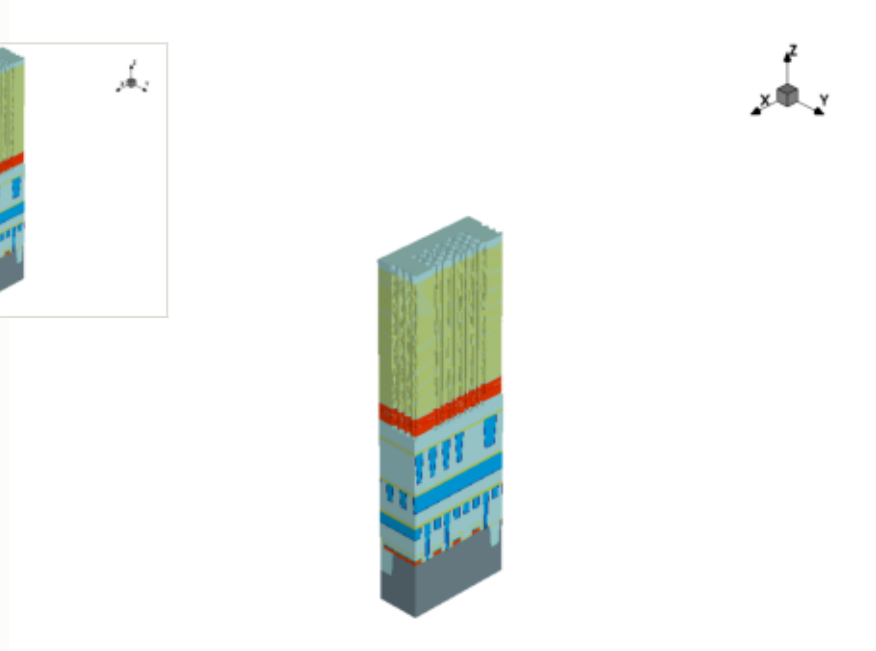
etch_isotropic · Wet_etch

0.169	401.5	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

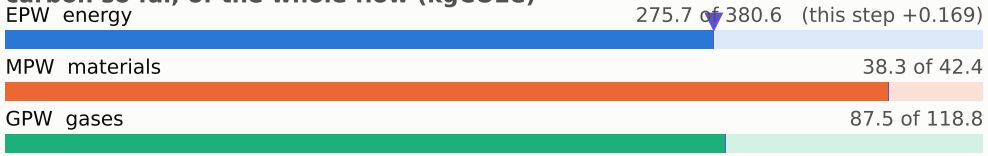




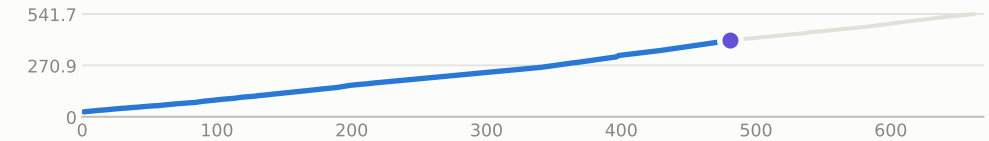
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

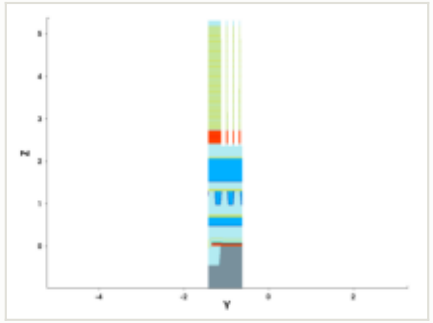
carbon so far, of the whole flow (kgCO2e)



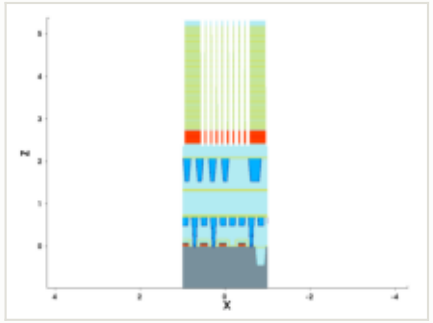
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



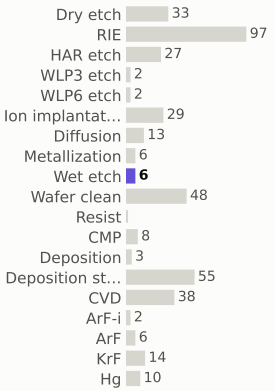
y cut at -1.032 µm



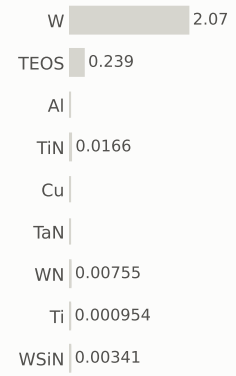
mask for this step

no mask for this step

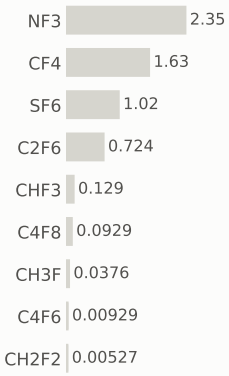
steps by process area



materials deposited (g)



gases emitted (g)



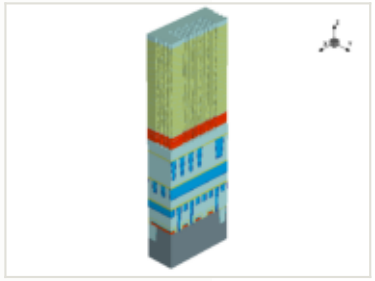
Step #4410: Wafer_clean

clean · Wafer_clean

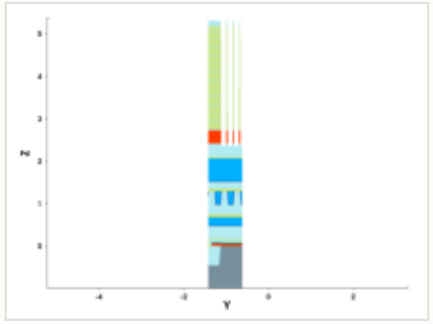
0.169	401.7	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

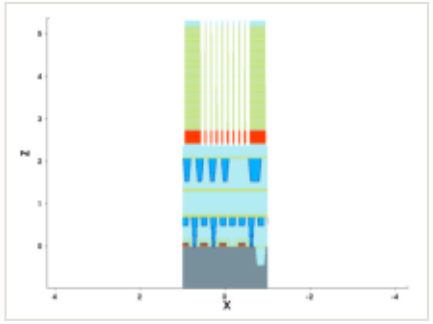
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

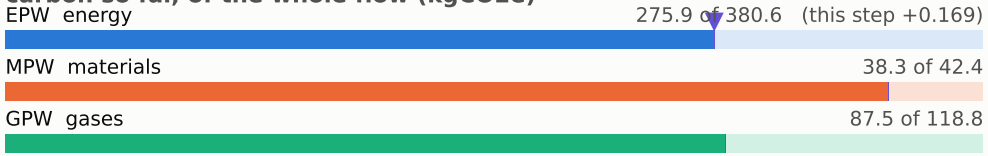
no mask for this step

this step

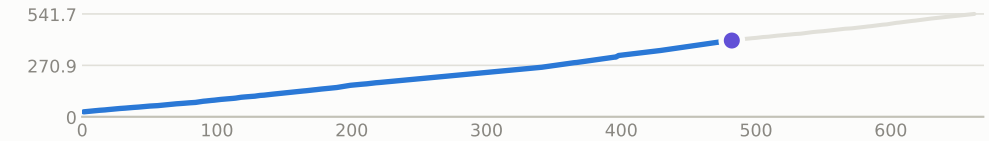
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

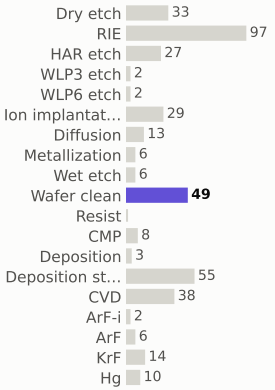
carbon so far, of the whole flow (kgCO2e)



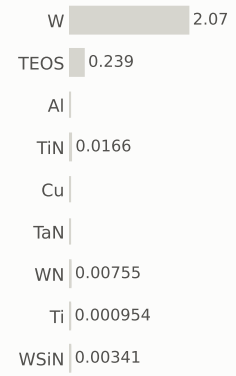
embodied carbon up to each step (kgCO2e per wafer)



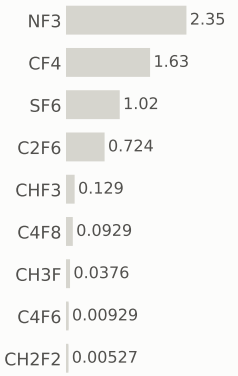
steps by process area



materials deposited (g)



gases emitted (g)



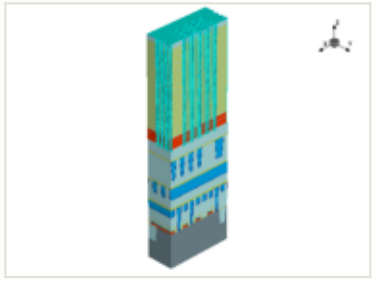
Step #4420: CVD

depo_isotropic · CVD

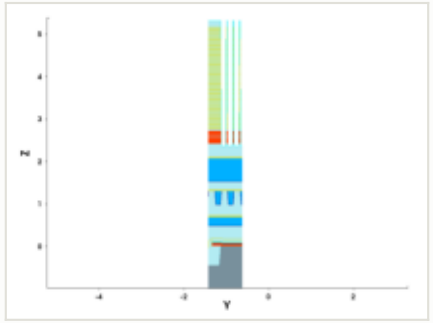
1.31	403.0	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

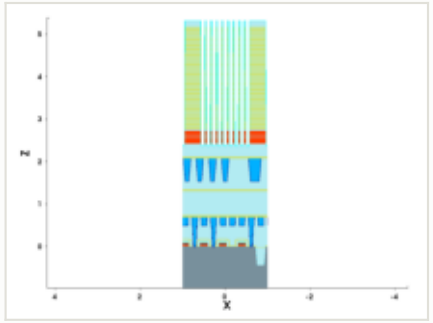
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

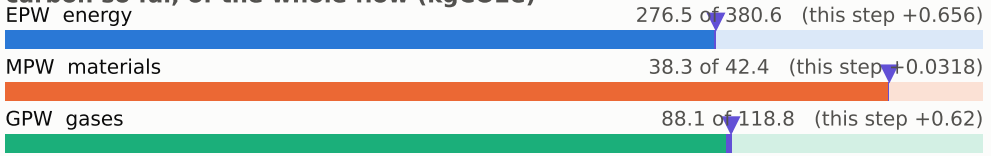
no mask for this step

this step

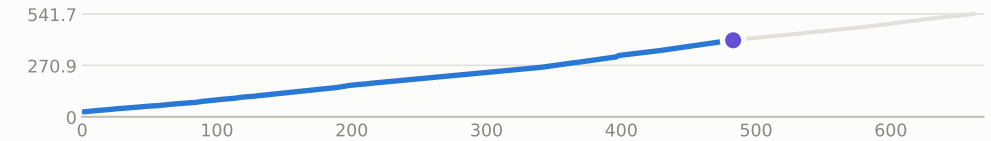
process area	CVD
material	—
thickness	0.008 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

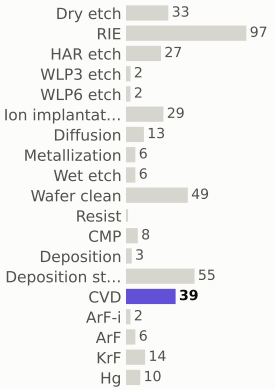
carbon so far, of the whole flow (kgCO2e)



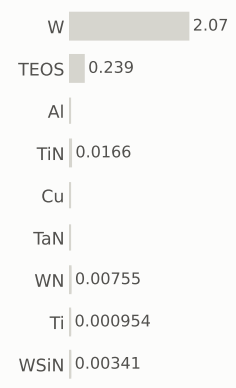
embodied carbon up to each step (kgCO2e per wafer)



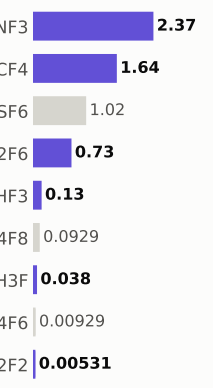
steps by process area



materials deposited (g)



gases emitted (g)



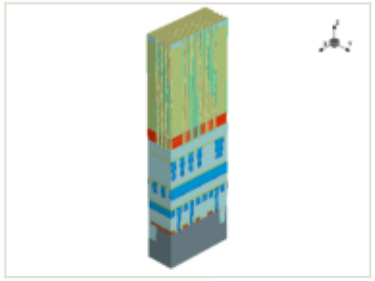
Step #4430: CVD

depo_isotropic · CVD

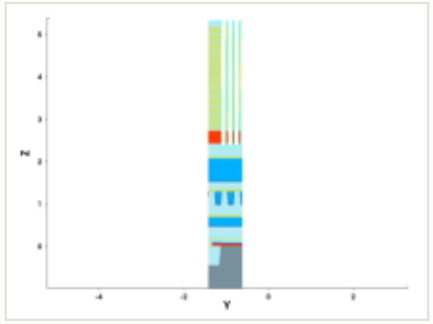
1.31	404.3	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

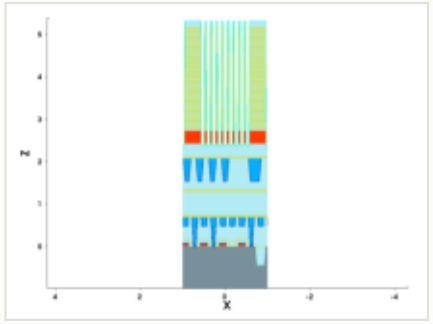
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

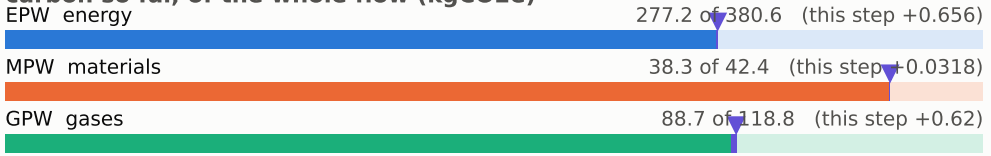
no mask for this step

this step

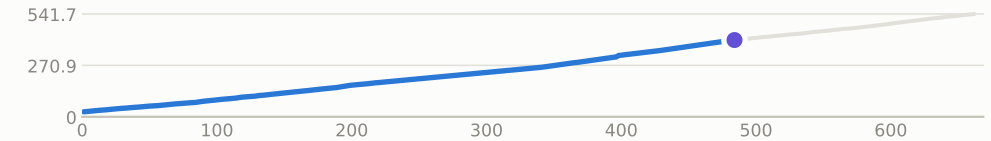
process area	CVD
material	—
thickness	0.006 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

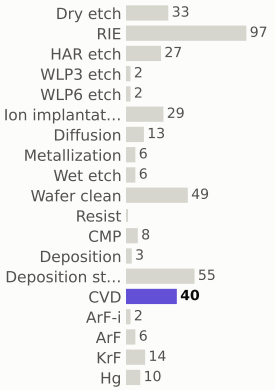
carbon so far, of the whole flow (kgCO2e)



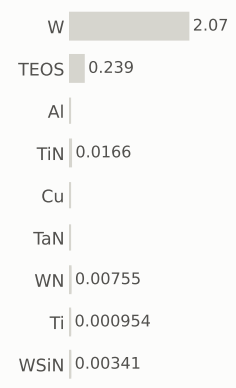
embodied carbon up to each step (kgCO2e per wafer)



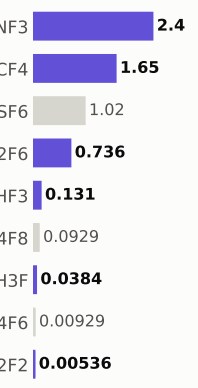
steps by process area



materials deposited (g)



gases emitted (g)



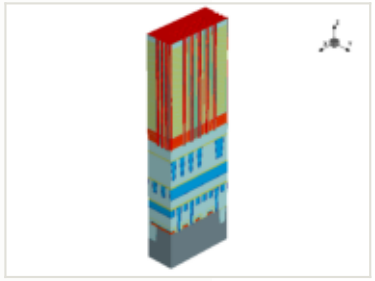
Step #4440: CVD

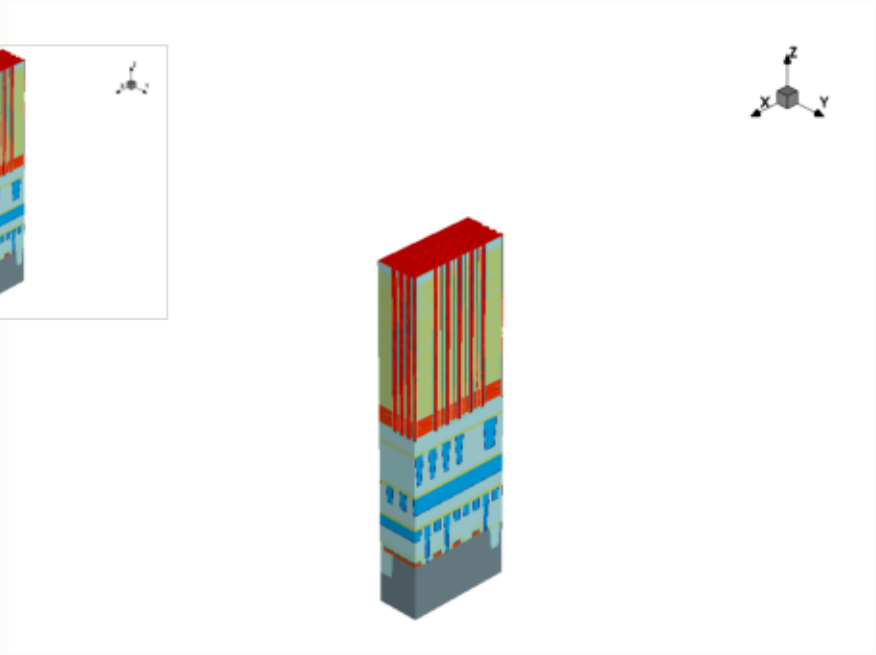
depo_isotropic · CVD

1.31	405.6	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

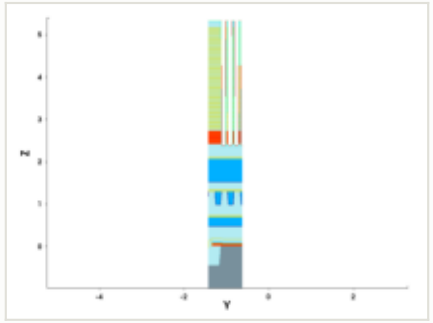
3D structure after this step

zoomed to fit

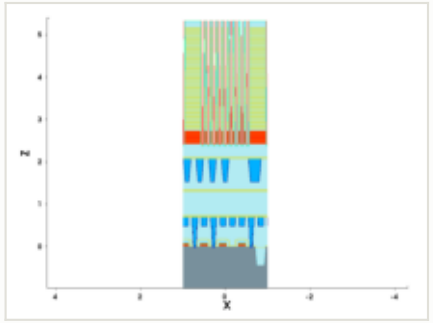




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

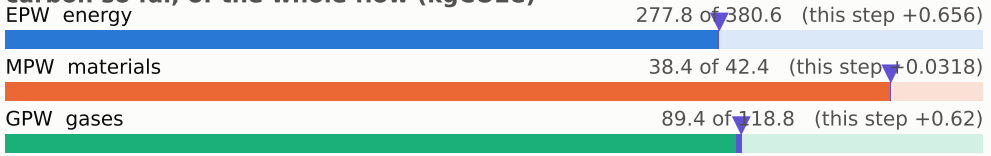
no mask for this step

this step

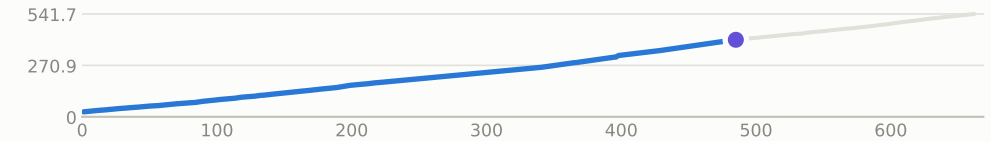
process area	CVD
material	—
thickness	0.004 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

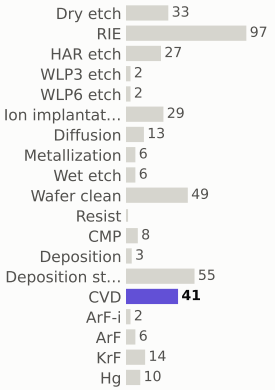
carbon so far, of the whole flow (kgCO2e)



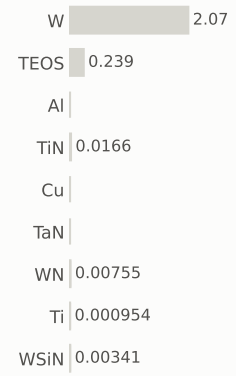
embodied carbon up to each step (kgCO2e per wafer)



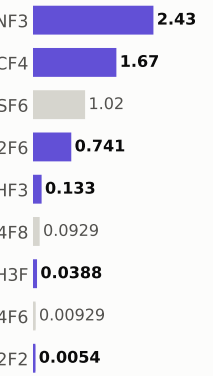
steps by process area



materials deposited (g)



gases emitted (g)



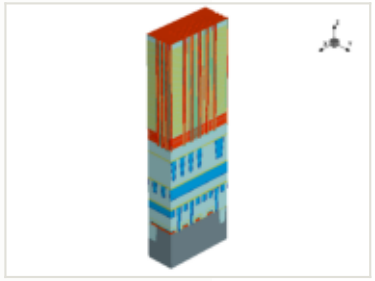
Step #4450: CVD

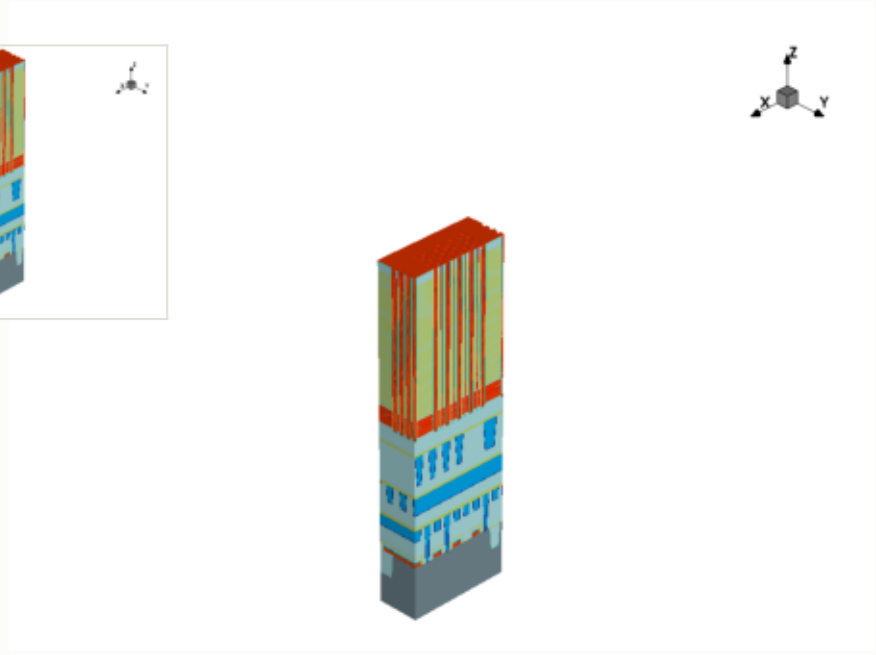
depo_isotropic · CVD

1.31	406.9	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

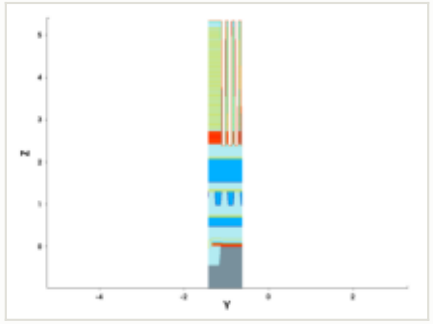
3D structure after this step

zoomed to fit

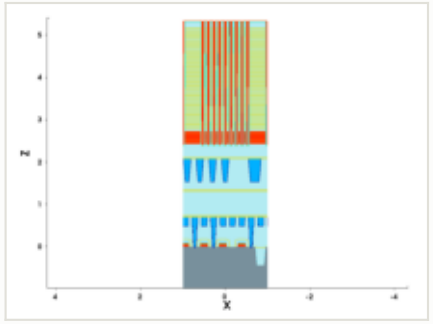




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

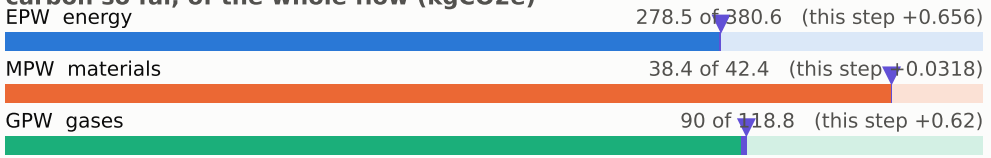
no mask for this step

this step

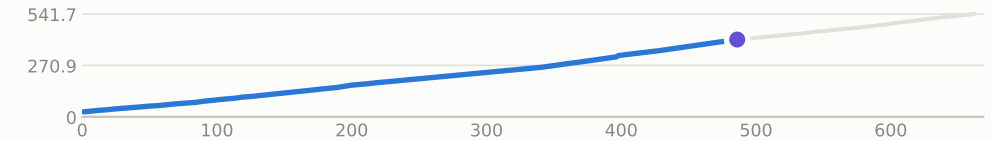
process area	CVD
material	—
thickness	0.008 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

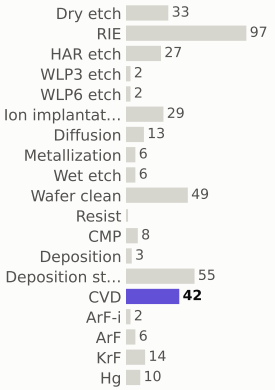
carbon so far, of the whole flow (kgCO2e)



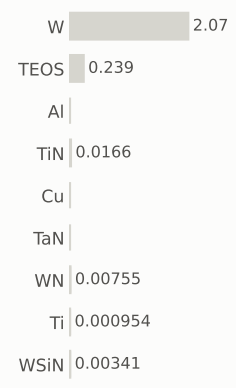
embodied carbon up to each step (kgCO2e per wafer)



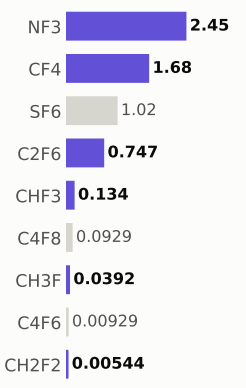
steps by process area



materials deposited (g)



gases emitted (g)



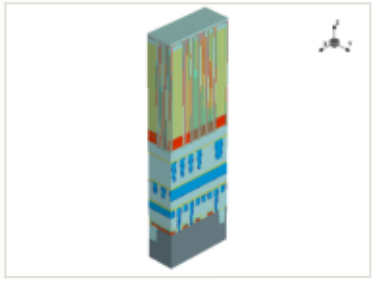
Step #4460: CVD

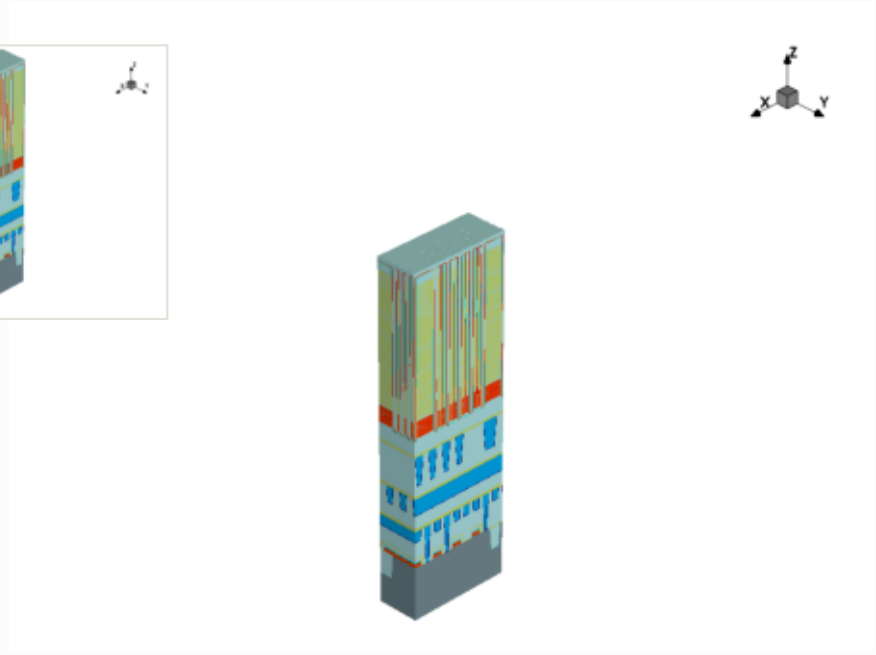
depo_isotropic · CVD

1.31	408.2	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

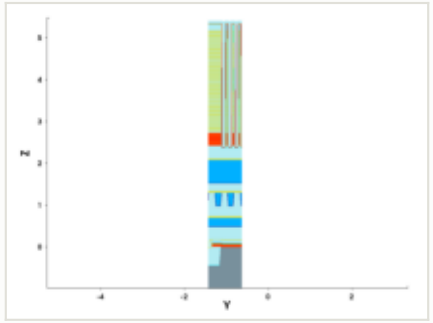
3D structure after this step

zoomed to fit

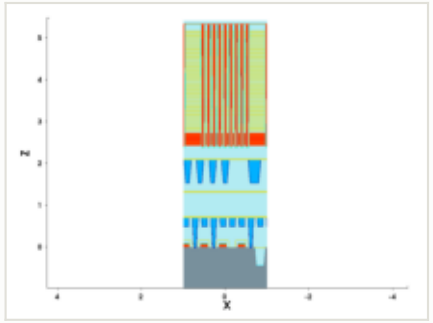




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

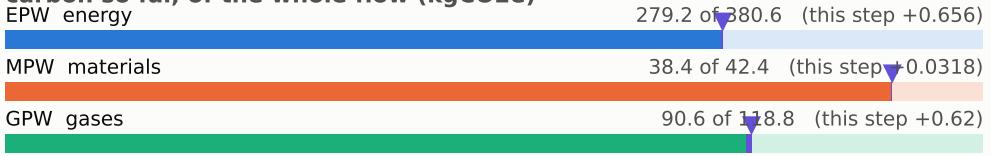
no mask for this step

this step

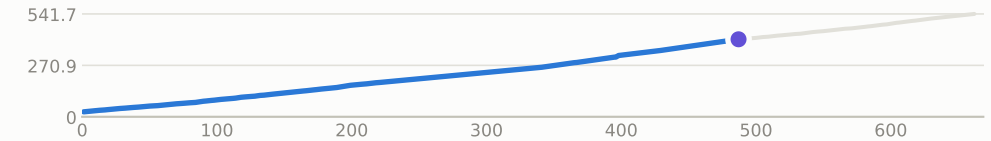
process area	CVD
material	—
thickness	0.07 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

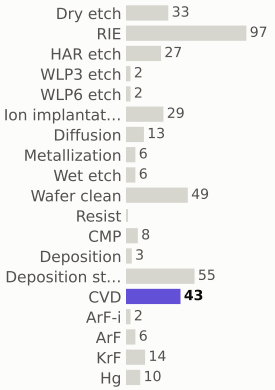
carbon so far, of the whole flow (kgCO2e)



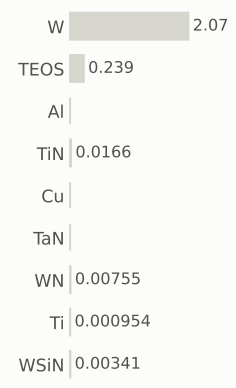
embodied carbon up to each step (kgCO2e per wafer)



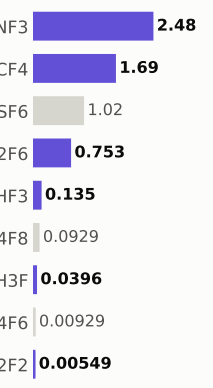
steps by process area



materials deposited (g)



gases emitted (g)



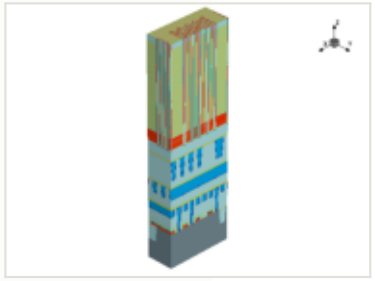
Step #4470: CMP

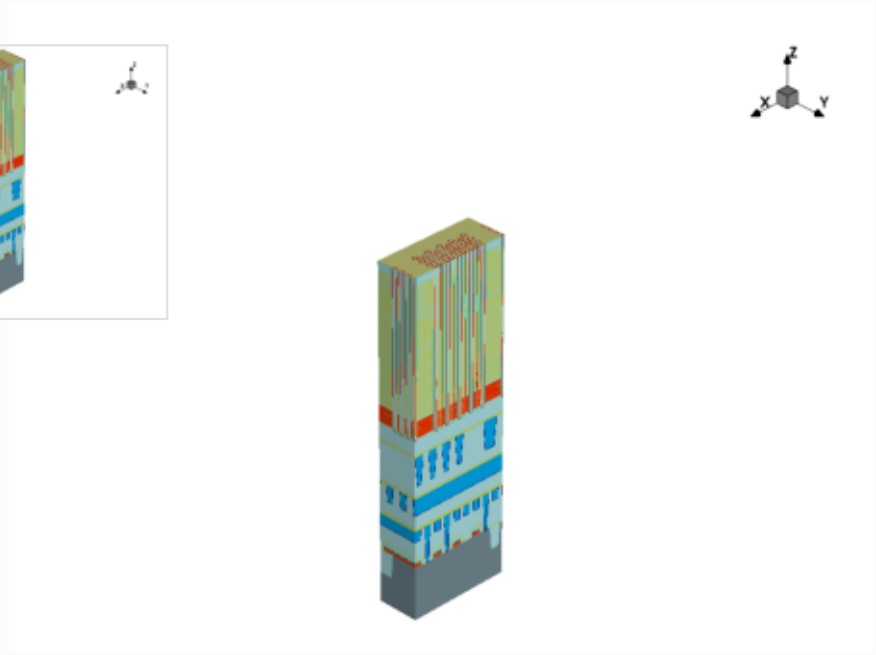
etch_cmp · CMP

0.169	408.4	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

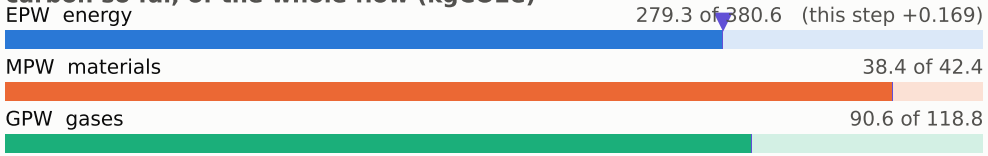




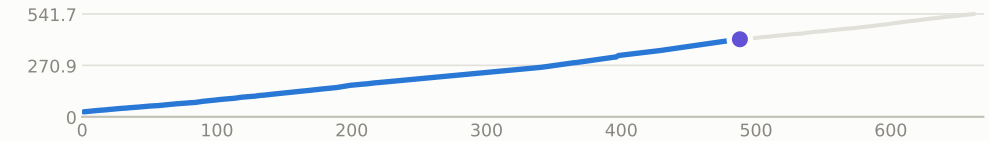
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

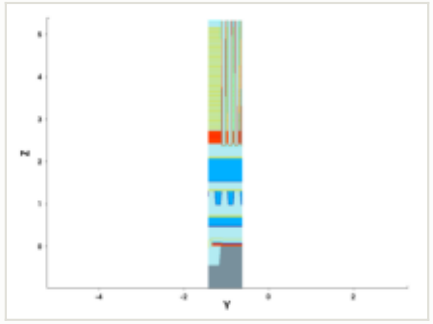
carbon so far, of the whole flow (kgCO2e)



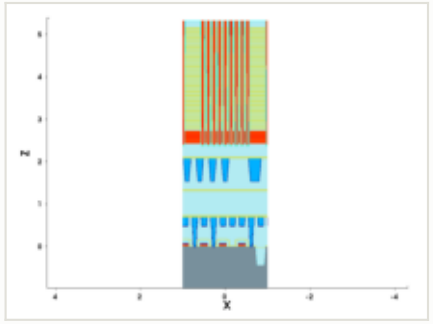
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



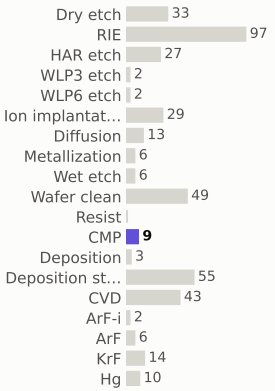
y cut at -1.032 µm



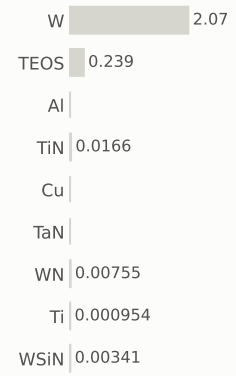
mask for this step

no mask for this step

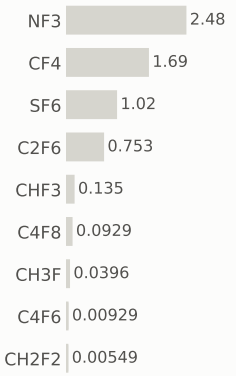
steps by process area



materials deposited (g)



gases emitted (g)



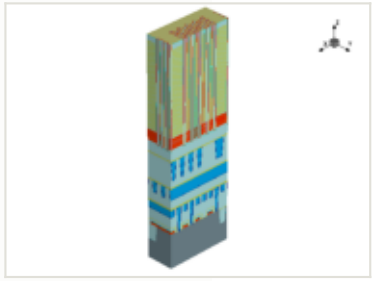
Step #4480: Wafer_clean

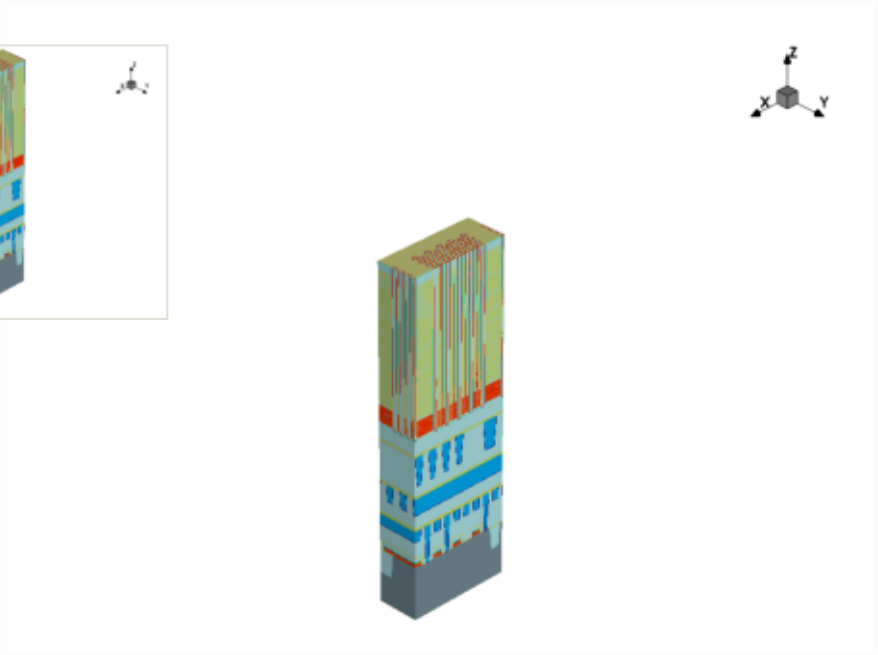
clean · Wafer_clean

0.169	408.5	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

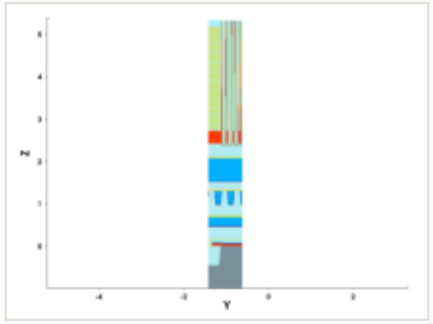
3D structure after this step

zoomed to fit

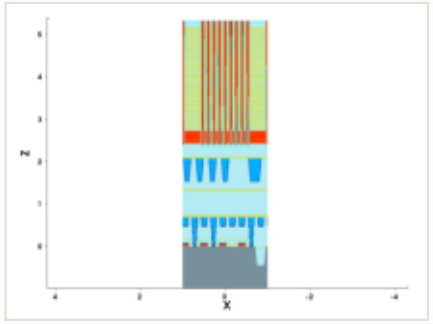




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

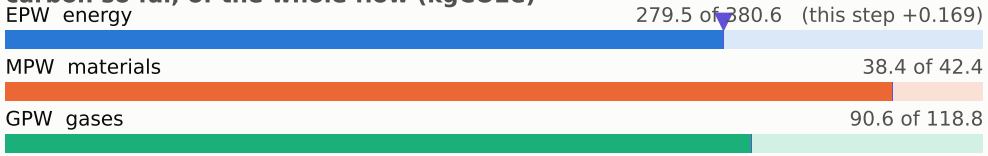
no mask for this step

this step

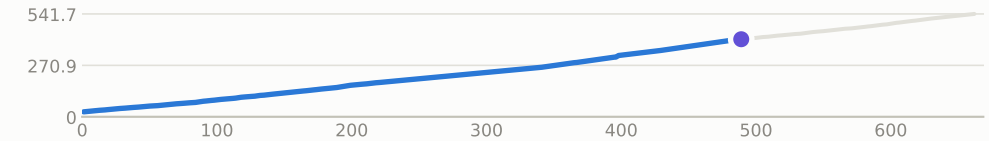
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

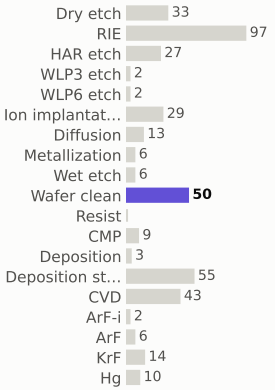
carbon so far, of the whole flow (kgCO2e)



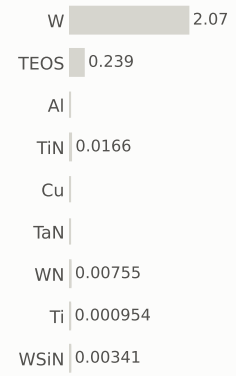
embodied carbon up to each step (kgCO2e per wafer)



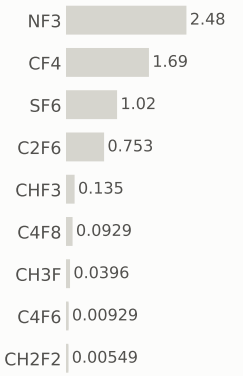
steps by process area



materials deposited (g)



gases emitted (g)



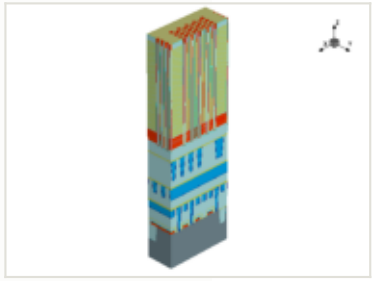
Step #4490: RIE

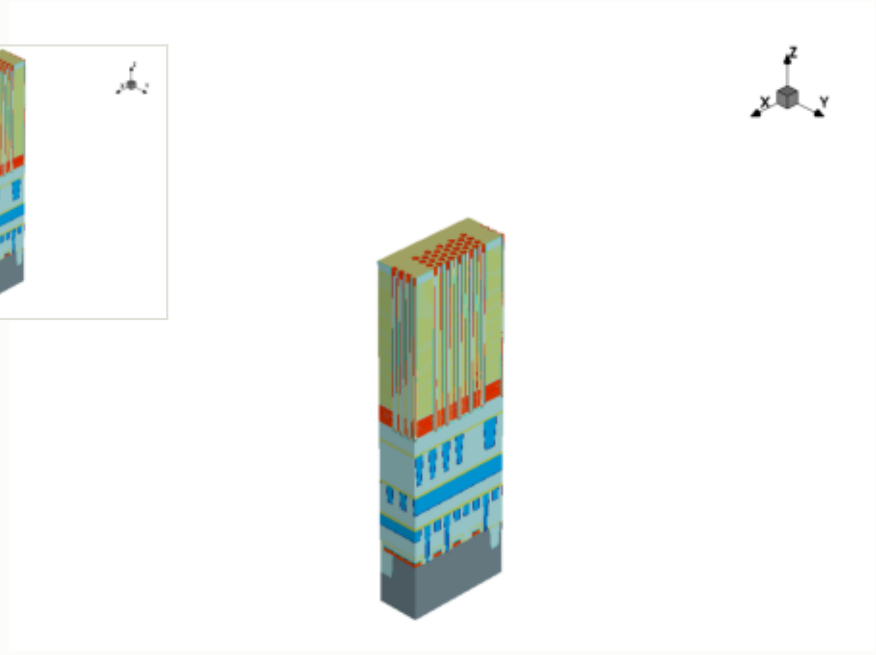
etch_anisotropic · RIE

1.04	409.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

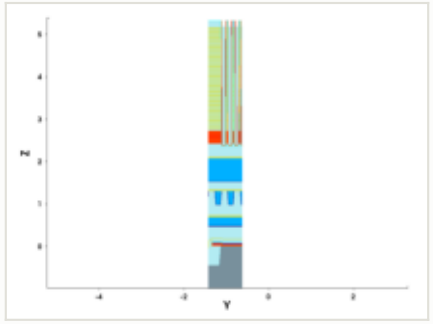
3D structure after this step

zoomed to fit

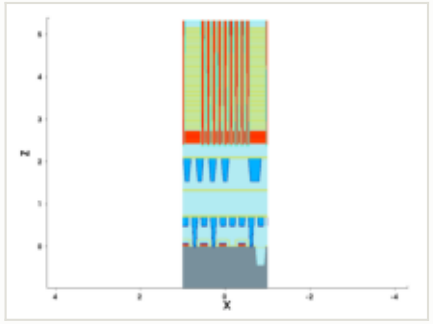




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

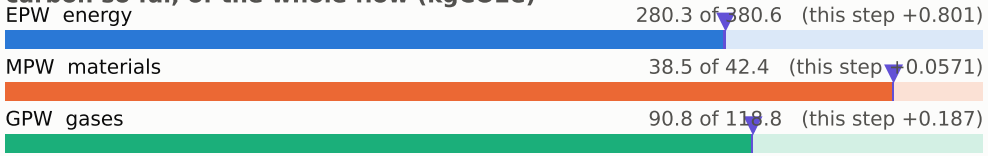
no mask for this step

this step

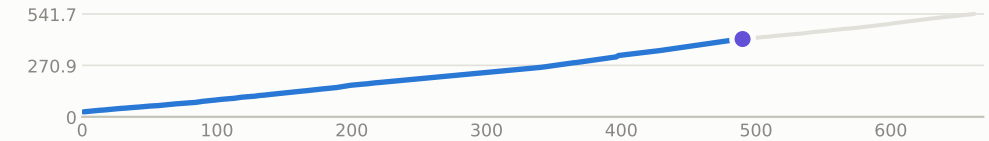
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

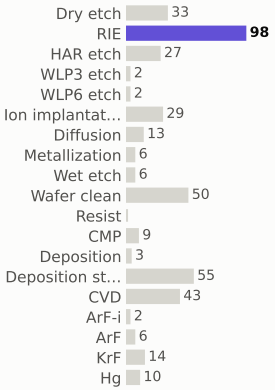
carbon so far, of the whole flow (kgCO2e)



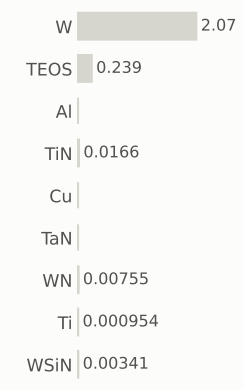
embodied carbon up to each step (kgCO2e per wafer)



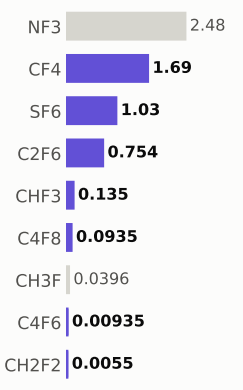
steps by process area



materials deposited (g)



gases emitted (g)



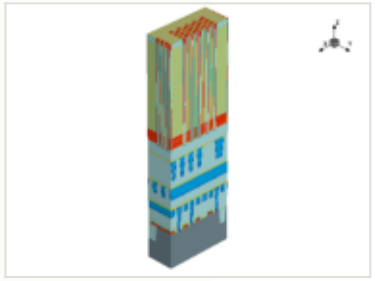
Step #4500: Wafer_clean

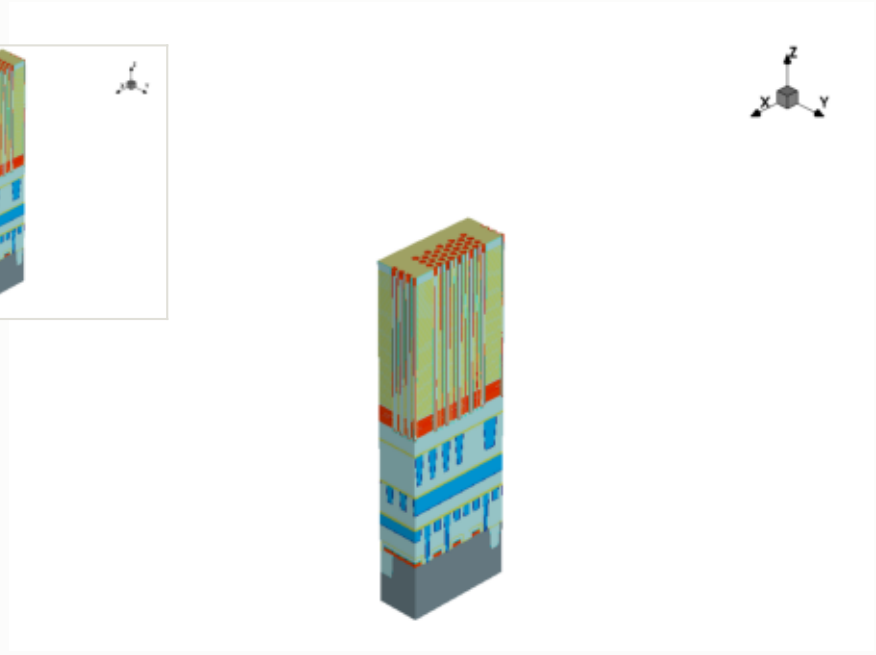
clean · Wafer_clean

0.169	409.8	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

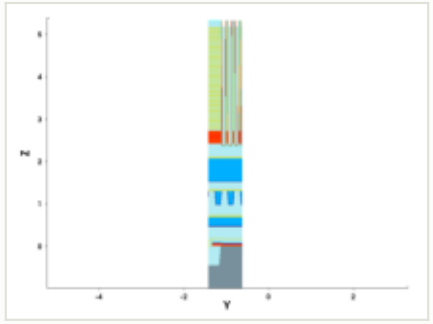
3D structure after this step

zoomed to fit

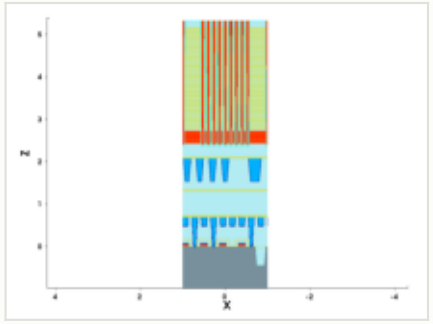




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

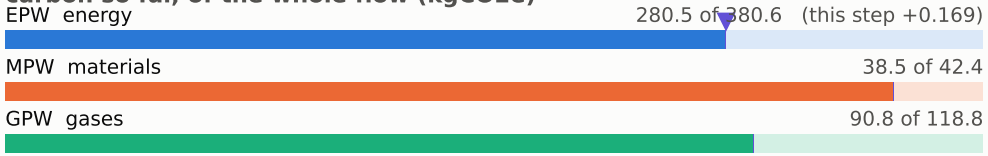
no mask for this step

this step

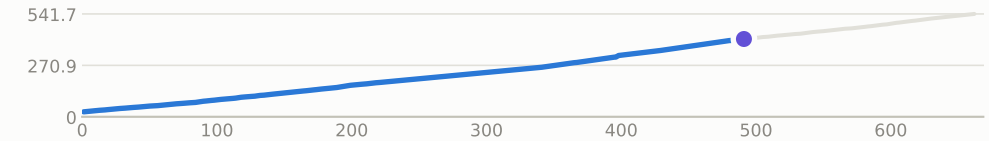
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

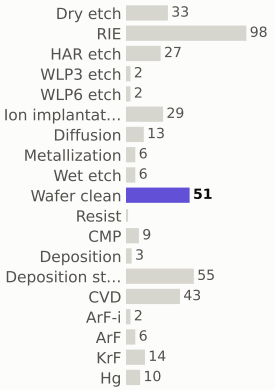
carbon so far, of the whole flow (kgCO2e)



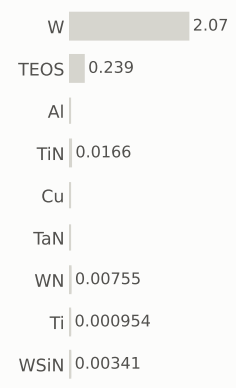
embodied carbon up to each step (kgCO2e per wafer)



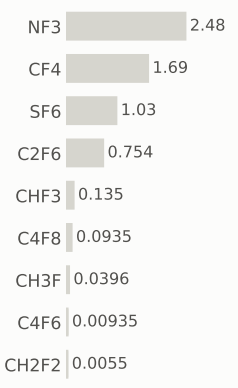
steps by process area



materials deposited (g)



gases emitted (g)



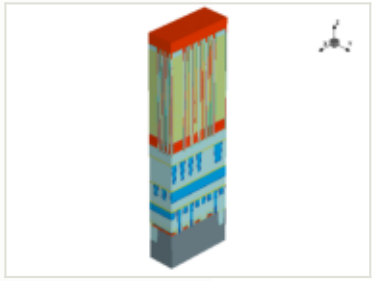
Step #4510: CVD

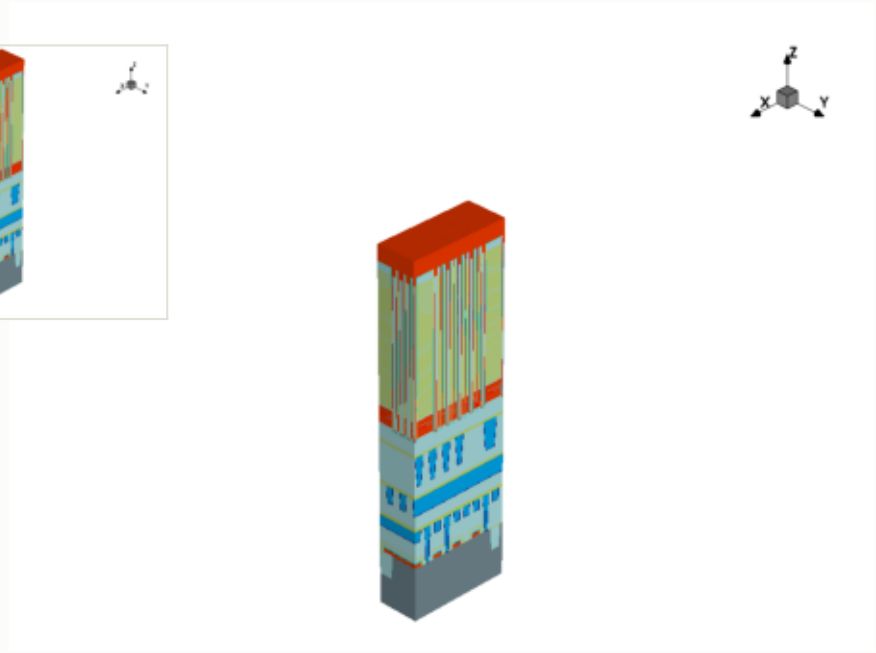
depo_isotropic · CVD

1.31	411.1	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

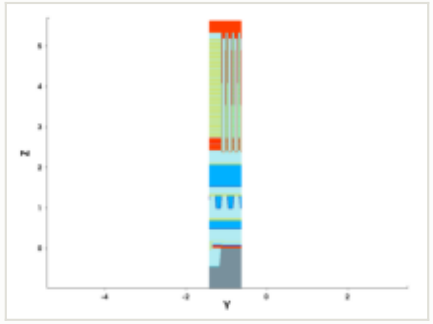
3D structure after this step

zoomed to fit

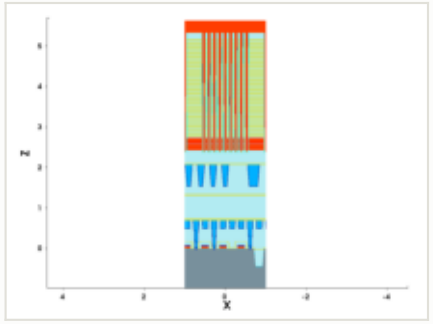




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

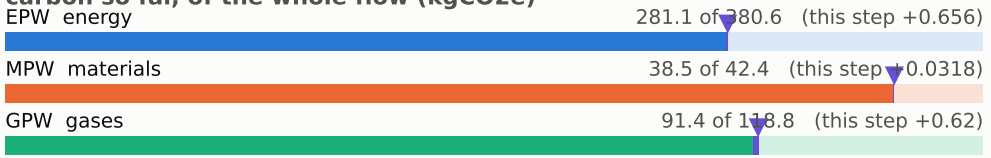
no mask for this step

this step

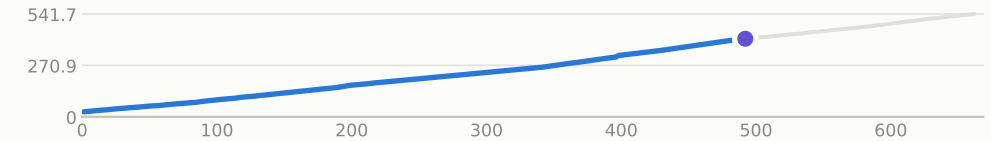
process area	CVD
material	—
thickness	0.3 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

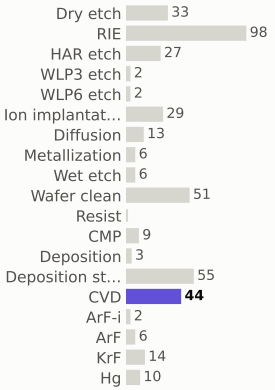
carbon so far, of the whole flow (kgCO2e)



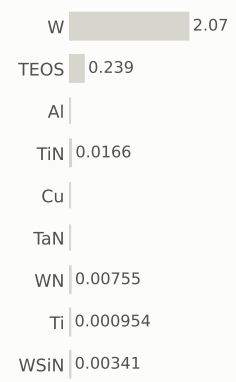
embodied carbon up to each step (kgCO2e per wafer)



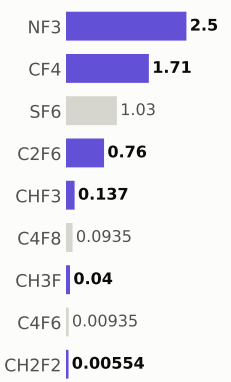
steps by process area



materials deposited (g)



gases emitted (g)



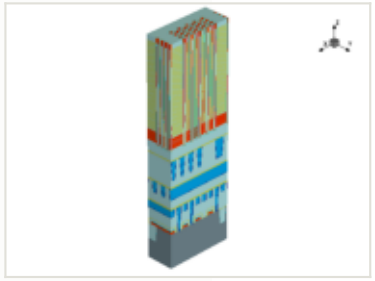
Step #4520: CMP

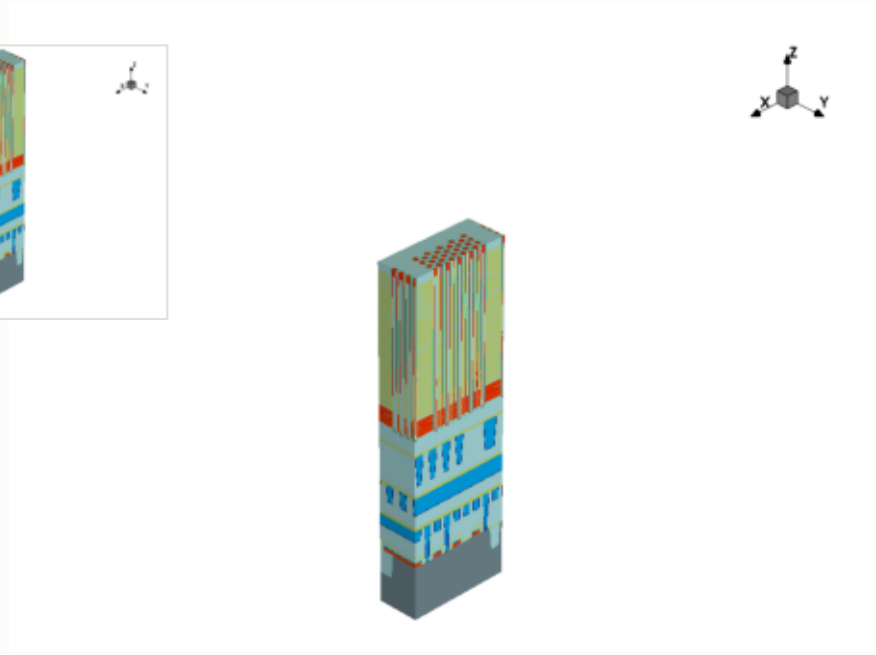
etch_cmp · CMP

0.169	411.2	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



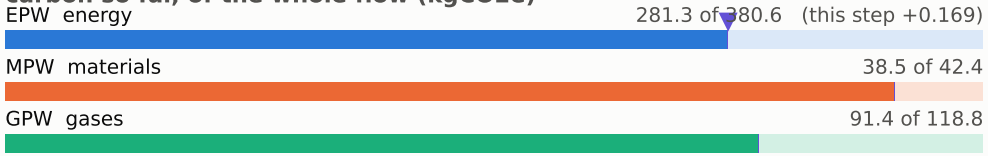


this step

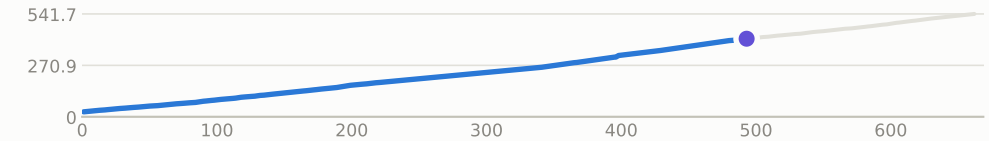
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

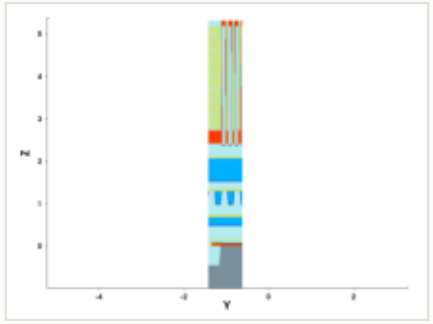
carbon so far, of the whole flow (kgCO2e)



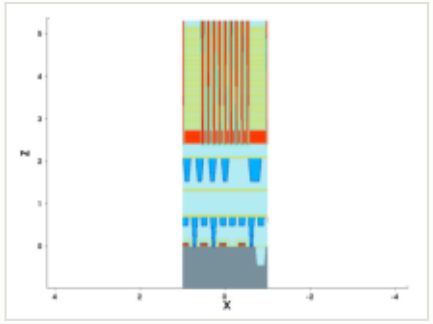
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



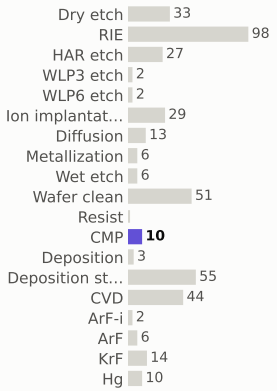
y cut at -1.032 µm



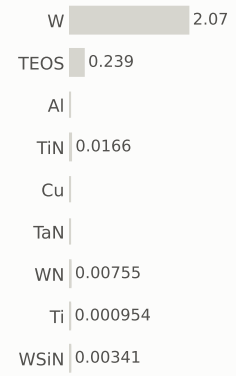
mask for this step

no mask for this step

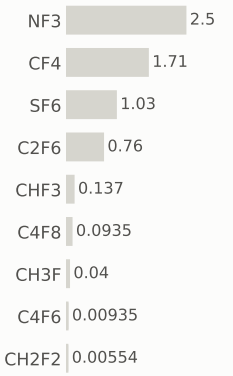
steps by process area



materials deposited (g)



gases emitted (g)



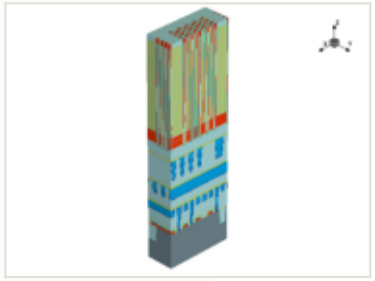
Step #4530: Wafer_clean

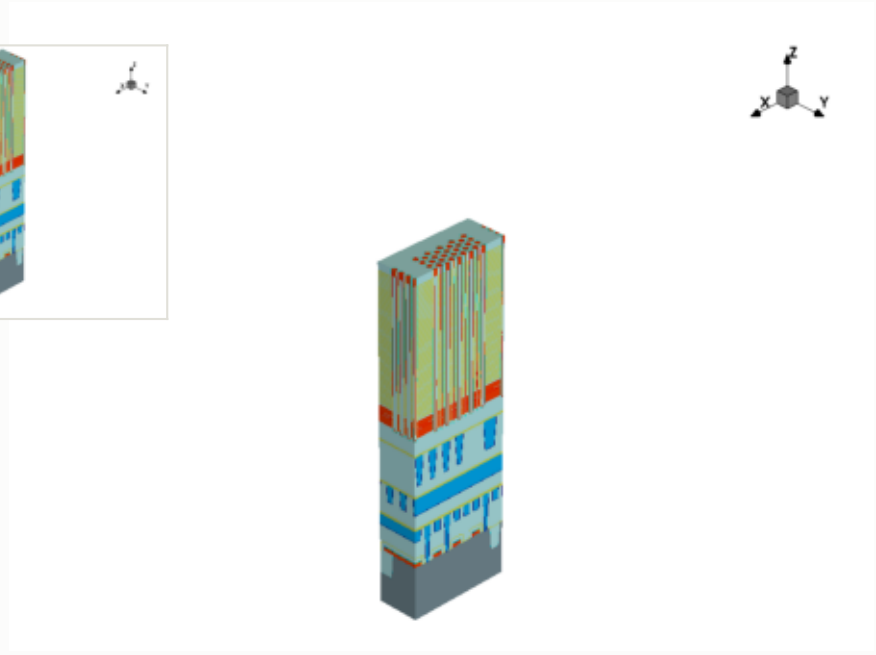
clean · Wafer_clean

0.169	411.4	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



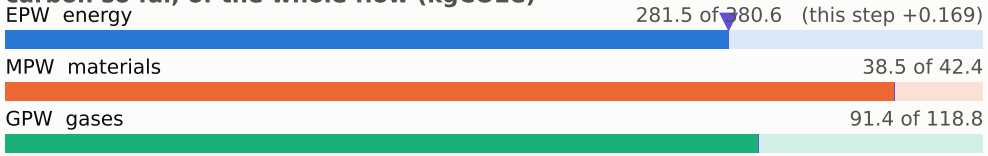


this step

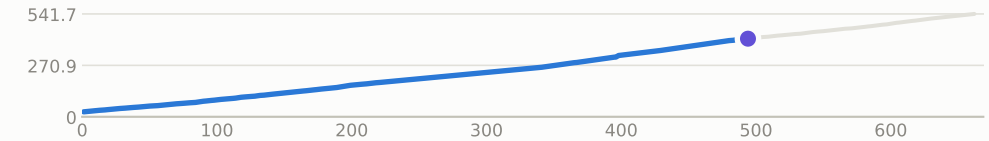
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

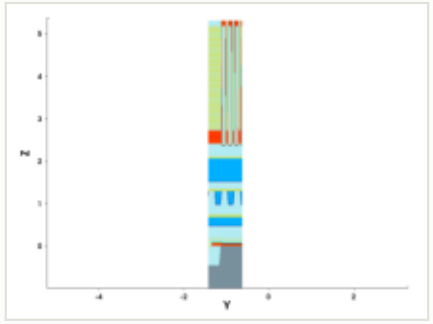
carbon so far, of the whole flow (kgCO2e)



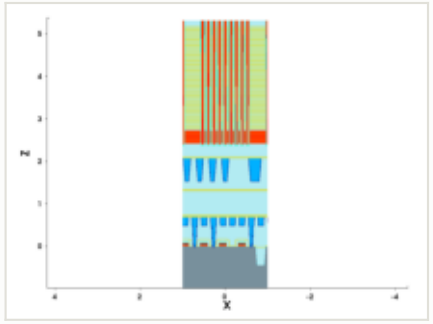
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



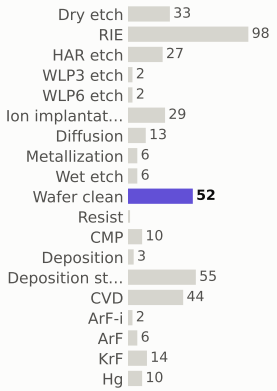
y cut at -1.032 µm



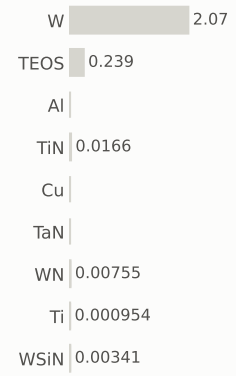
mask for this step

no mask for this step

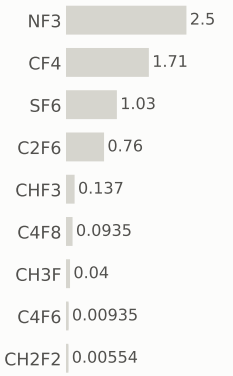
steps by process area



materials deposited (g)



gases emitted (g)



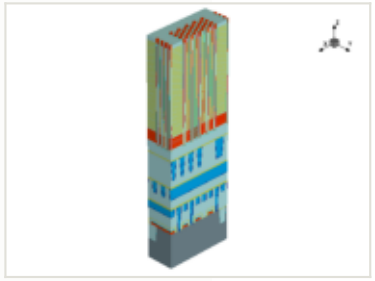
Step #4540: RIE

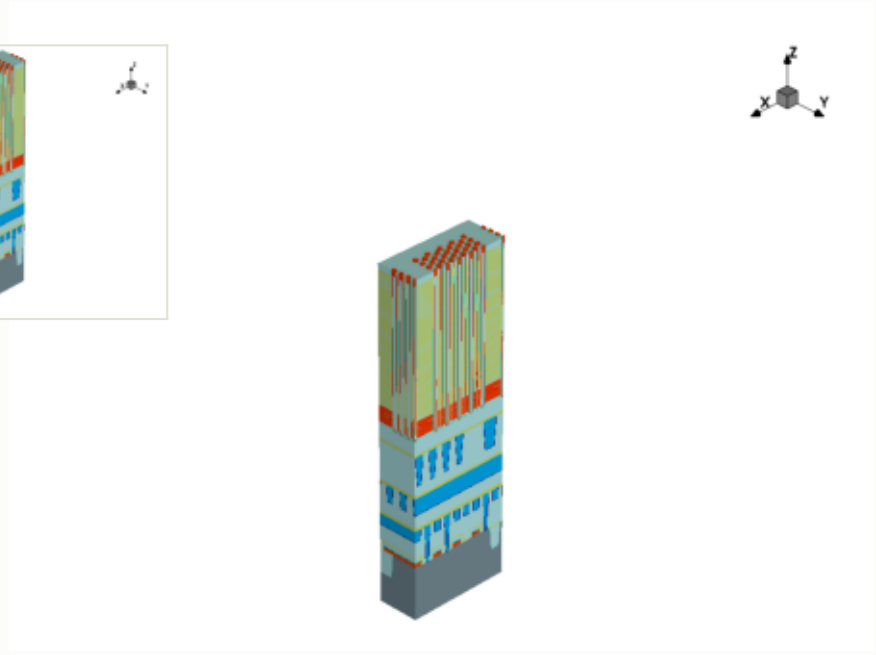
etch_isotropic · RIE

1.04	412.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



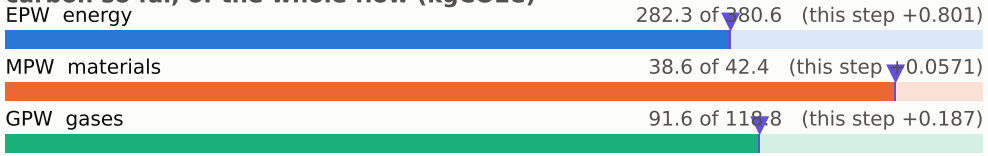


this step

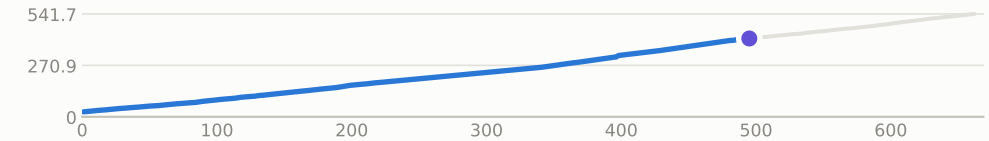
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

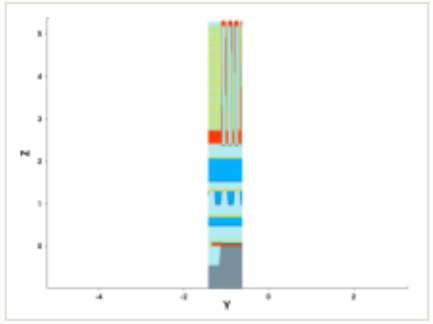
carbon so far, of the whole flow (kgCO2e)



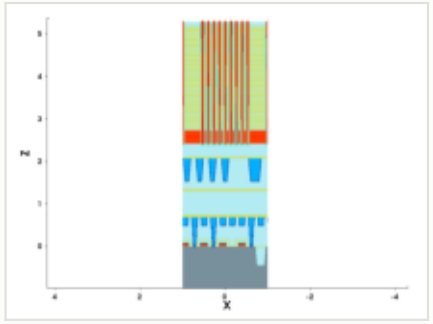
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



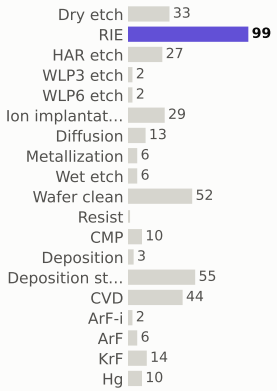
y cut at -1.032 µm



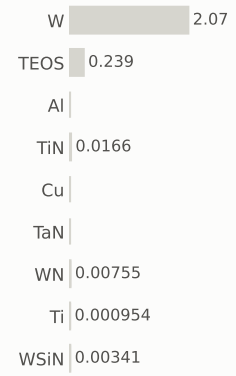
mask for this step

no mask for this step

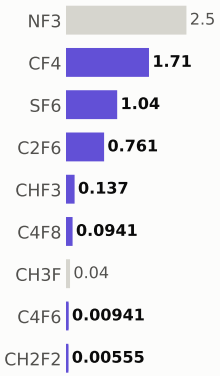
steps by process area



materials deposited (g)



gases emitted (g)



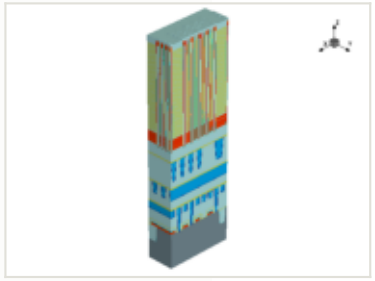
Step #4550: CVD

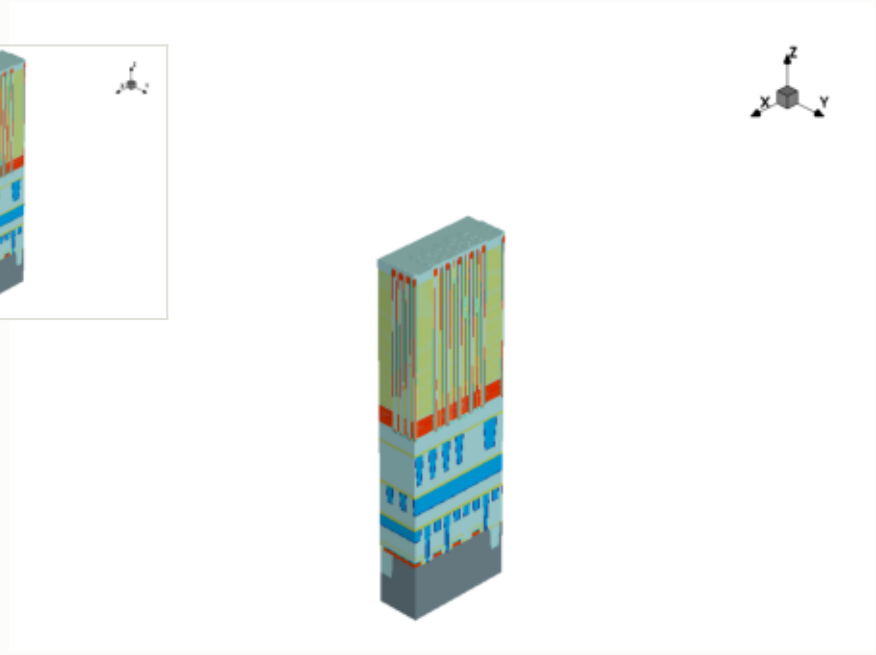
depo_isotropic · CVD

1.31	413.8	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

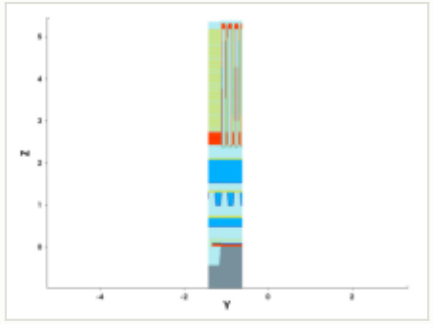
3D structure after this step

zoomed to fit

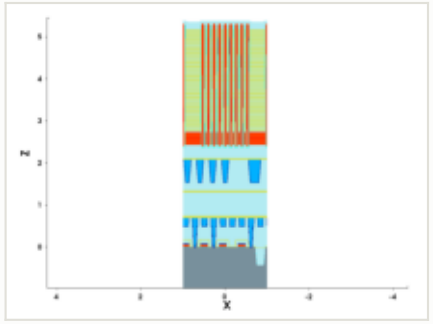




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

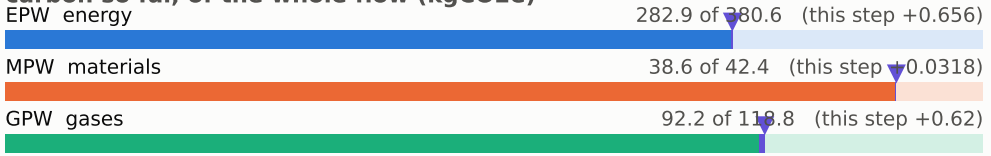
no mask for this step

this step

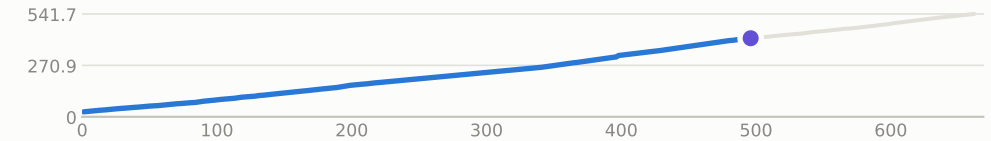
process area	CVD
material	—
thickness	0.07 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

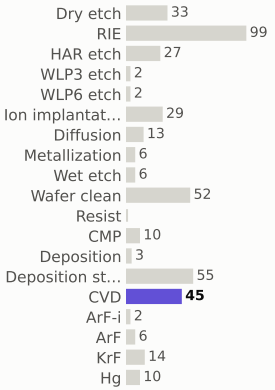
carbon so far, of the whole flow (kgCO2e)



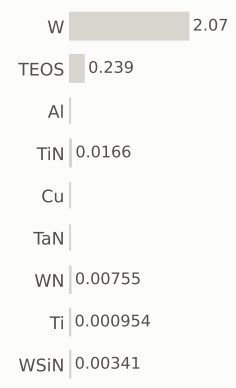
embodied carbon up to each step (kgCO2e per wafer)



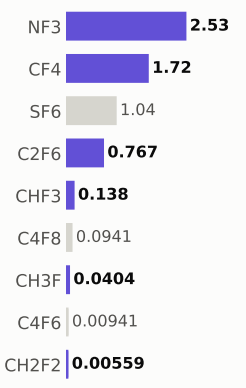
steps by process area



materials deposited (g)



gases emitted (g)



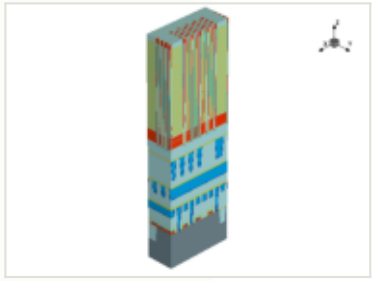
Step #4560: CMP

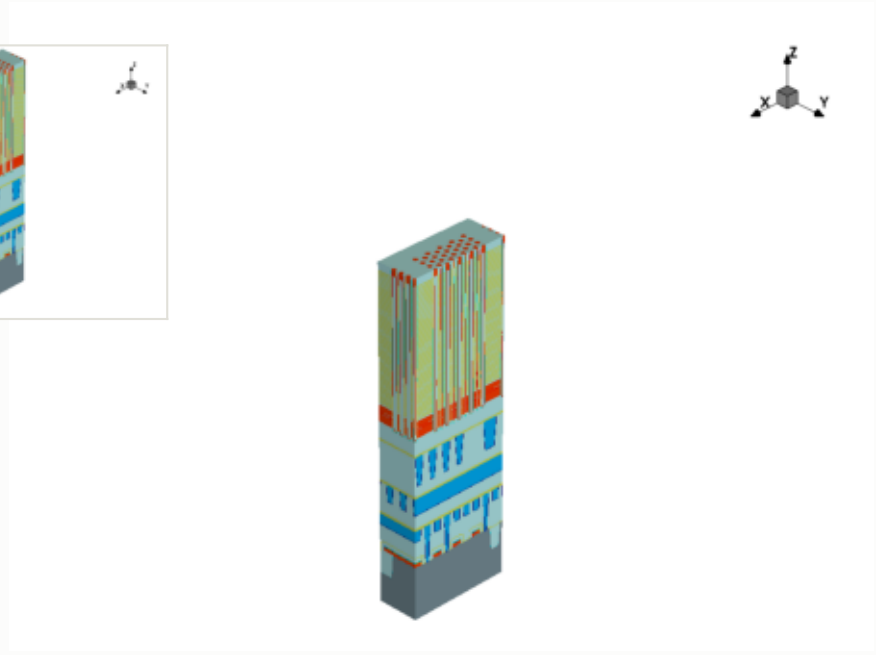
etch_cmp · CMP

0.169	413.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

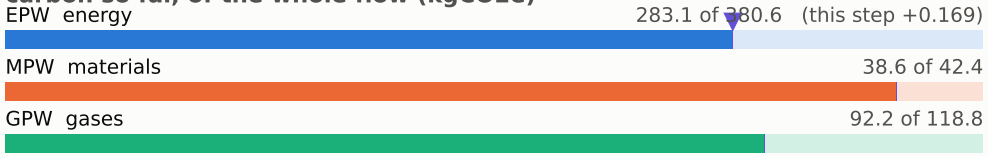




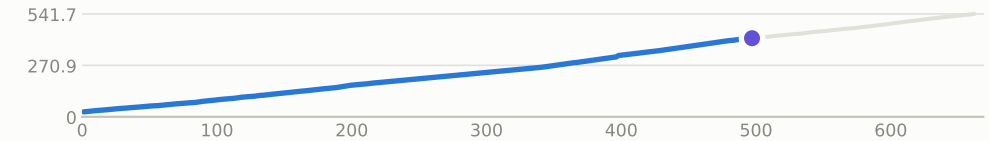
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

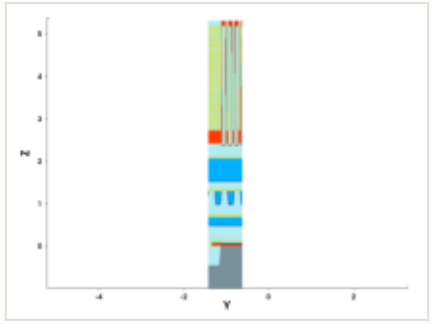
carbon so far, of the whole flow (kgCO2e)



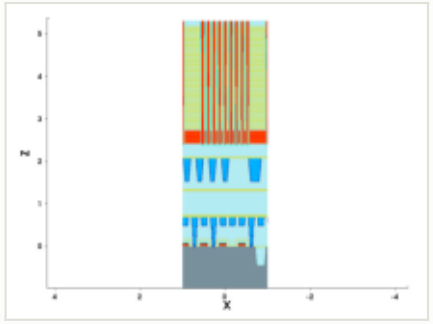
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



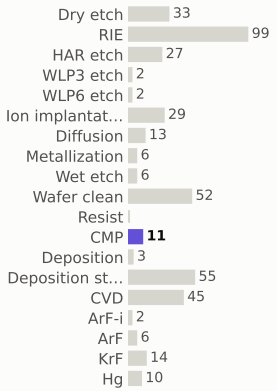
y cut at -1.032 µm



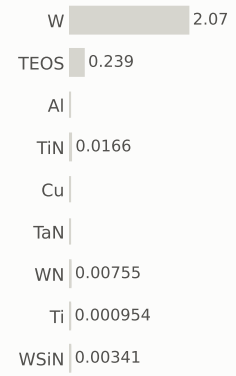
mask for this step

no mask for this step

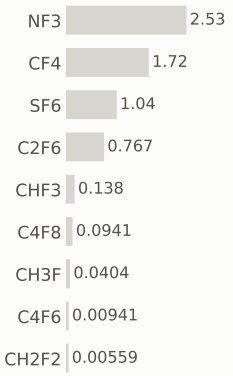
steps by process area



materials deposited (g)



gases emitted (g)



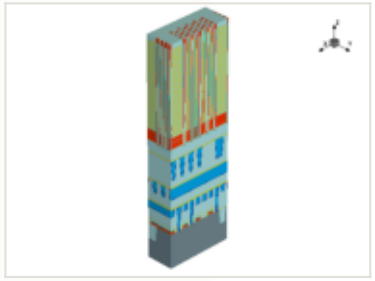
Step #4570: Wafer_clean

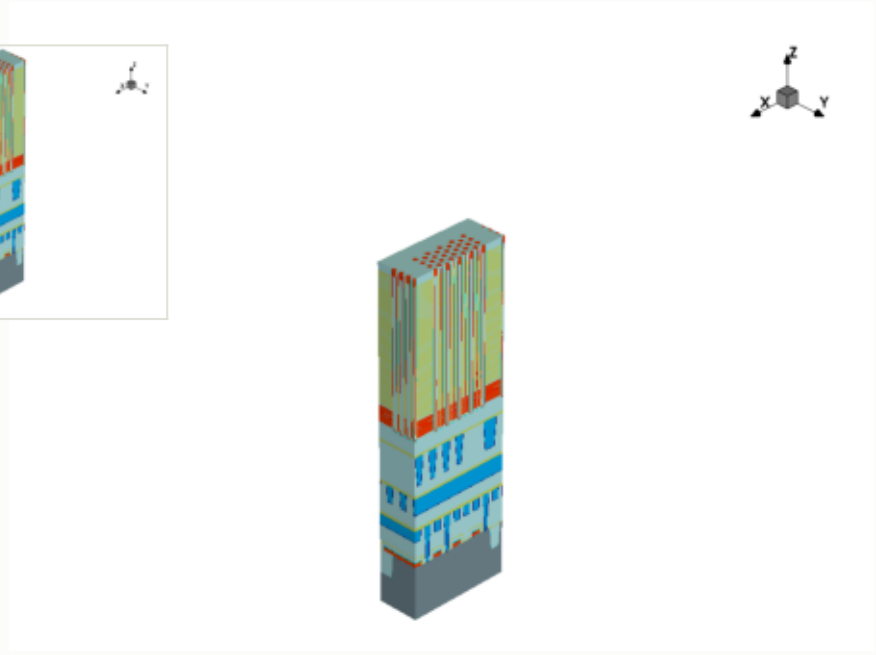
clean · Wafer_clean

0.169	414.1	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

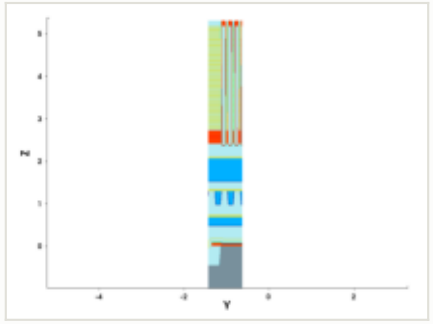
3D structure after this step

zoomed to fit

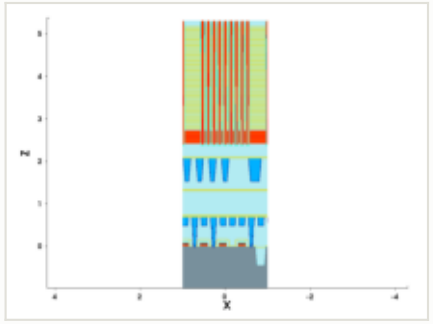




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

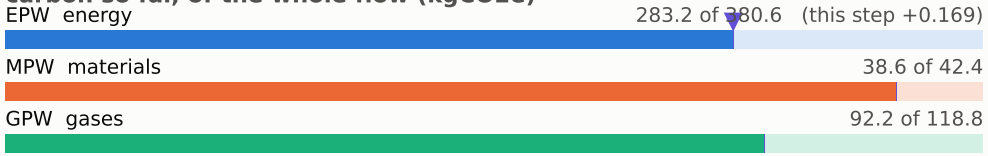
no mask for this step

this step

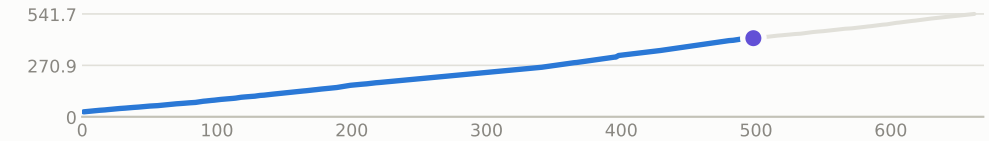
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

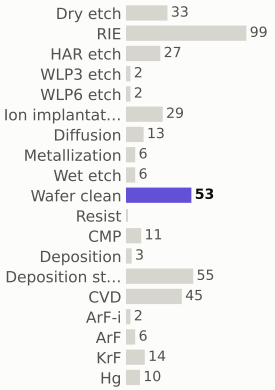
carbon so far, of the whole flow (kgCO2e)



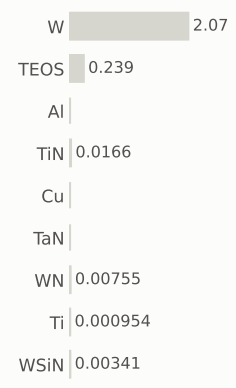
embodied carbon up to each step (kgCO2e per wafer)



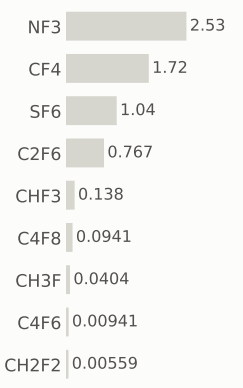
steps by process area



materials deposited (g)



gases emitted (g)



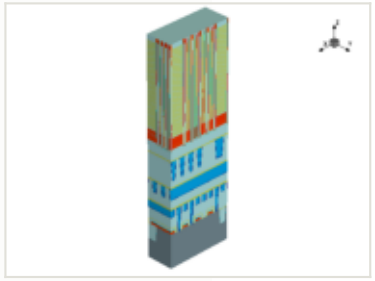
Step #4590: CVD

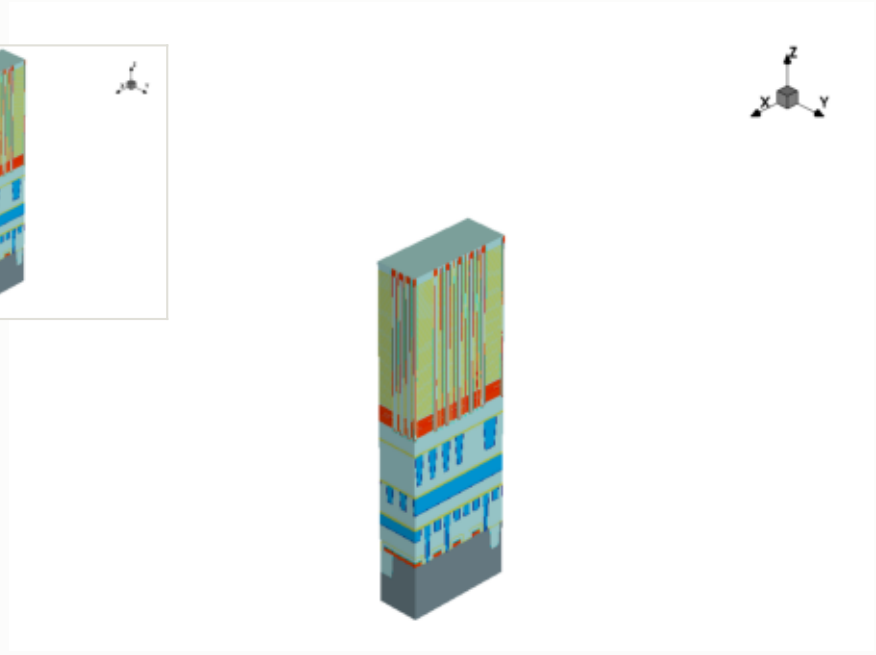
depo_isotropic · CVD

1.31	415.4	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

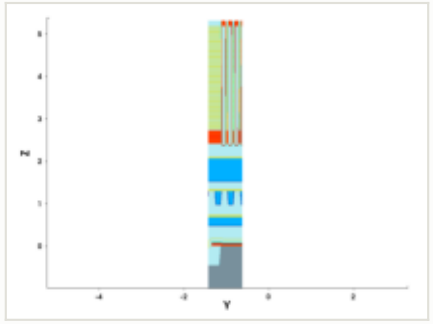
3D structure after this step

zoomed to fit

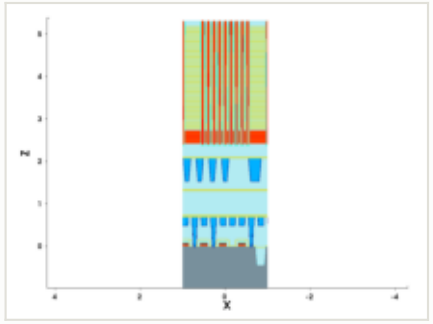




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

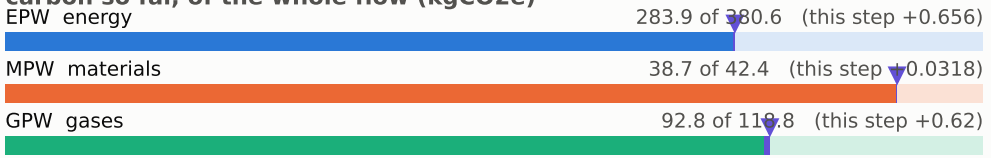
no mask for this step

this step

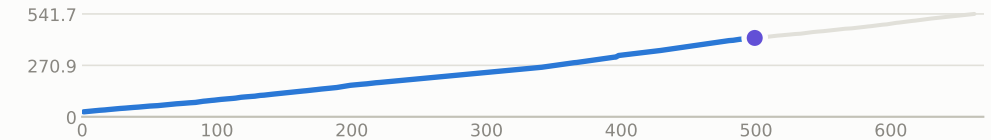
process area	CVD
material	—
thickness	0.003 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

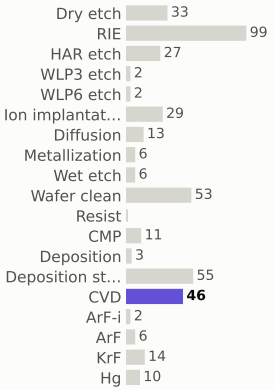
carbon so far, of the whole flow (kgCO2e)



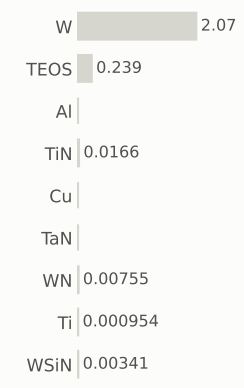
embodied carbon up to each step (kgCO2e per wafer)



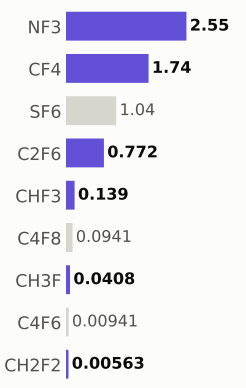
steps by process area



materials deposited (g)



gases emitted (g)



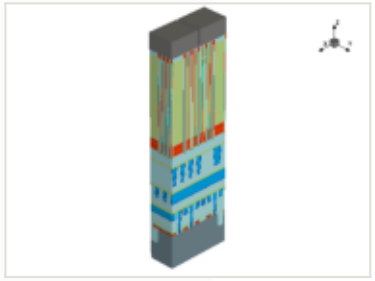
Step #4600: ArF

litho_patterning · ArF

0.199	415.6	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

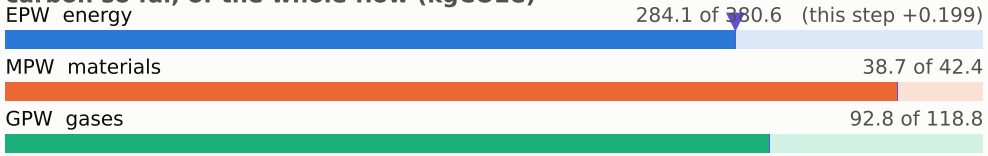
zoomed to fit



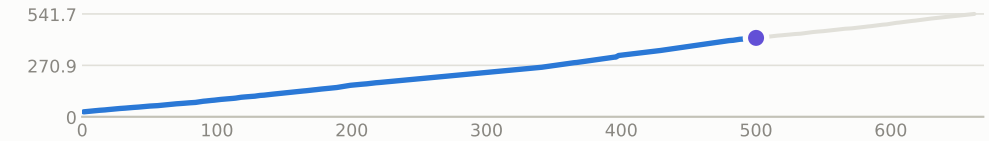
this step

process area	ArF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

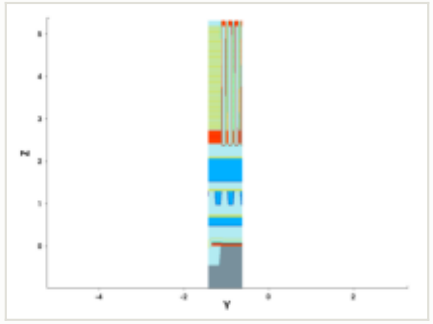
carbon so far, of the whole flow (kgCO2e)



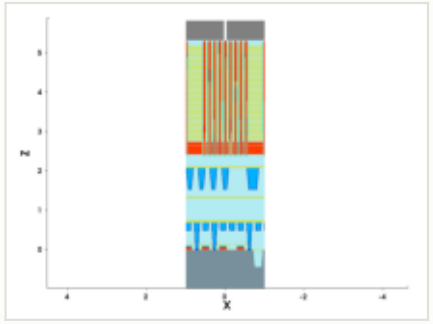
embodied carbon up to each step (kgCO2e per wafer)



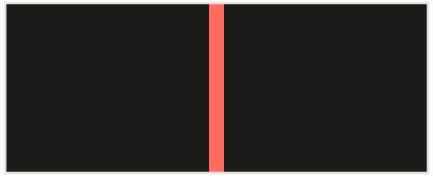
x cut at 0.000 µm



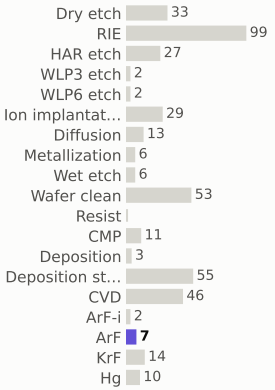
y cut at -1.032 µm



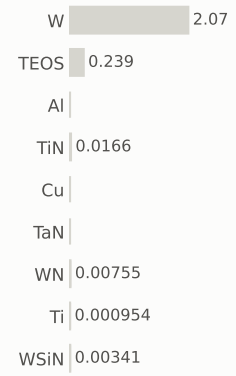
mask: layer 36:0 (OASIS)



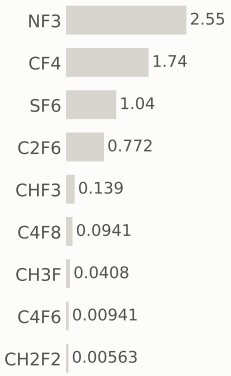
steps by process area



materials deposited (g)



gases emitted (g)



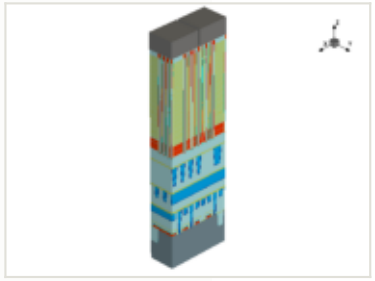
Step #4610: RIE

etch_anisotropic · RIE

1.04	416.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

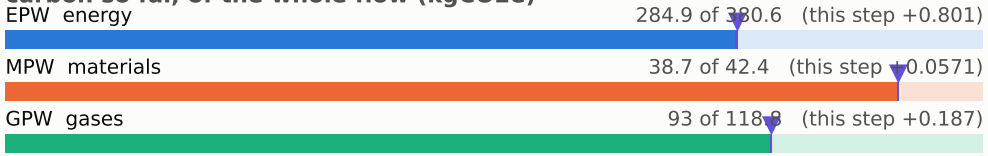


this step

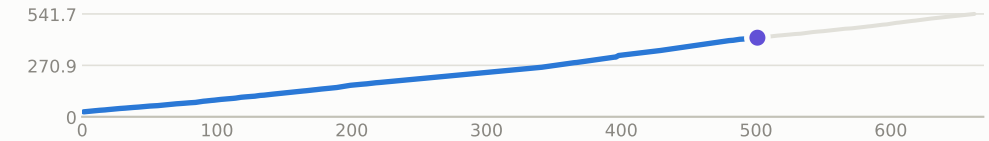
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

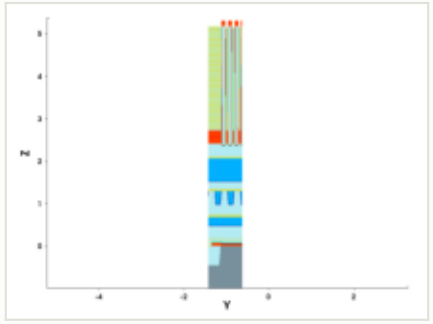
carbon so far, of the whole flow (kgCO2e)



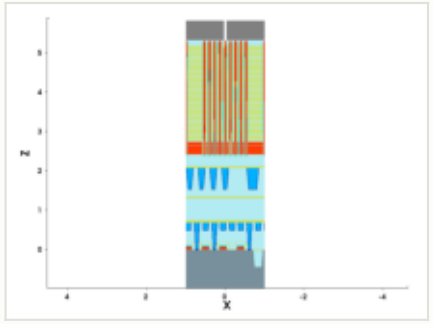
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



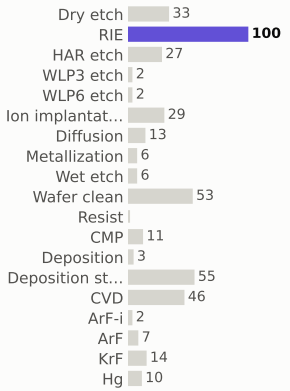
y cut at -1.032 µm



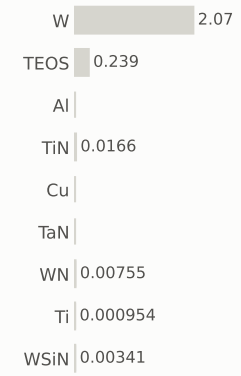
mask for this step

no mask for this step

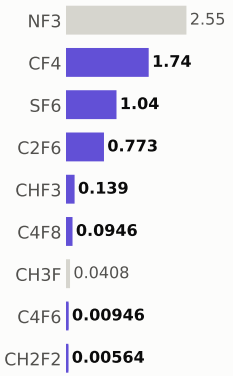
steps by process area



materials deposited (g)



gases emitted (g)



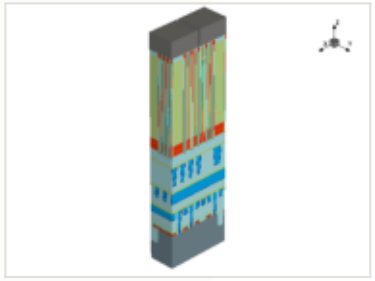
Step #4620: RIE

etch_anisotropic · RIE

1.04	417.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

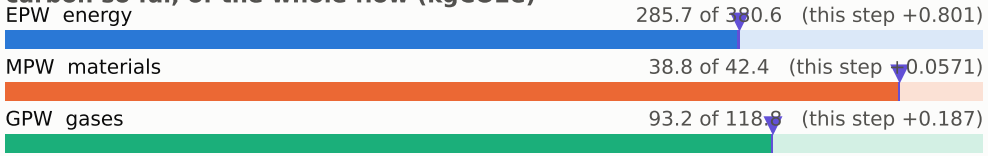


this step

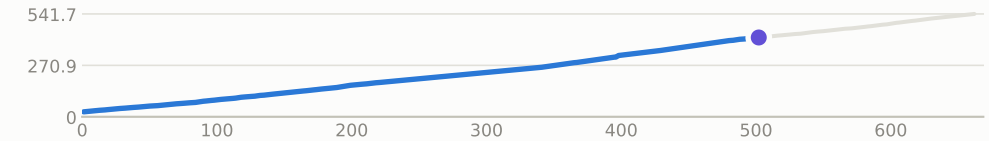
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

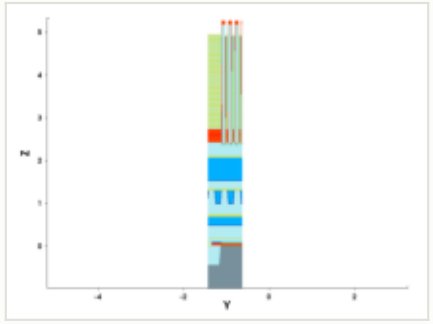
carbon so far, of the whole flow (kgCO2e)



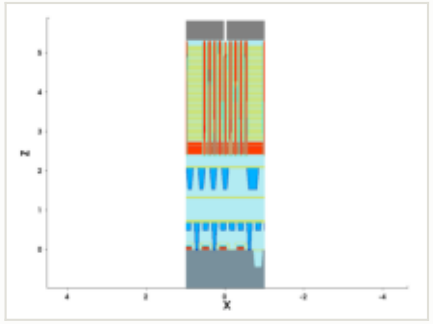
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



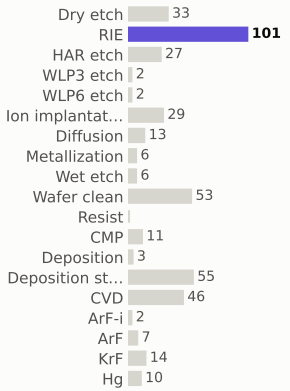
y cut at -1.032 µm



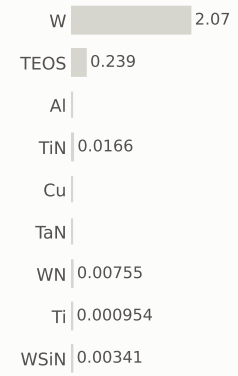
mask for this step

no mask for this step

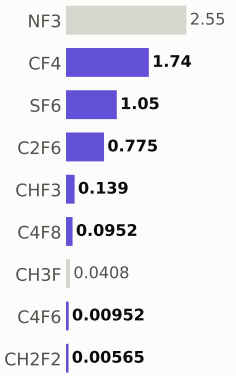
steps by process area



materials deposited (g)



gases emitted (g)



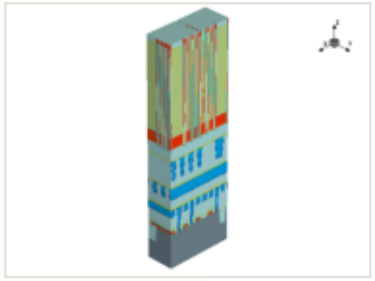
Step #4630: Dry_etch

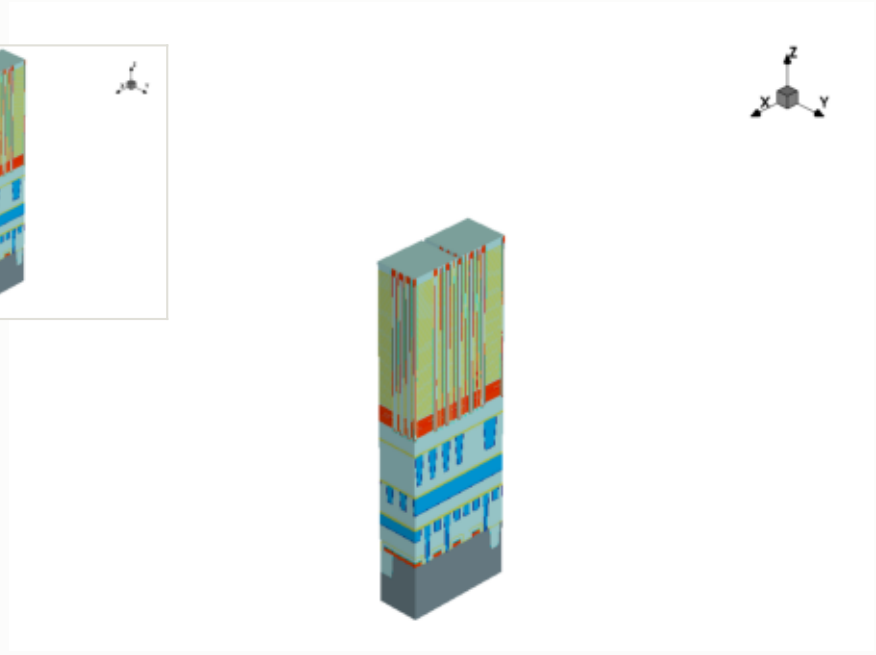
etch_strip · Dry_etch

0.801	418.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

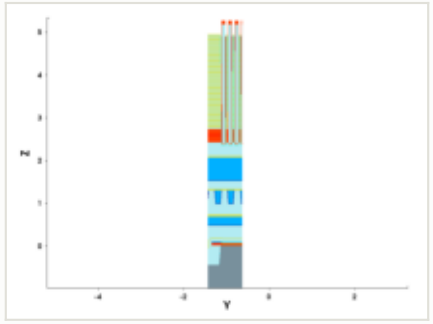
3D structure after this step

zoomed to fit

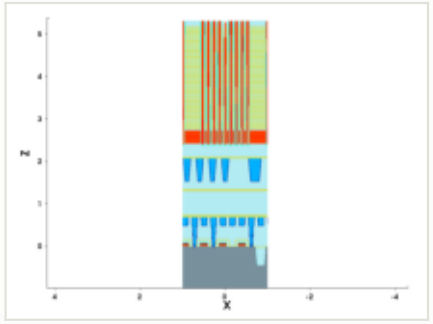




x cut at 0.000 µm



y cut at -1.032 µm



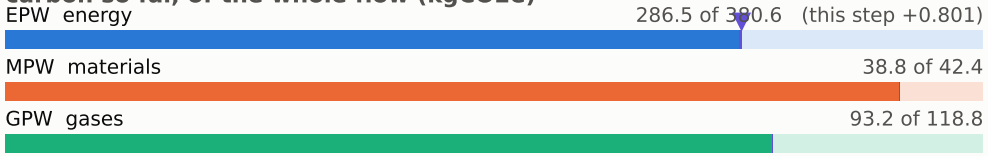
mask for this step

no mask for this step

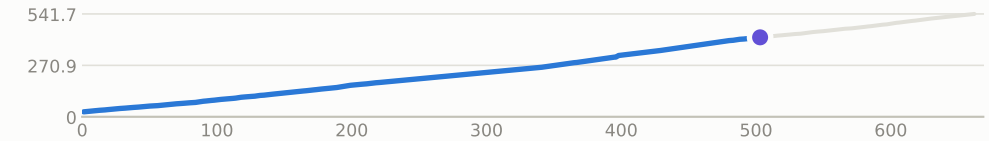
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

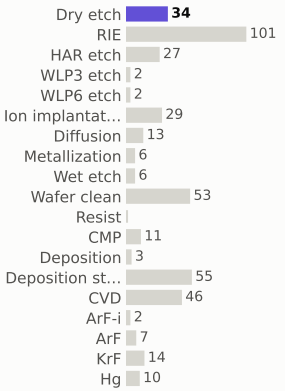
carbon so far, of the whole flow (kgCO2e)



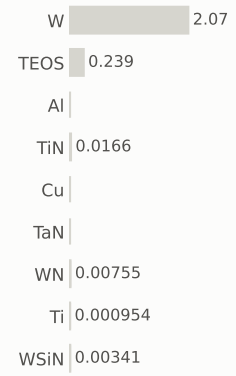
embodied carbon up to each step (kgCO2e per wafer)



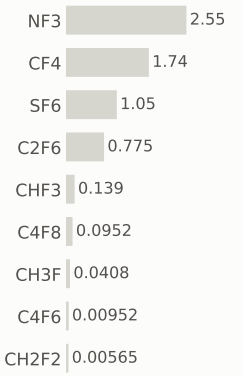
steps by process area



materials deposited (g)



gases emitted (g)



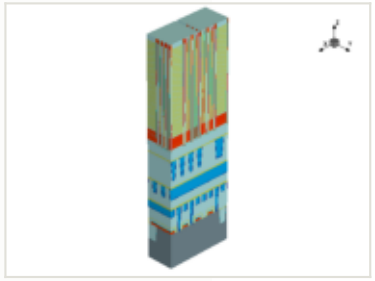
Step #4640: Wafer_clean

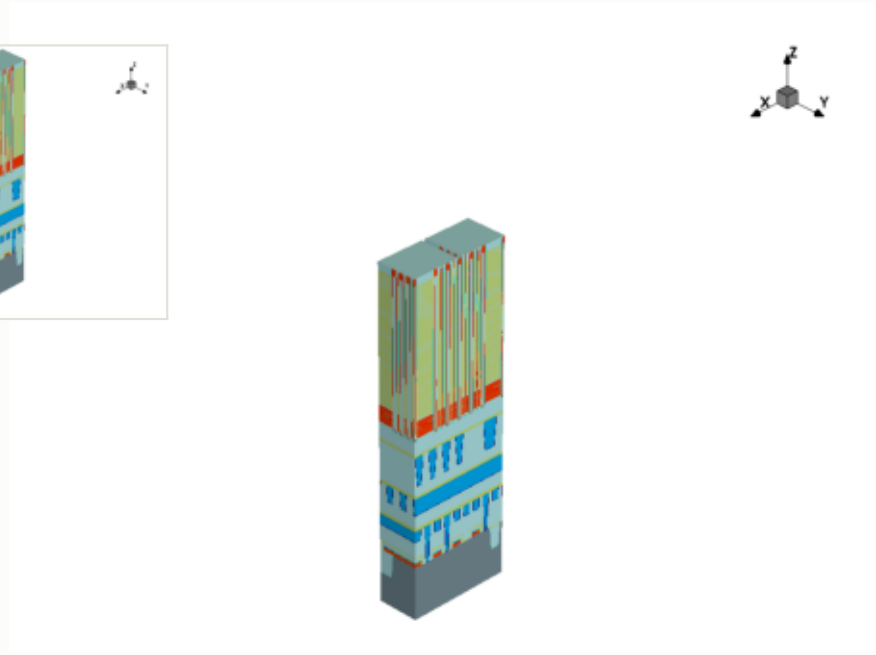
clean · Wafer_clean

0.169	418.7	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

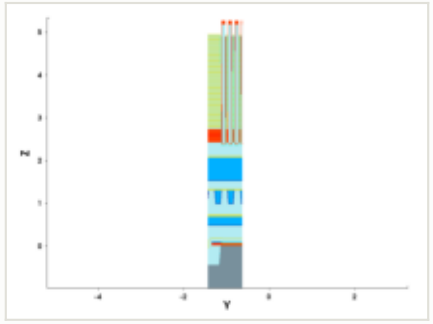
3D structure after this step

zoomed to fit

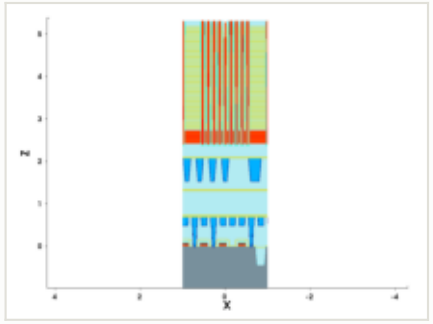




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

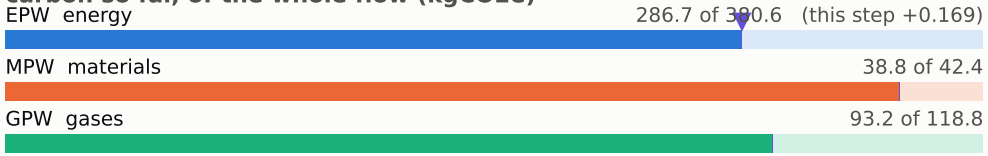
no mask for this step

this step

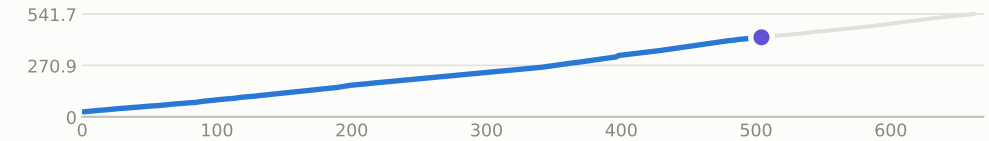
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

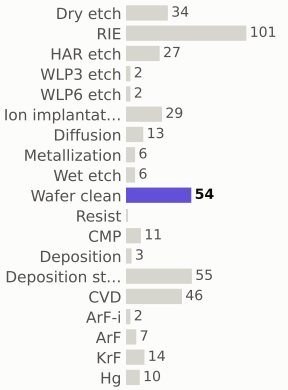
carbon so far, of the whole flow (kgCO2e)



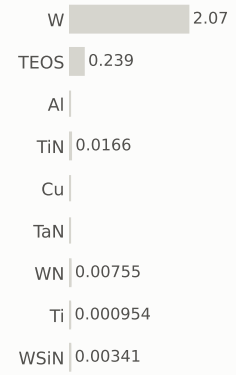
embodied carbon up to each step (kgCO2e per wafer)



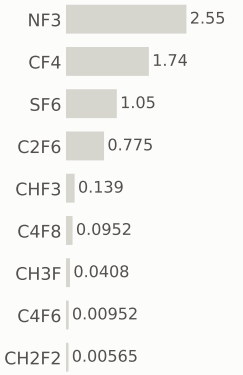
steps by process area



materials deposited (g)



gases emitted (g)



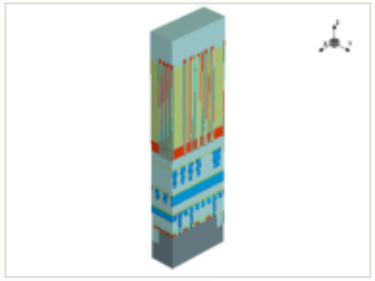
Step #4650: CVD

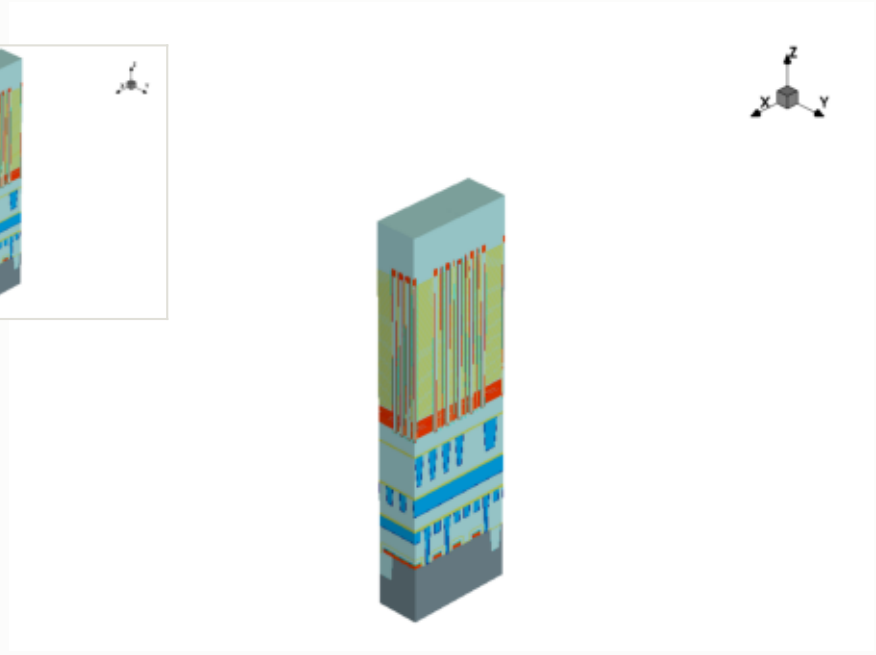
depo_isotropic · CVD

1.31	420.0	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

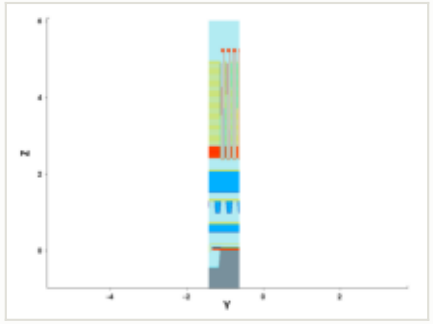
3D structure after this step

zoomed to fit

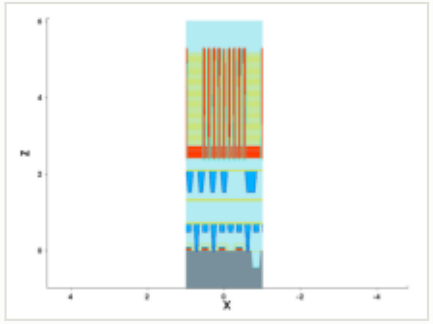




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

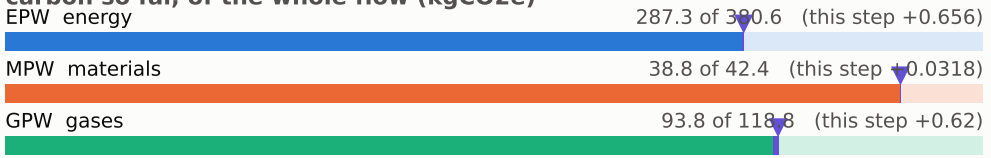
no mask for this step

this step

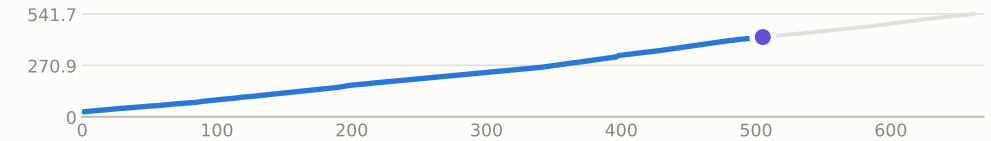
process area	CVD
material	—
thickness	0.7 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

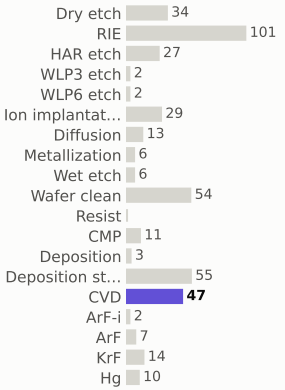
carbon so far, of the whole flow (kgCO2e)



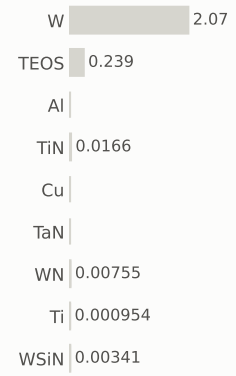
embodied carbon up to each step (kgCO2e per wafer)



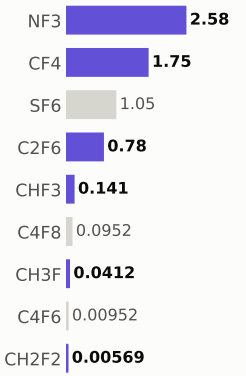
steps by process area



materials deposited (g)



gases emitted (g)



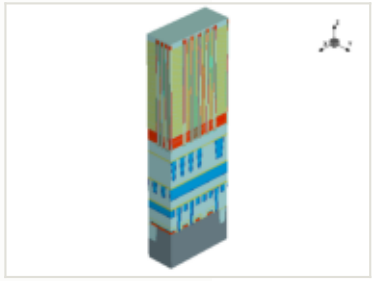
Step #4660: CMP

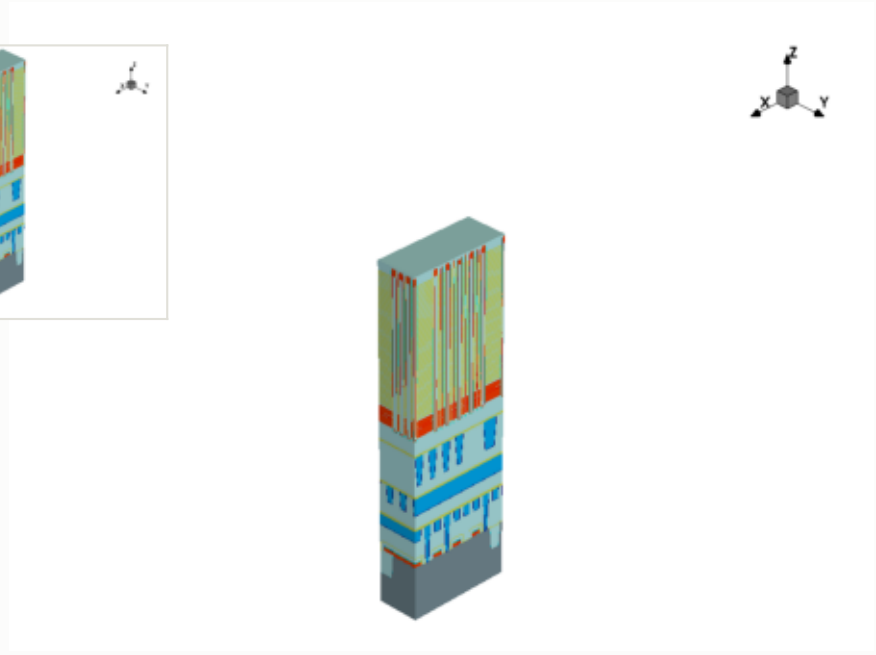
etch_cmp · CMP

0.169	420.1	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

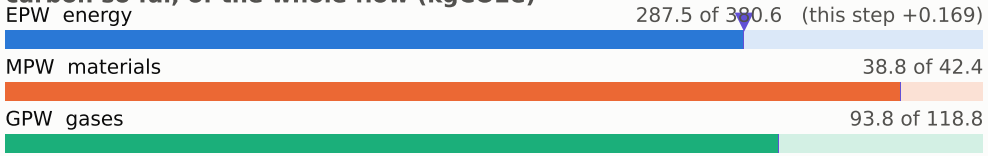




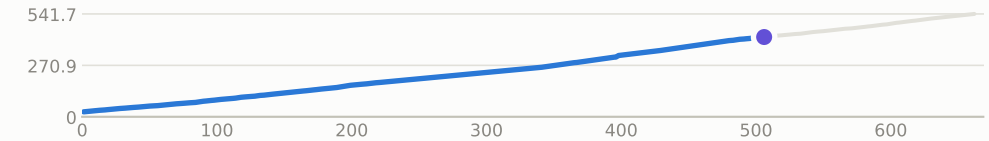
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

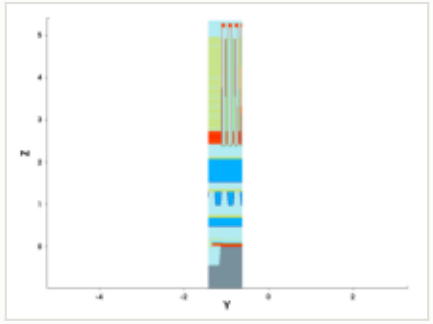
carbon so far, of the whole flow (kgCO2e)



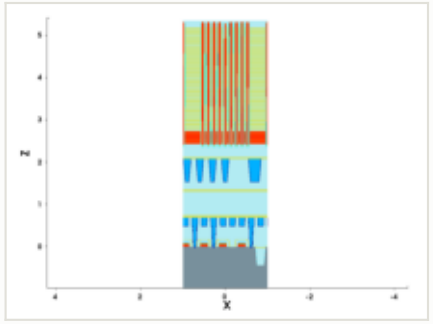
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



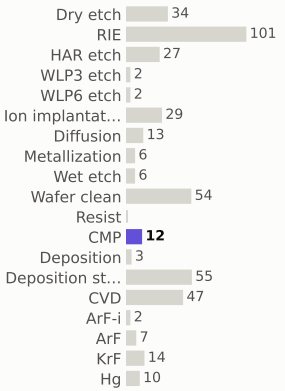
y cut at -1.032 µm



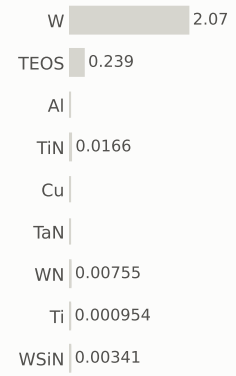
mask for this step

no mask for this step

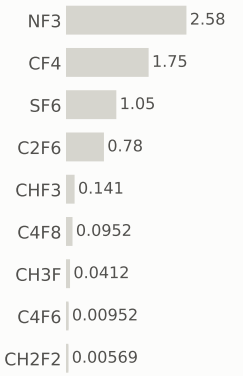
steps by process area



materials deposited (g)



gases emitted (g)



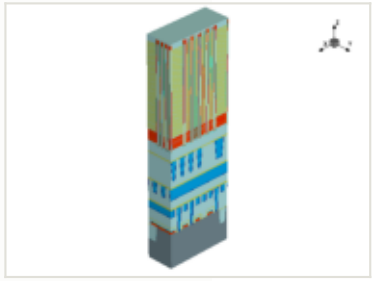
Step #4670: Wafer_clean

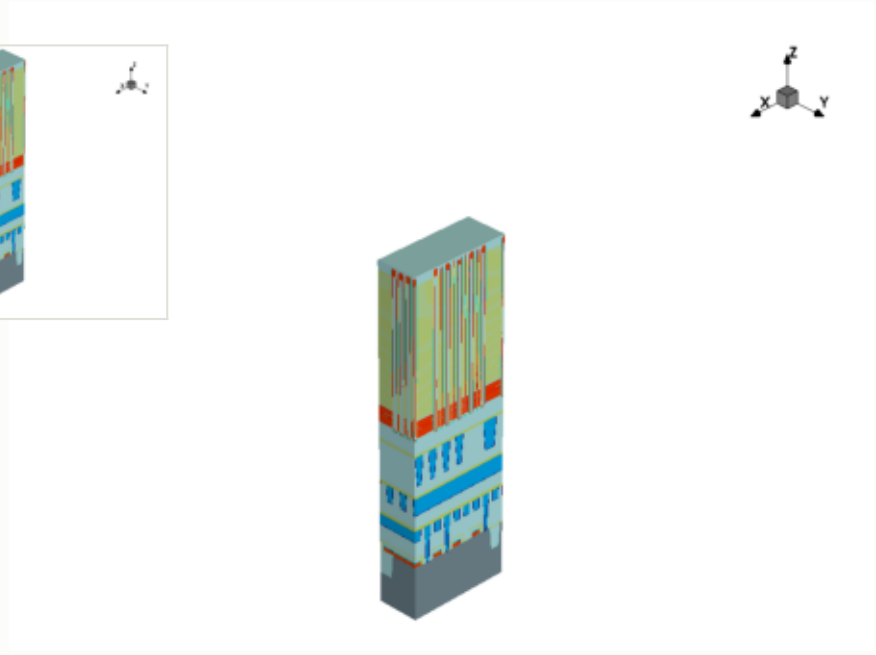
clean · Wafer_clean

0.169	420.3	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

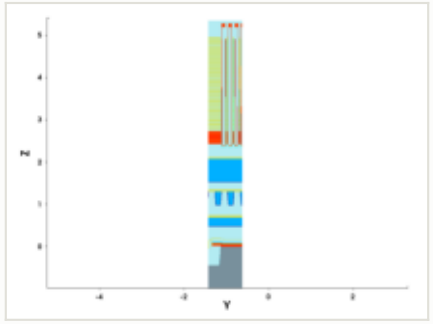
3D structure after this step

zoomed to fit

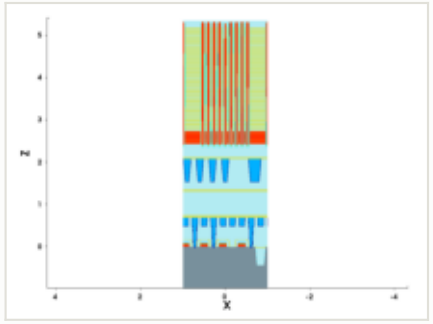




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

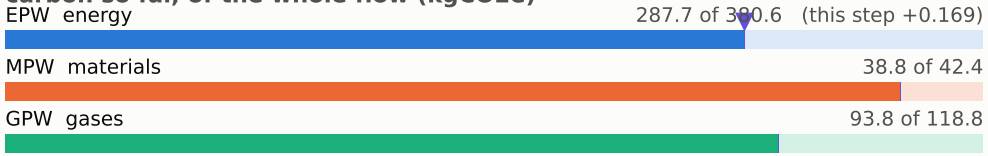
no mask for this step

this step

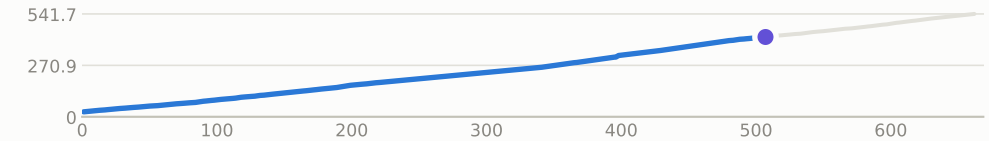
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

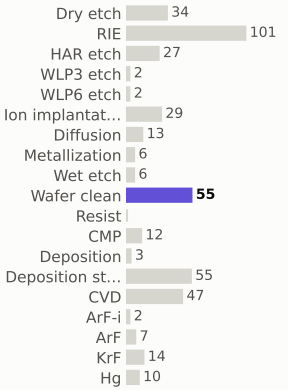
carbon so far, of the whole flow (kgCO2e)



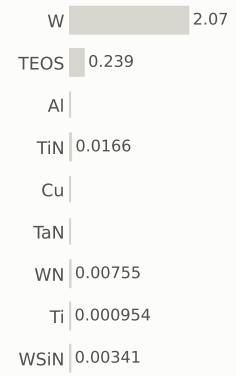
embodied carbon up to each step (kgCO2e per wafer)



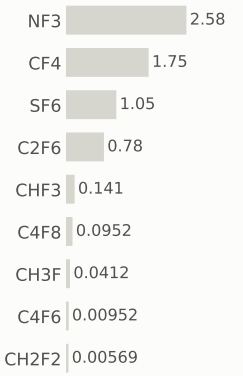
steps by process area



materials deposited (g)



gases emitted (g)



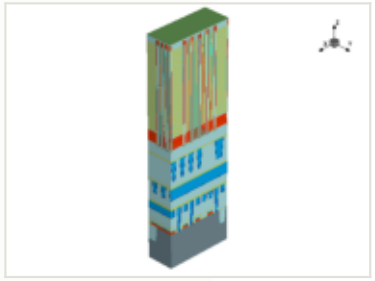
Step #4690: CVD

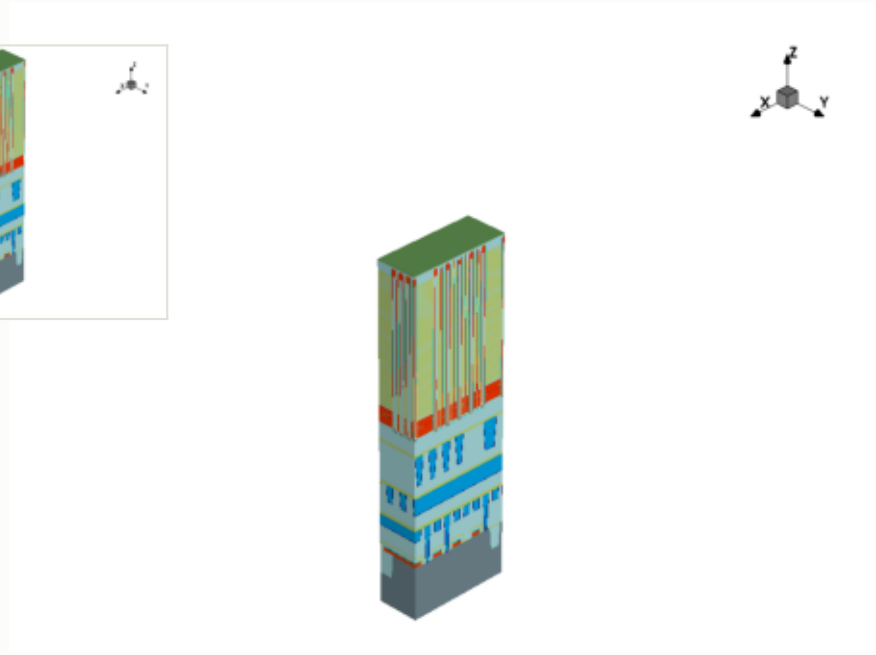
depo_isotropic · CVD

1.31	421.6	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



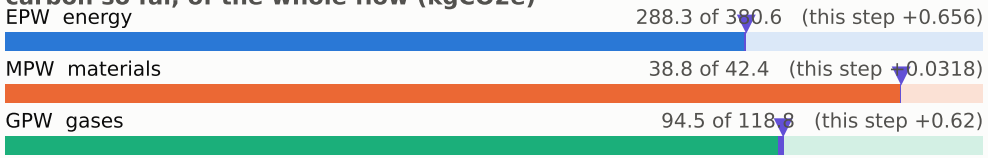


this step

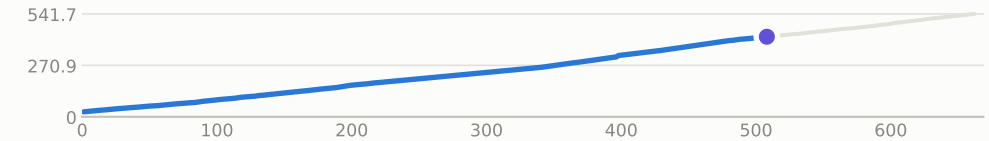
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

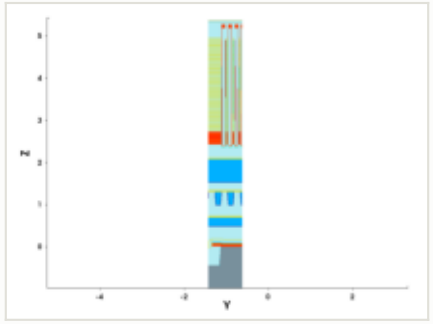
carbon so far, of the whole flow (kgCO2e)



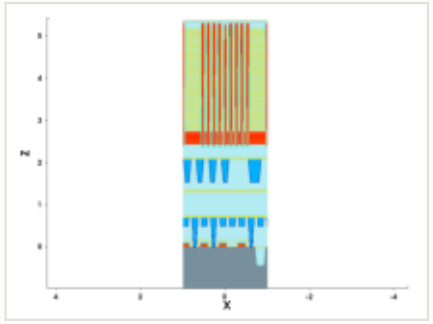
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



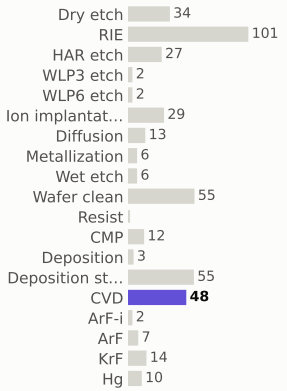
y cut at -1.032 µm



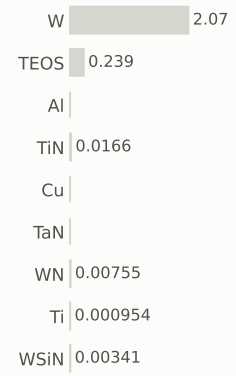
mask for this step

no mask for this step

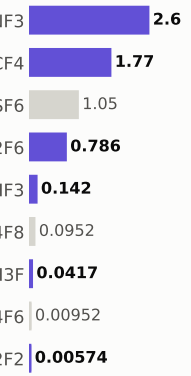
steps by process area



materials deposited (g)



gases emitted (g)



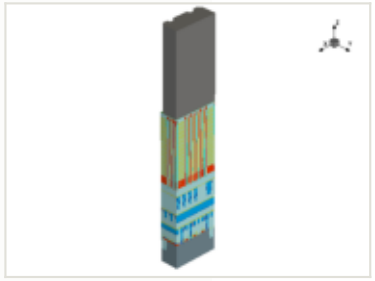
Step #4700: KrF

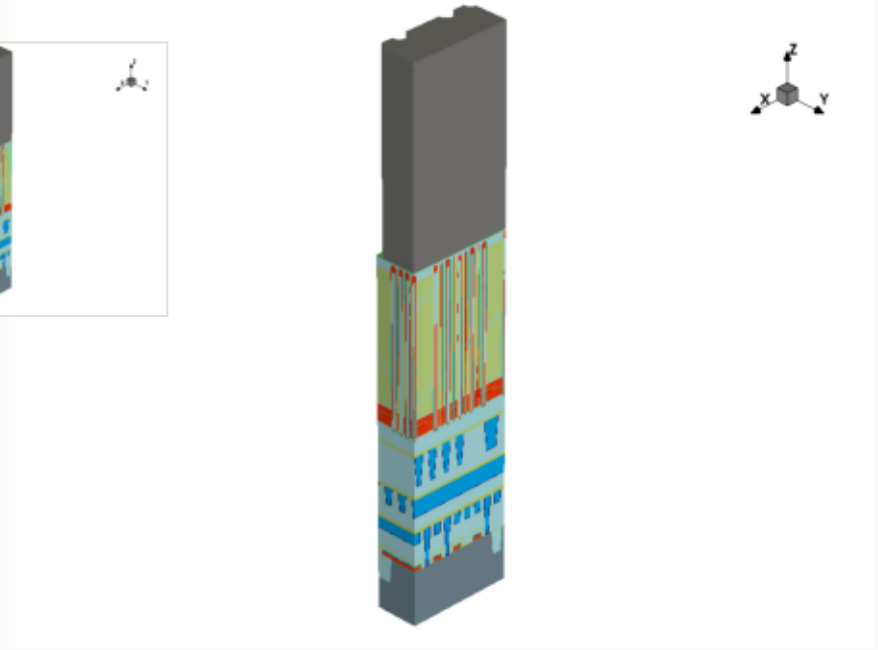
litho_patterning · KrF

0.199	421.8	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



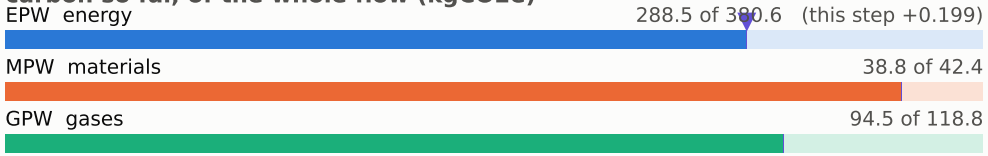


this step

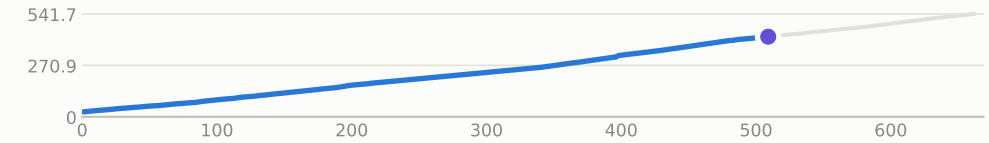
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

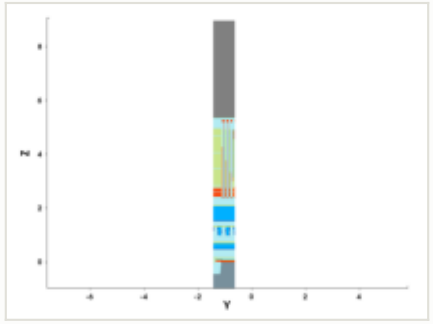
carbon so far, of the whole flow (kgCO2e)



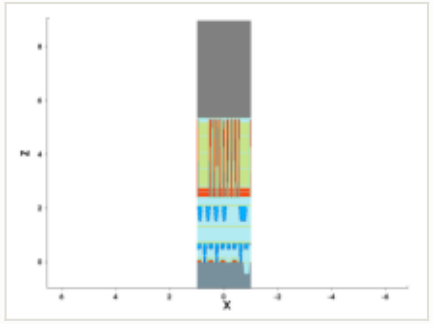
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

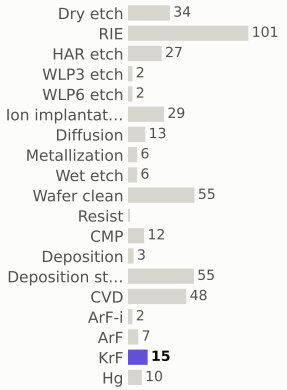


y cut at -1.032 µm

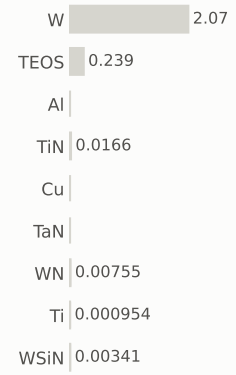


mask: layer 37:0 (OASIS)

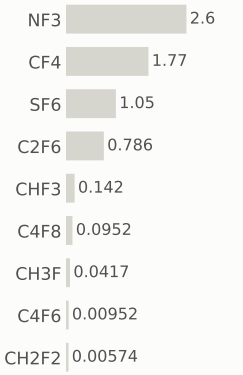
steps by process area



materials deposited (g)



gases emitted (g)



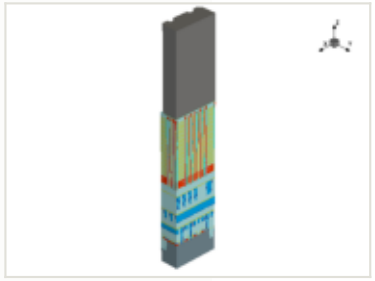
Step #4710: RIE

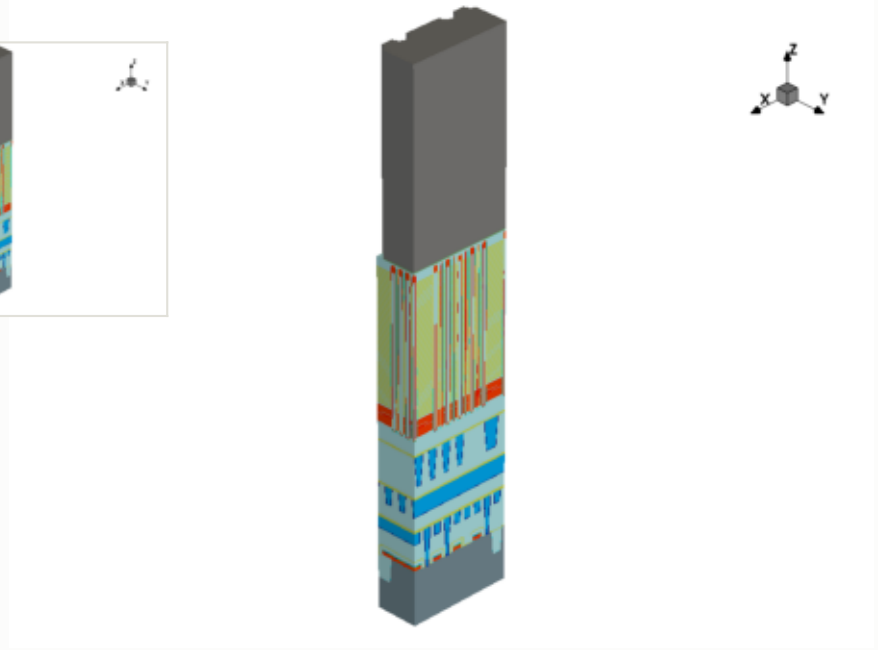
etch_anisotropic · RIE

1.04	422.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



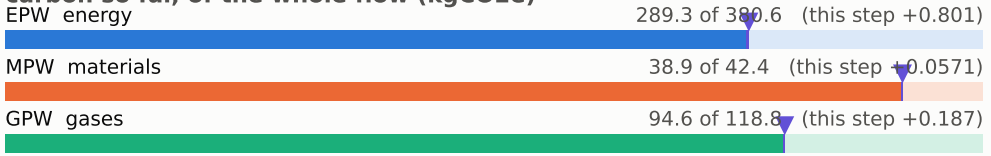


this step

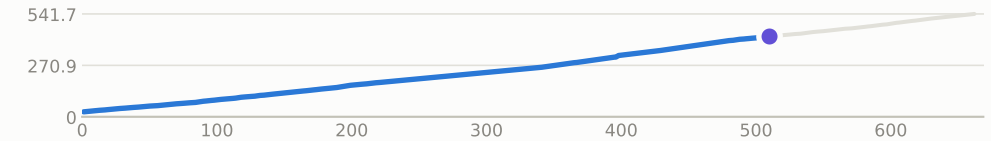
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

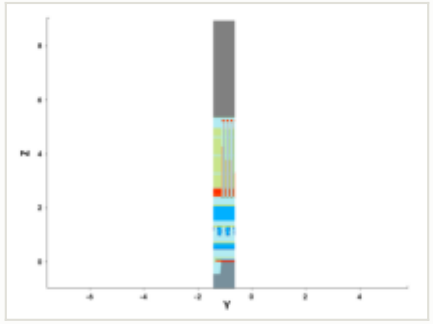
carbon so far, of the whole flow (kgCO2e)



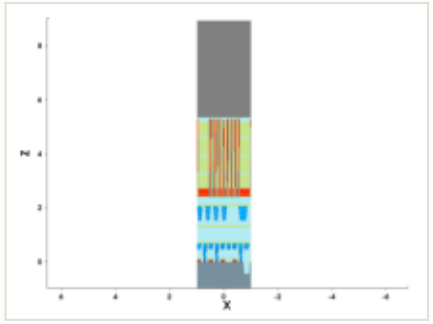
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



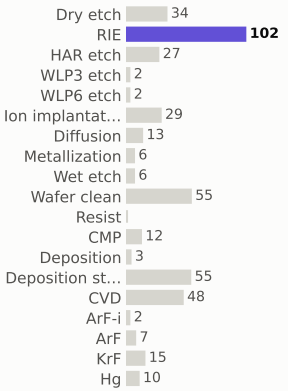
y cut at -1.032 µm



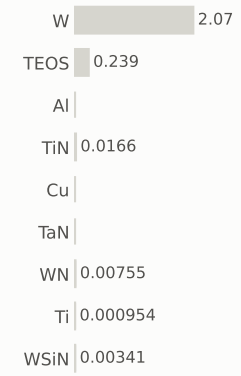
mask for this step

no mask for this step

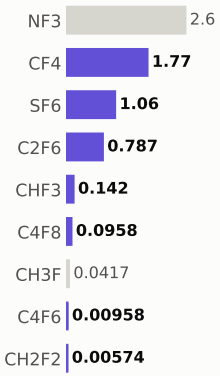
steps by process area



materials deposited (g)



gases emitted (g)



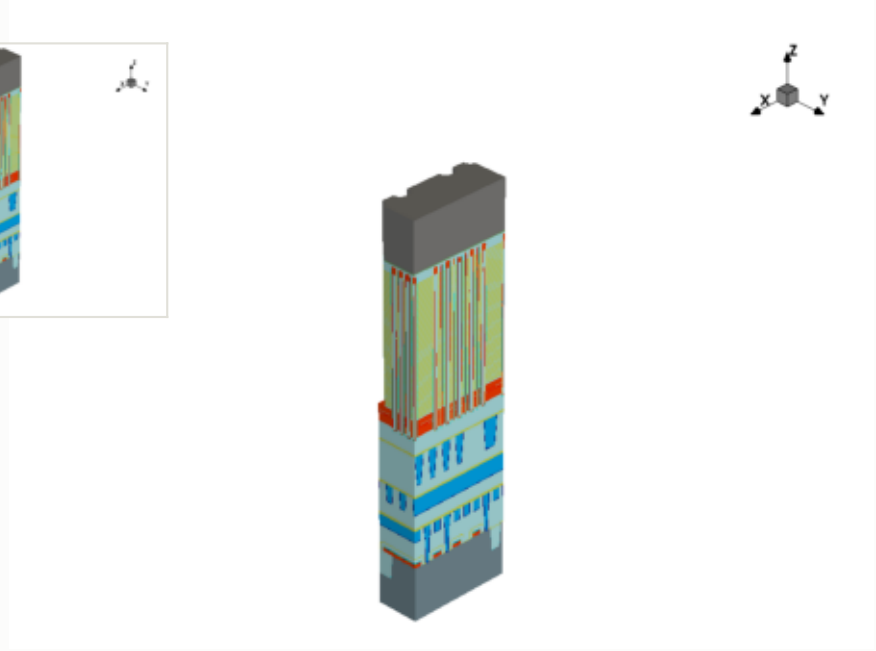
Step #4720: RIE

etch_anisotropic · RIE

1.04	423.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

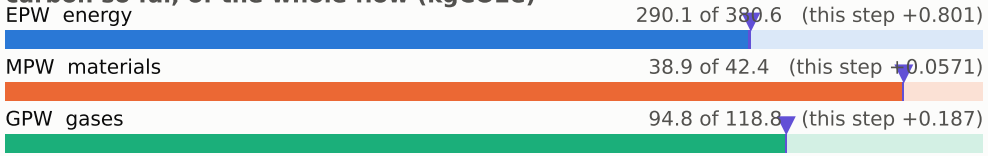


this step

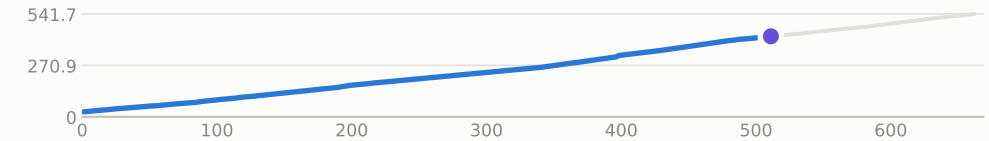
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

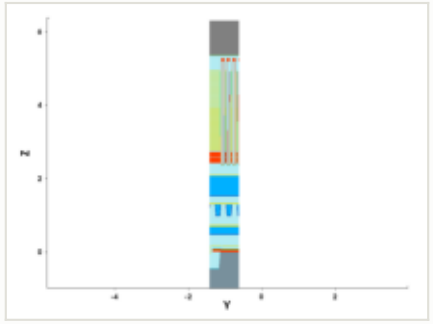
carbon so far, of the whole flow (kgCO2e)



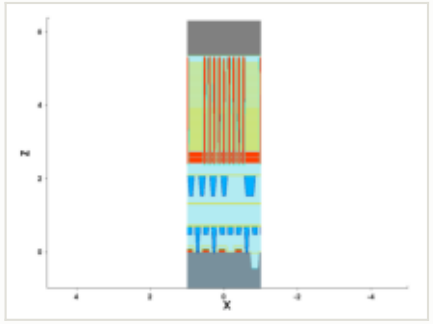
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



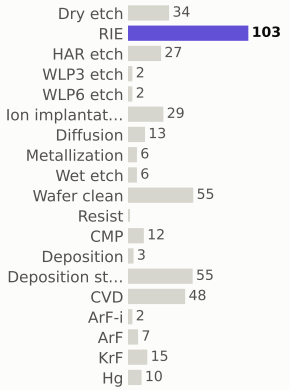
y cut at -1.032 µm



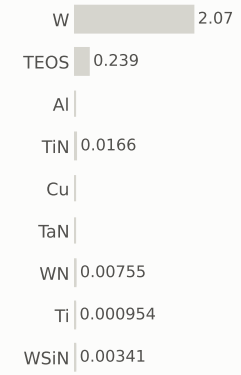
mask for this step

no mask for this step

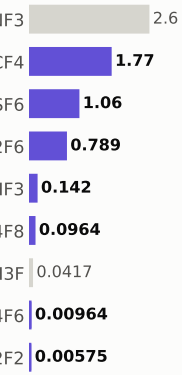
steps by process area



materials deposited (g)



gases emitted (g)



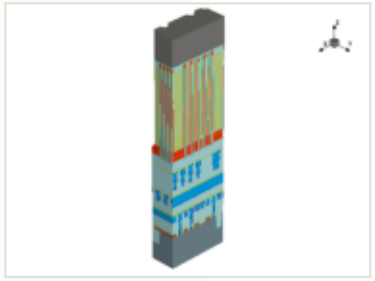
Step #4730: RIE

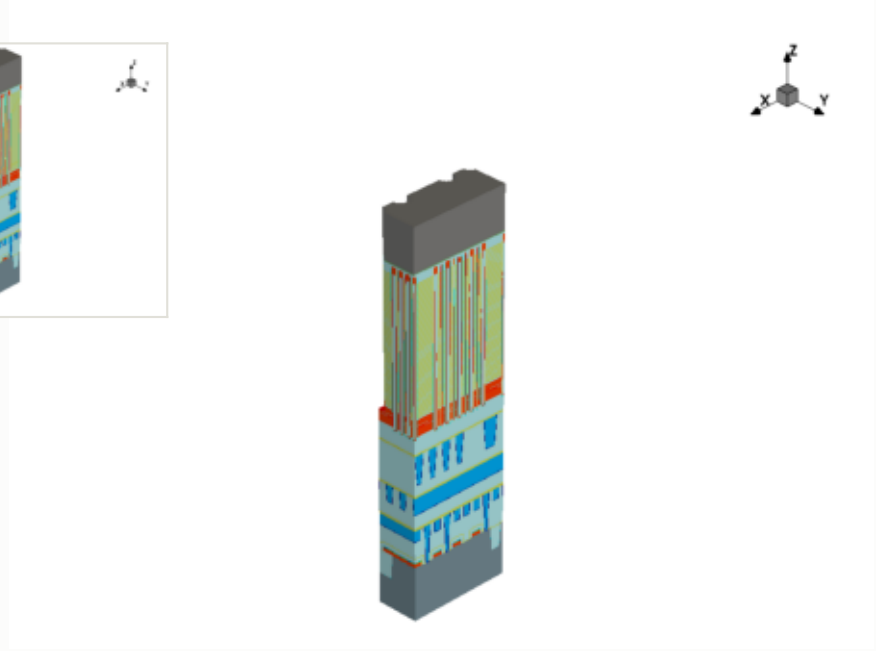
etch_anisotropic · RIE

1.04	424.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



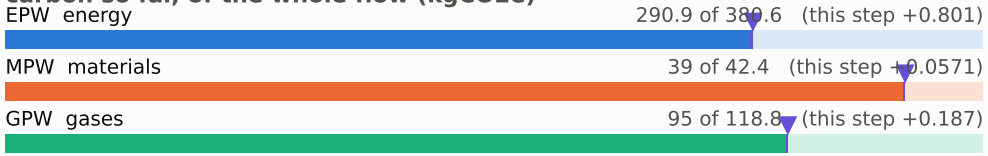


this step

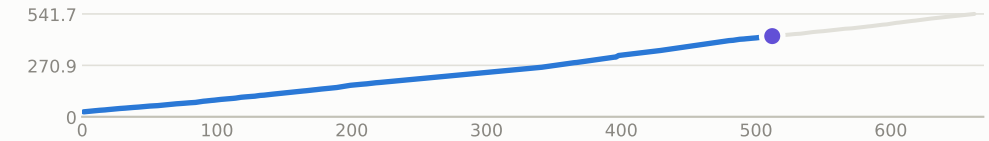
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

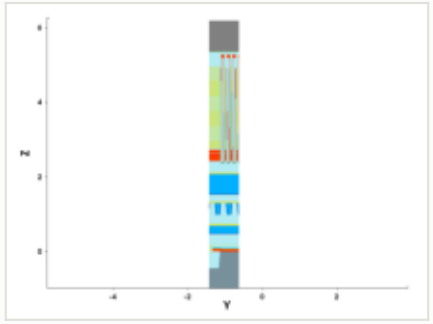
carbon so far, of the whole flow (kgCO2e)



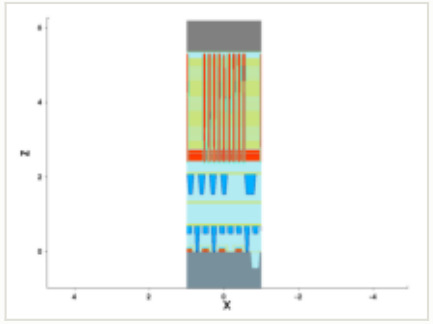
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



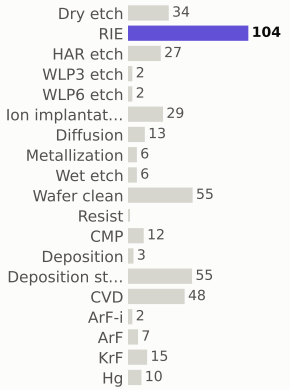
y cut at -1.032 µm



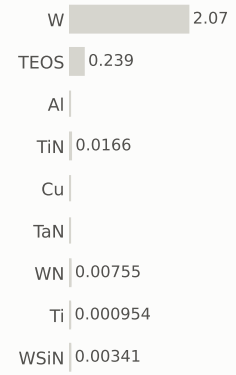
mask for this step

no mask for this step

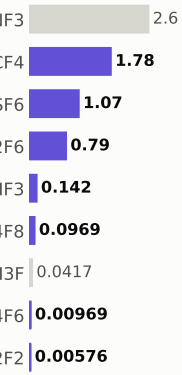
steps by process area



materials deposited (g)



gases emitted (g)



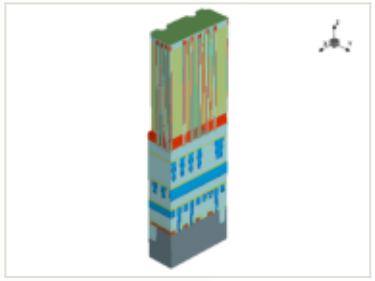
Step #4740: Dry_etch

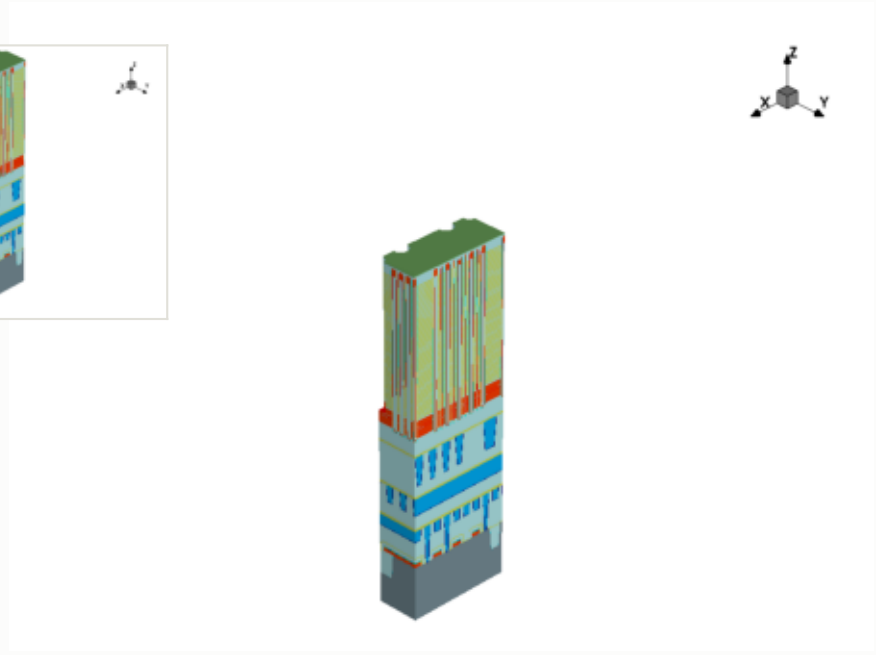
etch_strip · Dry_etch

0.801	425.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

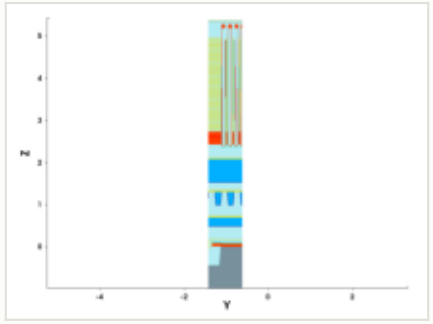
3D structure after this step

zoomed to fit

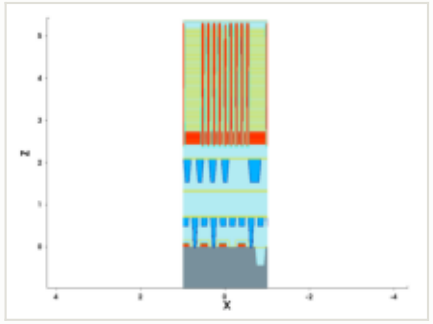




x cut at 0.000 µm



y cut at -1.032 µm



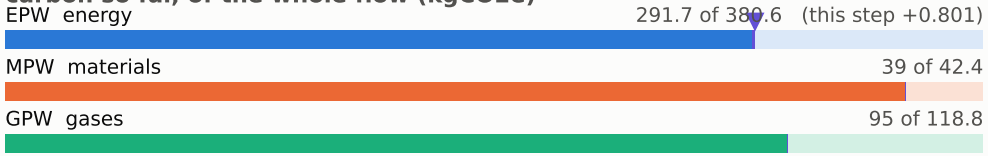
mask for this step

no mask for this step

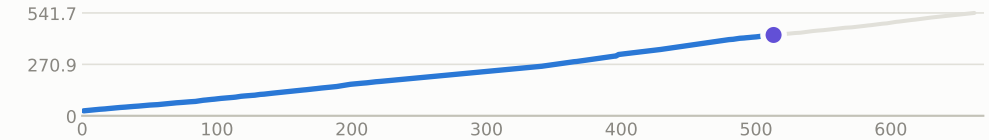
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

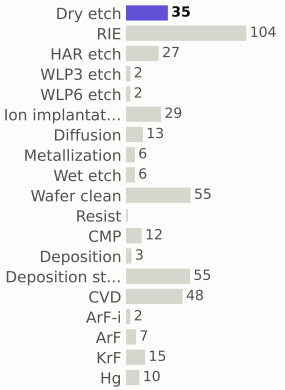
carbon so far, of the whole flow (kgCO2e)



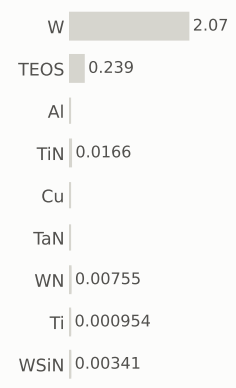
embodied carbon up to each step (kgCO2e per wafer)



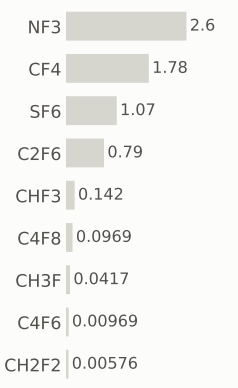
steps by process area



materials deposited (g)



gases emitted (g)



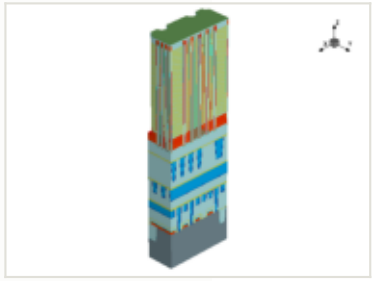
Step #4750: Wafer_clean

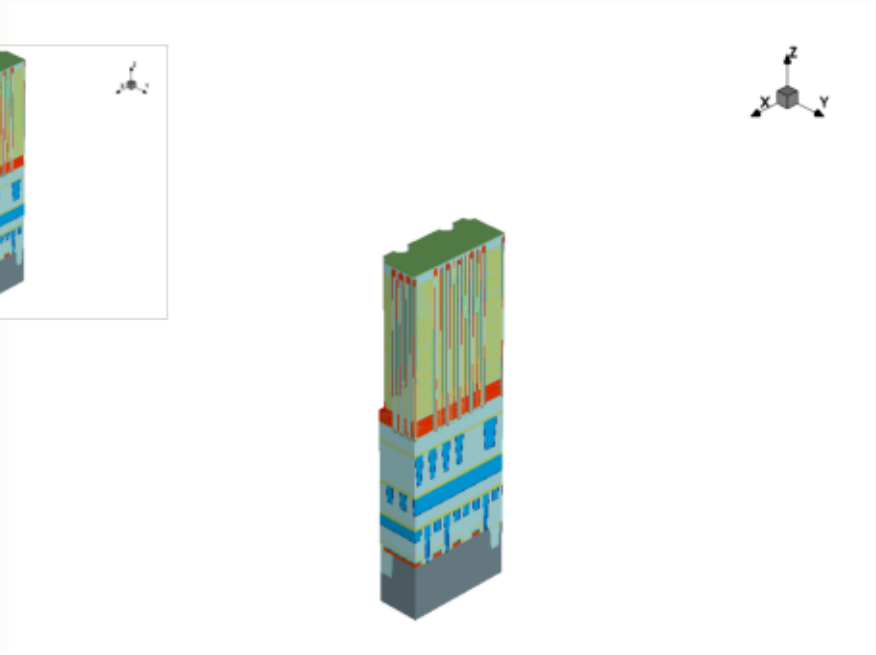
clean · Wafer_clean

0.169	425.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

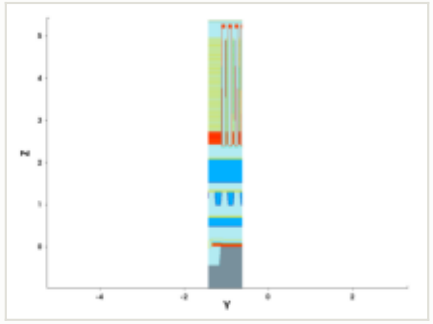
3D structure after this step

zoomed to fit

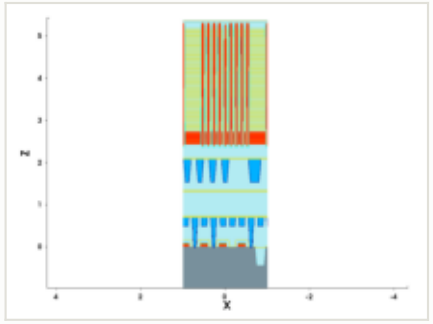




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

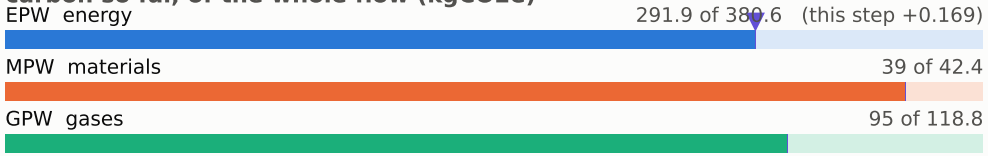
no mask for this step

this step

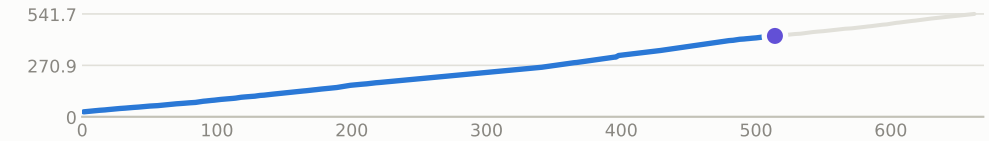
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

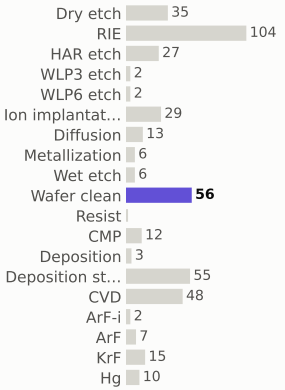
carbon so far, of the whole flow (kgCO2e)



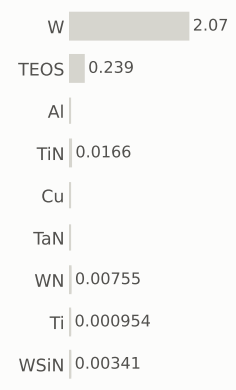
embodied carbon up to each step (kgCO2e per wafer)



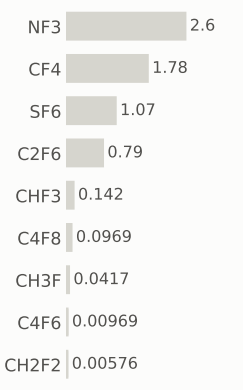
steps by process area



materials deposited (g)



gases emitted (g)



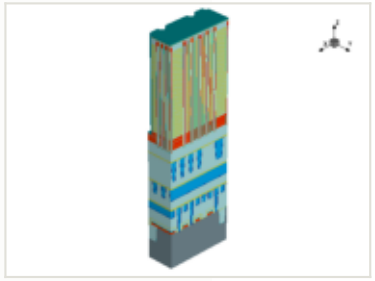
Step #4760: Deposition

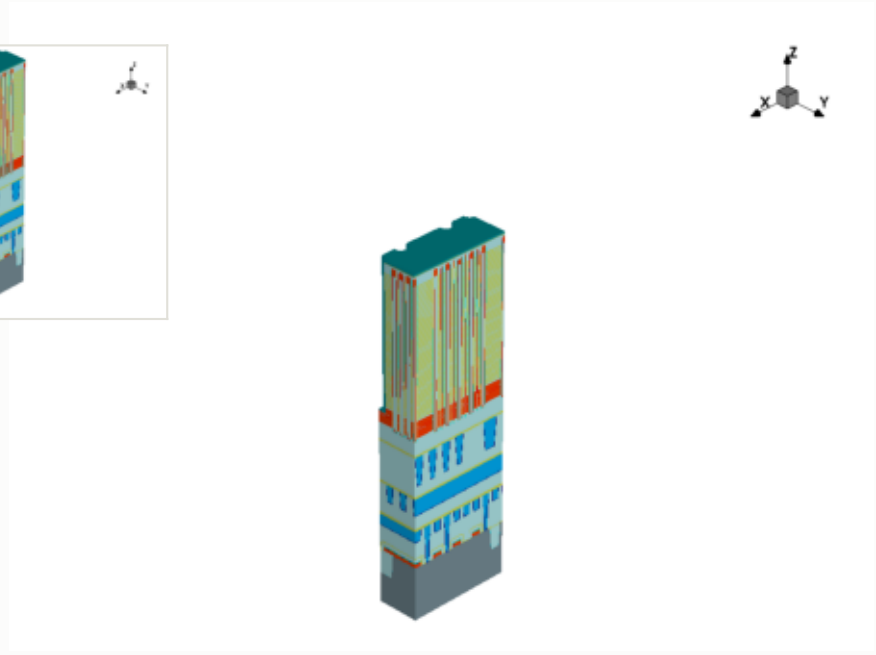
depo_isotropic · Deposition

0.656	426.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



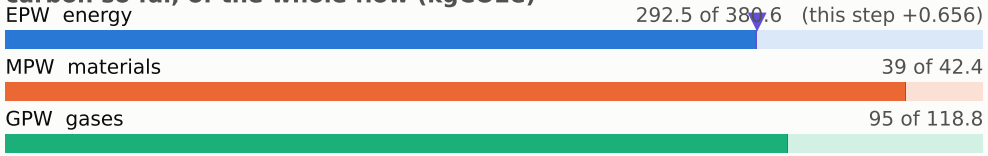


this step

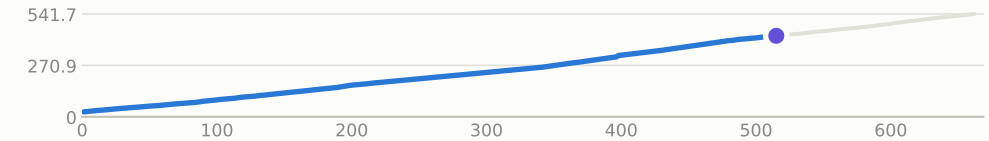
process area	Deposition
material	—
thickness	0.03 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

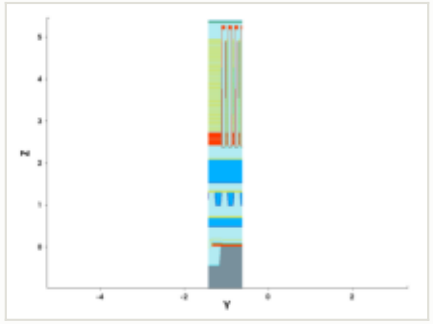
carbon so far, of the whole flow (kgCO2e)



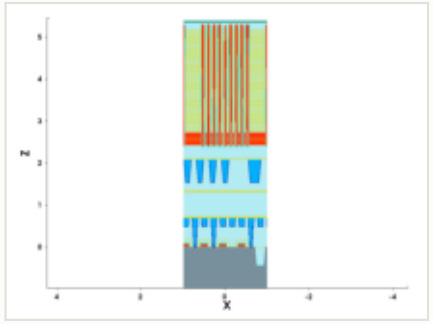
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



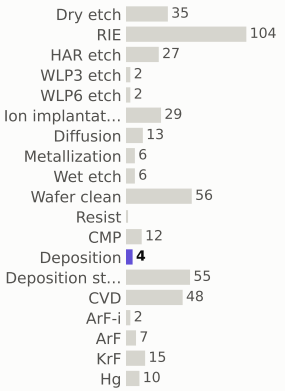
y cut at -1.032 µm



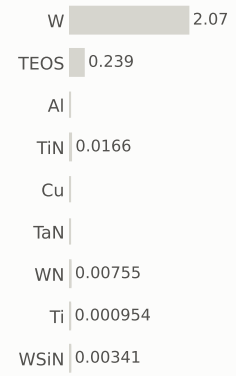
mask for this step

no mask for this step

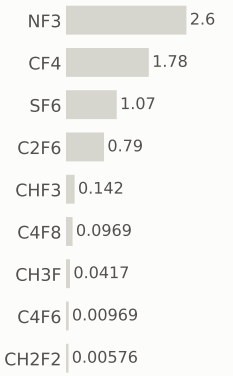
steps by process area



materials deposited (g)



gases emitted (g)



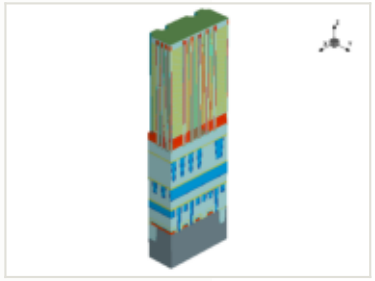
Step #4770: RIE

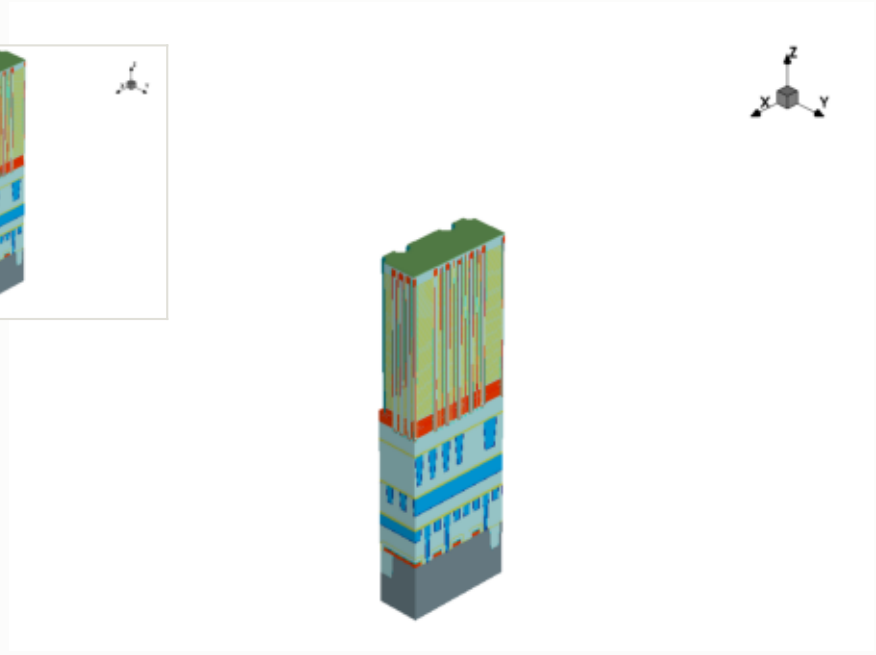
etch_anisotropic · RIE

1.04	427.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



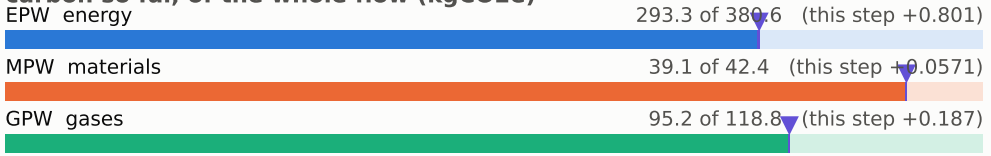


this step

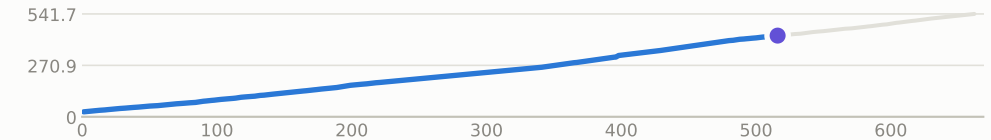
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

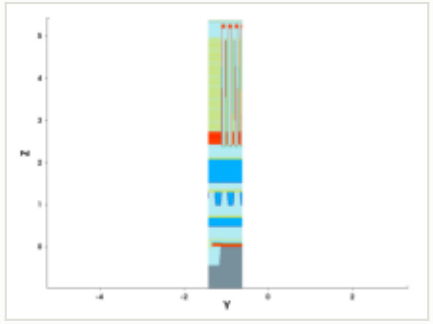
carbon so far, of the whole flow (kgCO2e)



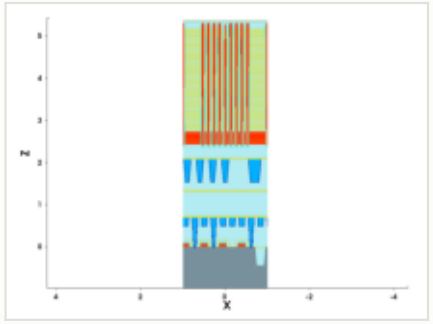
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



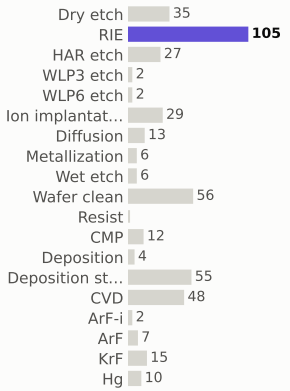
y cut at -1.032 µm



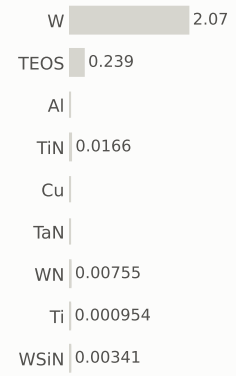
mask for this step

no mask for this step

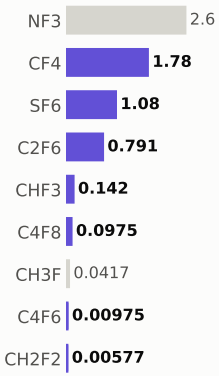
steps by process area



materials deposited (g)



gases emitted (g)



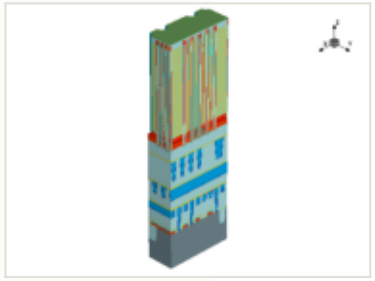
Step #4780: Wet_etch

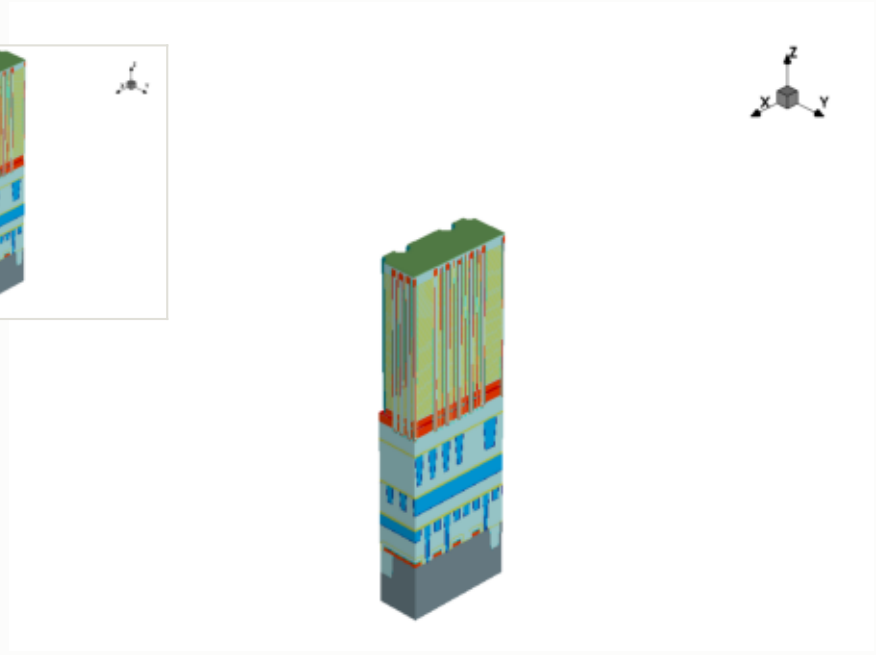
etch_isotropic · Wet_etch

0.169	427.8	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

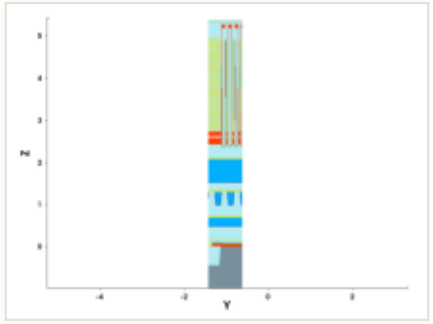
3D structure after this step

zoomed to fit

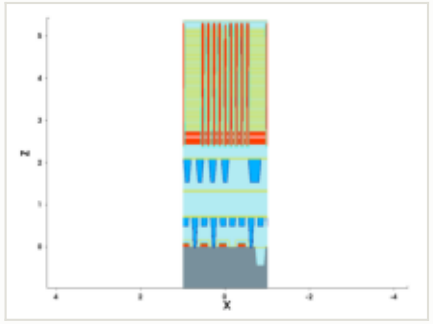




x cut at 0.000 µm



y cut at -1.032 µm



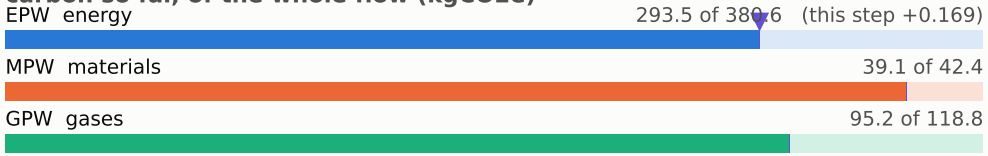
mask for this step

no mask for this step

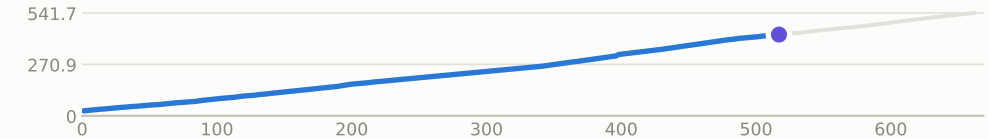
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

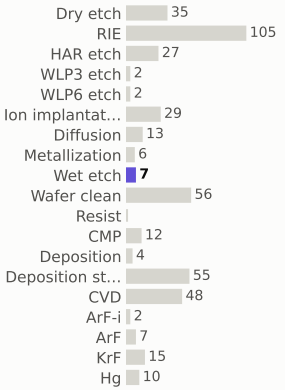
carbon so far, of the whole flow (kgCO2e)



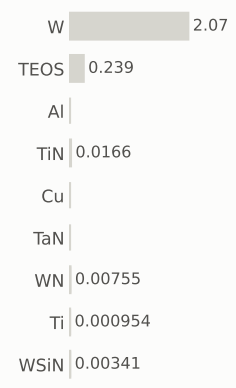
embodied carbon up to each step (kgCO2e per wafer)



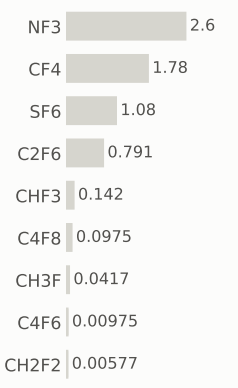
steps by process area



materials deposited (g)



gases emitted (g)



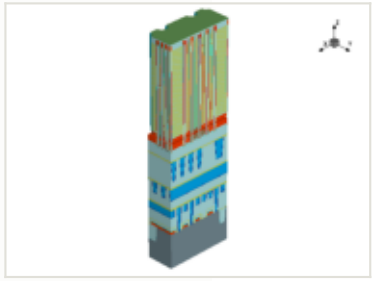
Step #4790: Wet_etch

etch_isotropic · Wet_etch

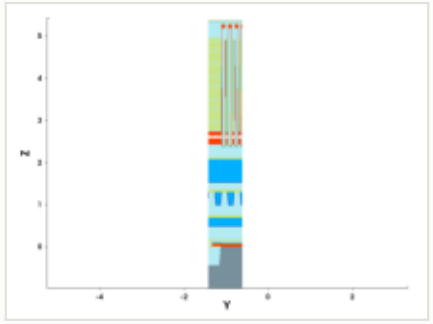
0.169	427.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

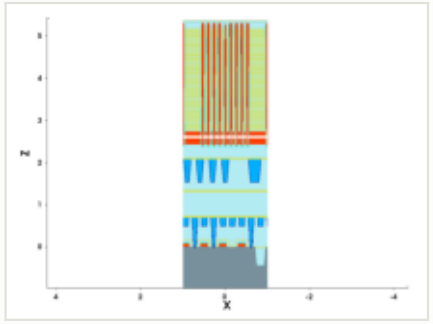
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



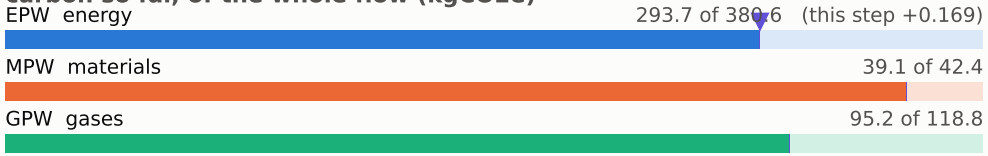
mask for this step

no mask for this step

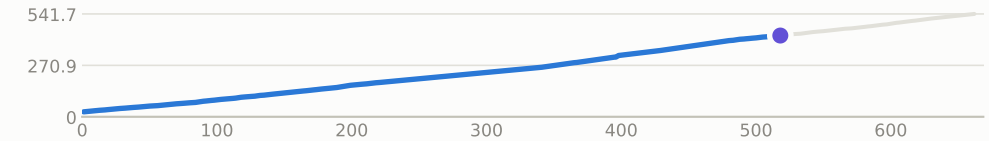
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

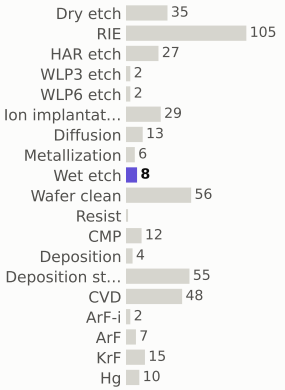
carbon so far, of the whole flow (kgCO2e)



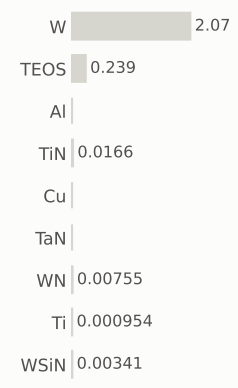
embodied carbon up to each step (kgCO2e per wafer)



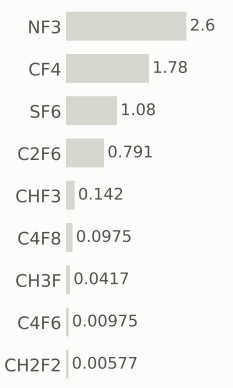
steps by process area



materials deposited (g)



gases emitted (g)



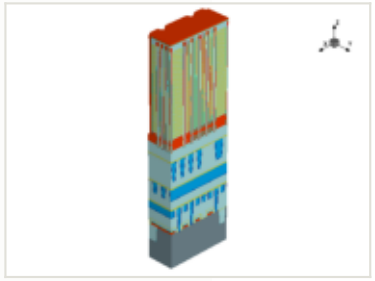
Step #4800: CVD

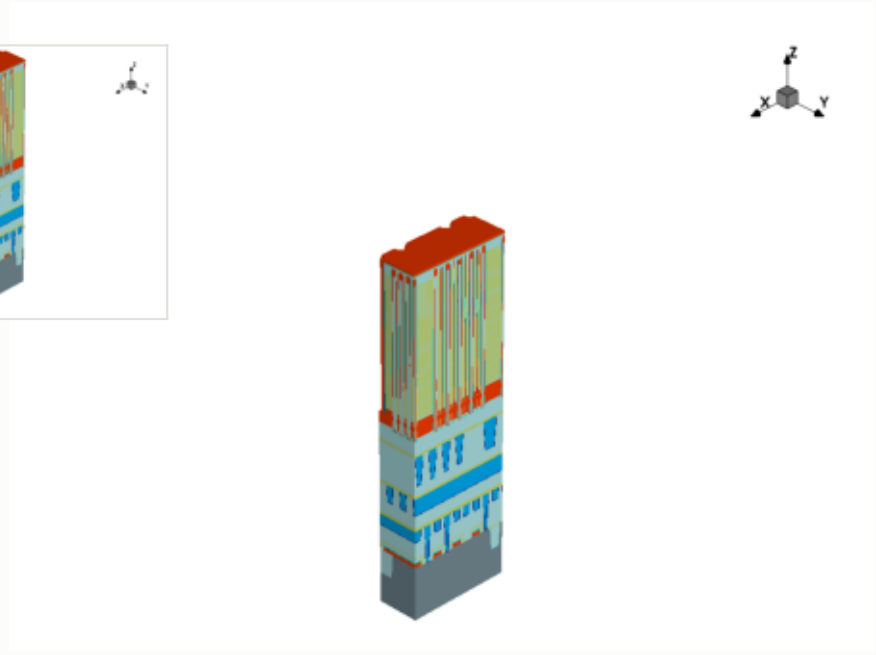
depo_isotropic · CVD

1.31	429.3	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



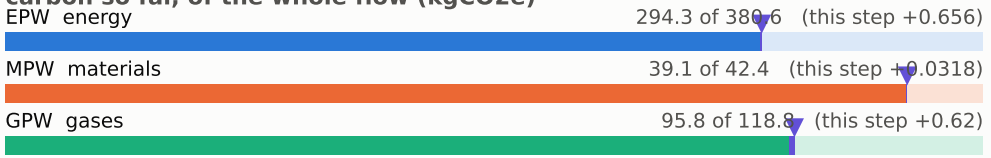


this step

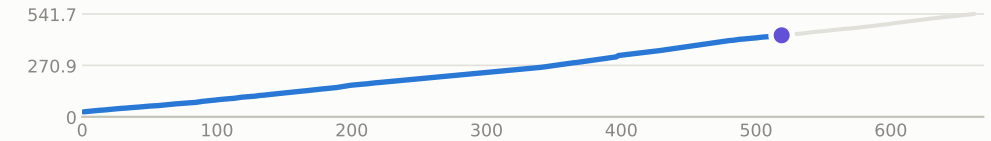
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

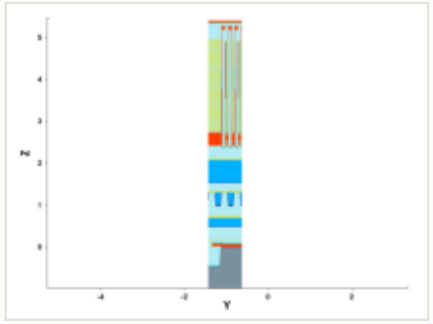
carbon so far, of the whole flow (kgCO2e)



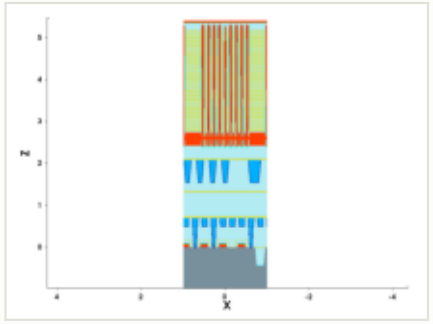
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



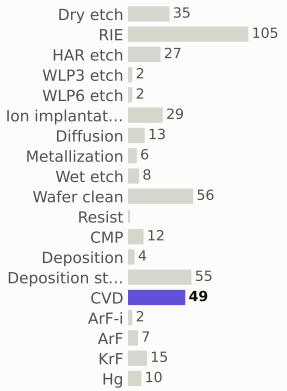
y cut at -1.032 µm



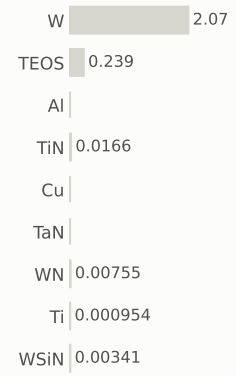
mask for this step

no mask for this step

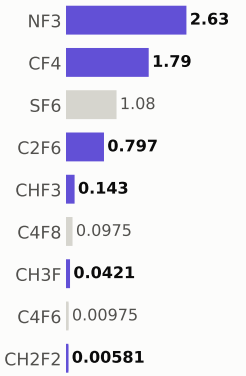
steps by process area



materials deposited (g)



gases emitted (g)



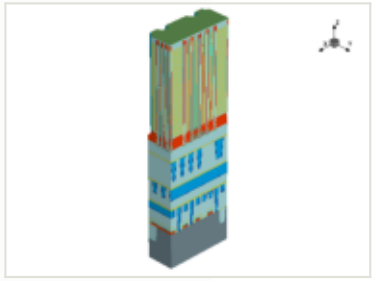
Step #4810: RIE

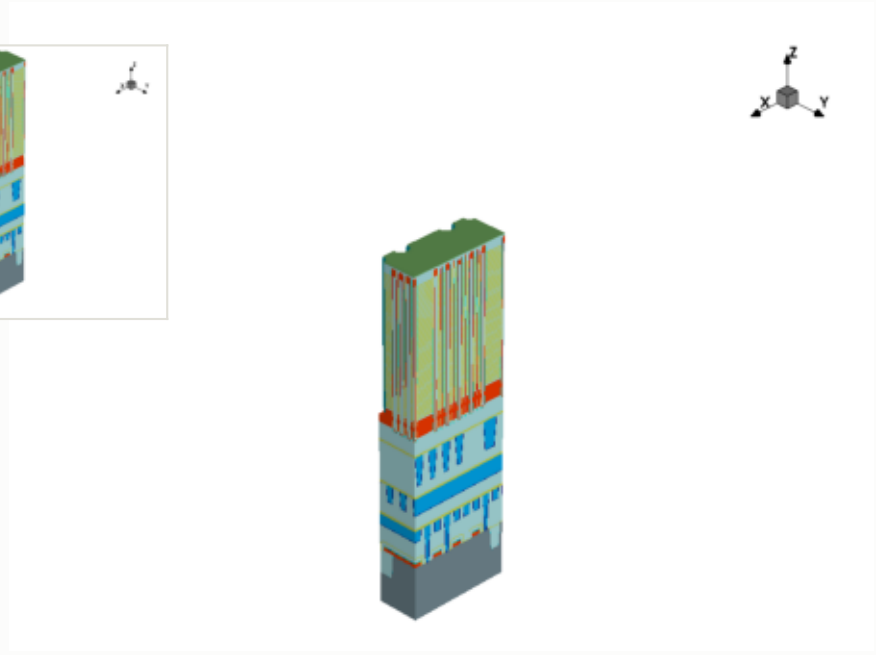
etch_isotropic · RIE

1.04	430.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

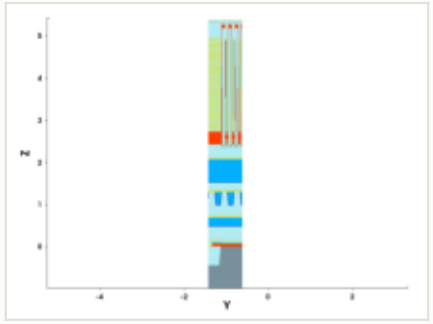
3D structure after this step

zoomed to fit

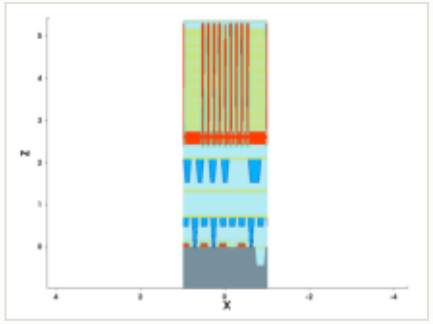




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

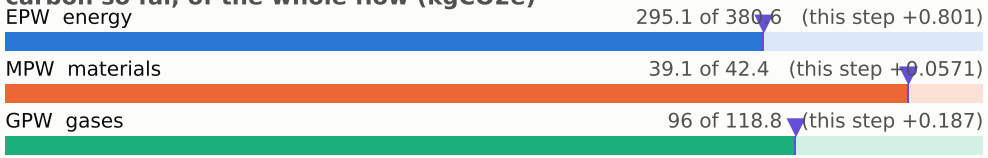
no mask for this step

this step

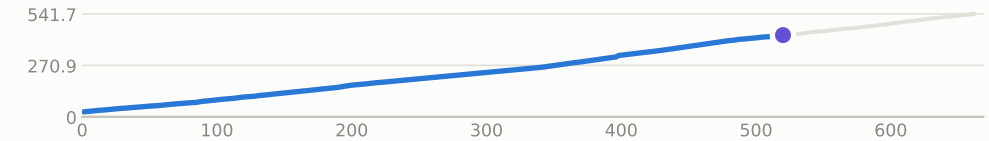
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

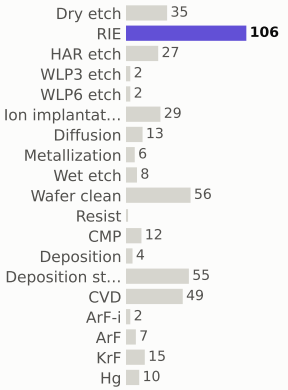
carbon so far, of the whole flow (kgCO2e)



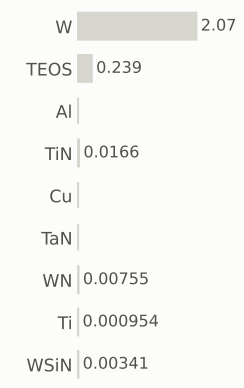
embodied carbon up to each step (kgCO2e per wafer)



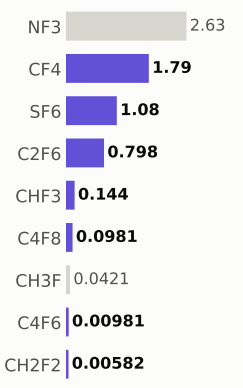
steps by process area



materials deposited (g)



gases emitted (g)



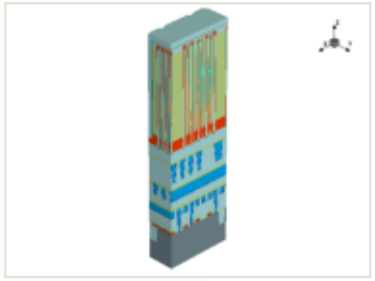
Step #4820: CVD

depo_isotropic · CVD

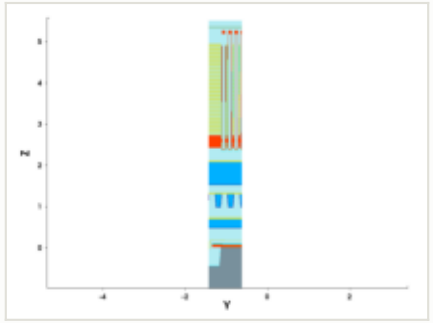
1.31	431.6	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

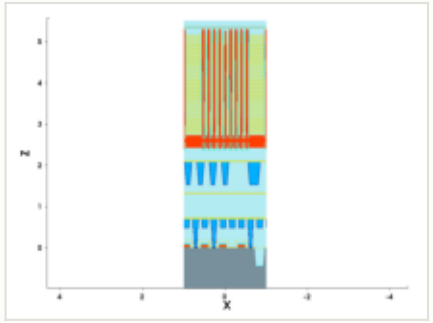
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

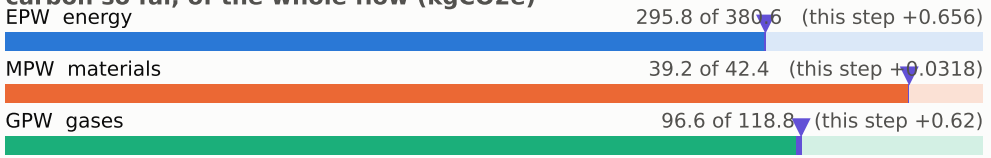
no mask for this step

this step

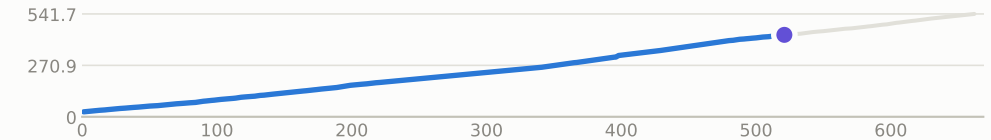
process area	CVD
material	—
thickness	0.15 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

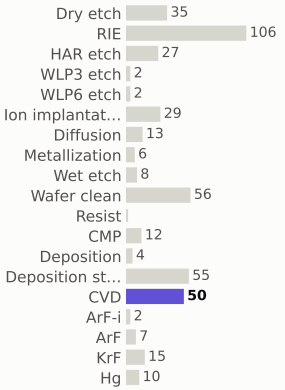
carbon so far, of the whole flow (kgCO2e)



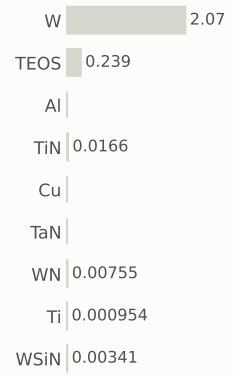
embodied carbon up to each step (kgCO2e per wafer)



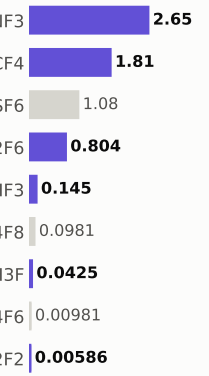
steps by process area



materials deposited (g)



gases emitted (g)



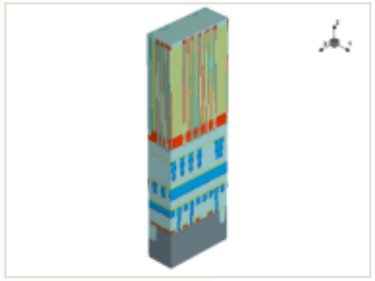
Step #4830: CMP

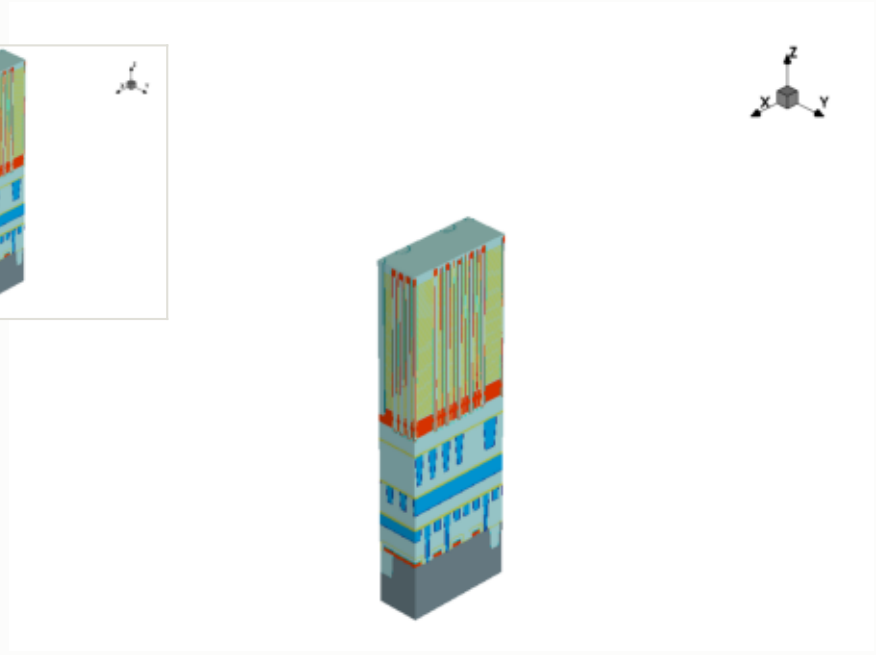
etch_cmp · CMP

0.169	431.8	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



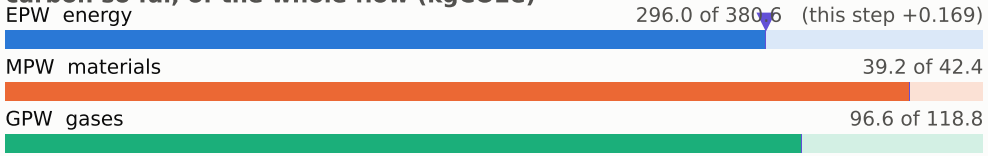


this step

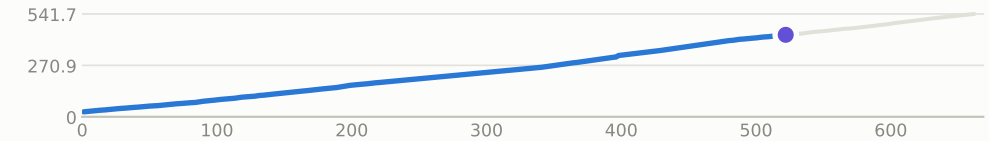
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

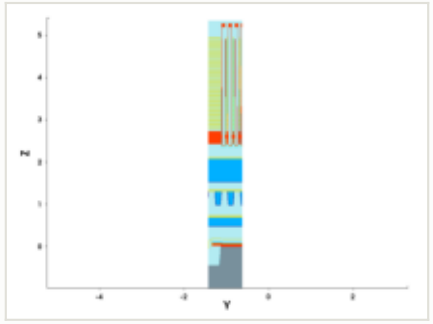
carbon so far, of the whole flow (kgCO2e)



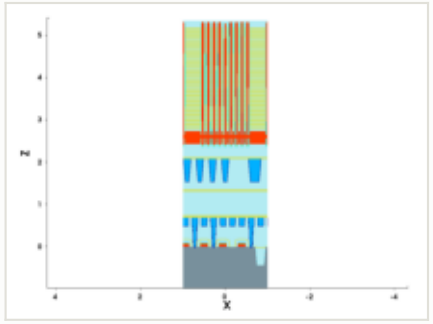
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



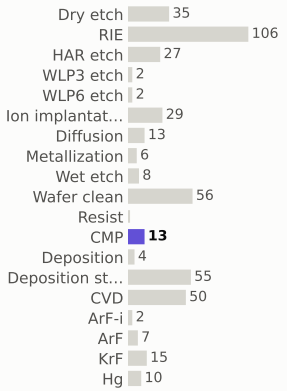
y cut at -1.032 µm



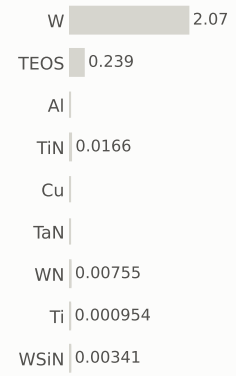
mask for this step

no mask for this step

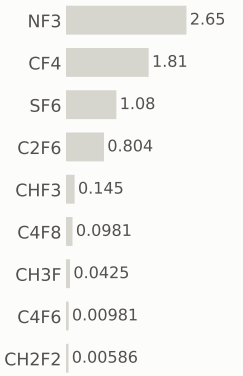
steps by process area



materials deposited (g)



gases emitted (g)



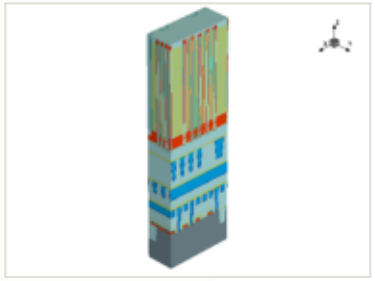
Step #4840: Wafer_clean

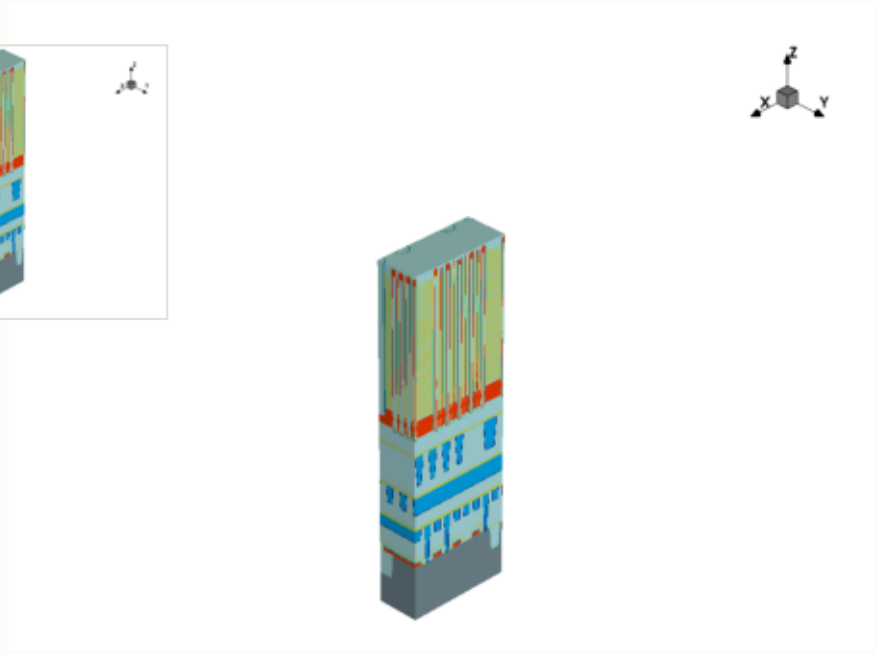
clean · Wafer_clean

0.169	431.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

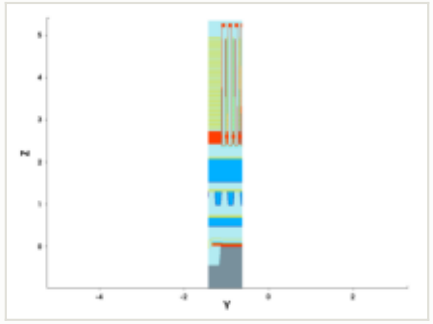
3D structure after this step

zoomed to fit

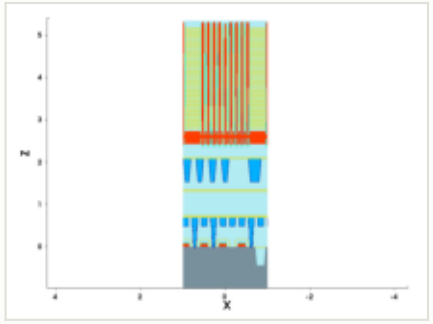




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

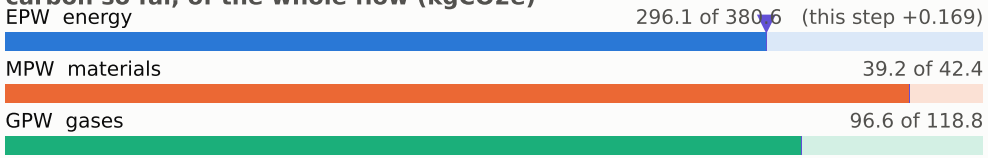
no mask for this step

this step

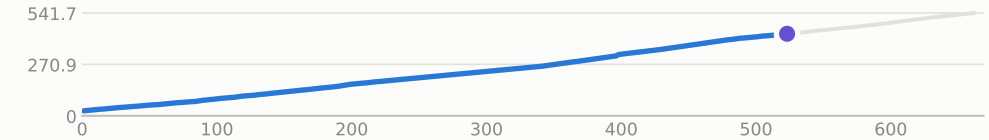
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

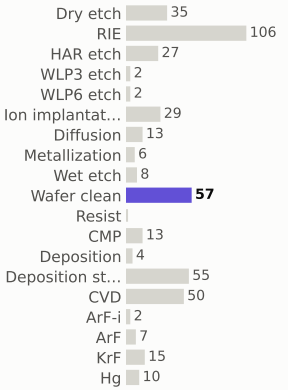
carbon so far, of the whole flow (kgCO2e)



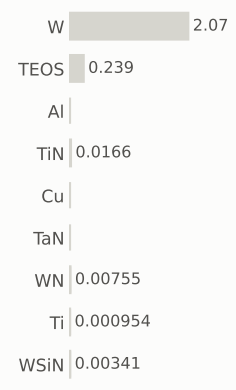
embodied carbon up to each step (kgCO2e per wafer)



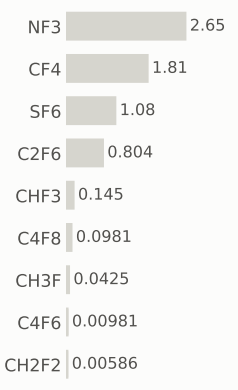
steps by process area



materials deposited (g)



gases emitted (g)



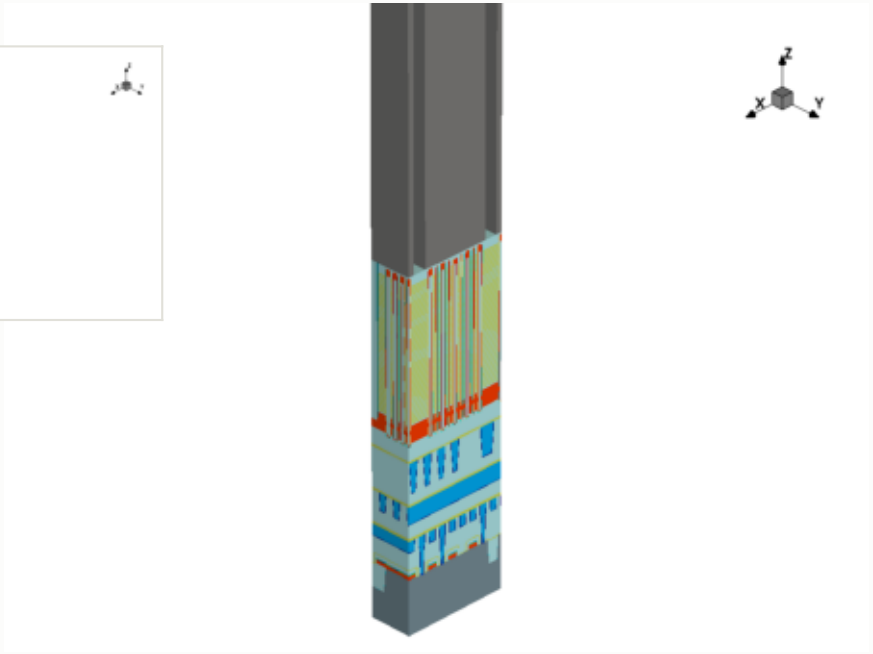
Step #4860: ArF

litho_patterning · ArF

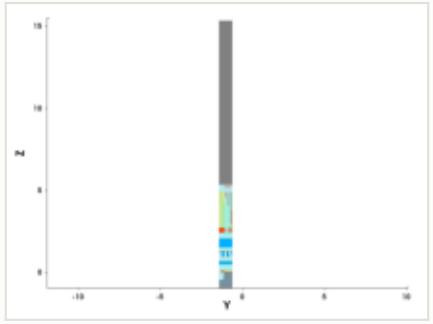
0.199	432.1	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

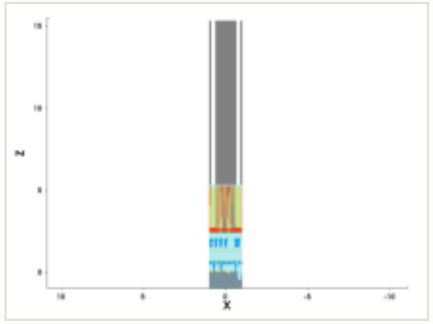
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



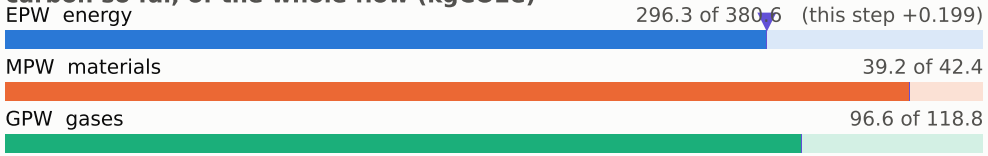
mask: layer 38:0 (OASIS)

this step

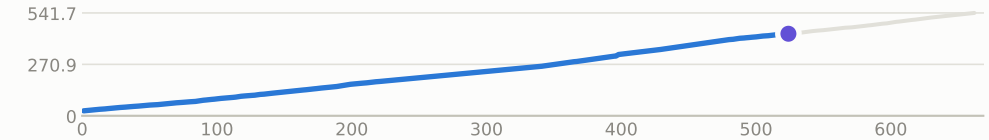
process area	ArF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

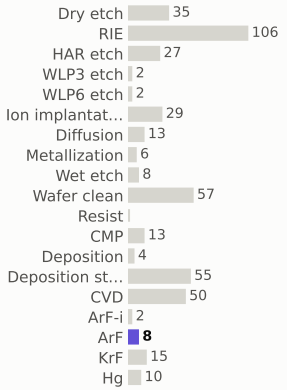
carbon so far, of the whole flow (kgCO2e)



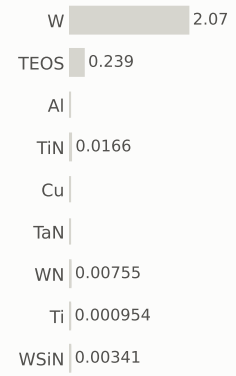
embodied carbon up to each step (kgCO2e per wafer)



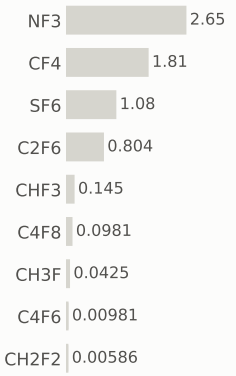
steps by process area



materials deposited (g)



gases emitted (g)



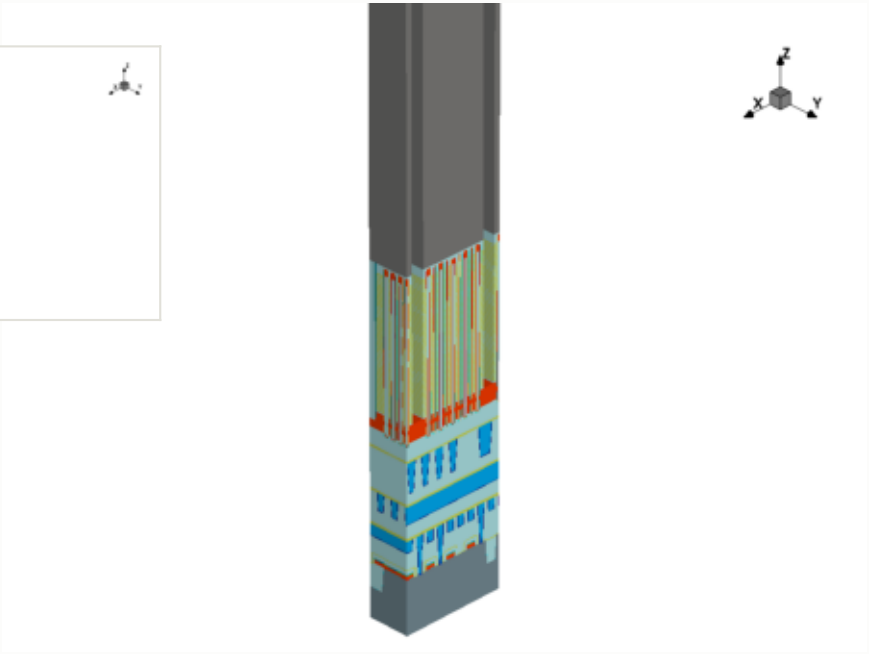
Step #4870: HAR_etch

etch_anisotropic · HAR_etch

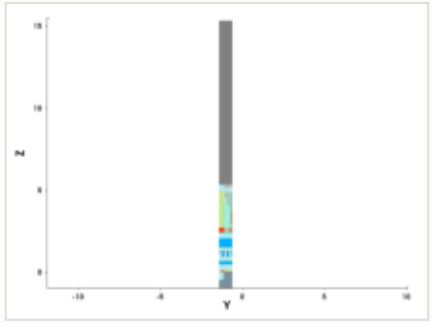
1.04	433.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

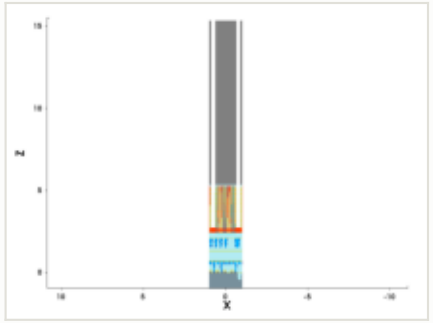
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

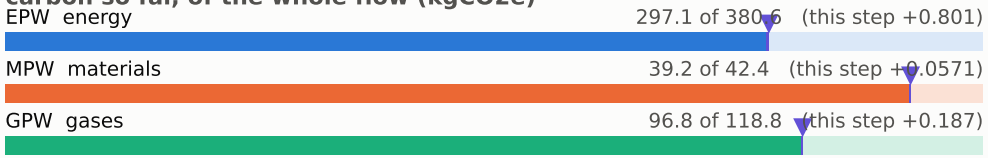
no mask for this step

this step

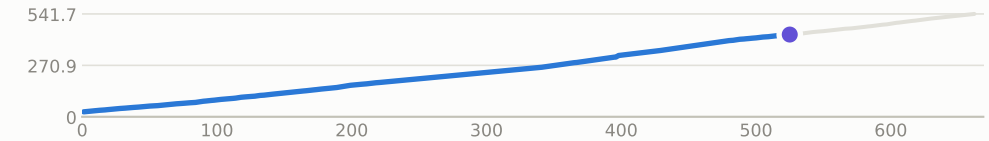
process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

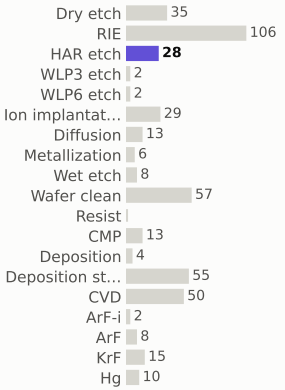
carbon so far, of the whole flow (kgCO2e)



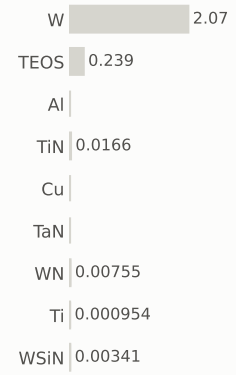
embodied carbon up to each step (kgCO2e per wafer)



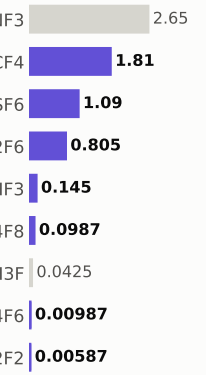
steps by process area



materials deposited (g)



gases emitted (g)



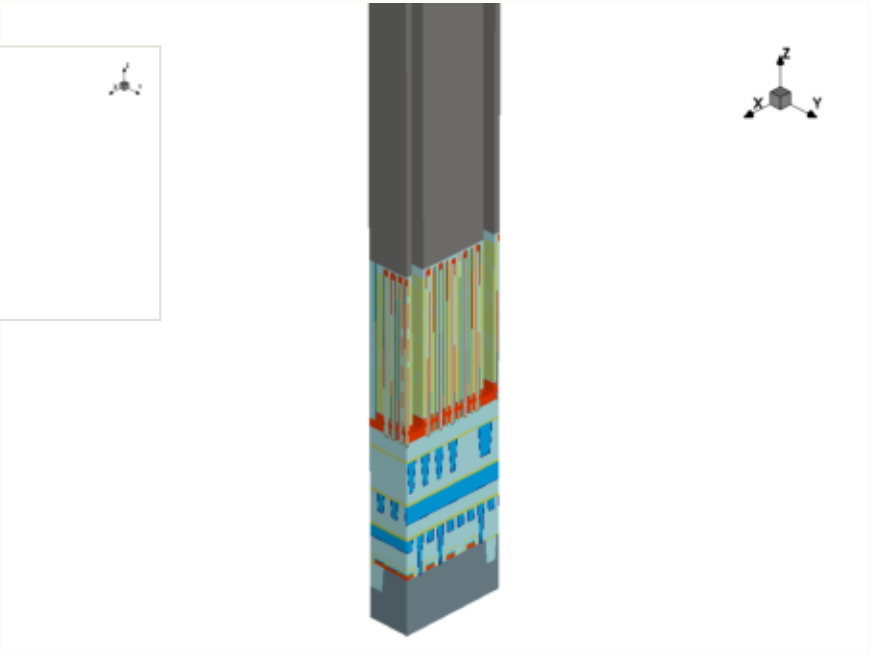
Step #4880: HAR_etch

etch_multi_offset · HAR_etch

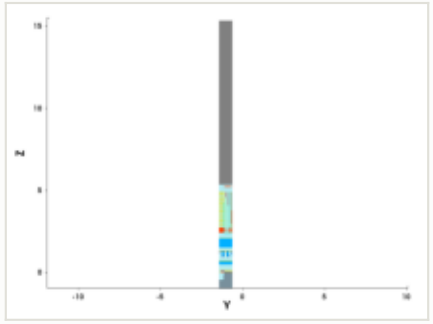
1.04	434.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

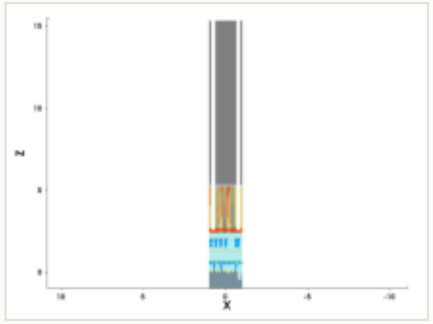
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

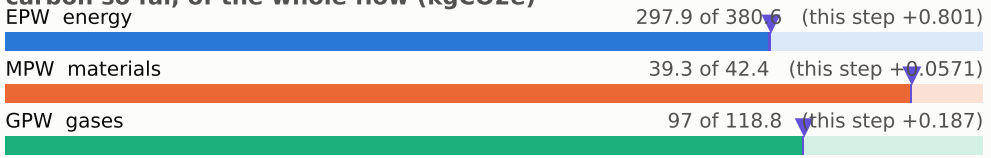
no mask for this step

this step

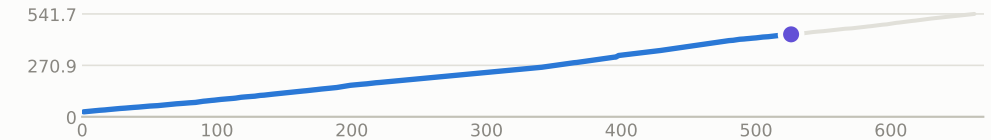
process area	HAR_etch
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

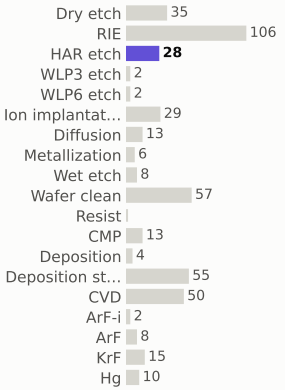
carbon so far, of the whole flow (kgCO2e)



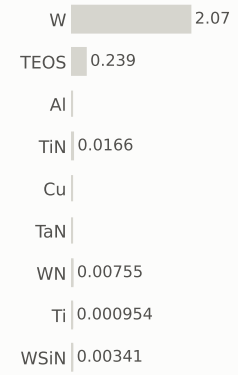
embodied carbon up to each step (kgCO2e per wafer)



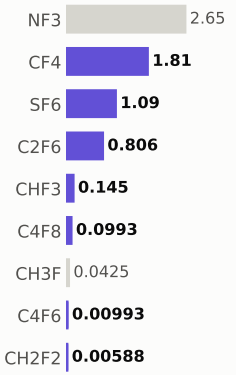
steps by process area



materials deposited (g)



gases emitted (g)



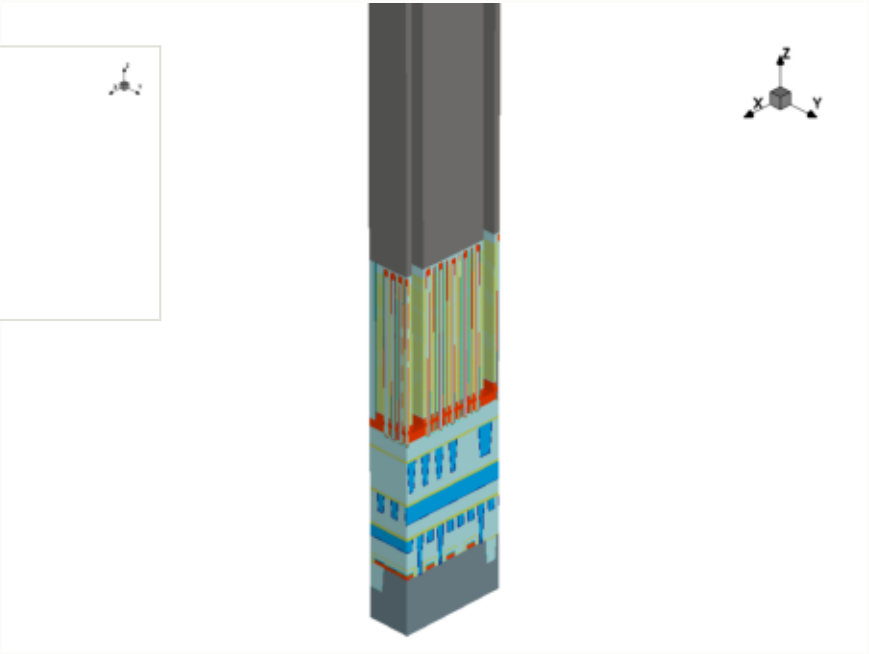
Step #4890: Ion_implantation

implant_ellipsoid · Ion_implantation

0.801	435.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

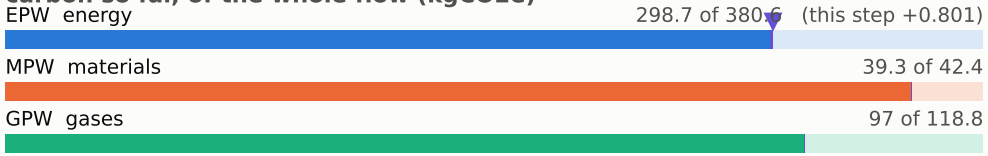


this step

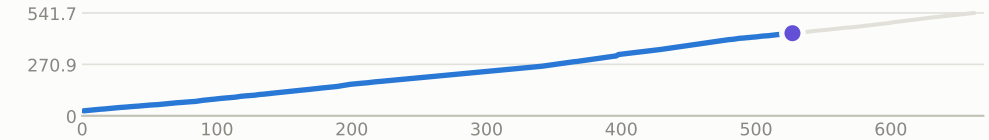
process area	Ion_implantation	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: AffectedMaterialsFilter, Depth, Lateral, MaterialFilter, Primary, RemoveEmptyRegions

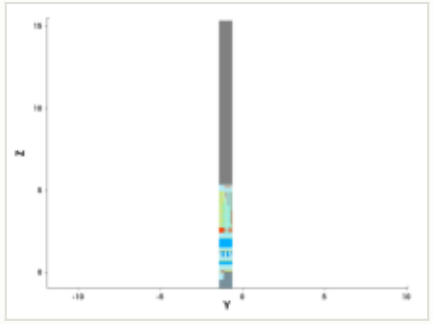
carbon so far, of the whole flow (kgCO2e)



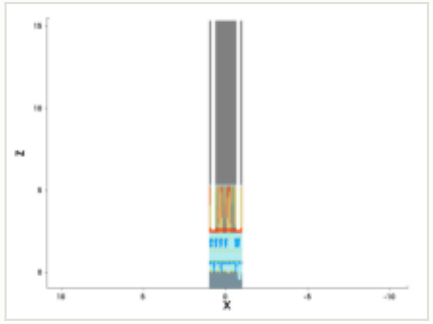
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



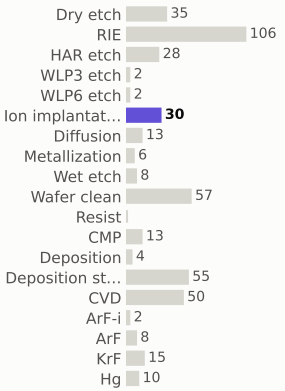
y cut at -1.032 µm



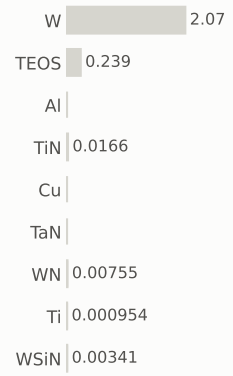
mask for this step

no mask for this step

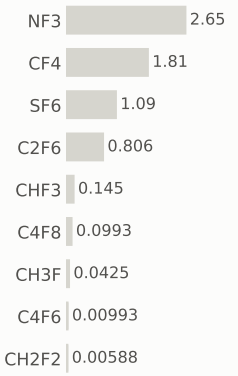
steps by process area



materials deposited (g)



gases emitted (g)



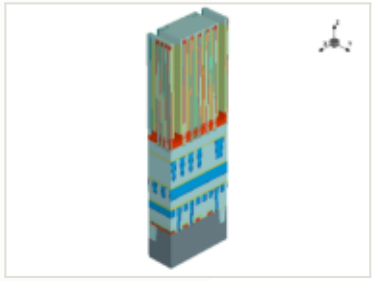
Step #4900: Dry_etch

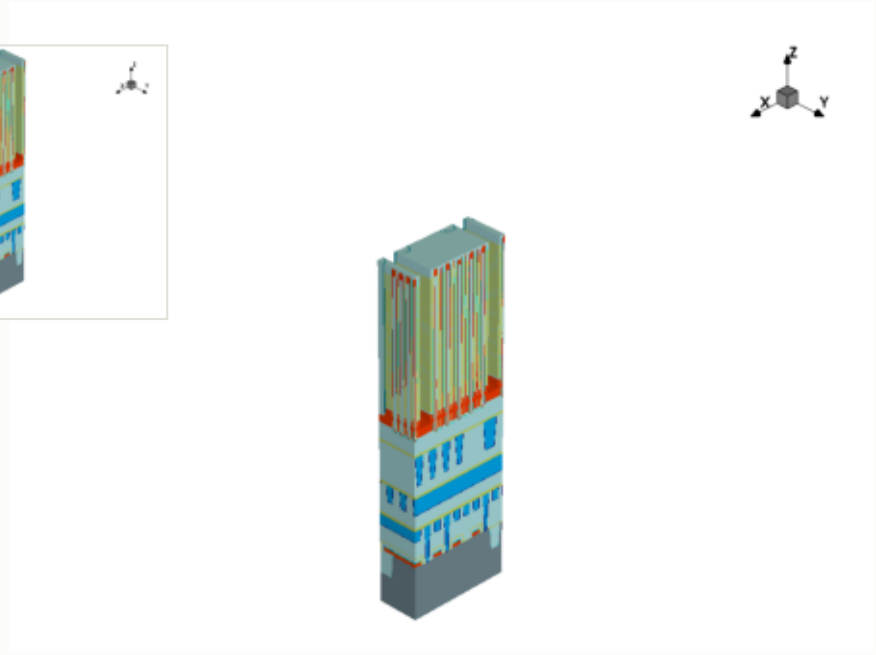
etch_strip · Dry_etch

0.801	435.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

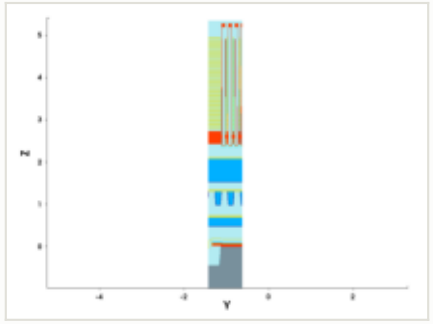
3D structure after this step

zoomed to fit

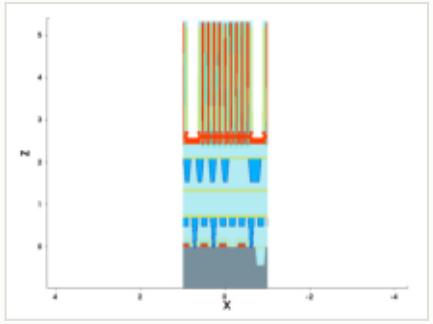




x cut at 0.000 µm



y cut at -1.032 µm



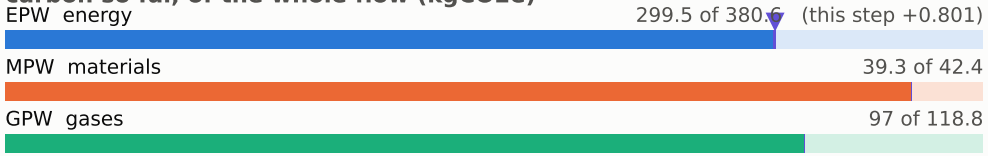
mask for this step

no mask for this step

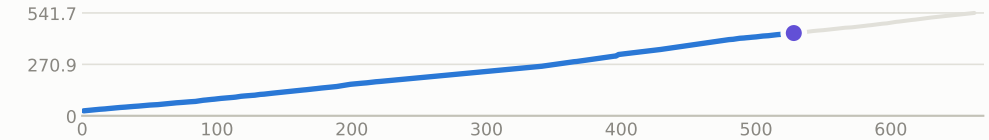
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

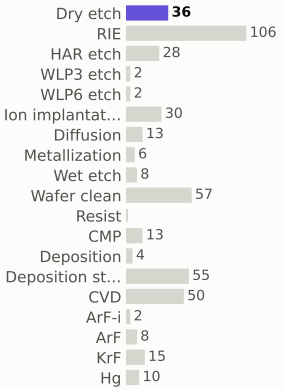
carbon so far, of the whole flow (kgCO2e)



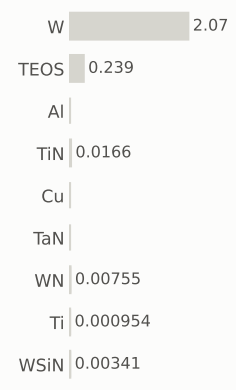
embodied carbon up to each step (kgCO2e per wafer)



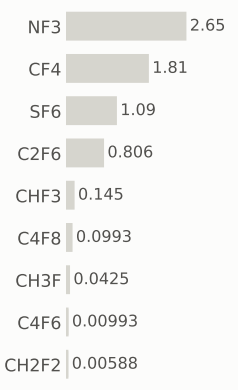
steps by process area



materials deposited (g)



gases emitted (g)



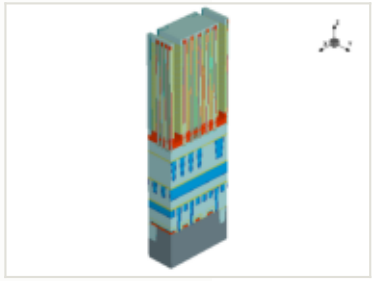
Step #4910: Wafer_clean

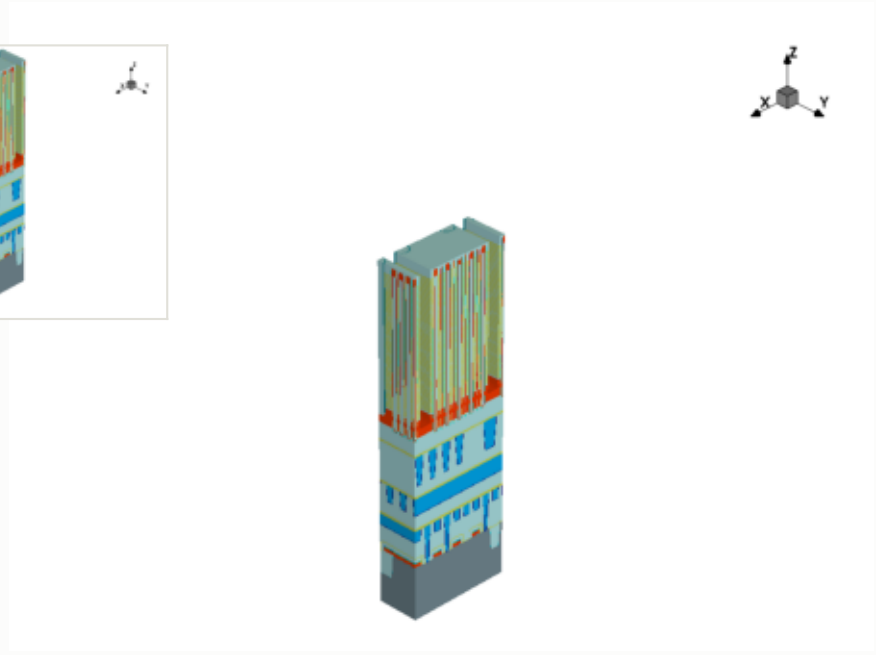
clean · Wafer_clean

0.169	436.0	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

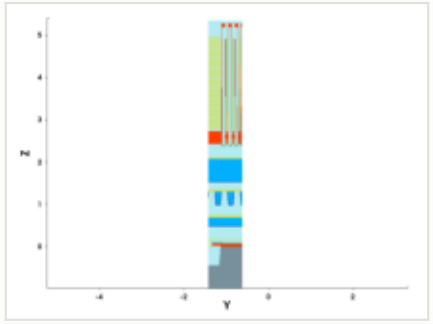
3D structure after this step

zoomed to fit

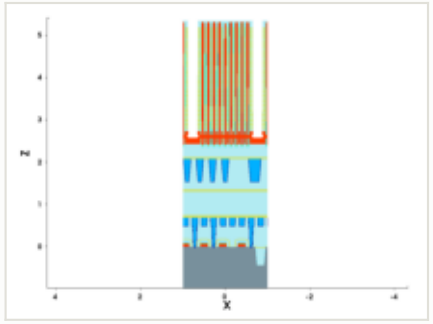




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

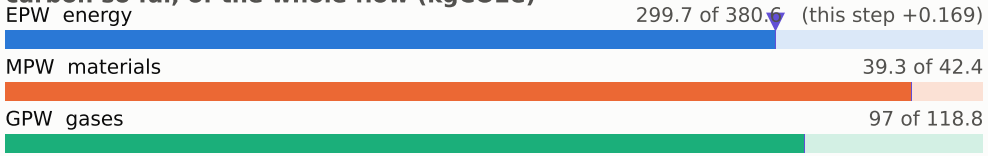
no mask for this step

this step

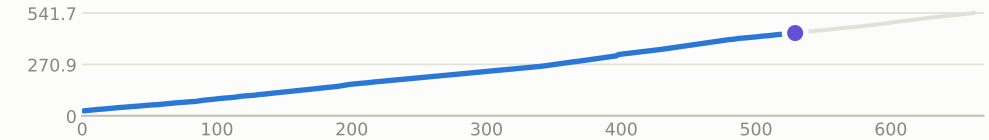
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

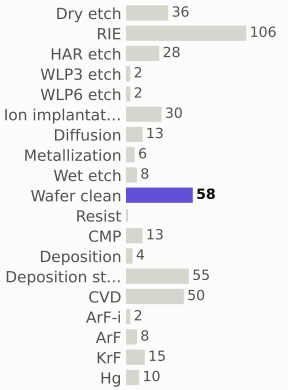
carbon so far, of the whole flow (kgCO2e)



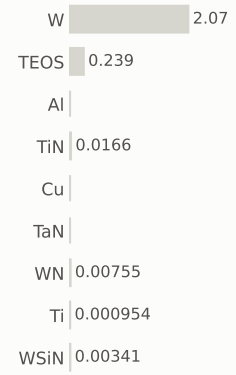
embodied carbon up to each step (kgCO2e per wafer)



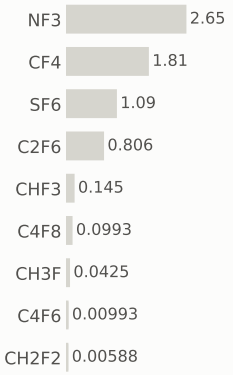
steps by process area



materials deposited (g)



gases emitted (g)



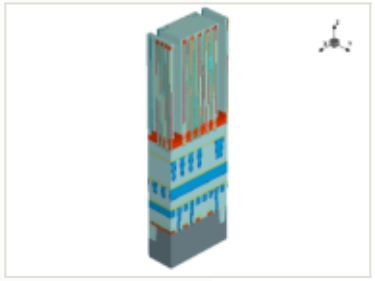
Step #4920: Wet_etch

etch_isotropic · Wet_etch

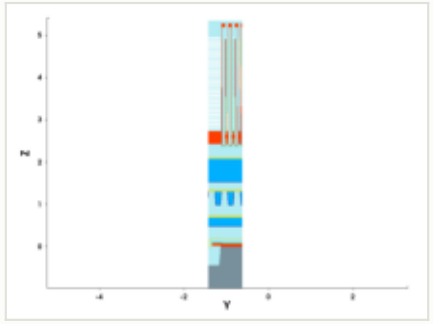
0.169	436.2	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

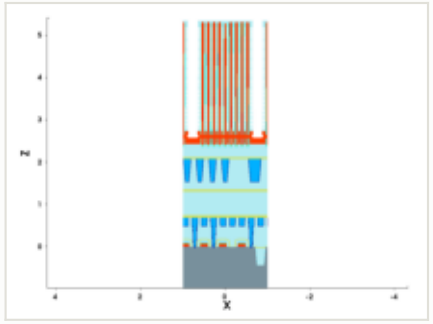
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



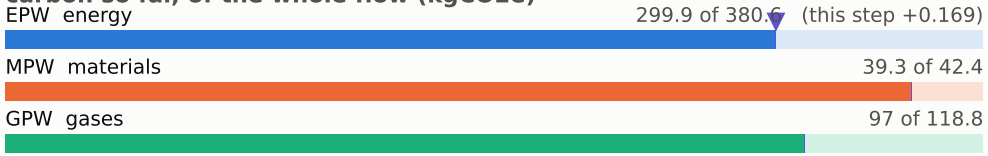
mask for this step

no mask for this step

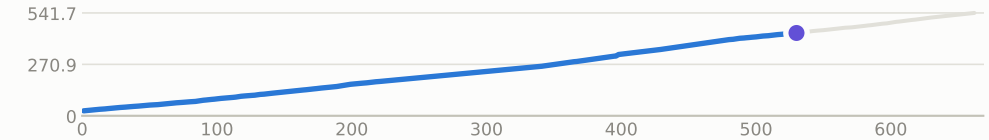
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

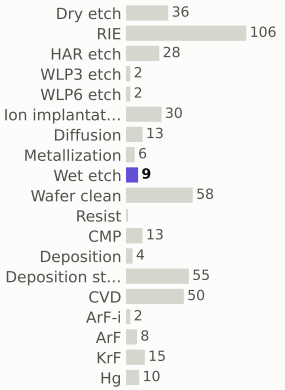
carbon so far, of the whole flow (kgCO2e)



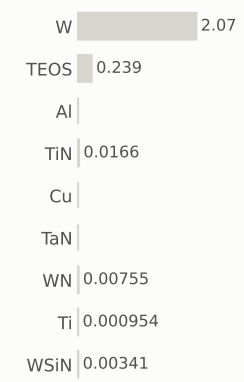
embodied carbon up to each step (kgCO2e per wafer)



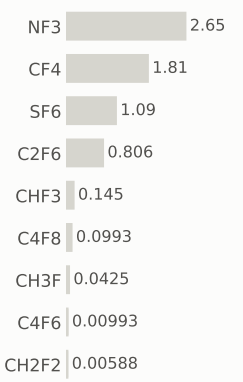
steps by process area



materials deposited (g)



gases emitted (g)



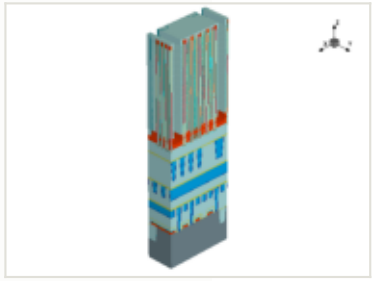
Step #4930: Spx_only

etch_isotropic · Spx_only

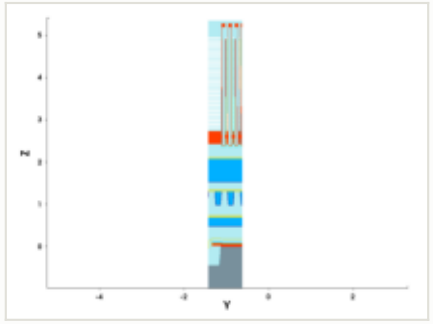
0	436.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

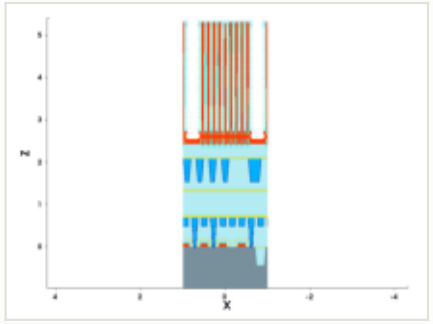
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

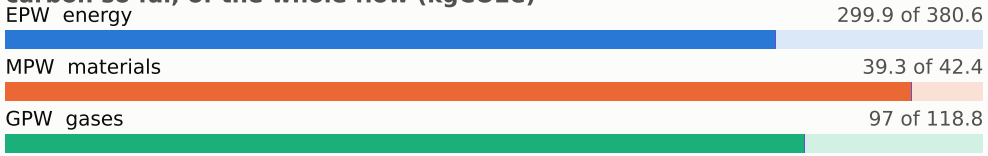
no mask for this step

this step

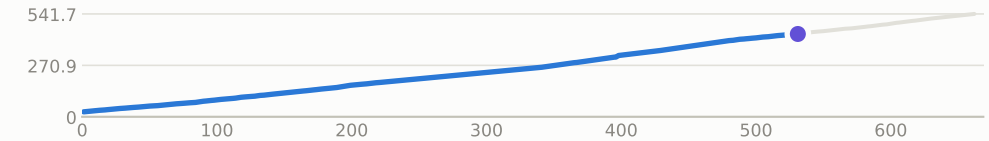
process area	Spx_only	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Spx_only step: not counted as a process step (uncounted_process_areas)

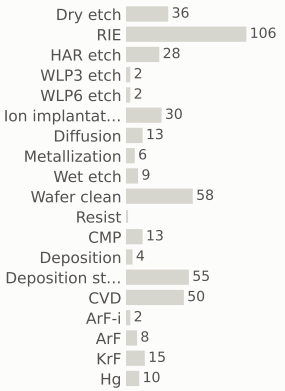
carbon so far, of the whole flow (kgCO2e)



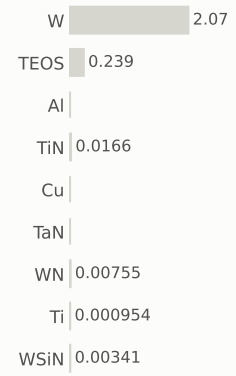
embodied carbon up to each step (kgCO2e per wafer)



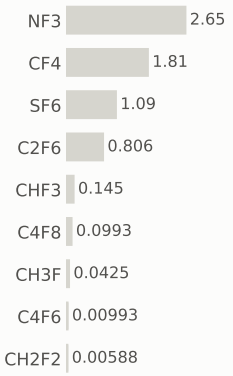
steps by process area



materials deposited (g)



gases emitted (g)



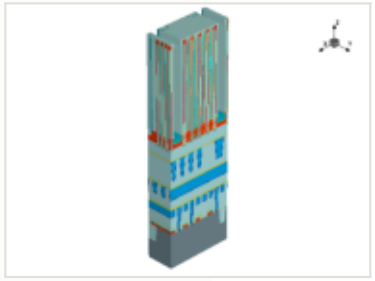
Step #4940: Diffusion

depo_isotropic · Diffusion

0.801	437.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

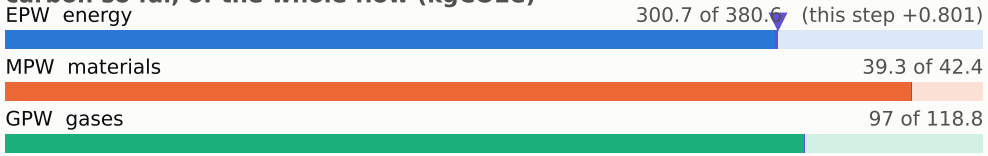


this step

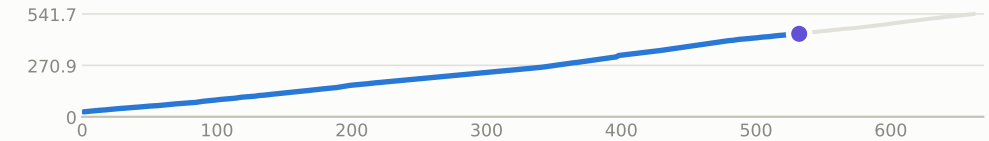
process area	Diffusion
material	—
thickness	0.03 µm
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

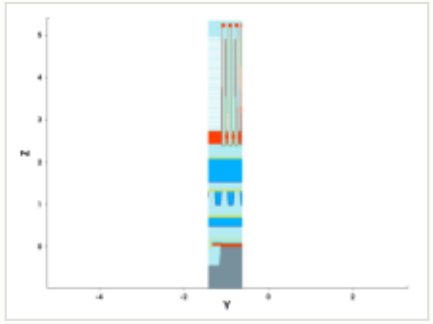
carbon so far, of the whole flow (kgCO2e)



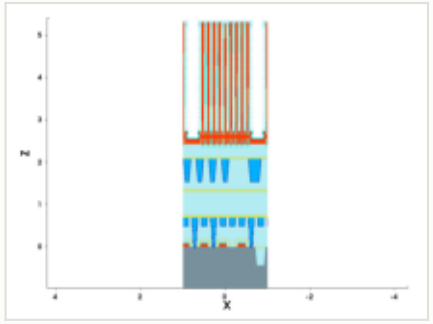
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



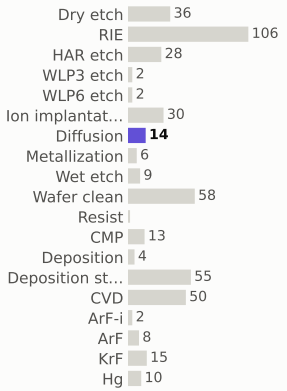
y cut at -1.032 µm



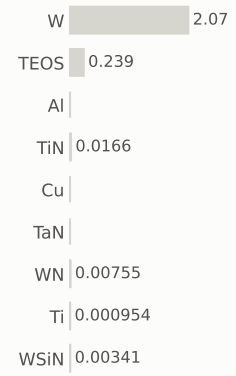
mask for this step

no mask for this step

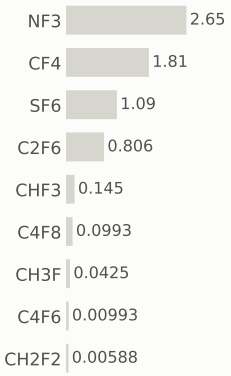
steps by process area



materials deposited (g)



gases emitted (g)



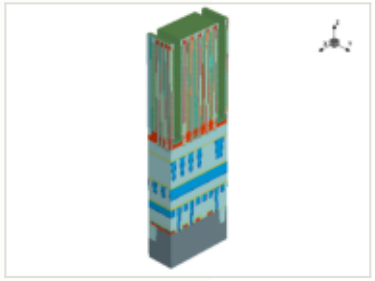
Step #4950: Deposition

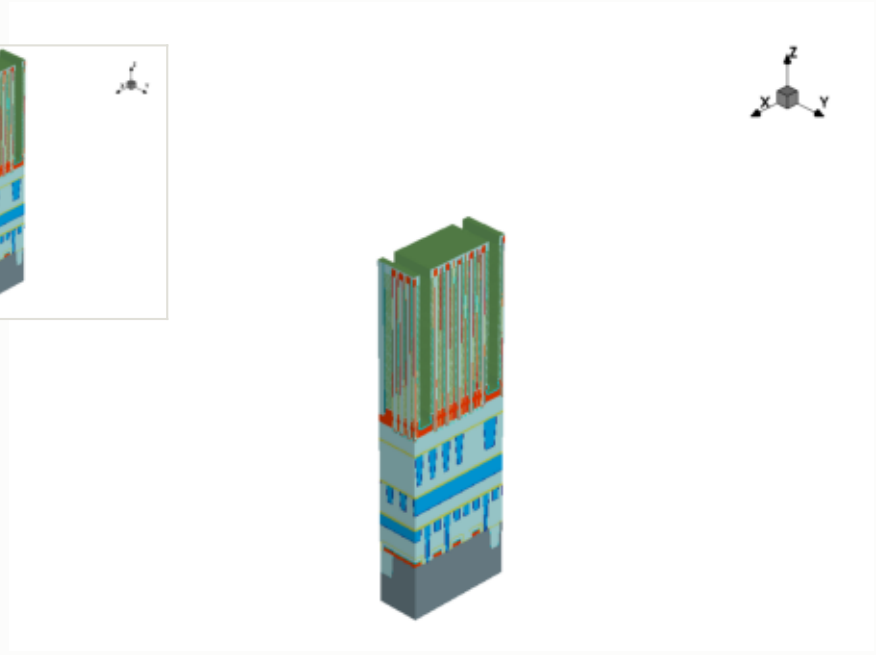
depo_isotropic · Deposition

0.656	437.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



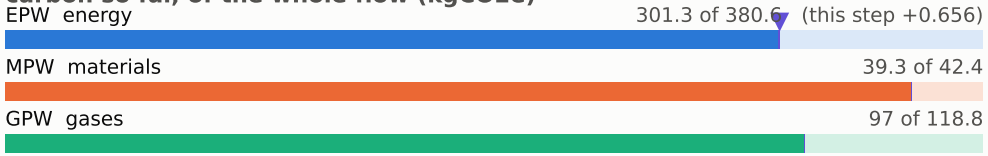


this step

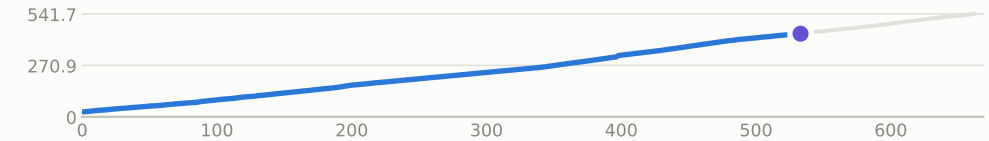
process area	Deposition
material	—
thickness	0.002 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

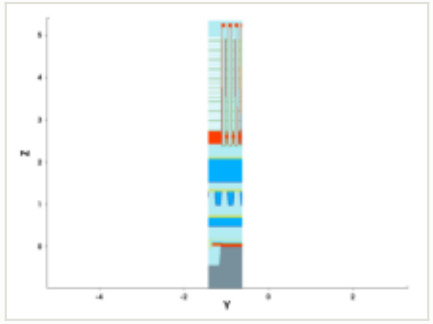
carbon so far, of the whole flow (kgCO2e)



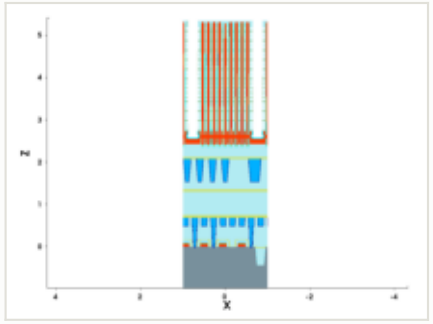
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



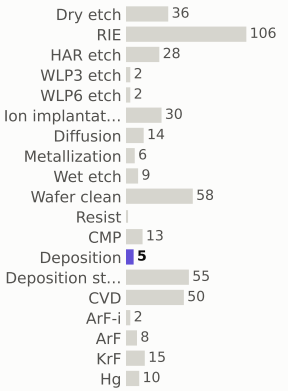
y cut at -1.032 µm



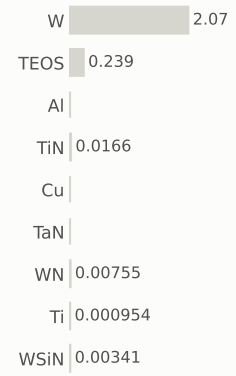
mask for this step

no mask for this step

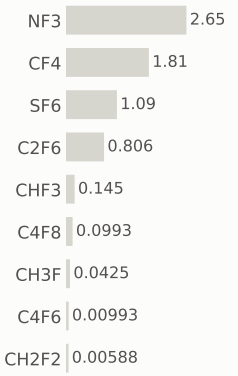
steps by process area



materials deposited (g)



gases emitted (g)



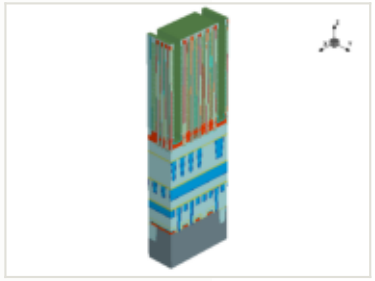
Step #4960: Diffusion

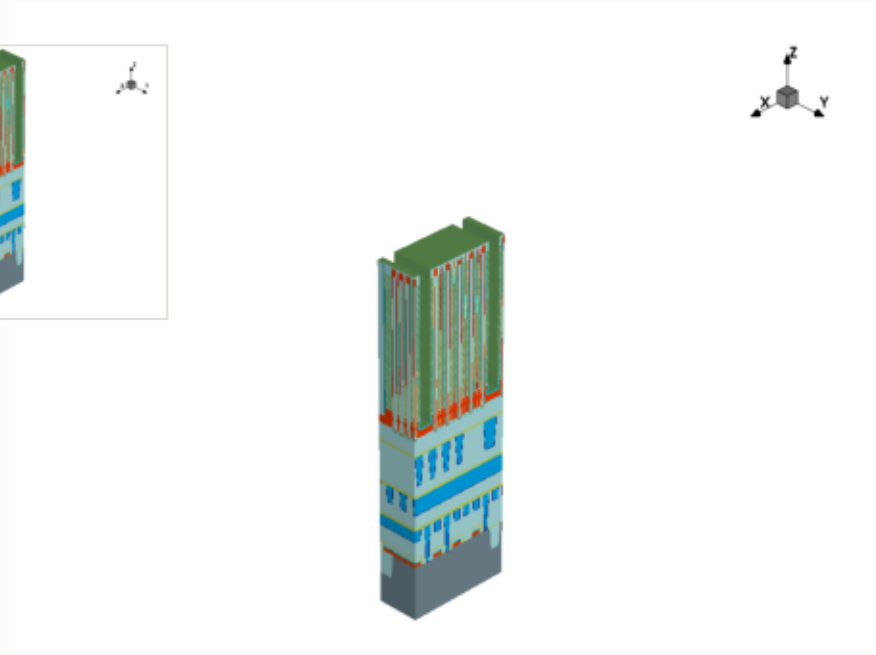
anneal · Diffusion

0.801	438.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

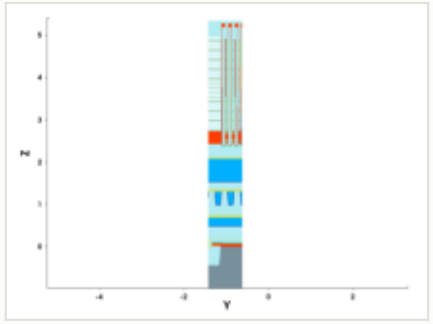
3D structure after this step

zoomed to fit

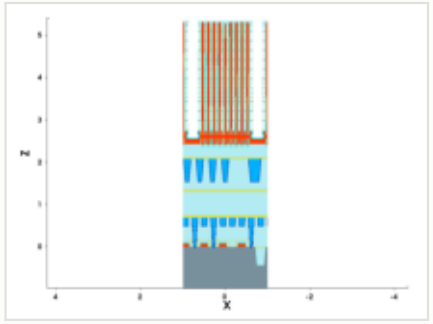




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

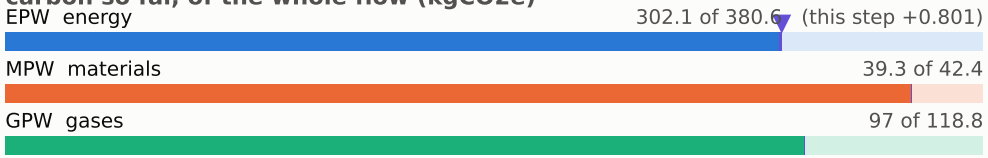
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

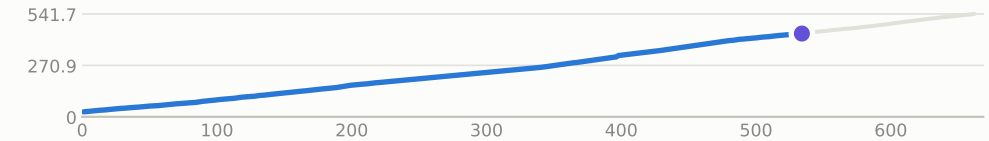
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

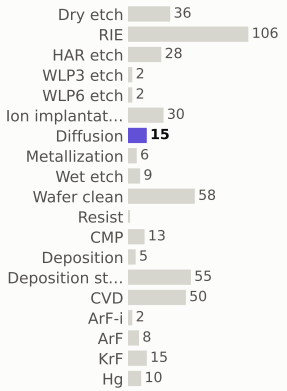
carbon so far, of the whole flow (kgCO2e)



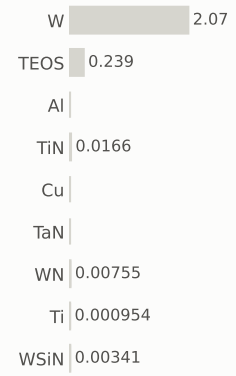
embodied carbon up to each step (kgCO2e per wafer)



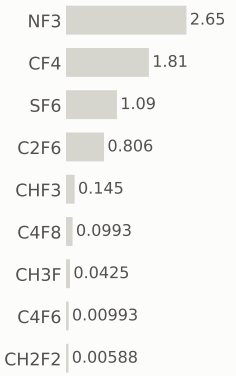
steps by process area



materials deposited (g)



gases emitted (g)



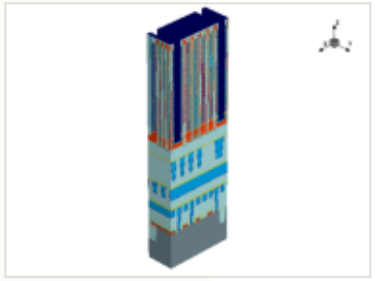
Step #4970: Deposition

depo_isotropic · Deposition

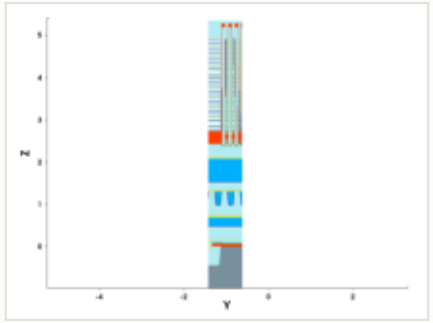
0.656	439.1	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

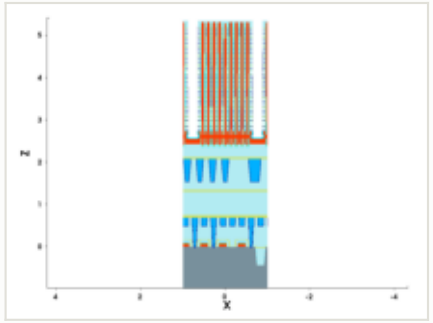
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



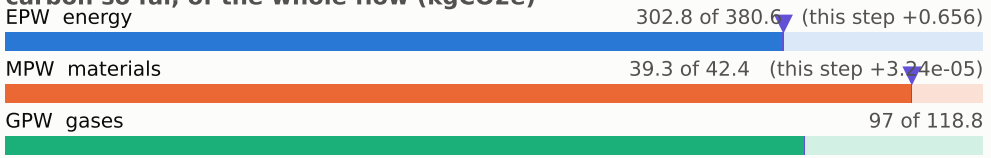
mask for this step

no mask for this step

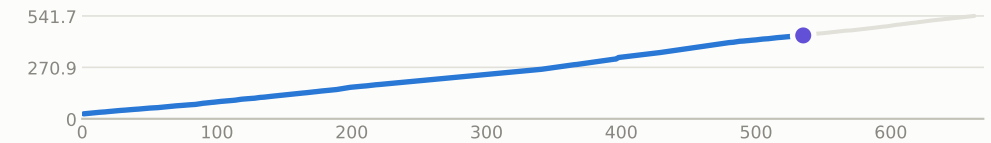
this step

process area	Deposition	EPW	0.656 kgCO2e
material	TiN	EPW tool	0.469 kgCO2e
thickness	0.002 µm	EPW overhead	0.187 kgCO2e
material GWP	43.9778 gCO2e/g	MPW	3.24e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

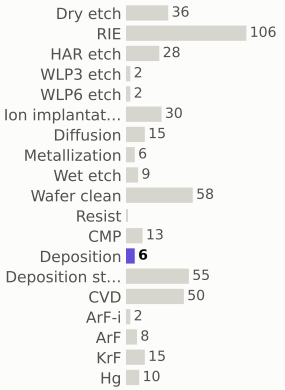
carbon so far, of the whole flow (kgCO2e)



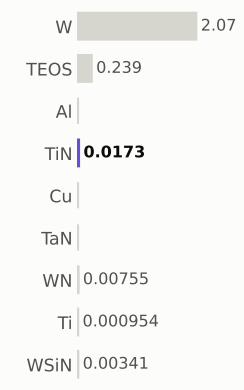
embodied carbon up to each step (kgCO2e per wafer)



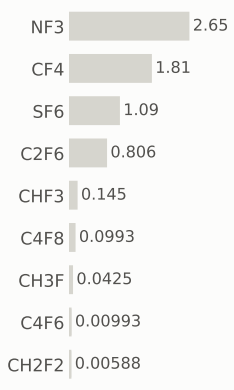
steps by process area



materials deposited (g)



gases emitted (g)



Step #4980: Metallization

depo_isotropic · Metallization

1.21	440.3	0.22%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

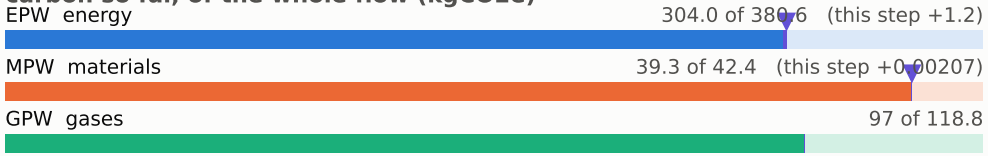
zoomed to fit

this step

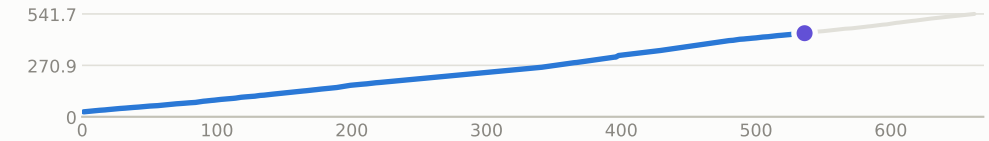
process area	Metallization
material	W
thickness	0.02 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00207 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

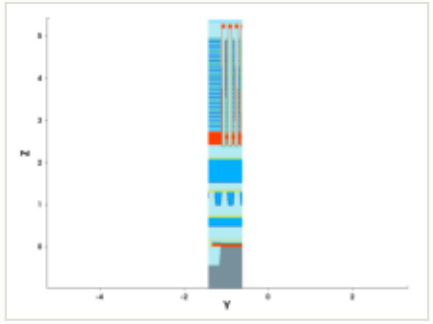
carbon so far, of the whole flow (kgCO2e)



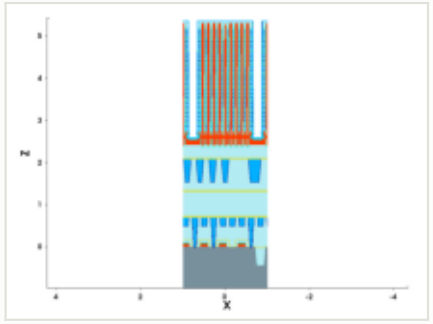
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



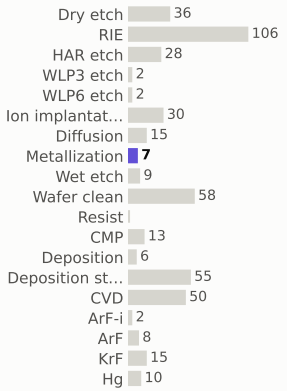
y cut at -1.032 µm



mask for this step

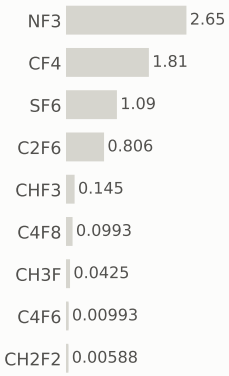
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



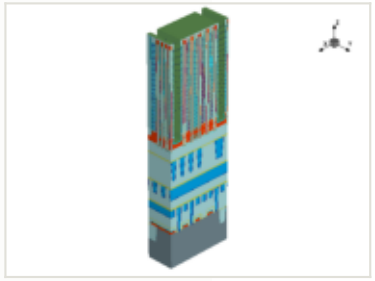
Step #4990: RIE

etch_isotropic · RIE

1.04	441.3	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

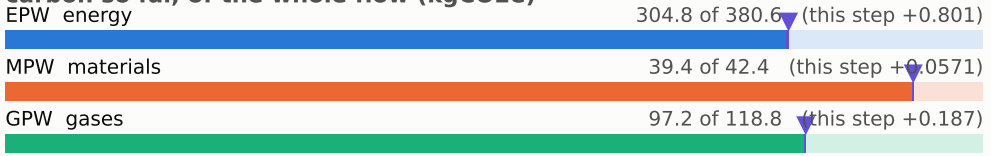


this step

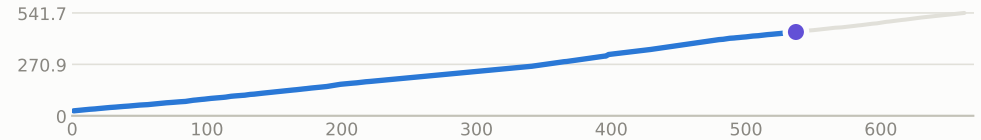
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

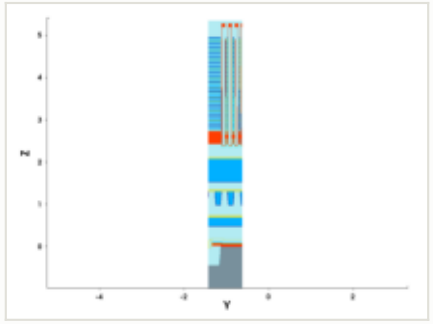
carbon so far, of the whole flow (kgCO2e)



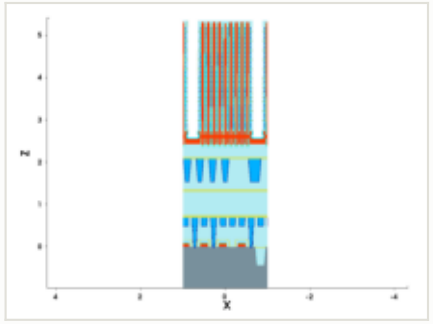
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



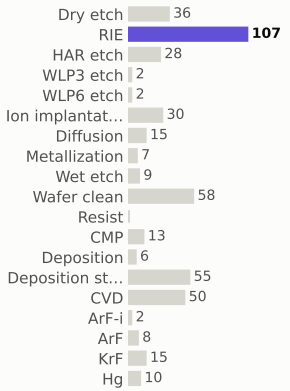
y cut at -1.032 µm



mask for this step

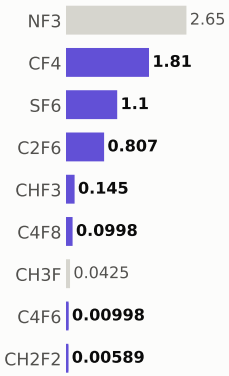
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



Step #5000: Deposition

depo_isotropic · Deposition

0.656	442.0	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

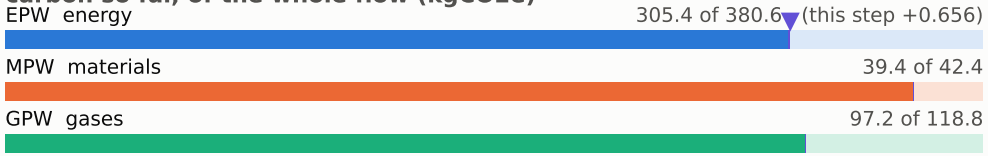
zoomed to fit

this step

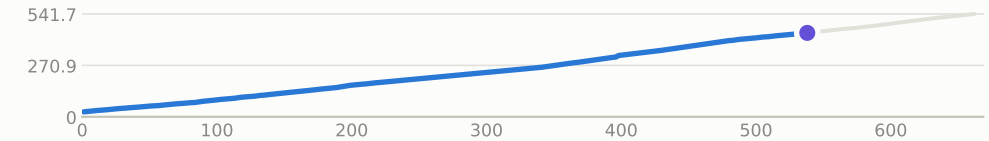
process area	Deposition
material	—
thickness	0.045 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

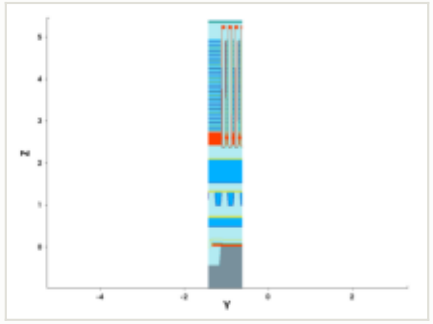
carbon so far, of the whole flow (kgCO2e)



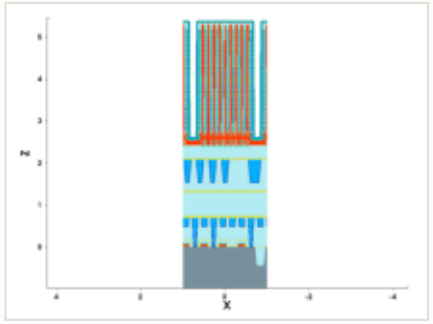
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



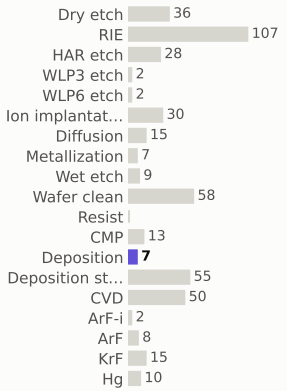
y cut at -1.032 µm



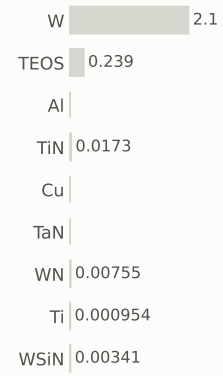
mask for this step

no mask for this step

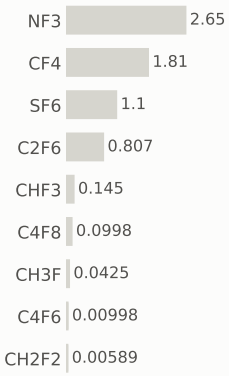
steps by process area



materials deposited (g)



gases emitted (g)



Step #5010: RIE

etch_multi_offset · RIE

1.04	443.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

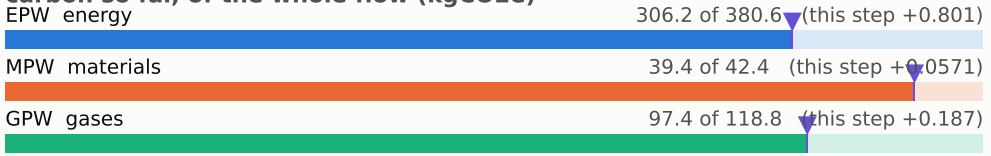
zoomed to fit

this step

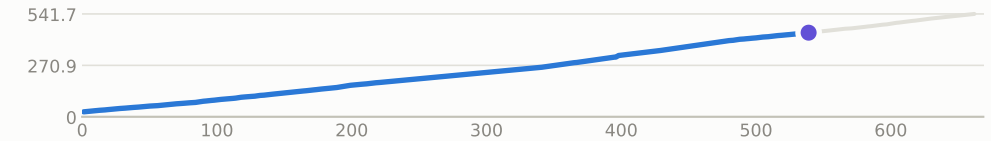
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

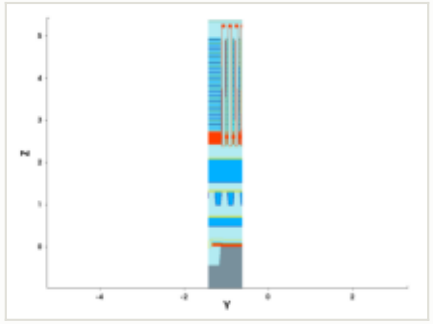
carbon so far, of the whole flow (kgCO2e)



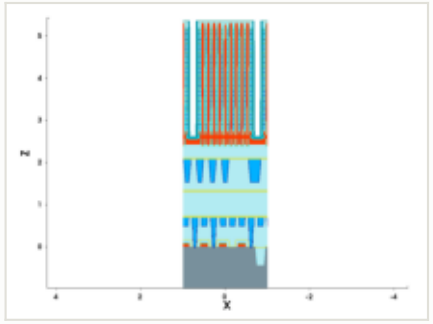
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



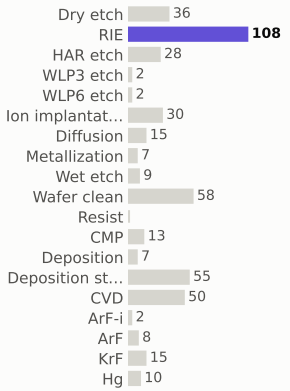
y cut at -1.032 µm



mask for this step

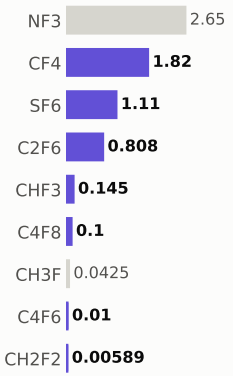
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



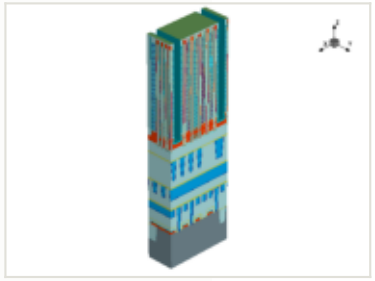
Step #5020: RIE

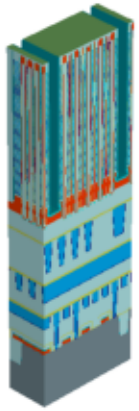
etch_isotropic · RIE

1.04	444.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



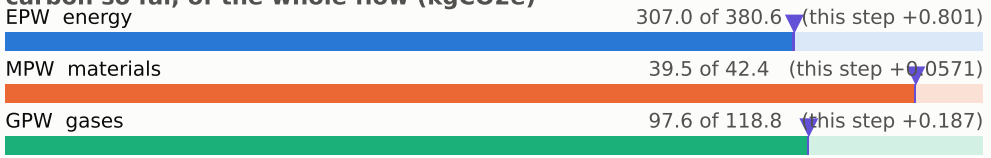


this step

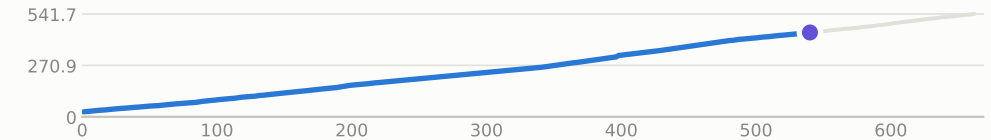
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

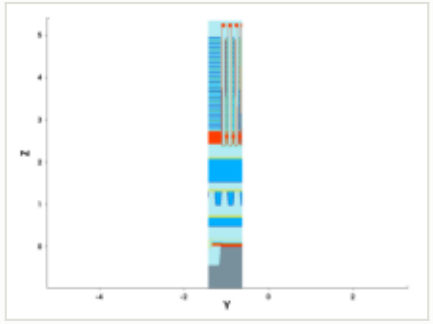
carbon so far, of the whole flow (kgCO2e)



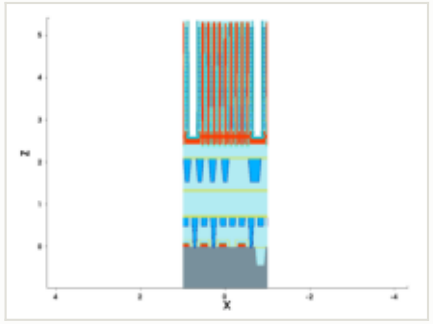
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



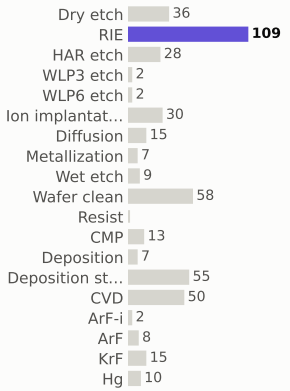
y cut at -1.032 µm



mask for this step

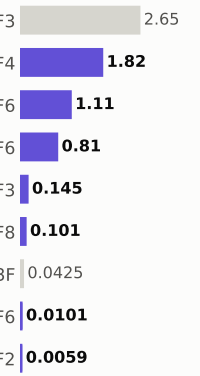
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



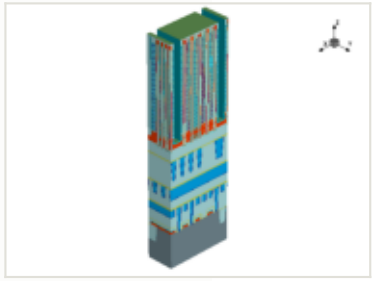
Step #5030: RIE

etch_anisotropic · RIE

1.04	445.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

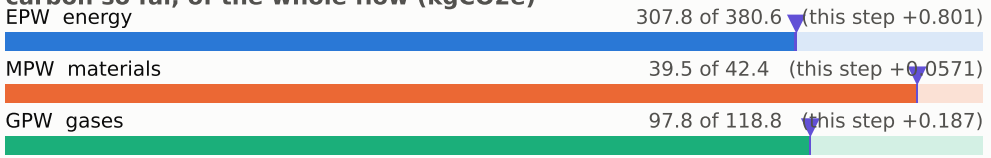


this step

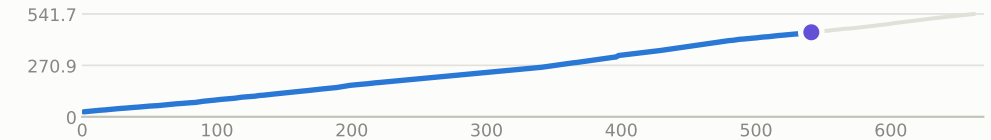
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

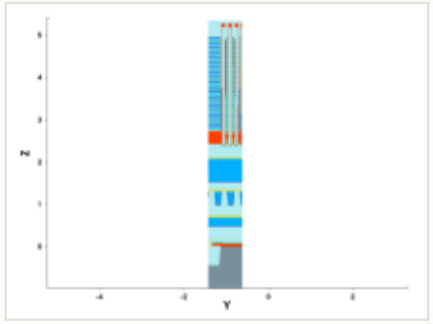
carbon so far, of the whole flow (kgCO2e)



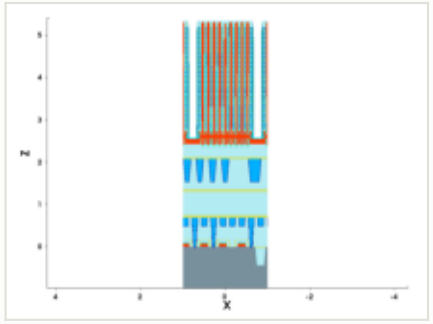
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



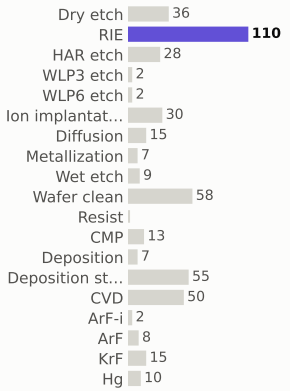
y cut at -1.032 µm



mask for this step

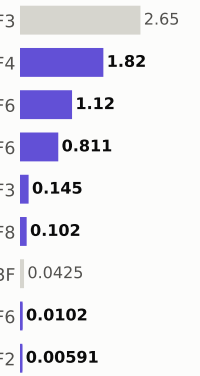
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



Step #5040: RIE

etch_anisotropic · RIE

1.04	446.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

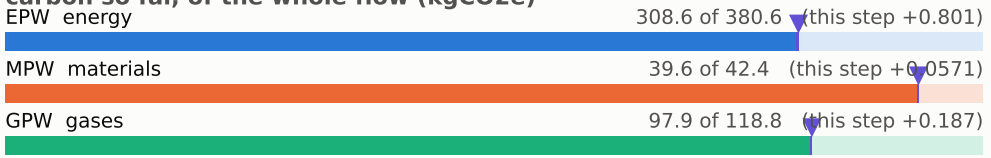
zoomed to fit

this step

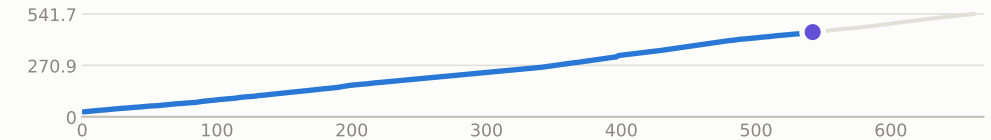
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

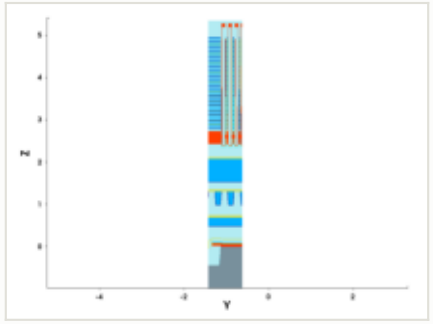
carbon so far, of the whole flow (kgCO2e)



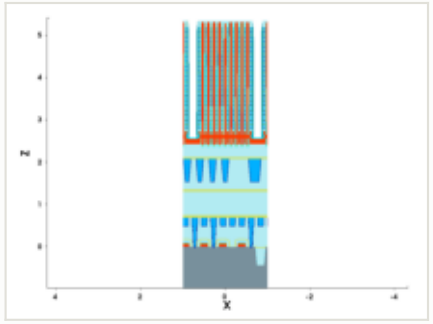
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



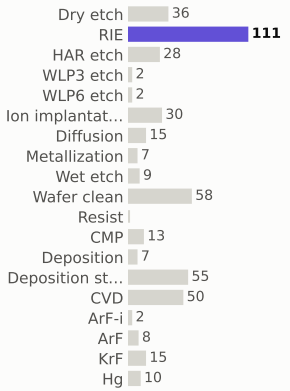
y cut at -1.032 µm



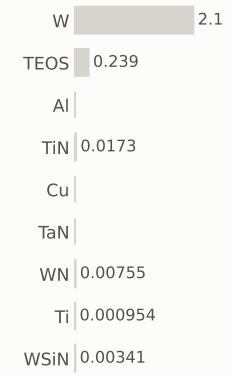
mask for this step

no mask for this step

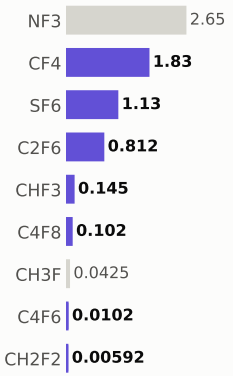
steps by process area



materials deposited (g)



gases emitted (g)



Step #5050: RIE

etch_anisotropic · RIE

1.04	447.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

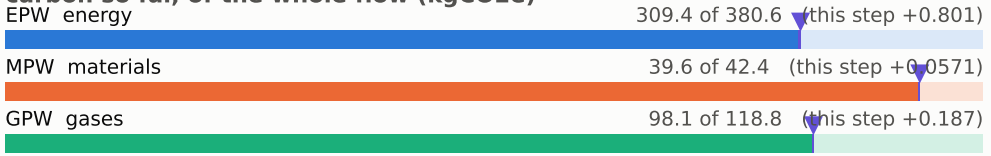
zoomed to fit

this step

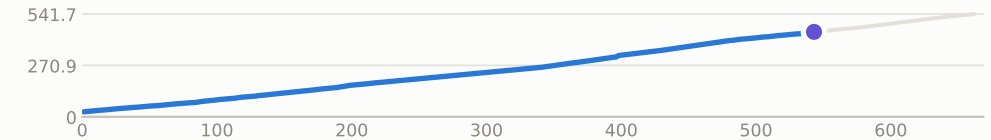
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

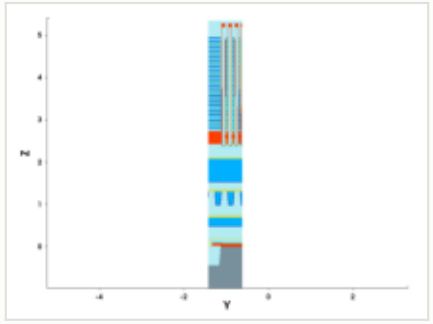
carbon so far, of the whole flow (kgCO2e)



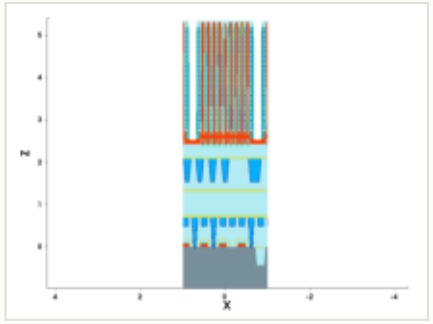
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



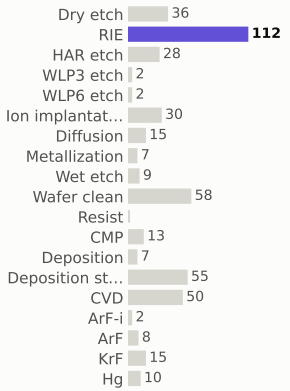
y cut at -1.032 µm



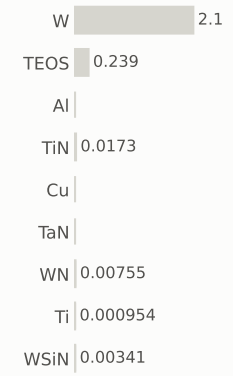
mask for this step

no mask for this step

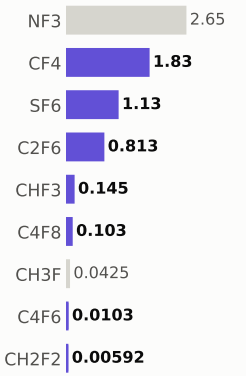
steps by process area



materials deposited (g)



gases emitted (g)



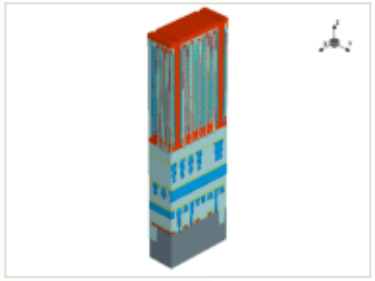
Step #5060: CVD

depo_isotropic · CVD

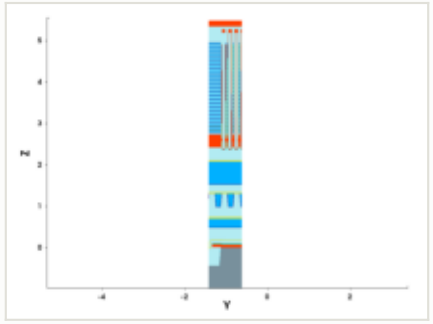
1.31	448.5	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

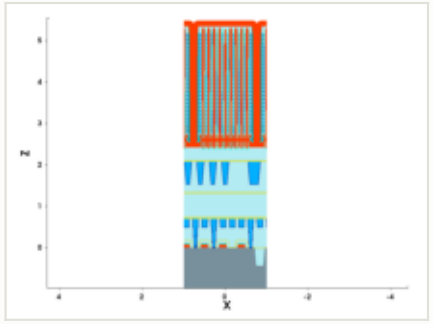
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

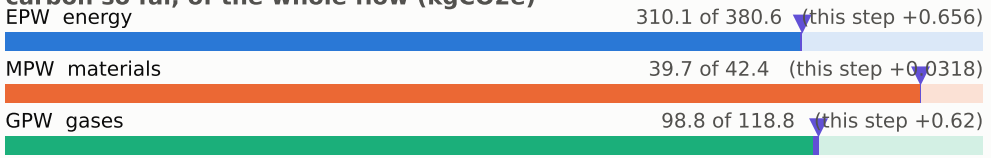
no mask for this step

this step

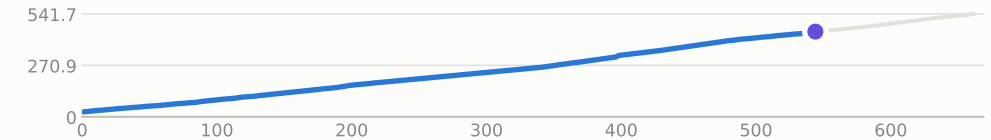
process area	CVD
material	—
thickness	0.14 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

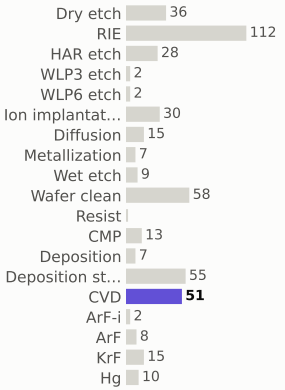
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

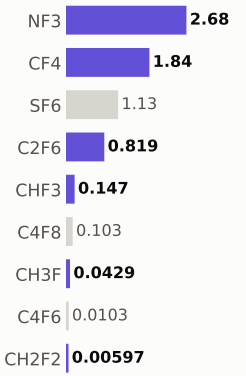


steps by process area



materials deposited (g)

gases emitted (g)



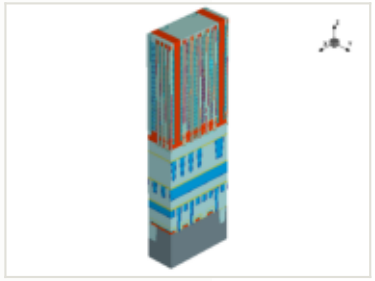
Step #5070: CMP

etch_cmp · CMP

0.169	448.7	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

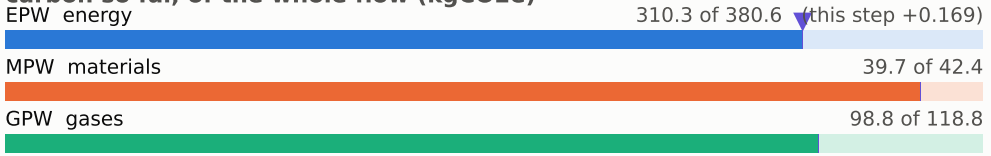
zoomed to fit



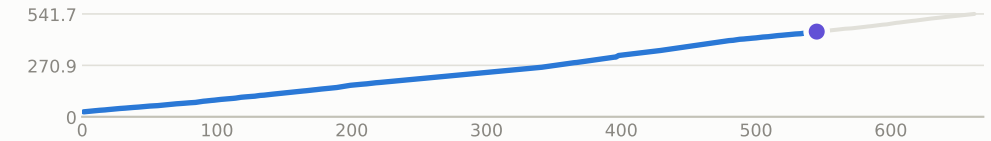
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

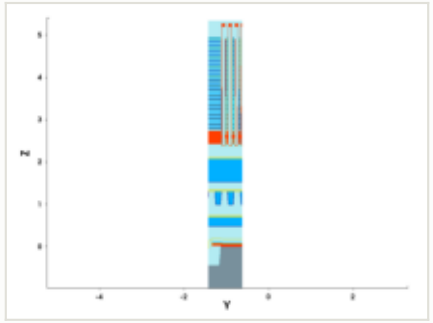
carbon so far, of the whole flow (kgCO2e)



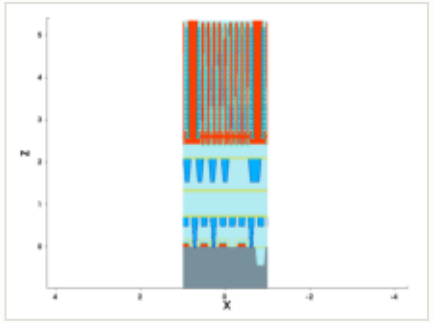
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



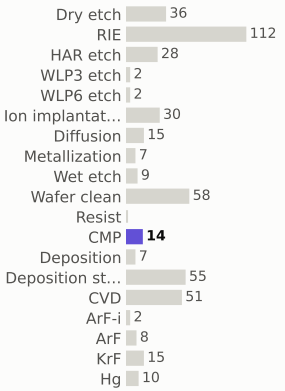
y cut at -1.032 µm



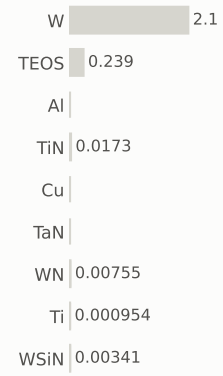
mask for this step

no mask for this step

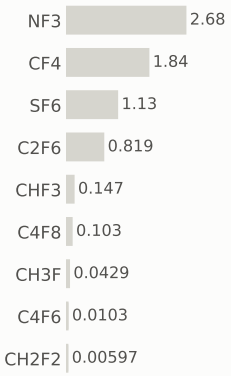
steps by process area



materials deposited (g)



gases emitted (g)



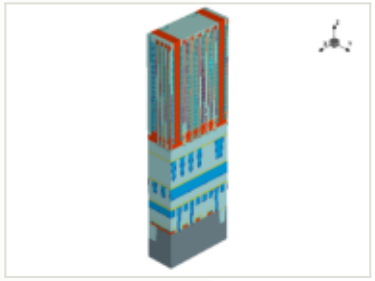
Step #5080: Wafer_clean

clean · Wafer_clean

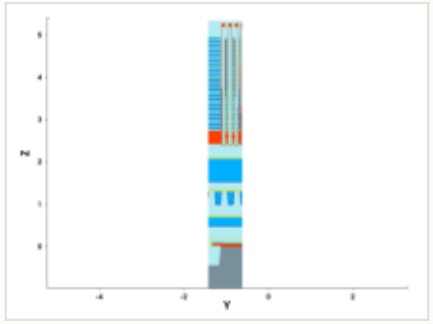
0.169	448.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

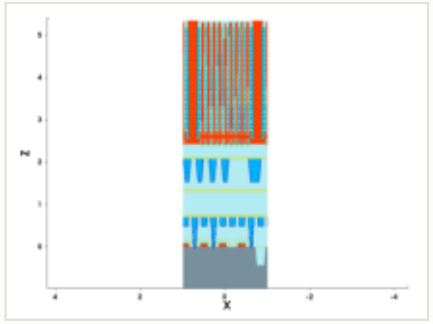
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

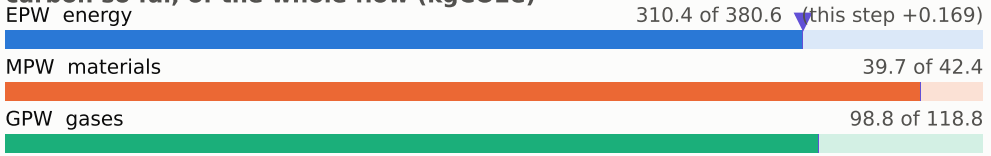
no mask for this step

this step

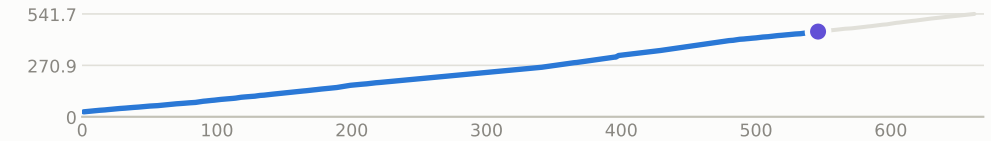
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

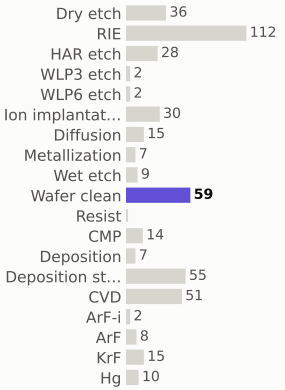
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

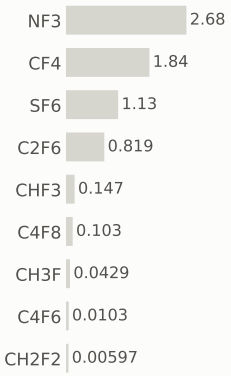


steps by process area



materials deposited (g)

gases emitted (g)



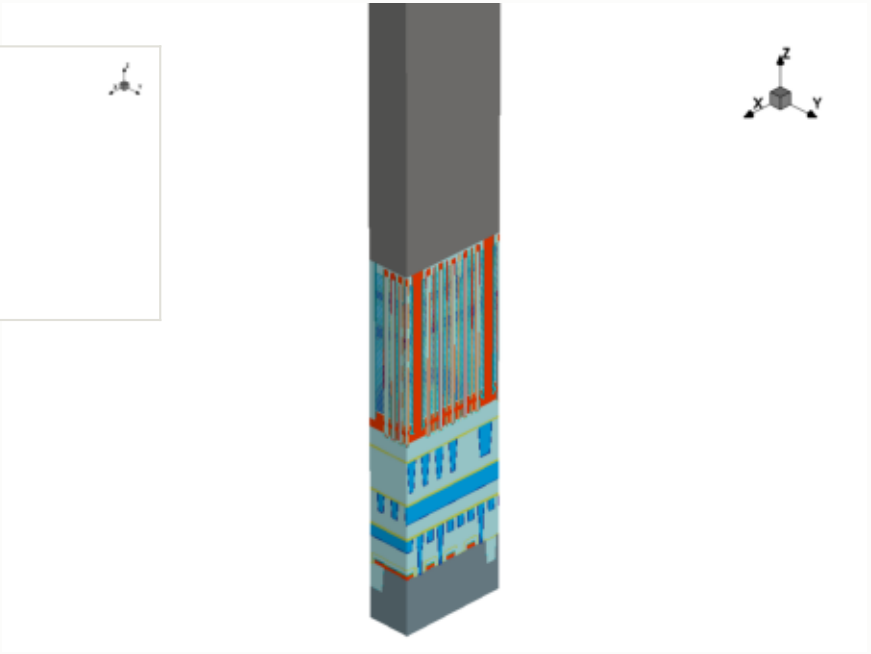
Step #5100: KrF

litho_patterning · KrF

0.199	449.1	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

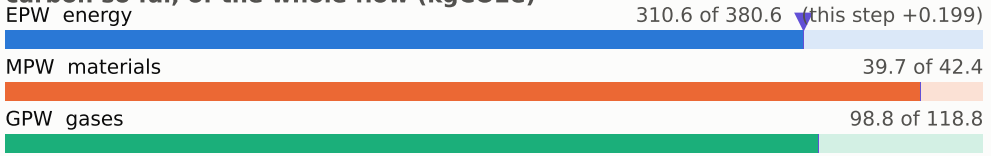
zoomed to fit



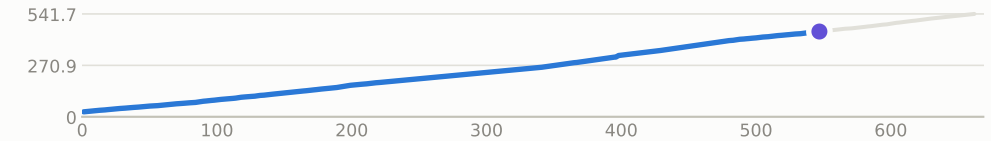
this step

process area	KrF	EPW	0.199 kgCO2e
material	—	EPW tool	0.142 kgCO2e
thickness	—	EPW overhead	0.0568 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.33 kWh + 0.132 overhead	embodied carbon	**0.199 kgCO2e**

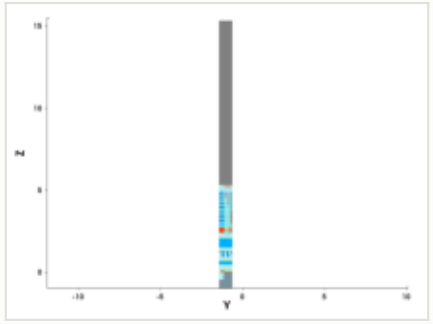
carbon so far, of the whole flow (kgCO2e)



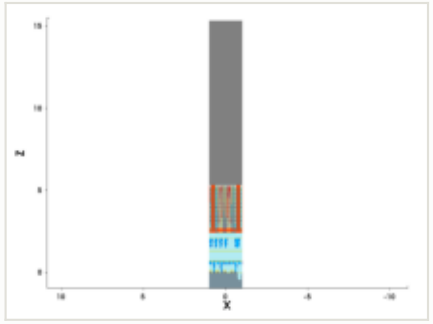
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



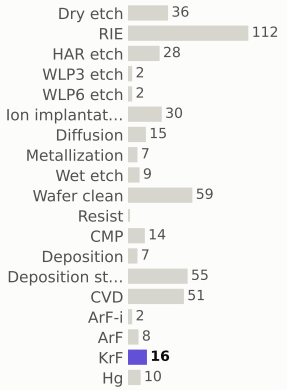
y cut at -1.032 µm



mask: layer 39:0 (OASIS)

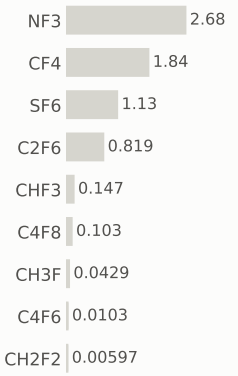
nothing in the clip window

steps by process area



materials deposited (g)

gases emitted (g)



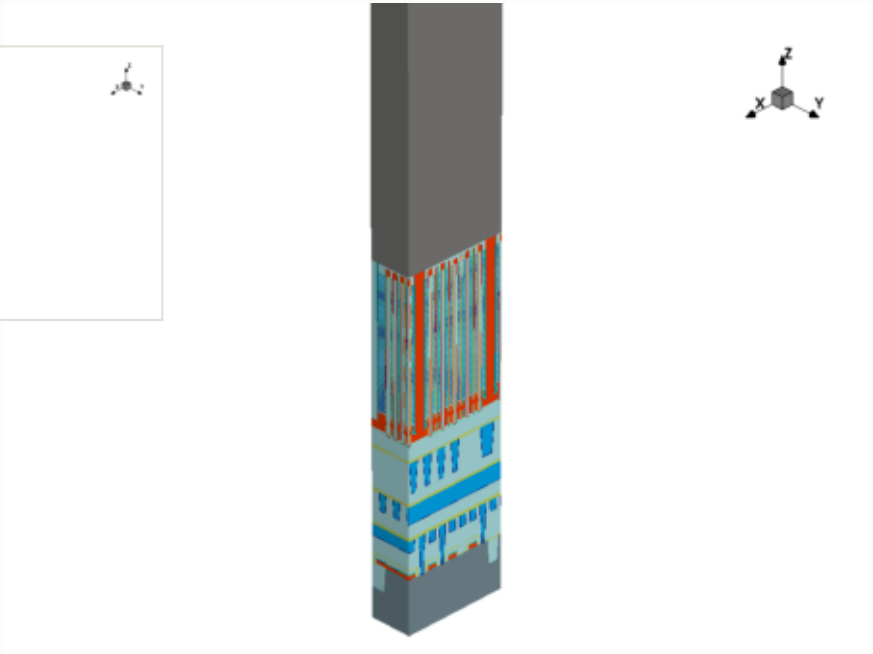
Step #5110: RIE

etch_anisotropic · RIE

1.04	450.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

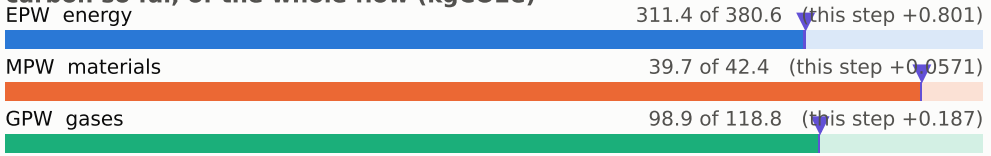


this step

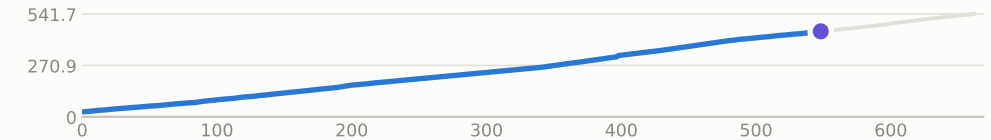
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

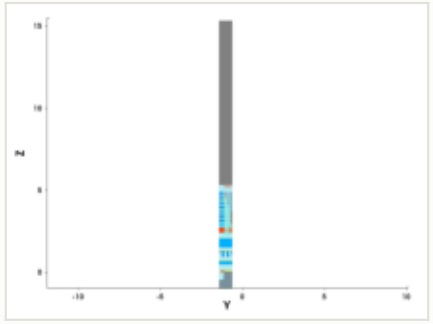
carbon so far, of the whole flow (kgCO2e)



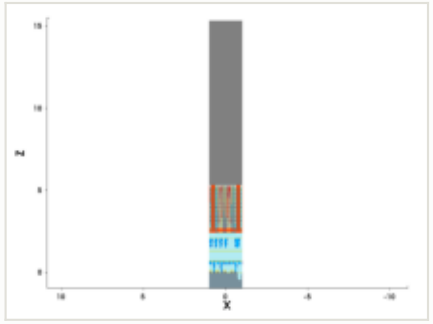
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



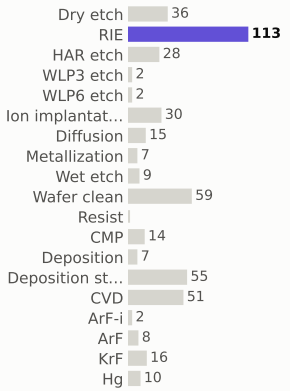
y cut at -1.032 µm



mask for this step

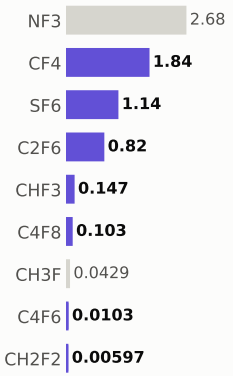
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



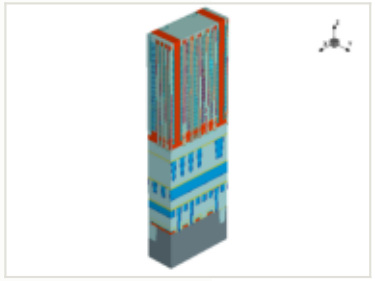
Step #5120: Dry_etch

etch_strip · Dry_etch

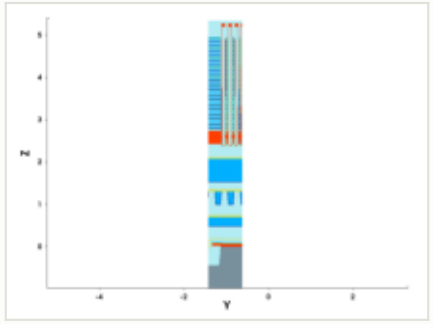
0.801	450.9	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

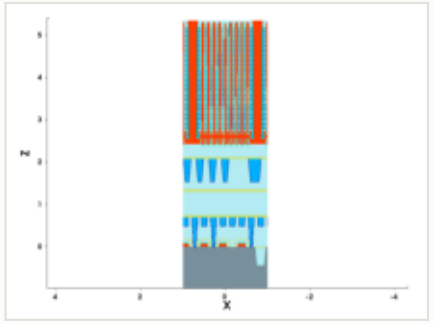
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

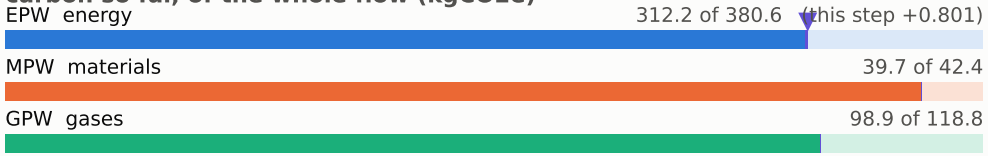
no mask for this step

this step

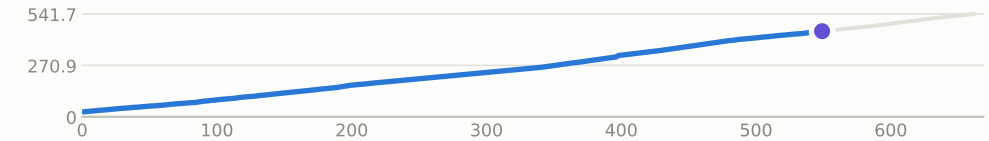
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

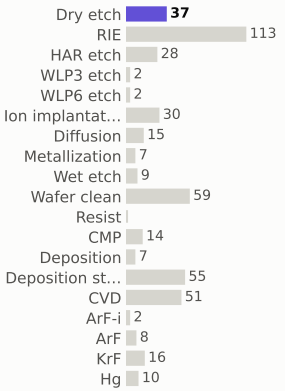
carbon so far, of the whole flow (kgCO2e)



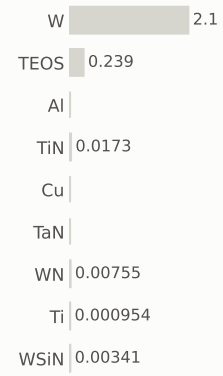
embodied carbon up to each step (kgCO2e per wafer)



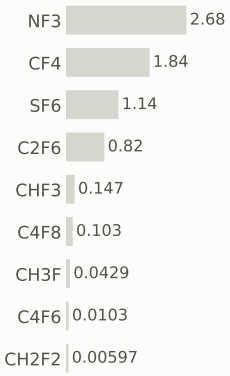
steps by process area



materials deposited (g)



gases emitted (g)



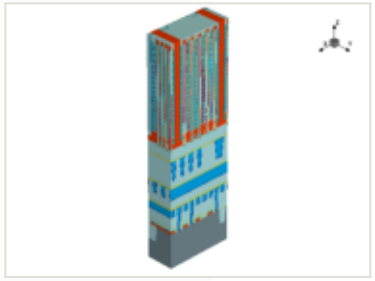
Step #5130: Wafer_clean

clean · Wafer_clean

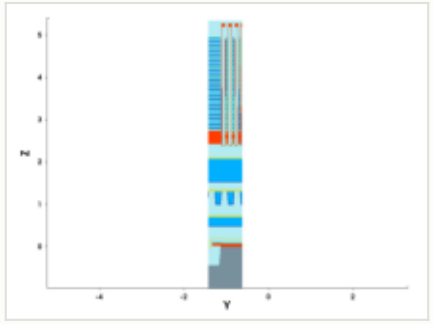
0.169	451.1	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

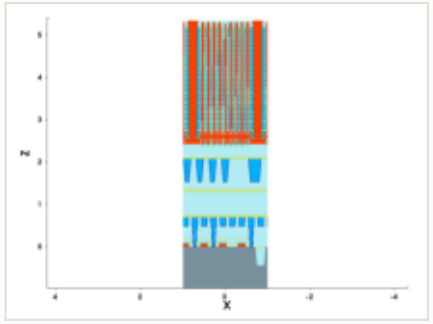
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

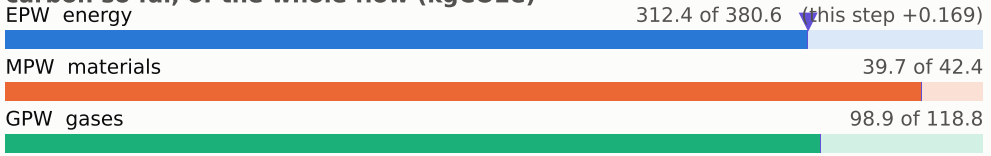
no mask for this step

this step

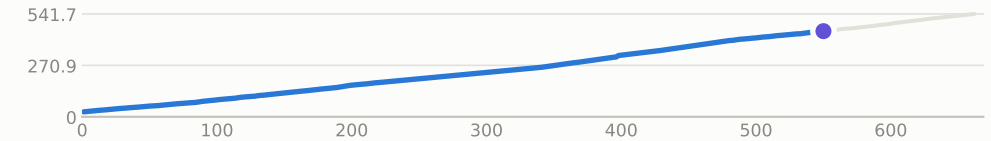
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

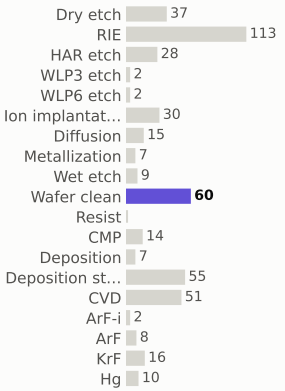
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

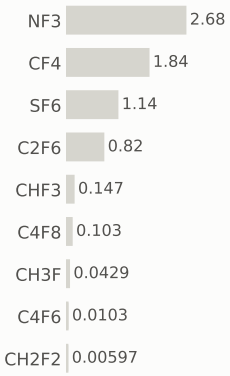


steps by process area



materials deposited (g)

gases emitted (g)



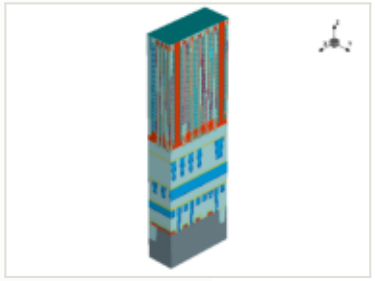
Step #5140: Deposition

depo_isotropic · Deposition

0.656	451.7	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

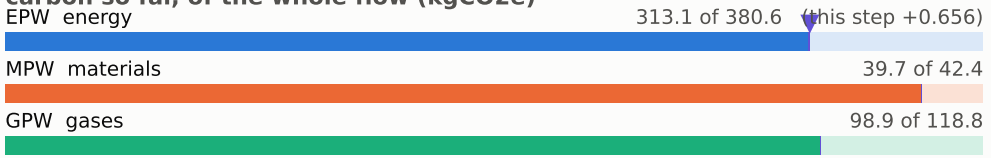


this step

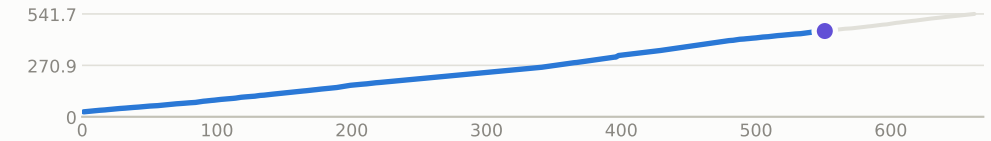
process area	Deposition
material	—
thickness	0.02 µm
material GWP	—
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

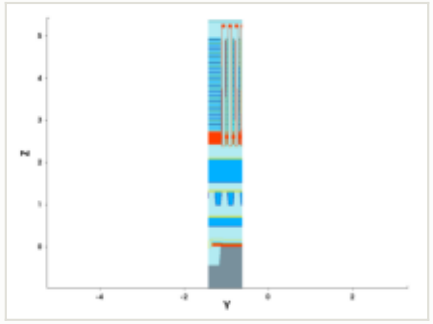
carbon so far, of the whole flow (kgCO2e)



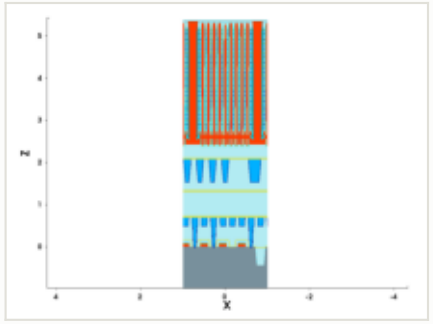
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



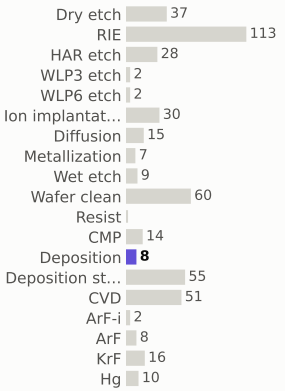
y cut at -1.032 µm



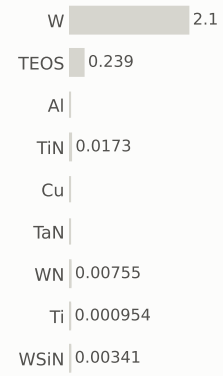
mask for this step

no mask for this step

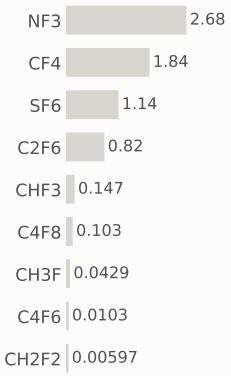
steps by process area



materials deposited (g)



gases emitted (g)



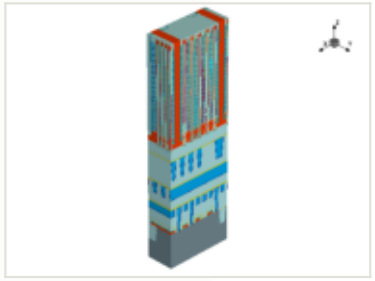
Step #5150: RIE

etch_anisotropic · RIE

1.04	452.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

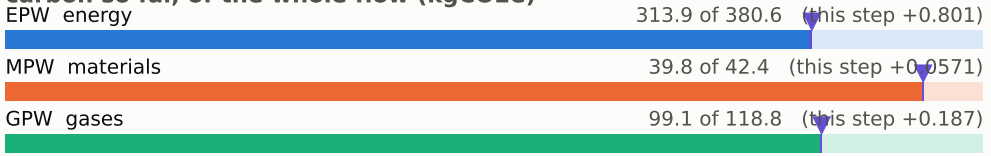


this step

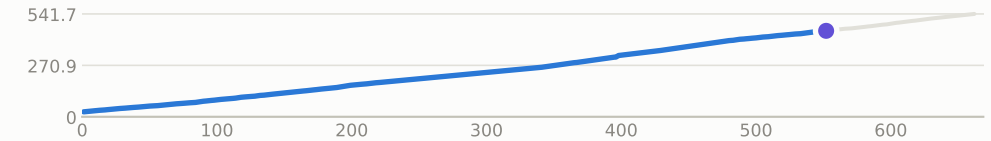
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

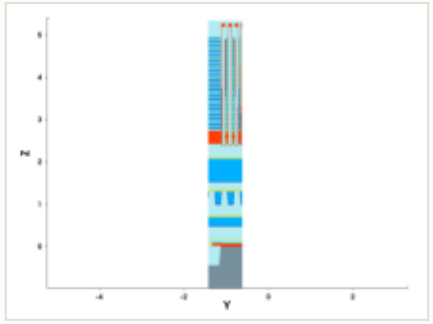
carbon so far, of the whole flow (kgCO2e)



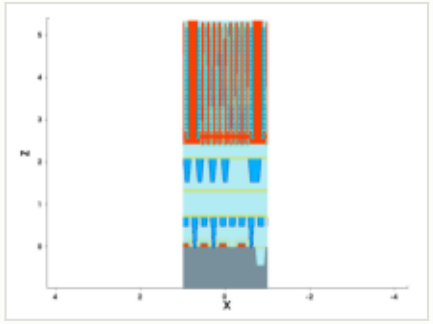
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



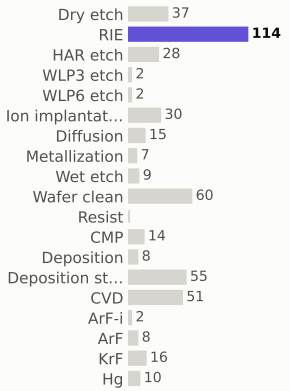
y cut at -1.032 µm



mask for this step

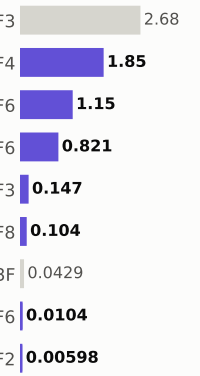
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



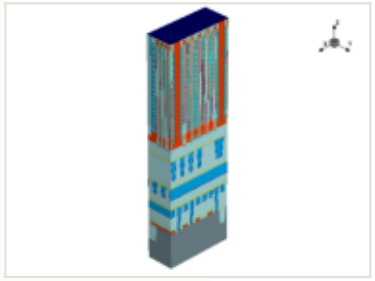
Step #5160: Deposition

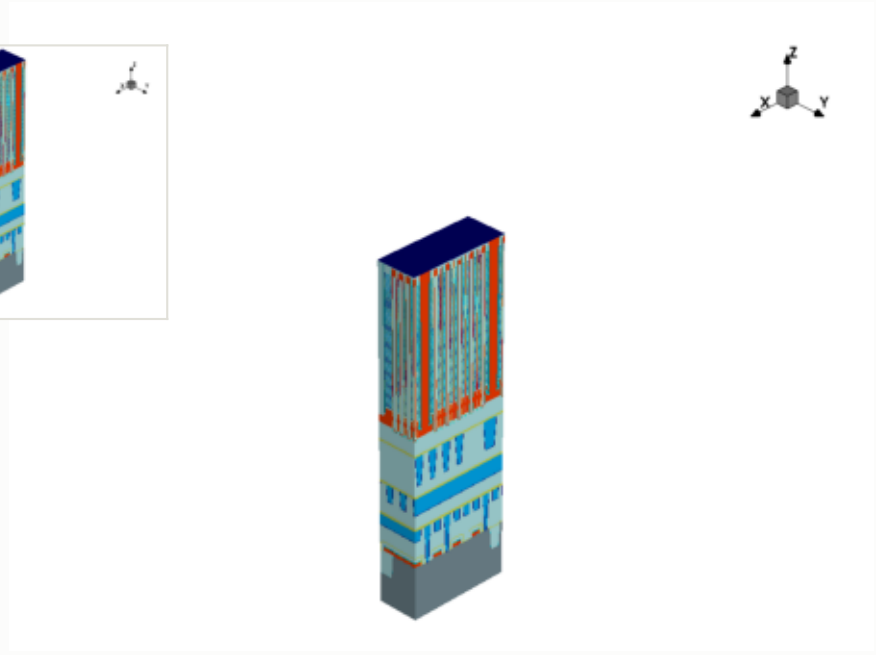
depo_isotropic · Deposition

0.656	453.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

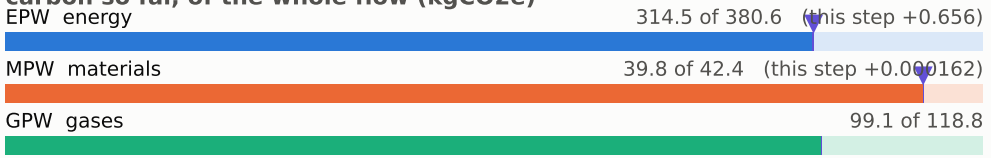




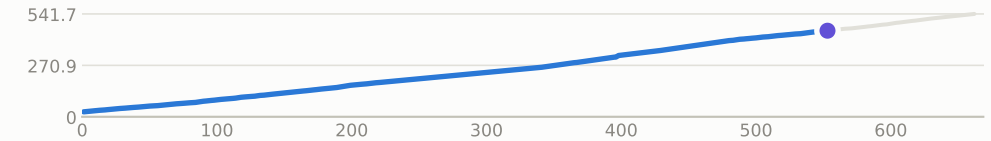
this step

process area	Deposition	EPW	0.656 kgCO2e
material	TiN	EPW tool	0.469 kgCO2e
thickness	0.01 µm	EPW overhead	0.187 kgCO2e
material GWP	43.9778 gCO2e/g	MPW	0.000162 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

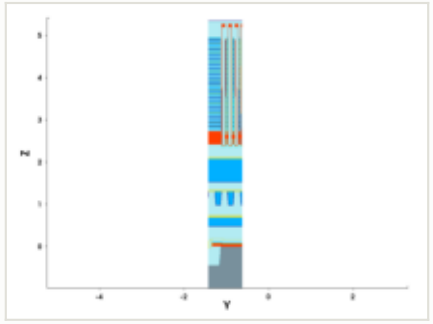
carbon so far, of the whole flow (kgCO2e)



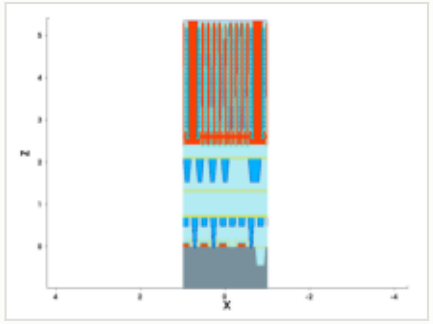
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



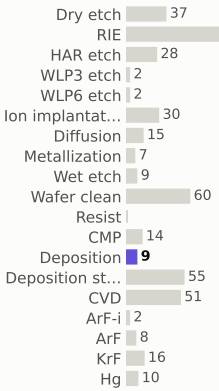
y cut at -1.032 µm



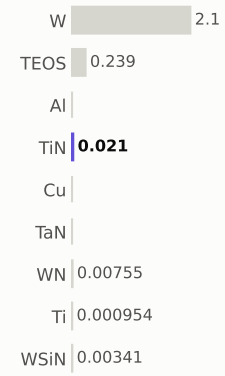
mask for this step

no mask for this step

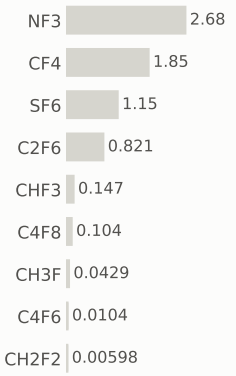
steps by process area



materials deposited (g)



gases emitted (g)



Step #5170: Metallization

depo_isotropic · Metallization

1.23	454.7	0.23%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

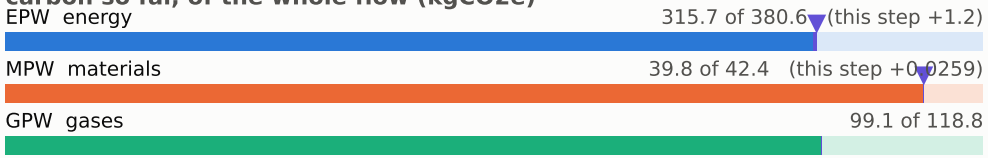
zoomed to fit

this step

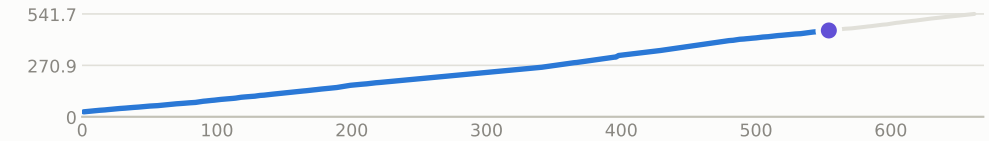
process area	Metallization
material	W
thickness	0.25 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0259 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.23 kgCO2e**

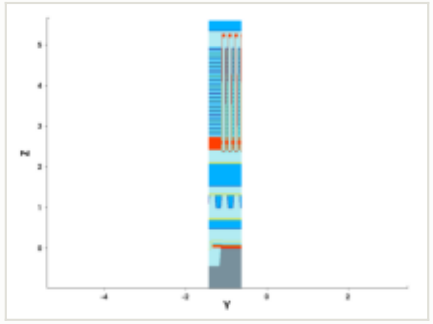
carbon so far, of the whole flow (kgCO2e)



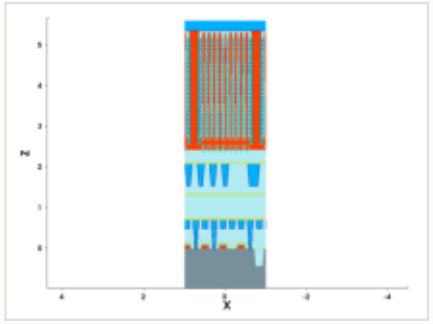
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



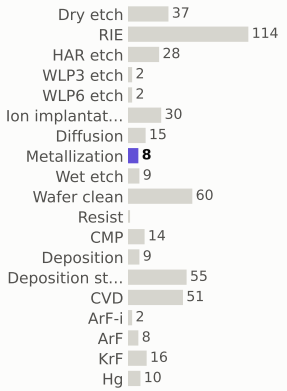
y cut at -1.032 µm



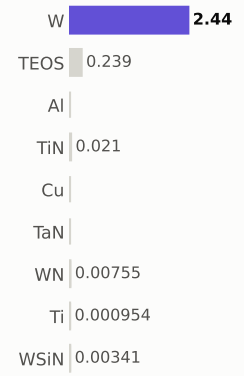
mask for this step

no mask for this step

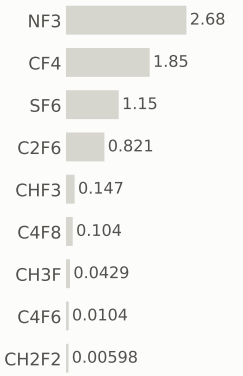
steps by process area



materials deposited (g)



gases emitted (g)



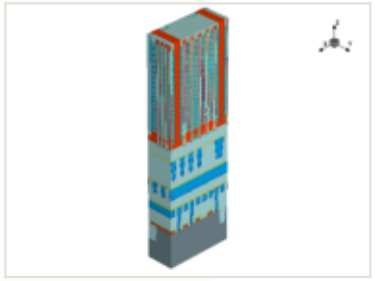
Step #5180: CMP

etch_cmp · CMP

0.169	454.8	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

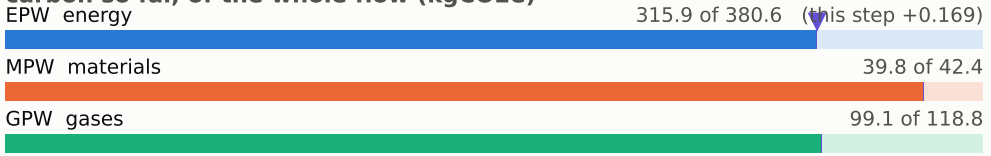
zoomed to fit



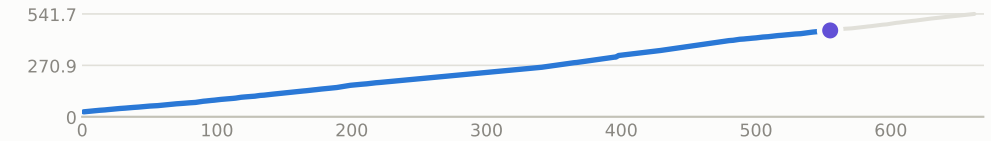
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

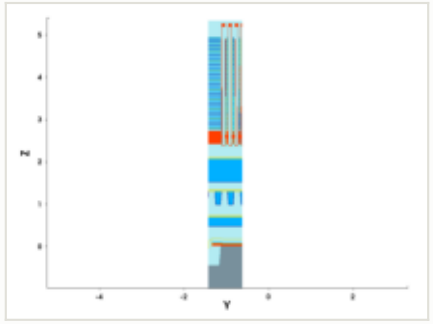
carbon so far, of the whole flow (kgCO2e)



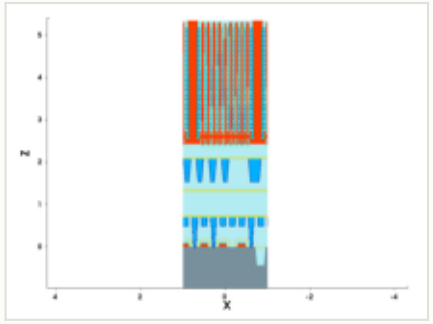
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



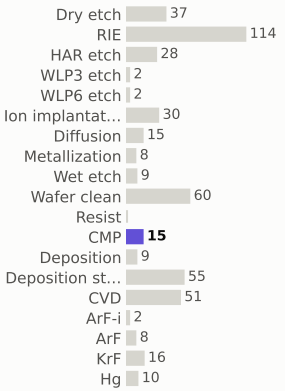
y cut at -1.032 µm



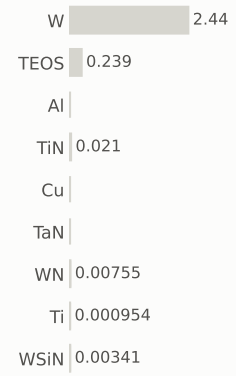
mask for this step

no mask for this step

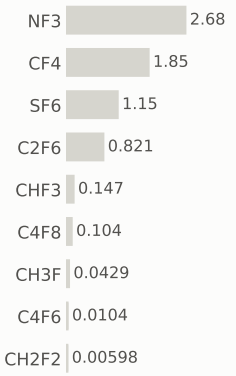
steps by process area



materials deposited (g)



gases emitted (g)



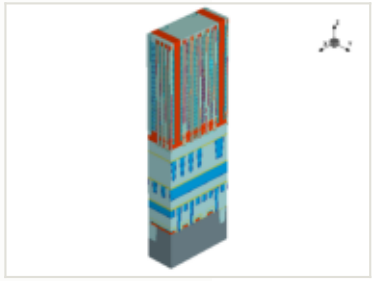
Step #5190: Wafer_clean

clean · Wafer_clean

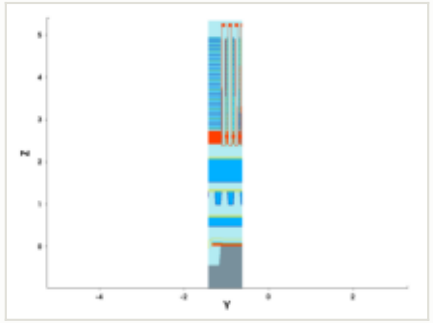
0.169	455.0	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

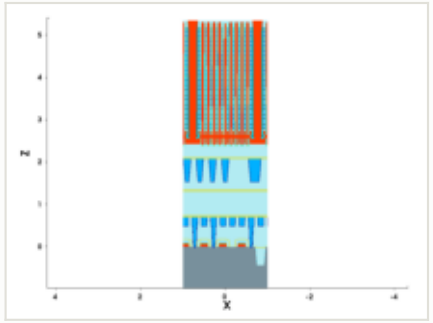
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

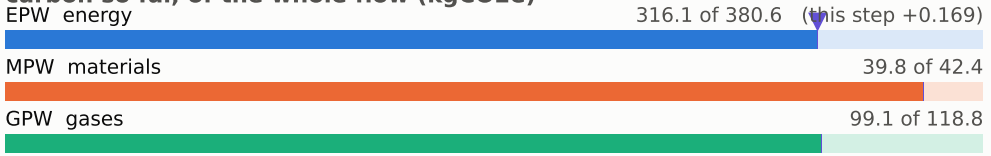
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

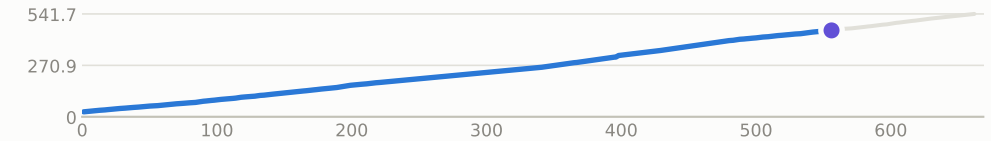
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

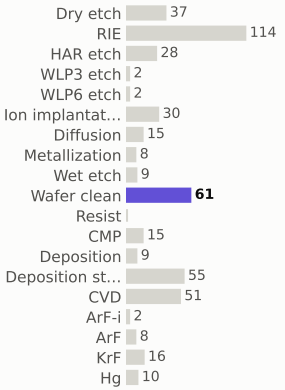
carbon so far, of the whole flow (kgCO2e)



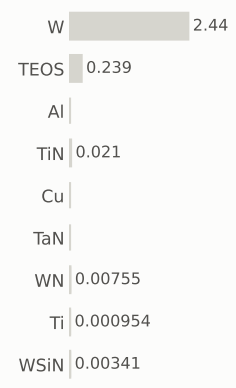
embodied carbon up to each step (kgCO2e per wafer)



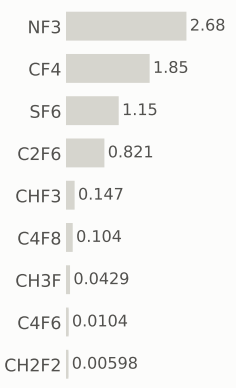
steps by process area



materials deposited (g)



gases emitted (g)



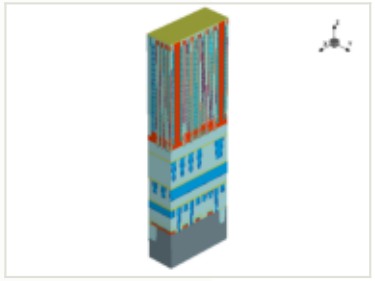
Step #5200: CVD

depo_isotropic · CVD

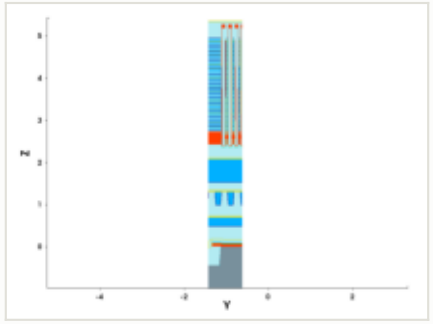
1.31	456.3	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

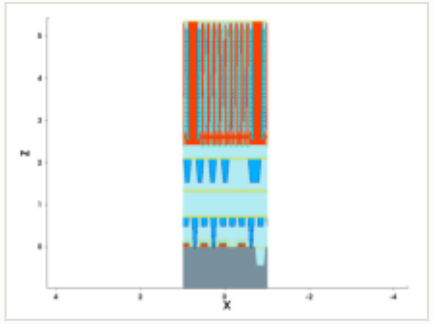
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

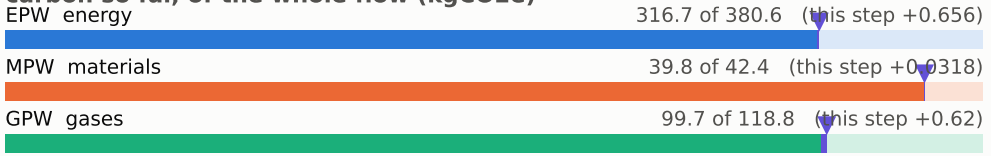
no mask for this step

this step

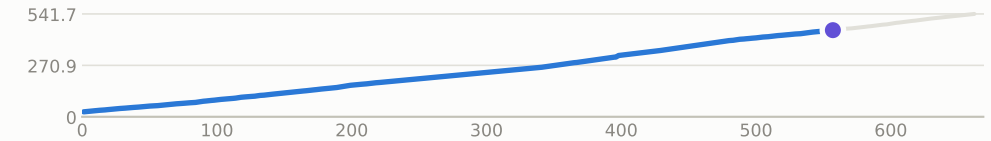
process area	CVD
material	—
thickness	0.027 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

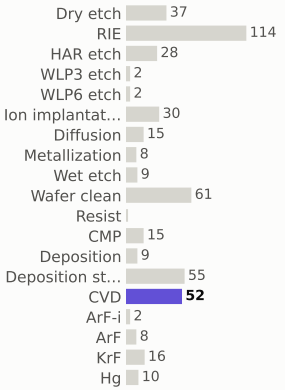
carbon so far, of the whole flow (kgCO2e)



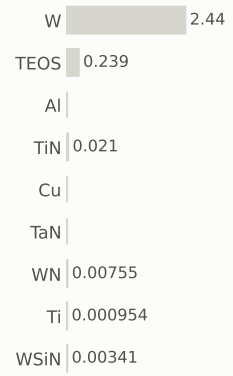
embodied carbon up to each step (kgCO2e per wafer)



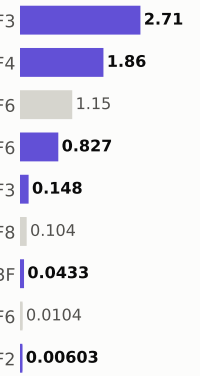
steps by process area



materials deposited (g)



gases emitted (g)



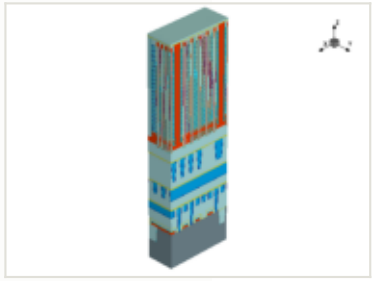
Step #5220: CVD

depo_isotropic · CVD

1.31	457.6	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

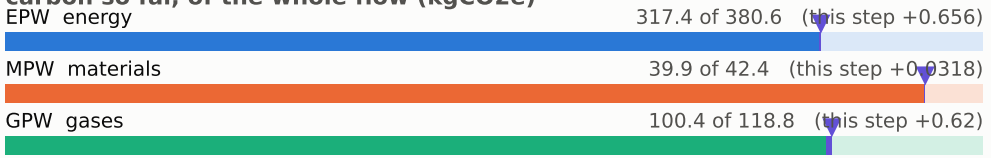


this step

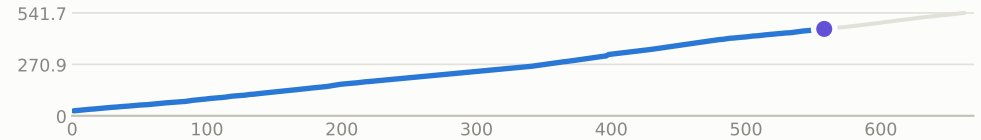
process area	CVD
material	—
thickness	0.045 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

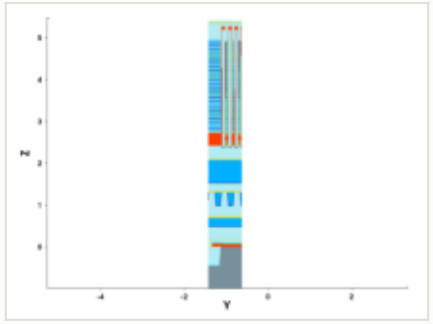
carbon so far, of the whole flow (kgCO2e)



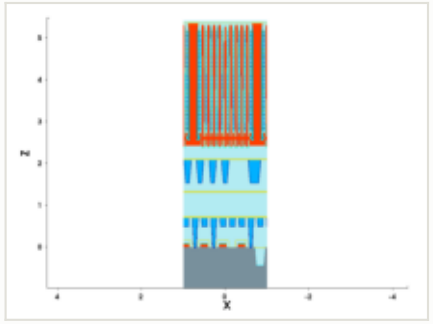
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



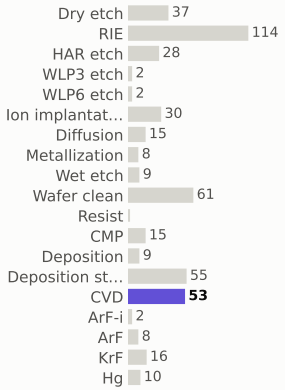
y cut at -1.032 µm



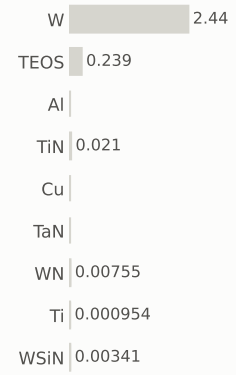
mask for this step

no mask for this step

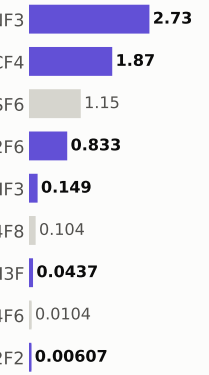
steps by process area



materials deposited (g)



gases emitted (g)



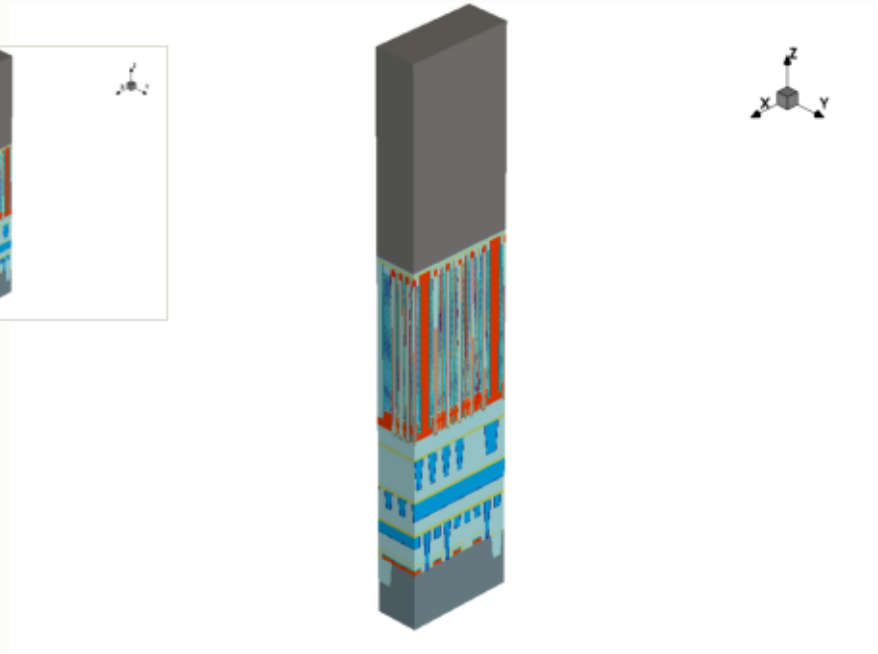
Step #5230: KrF

litho_patterning · KrF

0.199	457.8	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

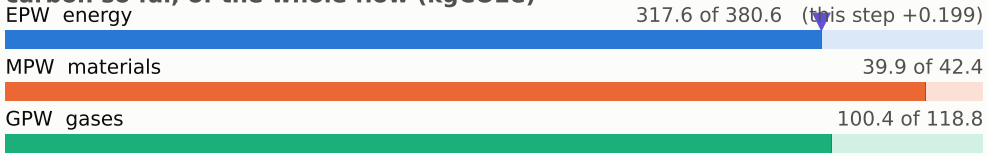


this step

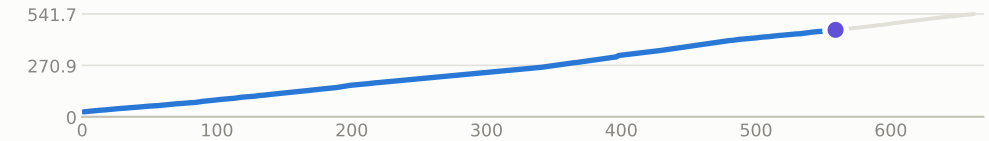
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

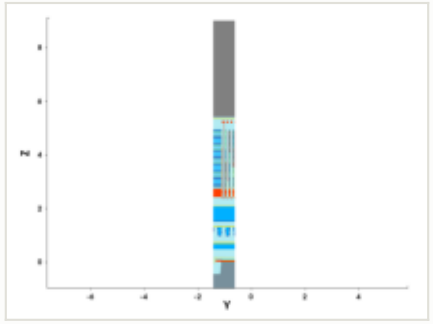
carbon so far, of the whole flow (kgCO2e)



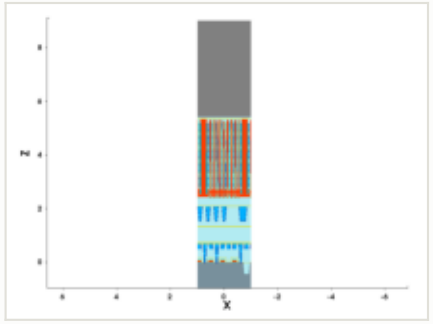
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

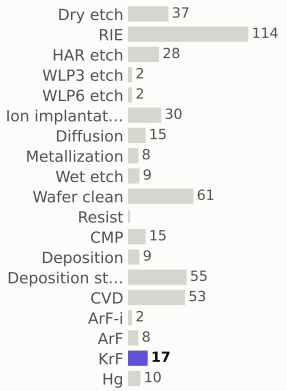


y cut at -1.032 µm

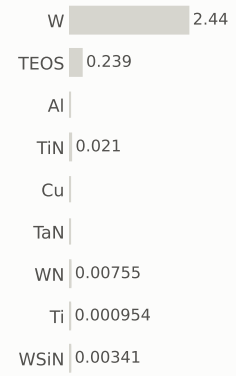


mask: layer 40:0, positive (OASIS)

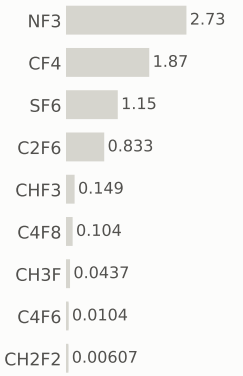
steps by process area



materials deposited (g)



gases emitted (g)



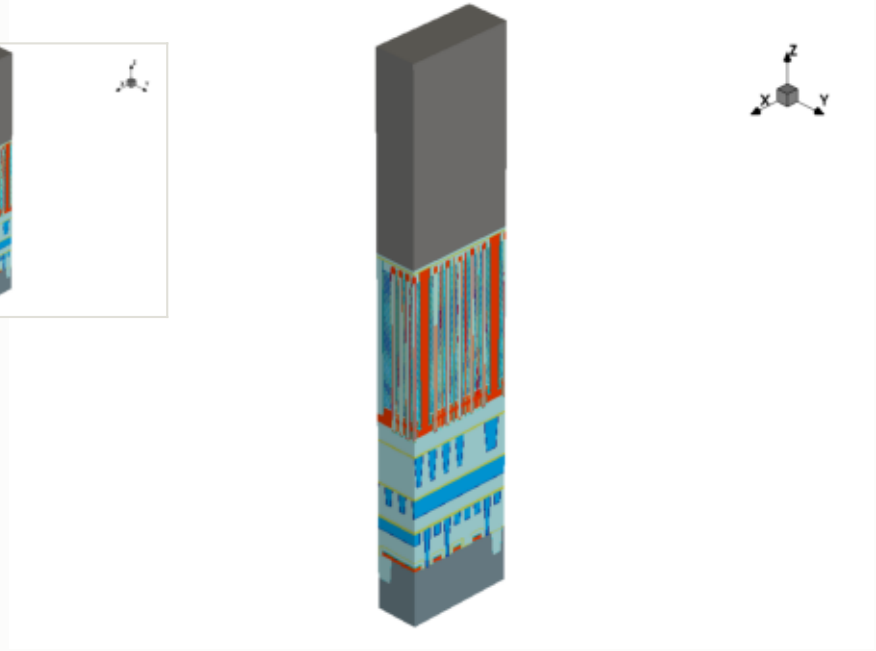
Step #5240: RIE

etch_anisotropic · RIE

1.04	458.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

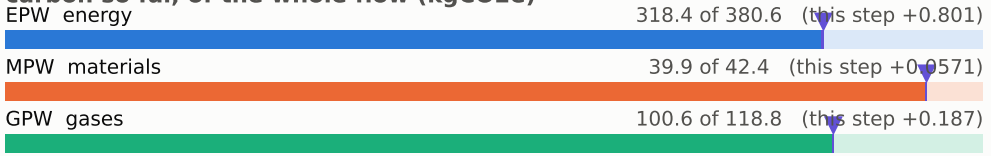


this step

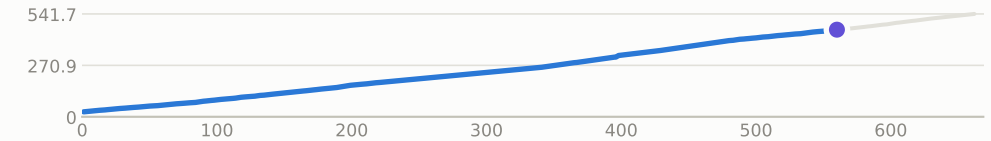
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

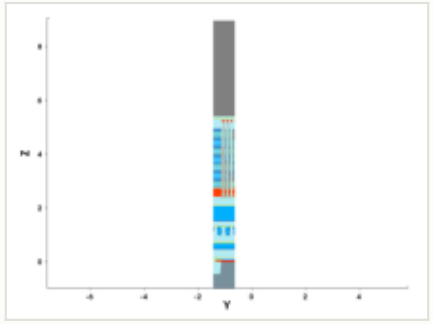
carbon so far, of the whole flow (kgCO2e)



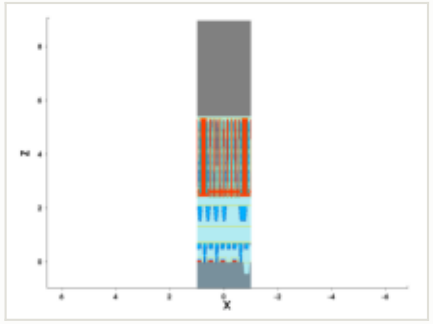
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



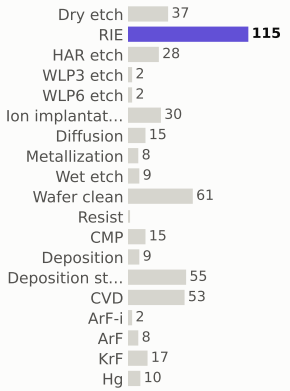
y cut at -1.032 µm



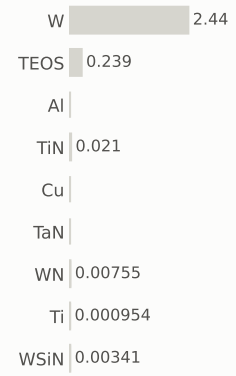
mask for this step

no mask for this step

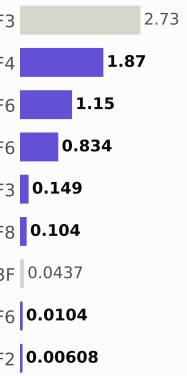
steps by process area



materials deposited (g)



gases emitted (g)



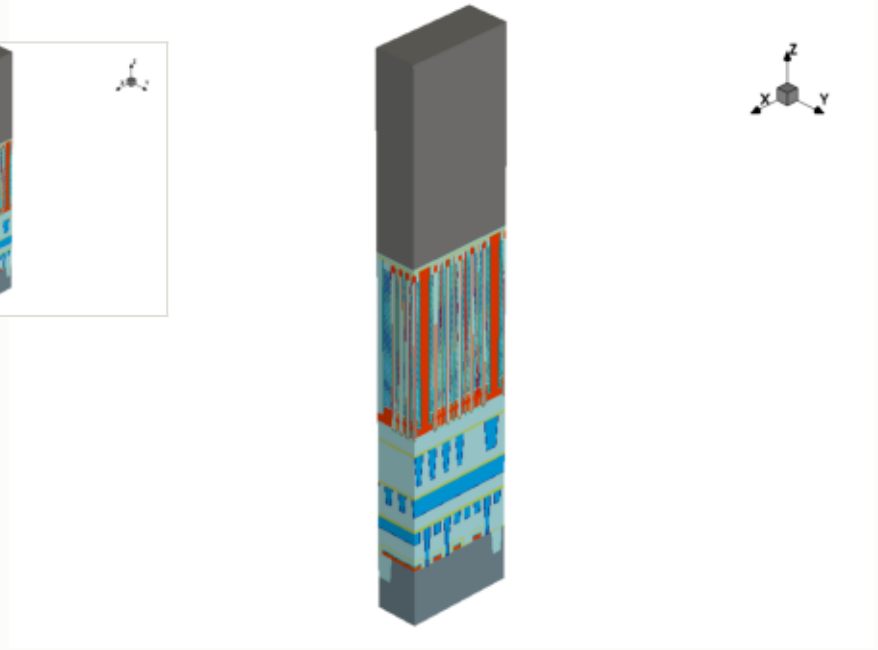
Step #5250: RIE

etch_anisotropic · RIE

1.04	459.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

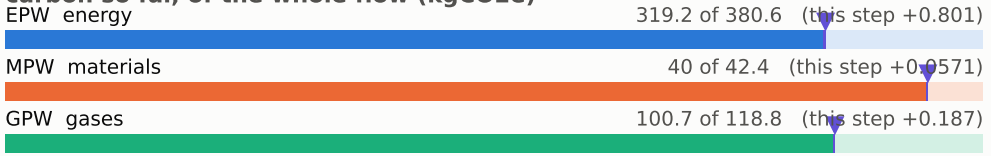


this step

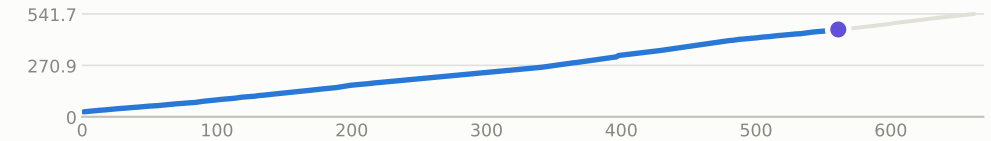
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

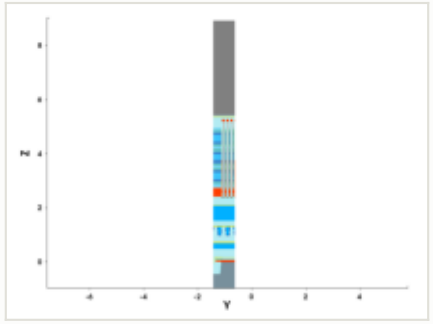
carbon so far, of the whole flow (kgCO2e)



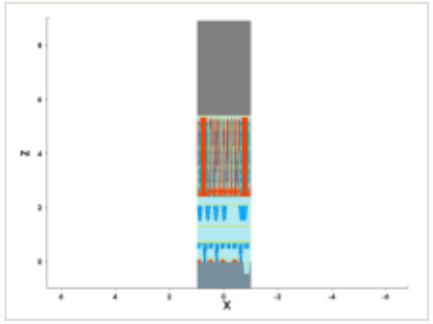
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



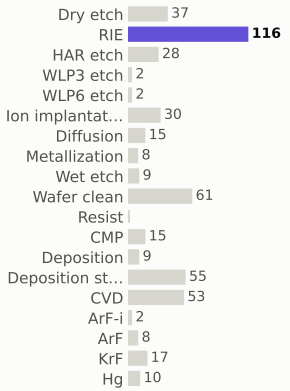
y cut at -1.032 µm



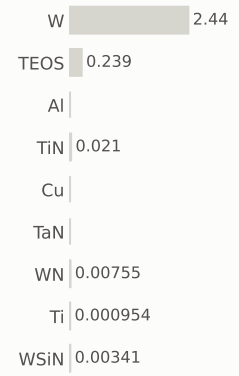
mask for this step

no mask for this step

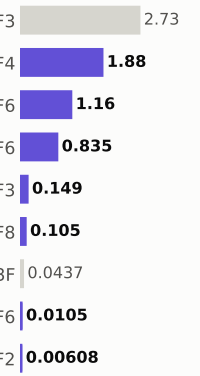
steps by process area



materials deposited (g)



gases emitted (g)



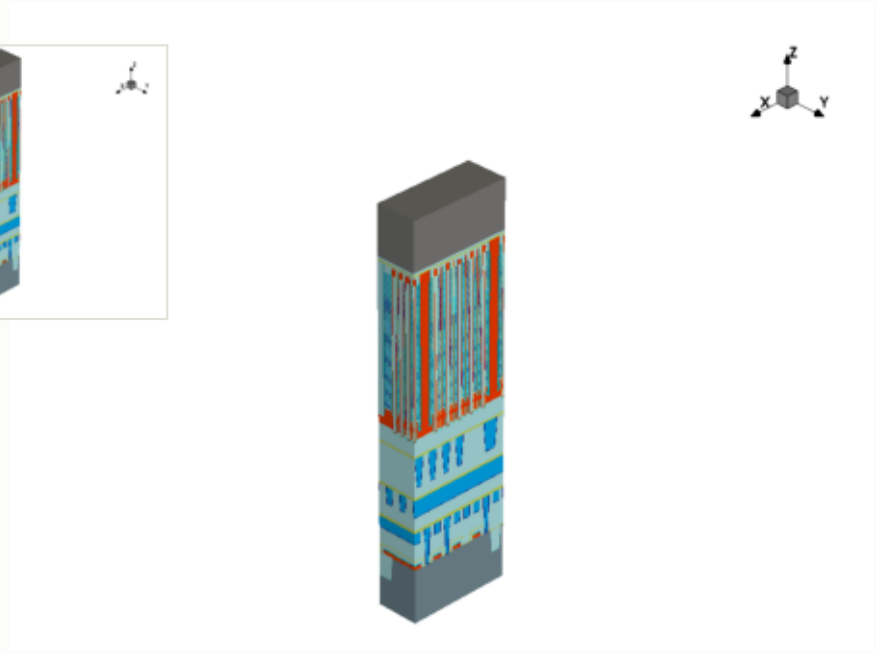
Step #5260: RIE

etch_anisotropic · RIE

1.04	461.0	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

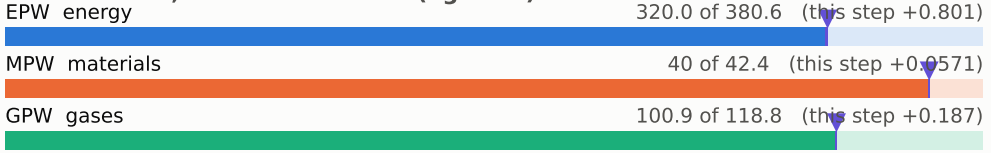


this step

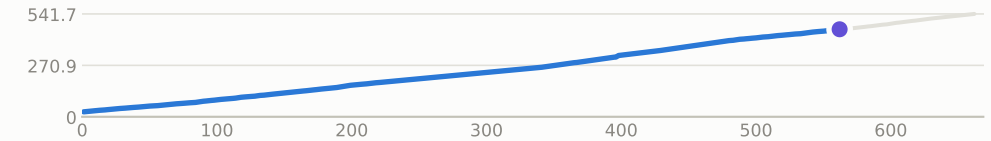
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

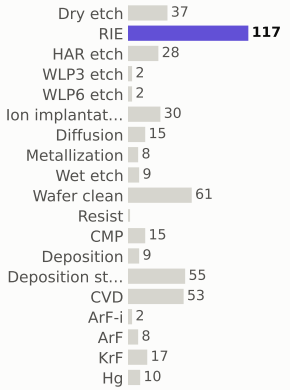
carbon so far, of the whole flow (kgCO2e)



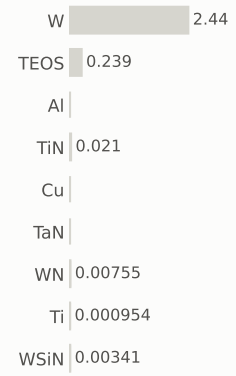
embodied carbon up to each step (kgCO2e per wafer)



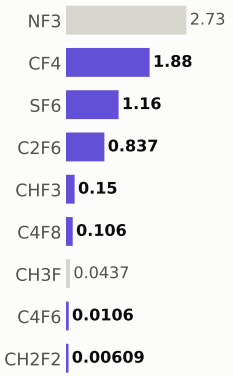
steps by process area



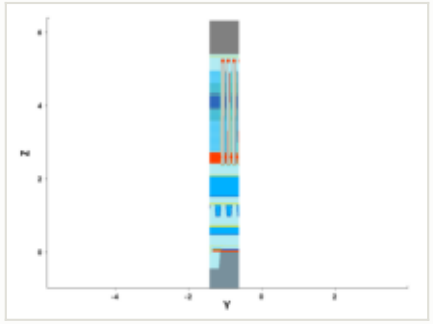
materials deposited (g)



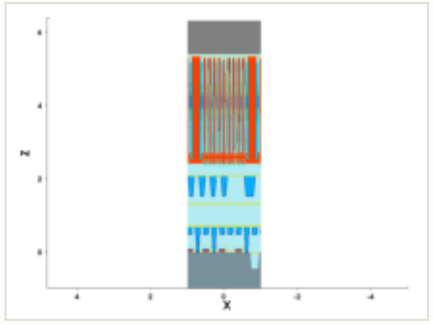
gases emitted (g)



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

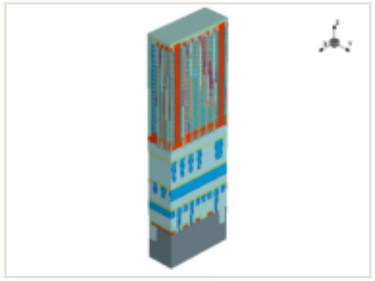
Step #5270: Dry_etch

etch_strip · Dry_etch

0.801	461.8	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

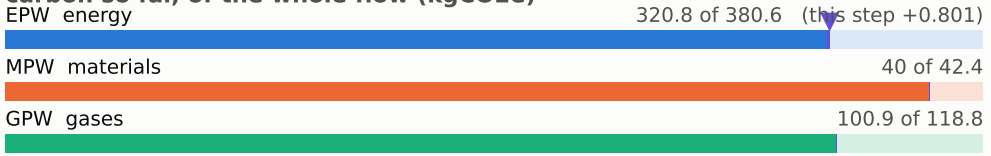
zoomed to fit



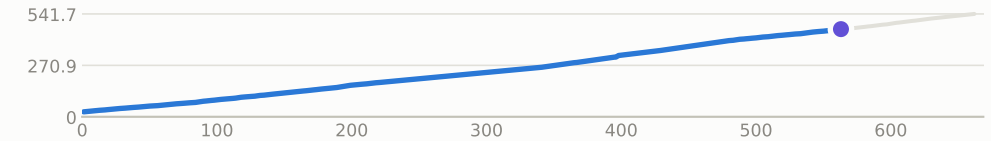
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

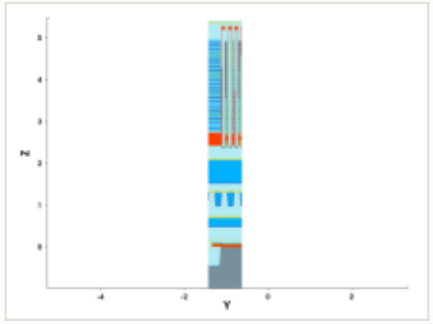
carbon so far, of the whole flow (kgCO2e)



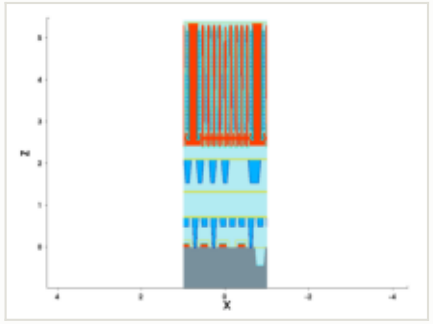
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



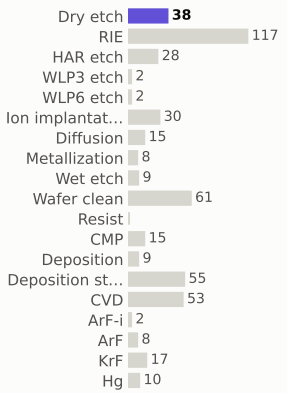
y cut at -1.032 µm



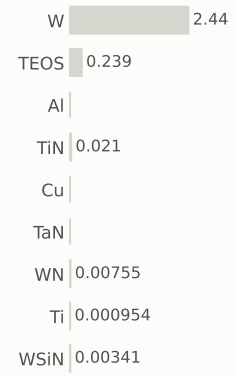
mask for this step

no mask for this step

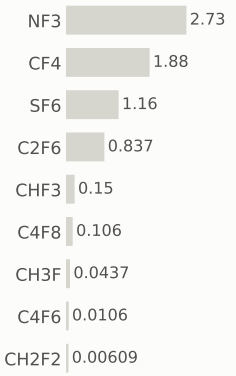
steps by process area



materials deposited (g)



gases emitted (g)



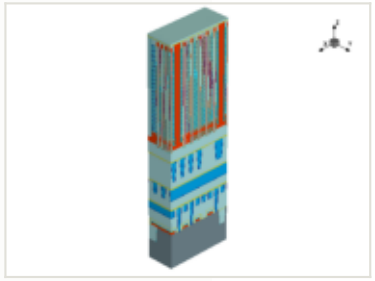
Step #5280: Wafer_clean

clean · Wafer_clean

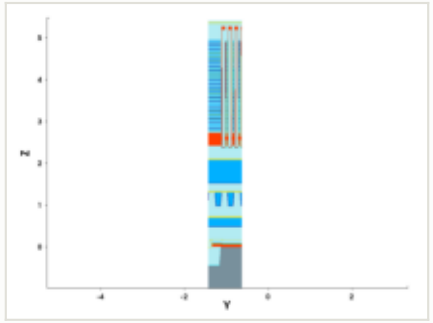
0.169	461.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

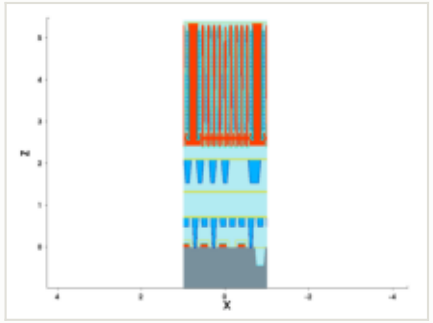
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

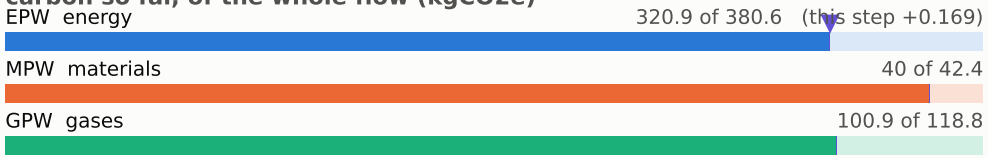
no mask for this step

this step

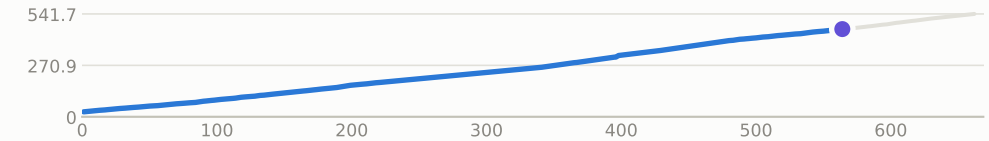
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

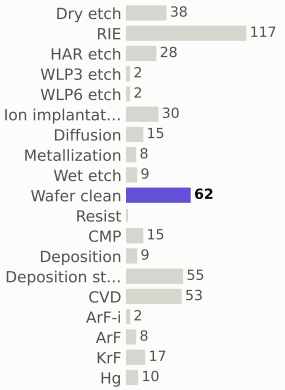
carbon so far, of the whole flow (kgCO2e)



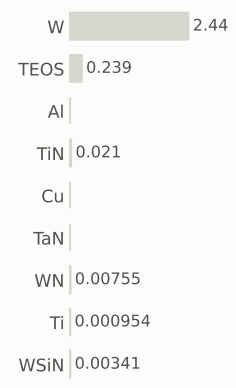
embodied carbon up to each step (kgCO2e per wafer)



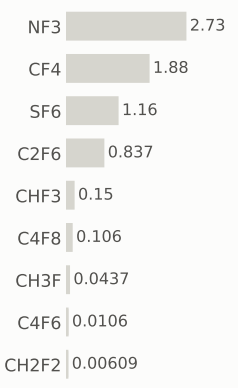
steps by process area



materials deposited (g)



gases emitted (g)



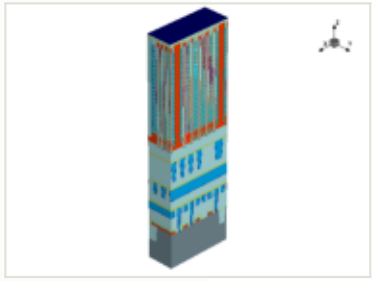
Step #5290: Deposition

depo_isotropic · Deposition

0.656	462.6	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

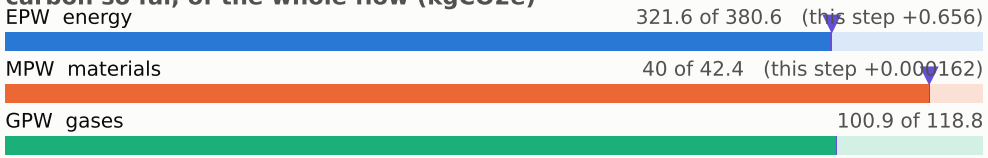
zoomed to fit



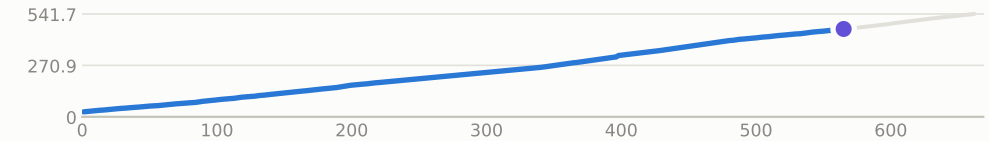
this step

process area	Deposition	EPW	0.656 kgCO2e
material	TiN	EPW tool	0.469 kgCO2e
thickness	0.01 µm	EPW overhead	0.187 kgCO2e
material GWP	43.9778 gCO2e/g	MPW	0.000162 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

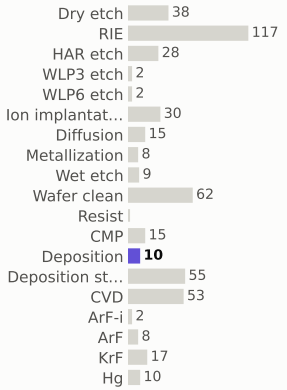
carbon so far, of the whole flow (kgCO2e)



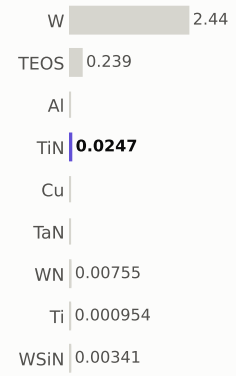
embodied carbon up to each step (kgCO2e per wafer)



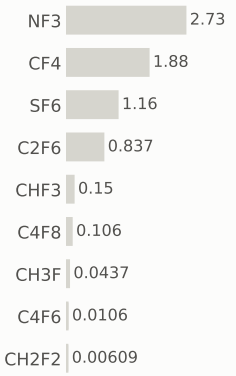
steps by process area



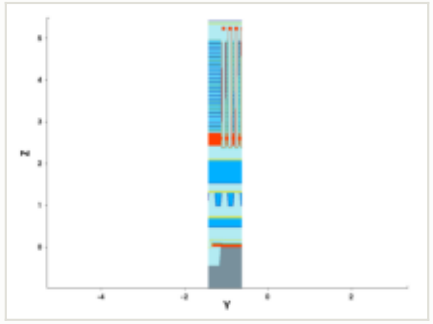
materials deposited (g)



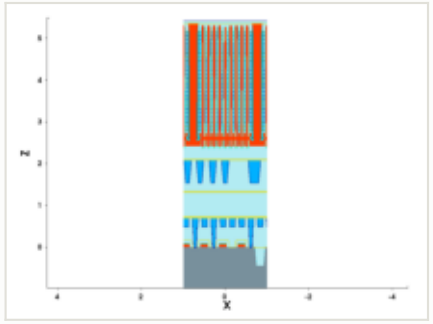
gases emitted (g)



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

Step #5300: Metallization

depo_isotropic · Metallization

1.23	463.8	0.23%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

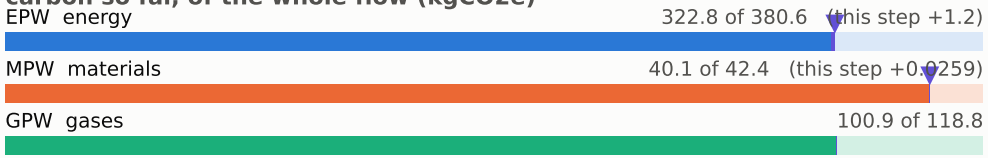
zoomed to fit

this step

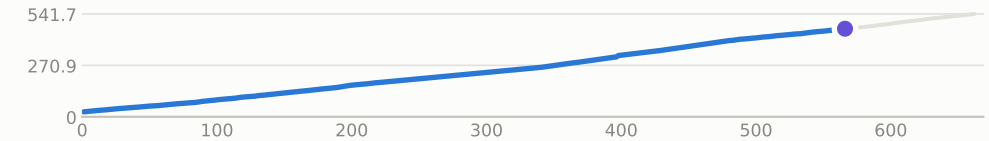
process area	Metallization
material	W
thickness	0.25 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0259 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.23 kgCO2e**

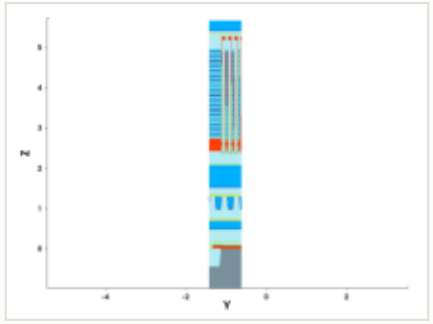
carbon so far, of the whole flow (kgCO2e)



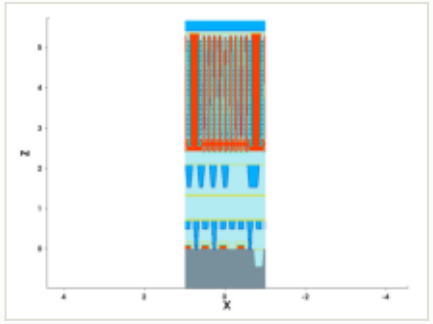
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



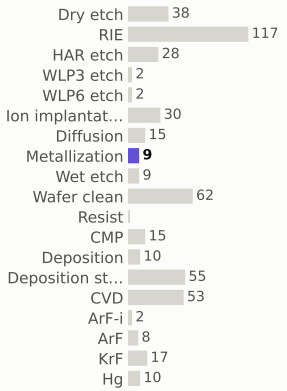
y cut at -1.032 µm



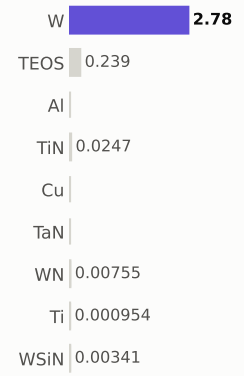
mask for this step

no mask for this step

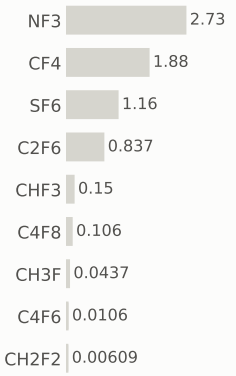
steps by process area



materials deposited (g)



gases emitted (g)



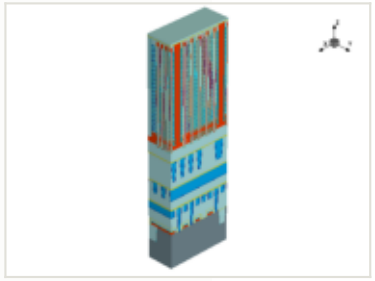
Step #5310: CMP

etch_cmp · CMP

0.169	464.0	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

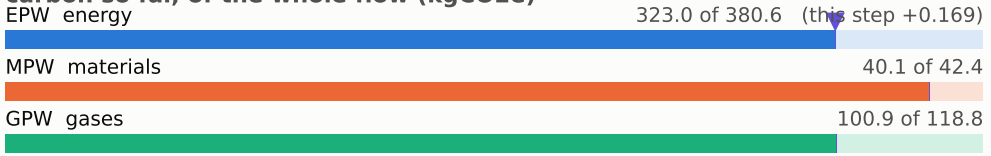
zoomed to fit



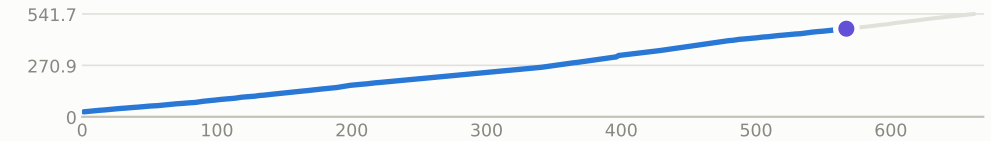
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

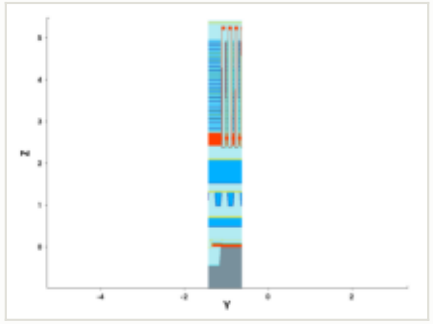
carbon so far, of the whole flow (kgCO2e)



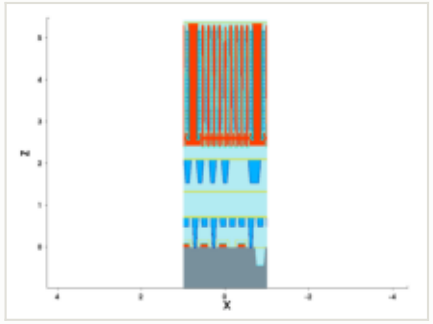
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



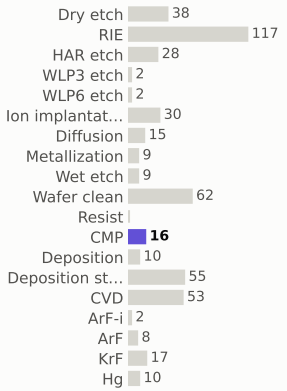
y cut at -1.032 µm



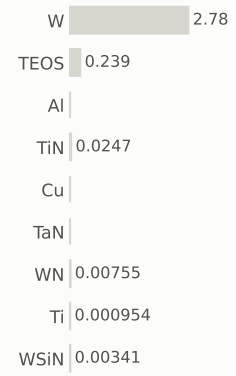
mask for this step

no mask for this step

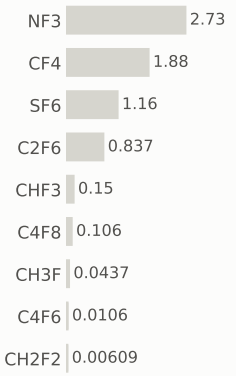
steps by process area



materials deposited (g)



gases emitted (g)



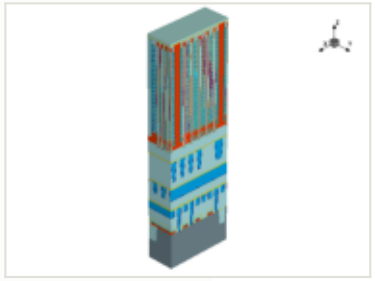
Step #5320: Wafer_clean

clean · Wafer_clean

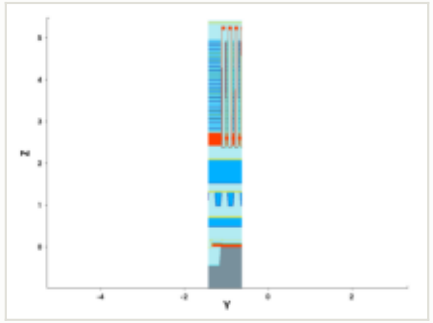
0.169	464.1	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

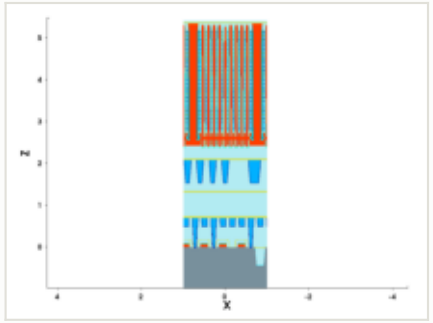
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

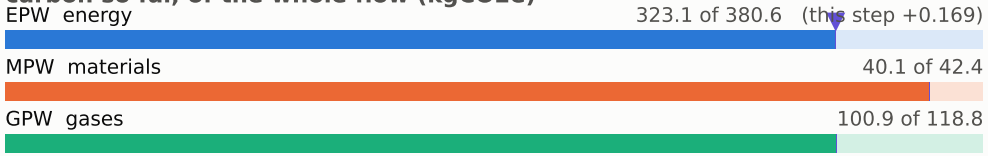
no mask for this step

this step

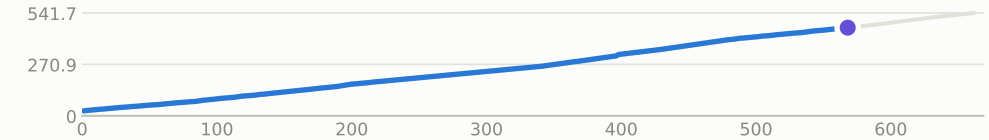
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

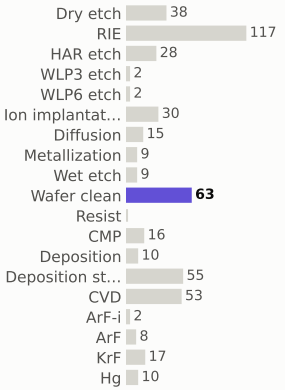
carbon so far, of the whole flow (kgCO2e)



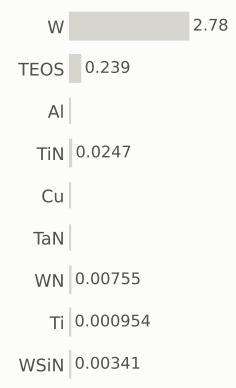
embodied carbon up to each step (kgCO2e per wafer)



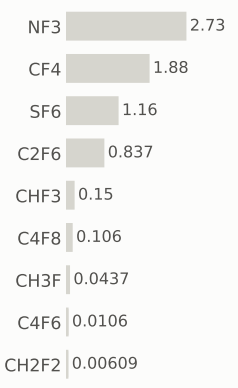
steps by process area



materials deposited (g)



gases emitted (g)



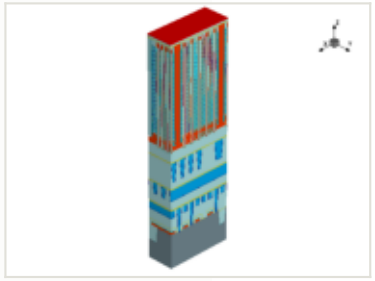
Step #5340: CVD

depo_isotropic · CVD

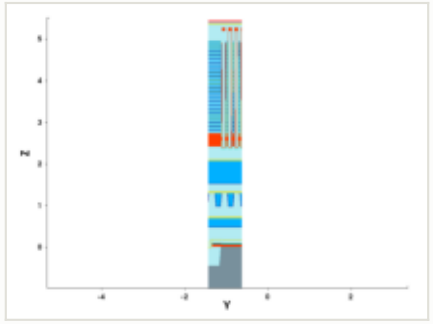
1.31	465.5	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

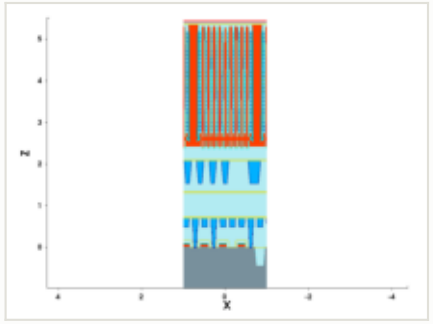
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

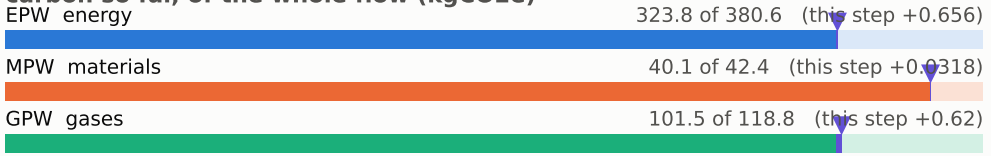
no mask for this step

this step

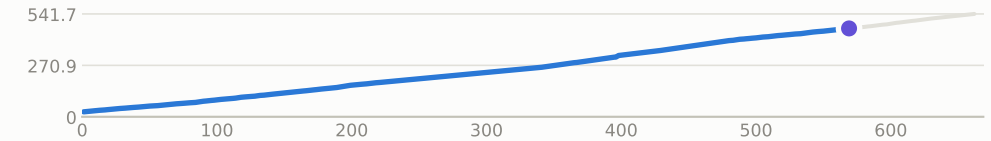
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

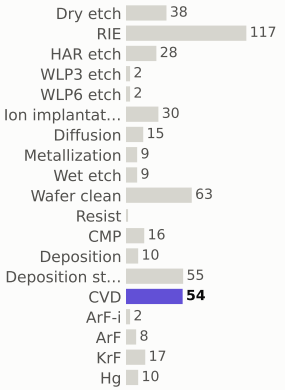
carbon so far, of the whole flow (kgCO2e)



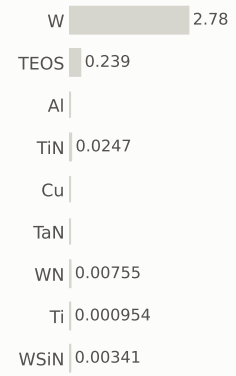
embodied carbon up to each step (kgCO2e per wafer)



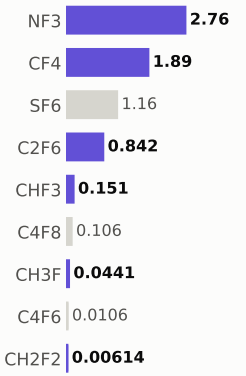
steps by process area



materials deposited (g)



gases emitted (g)



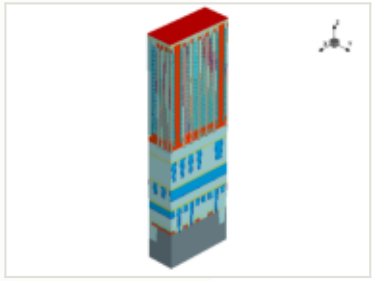
Step #5350: Wafer_clean

clean · Wafer_clean

0.169	465.6	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

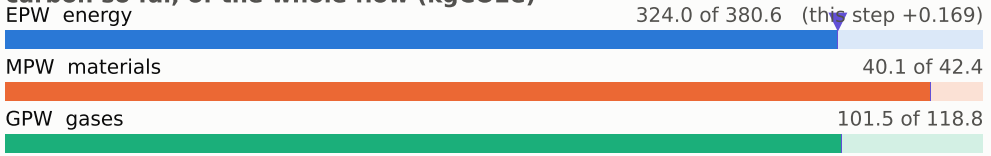


this step

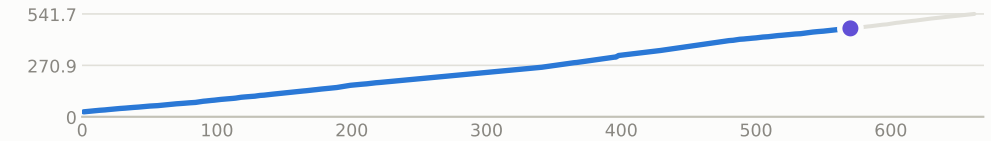
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

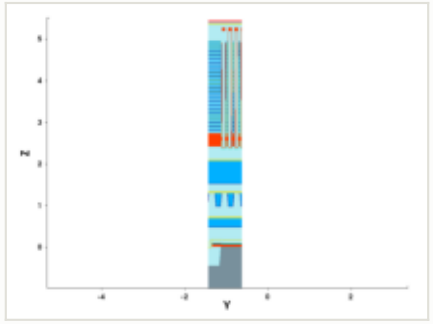
carbon so far, of the whole flow (kgCO2e)



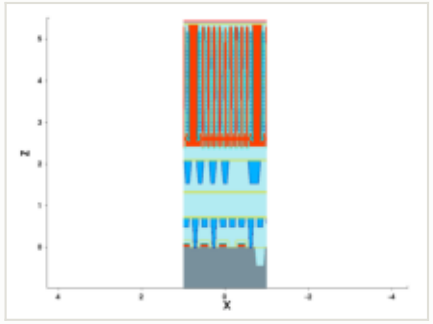
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



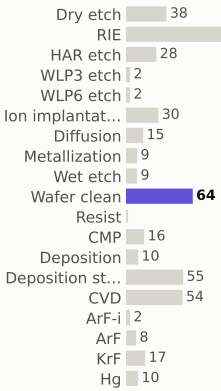
y cut at -1.032 µm



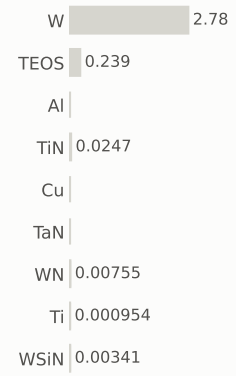
mask for this step

no mask for this step

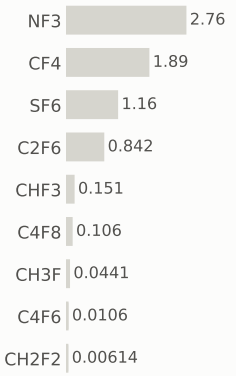
steps by process area



materials deposited (g)



gases emitted (g)



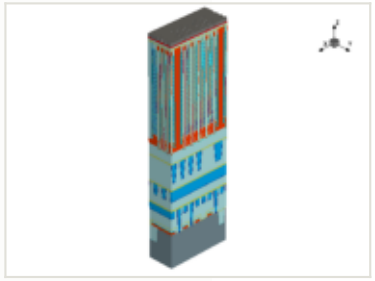
Step #5360: ArF-i

litho_patterning · ArF-i

0.217	465.8	0.04%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

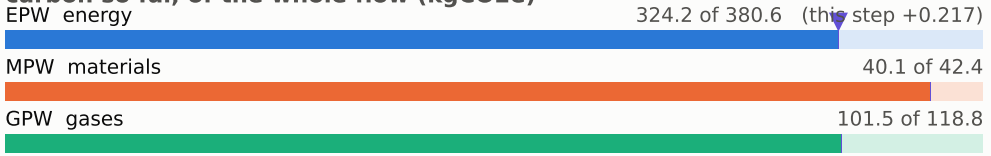
zoomed to fit



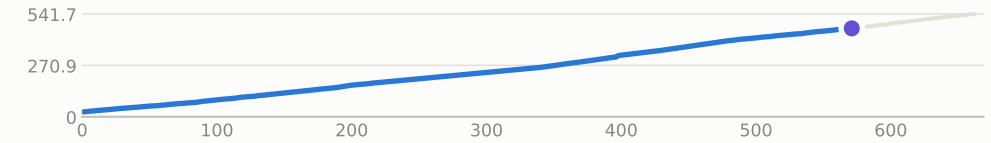
this step

process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

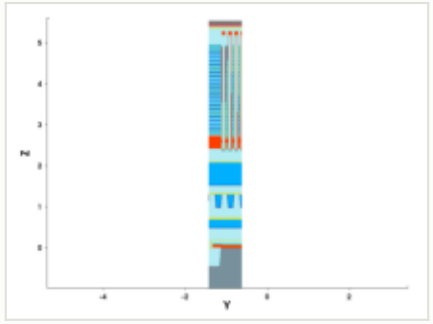
carbon so far, of the whole flow (kgCO2e)



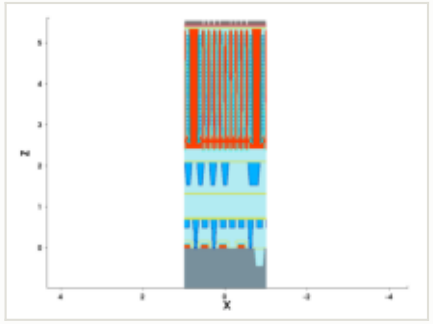
embodied carbon up to each step (kgCO2e per wafer)



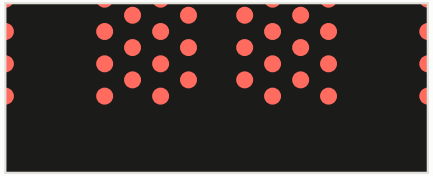
x cut at 0.000 µm



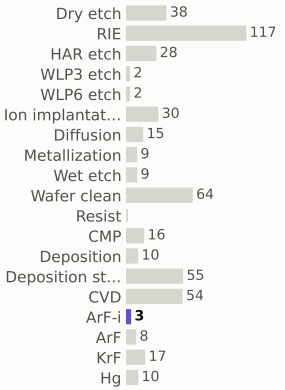
y cut at -1.032 µm



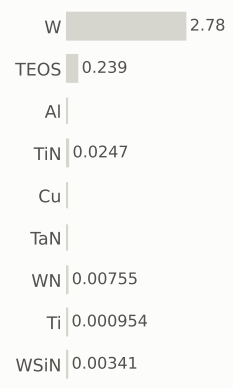
mask: layer 41:0 (OASIS)



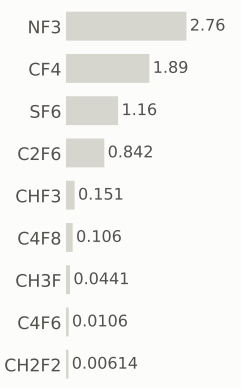
steps by process area



materials deposited (g)



gases emitted (g)



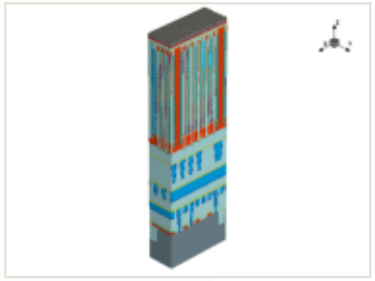
Step #5370: RIE

etch_anisotropic · RIE

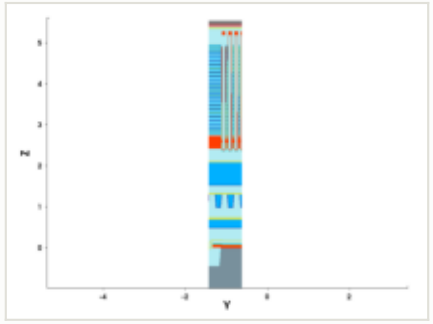
1.04	466.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

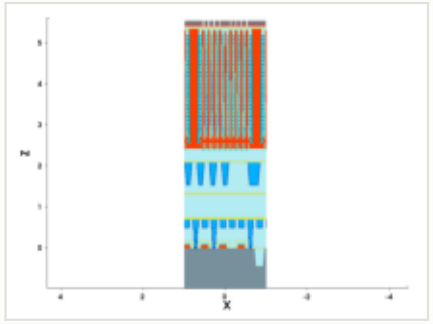
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

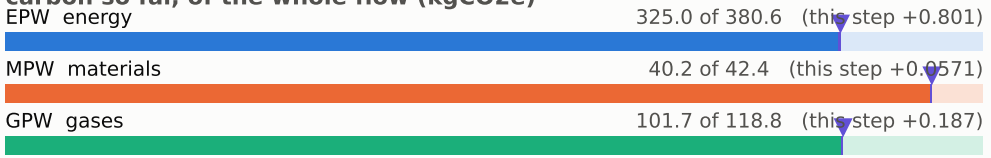
no mask for this step

this step

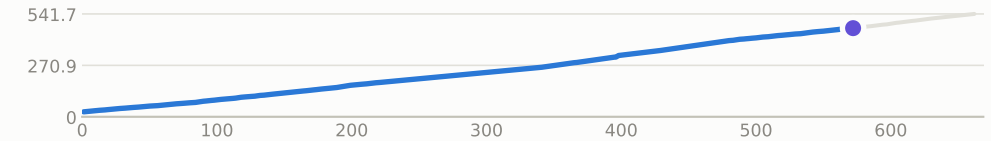
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

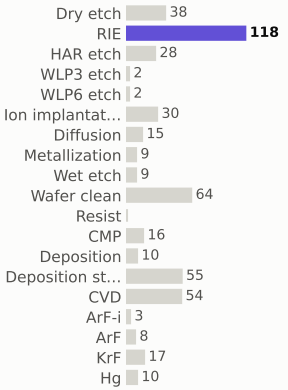
carbon so far, of the whole flow (kgCO2e)



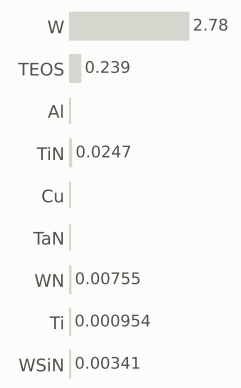
embodied carbon up to each step (kgCO2e per wafer)



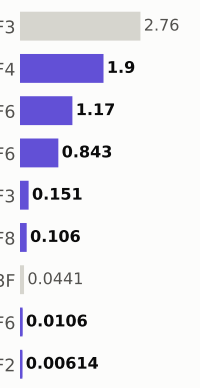
steps by process area



materials deposited (g)



gases emitted (g)



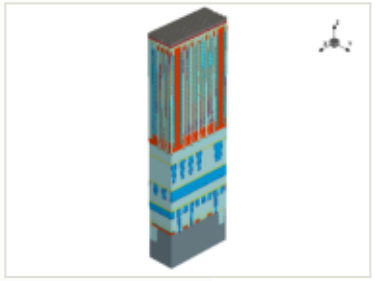
Step #5380: RIE

etch_anisotropic · RIE

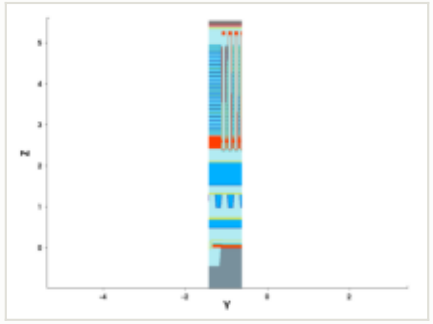
1.04	467.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

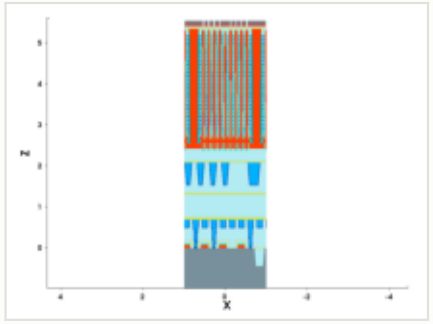
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

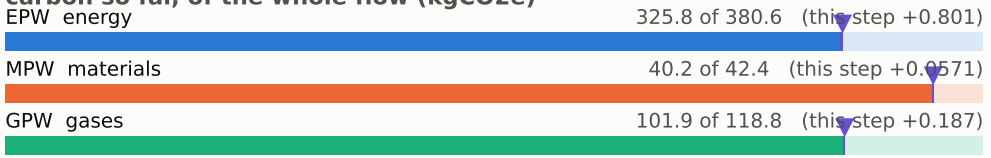
no mask for this step

this step

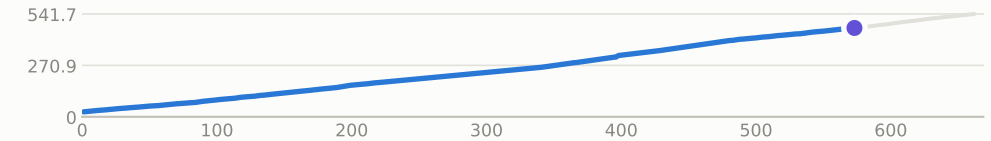
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

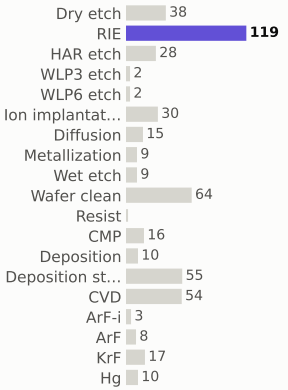
carbon so far, of the whole flow (kgCO2e)



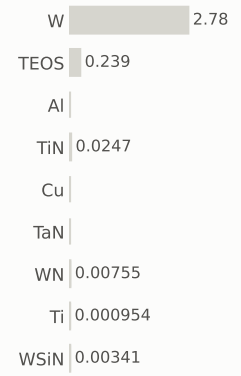
embodied carbon up to each step (kgCO2e per wafer)



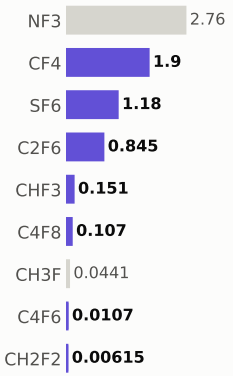
steps by process area



materials deposited (g)



gases emitted (g)



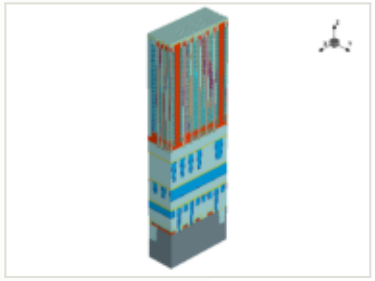
Step #5390: Dry_etch

etch_strip · Dry_etch

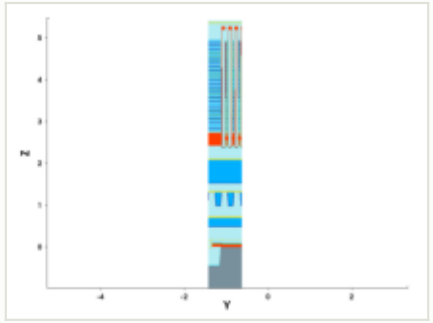
0.801	468.7	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

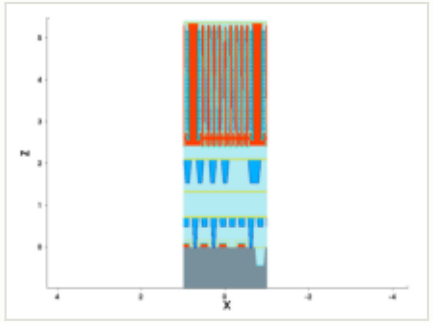
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

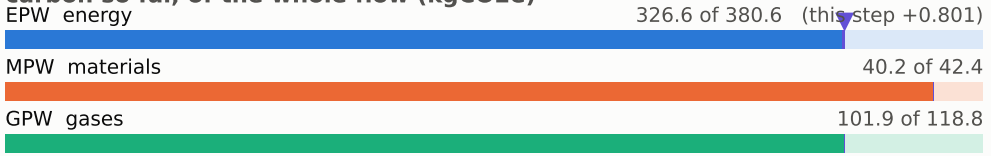
no mask for this step

this step

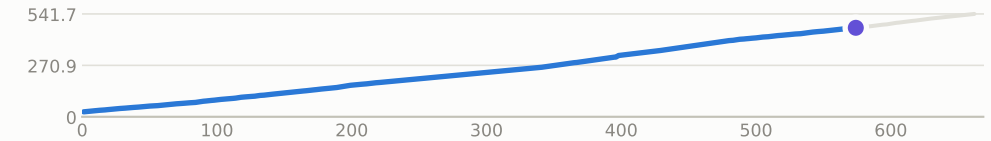
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

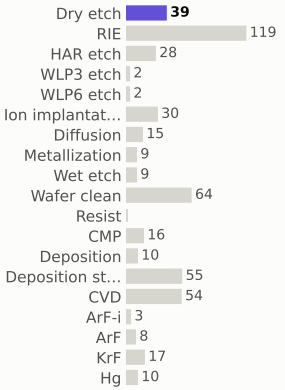
carbon so far, of the whole flow (kgCO2e)



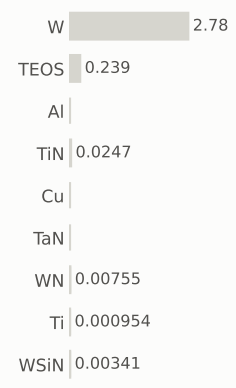
embodied carbon up to each step (kgCO2e per wafer)



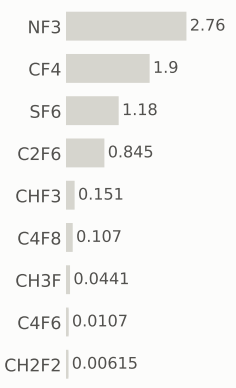
steps by process area



materials deposited (g)



gases emitted (g)



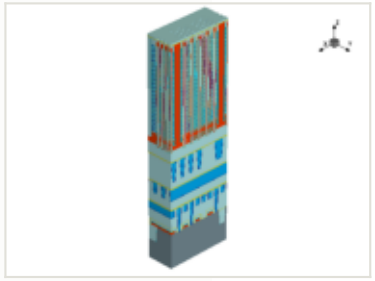
Step #5400: Wafer_clean

clean · Wafer_clean

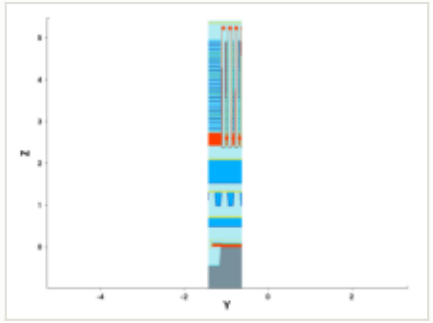
0.169	468.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

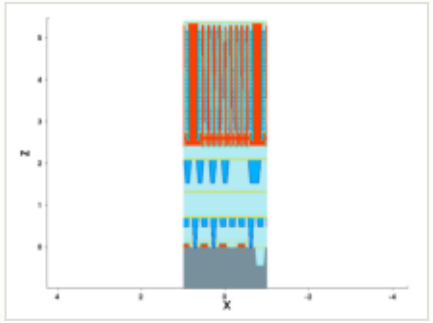
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

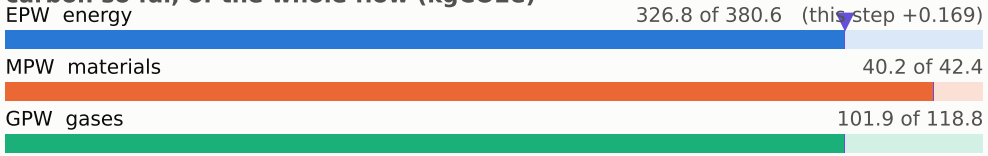
no mask for this step

this step

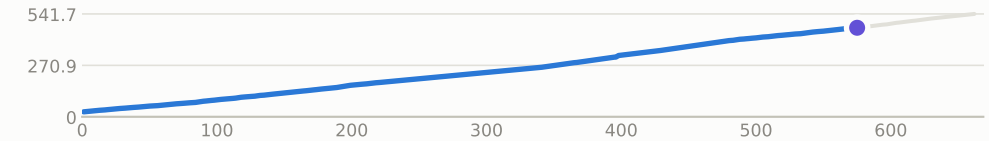
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

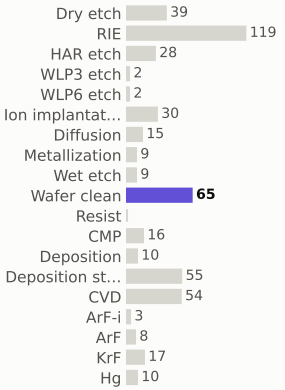
carbon so far, of the whole flow (kgCO2e)



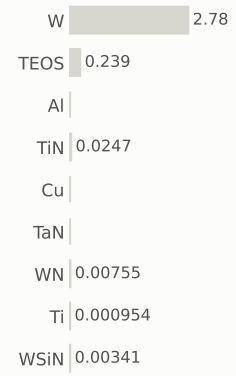
embodied carbon up to each step (kgCO2e per wafer)



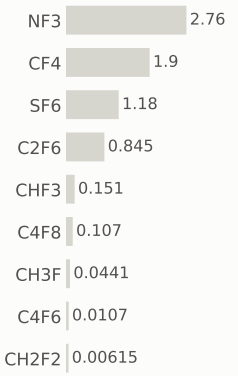
steps by process area



materials deposited (g)



gases emitted (g)



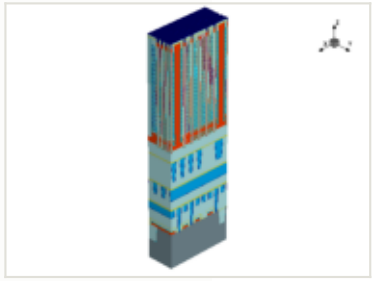
Step #5410: CVD

depo_ellipsoidal · CVD

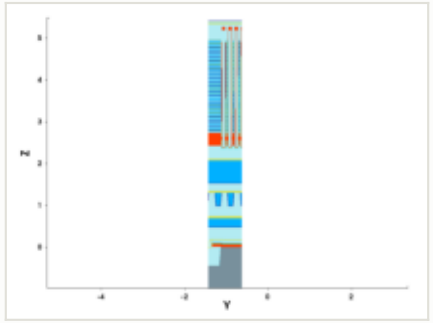
1.31	470.2	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

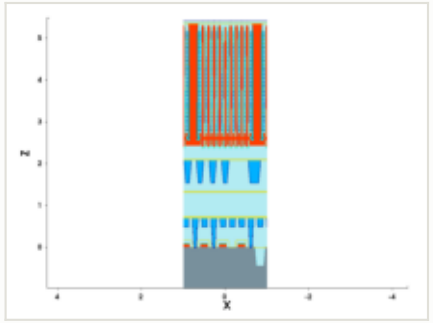
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

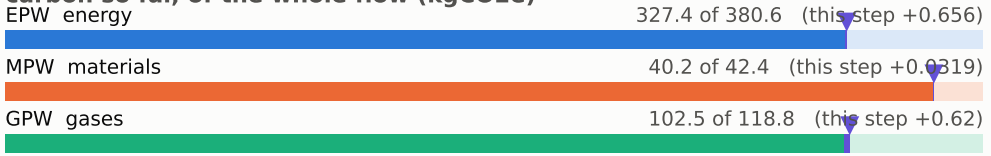
no mask for this step

this step

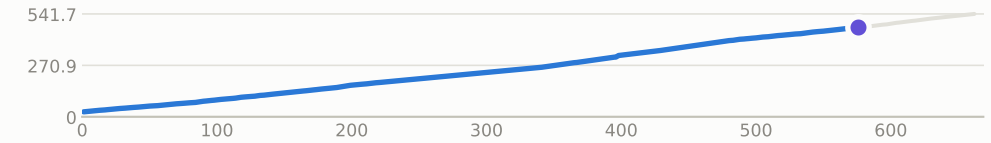
process area	CVD
material	TiN
thickness	0.009 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0319 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

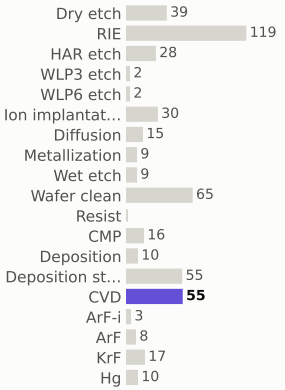
carbon so far, of the whole flow (kgCO2e)



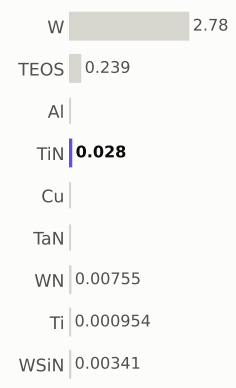
embodied carbon up to each step (kgCO2e per wafer)



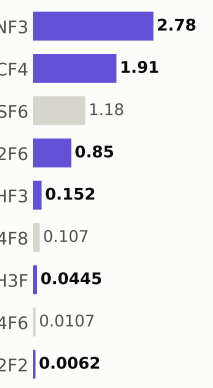
steps by process area



materials deposited (g)



gases emitted (g)



Step #5420: Metallization

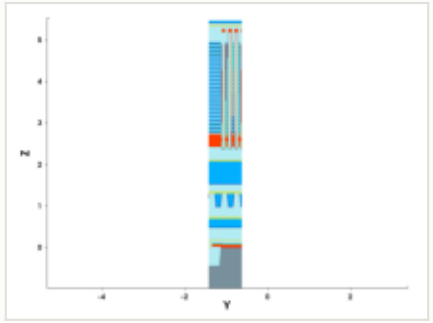
depo_isotropic · Metallization

1.21	471.4	0.22%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

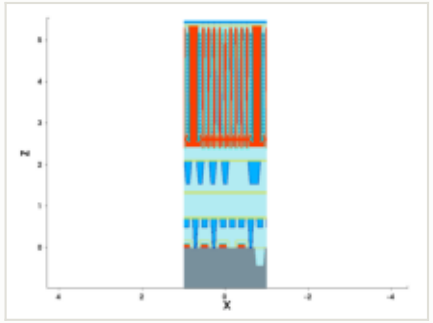
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

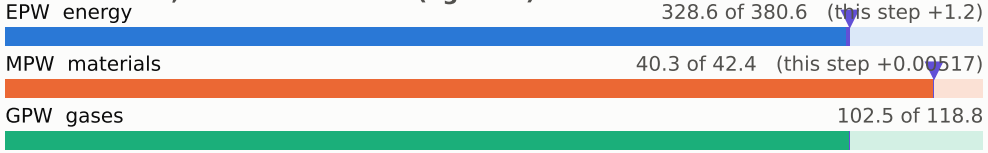
no mask for this step

this step

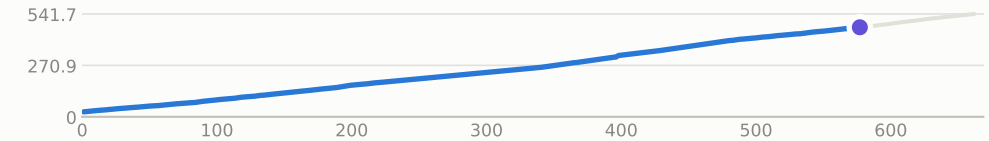
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

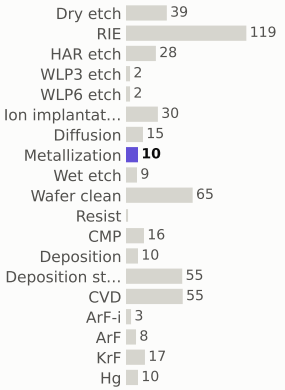
carbon so far, of the whole flow (kgCO2e)



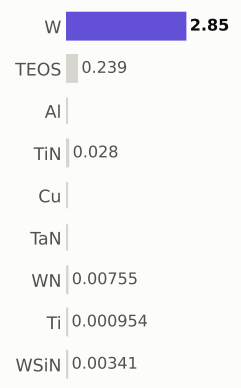
embodied carbon up to each step (kgCO2e per wafer)



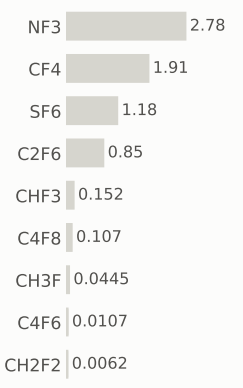
steps by process area



materials deposited (g)



gases emitted (g)



Step #5430: CMP

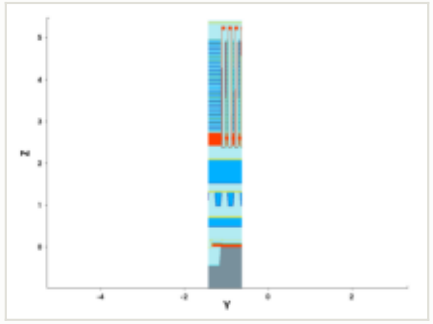
etch_cmp · CMP

0.169	471.6	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

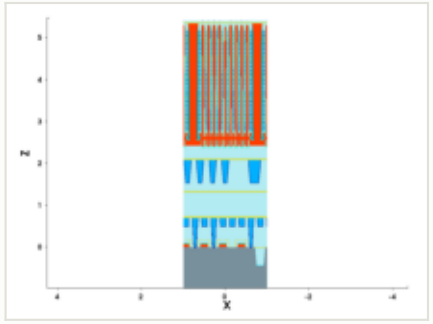
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

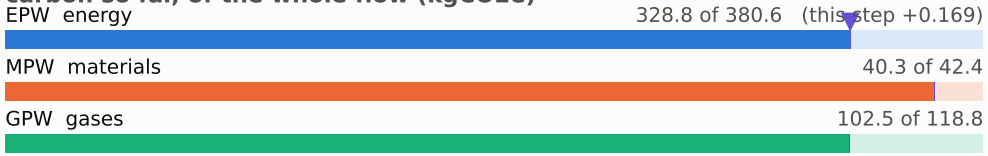
no mask for this step

this step

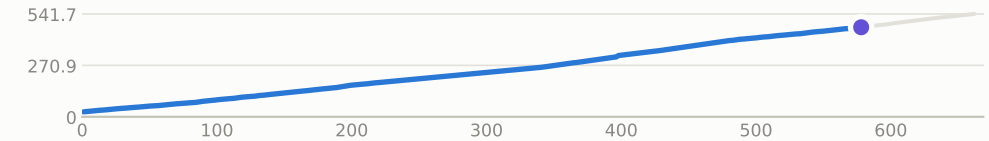
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

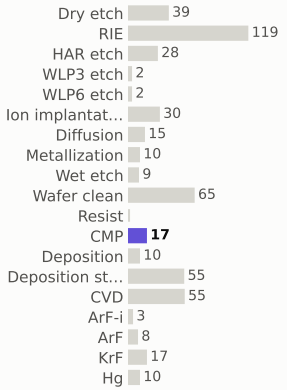
carbon so far, of the whole flow (kgCO2e)



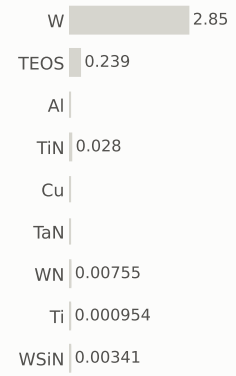
embodied carbon up to each step (kgCO2e per wafer)



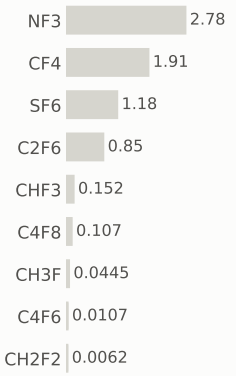
steps by process area



materials deposited (g)



gases emitted (g)



Step #5440: Wafer_clean

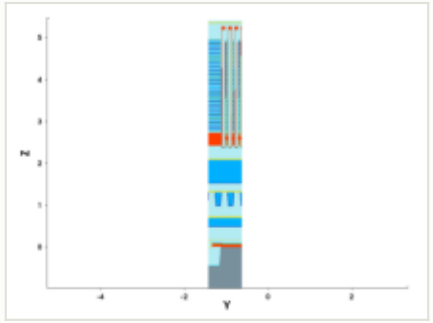
clean · Wafer_clean

0.169	471.8	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

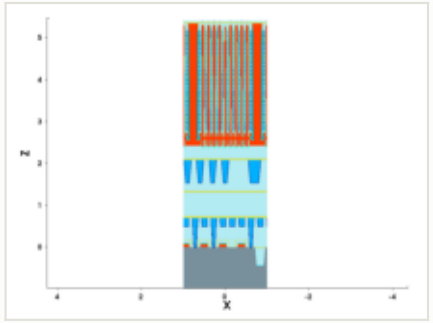
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

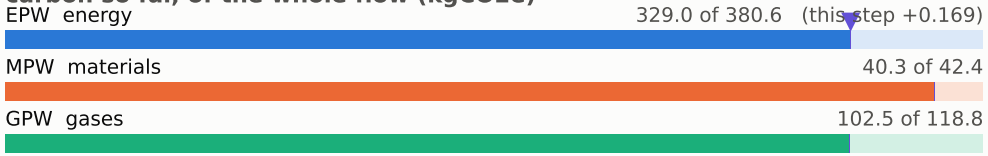
no mask for this step

this step

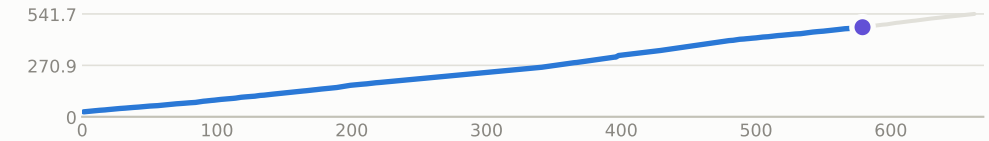
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

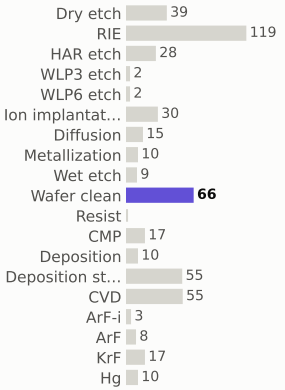
carbon so far, of the whole flow (kgCO2e)



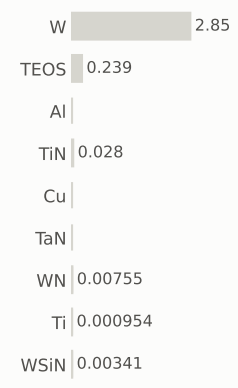
embodied carbon up to each step (kgCO2e per wafer)



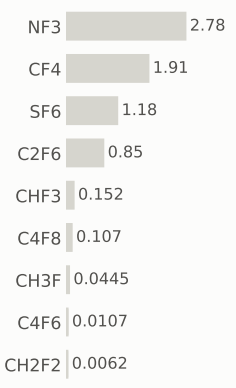
steps by process area



materials deposited (g)



gases emitted (g)



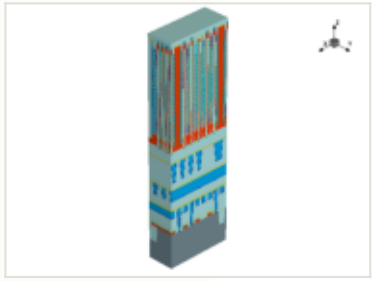
Step #5450: CVD

depo_isotropic · CVD

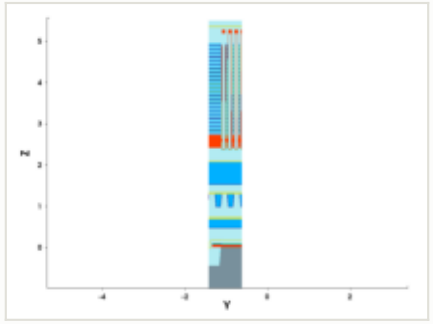
1.31	473.1	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

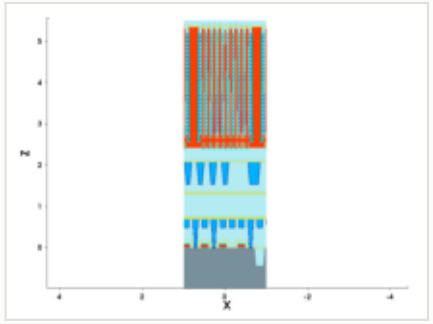
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

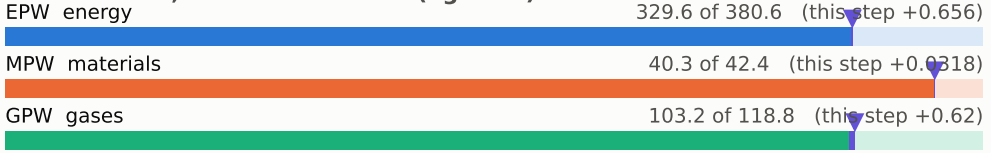
no mask for this step

this step

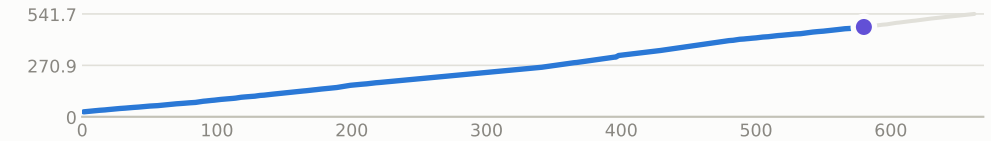
process area	CVD
material	—
thickness	0.09 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

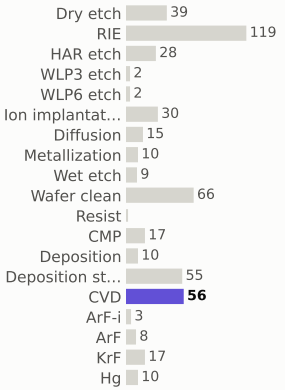
carbon so far, of the whole flow (kgCO2e)



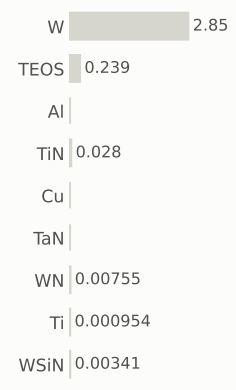
embodied carbon up to each step (kgCO2e per wafer)



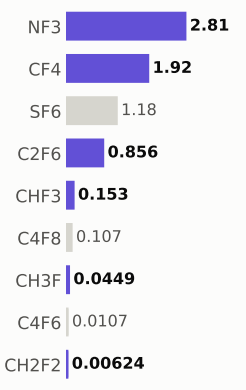
steps by process area



materials deposited (g)



gases emitted (g)



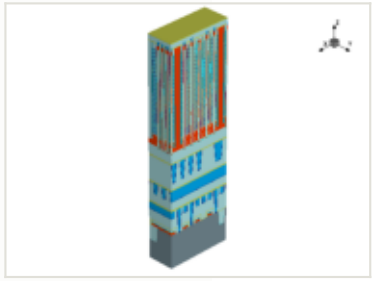
Step #5460: CVD

depo_isotropic · CVD

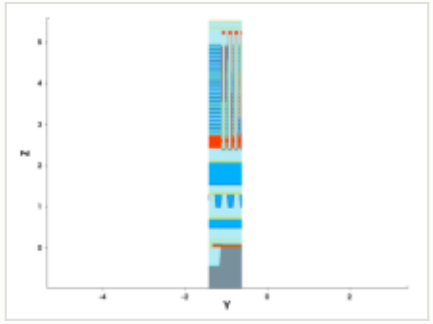
1.31	474.4	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

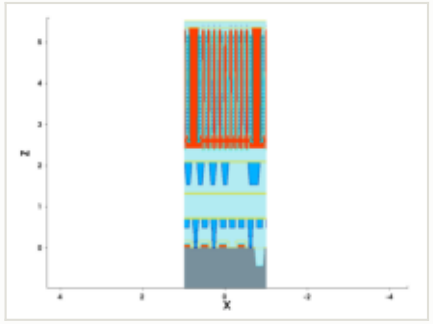
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

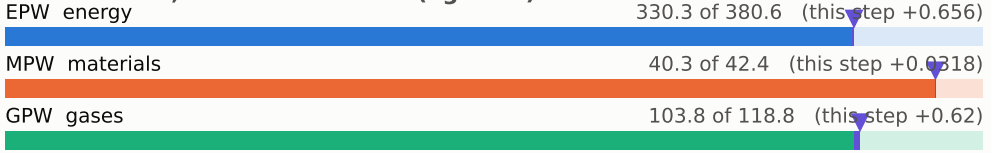
no mask for this step

this step

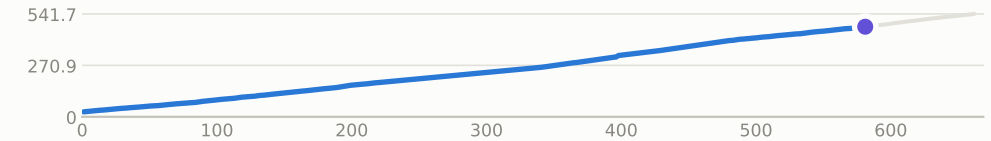
process area	CVD
material	—
thickness	0.024 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

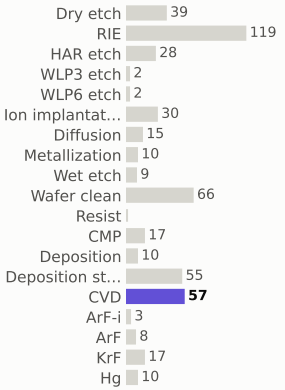
carbon so far, of the whole flow (kgCO2e)



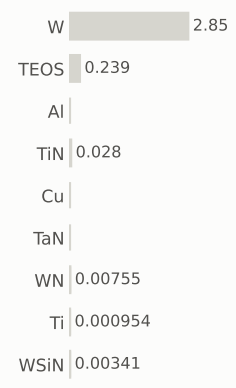
embodied carbon up to each step (kgCO2e per wafer)



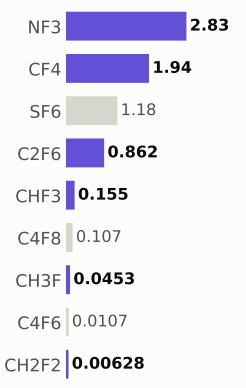
steps by process area



materials deposited (g)



gases emitted (g)



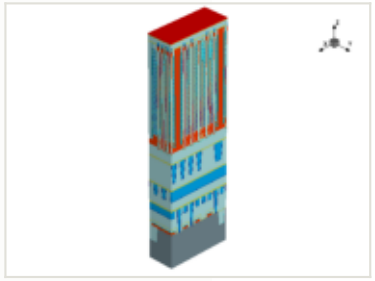
Step #5470: CVD

depo_isotropic · CVD

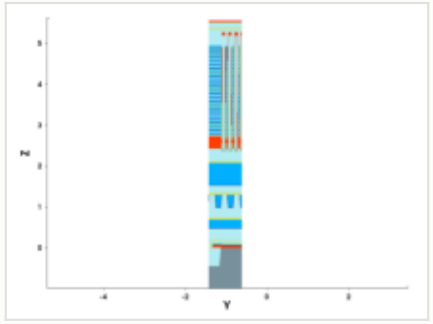
1.31	475.7	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

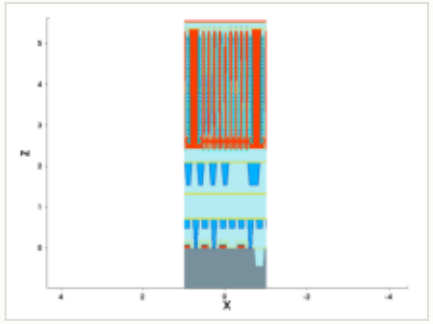
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

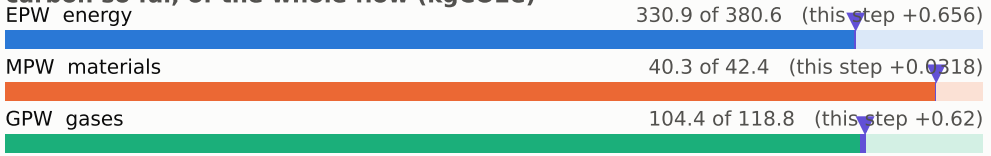
no mask for this step

this step

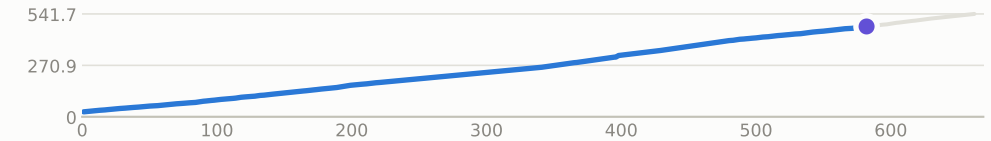
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

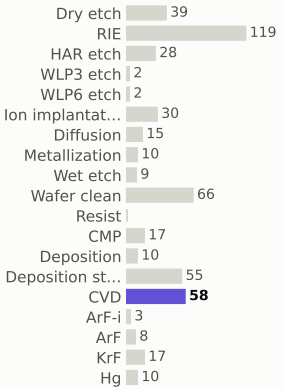
carbon so far, of the whole flow (kgCO2e)



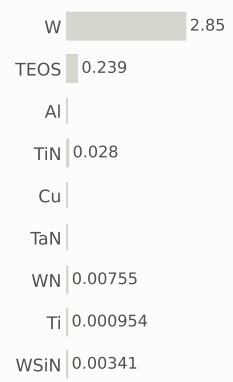
embodied carbon up to each step (kgCO2e per wafer)



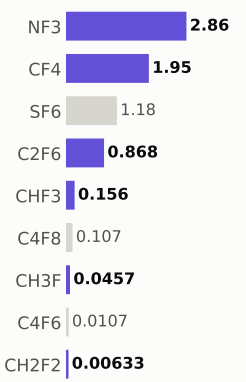
steps by process area



materials deposited (g)



gases emitted (g)



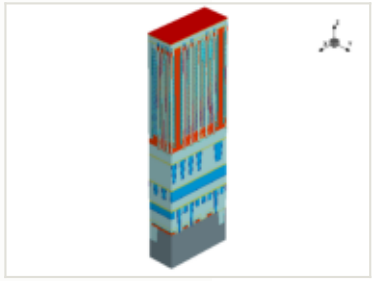
Step #5480: Wafer_clean

clean · Wafer_clean

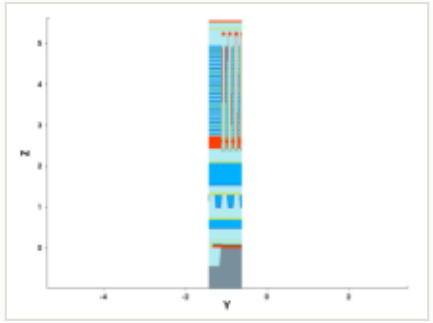
0.169	475.8	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

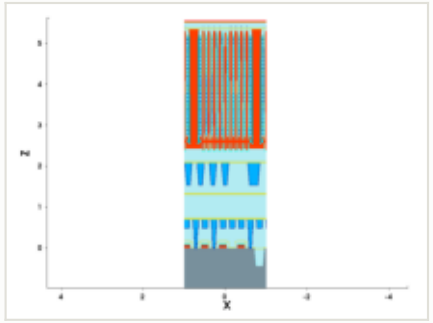
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

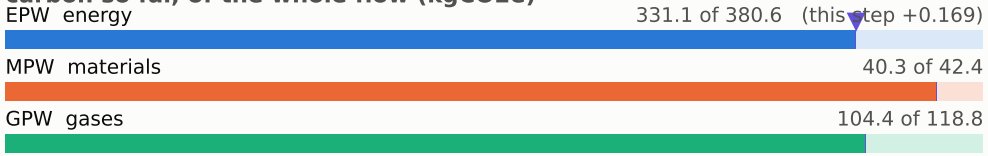
no mask for this step

this step

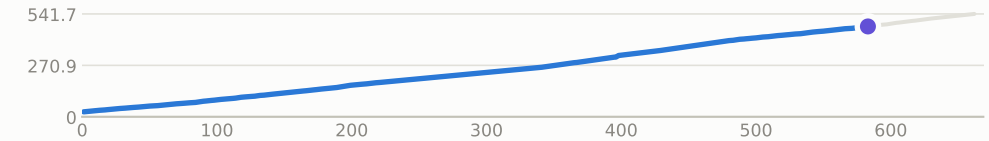
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

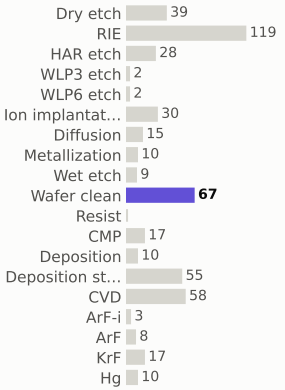
carbon so far, of the whole flow (kgCO2e)



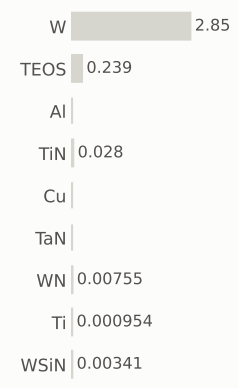
embodied carbon up to each step (kgCO2e per wafer)



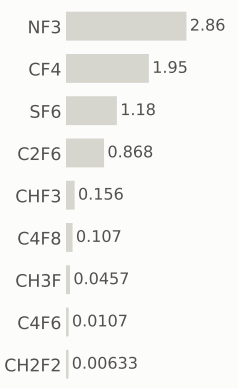
steps by process area



materials deposited (g)



gases emitted (g)



Step #5490: ArF-i

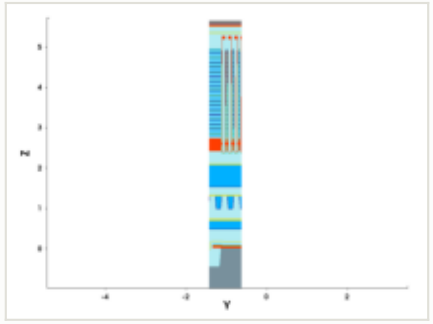
litho_patterning · ArF-i

0.217	476.1	0.04%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

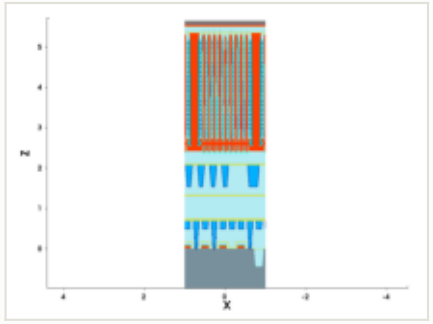
3D structure after this step

zoomed to fit

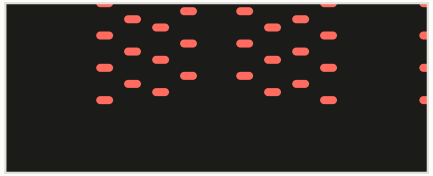
x cut at 0.000 µm



y cut at -1.032 µm



mask: layer 42:0 (OASIS)

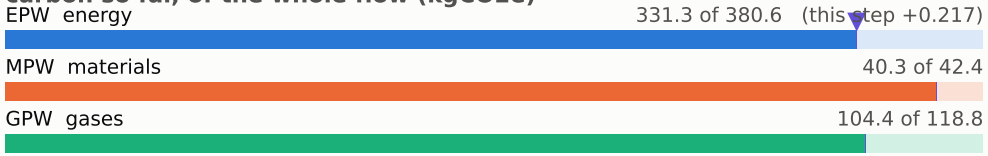


this step

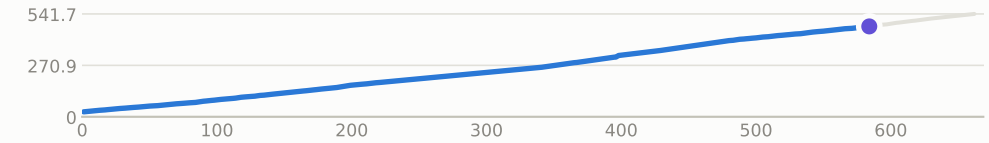
process area	ArF-i
material	—
thickness	—
material GWP	—
input gases	—
energy	0.36 kWh + 0.144 overhead

EPW	0.217 kgCO2e
EPW tool	0.155 kgCO2e
EPW overhead	0.0619 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.217 kgCO2e**

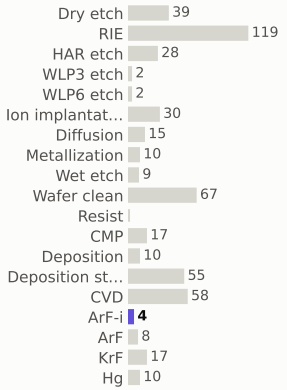
carbon so far, of the whole flow (kgCO2e)



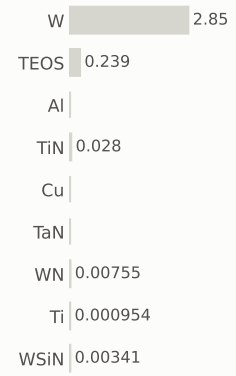
embodied carbon up to each step (kgCO2e per wafer)



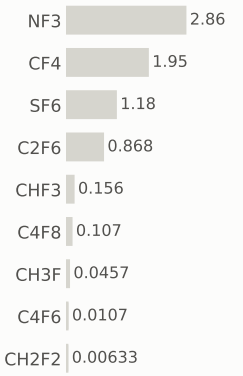
steps by process area



materials deposited (g)



gases emitted (g)



Step #5500: RIE

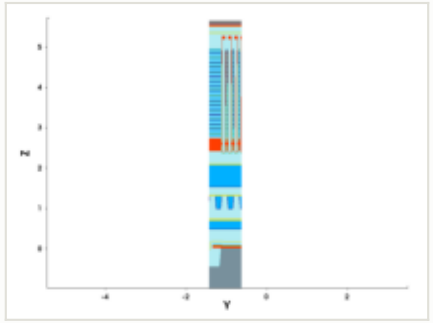
etch_anisotropic · RIE

1.04	477.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

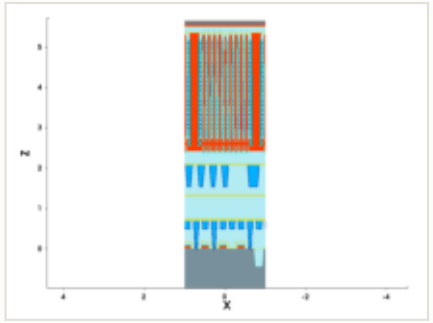
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

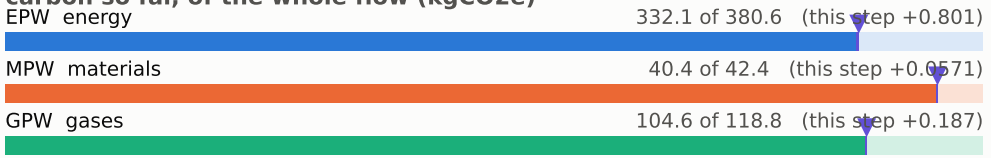
no mask for this step

this step

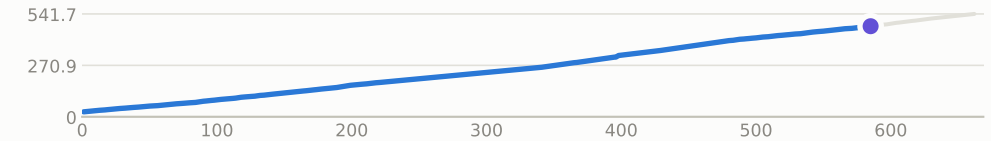
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

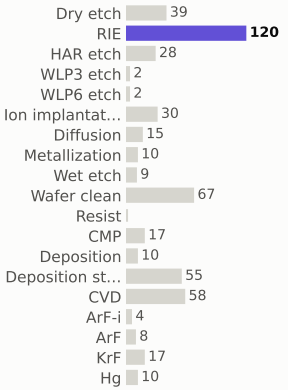
carbon so far, of the whole flow (kgCO2e)



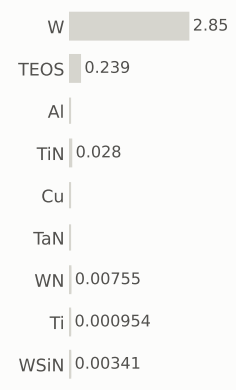
embodied carbon up to each step (kgCO2e per wafer)



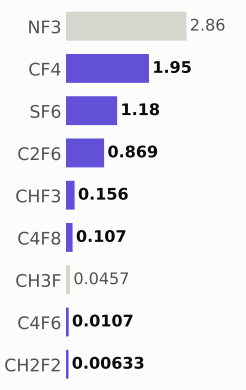
steps by process area



materials deposited (g)



gases emitted (g)



Step #5510: RIE

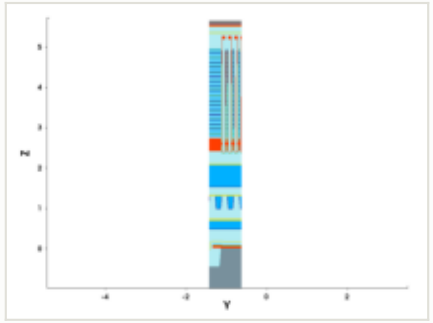
etch_anisotropic · RIE

1.04	478.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

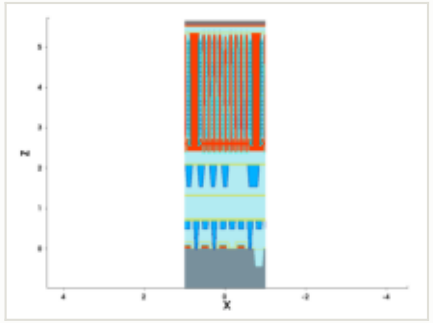
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

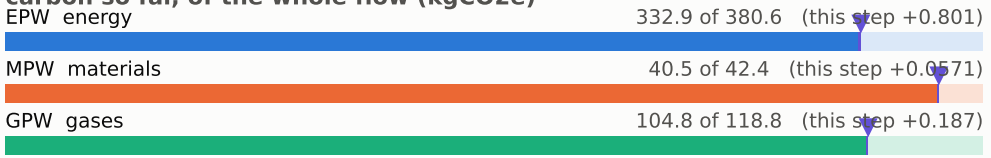
no mask for this step

this step

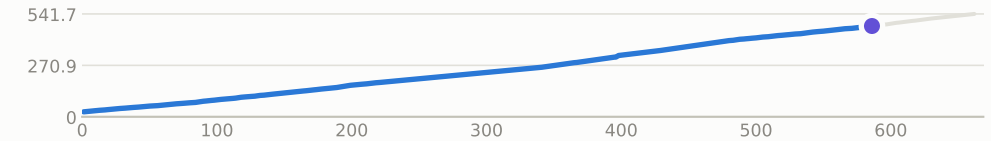
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

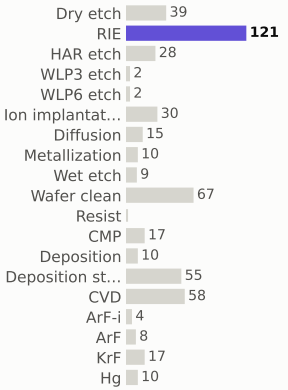
carbon so far, of the whole flow (kgCO2e)



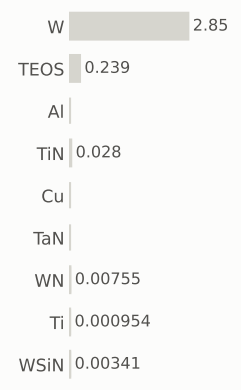
embodied carbon up to each step (kgCO2e per wafer)



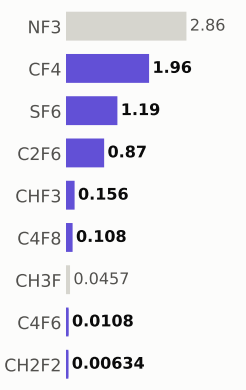
steps by process area



materials deposited (g)



gases emitted (g)



Step #5520: RIE

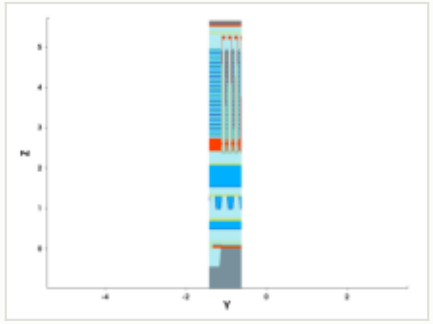
etch_isotropic · RIE

1.04	479.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

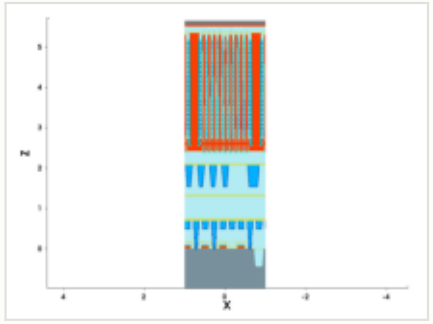
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

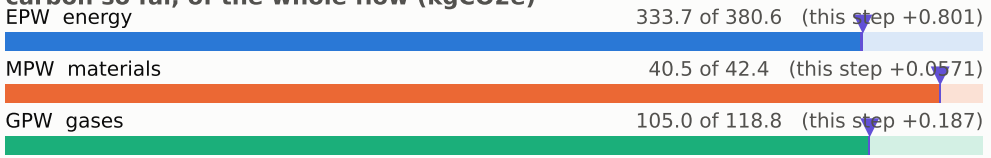
no mask for this step

this step

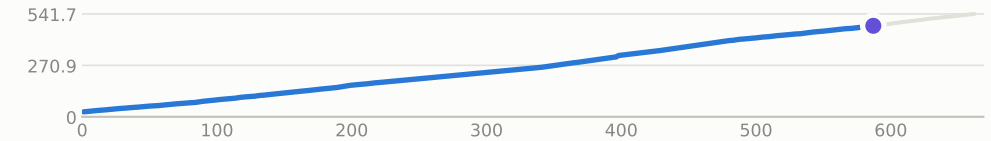
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

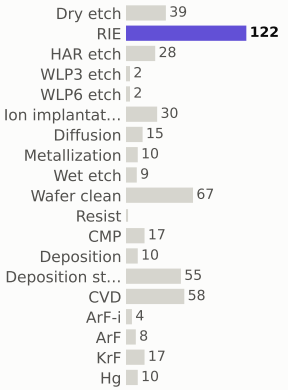
carbon so far, of the whole flow (kgCO2e)



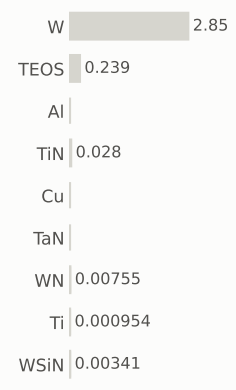
embodied carbon up to each step (kgCO2e per wafer)



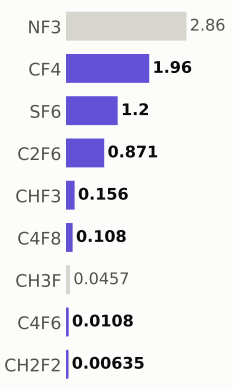
steps by process area



materials deposited (g)



gases emitted (g)



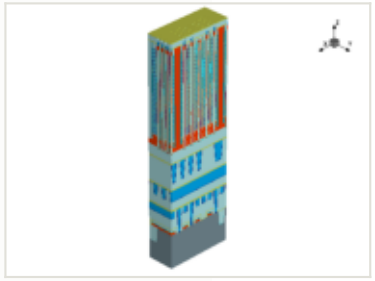
Step #5530: Dry_etch

etch_strip · Dry_etch

0.801	480.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

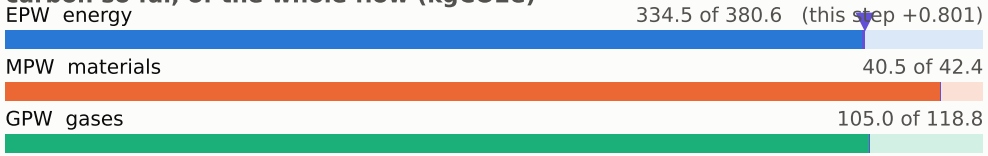


this step

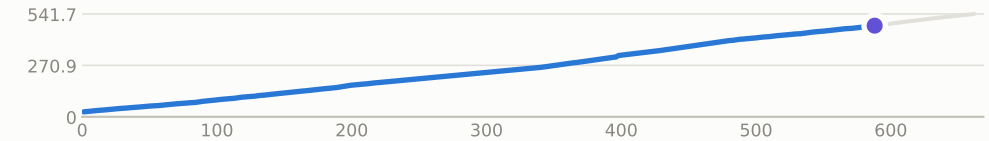
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

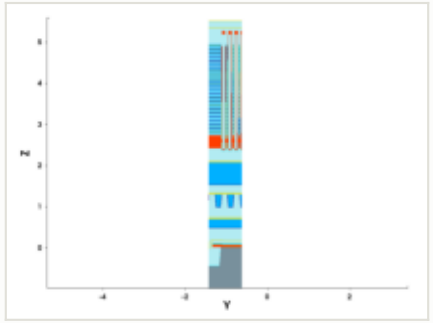
carbon so far, of the whole flow (kgCO2e)



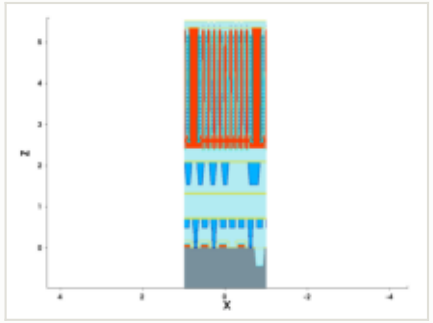
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



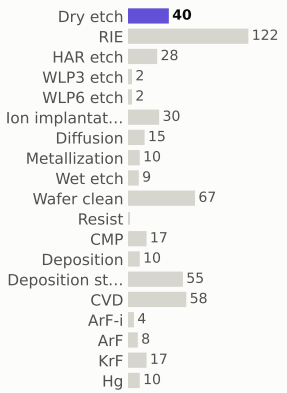
y cut at -1.032 µm



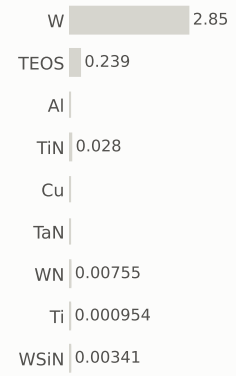
mask for this step

no mask for this step

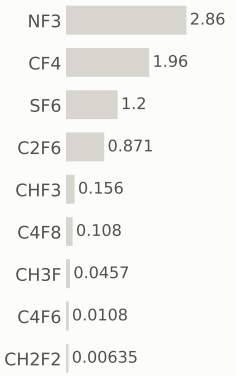
steps by process area



materials deposited (g)



gases emitted (g)



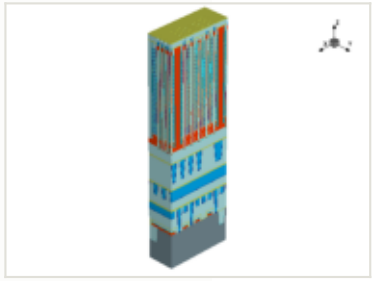
Step #5540: Wafer_clean

clean · Wafer_clean

0.169	480.2	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

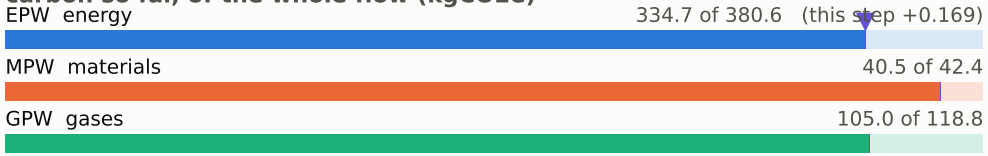


this step

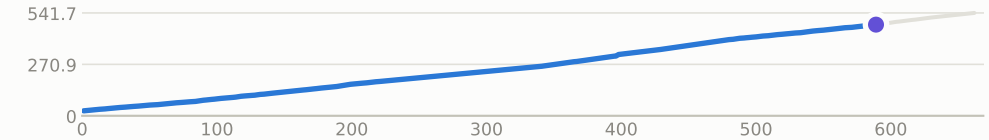
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

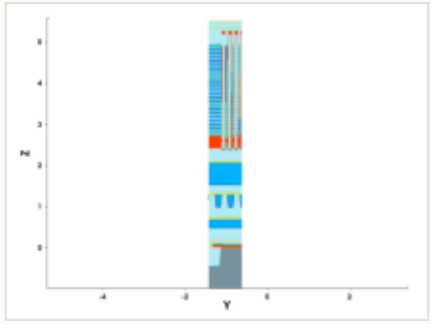
carbon so far, of the whole flow (kgCO2e)



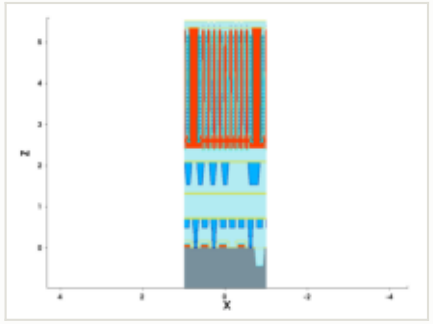
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



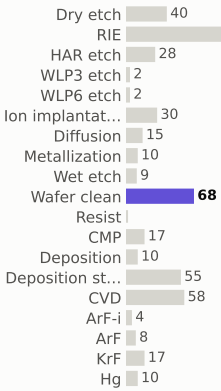
y cut at -1.032 µm



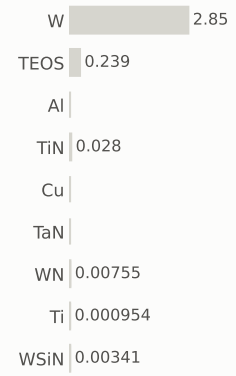
mask for this step

no mask for this step

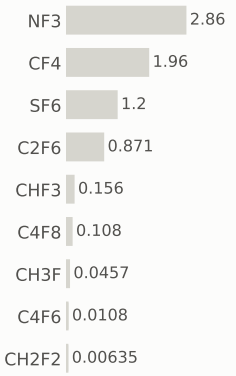
steps by process area



materials deposited (g)



gases emitted (g)



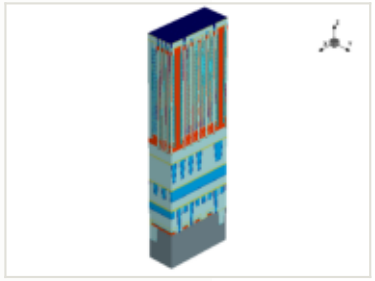
Step #5550: CVD

depo_ellipsoidal · CVD

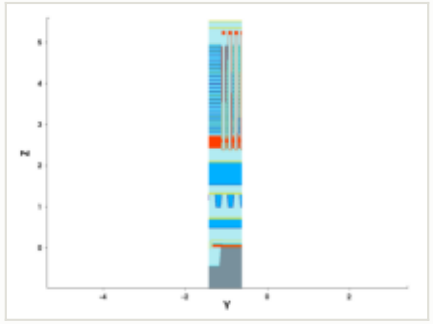
1.31	481.5	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

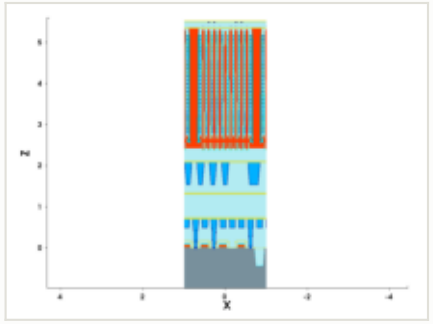
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

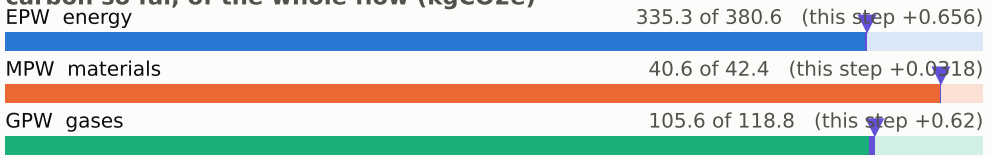
no mask for this step

this step

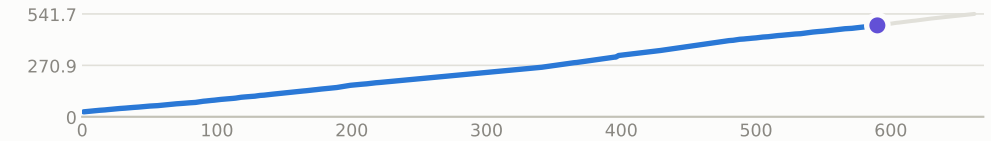
process area	CVD
material	TiN
thickness	0.005 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

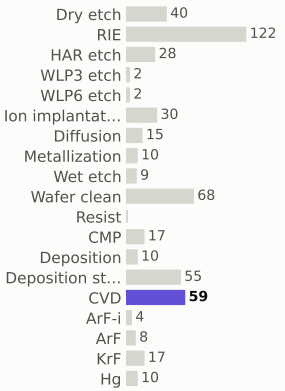
carbon so far, of the whole flow (kgCO2e)



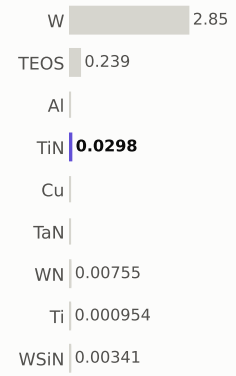
embodied carbon up to each step (kgCO2e per wafer)



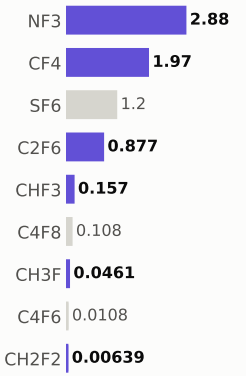
steps by process area



materials deposited (g)



gases emitted (g)



Step #5560: Metallization

depo_isotropic · Metallization

1.22	482.7	0.23%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

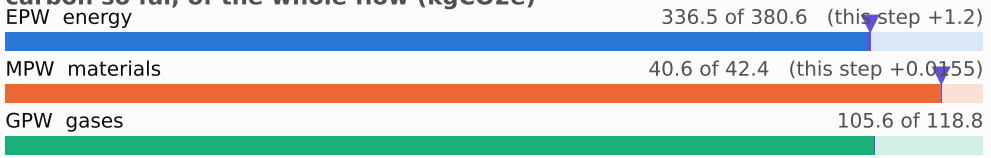
zoomed to fit

this step

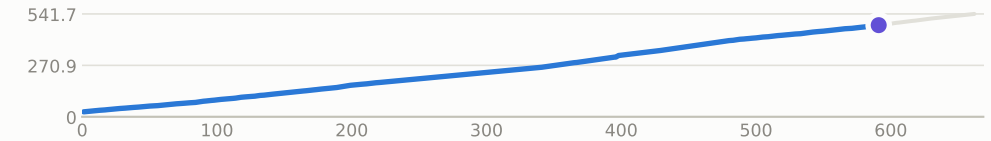
process area	Metallization
material	W
thickness	0.15 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0155 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.22 kgCO2e**

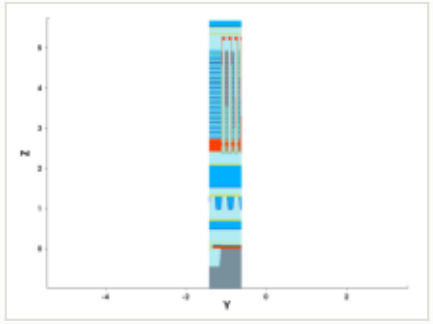
carbon so far, of the whole flow (kgCO2e)



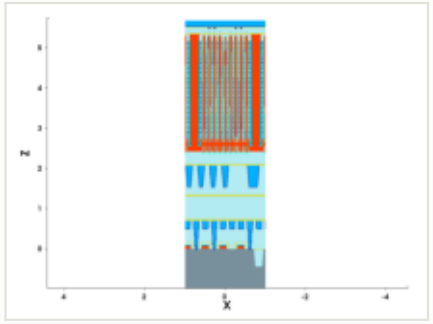
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



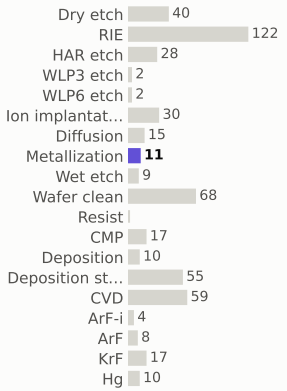
y cut at -1.032 µm



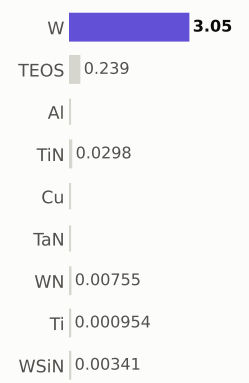
mask for this step

no mask for this step

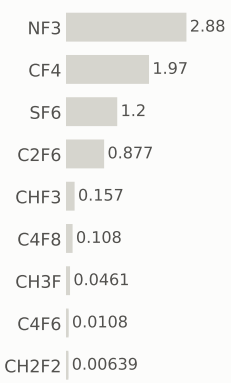
steps by process area



materials deposited (g)



gases emitted (g)



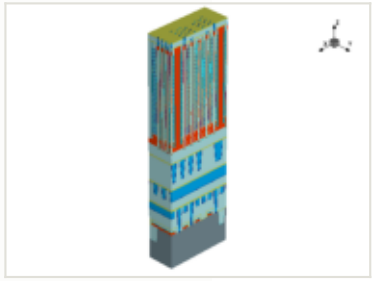
Step #5570: CMP

etch_cmp · CMP

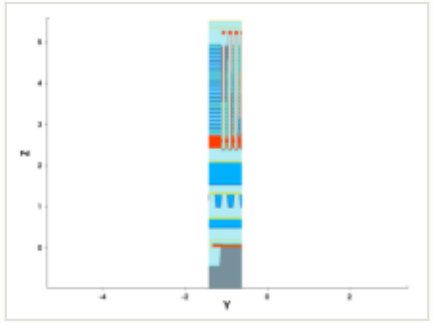
0.169	482.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

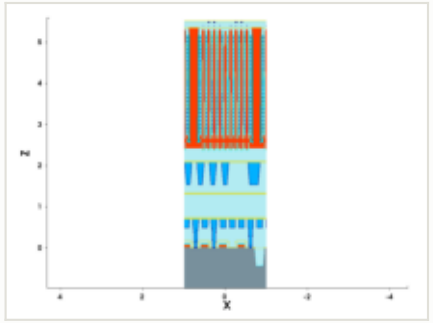
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



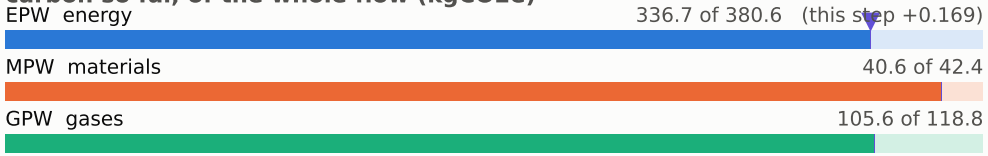
mask for this step

no mask for this step

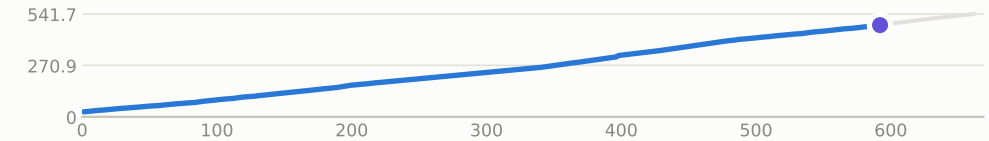
this step

process area	CMP	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

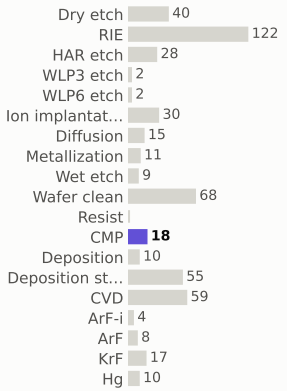
carbon so far, of the whole flow (kgCO2e)



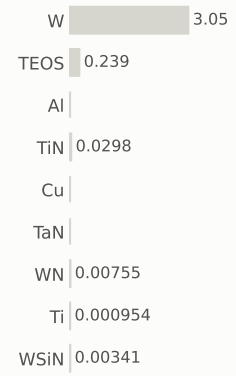
embodied carbon up to each step (kgCO2e per wafer)



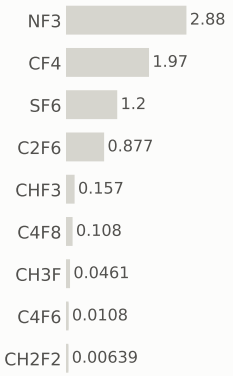
steps by process area



materials deposited (g)



gases emitted (g)



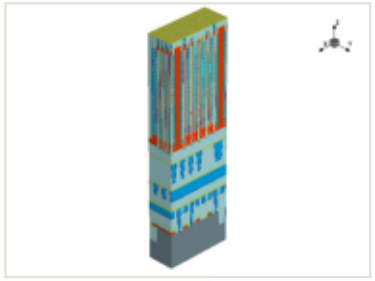
Step #5580: RIE

etch_anisotropic · RIE

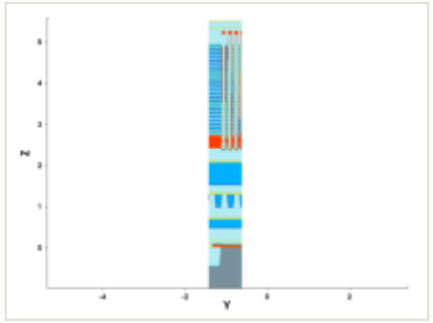
1.04	483.9	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

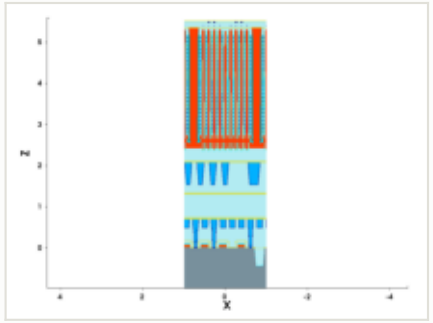
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

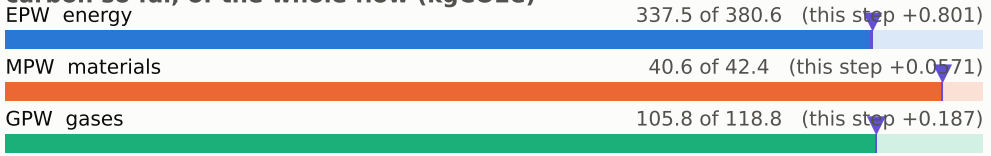
no mask for this step

this step

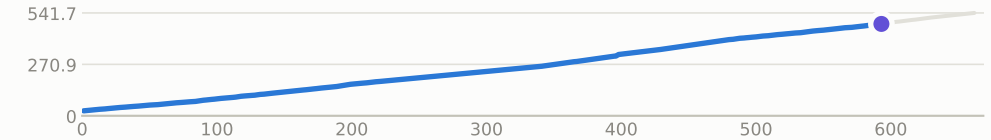
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

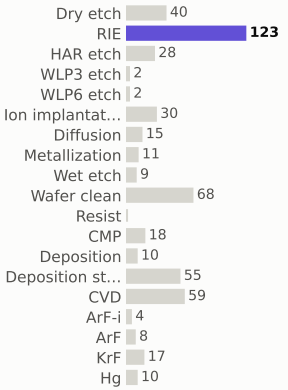
carbon so far, of the whole flow (kgCO2e)



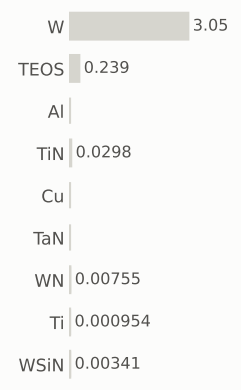
embodied carbon up to each step (kgCO2e per wafer)



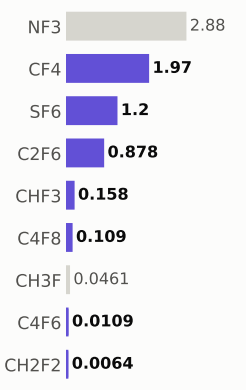
steps by process area



materials deposited (g)



gases emitted (g)



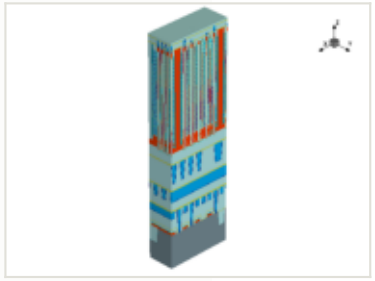
Step #5590: CVD

depo_isotropic · CVD

1.31	485.2	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

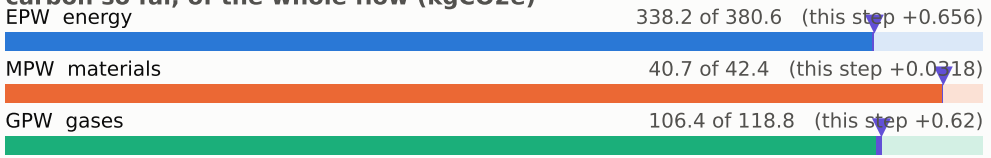


this step

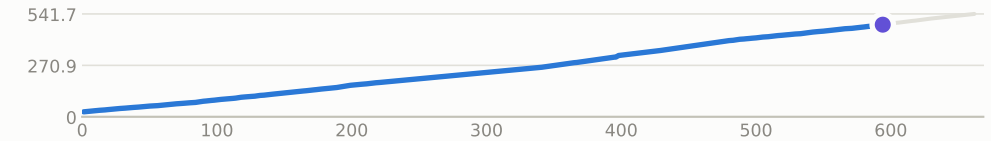
process area	CVD
material	—
thickness	0.05 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

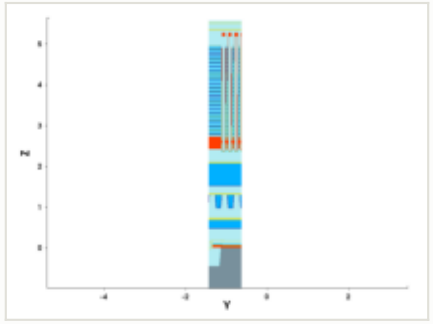
carbon so far, of the whole flow (kgCO2e)



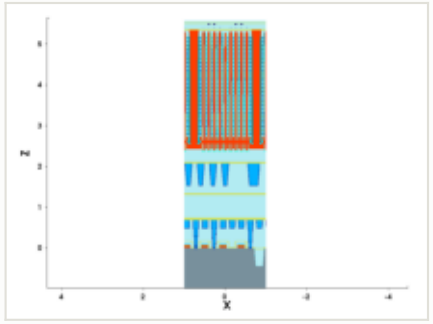
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



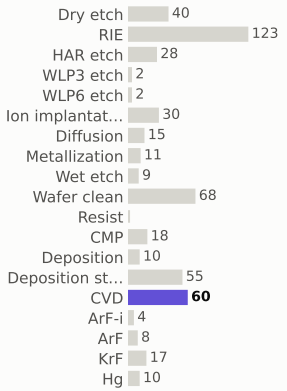
y cut at -1.032 µm



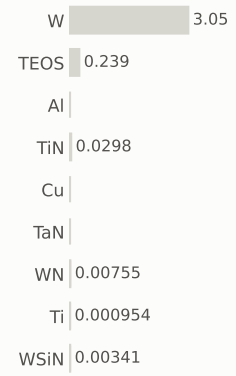
mask for this step

no mask for this step

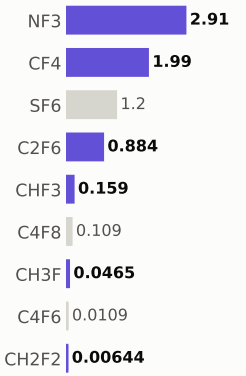
steps by process area



materials deposited (g)



gases emitted (g)



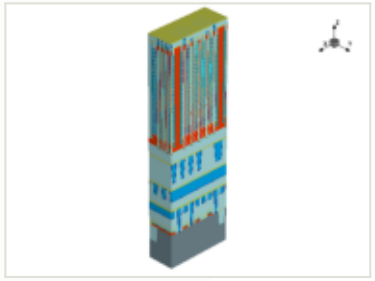
Step #5600: CMP

etch_cmp · CMP

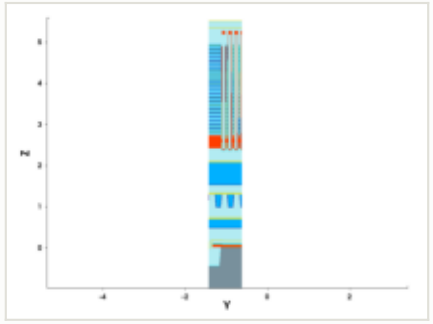
0.169	485.4	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

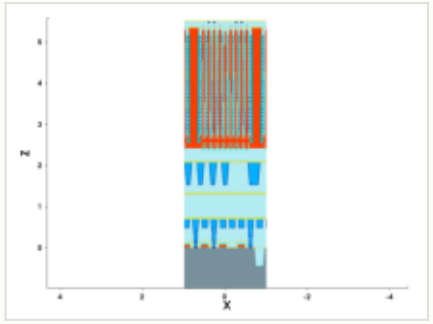
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

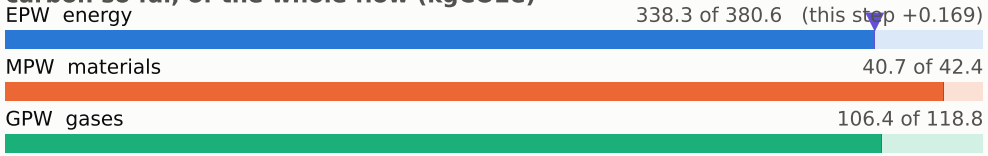
no mask for this step

this step

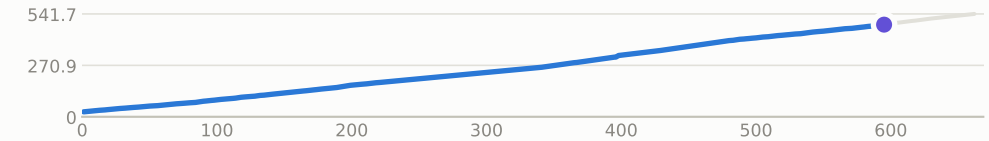
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

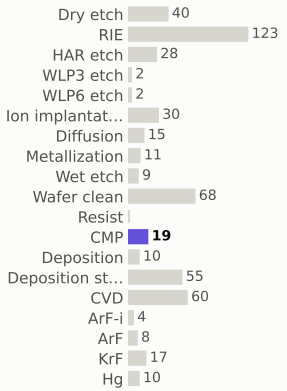
carbon so far, of the whole flow (kgCO2e)



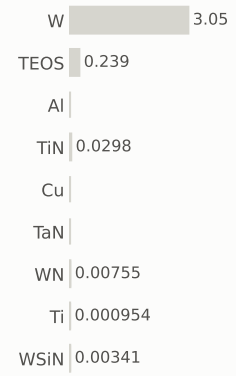
embodied carbon up to each step (kgCO2e per wafer)



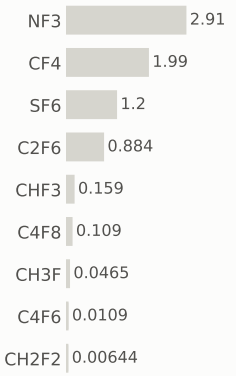
steps by process area



materials deposited (g)



gases emitted (g)



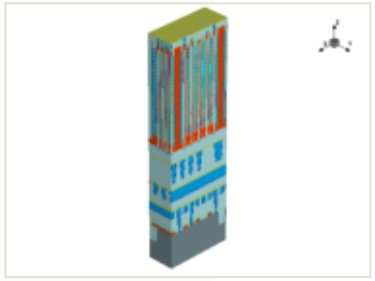
Step #5610: Wafer_clean

clean · Wafer_clean

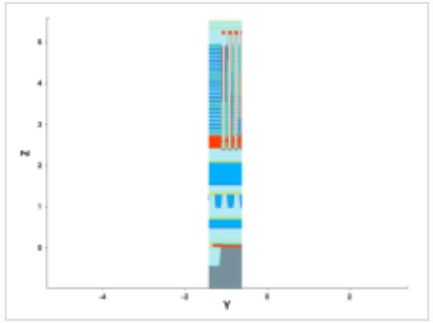
0.169	485.6	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

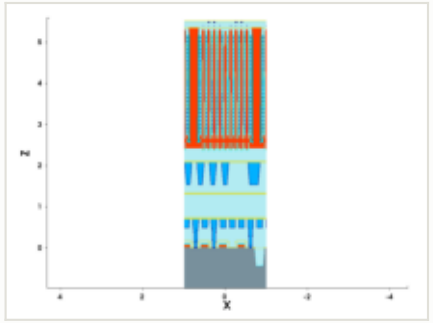
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

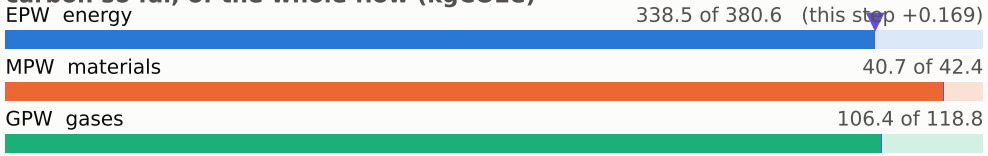
no mask for this step

this step

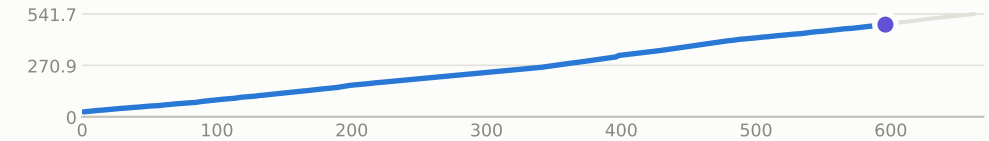
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

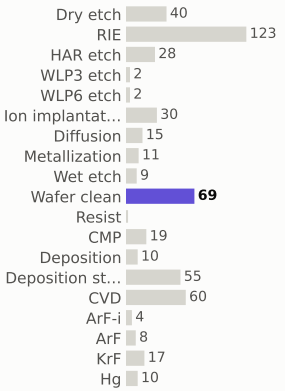
carbon so far, of the whole flow (kgCO2e)



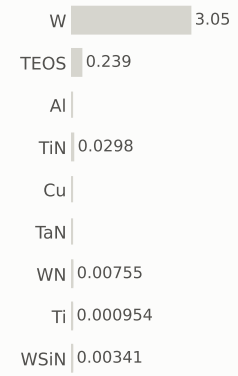
embodied carbon up to each step (kgCO2e per wafer)



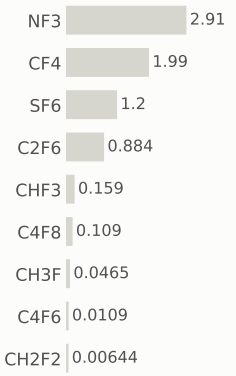
steps by process area



materials deposited (g)



gases emitted (g)



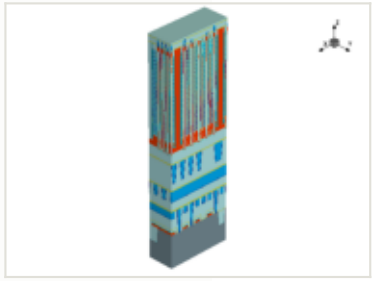
Step #5620: CVD

depo_isotropic · CVD

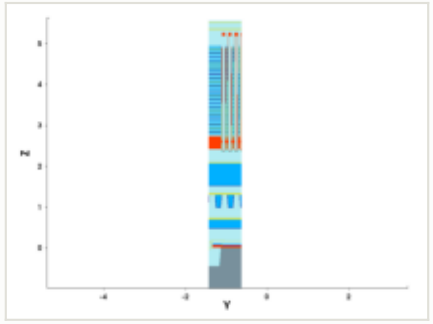
1.31	486.9	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

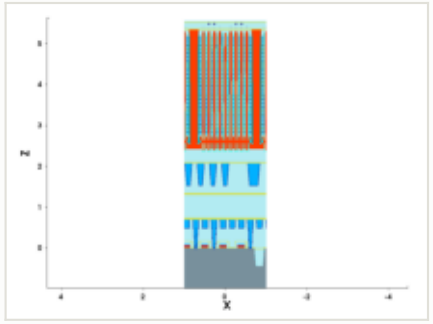
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

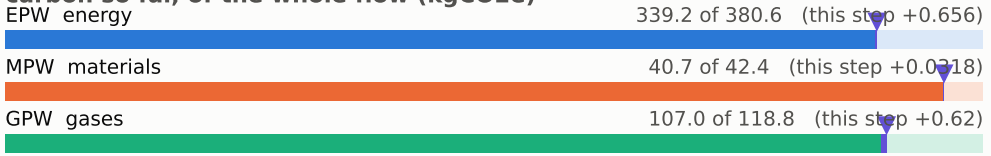
no mask for this step

this step

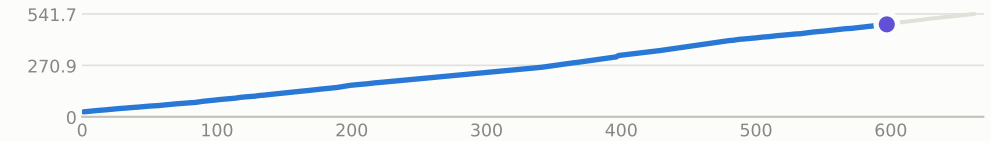
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

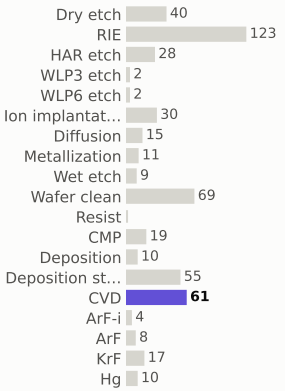
carbon so far, of the whole flow (kgCO2e)



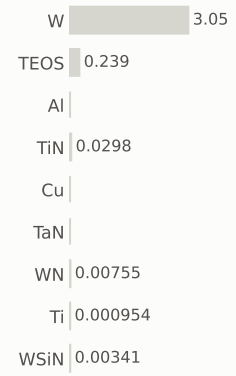
embodied carbon up to each step (kgCO2e per wafer)



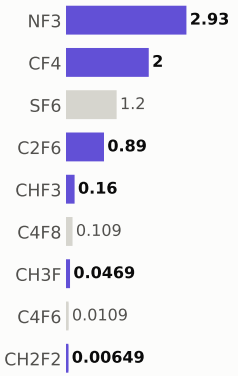
steps by process area



materials deposited (g)



gases emitted (g)



Step #5640: CVD

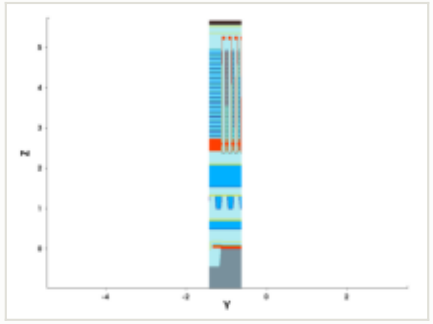
depo_isotropic · CVD

1.31	488.2	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

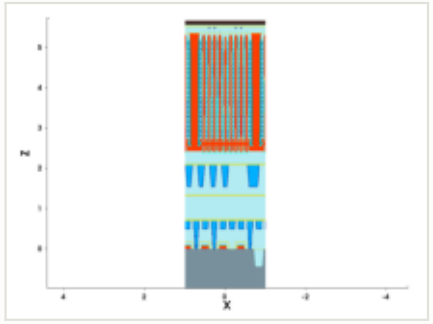
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

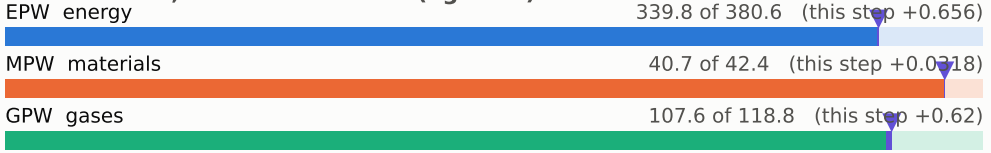
no mask for this step

this step

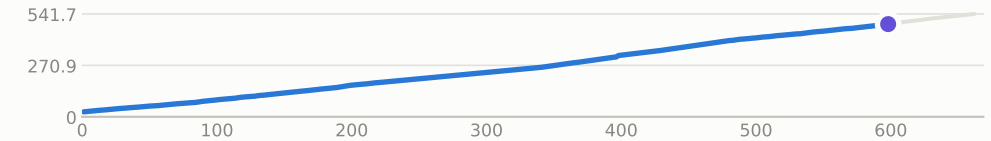
process area	CVD
material	—
thickness	0.1 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

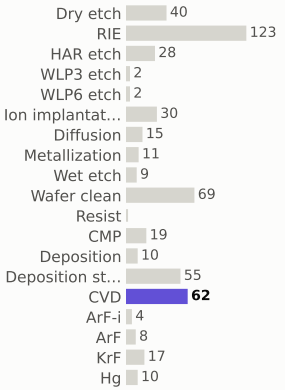
carbon so far, of the whole flow (kgCO2e)



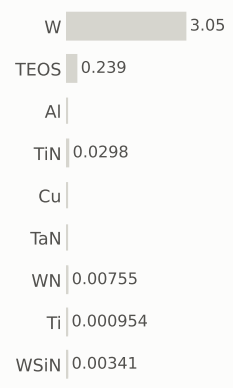
embodied carbon up to each step (kgCO2e per wafer)



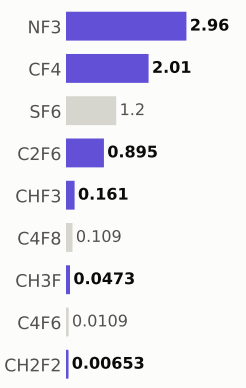
steps by process area



materials deposited (g)



gases emitted (g)



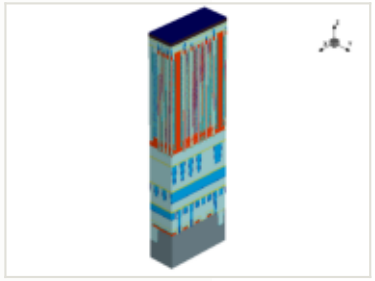
Step #5650: CVD

depo_isotropic · CVD

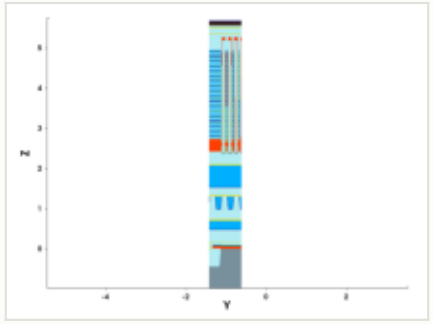
1.31	489.5	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

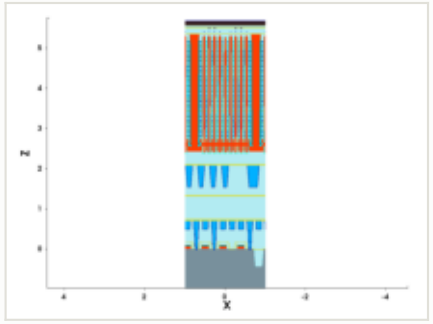
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

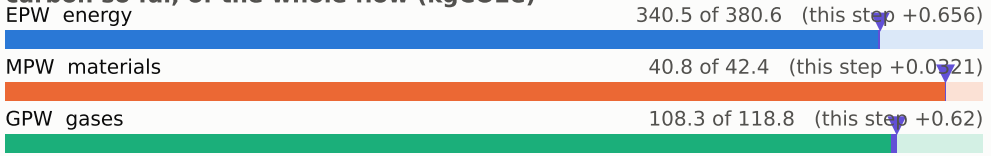
no mask for this step

this step

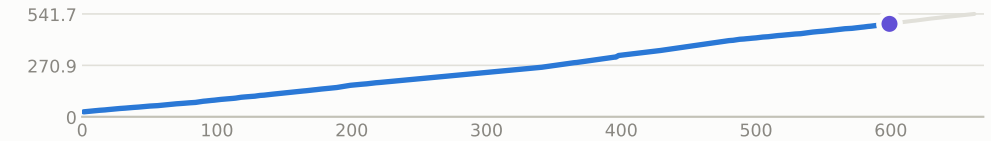
process area	CVD
material	TiN
thickness	0.02 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0321 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

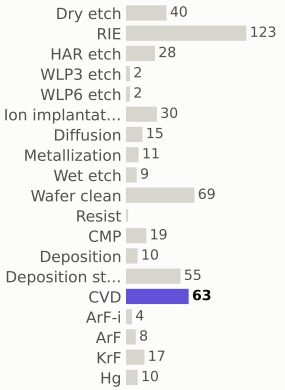
carbon so far, of the whole flow (kgCO2e)



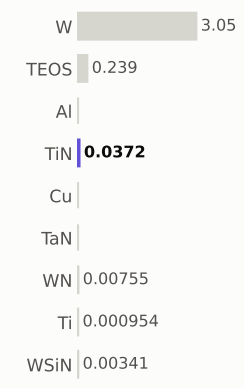
embodied carbon up to each step (kgCO2e per wafer)



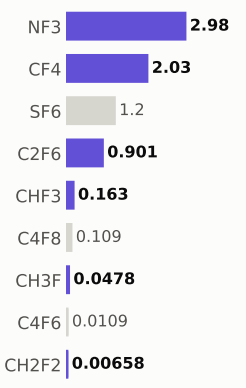
steps by process area



materials deposited (g)



gases emitted (g)



Step #5660: CVD

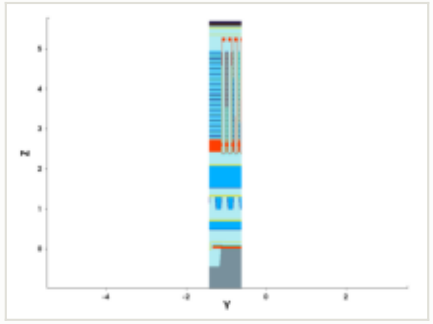
depo_isotropic · CVD

1.31	490.8	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

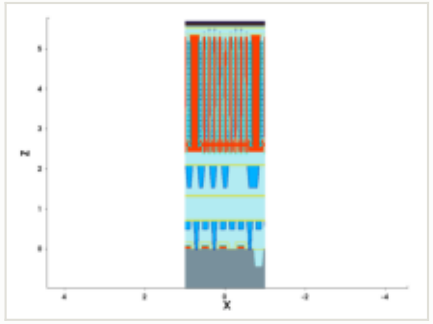
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

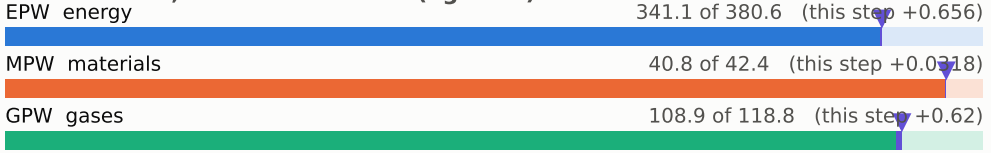
no mask for this step

this step

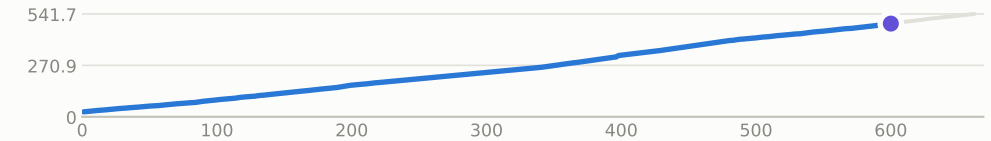
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

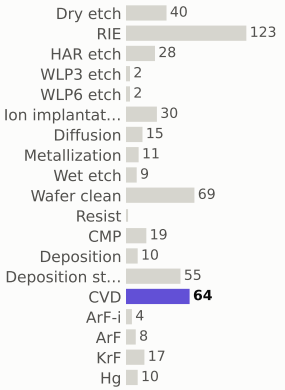
carbon so far, of the whole flow (kgCO2e)



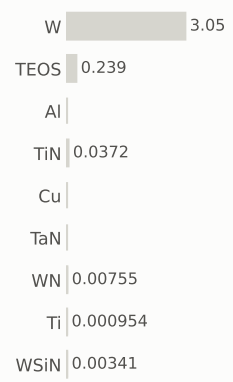
embodied carbon up to each step (kgCO2e per wafer)



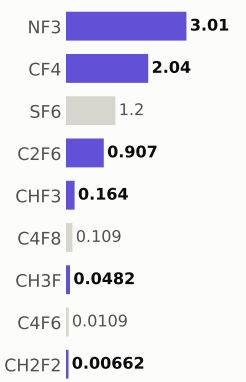
steps by process area



materials deposited (g)



gases emitted (g)



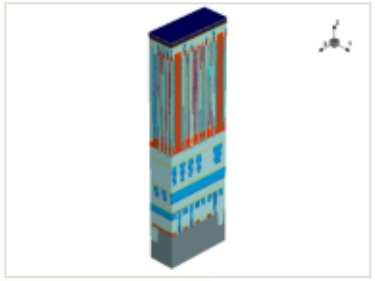
Step #5670: CVD

depo_isotropic · CVD

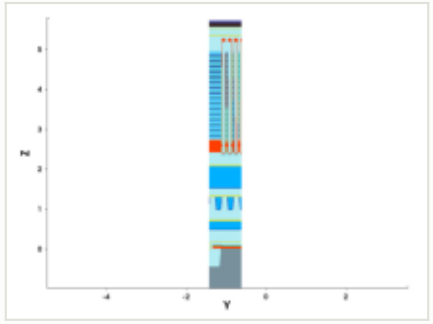
1.31	492.1	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

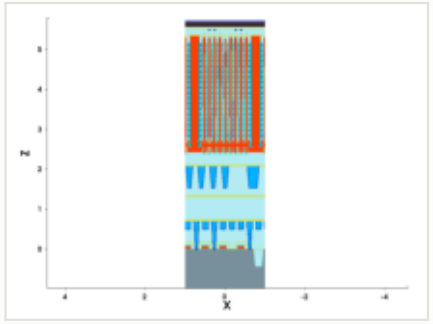
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

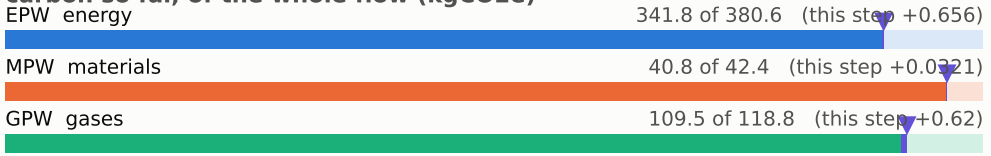
no mask for this step

this step

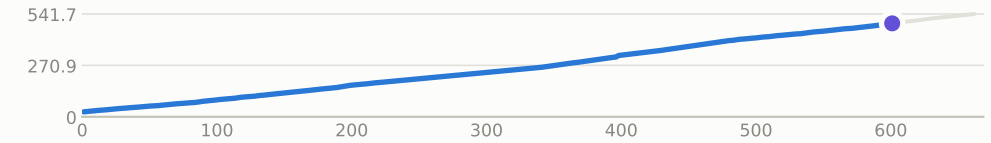
process area	CVD
material	TiN
thickness	0.02 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0321 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

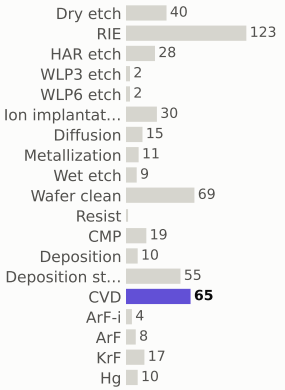
carbon so far, of the whole flow (kgCO2e)



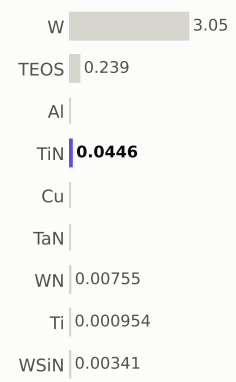
embodied carbon up to each step (kgCO2e per wafer)



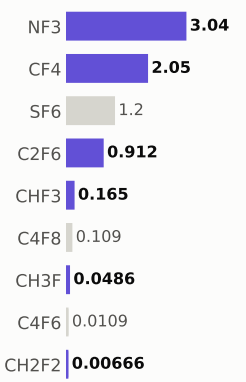
steps by process area



materials deposited (g)



gases emitted (g)



Step #5680: CVD

depo_isotropic · CVD

1.31	493.4	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

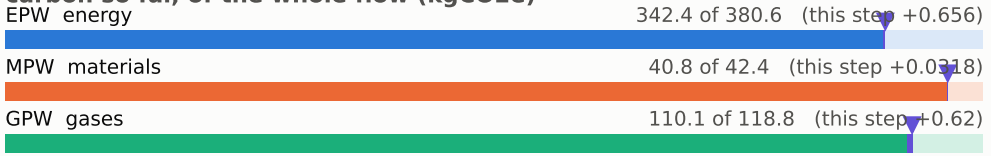
zoomed to fit

this step

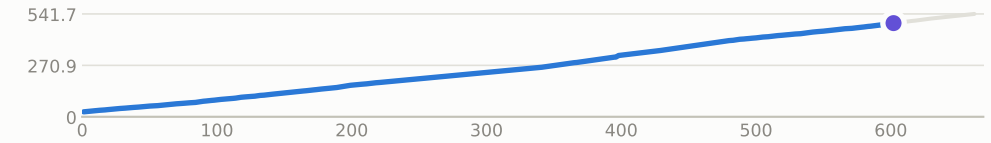
process area	CVD
material	—
thickness	0.07 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

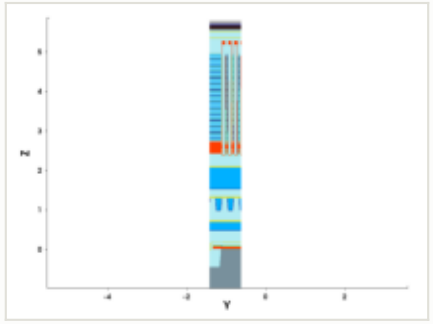
carbon so far, of the whole flow (kgCO2e)



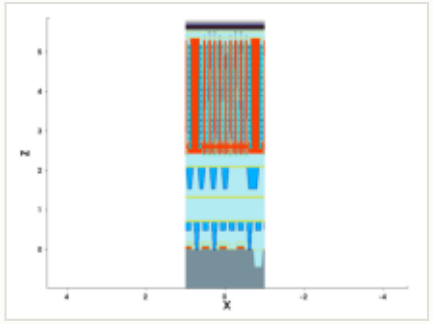
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



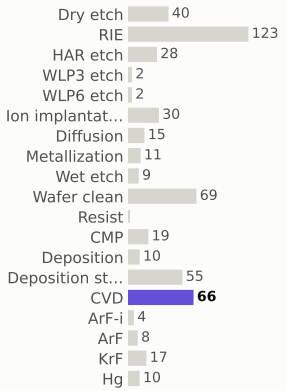
y cut at -1.032 µm



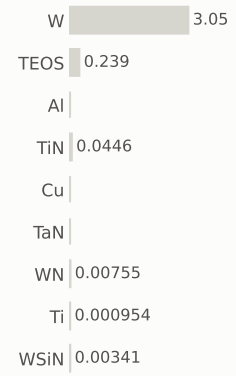
mask for this step

no mask for this step

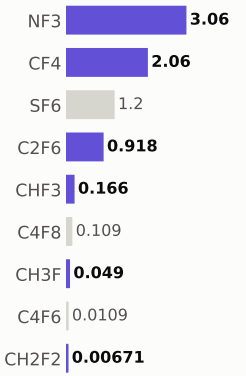
steps by process area



materials deposited (g)



gases emitted (g)



Step #5690: CVD

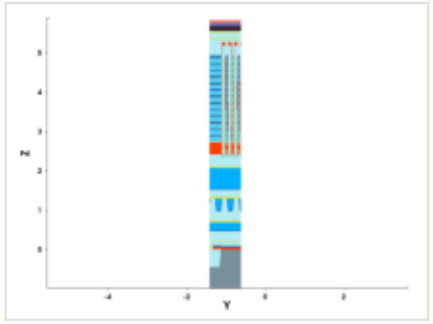
depo_isotropic · CVD

1.31	494.7	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

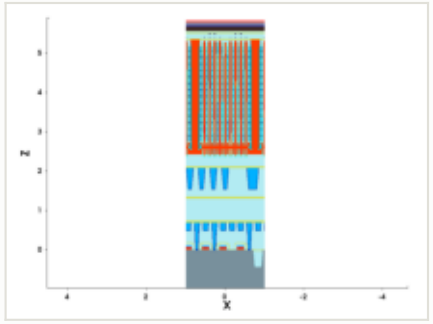
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

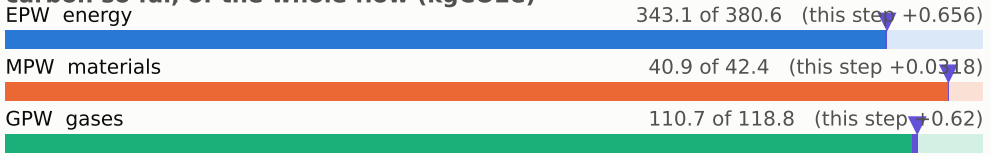
no mask for this step

this step

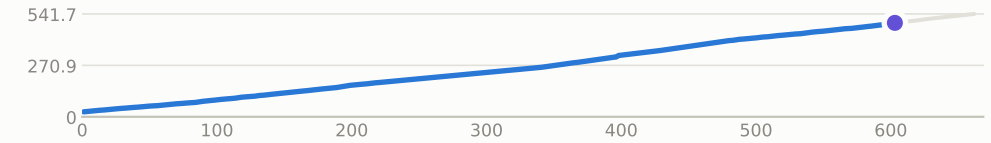
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

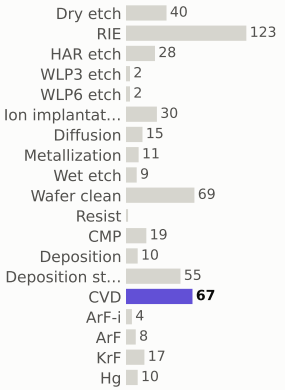
carbon so far, of the whole flow (kgCO2e)



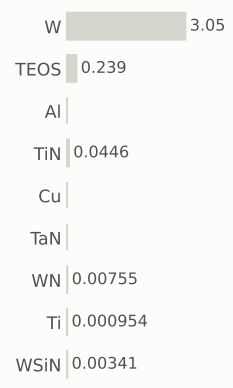
embodied carbon up to each step (kgCO2e per wafer)



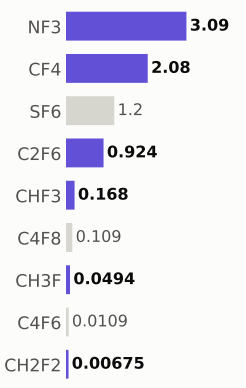
steps by process area



materials deposited (g)



gases emitted (g)



Step #5700: Wafer_clean

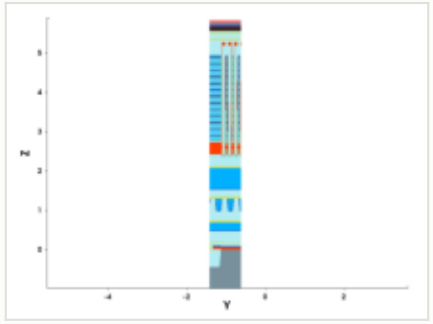
clean · Wafer_clean

0.169	494.9	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

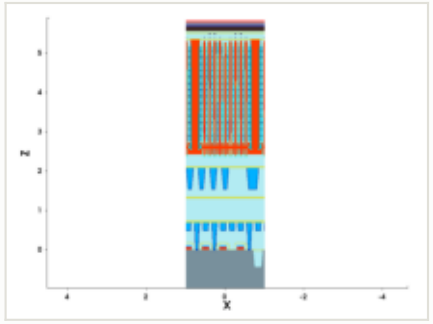
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

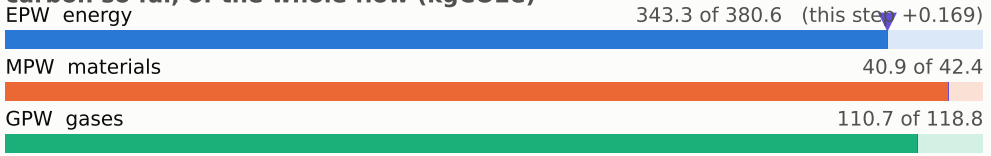
no mask for this step

this step

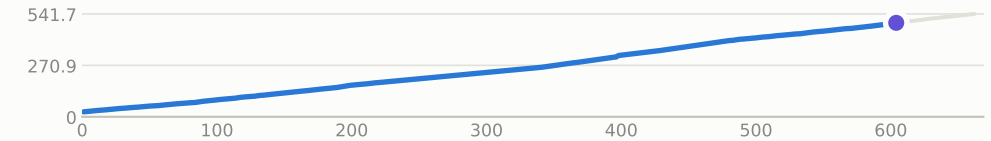
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

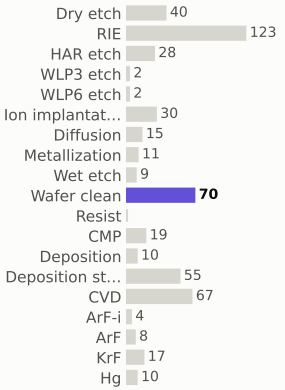
carbon so far, of the whole flow (kgCO2e)



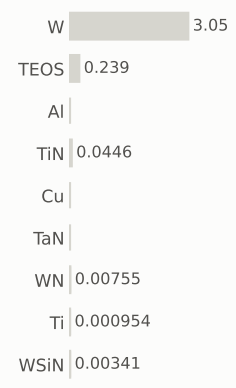
embodied carbon up to each step (kgCO2e per wafer)



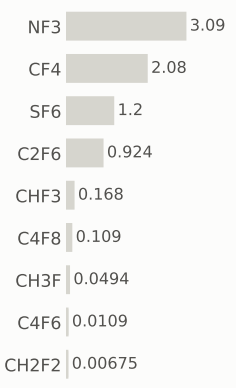
steps by process area



materials deposited (g)



gases emitted (g)



Step #5710: ArF-i

litho_patterning · ArF-i

0.217	495.1	0.04%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

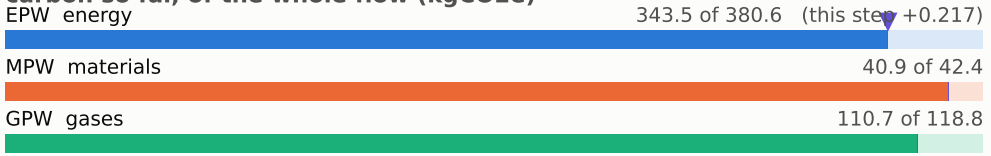
3D structure after this step

zoomed to fit

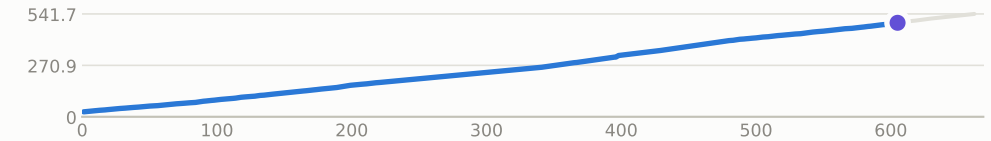
this step

process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

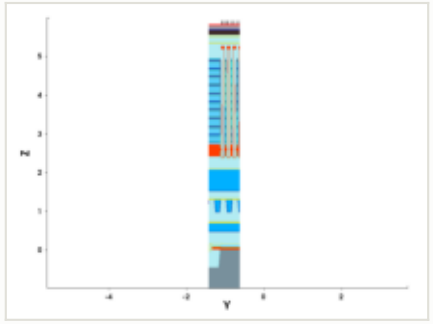
carbon so far, of the whole flow (kgCO2e)



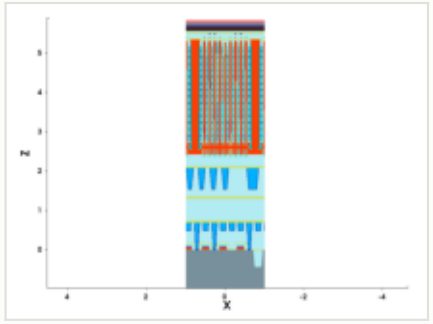
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

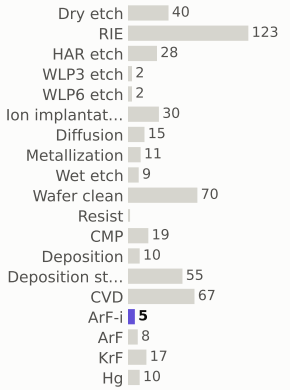


y cut at -1.032 µm

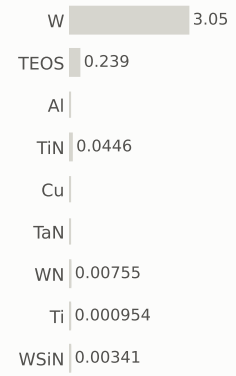


mask: layer 43:0, positive (OASIS)

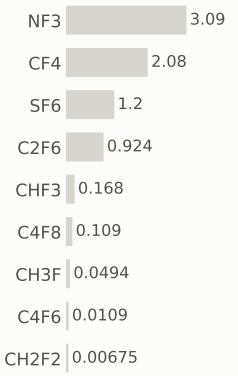
steps by process area



materials deposited (g)



gases emitted (g)



Step #5720: RIE

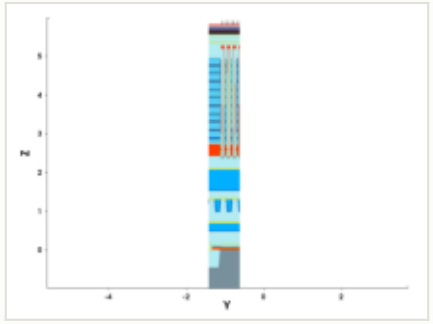
etch_isotropic · RIE

1.04	496.1	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

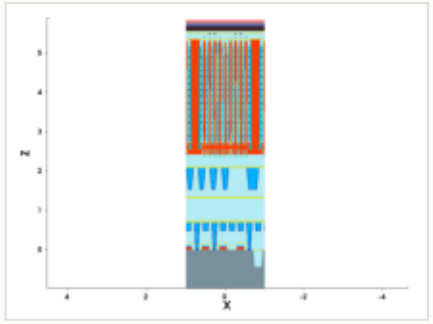
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

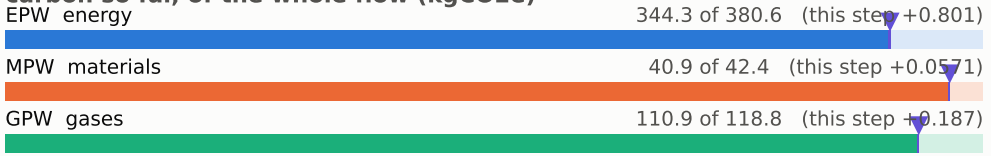
no mask for this step

this step

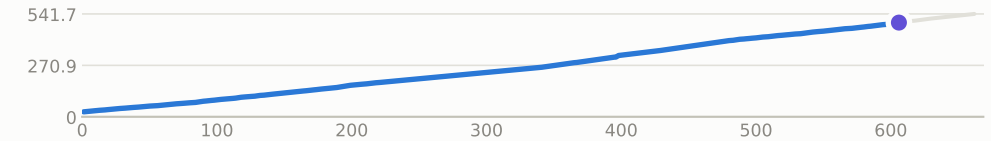
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

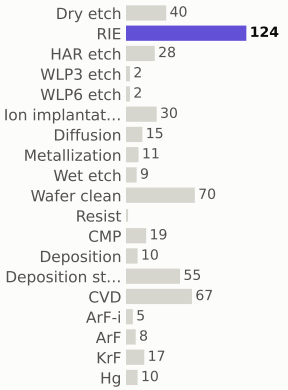
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

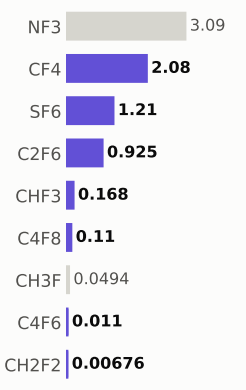


steps by process area



materials deposited (g)

gases emitted (g)



Step #5730: RIE

etch_anisotropic · RIE

1.04	497.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

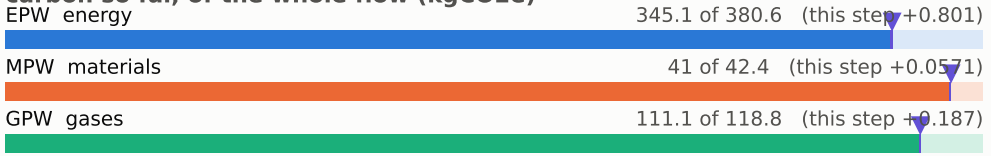
zoomed to fit

this step

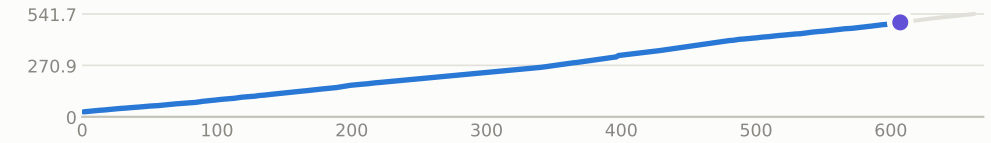
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

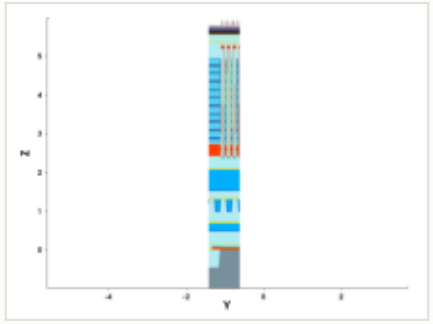
carbon so far, of the whole flow (kgCO2e)



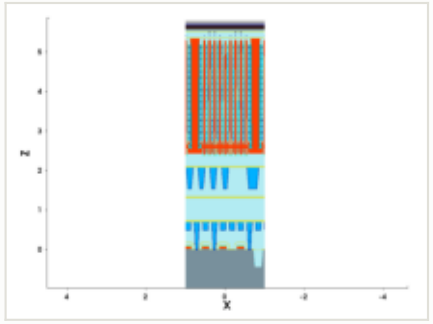
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



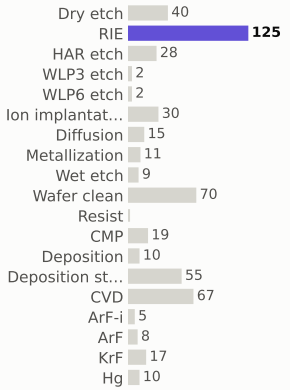
y cut at -1.032 µm



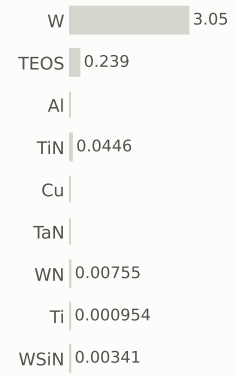
mask for this step

no mask for this step

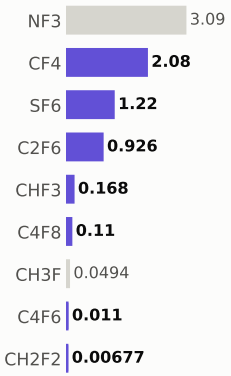
steps by process area



materials deposited (g)



gases emitted (g)



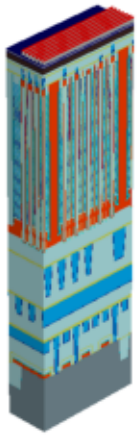
Step #5740: RIE

etch_anisotropic · RIE

1.04	498.2	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

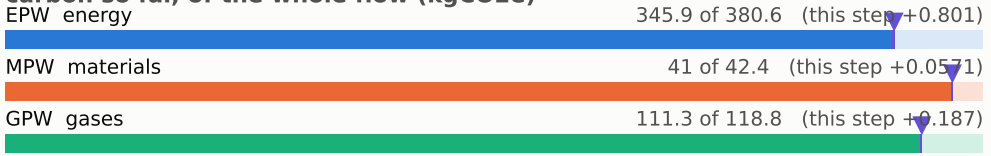


this step

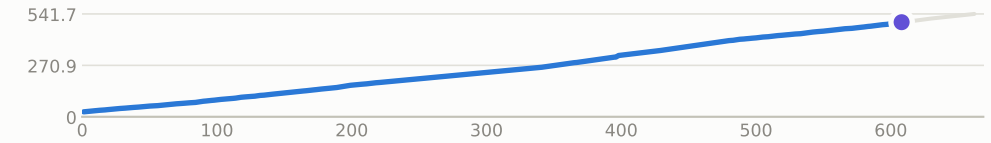
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

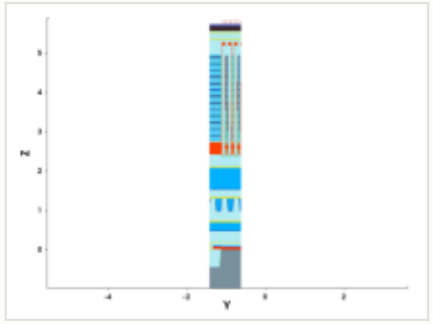
carbon so far, of the whole flow (kgCO2e)



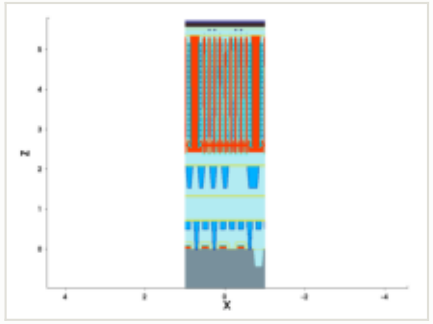
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



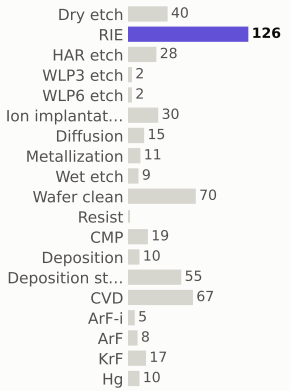
y cut at -1.032 µm



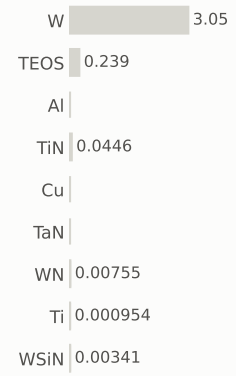
mask for this step

no mask for this step

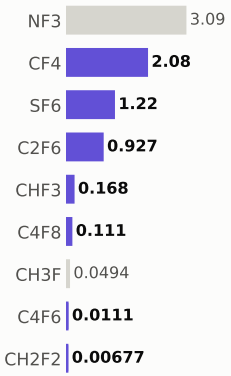
steps by process area



materials deposited (g)



gases emitted (g)



Step #5750: Dry_etch

etch_strip · Dry_etch

0.801	499.0	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

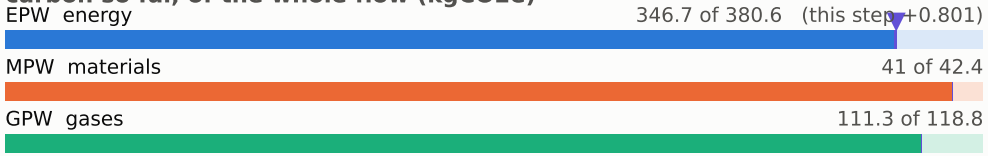
zoomed to fit

this step

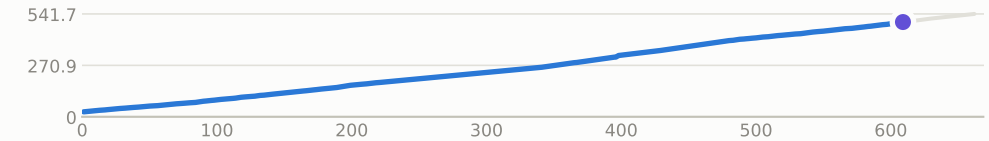
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

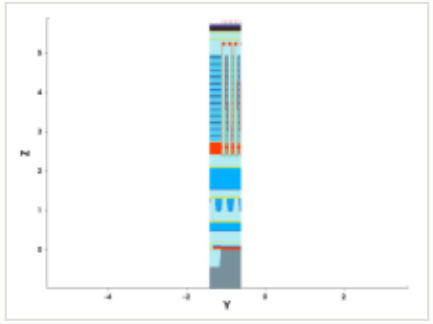
carbon so far, of the whole flow (kgCO2e)



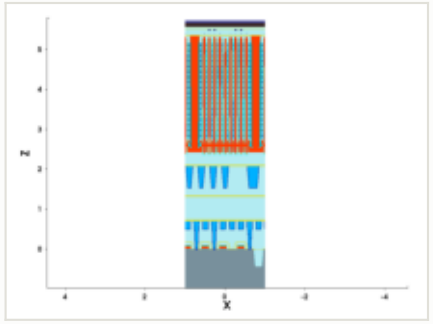
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



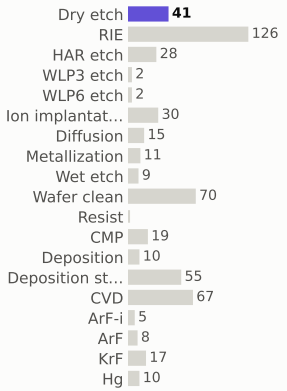
y cut at -1.032 µm



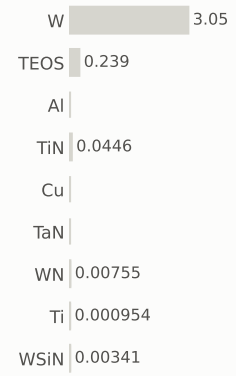
mask for this step

no mask for this step

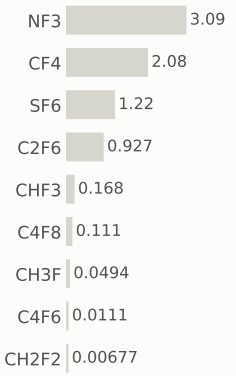
steps by process area



materials deposited (g)



gases emitted (g)



Step #5760: CVD

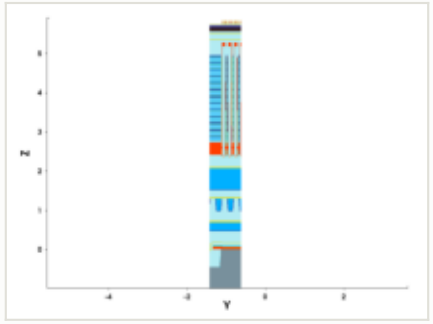
depo_isotropic · CVD

1.31	500.3	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

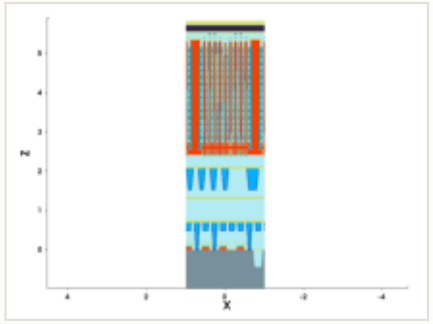
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

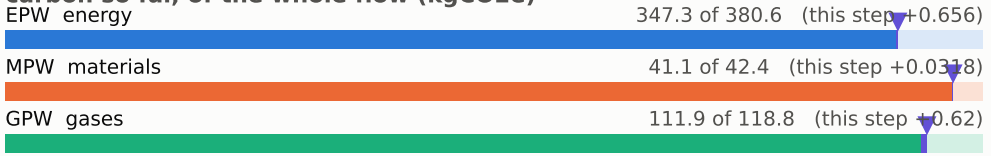
no mask for this step

this step

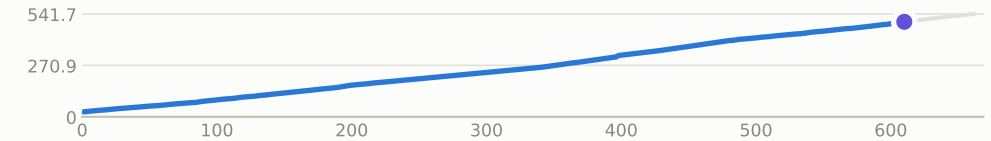
process area	CVD
material	—
thickness	0.016 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

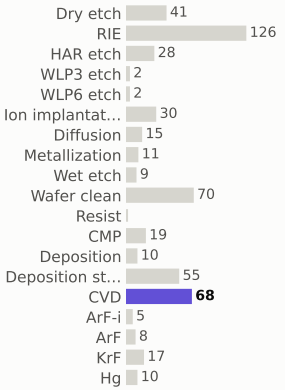
carbon so far, of the whole flow (kgCO2e)



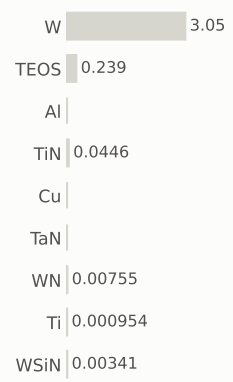
embodied carbon up to each step (kgCO2e per wafer)



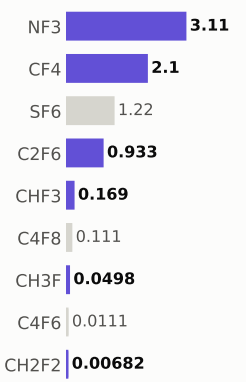
steps by process area



materials deposited (g)



gases emitted (g)



Step #5770: RIE

etch_anisotropic · RIE

1.04	501.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

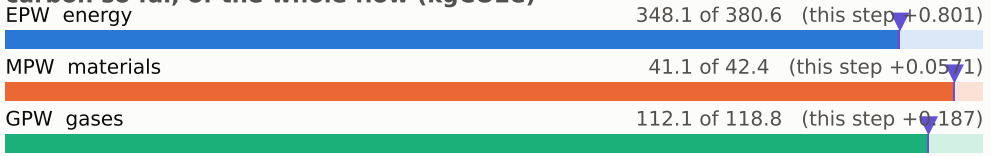
zoomed to fit

this step

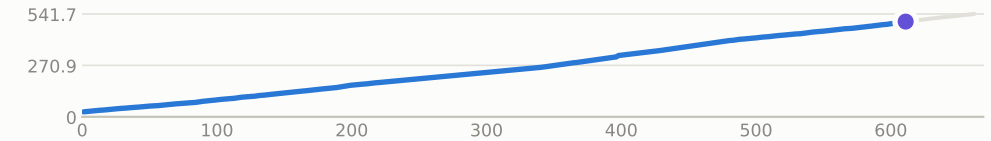
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

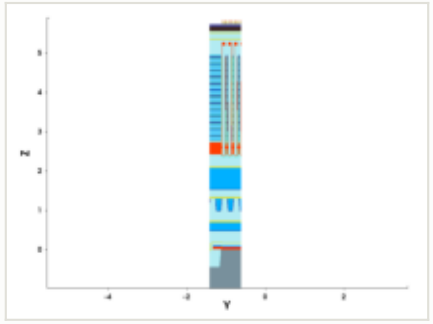
carbon so far, of the whole flow (kgCO2e)



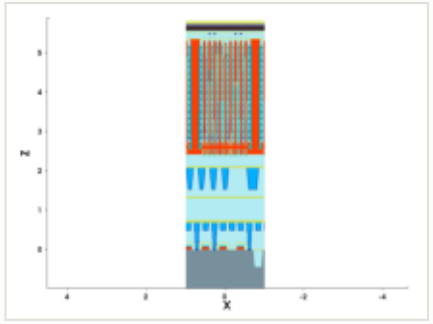
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



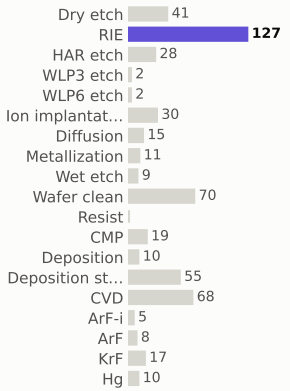
y cut at -1.032 µm



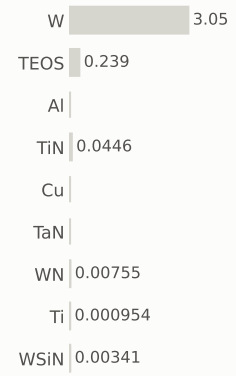
mask for this step

no mask for this step

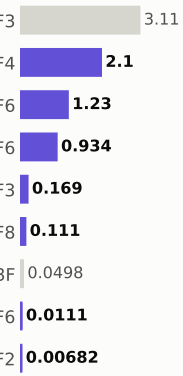
steps by process area



materials deposited (g)



gases emitted (g)



Step #5780: RIE

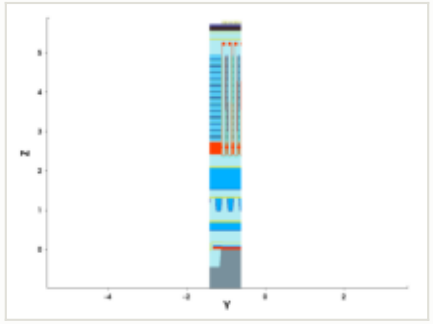
etch_anisotropic · RIE

1.04	502.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

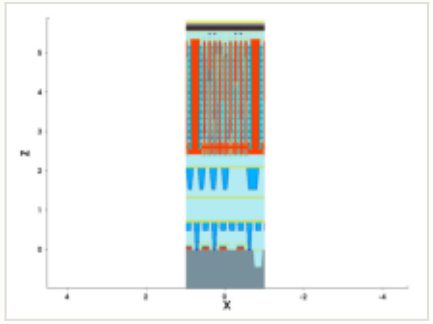
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

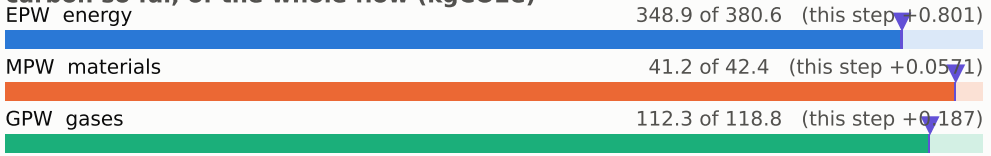
no mask for this step

this step

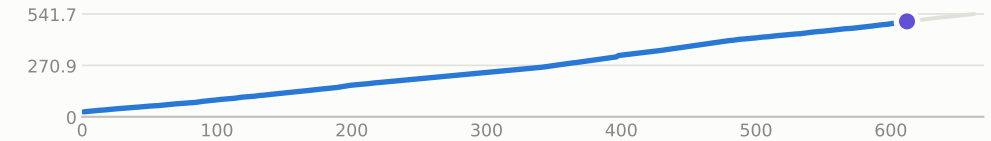
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

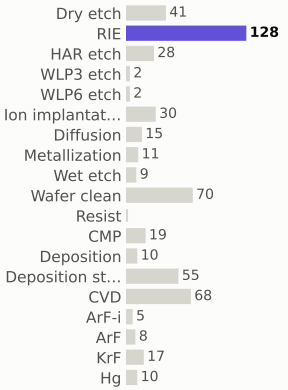
carbon so far, of the whole flow (kgCO2e)



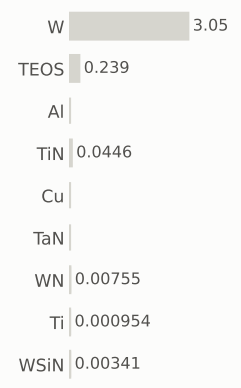
embodied carbon up to each step (kgCO2e per wafer)



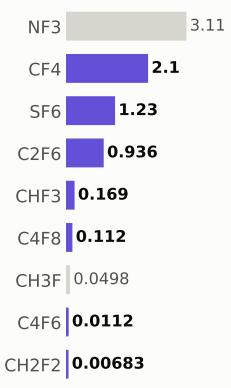
steps by process area



materials deposited (g)



gases emitted (g)



Step #5790: RIE

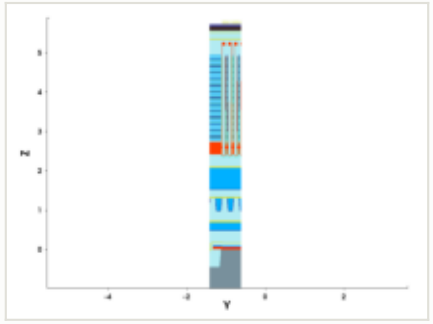
etch_anisotropic · RIE

1.04	503.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

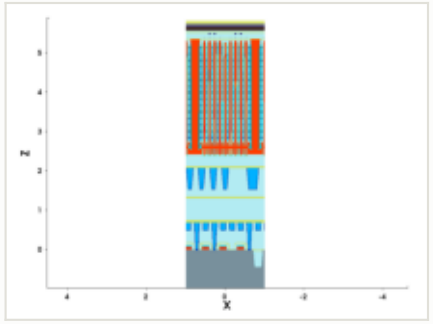
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

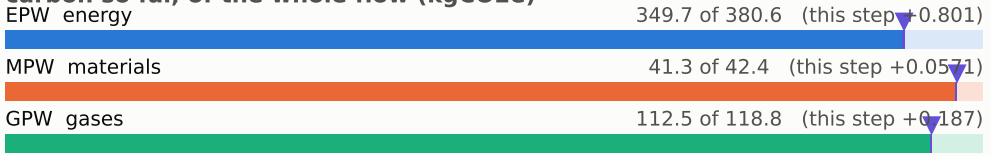
no mask for this step

this step

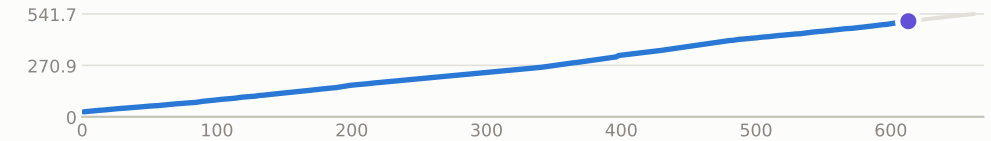
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

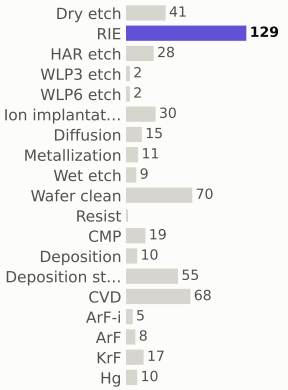
carbon so far, of the whole flow (kgCO2e)



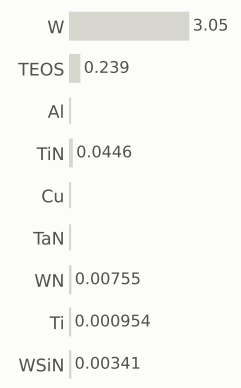
embodied carbon up to each step (kgCO2e per wafer)



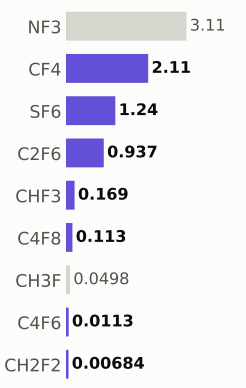
steps by process area



materials deposited (g)



gases emitted (g)



Step #5800: Resist

depo_sog · Resist

0.169	503.6	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

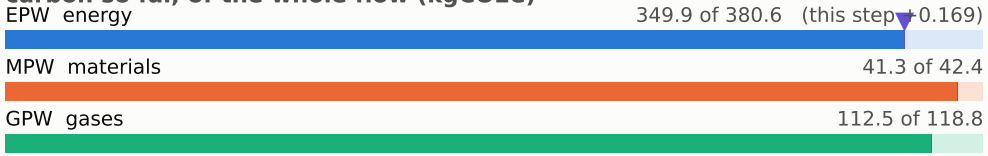
3D structure after this step

zoomed to fit

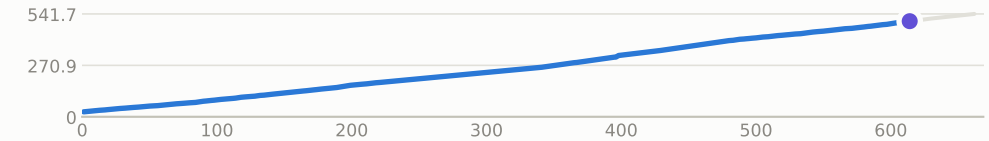
this step

process area	Resist	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	0.15 µm	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

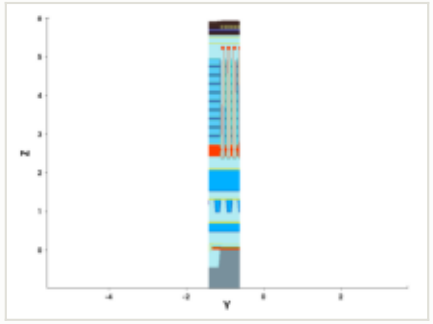
carbon so far, of the whole flow (kgCO2e)



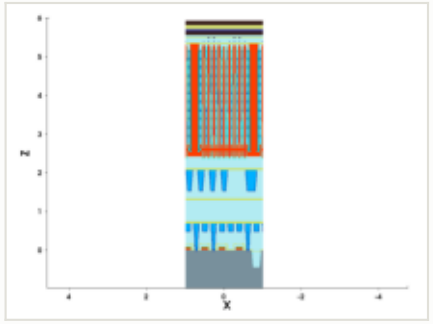
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



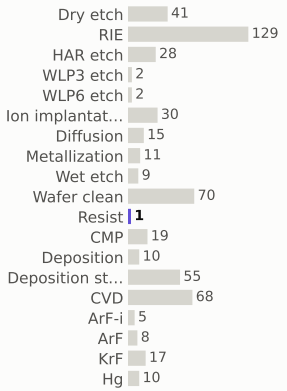
y cut at -1.032 µm



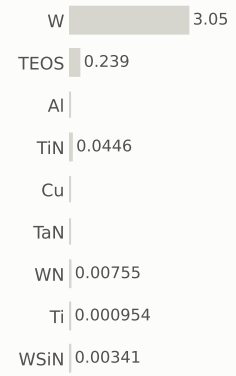
mask for this step

no mask for this step

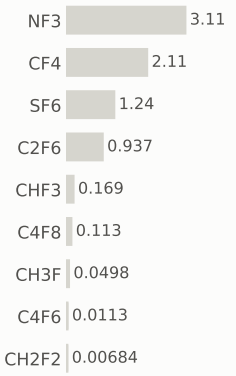
steps by process area



materials deposited (g)



gases emitted (g)



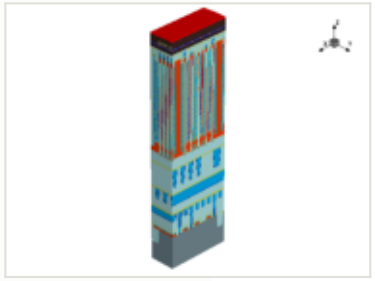
Step #5810: CVD

depo_isotropic · CVD

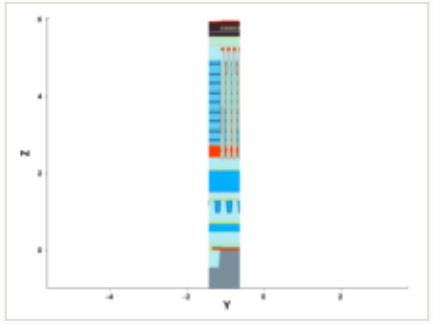
1.31	505.0	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

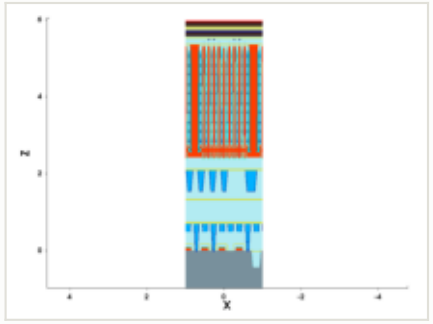
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

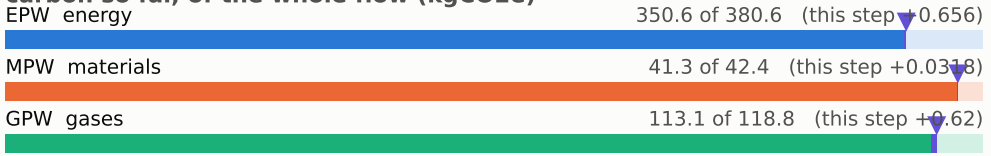
no mask for this step

this step

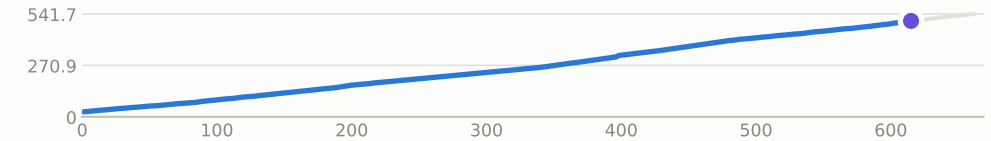
process area	CVD
material	—
thickness	0.03 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

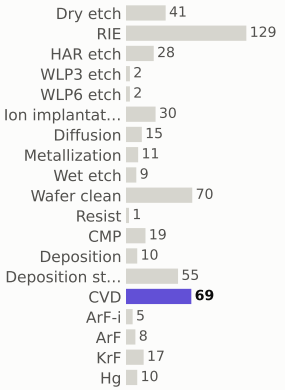
carbon so far, of the whole flow (kgCO2e)



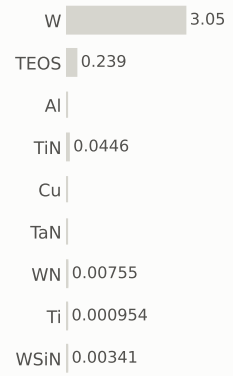
embodied carbon up to each step (kgCO2e per wafer)



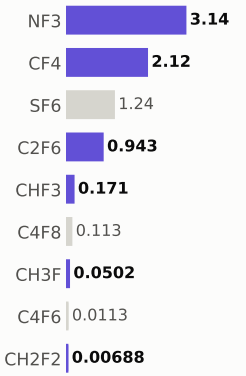
steps by process area



materials deposited (g)



gases emitted (g)



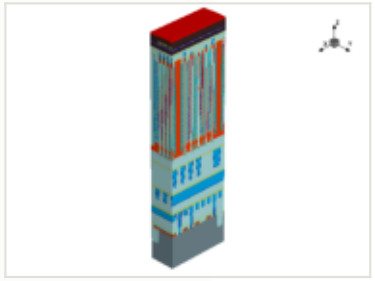
Step #5820: Wafer_clean

clean · Wafer_clean

0.169	505.1	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

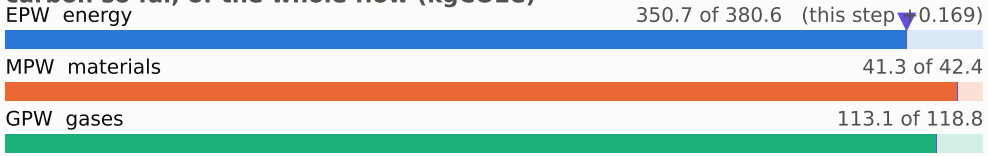


this step

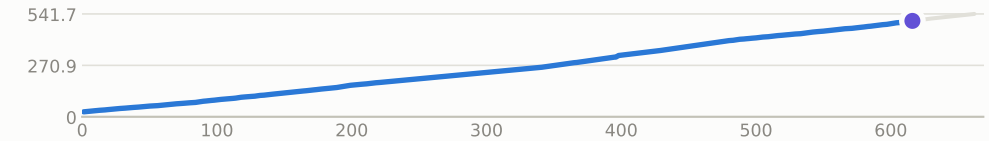
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

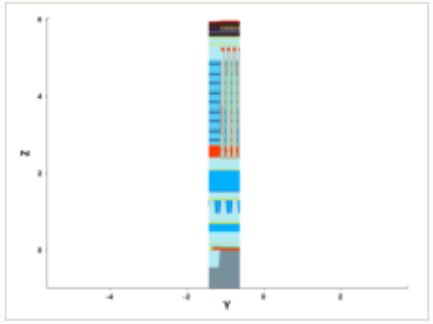
carbon so far, of the whole flow (kgCO2e)



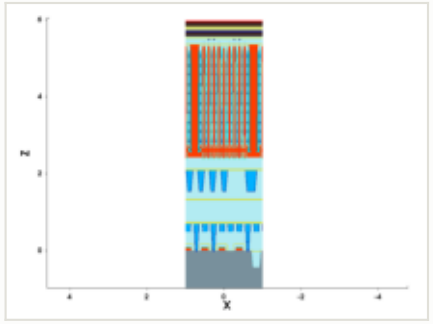
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



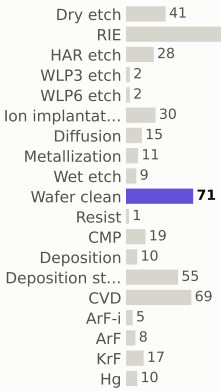
y cut at -1.032 µm



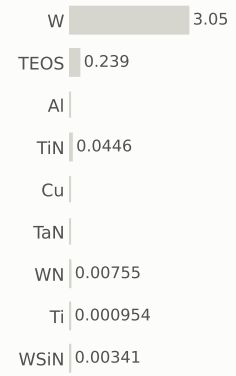
mask for this step

no mask for this step

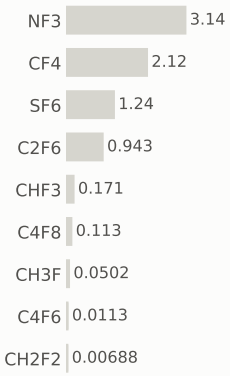
steps by process area



materials deposited (g)



gases emitted (g)



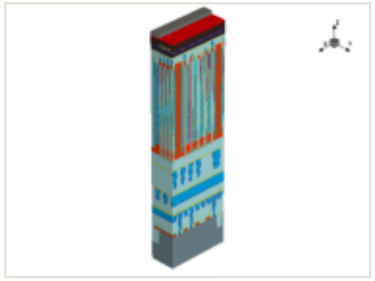
Step #5830: ArF-i

litho_patterning · ArF-i

0.217	505.3	0.04%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

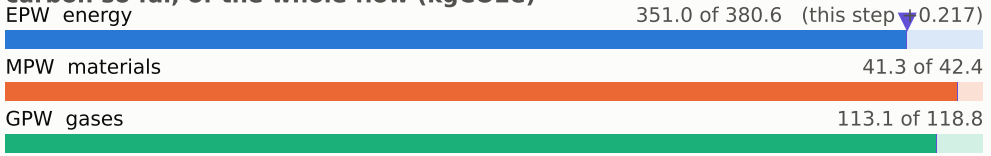
zoomed to fit



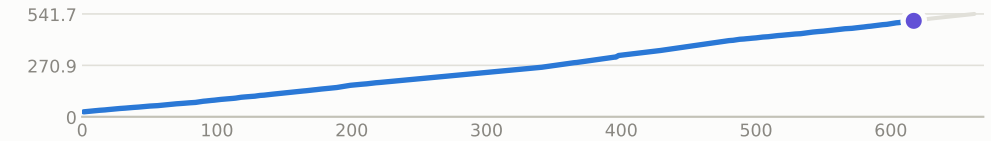
this step

process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

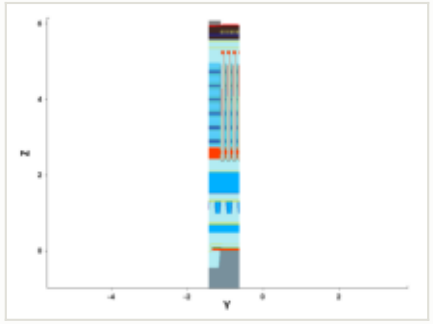
carbon so far, of the whole flow (kgCO2e)



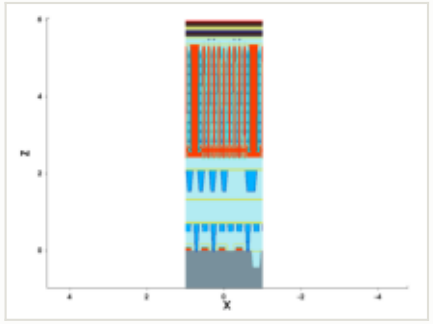
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

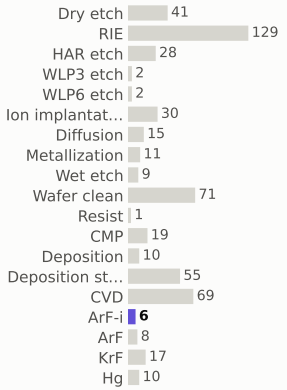


y cut at -1.032 µm

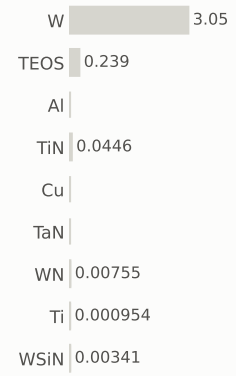


mask: layer 44:0 (OASIS)

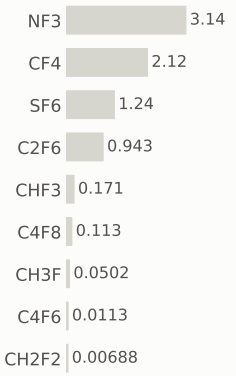
steps by process area



materials deposited (g)



gases emitted (g)



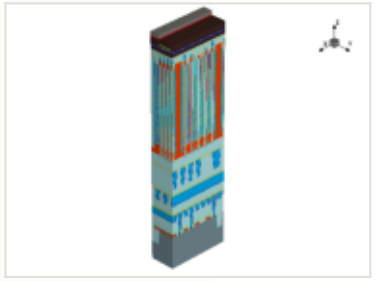
Step #5840: RIE

etch_anisotropic · RIE

1.04	506.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

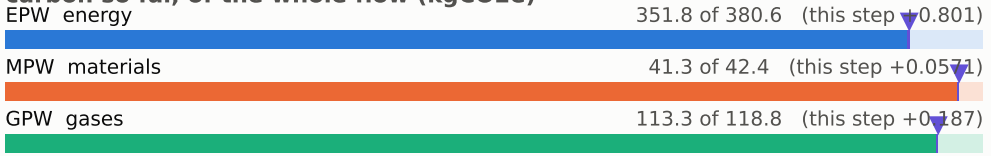


this step

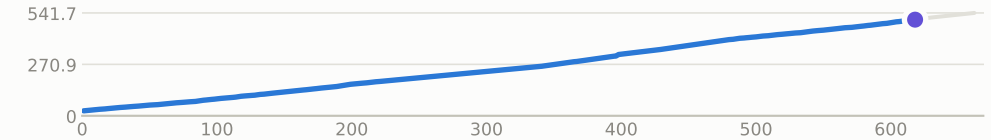
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

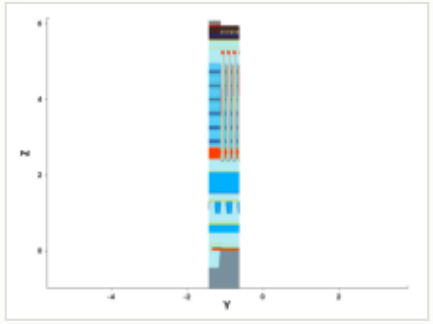
carbon so far, of the whole flow (kgCO2e)



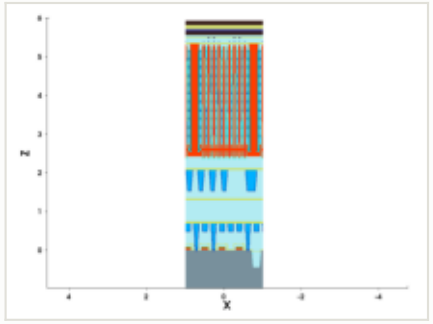
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



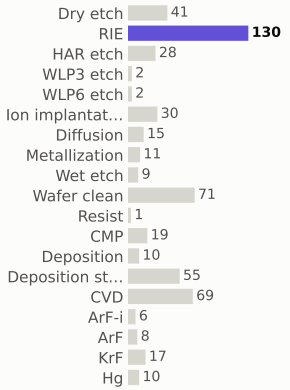
y cut at -1.032 µm



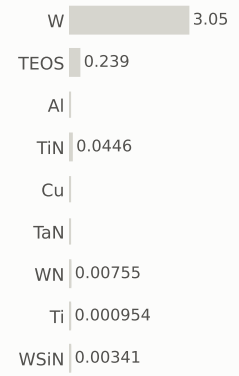
mask for this step

no mask for this step

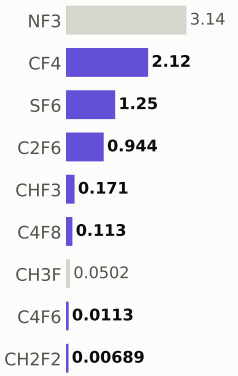
steps by process area



materials deposited (g)



gases emitted (g)



Step #5850: RIE

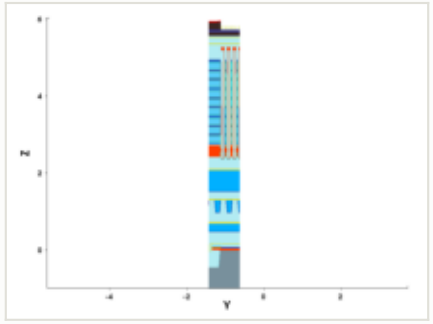
etch_anisotropic · RIE

1.04	507.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

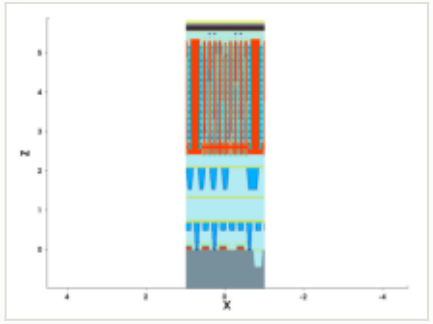
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

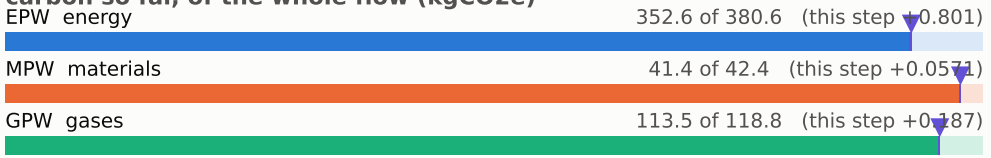
no mask for this step

this step

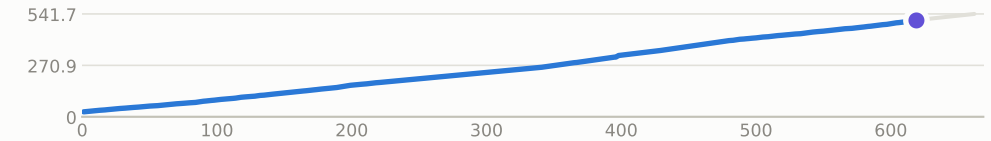
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

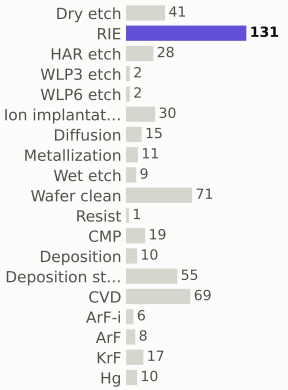
carbon so far, of the whole flow (kgCO2e)



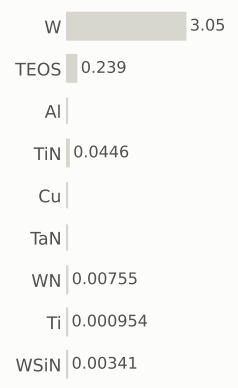
embodied carbon up to each step (kgCO2e per wafer)



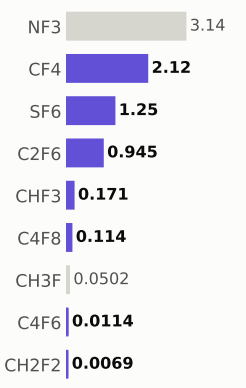
steps by process area



materials deposited (g)



gases emitted (g)



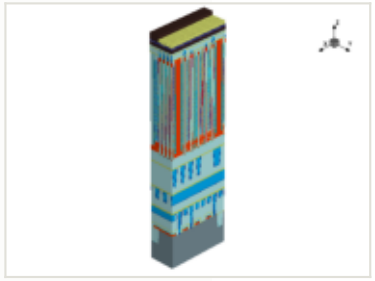
Step #5860: RIE

etch_anisotropic · RIE

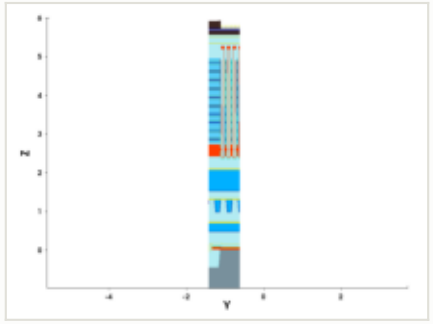
1.04	508.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

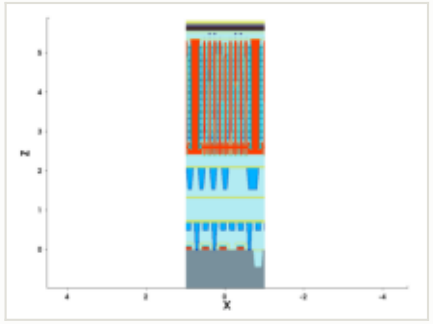
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

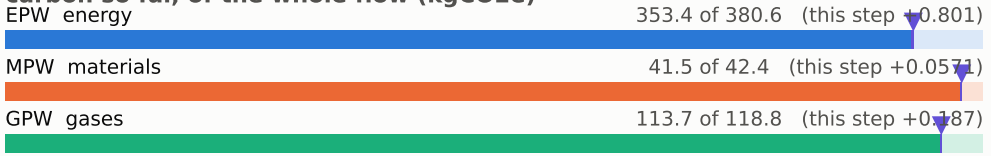
no mask for this step

this step

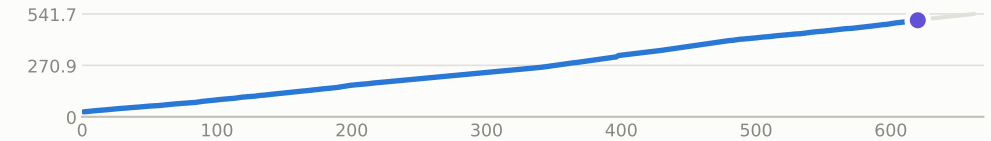
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

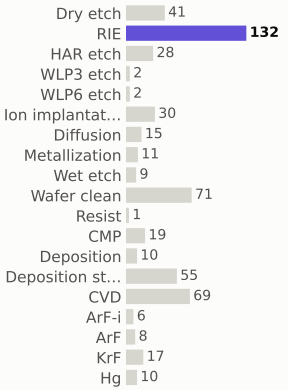
carbon so far, of the whole flow (kgCO2e)



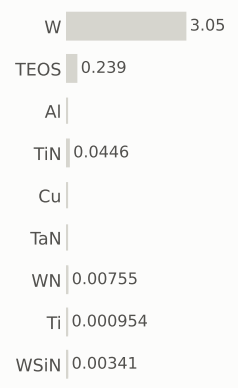
embodied carbon up to each step (kgCO2e per wafer)



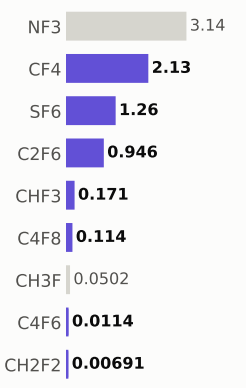
steps by process area



materials deposited (g)



gases emitted (g)



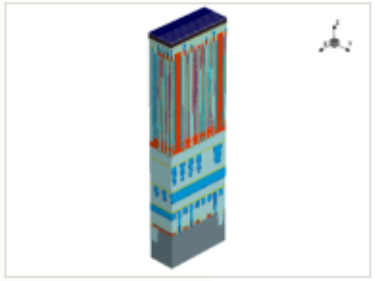
Step #5870: RIE

etch_anisotropic · RIE

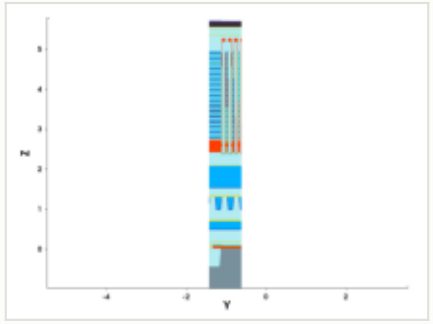
1.04	509.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

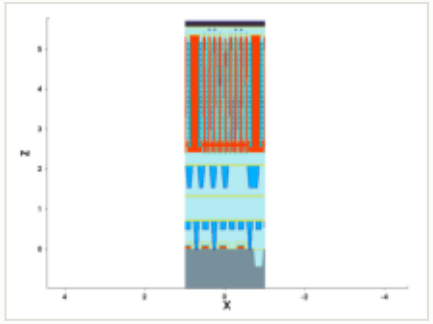
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

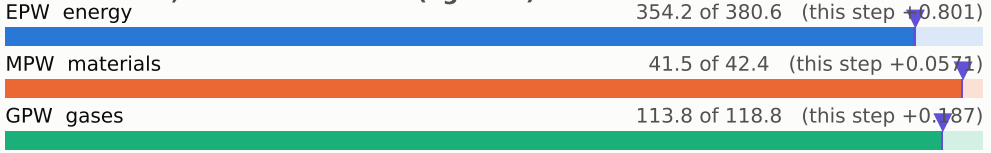
no mask for this step

this step

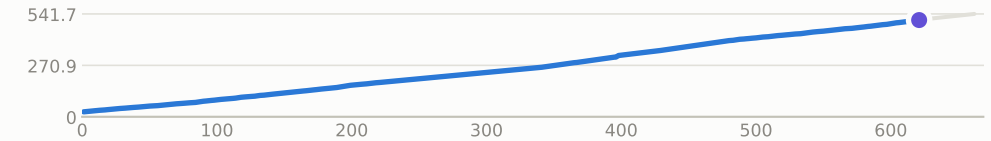
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

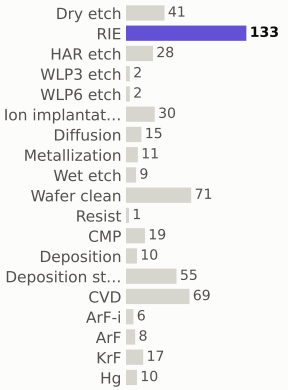
carbon so far, of the whole flow (kgCO2e)



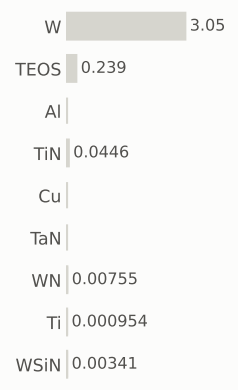
embodied carbon up to each step (kgCO2e per wafer)



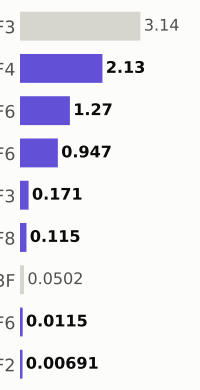
steps by process area



materials deposited (g)



gases emitted (g)



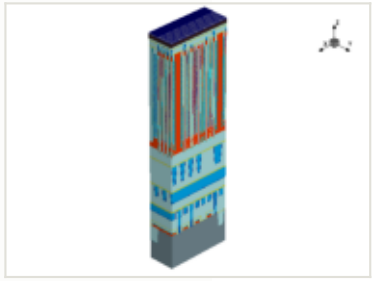
Step #5880: RIE

etch_anisotropic · RIE

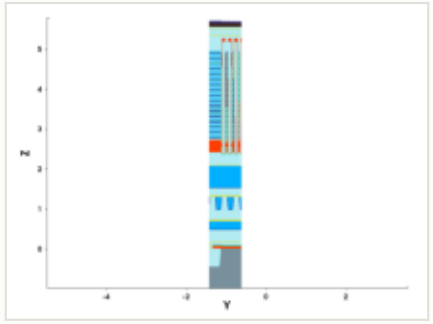
1.04	510.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

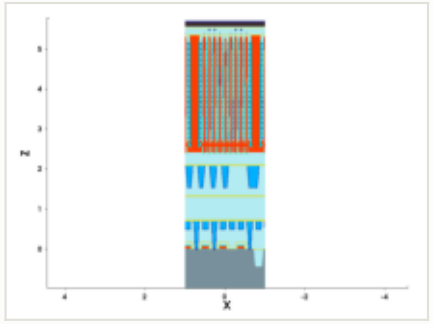
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

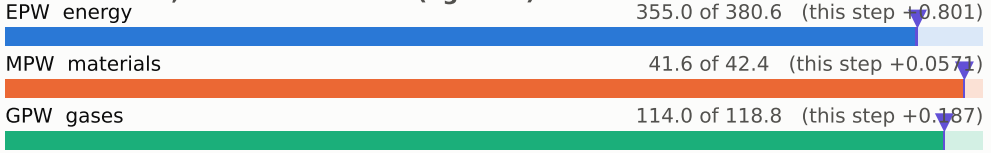
no mask for this step

this step

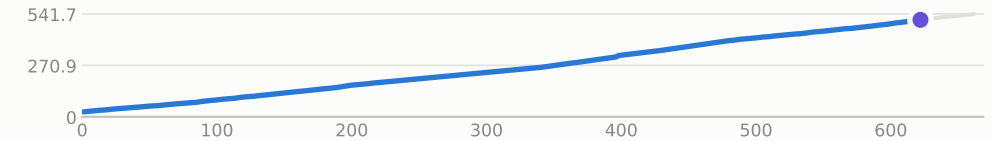
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

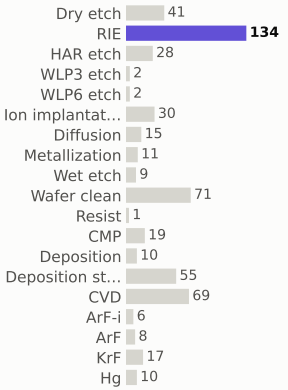
carbon so far, of the whole flow (kgCO2e)



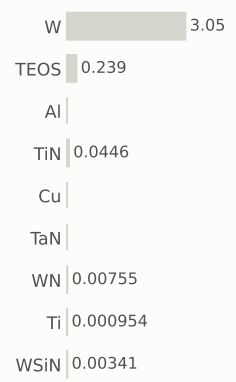
embodied carbon up to each step (kgCO2e per wafer)



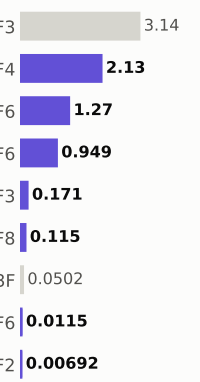
steps by process area



materials deposited (g)



gases emitted (g)



Step #5890: RIE

etch_anisotropic · RIE

1.04	511.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

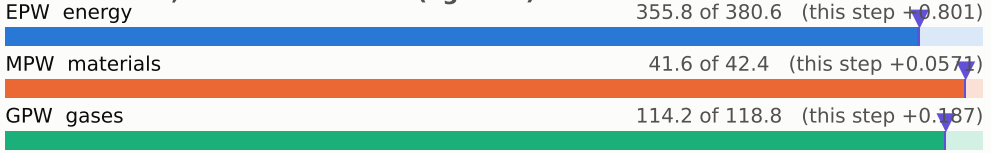
zoomed to fit

this step

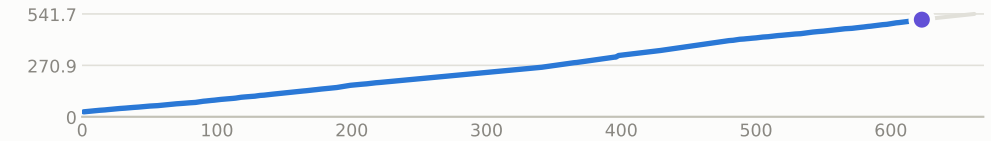
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

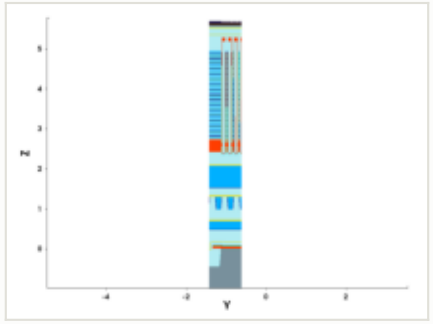
carbon so far, of the whole flow (kgCO2e)



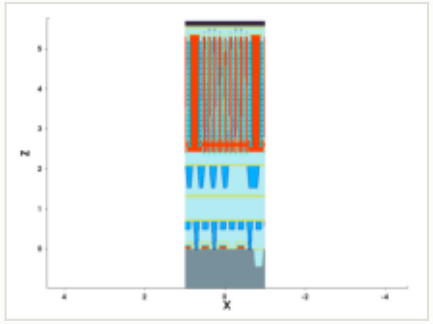
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



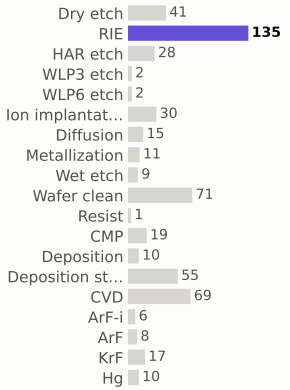
y cut at -1.032 µm



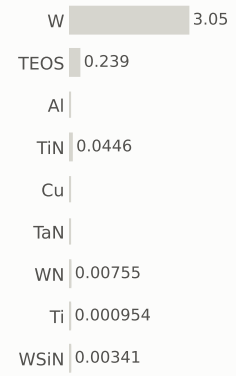
mask for this step

no mask for this step

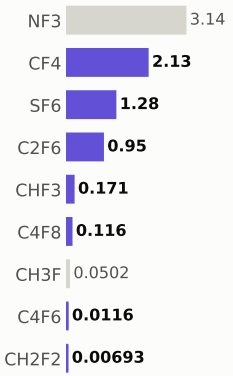
steps by process area



materials deposited (g)



gases emitted (g)



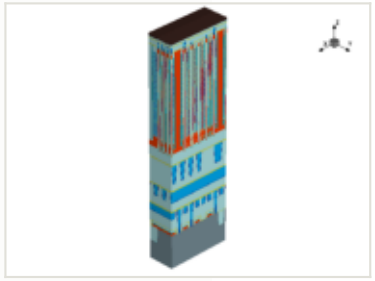
Step #5900: RIE

etch_anisotropic · RIE

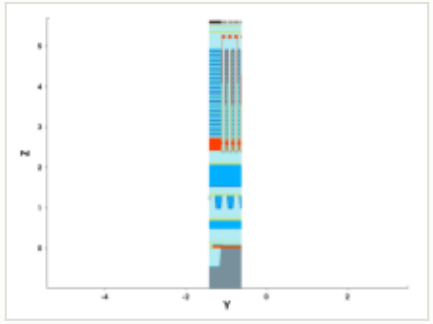
1.04	512.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

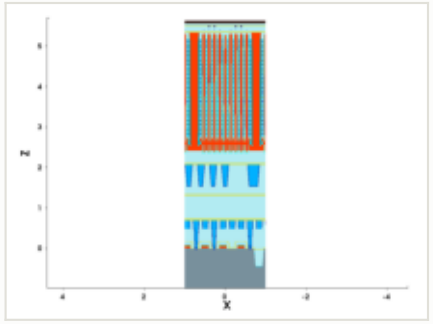
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

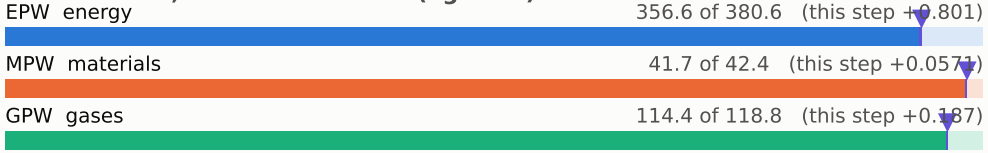
no mask for this step

this step

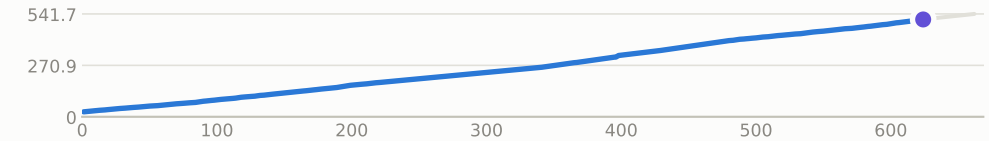
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

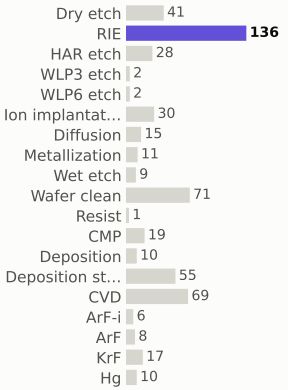
carbon so far, of the whole flow (kgCO2e)



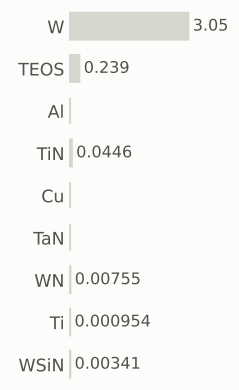
embodied carbon up to each step (kgCO2e per wafer)



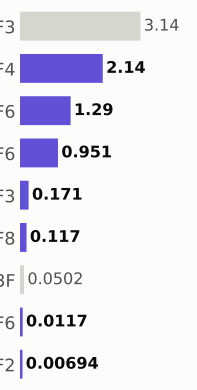
steps by process area



materials deposited (g)



gases emitted (g)



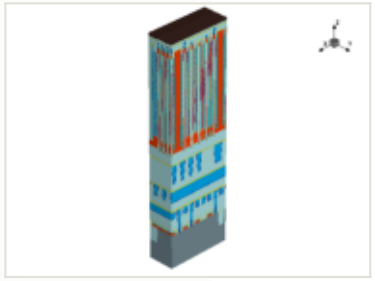
Step #5910: RIE

etch_anisotropic · RIE

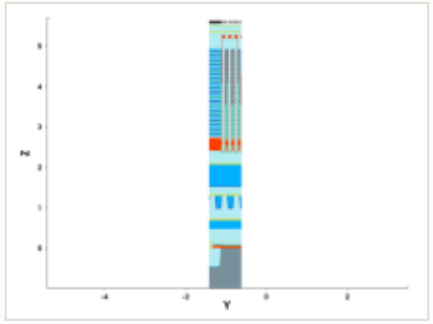
1.04	513.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

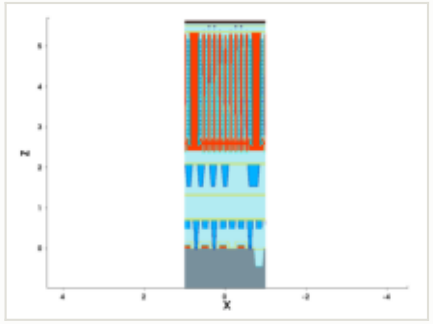
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

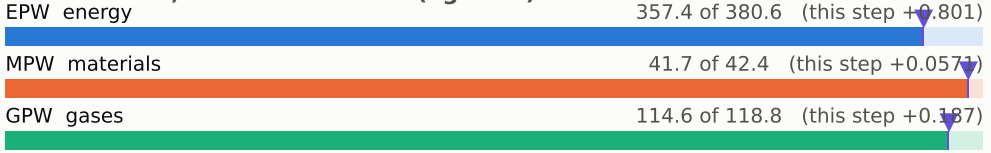
no mask for this step

this step

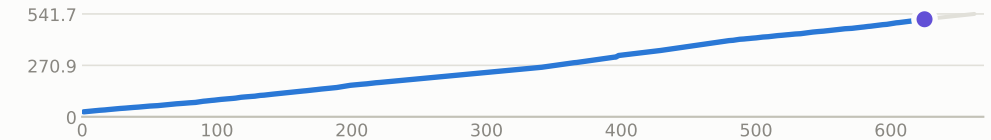
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

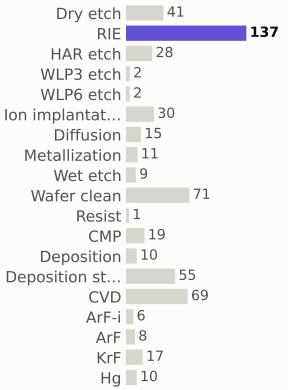
carbon so far, of the whole flow (kgCO2e)



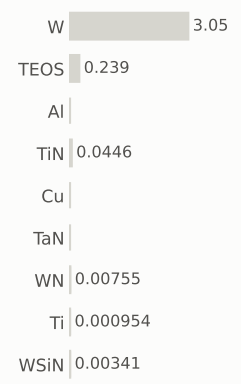
embodied carbon up to each step (kgCO2e per wafer)



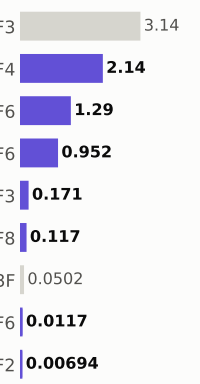
steps by process area



materials deposited (g)



gases emitted (g)



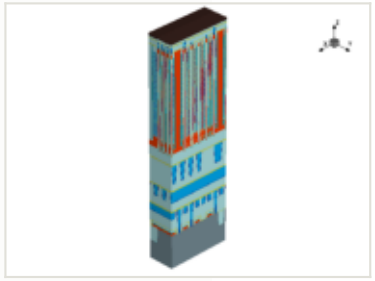
Step #5920: RIE

etch_anisotropic · RIE

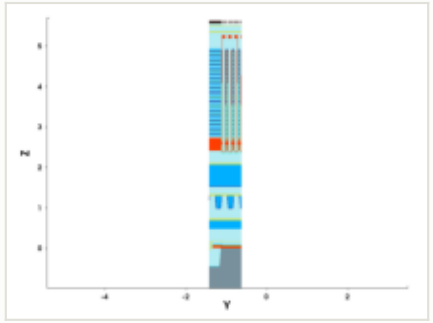
1.04	514.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

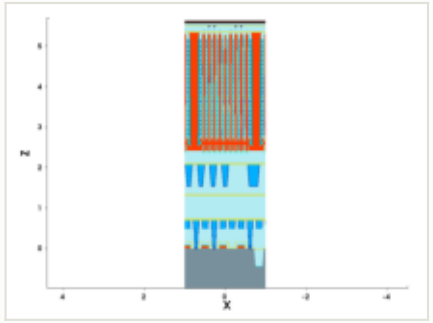
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

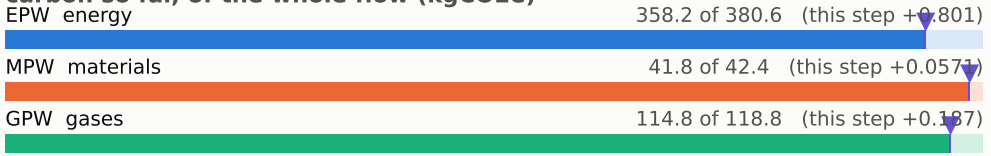
no mask for this step

this step

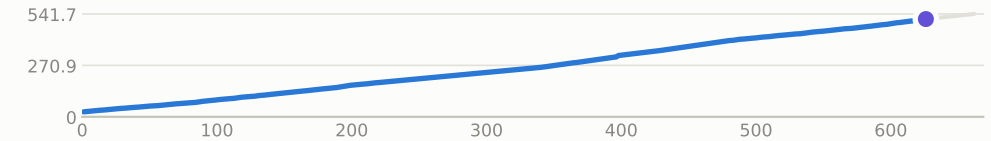
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

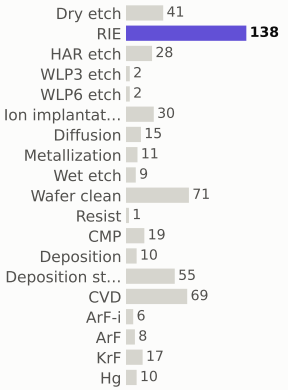
carbon so far, of the whole flow (kgCO2e)



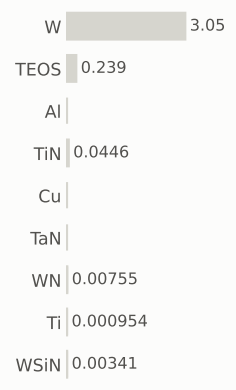
embodied carbon up to each step (kgCO2e per wafer)



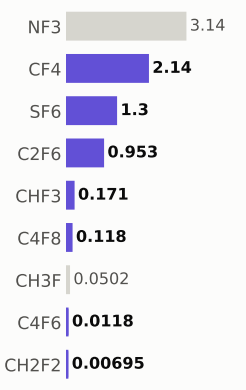
steps by process area



materials deposited (g)



gases emitted (g)



Step #5930: Dry_etch

etch_strip · Dry_etch

0.801	515.5	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

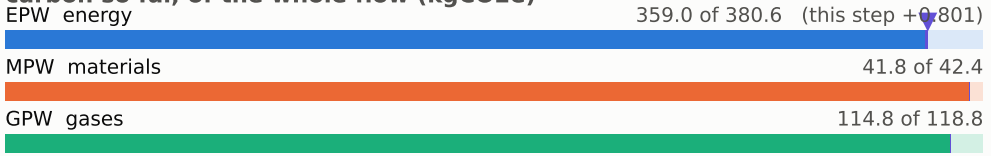
zoomed to fit

this step

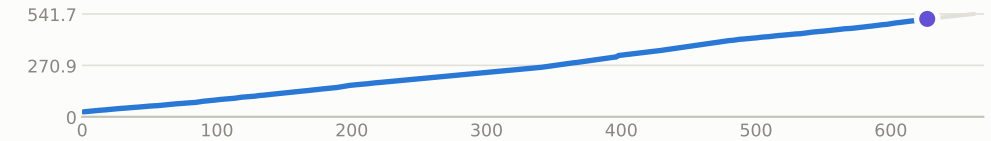
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

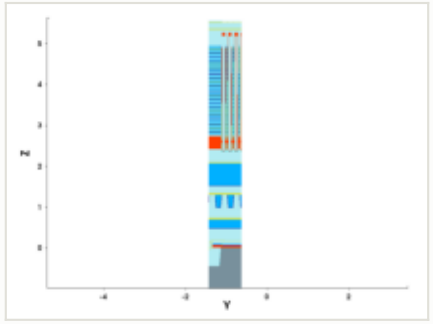
carbon so far, of the whole flow (kgCO2e)



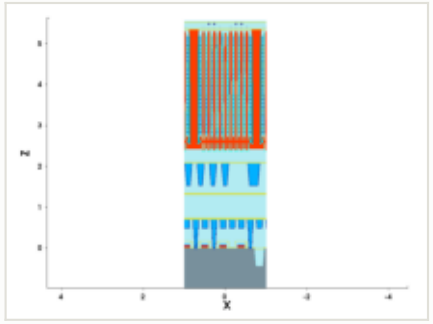
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



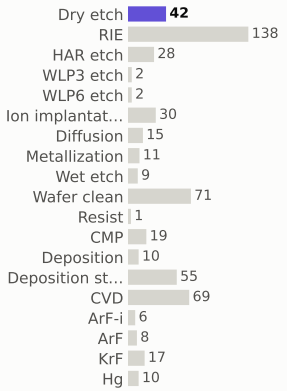
y cut at -1.032 µm



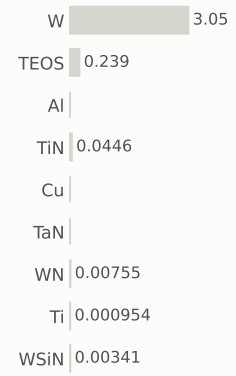
mask for this step

no mask for this step

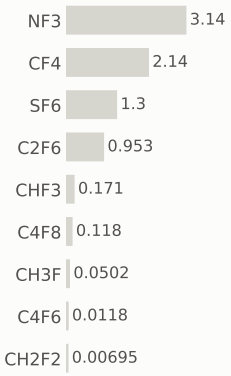
steps by process area



materials deposited (g)



gases emitted (g)



Step #5940: Wafer_clean

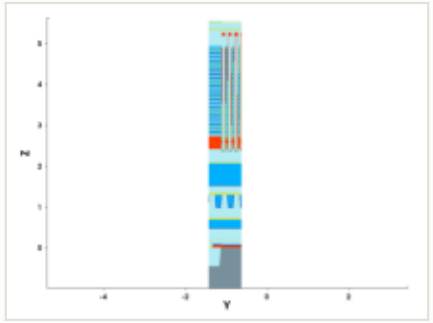
clean · Wafer_clean

0.169	515.7	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

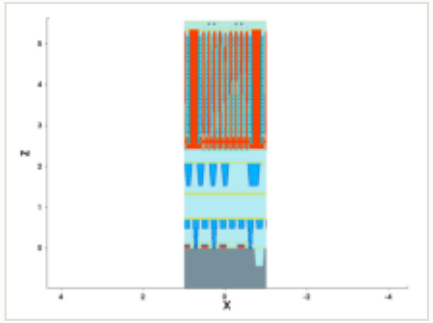
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

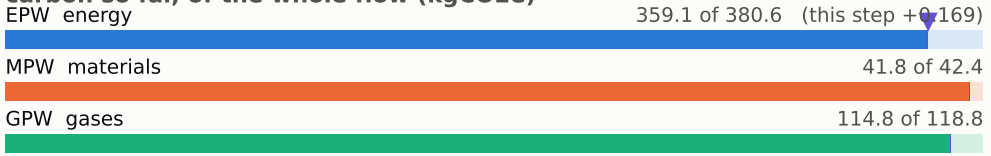
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

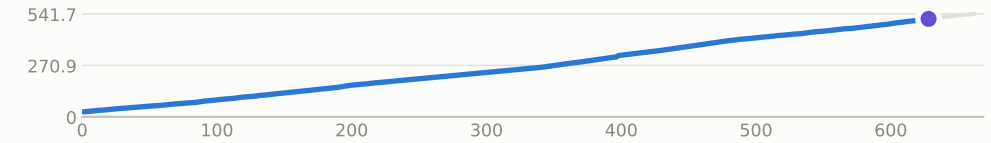
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

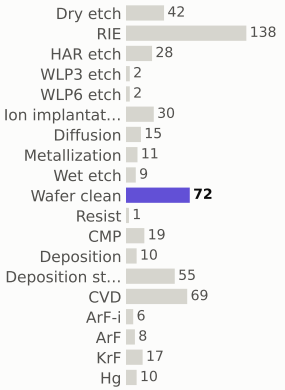
carbon so far, of the whole flow (kgCO2e)



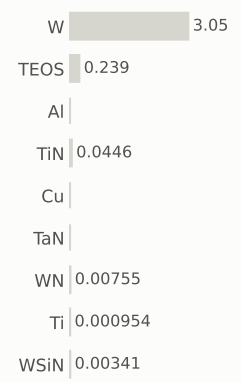
embodied carbon up to each step (kgCO2e per wafer)



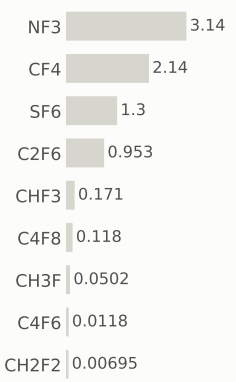
steps by process area



materials deposited (g)



gases emitted (g)



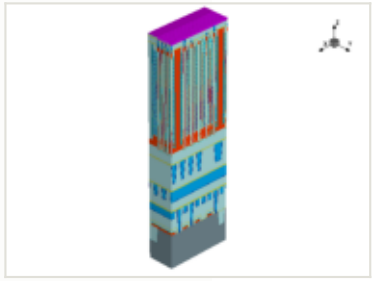
Step #5950: Deposition

depo_ellipsoidal · Deposition

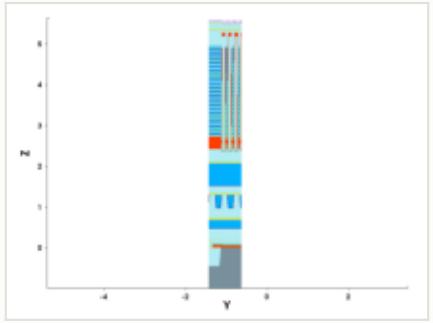
0.656	516.4	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

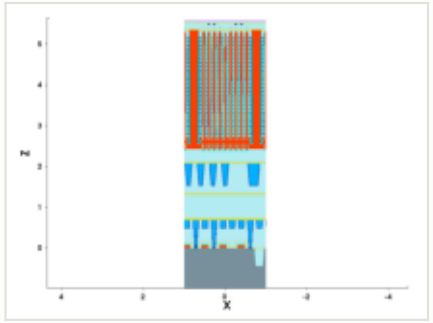
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

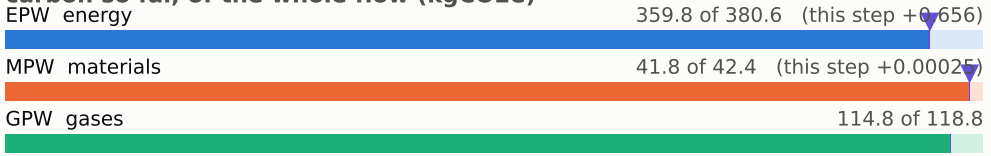
no mask for this step

this step

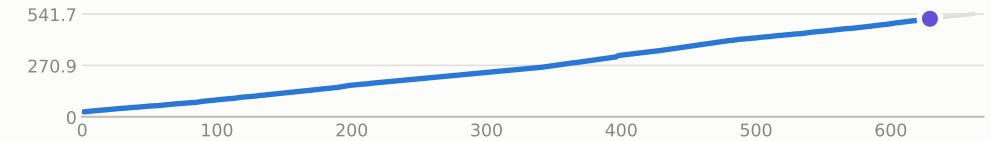
process area	Deposition
material	TaN
thickness	0.01 µm
material GWP	25.8159 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.00025 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

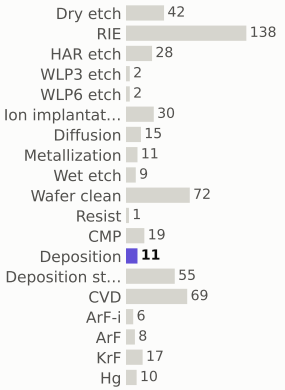
carbon so far, of the whole flow (kgCO2e)



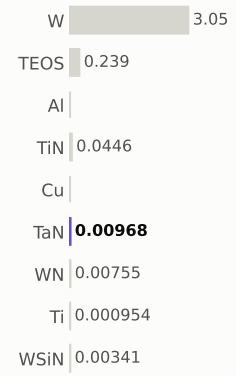
embodied carbon up to each step (kgCO2e per wafer)



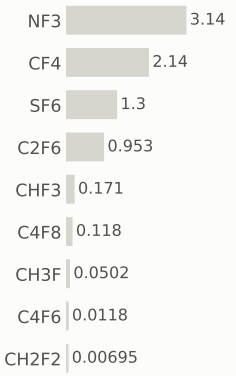
steps by process area



materials deposited (g)



gases emitted (g)



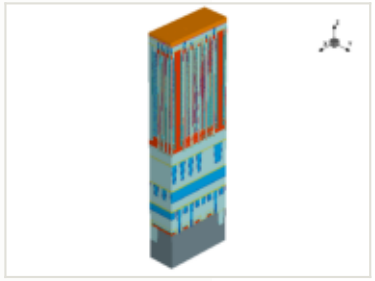
Step #5960: Metallization

depo_isotropic · Metallization

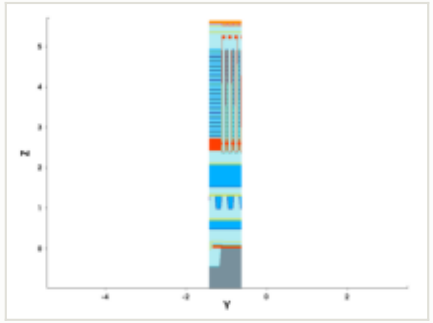
1.2	517.6	0.22%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

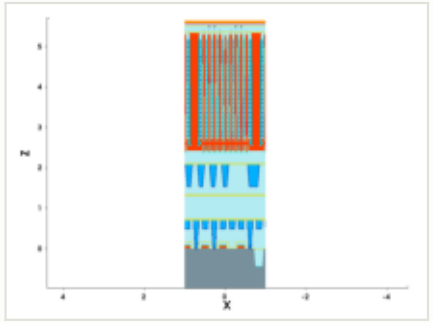
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

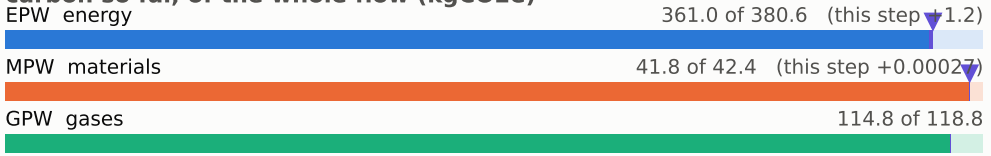
no mask for this step

this step

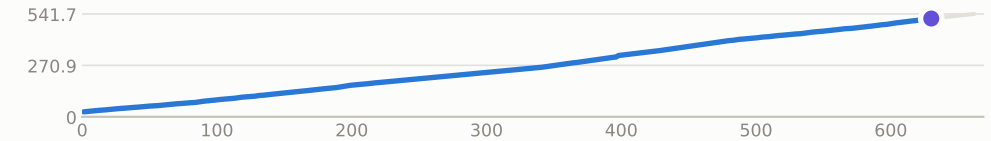
process area	Metallization
material	Cu
thickness	0.07 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00027 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

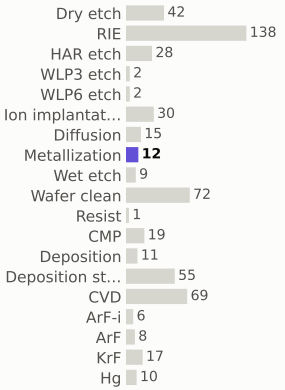
carbon so far, of the whole flow (kgCO2e)



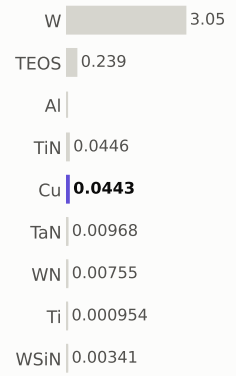
embodied carbon up to each step (kgCO2e per wafer)



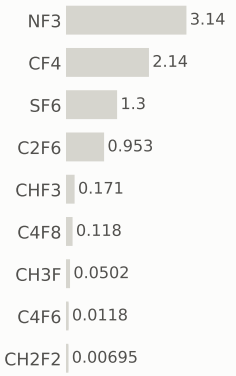
steps by process area



materials deposited (g)



gases emitted (g)



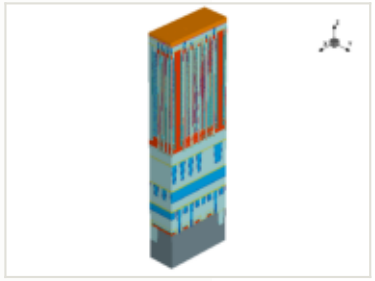
Step #5970: Diffusion

anneal · Diffusion

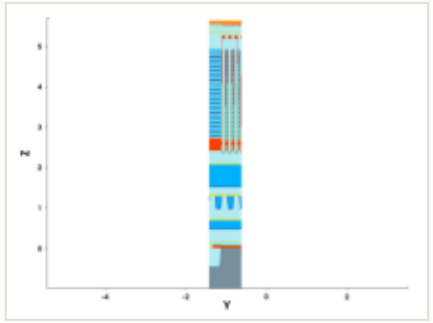
0.801	518.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

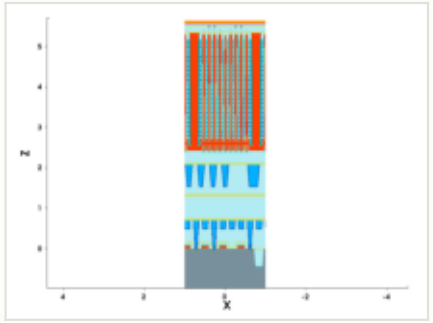
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

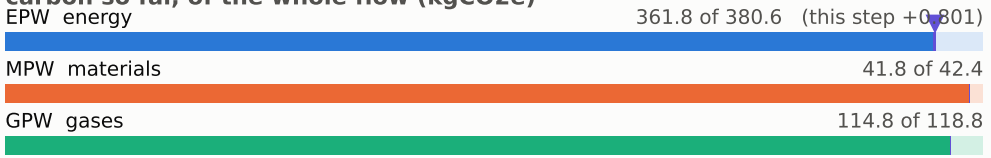
this step

process area	Diffusion
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

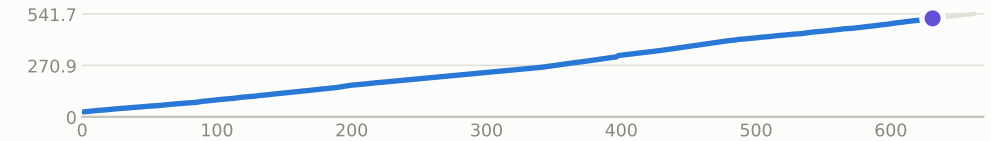
EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

spx ran with 0 values for: TemperatureRamp

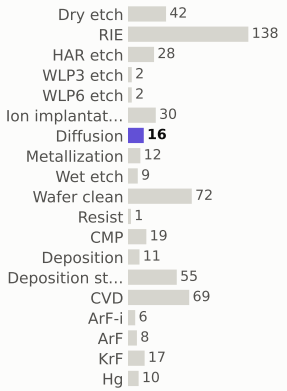
carbon so far, of the whole flow (kgCO2e)



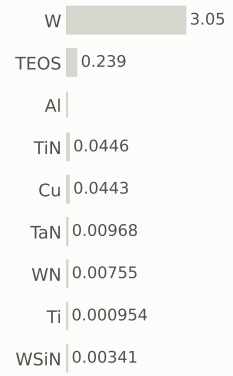
embodied carbon up to each step (kgCO2e per wafer)



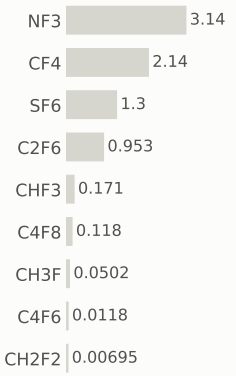
steps by process area



materials deposited (g)



gases emitted (g)



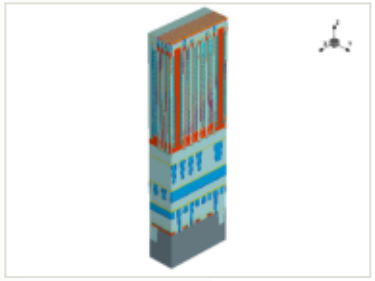
Step #5980: CMP

etch_cmp · CMP

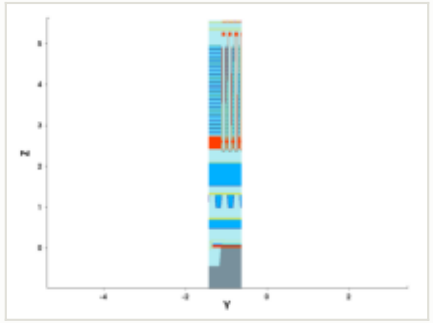
0.169	518.5	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

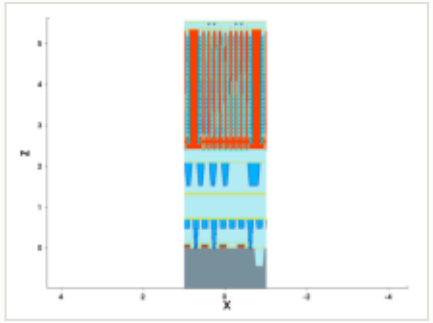
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

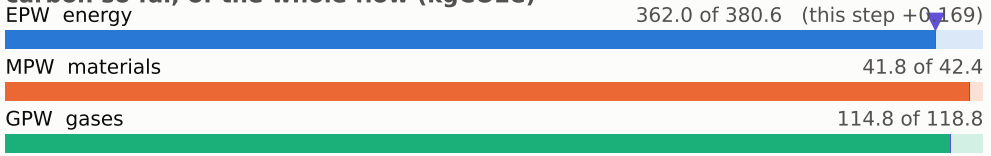
no mask for this step

this step

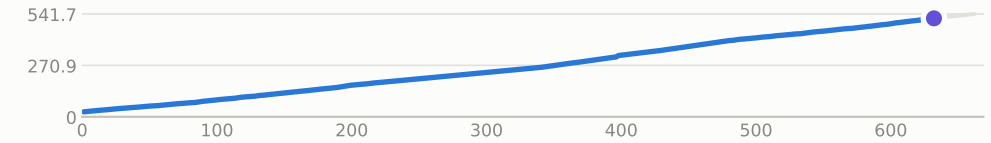
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

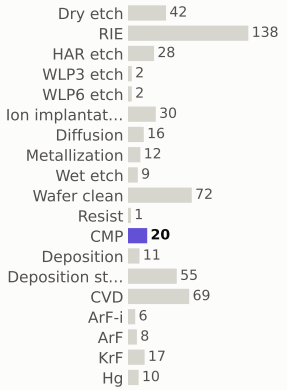
carbon so far, of the whole flow (kgCO2e)



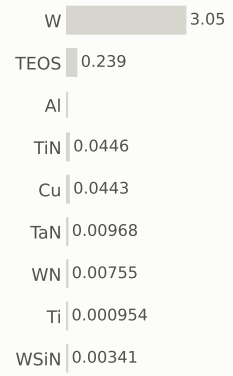
embodied carbon up to each step (kgCO2e per wafer)



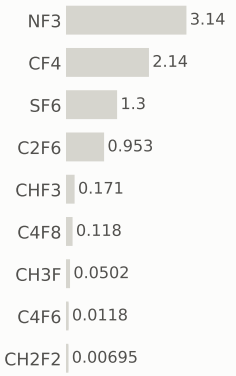
steps by process area



materials deposited (g)



gases emitted (g)



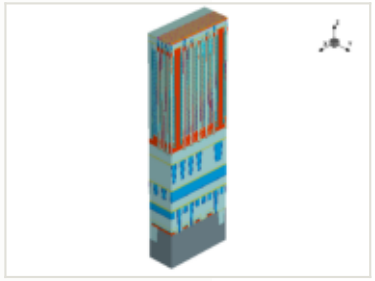
Step #5990: Wafer_clean

clean · Wafer_clean

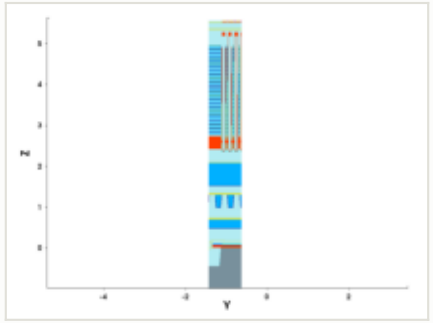
0.169	518.7	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

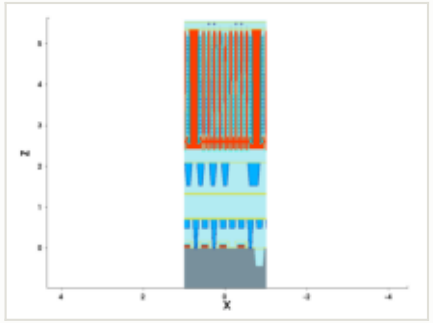
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

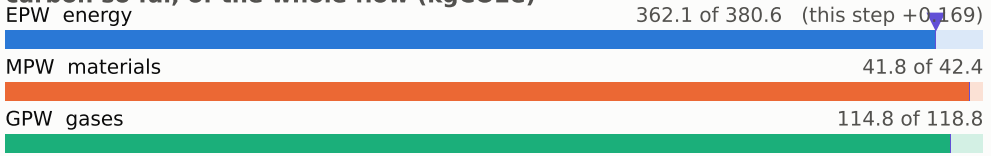
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

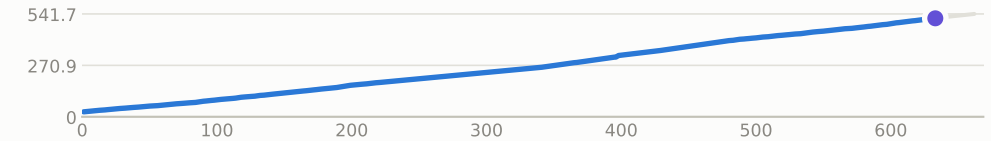
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

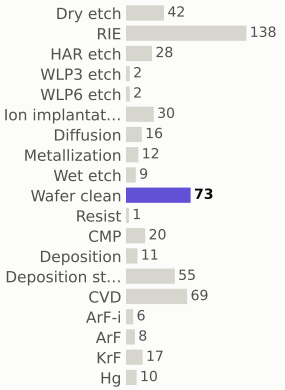
carbon so far, of the whole flow (kgCO2e)



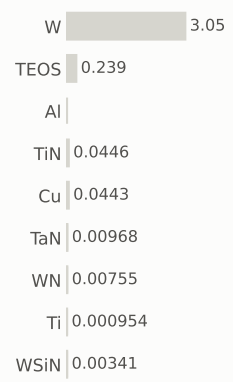
embodied carbon up to each step (kgCO2e per wafer)



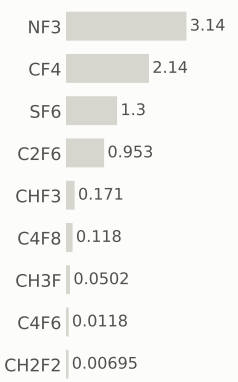
steps by process area



materials deposited (g)



gases emitted (g)



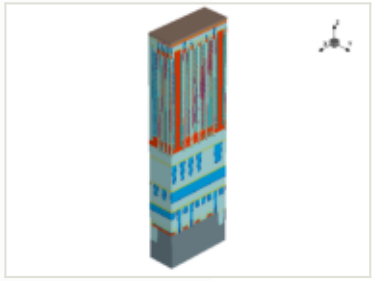
Step #6010: CVD

depo_isotropic · CVD

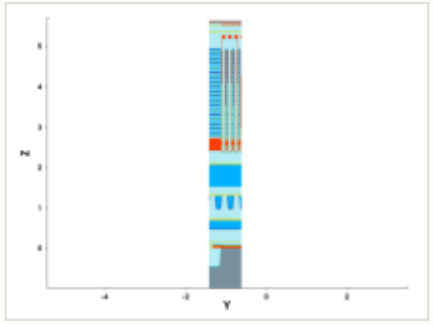
1.31	520.0	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

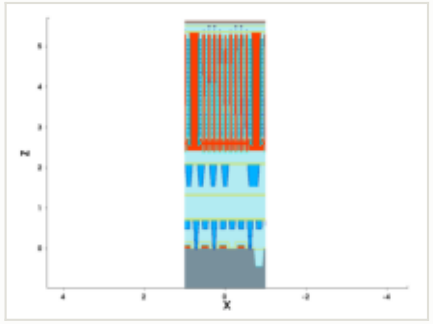
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

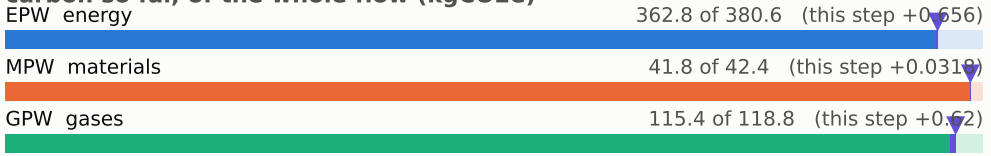
no mask for this step

this step

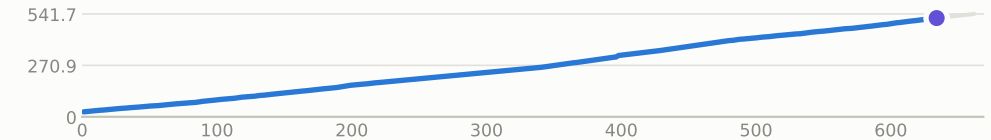
process area	CVD
material	—
thickness	0.075 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

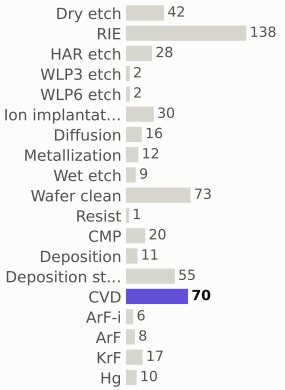
carbon so far, of the whole flow (kgCO2e)



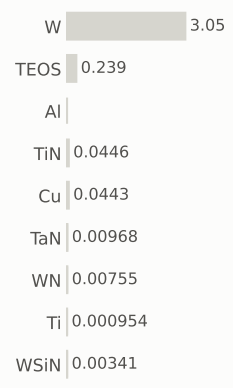
embodied carbon up to each step (kgCO2e per wafer)



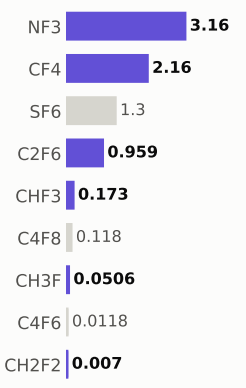
steps by process area



materials deposited (g)



gases emitted (g)



Step #6020: CVD

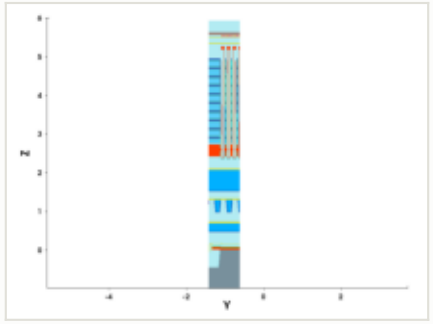
depo_isotropic · CVD

1.31	521.3	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

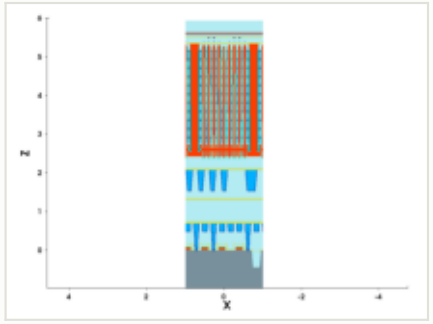
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

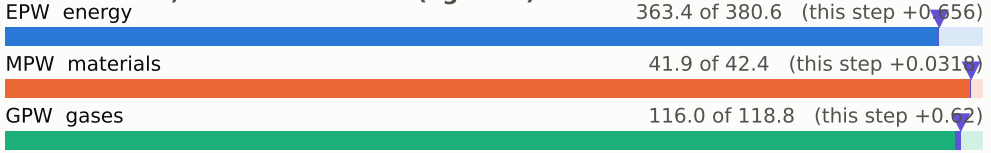
no mask for this step

this step

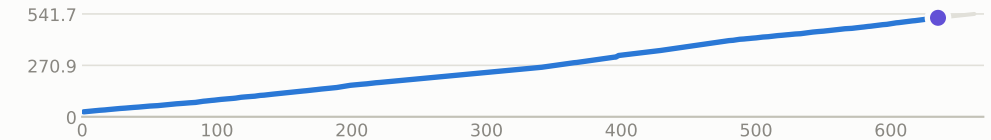
process area	CVD
material	—
thickness	0.3 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

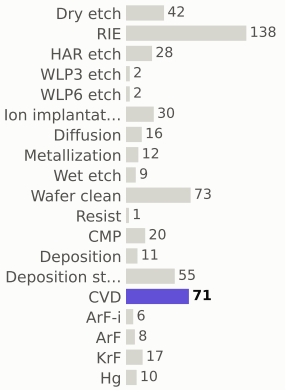
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

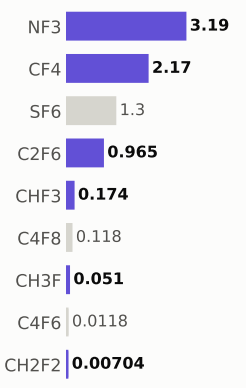


steps by process area



materials deposited (g)

gases emitted (g)



Step #6030: ArF

litho_patterning · ArF

0.199	521.5	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

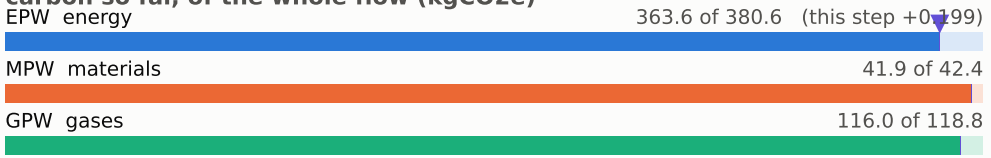
zoomed to fit

this step

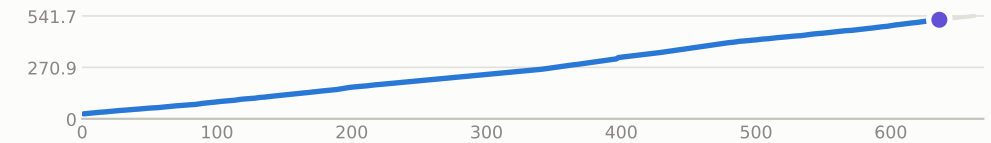
process area	ArF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

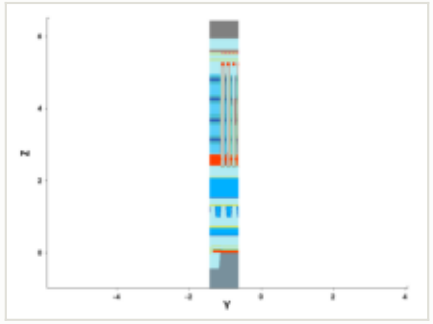
carbon so far, of the whole flow (kgCO2e)



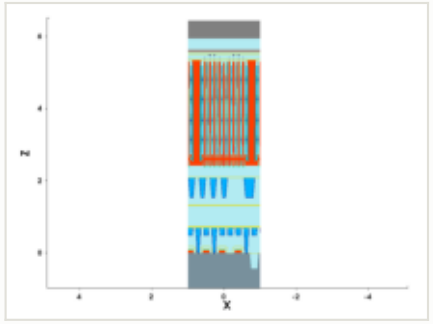
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

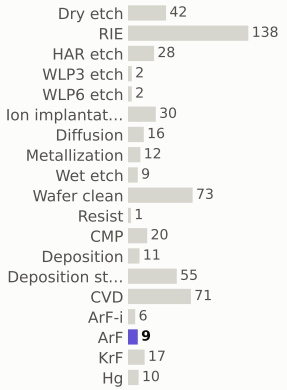


y cut at -1.032 µm

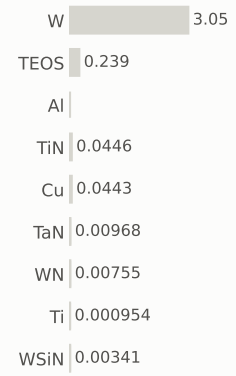


mask: layer 45:0, positive (OASIS)

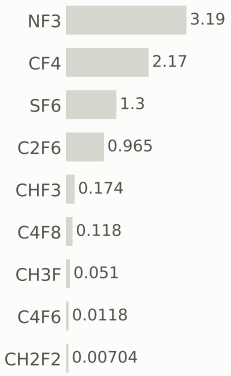
steps by process area



materials deposited (g)



gases emitted (g)



Step #6040: RIE

etch_anisotropic · RIE

1.04	522.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

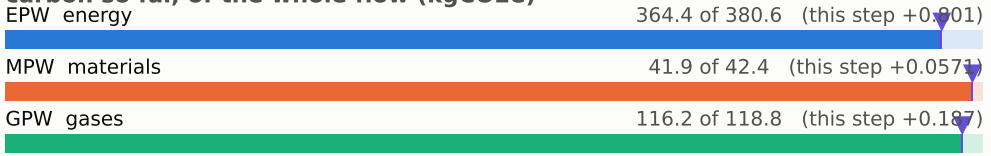
zoomed to fit

this step

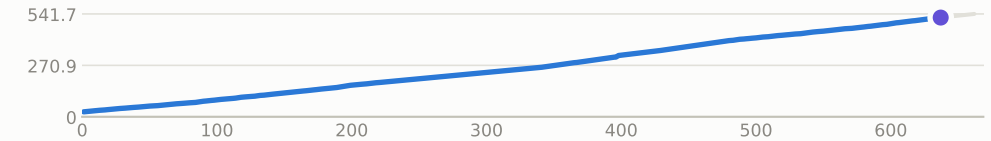
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

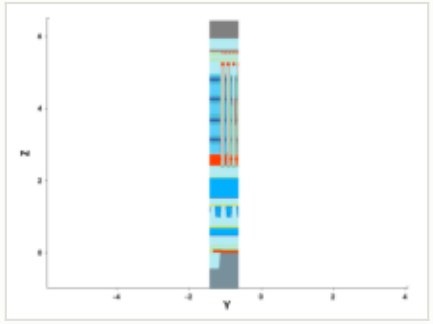
carbon so far, of the whole flow (kgCO2e)



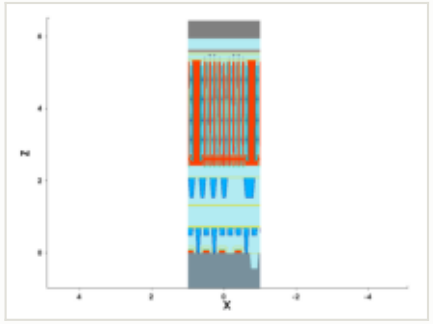
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



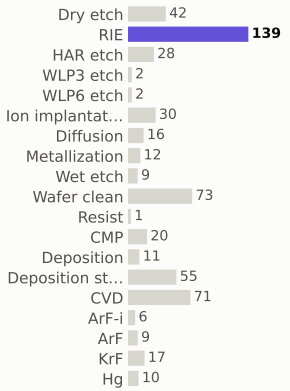
y cut at -1.032 µm



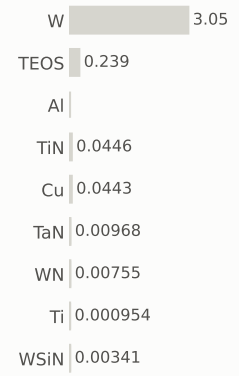
mask for this step

no mask for this step

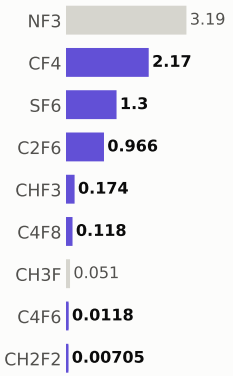
steps by process area



materials deposited (g)



gases emitted (g)



Step #6050: RIE

etch_anisotropic · RIE

1.04	523.6	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

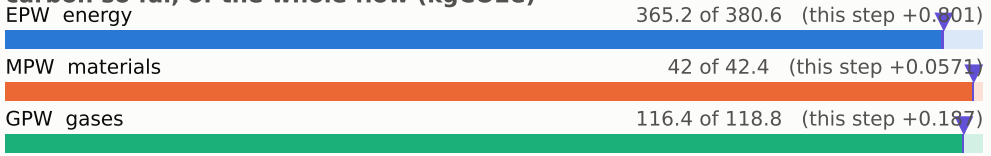
zoomed to fit

this step

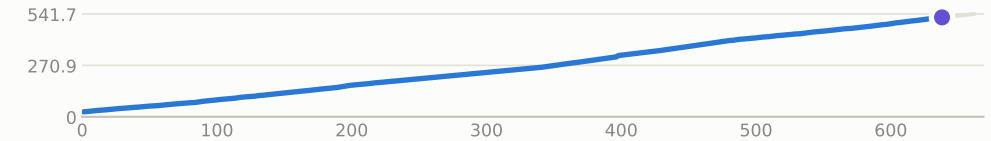
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

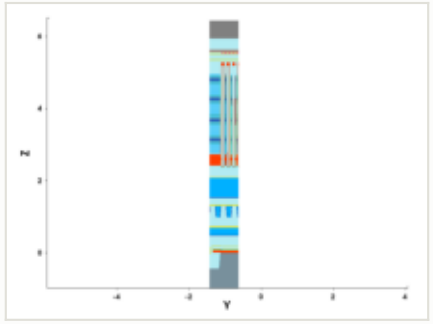
carbon so far, of the whole flow (kgCO2e)



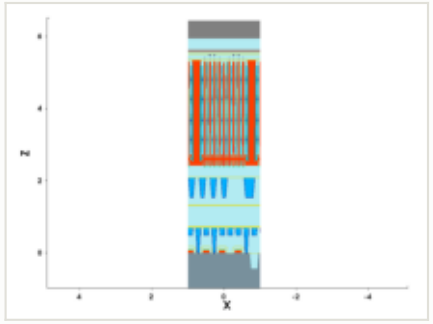
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



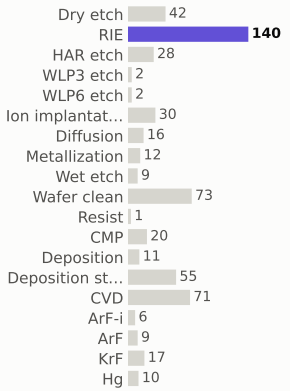
y cut at -1.032 µm



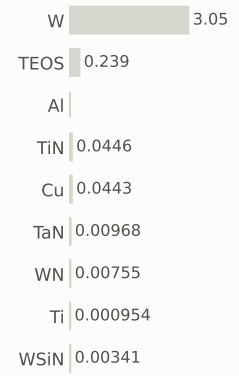
mask for this step

no mask for this step

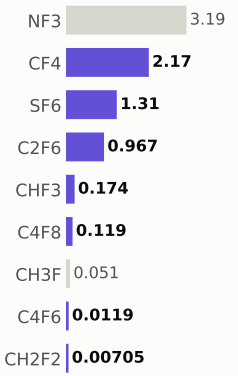
steps by process area



materials deposited (g)



gases emitted (g)



Step #6060: Dry_etch

etch_strip · Dry_etch

0.801	524.4	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

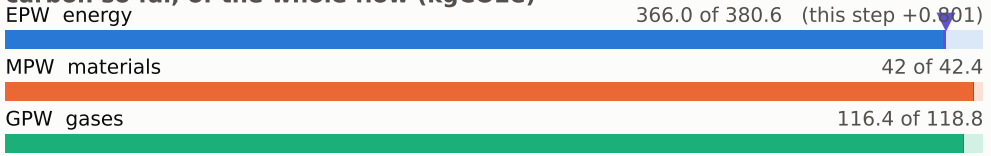
zoomed to fit

this step

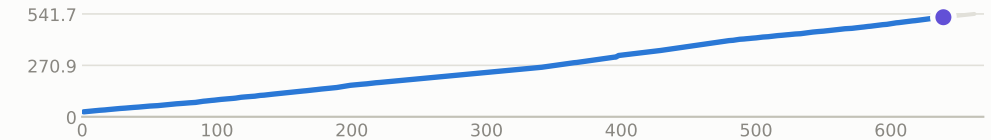
process area	Dry_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.801 kgCO2e**

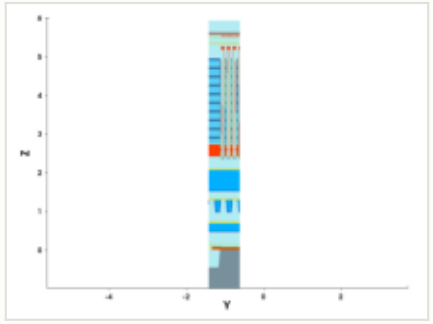
carbon so far, of the whole flow (kgCO2e)



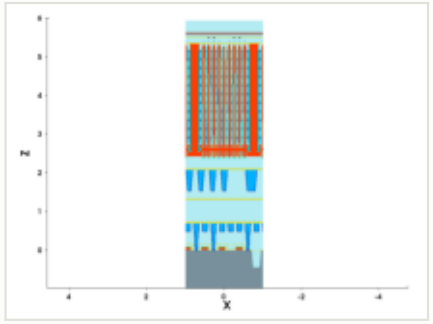
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



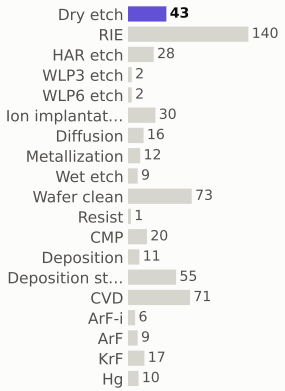
y cut at -1.032 µm



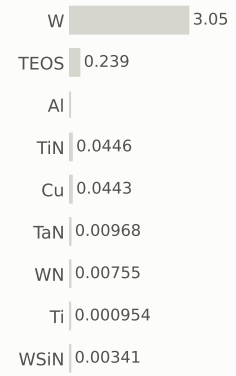
mask for this step

no mask for this step

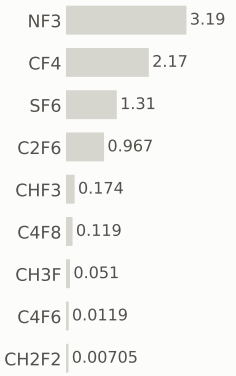
steps by process area



materials deposited (g)



gases emitted (g)



Step #6070: Wafer_clean

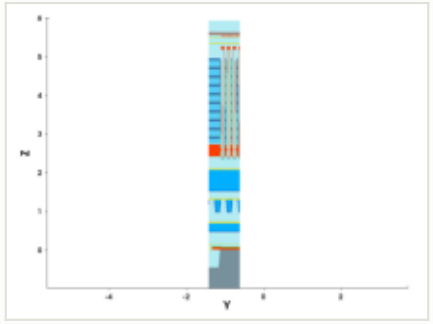
clean · Wafer_clean

0.169	524.6	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

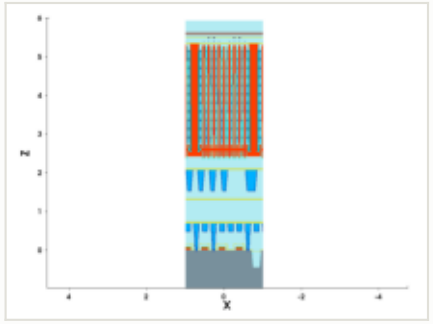
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

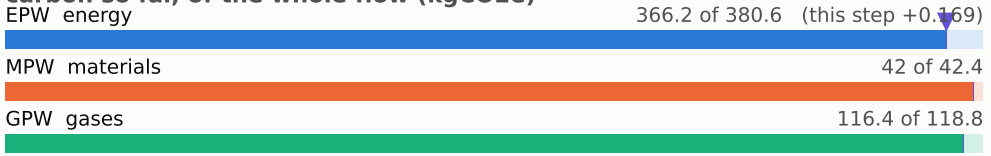
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

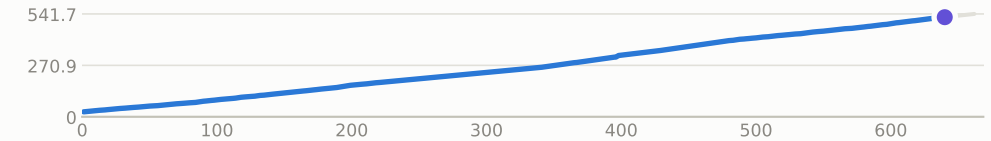
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

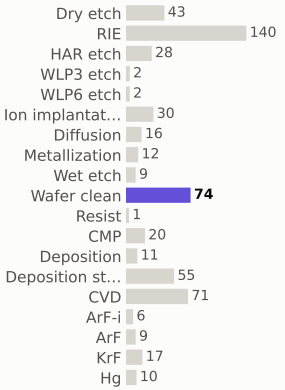
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

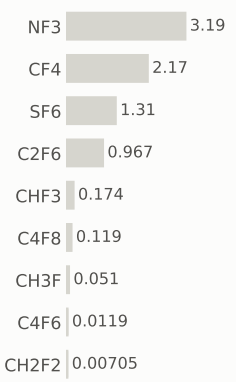


steps by process area



materials deposited (g)

gases emitted (g)



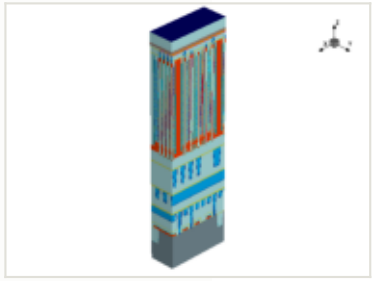
Step #6080: CVD

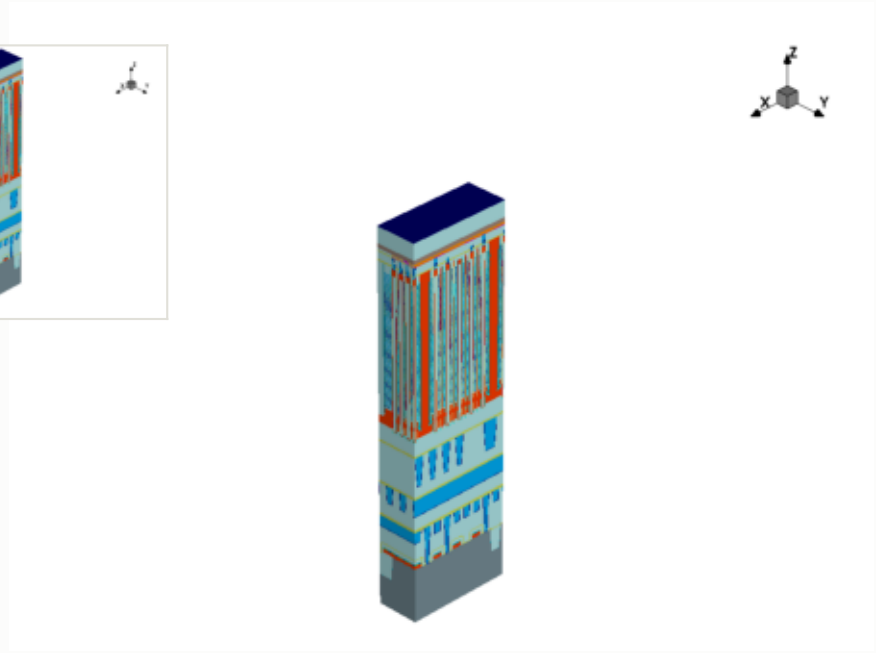
depo_isotropic · CVD

1.31	525.9	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

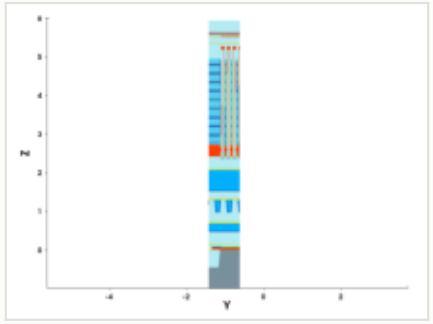
3D structure after this step

zoomed to fit

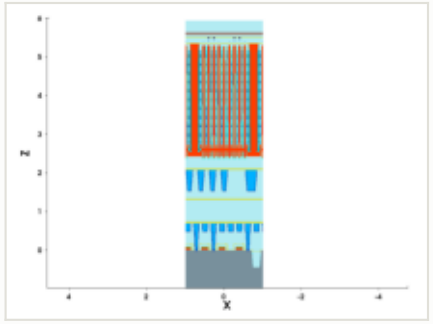




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

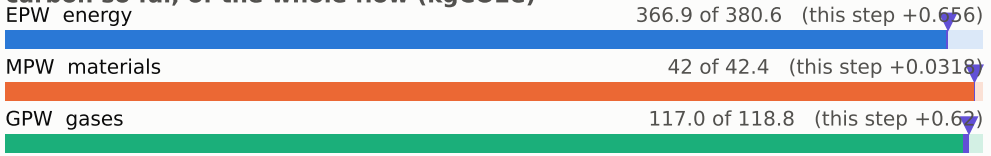
no mask for this step

this step

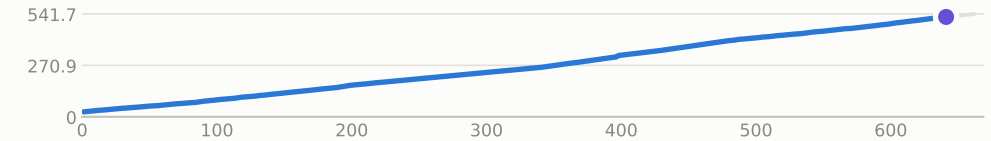
process area	CVD
material	TiN
thickness	0.005 µm
material GWP	43.9778 gCO2e/g
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

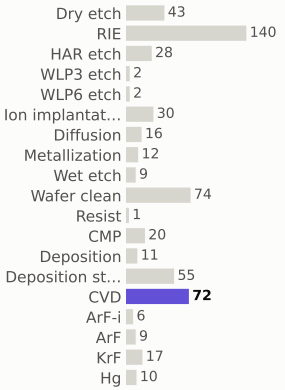
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

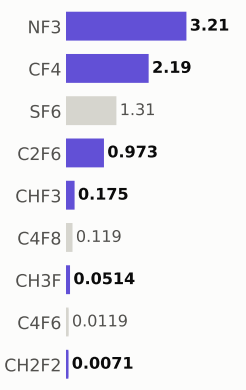


steps by process area



materials deposited (g)

gases emitted (g)



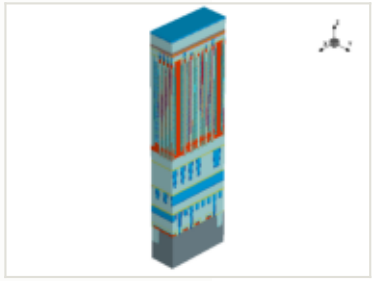
Step #6090: Metallization

depo_isotropic · Metallization

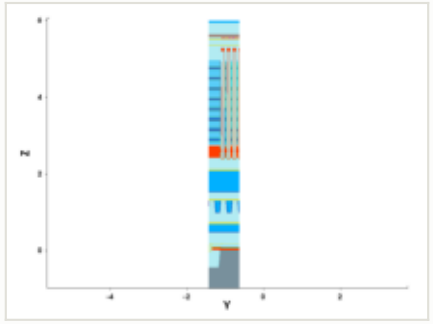
1.21	527.1	0.22%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

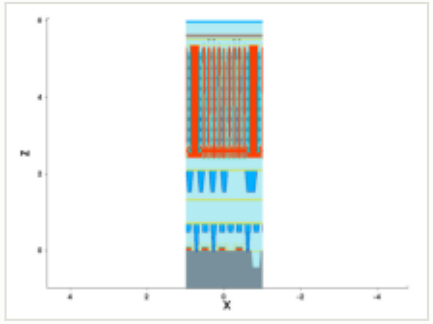
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

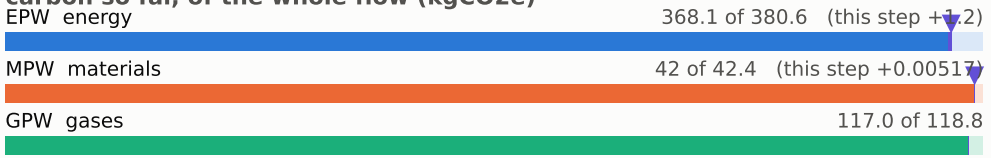
no mask for this step

this step

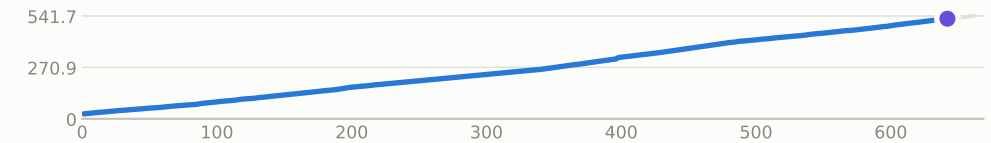
process area	Metallization
material	W
thickness	0.05 µm
material GWP	75.8149 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.00517 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

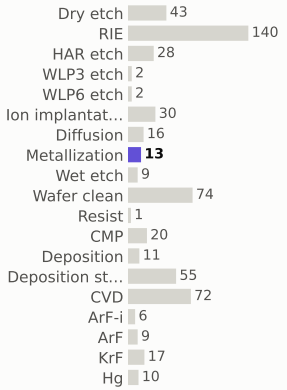
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

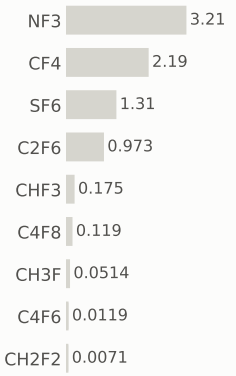


steps by process area



materials deposited (g)

gases emitted (g)



Step #6100: CMP

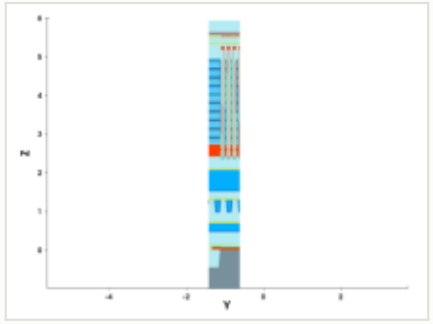
etch_cmp · CMP

0.169	527.3	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

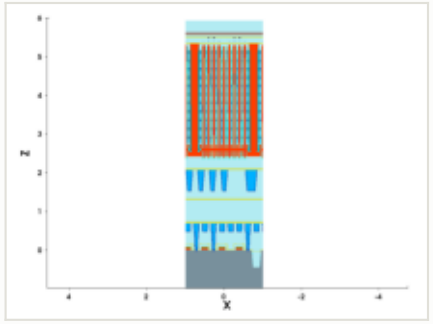
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

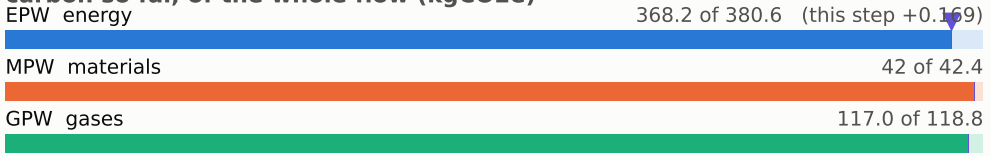
no mask for this step

this step

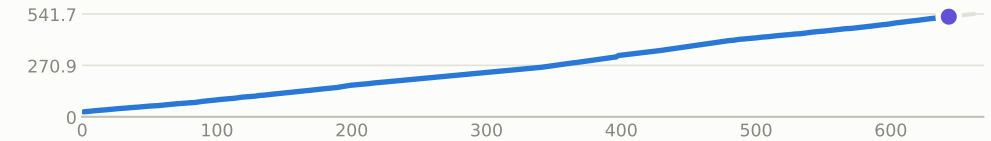
process area	CMP
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

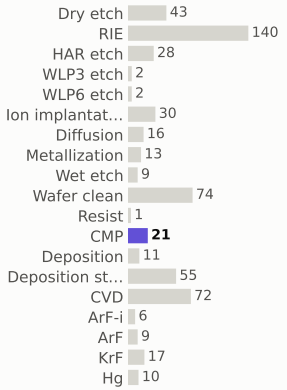
carbon so far, of the whole flow (kgCO2e)



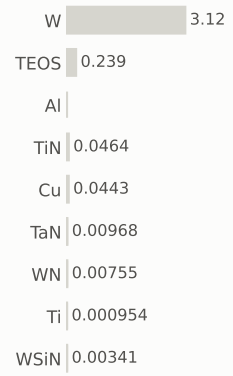
embodied carbon up to each step (kgCO2e per wafer)



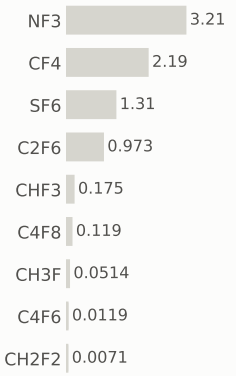
steps by process area



materials deposited (g)



gases emitted (g)



Step #6110: Wafer_clean

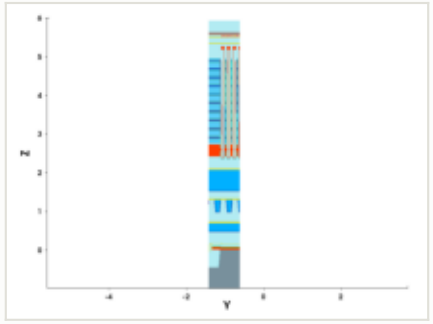
clean · Wafer_clean

0.169	527.4	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

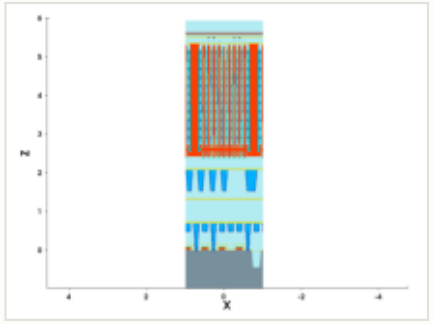
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

no mask for this step

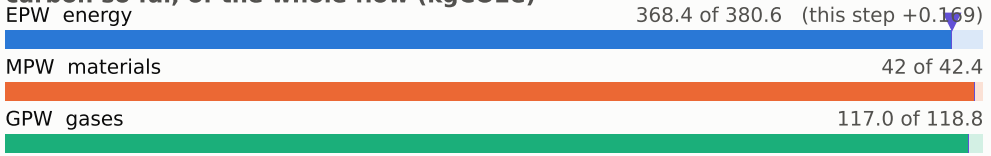
this step

process area	Wafer_clean
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

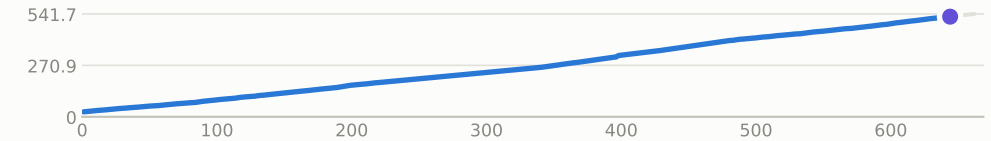
EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

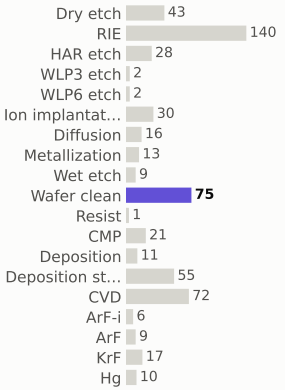
carbon so far, of the whole flow (kgCO2e)



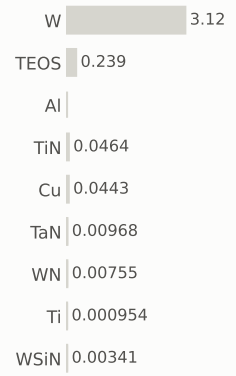
embodied carbon up to each step (kgCO2e per wafer)



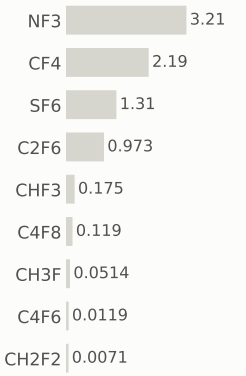
steps by process area



materials deposited (g)



gases emitted (g)



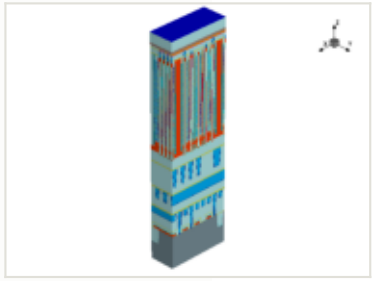
Step #6130: Metallization

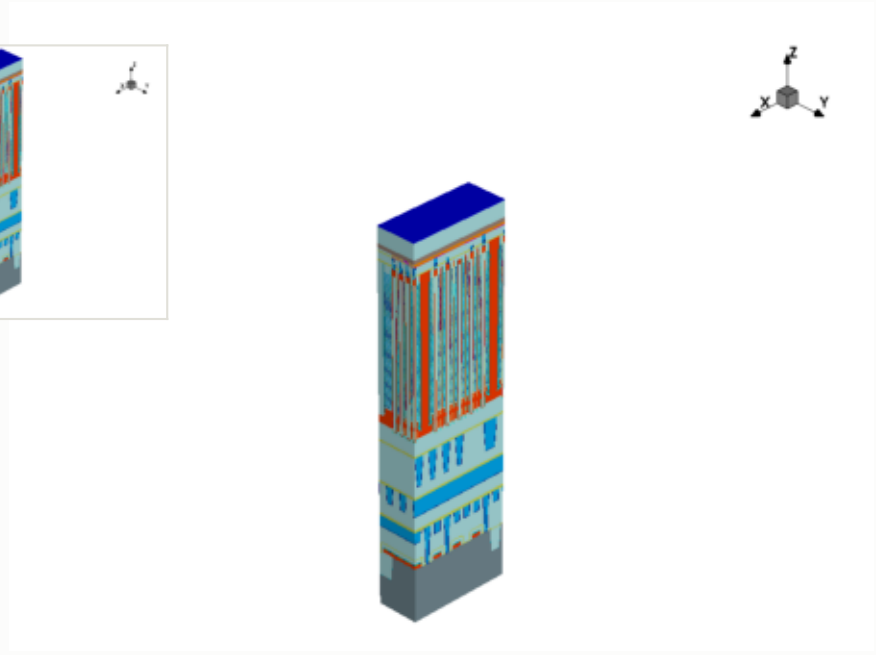
depo_isotropic · Metallization

1.2	528.6	0.22%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

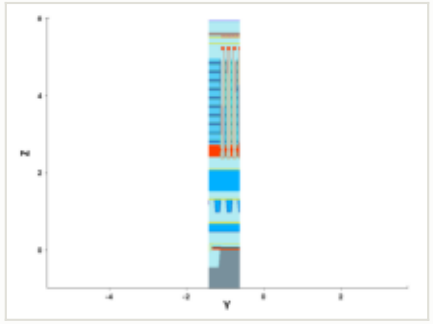
3D structure after this step

zoomed to fit

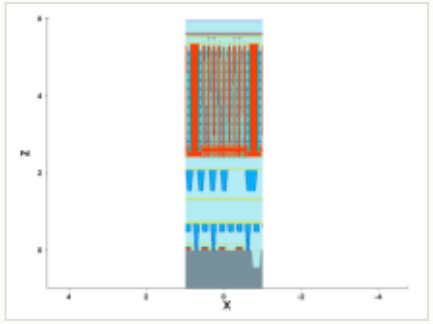




x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

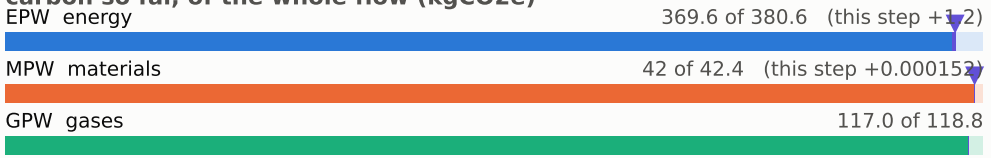
no mask for this step

this step

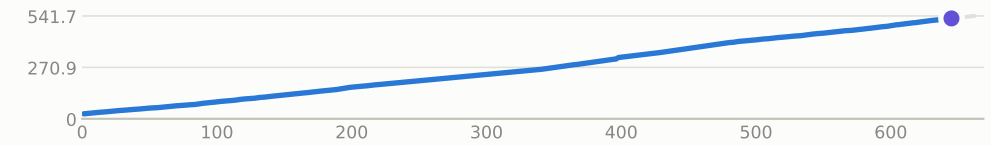
process area	Metallization
material	Ti
thickness	0.01 µm
material GWP	47.6334 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.000152 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

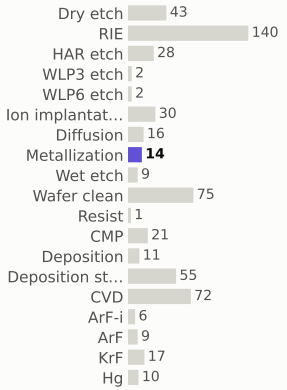
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

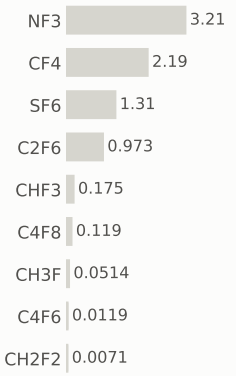


steps by process area



materials deposited (g)

gases emitted (g)



Step #6140: Metallization

depo_isotropic · Metallization

1.21	529.8	0.22%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

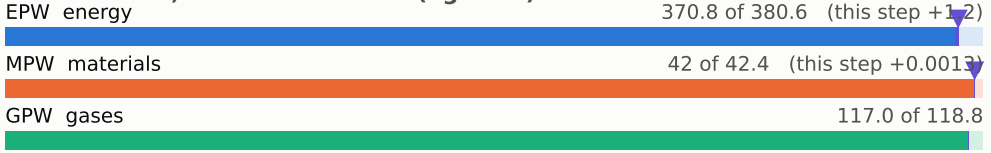
zoomed to fit

this step

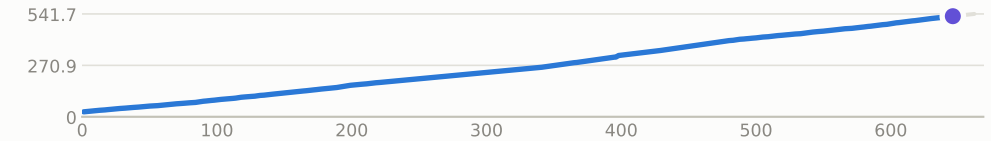
process area	Metallization
material	Al
thickness	0.635 µm
material GWP	10.7269 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	0.0013 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.21 kgCO2e**

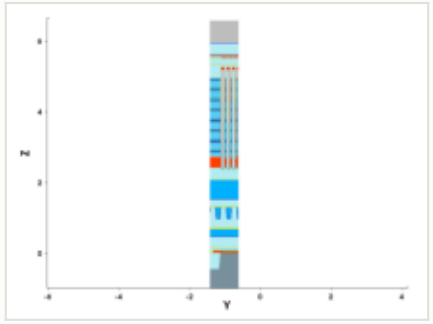
carbon so far, of the whole flow (kgCO2e)



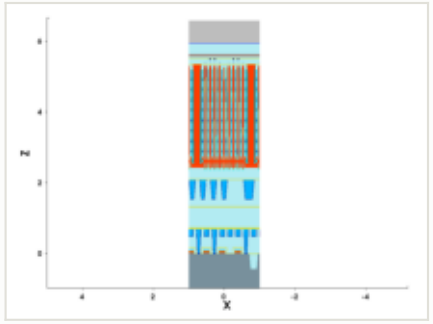
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



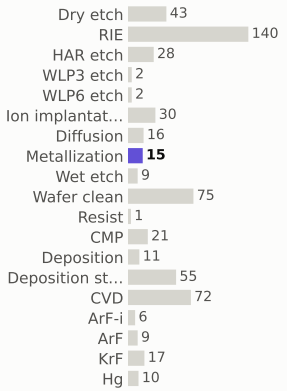
y cut at -1.032 µm



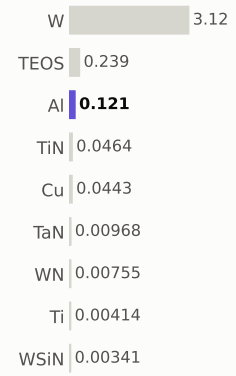
mask for this step

no mask for this step

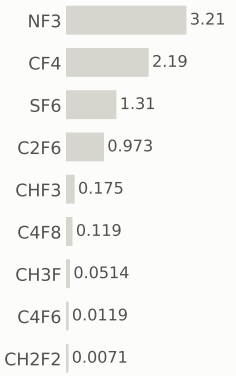
steps by process area



materials deposited (g)



gases emitted (g)



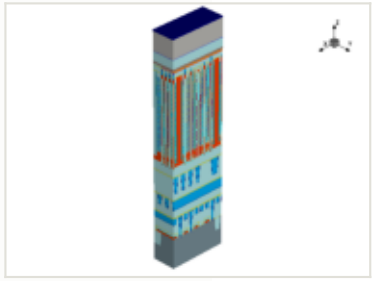
Step #6150: Deposition

depo_isotropic · Deposition

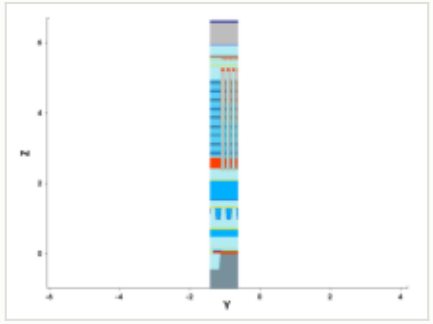
0.657	530.5	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

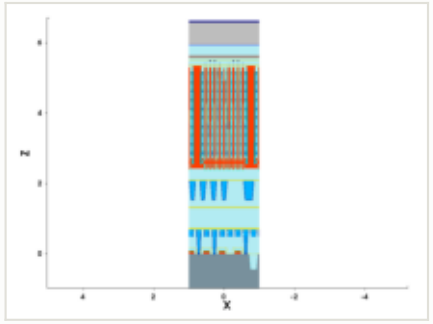
zoomed to fit



x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

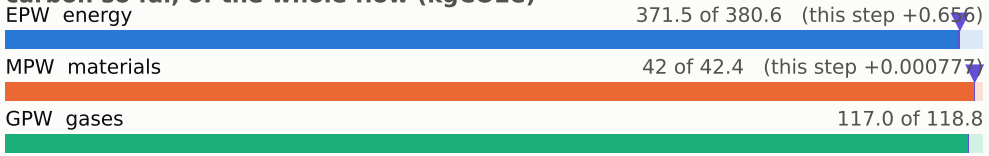
no mask for this step

this step

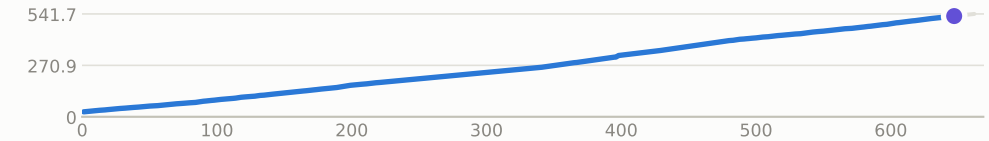
process area	Deposition
material	TiN
thickness	0.048 µm
material GWP	43.9778 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.000777 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.657 kgCO2e**

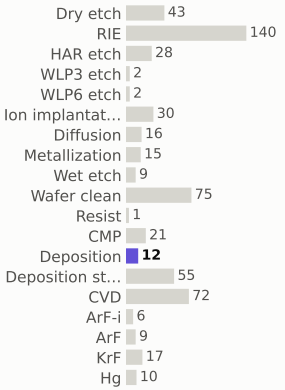
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

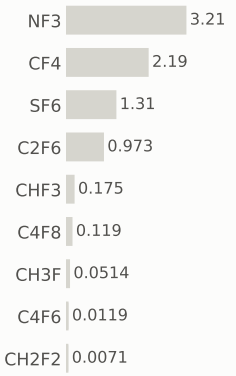


steps by process area



materials deposited (g)

gases emitted (g)



Step #6160: Deposition

depo_isotropic · Deposition

0.656	531.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

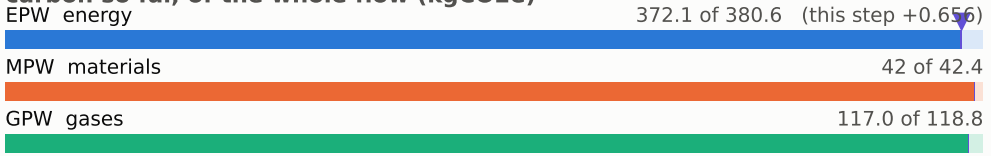
3D structure after this step

zoomed to fit

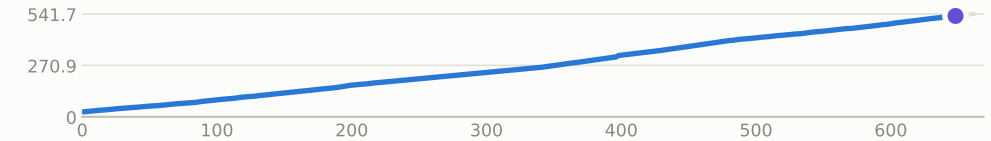
this step

process area	Deposition	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.007 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**0.656 kgCO2e**

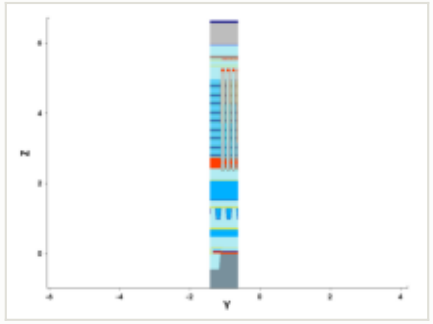
carbon so far, of the whole flow (kgCO2e)



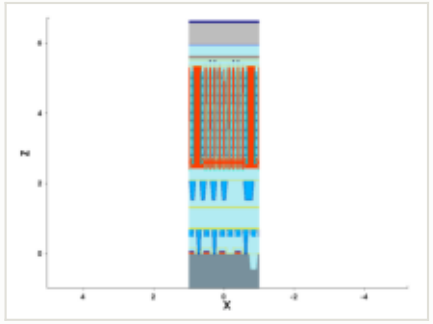
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



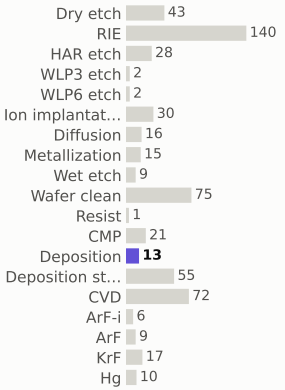
y cut at -1.032 µm



mask for this step

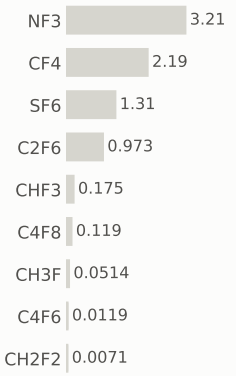
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



Step #6170: KrF

litho_patterning · KrF

0.199	531.4	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

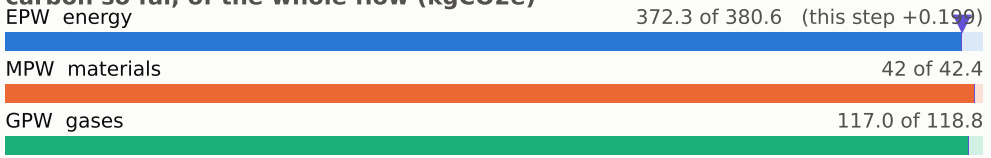
zoomed to fit

this step

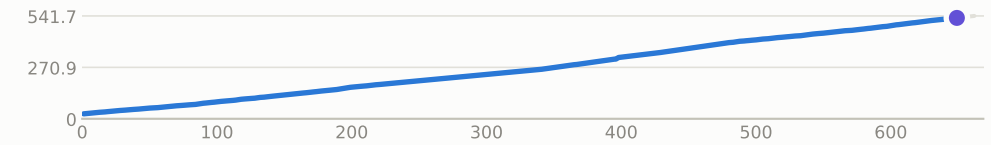
process area	KrF
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

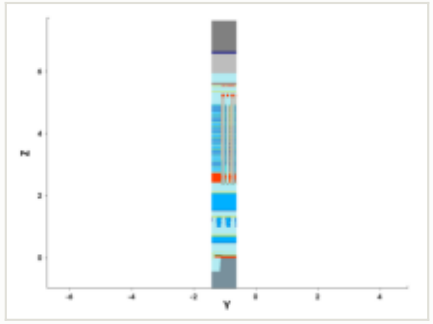
carbon so far, of the whole flow (kgCO2e)



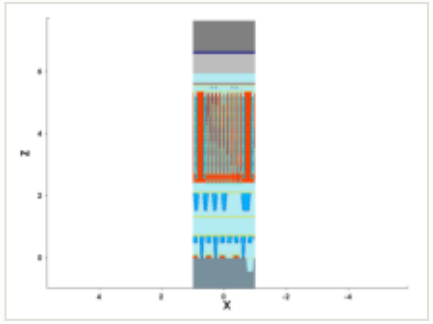
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

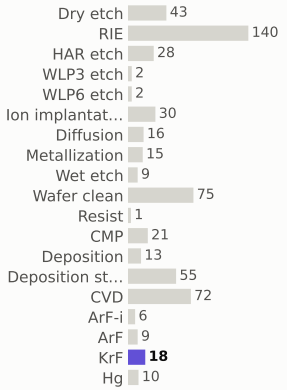


y cut at -1.032 µm

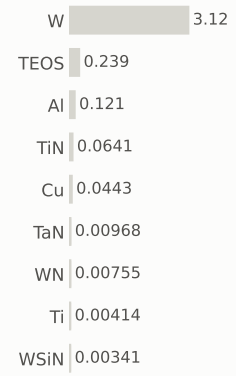


mask: layer 46:0, positive (OASIS)

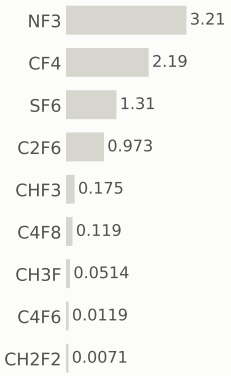
steps by process area



materials deposited (g)



gases emitted (g)



Step #6180: RIE

etch_anisotropic · RIE

1.04	532.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

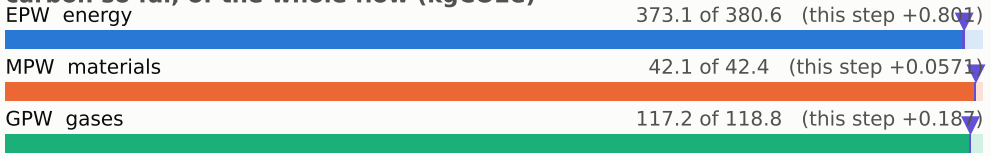
zoomed to fit

this step

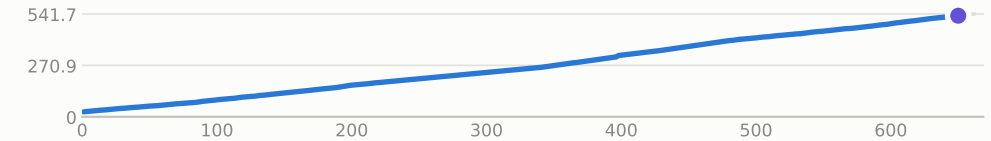
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

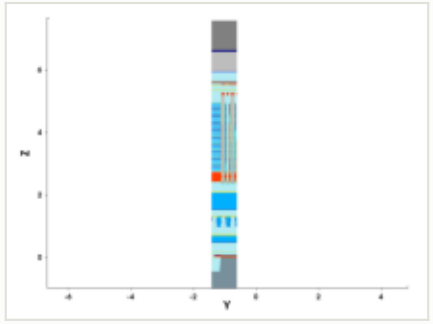
carbon so far, of the whole flow (kgCO2e)



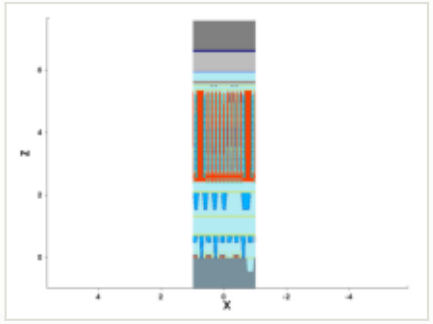
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



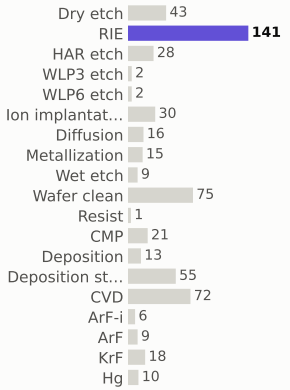
y cut at -1.032 µm



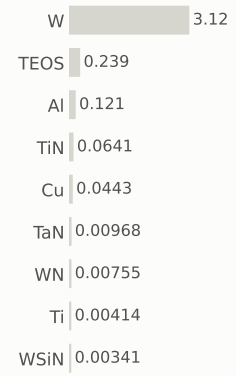
mask for this step

no mask for this step

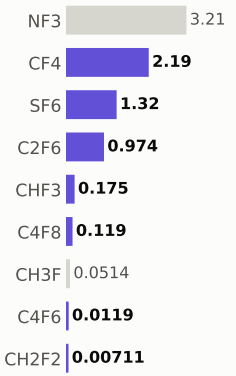
steps by process area



materials deposited (g)



gases emitted (g)



Step #6190: RIE

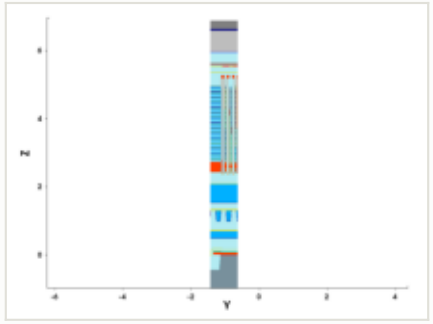
etch_anisotropic · RIE

1.04	533.4	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

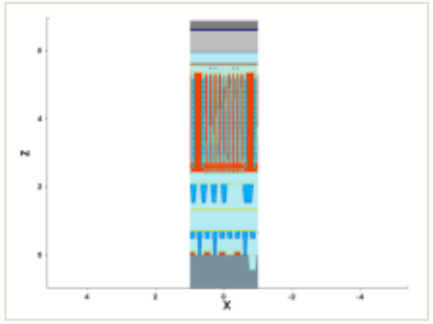
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

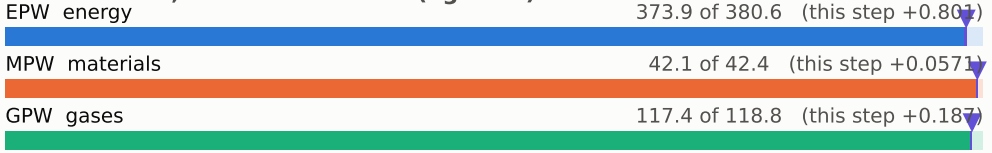
no mask for this step

this step

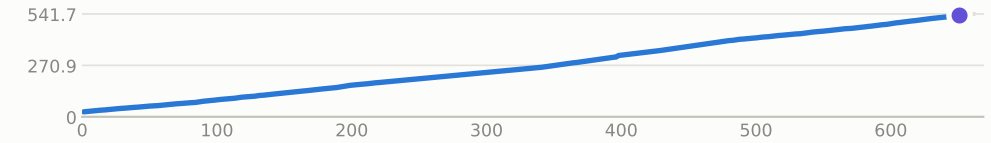
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

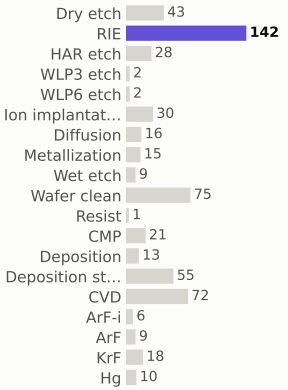
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)

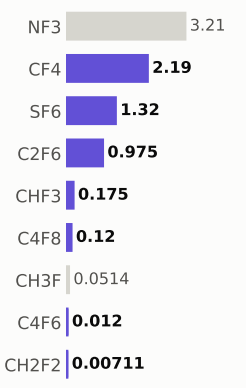


steps by process area



materials deposited (g)

gases emitted (g)



Step #6200: RIE

etch_anisotropic · RIE

1.04	534.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

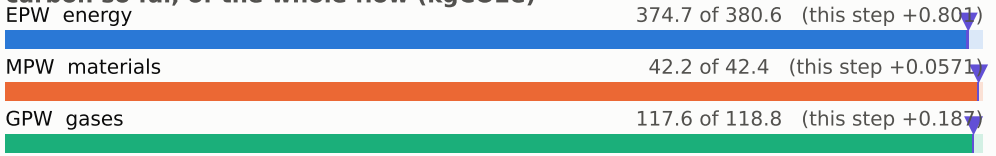
zoomed to fit

this step

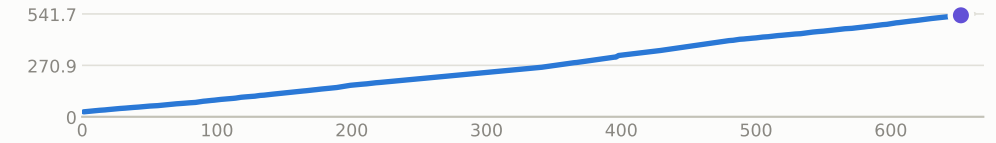
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

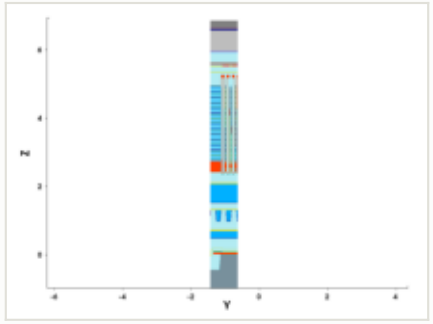
carbon so far, of the whole flow (kgCO2e)



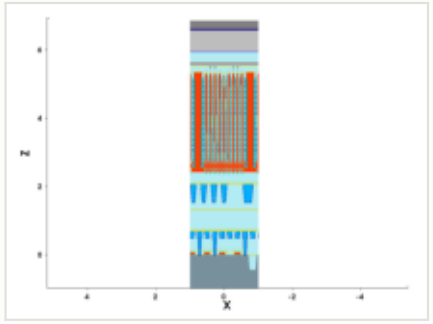
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



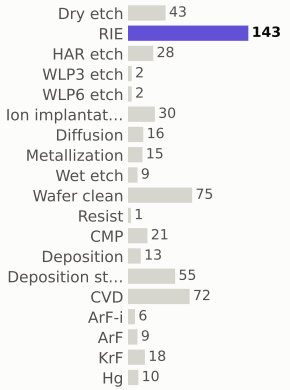
y cut at -1.032 µm



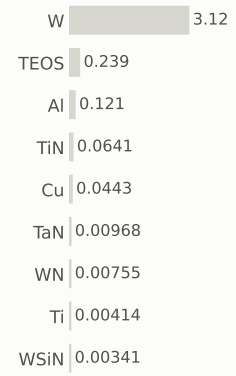
mask for this step

no mask for this step

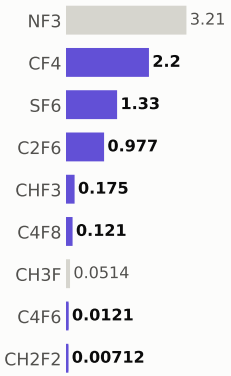
steps by process area



materials deposited (g)



gases emitted (g)



Step #6210: RIE

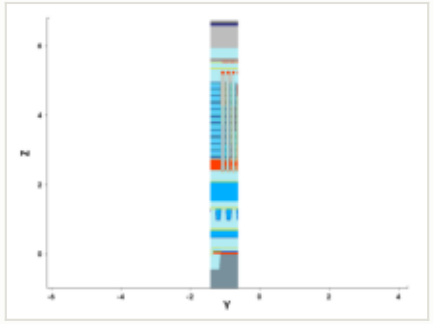
etch_anisotropic · RIE

1.04	535.5	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

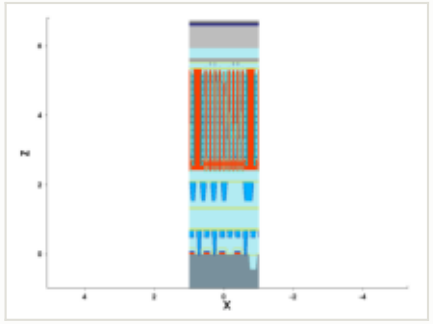
3D structure after this step

zoomed to fit

x cut at 0.000 µm



y cut at -1.032 µm



mask for this step

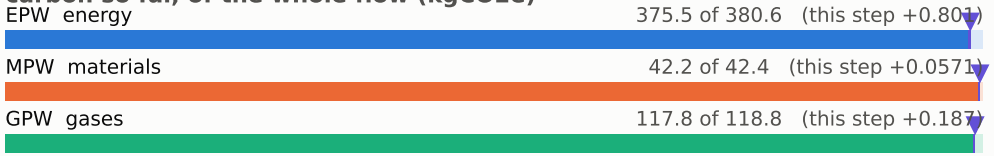
no mask for this step

this step

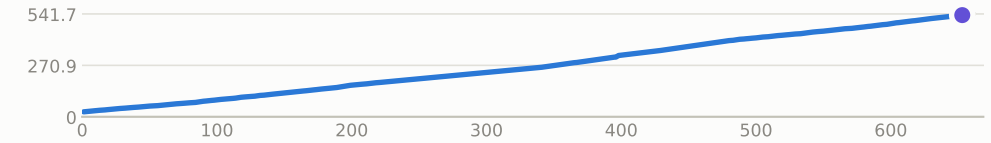
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

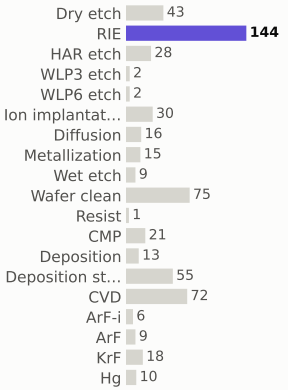
carbon so far, of the whole flow (kgCO2e)



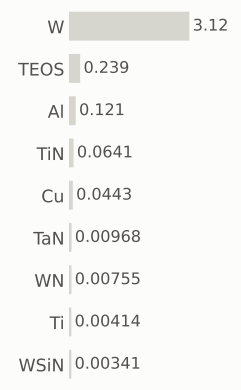
embodied carbon up to each step (kgCO2e per wafer)



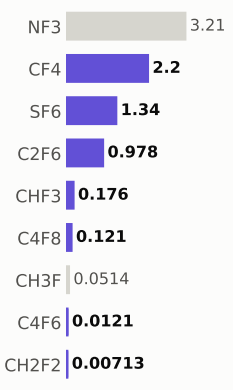
steps by process area



materials deposited (g)



gases emitted (g)



Step #6220: Dry_etch

etch_strip · Dry_etch

0.801	536.3	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

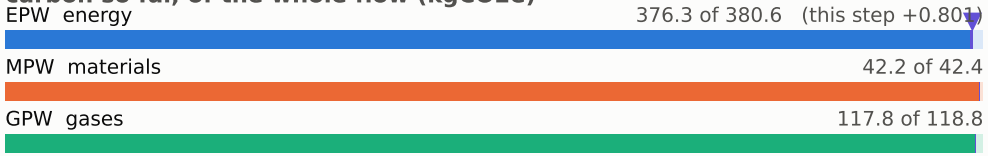
3D structure after this step

zoomed to fit

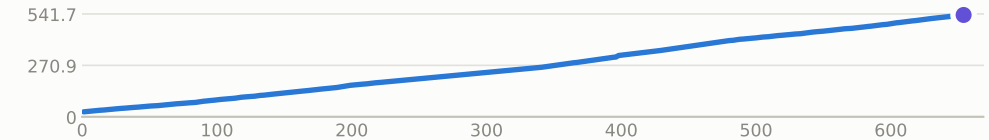
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

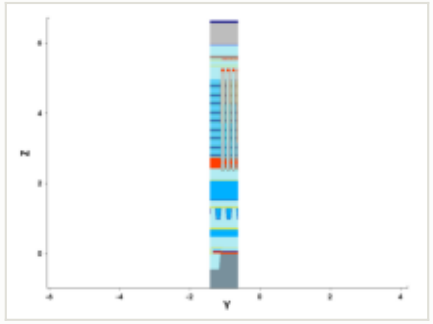
carbon so far, of the whole flow (kgCO2e)



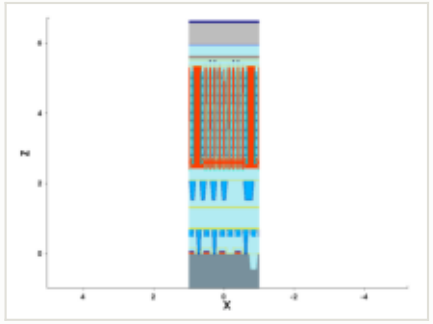
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



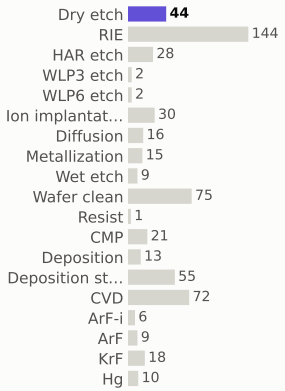
y cut at -1.032 µm



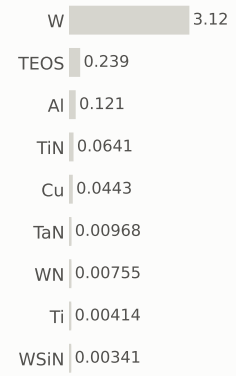
mask for this step

no mask for this step

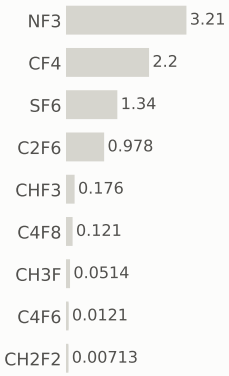
steps by process area



materials deposited (g)



gases emitted (g)



Step #6230: Wafer_clean

clean · Wafer_clean

0.169	536.5	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

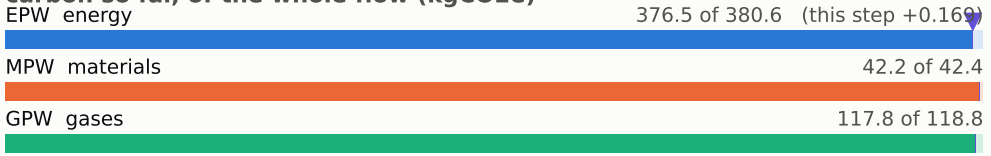
zoomed to fit

this step

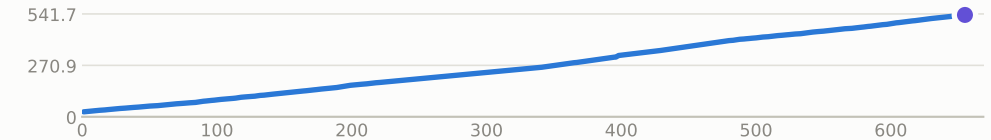
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

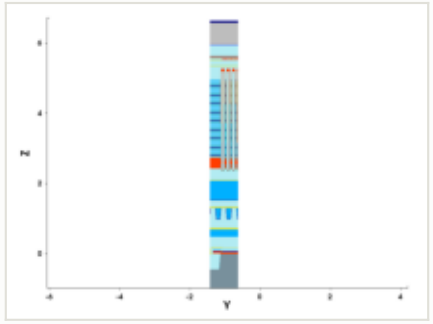
carbon so far, of the whole flow (kgCO2e)



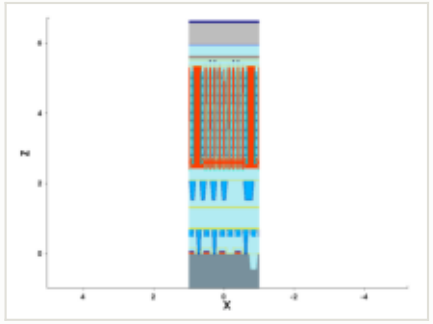
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



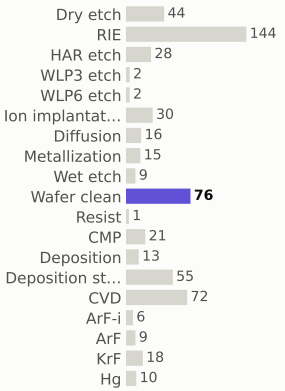
y cut at -1.032 µm



mask for this step

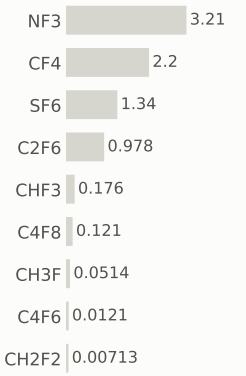
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



Step #6250: Deposition

depo_isotropic · Deposition

0.656	537.2	0.12%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

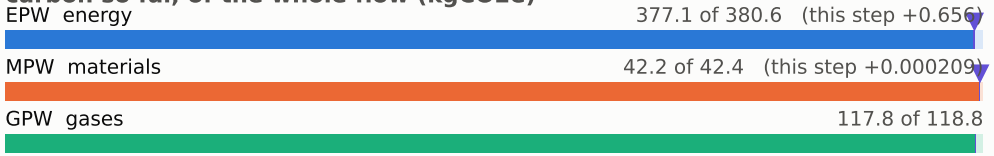
zoomed to fit

this step

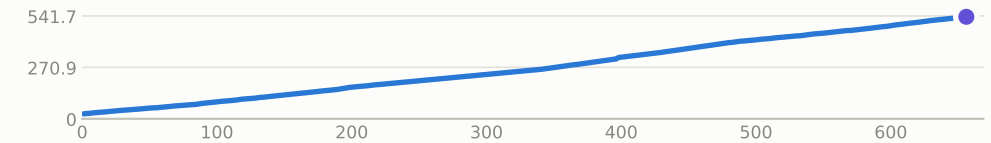
process area	Deposition
material	TEOS
thickness	1.1 µm
material GWP	2.85952 gCO2e/g
input gases	—
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.000209 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.656 kgCO2e**

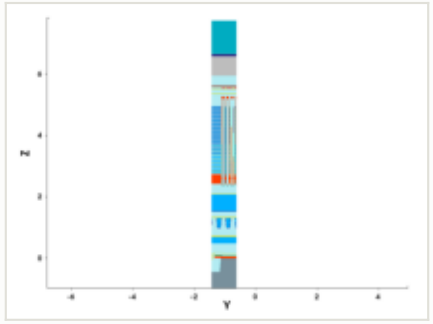
carbon so far, of the whole flow (kgCO2e)



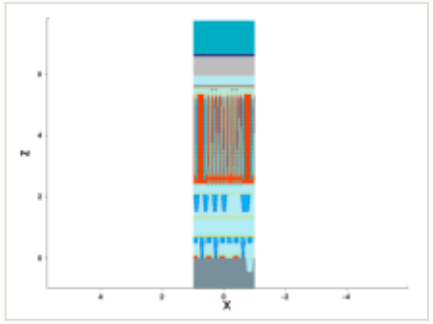
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



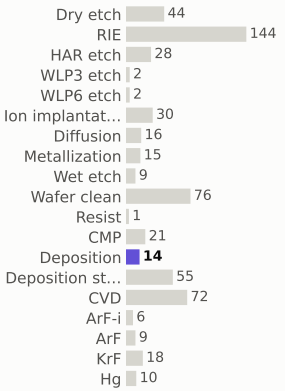
y cut at -1.032 µm



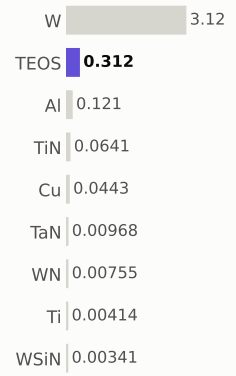
mask for this step

no mask for this step

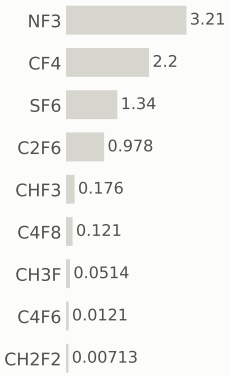
steps by process area



materials deposited (g)



gases emitted (g)



Step #6260: CVD

depo_isotropic · CVD

1.31	538.5	0.24%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

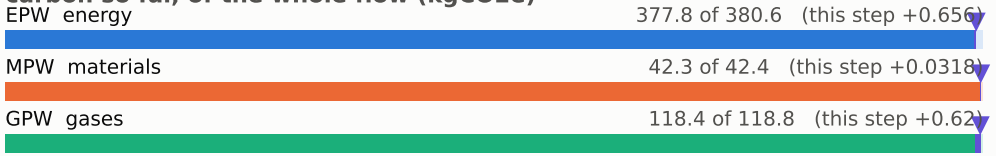
zoomed to fit

this step

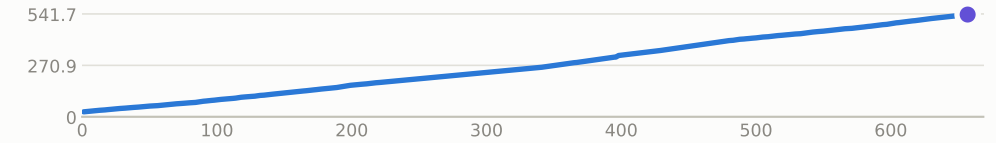
process area	CVD
material	—
thickness	0.3 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

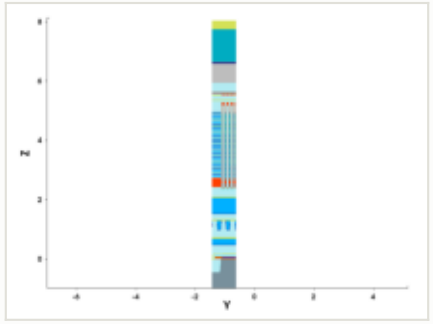
carbon so far, of the whole flow (kgCO2e)



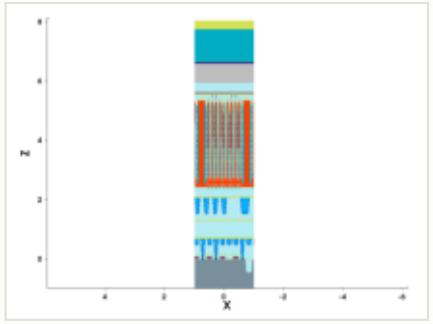
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



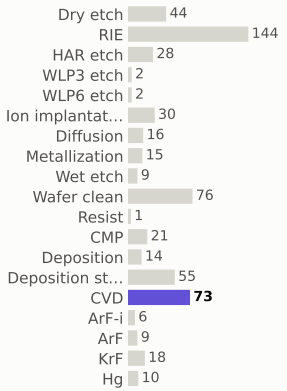
y cut at -1.032 µm



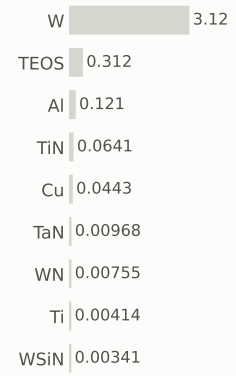
mask for this step

no mask for this step

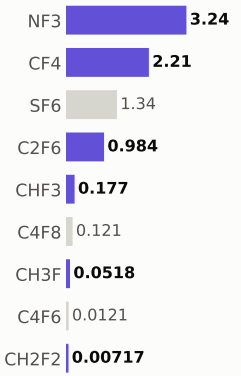
steps by process area



materials deposited (g)



gases emitted (g)



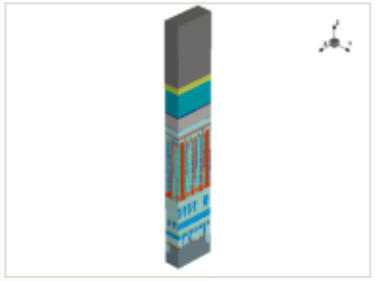
Step #6270: Hg

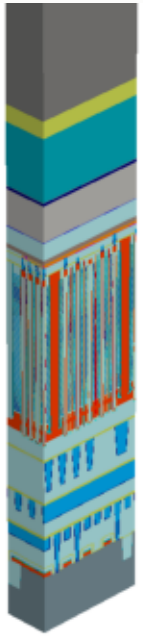
litho_patterning · Hg

0.199	538.7	0.037%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



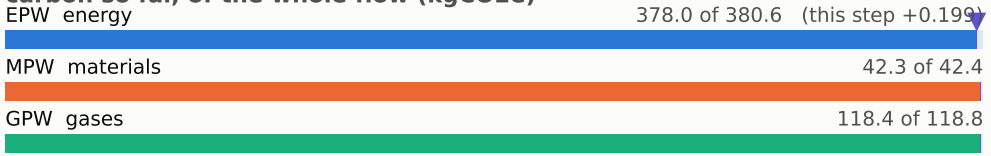


this step

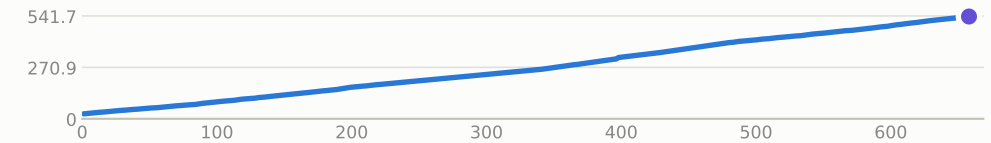
process area	Hg
material	—
thickness	—
material GWP	—
input gases	—
energy	0.33 kWh + 0.132 overhead

EPW	0.199 kgCO2e
EPW tool	0.142 kgCO2e
EPW overhead	0.0568 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.199 kgCO2e**

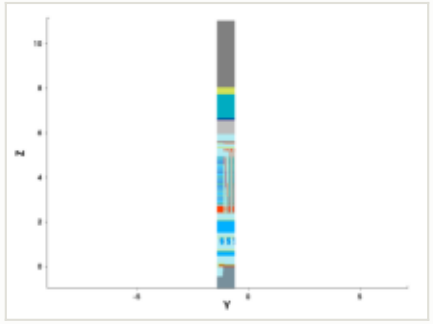
carbon so far, of the whole flow (kgCO2e)



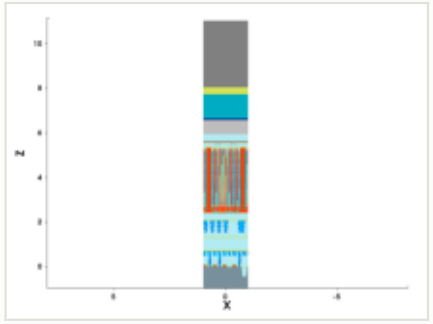
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm

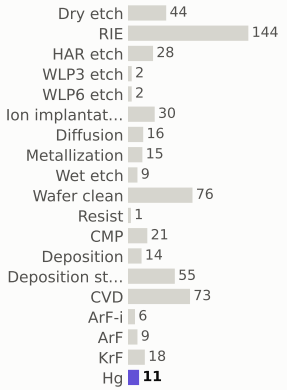


y cut at -1.032 µm

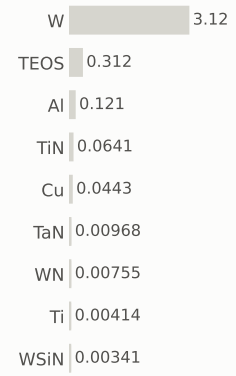


mask: layer 47:0, positive (OASIS)

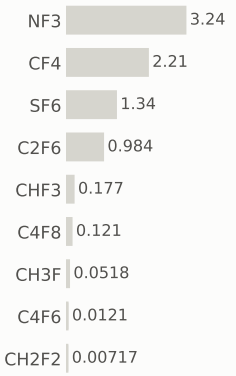
steps by process area



materials deposited (g)



gases emitted (g)



Step #6280: RIE

etch_anisotropic · RIE

1.04	539.7	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

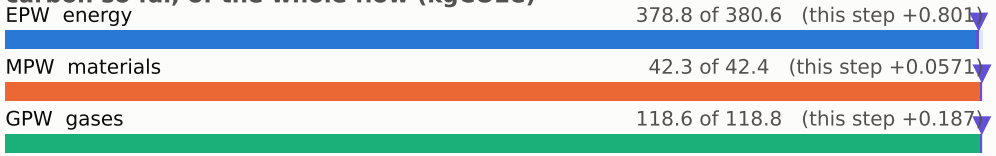
zoomed to fit

this step

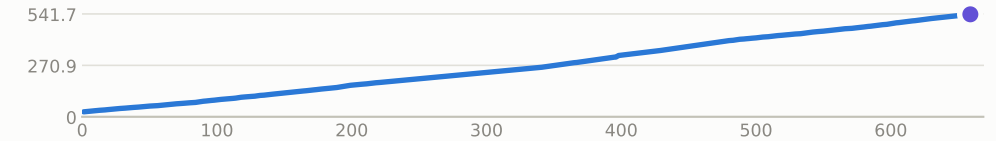
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

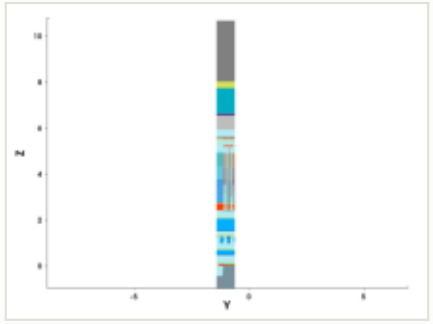
carbon so far, of the whole flow (kgCO2e)



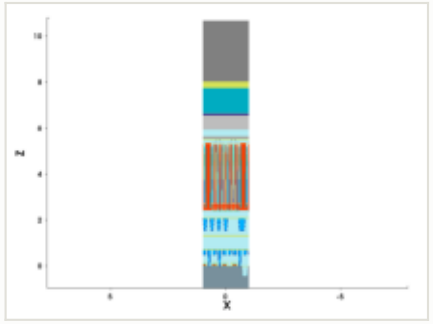
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



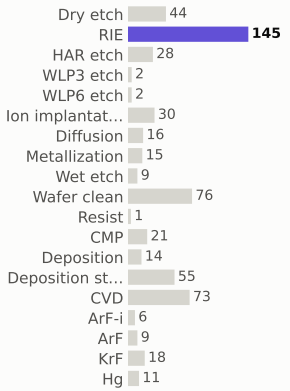
y cut at -1.032 µm



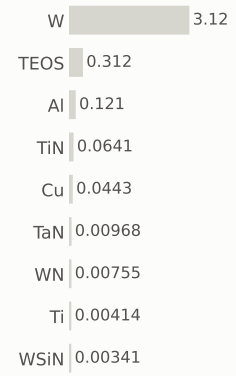
mask for this step

no mask for this step

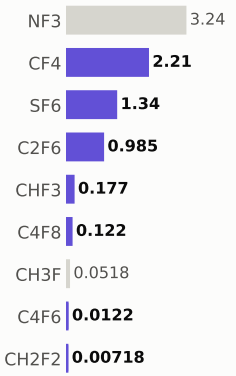
steps by process area



materials deposited (g)



gases emitted (g)



Step #6290: RIE

etch_anisotropic · RIE

1.04	540.8	0.19%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

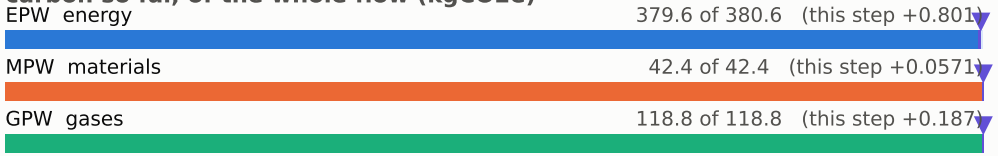
zoomed to fit

this step

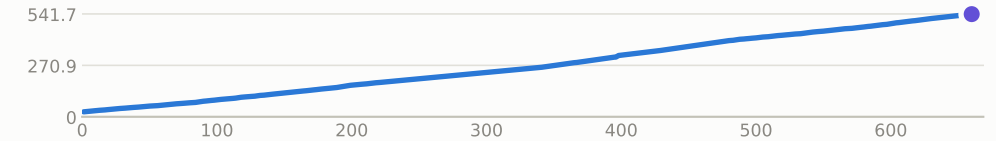
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

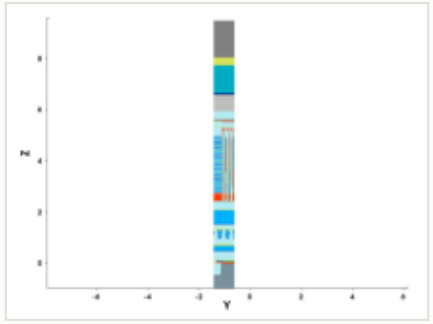
carbon so far, of the whole flow (kgCO2e)



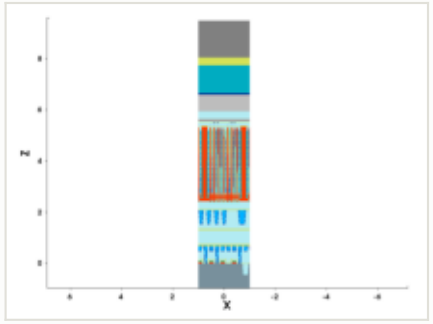
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



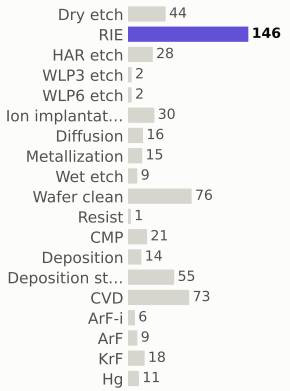
y cut at -1.032 µm



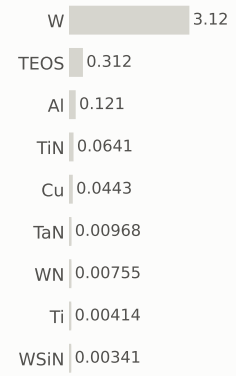
mask for this step

no mask for this step

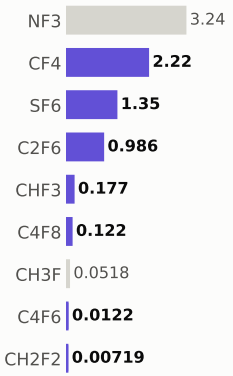
steps by process area



materials deposited (g)



gases emitted (g)



Step #6300: Dry_etch

etch_strip · Dry_etch

0.801	541.6	0.15%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

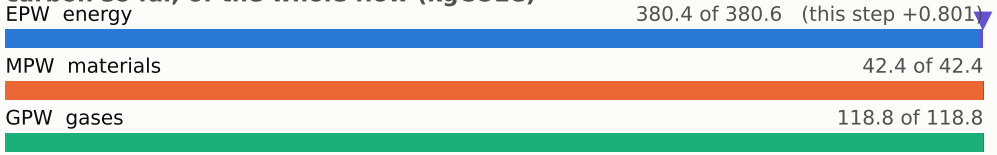
3D structure after this step

zoomed to fit

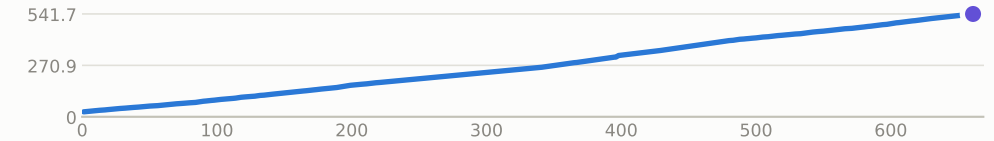
this step

process area	Dry_etch	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**0.801 kgCO2e**

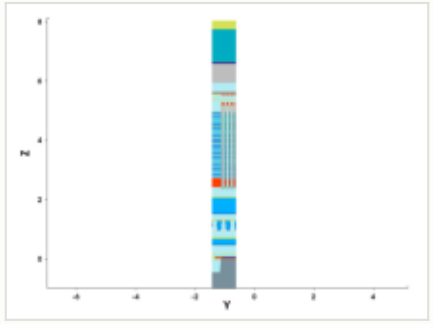
carbon so far, of the whole flow (kgCO2e)



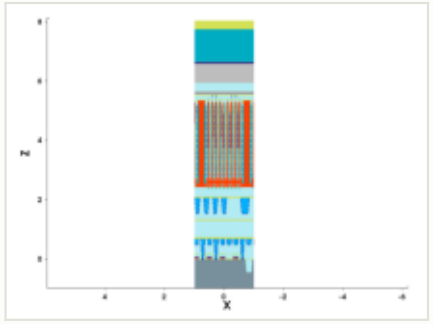
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



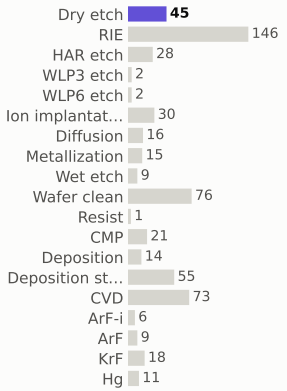
y cut at -1.032 µm



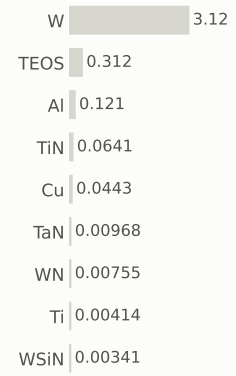
mask for this step

no mask for this step

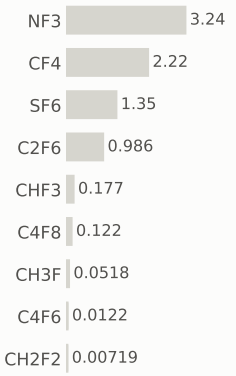
steps by process area



materials deposited (g)



gases emitted (g)



Step #6310: Wafer_clean

clean · Wafer_clean

0.169	541.7	0.031%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

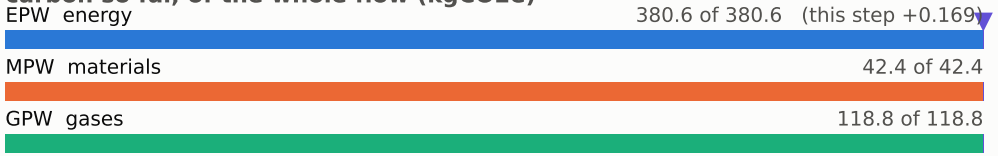
zoomed to fit

this step

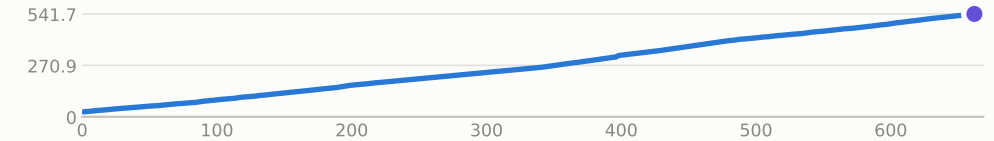
process area	Wafer_clean	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

carbon-only step: no spx operation (structure unchanged)

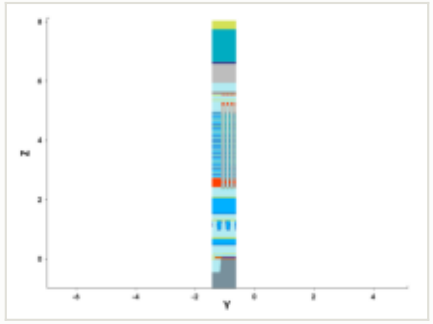
carbon so far, of the whole flow (kgCO2e)



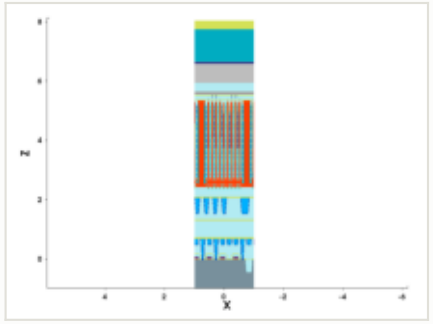
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.000 µm



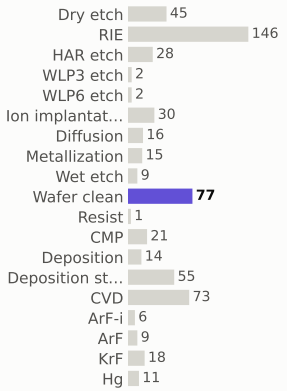
y cut at -1.032 µm



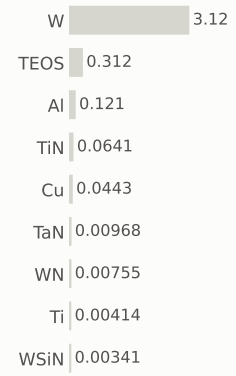
mask for this step

no mask for this step

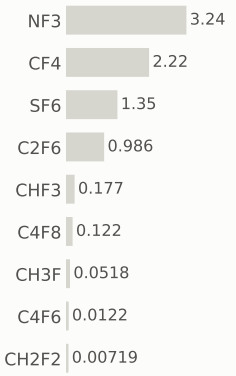
steps by process area



materials deposited (g)

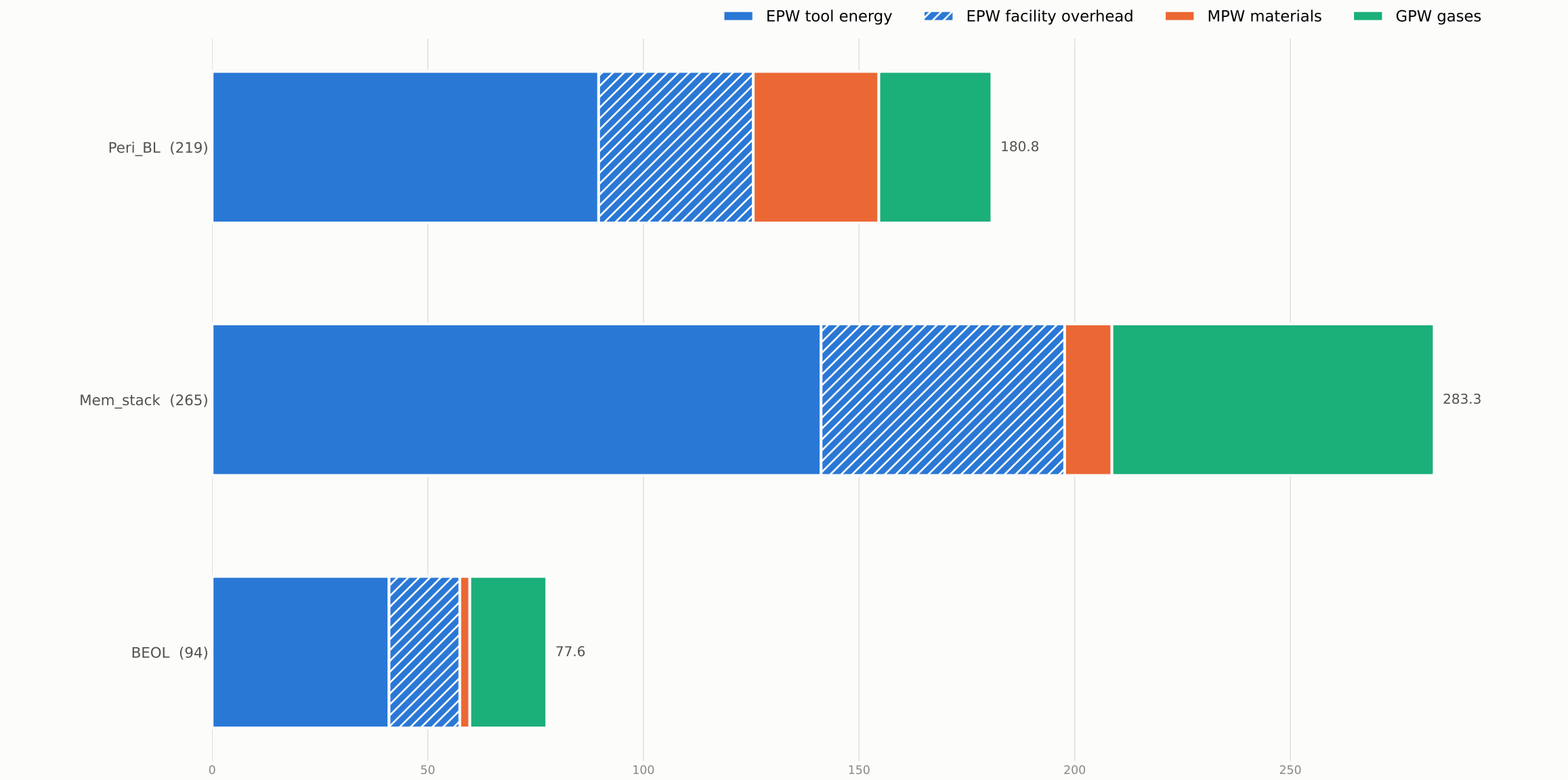


gases emitted (g)



Carbon by module

kgCO2e per wafer, in flow order (steps in brackets)



Totals by process area

kgCO2e per wafer; EPW = tool + facility overhead (overhead energy = tool energy x 0.4)

process area	steps	tool kWh	overhead kWh	EPW tool	EPW overhead	EPW	MPW	GPW	total
RIE	146	194.18	77.67	83.50	33.40	116.90	8.33	27.31	152.54
CVD	73	79.57	31.83	34.22	13.69	47.90	2.32	45.29	95.51
Deposition_stack	55	59.41	23.76	25.54	10.22	35.76	1.73	33.81	71.31
HAR_etch	28	74.48	29.79	32.03	12.81	44.84	3.20	10.47	58.51
Dry_etch	45	59.85	23.94	25.74	10.29	36.03	0.00	0.00	36.03
Ion_implantation	30	39.90	15.96	17.16	6.86	24.02	0.00	0.00	24.02
Metallization	15	30.00	12.00	12.90	5.16	18.06	0.24	0.00	18.30
Wafer_clean	77	21.56	8.62	9.27	3.71	12.98	0.00	0.00	12.98
Diffusion	16	21.28	8.51	9.15	3.66	12.81	0.00	0.00	12.81
Deposition	14	15.26	6.10	6.56	2.62	9.19	0.00	0.00	9.19
WLP3_etch	2	10.66	4.26	4.58	1.83	6.42	0.46	1.50	8.37
KrF	18	5.94	2.38	2.55	1.02	3.58	0.00	0.00	3.58
CMP	21	5.88	2.35	2.53	1.01	3.54	0.00	0.00	3.54
Hg	11	3.63	1.45	1.56	0.62	2.19	0.00	0.00	2.19
WLP6_etch	2	2.66	1.06	1.14	0.46	1.60	0.11	0.37	2.09
ArF	9	2.97	1.19	1.28	0.51	1.79	0.00	0.00	1.79
Wet_etch	9	2.52	1.01	1.08	0.43	1.52	0.00	0.00	1.52
ArF-i	6	2.16	0.86	0.93	0.37	1.30	0.00	0.00	1.30
Resist	1	0.28	0.11	0.12	0.05	0.17	0.00	0.00	0.17
Wafer		0.00	0.00	0.00	0.00	0.00	26.00	0.00	26.00
all	578	632.18	252.87	271.84	108.74	380.58	42.39	118.76	541.73